

3. Code fragments with comments

Ada (body)

driverspace_workspace/formal/ds-spark-core/src/ds_buddy_math.adb (line 6) - [internal]

```
Power_Of_Two

function Power_Of_Two (Exponent : Block_Size) return Phys_Addr is
begin
    return Phys_Addr (Shift_Left (1, Integer (Exponent)));
end Power_Of_Two;

function Is_Aligned (Addr : Phys_Addr; Size_Exp : Block_Size) return Boolean is
    Mask : constant Phys_Addr := Power_Of_Two (Size_Exp) - 1;
begin
    -- Udowodnienie, że (Addr mod 2^N) == 0 jest równoważne (Addr and (2^N - 1)) == 0
    return (Addr and Mask) = 0;
end Is_Aligned;

// ... (truncated, see source file for more)
```

driverspace_workspace/formal/ds-spark-core/src/ds_buddy_math.adb (line 11) - [internal]

```
Is_Aligned

function Is_Aligned (Addr : Phys_Addr; Size_Exp : Block_Size) return Boolean is
    Mask : constant Phys_Addr := Power_Of_Two (Size_Exp) - 1;
begin
    -- Udowodnienie, że (Addr mod 2^N) == 0 jest równoważne (Addr and (2^N - 1)) == 0
    return (Addr and Mask) = 0;
end Is_Aligned;

function Get_Buddy_Address (
    Addr      : Phys_Addr;
    Size_Exp  : Block_Size
) return Phys_Addr is
    -- Kluczowy bit maski dla alokatora Buddy
    // ... (truncated, see source file for more)
```

driverspace_workspace/formal/ds-spark-core/src/ds_buddy_math.adb (line 18) - [internal]

```
Get_Buddy_Address

function Get_Buddy_Address (
    Addr      : Phys_Addr;
    Size_Exp  : Block_Size
) return Phys_Addr is
    -- Kluczowy bit maski dla alokatora Buddy
    Bit_Mask : constant Phys_Addr := Power_Of_Two (Size_Exp);
begin
    -- XOR z maską odwraca dokładnie ten jeden bit, który definiuje pozycję buddy.
    -- SPARK udowodni, że ponieważ Addr jest wyrównane (Pre condition z .ads),
    -- to wynik również będzie wyrównany, a XOR nie spowoduje przepełnienia.
    return Addr xor Bit_Mask;
end Get_Buddy_Address;
// ... (truncated, see source file for more)
```

driverspace_workspace/formal/ds-spark-core/src/ds_buddy_math.adb (line 31) - [internal]

```
Get_Left_Child_Address

function Get_Left_Child_Address (
    Parent_Addr : Phys_Addr;
    Parent_Exp  : Block_Size
) return Phys_Addr is
begin
    -- Lewe dziecko zawsze zaczyna się tam, gdzie rodzic
    return Parent_Addr;
end Get_Left_Child_Address;

function Get_Right_Child_Address (
    Parent_Addr : Phys_Addr;
    Parent_Exp  : Block_Size
) return Phys_Addr is
    // ... (truncated, see source file for more)
```

driverspace_workspace/formal/ds-spark-core/src/ds_buddy_math.adb (line 40) - [internal]

Get_Right_Child_Address

```
function Get_Right_Child_Address (
  Parent_Addr : Phys_Addr;
  Parent_Exp  : Block_Size
) return Phys_Addr is
  -- Prawe dziecko to rodzic + połowa rozmiaru rodzica
  Half_Size : constant Phys_Addr := Power_Of_Two (Parent_Exp - 1);
begin
  return Parent_Addr + Half_Size;
end Get_Right_Child_Address;

function Safe_Calculate_DMA_End (
  Base_Addr : Phys_Addr;
  // ... (truncated, see source file for more)
```

driverspace_workspace/formal/ds-spark-core/src/ds_buddy_math.adb (line 50) - [internal]

Safe_Calculate_DMA_End

```
function Safe_Calculate_DMA_End (
  Base_Addr : Phys_Addr;
  Buffer_Len : Phys_Addr
) return Phys_Addr is
begin
  -- Ponieważ Pre condition w .ads gwarantuje, że Base + Len <= Last,
  -- kompilator SPARK wie, że to dodawanie nigdy nie "owinie" (wrap-around) pamięci.
  return Base_Addr + Buffer_Len;
end Safe_Calculate_DMA_End;

end DS_Buddy_Math;
```

kernel_Workspace/kernel/src/policy/ada/policy.adb (line 5) - [internal] Evaluate

```
function Evaluate (Ring  : Ring_Id;
                  Class  : Call_Class;
                  Op     : Unsigned_8;
                  Arg    : U64)
  return Decision
is
  pragma Unreferenced (Arg);
begin
  if Ring = Ring_Kernel then
    return Allow;
  end if;

  // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/policy/ada/policy_c.adb (line 5) - [internal] To_Ring

```
function To_Ring (V : Unsigned_8) return Ring_Id is
  (case V is
    when 0 => Ring_Kernel,
    when 1 => Ring_Driver,
    when others => Ring_User);

function To_Class (V : Unsigned_8) return Call_Class is
  (case V is
    when 0 => Cls_Sys,
    when 1 => Cls_Video,
    when 2 => Cls_Audio,
    when 3 => Cls_Input,
    // ... (truncated, see source file for more)
```

```

kernel_workspace/kernel/src/policy/ada/policy_c.adb (line 11) - [internal] To_Class
function To_Class (V : Unsigned_8) return Call_Class is
  (case V is
    when 0 => Cls_Sys,
    when 1 => Cls_Video,
    when 2 => Cls_Audio,
    when 3 => Cls_Input,
    when 4 => Cls_Block,
    when others => Cls_Net);

function policy_evaluate (ring : Unsigned_8;
                          cls : Unsigned_8;
                          op : Unsigned_8;
                          // ... (truncated, see source file for more)

kernel_workspace/kernel/src/policy/ada/policy_c.adb (line 20) - [internal] policy_evaluate
function policy_evaluate (ring : Unsigned_8;
                          cls : Unsigned_8;
                          op : Unsigned_8;
                          arg : Unsigned_64)
  return Unsigned_8
is
  D : constant Decision :=
    Evaluate (To_Ring (ring), To_Class (cls), op, U64 (arg));
begin
  return (case D is
    when Allow => 0,
    when Deny => 1);
  // ... (truncated, see source file for more)

```

Ada (spec)

```

driverspace_workspace/formal/ds-spark-core/src/ds_buddy_math.ads (line 4) - [public]
DS_Buddy_Math
package DS_Buddy_Math with
  SPARK_Mode,
  Pure
is
  -- Typy bazowe dla fizycznej pamieci (bare-metal)
  type Phys_Addr is new Unsigned_64;
  type Block_Size is new Unsigned_64;

  -- Minimalny i maksymalny rozmiar bloku (np. 4KB do 1GB)
  MIN_BLOCK_SIZE : constant Block_Size := 12; -- 2^12 = 4096 bytes (4KB)
  MAX_BLOCK_SIZE : constant Block_Size := 30; -- 2^30 = 1GB

  // ... (truncated, see source file for more)

driverspace_workspace/formal/ds-spark-core/src/ds_buddy_math.ads (line 17) - [public]
Power_Of_Two
  -- Funkcja pomocnicza: Obliczanie 2^N (Shift left)

  function Power_Of_Two (Exponent : Block_Size) return Phys_Addr with
    Pre => Exponent <= 63,
    -- Use a bit-shift to produce a value of type Unsigned_64 / Phys_Addr
    Post => Power_Of_Two'Result = Shift_Left (Phys_Addr (1), Integer (Exponent));

  -- Sprawdzenie, czy adres jest poprawnie wyrównany do rozmiaru bloku
  function Is_Aligned (Addr : Phys_Addr; Size_Exp : Block_Size) return Boolean with
    Post => Is_Aligned'Result = ((Addr mod Power_Of_Two (Size_Exp)) = 0);

  -- =====
  -- ALGORYTM 1: OBLICZANIE ADRESU BUDDY (PAROWANIE)
  -- =====
  // ... (truncated, see source file for more)

```

driverspace_workspace/formal/ds-spark-core/src/ds_buddy_math.ads (line 23) - [public]
Is_Aligned

-- Sprawdzenie, czy adres jest poprawnie wyrównany do rozmiaru bloku

```
function Is_Aligned (Addr : Phys_Addr; Size_Exp : Block_Size) return Boolean with
  Post => Is_Aligned'Result = ((Addr mod Power_Of_Two (Size_Exp)) = 0);
```

```
-- =====
-- ALGORYTM 1: OBLICZANIE ADRESU BUDDY (PAROWANIE)
-- =====
-- W alokatorze Buddy, "buddy" bloku to blok o tym samym rozmiarze,
-- który różni się tylko jednym bitem (na pozycji odpowiadającej rozmiarowi).
```

```
function Get_Buddy_Address (
  Addr      : Phys_Addr;
  Size_Exp  : Block_Size
  // ... (truncated, see source file for more)
```

driverspace_workspace/formal/ds-spark-core/src/ds_buddy_math.ads (line 32) - [public]
Get_Buddy_Address

```
-- =====
-- ALGORYTM 1: OBLICZANIE ADRESU BUDDY (PAROWANIE)
-- =====
-- W alokatorze Buddy, "buddy" bloku to blok o tym samym rozmiarze,
-- który różni się tylko jednym bitem (na pozycji odpowiadającej rozmiarowi).
```

```
function Get_Buddy_Address (
  Addr      : Phys_Addr;
  Size_Exp  : Block_Size
) return Phys_Addr with
  Pre => Size_Exp >= MIN_BLOCK_SIZE and then Size_Exp <= MAX_BLOCK_SIZE,
  Post => Is_Aligned (Get_Buddy_Address'Result, Size_Exp),
  Post => Get_Buddy_Address'Result /= Addr, -- Buddy NIGDY nie jest tym samym adresem
  Post => Shift_Right (Get_Buddy_Address'Result, Integer (Size_Exp)) =
    (Shift_Right (Addr, Integer (Size_Exp)) xor Phys_Addr (1)); -- Udowodniona różnica dok
-- =====
-- ALGORYTM 2: DZIELENIE BLOKU (SPLITTING)
// ... (truncated, see source file for more)
```

driverspace_workspace/formal/ds-spark-core/src/ds_buddy_math.ads (line 47) - [public]
Get_Left_Child_Address

```
-- =====
-- ALGORYTM 2: DZIELENIE BLOKU (SPLITTING)
-- =====
-- Dzielenie bloku o rozmiarze  $2^N$  na dwa bloki o rozmiarze  $2^{(N-1)}$ .
```

```
function Get_Left_Child_Address (
  Parent_Addr : Phys_Addr;
  Parent_Exp  : Block_Size
) return Phys_Addr with
  Pre => Parent_Exp > MIN_BLOCK_SIZE and then Is_Aligned (Parent_Addr, Parent_Exp),
  Post => Get_Left_Child_Address'Result = Parent_Addr,
  Post => Is_Aligned (Get_Left_Child_Address'Result, Parent_Exp - 1);
```

```
function Get_Right_Child_Address (
  Parent_Addr : Phys_Addr;
  Parent_Exp  : Block_Size
) return Phys_Addr with
  // ... (truncated, see source file for more)
```

driverspace_workspace/formal/ds-spark-core/src/ds_buddy_math.ads (line 55) - [public]

Get_Right_Child_Address

```
function Get_Right_Child_Address (
  Parent_Addr : Phys_Addr;
  Parent_Exp  : Block_Size
) return Phys_Addr with
  Pre  => Parent_Exp > MIN_BLOCK_SIZE and then Is_Aligned (Parent_Addr, Parent_Exp),
  Post => Get_Right_Child_Address'Result = Parent_Addr + Power_Of_Two (Parent_Exp - 1),
  Post => Is_Aligned (Get_Right_Child_Address'Result, Parent_Exp - 1),
  Post => Get_Right_Child_Address'Result > Parent_Addr; -- Prawe dziecko jest zawsze wyżej w pamięci

-- =====
-- ALGORYTM 3: BEZPIECZNE OBLICZANIE OFFSETU DMA
-- =====
// ... (truncated, see source file for more)
```

driverspace_workspace/formal/ds-spark-core/src/ds_buddy_math.ads (line 69) - [public]

Safe_Calculate_DMA_End

```
-- =====
-- ALGORYTM 3: BEZPIECZNE OBLICZANIE OFFSETU DMA
-- =====
-- Krytyczne dla sterowników: obliczanie adresu końcowego bufora bez przepełnienia (Overflow).

function Safe_Calculate_DMA_End (
  Base_Addr : Phys_Addr;
  Buffer_Len : Phys_Addr
) return Phys_Addr with
  Pre  => Base_Addr <= Phys_Addr'Last - Buffer_Len, -- Gwarancja braku przepełnienia UInt64
  Post => Safe_Calculate_DMA_End'Result = Base_Addr + Buffer_Len;

-- Eksport do C/Rust (FFI)
pragma Export (C, Get_Buddy_Address, "ds_spark_buddy_get_buddy");
pragma Export (C, Get_Left_Child_Address, "ds_spark_buddy_left_child");
pragma Export (C, Get_Right_Child_Address, "ds_spark_buddy_right_child");
pragma Export (C, Safe_Calculate_DMA_End, "ds_spark_safe_dma_end");
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/policy/ada/policy.ads (line 3) - [public] Policy

```
package Policy
with SPARK_Mode
is

  type Ring_Id is (Ring_Kernel, Ring_Driver, Ring_User)
    with Size => 8;

  type Call_Class is (Cls_Sys, Cls_Video, Cls_Audio,
    Cls_Input, Cls_Block, Cls_Net)
    with Size => 8;

  type Decision is (Allow, Deny)
  // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/policy/ada/policy.ads (line 21) - [public] Evaluate

```
function Evaluate (Ring : Ring_Id;
  Class : Call_Class;
  Op    : Unsigned_8;
  Arg   : U64)
  return Decision

with
  Global => null,
  Post  =>
    (if Ring = Ring_Kernel then Evaluate'Result = Allow)
    and then
    (if Ring = Ring_User
      and then Class = Cls_Block
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/policy/ada/policy_c.ads (line 4) - [public] Policy_C

```
package Policy_C is

    function policy_evaluate (ring : Unsigned_8;
                              cls  : Unsigned_8;
                              op   : Unsigned_8;
                              arg  : Unsigned_64)
        return Unsigned_8

    with Convention => C,
         Export,
         External_Name => "policy_evaluate";

end Policy_C;
```

kernel_workspace/kernel/src/policy/ada/policy_c.ads (line 6) - [public] policy_evaluate

```
function policy_evaluate (ring : Unsigned_8;
                          cls  : Unsigned_8;
                          op   : Unsigned_8;
                          arg  : Unsigned_64)
    return Unsigned_8

    with Convention => C,
         Export,
         External_Name => "policy_evaluate";

end Policy_C;
```

Assembly

comgrub/src/boot.asm (line 10) - [internal] mboot_header_start

```
mboot_header_start:
    dd MBOOT2_MAGIC
    dd MBOOT2_ARCH
    dd MBOOT2_LENGTH
    dd MBOOT2_CHECKSUM

    dw 0
    dw 0
    dd 8
mboot_header_end:

section .bss
    // ... (truncated, see source file for more)
```

comgrub/src/boot.asm (line 19) - [internal] mboot_header_end

```
mboot_header_end:

section .bss
    align 16
stack_bottom:
    resb 65536
stack_top:

section .text
    bits 32

global _start
    // ... (truncated, see source file for more)
```

comgrub/src/boot.asm (line 23) - [internal] stack_bottom

```
stack_bottom:
    resb 65536
stack_top:

section .text
bits 32

global _start
extern _rust_entry

_start:
    mov esp, stack_top
    // ... (truncated, see source file for more)
```

comgrub/src/boot.asm (line 25) - [internal] stack_top

```
stack_top:

section .text
bits 32

global _start
extern _rust_entry

_start:
    mov esp, stack_top

    push ebx                ; Multiboot2: EBX = adres struktury informacyjnej
    // ... (truncated, see source file for more)
```

comgrub/src/boot.asm (line 30) - [public] _start

```
global _start
extern _rust_entry

_start:
    mov esp, stack_top

    push ebx                ; Multiboot2: EBX = adres struktury informacyjnej
    push eax                ; Multiboot2: EAX = magic

    call _rust_entry

.hang:
    // ... (truncated, see source file for more)
```

comgrub/src/boot.asm (line 33) - [internal] _start

```
_start:
    mov esp, stack_top

    push ebx                ; Multiboot2: EBX = adres struktury informacyjnej
    push eax                ; Multiboot2: EAX = magic

    call _rust_entry

.hang:
    cli
    hlt
    jmp .hang
```

kernel_Workspace/kernel-bin/src/boot.asm (line 2) - [public] kernel_bin_entry

```
; kernel_Workspace/kernel-bin/src/boot.asm
```

```
global kernel_bin_entry
extern kernel_bin_main

section .text
bits 64

kernel_bin_entry:
    ; Wywołaj główną funkcję Rust w kernel-bin
    call kernel_bin_main

    ; Jeśli kernel_bin_main somehow wróci
.hang:
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel-bin/src/boot.asm (line 8) - [internal] kernel_bin_entry

```
kernel_bin_entry:
    ; Wywołaj główną funkcję Rust w kernel-bin
    call kernel_bin_main

    ; Jeśli kernel_bin_main somehow wróci
.hang:
    cli
    hlt
    jmp .hang
```

kernel_Workspace/kernel/src/cpu/trampoline.s (line 5) - [public] trampoline_start

```
.global trampoline_start
.global trampoline_end
.global trampoline_cr3
.global trampoline_entry
.global trampoline_stack
.global trampoline_arg

.code16
trampoline_start:
    cli
    cld

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/cpu/trampoline.s (line 6) - [public] trampoline_end

```
.global trampoline_end
.global trampoline_cr3
.global trampoline_entry
.global trampoline_stack
.global trampoline_arg

.code16
trampoline_start:
    cli
    cld

    mov %cs, %ax
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/cpu/trampoline.s (line 7) - [public] trampoline_cr3

```
.global trampoline_cr3
.global trampoline_entry
.global trampoline_stack
.global trampoline_arg

.code16
trampoline_start:
    cli
    cld

    mov %cs, %ax
    mov %ax, %ds
    // ... (truncated, see source file for more)
```


kernel_Workspace/kernel/src/cpu/trampoline.s (line 8) - [public] trampoline_entry

```
.global trampoline_entry
.global trampoline_stack
.global trampoline_arg

.code16
trampoline_start:
    cli
    cld

    mov %cs, %ax
    mov %ax, %ds
    mov %ax, %es
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/cpu/trampoline.s (line 9) - [public] trampoline_stack

```
.global trampoline_stack
.global trampoline_arg

.code16
trampoline_start:
    cli
    cld

    mov %cs, %ax
    mov %ax, %ds
    mov %ax, %es
    mov %ax, %ss
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/cpu/trampoline.s (line 10) - [public] trampoline_arg

```
.global trampoline_arg

.code16
trampoline_start:
    cli
    cld

    mov %cs, %ax
    mov %ax, %ds
    mov %ax, %es
    mov %ax, %ss

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/cpu/trampoline.s (line 13) - [internal] trampoline_start

```
trampoline_start:
    cli
    cld

    mov %cs, %ax
    mov %ax, %ds
    mov %ax, %es
    mov %ax, %ss

    lgdt %cs:(trampoline_gdtr - trampoline_start)

    mov %cr0, %eax
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/cpu/trampoline.s (line 31) - [internal] trampoline_32

```
trampoline_32:

    mov $0x10, %ax
    mov %ax, %ds
    mov %ax, %es
    mov %ax, %ss

    mov %cr4, %eax
    or $((1 << 5) | (1 << 9) | (1 << 10)), %eax
    mov %eax, %cr4

    movl $(TRAMPOLINE_BASE + (trampoline_cr3 - trampoline_start)), %eax
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/cpu/trampoline.s (line 59) - [internal] trampoline_64

```
trampoline_64:

    mov $0x20, %ax
    mov %ax, %ds
    mov %ax, %es
    mov %ax, %ss
    xor %eax, %eax
    mov %ax, %fs
    mov %ax, %gs

    movabs $(TRAMPOLINE_BASE + (trampoline_arg - trampoline_start)), %rcx
    mov (%rcx), %rdi
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/cpu/trampoline.s (line 81) - [internal] trampoline_gdt

```
trampoline_gdt:
    .quad 0x0000000000000000
    .quad 0x00CF9A000000FFFF
    .quad 0x00CF92000000FFFF
    .quad 0x00AF9A000000FFFF
    .quad 0x00AF92000000FFFF
trampoline_gdt_end:

.align 4
trampoline_gdtr:
    .word (trampoline_gdt_end - trampoline_gdt - 1)
    .long (TRAMPOLINE_BASE + (trampoline_gdt - trampoline_start))
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/cpu/trampoline.s (line 87) - [internal] trampoline_gdt_end

```
trampoline_gdt_end:

.align 4
trampoline_gdtr:
    .word (trampoline_gdt_end - trampoline_gdt - 1)
    .long (TRAMPOLINE_BASE + (trampoline_gdt - trampoline_start))

.align 8
trampoline_cr3:    .quad 0
trampoline_entry:  .quad 0
trampoline_stack:  .quad 0
trampoline_arg:    .quad 0
    // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/cpu/trampoline.s (line 90) - [internal] trampoline_gdtr
    trampoline_gdtr:
        .word (trampoline_gdt_end - trampoline_gdt - 1)
        .long (TRAMPOLINE_BASE + (trampoline_gdt - trampoline_start))

        .align 8
    trampoline_cr3:    .quad 0
    trampoline_entry:  .quad 0
    trampoline_stack:  .quad 0
    trampoline_arg:    .quad 0

    .global trampoline_end
    trampoline_end:
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/cpu/trampoline.s (line 95) - [internal] trampoline_cr3
    trampoline_cr3:    .quad 0
    trampoline_entry:  .quad 0
    trampoline_stack:  .quad 0
    trampoline_arg:    .quad 0

    .global trampoline_end
    trampoline_end:
        .quad 0

kernel_Workspace/kernel/src/cpu/trampoline.s (line 96) - [internal] trampoline_entry
    trampoline_entry:  .quad 0
    trampoline_stack:  .quad 0
    trampoline_arg:    .quad 0

    .global trampoline_end
    trampoline_end:
        .quad 0

kernel_Workspace/kernel/src/cpu/trampoline.s (line 97) - [internal] trampoline_stack
    trampoline_stack:  .quad 0
    trampoline_arg:    .quad 0

    .global trampoline_end
    trampoline_end:
        .quad 0

kernel_Workspace/kernel/src/cpu/trampoline.s (line 98) - [internal] trampoline_arg
    trampoline_arg:    .quad 0

    .global trampoline_end
    trampoline_end:
        .quad 0

kernel_Workspace/kernel/src/cpu/trampoline.s (line 100) - [public] trampoline_end
    .global trampoline_end
    trampoline_end:
        .quad 0

kernel_Workspace/kernel/src/cpu/trampoline.s (line 101) - [internal] trampoline_end
    trampoline_end:
        .quad 0

```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.s (line 3) - [public] tlb_asm_invlpg

```
.global tlb_asm_invlpg
.type tlb_asm_invlpg, @function
tlb_asm_invlpg:
    invlpg (%rdi)
    ret

.global tlb_asm_read_cr3
.type tlb_asm_read_cr3, @function
tlb_asm_read_cr3:
    mov %cr3, %rax
    ret

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.s (line 5) - [internal] tlb_asm_invlpg

```
tlb_asm_invlpg:
    invlpg (%rdi)
    ret

.global tlb_asm_read_cr3
.type tlb_asm_read_cr3, @function
tlb_asm_read_cr3:
    mov %cr3, %rax
    ret

.global tlb_asm_write_cr3
.type tlb_asm_write_cr3, @function
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.s (line 9) - [public] tlb_asm_read_cr3

```
.global tlb_asm_read_cr3
.type tlb_asm_read_cr3, @function
tlb_asm_read_cr3:
    mov %cr3, %rax
    ret

.global tlb_asm_write_cr3
.type tlb_asm_write_cr3, @function
tlb_asm_write_cr3:
    mov %rdi, %cr3
    ret

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.s (line 11) - [internal] tlb_asm_read_cr3

```
tlb_asm_read_cr3:
    mov %cr3, %rax
    ret

.global tlb_asm_write_cr3
.type tlb_asm_write_cr3, @function
tlb_asm_write_cr3:
    mov %rdi, %cr3
    ret

.global tlb_asm_read_cr4
.type tlb_asm_read_cr4, @function
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.s (line 15) - [public] tlb_asm_write_cr3

```
.global tlb_asm_write_cr3
.type tlb_asm_write_cr3, @function
tlb_asm_write_cr3:
    mov %rdi, %cr3
    ret
```

```
.global tlb_asm_read_cr4
.type tlb_asm_read_cr4, @function
tlb_asm_read_cr4:
    mov %cr4, %rax
    ret
```

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.s (line 17) - [internal] tlb_asm_write_cr3

```
tlb_asm_write_cr3:
    mov %rdi, %cr3
    ret
```

```
.global tlb_asm_read_cr4
.type tlb_asm_read_cr4, @function
tlb_asm_read_cr4:
    mov %cr4, %rax
    ret
```

```
.global tlb_asm_write_cr4
.type tlb_asm_write_cr4, @function
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.s (line 21) - [public] tlb_asm_read_cr4

```
.global tlb_asm_read_cr4
.type tlb_asm_read_cr4, @function
tlb_asm_read_cr4:
    mov %cr4, %rax
    ret
```

```
.global tlb_asm_write_cr4
.type tlb_asm_write_cr4, @function
tlb_asm_write_cr4:
    mov %rdi, %cr4
    ret
```

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.s (line 23) - [internal] tlb_asm_read_cr4

```
tlb_asm_read_cr4:
    mov %cr4, %rax
    ret
```

```
.global tlb_asm_write_cr4
.type tlb_asm_write_cr4, @function
tlb_asm_write_cr4:
    mov %rdi, %cr4
    ret
```

```
.global tlb_asm_invpclid
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.s (line 27) - [public] tlb_asm_write_cr4

```
.global tlb_asm_write_cr4
.type tlb_asm_write_cr4, @function
tlb_asm_write_cr4:
    mov %rdi, %cr4
    ret
```

```
.global tlb_asm_invpcid
.type tlb_asm_invpcid, @function
tlb_asm_invpcid:
    .byte 0x66, 0x0F, 0x38, 0x82, 0x3E
    ret
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.s (line 29) - [internal] tlb_asm_write_cr4

```
tlb_asm_write_cr4:
    mov %rdi, %cr4
    ret
```

```
.global tlb_asm_invpcid
.type tlb_asm_invpcid, @function
tlb_asm_invpcid:
    .byte 0x66, 0x0F, 0x38, 0x82, 0x3E
    ret
```

```
.global tlb_asm_wbinvd
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.s (line 34) - [public] tlb_asm_invpcid

```
.global tlb_asm_invpcid
.type tlb_asm_invpcid, @function
tlb_asm_invpcid:
    .byte 0x66, 0x0F, 0x38, 0x82, 0x3E
    ret
```

```
.global tlb_asm_wbinvd
.type tlb_asm_wbinvd, @function
tlb_asm_wbinvd:
    wbinvd
    ret
```

```
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.s (line 36) - [internal] tlb_asm_invpcid

```
tlb_asm_invpcid:
    .byte 0x66, 0x0F, 0x38, 0x82, 0x3E
    ret
```

```
.global tlb_asm_wbinvd
.type tlb_asm_wbinvd, @function
tlb_asm_wbinvd:
    wbinvd
    ret
```

```
.global tlb_asm_clflush
.type tlb_asm_clflush, @function
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.s (line 40) - [public] tlb_asm_wbinvd

```
.global tlb_asm_wbinvd
.type tlb_asm_wbinvd, @function
tlb_asm_wbinvd:
    wbinvd
    ret

.global tlb_asm_clflush
.type tlb_asm_clflush, @function
tlb_asm_clflush:
    clflush (%rdi)
    ret
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.s (line 42) - [internal] tlb_asm_wbinvd

```
tlb_asm_wbinvd:
    wbinvd
    ret

.global tlb_asm_clflush
.type tlb_asm_clflush, @function
tlb_asm_clflush:
    clflush (%rdi)
    ret
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.s (line 46) - [public] tlb_asm_clflush

```
.global tlb_asm_clflush
.type tlb_asm_clflush, @function
tlb_asm_clflush:
    clflush (%rdi)
    ret
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.s (line 48) - [internal] tlb_asm_clflush

```
tlb_asm_clflush:
    clflush (%rdi)
    ret
```

kernel_Workspace/kernel/src/trampoline_rings/arch/aarch64/trampoline_rings.s (line 4) -
[internal] vector_table

```
vector_table:
    .align 7
    b vector_default
    .align 7
    b vector_default
    .align 7
    b vector_default
    .align 7
    b vector_default

    .align 7
    b vector_default
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/trampoline_rings/arch/aarch64/trampoline_rings.s (line 41) -
[internal] vector_default

```
vector_default:
    eret

vector_hyper:
    sub sp, sp, #272

    stp x0, x1, [sp, #0]
    stp x2, x3, [sp, #16]
    stp x4, x5, [sp, #32]
    stp x6, x7, [sp, #48]
    stp x8, x9, [sp, #64]
    stp x10, x11, [sp, #80]
    // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/trampoline_rings/arch/aarch64/trampoline_rings.s (line 44) -
[internal] vector_hyper
    vector_hyper:
        sub sp, sp, #272

        stp x0, x1, [sp, #0]
        stp x2, x3, [sp, #16]
        stp x4, x5, [sp, #32]
        stp x6, x7, [sp, #48]
        stp x8, x9, [sp, #64]
        stp x10, x11, [sp, #80]
        stp x12, x13, [sp, #96]
        stp x14, x15, [sp, #112]
        stp x16, x17, [sp, #128]
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/trampoline_rings/arch/aarch64/trampoline_rings.s (line 102) -
[public] tr_init
    .global tr_init
    .type tr_init, @function
tr_init:
    adr x1, vector_table
    msr vbar_el1, x1
    isb
    ret

    .global tr_restore_ctx
    .type tr_restore_ctx, @function
tr_restore_ctx:
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/trampoline_rings/arch/aarch64/trampoline_rings.s (line 104) -
[internal] tr_init
    tr_init:
        adr x1, vector_table
        msr vbar_el1, x1
        isb
        ret

    .global tr_restore_ctx
    .type tr_restore_ctx, @function
tr_restore_ctx:
    ldr x9, [x0, #248]
    msr sp_el0, x9
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/trampoline_rings/arch/aarch64/trampoline_rings.s (line 111) -
[public] tr_restore_ctx
    .global tr_restore_ctx
    .type tr_restore_ctx, @function
tr_restore_ctx:
    ldr x9, [x0, #248]
    msr sp_el0, x9
    ldr x9, [x0, #256]
    msr elr_el1, x9
    ldr x9, [x0, #264]
    msr spsr_el1, x9

    ldr x30, [x0, #240]
    ldp x28, x29, [x0, #224]
    // ... (truncated, see source file for more)

```


kernel_Workspace/kernel/src/trampoline_rings/arch/aarch64/trampoline_rings.s (line 113) -
[internal] tr_restore_ctx

```
tr_restore_ctx:
    ldr x9, [x0, #248]
    msr sp_el0, x9
    ldr x9, [x0, #256]
    msr elr_el1, x9
    ldr x9, [x0, #264]
    msr spsr_el1, x9

    ldr x30, [x0, #240]
    ldp x28, x29, [x0, #224]
    ldp x26, x27, [x0, #208]
    ldp x24, x25, [x0, #192]
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/trampoline_rings/arch/risc-v/trampoline_rings.s (line 3) -
[internal] kstack_top

```
kstack_top:
    .dword 0

.section .text
.align 2

.global trap_entry
trap_entry:
    csrrw sp, sscratch, sp        # sp = kernel, sscratch = world sp
    addi sp, sp, -288

    sd x1, 8(sp)
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/trampoline_rings/arch/risc-v/trampoline_rings.s (line 9) -
[public] trap_entry

```
.global trap_entry
trap_entry:
    csrrw sp, sscratch, sp        # sp = kernel, sscratch = world sp
    addi sp, sp, -288

    sd x1, 8(sp)
    sd x3, 24(sp)
    sd x4, 32(sp)
    sd x5, 40(sp)
    sd x6, 48(sp)
    sd x7, 56(sp)
    sd x8, 64(sp)
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/trampoline_rings/arch/risc-v/trampoline_rings.s (line 10) -
[internal] trap_entry

```
trap_entry:
    csrrw sp, sscratch, sp        # sp = kernel, sscratch = world sp
    addi sp, sp, -288

    sd x1, 8(sp)
    sd x3, 24(sp)
    sd x4, 32(sp)
    sd x5, 40(sp)
    sd x6, 48(sp)
    sd x7, 56(sp)
    sd x8, 64(sp)
    sd x9, 72(sp)
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/trampoline_rings/arch/risc-v/trampoline_rings.s (line 98) -

```
[public] tr_init
    .global tr_init
    .type tr_init, @function
tr_init:
    la t0, kstack_top
    sd a0, 0(t0)

    csrw sscratch, a0

    la t0, trap_entry
    csrw stvec, t0

    li t0, (1 << 18)
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/trampoline_rings/arch/risc-v/trampoline_rings.s (line 100) -

```
[internal] tr_init
    tr_init:
        la t0, kstack_top
        sd a0, 0(t0)

        csrw sscratch, a0

        la t0, trap_entry
        csrw stvec, t0

        li t0, (1 << 18)
        csrs sstatus, t0

        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/trampoline_rings/arch/risc-v/trampoline_rings.s (line 114) -

```
[public] tr_restore_ctx
    .global tr_restore_ctx
    .type tr_restore_ctx, @function
tr_restore_ctx:
    ld t0, 264(a0)
    csrw sepc, t0
    ld t0, 272(a0)
    csrw sstatus, t0

    la t0, kstack_top
    ld t0, 0(t0)
    csrw sscratch, t0

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/trampoline_rings/arch/risc-v/trampoline_rings.s (line 116) -

```
[internal] tr_restore_ctx
    tr_restore_ctx:
        ld t0, 264(a0)
        csrw sepc, t0
        ld t0, 272(a0)
        csrw sstatus, t0

        la t0, kstack_top
        ld t0, 0(t0)
        csrw sscratch, t0

        ld sp, 256(a0)

        // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.s (line 3) -
[public] tr_gdt
    .global tr_gdt
    tr_gdt:
        .quad 0x0000000000000000
        .quad 0x00209A0000000000
        .quad 0x0000920000000000
        .quad 0x0020BA0000000000
        .quad 0x0000B20000000000
        .quad 0x0020FA0000000000
        .quad 0x0000F20000000000
    .global tr_tss_desc
    tr_tss_desc:
        .space 16
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.s (line 4) -
[internal] tr_gdt
    tr_gdt:
        .quad 0x0000000000000000
        .quad 0x00209A0000000000
        .quad 0x0000920000000000
        .quad 0x0020BA0000000000
        .quad 0x0000B20000000000
        .quad 0x0020FA0000000000
        .quad 0x0000F20000000000
    .global tr_tss_desc
    tr_tss_desc:
        .space 16

        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.s (line 12) -
[public] tr_tss_desc
    .global tr_tss_desc
    tr_tss_desc:
        .space 16

    gdt_ptr:
        .short (gdt_ptr - tr_gdt) - 1
        .quad tr_gdt

    .align 16
    .global tr_tss
    tr_tss:
        .space 104
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.s (line 13) -
[internal] tr_tss_desc
    tr_tss_desc:
        .space 16

    gdt_ptr:
        .short (gdt_ptr - tr_gdt) - 1
        .quad tr_gdt

    .align 16
    .global tr_tss
    tr_tss:
        .space 104

        // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.s (line 16) -
[internal] gdt_ptr
    gdt_ptr:
        .short (gdt_ptr - tr_gdt) - 1
        .quad tr_gdt

    .align 16
    .global tr_tss
tr_tss:
    .space 104

    .section .text

    .global tr_init
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.s (line 21) -
[public] tr_tss
    .global tr_tss
tr_tss:
    .space 104

    .section .text

    .global tr_init
    .type tr_init, @function
tr_init:
    mov %rdi, %rax
    mov %eax, (tr_tss + 4)
    shr $32, %rax
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.s (line 22) -
[internal] tr_tss
    tr_tss:
        .space 104

    .section .text

    .global tr_init
    .type tr_init, @function
tr_init:
    mov %rdi, %rax
    mov %eax, (tr_tss + 4)
    shr $32, %rax
    mov %eax, (tr_tss + 8)
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.s (line 27) -
[public] tr_init
    .global tr_init
    .type tr_init, @function
tr_init:
    mov %rdi, %rax
    mov %eax, (tr_tss + 4)
    shr $32, %rax
    mov %eax, (tr_tss + 8)

    lgdt gdt_ptr

    mov $0x10, %ax
    mov %ax, %ss
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.s (line 29) -
[internal] tr_init
    tr_init:
        mov %rdi, %rax
        mov %eax, (tr_tss + 4)
        shr $32, %rax
        mov %eax, (tr_tss + 8)

        lgdt gdt_ptr

        mov $0x10, %ax
        mov %ax, %ss
        mov %ax, %ds
        mov %ax, %es
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.s (line 54) -
[public] isr_hypercall
    .global isr_hypercall
    .type isr_hypercall, @function
isr_hypercall:
    push %rax
    push %rbx
    push %rcx
    push %rdx
    push %rsi
    push %rdi
    push %rbp
    push %r8
    push %r9
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.s (line 56) -
[internal] isr_hypercall
    isr_hypercall:
        push %rax
        push %rbx
        push %rcx
        push %rdx
        push %rsi
        push %rdi
        push %rbp
        push %r8
        push %r9
        push %r10
        push %r11
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.s (line 94) -
[public] isr_default
    .global isr_default
    .type isr_default, @function
isr_default:
    iretq

    .global tr_restore_ctx
    .type tr_restore_ctx, @function
tr_restore_ctx:
    mov %rdi, %rsp
    pop %r15
    pop %r14
    pop %r13
    // ... (truncated, see source file for more)

```

```

kernel_workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.s (line 96) -
[internal] isr_default
    isr_default:
        iretq

    .global tr_restore_ctx
    .type tr_restore_ctx, @function
tr_restore_ctx:
    mov %rdi, %rsp
    pop %r15
    pop %r14
    pop %r13
    pop %r12
    pop %r11
    // ... (truncated, see source file for more)

kernel_workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.s (line 99) -
[public] tr_restore_ctx
    .global tr_restore_ctx
    .type tr_restore_ctx, @function
tr_restore_ctx:
    mov %rdi, %rsp
    pop %r15
    pop %r14
    pop %r13
    pop %r12
    pop %r11
    pop %r10
    pop %r9
    pop %r8
    // ... (truncated, see source file for more)

kernel_workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.s (line 101) -
[internal] tr_restore_ctx
    tr_restore_ctx:
        mov %rdi, %rsp
        pop %r15
        pop %r14
        pop %r13
        pop %r12
        pop %r11
        pop %r10
        pop %r9
        pop %r8
        pop %rbp
        pop %rdi
        // ... (truncated, see source file for more)

triang-lang/out/x86_64/prog.asm (line 4) - [internal] buffer
    buffer: resb 16

    section .text
    global main
    main:
    fn_main:
        mov rax, 16
        mov rbx, 32
        mov rcx, rax
        add rcx, rbx
        lea r14, [rel buffer]
        mov r15, 16
        // ... (truncated, see source file for more)

```

triang-lang/out/x86_64/prog.asm (line 7) - [public] main

```
global main
main:
fn_main:
    mov rax, 16
    mov rbx, 32
    mov rcx, rax
    add rcx, rbx
    lea r14, [rel buffer]
    mov r15, 16
    mov r12, 0
fill_1:
    test r15, r15
    // ... (truncated, see source file for more)
```

triang-lang/out/x86_64/prog.asm (line 8) - [internal] main

```
main:
fn_main:
    mov rax, 16
    mov rbx, 32
    mov rcx, rax
    add rcx, rbx
    lea r14, [rel buffer]
    mov r15, 16
    mov r12, 0
fill_1:
    test r15, r15
    jz fill_end_2
    // ... (truncated, see source file for more)
```

triang-lang/out/x86_64/prog.asm (line 9) - [internal] fn_main

```
fn_main:
    mov rax, 16
    mov rbx, 32
    mov rcx, rax
    add rcx, rbx
    lea r14, [rel buffer]
    mov r15, 16
    mov r12, 0
fill_1:
    test r15, r15
    jz fill_end_2
    mov byte [r14], r12b
    // ... (truncated, see source file for more)
```

triang-lang/out/x86_64/prog.asm (line 17) - [internal] fill_1

```
fill_1:
    test r15, r15
    jz fill_end_2
    mov byte [r14], r12b
    inc r14
    dec r15
    jmp fill_1
fill_end_2:
    lea r14, [rel buffer]
    mov byte [r14 + 0], 65
    lea r14, [rel buffer]
    mov byte [r14 + 1], 66
    // ... (truncated, see source file for more)
```

```

triang-lang/out/x86_64/prog.asm (line 24) - [internal] fill_end_2
    fill_end_2:
        lea r14, [rel buffer]
        mov byte [r14 + 0], 65
        lea r14, [rel buffer]
        mov byte [r14 + 1], 66
        lea r14, [rel buffer]
        mov byte [r14 + 2], 67
        lea r14, [rel buffer]
        movzx rax, byte [r14 + 0]
        lea r14, [rel buffer]
        movzx rbx, byte [r14 + 1]
        mov rcx, rax
        // ... (truncated, see source file for more)

triang-lang/out/x86_64/prog.asm (line 41) - [internal] else_1
    else_1:
        mov rax, 0
    end_2:
    loop_3:
        cmp rax, 0
        je loop_end_4
        mov rbx, rax
        sub rbx, 1
        mov rax, rbx
        jmp loop_3
    loop_end_4:
        ret

triang-lang/out/x86_64/prog.asm (line 43) - [internal] end_2
    end_2:
    loop_3:
        cmp rax, 0
        je loop_end_4
        mov rbx, rax
        sub rbx, 1
        mov rax, rbx
        jmp loop_3
    loop_end_4:
        ret

triang-lang/out/x86_64/prog.asm (line 44) - [internal] loop_3
    loop_3:
        cmp rax, 0
        je loop_end_4
        mov rbx, rax
        sub rbx, 1
        mov rax, rbx
        jmp loop_3
    loop_end_4:
        ret

triang-lang/out/x86_64/prog.asm (line 51) - [internal] loop_end_4
    loop_end_4:
        ret

```

C


```

comlimine/src/main.c (line 6) - [public] kernel_main
extern void kernel_main(uint64_t magic, void *info);

LIMINE_BASE_REVISION(1);

static volatile struct limine_framebuffer_request framebuffer = {
    .id = LIMINE_FRAMEBUFFER_REQUEST,
    .revision = 0,
    .response = NULL,
};

static volatile struct limine_memmap_request memmap = {
    .id = LIMINE_MEMMAP_REQUEST,
    // ... (truncated, see source file for more)

comlimine/src/main.c (line 28) - [public] _start
void _start(void) {
    if (framebuffer.response == NULL ||
        memmap.response == NULL ||
        hhdm.response == NULL) {
        for (;;) { asm volatile ("hlt"); }
    }

    kernel_main(0, (void *)framebuffer.response);

    for (;;) { asm volatile ("hlt"); }
}

comlimine/src/main.c (line 35) - [public] kernel_main
kernel_main(0, (void *)framebuffer.response);

    for (;;) { asm volatile ("hlt"); }
}

drivers/vgpu/vgpu_driver.c (line 15) - [public] ds_poll
ds_poll();
ds_msg_t r;
ds_take(id, &r);
}

static uint64_t pci_val(uint32_t op, uint64_t a0, uint64_t a1, uint64_t a2)
{
    uint64_t id = ds_call(SVC_PCI, op, a0, a1, a2);
    ds_poll();
    ds_msg_t r;

    if (ds_take(id, &r) && r.status == 0) {
        // ... (truncated, see source file for more)

drivers/vgpu/vgpu_driver.c (line 17) - [public] ds_take
ds_take(id, &r);
}

static uint64_t pci_val(uint32_t op, uint64_t a0, uint64_t a1, uint64_t a2)
{
    uint64_t id = ds_call(SVC_PCI, op, a0, a1, a2);
    ds_poll();
    ds_msg_t r;

    if (ds_take(id, &r) && r.status == 0) {
        return r.arg0;
    }
    // ... (truncated, see source file for more)

```

```

drivers/vgpu/vgpu_driver.c (line 23) - [public] ds_poll
    ds_poll();
    ds_msg_t r;

    if (ds_take(id, &r) && r.status == 0) {
        return r.arg0;
    }

    return 0;
}

static void vbe_write(uint16_t idx, uint16_t val)
{
    // ... (truncated, see source file for more)
}

drivers/vgpu/vgpu_driver.c (line 26) - [public] if
    if (ds_take(id, &r) && r.status == 0) {
        return r.arg0;
    }

    return 0;
}

static void vbe_write(uint16_t idx, uint16_t val)
{
    pci_void(PORT_WRITE, VBE_INDEX, idx, 16);
    pci_void(PORT_WRITE, VBE_DATA, val, 16);
}

// ... (truncated, see source file for more)

drivers/vgpu/vgpu_driver.c (line 35) - [public] pci_void
    pci_void(PORT_WRITE, VBE_INDEX, idx, 16);
    pci_void(PORT_WRITE, VBE_DATA, val, 16);
}

bool vgpu_init(uint32_t w, uint32_t h)
{
    if (g_ready) {
        return true;
    }

    uint64_t id = ds_call(SVC_PCI, PCI_FIND, 0x0300, 0, 0);
    // ... (truncated, see source file for more)
}

drivers/vgpu/vgpu_driver.c (line 36) - [public] pci_void
    pci_void(PORT_WRITE, VBE_DATA, val, 16);
}

bool vgpu_init(uint32_t w, uint32_t h)
{
    if (g_ready) {
        return true;
    }

    uint64_t id = ds_call(SVC_PCI, PCI_FIND, 0x0300, 0, 0);
    ds_poll();
    // ... (truncated, see source file for more)
}

```

```
drivers/vgpu/vgpu_driver.c (line 42) - [public] if
    if (g_ready) {
        return true;
    }

    uint64_t id = ds_call(SVC_PCI, PCI_FIND, 0x0300, 0, 0);
    ds_poll();

    ds_msg_t r;

    if (!ds_take(id, &r) || r.status != 0) {
        return false;
    }
    // ... (truncated, see source file for more)
```

```
drivers/vgpu/vgpu_driver.c (line 47) - [public] ds_poll
    ds_poll();

    ds_msg_t r;

    if (!ds_take(id, &r) || r.status != 0) {
        return false;
    }

    uint64_t bdf = r.arg0;

    if (r.arg1 != (0x1234 | (0x1111 << 16))) {
        return false;
    }
    // ... (truncated, see source file for more)
```

```
drivers/vgpu/vgpu_driver.c (line 51) - [public] if
    if (!ds_take(id, &r) || r.status != 0) {
        return false;
    }

    uint64_t bdf = r.arg0;

    if (r.arg1 != (0x1234 | (0x1111 << 16))) {
        return false;
    }

    uint64_t fb_phys = pci_val(PCI_BAR, bdf, 0, 0);

    // ... (truncated, see source file for more)
```

```
drivers/vgpu/vgpu_driver.c (line 57) - [public] if
    if (r.arg1 != (0x1234 | (0x1111 << 16))) {
        return false;
    }

    uint64_t fb_phys = pci_val(PCI_BAR, bdf, 0, 0);

    if (fb_phys == 0) {
        return false;
    }

    pci_void(PCI_ENABLE, bdf, 0, 0);

    // ... (truncated, see source file for more)
```

drivers/vgpu/vgpu_driver.c (line 63) - [public] if

```
    if (fb_phys == 0) {
        return false;
    }

    pci_void(PCI_ENABLE, bdf, 0, 0);

    vbe_write(4, 0);
    vbe_write(0, 0xB0C4);
    vbe_write(1, w);
    vbe_write(2, h);
    vbe_write(3, 32);
    vbe_write(4, 0x01 | 0x40 | 0x80);
    // ... (truncated, see source file for more)
```

drivers/vgpu/vgpu_driver.c (line 67) - [public] pci_void

```
    pci_void(PCI_ENABLE, bdf, 0, 0);

    vbe_write(4, 0);
    vbe_write(0, 0xB0C4);
    vbe_write(1, w);
    vbe_write(2, h);
    vbe_write(3, 32);
    vbe_write(4, 0x01 | 0x40 | 0x80);

    uint64_t mid = ds_call(SVC_SYS, OP_MAPMMIO, fb_phys,
                           (uint64_t)w * h * 4, 0x45000000ULL);
    ds_poll();
    // ... (truncated, see source file for more)
```

drivers/vgpu/vgpu_driver.c (line 69) - [public] vbe_write

```
    vbe_write(4, 0);
    vbe_write(0, 0xB0C4);
    vbe_write(1, w);
    vbe_write(2, h);
    vbe_write(3, 32);
    vbe_write(4, 0x01 | 0x40 | 0x80);

    uint64_t mid = ds_call(SVC_SYS, OP_MAPMMIO, fb_phys,
                           (uint64_t)w * h * 4, 0x45000000ULL);
    ds_poll();

    ds_msg_t mr;
    // ... (truncated, see source file for more)
```

drivers/vgpu/vgpu_driver.c (line 70) - [public] vbe_write

```
    vbe_write(0, 0xB0C4);
    vbe_write(1, w);
    vbe_write(2, h);
    vbe_write(3, 32);
    vbe_write(4, 0x01 | 0x40 | 0x80);

    uint64_t mid = ds_call(SVC_SYS, OP_MAPMMIO, fb_phys,
                           (uint64_t)w * h * 4, 0x45000000ULL);
    ds_poll();

    ds_msg_t mr;

    // ... (truncated, see source file for more)
```

drivers/vgpu/vgpu_driver.c (line 71) - [public] vbe_write

```
vbe_write(1, w);
vbe_write(2, h);
vbe_write(3, 32);
vbe_write(4, 0x01 | 0x40 | 0x80);

uint64_t mid = ds_call(SVC_SYS, OP_MAPMMIO, fb_phys,
                      (uint64_t)w * h * 4, 0x45000000ULL);
ds_poll();

ds_msg_t mr;

if (!ds_take(mid, &mr) || mr.status != 0) {
    // ... (truncated, see source file for more)
```

drivers/vgpu/vgpu_driver.c (line 72) - [public] vbe_write

```
vbe_write(2, h);
vbe_write(3, 32);
vbe_write(4, 0x01 | 0x40 | 0x80);

uint64_t mid = ds_call(SVC_SYS, OP_MAPMMIO, fb_phys,
                      (uint64_t)w * h * 4, 0x45000000ULL);
ds_poll();

ds_msg_t mr;

if (!ds_take(mid, &mr) || mr.status != 0) {
    return false;
    // ... (truncated, see source file for more)
```

drivers/vgpu/vgpu_driver.c (line 73) - [public] vbe_write

```
vbe_write(3, 32);
vbe_write(4, 0x01 | 0x40 | 0x80);

uint64_t mid = ds_call(SVC_SYS, OP_MAPMMIO, fb_phys,
                      (uint64_t)w * h * 4, 0x45000000ULL);
ds_poll();

ds_msg_t mr;

if (!ds_take(mid, &mr) || mr.status != 0) {
    return false;
}
// ... (truncated, see source file for more)
```

drivers/vgpu/vgpu_driver.c (line 74) - [public] vbe_write

```
vbe_write(4, 0x01 | 0x40 | 0x80);

uint64_t mid = ds_call(SVC_SYS, OP_MAPMMIO, fb_phys,
                      (uint64_t)w * h * 4, 0x45000000ULL);
ds_poll();

ds_msg_t mr;

if (!ds_take(mid, &mr) || mr.status != 0) {
    return false;
}

// ... (truncated, see source file for more)
```

```

drivers/vgpu/vgpu_driver.c (line 78) - [public] ds_poll
    ds_poll();

    ds_msg_t mr;

    if (!ds_take(mid, &mr) || mr.status != 0) {
        return false;
    }

    g_info.width = w;
    g_info.height = h;
    g_info.bpp = 32;
    g_info.stride = w;
    // ... (truncated, see source file for more)

drivers/vgpu/vgpu_driver.c (line 82) - [public] if
    if (!ds_take(mid, &mr) || mr.status != 0) {
        return false;
    }

    g_info.width = w;
    g_info.height = h;
    g_info.bpp = 32;
    g_info.stride = w;
    g_info.fb_phys = fb_phys;
    g_info.fb = (void *)0x45000000ULL;

    g_ready = true;
    // ... (truncated, see source file for more)

drivers/vgpu/vgpu_driver.c (line 106) - [public] if
    if (!g_ready || x >= g_info.width || y >= g_info.height) {
        return;
    }

    ((uint32_t *)g_info.fb)[y * g_info.stride + x] = color;
}

void vgpu_clear(uint32_t color)
{
    if (!g_ready) {
        return;
    }
    // ... (truncated, see source file for more)

drivers/vgpu/vgpu_driver.c (line 115) - [public] if
    if (!g_ready) {
        return;
    }

    uint32_t *fb = (uint32_t *)g_info.fb;
    uint32_t n = g_info.width * g_info.height;

    for (uint32_t i = 0; i < n; i++) {
        fb[i] = color;
    }
}

// ... (truncated, see source file for more)

```

```

drivers/vgpu/vgpu_driver.c (line 133) - [public] ds_poll
    ds_poll();

    ds_msg_t ar;

    if (!ds_take(aid, &ar) || ar.status == 0 ? ar.arg0 == 0 : 1) {
        return -1;
    }

    uint64_t va = ar.arg0;

    uint64_t pid = ds_call(SVC_SYS, OP_PAGEPHYS, va, 0, 0);
    ds_poll();
    // ... (truncated, see source file for more)

```

```

drivers/vgpu/vgpu_driver.c (line 137) - [public] if
    if (!ds_take(aid, &ar) || ar.status == 0 ? ar.arg0 == 0 : 1) {
        return -1;
    }

    uint64_t va = ar.arg0;

    uint64_t pid = ds_call(SVC_SYS, OP_PAGEPHYS, va, 0, 0);
    ds_poll();

    ds_msg_t pr;

    if (!ds_take(pid, &pr) || pr.status != 0) {
        // ... (truncated, see source file for more)
    }

```

```

drivers/vgpu/vgpu_driver.c (line 144) - [public] ds_poll
    ds_poll();

    ds_msg_t pr;

    if (!ds_take(pid, &pr) || pr.status != 0) {
        return -1;
    }

    for (int32_t i = 0; i < VGPU_MAX_SURFACES; i++) {
        if (!g_surf[i].used) {
            g_surf[i].used = true;
            g_surf[i].phys = pr.arg0;
        }
        // ... (truncated, see source file for more)
    }

```

```

drivers/vgpu/vgpu_driver.c (line 148) - [public] if
    if (!ds_take(pid, &pr) || pr.status != 0) {
        return -1;
    }

    for (int32_t i = 0; i < VGPU_MAX_SURFACES; i++) {
        if (!g_surf[i].used) {
            g_surf[i].used = true;
            g_surf[i].phys = pr.arg0;
            g_surf[i].va = (void *)va;
            g_surf[i].w = w;
            g_surf[i].h = h;
        }
        // ... (truncated, see source file for more)
    }

```

```

drivers/vgpu/vgpu_driver.c (line 153) - [public] if
    if (!g_surf[i].used) {
        g_surf[i].used = true;
        g_surf[i].phys = pr.arg0;
        g_surf[i].va = (void *)va;
        g_surf[i].w = w;
        g_surf[i].h = h;

        *phys_out = pr.arg0;
        return i;
    }
}

// ... (truncated, see source file for more)

drivers/vgpu/vgpu_driver.c (line 170) - [public] if
if (id < 0 || id >= VGPU_MAX_SURFACES || !g_surf[id].used) {
    return false;
}

vgpu_surface_t *s = &g_surf[id];
uint32_t *src = (uint32_t *)s->va;

for (uint32_t yy = 0; yy < s->h && y + yy < g_info.height; yy++) {
    for (uint32_t xx = 0; xx < s->w && x + xx < g_info.width; xx++) {
        uint32_t c = src[yy * s->w + xx];

        if ((c & 0xFF000000) != 0) {
            // ... (truncated, see source file for more)
        }
    }
}

drivers/vgpu/vgpu_driver.c (line 181) - [public] if
    if ((c & 0xFF000000) != 0) {
        ((uint32_t *)g_info.fb)[(y + yy) * g_info.stride + (x + xx)] = c;
    }
}

return true;
}

void vgpu_process_ring(volatile vgpu_slot_t *ring)
{
    if (ring == (void *)0) {
        // ... (truncated, see source file for more)
    }
}

drivers/vgpu/vgpu_driver.c (line 192) - [public] if
if (ring == (void *)0) {
    return;
}

for (uint32_t i = 0; i < VGPU_RING_SLOTS; i++) {
    volatile vgpu_slot_t *s = &ring[i];

    if (s->id == 0 || s->done != 0) {
        continue;
    }

    switch (s->op) {
        // ... (truncated, see source file for more)
    }
}

```



```

drivers/vgpu/vgpu_driver.c (line 199) - [public] if
    if (s->id == 0 || s->done != 0) {
        continue;
    }

    switch (s->op) {
    case VGPU_INFO:
        s->r0 = ((uint64_t)g_info.width << 32) | g_info.height;
        s->r1 = g_info.fb_phys;
        s->status = 0;
        break;

        case VGPU_SURF_CREATE: {
// ... (truncated, see source file for more)

drivers/vgpu/vgpu_driver.c (line 203) - [public] switch
    switch (s->op) {
    case VGPU_INFO:
        s->r0 = ((uint64_t)g_info.width << 32) | g_info.height;
        s->r1 = g_info.fb_phys;
        s->status = 0;
        break;

        case VGPU_SURF_CREATE: {
            uint64_t phys = 0;
            int32_t sid = vgpu_surface_create((uint32_t)s->a0,
                                              (uint32_t)s->a1, &phys);

// ... (truncated, see source file for more)

drivers/vgpu/vgpu_driver.c (line 215) - [public] if
    if (sid < 0) {
        s->status = -1;
    } else {
        s->r0 = (uint64_t)(uint32_t)sid;
        s->r1 = phys;
        s->status = 0;
    }
    break;
}

    case VGPU_PRESENT:
        s->status = vgpu_surface_present((int32_t)s->a0,
// ... (truncated, see source file for more)

drivers/vgpu/vgpu_driver.c (line 248) - [public] ds_init
    ds_init(params_va);

    if (!initd) {
        if (vgpu_init(1024, 768)) {
            uint64_t rid = ds_call(SVC_SYS, OP_ALLOC, 1, 0, 0);
            ds_poll();

            ds_msg_t rr;

            if (ds_take(rid, &rr) && rr.status == 0) {
                g_ring = (volatile vgpu_slot_t *)rr.arg0;
            }

// ... (truncated, see source file for more)

```

```

drivers/vgpu/vgpu_driver.c (line 250) - [public] if
    if (!inited) {
        if (vgpu_init(1024, 768)) {
            uint64_t rid = ds_call(SVC_SYS, OP_ALLOC, 1, 0, 0);
            ds_poll();

            ds_msg_t rr;

            if (ds_take(rid, &rr) && rr.status == 0) {
                g_ring = (volatile vgpu_slot_t *)rr.arg0;

                uint64_t pid = ds_call(SVC_SYS, OP_PAGEPHYS, rr.arg0, 0, 0);
                ds_poll();
            }
            // ... (truncated, see source file for more)
        }
    }

drivers/vgpu/vgpu_driver.c (line 251) - [public] if
    if (vgpu_init(1024, 768)) {
        uint64_t rid = ds_call(SVC_SYS, OP_ALLOC, 1, 0, 0);
        ds_poll();

        ds_msg_t rr;

        if (ds_take(rid, &rr) && rr.status == 0) {
            g_ring = (volatile vgpu_slot_t *)rr.arg0;

            uint64_t pid = ds_call(SVC_SYS, OP_PAGEPHYS, rr.arg0, 0, 0);
            ds_poll();

            // ... (truncated, see source file for more)
        }
    }

drivers/vgpu/vgpu_driver.c (line 253) - [public] ds_poll
    ds_poll();

    ds_msg_t rr;

    if (ds_take(rid, &rr) && rr.status == 0) {
        g_ring = (volatile vgpu_slot_t *)rr.arg0;

        uint64_t pid = ds_call(SVC_SYS, OP_PAGEPHYS, rr.arg0, 0, 0);
        ds_poll();

        ds_msg_t pr;

        // ... (truncated, see source file for more)
    }

drivers/vgpu/vgpu_driver.c (line 257) - [public] if
    if (ds_take(rid, &rr) && rr.status == 0) {
        g_ring = (volatile vgpu_slot_t *)rr.arg0;

        uint64_t pid = ds_call(SVC_SYS, OP_PAGEPHYS, rr.arg0, 0, 0);
        ds_poll();

        ds_msg_t pr;

        if (ds_take(pid, &pr) && pr.status == 0) {
            ds_call(SVC_VGPU, VGPU_REGISTER, pr.arg0, 0, 0);
            ds_poll();

            // ... (truncated, see source file for more)
        }
    }

```

```

drivers/vgpu/vgpu_driver.c (line 261) - [public] ds_poll
    ds_poll();

    ds_msg_t pr;

    if (ds_take(pid, &pr) && pr.status == 0) {
        ds_call(SVC_VGPU, VGPU_REGISTER, pr.arg0, 0, 0);
        ds_poll();

        ds_msg_t zr;
        ds_take(0, &zr);
    }
}
// ... (truncated, see source file for more)

drivers/vgpu/vgpu_driver.c (line 265) - [public] if
    if (ds_take(pid, &pr) && pr.status == 0) {
        ds_call(SVC_VGPU, VGPU_REGISTER, pr.arg0, 0, 0);
        ds_poll();

        ds_msg_t zr;
        ds_take(0, &zr);
    }
}

    vgpu_clear(0xFF101020);
}

// ... (truncated, see source file for more)

drivers/vgpu/vgpu_driver.c (line 266) - [public] ds_call
    ds_call(SVC_VGPU, VGPU_REGISTER, pr.arg0, 0, 0);
    ds_poll();

    ds_msg_t zr;
    ds_take(0, &zr);
}

    vgpu_clear(0xFF101020);
}

    initied = true;
// ... (truncated, see source file for more)

drivers/vgpu/vgpu_driver.c (line 267) - [public] ds_poll
    ds_poll();

    ds_msg_t zr;
    ds_take(0, &zr);
}

    vgpu_clear(0xFF101020);
}

    initied = true;
}
// ... (truncated, see source file for more)

```

```
drivers/vgpu/vgpu_driver.c (line 270) - [public] ds_take
        ds_take(0, &zr);
    }
}

    vgpu_clear(0xFF101020);
}
```

```
    initied = true;
}
```

```
vgpu_process_ring(g_ring);
```

```
// ... (truncated, see source file for more)
```

```
drivers/vgpu/vgpu_driver.c (line 274) - [public] vgpu_clear
    vgpu_clear(0xFF101020);
}
```

```
    initied = true;
}
```

```
vgpu_process_ring(g_ring);
```

```
ds_yield();
```

```
}
```

```
drivers/vgpu/vgpu_driver.c (line 280) - [public] vgpu_process_ring
    vgpu_process_ring(g_ring);
```

```
    ds_yield();
```

```
}
```

```
drivers/vgpu/vgpu_driver.c (line 282) - [public] ds_yield
    ds_yield();
}
```

```
kernel_Workspace/kernel/src/battery/init.c (line 8) - [public] if
```

```
    if (is_ready) {
        return true;
    }
```

```
    is_ready = true;
```

```
    return true;
```

```
}
```

```
kernel_Workspace/kernel/src/battery/operation.c (line 23) - [public] if
```

```
    if (!model.present) {
        return false;
    }
```

```
    *out = model;
```

```
    return true;
```

```
}
```

```
static const battery_backend_t model_backend = {
    .present = model_present,
    .status = model_status,
};

// ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/battery/operation.c (line 40) - [public] if
    if (b != (void *)0 && b->status != (void *)0) {
        backend = b;
    }
}

bool battery_present(void)
{
    return backend->present();
}

bool battery_status(battery_status_t *out)
{
    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/battery/operation.c (line 52) - [public] if
    if (out == (void *)0) {
        return false;
    }

    return backend->status(out);
}

bool battery_status_packed(uint64_t *a0, uint64_t *a1, uint64_t *a2)
{
    battery_status_t s;

    if (!battery_status(&s)) {
        // ... (truncated, see source file for more)
    }

kernel_Workspace/kernel/src/battery/operation.c (line 63) - [public] if
    if (!battery_status(&s)) {
        return false;
    }

    *a0 = ((uint64_t)s.present << 32)
        | ((uint64_t)s.state << 24)
        | (s.percent & 0xFF);

    *a1 = ((uint64_t)s.voltage_mv << 16) | (s.temp_c & 0xFFFF);
    *a2 = s.health;

    return true;
    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/battery/operation.c (line 79) - [public] if
    if (low_pct > 50) {
        return false;
    }

    low_threshold = low_pct;
    low_pending = false;

    return true;
}

uint32_t battery_threshold(void)
{
    // ... (truncated, see source file for more)
}

```

kernel_Workspace/kernel/src/battery/operation.c (line 103) - [public] if

```
    if (!model.present || model.state == BATT_STATE_CHARGING) {
        return;
    }

    tick_counter++;

    if (tick_counter % 2000 == 0 && model.percent > 0) {
        model.percent--;

        if (model.percent <= low_threshold && !low_pending) {
            low_pending = true;
        }
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/battery/operation.c (line 109) - [public] if

```
    if (tick_counter % 2000 == 0 && model.percent > 0) {
        model.percent--;

        if (model.percent <= low_threshold && !low_pending) {
            low_pending = true;
        }
    }
}
```

kernel_Workspace/kernel/src/battery/operation.c (line 112) - [public] if

```
    if (model.percent <= low_threshold && !low_pending) {
        low_pending = true;
    }
}
```

kernel_Workspace/kernel/src/bluetooth/init.c (line 14) - [public] if

```
    if (ret_len != 0 && ret == NULL) {
        return false;
    }

    for (uint8_t i = 0; i < 32; i++) {
        if (!bt_event_poll(buf, EVT_MAX, &len)) {
            continue;
        }

        if (len < 7 || buf[0] != HCI_EVT_PKT || buf[1] != EVT_CMD_COMPLETE) {
            continue;
        }
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/bluetooth/init.c (line 19) - [public] if

```
    if (!bt_event_poll(buf, EVT_MAX, &len)) {
        continue;
    }

    if (len < 7 || buf[0] != HCI_EVT_PKT || buf[1] != EVT_CMD_COMPLETE) {
        continue;
    }

    if (buf[4] != (uint8_t)(opcode & 0xFF) || buf[5] != (uint8_t)(opcode >> 8)) {
        continue;
    }

    // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/bluetooth/init.c (line 23) - [public] if
    if (len < 7 || buf[0] != HCI_EVT_PKT || buf[1] != EVT_CMD_COMPLETE) {
        continue;
    }

    if (buf[4] != (uint8_t)(opcode & 0xFF) || buf[5] != (uint8_t)(opcode >> 8)) {
        continue;
    }

    if (buf[6] != 0 || len < (uint8_t)(7 + ret_len)) {
        return false;
    }

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/bluetooth/init.c (line 27) - [public] if
    if (buf[4] != (uint8_t)(opcode & 0xFF) || buf[5] != (uint8_t)(opcode >> 8)) {
        continue;
    }

    if (buf[6] != 0 || len < (uint8_t)(7 + ret_len)) {
        return false;
    }

    for (uint8_t k = 0; k < ret_len; k++) {
        ret[k] = buf[7 + k];
    }

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/bluetooth/init.c (line 31) - [public] if
    if (buf[6] != 0 || len < (uint8_t)(7 + ret_len)) {
        return false;
    }

    for (uint8_t k = 0; k < ret_len; k++) {
        ret[k] = buf[7 + k];
    }

    return true;
}

return false;
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/bluetooth/init.c (line 47) - [public] if
    if (bt_is_ready) {
        return true;
    }

    bt_op_model_init();

    if (!bt_hci_cmd(HCI_RESET, NULL, 0)) {
        return false;
    }

    uint8_t tmp[8];
    if (!wait_cmd_complete(HCI_RESET, tmp, 0)) {
        // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/bluetooth/init.c (line 51) - [public] bt_op_model_init
    bt_op_model_init();

    if (!bt_hci_cmd(HCI_RESET, NULL, 0)) {
        return false;
    }

    uint8_t tmp[8];
    if (!wait_cmd_complete(HCI_RESET, tmp, 0)) {
        return false;
    }

    if (!bt_hci_cmd(HCI_READ_LOCAL_VERSION, NULL, 0)) {
        // ... (truncated, see source file for more)
    }

kernel_Workspace/kernel/src/bluetooth/init.c (line 53) - [public] if
    if (!bt_hci_cmd(HCI_RESET, NULL, 0)) {
        return false;
    }

    uint8_t tmp[8];
    if (!wait_cmd_complete(HCI_RESET, tmp, 0)) {
        return false;
    }

    if (!bt_hci_cmd(HCI_READ_LOCAL_VERSION, NULL, 0)) {
        return false;
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/bluetooth/init.c (line 58) - [public] if
    if (!wait_cmd_complete(HCI_RESET, tmp, 0)) {
        return false;
    }

    if (!bt_hci_cmd(HCI_READ_LOCAL_VERSION, NULL, 0)) {
        return false;
    }

    if (!wait_cmd_complete(HCI_READ_LOCAL_VERSION, tmp, 1)) {
        return false;
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/bluetooth/init.c (line 62) - [public] if
    if (!bt_hci_cmd(HCI_READ_LOCAL_VERSION, NULL, 0)) {
        return false;
    }

    if (!wait_cmd_complete(HCI_READ_LOCAL_VERSION, tmp, 1)) {
        return false;
    }

    saved_ver = tmp[0];

    if (!bt_hci_cmd(HCI_READ_BD_ADDR, NULL, 0)) {
        return false;
    }
    // ... (truncated, see source file for more)

```



```

kernel_Workspace/kernel/src/bluetooth/init.c (line 66) - [public] if
    if (!wait_cmd_complete(HCI_READ_LOCAL_VERSION, tmp, 1)) {
        return false;
    }

    saved_ver = tmp[0];

    if (!bt_hci_cmd(HCI_READ_BD_ADDR, NULL, 0)) {
        return false;
    }

    if (!wait_cmd_complete(HCI_READ_BD_ADDR, saved_addr, 6)) {
        return false;
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/bluetooth/init.c (line 72) - [public] if
    if (!bt_hci_cmd(HCI_READ_BD_ADDR, NULL, 0)) {
        return false;
    }

    if (!wait_cmd_complete(HCI_READ_BD_ADDR, saved_addr, 6)) {
        return false;
    }

    bt_is_ready = true;
    return true;
}

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/bluetooth/init.c (line 76) - [public] if
    if (!wait_cmd_complete(HCI_READ_BD_ADDR, saved_addr, 6)) {
        return false;
    }

    bt_is_ready = true;
    return true;
}

bool bt_ready(void)
{
    return bt_is_ready;
}

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/bluetooth/init.c (line 91) - [public] if
    if (hci_ver != NULL) {
        *hci_ver = saved_ver;
    }

    if (bdaddr != NULL) {
        for (uint8_t i = 0; i < 6; i++) {
            bdaddr[i] = saved_addr[i];
        }
    }
}

kernel_Workspace/kernel/src/bluetooth/init.c (line 95) - [public] if
    if (bdaddr != NULL) {
        for (uint8_t i = 0; i < 6; i++) {
            bdaddr[i] = saved_addr[i];
        }
    }
}

```

```
kernel_Workspace/kernel/src/bluetooth/operation.c (line 18) - [public] if
    if (data == NULL || len == 0 || len > EVT_MAX) {
        return false;
    }

    uint32_t next = (evt_head + 1) % EVT_Q;
    if (next == evt_tail) {
        return false;
    }

    for (uint8_t i = 0; i < len; i++) {
        evt_q[evt_head][i] = data[i];
    }
    // ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/bluetooth/operation.c (line 23) - [public] if
    if (next == evt_tail) {
        return false;
    }

    for (uint8_t i = 0; i < len; i++) {
        evt_q[evt_head][i] = data[i];
    }

    evt_len[evt_head] = len;
    evt_head = next;
    return true;
}
// ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/bluetooth/operation.c (line 49) - [public] if
    if (len != 0 && params == NULL) {
        return false;
    }

    switch (opcode) {
    case HCI_RESET:
        ev[0] = HCI_EVT_PKT;
        ev[1] = EVT_CMD_COMPLETE;
        ev[2] = 4;
        ev[3] = 1;
        ev[4] = opcode & 0xFF;
        ev[5] = opcode >> 8;
        // ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/bluetooth/operation.c (line 53) - [public] switch
    switch (opcode) {
    case HCI_RESET:
        ev[0] = HCI_EVT_PKT;
        ev[1] = EVT_CMD_COMPLETE;
        ev[2] = 4;
        ev[3] = 1;
        ev[4] = opcode & 0xFF;
        ev[5] = opcode >> 8;
        ev[6] = 0;
        return evt_push(ev, 7);

    case HCI_READ_LOCAL_VERSION:
        // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/bluetooth/operation.c (line 62) - [public] evt_push
    return evt_push(ev, 7);

    case HCI_READ_LOCAL_VERSION:
        ev[0] = HCI_EVT_PKT;
        ev[1] = EVT_CMD_COMPLETE;
        ev[2] = 12;
        ev[3] = 1;
        ev[4] = opcode & 0xFF;
        ev[5] = opcode >> 8;
        ev[6] = 0;
        ev[7] = hci_version;
        ev[8] = 0;
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/bluetooth/operation.c (line 80) - [public] evt_push
    return evt_push(ev, 15);

    case HCI_READ_BD_ADDR:
        ev[0] = HCI_EVT_PKT;
        ev[1] = EVT_CMD_COMPLETE;
        ev[2] = 10;
        ev[3] = 1;
        ev[4] = opcode & 0xFF;
        ev[5] = opcode >> 8;
        ev[6] = 0;
        for (uint8_t i = 0; i < 6; i++) {
            ev[7 + i] = bdaddr[i];
        }
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/bluetooth/operation.c (line 93) - [public] evt_push
    return evt_push(ev, 13);

    default:
        ev[0] = HCI_EVT_PKT;
        ev[1] = EVT_CMD_COMPLETE;
        ev[2] = 4;
        ev[3] = 1;
        ev[4] = opcode & 0xFF;
        ev[5] = opcode >> 8;
        ev[6] = 0;
        return evt_push(ev, 7);
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/bluetooth/operation.c (line 103) - [public] evt_push
    return evt_push(ev, 7);
}

bool bt_hci_cmd(uint16_t opcode, const uint8_t *params, uint8_t len)
{
    if (!op_ready) {
        return false;
    }

    return bt_op_model_run_cmd(opcode, params, len);
}
// ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/bluetooth/operation.c (line 109) - [public] if

```
    if (!op_ready) {
        return false;
    }

    return bt_op_model_run_cmd(opcode, params, len);
}

bool bt_event_poll(uint8_t *buf, uint8_t cap, uint8_t *len)
{
    if (buf == NULL || len == NULL || evt_head == evt_tail) {
        return false;
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/bluetooth/operation.c (line 113) - [public] bt_op_model_run_cmd

```
    return bt_op_model_run_cmd(opcode, params, len);
}

bool bt_event_poll(uint8_t *buf, uint8_t cap, uint8_t *len)
{
    if (buf == NULL || len == NULL || evt_head == evt_tail) {
        return false;
    }

    uint8_t n = evt_len[evt_tail];
    if (n > cap) {
        return false;
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/bluetooth/operation.c (line 118) - [public] if

```
    if (buf == NULL || len == NULL || evt_head == evt_tail) {
        return false;
    }

    uint8_t n = evt_len[evt_tail];
    if (n > cap) {
        return false;
    }

    for (uint8_t i = 0; i < n; i++) {
        buf[i] = evt_q[evt_tail][i];
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/bluetooth/operation.c (line 123) - [public] if

```
    if (n > cap) {
        return false;
    }

    for (uint8_t i = 0; i < n; i++) {
        buf[i] = evt_q[evt_tail][i];
    }

    *len = n;
    evt_tail = (evt_tail + 1) % EVT_Q;
    return true;
}
// ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/bluetooth/operation.c (line 138) - [public] if
    if (!op_ready || (len != 0 && data == NULL) || len > ACL_MAX) {
        return false;
    }

    uint32_t next = (acl_head + 1) % ACL_Q;
    if (next == acl_tail) {
        return false;
    }

    for (uint16_t i = 0; i < len; i++) {
        acl_q[acl_head][i] = data[i];
    }
    // ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/bluetooth/operation.c (line 143) - [public] if
    if (next == acl_tail) {
        return false;
    }

    for (uint16_t i = 0; i < len; i++) {
        acl_q[acl_head][i] = data[i];
    }

    acl_len[acl_head] = len;
    acl_head = next;
    return true;
}
// ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/bluetooth/operation.c (line 158) - [public] if
    if (data == NULL || len == NULL || acl_head == acl_tail) {
        return false;
    }

    uint16_t n = acl_len[acl_tail];
    if (n > cap) {
        return false;
    }

    for (uint16_t i = 0; i < n; i++) {
        data[i] = acl_q[acl_tail][i];
    }
    // ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/bluetooth/operation.c (line 163) - [public] if
    if (n > cap) {
        return false;
    }

    for (uint16_t i = 0; i < n; i++) {
        data[i] = acl_q[acl_tail][i];
    }

    *len = n;
    acl_tail = (acl_tail + 1) % ACL_Q;
    return true;
}
```

```
kernel_Workspace/kernel/src/camera/init.c (line 7) - [public] if
    if (is_ready) {
        return true;
    }

    is_ready = true;

    return true;
}
```

```
kernel_Workspace/kernel/src/camera/operation.c (line 43) - [public] if
    if (((x + y + f * 8) & 0xFF) < 16) {
        r = g = b = 255;
    }

    uint32_t i = (y * CAM_W + x) * CAM_BPP;

    framebuf[i + 0] = r;
    framebuf[i + 1] = g;
    framebuf[i + 2] = b;
}
}
// ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/camera/operation.c (line 58) - [public] if
    if (!streaming) {
        return false;
    }

    gen_frame();

    uint32_t n = sizeof(framebuf);

    if (cap < n) {
        n = cap;
    }

    // ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/camera/operation.c (line 62) - [public] gen_frame
    gen_frame();

    uint32_t n = sizeof(framebuf);

    if (cap < n) {
        n = cap;
    }

    uint8_t *dst = (uint8_t *)buf;

    for (uint32_t i = 0; i < n; i++) {
        dst[i] = framebuf[i];
    }
    // ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/camera/operation.c (line 66) - [public] if
    if (cap < n) {
        n = cap;
    }

    uint8_t *dst = (uint8_t *)buf;

    for (uint32_t i = 0; i < n; i++) {
        dst[i] = framebuf[i];
    }

    frame_id++;

    // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/camera/operation.c (line 78) - [public] if
    if (fid) {
        *fid = frame_id;
    }

    return true;
}

static const camera_backend_t model_backend = {
    .present = model_present,
    .start = model_start,
    .stop = model_stop,
    .frame = model_frame,
    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/camera/operation.c (line 96) - [public] if
    if (b != (void *)0 && b->frame != (void *)0) {
        backend = b;
    }
}

bool camera_present(void)
{
    return backend->present();
}

bool camera_caps_get(camera_caps_t *out)
{
    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/camera/operation.c (line 108) - [public] if
    if (out == (void *)0 || !camera_present()) {
        return false;
    }

    out->present = 1;
    out->width = CAM_W;
    out->height = CAM_H;
    out->format = CAM_FMT_RGB888;
    out->fps = 30;

    return true;
}
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/camera/operation.c (line 123) - [public] if
    if (!camera_present()) {
        return false;
    }

    return backend->start();
}

bool camera_stop(void)
{
    return backend->stop();
}

// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/camera/operation.c (line 137) - [public] if
    if (buf == (void *)0) {
        return false;
    }

    return backend->frame(buf, cap, fid);
}

bool camera_frame_to_phys(uint64_t phys, uint32_t cap, uint64_t *fid)
{
    return backend->frame(arch_phys_to_virt(phys), cap, fid);
}

kernel_Workspace/kernel/src/core-lang/ast.c (line 14) - [public] if
    if (a->pos + aligned > a->cap) {
        return (void *)0;
    }

    void *p = a->buf + a->pos;

    a->pos += aligned;

    return p;
}

ast_node_t *cl_node(arena_t *ar, uint8_t kind, uint32_t line)
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/ast.c (line 29) - [public] if
    if (n == (void *)0) {
        return (void *)0;
    }

    n->kind = kind;
    n->type.bits = 0;
    n->type.neg = false;
    n->line = line;
    n->op = 0;
    n->pool_kind = 0;
    n->is_array = false;
    n->value = 0;
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/bridge.c (line 3) - [public] kprintf
extern void kprintf(const char *fmt, ...);
extern int32_t k_input_key(void);
extern int32_t k_fs_read(const char *path, void *buf, uint32_t cap);
extern uint32_t k_getpid(void);
extern uint64_t k_world_cr3(void);
extern uint64_t k_kernel_cr3(void);
extern int64_t k_spawn(const char *path, uint32_t parent, uint64_t cr3);
extern int32_t k_ipc_send(uint32_t dst, uint64_t a0, uint64_t a1);
extern int32_t k_ipc_recv(uint64_t *a0, uint64_t *a1);
extern uint64_t k_tick(void);

extern uint64_t hdmi_submit_fill(uint32_t color, uint32_t x, uint32_t y,
    // ... (truncated, see source file for more)

```


kernel_Workspace/kernel/src/core-lang/bridge.c (line 4) - [public] k_input_key

```
extern int32_t k_input_key(void);
extern int32_t k_fs_read(const char *path, void *buf, uint32_t cap);
extern uint32_t k_getpid(void);
extern uint64_t k_world_cr3(void);
extern uint64_t k_kernel_cr3(void);
extern int64_t k_spawn(const char *path, uint32_t parent, uint64_t cr3);
extern int32_t k_ipc_send(uint32_t dst, uint64_t a0, uint64_t a1);
extern int32_t k_ipc_recv(uint64_t *a0, uint64_t *a1);
extern uint64_t k_tick(void);

extern uint64_t hdmi_submit_fill(uint32_t color, uint32_t x, uint32_t y,
                                uint32_t w, uint32_t h);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 5) - [public] k_fs_read

```
extern int32_t k_fs_read(const char *path, void *buf, uint32_t cap);
extern uint32_t k_getpid(void);
extern uint64_t k_world_cr3(void);
extern uint64_t k_kernel_cr3(void);
extern int64_t k_spawn(const char *path, uint32_t parent, uint64_t cr3);
extern int32_t k_ipc_send(uint32_t dst, uint64_t a0, uint64_t a1);
extern int32_t k_ipc_recv(uint64_t *a0, uint64_t *a1);
extern uint64_t k_tick(void);

extern uint64_t hdmi_submit_fill(uint32_t color, uint32_t x, uint32_t y,
                                uint32_t w, uint32_t h);
extern bool battery_status_packed(uint64_t *a0, uint64_t *a1, uint64_t *a2);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 6) - [public] k_getpid

```
extern uint32_t k_getpid(void);
extern uint64_t k_world_cr3(void);
extern uint64_t k_kernel_cr3(void);
extern int64_t k_spawn(const char *path, uint32_t parent, uint64_t cr3);
extern int32_t k_ipc_send(uint32_t dst, uint64_t a0, uint64_t a1);
extern int32_t k_ipc_recv(uint64_t *a0, uint64_t *a1);
extern uint64_t k_tick(void);

extern uint64_t hdmi_submit_fill(uint32_t color, uint32_t x, uint32_t y,
                                uint32_t w, uint32_t h);
extern bool battery_status_packed(uint64_t *a0, uint64_t *a1, uint64_t *a2);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 7) - [public] k_world_cr3

```
extern uint64_t k_world_cr3(void);
extern uint64_t k_kernel_cr3(void);
extern int64_t k_spawn(const char *path, uint32_t parent, uint64_t cr3);
extern int32_t k_ipc_send(uint32_t dst, uint64_t a0, uint64_t a1);
extern int32_t k_ipc_recv(uint64_t *a0, uint64_t *a1);
extern uint64_t k_tick(void);

extern uint64_t hdmi_submit_fill(uint32_t color, uint32_t x, uint32_t y,
                                uint32_t w, uint32_t h);
extern bool battery_status_packed(uint64_t *a0, uint64_t *a1, uint64_t *a2);

static cl_vm_t *g_vm = (void *)0;
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 8) - [public] k_kernel_cr3

```
extern uint64_t k_kernel_cr3(void);
extern int64_t k_spawn(const char *path, uint32_t parent, uint64_t cr3);
extern int32_t k_ipc_send(uint32_t dst, uint64_t a0, uint64_t a1);
extern int32_t k_ipc_recv(uint64_t *a0, uint64_t *a1);
extern uint64_t k_tick(void);

extern uint64_t hdmi_submit_fill(uint32_t color, uint32_t x, uint32_t y,
                                uint32_t w, uint32_t h);
extern bool battery_status_packed(uint64_t *a0, uint64_t *a1, uint64_t *a2);

static cl_vm_t *g_vm = (void *)0;
static uint8_t g_ring = 0;
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 9) - [public] k_spawn

```
extern int64_t k_spawn(const char *path, uint32_t parent, uint64_t cr3);
extern int32_t k_ipc_send(uint32_t dst, uint64_t a0, uint64_t a1);
extern int32_t k_ipc_recv(uint64_t *a0, uint64_t *a1);
extern uint64_t k_tick(void);

extern uint64_t hdmi_submit_fill(uint32_t color, uint32_t x, uint32_t y,
                                uint32_t w, uint32_t h);
extern bool battery_status_packed(uint64_t *a0, uint64_t *a1, uint64_t *a2);

static cl_vm_t *g_vm = (void *)0;
static uint8_t g_ring = 0;

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 10) - [public] k_ipc_send

```
extern int32_t k_ipc_send(uint32_t dst, uint64_t a0, uint64_t a1);
extern int32_t k_ipc_recv(uint64_t *a0, uint64_t *a1);
extern uint64_t k_tick(void);

extern uint64_t hdmi_submit_fill(uint32_t color, uint32_t x, uint32_t y,
                                uint32_t w, uint32_t h);
extern bool battery_status_packed(uint64_t *a0, uint64_t *a1, uint64_t *a2);

static cl_vm_t *g_vm = (void *)0;
static uint8_t g_ring = 0;

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 11) - [public] k_ipc_recv

```
extern int32_t k_ipc_recv(uint64_t *a0, uint64_t *a1);
extern uint64_t k_tick(void);

extern uint64_t hdmi_submit_fill(uint32_t color, uint32_t x, uint32_t y,
                                uint32_t w, uint32_t h);
extern bool battery_status_packed(uint64_t *a0, uint64_t *a1, uint64_t *a2);

static cl_vm_t *g_vm = (void *)0;
static uint8_t g_ring = 0;

static uint64_t br_put(uint64_t v, uint64_t b, uint64_t c,
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 12) - [public] k_tick

```
extern uint64_t k_tick(void);

extern uint64_t hdmi_submit_fill(uint32_t color, uint32_t x, uint32_t y,
                                uint32_t w, uint32_t h);
extern bool battery_status_packed(uint64_t *a0, uint64_t *a1, uint64_t *a2);

static cl_vm_t *g_vm = (void *)0;
static uint8_t g_ring = 0;

static uint64_t br_put(uint64_t v, uint64_t b, uint64_t c,
                      uint64_t d, uint64_t e, uint64_t f)
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 16) - [public] battery_status_packed

```
extern bool battery_status_packed(uint64_t *a0, uint64_t *a1, uint64_t *a2);

static cl_vm_t *g_vm = (void *)0;
static uint8_t g_ring = 0;

static uint64_t br_put(uint64_t v, uint64_t b, uint64_t c,
                      uint64_t d, uint64_t e, uint64_t f)
{
    (void)b; (void)c; (void)d; (void)e; (void)f;
    kprintf("%llu\n", (unsigned long long)v);
    return 0;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 26) - [public] kprintf

```
kprintf("%llu\n", (unsigned long long)v);
return 0;
}

static uint64_t br_putc(uint64_t v, uint64_t b, uint64_t c,
                      uint64_t d, uint64_t e, uint64_t f)
{
    (void)b; (void)c; (void)d; (void)e; (void)f;
    kprintf("%c", (char)v);
    return 0;
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 34) - [public] kprintf

```
kprintf("%c", (char)v);
return 0;
}

static uint64_t br_log(uint64_t ptr, uint64_t b, uint64_t c,
                      uint64_t d, uint64_t e, uint64_t f)
{
    (void)b; (void)c; (void)d; (void)e; (void)f;

    char buf[128];

    if (g_ring == 0) {
        // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 45) - [public] if
    if (g_ring == 0) {
        const char *s = (const char *)ptr;
        int i = 0;

        while (s[i] && i < 127) {
            buf[i] = s[i];
            i++;
        }

        buf[i] = '\\0';
    } else {
        extern bool k_user_cstr(uint64_t ptr, char *buf, uint32_t cap);
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/bridge.c (line 49) - [public] while
        while (s[i] && i < 127) {
            buf[i] = s[i];
            i++;
        }

        buf[i] = '\\0';
    } else {
        extern bool k_user_cstr(uint64_t ptr, char *buf, uint32_t cap);

        if (!k_user_cstr(ptr, buf, sizeof(buf))) {
            return (uint64_t)-1;
        }
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/bridge.c (line 56) - [public] k_user_cstr
        extern bool k_user_cstr(uint64_t ptr, char *buf, uint32_t cap);

        if (!k_user_cstr(ptr, buf, sizeof(buf))) {
            return (uint64_t)-1;
        }
    }

    kprintf("%s\\n", buf);
    return 0;
}

static uint64_t br_key(uint64_t a, uint64_t b, uint64_t c,
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/bridge.c (line 58) - [public] if
        if (!k_user_cstr(ptr, buf, sizeof(buf))) {
            return (uint64_t)-1;
        }
    }

    kprintf("%s\\n", buf);
    return 0;
}

static uint64_t br_key(uint64_t a, uint64_t b, uint64_t c,
    uint64_t d, uint64_t e, uint64_t f)
{
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 63) - [public] kprintf
    kprintf("%s\n", buf);
    return 0;
}

static uint64_t br_key(uint64_t a, uint64_t b, uint64_t c,
                      uint64_t d, uint64_t e, uint64_t f)
{
    (void)a; (void)b; (void)c; (void)d; (void)e; (void)f;

    int32_t k = k_input_key();

    return k < 0 ? 0 : (uint64_t)k;
    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/core-lang/bridge.c (line 81) - [public] k_getpid
    return k_getpid();
}

static uint64_t br_tick(uint64_t a, uint64_t b, uint64_t c,
                      uint64_t d, uint64_t e, uint64_t f)
{
    (void)a; (void)b; (void)c; (void)d; (void)e; (void)f;
    return k_tick();
}

static uint64_t br_exit(uint64_t code, uint64_t b, uint64_t c,
                      uint64_t d, uint64_t e, uint64_t f)
    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/core-lang/bridge.c (line 88) - [public] k_tick
    return k_tick();
}

static uint64_t br_exit(uint64_t code, uint64_t b, uint64_t c,
                      uint64_t d, uint64_t e, uint64_t f)
{
    (void)b; (void)c; (void)d; (void)e; (void)f;

    if (g_vm != (void *)0) {
        g_vm->halt_req = true;
    }

    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/core-lang/bridge.c (line 96) - [public] if
    if (g_vm != (void *)0) {
        g_vm->halt_req = true;
    }

    return code;
}

static uint64_t br_spawn(uint64_t ptr, uint64_t b, uint64_t c,
                      uint64_t d, uint64_t e, uint64_t f)
{
    (void)b; (void)c; (void)d; (void)e; (void)f;

    // ... (truncated, see source file for more)
}

```

```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 108) - [public] if
    if (g_ring >= 3) {
        return (uint64_t)-1;
    }

    uint64_t cr3 = (g_ring == 0) ? k_kernel_cr3() : k_world_cr3();

    return (uint64_t)k_spawn((const char *)ptr, k_getpid(), cr3);
}

static uint64_t br_ipc_send(uint64_t dst, uint64_t a0, uint64_t a1,
                           uint64_t d, uint64_t e, uint64_t f)
{
    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/core-lang/bridge.c (line 114) - [public] return
    return (uint64_t)k_spawn((const char *)ptr, k_getpid(), cr3);
}

static uint64_t br_ipc_send(uint64_t dst, uint64_t a0, uint64_t a1,
                           uint64_t d, uint64_t e, uint64_t f)
{
    (void)d; (void)e; (void)f;
    return (uint64_t)(int64_t)k_ipc_send((uint32_t)dst, a0, a1);
}

static uint64_t br_ipc_recv(uint64_t a, uint64_t b, uint64_t c,
                           uint64_t d, uint64_t e, uint64_t f)
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/bridge.c (line 121) - [public] return
    return (uint64_t)(int64_t)k_ipc_send((uint32_t)dst, a0, a1);
}

static uint64_t br_ipc_recv(uint64_t a, uint64_t b, uint64_t c,
                           uint64_t d, uint64_t e, uint64_t f)
{
    (void)a; (void)b; (void)c; (void)d; (void)e; (void)f;

    uint64_t a0 = 0, a1 = 0;
    int32_t from = k_ipc_recv(&a0, &a1);

    return (uint64_t)(int64_t)from;
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/bridge.c (line 140) - [public] if
    if (g_ring >= 3) {
        return (uint64_t)-1;
    }

    return (uint64_t)(int64_t)k_fs_read((const char *)path,
                                       (void *)buf, (uint32_t)cap);
}

static uint64_t br_vfill(uint64_t color, uint64_t xy, uint64_t wh,
                        uint64_t d, uint64_t e, uint64_t f)
{
    (void)d; (void)e; (void)f;
    // ... (truncated, see source file for more)
}

```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 153) - [public] if

```
    if (g_ring >= 3) {
        return (uint64_t)-1;
    }

    return hdmi_submit_fill((uint32_t)color,
                           (uint32_t)(xy & 0xFFFF),
                           (uint32_t)(xy >> 16),
                           (uint32_t)(wh & 0xFFFF),
                           (uint32_t)(wh >> 16));
}

static uint64_t br_bat(uint64_t a, uint64_t b, uint64_t c,
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 171) - [public] if

```
    if (!battery_status_packed(&a0, &a1, &a2)) {
        return (uint64_t)-1;
    }

    return a0;
}
```

```
int cl_bridge_add(cl_vm_t *vm, const char *name, cl_ext_fn fn)
{
    return cl_vm_register_extern(vm, name, fn);
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 181) - [public] cl_vm_register_extern

```
    return cl_vm_register_extern(vm, name, fn);
}
```

```
int cl_bridge_init(cl_vm_t *vm, uint8_t ring)
{
    g_vm = vm;
    g_ring = ring;

    cl_bridge_add(vm, "put", br_put);
    cl_bridge_add(vm, "putc", br_putc);
    cl_bridge_add(vm, "log", br_log);
    cl_bridge_add(vm, "key", br_key);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 189) - [public] cl_bridge_add

```
cl_bridge_add(vm, "put", br_put);
cl_bridge_add(vm, "putc", br_putc);
cl_bridge_add(vm, "log", br_log);
cl_bridge_add(vm, "key", br_key);
cl_bridge_add(vm, "tick", br_tick);
cl_bridge_add(vm, "bat", br_bat);
cl_bridge_add(vm, "exit", br_exit);

if (ring <= 1) {
    cl_bridge_add(vm, "getpid", br_getpid);
    cl_bridge_add(vm, "spawn", br_spawn);
    cl_bridge_add(vm, "ipc_send", br_ipc_send);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 190) - [public] cl_bridge_add

```
cl_bridge_add(vm, "putc", br_putc);
cl_bridge_add(vm, "log", br_log);
cl_bridge_add(vm, "key", br_key);
cl_bridge_add(vm, "tick", br_tick);
cl_bridge_add(vm, "bat", br_bat);
cl_bridge_add(vm, "exit", br_exit);

if (ring <= 1) {
    cl_bridge_add(vm, "getpid", br_getpid);
    cl_bridge_add(vm, "spawn", br_spawn);
    cl_bridge_add(vm, "ipc_send", br_ipc_send);
    cl_bridge_add(vm, "ipc_recv", br_ipc_recv);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 191) - [public] cl_bridge_add

```
cl_bridge_add(vm, "log", br_log);
cl_bridge_add(vm, "key", br_key);
cl_bridge_add(vm, "tick", br_tick);
cl_bridge_add(vm, "bat", br_bat);
cl_bridge_add(vm, "exit", br_exit);

if (ring <= 1) {
    cl_bridge_add(vm, "getpid", br_getpid);
    cl_bridge_add(vm, "spawn", br_spawn);
    cl_bridge_add(vm, "ipc_send", br_ipc_send);
    cl_bridge_add(vm, "ipc_recv", br_ipc_recv);
    cl_bridge_add(vm, "fsread", br_fsread);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 192) - [public] cl_bridge_add

```
cl_bridge_add(vm, "key", br_key);
cl_bridge_add(vm, "tick", br_tick);
cl_bridge_add(vm, "bat", br_bat);
cl_bridge_add(vm, "exit", br_exit);

if (ring <= 1) {
    cl_bridge_add(vm, "getpid", br_getpid);
    cl_bridge_add(vm, "spawn", br_spawn);
    cl_bridge_add(vm, "ipc_send", br_ipc_send);
    cl_bridge_add(vm, "ipc_recv", br_ipc_recv);
    cl_bridge_add(vm, "fsread", br_fsread);
    cl_bridge_add(vm, "vfill", br_vfill);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 193) - [public] cl_bridge_add

```
cl_bridge_add(vm, "tick", br_tick);
cl_bridge_add(vm, "bat", br_bat);
cl_bridge_add(vm, "exit", br_exit);

if (ring <= 1) {
    cl_bridge_add(vm, "getpid", br_getpid);
    cl_bridge_add(vm, "spawn", br_spawn);
    cl_bridge_add(vm, "ipc_send", br_ipc_send);
    cl_bridge_add(vm, "ipc_recv", br_ipc_recv);
    cl_bridge_add(vm, "fsread", br_fsread);
    cl_bridge_add(vm, "vfill", br_vfill);
}
// ... (truncated, see source file for more)
```


kernel_Workspace/kernel/src/core-lang/bridge.c (line 194) - [public] cl_bridge_add

```
cl_bridge_add(vm, "bat", br_bat);
cl_bridge_add(vm, "exit", br_exit);

if (ring <= 1) {
    cl_bridge_add(vm, "getpid", br_getpid);
    cl_bridge_add(vm, "spawn", br_spawn);
    cl_bridge_add(vm, "ipc_send", br_ipc_send);
    cl_bridge_add(vm, "ipc_recv", br_ipc_recv);
    cl_bridge_add(vm, "fsread", br_fsread);
    cl_bridge_add(vm, "vfill", br_vfill);
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 195) - [public] cl_bridge_add

```
cl_bridge_add(vm, "exit", br_exit);

if (ring <= 1) {
    cl_bridge_add(vm, "getpid", br_getpid);
    cl_bridge_add(vm, "spawn", br_spawn);
    cl_bridge_add(vm, "ipc_send", br_ipc_send);
    cl_bridge_add(vm, "ipc_recv", br_ipc_recv);
    cl_bridge_add(vm, "fsread", br_fsread);
    cl_bridge_add(vm, "vfill", br_vfill);
}

return 0;
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 197) - [public] if

```
if (ring <= 1) {
    cl_bridge_add(vm, "getpid", br_getpid);
    cl_bridge_add(vm, "spawn", br_spawn);
    cl_bridge_add(vm, "ipc_send", br_ipc_send);
    cl_bridge_add(vm, "ipc_recv", br_ipc_recv);
    cl_bridge_add(vm, "fsread", br_fsread);
    cl_bridge_add(vm, "vfill", br_vfill);
}

return 0;
}
```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 198) - [public] cl_bridge_add

```
cl_bridge_add(vm, "getpid", br_getpid);
cl_bridge_add(vm, "spawn", br_spawn);
cl_bridge_add(vm, "ipc_send", br_ipc_send);
cl_bridge_add(vm, "ipc_recv", br_ipc_recv);
cl_bridge_add(vm, "fsread", br_fsread);
cl_bridge_add(vm, "vfill", br_vfill);
}

return 0;
}
```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 199) - [public] cl_bridge_add

```
cl_bridge_add(vm, "spawn", br_spawn);
cl_bridge_add(vm, "ipc_send", br_ipc_send);
cl_bridge_add(vm, "ipc_recv", br_ipc_recv);
cl_bridge_add(vm, "fsread", br_fsread);
cl_bridge_add(vm, "vfill", br_vfill);
}

return 0;
}
```

```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 200) - [public] cl_bridge_add
    cl_bridge_add(vm, "ipc_send", br_ipc_send);
    cl_bridge_add(vm, "ipc_recv", br_ipc_recv);
    cl_bridge_add(vm, "fsread", br_fsread);
    cl_bridge_add(vm, "vfill", br_vfill);
}

    return 0;
}

kernel_Workspace/kernel/src/core-lang/bridge.c (line 201) - [public] cl_bridge_add
    cl_bridge_add(vm, "ipc_recv", br_ipc_recv);
    cl_bridge_add(vm, "fsread", br_fsread);
    cl_bridge_add(vm, "vfill", br_vfill);
}

    return 0;
}

kernel_Workspace/kernel/src/core-lang/bridge.c (line 202) - [public] cl_bridge_add
    cl_bridge_add(vm, "fsread", br_fsread);
    cl_bridge_add(vm, "vfill", br_vfill);
}

    return 0;
}

kernel_Workspace/kernel/src/core-lang/bridge.c (line 203) - [public] cl_bridge_add
    cl_bridge_add(vm, "vfill", br_vfill);
}

    return 0;
}

kernel_Workspace/kernel/src/core-lang/codegen.c (line 22) - [public] if
    if (P->code_len >= P->code_cap) {
        return 0;
    }

    cl_insn_t in;

    in.op = op;
    in.bits = bits;
    in.a = a;
    in.b = b;
    in.c = c;

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 45) - [public] if
    if (P->gnames[i][k] != name[k]) {
        same = false;
        break;
    }
}

    if (same) return (int)i;
}

    return -1;
}

    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 63) - [public] if
    if (g->locals[i].name[k] != name[k]) {
        same = false;
        break;
    }
}

    if (same) return i;
}

return -1;
}

// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 79) - [public] if
    if (i >= 0) {
        return g->locals[i].slot;
    }

    if (g->nlocals >= 32) {
        return 0;
    }

    for (int k = 0; name[k] && k < 63; k++) {
        g->locals[g->nlocals].name[k] = name[k];
        g->locals[g->nlocals].name[k + 1] = '\0';
    }
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 83) - [public] if
    if (g->nlocals >= 32) {
        return 0;
    }

    for (int k = 0; name[k] && k < 63; k++) {
        g->locals[g->nlocals].name[k] = name[k];
        g->locals[g->nlocals].name[k + 1] = '\0';
    }

    g->locals[g->nlocals].bits = bits;
    g->locals[g->nlocals].slot = g->maxslots;

    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 112) - [public] if
    if (neg) {
        uint8_t z[64] = { 0 };
        uint8_t sub[64];
        cl_sub(sub, z, P->consts[P->nconsts].b, w);
        for (int i = 0; i < w; i++) P->consts[P->nconsts].b[i] = sub[i];
    }

    cl_mask(P->consts[P->nconsts].b, w, bits);

    P->const_bits[P->nconsts] = bits;

    return (uint32_t)P->nconsts++;
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 115) - [public] cl_sub
    cl_sub(sub, z, P->consts[P->nconsts].b, w);
    for (int i = 0; i < w; i++) P->consts[P->nconsts].b[i] = sub[i];
}

    cl_mask(P->consts[P->nconsts].b, w, bits);

    P->const_bits[P->nconsts] = bits;

    return (uint32_t)P->nconsts++;
}

static void gen_expr(cg_t *g, ast_node_t *n, uint8_t bits);
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 119) - [public] cl_mask
    cl_mask(P->consts[P->nconsts].b, w, bits);

    P->const_bits[P->nconsts] = bits;

    return (uint32_t)P->nconsts++;
}

static void gen_expr(cg_t *g, ast_node_t *n, uint8_t bits);

static void gen_expr(cg_t *g, ast_node_t *n, uint8_t bits)
{
    cl_prog_t *P = g->P;
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 126) - [public] gen_expr
static void gen_expr(cg_t *g, ast_node_t *n, uint8_t bits);

static void gen_expr(cg_t *g, ast_node_t *n, uint8_t bits)
{
    cl_prog_t *P = g->P;

    if (n == (void *)0) return;

    switch (n->kind) {
    case ND_NUM:
        emit(g, OP_CONST, bits, (uint16_t)add_const(g, n->value, bits, false), 0, 0);
        break;
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 134) - [public] switch
    switch (n->kind) {
    case ND_NUM:
        emit(g, OP_CONST, bits, (uint16_t)add_const(g, n->value, bits, false), 0, 0);
        break;

    case ND_VAR: {
        int li = find_local(g, n->name);

        if (li >= 0) {
            emit(g, OP_LOAD, bits, g->locals[li].slot, 0, 0);
        } else {
            int gi = find_global(P, n->name);
            // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 136) - [public] emit
    emit(g, OP_CONST, bits, (uint16_t)add_const(g, n->value, bits, false), 0, 0);
    break;

case ND_VAR: {
    int li = find_local(g, n->name);

    if (li >= 0) {
        emit(g, OP_LOAD, bits, g->locals[li].slot, 0, 0);
    } else {
        int gi = find_global(P, n->name);

        if (gi >= 0) {
            // ... (truncated, see source file for more)
        }
    }
}

kernel_Workspace/kernel/src/core-lang/codegen.c (line 142) - [public] if
    if (li >= 0) {
        emit(g, OP_LOAD, bits, g->locals[li].slot, 0, 0);
    } else {
        int gi = find_global(P, n->name);

        if (gi >= 0) {
            emit(g, OP_GLOAD, bits, (uint16_t)gi, 0, 0);
        } else {
            emit(g, OP_CONST, bits, (uint16_t)add_const(g, 0, bits, false), 0, 0);
        }
    }
    break;
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 143) - [public] emit
    emit(g, OP_LOAD, bits, g->locals[li].slot, 0, 0);
} else {
    int gi = find_global(P, n->name);

    if (gi >= 0) {
        emit(g, OP_GLOAD, bits, (uint16_t)gi, 0, 0);
    } else {
        emit(g, OP_CONST, bits, (uint16_t)add_const(g, 0, bits, false), 0, 0);
    }
}
break;
}
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 147) - [public] if
    if (gi >= 0) {
        emit(g, OP_GLOAD, bits, (uint16_t)gi, 0, 0);
    } else {
        emit(g, OP_CONST, bits, (uint16_t)add_const(g, 0, bits, false), 0, 0);
    }
}
break;
}

case ND_POOL:
    if (n->pool_kind == 0) {
        emit(g, OP_GLOAD, bits, (uint16_t)find_global(P, n->name), 0, 0);
    }
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 148) - [public] emit
    emit(g, OP_GLOAD, bits, (uint16_t)gi, 0, 0);
} else {
    emit(g, OP_CONST, bits, (uint16_t)add_const(g, 0, bits, false), 0, 0);
}
}
break;
}

case ND_POOL:
    if (n->pool_kind == 0) {
        emit(g, OP_GLOAD, bits, (uint16_t)find_global(P, n->name), 0, 0);
    } else if (n->pool_kind == 1) {
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 150) - [public] emit
    emit(g, OP_CONST, bits, (uint16_t)add_const(g, 0, bits, false), 0, 0);
}
}
break;
}

case ND_POOL:
    if (n->pool_kind == 0) {
        emit(g, OP_GLOAD, bits, (uint16_t)find_global(P, n->name), 0, 0);
    } else if (n->pool_kind == 1) {
        int gi = (uint16_t)find_global(P, n->name);
        emit(g, OP_GLOAD, bits, (uint16_t)gi, 0, 0);
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 157) - [public] if
    if (n->pool_kind == 0) {
        emit(g, OP_GLOAD, bits, (uint16_t)find_global(P, n->name), 0, 0);
    } else if (n->pool_kind == 1) {
        int gi = (uint16_t)find_global(P, n->name);
        emit(g, OP_GLOAD, bits, (uint16_t)gi, 0, 0);
        emit(g, OP_NEG, bits, 0, 0, 0);
        emit(g, OP_GSTORE, bits, (uint16_t)gi, 0, 0);
        emit(g, OP_GLOAD, bits, (uint16_t)gi, 0, 0);
    } else {
        int gi = (uint16_t)find_global(P, n->name);
        emit(g, OP_SHARED, bits, (uint16_t)gi, 0, 0);
        emit(g, OP_GLOAD, bits, (uint16_t)gi, 0, 0);
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 158) - [public] emit
    emit(g, OP_GLOAD, bits, (uint16_t)find_global(P, n->name), 0, 0);
} else if (n->pool_kind == 1) {
    int gi = (uint16_t)find_global(P, n->name);
    emit(g, OP_GLOAD, bits, (uint16_t)gi, 0, 0);
    emit(g, OP_NEG, bits, 0, 0, 0);
    emit(g, OP_GSTORE, bits, (uint16_t)gi, 0, 0);
    emit(g, OP_GLOAD, bits, (uint16_t)gi, 0, 0);
} else {
    int gi = (uint16_t)find_global(P, n->name);
    emit(g, OP_SHARED, bits, (uint16_t)gi, 0, 0);
    emit(g, OP_GLOAD, bits, (uint16_t)gi, 0, 0);
}
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 161) - [public] emit
    emit(g, OP_GLOAD, bits, (uint16_t)gi, 0, 0);
    emit(g, OP_NEG, bits, 0, 0, 0);
    emit(g, OP_GSTORE, bits, (uint16_t)gi, 0, 0);
    emit(g, OP_GLOAD, bits, (uint16_t)gi, 0, 0);
} else {
    int gi = (uint16_t)find_global(P, n->name);
    emit(g, OP_SHARED, bits, (uint16_t)gi, 0, 0);
    emit(g, OP_GLOAD, bits, (uint16_t)gi, 0, 0);
}
break;

case ND_CALLIDX: {
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 162) - [public] emit
    emit(g, OP_NEG, bits, 0, 0, 0);
    emit(g, OP_GSTORE, bits, (uint16_t)gi, 0, 0);
    emit(g, OP_GLOAD, bits, (uint16_t)gi, 0, 0);
} else {
    int gi = (uint16_t)find_global(P, n->name);
    emit(g, OP_SHARED, bits, (uint16_t)gi, 0, 0);
    emit(g, OP_GLOAD, bits, (uint16_t)gi, 0, 0);
}
break;

case ND_CALLIDX: {
    int li = find_local(g, n->name);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 163) - [public] emit
    emit(g, OP_GSTORE, bits, (uint16_t)gi, 0, 0);
    emit(g, OP_GLOAD, bits, (uint16_t)gi, 0, 0);
} else {
    int gi = (uint16_t)find_global(P, n->name);
    emit(g, OP_SHARED, bits, (uint16_t)gi, 0, 0);
    emit(g, OP_GLOAD, bits, (uint16_t)gi, 0, 0);
}
break;

case ND_CALLIDX: {
    int li = find_local(g, n->name);

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 164) - [public] emit
    emit(g, OP_GLOAD, bits, (uint16_t)gi, 0, 0);
} else {
    int gi = (uint16_t)find_global(P, n->name);
    emit(g, OP_SHARED, bits, (uint16_t)gi, 0, 0);
    emit(g, OP_GLOAD, bits, (uint16_t)gi, 0, 0);
}
break;

case ND_CALLIDX: {
    int li = find_local(g, n->name);

    if (li >= 0) {
        // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 167) - [public] emit
    emit(g, OP_SHARED, bits, (uint16_t)gi, 0, 0);
    emit(g, OP_GLOAD, bits, (uint16_t)gi, 0, 0);
}
break;

case ND_CALLIDX: {
    int li = find_local(g, n->name);

    if (li >= 0) {
        emit(g, OP_LOAD, bits, g->locals[li].slot, 0, 0);
    } else {
        emit(g, OP_GLOAD, bits, (uint16_t)find_global(P, n->name), 0, 0);
    }
    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/core-lang/codegen.c (line 168) - [public] emit
    emit(g, OP_GLOAD, bits, (uint16_t)gi, 0, 0);
}
break;

case ND_CALLIDX: {
    int li = find_local(g, n->name);

    if (li >= 0) {
        emit(g, OP_LOAD, bits, g->locals[li].slot, 0, 0);
    } else {
        emit(g, OP_GLOAD, bits, (uint16_t)find_global(P, n->name), 0, 0);
    }
    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/core-lang/codegen.c (line 175) - [public] if
    if (li >= 0) {
        emit(g, OP_LOAD, bits, g->locals[li].slot, 0, 0);
    } else {
        emit(g, OP_GLOAD, bits, (uint16_t)find_global(P, n->name), 0, 0);
    }
    break;
}

case ND_UN:
    gen_expr(g, n->a, bits);

    if (n->op == '-') {
        // ... (truncated, see source file for more)
    }

kernel_Workspace/kernel/src/core-lang/codegen.c (line 176) - [public] emit
    emit(g, OP_LOAD, bits, g->locals[li].slot, 0, 0);
} else {
    emit(g, OP_GLOAD, bits, (uint16_t)find_global(P, n->name), 0, 0);
}
break;
}

case ND_UN:
    gen_expr(g, n->a, bits);

    if (n->op == '-') {
        emit(g, OP_NEG, bits, 0, 0, 0);
    }
    // ... (truncated, see source file for more)
}

```



```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 178) - [public] emit
    emit(g, OP_GLOAD, bits, (uint16_t)find_global(P, n->name), 0, 0);
    }
    break;
}

case ND_UN:
    gen_expr(g, n->a, bits);

    if (n->op == '-') {
        emit(g, OP_NEG, bits, 0, 0, 0);
    } else {
        emit(g, OP_NOT, bits, 0, 0, 0);
    }
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 184) - [public] gen_expr
    gen_expr(g, n->a, bits);

    if (n->op == '-') {
        emit(g, OP_NEG, bits, 0, 0, 0);
    } else {
        emit(g, OP_NOT, bits, 0, 0, 0);
    }
    break;

case ND_BIN: {
    uint8_t op = 0;

    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 186) - [public] if
    if (n->op == '-') {
        emit(g, OP_NEG, bits, 0, 0, 0);
    } else {
        emit(g, OP_NOT, bits, 0, 0, 0);
    }
    break;

case ND_BIN: {
    uint8_t op = 0;

    switch (n->op) {
        case '+': op = OP_ADD; break;
    }
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 187) - [public] emit
    emit(g, OP_NEG, bits, 0, 0, 0);
    } else {
        emit(g, OP_NOT, bits, 0, 0, 0);
    }
    break;

case ND_BIN: {
    uint8_t op = 0;

    switch (n->op) {
        case '+': op = OP_ADD; break;
        case '-': op = OP_SUB; break;
    }
    // ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 189) - [public] emit

```
    emit(g, OP_NOT, bits, 0, 0, 0);
}
break;

case ND_BIN: {
    uint8_t op = 0;

    switch (n->op) {
        case '+': op = OP_ADD; break;
        case '-': op = OP_SUB; break;
        case '*': op = OP_MUL; break;
        case '/': op = OP_DIV; break;
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 196) - [public] switch

```
    switch (n->op) {
        case '+': op = OP_ADD; break;
        case '-': op = OP_SUB; break;
        case '*': op = OP_MUL; break;
        case '/': op = OP_DIV; break;
        case '%': op = OP_MOD; break;
        case '&': op = OP_AND; break;
        case '|': op = OP_OR; break;
        case '^': op = OP_XOR; break;
        case TK_SHL: op = OP_SHL; break;
        case TK_EQ: op = OP_EQ; break;
        case TK_NEQ: op = OP_NE; break;
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 215) - [public] gen_expr

```
    gen_expr(g, n->a, bits);
    gen_expr(g, n->b, bits);
    emit(g, op, bits, 0, 0, 0);
    break;
}

case ND_ASSIGN: {
    gen_expr(g, n->b, bits);

    if (n->a->kind == ND_VAR) {
        int li = find_local(g, n->a->name);

        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 216) - [public] gen_expr

```
    gen_expr(g, n->b, bits);
    emit(g, op, bits, 0, 0, 0);
    break;
}

case ND_ASSIGN: {
    gen_expr(g, n->b, bits);

    if (n->a->kind == ND_VAR) {
        int li = find_local(g, n->a->name);

        if (li >= 0) {
            // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 217) - [public] emit

```
    emit(g, op, bits, 0, 0, 0);
    break;
}

case ND_ASSIGN: {
    gen_expr(g, n->b, bits);

    if (n->a->kind == ND_VAR) {
        int li = find_local(g, n->a->name);

        if (li >= 0) {
            emit(g, OP_STORE, bits, g->locals[li].slot, 0, 0);
        }
        // ... (truncated, see source file for more)
    }
}
```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 222) - [public] gen_expr

```
gen_expr(g, n->b, bits);

if (n->a->kind == ND_VAR) {
    int li = find_local(g, n->a->name);

    if (li >= 0) {
        emit(g, OP_STORE, bits, g->locals[li].slot, 0, 0);
    } else {
        emit(g, OP_GSTORE, bits, (uint16_t)find_global(P, n->a->name), 0, 0);
    }
}
break;
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 224) - [public] if

```
if (n->a->kind == ND_VAR) {
    int li = find_local(g, n->a->name);

    if (li >= 0) {
        emit(g, OP_STORE, bits, g->locals[li].slot, 0, 0);
    } else {
        emit(g, OP_GSTORE, bits, (uint16_t)find_global(P, n->a->name), 0, 0);
    }
}
break;
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 227) - [public] if

```
if (li >= 0) {
    emit(g, OP_STORE, bits, g->locals[li].slot, 0, 0);
} else {
    emit(g, OP_GSTORE, bits, (uint16_t)find_global(P, n->a->name), 0, 0);
}
}
break;
}

case ND_MEMBER:
    gen_expr(g, n->a, bits);
    break;
// ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 228) - [public] emit
    emit(g, OP_STORE, bits, g->locals[li].slot, 0, 0);
    } else {
        emit(g, OP_GSTORE, bits, (uint16_t)find_global(P, n->a->name), 0, 0);
    }
    }
    break;
}

case ND_MEMBER:
    gen_expr(g, n->a, bits);
    break;

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 230) - [public] emit
    emit(g, OP_GSTORE, bits, (uint16_t)find_global(P, n->a->name), 0, 0);
    }
    }
    break;
}

case ND_MEMBER:
    gen_expr(g, n->a, bits);
    break;

case ND_CALL: {
    if (n->a != (void *)0 && n->a->kind == ND_VAR) {
        // ... (truncated, see source file for more)
    }

kernel_Workspace/kernel/src/core-lang/codegen.c (line 237) - [public] gen_expr
    gen_expr(g, n->a, bits);
    break;

case ND_CALL: {
    if (n->a != (void *)0 && n->a->kind == ND_VAR) {
        int ei = -1;

        for (size_t i = 0; i < P->nexts; i++) {
            bool same = true;

            for (int k = 0; P->ext_names[i][k] || n->a->name[k]; k++) {
                if (P->ext_names[i][k] != n->a->name[k]) {
                    // ... (truncated, see source file for more)
                }
            }
        }

kernel_Workspace/kernel/src/core-lang/codegen.c (line 241) - [public] if
    if (n->a != (void *)0 && n->a->kind == ND_VAR) {
        int ei = -1;

        for (size_t i = 0; i < P->nexts; i++) {
            bool same = true;

            for (int k = 0; P->ext_names[i][k] || n->a->name[k]; k++) {
                if (P->ext_names[i][k] != n->a->name[k]) {
                    same = false;
                    break;
                }
            }
        }

        // ... (truncated, see source file for more)
    }
}

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 248) - [public] if

```
        if (P->ext_names[i][k] != n->a->name[k]) {
            same = false;
            break;
        }
    }

    if (same) { ei = (int)i; break; }
}

for (size_t i = 0; i < n->nlist; i++) {
    gen_expr(g, n->list[i], 64);
}
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 254) - [public] if

```
    if (same) { ei = (int)i; break; }
}

for (size_t i = 0; i < n->nlist; i++) {
    gen_expr(g, n->list[i], 64);
}

if (ei >= 0) {
    emit(g, OP_EXTERN, 64, (uint16_t)ei, (uint16_t)n->nlist, 0);
} else {
    int fi = -1;

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 258) - [public] gen_expr

```
    gen_expr(g, n->list[i], 64);
}

if (ei >= 0) {
    emit(g, OP_EXTERN, 64, (uint16_t)ei, (uint16_t)n->nlist, 0);
} else {
    int fi = -1;

    for (size_t i = 0; i < P->nfns; i++) {
        bool same = true;

        for (int k = 0; P->fn_names[i][k] || n->a->name[k]; k++) {
            // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 261) - [public] if

```
    if (ei >= 0) {
        emit(g, OP_EXTERN, 64, (uint16_t)ei, (uint16_t)n->nlist, 0);
    } else {
        int fi = -1;

        for (size_t i = 0; i < P->nfns; i++) {
            bool same = true;

            for (int k = 0; P->fn_names[i][k] || n->a->name[k]; k++) {
                if (P->fn_names[i][k] != n->a->name[k]) {
                    same = false;
                    break;
                }
            }

            // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 262) - [public] emit
    emit(g, OP_EXTERN, 64, (uint16_t)ei, (uint16_t)n->nlist, 0);
} else {
    int fi = -1;

    for (size_t i = 0; i < P->nfns; i++) {
        bool same = true;

        for (int k = 0; P->fn_names[i][k] || n->a->name[k]; k++) {
            if (P->fn_names[i][k] != n->a->name[k]) {
                same = false;
                break;
            }
        }
        // ... (truncated, see source file for more)
    }

kernel_Workspace/kernel/src/core-lang/codegen.c (line 270) - [public] if
        if (P->fn_names[i][k] != n->a->name[k]) {
            same = false;
            break;
        }
    }

    if (same) { fi = (int)i; break; }
}

if (fi >= 0) {
    emit(g, OP_CALL, bits, (uint16_t)fi, (uint16_t)n->nlist, 0);
}
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 276) - [public] if
    if (same) { fi = (int)i; break; }
}

if (fi >= 0) {
    emit(g, OP_CALL, bits, (uint16_t)fi, (uint16_t)n->nlist, 0);
}
}
}
break;
}

default:
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 279) - [public] if
    if (fi >= 0) {
        emit(g, OP_CALL, bits, (uint16_t)fi, (uint16_t)n->nlist, 0);
    }
}
break;
}

default:
    break;
}
}
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 280) - [public] emit
        emit(g, OP_CALL, bits, (uint16_t)fi, (uint16_t)n->nlist, 0);
    }
}
break;
}

default:
    break;
}
}

// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 296) - [public] switch
switch (n->kind) {
case ND_LET: {
    uint16_t slot = add_local(g, n->name, n->type.bits);

    gen_expr(g, n->a, n->type.bits);
    emit(g, OP_STORE, n->type.bits, slot, 0, 0);
    break;
}

case ND_SET_FREE: {
    int li = find_local(g, n->name);

    // ... (truncated, see source file for more)
}
}

```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 300) - [public] gen_expr
    gen_expr(g, n->a, n->type.bits);
    emit(g, OP_STORE, n->type.bits, slot, 0, 0);
    break;
}

case ND_SET_FREE: {
    int li = find_local(g, n->name);

    if (li >= 0) {
        emit(g, OP_FREE, 0, 0, g->locals[li].slot, 0);
    } else {
        emit(g, OP_FREE, 0, 1, (uint16_t)find_global(P, n->name), 0);
    }
    // ... (truncated, see source file for more)
}

```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 301) - [public] emit
    emit(g, OP_STORE, n->type.bits, slot, 0, 0);
    break;
}

case ND_SET_FREE: {
    int li = find_local(g, n->name);

    if (li >= 0) {
        emit(g, OP_FREE, 0, 0, g->locals[li].slot, 0);
    } else {
        emit(g, OP_FREE, 0, 1, (uint16_t)find_global(P, n->name), 0);
    }
    // ... (truncated, see source file for more)
}

```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 308) - [public] if
    if (li >= 0) {
        emit(g, OP_FREE, 0, 0, g->locals[li].slot, 0);
    } else {
        emit(g, OP_FREE, 0, 1, (uint16_t)find_global(P, n->name), 0);
    }
    break;
}

case ND_IF: {
    if (n->a != (void *)0) {
        gen_expr(g, n->a, 1);

        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 309) - [public] emit
        emit(g, OP_FREE, 0, 0, g->locals[li].slot, 0);
    } else {
        emit(g, OP_FREE, 0, 1, (uint16_t)find_global(P, n->name), 0);
    }
    break;
}

case ND_IF: {
    if (n->a != (void *)0) {
        gen_expr(g, n->a, 1);

        uint32_t jz = emit(g, OP_JZ, 0, 0, 0, 0);
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 311) - [public] emit
        emit(g, OP_FREE, 0, 1, (uint16_t)find_global(P, n->name), 0);
    }
    break;
}

case ND_IF: {
    if (n->a != (void *)0) {
        gen_expr(g, n->a, 1);

        uint32_t jz = emit(g, OP_JZ, 0, 0, 0, 0);

        gen_stmt(g, n->b);
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 317) - [public] if
    if (n->a != (void *)0) {
        gen_expr(g, n->a, 1);

        uint32_t jz = emit(g, OP_JZ, 0, 0, 0, 0);

        gen_stmt(g, n->b);

        if (n->c != (void *)0) {
            uint32_t jmp = emit(g, OP_JMP, 0, 0, 0, 0);

            P->code[jz].c = (uint32_t)P->code_len;
            gen_stmt(g, n->c);
            // ... (truncated, see source file for more)

```



```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 318) - [public] gen_expr
    gen_expr(g, n->a, 1);

    uint32_t jz = emit(g, OP_JZ, 0, 0, 0, 0);

    gen_stmt(g, n->b);

    if (n->c != (void *)0) {
        uint32_t jmp = emit(g, OP_JMP, 0, 0, 0, 0);

        P->code[jz].c = (uint32_t)P->code_len;
        gen_stmt(g, n->c);
        P->code[jmp].c = (uint32_t)P->code_len;
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 322) - [public] gen_stmt
    gen_stmt(g, n->b);

    if (n->c != (void *)0) {
        uint32_t jmp = emit(g, OP_JMP, 0, 0, 0, 0);

        P->code[jz].c = (uint32_t)P->code_len;
        gen_stmt(g, n->c);
        P->code[jmp].c = (uint32_t)P->code_len;
    } else {
        P->code[jz].c = (uint32_t)P->code_len;
    }
} else {
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 324) - [public] if
    if (n->c != (void *)0) {
        uint32_t jmp = emit(g, OP_JMP, 0, 0, 0, 0);

        P->code[jz].c = (uint32_t)P->code_len;
        gen_stmt(g, n->c);
        P->code[jmp].c = (uint32_t)P->code_len;
    } else {
        P->code[jz].c = (uint32_t)P->code_len;
    }
} else {
    gen_stmt(g, n->b);
}
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 328) - [public] gen_stmt
    gen_stmt(g, n->c);
    P->code[jmp].c = (uint32_t)P->code_len;
} else {
    P->code[jz].c = (uint32_t)P->code_len;
}
} else {
    gen_stmt(g, n->b);
}
break;
}

case ND_WHEN: {
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 334) - [public] gen_stmt
    gen_stmt(g, n->b);
}
break;
}

case ND_WHEN: {
    uint16_t tslot = add_local(g, n->name, 64);

    int li2 = find_local(g, n->name);

    if (li2 >= 0 && g->locals[li2].slot != tslot) {
        emit(g, OP_LOAD, 64, g->locals[li2].slot, 0, 0);
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 344) - [public] if
    if (li2 >= 0 && g->locals[li2].slot != tslot) {
        emit(g, OP_LOAD, 64, g->locals[li2].slot, 0, 0);
        emit(g, OP_STORE, 64, tslot, 0, 0);
    } else if (li2 < 0) {
        emit(g, OP_GLOAD, 64, (uint16_t)find_global(P, n->name), 0, 0);
        emit(g, OP_STORE, 64, tslot, 0, 0);
    }

    for (size_t i = 0; i < n->nlist; i++) {
    }
    break;
}
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 345) - [public] emit
    emit(g, OP_LOAD, 64, g->locals[li2].slot, 0, 0);
    emit(g, OP_STORE, 64, tslot, 0, 0);
} else if (li2 < 0) {
    emit(g, OP_GLOAD, 64, (uint16_t)find_global(P, n->name), 0, 0);
    emit(g, OP_STORE, 64, tslot, 0, 0);
}

    for (size_t i = 0; i < n->nlist; i++) {
    }
    break;
}

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 346) - [public] emit
    emit(g, OP_STORE, 64, tslot, 0, 0);
} else if (li2 < 0) {
    emit(g, OP_GLOAD, 64, (uint16_t)find_global(P, n->name), 0, 0);
    emit(g, OP_STORE, 64, tslot, 0, 0);
}

    for (size_t i = 0; i < n->nlist; i++) {
    }
    break;
}

case ND_BLOCK:
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 348) - [public] emit
    emit(g, OP_GLOAD, 64, (uint16_t)find_global(P, n->name), 0, 0);
    emit(g, OP_STORE, 64, tslot, 0, 0);
}

    for (size_t i = 0; i < n->nlist; i++) {
    }
    break;
}

case ND_BLOCK:
    for (size_t i = 0; i < n->nlist; i++) {
        gen_stmt(g, n->list[i]);
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 349) - [public] emit
    emit(g, OP_STORE, 64, tslot, 0, 0);
}

    for (size_t i = 0; i < n->nlist; i++) {
    }
    break;
}

case ND_BLOCK:
    for (size_t i = 0; i < n->nlist; i++) {
        gen_stmt(g, n->list[i]);
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 359) - [public] gen_stmt
    gen_stmt(g, n->list[i]);
}
break;

default:
    gen_expr(g, n, 64);
    emit(g, OP_POP, 0, 0, 0, 0);
    break;
}
}

cl_prog_t *cl_codegen(ast_node_t *prog, arena_t *ar)
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 364) - [public] gen_expr
    gen_expr(g, n, 64);
    emit(g, OP_POP, 0, 0, 0, 0);
    break;
}
}

cl_prog_t *cl_codegen(ast_node_t *prog, arena_t *ar)
{
    cl_prog_t *P = (cl_prog_t *)cl_arena_alloc(ar, sizeof(cl_prog_t));

    if (P == (void *)0) return (void *)0;

    // ... (truncated, see source file for more)
}

```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 365) - [public] emit
    emit(g, OP_POP, 0, 0, 0, 0);
    break;
}
}

cl_prog_t *cl_codegen(ast_node_t *prog, arena_t *ar)
{
    cl_prog_t *P = (cl_prog_t *)cl_arena_alloc(ar, sizeof(cl_prog_t));

    if (P == (void *)0) return (void *)0;

    P->ar = ar;
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 387) - [public] if
    if (n->kind == ND_LET && P->nglobals < 64) {
        for (int k = 0; n->name[k] && k < 63; k++) {
            P->gnames[P->nglobals][k] = n->name[k];
            P->gnames[P->nglobals][k + 1] = '\0';
        }

        P->gbits[P->nglobals] = n->type.bits;
        P->gneg[P->nglobals] = n->type.neg;

        for (int k = 0; k < 8; k++) {
            P->ginit[P->nglobals].b[k] =
                (n->a != (void *)0 && n->a->kind == ND_NUM)
            // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 403) - [public] cl_wbytes
        cl_wbytes(n->type.bits, n->type.bits);

        P->nglobals++;
    } else if (n->kind == ND_EXTERN) {
        for (size_t k = 0; k < n->nlist && P->nexts < 16; k++) {
            for (int x = 0; n->list[k]->name[x] && x < 63; x++) {
                P->ext_names[P->nexts][x] = n->list[k]->name[x];
                P->ext_names[P->nexts][x + 1] = '\0';
            }
            P->nexts++;
        }
    } else if (n->kind == ND_FN && P->nfns < 16) {
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 436) - [public] if
        if (P->fn_names[k][x] != n->name[x]) {
            same = false;
            break;
        }
    }

    if (same) { fi = (int)k; break; }
}

cg_t g = { 0 };

g.P = P;
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 442) - [public] if
    if (same) { fi = (int)k; break; }
}

cg_t g = { 0 };

g.P = P;

for (size_t k = 0; k < n->nlist; k++) {
    add_local(&g, n->list[k]->name, n->list[k]->type.bits);
}

P->fns[fi].start = (uint32_t)P->code_len;
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 450) - [public] add_local
    add_local(&g, n->list[k]->name, n->list[k]->type.bits);
}

P->fns[fi].start = (uint32_t)P->code_len;

gen_stmt(&g, n->a);

emit(&g, OP_RET, 0, 0, 0, 0);

P->fns[fi].nslots = g.maxslots;

bool is_main = true;
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 455) - [public] gen_stmt
    gen_stmt(&g, n->a);

    emit(&g, OP_RET, 0, 0, 0, 0);

    P->fns[fi].nslots = g.maxslots;

    bool is_main = true;
    const char *m = "main";

    for (int k = 0; m[k] || n->name[k]; k++) {
        if (m[k] != n->name[k]) { is_main = false; break; }
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/codegen.c (line 457) - [public] emit
    emit(&g, OP_RET, 0, 0, 0, 0);

    P->fns[fi].nslots = g.maxslots;

    bool is_main = true;
    const char *m = "main";

    for (int k = 0; m[k] || n->name[k]; k++) {
        if (m[k] != n->name[k]) { is_main = false; break; }
    }

    if (is_main) {
// ... (truncated, see source file for more)

```

```
kernel_Workspace/kernel/src/core-lang/codegen.c (line 465) - [public] if
    if (m[k] != n->name[k]) { is_main = false; break; }
    }

    if (is_main) {
        P->main_fn = fi;
        P->code[P->code_len - 1].op = OP_HALT;
    }
}

return P;
}
```

```
kernel_Workspace/kernel/src/core-lang/codegen.c (line 468) - [public] if
    if (is_main) {
        P->main_fn = fi;
        P->code[P->code_len - 1].op = OP_HALT;
    }
}

return P;
}
```

```
kernel_Workspace/kernel/src/core-lang/lexer.c (line 49) - [public] if
    if (c == '\n') {
        l->line++;
        l->col = 1;
    } else {
        l->col++;
    }
}

return c;
}
```

```
static void emit(lexer_t *l, token_t *t, tok_kind_t kind,
    const char *start, uint32_t len,
    // ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/core-lang/lexer.c (line 77) - [public] if
    if (c == ' ' || c == '\t' || c == '\r') {
        take(l);
    } else if (c == '\n') {
        take(l);
    } else if (c == '/' && peek2(l) == '/') {
        while (l->pos < l->len && peek(l) != '\n') {
            take(l);
        }
    } else if (c == '/' && peek2(l) == '*') {
        take(l);
        take(l);
    }

    // ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/core-lang/lexer.c (line 78) - [public] take
    take(l);
} else if (c == '\n') {
    take(l);
} else if (c == '/' && peek2(l) == '/') {
    while (l->pos < l->len && peek(l) != '\n') {
        take(l);
    }
} else if (c == '/' && peek2(l) == '*') {
    take(l);
    take(l);
}

while (l->pos < l->len &&
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 80) - [public] take

```
    take(l);
} else if (c == '/' && peek2(l) == '/') {
    while (l->pos < l->len && peek(l) != '\n') {
        take(l);
    }
} else if (c == '/' && peek2(l) == '*') {
    take(l);
    take(l);

    while (l->pos < l->len &&
           !(peek(l) == '*' && peek2(l) == '/')) {
        take(l);
    }
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 82) - [public] while

```
    while (l->pos < l->len && peek(l) != '\n') {
        take(l);
    }
} else if (c == '/' && peek2(l) == '*') {
    take(l);
    take(l);

    while (l->pos < l->len &&
           !(peek(l) == '*' && peek2(l) == '/')) {
        take(l);
    }

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 83) - [public] take

```
    take(l);
}
} else if (c == '/' && peek2(l) == '*') {
    take(l);
    take(l);

    while (l->pos < l->len &&
           !(peek(l) == '*' && peek2(l) == '/')) {
        take(l);
    }

    if (l->pos < l->len) {
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 86) - [public] take

```
    take(l);
    take(l);

    while (l->pos < l->len &&
           !(peek(l) == '*' && peek2(l) == '/')) {
        take(l);
    }

    if (l->pos < l->len) {
        take(l);
        take(l);
    }
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 87) - [public] take

```
    take(l);

    while (l->pos < l->len &&
           !(peek(l) == '*' && peek2(l) == '/')) {
        take(l);
    }

    if (l->pos < l->len) {
        take(l);
        take(l);
    }
} else {
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 91) - [public] take

```
    take(l);
}

if (l->pos < l->len) {
    take(l);
    take(l);
}
} else {
    return;
}
}
}
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 94) - [public] if

```
    if (l->pos < l->len) {
        take(l);
        take(l);
    }
} else {
    return;
}
}
}
```

```
static bool is_id_start(char c)
{
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 95) - [public] take

```
    take(l);
    take(l);
}
} else {
    return;
}
}

static bool is_id_start(char c)
{
    return (c >= 'a' && c <= 'z') || (c >= 'A' && c <= 'Z') || c == '_';
    // ... (truncated, see source file for more)
```


kernel_Workspace/kernel/src/core-lang/lexer.c (line 96) - [public] take

```
        take(l);
    }
} else {
    return;
}
}

static bool is_id_start(char c)
{
    return (c >= 'a' && c <= 'z') || (c >= 'A' && c <= 'Z') || c == '_';
}
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 111) - [public] is_id_start

```
return is_id_start(c) || (c >= '0' && c <= '9');
}
```

```
static bool is_digit(char c)
{
    return c >= '0' && c <= '9';
}
```

```
static int hex_val(char c)
{
    if (c >= '0' && c <= '9') return c - '0';
    if (c >= 'a' && c <= 'f') return c - 'a' + 10;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 133) - [public] if

```
if (peek(l) == '0' && (peek2(l) == 'x' || peek2(l) == 'X')) {
    take(l);
    take(l);

    while (hex_val(peek(l)) >= 0) {
        val = (val << 4) | (uint64_t)hex_val(take(l));
    }
} else if (peek(l) == '0' && (peek2(l) == 'b' || peek2(l) == 'B')) {
    take(l);
    take(l);

    while (peek(l) == '0' || peek(l) == '1') {
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 134) - [public] take

```
    take(l);
    take(l);

    while (hex_val(peek(l)) >= 0) {
        val = (val << 4) | (uint64_t)hex_val(take(l));
    }
} else if (peek(l) == '0' && (peek2(l) == 'b' || peek2(l) == 'B')) {
    take(l);
    take(l);

    while (peek(l) == '0' || peek(l) == '1') {
        val = (val << 1) | (uint64_t)(take(l) - '0');
        // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 135) - [public] take
    take(l);

    while (hex_val(peek(l)) >= 0) {
        val = (val << 4) | (uint64_t)hex_val(take(l));
    }
} else if (peek(l) == '0' && (peek2(l) == 'b' || peek2(l) == 'B')) {
    take(l);
    take(l);

    while (peek(l) == '0' || peek(l) == '1') {
        val = (val << 1) | (uint64_t)(take(l) - '0');
    }
}
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 137) - [public] while
    while (hex_val(peek(l)) >= 0) {
        val = (val << 4) | (uint64_t)hex_val(take(l));
    }
} else if (peek(l) == '0' && (peek2(l) == 'b' || peek2(l) == 'B')) {
    take(l);
    take(l);

    while (peek(l) == '0' || peek(l) == '1') {
        val = (val << 1) | (uint64_t)(take(l) - '0');
    }
} else {
    while (is_digit(peek(l))) {
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 141) - [public] take
    take(l);
    take(l);

    while (peek(l) == '0' || peek(l) == '1') {
        val = (val << 1) | (uint64_t)(take(l) - '0');
    }
} else {
    while (is_digit(peek(l))) {
        val = val * 10 + (uint64_t)(take(l) - '0');
    }
}

// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 142) - [public] take
    take(l);

    while (peek(l) == '0' || peek(l) == '1') {
        val = (val << 1) | (uint64_t)(take(l) - '0');
    }
} else {
    while (is_digit(peek(l))) {
        val = val * 10 + (uint64_t)(take(l) - '0');
    }
}

emit(l, t, TK_NUM, start,
// ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 144) - [public] while

```
    while (peek(l) == '0' || peek(l) == '1') {
        val = (val << 1) | (uint64_t)(take(l) - '0');
    }
} else {
    while (is_digit(peek(l))) {
        val = val * 10 + (uint64_t)(take(l) - '0');
    }
}

emit(l, t, TK_NUM, start,
      (uint32_t)(l->src + l->pos - start), line, col);
t->num = val;
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 148) - [public] while

```
    while (is_digit(peek(l))) {
        val = val * 10 + (uint64_t)(take(l) - '0');
    }
}

emit(l, t, TK_NUM, start,
      (uint32_t)(l->src + l->pos - start), line, col);
t->num = val;
}
```

```
static void lex_ident(lexer_t *l, token_t *t,
                      uint32_t line, uint32_t col)
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 163) - [public] while

```
while (is_id_char(peek(l))) {
    take(l);
}

uint32_t len = (uint32_t)(l->src + l->pos - start);

for (size_t i = 0; i < sizeof(KEYWORDS) / sizeof(KEYWORDS[0]); i++) {
    const kw_t *k = &KEYWORDS[i];
    size_t kwlen = 0;

    while (k->kw[kwlen]) kwlen++;

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 164) - [public] take

```
    take(l);
}

uint32_t len = (uint32_t)(l->src + l->pos - start);

for (size_t i = 0; i < sizeof(KEYWORDS) / sizeof(KEYWORDS[0]); i++) {
    const kw_t *k = &KEYWORDS[i];
    size_t kwlen = 0;

    while (k->kw[kwlen]) kwlen++;

    if (kwlen == len) {
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 175) - [public] if

```
    if (kwlen == len) {
        bool same = true;

        for (size_t j = 0; j < len; j++) {
            if (start[j] != k->kw[j]) {
                same = false;
                break;
            }
        }

        if (same) {
            emit(l, t, k->kind, start, len, line, col);
        }
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 179) - [public] if

```
        if (start[j] != k->kw[j]) {
            same = false;
            break;
        }
    }

    if (same) {
        emit(l, t, k->kind, start, len, line, col);
        t->type_bits = k->bits;
        return;
    }
}
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 185) - [public] if

```
        if (same) {
            emit(l, t, k->kind, start, len, line, col);
            t->type_bits = k->bits;
            return;
        }
    }

    if (len == 1 && start[0] == '_') {
        emit(l, t, TK_UNDERSCORE, start, len, line, col);
        return;
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 186) - [public] emit

```
        emit(l, t, k->kind, start, len, line, col);
        t->type_bits = k->bits;
        return;
    }
}

    if (len == 1 && start[0] == '_') {
        emit(l, t, TK_UNDERSCORE, start, len, line, col);
        return;
    }

    // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 193) - [public] if
    if (len == 1 && start[0] == '_') {
        emit(l, t, TK_UNDERSCORE, start, len, line, col);
        return;
    }

    emit(l, t, TK_IDENT, start, len, line, col);
}

static void lex_string(lexer_t *l, token_t *t,
                      uint32_t line, uint32_t col)
{
    take(l);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/lexer.c (line 194) - [public] emit
    emit(l, t, TK_UNDERSCORE, start, len, line, col);
    return;
}

    emit(l, t, TK_IDENT, start, len, line, col);
}

static void lex_string(lexer_t *l, token_t *t,
                      uint32_t line, uint32_t col)
{
    take(l);

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/lexer.c (line 198) - [public] emit
    emit(l, t, TK_IDENT, start, len, line, col);
}

static void lex_string(lexer_t *l, token_t *t,
                      uint32_t line, uint32_t col)
{
    take(l);

    const char *start = l->src + l->pos;

    while (l->pos < l->len && peek(l) != '') {
        if (peek(l) == '\\') {
            // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/lexer.c (line 204) - [public] take
    take(l);

    const char *start = l->src + l->pos;

    while (l->pos < l->len && peek(l) != '') {
        if (peek(l) == '\\') {
            take(l);
        }

        take(l);
    }

    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 208) - [public] while
    while (l->pos < l->len && peek(l) != '') {
        if (peek(l) == '\\') {
            take(l);
        }

        take(l);
    }

    uint32_t len = (uint32_t)(l->src + l->pos - start);

    if (peek(l) == '') {
        take(l);
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/lexer.c (line 209) - [public] if
    if (peek(l) == '\\') {
        take(l);
    }

    take(l);
}

uint32_t len = (uint32_t)(l->src + l->pos - start);

if (peek(l) == '') {
    take(l);
} else {
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/lexer.c (line 210) - [public] take
    take(l);
}

    take(l);
}

uint32_t len = (uint32_t)(l->src + l->pos - start);

if (peek(l) == '') {
    take(l);
} else {
    emit(l, t, TK_ERR, start, len, line, col);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/lexer.c (line 213) - [public] take
    take(l);
}

uint32_t len = (uint32_t)(l->src + l->pos - start);

if (peek(l) == '') {
    take(l);
} else {
    emit(l, t, TK_ERR, start, len, line, col);
    return;
}

// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 218) - [public] if
    if (peek(l) == '') {
        take(l);
    } else {
        emit(l, t, TK_ERR, start, len, line, col);
        return;
    }

    emit(l, t, TK_STR, start, len, line, col);
}

static void lex_char(lexer_t *l, token_t *t,
                    uint32_t line, uint32_t col)
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/lexer.c (line 219) - [public] take
    take(l);
} else {
    emit(l, t, TK_ERR, start, len, line, col);
    return;
}

    emit(l, t, TK_STR, start, len, line, col);
}

static void lex_char(lexer_t *l, token_t *t,
                    uint32_t line, uint32_t col)
{
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/lexer.c (line 221) - [public] emit
    emit(l, t, TK_ERR, start, len, line, col);
    return;
}

    emit(l, t, TK_STR, start, len, line, col);
}

static void lex_char(lexer_t *l, token_t *t,
                    uint32_t line, uint32_t col)
{
    take(l);

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/lexer.c (line 225) - [public] emit
    emit(l, t, TK_STR, start, len, line, col);
}

static void lex_char(lexer_t *l, token_t *t,
                    uint32_t line, uint32_t col)
{
    take(l);

    uint64_t v = 0;

    if (peek(l) == '\\') {
        take(l);
        // ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 231) - [public] take

```
take(l);

uint64_t v = 0;

if (peek(l) == '\\') {
    take(l);

    char e = take(l);

    switch (e) {
        case 'n': v = '\\n'; break;
        case 't': v = '\\t'; break;
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 235) - [public] if

```
if (peek(l) == '\\') {
    take(l);

    char e = take(l);

    switch (e) {
        case 'n': v = '\\n'; break;
        case 't': v = '\\t'; break;
        case '0': v = 0; break;
        case '\\': v = '\\'; break;
        case '\\': v = '\\'; break;
        default: v = (uint64_t)e; break;
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 236) - [public] take

```
take(l);

char e = take(l);

switch (e) {
    case 'n': v = '\\n'; break;
    case 't': v = '\\t'; break;
    case '0': v = 0; break;
    case '\\': v = '\\'; break;
    case '\\': v = '\\'; break;
    default: v = (uint64_t)e; break;
}
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 240) - [public] switch

```
switch (e) {
    case 'n': v = '\\n'; break;
    case 't': v = '\\t'; break;
    case '0': v = 0; break;
    case '\\': v = '\\'; break;
    case '\\': v = '\\'; break;
    default: v = (uint64_t)e; break;
}
} else {
    v = (uint64_t)(unsigned char)take(l);
}

// ... (truncated, see source file for more)
```



```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 252) - [public] if
    if (peek(l) == '\\') {
        take(l);
    } else {
        emit(l, t, TK_ERR, l->src + l->pos, 1, line, col);
        return;
    }

    emit(l, t, TK_CHAR, l->src + l->pos - 1, 1, line, col);
    t->num = v;
}

bool cl_lex_next(lexer_t *l, token_t *t)
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/lexer.c (line 253) - [public] take
    take(l);
} else {
    emit(l, t, TK_ERR, l->src + l->pos, 1, line, col);
    return;
}

    emit(l, t, TK_CHAR, l->src + l->pos - 1, 1, line, col);
    t->num = v;
}

bool cl_lex_next(lexer_t *l, token_t *t)
{
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/lexer.c (line 255) - [public] emit
    emit(l, t, TK_ERR, l->src + l->pos, 1, line, col);
    return;
}

    emit(l, t, TK_CHAR, l->src + l->pos - 1, 1, line, col);
    t->num = v;
}

bool cl_lex_next(lexer_t *l, token_t *t)
{
    skip_ws(l);

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/lexer.c (line 259) - [public] emit
    emit(l, t, TK_CHAR, l->src + l->pos - 1, 1, line, col);
    t->num = v;
}

bool cl_lex_next(lexer_t *l, token_t *t)
{
    skip_ws(l);

    uint32_t line = l->line;
    uint32_t col = l->col;

    if (l->pos >= l->len) {
        // ... (truncated, see source file for more)
    }
}

```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 265) - [public] skip_ws

```
    skip_ws(l);

    uint32_t line = l->line;
    uint32_t col = l->col;

    if (l->pos >= l->len) {
        emit(l, t, TK_EOF, l->src + l->pos, 0, line, col);
        return true;
    }

    char c = peek(l);

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 270) - [public] if

```
    if (l->pos >= l->len) {
        emit(l, t, TK_EOF, l->src + l->pos, 0, line, col);
        return true;
    }

    char c = peek(l);

    if (is_digit(c)) {
        lex_number(l, t, line, col);
        return true;
    }

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 271) - [public] emit

```
        emit(l, t, TK_EOF, l->src + l->pos, 0, line, col);
        return true;
    }

    char c = peek(l);

    if (is_digit(c)) {
        lex_number(l, t, line, col);
        return true;
    }

    if (is_id_start(c)) {
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 277) - [public] if

```
    if (is_digit(c)) {
        lex_number(l, t, line, col);
        return true;
    }

    if (is_id_start(c)) {
        lex_ident(l, t, line, col);
        return true;
    }

    if (c == '"') {
        lex_string(l, t, line, col);
        // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 278) - [public] lex_number
    lex_number(l, t, line, col);
    return true;
}

if (is_id_start(c)) {
    lex_ident(l, t, line, col);
    return true;
}

if (c == '"') {
    lex_string(l, t, line, col);
    return t->kind != TK_ERR;
}
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/lexer.c (line 282) - [public] if
    if (is_id_start(c)) {
        lex_ident(l, t, line, col);
        return true;
    }

    if (c == '"') {
        lex_string(l, t, line, col);
        return t->kind != TK_ERR;
    }

    if (c == '\\') {
        lex_char(l, t, line, col);
        return t->kind != TK_ERR;
    }
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/lexer.c (line 283) - [public] lex_ident
    lex_ident(l, t, line, col);
    return true;
}

if (c == '"') {
    lex_string(l, t, line, col);
    return t->kind != TK_ERR;
}

if (c == '\\') {
    lex_char(l, t, line, col);
    return t->kind != TK_ERR;
}
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/lexer.c (line 287) - [public] if
    if (c == '"') {
        lex_string(l, t, line, col);
        return t->kind != TK_ERR;
    }

    if (c == '\\') {
        lex_char(l, t, line, col);
        return t->kind != TK_ERR;
    }

    const char *start = l->src + l->pos;

    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 288) - [public] lex_string
    lex_string(l, t, line, col);
    return t->kind != TK_ERR;
}

if (c == '\\') {
    lex_char(l, t, line, col);
    return t->kind != TK_ERR;
}

const char *start = l->src + l->pos;

if (c == '$') {
    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/core-lang/lexer.c (line 292) - [public] if
    if (c == '\\') {
        lex_char(l, t, line, col);
        return t->kind != TK_ERR;
    }

    const char *start = l->src + l->pos;

    if (c == '$') {
        take(l);

        if (peek(l) == '!') {
            take(l);
            // ... (truncated, see source file for more)
        }
    }

kernel_Workspace/kernel/src/core-lang/lexer.c (line 293) - [public] lex_char
    lex_char(l, t, line, col);
    return t->kind != TK_ERR;
}

const char *start = l->src + l->pos;

if (c == '$') {
    take(l);

    if (peek(l) == '!') {
        take(l);
        emit(l, t, TK_DOLLAR_BANG, start, 2, line, col);
        // ... (truncated, see source file for more)
    }
}

kernel_Workspace/kernel/src/core-lang/lexer.c (line 299) - [public] if
    if (c == '$') {
        take(l);

        if (peek(l) == '!') {
            take(l);
            emit(l, t, TK_DOLLAR_BANG, start, 2, line, col);
        } else if (peek(l) == '@') {
            take(l);
            emit(l, t, TK_DOLLAR_AT, start, 2, line, col);
        } else {
            emit(l, t, TK_DOLLAR, start, 1, line, col);
        }
    }
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 300) - [public] take
    take(l);

    if (peek(l) == '!') {
        take(l);
        emit(l, t, TK_DOLLAR_BANG, start, 2, line, col);
    } else if (peek(l) == '@') {
        take(l);
        emit(l, t, TK_DOLLAR_AT, start, 2, line, col);
    } else {
        emit(l, t, TK_DOLLAR, start, 1, line, col);
    }

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/lexer.c (line 302) - [public] if
    if (peek(l) == '!') {
        take(l);
        emit(l, t, TK_DOLLAR_BANG, start, 2, line, col);
    } else if (peek(l) == '@') {
        take(l);
        emit(l, t, TK_DOLLAR_AT, start, 2, line, col);
    } else {
        emit(l, t, TK_DOLLAR, start, 1, line, col);
    }

    return true;
}
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/lexer.c (line 303) - [public] take
    take(l);
    emit(l, t, TK_DOLLAR_BANG, start, 2, line, col);
} else if (peek(l) == '@') {
    take(l);
    emit(l, t, TK_DOLLAR_AT, start, 2, line, col);
} else {
    emit(l, t, TK_DOLLAR, start, 1, line, col);
}

    return true;
}

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/lexer.c (line 304) - [public] emit
    emit(l, t, TK_DOLLAR_BANG, start, 2, line, col);
} else if (peek(l) == '@') {
    take(l);
    emit(l, t, TK_DOLLAR_AT, start, 2, line, col);
} else {
    emit(l, t, TK_DOLLAR, start, 1, line, col);
}

    return true;
}

char d = peek2(l);
// ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 306) - [public] take

```
        take(l);
        emit(l, t, TK_DOLLAR_AT, start, 2, line, col);
    } else {
        emit(l, t, TK_DOLLAR, start, 1, line, col);
    }

    return true;
}

char d = peek2(l);

if (c == '=' && d == '=') { take(l); take(l); emit(l, t, TK_EQ, start, 2, line, col); return true; }
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 307) - [public] emit

```
        emit(l, t, TK_DOLLAR_AT, start, 2, line, col);
    } else {
        emit(l, t, TK_DOLLAR, start, 1, line, col);
    }

    return true;
}

char d = peek2(l);

if (c == '=' && d == '=') { take(l); take(l); emit(l, t, TK_EQ, start, 2, line, col); return true; }
if (c == '=' && d == '>') { take(l); take(l); emit(l, t, TK_ARROW, start, 2, line, col); return true; }
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 309) - [public] emit

```
        emit(l, t, TK_DOLLAR, start, 1, line, col);
    }

    return true;
}

char d = peek2(l);

if (c == '=' && d == '=') { take(l); take(l); emit(l, t, TK_EQ, start, 2, line, col); return true; }
if (c == '=' && d == '>') { take(l); take(l); emit(l, t, TK_ARROW, start, 2, line, col); return true; }
if (c == '=' && d == ':') { take(l); take(l); emit(l, t, TK_EQCOLON, start, 2, line, col); return true; }
if (c == '!' && d == '=') { take(l); take(l); emit(l, t, TK_NEQ, start, 2, line, col); return true; }
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 317) - [public] if

```
if (c == '=' && d == '=') { take(l); take(l); emit(l, t, TK_EQ, start, 2, line, col); return true; }
if (c == '=' && d == '>') { take(l); take(l); emit(l, t, TK_ARROW, start, 2, line, col); return true; }
if (c == '=' && d == ':') { take(l); take(l); emit(l, t, TK_EQCOLON, start, 2, line, col); return true; }
if (c == '!' && d == '=') { take(l); take(l); emit(l, t, TK_NEQ, start, 2, line, col); return true; }
if (c == '<' && d == '=') { take(l); take(l); emit(l, t, TK_LEQ, start, 2, line, col); return true; }
if (c == '>' && d == '=') { take(l); take(l); emit(l, t, TK_GEQ, start, 2, line, col); return true; }
if (c == '<' && d == '<') { take(l); take(l); emit(l, t, TK_SHL, start, 2, line, col); return true; }
if (c == '>' && d == '>') { take(l); take(l); emit(l, t, TK_SHR, start, 2, line, col); return true; }
if (c == ':' && d == ':') { take(l); take(l); emit(l, t, TK_DCOLON, start, 2, line, col); return true; }

/* jednoznaki */
take(l);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 318) - [public] if

```
if (c == '=' && d == '>') { take(l); take(l); emit(l, t, TK_ARROW, start, 2, line, col); return true; }
if (c == '=' && d == ':') { take(l); take(l); emit(l, t, TK_EQCOLON, start, 2, line, col); return true; }
if (c == '!' && d == '=') { take(l); take(l); emit(l, t, TK_NEQ, start, 2, line, col); return true; }
if (c == '<' && d == '=') { take(l); take(l); emit(l, t, TK_LEQ, start, 2, line, col); return true; }
if (c == '>' && d == '=') { take(l); take(l); emit(l, t, TK_GEQ, start, 2, line, col); return true; }
if (c == '<' && d == '<') { take(l); take(l); emit(l, t, TK_SHL, start, 2, line, col); return true; }
if (c == '>' && d == '>') { take(l); take(l); emit(l, t, TK_SHR, start, 2, line, col); return true; }
if (c == ':' && d == ':') { take(l); take(l); emit(l, t, TK_DCOLON, start, 2, line, col); return true; }

/* jednoznaki */
take(l);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 319) - [public] if

```
if (c == '=' && d == ':') { take(l); take(l); emit(l, t, TK_EQCOLON, start, 2, line, col); return true; }
if (c == '!' && d == '=') { take(l); take(l); emit(l, t, TK_NEQ, start, 2, line, col); return true; }
if (c == '<' && d == '=') { take(l); take(l); emit(l, t, TK_LEQ, start, 2, line, col); return true; }
if (c == '>' && d == '=') { take(l); take(l); emit(l, t, TK_GEQ, start, 2, line, col); return true; }
if (c == '<' && d == '<') { take(l); take(l); emit(l, t, TK_SHL, start, 2, line, col); return true; }
if (c == '>' && d == '>') { take(l); take(l); emit(l, t, TK_SHR, start, 2, line, col); return true; }
if (c == ':' && d == ':') { take(l); take(l); emit(l, t, TK_DCOLON, start, 2, line, col); return true; }

/* jednoznaki */
take(l);

switch (c) {
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 320) - [public] if

```
if (c == '!' && d == '=') { take(l); take(l); emit(l, t, TK_NEQ, start, 2, line, col); return true; }
if (c == '<' && d == '=') { take(l); take(l); emit(l, t, TK_LEQ, start, 2, line, col); return true; }
if (c == '>' && d == '=') { take(l); take(l); emit(l, t, TK_GEQ, start, 2, line, col); return true; }
if (c == '<' && d == '<') { take(l); take(l); emit(l, t, TK_SHL, start, 2, line, col); return true; }
if (c == '>' && d == '>') { take(l); take(l); emit(l, t, TK_SHR, start, 2, line, col); return true; }
if (c == ':' && d == ':') { take(l); take(l); emit(l, t, TK_DCOLON, start, 2, line, col); return true; }

/* jednoznaki */
take(l);

switch (c) {
case ';': emit(l, t, TK_SEMI, start, 1, line, col); return true;
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 321) - [public] if

```
if (c == '<' && d == '=') { take(l); take(l); emit(l, t, TK_LEQ, start, 2, line, col); return true; }
if (c == '>' && d == '=') { take(l); take(l); emit(l, t, TK_GEQ, start, 2, line, col); return true; }
if (c == '<' && d == '<') { take(l); take(l); emit(l, t, TK_SHL, start, 2, line, col); return true; }
if (c == '>' && d == '>') { take(l); take(l); emit(l, t, TK_SHR, start, 2, line, col); return true; }
if (c == ':' && d == ':') { take(l); take(l); emit(l, t, TK_DCOLON, start, 2, line, col); return true; }

/* jednoznaki */
take(l);

switch (c) {
case ';': emit(l, t, TK_SEMI, start, 1, line, col); return true;
case ',': emit(l, t, TK_COMMA, start, 1, line, col); return true;
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 322) - [public] if

```
if (c == '>' && d == '=') { take(l); take(l); emit(l, t, TK_GEQ, start, 2, line, col); return true; }
if (c == '<' && d == '<') { take(l); take(l); emit(l, t, TK_SHL, start, 2, line, col); return true; }
if (c == '>' && d == '>') { take(l); take(l); emit(l, t, TK_SHR, start, 2, line, col); return true; }
if (c == ':' && d == ':') { take(l); take(l); emit(l, t, TK_DCOLON, start, 2, line, col); return true; }

/* jednoznaki */
take(l);

switch (c) {
case ';': emit(l, t, TK_SEMI, start, 1, line, col); return true;
case ',': emit(l, t, TK_COMMA, start, 1, line, col); return true;
case '.': emit(l, t, TK_DOT, start, 1, line, col); return true;
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 323) - [public] if

```
if (c == '<' && d == '<') { take(l); take(l); emit(l, t, TK_SHL, start, 2, line, col); return true; }
if (c == '>' && d == '>') { take(l); take(l); emit(l, t, TK_SHR, start, 2, line, col); return true; }
if (c == ':' && d == ':') { take(l); take(l); emit(l, t, TK_DCOLON, start, 2, line, col); return true; }

/* jednoznaki */
take(l);

switch (c) {
case ';': emit(l, t, TK_SEMI, start, 1, line, col); return true;
case ',': emit(l, t, TK_COMMA, start, 1, line, col); return true;
case '.': emit(l, t, TK_DOT, start, 1, line, col); return true;
case ':': emit(l, t, TK_COLON, start, 1, line, col); return true;
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 324) - [public] if

```
if (c == '>' && d == '>') { take(l); take(l); emit(l, t, TK_SHR, start, 2, line, col); return true; }
if (c == ':' && d == ':') { take(l); take(l); emit(l, t, TK_DCOLON, start, 2, line, col); return true; }

/* jednoznaki */
take(l);

switch (c) {
case ';': emit(l, t, TK_SEMI, start, 1, line, col); return true;
case ',': emit(l, t, TK_COMMA, start, 1, line, col); return true;
case '.': emit(l, t, TK_DOT, start, 1, line, col); return true;
case ':': emit(l, t, TK_COLON, start, 1, line, col); return true;
case '=': emit(l, t, TK_ASSIGN, start, 1, line, col); return true;
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 325) - [public] if

```
if (c == ':' && d == ':') { take(l); take(l); emit(l, t, TK_DCOLON, start, 2, line, col); return true; }

/* jednoznaki */
take(l);

switch (c) {
case ';': emit(l, t, TK_SEMI, start, 1, line, col); return true;
case ',': emit(l, t, TK_COMMA, start, 1, line, col); return true;
case '.': emit(l, t, TK_DOT, start, 1, line, col); return true;
case ':': emit(l, t, TK_COLON, start, 1, line, col); return true;
case '=': emit(l, t, TK_ASSIGN, start, 1, line, col); return true;
case '<': emit(l, t, TK_LT, start, 1, line, col); return true;
// ... (truncated, see source file for more)
```


kernel_Workspace/kernel/src/core-lang/lexer.c (line 328) - [public] take

```
/* jednoznaki */
```

```
take(l);
```

```
switch (c) {
case ';': emit(l, t, TK_SEMI, start, 1, line, col); return true;
case ',': emit(l, t, TK_COMMA, start, 1, line, col); return true;
case '.': emit(l, t, TK_DOT, start, 1, line, col); return true;
case ':': emit(l, t, TK_COLON, start, 1, line, col); return true;
case '=': emit(l, t, TK_ASSIGN, start, 1, line, col); return true;
case '<': emit(l, t, TK_LT, start, 1, line, col); return true;
case '>': emit(l, t, TK_GT, start, 1, line, col); return true;
case '+': emit(l, t, TK_PLUS, start, 1, line, col); return true;
case '-': emit(l, t, TK_MINUS, start, 1, line, col); return true;
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 330) - [public] switch

```
switch (c) {
case ';': emit(l, t, TK_SEMI, start, 1, line, col); return true;
case ',': emit(l, t, TK_COMMA, start, 1, line, col); return true;
case '.': emit(l, t, TK_DOT, start, 1, line, col); return true;
case ':': emit(l, t, TK_COLON, start, 1, line, col); return true;
case '=': emit(l, t, TK_ASSIGN, start, 1, line, col); return true;
case '<': emit(l, t, TK_LT, start, 1, line, col); return true;
case '>': emit(l, t, TK_GT, start, 1, line, col); return true;
case '+': emit(l, t, TK_PLUS, start, 1, line, col); return true;
case '-': emit(l, t, TK_MINUS, start, 1, line, col); return true;
case '*': emit(l, t, TK_STAR, start, 1, line, col); return true;
case '/': emit(l, t, TK_SLASH, start, 1, line, col); return true;
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 355) - [public] emit

```
    emit(l, t, TK_ERR, start, 1, line, col);
    return false;
}
}
```

```
void cl_lexer_init(lexer_t *l, const char *src, size_t len)
```

```
{
    l->src = src;
    l->len = len;
    l->pos = 0;
    l->line = 1;
    l->col = 1;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 374) - [public] if

```
    if (n >= cap) {
        return -1;
    }

    bool ok = cl_lex_next(l, &buf[n]);

    if (!ok) {
        return -2;
    }

    tok_kind_t k = buf[n].kind;

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 380) - [public] if

```
    if (!ok) {
        return -2;
    }

    tok_kind_t k = buf[n].kind;

    n++;

    if (k == TK_EOF) {
        break;
    }
}
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 388) - [public] if

```
    if (k == TK_EOF) {
        break;
    }
}

*count = n;
return 0;
}
```

```
const char *cl_tok_name(tok_kind_t kind)
{
    switch (kind) {
        // ... (truncated, see source file for more)
    }
}
```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 399) - [public] switch

```
switch (kind) {
case TK_EOF: return "EOF";
case TK_ERR: return "ERR";
case TK_IDENT: return "IDENT";
case TK_TYPE: return "TYPE";
case TK_NUM: return "NUM";
case TK_STR: return "STR";
case TK_CHAR: return "CHAR";
case TK_IF: return "if";
case TK_ELSE: return "else";
case TK_WHEN: return "when";
case TK_CASE: return "case";
// ... (truncated, see source file for more)
}
```

kernel_Workspace/kernel/src/core-lang/loader.c (line 19) - [public] if

```
if (toks == (void *)0) {
    *err_line = 0;
    *err_msg = "arena full";
    return (void *)0;
}

cl_lexer_init(&l, src, len);

if (cl_lex_all(&l, toks, tcap, &n) != 0) {
    *err_line = l.line;
    *err_msg = "lex error";
    return (void *)0;
}
// ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/core-lang/loader.c (line 25) - [public] cl_lexer_init
    cl_lexer_init(&l, src, len);

    if (cl_lex_all(&l, toks, tcap, &n) != 0) {
        *err_line = l.line;
        *err_msg = "lex error";
        return (void *)0;
    }

    parser_t p;
    ast_node_t *prog = cl_parse(&p, toks, n, ar);

    if (prog == (void *)0) {
        // ... (truncated, see source file for more)
    }

kernel_Workspace/kernel/src/core-lang/loader.c (line 27) - [public] if
    if (cl_lex_all(&l, toks, tcap, &n) != 0) {
        *err_line = l.line;
        *err_msg = "lex error";
        return (void *)0;
    }

    parser_t p;
    ast_node_t *prog = cl_parse(&p, toks, n, ar);

    if (prog == (void *)0) {
        *err_line = p.err_line;
        *err_msg = p.err;
        // ... (truncated, see source file for more)
    }

kernel_Workspace/kernel/src/core-lang/loader.c (line 36) - [public] if
    if (prog == (void *)0) {
        *err_line = p.err_line;
        *err_msg = p.err;
        return (void *)0;
    }

    sema_ctx_t *s = cl_sema_new(ar);

    if (s == (void *)0) {
        *err_line = 0;
        *err_msg = "arena full";
        return (void *)0;
        // ... (truncated, see source file for more)
    }

kernel_Workspace/kernel/src/core-lang/loader.c (line 44) - [public] if
    if (s == (void *)0) {
        *err_line = 0;
        *err_msg = "arena full";
        return (void *)0;
    }

    cl_sema_run(s, prog);

    if (cl_sema_diag_count(s) > 0) {
        const cl_diag_t *d = cl_sema_diag(s, 0);

        *err_line = d->line;
        // ... (truncated, see source file for more)
    }

```

```

kernel_Workspace/kernel/src/core-lang/loader.c (line 50) - [public] cl_sema_run
    cl_sema_run(s, prog);

    if (cl_sema_diag_count(s) > 0) {
        const cl_diag_t *d = cl_sema_diag(s, 0);

        *err_line = d->line;
        *err_msg = d->msg;
        return (void *)0;
    }

    cl_prog_t *P = cl_codegen(prog, ar);

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/loader.c (line 52) - [public] if
    if (cl_sema_diag_count(s) > 0) {
        const cl_diag_t *d = cl_sema_diag(s, 0);

        *err_line = d->line;
        *err_msg = d->msg;
        return (void *)0;
    }

    cl_prog_t *P = cl_codegen(prog, ar);

    if (P == (void *)0) {
        *err_line = 0;
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/loader.c (line 62) - [public] if
    if (P == (void *)0) {
        *err_line = 0;
        *err_msg = "codegen failed";
        return (void *)0;
    }

    return P;
}

typedef struct {
    uint8_t *b;
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/loader.c (line 88) - [public] if
    if (!w->ok || w->o >= w->cap) {
        w->ok = false;
        return;
    }

    w->b[w->o++] = v;
}

static void wr_u32(wr_t *w, uint32_t v)
{
    for (int i = 0; i < 4; i++) {
        wr_u8(w, (uint8_t)(v >> (i * 8)));
        // ... (truncated, see source file for more)
    }
}

```

kernel_Workspace/kernel/src/core-lang/loader.c (line 99) - [public] wr_u8

```
    wr_u8(w, (uint8_t)(v >> (i * 8)));
}

static void wr_bytes(wr_t *w, const void *src, size_t n)
{
    for (size_t i = 0; i < n; i++) {
        wr_u8(w, ((const uint8_t *)src)[i]);
    }
}

static uint8_t rd_u8(rd_t *r)
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/loader.c (line 106) - [public] wr_u8

```
    wr_u8(w, ((const uint8_t *)src)[i]);
}

static uint8_t rd_u8(rd_t *r)
{
    if (!r->ok || r->o >= r->len) {
        r->ok = false;
        return 0;
    }

    return r->b[r->o++];
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/loader.c (line 112) - [public] if

```
    if (!r->ok || r->o >= r->len) {
        r->ok = false;
        return 0;
    }

    return r->b[r->o++];
}

static uint32_t rd_u32(rd_t *r)
{
    uint32_t v = 0;

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/loader.c (line 142) - [public] wr_bytes

```
    wr_bytes(&w, "CLBC1", 5);

    wr_u32(&w, (uint32_t)P->code_len);

    for (size_t i = 0; i < P->code_len; i++) {
        const cl_insn_t *in = &P->code[i];

        wr_u8(&w, in->op);
        wr_u8(&w, in->bits);
        wr_u8(&w, (uint8_t)(in->a & 0xFF));
        wr_u8(&w, (uint8_t)(in->a >> 8));
        wr_u8(&w, (uint8_t)(in->b & 0xFF));
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/loader.c (line 144) - [public] wr_u32

```
wr_u32(&w, (uint32_t)P->code_len);

for (size_t i = 0; i < P->code_len; i++) {
    const cl_insn_t *in = &P->code[i];

    wr_u8(&w, in->op);
    wr_u8(&w, in->bits);
    wr_u8(&w, (uint8_t)(in->a & 0xFF));
    wr_u8(&w, (uint8_t)(in->a >> 8));
    wr_u8(&w, (uint8_t)(in->b & 0xFF));
    wr_u8(&w, (uint8_t)(in->b >> 8));
    wr_u32(&w, in->c);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/loader.c (line 149) - [public] wr_u8

```
wr_u8(&w, in->op);
wr_u8(&w, in->bits);
wr_u8(&w, (uint8_t)(in->a & 0xFF));
wr_u8(&w, (uint8_t)(in->a >> 8));
wr_u8(&w, (uint8_t)(in->b & 0xFF));
wr_u8(&w, (uint8_t)(in->b >> 8));
wr_u32(&w, in->c);
}

wr_u32(&w, (uint32_t)P->nconsts);

for (size_t i = 0; i < P->nconsts; i++) {
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/loader.c (line 150) - [public] wr_u8

```
wr_u8(&w, in->bits);
wr_u8(&w, (uint8_t)(in->a & 0xFF));
wr_u8(&w, (uint8_t)(in->a >> 8));
wr_u8(&w, (uint8_t)(in->b & 0xFF));
wr_u8(&w, (uint8_t)(in->b >> 8));
wr_u32(&w, in->c);
}

wr_u32(&w, (uint32_t)P->nconsts);

for (size_t i = 0; i < P->nconsts; i++) {
    wr_bytes(&w, P->consts[i].b, 64);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/loader.c (line 151) - [public] wr_u8

```
wr_u8(&w, (uint8_t)(in->a & 0xFF));
wr_u8(&w, (uint8_t)(in->a >> 8));
wr_u8(&w, (uint8_t)(in->b & 0xFF));
wr_u8(&w, (uint8_t)(in->b >> 8));
wr_u32(&w, in->c);
}

wr_u32(&w, (uint32_t)P->nconsts);

for (size_t i = 0; i < P->nconsts; i++) {
    wr_bytes(&w, P->consts[i].b, 64);
    wr_u8(&w, P->const_bits[i]);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/loader.c (line 152) - [public] wr_u8

```
    wr_u8(&w, (uint8_t)(in->a >> 8));
    wr_u8(&w, (uint8_t)(in->b & 0xFF));
    wr_u8(&w, (uint8_t)(in->b >> 8));
    wr_u32(&w, in->c);
}

wr_u32(&w, (uint32_t)P->nconsts);

for (size_t i = 0; i < P->nconsts; i++) {
    wr_bytes(&w, P->consts[i].b, 64);
    wr_u8(&w, P->const_bits[i]);
}
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/loader.c (line 153) - [public] wr_u8

```
    wr_u8(&w, (uint8_t)(in->b & 0xFF));
    wr_u8(&w, (uint8_t)(in->b >> 8));
    wr_u32(&w, in->c);
}

wr_u32(&w, (uint32_t)P->nconsts);

for (size_t i = 0; i < P->nconsts; i++) {
    wr_bytes(&w, P->consts[i].b, 64);
    wr_u8(&w, P->const_bits[i]);
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/loader.c (line 154) - [public] wr_u8

```
    wr_u8(&w, (uint8_t)(in->b >> 8));
    wr_u32(&w, in->c);
}

wr_u32(&w, (uint32_t)P->nconsts);

for (size_t i = 0; i < P->nconsts; i++) {
    wr_bytes(&w, P->consts[i].b, 64);
    wr_u8(&w, P->const_bits[i]);
}

wr_u32(&w, (uint32_t)P->nfns);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/loader.c (line 155) - [public] wr_u32

```
    wr_u32(&w, in->c);
}

wr_u32(&w, (uint32_t)P->nconsts);

for (size_t i = 0; i < P->nconsts; i++) {
    wr_bytes(&w, P->consts[i].b, 64);
    wr_u8(&w, P->const_bits[i]);
}

wr_u32(&w, (uint32_t)P->nfns);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/loader.c (line 158) - [public] wr_u32

```
wr_u32(&w, (uint32_t)P->nconsts);

for (size_t i = 0; i < P->nconsts; i++) {
    wr_bytes(&w, P->consts[i].b, 64);
    wr_u8(&w, P->const_bits[i]);
}

wr_u32(&w, (uint32_t)P->nfns);

for (size_t i = 0; i < P->nfns; i++) {
    wr_u32(&w, P->fns[i].start);
    wr_u8(&w, (uint8_t)P->fns[i].nslots);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/loader.c (line 161) - [public] wr_bytes

```
    wr_bytes(&w, P->consts[i].b, 64);
    wr_u8(&w, P->const_bits[i]);
}

wr_u32(&w, (uint32_t)P->nfns);

for (size_t i = 0; i < P->nfns; i++) {
    wr_u32(&w, P->fns[i].start);
    wr_u8(&w, (uint8_t)P->fns[i].nslots);
    wr_u8(&w, (uint8_t)P->fns[i].nargs);
    wr_bytes(&w, P->fn_names[i], 64);
}
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/loader.c (line 162) - [public] wr_u8

```
    wr_u8(&w, P->const_bits[i]);
}

wr_u32(&w, (uint32_t)P->nfns);

for (size_t i = 0; i < P->nfns; i++) {
    wr_u32(&w, P->fns[i].start);
    wr_u8(&w, (uint8_t)P->fns[i].nslots);
    wr_u8(&w, (uint8_t)P->fns[i].nargs);
    wr_bytes(&w, P->fn_names[i], 64);
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/loader.c (line 165) - [public] wr_u32

```
wr_u32(&w, (uint32_t)P->nfns);

for (size_t i = 0; i < P->nfns; i++) {
    wr_u32(&w, P->fns[i].start);
    wr_u8(&w, (uint8_t)P->fns[i].nslots);
    wr_u8(&w, (uint8_t)P->fns[i].nargs);
    wr_bytes(&w, P->fn_names[i], 64);
}

wr_u32(&w, (uint32_t)P->nglobals);

for (size_t i = 0; i < P->nglobals; i++) {
    // ... (truncated, see source file for more)
```


kernel_Workspace/kernel/src/core-lang/loader.c (line 168) - [public] wr_u32

```
    wr_u32(&w, P->fns[i].start);
    wr_u8(&w, (uint8_t)P->fns[i].nslots);
    wr_u8(&w, (uint8_t)P->fns[i].nargs);
    wr_bytes(&w, P->fn_names[i], 64);
}

wr_u32(&w, (uint32_t)P->nglobals);

for (size_t i = 0; i < P->nglobals; i++) {
    wr_bytes(&w, P->gnames[i], 64);
    wr_u8(&w, P->gbits[i]);
    wr_u8(&w, P->gneg[i]);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/loader.c (line 169) - [public] wr_u8

```
    wr_u8(&w, (uint8_t)P->fns[i].nslots);
    wr_u8(&w, (uint8_t)P->fns[i].nargs);
    wr_bytes(&w, P->fn_names[i], 64);
}

wr_u32(&w, (uint32_t)P->nglobals);

for (size_t i = 0; i < P->nglobals; i++) {
    wr_bytes(&w, P->gnames[i], 64);
    wr_u8(&w, P->gbits[i]);
    wr_u8(&w, P->gneg[i]);
    wr_bytes(&w, P->ginit[i].b, 64);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/loader.c (line 170) - [public] wr_u8

```
    wr_u8(&w, (uint8_t)P->fns[i].nargs);
    wr_bytes(&w, P->fn_names[i], 64);
}

wr_u32(&w, (uint32_t)P->nglobals);

for (size_t i = 0; i < P->nglobals; i++) {
    wr_bytes(&w, P->gnames[i], 64);
    wr_u8(&w, P->gbits[i]);
    wr_u8(&w, P->gneg[i]);
    wr_bytes(&w, P->ginit[i].b, 64);
}
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/loader.c (line 171) - [public] wr_bytes

```
    wr_bytes(&w, P->fn_names[i], 64);
}

wr_u32(&w, (uint32_t)P->nglobals);

for (size_t i = 0; i < P->nglobals; i++) {
    wr_bytes(&w, P->gnames[i], 64);
    wr_u8(&w, P->gbits[i]);
    wr_u8(&w, P->gneg[i]);
    wr_bytes(&w, P->ginit[i].b, 64);
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/loader.c (line 174) - [public] wr_u32

```
wr_u32(&w, (uint32_t)P->nglobals);

for (size_t i = 0; i < P->nglobals; i++) {
    wr_bytes(&w, P->gnames[i], 64);
    wr_u8(&w, P->gbits[i]);
    wr_u8(&w, P->gneg[i]);
    wr_bytes(&w, P->ginit[i].b, 64);
}

wr_u32(&w, (uint32_t)P->nexts);

for (size_t i = 0; i < P->nexts; i++) {
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/loader.c (line 177) - [public] wr_bytes

```
    wr_bytes(&w, P->gnames[i], 64);
    wr_u8(&w, P->gbits[i]);
    wr_u8(&w, P->gneg[i]);
    wr_bytes(&w, P->ginit[i].b, 64);
}

wr_u32(&w, (uint32_t)P->nexts);

for (size_t i = 0; i < P->nexts; i++) {
    wr_bytes(&w, P->ext_names[i], 64);
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/loader.c (line 178) - [public] wr_u8

```
    wr_u8(&w, P->gbits[i]);
    wr_u8(&w, P->gneg[i]);
    wr_bytes(&w, P->ginit[i].b, 64);
}

wr_u32(&w, (uint32_t)P->nexts);

for (size_t i = 0; i < P->nexts; i++) {
    wr_bytes(&w, P->ext_names[i], 64);
}

wr_u32(&w, (uint32_t)P->main_fn);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/loader.c (line 179) - [public] wr_u8

```
    wr_u8(&w, P->gneg[i]);
    wr_bytes(&w, P->ginit[i].b, 64);
}

wr_u32(&w, (uint32_t)P->nexts);

for (size_t i = 0; i < P->nexts; i++) {
    wr_bytes(&w, P->ext_names[i], 64);
}

wr_u32(&w, (uint32_t)P->main_fn);

// ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/core-lang/loader.c (line 180) - [public] wr_bytes
    wr_bytes(&w, P->ginit[i].b, 64);
}

wr_u32(&w, (uint32_t)P->nexts);

for (size_t i = 0; i < P->nexts; i++) {
    wr_bytes(&w, P->ext_names[i], 64);
}

wr_u32(&w, (uint32_t)P->main_fn);

return w.ok ? w.o : 0;
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/loader.c (line 183) - [public] wr_u32
    wr_u32(&w, (uint32_t)P->nexts);

    for (size_t i = 0; i < P->nexts; i++) {
        wr_bytes(&w, P->ext_names[i], 64);
    }

    wr_u32(&w, (uint32_t)P->main_fn);

    return w.ok ? w.o : 0;
}

cl_prog_t *cl_load_bc(const uint8_t *buf, size_t len, arena_t *ar)
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/loader.c (line 186) - [public] wr_bytes
    wr_bytes(&w, P->ext_names[i], 64);
}

wr_u32(&w, (uint32_t)P->main_fn);

return w.ok ? w.o : 0;
}

cl_prog_t *cl_load_bc(const uint8_t *buf, size_t len, arena_t *ar)
{
    rd_t r = { buf, len, 0, true };

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/loader.c (line 189) - [public] wr_u32
    wr_u32(&w, (uint32_t)P->main_fn);

    return w.ok ? w.o : 0;
}

cl_prog_t *cl_load_bc(const uint8_t *buf, size_t len, arena_t *ar)
{
    rd_t r = { buf, len, 0, true };

    uint8_t magic[5];

    rd_bytes(&r, magic, 5);
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/loader.c (line 200) - [public] rd_bytes
    rd_bytes(&r, magic, 5);

    if (!r.ok || magic[0] != 'C' || magic[1] != 'L' ||
        magic[2] != 'B' || magic[3] != 'C' || magic[4] != '1') {
        return (void *)0;
    }

    cl_prog_t *P = (cl_prog_t *)cl_arena_alloc(ar, sizeof(cl_prog_t));

    if (P == (void *)0) {
        return (void *)0;
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/loader.c (line 209) - [public] if
    if (P == (void *)0) {
        return (void *)0;
    }

    P->ar = ar;
    P->code_cap = 4096;
    P->code = (cl_insn_t *)cl_arena_alloc(ar,
        P->code_cap * sizeof(cl_insn_t));

    if (P->code == (void *)0) {
        return (void *)0;
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/loader.c (line 218) - [public] if
    if (P->code == (void *)0) {
        return (void *)0;
    }

    uint32_t cl = rd_u32(&r);

    if (cl > P->code_cap) {
        return (void *)0;
    }

    P->code_len = cl;

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/loader.c (line 224) - [public] if
    if (cl > P->code_cap) {
        return (void *)0;
    }

    P->code_len = cl;

    for (size_t i = 0; i < P->code_len; i++) {
        cl_insn_t *in = &P->code[i];

        in->op = rd_u8(&r);
        in->bits = rd_u8(&r);
        in->a = (uint16_t)(rd_u8(&r) | ((uint16_t)rd_u8(&r) << 8));
    }
    // ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/core-lang/loader.c (line 242) - [public] if

```
    if (P->nconsts > 64) {
        return (void *)0;
    }

    for (size_t i = 0; i < P->nconsts; i++) {
        rd_bytes(&r, P->consts[i].b, 64);
        P->const_bits[i] = rd_u8(&r);
    }

    P->nfns = rd_u32(&r);

    if (P->nfns > 16) {
        // ... (truncated, see source file for more)
    }
```

kernel_Workspace/kernel/src/core-lang/loader.c (line 247) - [public] rd_bytes

```
        rd_bytes(&r, P->consts[i].b, 64);
        P->const_bits[i] = rd_u8(&r);
    }

    P->nfns = rd_u32(&r);

    if (P->nfns > 16) {
        return (void *)0;
    }

    for (size_t i = 0; i < P->nfns; i++) {
        P->fns[i].start = rd_u32(&r);
        // ... (truncated, see source file for more)
    }
```

kernel_Workspace/kernel/src/core-lang/loader.c (line 253) - [public] if

```
    if (P->nfns > 16) {
        return (void *)0;
    }

    for (size_t i = 0; i < P->nfns; i++) {
        P->fns[i].start = rd_u32(&r);
        P->fns[i].nslots = rd_u8(&r);
        P->fns[i].nargs = rd_u8(&r);
        rd_bytes(&r, P->fn_names[i], 64);
    }

    P->nglobals = rd_u32(&r);
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/kernel/src/core-lang/loader.c (line 261) - [public] rd_bytes

```
        rd_bytes(&r, P->fn_names[i], 64);
    }

    P->nglobals = rd_u32(&r);

    if (P->nglobals > 64) {
        return (void *)0;
    }

    for (size_t i = 0; i < P->nglobals; i++) {
        rd_bytes(&r, P->gnames[i], 64);
        P->gbits[i] = rd_u8(&r);
        // ... (truncated, see source file for more)
    }
```

```

kernel_Workspace/kernel/src/core-lang/loader.c (line 266) - [public] if
    if (P->nglobals > 64) {
        return (void *)0;
    }

    for (size_t i = 0; i < P->nglobals; i++) {
        rd_bytes(&r, P->gnames[i], 64);
        P->gbits[i] = rd_u8(&r);
        P->gneg[i] = rd_u8(&r);
        rd_bytes(&r, P->ginit[i].b, 64);
    }

    P->nexts = rd_u32(&r);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/loader.c (line 271) - [public] rd_bytes
    rd_bytes(&r, P->gnames[i], 64);
    P->gbits[i] = rd_u8(&r);
    P->gneg[i] = rd_u8(&r);
    rd_bytes(&r, P->ginit[i].b, 64);
}

    P->nexts = rd_u32(&r);

    if (P->nexts > 16) {
        return (void *)0;
    }

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/loader.c (line 274) - [public] rd_bytes
    rd_bytes(&r, P->ginit[i].b, 64);
}

    P->nexts = rd_u32(&r);

    if (P->nexts > 16) {
        return (void *)0;
    }

    for (size_t i = 0; i < P->nexts; i++) {
        rd_bytes(&r, P->ext_names[i], 64);
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/loader.c (line 279) - [public] if
    if (P->nexts > 16) {
        return (void *)0;
    }

    for (size_t i = 0; i < P->nexts; i++) {
        rd_bytes(&r, P->ext_names[i], 64);
    }

    P->main_fn = (int)rd_u32(&r);

    if (!r.ok) {
        return (void *)0;
    }
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/loader.c (line 284) - [public] rd_bytes
    rd_bytes(&r, P->ext_names[i], 64);
}

    P->main_fn = (int)rd_u32(&r);

    if (!r.ok) {
        return (void *)0;
    }

    return P;
}

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/loader.c (line 289) - [public] if
    if (!r.ok) {
        return (void *)0;
    }

    return P;
}

#ifdef CL_HOST
#include <stdio.h>

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/loader.c (line 305) - [public] if
    if (f == (void *)0) {
        return -1;
    }

    size_t n = fread(buf, 1, cap, f);

    fclose(f);

    return (long)n;
}
#else
extern int32_t k_fs_read(const char *path, void *buf, uint32_t cap);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/loader.c (line 311) - [public] fclose
    fclose(f);

    return (long)n;
}
#else
extern int32_t k_fs_read(const char *path, void *buf, uint32_t cap);

static long read_all(const char *path, uint8_t *buf, size_t cap)
{
    return (long)k_fs_read(path, buf, (uint32_t)cap);
}
#endif
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/loader.c (line 316) - [public] k_fs_read
extern int32_t k_fs_read(const char *path, void *buf, uint32_t cap);

static long read_all(const char *path, uint8_t *buf, size_t cap)
{
    return (long)k_fs_read(path, buf, (uint32_t)cap);
}
#endif

cl_module_t *cl_load_path(const char *path, arena_t *ar, bool try_native)
{
    static uint8_t io[64 * 1024];
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/loader.c (line 320) - [public] return
    return (long)k_fs_read(path, buf, (uint32_t)cap);
}
#endif

cl_module_t *cl_load_path(const char *path, arena_t *ar, bool try_native)
{
    static uint8_t io[64 * 1024];

    long n = read_all(path, io, sizeof(io));

    if (n <= 0) {
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/loader.c (line 331) - [public] if
        if (n <= 0) {
            return (void *)0;
        }

        cl_module_t *m = (cl_module_t *)cl_arena_alloc(ar, sizeof(cl_module_t));

        if (m == (void *)0) {
            return (void *)0;
        }

        m->native = (void *)0;

        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/loader.c (line 337) - [public] if
        if (m == (void *)0) {
            return (void *)0;
        }

        m->native = (void *)0;

        if (n > 5 && io[0] == 'C' && io[1] == 'L' && io[2] == 'B') {
            m->bc = cl_load_bc(io, (size_t)n, ar);
        } else {
            uint32_t el = 0;
            const char *em = (void *)0;

            // ... (truncated, see source file for more)

```



```
kernel_Workspace/kernel/src/core-lang/loader.c (line 343) - [public] if
    if (n > 5 && io[0] == 'C' && io[1] == 'L' && io[2] == 'B') {
        m->bc = cl_load_bc(io, (size_t)n, ar);
    } else {
        uint32_t el = 0;
        const char *em = (void *)0;

        m->bc = cl_compile_source((const char *)io, (size_t)n,
                                ar, &el, &em);
    }

    if (m->bc == (void *)0) {
        return (void *)0;
    }
    // ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/core-lang/loader.c (line 353) - [public] if
    if (m->bc == (void *)0) {
        return (void *)0;
    }

    if (try_native) {
        m->native = cl_native_compile(m->bc);
    }

    return m;
}
```

```
kernel_Workspace/kernel/src/core-lang/loader.c (line 357) - [public] if
    if (try_native) {
        m->native = cl_native_compile(m->bc);
    }

    return m;
}
```

```
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 27) - [public] nb_u8
    nb_u8(d, 0x48); nb_u8(d, 0xB8); nb_u64(d, v);
}

static void push_rax(nb_t *d) { nb_u8(d, 0x50); }
static void pop_rax(nb_t *d) { nb_u8(d, 0x58); }
static void pop_rcx(nb_t *d) { nb_u8(d, 0x59); }
static void push_rcx(nb_t *d) { nb_u8(d, 0x51); }

static void load_slot(nb_t *d, uint16_t a) /* rax=[rbx+a*8]; push */
{
    nb_u8(d, 0x48); nb_u8(d, 0x8B); nb_u8(d, 0x83); nb_u32(d, (uint32_t)a * 8);
    push_rax(d);
    // ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 30) - [internal] push_rax
static void push_rax(nb_t *d) { nb_u8(d, 0x50); }
static void pop_rax(nb_t *d) { nb_u8(d, 0x58); }
static void pop_rcx(nb_t *d) { nb_u8(d, 0x59); }
static void push_rcx(nb_t *d) { nb_u8(d, 0x51); }

static void load_slot(nb_t *d, uint16_t a) /* rax=[rbx+a*8]; push */
{
    nb_u8(d, 0x48); nb_u8(d, 0x8B); nb_u8(d, 0x83); nb_u32(d, (uint32_t)a * 8);
    push_rax(d);
}

static void store_slot(nb_t *d, uint16_t a) /* pop rax; [rbx+a*8]=rax */
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 31) - [internal] pop_rax

```
static void pop_rax(nb_t *d) { nb_u8(d, 0x58); }
static void pop_rcx(nb_t *d) { nb_u8(d, 0x59); }
static void push_rcx(nb_t *d) { nb_u8(d, 0x51); }

static void load_slot(nb_t *d, uint16_t a) /* rax=[rbx+a*8]; push */
{
    nb_u8(d, 0x48); nb_u8(d, 0x8B); nb_u8(d, 0x83); nb_u32(d, (uint32_t)a * 8);
    push_rax(d);
}

static void store_slot(nb_t *d, uint16_t a) /* pop rax; [rbx+a*8]=rax */
{
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 32) - [internal] pop_rcx

```
static void pop_rcx(nb_t *d) { nb_u8(d, 0x59); }
static void push_rcx(nb_t *d) { nb_u8(d, 0x51); }

static void load_slot(nb_t *d, uint16_t a) /* rax=[rbx+a*8]; push */
{
    nb_u8(d, 0x48); nb_u8(d, 0x8B); nb_u8(d, 0x83); nb_u32(d, (uint32_t)a * 8);
    push_rax(d);
}

static void store_slot(nb_t *d, uint16_t a) /* pop rax; [rbx+a*8]=rax */
{
    pop_rax(d);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 33) - [internal] push_rcx

```
static void push_rcx(nb_t *d) { nb_u8(d, 0x51); }

static void load_slot(nb_t *d, uint16_t a) /* rax=[rbx+a*8]; push */
{
    nb_u8(d, 0x48); nb_u8(d, 0x8B); nb_u8(d, 0x83); nb_u32(d, (uint32_t)a * 8);
    push_rax(d);
}

static void store_slot(nb_t *d, uint16_t a) /* pop rax; [rbx+a*8]=rax */
{
    pop_rax(d);
    nb_u8(d, 0x48); nb_u8(d, 0x89); nb_u8(d, 0x83); nb_u32(d, (uint32_t)a * 8);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 37) - [public] nb_u8

```
nb_u8(d, 0x48); nb_u8(d, 0x8B); nb_u8(d, 0x83); nb_u32(d, (uint32_t)a * 8);
push_rax(d);
}

static void store_slot(nb_t *d, uint16_t a) /* pop rax; [rbx+a*8]=rax */
{
    pop_rax(d);
    nb_u8(d, 0x48); nb_u8(d, 0x89); nb_u8(d, 0x83); nb_u32(d, (uint32_t)a * 8);
}

static void load_glob(nb_t *d, uint16_t a)
{
    // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 38) - [public] push_rax
    push_rax(d);
}

static void store_slot(nb_t *d, uint16_t a) /* pop rax; [rbx+a*8]=rax */
{
    pop_rax(d);
    nb_u8(d, 0x48); nb_u8(d, 0x89); nb_u8(d, 0x83); nb_u32(d, (uint32_t)a * 8);
}

static void load_glob(nb_t *d, uint16_t a)
{
    nb_u8(d, 0x48); nb_u8(d, 0x8B); nb_u8(d, 0x85); nb_u32(d, (uint32_t)a * 8);
    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 43) - [public] pop_rax
    pop_rax(d);
    nb_u8(d, 0x48); nb_u8(d, 0x89); nb_u8(d, 0x83); nb_u32(d, (uint32_t)a * 8);
}

static void load_glob(nb_t *d, uint16_t a)
{
    nb_u8(d, 0x48); nb_u8(d, 0x8B); nb_u8(d, 0x85); nb_u32(d, (uint32_t)a * 8);
    push_rax(d);
}

static void store_glob(nb_t *d, uint16_t a)
{
    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 44) - [public] nb_u8
    nb_u8(d, 0x48); nb_u8(d, 0x89); nb_u8(d, 0x83); nb_u32(d, (uint32_t)a * 8);
}

static void load_glob(nb_t *d, uint16_t a)
{
    nb_u8(d, 0x48); nb_u8(d, 0x8B); nb_u8(d, 0x85); nb_u32(d, (uint32_t)a * 8);
    push_rax(d);
}

static void store_glob(nb_t *d, uint16_t a)
{
    pop_rax(d);
    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 49) - [public] nb_u8
    nb_u8(d, 0x48); nb_u8(d, 0x8B); nb_u8(d, 0x85); nb_u32(d, (uint32_t)a * 8);
    push_rax(d);
}

static void store_glob(nb_t *d, uint16_t a)
{
    pop_rax(d);
    nb_u8(d, 0x48); nb_u8(d, 0x89); nb_u8(d, 0x85); nb_u32(d, (uint32_t)a * 8);
}

typedef struct {
    uint32_t insn_idx;
    // ... (truncated, see source file for more)
}

```

```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 50) - [public] push_rax
    push_rax(d);
}

static void store_glob(nb_t *d, uint16_t a)
{
    pop_rax(d);
    nb_u8(d, 0x48); nb_u8(d, 0x89); nb_u8(d, 0x85); nb_u32(d, (uint32_t)a * 8);
}

typedef struct {
    uint32_t insn_idx;
    size_t patch_off;
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 55) - [public] pop_rax
    pop_rax(d);
    nb_u8(d, 0x48); nb_u8(d, 0x89); nb_u8(d, 0x85); nb_u32(d, (uint32_t)a * 8);
}

typedef struct {
    uint32_t insn_idx;
    size_t patch_off;
} nb_patch_t;

static uint8_t g_nb_mem[16384];

void *cl_native_compile(cl_prog_t *P)
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 56) - [public] nb_u8
    nb_u8(d, 0x48); nb_u8(d, 0x89); nb_u8(d, 0x85); nb_u32(d, (uint32_t)a * 8);
}

typedef struct {
    uint32_t insn_idx;
    size_t patch_off;
} nb_patch_t;

static uint8_t g_nb_mem[16384];

void *cl_native_compile(cl_prog_t *P)
{
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 76) - [public] nb_u8
/* prolog: rbx=slots, rbp=globals, r10=externs */

    nb_u8(&d, 0x48); nb_u8(&d, 0x89); nb_u8(&d, 0xFB);
    nb_u8(&d, 0x48); nb_u8(&d, 0x89); nb_u8(&d, 0xF5);
    nb_u8(&d, 0x49); nb_u8(&d, 0x89); nb_u8(&d, 0xD2);

    uint32_t pc = P->fns[P->main_fn].start;

    for (;;) {
        if (pc >= P->code_len) return (void *)0;

        off_by_insn[pc] = d.len;

        cl_insn_t in = P->code[pc++];
        // ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 77) - [public] nb_u8

```
nb_u8(&d, 0x48); nb_u8(&d, 0x89); nb_u8(&d, 0xF5);
nb_u8(&d, 0x49); nb_u8(&d, 0x89); nb_u8(&d, 0xD2);
```

```
uint32_t pc = P->fns[P->main_fn].start;
```

```
for (;;) {
    if (pc >= P->code_len) return (void *)0;
```

```
    off_by_insn[pc] = d.len;
```

```
    cl_insn_t in = P->code[pc++];
```

```
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 78) - [public] nb_u8

```
nb_u8(&d, 0x49); nb_u8(&d, 0x89); nb_u8(&d, 0xD2);
```

```
uint32_t pc = P->fns[P->main_fn].start;
```

```
for (;;) {
    if (pc >= P->code_len) return (void *)0;
```

```
    off_by_insn[pc] = d.len;
```

```
    cl_insn_t in = P->code[pc++];
```

```
    switch (in.op) {
```

```
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 89) - [public] switch

```
switch (in.op) {
```

```
case OP_CONST: {
```

```
    uint64_t v = 0;
```

```
    for (int i = 7; i >= 0; i--) {
        v = (v << 8) | P->consts[in.a].b[i];
    }
```

```
    mov_rax_u64(&d, v);
```

```
    push_rax(&d);
```

```
    break;
```

```
}
```

```
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 97) - [public] mov_rax_u64

```
mov_rax_u64(&d, v);
```

```
push_rax(&d);
```

```
break;
```

```
}
```

```
case OP_LOAD: load_slot(&d, in.a); break;
```

```
case OP_STORE: store_slot(&d, in.a); break;
```

```
case OP_GLOAD: load_glob(&d, in.a); break;
```

```
case OP_GSTORE: store_glob(&d, in.a); break;
```

```
case OP_ADD:
```

```
    pop_rax(&d); pop_rcx(&d);
```

```
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 98) - [public] push_rax

```
    push_rax(&d);
    break;
}

case OP_LOAD: load_slot(&d, in.a); break;
case OP_STORE: store_slot(&d, in.a); break;
case OP_GLOAD: load_glob(&d, in.a); break;
case OP_GSTORE: store_glob(&d, in.a); break;

case OP_ADD:
    pop_rax(&d); pop_rcx(&d);
    nb_u8(&d, 0x48); nb_u8(&d, 0x01); nb_u8(&d, 0xC1);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 102) - [public] load_slot

```
case OP_LOAD: load_slot(&d, in.a); break;
case OP_STORE: store_slot(&d, in.a); break;
case OP_GLOAD: load_glob(&d, in.a); break;
case OP_GSTORE: store_glob(&d, in.a); break;

case OP_ADD:
    pop_rax(&d); pop_rcx(&d);
    nb_u8(&d, 0x48); nb_u8(&d, 0x01); nb_u8(&d, 0xC1);
    push_rcx(&d);
    break;

case OP_SUB:
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 103) - [public] store_slot

```
case OP_STORE: store_slot(&d, in.a); break;
case OP_GLOAD: load_glob(&d, in.a); break;
case OP_GSTORE: store_glob(&d, in.a); break;

case OP_ADD:
    pop_rax(&d); pop_rcx(&d);
    nb_u8(&d, 0x48); nb_u8(&d, 0x01); nb_u8(&d, 0xC1);
    push_rcx(&d);
    break;

case OP_SUB:
    pop_rax(&d); pop_rcx(&d);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 104) - [public] load_glob

```
case OP_GLOAD: load_glob(&d, in.a); break;
case OP_GSTORE: store_glob(&d, in.a); break;

case OP_ADD:
    pop_rax(&d); pop_rcx(&d);
    nb_u8(&d, 0x48); nb_u8(&d, 0x01); nb_u8(&d, 0xC1);
    push_rcx(&d);
    break;

case OP_SUB:
    pop_rax(&d); pop_rcx(&d);
    nb_u8(&d, 0x48); nb_u8(&d, 0x29); nb_u8(&d, 0xC1);
// ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 105) - [public] store_glob
    case OP_GSTORE: store_glob(&d, in.a); break;

    case OP_ADD:
        pop_rax(&d); pop_rcx(&d);
        nb_u8(&d, 0x48); nb_u8(&d, 0x01); nb_u8(&d, 0xC1);
        push_rcx(&d);
        break;

    case OP_SUB:
        pop_rax(&d); pop_rcx(&d);
        nb_u8(&d, 0x48); nb_u8(&d, 0x29); nb_u8(&d, 0xC1);
        push_rcx(&d);
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 108) - [public] pop_rax
    pop_rax(&d); pop_rcx(&d);
    nb_u8(&d, 0x48); nb_u8(&d, 0x01); nb_u8(&d, 0xC1);
    push_rcx(&d);
    break;

    case OP_SUB:
        pop_rax(&d); pop_rcx(&d);
        nb_u8(&d, 0x48); nb_u8(&d, 0x29); nb_u8(&d, 0xC1);
        push_rcx(&d);
        break;

    case OP_MUL:
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 109) - [public] nb_u8
    nb_u8(&d, 0x48); nb_u8(&d, 0x01); nb_u8(&d, 0xC1);
    push_rcx(&d);
    break;

    case OP_SUB:
        pop_rax(&d); pop_rcx(&d);
        nb_u8(&d, 0x48); nb_u8(&d, 0x29); nb_u8(&d, 0xC1);
        push_rcx(&d);
        break;

    case OP_MUL:
        pop_rax(&d); pop_rcx(&d);
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 110) - [public] push_rcx
    push_rcx(&d);
    break;

    case OP_SUB:
        pop_rax(&d); pop_rcx(&d);
        nb_u8(&d, 0x48); nb_u8(&d, 0x29); nb_u8(&d, 0xC1);
        push_rcx(&d);
        break;

    case OP_MUL:
        pop_rax(&d); pop_rcx(&d);
        nb_u8(&d, 0x48); nb_u8(&d, 0x0F); nb_u8(&d, 0xAF); nb_u8(&d, 0xC8);
// ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 114) - [public] pop_rax

```
    pop_rax(&d); pop_rcx(&d);
    nb_u8(&d, 0x48); nb_u8(&d, 0x29); nb_u8(&d, 0xC1);
    push_rcx(&d);
    break;
```

```
case OP_MUL:
    pop_rax(&d); pop_rcx(&d);
    nb_u8(&d, 0x48); nb_u8(&d, 0x0F); nb_u8(&d, 0xAF); nb_u8(&d, 0xC8);
    push_rcx(&d);
    break;
```

```
case OP_EXTERN: {
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 115) - [public] nb_u8

```
    nb_u8(&d, 0x48); nb_u8(&d, 0x29); nb_u8(&d, 0xC1);
    push_rcx(&d);
    break;
```

```
case OP_MUL:
    pop_rax(&d); pop_rcx(&d);
    nb_u8(&d, 0x48); nb_u8(&d, 0x0F); nb_u8(&d, 0xAF); nb_u8(&d, 0xC8);
    push_rcx(&d);
    break;
```

```
case OP_EXTERN: {
    int n = in.b > 6 ? 6 : in.b;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 116) - [public] push_rcx

```
    push_rcx(&d);
    break;
```

```
case OP_MUL:
    pop_rax(&d); pop_rcx(&d);
    nb_u8(&d, 0x48); nb_u8(&d, 0x0F); nb_u8(&d, 0xAF); nb_u8(&d, 0xC8);
    push_rcx(&d);
    break;
```

```
case OP_EXTERN: {
    int n = in.b > 6 ? 6 : in.b;
```

```
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 120) - [public] pop_rax

```
    pop_rax(&d); pop_rcx(&d);
    nb_u8(&d, 0x48); nb_u8(&d, 0x0F); nb_u8(&d, 0xAF); nb_u8(&d, 0xC8);
    push_rcx(&d);
    break;
```

```
case OP_EXTERN: {
    int n = in.b > 6 ? 6 : in.b;

    for (int i = n - 1; i >= 0; i--) {
        pop_rax(&d);
```

```
        switch (i) {
    // ... (truncated, see source file for more)
```



```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 121) - [public] nb_u8
    nb_u8(&d, 0x48); nb_u8(&d, 0x0F); nb_u8(&d, 0xAF); nb_u8(&d, 0xC8);
    push_rcx(&d);
    break;

    case OP_EXTERN: {
        int n = in.b > 6 ? 6 : in.b;

        for (int i = n - 1; i >= 0; i--) {
            pop_rax(&d);

            switch (i) {
                case 0: nb_u8(&d, 0x48); nb_u8(&d, 0x89); nb_u8(&d, 0xC7); break;
            }
            // ... (truncated, see source file for more)
        }
    }

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 122) - [public] push_rcx
    push_rcx(&d);
    break;

    case OP_EXTERN: {
        int n = in.b > 6 ? 6 : in.b;

        for (int i = n - 1; i >= 0; i--) {
            pop_rax(&d);

            switch (i) {
                case 0: nb_u8(&d, 0x48); nb_u8(&d, 0x89); nb_u8(&d, 0xC7); break;
                case 1: nb_u8(&d, 0x48); nb_u8(&d, 0x89); nb_u8(&d, 0xC6); break;
            }
            // ... (truncated, see source file for more)
        }
    }

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 129) - [public] pop_rax
    pop_rax(&d);

    switch (i) {
        case 0: nb_u8(&d, 0x48); nb_u8(&d, 0x89); nb_u8(&d, 0xC7); break;
        case 1: nb_u8(&d, 0x48); nb_u8(&d, 0x89); nb_u8(&d, 0xC6); break;
        case 2: nb_u8(&d, 0x48); nb_u8(&d, 0x89); nb_u8(&d, 0xC2); break;
        case 3: nb_u8(&d, 0x48); nb_u8(&d, 0x89); nb_u8(&d, 0xC1); break;
        case 4: nb_u8(&d, 0x49); nb_u8(&d, 0x89); nb_u8(&d, 0xC0); break;
        case 5: nb_u8(&d, 0x49); nb_u8(&d, 0x89); nb_u8(&d, 0xC1); break;
    }
}

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 131) - [public] switch
    switch (i) {
        case 0: nb_u8(&d, 0x48); nb_u8(&d, 0x89); nb_u8(&d, 0xC7); break;
        case 1: nb_u8(&d, 0x48); nb_u8(&d, 0x89); nb_u8(&d, 0xC6); break;
        case 2: nb_u8(&d, 0x48); nb_u8(&d, 0x89); nb_u8(&d, 0xC2); break;
        case 3: nb_u8(&d, 0x48); nb_u8(&d, 0x89); nb_u8(&d, 0xC1); break;
        case 4: nb_u8(&d, 0x49); nb_u8(&d, 0x89); nb_u8(&d, 0xC0); break;
        case 5: nb_u8(&d, 0x49); nb_u8(&d, 0x89); nb_u8(&d, 0xC1); break;
    }
}

    /* rax = externs[a]; call rax; push rax */
    nb_u8(&d, 0x49); nb_u8(&d, 0x8B); nb_u8(&d, 0x82);
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 132) - [public] nb_u8
    case 0: nb_u8(&d,0x48); nb_u8(&d,0x89); nb_u8(&d,0xC7); break;
    case 1: nb_u8(&d,0x48); nb_u8(&d,0x89); nb_u8(&d,0xC6); break;
    case 2: nb_u8(&d,0x48); nb_u8(&d,0x89); nb_u8(&d,0xC2); break;
    case 3: nb_u8(&d,0x48); nb_u8(&d,0x89); nb_u8(&d,0xC1); break;
    case 4: nb_u8(&d,0x49); nb_u8(&d,0x89); nb_u8(&d,0xC0); break;
    case 5: nb_u8(&d,0x49); nb_u8(&d,0x89); nb_u8(&d,0xC1); break;
    }
}

/* rax = externs[a]; call rax; push rax */
nb_u8(&d, 0x49); nb_u8(&d, 0x8B); nb_u8(&d, 0x82);
nb_u32(&d, (uint32_t)in.a * 8);
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 133) - [public] nb_u8
    case 1: nb_u8(&d,0x48); nb_u8(&d,0x89); nb_u8(&d,0xC6); break;
    case 2: nb_u8(&d,0x48); nb_u8(&d,0x89); nb_u8(&d,0xC2); break;
    case 3: nb_u8(&d,0x48); nb_u8(&d,0x89); nb_u8(&d,0xC1); break;
    case 4: nb_u8(&d,0x49); nb_u8(&d,0x89); nb_u8(&d,0xC0); break;
    case 5: nb_u8(&d,0x49); nb_u8(&d,0x89); nb_u8(&d,0xC1); break;
    }
}

/* rax = externs[a]; call rax; push rax */
nb_u8(&d, 0x49); nb_u8(&d, 0x8B); nb_u8(&d, 0x82);
nb_u32(&d, (uint32_t)in.a * 8);
nb_u8(&d, 0xFF); nb_u8(&d, 0xD0);
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 134) - [public] nb_u8
    case 2: nb_u8(&d,0x48); nb_u8(&d,0x89); nb_u8(&d,0xC2); break;
    case 3: nb_u8(&d,0x48); nb_u8(&d,0x89); nb_u8(&d,0xC1); break;
    case 4: nb_u8(&d,0x49); nb_u8(&d,0x89); nb_u8(&d,0xC0); break;
    case 5: nb_u8(&d,0x49); nb_u8(&d,0x89); nb_u8(&d,0xC1); break;
    }
}

/* rax = externs[a]; call rax; push rax */
nb_u8(&d, 0x49); nb_u8(&d, 0x8B); nb_u8(&d, 0x82);
nb_u32(&d, (uint32_t)in.a * 8);
nb_u8(&d, 0xFF); nb_u8(&d, 0xD0);
push_rax(&d);
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 135) - [public] nb_u8
    case 3: nb_u8(&d,0x48); nb_u8(&d,0x89); nb_u8(&d,0xC1); break;
    case 4: nb_u8(&d,0x49); nb_u8(&d,0x89); nb_u8(&d,0xC0); break;
    case 5: nb_u8(&d,0x49); nb_u8(&d,0x89); nb_u8(&d,0xC1); break;
    }
}

/* rax = externs[a]; call rax; push rax */
nb_u8(&d, 0x49); nb_u8(&d, 0x8B); nb_u8(&d, 0x82);
nb_u32(&d, (uint32_t)in.a * 8);
nb_u8(&d, 0xFF); nb_u8(&d, 0xD0);
push_rax(&d);
break;
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 136) - [public] nb_u8
        case 4: nb_u8(&d, 0x49); nb_u8(&d, 0x89); nb_u8(&d, 0xC0); break;
        case 5: nb_u8(&d, 0x49); nb_u8(&d, 0x89); nb_u8(&d, 0xC1); break;
    }
}

/* rax = externs[a]; call rax; push rax */
nb_u8(&d, 0x49); nb_u8(&d, 0x8B); nb_u8(&d, 0x82);
nb_u32(&d, (uint32_t)in.a * 8);
nb_u8(&d, 0xFF); nb_u8(&d, 0xD0);
push_rax(&d);
break;
}
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 137) - [public] nb_u8
        case 5: nb_u8(&d, 0x49); nb_u8(&d, 0x89); nb_u8(&d, 0xC1); break;
    }
}

/* rax = externs[a]; call rax; push rax */
nb_u8(&d, 0x49); nb_u8(&d, 0x8B); nb_u8(&d, 0x82);
nb_u32(&d, (uint32_t)in.a * 8);
nb_u8(&d, 0xFF); nb_u8(&d, 0xD0);
push_rax(&d);
break;
}
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 142) - [public] nb_u8
/* rax = externs[a]; call rax; push rax */
    nb_u8(&d, 0x49); nb_u8(&d, 0x8B); nb_u8(&d, 0x82);
    nb_u32(&d, (uint32_t)in.a * 8);
    nb_u8(&d, 0xFF); nb_u8(&d, 0xD0);
    push_rax(&d);
    break;
}

case OP_POP: pop_rax(&d); break;

case OP_JMP:
    nb_u8(&d, 0xE9);
    patches[npatches].insn_idx = in.c;
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 143) - [public] nb_u32
    nb_u32(&d, (uint32_t)in.a * 8);
    nb_u8(&d, 0xFF); nb_u8(&d, 0xD0);
    push_rax(&d);
    break;
}

case OP_POP: pop_rax(&d); break;

case OP_JMP:
    nb_u8(&d, 0xE9);
    patches[npatches].insn_idx = in.c;
    patches[npatches].patch_off = d.len;
// ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 144) - [public] nb_u8

```
        nb_u8(&d, 0xFF); nb_u8(&d, 0xD0);
        push_rax(&d);
        break;
    }

    case OP_POP: pop_rax(&d); break;

    case OP_JMP:
        nb_u8(&d, 0xE9);
        patches[npatches].insn_idx = in.c;
        patches[npatches].patch_off = d.len;
        npatches++;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 145) - [public] push_rax

```
        push_rax(&d);
        break;
    }

    case OP_POP: pop_rax(&d); break;

    case OP_JMP:
        nb_u8(&d, 0xE9);
        patches[npatches].insn_idx = in.c;
        patches[npatches].patch_off = d.len;
        npatches++;
        nb_u32(&d, 0);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 149) - [public] pop_rax

```
    case OP_POP: pop_rax(&d); break;

    case OP_JMP:
        nb_u8(&d, 0xE9);
        patches[npatches].insn_idx = in.c;
        patches[npatches].patch_off = d.len;
        npatches++;
        nb_u32(&d, 0);
        break;

    case OP_JZ:
        pop_rax(&d);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 152) - [public] nb_u8

```
        nb_u8(&d, 0xE9);
        patches[npatches].insn_idx = in.c;
        patches[npatches].patch_off = d.len;
        npatches++;
        nb_u32(&d, 0);
        break;

    case OP_JZ:
        pop_rax(&d);
        nb_u8(&d, 0x48); nb_u8(&d, 0x85); nb_u8(&d, 0xC0);
        nb_u8(&d, 0x0F); nb_u8(&d, 0x84);
        patches[npatches].insn_idx = in.c;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 156) - [public] nb_u32

```
    nb_u32(&d, 0);
    break;

    case OP_JZ:
        pop_rax(&d);
        nb_u8(&d, 0x48); nb_u8(&d, 0x85); nb_u8(&d, 0xC0);
        nb_u8(&d, 0x0F); nb_u8(&d, 0x84);
        patches[npatches].insn_idx = in.c;
        patches[npatches].patch_off = d.len;
        npatches++;
        nb_u32(&d, 0);
        break;
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 160) - [public] pop_rax

```
    pop_rax(&d);
    nb_u8(&d, 0x48); nb_u8(&d, 0x85); nb_u8(&d, 0xC0);
    nb_u8(&d, 0x0F); nb_u8(&d, 0x84);
    patches[npatches].insn_idx = in.c;
    patches[npatches].patch_off = d.len;
    npatches++;
    nb_u32(&d, 0);
    break;

    case OP_RET:
    case OP_HALT:
        nb_u8(&d, 0x48); nb_u8(&d, 0x31); nb_u8(&d, 0xC0);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 161) - [public] nb_u8

```
    nb_u8(&d, 0x48); nb_u8(&d, 0x85); nb_u8(&d, 0xC0);
    nb_u8(&d, 0x0F); nb_u8(&d, 0x84);
    patches[npatches].insn_idx = in.c;
    patches[npatches].patch_off = d.len;
    npatches++;
    nb_u32(&d, 0);
    break;

    case OP_RET:
    case OP_HALT:
        nb_u8(&d, 0x48); nb_u8(&d, 0x31); nb_u8(&d, 0xC0);
        nb_u8(&d, 0xC3);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 162) - [public] nb_u8

```
    nb_u8(&d, 0x0F); nb_u8(&d, 0x84);
    patches[npatches].insn_idx = in.c;
    patches[npatches].patch_off = d.len;
    npatches++;
    nb_u32(&d, 0);
    break;

    case OP_RET:
    case OP_HALT:
        nb_u8(&d, 0x48); nb_u8(&d, 0x31); nb_u8(&d, 0xC0);
        nb_u8(&d, 0xC3);
        goto done;
// ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 166) - [public] nb_u32
    nb_u32(&d, 0);
    break;

    case OP_RET:
    case OP_HALT:
        nb_u8(&d, 0x48); nb_u8(&d, 0x31); nb_u8(&d, 0xC0);
        nb_u8(&d, 0xC3);
        goto done;

    default:
        return (void *)0; /* subset v1 */
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 171) - [public] nb_u8
    nb_u8(&d, 0x48); nb_u8(&d, 0x31); nb_u8(&d, 0xC0);
    nb_u8(&d, 0xC3);
    goto done;

    default:
        return (void *)0; /* subset v1 */
    }
}

done:
for (size_t i = 0; i < npatches; i++) {
    uint32_t target = (uint32_t)off_by_insn[patches[i].insn_idx];
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 172) - [public] nb_u8
    nb_u8(&d, 0xC3);
    goto done;

    default:
        return (void *)0; /* subset v1 */
    }
}

done:
for (size_t i = 0; i < npatches; i++) {
    uint32_t target = (uint32_t)off_by_insn[patches[i].insn_idx];
    uint32_t rel = (uint32_t)(target - (patches[i].patch_off + 4));
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 197) - [public] sizeof
    return sizeof(g_nb_mem);
}

#else /* aarch64 / riscv64 – na razie NULL, VM przejmuje */

void *cl_native_compile(cl_prog_t *P)
{
    (void)P;
    return (void *)0;
}

size_t cl_native_size(void)
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 7) - [public] if
    if (n >= cap) {
        n = cap - 1;
    }

    for (size_t i = 0; i < n; i++) {
        dst[i] = t->start[i];
    }

    dst[n] = '\0';
}

static token_t *peek(parser_t *p)
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 27) - [public] if
    if (i >= p->n) {
        i = p->n - 1;
    }

    return &p->toks[i];
}

static token_t *advance(parser_t *p)
{
    token_t *t = &p->toks[p->pos];

    if (p->pos + 1 < p->n) {
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 38) - [public] if
    if (p->pos + 1 < p->n) {
        p->pos++;
    }

    return t;
}

static bool check(parser_t *p, tok_kind_t k)
{
    return peek(p)->kind == k;
}

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 52) - [public] if
    if (check(p, k)) {
        advance(p);
        return true;
    }

    return false;
}

static void perr(parser_t *p, const char *m)
{
    if (!p->failed) {
        p->failed = true;
        // ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 53) - [public] advance

```
    advance(p);
    return true;
}

return false;
}

static void perr(parser_t *p, const char *m)
{
    if (!p->failed) {
        p->failed = true;
        p->err = m;
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 62) - [public] if

```
    if (!p->failed) {
        p->failed = true;
        p->err = m;
        p->err_line = peek(p)->line;
    }
}

static token_t *expect(parser_t *p, tok_kind_t k, const char *m)
{
    if (!check(p, k)) {
        perr(p, m);
        return peek(p);
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 71) - [public] if

```
    if (!check(p, k)) {
        perr(p, m);
        return peek(p);
    }

    return advance(p);
}

static ast_node_t *new_node(parser_t *p, uint8_t kind)
{
    return cl_node(p->ar, kind, peek(p)->line);
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 72) - [public] perr

```
    perr(p, m);
    return peek(p);
}

return advance(p);
}

static ast_node_t *new_node(parser_t *p, uint8_t kind)
{
    return cl_node(p->ar, kind, peek(p)->line);
}

// ... (truncated, see source file for more)
```


kernel_Workspace/kernel/src/core-lang/parser.c (line 73) - [public] peek

```
    return peek(p);
}

    return advance(p);
}

static ast_node_t *new_node(parser_t *p, uint8_t kind)
{
    return cl_node(p->ar, kind, peek(p)->line);
}

static ast_node_t *parse_expr(parser_t *p);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 76) - [public] advance

```
    return advance(p);
}

static ast_node_t *new_node(parser_t *p, uint8_t kind)
{
    return cl_node(p->ar, kind, peek(p)->line);
}

static ast_node_t *parse_expr(parser_t *p);
static ast_node_t *parse_block(parser_t *p);

/* ---- lista node'ów (rozmie przez arena-realloc nie jest możliwy,
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 81) - [public] cl_node

```
    return cl_node(p->ar, kind, peek(p)->line);
}

static ast_node_t *parse_expr(parser_t *p);
static ast_node_t *parse_block(parser_t *p);

/* ---- lista node'ów (rozmie przez arena-realloc nie jest możliwy,
    więc zbieramy do tymczasowej tablicy na stacku i kopiujemy) ---- */

#define LIST_MAX 128

typedef struct {
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 99) - [public] if

```
    if (l->n < LIST_MAX) {
        l->v[l->n++] = x;
    } else {
        perr(p, "list too long");
    }
}

static void nl_commit(parser_t *p, nlist_t *l, ast_node_t *parent)
{
    ast_node_t **arr = (ast_node_t **)cl_arena_alloc(
        p->ar, l->n * sizeof(ast_node_t *));

    // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 102) - [public] perr
    perr(p, "list too long");
}
}

static void nl_commit(parser_t *p, nlist_t *l, ast_node_t *parent)
{
    ast_node_t **arr = (ast_node_t **)cl_arena_alloc(
        p->ar, l->n * sizeof(ast_node_t *));

    if (arr == (void *)0) {
        perr(p, "arena full");
        return;
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 111) - [public] if
    if (arr == (void *)0) {
        perr(p, "arena full");
        return;
    }

    for (size_t i = 0; i < l->n; i++) {
        arr[i] = l->v[i];
    }

    parent->list = arr;
    parent->nlist = l->n;
}
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 112) - [public] perr
    perr(p, "arena full");
    return;
}

    for (size_t i = 0; i < l->n; i++) {
        arr[i] = l->v[i];
    }

    parent->list = arr;
    parent->nlist = l->n;
}

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 128) - [public] if
    if (check(p, TK_TYPE)) {
        out->bits = peek(p)->type_bits;
        out->neg = false;
        advance(p);
        return true;
    }

    perr(p, "expected type (u4..u512/str)");
    return false;
}

/* ---- deklaracje ---- */
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 131) - [public] advance
    advance(p);
    return true;
}

    perr(p, "expected type (u4..u512/str)");
    return false;
}

/* ---- deklaracje ---- */

static ast_node_t *parse_let(parser_t *p)
{
    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/core-lang/parser.c (line 135) - [public] perr
    perr(p, "expected type (u4..u512/str)");
    return false;
}

/* ---- deklaracje ---- */

static ast_node_t *parse_let(parser_t *p)
{
    ast_node_t *n = new_node(p, ND_LET);

    expect(p, TK_LET, "let");

    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/core-lang/parser.c (line 145) - [public] expect
    expect(p, TK_LET, "let");

    parse_type(p, &n->type);

    copy_name(n->name, sizeof(n->name), expect(p, TK_IDENT, "let name"));

    expect(p, TK_ASSIGN, "let =");

    n->a = parse_expr(p);

    expect(p, TK_SEMI, "let ;");

    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/core-lang/parser.c (line 147) - [public] parse_type
    parse_type(p, &n->type);

    copy_name(n->name, sizeof(n->name), expect(p, TK_IDENT, "let name"));

    expect(p, TK_ASSIGN, "let =");

    n->a = parse_expr(p);

    expect(p, TK_SEMI, "let ;");

    /* opcjonalny zakres ujemny: _=>-1; */
    if (match(p, TK_UNDERSCORE)) {
        // ... (truncated, see source file for more)
    }
}

```

```
kernel_Workspace/kernel/src/core-lang/parser.c (line 149) - [public] copy_name
    copy_name(n->name, sizeof(n->name), expect(p, TK_IDENT, "let name"));

    expect(p, TK_ASSIGN, "let =");

    n->a = parse_expr(p);

    expect(p, TK_SEMI, "let ;");

    /* opcjonalny zakres ujemny: _=>-1; */
    if (match(p, TK_UNDERSCORE)) {
        expect(p, TK_ARROW, "_=>");
        ast_node_t *r = parse_expr(p);
        // ... (truncated, see source file for more)
    }
}
```

```
kernel_Workspace/kernel/src/core-lang/parser.c (line 151) - [public] expect
    expect(p, TK_ASSIGN, "let =");

    n->a = parse_expr(p);

    expect(p, TK_SEMI, "let ;");

    /* opcjonalny zakres ujemny: _=>-1; */
    if (match(p, TK_UNDERSCORE)) {
        expect(p, TK_ARROW, "_=>");
        ast_node_t *r = parse_expr(p);
        expect(p, TK_SEMI, "range ;");

        // ... (truncated, see source file for more)
    }
}
```

```
kernel_Workspace/kernel/src/core-lang/parser.c (line 158) - [public] if
    /* opcjonalny zakres ujemny: _=>-1; */
    if (match(p, TK_UNDERSCORE)) {
        expect(p, TK_ARROW, "_=>");
        ast_node_t *r = parse_expr(p);
        expect(p, TK_SEMI, "range ;");

        n->type.neg = true;
        n->b = r;
    }

    return n;
}

// ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/core-lang/parser.c (line 159) - [public] expect
    expect(p, TK_ARROW, "_=>");
    ast_node_t *r = parse_expr(p);
    expect(p, TK_SEMI, "range ;");

    n->type.neg = true;
    n->b = r;
}

return n;
}

static ast_node_t *parse_static(parser_t *p)
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 174) - [public] expect

```
expect(p, TK_STATIC, "static");
expect(p, TK_DATA, "static data");
expect(p, TK_LBRACK, "[");

parse_type(p, &n->type);

expect(p, TK_COLON, ":");

n->arr_count = expect(p, TK_NUM, "count")->num;

expect(p, TK_SEMI, ";");

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 175) - [public] expect

```
expect(p, TK_DATA, "static data");
expect(p, TK_LBRACK, "[");

parse_type(p, &n->type);

expect(p, TK_COLON, ":");

n->arr_count = expect(p, TK_NUM, "count")->num;

expect(p, TK_SEMI, ";");

n->arr_init = expect(p, TK_NUM, "init")->num;
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 176) - [public] expect

```
expect(p, TK_LBRACK, "[");

parse_type(p, &n->type);

expect(p, TK_COLON, ":");

n->arr_count = expect(p, TK_NUM, "count")->num;

expect(p, TK_SEMI, ";");

n->arr_init = expect(p, TK_NUM, "init")->num;

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 178) - [public] parse_type

```
parse_type(p, &n->type);

expect(p, TK_COLON, ":");

n->arr_count = expect(p, TK_NUM, "count")->num;

expect(p, TK_SEMI, ";");

n->arr_init = expect(p, TK_NUM, "init")->num;

expect(p, TK_RBRACK, "]);
expect(p, TK_SEMI, "static ;");
// ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/core-lang/parser.c (line 180) - [public] expect
    expect(p, TK_COLON, ":");

    n->arr_count = expect(p, TK_NUM, "count")->num;

    expect(p, TK_SEMI, ";");

    n->arr_init = expect(p, TK_NUM, "init")->num;

    expect(p, TK_RBRACK, "]);
    expect(p, TK_SEMI, "static ;");

    n->is_array = true;
    // ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/core-lang/parser.c (line 188) - [public] expect
    expect(p, TK_RBRACK, "]);
    expect(p, TK_SEMI, "static ;");

    n->is_array = true;

    return n;
}

static ast_node_t *parse_fn(parser_t *p)
{
    ast_node_t *n = new_node(p, ND_FN);

    // ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/core-lang/parser.c (line 200) - [public] expect
    expect(p, TK_FN, "fn");

    copy_name(n->name, sizeof(n->name), expect(p, TK_IDENT, "fn name"));

    expect(p, TK_LPAREN, "fn (");

    nlist_t params = { { 0 }, 0 };

    while (!check(p, TK_RPAREN) && !check(p, TK_EOF)) {
        ast_node_t *arg = new_node(p, ND_LET);

        parse_type(p, &arg->type);
        // ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/core-lang/parser.c (line 202) - [public] copy_name
    copy_name(n->name, sizeof(n->name), expect(p, TK_IDENT, "fn name"));

    expect(p, TK_LPAREN, "fn (");

    nlist_t params = { { 0 }, 0 };

    while (!check(p, TK_RPAREN) && !check(p, TK_EOF)) {
        ast_node_t *arg = new_node(p, ND_LET);

        parse_type(p, &arg->type);
        copy_name(arg->name, sizeof(arg->name),
            expect(p, TK_IDENT, "param name"));
        // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 204) - [public] expect
    expect(p, TK_LPAREN, "fn (");

    nlist_t params = { { 0 }, 0 };

    while (!check(p, TK_RPAREN) && !check(p, TK_EOF)) {
        ast_node_t *arg = new_node(p, ND_LET);

        parse_type(p, &arg->type);
        copy_name(arg->name, sizeof(arg->name),
            expect(p, TK_IDENT, "param name"));

        nl_push(p, &params, arg);
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 208) - [public] while
    while (!check(p, TK_RPAREN) && !check(p, TK_EOF)) {
        ast_node_t *arg = new_node(p, ND_LET);

        parse_type(p, &arg->type);
        copy_name(arg->name, sizeof(arg->name),
            expect(p, TK_IDENT, "param name"));

        nl_push(p, &params, arg);

        if (!match(p, TK_COMMA)) {
            break;
        }
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 211) - [public] parse_type
    parse_type(p, &arg->type);
    copy_name(arg->name, sizeof(arg->name),
        expect(p, TK_IDENT, "param name"));

    nl_push(p, &params, arg);

    if (!match(p, TK_COMMA)) {
        break;
    }
}

expect(p, TK_RPAREN, "fn ");
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 213) - [public] expect
    expect(p, TK_IDENT, "param name"));

    nl_push(p, &params, arg);

    if (!match(p, TK_COMMA)) {
        break;
    }
}

expect(p, TK_RPAREN, "fn ");

nl_commit(p, &params, n);
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 215) - [public] nl_push
    nl_push(p, &params, arg);

    if (!match(p, TK_COMMA)) {
        break;
    }
}

expect(p, TK_RPAREN, "fn ");

nl_commit(p, &params, n);

n->a = parse_block(p);
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 217) - [public] if
    if (!match(p, TK_COMMA)) {
        break;
    }
}

expect(p, TK_RPAREN, "fn ");

nl_commit(p, &params, n);

n->a = parse_block(p);

return n;
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 222) - [public] expect
    expect(p, TK_RPAREN, "fn ");

    nl_commit(p, &params, n);

    n->a = parse_block(p);

    return n;
}

static ast_node_t *parse_enum(parser_t *p)
{
    ast_node_t *n = new_node(p, ND_ENUM);
    // ... (truncated, see source file for more)
}

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 224) - [public] nl_commit
    nl_commit(p, &params, n);

    n->a = parse_block(p);

    return n;
}

static ast_node_t *parse_enum(parser_t *p)
{
    ast_node_t *n = new_node(p, ND_ENUM);

    expect(p, TK_ENUM, "enum");
    // ... (truncated, see source file for more)
}

```



```

kernel_Workspace/kernel/src/core-lang/parser.c (line 235) - [public] expect
    expect(p, TK_ENUM, "enum");

    copy_name(n->name, sizeof(n->name), expect(p, TK_IDENT, "enum name"));

    expect(p, TK_LBRACE, "enum {");

    nlist_t items = { { 0 }, 0 };
    uint64_t val = 0;

    while (!check(p, TK_RBACE) && !check(p, TK_EOF)) {
        ast_node_t *it = new_node(p, ND_VAR);

        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 237) - [public] copy_name
    copy_name(n->name, sizeof(n->name), expect(p, TK_IDENT, "enum name"));

    expect(p, TK_LBRACE, "enum {");

    nlist_t items = { { 0 }, 0 };
    uint64_t val = 0;

    while (!check(p, TK_RBACE) && !check(p, TK_EOF)) {
        ast_node_t *it = new_node(p, ND_VAR);

        copy_name(it->name, sizeof(it->name),
            expect(p, TK_IDENT, "enum item"));
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 244) - [public] while
    while (!check(p, TK_RBACE) && !check(p, TK_EOF)) {
        ast_node_t *it = new_node(p, ND_VAR);

        copy_name(it->name, sizeof(it->name),
            expect(p, TK_IDENT, "enum item"));

        it->value = val++;

        nl_push(p, &items, it);

        if (!match(p, TK_COMMA)) {
            break;
        }
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 248) - [public] expect
        expect(p, TK_IDENT, "enum item"));

        it->value = val++;

        nl_push(p, &items, it);

        if (!match(p, TK_COMMA)) {
            break;
        }
    }

    expect(p, TK_RBACE, "enum }");
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 252) - [public] nl_push
    nl_push(p, &items, it);

    if (!match(p, TK_COMMA)) {
        break;
    }
}

expect(p, TK_RBRACE, "enum ");

nl_commit(p, &items, n);

return n;
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 254) - [public] if
    if (!match(p, TK_COMMA)) {
        break;
    }
}

expect(p, TK_RBRACE, "enum ");

nl_commit(p, &items, n);

return n;
}

// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 259) - [public] expect
    expect(p, TK_RBRACE, "enum ");

    nl_commit(p, &items, n);

    return n;
}

static ast_node_t *parse_extern(parser_t *p)
{
    ast_node_t *n = new_node(p, ND_EXTERN);

    expect(p, TK_EXTERN, "extern");
    // ... (truncated, see source file for more)
}

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 261) - [public] nl_commit
    nl_commit(p, &items, n);

    return n;
}

static ast_node_t *parse_extern(parser_t *p)
{
    ast_node_t *n = new_node(p, ND_EXTERN);

    expect(p, TK_EXTERN, "extern");

    token_t *abi = expect(p, TK_CHAR, "extern 'C'");
    // ... (truncated, see source file for more)
}

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 270) - [public] expect
    expect(p, TK_EXTERN, "extern");

    token_t *abi = expect(p, TK_CHAR, "extern 'C'");

    n->name[0] = (char)abi->num;
    n->name[1] = '\0';

    expect(p, TK_LBRACE, "extern {");

    nlist_t fns = { { 0 }, 0 };

    while (!check(p, TK_RBACE) && !check(p, TK_EOF)) {
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 281) - [public] while
    while (!check(p, TK_RBACE) && !check(p, TK_EOF)) {
        ast_node_t *f = new_node(p, ND_VAR);

        copy_name(f->name, sizeof(f->name),
            expect(p, TK_IDENT, "extern fn"));

        nl_push(p, &fns, f);

        if (!match(p, TK_COMMA)) {
            break;
        }
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 285) - [public] expect
        expect(p, TK_IDENT, "extern fn"));

        nl_push(p, &fns, f);

        if (!match(p, TK_COMMA)) {
            break;
        }
    }

    expect(p, TK_RBACE, "extern }");

    nl_commit(p, &fns, n);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 287) - [public] nl_push
        nl_push(p, &fns, f);

        if (!match(p, TK_COMMA)) {
            break;
        }
    }

    expect(p, TK_RBACE, "extern }");

    nl_commit(p, &fns, n);

    return n;
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 289) - [public] if
    if (!match(p, TK_COMMA)) {
        break;
    }
}

expect(p, TK_RBRACE, "extern ");

nl_commit(p, &fns, n);

return n;
}

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 294) - [public] expect
expect(p, TK_RBRACE, "extern ");

nl_commit(p, &fns, n);

return n;
}

static ast_node_t *parse_dataimpl(parser_t *p)
{
    ast_node_t *n = new_node(p, ND_DATAIMPL);

    expect(p, TK_DATAIMPL, "dataimpl");
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 296) - [public] nl_commit
    nl_commit(p, &fns, n);

    return n;
}

static ast_node_t *parse_dataimpl(parser_t *p)
{
    ast_node_t *n = new_node(p, ND_DATAIMPL);

    expect(p, TK_DATAIMPL, "dataimpl");

    expect(p, TK_LBRACE, "dataimpl {");
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 305) - [public] expect
    expect(p, TK_DATAIMPL, "dataimpl");

    expect(p, TK_LBRACE, "dataimpl {");

    /* baza: data: ... a■ do } */
    nlist_t base = { { 0 }, 0 };

    while (!check(p, TK_RBRACE) && !check(p, TK_EOF)) {
        if (match(p, TK_DATA)) {
            expect(p, TK_COLON, "data:");
            continue;
        }
    }
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 312) - [public] while
    while (!check(p, TK_RBACE) && !check(p, TK_EOF)) {
        if (match(p, TK_DATA)) {
            expect(p, TK_COLON, "data:");
            continue;
        }

        nl_push(p, &base, parse_expr(p));

        if (!match(p, TK_COMMA)) {
            break;
        }
    }
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 313) - [public] if
    if (match(p, TK_DATA)) {
        expect(p, TK_COLON, "data:");
        continue;
    }

    nl_push(p, &base, parse_expr(p));

    if (!match(p, TK_COMMA)) {
        break;
    }
}

// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 314) - [public] expect
    expect(p, TK_COLON, "data:");
    continue;
}

    nl_push(p, &base, parse_expr(p));

    if (!match(p, TK_COMMA)) {
        break;
    }
}

    expect(p, TK_RBACE, "dataimpl }");
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 318) - [public] nl_push
    nl_push(p, &base, parse_expr(p));

    if (!match(p, TK_COMMA)) {
        break;
    }
}

    expect(p, TK_RBACE, "dataimpl }");

    expect(p, TK_ARROW, "=>");

    ast_node_t *baseblk = new_node(p, ND_BLOCK);
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 320) - [public] if
    if (!match(p, TK_COMMA)) {
        break;
    }
}

expect(p, TK_RBRACE, "dataimpl }");

expect(p, TK_ARROW, "=>");

ast_node_t *baseblk = new_node(p, ND_BLOCK);
nl_commit(p, &base, baseblk);

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 325) - [public] expect
expect(p, TK_RBRACE, "dataimpl }");

expect(p, TK_ARROW, "=>");

ast_node_t *baseblk = new_node(p, ND_BLOCK);
nl_commit(p, &base, baseblk);

n->a = baseblk;
n->b = parse_block(p);

return n;
}
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 327) - [public] expect
expect(p, TK_ARROW, "=>");

ast_node_t *baseblk = new_node(p, ND_BLOCK);
nl_commit(p, &base, baseblk);

n->a = baseblk;
n->b = parse_block(p);

return n;
}

/* ---- statementy ---- */
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 330) - [public] nl_commit
    nl_commit(p, &base, baseblk);

    n->a = baseblk;
    n->b = parse_block(p);

    return n;
}

/* ---- statementy ---- */

static ast_node_t *parse_block(parser_t *p)
{
    // ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 348) - [public] while

```
while (!check(p, TK_RBACE) && !check(p, TK_EOF)) {
    ast_node_t *s = (void *)0;

    switch (peek(p)->kind) {
    case TK_LET:      s = parse_let(p); break;
    case TK_STATIC:   s = parse_static(p); break;
    case TK_IF:       s = (void *)1; break; /* ni■ej */
    case TK_WHEN:     s = (void *)1; break;
    case TK_SET_FREE: {
        s = new_node(p, ND_SET_FREE);
        advance(p);
        expect(p, TK_LBRACE, "set_free {");
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 351) - [public] switch

```
switch (peek(p)->kind) {
case TK_LET:      s = parse_let(p); break;
case TK_STATIC:   s = parse_static(p); break;
case TK_IF:       s = (void *)1; break; /* ni■ej */
case TK_WHEN:     s = (void *)1; break;
case TK_SET_FREE: {
    s = new_node(p, ND_SET_FREE);
    advance(p);
    expect(p, TK_LBRACE, "set_free {");
    copy_name(s->name, sizeof(s->name),
               expect(p, TK_IDENT, "set_free var"));
    expect(p, TK_RBACE, "set_free }");
}
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 358) - [public] advance

```
advance(p);
expect(p, TK_LBRACE, "set_free {");
copy_name(s->name, sizeof(s->name),
           expect(p, TK_IDENT, "set_free var"));
expect(p, TK_RBACE, "set_free }");
expect(p, TK_SEMI, "set_free ;");
break;
}
default:
    s = (void *)1;
    break;
}
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 361) - [public] expect

```
expect(p, TK_IDENT, "set_free var"));
expect(p, TK_RBACE, "set_free }");
expect(p, TK_SEMI, "set_free ;");
break;
}
default:
    s = (void *)1;
    break;
}

if (s == (void *)1) {
    if (check(p, TK_IF)) {
// ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 362) - [public] expect
    expect(p, TK_RBRACE, "set_free }");
    expect(p, TK_SEMI, "set_free ;");
    break;
}
default:
    s = (void *)1;
    break;
}

    if (s == (void *)1) {
        if (check(p, TK_IF)) {
            extern ast_node_t *cl_parse_if(parser_t *p);
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 371) - [public] if
    if (s == (void *)1) {
        if (check(p, TK_IF)) {
            extern ast_node_t *cl_parse_if(parser_t *p);
            s = cl_parse_if(p);
        } else if (check(p, TK_WHEN)) {
            extern ast_node_t *cl_parse_when(parser_t *p);
            s = cl_parse_when(p);
        } else {
            s = parse_expr(p);
            expect(p, TK_SEMI, "stmt ;");
        }
    }
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 372) - [public] if
    if (check(p, TK_IF)) {
        extern ast_node_t *cl_parse_if(parser_t *p);
        s = cl_parse_if(p);
    } else if (check(p, TK_WHEN)) {
        extern ast_node_t *cl_parse_when(parser_t *p);
        s = cl_parse_when(p);
    } else {
        s = parse_expr(p);
        expect(p, TK_SEMI, "stmt ;");
    }
}

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 384) - [public] nl_push
    nl_push(p, &stmts, s);
}

    expect(p, TK_RBRACE, "}");

    nl_commit(p, &stmts, n);

    return n;
}

ast_node_t *cl_parse_if(parser_t *p)
{
    // ... (truncated, see source file for more)

```



```

kernel_Workspace/kernel/src/core-lang/parser.c (line 387) - [public] expect
    expect(p, TK_RBRACE, "}");

    nl_commit(p, &stmts, n);

    return n;
}

ast_node_t *cl_parse_if(parser_t *p)
{
    ast_node_t *n = new_node(p, ND_IF);

    expect(p, TK_IF, "if");
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 389) - [public] nl_commit
    nl_commit(p, &stmts, n);

    return n;
}

ast_node_t *cl_parse_if(parser_t *p)
{
    ast_node_t *n = new_node(p, ND_IF);

    expect(p, TK_IF, "if");

    if (!check(p, TK_LBRACE)) {
        // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 398) - [public] expect
    expect(p, TK_IF, "if");

    if (!check(p, TK_LBRACE)) {
        n->a = parse_expr(p);
    }

    n->b = parse_block(p);

    /* else / else if / if else */
    if (match(p, TK_ELSE)) {
        if (check(p, TK_IF) && peek_at(p, 1)->kind != TK_ELSE) {
            advance(p);
        }
        // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 400) - [public] if
    if (!check(p, TK_LBRACE)) {
        n->a = parse_expr(p);
    }

    n->b = parse_block(p);

    /* else / else if / if else */
    if (match(p, TK_ELSE)) {
        if (check(p, TK_IF) && peek_at(p, 1)->kind != TK_ELSE) {
            advance(p);
        }

        ast_node_t *inner = new_node(p, ND_IF);
        // ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 407) - [public] if

/ else / else if / if else */*

```
    if (match(p, TK_ELSE)) {
        if (check(p, TK_IF) && peek_at(p, 1)->kind != TK_ELSE) {
            advance(p);

            ast_node_t *inner = new_node(p, ND_IF);

            if (!check(p, TK_LBRACE)) {
                inner->a = parse_expr(p);
            }

            inner->b = parse_block(p);
            n->c = inner;
        }
        // ... (truncated, see source file for more)
    }
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 408) - [public] if

```
    if (check(p, TK_IF) && peek_at(p, 1)->kind != TK_ELSE) {
        advance(p);

        ast_node_t *inner = new_node(p, ND_IF);

        if (!check(p, TK_LBRACE)) {
            inner->a = parse_expr(p);
        }

        inner->b = parse_block(p);
        n->c = inner;
    } else {
        // ... (truncated, see source file for more)
    }
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 409) - [public] advance

```
    advance(p);

    ast_node_t *inner = new_node(p, ND_IF);

    if (!check(p, TK_LBRACE)) {
        inner->a = parse_expr(p);
    }

    inner->b = parse_block(p);
    n->c = inner;
} else {
    n->c = parse_block(p);
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 413) - [public] if

```
    if (!check(p, TK_LBRACE)) {
        inner->a = parse_expr(p);
    }

    inner->b = parse_block(p);
    n->c = inner;
} else {
    n->c = parse_block(p);
}
} else if (check(p, TK_IF) && peek_at(p, 1)->kind == TK_ELSE) {
    advance(p);
    advance(p);
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 423) - [public] advance

```
    advance(p);
    advance(p);
    n->c = parse_block(p);
}

return n;
}

ast_node_t *cl_parse_when(parser_t *p)
{
    ast_node_t *n = new_node(p, ND_WHEN);

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 424) - [public] advance

```
    advance(p);
    n->c = parse_block(p);
}

return n;
}

ast_node_t *cl_parse_when(parser_t *p)
{
    ast_node_t *n = new_node(p, ND_WHEN);

    expect(p, TK_WHEN, "when");
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 435) - [public] expect

```
    expect(p, TK_WHEN, "when");

    copy_name(n->name, sizeof(n->name), expect(p, TK_IDENT, "when target"));

    expect(p, TK_EQCOLON, "=:");
    expect(p, TK_LBRACE, "when {");

    nlist_t cases = { { 0 }, 0 };

    while (!check(p, TK_RBRACE) && !check(p, TK_EOF)) {
        ast_node_t *c = new_node(p, ND_CASE);

        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 437) - [public] copy_name

```
    copy_name(n->name, sizeof(n->name), expect(p, TK_IDENT, "when target"));

    expect(p, TK_EQCOLON, "=:");
    expect(p, TK_LBRACE, "when {");

    nlist_t cases = { { 0 }, 0 };

    while (!check(p, TK_RBRACE) && !check(p, TK_EOF)) {
        ast_node_t *c = new_node(p, ND_CASE);

        c->a = parse_expr(p);
        c->b = parse_block(p);
        // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 439) - [public] expect
    expect(p, TK_EQCOLON, "=:");
    expect(p, TK_LBRACE, "when {");

    nlist_t cases = { { 0 }, 0 };

    while (!check(p, TK_RBRACE) && !check(p, TK_EOF)) {
        ast_node_t *c = new_node(p, ND_CASE);

        c->a = parse_expr(p);
        c->b = parse_block(p);

        nl_push(p, &cases, c);
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 444) - [public] while
    while (!check(p, TK_RBRACE) && !check(p, TK_EOF)) {
        ast_node_t *c = new_node(p, ND_CASE);

        c->a = parse_expr(p);
        c->b = parse_block(p);

        nl_push(p, &cases, c);
    }

    expect(p, TK_RBRACE, "when }");

    nl_commit(p, &cases, n);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 450) - [public] nl_push
    nl_push(p, &cases, c);
}

    expect(p, TK_RBRACE, "when }");

    nl_commit(p, &cases, n);

    return n;
}

/* ---- wyrażenia ---- */

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 453) - [public] expect
    expect(p, TK_RBRACE, "when }");

    nl_commit(p, &cases, n);

    return n;
}

/* ---- wyrażenia ---- */

static ast_node_t *parse_primary(parser_t *p)
{
    ast_node_t *n;
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 455) - [public] nl_commit
    nl_commit(p, &cases, n);

    return n;
}

/* ---- wyrażenia ---- */

static ast_node_t *parse_primary(parser_t *p)
{
    ast_node_t *n;

    if (check(p, TK_NUM)) {
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 466) - [public] if
    if (check(p, TK_NUM)) {
        n = new_node(p, ND_NUM);
        n->value = advance(p)->num;
        return n;
    }

    if (check(p, TK_STR)) {
        n = new_node(p, ND_STR);
        token_t *t = advance(p);
        copy_name(n->name, sizeof(n->name), t);
        return n;
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 472) - [public] if
    if (check(p, TK_STR)) {
        n = new_node(p, ND_STR);
        token_t *t = advance(p);
        copy_name(n->name, sizeof(n->name), t);
        return n;
    }

    if (check(p, TK_CHAR)) {
        n = new_node(p, ND_CHAR);
        n->value = advance(p)->num;
        return n;
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 475) - [public] copy_name
    copy_name(n->name, sizeof(n->name), t);
    return n;
}

    if (check(p, TK_CHAR)) {
        n = new_node(p, ND_CHAR);
        n->value = advance(p)->num;
        return n;
    }

    if (check(p, TK_DOLLAR) || check(p, TK_DOLLAR_BANG) ||
        check(p, TK_DOLLAR_AT)) {
        // ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 479) - [public] if

```
if (check(p, TK_CHAR)) {
    n = new_node(p, ND_CHAR);
    n->value = advance(p)->num;
    return n;
}

if (check(p, TK_DOLLAR) || check(p, TK_DOLLAR_BANG) ||
    check(p, TK_DOLLAR_AT)) {
    n = new_node(p, ND_POOL);
    token_t *t = advance(p);

    n->pool_kind = (t->kind == TK_DOLLAR) ? 0
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 486) - [public] check

```
check(p, TK_DOLLAR_AT)) {
    n = new_node(p, ND_POOL);
    token_t *t = advance(p);

    n->pool_kind = (t->kind == TK_DOLLAR) ? 0
                  : (t->kind == TK_DOLLAR_BANG) ? 1 : 2;

    copy_name(n->name, sizeof(n->name),
              expect(p, TK_IDENT, "$var"));
    return n;
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 494) - [public] expect

```
expect(p, TK_IDENT, "$var"));
return n;
}

if (check(p, TK_CALL) && peek_at(p, 1)->kind == TK_LBRACK) {
    n = new_node(p, ND_CALLIDX);
    advance(p);
    expect(p, TK_LBRACK, "call[");
    copy_name(n->name, sizeof(n->name),
              expect(p, TK_IDENT, "call[x]"));
    expect(p, TK_RBRACK, "]);
    return n;
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 498) - [public] if

```
if (check(p, TK_CALL) && peek_at(p, 1)->kind == TK_LBRACK) {
    n = new_node(p, ND_CALLIDX);
    advance(p);
    expect(p, TK_LBRACK, "call[");
    copy_name(n->name, sizeof(n->name),
              expect(p, TK_IDENT, "call[x]"));
    expect(p, TK_RBRACK, "]);
    return n;
}

if (match(p, TK_LPAREN)) {
    n = parse_expr(p);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 500) - [public] advance

```
    advance(p);
    expect(p, TK_LBRACK, "call[");
    copy_name(n->name, sizeof(n->name),
              expect(p, TK_IDENT, "call[x]"));
    expect(p, TK_RBRACK, "]);");
    return n;
}

if (match(p, TK_LPAREN)) {
    n = parse_expr(p);
    expect(p, TK_RPAREN, "));");
    return n;
}
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 501) - [public] expect

```
    expect(p, TK_LBRACK, "call[");
    copy_name(n->name, sizeof(n->name),
              expect(p, TK_IDENT, "call[x]"));
    expect(p, TK_RBRACK, "]);");
    return n;
}

if (match(p, TK_LPAREN)) {
    n = parse_expr(p);
    expect(p, TK_RPAREN, "));");
    return n;
}
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 503) - [public] expect

```
    expect(p, TK_IDENT, "call[x]"));
    expect(p, TK_RBRACK, "]);");
    return n;
}

if (match(p, TK_LPAREN)) {
    n = parse_expr(p);
    expect(p, TK_RPAREN, "));");
    return n;
}

if (check(p, TK_IDENT)) {
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 504) - [public] expect

```
    expect(p, TK_RBRACK, "]);");
    return n;
}

if (match(p, TK_LPAREN)) {
    n = parse_expr(p);
    expect(p, TK_RPAREN, "));");
    return n;
}

if (check(p, TK_IDENT)) {
    n = new_node(p, ND_VAR);
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 508) - [public] if

```
    if (match(p, TK_LPAREN)) {
        n = parse_expr(p);
        expect(p, TK_RPAREN, "");
        return n;
    }

    if (check(p, TK_IDENT)) {
        n = new_node(p, ND_VAR);
        copy_name(n->name, sizeof(n->name), advance(p));
        return n;
    }

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 510) - [public] expect

```
    expect(p, TK_RPAREN, "");
    return n;
}

if (check(p, TK_IDENT)) {
    n = new_node(p, ND_VAR);
    copy_name(n->name, sizeof(n->name), advance(p));
    return n;
}

perr(p, "unexpected token in expr");
advance(p);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 514) - [public] if

```
    if (check(p, TK_IDENT)) {
        n = new_node(p, ND_VAR);
        copy_name(n->name, sizeof(n->name), advance(p));
        return n;
    }

    perr(p, "unexpected token in expr");
    advance(p);

    return new_node(p, ND_NUM);
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 516) - [public] copy_name

```
    copy_name(n->name, sizeof(n->name), advance(p));
    return n;
}

perr(p, "unexpected token in expr");
advance(p);

return new_node(p, ND_NUM);
}

static ast_node_t *parse_postfix(parser_t *p)
{
    // ... (truncated, see source file for more)
```



```
kernel_Workspace/kernel/src/core-lang/parser.c (line 520) - [public] perr
    perr(p, "unexpected token in expr");
    advance(p);

    return new_node(p, ND_NUM);
}
```

```
static ast_node_t *parse_postfix(parser_t *p)
{
    ast_node_t *n = parse_primary(p);

    for (;;) {
        if (match(p, TK_DOT)) {
            // ... (truncated, see source file for more)

```

```
kernel_Workspace/kernel/src/core-lang/parser.c (line 521) - [public] advance
    advance(p);

    return new_node(p, ND_NUM);
}
```

```
static ast_node_t *parse_postfix(parser_t *p)
{
    ast_node_t *n = parse_primary(p);

    for (;;) {
        if (match(p, TK_DOT)) {
            ast_node_t *m = new_node(p, ND_MEMBER);
            // ... (truncated, see source file for more)

```

```
kernel_Workspace/kernel/src/core-lang/parser.c (line 523) - [public] new_node
    return new_node(p, ND_NUM);
}
```

```
static ast_node_t *parse_postfix(parser_t *p)
{
    ast_node_t *n = parse_primary(p);

    for (;;) {
        if (match(p, TK_DOT)) {
            ast_node_t *m = new_node(p, ND_MEMBER);

            m->a = n;
            // ... (truncated, see source file for more)

```

```
kernel_Workspace/kernel/src/core-lang/parser.c (line 531) - [public] if
    if (match(p, TK_DOT)) {
        ast_node_t *m = new_node(p, ND_MEMBER);

        m->a = n;
        copy_name(m->name, sizeof(m->name),
            expect(p, TK_IDENT, "member"));
        n = m;
    } else if (check(p, TK_DCOLON)) {
        advance(p);

        ast_node_t *m = new_node(p, ND_SCOPECALL);

        // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 536) - [public] expect
        expect(p, TK_IDENT, "member"));
        n = m;
    } else if (check(p, TK_DCOLON)) {
        advance(p);

        ast_node_t *m = new_node(p, ND_SCOPECALL);

        for (size_t i = 0; i < sizeof(m->name); i++) {
            m->name[i] = 0;
        }

        if (n->kind == ND_VAR) {
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 539) - [public] advance
        advance(p);

        ast_node_t *m = new_node(p, ND_SCOPECALL);

        for (size_t i = 0; i < sizeof(m->name); i++) {
            m->name[i] = 0;
        }

        if (n->kind == ND_VAR) {
            for (size_t i = 0; n->name[i] && i < 63; i++) {
                m->name[i] = n->name[i];
            }
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 547) - [public] if
        if (n->kind == ND_VAR) {
            for (size_t i = 0; n->name[i] && i < 63; i++) {
                m->name[i] = n->name[i];
            }
        }

        copy_name(m->name2, sizeof(m->name2),
            expect(p, TK_IDENT, "scope fn"));

        expect(p, TK_LPAREN, "scope (");

        nlist_t args = { { 0 }, 0 };
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 554) - [public] expect
        expect(p, TK_IDENT, "scope fn"));

        expect(p, TK_LPAREN, "scope (");

        nlist_t args = { { 0 }, 0 };

        while (!check(p, TK_RPAREN) && !check(p, TK_EOF)) {
            nl_push(p, &args, parse_expr(p));

            if (!match(p, TK_COMMA)) {
                break;
            }
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 556) - [public] expect
    expect(p, TK_LPAREN, "scope (");

    nlist_t args = { { 0 }, 0 };

    while (!check(p, TK_RPAREN) && !check(p, TK_EOF)) {
        nl_push(p, &args, parse_expr(p));

        if (!match(p, TK_COMMA)) {
            break;
        }
    }

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 560) - [public] while
    while (!check(p, TK_RPAREN) && !check(p, TK_EOF)) {
        nl_push(p, &args, parse_expr(p));

        if (!match(p, TK_COMMA)) {
            break;
        }
    }

    expect(p, TK_RPAREN, "::");

    nl_commit(p, &args, m);

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 561) - [public] nl_push
    nl_push(p, &args, parse_expr(p));

    if (!match(p, TK_COMMA)) {
        break;
    }
}

    expect(p, TK_RPAREN, "::");

    nl_commit(p, &args, m);

    n = m;
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 563) - [public] if
    if (!match(p, TK_COMMA)) {
        break;
    }
}

    expect(p, TK_RPAREN, "::");

    nl_commit(p, &args, m);

    n = m;
} else if (check(p, TK_LPAREN)) {
    advance(p);
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 568) - [public] expect
    expect(p, TK_RPAREN, "::-");

    nl_commit(p, &args, m);

    n = m;
} else if (check(p, TK_LPAREN)) {
    advance(p);

    ast_node_t *m = new_node(p, ND_CALL);

    m->a = n;

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 570) - [public] nl_commit
    nl_commit(p, &args, m);

    n = m;
} else if (check(p, TK_LPAREN)) {
    advance(p);

    ast_node_t *m = new_node(p, ND_CALL);

    m->a = n;

    nlist_t args = { { 0 }, 0 };

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 574) - [public] advance
    advance(p);

    ast_node_t *m = new_node(p, ND_CALL);

    m->a = n;

    nlist_t args = { { 0 }, 0 };

    while (!check(p, TK_RPAREN) && !check(p, TK_EOF)) {
        nl_push(p, &args, parse_expr(p));

        if (!match(p, TK_COMMA)) {
            // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 582) - [public] while
    while (!check(p, TK_RPAREN) && !check(p, TK_EOF)) {
        nl_push(p, &args, parse_expr(p));

        if (!match(p, TK_COMMA)) {
            break;
        }
    }

    expect(p, TK_RPAREN, "call ");

    nl_commit(p, &args, m);

    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 583) - [public] nl_push
    nl_push(p, &args, parse_expr(p));

    if (!match(p, TK_COMMA)) {
        break;
    }
}

expect(p, TK_RPAREN, "call ");

nl_commit(p, &args, m);

n = m;
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 585) - [public] if
    if (!match(p, TK_COMMA)) {
        break;
    }
}

expect(p, TK_RPAREN, "call ");

nl_commit(p, &args, m);

n = m;
} else {
    break;
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 590) - [public] expect
    expect(p, TK_RPAREN, "call ");

    nl_commit(p, &args, m);

    n = m;
} else {
    break;
}

return n;
}
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 592) - [public] nl_commit
    nl_commit(p, &args, m);

    n = m;
} else {
    break;
}

return n;
}

static ast_node_t *parse_unary(parser_t *p)
// ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 606) - [public] check

```
    check(p, TK_TILDE)) {
        ast_node_t *n = new_node(p, ND_UN);

        n->op = advance(p)->kind;
        n->a = parse_unary(p);

        return n;
    }

    return parse_postfix(p);
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 615) - [public] parse_postfix

```
    return parse_postfix(p);
}

static ast_node_t *parse_bin(parser_t *p, int level)
{
    ast_node_t *lhs = parse_unary(p);

    for (;;) {
        tok_kind_t k = peek(p)->kind;
        int lv = 0;

        switch (k) {
            // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 626) - [public] switch

```
        switch (k) {
            case TK_STAR: case TK_SLASH: case TK_PERCENT: lv = 1; break;
            case TK_PLUS: case TK_MINUS: lv = 2; break;
            case TK_SHL: case TK_SHR: lv = 3; break;
            case TK_LT: case TK_GT: case TK_LEQ: case TK_GEQ: lv = 4; break;
            case TK_EQ: case TK_NEQ: lv = 5; break;
            case TK_AMP: lv = 6; break;
            case TK_CARET: lv = 7; break;
            case TK_PIPE: lv = 8; break;
            default: return lhs;
        }

        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 638) - [public] if

```
        if (lv > level) {
            return lhs;
        }

        advance(p);

        ast_node_t *rhs = parse_bin(p, lv + 1 > 8 ? 8 : lv + 1);

        ast_node_t *b = new_node(p, ND_BIN);

        b->op = k;
        b->a = lhs;
        // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 642) - [public] advance
    advance(p);

    ast_node_t *rhs = parse_bin(p, lv + 1 > 8 ? 8 : lv + 1);

    ast_node_t *b = new_node(p, ND_BIN);

    b->op = k;
    b->a = lhs;
    b->b = rhs;

    lhs = b;

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 662) - [public] if
    if (match(p, TK_ASSIGN)) {
        ast_node_t *n = new_node(p, ND_ASSIGN);

        n->a = lhs;
        n->b = parse_assign(p);

        return n;
    }

    return lhs;
}

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 676) - [public] parse_assign
    return parse_assign(p);
}

/* ---- top ---- */

ast_node_t *cl_parse(parser_t *p, token_t *toks, size_t n, arena_t *ar)
{
    p->toks = toks;
    p->n = n;
    p->pos = 0;
    p->ar = ar;
    p->failed = false;
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/parser.c (line 694) - [public] while
    while (!check(p, TK_EOF)) {
        ast_node_t *d;

        switch (peek(p)->kind) {
            case TK_FN:      d = parse_fn(p); break;
            case TK_LET:     d = parse_let(p); break;
            case TK_STATIC:  d = parse_static(p); break;
            case TK_ENUM:    d = parse_enum(p); break;
            case TK_EXTERN:  d = parse_extern(p); break;
            case TK_DATAIMPL: d = parse_dataimpl(p); break;
            default:
                perr(p, "expected top-level decl");
        }
        // ... (truncated, see source file for more)
    }
}

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 697) - [public] switch

```
    switch (peek(p)->kind) {
    case TK_FN:      d = parse_fn(p); break;
    case TK_LET:     d = parse_let(p); break;
    case TK_STATIC:  d = parse_static(p); break;
    case TK_ENUM:    d = parse_enum(p); break;
    case TK_EXTERN:  d = parse_extern(p); break;
    case TK_DATAIMPL: d = parse_dataimpl(p); break;
    default:
        perr(p, "expected top-level decl");
        advance(p);
        continue;
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 705) - [public] perr

```
    perr(p, "expected top-level decl");
    advance(p);
    continue;
}

nl_push(p, &tops, d);

if (p->failed) {
    break;
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 706) - [public] advance

```
    advance(p);
    continue;
}

nl_push(p, &tops, d);

if (p->failed) {
    break;
}

nl_commit(p, &tops, prog);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/parser.c (line 710) - [public] nl_push

```
    nl_push(p, &tops, d);

    if (p->failed) {
        break;
    }

    nl_commit(p, &tops, prog);

    if (p->failed) {
        return (void *)0;
    }
    // ... (truncated, see source file for more)
```



```
kernel_Workspace/kernel/src/core-lang/parser.c (line 712) - [public] if
    if (p->failed) {
        break;
    }
}

nl_commit(p, &tops, prog);

if (p->failed) {
    return (void *)0;
}

return prog;
// ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/core-lang/parser.c (line 717) - [public] nl_commit
    nl_commit(p, &tops, prog);

    if (p->failed) {
        return (void *)0;
    }

    return prog;
}
```

```
kernel_Workspace/kernel/src/core-lang/parser.c (line 719) - [public] if
    if (p->failed) {
        return (void *)0;
    }

    return prog;
}
```

```
kernel_Workspace/kernel/src/core-lang/run.c (line 4) - [public] cl_kernel_read
extern long cl_kernel_read(const char *path, uint8_t *buf, size_t cap);
extern void *cl_native_compile(cl_prog_t *P);

int cl_run_script(const char *path, uint8_t ring, arena_t *ar)
{
    static uint8_t io[64 * 1024];

    long n = cl_kernel_read(path, io, sizeof(io));

    if (n <= 0) {
        return -1;
    }
    // ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/core-lang/run.c (line 13) - [public] if
    if (n <= 0) {
        return -1;
    }

    uint32_t el = 0;
    const char *em = (void *)0;

    cl_prog_t *P = cl_compile_source((const char *)io, (size_t)n, ar, &el, &em);

    if (P == (void *)0) {
        extern void kprintf(const char *, ...);
        kprintf("core-lang: %s line %u: %s\n", path, el, em);
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/run.c (line 22) - [public] if

```
    if (P == (void *)0) {
        extern void kprintf(const char *, ...);
        kprintf("core-lang: %s line %u: %s\n", path, el, em);
        return -1;
    }

    cl_vm_t vm;

    cl_vm_init(&vm, P);
    cl_bridge_init(&vm, ring);

    void *code = cl_native_compile(P);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/run.c (line 23) - [public] kprintf

```
    extern void kprintf(const char *, ...);
    kprintf("core-lang: %s line %u: %s\n", path, el, em);
    return -1;
}

cl_vm_t vm;

cl_vm_init(&vm, P);
cl_bridge_init(&vm, ring);

void *code = cl_native_compile(P);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/run.c (line 24) - [public] kprintf

```
    kprintf("core-lang: %s line %u: %s\n", path, el, em);
    return -1;
}

cl_vm_t vm;

cl_vm_init(&vm, P);
cl_bridge_init(&vm, ring);

void *code = cl_native_compile(P);

if (code != (void *)0) {
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/run.c (line 30) - [public] cl_vm_init

```
    cl_vm_init(&vm, P);
    cl_bridge_init(&vm, ring);

    void *code = cl_native_compile(P);

    if (code != (void *)0) {
        extern void cl_make_exec(void *p, size_t len);
        cl_make_exec(code, 16384);

        static uint64_t slots[64], globs[64];
        static cl_ext_fn exts[16];

        // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/core-lang/run.c (line 31) - [public] cl_bridge_init
    cl_bridge_init(&vm, ring);

    void *code = cl_native_compile(P);

    if (code != (void *)0) {
        extern void cl_make_exec(void *p, size_t len);
        cl_make_exec(code, 16384);

        static uint64_t slots[64], globs[64];
        static cl_ext_fn exts[16];

        for (size_t i = 0; i < 64; i++) {
            // ... (truncated, see source file for more)
        }
    }

kernel_Workspace/kernel/src/core-lang/run.c (line 35) - [public] if
    if (code != (void *)0) {
        extern void cl_make_exec(void *p, size_t len);
        cl_make_exec(code, 16384);

        static uint64_t slots[64], globs[64];
        static cl_ext_fn exts[16];

        for (size_t i = 0; i < 64; i++) {
            globs[i] = 0;

            for (int k = 0; k < 8; k++) {
                globs[i] |= (uint64_t)P->ginit[i].b[k] << (k * 8);
            }
            // ... (truncated, see source file for more)
        }
    }

kernel_Workspace/kernel/src/core-lang/run.c (line 36) - [public] cl_make_exec
    extern void cl_make_exec(void *p, size_t len);
    cl_make_exec(code, 16384);

    static uint64_t slots[64], globs[64];
    static cl_ext_fn exts[16];

    for (size_t i = 0; i < 64; i++) {
        globs[i] = 0;

        for (int k = 0; k < 8; k++) {
            globs[i] |= (uint64_t)P->ginit[i].b[k] << (k * 8);
        }
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/run.c (line 37) - [public] cl_make_exec
    cl_make_exec(code, 16384);

    static uint64_t slots[64], globs[64];
    static cl_ext_fn exts[16];

    for (size_t i = 0; i < 64; i++) {
        globs[i] = 0;

        for (int k = 0; k < 8; k++) {
            globs[i] |= (uint64_t)P->ginit[i].b[k] << (k * 8);
        }
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/run.c (line 57) - [public] cl_vm_run
    cl_vm_run(&vm);
}

return 0;
}

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 45) - [public] if

```
    if (s == (void *)0) {
        return (void *)0;
    }

    s->ar = ar;
    s->top = 0;

    return s;
}

static void diag(sema_ctx_t *s, uint32_t line, const char *msg,
                const char *name, bool warn)
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/sema.c (line 58) - [public] if

```
    if (s->ndiags < 64) {
        cl_diag_t *d = &s->diags[s->ndiags++];

        d->line = line;
        d->msg = msg;
        d->is_warn = warn;
        d->name[0] = '\0';

        if (name != (void *)0) {
            for (int i = 0; name[i] && i < 63; i++) {
                d->name[i] = name[i];
                d->name[i + 1] = '\0';
            }
        }
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/sema.c (line 66) - [public] if

```
        if (name != (void *)0) {
            for (int i = 0; name[i] && i < 63; i++) {
                d->name[i] = name[i];
                d->name[i + 1] = '\0';
            }
        }
    }
}

static void push_scope(sema_ctx_t *s)
{
    if (s->top < 15) {
        // ... (truncated, see source file for more)
    }
}
```

kernel_Workspace/kernel/src/core-lang/sema.c (line 77) - [public] if

```
    if (s->top < 15) {
        s->top++;
        s->scopes[s->top].head = (void *)0;
    }
}

static void pop_scope(sema_ctx_t *s)
{
    for (symbol_t *sym = s->scopes[s->top].head; sym; sym = sym->next) {
        if (sym->kind == SYM_VAR && sym->live) {
            diag(s, 0, "warning: variable left alive without set_free",
                sym->name, true);
        }
// ... (truncated, see source file for more)
    }
}
```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 86) - [public] if
    if (sym->kind == SYM_VAR && sym->live) {
        diag(s, 0, "warning: variable left alive without set_free",
            sym->name, true);
    }
}

if (s->top > 0) {
    s->top--;
}

static symbol_t *declare(sema_ctx_t *s, const char *name, uint8_t kind,
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 92) - [public] if
    if (s->top > 0) {
        s->top--;
    }
}

static symbol_t *declare(sema_ctx_t *s, const char *name, uint8_t kind,
    cl_type_t type, uint32_t line)
{
    for (symbol_t *sym = s->scopes[s->top].head; sym; sym = sym->next) {
        bool same = true;

        for (int i = 0; name[i] || sym->name[i]; i++) {
            // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 104) - [public] if
        if (name[i] != sym->name[i]) {
            same = false;
            break;
        }
    }

    if (same) {
        diag(s, line, "redeclaration in same scope", name, false);
        return sym;
    }
}

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 110) - [public] if
    if (same) {
        diag(s, line, "redeclaration in same scope", name, false);
        return sym;
    }
}

symbol_t *sym = (symbol_t *)cl_arena_alloc(s->ar, sizeof(symbol_t));

if (sym == (void *)0) {
    return (void *)0;
}

// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 111) - [public] diag
    diag(s, line, "redeclaration in same scope", name, false);
    return sym;
}
}

symbol_t *sym = (symbol_t *)cl_arena_alloc(s->ar, sizeof(symbol_t));

if (sym == (void *)0) {
    return (void *)0;
}

for (int i = 0; name[i] && i < 63; i++) {
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 118) - [public] if
    if (sym == (void *)0) {
        return (void *)0;
    }

    for (int i = 0; name[i] && i < 63; i++) {
        sym->name[i] = name[i];
        sym->name[i + 1] = '\0';
    }

    sym->kind = kind;
    sym->type = type;
    sym->live = true;
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 149) - [public] if
    if (name[k] != sym->name[k]) {
        same = false;
        break;
    }
}

if (same) {
    return sym;
}
}

// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 155) - [public] if
    if (same) {
        return sym;
    }
}

return (void *)0;
}

```

```

static bool value_fits(uint64_t v, bool negative, cl_type_t t)
{
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 167) - [public] if
    if (t.bits == 0) {
        return true;
    }

    if (t.bits >= 64) {
        return true;
    }

    if (negative) {
        if (!t.neg) {
            return false;
        }
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 171) - [public] if
    if (t.bits >= 64) {
        return true;
    }

    if (negative) {
        if (!t.neg) {
            return false;
        }
    }

    return v <= (1ULL << (t.bits - 1));
}

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 175) - [public] if
    if (negative) {
        if (!t.neg) {
            return false;
        }
    }

    return v <= (1ULL << (t.bits - 1));
}

    if (t.neg) {
        return v <= (1ULL << (t.bits - 1)) - 1;
    }

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 176) - [public] if
    if (!t.neg) {
        return false;
    }

    return v <= (1ULL << (t.bits - 1));
}

    if (t.neg) {
        return v <= (1ULL << (t.bits - 1)) - 1;
    }

    return v < (1ULL << t.bits);
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 183) - [public] if
    if (t.neg) {
        return v <= (1ULL << (t.bits - 1)) - 1;
    }

    return v < (1ULL << t.bits);
}

typedef struct {
    cl_type_t t;
    bool is_const;
    uint64_t cval;
    bool cneg;
    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/core-lang/sema.c (line 197) - [public] check_expr
static etype_t check_expr(sema_ctx_t *s, ast_node_t *n);

static void shared_note(sema_ctx_t *s, const char *name, cl_type_t t,
                        uint32_t line)
{
    if (s->sh_count < 2) {
        for (int i = 0; name[i] && i < 63; i++) {
            s->sh_name[s->sh_count][i] = name[i];
            s->sh_name[s->sh_count][i + 1] = '\0';
        }

        s->sh_type[s->sh_count] = t;
        // ... (truncated, see source file for more)
    }
}

kernel_Workspace/kernel/src/core-lang/sema.c (line 202) - [public] if
    if (s->sh_count < 2) {
        for (int i = 0; name[i] && i < 63; i++) {
            s->sh_name[s->sh_count][i] = name[i];
            s->sh_name[s->sh_count][i + 1] = '\0';
        }

        s->sh_type[s->sh_count] = t;
        s->sh_line[s->sh_count] = line;
        s->sh_count++;
    }
}

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 216) - [public] if
    if (s->sh_count == 1) {
        diag(s, s->sh_line[0], "$@ requires two participating variables",
            s->sh_name[0], false);
    } else if (s->sh_count == 2) {
        if (s->sh_type[0].bits != s->sh_type[1].bits) {
            diag(s, s->sh_line[1], "$@ pair must share the same range",
                s->sh_name[1], false);
        }
    }

    s->sh_count = 0;
}

// ... (truncated, see source file for more)

```



```

kernel_Workspace/kernel/src/core-lang/sema.c (line 220) - [public] if
    if (s->sh_type[0].bits != s->sh_type[1].bits) {
        diag(s, s->sh_line[1], "$@ pair must share the same range",
            s->sh_name[1], false);
    }
}

s->sh_count = 0;
}

static etype_t check_expr(sema_ctx_t *s, ast_node_t *n)
{
    etype_t r = { { 0, false }, false, 0, false };
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 233) - [public] if
    if (n == (void *)0) {
        return r;
    }

    switch (n->kind) {
    case ND_NUM:
        r.is_const = true;
        r.cval = n->value;
        return r;

    case ND_CHAR:
        r.is_const = true;
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 237) - [public] switch
    switch (n->kind) {
    case ND_NUM:
        r.is_const = true;
        r.cval = n->value;
        return r;

    case ND_CHAR:
        r.is_const = true;
        r.cval = n->value;
        r.t.bits = 64;
        return r;

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 255) - [public] if
    if (n->op == 45 ) {
        if (r.is_const) {
            r.cneg = true;
        } else if (r.t.bits != 0 && !r.t.neg) {
            diag(s, n->line, "negation needs _=> range", (void *)0, false);
        }
    }

    return r;

    case ND_VAR: {
        symbol_t *sym = lookup(s, n->name);
        // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 256) - [public] if
    if (r.is_const) {
        r.cneg = true;
    } else if (r.t.bits != 0 && !r.t.neg) {
        diag(s, n->line, "negation needs _=> range", (void *)0, false);
    }
}

return r;

case ND_VAR: {
    symbol_t *sym = lookup(s, n->name);

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 259) - [public] diag
    diag(s, n->line, "negation needs _=> range", (void *)0, false);
}

return r;

case ND_VAR: {
    symbol_t *sym = lookup(s, n->name);

    if (sym == (void *)0) {
        diag(s, n->line, "undeclared identifier", n->name, false);
        return r;
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 268) - [public] if
    if (sym == (void *)0) {
        diag(s, n->line, "undeclared identifier", n->name, false);
        return r;
    }

    if (sym->kind == SYM_VAR || sym->kind == SYM_STATIC) {
        if (!sym->live) {
            diag(s, n->line, "use after set_free", n->name, false);
        }

        r.t = sym->type;
    } else if (sym->kind == SYM_ENUM) {
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 269) - [public] diag
    diag(s, n->line, "undeclared identifier", n->name, false);
    return r;
}

    if (sym->kind == SYM_VAR || sym->kind == SYM_STATIC) {
        if (!sym->live) {
            diag(s, n->line, "use after set_free", n->name, false);
        }

        r.t = sym->type;
    } else if (sym->kind == SYM_ENUM) {
        r.t.bits = 64;
        // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 273) - [public] if
    if (sym->kind == SYM_VAR || sym->kind == SYM_STATIC) {
        if (!sym->live) {
            diag(s, n->line, "use after set_free", n->name, false);
        }

        r.t = sym->type;
    } else if (sym->kind == SYM_ENUM) {
        r.t.bits = 64;
    } else {
        diag(s, n->line, "not a value", n->name, false);
    }

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 274) - [public] if
    if (!sym->live) {
        diag(s, n->line, "use after set_free", n->name, false);
    }

    r.t = sym->type;
} else if (sym->kind == SYM_ENUM) {
    r.t.bits = 64;
} else {
    diag(s, n->line, "not a value", n->name, false);
}

    return r;
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 275) - [public] diag
    diag(s, n->line, "use after set_free", n->name, false);
}

    r.t = sym->type;
} else if (sym->kind == SYM_ENUM) {
    r.t.bits = 64;
} else {
    diag(s, n->line, "not a value", n->name, false);
}

    return r;
}
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 282) - [public] diag
    diag(s, n->line, "not a value", n->name, false);
}

    return r;
}

case ND_POOL: {
    symbol_t *sym = lookup(s, n->name);

    if (sym == (void *)0) {
        diag(s, n->line, "undeclared identifier", n->name, false);
        return r;
    }
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 291) - [public] if
    if (sym == (void *)0) {
        diag(s, n->line, "undeclared identifier", n->name, false);
        return r;
    }

    if (sym->depth != 0 || sym->kind != SYM_VAR) {
        diag(s, n->line, "$ works only on base pool variables",
            n->name, false);
    }

    if (!sym->live) {
        diag(s, n->line, "use after set_free", n->name, false);
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 292) - [public] diag
    diag(s, n->line, "undeclared identifier", n->name, false);
    return r;
}

    if (sym->depth != 0 || sym->kind != SYM_VAR) {
        diag(s, n->line, "$ works only on base pool variables",
            n->name, false);
    }

    if (!sym->live) {
        diag(s, n->line, "use after set_free", n->name, false);
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 296) - [public] if
    if (sym->depth != 0 || sym->kind != SYM_VAR) {
        diag(s, n->line, "$ works only on base pool variables",
            n->name, false);
    }

    if (!sym->live) {
        diag(s, n->line, "use after set_free", n->name, false);
    }

    sym->pool_ref = true;

    if (n->pool_kind == 1 && !sym->type.neg) {
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 301) - [public] if
    if (!sym->live) {
        diag(s, n->line, "use after set_free", n->name, false);
    }

    sym->pool_ref = true;

    if (n->pool_kind == 1 && !sym->type.neg) {
        diag(s, n->line, "$! needs neg range (=>-1)", n->name, false);
    }

    if (n->pool_kind == 2) {
        sym->live = true;
    }
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 302) - [public] diag
    diag(s, n->line, "use after set_free", n->name, false);
}

    sym->pool_ref = true;

    if (n->pool_kind == 1 && !sym->type.neg) {
        diag(s, n->line, "$! needs neg range (=>-1)", n->name, false);
    }

    if (n->pool_kind == 2) {
        sym->live = true;
        shared_note(s, n->name, sym->type, n->line);
    }
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 307) - [public] if
    if (n->pool_kind == 1 && !sym->type.neg) {
        diag(s, n->line, "$! needs neg range (=>-1)", n->name, false);
    }

    if (n->pool_kind == 2) {
        sym->live = true;
        shared_note(s, n->name, sym->type, n->line);
    }

    r.t = sym->type;
    return r;
}
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 308) - [public] diag
    diag(s, n->line, "$! needs neg range (=>-1)", n->name, false);
}

    if (n->pool_kind == 2) {
        sym->live = true;
        shared_note(s, n->name, sym->type, n->line);
    }

    r.t = sym->type;
    return r;
}

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 311) - [public] if
    if (n->pool_kind == 2) {
        sym->live = true;
        shared_note(s, n->name, sym->type, n->line);
    }

    r.t = sym->type;
    return r;
}

case ND_CALLIDX: {
    symbol_t *sym = lookup(s, n->name);

// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 313) - [public] shared_note
    shared_note(s, n->name, sym->type, n->line);
}

    r.t = sym->type;
    return r;
}

case ND_CALLIDX: {
    symbol_t *sym = lookup(s, n->name);

    if (sym == (void *)0) {
        diag(s, n->line, "call[] on undeclared", n->name, false);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 323) - [public] if
    if (sym == (void *)0) {
        diag(s, n->line, "call[] on undeclared", n->name, false);
        return r;
    }

    if ((sym->kind == SYM_VAR || sym->kind == SYM_STATIC) && !sym->live) {
        diag(s, n->line, "use after set_free", n->name, false);
    }

    r.t = (sym->kind == SYM_FN || sym->kind == SYM_EXTERN)
        ? (cl_type_t){ 64, false } : sym->type;
    return r;
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 324) - [public] diag
    diag(s, n->line, "call[] on undeclared", n->name, false);
    return r;
}

    if ((sym->kind == SYM_VAR || sym->kind == SYM_STATIC) && !sym->live) {
        diag(s, n->line, "use after set_free", n->name, false);
    }

    r.t = (sym->kind == SYM_FN || sym->kind == SYM_EXTERN)
        ? (cl_type_t){ 64, false } : sym->type;
    return r;
}
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 328) - [public] if
    if ((sym->kind == SYM_VAR || sym->kind == SYM_STATIC) && !sym->live) {
        diag(s, n->line, "use after set_free", n->name, false);
    }

    r.t = (sym->kind == SYM_FN || sym->kind == SYM_EXTERN)
        ? (cl_type_t){ 64, false } : sym->type;
    return r;
}

case ND_SCOPECALL: {
    symbol_t *sc = lookup(s, n->name);

    // ... (truncated, see source file for more)

```

```
kernel_Workspace/kernel/src/core-lang/sema.c (line 329) - [public] diag
    diag(s, n->line, "use after set_free", n->name, false);
}

r.t = (sym->kind == SYM_FN || sym->kind == SYM_EXTERN)
    ? (cl_type_t){ 64, false } : sym->type;
return r;
}
```

```
case ND_SCOPECALL: {
    symbol_t *sc = lookup(s, n->name);

    if (sc == (void *)0) {
// ... (truncated, see source file for more)
}
```

```
kernel_Workspace/kernel/src/core-lang/sema.c (line 340) - [public] if
    if (sc == (void *)0) {
        diag(s, n->line, "unknown scope", n->name, false);
        return r;
    }
```

```
    if (sc->kind == SYM_ENUM) {
        bool found = false;

        if (sc->node != (void *)0) {
            for (size_t i = 0; i < sc->node->nlist; i++) {
                ast_node_t *it = sc->node->list[i];

// ... (truncated, see source file for more)
}
```

```
kernel_Workspace/kernel/src/core-lang/sema.c (line 341) - [public] diag
    diag(s, n->line, "unknown scope", n->name, false);
    return r;
}
```

```
    if (sc->kind == SYM_ENUM) {
        bool found = false;

        if (sc->node != (void *)0) {
            for (size_t i = 0; i < sc->node->nlist; i++) {
                ast_node_t *it = sc->node->list[i];

                bool same = true;
// ... (truncated, see source file for more)
}
```

```
kernel_Workspace/kernel/src/core-lang/sema.c (line 345) - [public] if
    if (sc->kind == SYM_ENUM) {
        bool found = false;

        if (sc->node != (void *)0) {
            for (size_t i = 0; i < sc->node->nlist; i++) {
                ast_node_t *it = sc->node->list[i];

                bool same = true;

                for (int k = 0; it->name[k] || n->name2[k]; k++) {
                    if (it->name[k] != n->name2[k]) {
                        same = false;
// ... (truncated, see source file for more)
}
```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 348) - [public] if
    if (sc->node != (void *)0) {
        for (size_t i = 0; i < sc->node->nlist; i++) {
            ast_node_t *it = sc->node->list[i];

            bool same = true;

            for (int k = 0; it->name[k] || n->name2[k]; k++) {
                if (it->name[k] != n->name2[k]) {
                    same = false;
                    break;
                }
            }
            // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 355) - [public] if
        if (it->name[k] != n->name2[k]) {
            same = false;
            break;
        }

        if (same) {
            found = true;
            r.is_const = true;
            r.cval = it->value;
            r.t.bits = 64;
        }
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 361) - [public] if
        if (same) {
            found = true;
            r.is_const = true;
            r.cval = it->value;
            r.t.bits = 64;
        }
    }

    if (!found) {
        diag(s, n->line, "no such enum item", n->name2, false);
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 370) - [public] if
    if (!found) {
        diag(s, n->line, "no such enum item", n->name2, false);
    }

    return r;
}

if (sc->kind == SYM_DATA) {
    bool found = false;

    if (sc->node != (void *)0 && sc->node->a != (void *)0) {
        ast_node_t *base = sc->node->a;
        // ... (truncated, see source file for more)

```



```

kernel_Workspace/kernel/src/core-lang/sema.c (line 371) - [public] diag
    diag(s, n->line, "no such enum item", n->name2, false);
}

    return r;
}

if (sc->kind == SYM_DATA) {
    bool found = false;

    if (sc->node != (void *)0 && sc->node->a != (void *)0) {
        ast_node_t *base = sc->node->a;

        // ... (truncated, see source file for more)
kernel_Workspace/kernel/src/core-lang/sema.c (line 377) - [public] if
    if (sc->kind == SYM_DATA) {
        bool found = false;

        if (sc->node != (void *)0 && sc->node->a != (void *)0) {
            ast_node_t *base = sc->node->a;

            for (size_t i = 0; i < base->nlist; i++) {
                ast_node_t *e = base->list[i];

                if (e->kind == ND_VAR) {
                    bool same = true;

                    // ... (truncated, see source file for more)
kernel_Workspace/kernel/src/core-lang/sema.c (line 380) - [public] if
    if (sc->node != (void *)0 && sc->node->a != (void *)0) {
        ast_node_t *base = sc->node->a;

        for (size_t i = 0; i < base->nlist; i++) {
            ast_node_t *e = base->list[i];

            if (e->kind == ND_VAR) {
                bool same = true;

                for (int k = 0; e->name[k] || n->name2[k]; k++) {
                    if (e->name[k] != n->name2[k]) {
                        same = false;
                    }
                }

                // ... (truncated, see source file for more)
kernel_Workspace/kernel/src/core-lang/sema.c (line 386) - [public] if
    if (e->kind == ND_VAR) {
        bool same = true;

        for (int k = 0; e->name[k] || n->name2[k]; k++) {
            if (e->name[k] != n->name2[k]) {
                same = false;
                break;
            }
        }

        if (same) {
            found = true;
        }

        // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 390) - [public] if
        if (e->name[k] != n->name2[k]) {
            same = false;
            break;
        }
    }

    if (same) {
        found = true;
        r.t.bits = 64;
    }
}
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 396) - [public] if
        if (same) {
            found = true;
            r.t.bits = 64;
        }
    }
}

    if (!found) {
        diag(s, n->line, "no such data entry", n->name2, false);
    }

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 404) - [public] if
    if (!found) {
        diag(s, n->line, "no such data entry", n->name2, false);
    }

    return r;
}

    diag(s, n->line, "scope is not enum/data", n->name, false);
    return r;
}

case ND_CALL: {
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 405) - [public] diag
        diag(s, n->line, "no such data entry", n->name2, false);
    }

    return r;
}

    diag(s, n->line, "scope is not enum/data", n->name, false);
    return r;
}

case ND_CALL: {
    if (n->a == (void *)0 || n->a->kind != ND_VAR) {
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 411) - [public] diag
    diag(s, n->line, "scope is not enum/data", n->name, false);
    return r;
}

case ND_CALL: {
    if (n->a == (void *)0 || n->a->kind != ND_VAR) {
        diag(s, n->line, "bad call target", (void *)0, false);
        return r;
    }

    symbol_t *fn = lookup(s, n->a->name);

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 416) - [public] if
    if (n->a == (void *)0 || n->a->kind != ND_VAR) {
        diag(s, n->line, "bad call target", (void *)0, false);
        return r;
    }

    symbol_t *fn = lookup(s, n->a->name);

    if (fn == (void *)0) {
        diag(s, n->line, "call of undeclared", n->a->name, false);
        return r;
    }

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 417) - [public] diag
    diag(s, n->line, "bad call target", (void *)0, false);
    return r;
}

    symbol_t *fn = lookup(s, n->a->name);

    if (fn == (void *)0) {
        diag(s, n->line, "call of undeclared", n->a->name, false);
        return r;
    }

    if (fn->kind == SYM_FN) {
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 423) - [public] if
    if (fn == (void *)0) {
        diag(s, n->line, "call of undeclared", n->a->name, false);
        return r;
    }

    if (fn->kind == SYM_FN) {
        if (fn->nparams != n->nlist) {
            diag(s, n->line, "arity mismatch", fn->name, false);
        }

        for (size_t i = 0; i < n->nlist && i < fn->nparams; i++) {
            etype_t a = check_expr(s, n->list[i]);
        }
    }

    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 424) - [public] diag
    diag(s, n->line, "call of undeclared", n->a->name, false);
    return r;
}

if (fn->kind == SYM_FN) {
    if (fn->nparams != n->nlist) {
        diag(s, n->line, "arity mismatch", fn->name, false);
    }

    for (size_t i = 0; i < n->nlist && i < fn->nparams; i++) {
        etype_t a = check_expr(s, n->list[i]);

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 428) - [public] if
    if (fn->kind == SYM_FN) {
        if (fn->nparams != n->nlist) {
            diag(s, n->line, "arity mismatch", fn->name, false);
        }

        for (size_t i = 0; i < n->nlist && i < fn->nparams; i++) {
            etype_t a = check_expr(s, n->list[i]);

            if (a.is_const &&
                !value_fits(a.cval, a.cneg, fn->params[i])) {
                diag(s, n->line, "arg out of range", fn->name, false);
            }
        }
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 429) - [public] if
    if (fn->nparams != n->nlist) {
        diag(s, n->line, "arity mismatch", fn->name, false);
    }

    for (size_t i = 0; i < n->nlist && i < fn->nparams; i++) {
        etype_t a = check_expr(s, n->list[i]);

        if (a.is_const &&
            !value_fits(a.cval, a.cneg, fn->params[i])) {
            diag(s, n->line, "arg out of range", fn->name, false);
        }
    }
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 430) - [public] diag
    diag(s, n->line, "arity mismatch", fn->name, false);
}

for (size_t i = 0; i < n->nlist && i < fn->nparams; i++) {
    etype_t a = check_expr(s, n->list[i]);

    if (a.is_const &&
        !value_fits(a.cval, a.cneg, fn->params[i])) {
        diag(s, n->line, "arg out of range", fn->name, false);
    }
}

// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 438) - [public] diag
        diag(s, n->line, "arg out of range", fn->name, false);
    }
}

    r.t.bits = 64;
    return r;
}

    if (fn->kind == SYM_EXTERN) {
        for (size_t i = 0; i < n->nlist; i++) {
            check_expr(s, n->list[i]);
        }
    }
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 446) - [public] if
    if (fn->kind == SYM_EXTERN) {
        for (size_t i = 0; i < n->nlist; i++) {
            check_expr(s, n->list[i]);
        }

        r.t.bits = 64;
        return r;
    }

    diag(s, n->line, "not callable", n->a->name, false);
    return r;
}
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 448) - [public] check_expr
        check_expr(s, n->list[i]);
    }

    r.t.bits = 64;
    return r;
}

    diag(s, n->line, "not callable", n->a->name, false);
    return r;
}

case ND_MEMBER:
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 455) - [public] diag
    diag(s, n->line, "not callable", n->a->name, false);
    return r;
}

case ND_MEMBER:
    r = check_expr(s, n->a);
    return r;

case ND_ASSIGN: {
    etype_t l = check_expr(s, n->a);
    etype_t rhs = check_expr(s, n->b);

// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 467) - [public] if
    if (l.t.bits != 0) {
        if (rhs.is_const) {
            if (!value_fits(rhs.cval, rhs.cneg, l.t)) {
                diag(s, n->line, "value out of range", (void *)0, false);
            }
        }
        } else if (rhs.t.bits != 0) {
            if (rhs.t.bits > l.t.bits) {
                diag(s, n->line, "narrowing assignment", (void *)0, false);
            } else if (rhs.t.bits == l.t.bits &&
                rhs.t.neg != l.t.neg) {
                diag(s, n->line, "signed/unsigned mismatch",
                    (void *)0, false);
            }
        }
    }
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 468) - [public] if
    if (rhs.is_const) {
        if (!value_fits(rhs.cval, rhs.cneg, l.t)) {
            diag(s, n->line, "value out of range", (void *)0, false);
        }
    }
    } else if (rhs.t.bits != 0) {
        if (rhs.t.bits > l.t.bits) {
            diag(s, n->line, "narrowing assignment", (void *)0, false);
        } else if (rhs.t.bits == l.t.bits &&
            rhs.t.neg != l.t.neg) {
            diag(s, n->line, "signed/unsigned mismatch",
                (void *)0, false);
        }
    }
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 469) - [public] if
    if (!value_fits(rhs.cval, rhs.cneg, l.t)) {
        diag(s, n->line, "value out of range", (void *)0, false);
    }
    } else if (rhs.t.bits != 0) {
        if (rhs.t.bits > l.t.bits) {
            diag(s, n->line, "narrowing assignment", (void *)0, false);
        } else if (rhs.t.bits == l.t.bits &&
            rhs.t.neg != l.t.neg) {
            diag(s, n->line, "signed/unsigned mismatch",
                (void *)0, false);
        }
    }
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 470) - [public] diag
    diag(s, n->line, "value out of range", (void *)0, false);
    }
    } else if (rhs.t.bits != 0) {
        if (rhs.t.bits > l.t.bits) {
            diag(s, n->line, "narrowing assignment", (void *)0, false);
        } else if (rhs.t.bits == l.t.bits &&
            rhs.t.neg != l.t.neg) {
            diag(s, n->line, "signed/unsigned mismatch",
                (void *)0, false);
        }
    }
    }
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 473) - [public] if
    if (rhs.t.bits > l.t.bits) {
        diag(s, n->line, "narrowing assignment", (void *)0, false);
    } else if (rhs.t.bits == l.t.bits &&
        rhs.t.neg != l.t.neg) {
        diag(s, n->line, "signed/unsigned mismatch",
            (void *)0, false);
    }
}
}
return l;
}
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 474) - [public] diag
    diag(s, n->line, "narrowing assignment", (void *)0, false);
    } else if (rhs.t.bits == l.t.bits &&
        rhs.t.neg != l.t.neg) {
        diag(s, n->line, "signed/unsigned mismatch",
            (void *)0, false);
    }
}
}
return l;
}
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 493) - [public] if
    if (cmp) {
        r.t.bits = 1;
        return r;
    }

    if (a.t.bits == 0 && b.t.bits != 0) {
        if (a.is_const && !value_fits(a.cval, a.cneg, b.t)) {
            diag(s, n->line, "value out of range", (void *)0, false);
        }

        r.t = b.t;
    } else if (b.t.bits == 0 && a.t.bits != 0) {
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 498) - [public] if
    if (a.t.bits == 0 && b.t.bits != 0) {
        if (a.is_const && !value_fits(a.cval, a.cneg, b.t)) {
            diag(s, n->line, "value out of range", (void *)0, false);
        }

        r.t = b.t;
    } else if (b.t.bits == 0 && a.t.bits != 0) {
        if (b.is_const && !value_fits(b.cval, b.cneg, a.t)) {
            diag(s, n->line, "value out of range", (void *)0, false);
        }

        r.t = a.t;
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 499) - [public] if
    if (a.is_const && !value_fits(a.cval, a.cneg, b.t)) {
        diag(s, n->line, "value out of range", (void *)0, false);
    }

    r.t = b.t;
} else if (b.t.bits == 0 && a.t.bits != 0) {
    if (b.is_const && !value_fits(b.cval, b.cneg, a.t)) {
        diag(s, n->line, "value out of range", (void *)0, false);
    }

    r.t = a.t;
} else {
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 500) - [public] diag
    diag(s, n->line, "value out of range", (void *)0, false);
}

    r.t = b.t;
} else if (b.t.bits == 0 && a.t.bits != 0) {
    if (b.is_const && !value_fits(b.cval, b.cneg, a.t)) {
        diag(s, n->line, "value out of range", (void *)0, false);
    }

    r.t = a.t;
} else {
    r.t.bits = a.t.bits > b.t.bits ? a.t.bits : b.t.bits;
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 505) - [public] if
    if (b.is_const && !value_fits(b.cval, b.cneg, a.t)) {
        diag(s, n->line, "value out of range", (void *)0, false);
    }

    r.t = a.t;
} else {
    r.t.bits = a.t.bits > b.t.bits ? a.t.bits : b.t.bits;
    r.t.neg = a.t.neg || b.t.neg;
}

    return r;
}
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 506) - [public] diag
    diag(s, n->line, "value out of range", (void *)0, false);
}

    r.t = a.t;
} else {
    r.t.bits = a.t.bits > b.t.bits ? a.t.bits : b.t.bits;
    r.t.neg = a.t.neg || b.t.neg;
}

    return r;
}
// ... (truncated, see source file for more)

```



```

kernel_Workspace/kernel/src/core-lang/sema.c (line 524) - [public] check_block
static void check_block(sema_ctx_t *s, ast_node_t *n);

static void check_stmt(sema_ctx_t *s, ast_node_t *n)
{
    s->sh_count = 0;

    switch (n->kind) {
    case ND_LET: {
        etype_t init = check_expr(s, n->a);

        if (init.is_const &&
            !value_fits(init.cval, init.cneg, n->type)) {
            // ... (truncated, see source file for more)
        }
    }
}

kernel_Workspace/kernel/src/core-lang/sema.c (line 530) - [public] switch
switch (n->kind) {
case ND_LET: {
    etype_t init = check_expr(s, n->a);

    if (init.is_const &&
        !value_fits(init.cval, init.cneg, n->type)) {
        diag(s, n->line, "literal out of type range", n->name, false);
    }

    if (init.t.bits != 0 && init.t.bits > n->type.bits) {
        diag(s, n->line, "narrowing assignment", n->name, false);
    }
    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/core-lang/sema.c (line 536) - [public] diag
diag(s, n->line, "literal out of type range", n->name, false);
}

if (init.t.bits != 0 && init.t.bits > n->type.bits) {
    diag(s, n->line, "narrowing assignment", n->name, false);
}

symbol_t *sym = declare(s, n->name, SYM_VAR, n->type, n->line);

if (sym != (void *)0 && n->type.neg) {
    sym->type.neg = true;
}
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 539) - [public] if
if (init.t.bits != 0 && init.t.bits > n->type.bits) {
    diag(s, n->line, "narrowing assignment", n->name, false);
}

symbol_t *sym = declare(s, n->name, SYM_VAR, n->type, n->line);

if (sym != (void *)0 && n->type.neg) {
    sym->type.neg = true;
}

break;
}
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 540) - [public] diag
    diag(s, n->line, "narrowing assignment", n->name, false);
}

symbol_t *sym = declare(s, n->name, SYM_VAR, n->type, n->line);

if (sym != (void *)0 && n->type.neg) {
    sym->type.neg = true;
}

break;
}

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 545) - [public] if
    if (sym != (void *)0 && n->type.neg) {
        sym->type.neg = true;
    }

    break;
}

case ND_STATIC:
    declare(s, "data", SYM_STATIC, n->type, n->line);
    break;

case ND_SET_FREE: {
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 553) - [public] declare
    declare(s, "data", SYM_STATIC, n->type, n->line);
    break;

case ND_SET_FREE: {
    symbol_t *sym = lookup(s, n->name);

    if (sym == (void *)0) {
        diag(s, n->line, "set_free on undeclared", n->name, false);
    } else if (sym->freed) {
        diag(s, n->line, "double set_free", n->name, false);
    } else {
        sym->live = false;
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 559) - [public] if
    if (sym == (void *)0) {
        diag(s, n->line, "set_free on undeclared", n->name, false);
    } else if (sym->freed) {
        diag(s, n->line, "double set_free", n->name, false);
    } else {
        sym->live = false;
        sym->freed = true;
    }

    break;
}

// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 560) - [public] diag
    diag(s, n->line, "set_free on undeclared", n->name, false);
} else if (sym->freed) {
    diag(s, n->line, "double set_free", n->name, false);
} else {
    sym->live = false;
    sym->freed = true;
}

break;
}

case ND_IF:
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 562) - [public] diag
    diag(s, n->line, "double set_free", n->name, false);
} else {
    sym->live = false;
    sym->freed = true;
}

break;
}

case ND_IF:
    if (n->a != (void *)0) {
        check_expr(s, n->a);
    }
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 572) - [public] if
    if (n->a != (void *)0) {
        check_expr(s, n->a);
    }

    check_block(s, n->b);

    if (n->c != (void *)0) {
        if (n->c->kind == ND_BLOCK) {
            check_block(s, n->c);
        } else {
            check_stmt(s, n->c);
        }
    }
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 573) - [public] check_expr
    check_expr(s, n->a);
}

check_block(s, n->b);

if (n->c != (void *)0) {
    if (n->c->kind == ND_BLOCK) {
        check_block(s, n->c);
    } else {
        check_stmt(s, n->c);
    }
}
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 576) - [public] check_block
    check_block(s, n->b);

    if (n->c != (void *)0) {
        if (n->c->kind == ND_BLOCK) {
            check_block(s, n->c);
        } else {
            check_stmt(s, n->c);
        }
    }

    break;

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 578) - [public] if
    if (n->c != (void *)0) {
        if (n->c->kind == ND_BLOCK) {
            check_block(s, n->c);
        } else {
            check_stmt(s, n->c);
        }
    }

    break;

case ND_WHEN: {
    symbol_t *tgt = lookup(s, n->name);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 579) - [public] if
    if (n->c->kind == ND_BLOCK) {
        check_block(s, n->c);
    } else {
        check_stmt(s, n->c);
    }
}

break;

case ND_WHEN: {
    symbol_t *tgt = lookup(s, n->name);

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 580) - [public] check_block
    check_block(s, n->c);
} else {
    check_stmt(s, n->c);
}

break;

case ND_WHEN: {
    symbol_t *tgt = lookup(s, n->name);

    if (tgt == (void *)0) {
        // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 582) - [public] check_stmt
    check_stmt(s, n->c);
    }
}

break;

case ND_WHEN: {
    symbol_t *tgt = lookup(s, n->name);

    if (tgt == (void *)0) {
        diag(s, n->line, "when on undeclared", n->name, false);
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 591) - [public] if
    if (tgt == (void *)0) {
        diag(s, n->line, "when on undeclared", n->name, false);
    }

    for (size_t i = 0; i < n->nlist; i++) {
        ast_node_t *c = n->list[i];

        if (c->a != (void *)0 && c->a->kind == ND_BIN &&
            c->a->a != (void *)0 && c->a->a->kind == ND_MEMBER &&
            c->a->a->a != (void *)0 &&
            c->a->a->a->kind == ND_VAR) {
            const char *cn = c->a->a->a->name;
            // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 592) - [public] diag
        diag(s, n->line, "when on undeclared", n->name, false);
    }

    for (size_t i = 0; i < n->nlist; i++) {
        ast_node_t *c = n->list[i];

        if (c->a != (void *)0 && c->a->kind == ND_BIN &&
            c->a->a != (void *)0 && c->a->a->kind == ND_MEMBER &&
            c->a->a->a != (void *)0 &&
            c->a->a->a->kind == ND_VAR) {
            const char *cn = c->a->a->a->name;
            bool same = true;
            // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 606) - [public] if
        if (cn[k] != n->name[k]) {
            same = false;
            break;
        }
    }

    if (!same) {
        diag(s, c->line, "case target mismatch", cn, false);
    }

    if (tgt != (void *)0 && c->a->b != (void *)0 &&
        c->a->b->kind == ND_NUM &&
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 612) - [public] if
    if (!same) {
        diag(s, c->line, "case target mismatch", cn, false);
    }

    if (tgt != (void *)0 && c->a->b != (void *)0 &&
        c->a->b->kind == ND_NUM &&
        !value_fits(c->a->b->value, false, tgt->type)) {
        diag(s, c->line, "case value out of range",
            n->name, false);
    }
}

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 613) - [public] diag
    diag(s, c->line, "case target mismatch", cn, false);
}

    if (tgt != (void *)0 && c->a->b != (void *)0 &&
        c->a->b->kind == ND_NUM &&
        !value_fits(c->a->b->value, false, tgt->type)) {
        diag(s, c->line, "case value out of range",
            n->name, false);
    }
}

    check_block(s, c->b);
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 624) - [public] check_block
    check_block(s, c->b);
}

    break;
}

case ND_BLOCK:
    check_block(s, n);
    break;

default:
    check_expr(s, n);
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 631) - [public] check_block
    check_block(s, n);
    break;

default:
    check_expr(s, n);
    shared_validate(s);
    break;
}

if (n->kind != ND_BLOCK) {
    shared_validate(s);
}
// ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 635) - [public] check_expr

```
        check_expr(s, n);
        shared_validate(s);
        break;
    }

    if (n->kind != ND_BLOCK) {
        shared_validate(s);
    }
}

static void check_block(sema_ctx_t *s, ast_node_t *n)
{
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/sema.c (line 636) - [public] shared_validate

```
        shared_validate(s);
        break;
    }

    if (n->kind != ND_BLOCK) {
        shared_validate(s);
    }
}

static void check_block(sema_ctx_t *s, ast_node_t *n)
{
    push_scope(s);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/sema.c (line 640) - [public] if

```
    if (n->kind != ND_BLOCK) {
        shared_validate(s);
    }
}

static void check_block(sema_ctx_t *s, ast_node_t *n)
{
    push_scope(s);

    for (size_t i = 0; i < n->nlist; i++) {
        check_stmt(s, n->list[i]);
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/sema.c (line 641) - [public] shared_validate

```
        shared_validate(s);
    }
}

static void check_block(sema_ctx_t *s, ast_node_t *n)
{
    push_scope(s);

    for (size_t i = 0; i < n->nlist; i++) {
        check_stmt(s, n->list[i]);
    }

    // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 647) - [public] push_scope
    push_scope(s);

    for (size_t i = 0; i < n->nlist; i++) {
        check_stmt(s, n->list[i]);
    }

    pop_scope(s);
}

static void declare_top(sema_ctx_t *s, ast_node_t *n)
{
    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/core-lang/sema.c (line 650) - [public] check_stmt
    check_stmt(s, n->list[i]);
}

    pop_scope(s);
}

static void declare_top(sema_ctx_t *s, ast_node_t *n)
{
    switch (n->kind) {
    case ND_FN: {
        cl_type_t t = { 0, false };
        // ... (truncated, see source file for more)
    }

kernel_Workspace/kernel/src/core-lang/sema.c (line 653) - [public] pop_scope
    pop_scope(s);
}

static void declare_top(sema_ctx_t *s, ast_node_t *n)
{
    switch (n->kind) {
    case ND_FN: {
        cl_type_t t = { 0, false };
        symbol_t *sym = declare(s, n->name, SYM_FN, t, n->line);

        if (sym == (void *)0) {
            // ... (truncated, see source file for more)
        }

kernel_Workspace/kernel/src/core-lang/sema.c (line 659) - [public] switch
    switch (n->kind) {
    case ND_FN: {
        cl_type_t t = { 0, false };
        symbol_t *sym = declare(s, n->name, SYM_FN, t, n->line);

        if (sym == (void *)0) {
            return;
        }

        for (size_t i = 0; i < n->nlist && i < 8; i++) {
            sym->params[sym->nparams++] = n->list[i]->type;
        }
        // ... (truncated, see source file for more)
    }
}

```



```

kernel_Workspace/kernel/src/core-lang/sema.c (line 664) - [public] if
    if (sym == (void *)0) {
        return;
    }

    for (size_t i = 0; i < n->nlist && i < 8; i++) {
        sym->params[sym->nparams++] = n->list[i]->type;
    }

    sym->node = n;
    break;
}

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 677) - [public] declare
    declare(s, n->name, SYM_VAR, n->type, n->line);
    break;

case ND_STATIC:
    declare(s, "data", SYM_STATIC, n->type, n->line);
    break;

case ND_ENUM: {
    cl_type_t t = { 64, false };
    symbol_t *sym = declare(s, n->name, SYM_ENUM, t, n->line);

    if (sym != (void *)0) {
        // ... (truncated, see source file for more)
    }

kernel_Workspace/kernel/src/core-lang/sema.c (line 681) - [public] declare
    declare(s, "data", SYM_STATIC, n->type, n->line);
    break;

case ND_ENUM: {
    cl_type_t t = { 64, false };
    symbol_t *sym = declare(s, n->name, SYM_ENUM, t, n->line);

    if (sym != (void *)0) {
        sym->node = n;
    }

    break;
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 688) - [public] if
    if (sym != (void *)0) {
        sym->node = n;
    }

    break;
}

case ND_EXTERN:
    for (size_t i = 0; i < n->nlist; i++) {
        cl_type_t t = { 64, false };
        declare(s, n->list[i]->name, SYM_EXTERN, t, n->line);
    }
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 698) - [public] declare
    declare(s, n->list[i]->name, SYM_EXTERN, t, n->line);
}

break;

case ND_DATAIMPL: {
    cl_type_t t = { 0, false };
    symbol_t *sym = declare(s, "data", SYM_DATA, t, n->line);

    if (sym != (void *)0) {
        sym->node = n;
    }
}
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 707) - [public] if
    if (sym != (void *)0) {
        sym->node = n;
    }

    break;
}

default:
    break;
}

}

// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 721) - [public] if
    if (prog == (void *)0) {
        return -1;
    }

    for (size_t i = 0; i < prog->nlist; i++) {
        declare_top(s, prog->list[i]);
    }

    for (size_t i = 0; i < prog->nlist; i++) {
        ast_node_t *n = prog->list[i];

        if (n->kind != ND_FN) {
            // ... (truncated, see source file for more)
        }
    }
}

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 726) - [public] declare_top
    declare_top(s, prog->list[i]);
}

for (size_t i = 0; i < prog->nlist; i++) {
    ast_node_t *n = prog->list[i];

    if (n->kind != ND_FN) {
        continue;
    }

    push_scope(s);

    // ... (truncated, see source file for more)
}

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 732) - [public] if
    if (n->kind != ND_FN) {
        continue;
    }

    push_scope(s);

    for (size_t k = 0; k < n->nlist; k++) {
        declare(s, n->list[k]->name, SYM_VAR, n->list[k]->type, n->line);
    }

    check_block(s, n->a);

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 736) - [public] push_scope
    push_scope(s);

    for (size_t k = 0; k < n->nlist; k++) {
        declare(s, n->list[k]->name, SYM_VAR, n->list[k]->type, n->line);
    }

    check_block(s, n->a);

    pop_scope(s);
}

    return (int)s->ndiags;
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 739) - [public] declare
    declare(s, n->list[k]->name, SYM_VAR, n->list[k]->type, n->line);
}

    check_block(s, n->a);

    pop_scope(s);
}

    return (int)s->ndiags;
}

size_t cl_sema_diag_count(const sema_ctx_t *s)
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/sema.c (line 742) - [public] check_block
    check_block(s, n->a);

    pop_scope(s);
}

    return (int)s->ndiags;
}

size_t cl_sema_diag_count(const sema_ctx_t *s)
{
    return s->ndiags;
}

    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 744) - [public] pop_scope
    pop_scope(s);
}

    return (int)s->ndiags;
}

size_t cl_sema_diag_count(const sema_ctx_t *s)
{
    return s->ndiags;
}

const cl_diag_t *cl_sema_diag(const sema_ctx_t *s, size_t i)
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/vm.c (line 19) - [public] for
    for (int i = full + (rem ? 1 : 0); i < w; i++) {
        r[i] = 0;
    }

    if (rem) {
        r[full] &= (uint8_t)((1 << rem) - 1);
    }
}

void cl_add(uint8_t *r, const uint8_t *a, const uint8_t *b, int w)
{
    int c = 0;
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/vm.c (line 23) - [public] if
    if (rem) {
        r[full] &= (uint8_t)((1 << rem) - 1);
    }
}

void cl_add(uint8_t *r, const uint8_t *a, const uint8_t *b, int w)
{
    int c = 0;

    for (int i = 0; i < w; i++) {
        int s = a[i] + b[i] + c;

        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/vm.c (line 85) - [public] if
    if (a[i] != b[i]) {
        return a[i] > b[i] ? 1 : -1;
    }
}

    return 0;
}

int cl_cmp(const uint8_t *a, const uint8_t *b, int w, bool sign)
{
    if (!sign) {
        return cmp_u(a, b, w);
        // ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/core-lang/vm.c (line 95) - [public] if

```
    if (!sign) {
        return cmp_u(a, b, w);
    }

    uint8_t x[64], y[64];

    for (int i = 0; i < w; i++) {
        x[i] = a[i];
        y[i] = b[i];
    }

    int sb = w * 8 - 1;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/vm.c (line 96) - [public] cmp_u

```
    return cmp_u(a, b, w);
}

uint8_t x[64], y[64];

for (int i = 0; i < w; i++) {
    x[i] = a[i];
    y[i] = b[i];
}

int sb = w * 8 - 1;

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/vm.c (line 113) - [public] cmp_u

```
    return cmp_u(x, y, w);
}

void cl_shl(uint8_t *r, const uint8_t *a, uint32_t n, int w)
{
    uint8_t t[64] = { 0 };

    for (int i = 0; i < w * 8; i++) {
        if (bit_get(a, i) && (int)(i + n) < w * 8) {
            bit_set(t, i + (int)n);
        }
    }

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/vm.c (line 121) - [public] if

```
    if (bit_get(a, i) && (int)(i + n) < w * 8) {
        bit_set(t, i + (int)n);
    }
}

for (int i = 0; i < w; i++) {
    r[i] = t[i];
}

void cl_shr(uint8_t *r, const uint8_t *a, uint32_t n, int w)
{
    // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/core-lang/vm.c (line 122) - [public] bit_set
    bit_set(t, i + (int)n);
    }
}

for (int i = 0; i < w; i++) {
    r[i] = t[i];
}

}

void cl_shr(uint8_t *r, const uint8_t *a, uint32_t n, int w)
{
    uint8_t t[64] = { 0 };
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/vm.c (line 136) - [public] if
    if (bit_get(a, i) && i >= (int)n) {
        bit_set(t, i - (int)n);
    }
}

for (int i = 0; i < w; i++) {
    r[i] = t[i];
}

}

void cl_divmod(const uint8_t *n, const uint8_t *d, int w,
               uint8_t *q, uint8_t *m)
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/vm.c (line 137) - [public] bit_set
    bit_set(t, i - (int)n);
    }
}

for (int i = 0; i < w; i++) {
    r[i] = t[i];
}

}

void cl_divmod(const uint8_t *n, const uint8_t *d, int w,
               uint8_t *q, uint8_t *m)
{
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/vm.c (line 154) - [public] cl_shl
    cl_shl(shl, rem, 1, w);

    for (int k = 0; k < w; k++) {
        rem[k] = shl[k];
    }

    if (bit_get(n, i)) {
        rem[0] |= 1;
    }

    if (cmp_u(rem, d, w) >= 0) {
        uint8_t sub[64];
        // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/vm.c (line 160) - [public] if
    if (bit_get(n, i)) {
        rem[0] |= 1;
    }

    if (cmp_u(rem, d, w) >= 0) {
        uint8_t sub[64];
        cl_sub(sub, rem, d, w);

        for (int k = 0; k < w; k++) {
            rem[k] = sub[k];
        }

        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/vm.c (line 164) - [public] if
    if (cmp_u(rem, d, w) >= 0) {
        uint8_t sub[64];
        cl_sub(sub, rem, d, w);

        for (int k = 0; k < w; k++) {
            rem[k] = sub[k];
        }

        bit_set(quo, i);
    }
}

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/vm.c (line 166) - [public] cl_sub
    cl_sub(sub, rem, d, w);

    for (int k = 0; k < w; k++) {
        rem[k] = sub[k];
    }

    bit_set(quo, i);
}

for (int i = 0; i < w; i++) {
    q[i] = quo[i];
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/vm.c (line 172) - [public] bit_set
    bit_set(quo, i);
}

for (int i = 0; i < w; i++) {
    q[i] = quo[i];
    m[i] = rem[i];
}

/* ---- VM ---- */

// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/vm.c (line 207) - [public] if
    if (vm->prog->ext_names[i][k] != name[k]) {
        same = false;
        break;
    }
}

    if (same) {
        vm->externs[i] = fn;
        return 0;
    }
}

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/vm.c (line 213) - [public] if
    if (same) {
        vm->externs[i] = fn;
        return 0;
    }
}

    return -1;
}

static void push(cl_vm_t *vm, const cl_val_t *v)
{
    if (vm->sp < 256) {
        // ... (truncated, see source file for more)
    }
}

kernel_Workspace/kernel/src/core-lang/vm.c (line 224) - [public] if
    if (vm->sp < 256) {
        vm->stack[vm->sp++] = *v;
    } else {
        vm->err = CL_ERR_STACK;
    }
}

static cl_val_t pop(cl_vm_t *vm)
{
    if (vm->sp > 0) {
        return vm->stack[--vm->sp];
    }
    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/core-lang/vm.c (line 233) - [public] if
    if (vm->sp > 0) {
        return vm->stack[--vm->sp];
    }

    vm->err = CL_ERR_STACK;

    cl_val_t z = { { 0 } };
    return z;
}

static uint64_t val_u64(const cl_val_t *v)
{
    // ... (truncated, see source file for more)
}

```


kernel_Workspace/kernel/src/core-lang/vm.c (line 260) - [public] if

```
    if (P->main_fn < 0) {
        return CL_ERR_BADOP;
    }

    vm->frames[0].pc = P->fns[P->main_fn].start;
    vm->frames[0].fn_idx = (uint16_t)P->main_fn;

    for (int i = 0; i < 64; i++) {
        vm->frames[0].slive[i] = 0;
    }

    vm->depth = 1;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/vm.c (line 273) - [public] while

```
    while (vm->depth > 0 && vm->err == CL_OK && !vm->halt_req) {
        vm_frame_t *f = &vm->frames[vm->depth - 1];

        if (f->pc >= P->code_len) {
            break;
        }

        cl_insn_t in = P->code[f->pc++];

        int w = cl_wbytes(in.bits);
        bool sign = false;

        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/vm.c (line 276) - [public] if

```
        if (f->pc >= P->code_len) {
            break;
        }

        cl_insn_t in = P->code[f->pc++];

        int w = cl_wbytes(in.bits);
        bool sign = false;

        cl_val_t a, b, r;

        switch (in.op) {
            // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/vm.c (line 287) - [public] switch

```
        switch (in.op) {
            case OP_NOP: break;

            case OP_CONST:
                r = P->consts[in.a];
                cl_mask(r.b, w, in.bits);
                push(vm, &r);
                break;

            case OP_LOAD:
                if (!f->slive[in.a]) {
                    vm->err = CL_ERR_USE_AFTER_FREE;
                }
                // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/vm.c (line 292) - [public] cl_mask

```
    cl_mask(r.b, w, in.bits);
    push(vm, &r);
    break;

    case OP_LOAD:
        if (!f->slive[in.a]) {
            vm->err = CL_ERR_USE_AFTER_FREE;
            break;
        }
        push(vm, &f->slots[in.a]);
        break;
```

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/vm.c (line 293) - [public] push

```
    push(vm, &r);
    break;

    case OP_LOAD:
        if (!f->slive[in.a]) {
            vm->err = CL_ERR_USE_AFTER_FREE;
            break;
        }
        push(vm, &f->slots[in.a]);
        break;
```

case OP_STORE:

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/vm.c (line 297) - [public] if

```
    if (!f->slive[in.a]) {
        vm->err = CL_ERR_USE_AFTER_FREE;
        break;
    }
    push(vm, &f->slots[in.a]);
    break;
```

case OP_STORE:

```
    a = pop(vm);
    cl_mask(a.b, w, in.bits);
    f->slots[in.a] = a;
    f->slive[in.a] = 1;
```

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/vm.c (line 301) - [public] push

```
    push(vm, &f->slots[in.a]);
    break;
```

case OP_STORE:

```
    a = pop(vm);
    cl_mask(a.b, w, in.bits);
    f->slots[in.a] = a;
    f->slive[in.a] = 1;
    break;
```

case OP_GLOAD:

```
    if (!vm->glive[in.a]) {
```

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/vm.c (line 306) - [public] cl_mask

```
    cl_mask(a.b, w, in.bits);
    f->slots[in.a] = a;
    f->slive[in.a] = 1;
    break;
```

```
case OP_GLOAD:
    if (!vm->glive[in.a]) {
        vm->err = CL_ERR_USE_AFTER_FREE;
        break;
    }
    push(vm, &vm->globals[in.a]);
    break;
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/vm.c (line 312) - [public] if

```
    if (!vm->glive[in.a]) {
        vm->err = CL_ERR_USE_AFTER_FREE;
        break;
    }
    push(vm, &vm->globals[in.a]);
    break;
```

```
case OP_GSTORE:
    a = pop(vm);
    cl_mask(a.b, w, in.bits);
    vm->globals[in.a] = a;
    vm->glive[in.a] = 1;
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/vm.c (line 316) - [public] push

```
    push(vm, &vm->globals[in.a]);
    break;
```

```
case OP_GSTORE:
    a = pop(vm);
    cl_mask(a.b, w, in.bits);
    vm->globals[in.a] = a;
    vm->glive[in.a] = 1;
    break;
```

```
case OP_ADD: case OP_SUB: case OP_MUL:
case OP_DIV: case OP_MOD:
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/vm.c (line 321) - [public] cl_mask

```
    cl_mask(a.b, w, in.bits);
    vm->globals[in.a] = a;
    vm->glive[in.a] = 1;
    break;
```

```
case OP_ADD: case OP_SUB: case OP_MUL:
case OP_DIV: case OP_MOD:
case OP_AND: case OP_OR: case OP_XOR:
case OP_SHL: case OP_SHR:
    b = pop(vm);
    a = pop(vm);
```

```
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/vm.c (line 335) - [public] switch

```
switch (in.op) {
case OP_ADD: cl_add(r.b, a.b, b.b, w); break;
case OP_SUB: cl_sub(r.b, a.b, b.b, w); break;
case OP_MUL: cl_mul(r.b, a.b, b.b, w); break;
case OP_AND: for (int i = 0; i < w; i++) r.b[i] = a.b[i] & b.b[i]; break;
case OP_OR:  for (int i = 0; i < w; i++) r.b[i] = a.b[i] | b.b[i]; break;
case OP_XOR: for (int i = 0; i < w; i++) r.b[i] = a.b[i] ^ b.b[i]; break;
case OP_SHL: cl_shl(r.b, a.b, (uint32_t)b.b[0], w); break;
case OP_SHR: cl_shr(r.b, a.b, (uint32_t)b.b[0], w); break;
case OP_DIV: case OP_MOD: {
    uint8_t q[64], m[64];
    bool zero = true;
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/vm.c (line 336) - [public] cl_add

```
case OP_ADD: cl_add(r.b, a.b, b.b, w); break;
case OP_SUB: cl_sub(r.b, a.b, b.b, w); break;
case OP_MUL: cl_mul(r.b, a.b, b.b, w); break;
case OP_AND: for (int i = 0; i < w; i++) r.b[i] = a.b[i] & b.b[i]; break;
case OP_OR:  for (int i = 0; i < w; i++) r.b[i] = a.b[i] | b.b[i]; break;
case OP_XOR: for (int i = 0; i < w; i++) r.b[i] = a.b[i] ^ b.b[i]; break;
case OP_SHL: cl_shl(r.b, a.b, (uint32_t)b.b[0], w); break;
case OP_SHR: cl_shr(r.b, a.b, (uint32_t)b.b[0], w); break;
case OP_DIV: case OP_MOD: {
    uint8_t q[64], m[64];
    bool zero = true;

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/vm.c (line 337) - [public] cl_sub

```
case OP_SUB: cl_sub(r.b, a.b, b.b, w); break;
case OP_MUL: cl_mul(r.b, a.b, b.b, w); break;
case OP_AND: for (int i = 0; i < w; i++) r.b[i] = a.b[i] & b.b[i]; break;
case OP_OR:  for (int i = 0; i < w; i++) r.b[i] = a.b[i] | b.b[i]; break;
case OP_XOR: for (int i = 0; i < w; i++) r.b[i] = a.b[i] ^ b.b[i]; break;
case OP_SHL: cl_shl(r.b, a.b, (uint32_t)b.b[0], w); break;
case OP_SHR: cl_shr(r.b, a.b, (uint32_t)b.b[0], w); break;
case OP_DIV: case OP_MOD: {
    uint8_t q[64], m[64];
    bool zero = true;

    for (int i = 0; i < w; i++) {
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/vm.c (line 338) - [public] cl_mul

```
case OP_MUL: cl_mul(r.b, a.b, b.b, w); break;
case OP_AND: for (int i = 0; i < w; i++) r.b[i] = a.b[i] & b.b[i]; break;
case OP_OR:  for (int i = 0; i < w; i++) r.b[i] = a.b[i] | b.b[i]; break;
case OP_XOR: for (int i = 0; i < w; i++) r.b[i] = a.b[i] ^ b.b[i]; break;
case OP_SHL: cl_shl(r.b, a.b, (uint32_t)b.b[0], w); break;
case OP_SHR: cl_shr(r.b, a.b, (uint32_t)b.b[0], w); break;
case OP_DIV: case OP_MOD: {
    uint8_t q[64], m[64];
    bool zero = true;

    for (int i = 0; i < w; i++) {
        if (b.b[i]) zero = false;
// ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/core-lang/vm.c (line 342) - [public] cl_shl
    case OP_SHL: cl_shl(r.b, a.b, (uint32_t)b.b[0], w); break;
    case OP_SHR: cl_shr(r.b, a.b, (uint32_t)b.b[0], w); break;
    case OP_DIV: case OP_MOD: {
        uint8_t q[64], m[64];
        bool zero = true;

        for (int i = 0; i < w; i++) {
            if (b.b[i]) zero = false;
        }

        if (zero) {
            vm->err = CL_ERR_DIV_ZERO;
        }
        // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/vm.c (line 343) - [public] cl_shr
    case OP_SHR: cl_shr(r.b, a.b, (uint32_t)b.b[0], w); break;
    case OP_DIV: case OP_MOD: {
        uint8_t q[64], m[64];
        bool zero = true;

        for (int i = 0; i < w; i++) {
            if (b.b[i]) zero = false;
        }

        if (zero) {
            vm->err = CL_ERR_DIV_ZERO;
            break;
        }
        // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/vm.c (line 352) - [public] if
    if (zero) {
        vm->err = CL_ERR_DIV_ZERO;
        break;
    }

    cl_divmod(a.b, b.b, w, q, m);

    for (int i = 0; i < w; i++) {
        r.b[i] = (in.op == OP_DIV) ? q[i] : m[i];
    }
    break;
}
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/core-lang/vm.c (line 357) - [public] cl_divmod
    cl_divmod(a.b, b.b, w, q, m);

    for (int i = 0; i < w; i++) {
        r.b[i] = (in.op == OP_DIV) ? q[i] : m[i];
    }
    break;
}
}

    cl_mask(r.b, w, in.bits);
    push(vm, &r);
    break;
// ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/core-lang/vm.c (line 366) - [public] cl_mask

```
    cl_mask(r.b, w, in.bits);
    push(vm, &r);
    break;

    case OP_NEG:
        a = pop(vm);
        {
            uint8_t z[64] = { 0 };
            cl_sub(r.b, z, a.b, w);
            cl_mask(r.b, w, in.bits);
        }
        push(vm, &r);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/vm.c (line 367) - [public] push

```
    push(vm, &r);
    break;

    case OP_NEG:
        a = pop(vm);
        {
            uint8_t z[64] = { 0 };
            cl_sub(r.b, z, a.b, w);
            cl_mask(r.b, w, in.bits);
        }
        push(vm, &r);
        break;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/vm.c (line 374) - [public] cl_sub

```
    cl_sub(r.b, z, a.b, w);
    cl_mask(r.b, w, in.bits);
}
push(vm, &r);
break;

    case OP_NOT:
        a = pop(vm);
        for (int i = 0; i < w; i++) r.b[i] = ~a.b[i];
        cl_mask(r.b, w, in.bits);
        push(vm, &r);
        break;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/vm.c (line 375) - [public] cl_mask

```
    cl_mask(r.b, w, in.bits);
}
push(vm, &r);
break;

    case OP_NOT:
        a = pop(vm);
        for (int i = 0; i < w; i++) r.b[i] = ~a.b[i];
        cl_mask(r.b, w, in.bits);
        push(vm, &r);
        break;

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/vm.c (line 377) - [public] push

```
    push(vm, &r);
    break;

    case OP_NOT:
        a = pop(vm);
        for (int i = 0; i < w; i++) r.b[i] = ~a.b[i];
        cl_mask(r.b, w, in.bits);
        push(vm, &r);
        break;

    case OP_EQ: case OP_NE: case OP_LT: case OP_GT:
    case OP_LE: case OP_GE:
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/vm.c (line 383) - [public] cl_mask

```
    cl_mask(r.b, w, in.bits);
    push(vm, &r);
    break;

    case OP_EQ: case OP_NE: case OP_LT: case OP_GT:
    case OP_LE: case OP_GE:
        b = pop(vm);
        a = pop(vm);
        {
            int c = cl_cmp(a.b, b.b, w, sign);
            bool t = false;

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/vm.c (line 384) - [public] push

```
    push(vm, &r);
    break;

    case OP_EQ: case OP_NE: case OP_LT: case OP_GT:
    case OP_LE: case OP_GE:
        b = pop(vm);
        a = pop(vm);
        {
            int c = cl_cmp(a.b, b.b, w, sign);
            bool t = false;

            switch (in.op) {
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/vm.c (line 395) - [public] switch

```
        switch (in.op) {
            case OP_EQ: t = (c == 0); break;
            case OP_NE: t = (c != 0); break;
            case OP_LT: t = (c < 0); break;
            case OP_GT: t = (c > 0); break;
            case OP_LE: t = (c <= 0); break;
            case OP_GE: t = (c >= 0); break;
        }

        for (int i = 0; i < 64; i++) r.b[i] = 0;
        r.b[0] = t ? 1 : 0;
    }
// ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/core-lang/vm.c (line 407) - [public] push
    push(vm, &r);
    break;

    case OP_JMP:
        f->pc = in.c;
        break;

    case OP_JZ:
        a = pop(vm);
        if (a.b[0] == 0) f->pc = in.c;
        break;

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/vm.c (line 425) - [public] pop
    pop(vm);
    break;

    case OP_FREE:
        if (in.a == 1) {
            vm->glive[in.b] = 0;
            for (int i = 0; i < 64; i++) vm->globals[in.b].b[i] = 0;
        } else {
            f->slive[in.b] = 0;
            for (int i = 0; i < 64; i++) f->slots[in.b].b[i] = 0;
        }
        break;
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/vm.c (line 429) - [public] if
    if (in.a == 1) {
        vm->glive[in.b] = 0;
        for (int i = 0; i < 64; i++) vm->globals[in.b].b[i] = 0;
    } else {
        f->slive[in.b] = 0;
        for (int i = 0; i < 64; i++) f->slots[in.b].b[i] = 0;
    }
    break;

    case OP_SHARED:
        if (!vm->glive[in.a]) {
            vm->err = CL_ERR_USE_AFTER_FREE;
        }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/vm.c (line 439) - [public] if
    if (!vm->glive[in.a]) {
        vm->err = CL_ERR_USE_AFTER_FREE;
    }
    break;

    case OP_EXTERN: {
        if (in.a >= 16 || vm->externs[in.a] == (cl_ext_fn)0) {
            vm->err = CL_ERR_EXTERN;
            break;
        }

        uint64_t args[6] = { 0 };
    // ... (truncated, see source file for more)

```



```

kernel_Workspace/kernel/src/core-lang/vm.c (line 445) - [public] if
    if (in.a >= 16 || vm->externs[in.a] == (cl_ext_fn)0) {
        vm->err = CL_ERR_EXTERN;
        break;
    }

    uint64_t args[6] = { 0 };
    int n = in.b > 6 ? 6 : in.b;

    for (int i = n - 1; i >= 0; i--) {
        cl_val_t v = pop(vm);
        args[i] = val_u64(&v);
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/vm.c (line 464) - [public] push
    push(vm, &r);
    break;
}

case OP_CALL: {
    if (vm->depth >= 8) {
        vm->err = CL_ERR_DEPTH;
        break;
    }

    cl_val_t argv[8];
    int n = in.b > 8 ? 8 : in.b;
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/core-lang/vm.c (line 469) - [public] if
    if (vm->depth >= 8) {
        vm->err = CL_ERR_DEPTH;
        break;
    }

    cl_val_t argv[8];
    int n = in.b > 8 ? 8 : in.b;

    for (int i = n - 1; i >= 0; i--) {
        argv[i] = pop(vm);
    }

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/displayport/aux.c (line 50) - [public] if
    if (addr + len > 0x400) {
        return false;
    }

    uint8_t *out = (uint8_t *)buf;

    for (uint32_t i = 0; i < len; i++) {
        out[i] = dpd[addr + i];
    }

    return true;
}
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/displayport/aux.c (line 65) - [public] if
    if (addr + len > 0x400) {
        return false;
    }

    const uint8_t *in = (const uint8_t *)buf;

    for (uint32_t i = 0; i < len; i++) {
        dpcd[addr + i] = in[i];
    }

    if (addr <= DPCD_TRAINING_SET &&
        DPCD_TRAINING_SET < addr + len) {
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/displayport/aux.c (line 79) - [public] if
    if (pat == 1) {
        dpcd[DPCD_LANE_STATUS] = 0x11;
        dpcd[DPCD_LANE_STATUS + 1] = 0x11;
    } else if (pat == 2) {
        train_step++;

        if (train_step >= 2) {
            dpcd[DPCD_LANE_STATUS] = 0x77;
            dpcd[DPCD_LANE_STATUS + 1] = 0x77;
            dpcd[DPCD_SYMBOL_LOCK] = 0x01;
        }
    } else {
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/displayport/aux.c (line 85) - [public] if
    if (train_step >= 2) {
        dpcd[DPCD_LANE_STATUS] = 0x77;
        dpcd[DPCD_LANE_STATUS + 1] = 0x77;
        dpcd[DPCD_SYMBOL_LOCK] = 0x01;
    }
    } else {
        train_step = 0;
    }
}

return true;
}
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/displayport/aux.c (line 100) - [public] if
    if (len > 128) {
        len = 128;
    }

    uint8_t *out = (uint8_t *)buf;

    for (uint32_t i = 0; i < len; i++) {
        out[i] = EDID[i];
    }

    return true;
}

```

kernel_Workspace/kernel/src/displayport/edid.c (line 8) - [public] if

```
*count = 0;

if (b[0] != 0x00 || b[1] != 0xFF || b[7] != 0x00) {
    return false;
}

for (uint32_t d = 0; d < 4; d++) {
    const uint8_t *p = &b[54 + d * 18];

    uint32_t clock = (uint32_t)p[0] | ((uint32_t)p[1] << 8);

    if (clock == 0) {
        continue;
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/displayport/edid.c (line 17) - [public] if

```
if (clock == 0) {
    continue;
}

uint32_t hact = p[2] | (((uint32_t)p[4] >> 4) << 8);
uint32_t hblk = p[3] | (((uint32_t)p[4] & 0xF) << 8);
uint32_t vact = p[5] | (((uint32_t)p[7] >> 4) << 8);
uint32_t vblk = p[6] | (((uint32_t)p[7] & 0xF) << 8);

uint32_t hsync = p[9] | (((uint32_t)p[10] >> 4) & 0xF) << 8);
uint32_t vsync = (uint32_t)p[10] & 0xF;
uint32_t hoff = p[8] | (((uint32_t)p[11] >> 6) << 8);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/displayport/edid.c (line 32) - [public] if

```
if (hact == 0 || vact == 0) {
    continue;
}

if (*count >= max) {
    break;
}

uint32_t htot = hact + hblk;
uint32_t vtot = vact + vblk;

uint32_t refresh = (clock * 1000u) / (htot * vtot);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/displayport/edid.c (line 36) - [public] if

```
if (*count >= max) {
    break;
}

uint32_t htot = hact + hblk;
uint32_t vtot = vact + vblk;

uint32_t refresh = (clock * 1000u) / (htot * vtot);

hdmi_mode_t *m = &out[*count];

m->id = 100 + *count;
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/displayport/init.c (line 13) - [public] if

```
    if (dp_is_ready) {
        return true;
    }

    dp_aux_sink_init();

    if (!dp_link_train()) {
        return false;
    }

    uint8_t raw[128];

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/displayport/init.c (line 17) - [public] dp_aux_sink_init

```
dp_aux_sink_init();

    if (!dp_link_train()) {
        return false;
    }

    uint8_t raw[128];

    if (!dp_aux_read_edid(raw, 128)) {
        return false;
    }

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/displayport/init.c (line 19) - [public] if

```
    if (!dp_link_train()) {
        return false;
    }

    uint8_t raw[128];

    if (!dp_aux_read_edid(raw, 128)) {
        return false;
    }

    if (!edid_parse(raw, MODES, 8, &MODE_COUNT)) {
        return false;
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/displayport/init.c (line 25) - [public] if

```
    if (!dp_aux_read_edid(raw, 128)) {
        return false;
    }

    if (!edid_parse(raw, MODES, 8, &MODE_COUNT)) {
        return false;
    }

    dp_is_ready = true;

    return true;
}
// ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/displayport/init.c (line 29) - [public] if
    if (!edid_parse(raw, MODES, 8, &MODE_COUNT)) {
        return false;
    }

    dp_is_ready = true;

    return true;
}

bool dp_ready(void)
{
    return dp_is_ready;
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/displayport/init.c (line 51) - [public] if
    if (i >= MODE_COUNT) {
        return false;
    }

    *id = MODES[i].id;
    *w = MODES[i].w;
    *h = MODES[i].h;
    *r = MODES[i].refresh;

    return true;
}

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/displayport/init.c (line 66) - [public] if
    if (MODES[i].id == id) {
        dp_op_set_fb(0, MODES[i].w, MODES[i].h, MODES[i].w);
        return true;
    }

    return false;
}

void dp_caps(uint64_t *fb_phys, uint32_t *w, uint32_t *h, uint32_t *stride)
{
    dp_op_state(fb_phys, w, h, stride);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/displayport/init.c (line 67) - [public] dp_op_set_fb
    dp_op_set_fb(0, MODES[i].w, MODES[i].h, MODES[i].w);
    return true;
}

    return false;
}

void dp_caps(uint64_t *fb_phys, uint32_t *w, uint32_t *h, uint32_t *stride)
{
    dp_op_state(fb_phys, w, h, stride);
}

    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/displayport/init.c (line 77) - [public] dp_op_state
    dp_op_state(fb_phys, w, h, stride);
}

bool dp_fb_set(uint64_t phys, uint32_t w, uint32_t h, uint32_t stride)
{
    dp_op_set_fb(phys, w, h, stride);
    return true;
}

uint64_t dp_submit_fill(uint32_t color, uint32_t x, uint32_t y,
                        uint32_t w, uint32_t h)
{
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/displayport/init.c (line 82) - [public] dp_op_set_fb
    dp_op_set_fb(phys, w, h, stride);
    return true;
}

uint64_t dp_submit_fill(uint32_t color, uint32_t x, uint32_t y,
                        uint32_t w, uint32_t h)
{
    static uint64_t seq = 0;

    dp_op_fill(color, x, y, w, h);

    return ++seq;
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/displayport/init.c (line 91) - [public] dp_op_fill
    dp_op_fill(color, x, y, w, h);

    return ++seq;
}

```

```

kernel_Workspace/kernel/src/displayport/link.c (line 12) - [public] dp_aux_read
    dp_aux_read(DPCD_MAX_LINK_RATE, &max_rate, 1);
    dp_aux_read(DPCD_MAX_LANE_COUNT, &max_lanes, 1);

    uint8_t bw = max_rate < 0x14 ? max_rate : 0x14;
    uint8_t lanes = max_lanes & 0x1F;

    if (lanes > 4) {
        lanes = 4;
    }

    dp_aux_write(DPCD_LINK_BW_SET, &bw, 1);
    dp_aux_write(DPCD_LANE_COUNT_SET, &lanes, 1);
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/displayport/link.c (line 13) - [public] dp_aux_read
    dp_aux_read(DPCD_MAX_LANE_COUNT, &max_lanes, 1);

    uint8_t bw = max_rate < 0x14 ? max_rate : 0x14;
    uint8_t lanes = max_lanes & 0x1F;

    if (lanes > 4) {
        lanes = 4;
    }

    dp_aux_write(DPCD_LINK_BW_SET, &bw, 1);
    dp_aux_write(DPCD_LANE_COUNT_SET, &lanes, 1);

    // ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/displayport/link.c (line 18) - [public] if

```
    if (lanes > 4) {
        lanes = 4;
    }

    dp_aux_write(DPCD_LINK_BW_SET, &bw, 1);
    dp_aux_write(DPCD_LANE_COUNT_SET, &lanes, 1);

    uint8_t tp = 1;
    dp_aux_write(DPCD_TRAINING_SET, &tp, 1);

    uint8_t st[2] = { 0, 0 };

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/displayport/link.c (line 22) - [public] dp_aux_write

```
    dp_aux_write(DPCD_LINK_BW_SET, &bw, 1);
    dp_aux_write(DPCD_LANE_COUNT_SET, &lanes, 1);

    uint8_t tp = 1;
    dp_aux_write(DPCD_TRAINING_SET, &tp, 1);

    uint8_t st[2] = { 0, 0 };

    for (int i = 0; i < 5; i++) {
        dp_aux_read(DPCD_LANE_STATUS, st, 2);

        if ((st[0] & 0x11) == 0x11 && (st[1] & 0x11) == 0x11) {
            // ... (truncated, see source file for more)
        }
    }
```

kernel_Workspace/kernel/src/displayport/link.c (line 23) - [public] dp_aux_write

```
    dp_aux_write(DPCD_LANE_COUNT_SET, &lanes, 1);

    uint8_t tp = 1;
    dp_aux_write(DPCD_TRAINING_SET, &tp, 1);

    uint8_t st[2] = { 0, 0 };

    for (int i = 0; i < 5; i++) {
        dp_aux_read(DPCD_LANE_STATUS, st, 2);

        if ((st[0] & 0x11) == 0x11 && (st[1] & 0x11) == 0x11) {
            break;
        }
        // ... (truncated, see source file for more)
    }
```

kernel_Workspace/kernel/src/displayport/link.c (line 26) - [public] dp_aux_write

```
    dp_aux_write(DPCD_TRAINING_SET, &tp, 1);

    uint8_t st[2] = { 0, 0 };

    for (int i = 0; i < 5; i++) {
        dp_aux_read(DPCD_LANE_STATUS, st, 2);

        if ((st[0] & 0x11) == 0x11 && (st[1] & 0x11) == 0x11) {
            break;
        }
    }

    // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/displayport/link.c (line 31) - [public] dp_aux_read
    dp_aux_read(DPCD_LANE_STATUS, st, 2);

    if ((st[0] & 0x11) == 0x11 && (st[1] & 0x11) == 0x11) {
        break;
    }
}

tp = 2;
dp_aux_write(DPCD_TRAINING_SET, &tp, 1);

for (int i = 0; i < 5; i++) {
    dp_aux_read(DPCD_LANE_STATUS, st, 2);
    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/displayport/link.c (line 33) - [public] if
    if ((st[0] & 0x11) == 0x11 && (st[1] & 0x11) == 0x11) {
        break;
    }
}

tp = 2;
dp_aux_write(DPCD_TRAINING_SET, &tp, 1);

for (int i = 0; i < 5; i++) {
    dp_aux_read(DPCD_LANE_STATUS, st, 2);

    if ((st[0] & 0x66) == 0x66 && (st[1] & 0x66) == 0x66) {
        // ... (truncated, see source file for more)
    }
}

kernel_Workspace/kernel/src/displayport/link.c (line 39) - [public] dp_aux_write
    dp_aux_write(DPCD_TRAINING_SET, &tp, 1);

    for (int i = 0; i < 5; i++) {
        dp_aux_read(DPCD_LANE_STATUS, st, 2);

        if ((st[0] & 0x66) == 0x66 && (st[1] & 0x66) == 0x66) {
            break;
        }
    }

    uint8_t lock = 0;
    dp_aux_read(DPCD_SYMBOL_LOCK, &lock, 1);
    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/displayport/link.c (line 42) - [public] dp_aux_read
    dp_aux_read(DPCD_LANE_STATUS, st, 2);

    if ((st[0] & 0x66) == 0x66 && (st[1] & 0x66) == 0x66) {
        break;
    }
}

uint8_t lock = 0;
dp_aux_read(DPCD_SYMBOL_LOCK, &lock, 1);

if (lock != 1) {
    return false;
    // ... (truncated, see source file for more)
}

```



```

kernel_Workspace/kernel/src/displayport/link.c (line 44) - [public] if
    if ((st[0] & 0x66) == 0x66 && (st[1] & 0x66) == 0x66) {
        break;
    }
}

uint8_t lock = 0;
dp_aux_read(DPCD_SYMBOL_LOCK, &lock, 1);

if (lock != 1) {
    return false;
}

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/displayport/link.c (line 50) - [public] dp_aux_read
dp_aux_read(DPCD_SYMBOL_LOCK, &lock, 1);

if (lock != 1) {
    return false;
}

tp = 0;
dp_aux_write(DPCD_TRAINING_SET, &tp, 1);

switch (bw) {
case 0x0A: cur_rate = 2700; break;
case 0x14: cur_rate = 5400; break;
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/displayport/link.c (line 52) - [public] if
if (lock != 1) {
    return false;
}

tp = 0;
dp_aux_write(DPCD_TRAINING_SET, &tp, 1);

switch (bw) {
case 0x0A: cur_rate = 2700; break;
case 0x14: cur_rate = 5400; break;
default:   cur_rate = 1620; break;
}
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/displayport/link.c (line 57) - [public] dp_aux_write
dp_aux_write(DPCD_TRAINING_SET, &tp, 1);

switch (bw) {
case 0x0A: cur_rate = 2700; break;
case 0x14: cur_rate = 5400; break;
default:   cur_rate = 1620; break;
}

cur_lanes = lanes;

return true;
}
// ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/displayport/link.c (line 59) - [public] switch

```
    switch (bw) {
        case 0x0A: cur_rate = 2700; break;
        case 0x14: cur_rate = 5400; break;
        default:   cur_rate = 1620; break;
    }

    cur_lanes = lanes;

    return true;
}

void dp_link_info(uint32_t *rate_mbps, uint32_t *lanes)
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/displayport/operation.c (line 27) - [public] if

```
    if (fb_phys == 0) {
        return;
    }

    uint32_t *dst = (uint32_t *)arch_phys_to_virt(fb_phys);

    for (uint32_t yy = y; yy < y + fh && yy < h; yy++) {
        for (uint32_t xx = x; xx < x + fw && xx < w; xx++) {
            dst[yy * stride + xx] = color;
        }
    }
}
```

kernel_Workspace/kernel/src/editor/editor.c (line 11) - [public] console_set_enabled

```
extern void console_set_enabled(int on);
extern int32_t k_fs_read(const char *path, void *buf, uint32_t cap);
extern uint32_t k_input_keycode(void);
```

```
#define TOP_BAR 24
#define BOT_BAR 24
#define GUTTER 64
#define CHAR_W 16
#define CHAR_H 16
#define SCALE 2
```

```
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 12) - [public] k_fs_read

```
extern int32_t k_fs_read(const char *path, void *buf, uint32_t cap);
extern uint32_t k_input_keycode(void);
```

```
#define TOP_BAR 24
#define BOT_BAR 24
#define GUTTER 64
#define CHAR_W 16
#define CHAR_H 16
#define SCALE 2
```

```
#define COL_BG      0xFF1E1E1E
    // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/editor/editor.c (line 13) - [public] k_input_keycode
extern uint32_t k_input_keycode(void);

#define TOP_BAR 24
#define BOT_BAR 24
#define GUTTER 64
#define CHAR_W 16
#define CHAR_H 16
#define SCALE 2

#define COL_BG 0xFF1E1E1E
#define COL_FG 0xFFD4D4D4
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/editor.c (line 88) - [public] if
    if (bits & (0x80 >> rx)) {
        uint32_t px = x + (uint32_t)rx * SCALE;
        uint32_t py = y + (uint32_t)ry * SCALE;

        ed_fill(px, py, SCALE, SCALE, color);
    }
}

static void ed_text(uint32_t x, uint32_t y, const char *s, uint32_t color)
{
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/editor.c (line 92) - [public] ed_fill
    ed_fill(px, py, SCALE, SCALE, color);
}
}

static void ed_text(uint32_t x, uint32_t y, const char *s, uint32_t color)
{
    while (*s) {
        ed_char(x, y, *s, color);
        x += CHAR_W;
        s++;
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/editor.c (line 100) - [public] while
    while (*s) {
        ed_char(x, y, *s, color);
        x += CHAR_W;
        s++;
    }
}

static void ed_num(char *buf, int v)
{
    int i = 0;

    if (v == 0) { buf[0] = '0'; buf[1] = 0; return; }
    // ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/editor/editor.c (line 101) - [public] ed_char

```
    ed_char(x, y, *s, color);
    x += CHAR_W;
    s++;
}
```

```
static void ed_num(char *buf, int v)
{
```

```
    int i = 0;

    if (v == 0) { buf[0] = '0'; buf[1] = 0; return; }

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 111) - [public] if

```
    if (v == 0) { buf[0] = '0'; buf[1] = 0; return; }

    char tmp[12];
    int n = 0;

    while (v > 0 && n < 11) { tmp[n++] = (char)('0' + v % 10); v /= 10; }

    while (n > 0) buf[i++] = tmp[--n];
    buf[i] = 0;
}
```

```
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 116) - [public] while

```
    while (v > 0 && n < 11) { tmp[n++] = (char)('0' + v % 10); v /= 10; }

    while (n > 0) buf[i++] = tmp[--n];
    buf[i] = 0;
}
```

```
static void log_append(const char *s)
{
```

```
    for (int i = 0; i < 7; i++) {
        for (int k = 0; k < ED_LINE_LEN; k++) {
            log_lines[i][k] = log_lines[i + 1][k];
        }
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 147) - [public] if

```
    if (v == 0) { tmp[n++] = '0'; }

    while (v > 0 && n < 20) { tmp[n++] = (char)('0' + v % 10); v /= 10; }

    while (n > 0) buf[i++] = tmp[--n];
    buf[i] = 0;

    log_append(buf);

    return 0;
}
```

```
    // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/editor/editor.c (line 149) - [public] while
    while (v > 0 && n < 20) { tmp[n++] = (char)('0' + v % 10); v /= 10; }

    while (n > 0) buf[i++] = tmp[--n];
    buf[i] = 0;

    log_append(buf);

    return 0;
}

static uint64_t ed_putc(uint64_t v, uint64_t b, uint64_t c,
                        uint64_t d, uint64_t e, uint64_t f)
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/editor.c (line 154) - [public] log_append
    log_append(buf);

    return 0;
}

static uint64_t ed_putc(uint64_t v, uint64_t b, uint64_t c,
                        uint64_t d, uint64_t e, uint64_t f)
{
    (void)b; (void)c; (void)d; (void)e; (void)f;

    char buf[2] = { (char)v, 0 };
    log_append(buf);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/editor.c (line 165) - [public] log_append
    log_append(buf);
    return 0;
}

static uint32_t tok_color(tok_kind_t k)
{
    switch (k) {
        case TK_IF: case TK_ELSE: case TK_WHEN: case TK_CASE:
        case TK_FN: case TK_LET: case TK_STATIC: case TK_DATA:
        case TK_DATAIMPL: case TK_ENUM: case TK_EXTERN:
        case TK_CALL: case TK_SET_FREE:
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/editor.c (line 172) - [public] switch
    switch (k) {
        case TK_IF: case TK_ELSE: case TK_WHEN: case TK_CASE:
        case TK_FN: case TK_LET: case TK_STATIC: case TK_DATA:
        case TK_DATAIMPL: case TK_ENUM: case TK_EXTERN:
        case TK_CALL: case TK_SET_FREE:
            return COL_KEY;
        case TK_TYPE:    return COL_TYPE;
        case TK_NUM:     return COL_NUM;
        case TK_STR:
        case TK_CHAR:    return COL_STR;
        case TK_IDENT:   return COL_ID;
        default:         return COL_OP;
        // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/editor/editor.c (line 201) - [public] if
    if (tmp[k] == '/' && tmp[k + 1] == '/') { com_at = k; break; }
}

lexer_t l;
token_t toks[64];
size_t n = 0;

cl_lexer_init(&l, tmp, ed_strlen(tmp));

if (com_at >= 0) {
    char cut[ED_LINE_LEN + 1];

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/editor.c (line 208) - [public] cl_lexer_init
cl_lexer_init(&l, tmp, ed_strlen(tmp));

if (com_at >= 0) {
    char cut[ED_LINE_LEN + 1];

    for (int k = 0; k < com_at; k++) cut[k] = tmp[k];
    cut[com_at] = 0;

    cl_lexer_init(&l, cut, (size_t)com_at);
}

int pos = 0;
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/editor.c (line 210) - [public] if
if (com_at >= 0) {
    char cut[ED_LINE_LEN + 1];

    for (int k = 0; k < com_at; k++) cut[k] = tmp[k];
    cut[com_at] = 0;

    cl_lexer_init(&l, cut, (size_t)com_at);
}

int pos = 0;

if (cl_lex_all(&l, toks, 64, &n) == 0) {
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/editor.c (line 216) - [public] cl_lexer_init
    cl_lexer_init(&l, cut, (size_t)com_at);
}

int pos = 0;

if (cl_lex_all(&l, toks, 64, &n) == 0) {
    for (size_t t = 0; t < n && toks[t].kind != TK_EOF; t++) {
        int ts = (int)(toks[t].start - l.src);

        while (pos < ts) {
            ed_char(x + (uint32_t)pos * CHAR_W, y, ' ', COL_FG);
            pos++;
        }
        // ... (truncated, see source file for more)
    }
}

```

```

kernel_Workspace/kernel/src/editor/editor.c (line 221) - [public] if
    if (cl_lex_all(&l, toks, 64, &n) == 0) {
        for (size_t t = 0; t < n && toks[t].kind != TK_EOF; t++) {
            int ts = (int)(toks[t].start - l.src);

            while (pos < ts) {
                ed_char(x + (uint32_t)pos * CHAR_W, y, ' ', COL_FG);
                pos++;
            }

            uint32_t col = tok_color(toks[t].kind);

            for (uint32_t k = 0; k < toks[t].len; k++) {
                // ... (truncated, see source file for more)
            }
        }
    }

kernel_Workspace/kernel/src/editor/editor.c (line 225) - [public] while
    while (pos < ts) {
        ed_char(x + (uint32_t)pos * CHAR_W, y, ' ', COL_FG);
        pos++;
    }

    uint32_t col = tok_color(toks[t].kind);

    for (uint32_t k = 0; k < toks[t].len; k++) {
        ed_char(x + (uint32_t)pos * CHAR_W, y,
            toks[t].start[k], col);
        pos++;
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/editor.c (line 226) - [public] ed_char
    ed_char(x + (uint32_t)pos * CHAR_W, y, ' ', COL_FG);
    pos++;
}

    uint32_t col = tok_color(toks[t].kind);

    for (uint32_t k = 0; k < toks[t].len; k++) {
        ed_char(x + (uint32_t)pos * CHAR_W, y,
            toks[t].start[k], col);
        pos++;
    }
}
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/editor.c (line 240) - [public] while
    while (tmp[pos] && (com_at < 0 || pos < com_at)) {
        ed_char(x + (uint32_t)pos * CHAR_W, y, tmp[pos], COL_FG);
        pos++;
    }

    if (com_at >= 0) {
        for (int k = com_at; tmp[k]; k++) {
            ed_char(x + (uint32_t)k * CHAR_W, y, tmp[k], COL_COM);
        }
    }
}

// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/editor/editor.c (line 241) - [public] ed_char
    ed_char(x + (uint32_t)pos * CHAR_W, y, tmp[pos], COL_FG);
    pos++;
}

    if (com_at >= 0) {
        for (int k = com_at; tmp[k]; k++) {
            ed_char(x + (uint32_t)k * CHAR_W, y, tmp[k], COL_COM);
        }
    }
}

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/editor.c (line 245) - [public] if
    if (com_at >= 0) {
        for (int k = com_at; tmp[k]; k++) {
            ed_char(x + (uint32_t)k * CHAR_W, y, tmp[k], COL_COM);
        }
    }
}

static const char *CURSOR_SPRITE[] = {
    "X.....",
    "XX.....",
    "XoX.....",
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/editor.c (line 247) - [public] ed_char
    ed_char(x + (uint32_t)k * CHAR_W, y, tmp[k], COL_COM);
}
}

static const char *CURSOR_SPRITE[] = {
    "X.....",
    "XX.....",
    "XoX.....",
    "XooX.....",
    "XoooX.....",
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/editor.c (line 285) - [public] if
    if (sx >= 0 && sx < (int)fw && sy >= 0 && sy < (int)fh) {
        int row = ed_flip_rows ? (int)fh - 1 - sy : sy;
        fb[row * fstride + sx] = col;
    }
}
}

static void render(void)
{
    ed_fill(0, 0, fw, fh, COL_BG);
    // ... (truncated, see source file for more)

```



```

kernel_Workspace/kernel/src/editor/editor.c (line 296) - [public] ed_fill
    ed_fill(0, 0, fw, fh, COL_BG);

    ed_fill(0, 0, fw, TOP_BAR, COL_BAR);
    ed_text(8, 4, "TrangEdit [core-lang]", COL_TYPE);
    ed_text(GUTTER + 160, 4, msg, COL_GUTTER);

    int rows = (int)((fh - TOP_BAR - BOT_BAR) / CHAR_H);
    int cols = (int)((fw - GUTTER) / CHAR_W);

    for (int r = 0; r < rows; r++) {
        int row = top_line + r;

        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/editor.c (line 298) - [public] ed_fill
    ed_fill(0, 0, fw, TOP_BAR, COL_BAR);
    ed_text(8, 4, "TrangEdit [core-lang]", COL_TYPE);
    ed_text(GUTTER + 160, 4, msg, COL_GUTTER);

    int rows = (int)((fh - TOP_BAR - BOT_BAR) / CHAR_H);
    int cols = (int)((fw - GUTTER) / CHAR_W);

    for (int r = 0; r < rows; r++) {
        int row = top_line + r;

        if (row >= nlines) break;

        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/editor.c (line 299) - [public] ed_text
    ed_text(8, 4, "TrangEdit [core-lang]", COL_TYPE);
    ed_text(GUTTER + 160, 4, msg, COL_GUTTER);

    int rows = (int)((fh - TOP_BAR - BOT_BAR) / CHAR_H);
    int cols = (int)((fw - GUTTER) / CHAR_W);

    for (int r = 0; r < rows; r++) {
        int row = top_line + r;

        if (row >= nlines) break;

        uint32_t y = TOP_BAR + (uint32_t)r * CHAR_H;
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/editor.c (line 300) - [public] ed_text
    ed_text(GUTTER + 160, 4, msg, COL_GUTTER);

    int rows = (int)((fh - TOP_BAR - BOT_BAR) / CHAR_H);
    int cols = (int)((fw - GUTTER) / CHAR_W);

    for (int r = 0; r < rows; r++) {
        int row = top_line + r;

        if (row >= nlines) break;

        uint32_t y = TOP_BAR + (uint32_t)r * CHAR_H;

        // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/editor/editor.c (line 312) - [public] if
    if (row == caret_l) {
        ed_fill(GUTTER, y, (uint32_t)cols * CHAR_W, CHAR_H, 0xFF2D2D2D);
    }

    char nb[12];
    ed_num(nb, row + 1);
    ed_text(8, y, nb, COL_GUTTER);

    draw_hl(row, GUTTER + 4, y);
}

{
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/editor/editor.c (line 313) - [public] ed_fill
    ed_fill(GUTTER, y, (uint32_t)cols * CHAR_W, CHAR_H, 0xFF2D2D2D);
}

    char nb[12];
    ed_num(nb, row + 1);
    ed_text(8, y, nb, COL_GUTTER);

    draw_hl(row, GUTTER + 4, y);
}

{
    int r = caret_l - top_line;
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/editor/editor.c (line 317) - [public] ed_num
    ed_num(nb, row + 1);
    ed_text(8, y, nb, COL_GUTTER);

    draw_hl(row, GUTTER + 4, y);
}

{
    int r = caret_l - top_line;

    if (r >= 0 && r < rows) {
        uint32_t cx = GUTTER + 4 + (uint32_t)caret_c * CHAR_W;
        uint32_t cy = TOP_BAR + (uint32_t)r * CHAR_H;
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/editor/editor.c (line 318) - [public] ed_text
    ed_text(8, y, nb, COL_GUTTER);

    draw_hl(row, GUTTER + 4, y);
}

{
    int r = caret_l - top_line;

    if (r >= 0 && r < rows) {
        uint32_t cx = GUTTER + 4 + (uint32_t)caret_c * CHAR_W;
        uint32_t cy = TOP_BAR + (uint32_t)r * CHAR_H;

// ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/editor/editor.c (line 320) - [public] draw_hl

```
    draw_hl(row, GUTTER + 4, y);
}

{
    int r = caret_l - top_line;

    if (r >= 0 && r < rows) {
        uint32_t cx = GUTTER + 4 + (uint32_t)caret_c * CHAR_W;
        uint32_t cy = TOP_BAR + (uint32_t)r * CHAR_H;

        ed_fill(cx, cy, 2, CHAR_H, COL_CARET);
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 326) - [public] if

```
    if (r >= 0 && r < rows) {
        uint32_t cx = GUTTER + 4 + (uint32_t)caret_c * CHAR_W;
        uint32_t cy = TOP_BAR + (uint32_t)r * CHAR_H;

        ed_fill(cx, cy, 2, CHAR_H, COL_CARET);
    }
}

if (show_log) {
    uint32_t ly = fh - BOT_BAR - 8 * CHAR_H;

    ed_fill(0, ly, fw, 8 * CHAR_H, 0xFF101010);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 330) - [public] ed_fill

```
    ed_fill(cx, cy, 2, CHAR_H, COL_CARET);
}

}

if (show_log) {
    uint32_t ly = fh - BOT_BAR - 8 * CHAR_H;

    ed_fill(0, ly, fw, 8 * CHAR_H, 0xFF101010);

    for (int i = 0; i < 8; i++) {
        ed_text(8, ly + (uint32_t)i * CHAR_H, log_lines[i], COL_NUM);
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 334) - [public] if

```
    if (show_log) {
        uint32_t ly = fh - BOT_BAR - 8 * CHAR_H;

        ed_fill(0, ly, fw, 8 * CHAR_H, 0xFF101010);

        for (int i = 0; i < 8; i++) {
            ed_text(8, ly + (uint32_t)i * CHAR_H, log_lines[i], COL_NUM);
        }
    }

    ed_fill(0, fh - BOT_BAR, fw, BOT_BAR, COL_BAR);

    // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/editor/editor.c (line 337) - [public] ed_fill
    ed_fill(0, ly, fw, 8 * CHAR_H, 0xFF101010);

    for (int i = 0; i < 8; i++) {
        ed_text(8, ly + (uint32_t)i * CHAR_H, log_lines[i], COL_NUM);
    }
}

ed_fill(0, fh - BOT_BAR, fw, BOT_BAR, COL_BAR);

char st[64];
int i = 0;
const char *p1 = "ln ";
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/editor/editor.c (line 340) - [public] ed_text
    ed_text(8, ly + (uint32_t)i * CHAR_H, log_lines[i], COL_NUM);
}
}

ed_fill(0, fh - BOT_BAR, fw, BOT_BAR, COL_BAR);

char st[64];
int i = 0;
const char *p1 = "ln ";
while (*p1) st[i++] = *p1++;
char tmp[12];
ed_num(tmp, caret_l + 1);
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/editor/editor.c (line 344) - [public] ed_fill
    ed_fill(0, fh - BOT_BAR, fw, BOT_BAR, COL_BAR);

char st[64];
int i = 0;
const char *p1 = "ln ";
while (*p1) st[i++] = *p1++;
char tmp[12];
ed_num(tmp, caret_l + 1);
for (int k = 0; tmp[k]; k++) st[i++] = tmp[k];
const char *p2 = " col ";
while (*p2) st[i++] = *p2++;
ed_num(tmp, caret_c + 1);
// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/editor/editor.c (line 351) - [public] ed_num
    ed_num(tmp, caret_l + 1);
    for (int k = 0; tmp[k]; k++) st[i++] = tmp[k];
    const char *p2 = " col ";
    while (*p2) st[i++] = *p2++;
    ed_num(tmp, caret_c + 1);
    for (int k = 0; tmp[k]; k++) st[i++] = tmp[k];
    st[i] = 0;

    ed_text(8, fh - BOT_BAR + 4, st, COL_FG);
    ed_text(GUTTER + 200, fh - BOT_BAR + 4, msg, COL_GUTTER);

    draw_mouse();
    // ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/editor/editor.c (line 355) - [public] ed_num

```
    ed_num(tmp, caret_c + 1);
    for (int k = 0; tmp[k]; k++) st[i++] = tmp[k];
    st[i] = 0;

    ed_text(8, fh - BOT_BAR + 4, st, COL_FG);
    ed_text(GUTTER + 200, fh - BOT_BAR + 4, msg, COL_GUTTER);

    draw_mouse();
}
```

```
static void ins_char(char ch)
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 359) - [public] ed_text

```
    ed_text(8, fh - BOT_BAR + 4, st, COL_FG);
    ed_text(GUTTER + 200, fh - BOT_BAR + 4, msg, COL_GUTTER);

    draw_mouse();
}
```

```
static void ins_char(char ch)
{
    char *ln = lines[caret_l];
    int len = (int)ed_strlen(ln);

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 360) - [public] ed_text

```
    ed_text(GUTTER + 200, fh - BOT_BAR + 4, msg, COL_GUTTER);

    draw_mouse();
}
```

```
static void ins_char(char ch)
{
    char *ln = lines[caret_l];
    int len = (int)ed_strlen(ln);

    if (len >= ED_LINE_LEN - 1 || caret_c > len) return;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 362) - [public] draw_mouse

```
    draw_mouse();
}
```

```
static void ins_char(char ch)
{
    char *ln = lines[caret_l];
    int len = (int)ed_strlen(ln);

    if (len >= ED_LINE_LEN - 1 || caret_c > len) return;

    for (int i = len; i >= caret_c; i--) {
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 413) - [public] if

```
    if (caret_c > 0) {
        int len = (int)ed_strlen(ln);

        for (int i = caret_c - 1; i < len; i++) {
            ln[i] = ln[i + 1];
        }

        caret_c--;
    } else if (caret_l > 0) {
        int plen = (int)ed_strlen(lines[caret_l - 1]);
        int slen = (int)ed_strlen(ln);

        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 425) - [public] if

```
    if (plen + slen < ED_LINE_LEN - 1) {
        for (int k = 0; k <= slen; k++) {
            lines[caret_l - 1][plen + k] = ln[k];
        }

        for (int i = caret_l; i < nlines - 1; i++) {
            for (int k = 0; k < ED_LINE_LEN; k++) {
                lines[i][k] = lines[i + 1][k];
            }
        }

        nlines--;
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 448) - [public] if

```
    if (caret_c < len) {
        for (int i = caret_c; i < len; i++) {
            ln[i] = ln[i + 1];
        }
    }

    static void clamp_scroll(int rows)
    {
        if (caret_l < top_line) top_line = caret_l;
        if (caret_l >= top_line + rows) top_line = caret_l - rows + 1;
        if (top_line < 0) top_line = 0;
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 485) - [public] render

```
    render();

    ed_arena.pos = 0;

    int n = flatten();

    uint32_t el = 0;
    const char *em = (void *)0;

    cl_prog_t *P = cl_compile_source(srcbuf, (size_t)n, &ed_arena, &el, &em);

    if (P == (void *)0) {
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 496) - [public] if

```
if (P == (void *)0) {
    const char *pre = "ERR ln ";
    int i = 0;
    while (pre[i]) { msg[i] = pre[i]; i++; }
    char tmp[12];
    ed_num(tmp, (int)el);
    for (int k = 0; tmp[k]; k++) msg[i++] = tmp[k];
    msg[i++] = ' ';
    if (em) for (int k = 0; em[k] && i < ED_LINE_LEN - 1; k++) msg[i++] = em[k];
    msg[i] = 0;

    if (em) log_append(em);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 499) - [public] while

```
while (pre[i]) { msg[i] = pre[i]; i++; }
char tmp[12];
ed_num(tmp, (int)el);
for (int k = 0; tmp[k]; k++) msg[i++] = tmp[k];
msg[i++] = ' ';
if (em) for (int k = 0; em[k] && i < ED_LINE_LEN - 1; k++) msg[i++] = em[k];
msg[i] = 0;

if (em) log_append(em);
return;
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 501) - [public] ed_num

```
ed_num(tmp, (int)el);
for (int k = 0; tmp[k]; k++) msg[i++] = tmp[k];
msg[i++] = ' ';
if (em) for (int k = 0; em[k] && i < ED_LINE_LEN - 1; k++) msg[i++] = em[k];
msg[i] = 0;

if (em) log_append(em);
return;
}

cl_vm_init(&ed_vm, P);
cl_bridge_init(&ed_vm, 0);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 507) - [public] if

```
if (em) log_append(em);
return;
}

cl_vm_init(&ed_vm, P);
cl_bridge_init(&ed_vm, 0);

cl_vm_register_extern(&ed_vm, "put", ed_put);
cl_vm_register_extern(&ed_vm, "putc", ed_putc);

log_append("== run ==");

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 511) - [public] cl_vm_init

```
cl_vm_init(&ed_vm, P);
cl_bridge_init(&ed_vm, 0);

cl_vm_register_extern(&ed_vm, "put", ed_put);
cl_vm_register_extern(&ed_vm, "putc", ed_putc);

log_append("== run ==");

cl_vm_err_t e = cl_vm_run(&ed_vm);

const char *ok = (e == CL_OK) ? "run: ok" : "run: vm error";
int i = 0;
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 512) - [public] cl_bridge_init

```
cl_bridge_init(&ed_vm, 0);

cl_vm_register_extern(&ed_vm, "put", ed_put);
cl_vm_register_extern(&ed_vm, "putc", ed_putc);

log_append("== run ==");

cl_vm_err_t e = cl_vm_run(&ed_vm);

const char *ok = (e == CL_OK) ? "run: ok" : "run: vm error";
int i = 0;
while (ok[i]) { msg[i] = ok[i]; i++; }
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 514) - [public] cl_vm_register_extern

```
cl_vm_register_extern(&ed_vm, "put", ed_put);
cl_vm_register_extern(&ed_vm, "putc", ed_putc);

log_append("== run ==");

cl_vm_err_t e = cl_vm_run(&ed_vm);

const char *ok = (e == CL_OK) ? "run: ok" : "run: vm error";
int i = 0;
while (ok[i]) { msg[i] = ok[i]; i++; }
msg[i] = 0;

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 515) - [public] cl_vm_register_extern

```
cl_vm_register_extern(&ed_vm, "putc", ed_putc);

log_append("== run ==");

cl_vm_err_t e = cl_vm_run(&ed_vm);

const char *ok = (e == CL_OK) ? "run: ok" : "run: vm error";
int i = 0;
while (ok[i]) { msg[i] = ok[i]; i++; }
msg[i] = 0;

show_log = 1;
// ... (truncated, see source file for more)
```



```

kernel_Workspace/kernel/src/editor/editor.c (line 517) - [public] log_append
    log_append("== run ==");

    cl_vm_err_t e = cl_vm_run(&ed_vm);

    const char *ok = (e == CL_OK) ? "run: ok" : "run: vm error";
    int i = 0;
    while (ok[i]) { msg[i] = ok[i]; i++; }
    msg[i] = 0;

    show_log = 1;
}

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/editor.c (line 523) - [public] while
    while (ok[i]) { msg[i] = ok[i]; i++; }
    msg[i] = 0;

    show_log = 1;
}

int editor_run(const char *path)
{
    uint32_t w = 0, h = 0, s = 0;
    uint64_t base = 0;
    int32_t fliprows = 0;
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/editor.c (line 536) - [public] if
    if (gfx_fb_info_raw(&w, &h, &s, &base, &fliprows) != 0 || base == 0) {
        return -1;
    }

    fb = (uint32_t *) (uintptr_t) base;
    fw = w; fh = h; fstride = s;
    ed_flip_rows = fliprows;

    nlines = 1;
    lines[0][0] = 0;
    caret_l = caret_c = 0;
    top_line = 0;
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/editor.c (line 549) - [public] if
    if (path != (void *) 0) {
        static char rbuf[ED_MAX_LINES * ED_LINE_LEN];

        int32_t n = k_fs_read(path, rbuf, sizeof(rbuf) - 1);

        if (n > 0) {
            rbuf[n] = 0;

            nlines = 0;
            int col = 0;

            for (int i = 0; i < n && nlines < ED_MAX_LINES; i++) {
                // ... (truncated, see source file for more)
            }
        }
    }
}

```

kernel_Workspace/kernel/src/editor/editor.c (line 554) - [public] if

```
    if (n > 0) {
        rbuf[n] = 0;

        nlines = 0;
        int col = 0;

        for (int i = 0; i < n && nlines < ED_MAX_LINES; i++) {
            char c = rbuf[i];

            if (c == '\n') {
                lines[nlines][col] = 0;
                nlines++;
            }
        }
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 563) - [public] if

```
        if (c == '\n') {
            lines[nlines][col] = 0;
            nlines++;
            col = 0;
        } else if (col < ED_LINE_LEN - 1) {
            lines[nlines][col++] = c;
        }
    }

    if (col > 0 || nlines == 0) {
        lines[nlines][col] = 0;
        nlines++;
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 572) - [public] if

```
        if (col > 0 || nlines == 0) {
            lines[nlines][col] = 0;
            nlines++;
        }
    }
}
```

```
console_set_enabled(0);
mouse_init();
```

```
int running = 1;
```

```
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 579) - [public] console_set_enabled

```
console_set_enabled(0);
mouse_init();
```

```
int running = 1;
```

```
while (running) {
    int dirty = 0;
    int rows = (int)((fh - TOP_BAR - BOT_BAR) / CHAR_H);
```

```
    int dx = 0, dy = 0, dz = 0, btn = 0;
```

```
    if (mouse_poll(&dx, &dy, &dz, &btn)) {
        // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/editor/editor.c (line 580) - [public] mouse_init
    mouse_init();

    int running = 1;

    while (running) {
        int dirty = 0;
        int rows = (int)((fh - TOP_BAR - BOT_BAR) / CHAR_H);

        int dx = 0, dy = 0, dz = 0, btn = 0;

        if (mouse_poll(&dx, &dy, &dz, &btn)) {
            mx += dx;
            // ... (truncated, see source file for more)
        }

kernel_Workspace/kernel/src/editor/editor.c (line 584) - [public] while
    while (running) {
        int dirty = 0;
        int rows = (int)((fh - TOP_BAR - BOT_BAR) / CHAR_H);

        int dx = 0, dy = 0, dz = 0, btn = 0;

        if (mouse_poll(&dx, &dy, &dz, &btn)) {
            mx += dx;
            my += dy;

            if (mx < 0) mx = 0;
            if (my < 0) my = 0;
            // ... (truncated, see source file for more)
        }

kernel_Workspace/kernel/src/editor/editor.c (line 590) - [public] if
        if (mouse_poll(&dx, &dy, &dz, &btn)) {
            mx += dx;
            my += dy;

            if (mx < 0) mx = 0;
            if (my < 0) my = 0;
            if (mx >= (int)fw) mx = (int)fw - 1;
            if (my >= (int)fh) my = (int)fh - 1;

            if (dz != 0) {
                top_line += dz;
                clamp_scroll(rows);
                // ... (truncated, see source file for more)
            }

kernel_Workspace/kernel/src/editor/editor.c (line 599) - [public] if
            if (dz != 0) {
                top_line += dz;
                clamp_scroll(rows);
                dirty = 1;
            }

            if (btn & 1 && !(mbuttons & 1)) {
                if (my >= (int)TOP_BAR && my < (int)(fh - BOT_BAR)) {
                    int row = top_line + (my - (int)TOP_BAR) / CHAR_H;
                    int col = (mx - (int)GUTTER - 4) / CHAR_W;

                    if (row >= 0 && row < nlines) caret_l = row;
                    // ... (truncated, see source file for more)
                }
            }
        }
    }
}

```

```

kernel_Workspace/kernel/src/editor/editor.c (line 601) - [public] clamp_scroll
    clamp_scroll(rows);
    dirty = 1;
}

    if (btn & 1 && !(mbuttons & 1)) {
        if (my >= (int)TOP_BAR && my < (int)(fh - BOT_BAR)) {
            int row = top_line + (my - (int)TOP_BAR) / CHAR_H;
            int col = (mx - (int)GUTTER - 4) / CHAR_W;

            if (row >= 0 && row < nlines) caret_l = row;

            int len = (int)ed_strlen(lines[caret_l]);
            // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/editor.c (line 605) - [public] if
    if (btn & 1 && !(mbuttons & 1)) {
        if (my >= (int)TOP_BAR && my < (int)(fh - BOT_BAR)) {
            int row = top_line + (my - (int)TOP_BAR) / CHAR_H;
            int col = (mx - (int)GUTTER - 4) / CHAR_W;

            if (row >= 0 && row < nlines) caret_l = row;

            int len = (int)ed_strlen(lines[caret_l]);

            if (col < 0) col = 0;
            if (col > len) col = len;

            // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/editor.c (line 606) - [public] if
    if (my >= (int)TOP_BAR && my < (int)(fh - BOT_BAR)) {
        int row = top_line + (my - (int)TOP_BAR) / CHAR_H;
        int col = (mx - (int)GUTTER - 4) / CHAR_W;

        if (row >= 0 && row < nlines) caret_l = row;

        int len = (int)ed_strlen(lines[caret_l]);

        if (col < 0) col = 0;
        if (col > len) col = len;

        caret_c = col;
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/editor.c (line 618) - [public] clamp_scroll
    clamp_scroll(rows);
    dirty = 1;
    }
}

    mbuttons = btn;
    dirty = 1;
}

    uint32_t k = k_input_keycode();

    if (k != 0) {
        // ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/editor/editor.c (line 629) - [public] if

```
if (k != 0) {
    switch (k) {
        case EDK_ESC:      running = 0; break;
        case EDK_ENTER:    do_enter(); dirty = 1; break;
        case EDK_BACKSPACE: do_backspace(); dirty = 1; break;
        case EDK_DELETE:   do_delete(); dirty = 1; break;
        case EDK_TAB:      ins_char(' '); ins_char(' ');
                          ins_char(' '); ins_char(' '); dirty = 1; break;
        case EDK_LEFT:     if (caret_c > 0) caret_c--; dirty = 1; break;
        case EDK_RIGHT:    caret_c++; dirty = 1; break;
        case EDK_UP:       if (caret_l > 0) caret_l--; dirty = 1; break;
        case EDK_DOWN:     if (caret_l < nlines - 1) caret_l++; dirty = 1; break;
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 630) - [public] switch

```
switch (k) {
    case EDK_ESC:      running = 0; break;
    case EDK_ENTER:    do_enter(); dirty = 1; break;
    case EDK_BACKSPACE: do_backspace(); dirty = 1; break;
    case EDK_DELETE:   do_delete(); dirty = 1; break;
    case EDK_TAB:      ins_char(' '); ins_char(' ');
                      ins_char(' '); ins_char(' '); dirty = 1; break;
    case EDK_LEFT:     if (caret_c > 0) caret_c--; dirty = 1; break;
    case EDK_RIGHT:    caret_c++; dirty = 1; break;
    case EDK_UP:       if (caret_l > 0) caret_l--; dirty = 1; break;
    case EDK_DOWN:     if (caret_l < nlines - 1) caret_l++; dirty = 1; break;
    case EDK_HOME:     caret_c = 0; dirty = 1; break;
}
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 632) - [public] do_enter

```
case EDK_ENTER:    do_enter(); dirty = 1; break;
case EDK_BACKSPACE: do_backspace(); dirty = 1; break;
case EDK_DELETE:   do_delete(); dirty = 1; break;
case EDK_TAB:      ins_char(' '); ins_char(' ');
                  ins_char(' '); ins_char(' '); dirty = 1; break;
case EDK_LEFT:     if (caret_c > 0) caret_c--; dirty = 1; break;
case EDK_RIGHT:    caret_c++; dirty = 1; break;
case EDK_UP:       if (caret_l > 0) caret_l--; dirty = 1; break;
case EDK_DOWN:     if (caret_l < nlines - 1) caret_l++; dirty = 1; break;
case EDK_HOME:     caret_c = 0; dirty = 1; break;
case EDK_END:      caret_c = (int)ed_strlen(lines[caret_l]); dirty = 1; break;
case EDK_F5:       ed_compile_run(); dirty = 1; break;
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 633) - [public] do_backspace

```
case EDK_BACKSPACE: do_backspace(); dirty = 1; break;
case EDK_DELETE:   do_delete(); dirty = 1; break;
case EDK_TAB:      ins_char(' '); ins_char(' ');
                  ins_char(' '); ins_char(' '); dirty = 1; break;
case EDK_LEFT:     if (caret_c > 0) caret_c--; dirty = 1; break;
case EDK_RIGHT:    caret_c++; dirty = 1; break;
case EDK_UP:       if (caret_l > 0) caret_l--; dirty = 1; break;
case EDK_DOWN:     if (caret_l < nlines - 1) caret_l++; dirty = 1; break;
case EDK_HOME:     caret_c = 0; dirty = 1; break;
case EDK_END:      caret_c = (int)ed_strlen(lines[caret_l]); dirty = 1; break;
case EDK_F5:       ed_compile_run(); dirty = 1; break;
case EDK_F8:       show_log = !show_log; dirty = 1; break;
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 634) - [public] do_delete

```
case EDK_DELETE:    do_delete(); dirty = 1; break;
case EDK_TAB:        ins_char(' '); ins_char(' ');
                    ins_char(' '); ins_char(' '); dirty = 1; break;
case EDK_LEFT:      if (caret_c > 0) caret_c--; dirty = 1; break;
case EDK_RIGHT:     caret_c++; dirty = 1; break;
case EDK_UP:        if (caret_l > 0) caret_l--; dirty = 1; break;
case EDK_DOWN:      if (caret_l < nlines - 1) caret_l++; dirty = 1; break;
case EDK_HOME:      caret_c = 0; dirty = 1; break;
case EDK_END:       caret_c = (int)ed_strlen(lines[caret_l]); dirty = 1; break;
case EDK_F5:        ed_compile_run(); dirty = 1; break;
case EDK_F8:        show_log = !show_log; dirty = 1; break;
default:
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 635) - [public] ins_char

```
case EDK_TAB:        ins_char(' '); ins_char(' ');
                    ins_char(' '); ins_char(' '); dirty = 1; break;
case EDK_LEFT:      if (caret_c > 0) caret_c--; dirty = 1; break;
case EDK_RIGHT:     caret_c++; dirty = 1; break;
case EDK_UP:        if (caret_l > 0) caret_l--; dirty = 1; break;
case EDK_DOWN:      if (caret_l < nlines - 1) caret_l++; dirty = 1; break;
case EDK_HOME:      caret_c = 0; dirty = 1; break;
case EDK_END:       caret_c = (int)ed_strlen(lines[caret_l]); dirty = 1; break;
case EDK_F5:        ed_compile_run(); dirty = 1; break;
case EDK_F8:        show_log = !show_log; dirty = 1; break;
default:
    if (k >= 32 && k < 0x100) {
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 636) - [public] ins_char

```
                    ins_char(' '); ins_char(' '); dirty = 1; break;
case EDK_LEFT:      if (caret_c > 0) caret_c--; dirty = 1; break;
case EDK_RIGHT:     caret_c++; dirty = 1; break;
case EDK_UP:        if (caret_l > 0) caret_l--; dirty = 1; break;
case EDK_DOWN:      if (caret_l < nlines - 1) caret_l++; dirty = 1; break;
case EDK_HOME:      caret_c = 0; dirty = 1; break;
case EDK_END:       caret_c = (int)ed_strlen(lines[caret_l]); dirty = 1; break;
case EDK_F5:        ed_compile_run(); dirty = 1; break;
case EDK_F8:        show_log = !show_log; dirty = 1; break;
default:
    if (k >= 32 && k < 0x100) {
        ins_char((char)k);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 643) - [public] ed_compile_run

```
case EDK_F5:        ed_compile_run(); dirty = 1; break;
case EDK_F8:        show_log = !show_log; dirty = 1; break;
default:
    if (k >= 32 && k < 0x100) {
        ins_char((char)k);
        dirty = 1;
    }
    break;
}

clamp_scroll(rows);
}
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 646) - [public] if

```
        if (k >= 32 && k < 0x100) {
            ins_char((char)k);
            dirty = 1;
        }
        break;
    }

    clamp_scroll(rows);
}

if (dirty) {
    render();
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 647) - [public] ins_char

```
        ins_char((char)k);
        dirty = 1;
    }
    break;
}

    clamp_scroll(rows);
}

if (dirty) {
    render();
}
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 653) - [public] clamp_scroll

```
    clamp_scroll(rows);
}

if (dirty) {
    render();
}

    __asm__ volatile("pause");
}

console_set_enabled(1);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/editor.c (line 656) - [public] if

```
    if (dirty) {
        render();
    }

    __asm__ volatile("pause");
}

console_set_enabled(1);

return 0;
}
```

kernel_Workspace/kernel/src/editor/editor.c (line 657) - [public] render

```
    render();
}

    __asm__ volatile("pause");
}

console_set_enabled(1);

return 0;
}
```

kernel_Workspace/kernel/src/editor/editor.c (line 660) - [public] volatile

```
    __asm__ volatile("pause");
}

console_set_enabled(1);

return 0;
}
```

kernel_Workspace/kernel/src/editor/editor.c (line 663) - [public] console_set_enabled

```
    console_set_enabled(1);

    return 0;
}
```

kernel_Workspace/kernel/src/editor/mouse.c (line 5) - [public] volatile

```
    __asm__ volatile("outb %0, %1" :: "a"(val), "Nd"(port));
}

static inline uint8_t inb(uint16_t port)
{
    uint8_t v;
    __asm__ volatile("inb %1, %0" : "=a"(v) : "Nd"(port));
    return v;
}

static void wait_write(void)
{
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/kernel/src/editor/mouse.c (line 11) - [public] volatile

```
    __asm__ volatile("inb %1, %0" : "=a"(v) : "Nd"(port));
    return v;
}

static void wait_write(void)
{
    while (inb(0x64) & 0x02) { }
}

static void wait_read(void)
{
    while (!(inb(0x64) & 0x01)) { }
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/kernel/src/editor/mouse.c (line 17) - [public] while

```
    while (inb(0x64) & 0x02) { }
}

static void wait_read(void)
{
    while (!(inb(0x64) & 0x01)) { }
}

static void mouse_cmd(uint8_t cmd)
{
    wait_write(); outb(0x64, 0xD4);
    wait_write(); outb(0x60, cmd);
    // ... (truncated, see source file for more)
}
```


kernel_Workspace/kernel/src/editor/mouse.c (line 22) - [public] while

```
    while (!(inb(0x64) & 0x01)) { }  
}  
  
static void mouse_cmd(uint8_t cmd)  
{  
    wait_write(); outb(0x64, 0xD4);  
    wait_write(); outb(0x60, cmd);  
    wait_read();  inb(0x60);  
}  
  
static void mouse_rate(uint8_t r)  
{  
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/mouse.c (line 27) - [public] wait_write

```
    wait_write(); outb(0x64, 0xD4);  
    wait_write(); outb(0x60, cmd);  
    wait_read();  inb(0x60);  
}  
  
static void mouse_rate(uint8_t r)  
{  
    wait_write(); outb(0x64, 0xD4);  
    wait_write(); outb(0x60, 0xF3);  
    wait_read();  inb(0x60);  
  
    wait_write(); outb(0x64, 0xD4);  
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/mouse.c (line 28) - [public] wait_write

```
    wait_write(); outb(0x60, cmd);  
    wait_read();  inb(0x60);  
}  
  
static void mouse_rate(uint8_t r)  
{  
    wait_write(); outb(0x64, 0xD4);  
    wait_write(); outb(0x60, 0xF3);  
    wait_read();  inb(0x60);  
  
    wait_write(); outb(0x64, 0xD4);  
    wait_write(); outb(0x60, r);  
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/mouse.c (line 29) - [public] wait_read

```
    wait_read();  inb(0x60);  
}  
  
static void mouse_rate(uint8_t r)  
{  
    wait_write(); outb(0x64, 0xD4);  
    wait_write(); outb(0x60, 0xF3);  
    wait_read();  inb(0x60);  
  
    wait_write(); outb(0x64, 0xD4);  
    wait_write(); outb(0x60, r);  
    wait_read();  inb(0x60);  
    // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/editor/mouse.c (line 34) - [public] wait_write
    wait_write(); outb(0x64, 0xD4);
    wait_write(); outb(0x60, 0xF3);
    wait_read();  inb(0x60);

    wait_write(); outb(0x64, 0xD4);
    wait_write(); outb(0x60, r);
    wait_read();  inb(0x60);
}

void mouse_init(void)
{
    wait_write(); outb(0x64, 0xA8);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/mouse.c (line 35) - [public] wait_write
    wait_write(); outb(0x60, 0xF3);
    wait_read();  inb(0x60);

    wait_write(); outb(0x64, 0xD4);
    wait_write(); outb(0x60, r);
    wait_read();  inb(0x60);
}

void mouse_init(void)
{
    wait_write(); outb(0x64, 0xA8);

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/mouse.c (line 36) - [public] wait_read
    wait_read();  inb(0x60);

    wait_write(); outb(0x64, 0xD4);
    wait_write(); outb(0x60, r);
    wait_read();  inb(0x60);
}

void mouse_init(void)
{
    wait_write(); outb(0x64, 0xA8);

    mouse_cmd(0xF6);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/mouse.c (line 38) - [public] wait_write
    wait_write(); outb(0x64, 0xD4);
    wait_write(); outb(0x60, r);
    wait_read();  inb(0x60);
}

void mouse_init(void)
{
    wait_write(); outb(0x64, 0xA8);

    mouse_cmd(0xF6);

    mouse_rate(200);
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/editor/mouse.c (line 39) - [public] wait_write
    wait_write(); outb(0x60, r);
    wait_read();  inb(0x60);
}

void mouse_init(void)
{
    wait_write(); outb(0x64, 0xA8);

    mouse_cmd(0xF6);

    mouse_rate(200);
    mouse_rate(100);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/mouse.c (line 40) - [public] wait_read
    wait_read();  inb(0x60);
}

void mouse_init(void)
{
    wait_write(); outb(0x64, 0xA8);

    mouse_cmd(0xF6);

    mouse_rate(200);
    mouse_rate(100);
    mouse_rate(80);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/mouse.c (line 45) - [public] wait_write
    wait_write(); outb(0x64, 0xA8);

    mouse_cmd(0xF6);

    mouse_rate(200);
    mouse_rate(100);
    mouse_rate(80);

    mouse_cmd(0xF4);
}

int mouse_poll(int *dx, int *dy, int *dz, int *buttons)
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/mouse.c (line 47) - [public] mouse_cmd
    mouse_cmd(0xF6);

    mouse_rate(200);
    mouse_rate(100);
    mouse_rate(80);

    mouse_cmd(0xF4);
}

int mouse_poll(int *dx, int *dy, int *dz, int *buttons)
{
    static uint8_t pkt[4];
    // ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/editor/mouse.c (line 49) - [public] mouse_rate

```
    mouse_rate(200);
    mouse_rate(100);
    mouse_rate(80);

    mouse_cmd(0xF4);
}

int mouse_poll(int *dx, int *dy, int *dz, int *buttons)
{
    static uint8_t pkt[4];
    static int idx = 0;
    int got = 0;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/mouse.c (line 50) - [public] mouse_rate

```
    mouse_rate(100);
    mouse_rate(80);

    mouse_cmd(0xF4);
}

int mouse_poll(int *dx, int *dy, int *dz, int *buttons)
{
    static uint8_t pkt[4];
    static int idx = 0;
    int got = 0;

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/mouse.c (line 51) - [public] mouse_rate

```
    mouse_rate(80);

    mouse_cmd(0xF4);
}

int mouse_poll(int *dx, int *dy, int *dz, int *buttons)
{
    static uint8_t pkt[4];
    static int idx = 0;
    int got = 0;

    while (inb(0x64) & 0x01) {
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/editor/mouse.c (line 53) - [public] mouse_cmd

```
    mouse_cmd(0xF4);
}

int mouse_poll(int *dx, int *dy, int *dz, int *buttons)
{
    static uint8_t pkt[4];
    static int idx = 0;
    int got = 0;

    while (inb(0x64) & 0x01) {
        uint8_t st = inb(0x64);
        uint8_t d = inb(0x60);
        // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/editor/mouse.c (line 62) - [public] while
    while (inb(0x64) & 0x01) {
        uint8_t st = inb(0x64);
        uint8_t d = inb(0x60);

        if (!(st & 0x20)) {
            continue;
        }

        if (idx == 0 && !(d & 0x08)) {
            continue;
        }

        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/mouse.c (line 66) - [public] if
    if (!(st & 0x20)) {
        continue;
    }

    if (idx == 0 && !(d & 0x08)) {
        continue;
    }

    pkt[idx++] = d;

    if (idx == 4) {
        idx = 0;
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/mouse.c (line 70) - [public] if
    if (idx == 0 && !(d & 0x08)) {
        continue;
    }

    pkt[idx++] = d;

    if (idx == 4) {
        idx = 0;

        *buttons = pkt[0] & 0x07;

        *dx = (int)pkt[1] - (int)((pkt[0] << 4) & 0x100);
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/editor/mouse.c (line 76) - [public] if
    if (idx == 4) {
        idx = 0;

        *buttons = pkt[0] & 0x07;

        *dx = (int)pkt[1] - (int)((pkt[0] << 4) & 0x100);
        *dy = (int)pkt[2] - (int)((pkt[0] << 3) & 0x100);
        *dz = (int)(int8_t)pkt[3];

        got = 1;
    }
}
// ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/hdmi/init.c (line 8) - [public] if

```
    if (hdmi_is_ready) {
        return true;
    }

    if (fb_phys == 0 || w == 0 || h == 0 || stride < w) {
        return false;
    }

    hdmi_op_set_fb(fb_phys, w, h, stride);
    if (!hdmi_op_ready()) {
        return false;
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/init.c (line 12) - [public] if

```
    if (fb_phys == 0 || w == 0 || h == 0 || stride < w) {
        return false;
    }

    hdmi_op_set_fb(fb_phys, w, h, stride);
    if (!hdmi_op_ready()) {
        return false;
    }

    const hdmi_mode_t *mode = hdmi_mode_find(w, h, 60);
    if (mode != NULL && !hdmi_mode_apply(mode)) {
        return false;
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/init.c (line 16) - [public] hdmi_op_set_fb

```
    hdmi_op_set_fb(fb_phys, w, h, stride);
    if (!hdmi_op_ready()) {
        return false;
    }

    const hdmi_mode_t *mode = hdmi_mode_find(w, h, 60);
    if (mode != NULL && !hdmi_mode_apply(mode)) {
        return false;
    }

    hdmi_is_ready = true;
    return true;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/init.c (line 17) - [public] if

```
    if (!hdmi_op_ready()) {
        return false;
    }

    const hdmi_mode_t *mode = hdmi_mode_find(w, h, 60);
    if (mode != NULL && !hdmi_mode_apply(mode)) {
        return false;
    }

    hdmi_is_ready = true;
    return true;
}
// ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/hdmi/init.c (line 22) - [public] if
    if (mode != NULL && !hdmi_mode_apply(mode)) {
        return false;
    }

    hdmi_is_ready = true;
    return true;
}

bool hdmi_ready(void)
{
    return hdmi_is_ready && hdmi_op_ready();
}
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/hdmi/init.c (line 32) - [public] hdmi_op_ready
    return hdmi_is_ready && hdmi_op_ready();
}

void hdmi_caps(hdmi_caps_t *out)
{
    hdmi_op_state(out);
}

bool hdmi_fb_set(uint64_t phys, uint32_t w, uint32_t h, uint32_t stride)
{
    if (phys == 0 || w == 0 || h == 0 || stride < w) {
        return false;
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/hdmi/init.c (line 37) - [public] hdmi_op_state
    hdmi_op_state(out);
}

bool hdmi_fb_set(uint64_t phys, uint32_t w, uint32_t h, uint32_t stride)
{
    if (phys == 0 || w == 0 || h == 0 || stride < w) {
        return false;
    }

    hdmi_op_set_fb(phys, w, h, stride);
    hdmi_is_ready = hdmi_op_ready();
    return hdmi_is_ready;
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/hdmi/init.c (line 42) - [public] if
    if (phys == 0 || w == 0 || h == 0 || stride < w) {
        return false;
    }

    hdmi_op_set_fb(phys, w, h, stride);
    hdmi_is_ready = hdmi_op_ready();
    return hdmi_is_ready;
}

bool hdmi_mode_set_by_id(uint32_t id)
{
    if (!hdmi_ready()) {
        // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/hdmi/init.c (line 46) - [public] hdmi_op_set_fb
    hdmi_op_set_fb(phys, w, h, stride);
    hdmi_is_ready = hdmi_op_ready();
    return hdmi_is_ready;
}

bool hdmi_mode_set_by_id(uint32_t id)
{
    if (!hdmi_ready()) {
        return false;
    }

    const hdmi_mode_t *mode = hdmi_mode_by_id(id);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/hdmi/init.c (line 53) - [public] if
    if (!hdmi_ready()) {
        return false;
    }

    const hdmi_mode_t *mode = hdmi_mode_by_id(id);
    hdmi_caps_t caps;
    hdmi_op_state(&caps);
    if (mode == NULL || mode->w != caps.w || mode->h != caps.h) {
        return false;
    }

    return hdmi_mode_apply(mode);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/hdmi/init.c (line 59) - [public] hdmi_op_state
    hdmi_op_state(&caps);
    if (mode == NULL || mode->w != caps.w || mode->h != caps.h) {
        return false;
    }

    return hdmi_mode_apply(mode);
}

bool hdmi_mode_current_raw(uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r)
{
    const hdmi_mode_t *mode = hdmi_mode_current();
    if (mode == NULL || id == NULL || w == NULL || h == NULL || r == NULL) {
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/hdmi/init.c (line 60) - [public] if
    if (mode == NULL || mode->w != caps.w || mode->h != caps.h) {
        return false;
    }

    return hdmi_mode_apply(mode);
}

bool hdmi_mode_current_raw(uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r)
{
    const hdmi_mode_t *mode = hdmi_mode_current();
    if (mode == NULL || id == NULL || w == NULL || h == NULL || r == NULL) {
        return false;
    }
    // ... (truncated, see source file for more)

```


kernel_Workspace/kernel/src/hdmi/init.c (line 64) - [public] hdmi_mode_apply

```
    return hdmi_mode_apply(mode);
}

bool hdmi_mode_current_raw(uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r)
{
    const hdmi_mode_t *mode = hdmi_mode_current();
    if (mode == NULL || id == NULL || w == NULL || h == NULL || r == NULL) {
        return false;
    }

    *id = mode->id;
    *w = mode->w;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/init.c (line 70) - [public] if

```
    if (mode == NULL || id == NULL || w == NULL || h == NULL || r == NULL) {
        return false;
    }

    *id = mode->id;
    *w = mode->w;
    *h = mode->h;
    *r = mode->refresh;
    return true;
}
```

kernel_Workspace/kernel/src/hdmi/interface.c (line 9) - [public] if

```
    if (owner == 0) {
        return false;
    }

    uint32_t expected = 0;
    if (atomic_compare_exchange_strong(&port_owner, &expected, owner)) {
        return true;
    }

    return expected == owner;
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/interface.c (line 14) - [public] if

```
    if (atomic_compare_exchange_strong(&port_owner, &expected, owner)) {
        return true;
    }

    return expected == owner;
}

bool hdmi_iface_release(uint32_t owner)
{
    uint32_t expected = owner;
    return owner != 0 && atomic_compare_exchange_strong(&port_owner, &expected, 0);
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/interface.c (line 29) - [public] atomic_load

```
    return atomic_load(&port_owner);
}

bool hdmi_iface_check(uint32_t owner)
{
    return owner != 0 && atomic_load(&port_owner) == owner;
}

bool hdmi_iface_init(uint32_t owner, uint64_t fb_phys, uint32_t w, uint32_t h, uint32_t stride)
{
    if (owner != 0 && !hdmi_iface_check(owner)) {
        return false;
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/interface.c (line 39) - [public] if

```
    if (owner != 0 && !hdmi_iface_check(owner)) {
        return false;
    }

    return hdmi_init_with(fb_phys, w, h, stride);
}

bool hdmi_iface_ready(void)
{
    return hdmi_ready();
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/interface.c (line 43) - [public] hdmi_init_with

```
    return hdmi_init_with(fb_phys, w, h, stride);
}

bool hdmi_iface_ready(void)
{
    return hdmi_ready();
}

bool hdmi_iface_mode_set(uint32_t owner, uint32_t id)
{
    if (!hdmi_iface_check(owner)) {
        return false;
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/interface.c (line 48) - [public] hdmi_ready

```
    return hdmi_ready();
}

bool hdmi_iface_mode_set(uint32_t owner, uint32_t id)
{
    if (!hdmi_iface_check(owner)) {
        return false;
    }

    return hdmi_mode_set_by_id(id);
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/interface.c (line 53) - [public] if

```
    if (!hdmi_iface_check(owner)) {
        return false;
    }

    return hdmi_mode_set_by_id(id);
}

bool hdmi_iface_mode_current(uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r)
{
    return hdmi_mode_current_raw(id, w, h, r);
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/interface.c (line 57) - [public] hdmi_mode_set_by_id

```
    return hdmi_mode_set_by_id(id);
}

bool hdmi_iface_mode_current(uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r)
{
    return hdmi_mode_current_raw(id, w, h, r);
}

bool hdmi_iface_mode_at(uint32_t i, uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r)
{
    return hdmi_mode_at_raw(i, id, w, h, r);
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/interface.c (line 62) - [public] hdmi_mode_current_raw

```
    return hdmi_mode_current_raw(id, w, h, r);
}

bool hdmi_iface_mode_at(uint32_t i, uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r)
{
    return hdmi_mode_at_raw(i, id, w, h, r);
}

uint64_t hdmi_iface_submit_fill(uint32_t owner, uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h)
{
    if (!hdmi_iface_check(owner)) {
        return 0;
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/interface.c (line 67) - [public] hdmi_mode_at_raw

```
    return hdmi_mode_at_raw(i, id, w, h, r);
}

uint64_t hdmi_iface_submit_fill(uint32_t owner, uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h)
{
    if (!hdmi_iface_check(owner)) {
        return 0;
    }

    return hdmi_submit_fill(color, x, y, w, h);
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/interface.c (line 72) - [public] if

```
    if (!hdmi_iface_check(owner)) {
        return 0;
    }

    return hdmi_submit_fill(color, x, y, w, h);
}

bool hdmi_iface_poll(uint32_t owner, uint64_t *out_seq)
{
    if (!hdmi_iface_check(owner)) {
        return false;
    }
    // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/hdmi/interface.c (line 76) - [public] hdmi_submit_fill
    return hdmi_submit_fill(color, x, y, w, h);
}

bool hdmi_iface_poll(uint32_t owner, uint64_t *out_seq)
{
    if (!hdmi_iface_check(owner)) {
        return false;
    }

    return hdmi_poll(out_seq);
}

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/hdmi/interface.c (line 81) - [public] if
    if (!hdmi_iface_check(owner)) {
        return false;
    }

    return hdmi_poll(out_seq);
}

void hdmi_iface_caps(uint32_t *w, uint32_t *h, uint32_t *s, uint64_t *phys)
{
    hdmi_caps_raw(w, h, s, phys);
}

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/hdmi/interface.c (line 85) - [public] hdmi_poll
    return hdmi_poll(out_seq);
}

void hdmi_iface_caps(uint32_t *w, uint32_t *h, uint32_t *s, uint64_t *phys)
{
    hdmi_caps_raw(w, h, s, phys);
}

bool hdmi_iface_fb_grant(uint32_t owner, uint64_t *phys, uint32_t *w, uint32_t *h, uint32_t *s)
{
    if (!hdmi_iface_check(owner) || phys == NULL || w == NULL || h == NULL || s == NULL) {
        return false;
    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/hdmi/interface.c (line 90) - [public] hdmi_caps_raw
    hdmi_caps_raw(w, h, s, phys);
}

bool hdmi_iface_fb_grant(uint32_t owner, uint64_t *phys, uint32_t *w, uint32_t *h, uint32_t *s)
{
    if (!hdmi_iface_check(owner) || phys == NULL || w == NULL || h == NULL || s == NULL) {
        return false;
    }

    uint64_t p;
    uint32_t ww, hh, ss;
    hdmi_caps_raw(&ww, &hh, &ss, &p);
    // ... (truncated, see source file for more)

```

```
kernel_Workspace/kernel/src/hdmi/interface.c (line 95) - [public] if
    if (!hdmi_iface_check(owner) || phys == NULL || w == NULL || h == NULL || s == NULL) {
        return false;
    }

    uint64_t p;
    uint32_t ww, hh, ss;
    hdmi_caps_raw(&ww, &hh, &ss, &p);

    if (p == 0 || !hdmi_fb_grant()) {
        return false;
    }

    // ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/hdmi/interface.c (line 101) - [public] hdmi_caps_raw
    hdmi_caps_raw(&ww, &hh, &ss, &p);

    if (p == 0 || !hdmi_fb_grant()) {
        return false;
    }

    *phys = p;
    *w = ww;
    *h = hh;
    *s = ss;
    return true;
}

// ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/hdmi/interface.c (line 103) - [public] if
    if (p == 0 || !hdmi_fb_grant()) {
        return false;
    }

    *phys = p;
    *w = ww;
    *h = hh;
    *s = ss;
    return true;
}

bool hdmi_iface_fb_revoke(uint32_t owner)
// ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/hdmi/interface.c (line 116) - [public] if
    if (!hdmi_iface_check(owner)) {
        return false;
    }

    hdmi_fb_revoke();
    return true;
}
```

```
kernel_Workspace/kernel/src/hdmi/interface.c (line 120) - [public] hdmi_fb_revoke
    hdmi_fb_revoke();
    return true;
}
```

kernel_Workspace/kernel/src/hdmi/mode.c (line 17) - [public] return

```
    return (uint32_t)(sizeof(MODES) / sizeof(MODES[0]));
}

const hdmi_mode_t *hdmi_mode_at(uint32_t i)
{
    if (i >= hdmi_mode_count()) {
        return NULL;
    }

    return &MODES[i];
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/mode.c (line 22) - [public] if

```
    if (i >= hdmi_mode_count()) {
        return NULL;
    }

    return &MODES[i];
}

const hdmi_mode_t *hdmi_mode_find(uint32_t w, uint32_t h, uint32_t refresh)
{
    for (uint32_t i = 0; i < hdmi_mode_count(); i++) {
        if (MODES[i].w == w && MODES[i].h == h && MODES[i].refresh == refresh) {
            return &MODES[i];
        }
    }

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/mode.c (line 32) - [public] if

```
    if (MODES[i].w == w && MODES[i].h == h && MODES[i].refresh == refresh) {
        return &MODES[i];
    }

    return NULL;
}

const hdmi_mode_t *hdmi_mode_by_id(uint32_t id)
{
    for (uint32_t i = 0; i < hdmi_mode_count(); i++) {
        if (MODES[i].id == id) {
            // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/mode.c (line 43) - [public] if

```
    if (MODES[i].id == id) {
        return &MODES[i];
    }

    return NULL;
}

bool hdmi_mode_valid(const hdmi_mode_t *m)
{
    if (m == NULL || m->w == 0 || m->h == 0 || m->refresh == 0 || m->clock_khz == 0) {
        return false;
    }

    // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/hdmi/mode.c (line 53) - [public] if
    if (m == NULL || m->w == 0 || m->h == 0 || m->refresh == 0 || m->clock_khz == 0) {
        return false;
    }

    hdmi_caps_t caps;
    hdmi_op_state(&caps);
    return caps.ready != 0 && caps.fb_phys != 0;
}

bool hdmi_mode_apply(const hdmi_mode_t *m)
{
    if (!hdmi_mode_valid(m)) {
        // ... (truncated, see source file for more)
    }
}

kernel_Workspace/kernel/src/hdmi/mode.c (line 58) - [public] hdmi_op_state
    hdmi_op_state(&caps);
    return caps.ready != 0 && caps.fb_phys != 0;
}

bool hdmi_mode_apply(const hdmi_mode_t *m)
{
    if (!hdmi_mode_valid(m)) {
        return false;
    }

    hdmi_op_set_mode(m);
    current_mode = m;
    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/hdmi/mode.c (line 64) - [public] if
    if (!hdmi_mode_valid(m)) {
        return false;
    }

    hdmi_op_set_mode(m);
    current_mode = m;
    return true;
}

const hdmi_mode_t *hdmi_mode_current(void)
{
    return current_mode;
    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/hdmi/mode.c (line 68) - [public] hdmi_op_set_mode
    hdmi_op_set_mode(m);
    current_mode = m;
    return true;
}

const hdmi_mode_t *hdmi_mode_current(void)
{
    return current_mode;
}

bool hdmi_mode_at_raw(uint32_t i, uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r)
{
    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/hdmi/mode.c (line 81) - [public] if
    if (m == NULL || id == NULL || w == NULL || h == NULL || r == NULL) {
        return false;
    }

    *id = m->id;
    *w = m->w;
    *h = m->h;
    *r = m->refresh;
    return true;
}

```

kernel_Workspace/kernel/src/hdmi/operation.c (line 7) - [public] return

```
    return (void *)(DIRECT_BASE + phys);
}

static hdmi_caps_t cur;
static hdmi_mode_t cur_mode;
static bool op_ready;
static hdmi_transfer_t slots[HDMI_MAX_TRANSFERS];
static uint64_t slot_seq[HDMI_MAX_TRANSFERS];
static bool slot_used[HDMI_MAX_TRANSFERS];
static bool slot_done[HDMI_MAX_TRANSFERS];
static uint64_t next_seq = 1;
static uint32_t pending;
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/operation.c (line 22) - [public] if

```
    if (out != NULL) {
        *out = cur;
    }
}

bool hdmi_op_ready(void)
{
    return op_ready;
}

void hdmi_op_set_fb(uint64_t phys, uint32_t w, uint32_t h, uint32_t stride)
{
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/operation.c (line 44) - [public] if

```
    if (t == NULL || !op_ready) {
        return;
    }

    uint32_t stride = t->stride != 0 ? t->stride : cur.stride;
    uint64_t dst_phys = t->dst_phys != 0 ? t->dst_phys : cur.fb_phys;
    if (stride < cur.w || dst_phys == 0) {
        return;
    }

    uint32_t *dst = (uint32_t *)phys_to_virt(dst_phys);
    if (dst == NULL) {
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/operation.c (line 50) - [public] if

```
    if (stride < cur.w || dst_phys == 0) {
        return;
    }

    uint32_t *dst = (uint32_t *)phys_to_virt(dst_phys);
    if (dst == NULL) {
        return;
    }

    if (t->kind == HDMI_TR_FILL) {
        if (t->x >= cur.w || t->y >= cur.h) {
            return;
        }
        // ... (truncated, see source file for more)
```


kernel_Workspace/kernel/src/hdmi/operation.c (line 55) - [public] if

```
    if (dst == NULL) {
        return;
    }

    if (t->kind == HDMI_TR_FILL) {
        if (t->x >= cur.w || t->y >= cur.h) {
            return;
        }
        uint32_t width = t->w < cur.w - t->x ? t->w : cur.w - t->x;
        uint32_t height = t->h < cur.h - t->y ? t->h : cur.h - t->y;
        for (uint32_t y = 0; y < height; y++) {
            for (uint32_t x = 0; x < width; x++) {
                // ... (truncated, see source file for more)
            }
        }
    }
}
```

kernel_Workspace/kernel/src/hdmi/operation.c (line 59) - [public] if

```
    if (t->kind == HDMI_TR_FILL) {
        if (t->x >= cur.w || t->y >= cur.h) {
            return;
        }
        uint32_t width = t->w < cur.w - t->x ? t->w : cur.w - t->x;
        uint32_t height = t->h < cur.h - t->y ? t->h : cur.h - t->y;
        for (uint32_t y = 0; y < height; y++) {
            for (uint32_t x = 0; x < width; x++) {
                dst[(t->y + y) * stride + t->x + x] = t->color;
            }
        }
        return;
    }
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/kernel/src/hdmi/operation.c (line 60) - [public] if

```
    if (t->x >= cur.w || t->y >= cur.h) {
        return;
    }
    uint32_t width = t->w < cur.w - t->x ? t->w : cur.w - t->x;
    uint32_t height = t->h < cur.h - t->y ? t->h : cur.h - t->y;
    for (uint32_t y = 0; y < height; y++) {
        for (uint32_t x = 0; x < width; x++) {
            dst[(t->y + y) * stride + t->x + x] = t->color;
        }
    }
    return;
}
// ... (truncated, see source file for more)
}
```

kernel_Workspace/kernel/src/hdmi/operation.c (line 73) - [public] if

```
    if (t->kind == HDMI_TR_BLIT) {
        if (t->src_phys == 0 || t->stride == 0 || t->x >= cur.w || t->y >= cur.h) {
            return;
        }
        uint32_t *src = (uint32_t *)phys_to_virt(t->src_phys);
        if (src == NULL) {
            return;
        }
        uint32_t width = t->w < cur.w - t->x ? t->w : cur.w - t->x;
        uint32_t height = t->h < cur.h - t->y ? t->h : cur.h - t->y;
        for (uint32_t y = 0; y < height; y++) {
            for (uint32_t x = 0; x < width; x++) {
                // ... (truncated, see source file for more)
            }
        }
    }
}
```

```
kernel_Workspace/kernel/src/hdmi/operation.c (line 74) - [public] if
    if (t->src_phys == 0 || t->stride == 0 || t->x >= cur.w || t->y >= cur.h) {
        return;
    }
    uint32_t *src = (uint32_t *)phys_to_virt(t->src_phys);
    if (src == NULL) {
        return;
    }
    uint32_t width = t->w < cur.w - t->x ? t->w : cur.w - t->x;
    uint32_t height = t->h < cur.h - t->y ? t->h : cur.h - t->y;
    for (uint32_t y = 0; y < height; y++) {
        for (uint32_t x = 0; x < width; x++) {
            dst[(t->y + y) * stride + t->x + x] = src[y * t->stride + x];
        }
    }
    // ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/hdmi/operation.c (line 78) - [public] if
    if (src == NULL) {
        return;
    }
    uint32_t width = t->w < cur.w - t->x ? t->w : cur.w - t->x;
    uint32_t height = t->h < cur.h - t->y ? t->h : cur.h - t->y;
    for (uint32_t y = 0; y < height; y++) {
        for (uint32_t x = 0; x < width; x++) {
            dst[(t->y + y) * stride + t->x + x] = src[y * t->stride + x];
        }
    }
    return;
}
// ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/hdmi/operation.c (line 91) - [public] if
    if (t->kind == HDMI_TR_FLIP && t->dst_phys != 0) {
        cur.fb_phys = t->dst_phys;
    }
}

uint64_t hdmi_submit(const hdmi_transfer_t *t)
{
    if (t == NULL || !op_ready || pending >= HDMI_MAX_TRANSFERS) {
        return 0;
    }

    if (t->kind != HDMI_TR_FILL && t->kind != HDMI_TR_BLIT && t->kind != HDMI_TR_FLIP) {
        // ... (truncated, see source file for more)
    }
}
```

```
kernel_Workspace/kernel/src/hdmi/operation.c (line 98) - [public] if
    if (t == NULL || !op_ready || pending >= HDMI_MAX_TRANSFERS) {
        return 0;
    }

    if (t->kind != HDMI_TR_FILL && t->kind != HDMI_TR_BLIT && t->kind != HDMI_TR_FLIP) {
        return 0;
    }

    for (uint32_t i = 0; i < HDMI_MAX_TRANSFERS; i++) {
        if (!slot_used[i]) {
            slots[i] = *t;
            slot_seq[i] = next_seq++;
        }
    }
    // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/hdmi/operation.c (line 102) - [public] if
    if (t->kind != HDMI_TR_FILL && t->kind != HDMI_TR_BLIT && t->kind != HDMI_TR_FLIP) {
        return 0;
    }

    for (uint32_t i = 0; i < HDMI_MAX_TRANSFERS; i++) {
        if (!slot_used[i]) {
            slots[i] = *t;
            slot_seq[i] = next_seq++;
            if (next_seq == 0) {
                next_seq = 1;
            }
            slot_used[i] = true;
            // ... (truncated, see source file for more)
        }
    }

kernel_Workspace/kernel/src/hdmi/operation.c (line 107) - [public] if
    if (!slot_used[i]) {
        slots[i] = *t;
        slot_seq[i] = next_seq++;
        if (next_seq == 0) {
            next_seq = 1;
        }
        slot_used[i] = true;
        slot_done[i] = false;
        pending++;
        hdmi_op_exec(&slots[i]);
        slot_done[i] = true;
        return slot_seq[i];
        // ... (truncated, see source file for more)
    }

kernel_Workspace/kernel/src/hdmi/operation.c (line 110) - [public] if
    if (next_seq == 0) {
        next_seq = 1;
    }
    slot_used[i] = true;
    slot_done[i] = false;
    pending++;
    hdmi_op_exec(&slots[i]);
    slot_done[i] = true;
    return slot_seq[i];
}

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/hdmi/operation.c (line 116) - [public] hdmi_op_exec
    hdmi_op_exec(&slots[i]);
    slot_done[i] = true;
    return slot_seq[i];
}

return 0;
}

bool hdmi_poll(uint64_t *out_seq)
{
    if (out_seq == NULL) {
        // ... (truncated, see source file for more)
    }
}

```

kernel_Workspace/kernel/src/hdmi/operation.c (line 127) - [public] if

```
    if (out_seq == NULL) {
        return false;
    }

    for (uint32_t i = 0; i < HDMI_MAX_TRANSFERS; i++) {
        if (slot_used[i] && slot_done[i]) {
            *out_seq = slot_seq[i];
            slot_used[i] = false;
            slot_done[i] = false;
            pending--;
            return true;
        }
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/operation.c (line 132) - [public] if

```
        if (slot_used[i] && slot_done[i]) {
            *out_seq = slot_seq[i];
            slot_used[i] = false;
            slot_done[i] = false;
            pending--;
            return true;
        }
    }

    return false;
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/operation.c (line 159) - [public] hdmi_submit

```
    return hdmi_submit(&t);
}

void hdmi_op_set_mode(const hdmi_mode_t *m)
{
    if (m == NULL) {
        return;
    }
    cur_mode = *m;
    cur.w = m->w;
    cur.h = m->h;
    cur.ready = (cur.fb_phys != 0 && cur.w != 0 && cur.h != 0 && cur.stride >= cur.w) ? 1 : 0;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/operation.c (line 164) - [public] if

```
    if (m == NULL) {
        return;
    }
    cur_mode = *m;
    cur.w = m->w;
    cur.h = m->h;
    cur.ready = (cur.fb_phys != 0 && cur.w != 0 && cur.h != 0 && cur.stride >= cur.w) ? 1 : 0;
    op_ready = cur.ready != 0;
}

static bool fb_granted = false;

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/operation.c (line 178) - [public] if

```
    if (cur.fb_phys == 0) {
        return false;
    }

    fb_granted = true;
    return true;
}
```

```
void hdmi_fb_revoke(void)
{
    fb_granted = false;
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/operation.c (line 198) - [public] if

```
    if (w == NULL || h == NULL || s == NULL || phys == NULL) {
        return;
    }
    *w = cur.w;
    *h = cur.h;
    *s = cur.stride;
    *phys = cur.fb_phys;
}
```

kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 8) - [public] kprintf

```
extern void kprintf(const char *fmt, ...);

static void k_memset(const void *dst, const uint8 value, const size_t n) {
    ■uint8_t *p = (uint8_t *)dst;
    ■for (size_t i = 0; i < n; i++) p[i] = value;
}

static void k_memcpy(const void *dst, const void *src, const size_t n) {
    ■uint8_t *d = (uint8_t *)dst;
    ■const uint8_t *s = (const uint8_t *)src;
    ■for (size_t i = 0; i < n; i++) d[i] = s[i];
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 10) - [internal] k_memset

```
static void k_memset(const void *dst, const uint8 value, const size_t n) {
    ■uint8_t *p = (uint8_t *)dst;
    ■for (size_t i = 0; i < n; i++) p[i] = value;
}

static void k_memcpy(const void *dst, const void *src, const size_t n) {
    ■uint8_t *d = (uint8_t *)dst;
    ■const uint8_t *s = (const uint8_t *)src;
    ■for (size_t i = 0; i < n; i++) d[i] = s[i];
}

static size_t k_strlen(const char *s) {
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 15) - [internal] k_memcpy

```
static void k_memcpy(const void *dst, const void *src, const size_t n) {
    ■uint8_t *d = (uint8_t *)dst;
    ■const uint8_t *s = (const uint8_t *)src;
    ■for (size_t i = 0; i < n; i++) d[i] = s[i];
}

static size_t k_strlen(const char *s) {
    ■size_t n = 0;
    ■while (s[n] != '\0') n++;
    ■return n;
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 21) - [internal] k_strlen

```
static size_t k_strlen(const char *s) {
    size_t n = 0;
    while (s[n] != '\0') n++;
    return n;
}

static size_t k_usable_size(void *ptr) {
#ifdef ALLOC_DEBUG
    size_t s = dbg_usable_size(ptr);
    if (s != 0) return s;
#endif
    return heap_usable_size(ptr);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 27) - [internal] k_usable_size

```
static size_t k_usable_size(void *ptr) {
#ifdef ALLOC_DEBUG
    size_t s = dbg_usable_size(ptr);
    if (s != 0) return s;
#endif
    return heap_usable_size(ptr);
}

void *kmalloc(const size_t size) {
    if (size == 0) return NULL;
#ifdef ALLOC_DEBUG
    return dbg_alloc(size);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 32) - [public] heap_usable_size

```
return heap_usable_size(ptr);
}

void *kmalloc(const size_t size) {
    if (size == 0) return NULL;
#ifdef ALLOC_DEBUG
    return dbg_alloc(size);
#else
    return heap_alloc(size);
#endif
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 38) - [public] dbg_alloc

```
return dbg_alloc(size);
#else
return heap_alloc(size);
#endif
}

void kfree(void *ptr) {
    if (ptr == NULL) return;
#ifdef ALLOC_DEBUG
    dbg_free(ptr);
#else
    heap_free(ptr);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 40) - [public] heap_alloc

```
    return heap_alloc(size);
#endif
}

void kfree(void *ptr) {
    if (ptr == NULL) return;
#ifdef ALLOC_DEBUG
    dbg_free(ptr);
#else
    heap_free(ptr);
#endif
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 44) - [public] kfree

```
void kfree(void *ptr) {
    if (ptr == NULL) return;
#ifdef ALLOC_DEBUG
    dbg_free(ptr);
#else
    heap_free(ptr);
#endif
}

void *kzalloc(const size_t size) {
    void *ptr = kmalloc(size);
    if (ptr == NULL) return NULL;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 63) - [public] kzalloc

```
    return kzalloc(count * size);
}

void *kmalloc_aligned(const size_t size, const size_t align) {
    if (size == 0) return NULL;

    if (!size_is_pow2(align)) return NULL;
    return heap_alloc_aligned(size, align);
}

size_t kmalloc_usable_size(void *ptr) { return k_usable_size(ptr); }
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 70) - [public] heap_alloc_aligned

```
    return heap_alloc_aligned(size, align);
}

size_t kmalloc_usable_size(void *ptr) { return k_usable_size(ptr); }

void *krealloc(void *ptr, const size_t new_size) {
    if (ptr == NULL) return kmalloc(new_size);
    if (new_size == 0) {
        kfree(ptr);
        return NULL;
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 74) - [public] kmalloc_usable_size

```
size_t kmalloc_usable_size(void *ptr) { return k_usable_size(ptr); }
```

```
void *krealloc(void *ptr, const size_t new_size) {  
    if (ptr == NULL) return kmalloc(new_size);  
    if (new_size == 0) {  
        kfree(ptr);  
        return NULL;  
    }  
    const size_t old_size = k_usable_size(ptr);  
    if (old_size == 0) return NULL;  
    if (old_size >= new_size) return ptr;  
    void *new_ptr = kmalloc(new_size);  
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 79) - [public] kfree

```
    kfree(ptr);  
    return NULL;  
}  
const size_t old_size = k_usable_size(ptr);  
if (old_size == 0) return NULL;  
if (old_size >= new_size) return ptr;  
void *new_ptr = kmalloc(new_size);  
if (new_ptr == NULL) return NULL;  
k_memcpy(new_ptr, ptr, old_size);  
kfree(ptr);  
return new_ptr;  
}  
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 95) - [public] heap_alloc

```
return heap_alloc(pages * ARCH_PAGE_SIZE);  
}
```

```
void kfree_pages(void *ptr, const size_t pages) {  
    if (ptr == NULL) return;  
    size_t requested_bytes = 0;  
    const size_t usable = heap_usable_size(ptr);  
    if (!kfree_pages_validate(pages, ARCH_PAGE_SIZE, usable, &requested_bytes)) {  
        kprintf("kfree_pages: nieprawidlowy parametr pages=%zu (pojemnosc bloku=%zu B) dla ptr=%p – odmowa zwo  
        return;  
    }  
    heap_free(ptr);  
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 98) - [public] kfree_pages

```
void kfree_pages(void *ptr, const size_t pages) {  
    if (ptr == NULL) return;  
    size_t requested_bytes = 0;  
    const size_t usable = heap_usable_size(ptr);  
    if (!kfree_pages_validate(pages, ARCH_PAGE_SIZE, usable, &requested_bytes)) {  
        kprintf("kfree_pages: nieprawidlowy parametr pages=%zu (pojemnosc bloku=%zu B) dla ptr=%p – odmowa zwo  
        return;  
    }  
    heap_free(ptr);  
}
```

```
char *kstrdup(const char *s) {  
    // ... (truncated, see source file for more)
```


kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 103) - [public] kprintf

```
    kprintf("kfree_pages: nieprawidlowy parametr pages=%zu (pojemnosc bloku=%zu B) dla ptr=%p - odmowa zwolnienia\n", pages, block_size, ptr);
    return;
}

heap_free(ptr);
}

char *kstrdup(const char *s) {
    if (s == NULL) return NULL;
    const size_t n = k_strlen(s) + 1;
    char *ptr = (char *)kmalloc(n);
    if (ptr == NULL) return NULL;
    k_memcpy(ptr, s, n);
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 118) - [public] kvirt_to_phys

```
uint64_t kvirt_to_phys(void *ptr) {
    if (ptr == NULL) return UINT64_MAX;
    const uint64_t va = (uint64_t)(uintptr_t)ptr;
    phys = mapping_translate(MAPPING_KERNEL, va);
    if (phys != UINT64_MAX) return phys;
    return arch_virt_to_phys(ptr);
}

void kalloc_dump(void) {
#ifdef ALLOC_DEBUG
    mm_debug_dump();
#else
    // ... (truncated, see source file for more)
}
}
```

kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 123) - [public] arch_virt_to_phys

```
return arch_virt_to_phys(ptr);
}

void kalloc_dump(void) {
#ifdef ALLOC_DEBUG
    mm_debug_dump();
#else
    heap_dump();
#endif
}
}
```

kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 126) - [public] kalloc_dump

```
void kalloc_dump(void) {
#ifdef ALLOC_DEBUG
    mm_debug_dump();
#else
    heap_dump();
#endif
}
}
```

kernel_Workspace/kernel/src/mm/alloc/debug/alloc_debug.c (line 11) - [public] kprintf

```
extern void kprintf(const char *fmt, ...);

#define DBG_MAGIC 0x444247414C4C4F43ULL
#define DBG_POISON_ALLOC 0xAA
#define DBG_POISON_FREE 0xFF

typedef struct dbg_header {
    uint64_t magic;
    uint64_t size;
    uint64_t caller;
    uint64_t tracked;
    uint64_t alloc_id;
    // ... (truncated, see source file for more)
}
```

kernel_workspace/kernel/src/mm/alloc/debug/alloc_debug.c (line 25) - [internal]

dbg_tail_canary

```
static uint64_t dbg_tail_canary(size_t size) { return ~((uint64_t)size) ^ DBG_MAGIC; }
static uint64_t *dbg_tail_ptr(void *user, size_t size) { return (uint64_t *)((uint8_t *)user + size); }

void *dbg_alloc(size_t size) {
    if (size == 0) return NULL;

    uint64_t caller = (uint64_t)(uintptr_t)__builtin_return_address(0);
    uint8_t *real = (uint8_t *)heap_alloc(sizeof(dbg_header_t) + size + sizeof(uint64_t));

    if (real == NULL) return NULL;

    dbg_header_t *h = (dbg_header_t *)real;
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/debug/alloc_debug.c (line 56) - [public] dbg_free

```
void dbg_free(void *ptr) {
    if (ptr == NULL) return;

    dbg_header_t *h = (dbg_header_t *)((uint8_t *)ptr - sizeof(dbg_header_t));
    if (h->magic != DBG_MAGIC) {
        kprintf("dbg_free: bad magic at %p\n", ptr);
        return;
    }

    size_t size = h->size;
    if (*dbg_tail_ptr(ptr, size) != dbg_tail_canary(size)) kprintf("dbg_free: buffer overflow at %p caller 0\n", ptr);
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/debug/alloc_debug.c (line 61) - [public] kprintf

```
kprintf("dbg_free: bad magic at %p\n", ptr);
return;
}

size_t size = h->size;
if (*dbg_tail_ptr(ptr, size) != dbg_tail_canary(size)) kprintf("dbg_free: buffer overflow at %p caller 0\n", ptr);

if (h->tracked) {
    size_t leaked_size = 0;
    uint64_t alloc_caller = 0;
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/debug/alloc_debug.c (line 73) - [public] if

```
if (!leak_untrack(ptr, &leaked_size, &alloc_caller)) {
    kprintf("dbg_free: double free or unknown ptr %p\n", ptr);
    return;
}

alloc_stats_note_free(size);

h->magic = 0;

uint8_t *real = (uint8_t *)h;
size_t total = sizeof(dbg_header_t) + size + sizeof(uint64_t);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/debug/alloc_debug.c (line 74) - [public] kprintf

```
    kprintf("dbg_free: double free or unknown ptr %p\n", ptr);
    return;
}

alloc_stats_note_free(size);

h->magic = 0;

uint8_t *real = (uint8_t *)h;
size_t total = sizeof(dbg_header_t) + size + sizeof(uint64_t);
for (size_t i = 0; i < total; i++) real[i] = DBG_POISON_FREE;
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/debug/alloc_debug.c (line 90) - [public] dbg_usable_size

```
size_t dbg_usable_size(void *ptr) {
    if (ptr == NULL) return 0;
    dbg_header_t *h = (dbg_header_t *)((uint8_t *)ptr - sizeof(dbg_header_t));
    if (h->magic != DBG_MAGIC) return 0;
    return h->size;
}

bool dbg_verify(void *ptr) {
    if (ptr == NULL) return false;
    dbg_header_t *h = (dbg_header_t *)((uint8_t *)ptr - sizeof(dbg_header_t));
    if (h->magic != DBG_MAGIC) return false;
    return *dbg_tail_ptr(ptr, h->size) == dbg_tail_canary(h->size);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/debug/alloc_debug.c (line 97) - [public] dbg_verify

```
bool dbg_verify(void *ptr) {
    if (ptr == NULL) return false;
    dbg_header_t *h = (dbg_header_t *)((uint8_t *)ptr - sizeof(dbg_header_t));
    if (h->magic != DBG_MAGIC) return false;
    return *dbg_tail_ptr(ptr, h->size) == dbg_tail_canary(h->size);
}

void mm_debug_dump(void) {
    alloc_stats_dump();
    leak_dump();
    pmm_dump();
    vmm_dump();
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/debug/alloc_debug.c (line 104) - [public] mm_debug_dump

```
void mm_debug_dump(void) {
    alloc_stats_dump();
    leak_dump();
    pmm_dump();
    vmm_dump();
    buddy_dump();
    slab_dump();
    cache_dump();
}
```

kernel_Workspace/kernel/src/mm/alloc/debug/leak.c (line 4) - [public] kprintf

```
extern void kprintf(const char *fmt, ...);

#define LEAK_MAX 2048

typedef struct leak_entry {
    void *ptr;
    size_t size;
    uint64_t caller, alloc_id;
    int64_t cpu_id;
    bool used;
} leak_entry_t;

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/debug/leak.c (line 22) - [public] leak_track_id

```
bool leak_track_id(void *ptr, size_t size, uint64_t caller, uint64_t *out_id) {
    smp_lock_acquire(&leak_smp_lock);
    for (size_t i = 0; i < LEAK_MAX; i++) {
        if (!leak_table[i].used) {
            uint64_t id = next_alloc_id++;
            leak_table[i].ptr = ptr;
            leak_table[i].size = size;
            leak_table[i].caller = caller;
            leak_table[i].alloc_id = id;
            leak_table[i].cpu_id = smp_current_cpu_id();
            leak_table[i].used = true;
            smp_lock_release(&leak_smp_lock);
            // ... (truncated, see source file for more)
        }
    }
}
```

kernel_Workspace/kernel/src/mm/alloc/debug/leak.c (line 25) - [public] if

```
    if (!leak_table[i].used) {
        uint64_t id = next_alloc_id++;
        leak_table[i].ptr = ptr;
        leak_table[i].size = size;
        leak_table[i].caller = caller;
        leak_table[i].alloc_id = id;
        leak_table[i].cpu_id = smp_current_cpu_id();
        leak_table[i].used = true;
        smp_lock_release(&leak_smp_lock);
        if (out_id != NULL) *out_id = id;
        return true;
    }
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/kernel/src/mm/alloc/debug/leak.c (line 33) - [public] smp_lock_release

```
    smp_lock_release(&leak_smp_lock);
    if (out_id != NULL) *out_id = id;
    return true;
}
}
smp_lock_release(&leak_smp_lock);
return false;
}
bool leak_track(void *ptr, size_t size, uint64_t caller) { return leak_track_id(ptr, size, caller, NULL); }
bool leak_untrack(void *ptr, size_t *out_size, uint64_t *out_caller) {
    smp_lock_acquire(&leak_smp_lock);
    for (size_t i = 0; i < LEAK_MAX; i++) {
        // ... (truncated, see source file for more)
    }
}
```

kernel_Workspace/kernel/src/mm/alloc/debug/leak.c (line 41) - [public] leak_track

```
bool leak_track(void *ptr, size_t size, uint64_t caller) { return leak_track_id(ptr, size, caller, NULL); }
bool leak_untrack(void *ptr, size_t *out_size, uint64_t *out_caller) {
    smp_lock_acquire(&leak_smp_lock);
    for (size_t i = 0; i < LEAK_MAX; i++) {
        if (leak_table[i].used && leak_table[i].ptr == ptr) {
            if (out_size != NULL) *out_size = leak_table[i].size;
            if (out_caller != NULL) *out_caller = leak_table[i].caller;
            leak_table[i].used = false;
            smp_lock_release(&leak_smp_lock);
            return true;
        }
    }
}
// ... (truncated, see source file for more)
}
```

kernel_Workspace/kernel/src/mm/alloc/debug/leak.c (line 42) - [public] leak_untrack

```
bool leak_untrack(void *ptr, size_t *out_size, uint64_t *out_caller) {
    ■smp_lock_acquire(&leak_smp_lock);
    ■for (size_t i = 0; i < LEAK_MAX; i++) {
    ■    if (leak_table[i].used && leak_table[i].ptr == ptr) {
    ■        if (out_size != NULL) *out_size = leak_table[i].size;
    ■        if (out_caller != NULL) *out_caller = leak_table[i].caller;
    ■        leak_table[i].used = false;
    ■        smp_lock_release(&leak_smp_lock);
    ■        return true;
    ■    }
    ■}
    ■smp_lock_release(&leak_smp_lock);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/debug/leak.c (line 45) - [public] if

```
    ■    if (leak_table[i].used && leak_table[i].ptr == ptr) {
    ■        if (out_size != NULL) *out_size = leak_table[i].size;
    ■        if (out_caller != NULL) *out_caller = leak_table[i].caller;
    ■        leak_table[i].used = false;
    ■        smp_lock_release(&leak_smp_lock);
    ■        return true;
    ■    }
    ■}
    ■smp_lock_release(&leak_smp_lock);
    ■return false;
    }
    bool leak_contains(void *ptr) {
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/debug/leak.c (line 49) - [public] smp_lock_release

```
    ■        smp_lock_release(&leak_smp_lock);
    ■        return true;
    ■    }
    ■}
    ■smp_lock_release(&leak_smp_lock);
    ■return false;
    }
    bool leak_contains(void *ptr) {
    ■smp_lock_acquire(&leak_smp_lock);
    ■for (size_t i = 0; i < LEAK_MAX; i++) {
    ■    if (leak_table[i].used && leak_table[i].ptr == ptr) {
    ■        smp_lock_release(&leak_smp_lock);
    ■        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/debug/leak.c (line 56) - [public] leak_contains

```
    bool leak_contains(void *ptr) {
    ■smp_lock_acquire(&leak_smp_lock);
    ■for (size_t i = 0; i < LEAK_MAX; i++) {
    ■    if (leak_table[i].used && leak_table[i].ptr == ptr) {
    ■        smp_lock_release(&leak_smp_lock);
    ■        return true;
    ■    }
    ■}
    ■smp_lock_release(&leak_smp_lock);
    ■return false;
    }
    size_t leak_count(void) {
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/debug/leak.c (line 59) - [public] if

```
■    if (leak_table[i].used && leak_table[i].ptr == ptr) {
■        smp_lock_release(&leak_smp_lock);
■        return true;
■    }
■}
■smp_lock_release(&leak_smp_lock);
■return false;
}
size_t leak_count(void) {
■smp_lock_acquire(&leak_smp_lock);
■size_t n = 0;
■for (size_t i = 0; i < LEAK_MAX; i++) if (leak_table[i].used) n++;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/debug/leak.c (line 60) - [public] smp_lock_release

```
■        smp_lock_release(&leak_smp_lock);
■        return true;
■    }
■}
■smp_lock_release(&leak_smp_lock);
■return false;
}
size_t leak_count(void) {
■smp_lock_acquire(&leak_smp_lock);
■size_t n = 0;
■for (size_t i = 0; i < LEAK_MAX; i++) if (leak_table[i].used) n++;
■smp_lock_release(&leak_smp_lock);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/debug/leak.c (line 67) - [public] leak_count

```
size_t leak_count(void) {
■smp_lock_acquire(&leak_smp_lock);
■size_t n = 0;
■for (size_t i = 0; i < LEAK_MAX; i++) if (leak_table[i].used) n++;
■smp_lock_release(&leak_smp_lock);
■return n;
}
void leak_dump(void) {
■smp_lock_acquire(&leak_smp_lock);
■size_t n = 0;
■for (size_t i = 0; i < LEAK_MAX; i++) if (leak_table[i].used) n++;
■kprintf("LEAKS: %llu live\n", (unsigned long long)n);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/debug/leak.c (line 74) - [public] leak_dump

```
void leak_dump(void) {
■smp_lock_acquire(&leak_smp_lock);
■size_t n = 0;
■for (size_t i = 0; i < LEAK_MAX; i++) if (leak_table[i].used) n++;
■kprintf("LEAKS: %llu live\n", (unsigned long long)n);
■for (size_t i = 0; i < LEAK_MAX; i++) {
■    if (!leak_table[i].used) continue;
■    kprintf("    leak: %p size %llu caller 0x%llx id %llu cpu %lld\n",
■        leak_table[i].ptr,
■        (unsigned long long)leak_table[i].size,
■        (unsigned long long)leak_table[i].caller,
■        (unsigned long long)leak_table[i].alloc_id,
■        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/debug/stats.c (line 4) - [public] kprintf

```
extern void kprintf(const char *fmt, ...);

static size_t
■stat_live_count = 0,
■stat_live_bytes = 0,
■stat_peak_bytes = 0,
■stat_total_allocs = 0,
■stat_total_frees = 0;

static smp_ticket_lock_t stats_smp_lock = SMP_TICKET_LOCK_INIT;

void alloc_stats_note_alloc(size_t size) {
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/debug/stats.c (line 15) - [public] alloc_stats_note_alloc

```
void alloc_stats_note_alloc(size_t size) {
    ■smp_lock_acquire(&stats_smp_lock);
    ■stat_live_count++;
    ■stat_live_bytes += size;
    ■stat_total_allocs++;
    ■if (stat_live_bytes > stat_peak_bytes) stat_peak_bytes = stat_live_bytes;
    ■smp_lock_release(&stats_smp_lock);
}

void alloc_stats_note_free(size_t size) {
    ■smp_lock_acquire(&stats_smp_lock);
    ■if (stat_live_count > 0) stat_live_count--;
    ■if (stat_live_bytes >= size) stat_live_bytes -= size;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/debug/stats.c (line 23) - [public] alloc_stats_note_free

```
void alloc_stats_note_free(size_t size) {
    ■smp_lock_acquire(&stats_smp_lock);
    ■if (stat_live_count > 0) stat_live_count--;
    ■if (stat_live_bytes >= size) stat_live_bytes -= size;
    ■else stat_live_bytes = 0;
    ■stat_total_frees++;
    ■smp_lock_release(&stats_smp_lock);
}

size_t alloc_stats_live_count(void) {
    ■smp_lock_acquire(&stats_smp_lock);
    ■size_t v = stat_live_count;
    ■smp_lock_release(&stats_smp_lock);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/debug/stats.c (line 31) - [public] alloc_stats_live_count

```
size_t alloc_stats_live_count(void) {
    ■smp_lock_acquire(&stats_smp_lock);
    ■size_t v = stat_live_count;
    ■smp_lock_release(&stats_smp_lock);
    ■return v;
}

size_t alloc_stats_live_bytes(void) {
    ■smp_lock_acquire(&stats_smp_lock);
    ■size_t v = stat_live_bytes;
    ■smp_lock_release(&stats_smp_lock);
    ■return v;
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/debug/stats.c (line 37) - [public] alloc_stats_live_bytes

```
size_t alloc_stats_live_bytes(void) {
    ■smp_lock_acquire(&stats_smp_lock);
    ■size_t v = stat_live_bytes;
    ■smp_lock_release(&stats_smp_lock);
    ■return v;
}
size_t alloc_stats_peak_bytes(void) {
    ■smp_lock_acquire(&stats_smp_lock);
    ■size_t v = stat_peak_bytes;
    ■smp_lock_release(&stats_smp_lock);
    ■return v;
}
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/debug/stats.c (line 43) - [public] alloc_stats_peak_bytes

```
size_t alloc_stats_peak_bytes(void) {
    ■smp_lock_acquire(&stats_smp_lock);
    ■size_t v = stat_peak_bytes;
    ■smp_lock_release(&stats_smp_lock);
    ■return v;
}
size_t alloc_stats_total_allocs(void) {
    ■smp_lock_acquire(&stats_smp_lock);
    ■size_t v = stat_total_allocs;
    ■smp_lock_release(&stats_smp_lock);
    ■return v;
}
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/debug/stats.c (line 49) - [public]
alloc_stats_total_allocs

```
size_t alloc_stats_total_allocs(void) {
    ■smp_lock_acquire(&stats_smp_lock);
    ■size_t v = stat_total_allocs;
    ■smp_lock_release(&stats_smp_lock);
    ■return v;
}
size_t alloc_stats_total_frees(void) {
    ■smp_lock_acquire(&stats_smp_lock);
    ■size_t v = stat_total_frees;
    ■smp_lock_release(&stats_smp_lock);
    ■return v;
}
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/debug/stats.c (line 55) - [public]
alloc_stats_total_frees

```
size_t alloc_stats_total_frees(void) {
    ■smp_lock_acquire(&stats_smp_lock);
    ■size_t v = stat_total_frees;
    ■smp_lock_release(&stats_smp_lock);
    ■return v;
}
void alloc_stats_dump(void) {
    ■smp_lock_acquire(&stats_smp_lock);
    ■size_t
    ■■live_count = stat_live_count,
    ■■live_bytes = stat_live_bytes,
    ■■peak_bytes = stat_peak_bytes,
    // ... (truncated, see source file for more)
```


kernel_Workspace/kernel/src/mm/alloc/debug/stats.c (line 61) - [public] alloc_stats_dump

```
void alloc_stats_dump(void) {
    ■smp_lock_acquire(&stats_smp_lock);
    ■size_t
    ■■live_count = stat_live_count,
    ■■live_bytes = stat_live_bytes,
    ■■peak_bytes = stat_peak_bytes,
    ■■total_allocs = stat_total_allocs,
    ■■total_frees = stat_total_frees;
    ■smp_lock_release(&stats_smp_lock);
    ■kprintf("ALLOC STATS:\n");
    ■kprintf("  live: %llu objs, %llu B\n",
    ■■■(unsigned long long)live_count,
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 6) - [public] kprintf

```
extern void kprintf(const char *fmt, ...);

#define BUDDY_STATE_FREE 0
#define BUDDY_STATE_USED 1
#define BUDDY_STATE_TAIL 2

static uint64_t buddy_base = 0;
static size_t buddy_pages = 0, buddy_max_order = 0;

static uint8_t *buddy_order = NULL, *buddy_state = NULL;
static int32_t *buddy_next = NULL, *buddy_heads = NULL;

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 25) - [internal] order_size

```
static inline uint64_t order_size(size_t o) { return ARCH_PAGE_SIZE << o; }
static size_t page_of(uint64_t va) { return (size_t)((va - buddy_base) / ARCH_PAGE_SIZE); }
static uint64_t va_of(size_t page) { return buddy_base + (uint64_t)page * ARCH_PAGE_SIZE; }
static size_t size_to_order(size_t size) {
    ■size_t o = 0, s = ARCH_PAGE_SIZE;
    ■while (s < size && o < buddy_max_order) {
    ■    s <= 1;
    ■    o++;
    ■}
    ■return o;
}
static void fl_push(size_t o, size_t idx) {
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 26) - [internal] page_of

```
static size_t page_of(uint64_t va) { return (size_t)((va - buddy_base) / ARCH_PAGE_SIZE); }
static uint64_t va_of(size_t page) { return buddy_base + (uint64_t)page * ARCH_PAGE_SIZE; }
static size_t size_to_order(size_t size) {
    ■size_t o = 0, s = ARCH_PAGE_SIZE;
    ■while (s < size && o < buddy_max_order) {
    ■    s <= 1;
    ■    o++;
    ■}
    ■return o;
}
static void fl_push(size_t o, size_t idx) {
    ■buddy_next[idx] = buddy_heads[o];
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 27) - [internal] va_of

```
static uint64_t va_of(size_t page) { return buddy_base + (uint64_t)page * ARCH_PAGE_SIZE; }
static size_t size_to_order(size_t size) {
    size_t o = 0, s = ARCH_PAGE_SIZE;
    while (s < size && o < buddy_max_order) {
        s <= 1;
        o++;
    }
    return o;
}
static void fl_push(size_t o, size_t idx) {
    buddy_next[idx] = buddy_heads[o];
    buddy_heads[o] = (int32_t)idx;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 28) - [internal] size_to_order

```
static size_t size_to_order(size_t size) {
    size_t o = 0, s = ARCH_PAGE_SIZE;
    while (s < size && o < buddy_max_order) {
        s <= 1;
        o++;
    }
    return o;
}
static void fl_push(size_t o, size_t idx) {
    buddy_next[idx] = buddy_heads[o];
    buddy_heads[o] = (int32_t)idx;
}
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 36) - [internal] fl_push

```
static void fl_push(size_t o, size_t idx) {
    buddy_next[idx] = buddy_heads[o];
    buddy_heads[o] = (int32_t)idx;
}
static size_t fl_pop(size_t o) {
    int32_t i = buddy_heads[o];
    if (i < 0) return (size_t)-1;
    buddy_heads[o] = buddy_next[i];
    return (size_t)i;
}
static void fl_remove(size_t o, size_t idx) {
    int32_t *p = &buddy_heads[o];
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 40) - [internal] fl_pop

```
static size_t fl_pop(size_t o) {
    int32_t i = buddy_heads[o];
    if (i < 0) return (size_t)-1;
    buddy_heads[o] = buddy_next[i];
    return (size_t)i;
}
static void fl_remove(size_t o, size_t idx) {
    int32_t *p = &buddy_heads[o];
    while (*p >= 0) {
        if ((size_t)*p == idx) {
            *p = buddy_next[*p];
            return;
        }
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 46) - [internal] fl_remove

```
static void fl_remove(size_t o, size_t idx) {
    int32_t *p = &buddy_heads[o];
    while (*p >= 0) {
        if ((size_t)*p == idx) {
            *p = buddy_next[*p];
            return;
        }
        p = &buddy_next[*p];
    }
}

bool buddy_init(uint64_t base, size_t size, buddy_map_cb map_cb, buddy_unmap_cb unmap_cb) {
    if (buddy_initialized) return true;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 49) - [public] if

```
    if ((size_t)*p == idx) {
        *p = buddy_next[*p];
        return;
    }
    p = &buddy_next[*p];
}

bool buddy_init(uint64_t base, size_t size, buddy_map_cb map_cb, buddy_unmap_cb unmap_cb) {
    if (buddy_initialized) return true;
    if (map_cb == NULL || unmap_cb == NULL) return false;
    if (size == 0 || (size & (size - 1)) != 0) return false;
    if (size < ARCH_PAGE_SIZE) return false;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 56) - [public] buddy_init

```
bool buddy_init(uint64_t base, size_t size, buddy_map_cb map_cb, buddy_unmap_cb unmap_cb) {
    if (buddy_initialized) return true;
    if (map_cb == NULL || unmap_cb == NULL) return false;
    if (size == 0 || (size & (size - 1)) != 0) return false;
    if (size < ARCH_PAGE_SIZE) return false;
    if (!arch_is_page_aligned(base)) return false;
    if ((base & (size - 1)) != 0) return false;
    buddy_base = base;
    buddy_pages = size / ARCH_PAGE_SIZE;
    buddy_max_order = 0;
    while (order_size(buddy_max_order) < size) buddy_max_order++;
    size_t
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 95) - [public] buddy_ready

```
bool buddy_ready(void) { return buddy_initialized; }

void *buddy_alloc(size_t size) {
    if (!buddy_initialized || size == 0) return NULL;
    size_t o = size_to_order(size);
    if (order_size(o) < size || o > buddy_max_order) return NULL;
    size_t cur = 0;
    while (buddy_heads[cur] < 0 && cur < buddy_max_order) cur++;
    if (buddy_heads[cur] < 0) return NULL;
    size_t idx = fl_pop(cur);
    while (cur > 0) {
        cur--;
        size_t half_pages = (size_t)1 << cur, buddy_idx = idx + half_pages;
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 109) - [public] fl_push

```
    fl_push(cur, buddy_idx);
}
buddy_order[idx] = (uint8_t)o;
buddy_state[idx] = BUDDY_STATE_USED;
size_t bytes = order_size(o);
if (!buddy_map(va_of(idx), bytes)) {
    buddy_state[idx] = BUDDY_STATE_FREE;
    fl_push(o, idx);
    return NULL;
}
buddy_used_bytes += bytes;
buddy_free_bytes -= bytes;
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 116) - [public] fl_push

```
    fl_push(o, idx);
    return NULL;
}
buddy_used_bytes += bytes;
buddy_free_bytes -= bytes;
return (void *) (uintptr_t) va_of(idx);
}
```

```
void *buddy_alloc_aligned(size_t size, size_t align) {
    if (!buddy_initialized || size == 0) return NULL;
    if (!size_is_pow2(align)) return NULL;
    if (align <= ARCH_PAGE_SIZE) return buddy_alloc(size);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 133) - [public] buddy_alloc

```
return buddy_alloc(p);
}
```

```
void buddy_free(void *ptr) {
    if (!buddy_initialized || ptr == NULL) return;
    uint64_t va = (uint64_t)(uintptr_t)ptr;
    if (va < buddy_base || va >= buddy_base + (uint64_t)buddy_pages * ARCH_PAGE_SIZE) return;
    size_t idx = page_of(va);
    if (buddy_state[idx] != BUDDY_STATE_USED) return;
    size_t o = buddy_order[idx], bytes = order_size(o);
    buddy_unmap(va, bytes);
    buddy_state[idx] = BUDDY_STATE_FREE;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 136) - [public] buddy_free

```
void buddy_free(void *ptr) {
    if (!buddy_initialized || ptr == NULL) return;
    uint64_t va = (uint64_t)(uintptr_t)ptr;
    if (va < buddy_base || va >= buddy_base + (uint64_t)buddy_pages * ARCH_PAGE_SIZE) return;
    size_t idx = page_of(va);
    if (buddy_state[idx] != BUDDY_STATE_USED) return;
    size_t o = buddy_order[idx], bytes = order_size(o);
    buddy_unmap(va, bytes);
    buddy_state[idx] = BUDDY_STATE_FREE;
    buddy_order[idx] = (uint8_t)o;
    size_t cur = o;
    while (cur < buddy_max_order) {
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 153) - [public] fl_remove

```
■ fl_remove(cur, buddy_idx);
■ buddy_state[buddy_idx] = BUDDY_STATE_TAIL;
■ if (buddy_idx < idx) idx = buddy_idx;
■ cur++;
■ buddy_order[idx] = (uint8_t)cur;
■ }
■ fl_push(cur, idx);
■ buddy_used_bytes -= bytes;
■ buddy_free_bytes += bytes;
}
size_t buddy_block_size(void *ptr) {
■ if (!buddy_initialized || ptr == NULL) return 0;
  // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 163) - [public] buddy_block_size

```
size_t buddy_block_size(void *ptr) {
■ if (!buddy_initialized || ptr == NULL) return 0;
■ uint64_t va = (uint64_t)(uintptr_t)ptr;
■ if (va < buddy_base || va >= buddy_base + (uint64_t)buddy_pages * ARCH_PAGE_SIZE) return 0;
■ size_t idx = page_of(va);
■ if (buddy_state[idx] != BUDDY_STATE_USED) return 0;
■ return (size_t)order_size(buddy_order[idx]);
}
size_t buddy_stat_used_bytes(void) { return buddy_used_bytes; }
size_t buddy_stat_free_bytes(void) { return buddy_free_bytes; }
void buddy_dump(void) {
■ if (!buddy_initialized) {
  // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 171) - [public] buddy_stat_used_bytes

```
size_t buddy_stat_used_bytes(void) { return buddy_used_bytes; }
size_t buddy_stat_free_bytes(void) { return buddy_free_bytes; }
void buddy_dump(void) {
■ if (!buddy_initialized) {
  kprintf("BUDDY: not initialized\n");
  return;
■ }
■ kprintf("BUDDY:\n");
■ kprintf(" base: 0x%llx\n", (unsigned long long)buddy_base);
■ kprintf(" max order: %llu\n", (unsigned long long)buddy_max_order);
■ kprintf(" used: %llu KiB\n", (unsigned long long)(buddy_used_bytes >> 10));
■ kprintf(" free: %llu KiB\n", (unsigned long long)(buddy_free_bytes >> 10));
  // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 172) - [public] buddy_stat_free_bytes

```
size_t buddy_stat_free_bytes(void) { return buddy_free_bytes; }
void buddy_dump(void) {
■ if (!buddy_initialized) {
  kprintf("BUDDY: not initialized\n");
  return;
■ }
■ kprintf("BUDDY:\n");
■ kprintf(" base: 0x%llx\n", (unsigned long long)buddy_base);
■ kprintf(" max order: %llu\n", (unsigned long long)buddy_max_order);
■ kprintf(" used: %llu KiB\n", (unsigned long long)(buddy_used_bytes >> 10));
■ kprintf(" free: %llu KiB\n", (unsigned long long)(buddy_free_bytes >> 10));
}
```

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 173) - [public] buddy_dump

```
void buddy_dump(void) {
■ if (!buddy_initialized) {
  kprintf("BUDDY: not initialized\n");
  return;
■ }
■ kprintf("BUDDY:\n");
■ kprintf(" base: 0x%llx\n", (unsigned long long)buddy_base);
■ kprintf(" max order: %llu\n", (unsigned long long)buddy_max_order);
■ kprintf(" used: %llu KiB\n", (unsigned long long)(buddy_used_bytes >> 10));
■ kprintf(" free: %llu KiB\n", (unsigned long long)(buddy_free_bytes >> 10));
}
```

kernel_workspace/kernel/src/mm/alloc/heap/buddy.c (line 175) - [public] kprintf

```
■ kprintf("BUDDY: not initialized\n");
■ return;
■}
■kprintf("BUDDY:\n");
■kprintf("  base: 0x%llx\n", (unsigned long long)buddy_base);
■kprintf("  max order: %llu\n", (unsigned long long)buddy_max_order);
■kprintf("  used: %llu KiB\n", (unsigned long long)(buddy_used_bytes >> 10));
■kprintf("  free: %llu KiB\n", (unsigned long long)(buddy_free_bytes >> 10));
■}
```

kernel_workspace/kernel/src/mm/alloc/heap/heap.c (line 32) - [public] if

```
■ if (!pmm_alloc_frame(&phys)) {
■     for (size_t j = 0; j < i; j++) {
■         uint64_t v = virt + j * ARCH_PAGE_SIZE;
■         uint64_t p = mapping_translate(MAPPING_KERNEL, v);

■         mapping_unmap_range(MAPPING_KERNEL, v, ARCH_PAGE_SIZE);

■         if (p != UINT64_MAX) {
■             pmm_free_frame(p);
■         }
■     }

    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/heap/heap.c (line 37) - [public] mapping_unmap_range

```
■         mapping_unmap_range(MAPPING_KERNEL, v, ARCH_PAGE_SIZE);

■         if (p != UINT64_MAX) {
■             pmm_free_frame(p);
■         }
■     }

    return false;
■ }

■ if (!mapping_map_range(MAPPING_KERNEL,
■                       virt + i * ARCH_PAGE_SIZE,
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/heap/heap.c (line 39) - [public] if

```
■         if (p != UINT64_MAX) {
■             pmm_free_frame(p);
■         }
■     }

    return false;
■ }

■ if (!mapping_map_range(MAPPING_KERNEL,
■                       virt + i * ARCH_PAGE_SIZE,
■                       phys,
■                       ARCH_PAGE_SIZE,
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/heap/heap.c (line 40) - [public] pmm_free_frame

```
■         pmm_free_frame(p);
■     }
■ }

    return false;
■ }

■ if (!mapping_map_range(MAPPING_KERNEL,
■                       virt + i * ARCH_PAGE_SIZE,
■                       phys,
■                       ARCH_PAGE_SIZE,
■                       PTE_PRESENT | PTE_WRITABLE | PTE_NX)) {
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 52) - [public] pmm_free_frame

```
■      pmm_free_frame(phys);

■      for (size_t j = 0; j < i; j++) {
■          uint64_t v = virt + j * ARCH_PAGE_SIZE;
■          uint64_t p = mapping_translate(MAPPING_KERNEL, v);

■          mapping_unmap_range(MAPPING_KERNEL, v, ARCH_PAGE_SIZE);

■          if (p != UINT64_MAX) {
■              pmm_free_frame(p);
■          }
■      }
      // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 58) - [public] mapping_unmap_range

```
■      mapping_unmap_range(MAPPING_KERNEL, v, ARCH_PAGE_SIZE);

■      if (p != UINT64_MAX) {
■          pmm_free_frame(p);
■      }

■      return false;
■  }
■}

■return true;
      // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 60) - [public] if

```
■      if (p != UINT64_MAX) {
■          pmm_free_frame(p);
■      }

■      return false;
■  }
■}

■return true;
}

      // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 61) - [public] pmm_free_frame

```
■          pmm_free_frame(p);
■      }

■      return false;
■  }
■}

■return true;
}

static void heap_unmap_cb(uint64_t virt, size_t size)
      // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/heap/heap.c (line 80) - [public] if

```
■ if (p != UINT64_MAX) {
■     mapping_unmap_range(MAPPING_KERNEL, v, ARCH_PAGE_SIZE);
■     pmm_free_frame(p);
■ }
■}
}
```

```
static void heap_memset(void *dst, uint8_t value, size_t n)
{
■uint8_t *p = (uint8_t *)dst;

■for (size_t i = 0; i < n; i++) {
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/heap/heap.c (line 81) - [public] mapping_unmap_range

```
■     mapping_unmap_range(MAPPING_KERNEL, v, ARCH_PAGE_SIZE);
■     pmm_free_frame(p);
■ }
■}
}
```

```
static void heap_memset(void *dst, uint8_t value, size_t n)
{
■uint8_t *p = (uint8_t *)dst;

■for (size_t i = 0; i < n; i++) {
■    p[i] = value;
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/heap/heap.c (line 82) - [public] pmm_free_frame

```
■     pmm_free_frame(p);
■ }
■}
}
```

```
static void heap_memset(void *dst, uint8_t value, size_t n)
{
■uint8_t *p = (uint8_t *)dst;

■for (size_t i = 0; i < n; i++) {
■    p[i] = value;
■}

    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/heap/heap.c (line 147) - [public] slab_alloc

```
■     return slab_alloc(size);
■}
#endif
```

```
■return buddy_alloc(size);
}
```

```
void *heap_alloc_aligned(size_t size, size_t align)
{
■if (!heap_initialized || size == 0) {
■    return NULL;
    // ... (truncated, see source file for more)
```


kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 151) - [public] buddy_alloc

```
    return buddy_alloc(size);
}

void *heap_alloc_aligned(size_t size, size_t align)
{
    if (!heap_initialized || size == 0) {
        return NULL;
    }

    if (!size_is_pow2(align)) {
        return NULL;
        // ... (truncated, see source file for more)
    }
}
```

kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 167) - [public] slab_alloc

```
    return slab_alloc(size);
}
#endif

return buddy_alloc_aligned(size, align);
}

void *heap_zalloc(size_t size)
{
    void *ptr = heap_alloc(size);

    if (ptr == NULL) {
        // ... (truncated, see source file for more)
    }
}
```

kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 171) - [public] buddy_alloc_aligned

```
return buddy_alloc_aligned(size, align);
}

void *heap_zalloc(size_t size)
{
    void *ptr = heap_alloc(size);

    if (ptr == NULL) {
        return NULL;
    }

    heap_memset(ptr, 0, size);
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 197) - [public] slab_free

```
    slab_free(ptr);
    return;
}
#endif

if (heap_in_buddy(va)) {
    buddy_free(ptr);
}

size_t heap_usable_size(void *ptr)
{
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 203) - [public] buddy_free

```
    buddy_free(ptr);
}

size_t heap_usable_size(void *ptr)
{
    if (!heap_initialized || ptr == NULL) {
        return 0;
    }

    uint64_t va = (uint64_t)(uintptr_t)ptr;

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 217) - [public] slab_usable_size

```
    return slab_usable_size(ptr);
}
#endif

if (heap_in_buddy(va)) {
    return buddy_block_size(ptr);
}

return 0;
}

void heap_dump(void)
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 222) - [public] buddy_block_size

```
    return buddy_block_size(ptr);
}

return 0;
}

void heap_dump(void)
{
    if (!heap_initialized) {
        return;
    }

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 238) - [public] slab_dump

```
    slab_dump();
}
#endif
}
```

kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 9) - [public] kprintf

```
extern void kprintf(const char *fmt, ...);

#define SLAB_CACHE_COUNT 8

static const size_t slab_classes[SLAB_CACHE_COUNT] = {
    16, 32, 64, 128, 256, 512, 1024, 2048
};

#define SLAB_MAX_OBJECTS_PER_SLAB 256
#define SLAB_BITMAP_WORDS (SLAB_MAX_OBJECTS_PER_SLAB / 32)

typedef struct slab_desc {
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 92) - [public] if

```
■   if ((uint32_t)*p == page) {
■       *p = slab_descs[*p].next_partial;
■       return;
■   }

■   p = &slab_descs[*p].next_partial;
■ }
}

static bool slab_grow(uint32_t cache_id)
{
■ slab_cache_t *c = &slab_caches[cache_id];
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 116) - [public] bitmap_free

```
■   bitmap_free(&slab_page_bitmap, page);
■   return false;
■ }

■ if (!mapping_map_range(MAPPING_KERNEL,
■                         va,
■                         phys,
■                         ARCH_PAGE_SIZE,
■                         PTE_PRESENT | PTE_WRITABLE | PTE_NX)) {
■     pmm_free_frame(phys);
■     bitmap_free(&slab_page_bitmap, page);
■     return false;
■ }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 125) - [public] pmm_free_frame

```
■   pmm_free_frame(phys);
■   bitmap_free(&slab_page_bitmap, page);
■   return false;
■ }

■ slab_desc_t *d = &slab_descs[page];

■ d->cache_id = cache_id;
■ d->state = 1;
■ d->free_count = c->objects_per_slab;
■ d->free_head = 0;

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 126) - [public] bitmap_free

```
■   bitmap_free(&slab_page_bitmap, page);
■   return false;
■ }

■ slab_desc_t *d = &slab_descs[page];

■ d->cache_id = cache_id;
■ d->state = 1;
■ d->free_count = c->objects_per_slab;
■ d->free_head = 0;

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 207) - [public] slab_bitmap_clear_all

```
■    slab_bitmap_clear_all(&slab_descs[i]);
■}

■for (size_t i = 0; i < SLAB_CACHE_COUNT; i++) {
■    slab_caches[i].object_size = slab_classes[i];
■    slab_caches[i].objects_per_slab =
■        (uint32_t)(ARCH_PAGE_SIZE / slab_classes[i]);
■    slab_caches[i].partial_head = -1;
■    slab_caches[i].total_slabs = 0;

■    if (slab_caches[i].objects_per_slab > SLAB_MAX_OBJECTS_PER_SLAB) {
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 218) - [public] if

```
■    if (slab_caches[i].objects_per_slab > SLAB_MAX_OBJECTS_PER_SLAB) {
■        return false;
■    }
■}
```

```
■slab_initialized = true;
```

```
■return true;
■}
```

```
bool slab_ready(void)
{
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 255) - [public] if

```
■    if (!slab_grow(cache_id)) {
■        slab_unlock();
■        return NULL;
■    }
■}
```

```
■uint32_t page = (uint32_t)c->partial_head;
■slab_desc_t *d = &slab_descs[page];
```

```
■uint32_t idx = d->free_head;
```

```
■uint64_t va = slab_page_va(page);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 256) - [public] slab_unlock

```
■    slab_unlock();
■    return NULL;
■    }
■}
```

```
■uint32_t page = (uint32_t)c->partial_head;
■slab_desc_t *d = &slab_descs[page];
```

```
■uint32_t idx = d->free_head;
```

```
■uint64_t va = slab_page_va(page);
■uint8_t *obj =
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/heap/slab.c (line 328) - [public] slab_unlock

```
■    slab_unlock();
■    return;
■}

■slab_bit_clear(d, idx);

■bool was_full = (d->free_count == 0);

■*(uint32_t *) (uintptr_t)va = d->free_head;
■d->free_head = idx;
■d->free_count++;

    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/heap/slab.c (line 346) - [public] partial_remove

```
■    partial_remove(c, page);

■    uint64_t phys = mapping_translate(MAPPING_KERNEL, slab_va);

■    mapping_unmap_range(MAPPING_KERNEL, slab_va, ARCH_PAGE_SIZE);

■    if (phys != UINT64_MAX) {
■        pmm_free_frame(phys);
■    }

■    bitmap_free(&slab_page_bitmap, page);

    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/heap/slab.c (line 350) - [public] mapping_unmap_range

```
■    mapping_unmap_range(MAPPING_KERNEL, slab_va, ARCH_PAGE_SIZE);

■    if (phys != UINT64_MAX) {
■        pmm_free_frame(phys);
■    }

■    bitmap_free(&slab_page_bitmap, page);

■    d->state = 0;
■    c->total_slabs--;
■}

    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/heap/slab.c (line 352) - [public] if

```
■    if (phys != UINT64_MAX) {
■        pmm_free_frame(phys);
■    }

■    bitmap_free(&slab_page_bitmap, page);

■    d->state = 0;
■    c->total_slabs--;
■}

■slab_unlock();
}

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 353) - [public] pmm_free_frame

```
■    pmm_free_frame(phys);
■    }

■    bitmap_free(&slab_page_bitmap, page);

■    d->state = 0;
■    c->total_slabs--;
■}

■slab_unlock();
}

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 356) - [public] bitmap_free

```
■    bitmap_free(&slab_page_bitmap, page);

■    d->state = 0;
■    c->total_slabs--;
■}

■slab_unlock();
}

size_t slab_usable_size(void *ptr)
{
■if (!slab_initialized || ptr == NULL) {
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 400) - [public] if

```
■    if (d->state != 1) {
■        continue;
■    }

■    const slab_cache_t *c = &slab_caches[d->cache_id];

■    used += (size_t)(c->objects_per_slab - d->free_count) *
■        c->object_size;
■}

■return used;
}

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 424) - [public] if

```
■    if (d->state != 1) {
■        continue;
■    }

■    const slab_cache_t *c = &slab_caches[d->cache_id];

■    free_bytes += (size_t)d->free_count * c->object_size;
■}

■return free_bytes;
}

    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/heap/slab.c (line 439) - [public] kprintf

```
■ kprintf("SLAB: not initialized\n");
■ return;
■}

■kprintf("SLAB:\n");

■for (size_t i = 0; i < SLAB_CACHE_COUNT; i++) {
■ kprintf(" size %4u: slabs %u\n",
■         (unsigned int)slab_caches[i].object_size,
■         (unsigned int)slab_caches[i].total_slabs);
■}

// ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/physical/bitmap.c (line 65) - [public]
bitmap_words_for_bits

```
■return bitmap_words_for_bits(bit_count) * sizeof(uint64_t);
}

static void bitmap_mark_tail_used(bitmap_t *bm)
{
■if (bm->word_count == 0) {
■ return;
■}

■size_t rem = bm->bit_count & 63;

■if (rem == 0) {
// ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/physical/bitmap.c (line 132) - [public]
bitmap_mark_tail_used

```
■ bitmap_mark_tail_used(bm);
■}
}

void bitmap_set(bitmap_t *bm, size_t bit)
{
■if (bm == NULL || bm->bits == NULL || bit >= bm->bit_count) {
■ return;
■}

■size_t word = bit / 64;
// ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/physical/bitmap.c (line 298) - [public] if

```
■ if (bm->bits[i] != 0) {
■ return false;
■ }
■}

■if ((bm->bits[last_word] & last_mask) != 0) {
■ return false;
■}

■return true;
}

// ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/physical/bitmap.c (line 329) - [public] if

```
■   if (idx < bm->bit_count) {
■       return idx;
■   }

■   return BITMAP_INVALID;
■}

■for (size_t i = word + 1; i < bm->word_count; i++) {
■   w = bm->bits[i];

■   if (~w != 0) {
■       size_t found_bit = (size_t)__builtin_ctzll(~w);
■       // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/physical/bitmap.c (line 339) - [public] if

```
■   if (~w != 0) {
■       size_t found_bit = (size_t)__builtin_ctzll(~w);
■       size_t idx = i * 64 + found_bit;

■       if (idx < bm->bit_count) {
■           return idx;
■       }

■       return BITMAP_INVALID;
■   }
■}

// ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/physical/bitmap.c (line 343) - [public] if

```
■       if (idx < bm->bit_count) {
■           return idx;
■       }

■       return BITMAP_INVALID;
■   }
■}

■return BITMAP_INVALID;
■}

static size_t find_next_used_bit(const bitmap_t *bm, size_t start, size_t limit)
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/physical/bitmap.c (line 377) - [public] if

```
■   if (i == start_word && start_bit != 0) {
■       w &= ~lower_bits_mask(start_bit);
■   }

■   if (i == end_word) {
■       w &= mask_to_bit(end_bit);
■   }

■   if (w != 0) {
■       size_t found_bit = (size_t)__builtin_ctzll(w);
■       size_t idx = i * 64 + found_bit;

■       // ... (truncated, see source file for more)
```


kernel_workspace/kernel/src/mm/alloc/physical/bitmap.c (line 381) - [public] if

```
■ if (i == end_word) {
■     w &= mask_to_bit(end_bit);
■ }

■ if (w != 0) {
■     size_t found_bit = (size_t)__builtin_ctzll(w);
■     size_t idx = i * 64 + found_bit;

■     if (idx >= start && idx < limit) {
■         return idx;
■     }
■ }
// ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/physical/bitmap.c (line 385) - [public] if

```
■ if (w != 0) {
■     size_t found_bit = (size_t)__builtin_ctzll(w);
■     size_t idx = i * 64 + found_bit;

■     if (idx >= start && idx < limit) {
■         return idx;
■     }
■ }

■return BITMAP_INVALID;
}
// ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/physical/bitmap.c (line 389) - [public] if

```
■ if (idx >= start && idx < limit) {
■     return idx;
■ }
■ }

■return BITMAP_INVALID;
}

size_t bitmap_alloc(bitmap_t *bm)
{
■if (bm == NULL || bm->bits == NULL || bm->bit_count == 0) {
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/physical/bitmap.c (line 474) - [public] bitmap_alloc

```
■ return bitmap_alloc(bm);
■}

■size_t candidate = 0;

■while (candidate <= bm->bit_count - count) {
■    candidate = align_up_size(candidate, align);

■    if (candidate == BITMAP_INVALID ||
■        candidate > bm->bit_count - count) {
■        break;
■    }
■ }
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.c (line 489) - [public] if

```
■   if (free_bit == BITMAP_INVALID) {
■       break;
■   }

■   candidate = align_up_size(free_bit, align);

■   if (candidate == BITMAP_INVALID ||
■       candidate > bm->bit_count - count) {
■       break;
■   }

■   if (bitmap_test_range_free(bm, candidate, count)) {
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.c (line 500) - [public] if

```
■   if (bitmap_test_range_free(bm, candidate, count)) {
■       bitmap_set_range(bm, candidate, count);
■       bm->alloc_hint = candidate / 64;
■       return candidate;
■   }

■   size_t used = find_next_used_bit(bm, candidate, candidate + count);

■   if (used == BITMAP_INVALID) {
■       candidate++;
■   } else {
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.c (line 501) - [public] bitmap_set_range

```
■       bitmap_set_range(bm, candidate, count);
■       bm->alloc_hint = candidate / 64;
■       return candidate;
■   }

■   size_t used = find_next_used_bit(bm, candidate, candidate + count);

■   if (used == BITMAP_INVALID) {
■       candidate++;
■   } else {
■       candidate = used + 1;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.c (line 509) - [public] if

```
■   if (used == BITMAP_INVALID) {
■       candidate++;
■   } else {
■       candidate = used + 1;
■   }
■}

■return BITMAP_INVALID;
}

void bitmap_free(bitmap_t *bm, size_t bit)
{
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/frame.c (line 64) - [public] arch_phys_to_virt

```
■return arch_phys_to_virt(frame);
}

bool frame_is_valid(frame_t frame)
{
■if (!frame_initialized) {
■    return false;
■}

■if (!arch_is_page_aligned(frame)) {
■    return false;
■}

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/frame.c (line 159) - [public] if

```
■    if (r->type != ARCH_MEM_TYPE_USABLE) {
■        continue;
■    }

■    if (!arch_is_page_aligned(r->base) ||
■        !arch_is_page_aligned(r->len)) {
■        continue;
■    }

■    size_t pfn_start = (size_t)(r->base / ARCH_PAGE_SIZE);
■    size_t frame_count = (size_t)(r->len / ARCH_PAGE_SIZE);

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/frame.c (line 171) - [public] if

```
■    if (frame_count == 0) {
■        continue;
■    }

■    bitmap_clear_range(&frame_bitmap, pfn_start, frame_count);

■    if (allocated_frames >= frame_count) {
■        allocated_frames -= frame_count;
■    } else {
■        allocated_frames = 0;
■    }

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/frame.c (line 175) - [public] bitmap_clear_range

```
■    bitmap_clear_range(&frame_bitmap, pfn_start, frame_count);

■    if (allocated_frames >= frame_count) {
■        allocated_frames -= frame_count;
■    } else {
■        allocated_frames = 0;
■    }

■    dma32_account_free(pfn_start, frame_count);
■}

■return true;
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/physical/frame.c (line 177) - [public] if

```
■    if (allocated_frames >= frame_count) {
■        allocated_frames -= frame_count;
■    } else {
■        allocated_frames = 0;
■    }

■    dma32_account_free(pfn_start, frame_count);
■}

■return true;
}

    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/physical/frame.c (line 183) - [public] dma32_account_free

```
■    dma32_account_free(pfn_start, frame_count);
■}

■return true;
}

bool frame_ready(void)
{
■return frame_initialized;
}

bool frame_alloc(frame_t *out)
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/physical/frame.c (line 230) - [public] frame_free

```
■    frame_free(frame);
■    return false;
■}

■*out = frame;

■return true;
}

bool frame_alloc_contiguous(size_t count,
■                        size_t align_frames,
■                        frame_t *out)
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/physical/frame.c (line 345) - [public] volatile

```
■__asm__ volatile("" ::: "memory");

■return true;
}

bool frame_alloc_below(size_t count,
■                        size_t align_frames,
■                        uint64_t max_phys,
■                        frame_t *out)
{
■if (!frame_initialized || count == 0 || out == NULL) {
■    return false;
■    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/frame.c (line 377) - [public] if

```
■   if (aligned > limit - count) {
■       break;
■   }

■   if (bitmap_test_range_free(&frame_bitmap, aligned, count)) {
■       bitmap_set_range(&frame_bitmap, aligned, count);

■       allocated_frames += count;
■       dma32_account_alloc(aligned, count);

■       *out = frame_from_pfn(aligned);

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/frame.c (line 381) - [public] if

```
■   if (bitmap_test_range_free(&frame_bitmap, aligned, count)) {
■       bitmap_set_range(&frame_bitmap, aligned, count);

■       allocated_frames += count;
■       dma32_account_alloc(aligned, count);

■       *out = frame_from_pfn(aligned);

■       return true;
■   }

■   pfn = aligned + 1;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/frame.c (line 382) - [public] bitmap_set_range

```
■       bitmap_set_range(&frame_bitmap, aligned, count);

■       allocated_frames += count;
■       dma32_account_alloc(aligned, count);

■       *out = frame_from_pfn(aligned);

■       return true;
■   }

■   pfn = aligned + 1;
■}

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/frame.c (line 385) - [public]
dma32_account_alloc

```
■       dma32_account_alloc(aligned, count);

■       *out = frame_from_pfn(aligned);

■       return true;
■   }

■   pfn = aligned + 1;
■}

■return false;
}

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 6) - [public] kprintf

```
extern void kprintf(const char *fmt, ...);

static void pmm_panic(const char *msg) __attribute__((noreturn));
static void pmm_panic(const char *msg)
{
    kprintf("pmm panic: %s\n", msg);

    for (;;) {
        __asm__ volatile("cli; hlt");
    }

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 8) - [public] pmm_panic

```
static void pmm_panic(const char *msg) __attribute__((noreturn));
static void pmm_panic(const char *msg)
{
    kprintf("pmm panic: %s\n", msg);

    for (;;) {
        __asm__ volatile("cli; hlt");
    }

    static bool pmm_initialized = false;

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 31) - [public] pmm_panic

```
    pmm_panic("pmm_unlock without lock");
}

#define PMM_SIZE_MAX ((size_t)-1)

static size_t pmm_bytes_to_frames(size_t bytes)
{
    if (bytes == 0) {
        return 0;
    }

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 80) - [public] if

```
    if (!frame_zero(phys)) {
        return false;
    }

    return true;
}

bool pmm_init(void)
{
    pmm_lock();

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 93) - [public] pmm_unlock

```
■ pmm_unlock();
■ return true;
■}

■if (!arch_memory_ready()) {
■ pmm_unlock();
■ kprintf("pmm_init: arch memory not initialized\n");
■ return false;
■}

■if (!frame_init_from_memory()) {
■ pmm_unlock();
■ // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 98) - [public] pmm_unlock

```
■ pmm_unlock();
■ kprintf("pmm_init: arch memory not initialized\n");
■ return false;
■}

■if (!frame_init_from_memory()) {
■ pmm_unlock();
■ kprintf("pmm_init: frame allocator initialization failed\n");
■ return false;
■}

■pmm_initialized = true;
■ // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 99) - [public] kprintf

```
■ kprintf("pmm_init: arch memory not initialized\n");
■ return false;
■}

■if (!frame_init_from_memory()) {
■ pmm_unlock();
■ kprintf("pmm_init: frame allocator initialization failed\n");
■ return false;
■}

■pmm_initialized = true;

■ // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 104) - [public] pmm_unlock

```
■ pmm_unlock();
■ kprintf("pmm_init: frame allocator initialization failed\n");
■ return false;
■}

■pmm_initialized = true;

■pmm_unlock();

■return true;
}

■ // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 105) - [public] kprintf

```
■ kprintf("pmm_init: frame allocator initialization failed\n");  
■ return false;  
■ }
```

```
■ pmm_initialized = true;
```

```
■ pmm_unlock();
```

```
■ return true;  
}
```

```
bool pmm_ready(void)  
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 220) - [public] pmm_alloc_frames

```
■ return pmm_alloc_frames(frames, out_phys);  
}
```

```
bool pmm_alloc_zero_bytes(size_t bytes, uint64_t *out_phys)  
{  
■ if (!pmm_initialized || out_phys == NULL || bytes == 0) {  
■     return false;  
■ }
```

```
■ size_t frames = pmm_bytes_to_frames(bytes);
```

```
■ if (frames == 0) {  
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 236) - [public] pmm_alloc_zero_frame

```
■ return pmm_alloc_zero_frame(out_phys);  
■ }
```

```
■ pmm_lock();
```

```
■ frame_t frame = FRAME_INVALID;  
■ bool ok = frame_alloc_contiguous(frames, 1, &frame);
```

```
■ pmm_unlock();
```

```
■ if (!ok) {  
■     return false;  
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 251) - [public] pmm_free_frames

```
■ pmm_free_frames(frame, frames);  
■ return false;  
■ }
```

```
■ *out_phys = frame;
```

```
■ return true;  
}
```

```
bool pmm_alloc_contiguous_bytes(size_t bytes,  
■                               size_t align_bytes,  
■                               uint64_t *out_phys)  
    // ... (truncated, see source file for more)
```


kernel_workspace/kernel/src/mm/alloc/physical/pmm.c (line 275) - [public]

pmm_alloc_frames_aligned

```
■return pmm_alloc_frames_aligned(frames, align_frames, out_phys);
}
```

```
bool pmm_free_frame(uint64_t phys)
```

```
{
■if (!pmm_initialized) {
■    return false;
■}
}
```

```
■pmm_lock();
```

```
■bool ok = frame_free(phys);
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/physical/pmm.c (line 328) - [public] pmm_free_frames

```
■return pmm_free_frames(phys, frames);
}
```

```
size_t pmm_stat_total_frames(void)
```

```
{
■if (!pmm_initialized) {
■    return 0;
■}
}
```

```
■return frame_total();
}
```

```
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/physical/pmm.c (line 337) - [public] frame_total

```
■return frame_total();
}
```

```
size_t pmm_stat_free_frames(void)
```

```
{
■if (!pmm_initialized) {
■    return 0;
■}
}
```

```
■return frame_free_count();
}
```

```
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/physical/pmm.c (line 346) - [public] frame_free_count

```
■return frame_free_count();
}
```

```
size_t pmm_stat_allocated_frames(void)
```

```
{
■if (!pmm_initialized) {
■    return 0;
■}
}
```

```
■return frame_allocated();
}
```

```
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/physical/pmm.c (line 355) - [public] frame_allocated

```
■return frame_allocated();
}

uint64_t pmm_stat_total_bytes(void)
{
■if (!pmm_initialized) {
■    return 0;
■}

■uint64_t frames = frame_total();

■if (frames > UINT64_MAX / ARCH_PAGE_SIZE) {
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/physical/pmm.c (line 394) - [public]

frame_zone_dma32_total

```
■return frame_zone_dma32_total();
}

size_t pmm_stat_dma32_free_frames(void)
{
■if (!pmm_initialized) {
■    return 0;
■}

■return frame_zone_dma32_free();
}

// ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/physical/pmm.c (line 403) - [public]

frame_zone_dma32_free

```
■return frame_zone_dma32_free();
}

size_t pmm_stat_normal_total_frames(void)
{
■if (!pmm_initialized) {
■    return 0;
■}

■return frame_zone_normal_total();
}

// ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/physical/pmm.c (line 412) - [public]

frame_zone_normal_total

```
■return frame_zone_normal_total();
}

size_t pmm_stat_normal_free_frames(void)
{
■if (!pmm_initialized) {
■    return 0;
■}

■return frame_zone_normal_free();
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 421) - [public]

frame_zone_normal_free

```
    return frame_zone_normal_free();
}

bool pmm_alloc_frames_below(size_t count,
    size_t align_frames,
    uint64_t max_phys,
    uint64_t *out_phys)
{
    if (!pmm_initialized || out_phys == NULL || count == 0) {
        return false;
    }

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 452) - [public] kprintf

```
    kprintf("PMM: not initialized\n");
    return;
}

uint64_t total_frames = pmm_stat_total_frames();
uint64_t free_frames = pmm_stat_free_frames();
uint64_t allocated_frames = pmm_stat_allocated_frames();

uint64_t total_mb = pmm_stat_total_bytes() >> 20;
uint64_t free_mb = pmm_stat_free_bytes() >> 20;

kprintf("PMM:\n");
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 490) - [public] if

```
    if (!pmm_alloc_zero_frame(&frames[i])) {
        kprintf("pmm_self_test: single frame alloc failed\n");
        return false;
    }

    if (!arch_is_page_aligned(frames[i])) {
        kprintf("pmm_self_test: frame not aligned\n");
        return false;
    }

    volatile uint64_t *p =
        (volatile uint64_t *)arch_phys_to_virt(frames[i]);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 491) - [public] kprintf

```
    kprintf("pmm_self_test: single frame alloc failed\n");
    return false;
}

    if (!arch_is_page_aligned(frames[i])) {
        kprintf("pmm_self_test: frame not aligned\n");
        return false;
    }

    volatile uint64_t *p =
        (volatile uint64_t *)arch_phys_to_virt(frames[i]);

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 495) - [public] if

```
■   if (!arch_is_page_aligned(frames[i])) {
■       kprintf("pmm_self_test: frame not aligned\n");
■       return false;
■   }

■   volatile uint64_t *p =
■       (volatile uint64_t *)arch_phys_to_virt(frames[i]);

■   for (size_t j = 0; j < (ARCH_PAGE_SIZE / sizeof(uint64_t)); j++) {
■       if (p[j] != 0) {
■           kprintf("pmm_self_test: zero frame not zero\n");
■           return false;
■       }
■   }
■   // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 496) - [public] kprintf

```
■       kprintf("pmm_self_test: frame not aligned\n");
■       return false;
■   }

■   volatile uint64_t *p =
■       (volatile uint64_t *)arch_phys_to_virt(frames[i]);

■   for (size_t j = 0; j < (ARCH_PAGE_SIZE / sizeof(uint64_t)); j++) {
■       if (p[j] != 0) {
■           kprintf("pmm_self_test: zero frame not zero\n");
■           return false;
■       }
■   }
■   // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 504) - [public] if

```
■       if (p[j] != 0) {
■           kprintf("pmm_self_test: zero frame not zero\n");
■           return false;
■       }
■   }
■ }

■ for (size_t i = 0; i < 8; i++) {
■     if (!pmm_free_frame(frames[i])) {
■         kprintf("pmm_self_test: single frame free failed\n");
■         return false;
■     }
■ }
■ // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 505) - [public] kprintf

```
■       kprintf("pmm_self_test: zero frame not zero\n");
■       return false;
■   }
■ }

■ for (size_t i = 0; i < 8; i++) {
■     if (!pmm_free_frame(frames[i])) {
■         kprintf("pmm_self_test: single frame free failed\n");
■         return false;
■     }
■ }
■ // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 512) - [public] if

```
■    if (!pmm_free_frame(frames[i])) {  
■        kprintf("pmm_self_test: single frame free failed\n");  
■        return false;  
■    }  
■}
```

```
■uint64_t contiguous = PMM_INVALID_FRAME;
```

```
■if (!pmm_alloc_frames(16, &contiguous)) {  
■    kprintf("pmm_self_test: contiguous alloc failed\n");  
■    return false;  
■    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 513) - [public] kprintf

```
■        kprintf("pmm_self_test: single frame free failed\n");  
■        return false;  
■    }  
■}
```

```
■uint64_t contiguous = PMM_INVALID_FRAME;
```

```
■if (!pmm_alloc_frames(16, &contiguous)) {  
■    kprintf("pmm_self_test: contiguous alloc failed\n");  
■    return false;  
■}  
■    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 522) - [public] kprintf

```
■    kprintf("pmm_self_test: contiguous alloc failed\n");  
■    return false;  
■}
```

```
■if (!arch_is_page_aligned(contiguous)) {  
■    kprintf("pmm_self_test: contiguous not aligned\n");  
■    return false;  
■}
```

```
■if (!pmm_free_frames(contiguous, 16)) {  
■    kprintf("pmm_self_test: contiguous free failed\n");  
■    return false;  
■    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 527) - [public] kprintf

```
■    kprintf("pmm_self_test: contiguous not aligned\n");  
■    return false;  
■}
```

```
■if (!pmm_free_frames(contiguous, 16)) {  
■    kprintf("pmm_self_test: contiguous free failed\n");  
■    return false;  
■}
```

```
■uint64_t aligned2m = PMM_INVALID_FRAME;
```

```
■    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 532) - [public] kprintf

```
■ kprintf("pmm_self_test: contiguous free failed\n");
■ return false;
■ }
```

```
■ uint64_t aligned2m = PMM_INVALID_FRAME;
```

```
■ if (!pmm_alloc_frames_aligned(512, 512, &aligned2m)) {
■ kprintf("pmm_self_test: 2M aligned alloc failed\n");
■ return false;
■ }
```

```
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 540) - [public] kprintf

```
■ kprintf("pmm_self_test: 2M aligned alloc failed\n");
■ return false;
■ }
```

```
■ uint64_t align2m_mask = (512ULL * ARCH_PAGE_SIZE) - 1;
```

```
■ if ((aligned2m & align2m_mask) != 0) {
■ kprintf("pmm_self_test: 2M alignment failed\n");
■ return false;
■ }
```

```
■ if (!pmm_free_frames(aligned2m, 512)) {
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 547) - [public] kprintf

```
■ kprintf("pmm_self_test: 2M alignment failed\n");
■ return false;
■ }
```

```
■ if (!pmm_free_frames(aligned2m, 512)) {
■ kprintf("pmm_self_test: 2M aligned free failed\n");
■ return false;
■ }
```

```
■ uint64_t bytes_alloc = PMM_INVALID_FRAME;
```

```
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 552) - [public] kprintf

```
■ kprintf("pmm_self_test: 2M aligned free failed\n");
■ return false;
■ }
```

```
■ uint64_t bytes_alloc = PMM_INVALID_FRAME;
```

```
■ if (!pmm_alloc_zero_bytes(10000, &bytes_alloc)) {
■ kprintf("pmm_self_test: bytes alloc failed\n");
■ return false;
■ }
```

```
// ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 560) - [public] kprintf
    ■ kprintf("pmm_self_test: bytes alloc failed\n");
    ■ return false;
    ■}

    ■size_t bytes_frames = pmm_bytes_to_frames(10000);

    ■volatile uint64_t *p =
    ■ (volatile uint64_t *)arch_phys_to_virt(bytes_alloc);

    ■size_t words = bytes_frames * (ARCH_PAGE_SIZE / sizeof(uint64_t));

    ■for (size_t i = 0; i < words; i++) {
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 572) - [public] if
    ■ if (p[i] != 0) {
    ■     kprintf("pmm_self_test: bytes alloc not zero\n");
    ■     return false;
    ■ }
    ■}

    ■if (!pmm_free_bytes(bytes_alloc, 10000)) {
    ■ kprintf("pmm_self_test: bytes free failed\n");
    ■ return false;
    ■}

    ■kprintf("pmm_self_test: OK\n");
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 573) - [public] kprintf
    ■ kprintf("pmm_self_test: bytes alloc not zero\n");
    ■ return false;
    ■ }
    ■}

    ■if (!pmm_free_bytes(bytes_alloc, 10000)) {
    ■ kprintf("pmm_self_test: bytes free failed\n");
    ■ return false;
    ■}

    ■kprintf("pmm_self_test: OK\n");

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 579) - [public] kprintf
    ■ kprintf("pmm_self_test: bytes free failed\n");
    ■ return false;
    ■}

    ■kprintf("pmm_self_test: OK\n");

    ■return true;
    }

#endif

```

kernel_workspace/kernel/src/mm/alloc/special/contiguous.c (line 9) - [public]

size_bytes_to_pages_checked

```
■return size_bytes_to_pages_checked(bytes, ARCH_PAGE_SIZE, out_frames);
}

bool contig_alloc(size_t bytes,
■          size_t align,
■          uint64_t *out_phys,
■          void **out_virt)
{
■if (bytes == 0) {
■    return false;
■}

    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/special/dma.c (line 10) - [public]

size_bytes_to_pages_checked

```
■return size_bytes_to_pages_checked(bytes, ARCH_PAGE_SIZE, out_frames);
}

bool dma_alloc_coherent(size_t bytes,
■          uint64_t zone_max,
■          uint64_t *out_phys,
■          void **out_virt)
{
■if (bytes == 0 || out_phys == NULL || out_virt == NULL) {
■    return false;
■}

    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/special/dma.c (line 47) - [public] pmm_free_frames

```
■    pmm_free_frames(phys, frames);
■    return false;
■}

■*out_phys = phys;
■*out_virt = (void *) (uintptr_t) virt;

■return true;
}

void dma_free_coherent(uint64_t phys, void *virt, size_t bytes)
{
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/special/dma.c (line 76) - [public] vmm_unmap_device

```
■    vmm_unmap_device((uint64_t) (uintptr_t) virt, len);
■}

■pmm_free_frames(phys, frames);
}

static void clflush_range(void *addr, size_t len)
{
■uintptr_t start = (uintptr_t) addr & ~(uintptr_t) 63;
■uintptr_t end = (uintptr_t) addr + len;

■for (uintptr_t a = start; a < end; a += 64) {
    // ... (truncated, see source file for more)
```


kernel_workspace/kernel/src/mm/alloc/special/dma.c (line 88) - [public] volatile

```
■ __asm__ volatile("clflush (%0)" :: "r"(a) : "memory");
■}

■__asm__ volatile("" ::: "memory");
}

void dma_sync_for_device(void *virt, size_t len)
{
■if (virt == NULL || len == 0) {
■    return;
■}

    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/special/dma.c (line 91) - [public] volatile

```
■__asm__ volatile("" ::: "memory");
}

void dma_sync_for_device(void *virt, size_t len)
{
■if (virt == NULL || len == 0) {
■    return;
■}

■clflush_range(virt, len);
}

    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/virtual/mapping.c (line 27) - [public] mapping_init

```
■    mapping_init();
■}

■return mapping_kernel_pml4;
}

static bool page_range_ok(uint64_t addr, size_t len)
{
■uint64_t start, end;

■if (!range_from_addr_len(addr, (uint64_t)len, ARCH_PAGE_SIZE,
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/virtual/mapping.c (line 70) - [public] if

```
■    if (!paging_map_page_in(pml4, v, p, pte_flags)) {
■        for (size_t j = 0; j < mapped; j++) {
■            paging_unmap_page_in(pml4,
■                                virt + (uint64_t)j * ARCH_PAGE_SIZE);
■        }

■        return false;
■    }

■    mapped++;
■}

    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/virtual/mapping.c (line 102) - [public]
paging_unmap_page_in

```
■    paging_unmap_page_in(pml4, virt + (uint64_t)i * ARCH_PAGE_SIZE);
■}

■return true;
}

bool mapping_protect_range(mapping_space_t space,
■    uint64_t virt,
■    size_t len,
■    uint64_t pte_flags)
{
■if (len == 0) {
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/virtual/mapping.c (line 128) - [public] if

```
■    if (!paging_is_mapped_in(pml4, v)) {
■        return false;
■    }
■}

■for (size_t i = 0; i < pages; i++) {
■    uint64_t v = virt + (uint64_t)i * ARCH_PAGE_SIZE;

■    if (!paging_set_flags_in(pml4, v, pte_flags)) {

■        return false;
■    }
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/virtual/mapping.c (line 136) - [public] if

```
■    if (!paging_set_flags_in(pml4, v, pte_flags)) {

■        return false;
■    }
■}

■return true;
}

uint64_t mapping_translate(mapping_space_t space, uint64_t virt)
{
■return paging_translate_in(resolve_space(space), virt);
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/virtual/mapping.c (line 147) - [public]
paging_translate_in

```
■return paging_translate_in(resolve_space(space), virt);
}

bool mapping_is_mapped(mapping_space_t space, uint64_t virt)
{
■return paging_is_mapped_in(resolve_space(space), virt);
}

static void copy_page_by_phys(uint64_t dst_phys, uint64_t src_phys)
{
■uint64_t *dst = (uint64_t *)arch_phys_to_virt(dst_phys);
■const uint64_t *src = (const uint64_t *)arch_phys_to_virt(src_phys);
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/virtual/mapping.c (line 152) - [public] paging_is_mapped_in

```
    return paging_is_mapped_in(resolve_space(space), virt);
}

static void copy_page_by_phys(uint64_t dst_phys, uint64_t src_phys)
{
    uint64_t *dst = (uint64_t *)arch_phys_to_virt(dst_phys);
    const uint64_t *src = (const uint64_t *)arch_phys_to_virt(src_phys);

    for (size_t i = 0; i < ARCH_PAGE_SIZE / sizeof(uint64_t); i++) {
        dst[i] = src[i];
    }

    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/virtual/mapping.c (line 164) - [public] volatile

```
    __asm__ volatile("" ::: "memory");
}

bool mapping_copy_range(mapping_space_t dst,
    uint64_t dst_virt,
    mapping_space_t src,
    uint64_t src_virt,
    size_t len)
{
    if (len == 0) {
        return true;
    }

    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/virtual/mapping.c (line 199) - [public] copy_page_by_phys

```
    copy_page_by_phys(dp, sp);
}

return true;
}
```

kernel_workspace/kernel/src/mm/alloc/virtual/page.c (line 5) - [public] kprintf

```
extern void kprintf(const char *fmt, ...);

static page_t *page_descs = NULL;
static size_t page_desc_count = 0;
static bool page_initialized = false;

static size_t page_lock_depth = 0;
static uint64_t page_lock_flags = 0;

static void page_lock(void)
{
    uint64_t flags;
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/virtual/page.c (line 78) - [public] kprintf

```
    kprintf("page_init: pmm not ready\n");
    return false;
}

const arch_mem_info_t *info = arch_memory_get();

if (info == NULL) {
    kprintf("page_init: no arch mem info\n");
    return false;
}

uint64_t max_pfn = info->max_usable_address / ARCH_PAGE_SIZE;
// ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/virtual/page.c (line 85) - [public] kprintf

```
■ kprintf("page_init: no arch mem info\n");
■ return false;
■}

■uint64_t max_pfn = info->max_usable_address / ARCH_PAGE_SIZE;

■page_desc_count = (size_t)(max_pfn + 1);

■size_t bytes = page_desc_count * sizeof(page_t);

■kprintf("page_init: max_usable=0x%llx max_addr=0x%llx desc_count=%llu bytes=%llu\n",
■ (unsigned long long)info->max_usable_address,
■ // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/virtual/page.c (line 141) - [public] page_unlock

```
■ page_unlock();
■ return NULL;
■}

■page_t *p = &page_descs[page_phys_to_pfn(phys)];

■p->refcount = 1;
■p->type = type;
■p->flags = 0;
■p->reserved = 0;

■page_unlock();
■ // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/virtual/page.c (line 168) - [public] page_unlock

```
■ page_unlock();
■ return NULL;
■}

■page_t *p = &page_descs[page_phys_to_pfn(phys)];

■p->refcount = 1;
■p->type = type;
■p->flags = 0;
■p->reserved = 0;

■page_unlock();
■ // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/virtual/page.c (line 205) - [public] page_unlock

```
■ page_unlock();
■ return false;
■}

■p->refcount = 1;
■p->type = type;

■page_unlock();

■return true;
}

■ // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/virtual/page.c (line 226) - [public] page_unlock

```
■ page_unlock();  
■ return false;  
■ }
```

```
■ page->refcount++;
```

```
■ page_unlock();
```

```
■ return true;  
}
```

```
bool page_put(page_t *page)  
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/virtual/page.c (line 246) - [public] page_unlock

```
■ page_unlock();  
■ return false;  
■ }
```

```
■ page->refcount--;
```

```
■ if (page->refcount != 0) {  
■     page_unlock();  
■     return false;  
■ }
```

```
■ size_t idx = (size_t)(page - page_descs);  
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/virtual/page.c (line 253) - [public] page_unlock

```
■ page_unlock();  
■ return false;  
■ }
```

```
■ size_t idx = (size_t)(page - page_descs);  
■ uint64_t phys = (uint64_t)idx * ARCH_PAGE_SIZE;
```

```
■ page->type = PAGE_TYPE_FREE;  
■ page->flags = 0;
```

```
■ page_unlock();
```

```
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/virtual/page.c (line 289) - [public] arch_phys_to_virt

```
■ return arch_phys_to_virt(phys);  
}
```

```
size_t page_pfn(const page_t *page)
```

```
{  
■ if (!page_ptr_valid(page)) {  
■     return (size_t)-1;  
■ }
```

```
■ return (size_t)(page - page_descs);  
}
```

```
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/page.c (line 331) - [public] page_from_phys

```
■return page_from_phys(arch_virt_to_phys(va));  
}
```

```
uint32_t page_refcount(const page_t *page)  
{  
■if (!page_ptr_valid(page)) {  
■    return 0;  
■}  
  
■return page->refcount;  
}
```

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/alloc/virtual/page.c (line 379) - [public] if

```
■    if (page_descs[i].refcount != 0) {  
■        count++;  
■    }  
■}
```

```
■return count;  
}
```

```
void page_dump(void)  
{  
■if (!page_initialized) {  
■    kprintf("PAGE: not initialized\n");  
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/page.c (line 390) - [public] kprintf

```
■    kprintf("PAGE: not initialized\n");  
■    return;  
■}  
  
■kprintf("PAGE:\n");  
■kprintf("  total pages: %llu\n", (unsigned long long)page_stat_total());  
■kprintf("  referenced: %llu\n", (unsigned long long)page_stat_referenced());  
}
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 7) - [public] kprintf

```
extern void kprintf(const char *fmt, ...);  
  
#define VMM_VMALLOC_BASE 0xFFFFFA0000000000ULL  
#define VMM_VMALLOC_SIZE (128ULL << 30)  
#define VMM_SIZE_MAX ((size_t)-1)  
  
static bitmap_t vmm_bitmap;  
static bool vmm_initialized = false;  
static size_t vmm_total_pages = 0;  
static size_t vmm_allocated_pages = 0;  
  
static size_t vmm_lock_depth = 0;  
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 201) - [public] vmm_unlock

```
■    vmm_unlock();  
■    return false;  
■}  
  
■size_t mapped = 0;  
■bool ok = true;  
  
■for (size_t i = 0; i < pages; i++) {  
■    uint64_t phys = 0;  
  
■    bool got = (flags & VMM_FLAG_ZERO)  
■        ? pmm_alloc_zero_frame(&phys)  
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 213) - [public] pmm_alloc_frame

```
■          : pmm_alloc_frame(&phys);

■  if (!got) {
■      ok = false;
■      break;
■  }

■  uint64_t virt = vmm_page_virt(start + i);

■  if (!paging_map_page(virt, phys, pte)) {
■      pmm_free_frame(phys);
■      ok = false;
■      // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 215) - [public] if

```
■  if (!got) {
■      ok = false;
■      break;
■  }

■  uint64_t virt = vmm_page_virt(start + i);

■  if (!paging_map_page(virt, phys, pte)) {
■      pmm_free_frame(phys);
■      ok = false;
■      break;
■  }
■  // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 222) - [public] if

```
■  if (!paging_map_page(virt, phys, pte)) {
■      pmm_free_frame(phys);
■      ok = false;
■      break;
■  }

■  mapped++;
■}

■if (!ok) {
■  for (size_t i = 0; i < mapped; i++) {
■      uint64_t virt = vmm_page_virt(start + i);
■      // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 223) - [public] pmm_free_frame

```
■      pmm_free_frame(phys);
■      ok = false;
■      break;
■  }

■  mapped++;
■}

■if (!ok) {
■  for (size_t i = 0; i < mapped; i++) {
■      uint64_t virt = vmm_page_virt(start + i);

■      // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 235) - [public] if

```
■      if (paging_is_mapped(virt)) {  
■          uint64_t phys = paging_translate(virt);  
■          paging_unmap_page(virt);  
■          pmm_free_frame(phys);  
■      }  
■  
■      bitmap_free_range(&vmm_bitmap, start, pages);  
■      vmm_unlock();  
■      return false;  
■}  
  
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 237) - [public] paging_unmap_page

```
■          paging_unmap_page(virt);  
■          pmm_free_frame(phys);  
■      }  
■  
■      bitmap_free_range(&vmm_bitmap, start, pages);  
■      vmm_unlock();  
■      return false;  
■}  
  
vmm_allocated_pages += pages;  
  
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 238) - [public] pmm_free_frame

```
■          pmm_free_frame(phys);  
■      }  
■  
■      bitmap_free_range(&vmm_bitmap, start, pages);  
■      vmm_unlock();  
■      return false;  
■}  
  
vmm_allocated_pages += pages;  
  
vmm_unlock();  
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 242) - [public] bitmap_free_range

```
■      bitmap_free_range(&vmm_bitmap, start, pages);  
■      vmm_unlock();  
■      return false;  
■}  
  
vmm_allocated_pages += pages;  
  
vmm_unlock();  
  
*out_virt = vmm_page_virt(start);  
  
return true;  
    // ... (truncated, see source file for more)
```


kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 243) - [public] vmm_unlock

```
■    vmm_unlock();
■    return false;
■}

■vmm_allocated_pages += pages;

■vmm_unlock();

■*out_virt = vmm_page_virt(start);

■return true;
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 258) - [public] vmm_alloc_aligned

```
■return vmm_alloc_aligned(bytes, ARCH_PAGE_SIZE, flags, out_virt);
}

bool vmm_map_device(uint64_t phys, size_t len, uint64_t *out_virt)
{
■if (!vmm_initialized || out_virt == NULL || len == 0) {
■    return false;
■}

■if (!arch_is_page_aligned(phys) || !arch_is_page_aligned(len)) {
■    return false;
■}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 284) - [public] vmm_unlock

```
■    vmm_unlock();
■    return false;
■}

■size_t mapped = 0;
■bool ok = true;

■for (size_t i = 0; i < pages; i++) {
■    uint64_t virt = vmm_page_virt(start + i);
■    uint64_t frame_phys = phys + (uint64_t)i * ARCH_PAGE_SIZE;

■    if (!paging_map_page(virt, frame_phys, pte)) {
■        // ... (truncated, see source file for more)
■    }
■}
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 295) - [public] if

```
■    if (!paging_map_page(virt, frame_phys, pte)) {
■        ok = false;
■        break;
■    }

■    mapped++;
■}

■if (!ok) {
■    for (size_t i = 0; i < mapped; i++) {
■        paging_unmap_page(vmm_page_virt(start + i));
■    }
■    // ... (truncated, see source file for more)
■}
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 305) - [public] paging_unmap_page

```
■    paging_unmap_page(vmm_page_virt(start + i));  
■    }  
  
■    bitmap_free_range(&vmm_bitmap, start, pages);  
■    vmm_unlock();  
■    return false;  
■}  
  
■vmm_allocated_pages += pages;  
  
■vmm_unlock();  
  
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 308) - [public] bitmap_free_range

```
■    bitmap_free_range(&vmm_bitmap, start, pages);  
■    vmm_unlock();  
■    return false;  
■}  
  
■vmm_allocated_pages += pages;  
  
■vmm_unlock();  
  
■*out_virt = vmm_page_virt(start);  
  
■return true;  
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 309) - [public] vmm_unlock

```
■    vmm_unlock();  
■    return false;  
■}  
  
■vmm_allocated_pages += pages;  
  
■vmm_unlock();  
  
■*out_virt = vmm_page_virt(start);  
  
■return true;  
}  
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 349) - [public] if

```
■    if (paging_is_mapped(v)) {  
■        uint64_t phys = paging_translate(v);  
■        paging_unmap_page(v);  
■        pmm_free_frame(phys);  
■    }  
■}  
  
■bitmap_free_range(&vmm_bitmap, start, pages);  
  
■if (vmm_allocated_pages >= pages) {  
■    vmm_allocated_pages -= pages;  
■} else {  
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 351) - [public] paging_unmap_page

```
■    paging_unmap_page(v);
■    pmm_free_frame(phys);
■    }
■}

■bitmap_free_range(&vmm_bitmap, start, pages);

■if (vmm_allocated_pages >= pages) {
■    vmm_allocated_pages -= pages;
■} else {
■    vmm_allocated_pages = 0;
■}

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 352) - [public] pmm_free_frame

```
■    pmm_free_frame(phys);
■    }
■}

■bitmap_free_range(&vmm_bitmap, start, pages);

■if (vmm_allocated_pages >= pages) {
■    vmm_allocated_pages -= pages;
■} else {
■    vmm_allocated_pages = 0;
■}

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 395) - [public] if

```
■    if (paging_is_mapped(v)) {
■        paging_unmap_page(v);
■    }
■}

■bitmap_free_range(&vmm_bitmap, start, pages);

■if (vmm_allocated_pages >= pages) {
■    vmm_allocated_pages -= pages;
■} else {
■    vmm_allocated_pages = 0;
■}

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 396) - [public] paging_unmap_page

```
■    paging_unmap_page(v);
■    }
■}

■bitmap_free_range(&vmm_bitmap, start, pages);

■if (vmm_allocated_pages >= pages) {
■    vmm_allocated_pages -= pages;
■} else {
■    vmm_allocated_pages = 0;
■}

    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/alloc/virtual/vmm.c (line 419) - [public] paging_translate

```
■return paging_translate(virt);
}

size_t vmm_stat_total_pages(void)
{
■if (!vmm_initialized) {
■    return 0;
■}

■return vmm_total_pages;
}
```

// ... (truncated, see source file for more)

kernel_workspace/kernel/src/mm/alloc/virtual/vmm.c (line 466) - [public] kprintf

```
■    kprintf("VMM: not initialized\n");
■    return;
■}

■kprintf("VMM:\n");
■kprintf("  base: 0x%llx\n", (unsigned long long)VMM_VMALLOC_BASE);
■kprintf("  total pages: %llu\n", (unsigned long long)vmm_stat_total_pages());
■kprintf("  free pages: %llu\n", (unsigned long long)vmm_stat_free_pages());
■kprintf("  allocated pages: %llu\n", (unsigned long long)vmm_stat_allocated_pages());
■kprintf("  total: %llu MiB\n", (unsigned long long)(vmm_stat_total_bytes() >> 20));
■kprintf("  free: %llu MiB\n", (unsigned long long)(vmm_stat_free_bytes() >> 20));
}
```

kernel_workspace/kernel/src/mm/arch/aarch64/paging.c (line 115) - [public]

arch_memory_boot_alloc

```
■    arch_memory_boot_alloc(4096, 4096, &phys);

■    uint64_t *t = tbl(phys);

■    for (int i = 0; i < 512; i++) {
■        t[i] = 0;
■    }
■}

■*entry = phys | HW_TABLE;

■return phys;
// ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/aarch64/paging.c (line 139) - [public] tlb_flush_page_addr

```
■    tlb_flush_page_addr(virt);
■    return true;
■}

■uint64_t l2 = ensure_table(&l1t[IDX(virt, L1_SHIFT)], false);
■uint64_t *l2t = tbl(l2);

■uint64_t l3 = ensure_table(&l2t[IDX(virt, L2_SHIFT)], false);
■uint64_t *l3t = tbl(l3);

■l3t[IDX(virt, L3_SHIFT)] = (phys & PAGING_ADDR_MASK) | hw_bits(flags, 3);

// ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 172) - [public] tlb_flush_page_addr
    ■ tlb_flush_page_addr(virt);
    ■ return true;
    ■}

    ■uint64_t *l2t = tbl(l1t[IDX(virt, L1_SHIFT)] & PAGING_ADDR_MASK);

    ■if (!(l2t[IDX(virt, L2_SHIFT)] & HW_VALID)) {
    ■    return false;
    ■}

    ■if ((l2t[IDX(virt, L2_SHIFT)] & 3) == HW_BLOCK) {
    ■    l2t[IDX(virt, L2_SHIFT)] = 0;
    ■    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 184) - [public] tlb_flush_page_addr
    ■ tlb_flush_page_addr(virt);
    ■ return true;
    ■}

    ■uint64_t *l3t = tbl(l2t[IDX(virt, L2_SHIFT)] & PAGING_ADDR_MASK);

    ■if (!(l3t[IDX(virt, L3_SHIFT)] & HW_VALID)) {
    ■    return false;
    ■}

    ■l3t[IDX(virt, L3_SHIFT)] = 0;
    ■tlb_flush_page_addr(virt);
    ■    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 216) - [public] return
    ■    return (e1 & 0x0000FFFFFFFFE0000ULL) | (virt & 0x1FFFFFFULL);
    ■}

    ■uint64_t *l2t = tbl(e1 & PAGING_ADDR_MASK);
    ■uint64_t e2 = l2t[IDX(virt, L2_SHIFT)];

    ■if (!(e2 & HW_VALID)) {
    ■    return UINT64_MAX;
    ■}

    ■if ((e2 & 3) == HW_BLOCK) {
    ■    return (e2 & 0x0000FFFFFFFFF0000ULL) | (virt & 0xFFFFFULL);
    ■    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 227) - [public] return
    ■    return (e2 & 0x0000FFFFFFFFF0000ULL) | (virt & 0xFFFFFULL);
    ■}

    ■uint64_t *l3t = tbl(e2 & PAGING_ADDR_MASK);
    ■uint64_t e3 = l3t[IDX(virt, L3_SHIFT)];

    ■if (!(e3 & HW_VALID)) {
    ■    return UINT64_MAX;
    ■}

    ■return (e3 & PAGING_ADDR_MASK) | (virt & 0xFFFFFULL);
    }
    // ... (truncated, see source file for more)

```

```

kernel_workspace/kernel/src/mm/arch/aarch64/paging.c (line 273) - [public] volatile
    ■__asm__ volatile("mrs %0, ttbr1_el1" : "=r"(cur));

    ■uint64_t *boot_l0 = tbl_early(cur & PAGING_ADDR_MASK);

    ■kernel_l0 = alloc_table_early();

    ■uint64_t *l0 = tbl_early(kernel_l0);

    ■for (int i = 0; i < 512; i++) {
    ■    l0[i] = boot_l0[i];
    ■}

    // ... (truncated, see source file for more)

kernel_workspace/kernel/src/mm/arch/aarch64/paging.c (line 297) - [public] if
    ■    if (idx == 0 && pa >= 0x200000 * 512) {
    ■        break;
    ■    }

    ■    l2t[idx] = pa | HW_BLOCK | HW_AF | HW_SH | ATTR_WB | HW_PXN;
    ■}

    ■uint64_t mair = 0x00ULL | (0xFFULL << 8) | (0x44ULL << 16);

    ■__asm__ volatile("msr mair_el1, %0" :: "r"(mair));

    ■uint64_t tcr = (16ULL << 0)
    // ... (truncated, see source file for more)

kernel_workspace/kernel/src/mm/arch/aarch64/paging.c (line 306) - [public] volatile
    ■__asm__ volatile("msr mair_el1, %0" :: "r"(mair));

    ■uint64_t tcr = (16ULL << 0)
    ■    | (16ULL << 16)
    ■    | (0b000ULL << 14)
    ■    | (0b100ULL << 30)
    ■    | (0b010ULL << 8)
    ■    | (0b010ULL << 10)
    ■    | (0b110ULL << 12)
    ■    | (0b010ULL << 24)
    ■    | (0b010ULL << 26)
    ■    | (0b110ULL << 28)
    // ... (truncated, see source file for more)

kernel_workspace/kernel/src/mm/arch/aarch64/paging.c (line 320) - [public] volatile
    ■__asm__ volatile("msr tcr_el1, %0" :: "r"(tcr));

    ■__asm__ volatile("msr ttbr0_el1, %0" :: "r"(kernel_l0));
    ■__asm__ volatile("msr ttbr1_el1, %0" :: "r"(kernel_l0));

    ■__asm__ volatile("isb");
    ■__asm__ volatile("tlbi vmalle1");
    ■__asm__ volatile("dsb sy");
    ■__asm__ volatile("isb");
    ■}

    bool paging_enable_nx(void)
    // ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 322) - [public] volatile

```
    __asm__ volatile("msr ttbr0_el1, %0" :: "r"(kernel_l0));
    __asm__ volatile("msr ttbr1_el1, %0" :: "r"(kernel_l0));

    __asm__ volatile("isb");
    __asm__ volatile("tlbi vmalle1");
    __asm__ volatile("dsb sy");
    __asm__ volatile("isb");
}

bool paging_enable_nx(void)
{
    return true;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 323) - [public] volatile

```
    __asm__ volatile("msr ttbr1_el1, %0" :: "r"(kernel_l0));

    __asm__ volatile("isb");
    __asm__ volatile("tlbi vmalle1");
    __asm__ volatile("dsb sy");
    __asm__ volatile("isb");
}

bool paging_enable_nx(void)
{
    return true;
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 325) - [public] volatile

```
    __asm__ volatile("isb");
    __asm__ volatile("tlbi vmalle1");
    __asm__ volatile("dsb sy");
    __asm__ volatile("isb");
}

bool paging_enable_nx(void)
{
    return true;
}

uint64_t paging_read_cr3(void)
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 326) - [public] volatile

```
    __asm__ volatile("tlbi vmalle1");
    __asm__ volatile("dsb sy");
    __asm__ volatile("isb");
}

bool paging_enable_nx(void)
{
    return true;
}

uint64_t paging_read_cr3(void)
{
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 327) - [public] volatile

```
    __asm__ volatile("dsb sy");
    __asm__ volatile("isb");
}

bool paging_enable_nx(void)
{
    return true;
}

uint64_t paging_read_cr3(void)
{
    uint64_t v;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 328) - [public] volatile

```
    __asm__ volatile("isb");
}

bool paging_enable_nx(void)
{
    return true;
}

uint64_t paging_read_cr3(void)
{
    uint64_t v;

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 340) - [public] volatile

```
    __asm__ volatile("mrs %0, ttbr1_el1" : "=r"(v));

    return v & PAGING_ADDR_MASK;
}

void paging_write_cr3(uint64_t pml4_phys)
{
    __asm__ volatile("msr ttbr0_el1, %0" :: "r"(pml4_phys));
    __asm__ volatile("msr ttbr1_el1, %0" :: "r"(pml4_phys));
    __asm__ volatile("isb");

    tlb_flush_all();
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 347) - [public] volatile

```
    __asm__ volatile("msr ttbr0_el1, %0" :: "r"(pml4_phys));
    __asm__ volatile("msr ttbr1_el1, %0" :: "r"(pml4_phys));
    __asm__ volatile("isb");

    tlb_flush_all();
}
```

kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 348) - [public] volatile

```
    __asm__ volatile("msr ttbr1_el1, %0" :: "r"(pml4_phys));
    __asm__ volatile("isb");

    tlb_flush_all();
}
```

kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 349) - [public] volatile

```
    __asm__ volatile("isb");

    tlb_flush_all();
}
```


kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.c (line 18) - [public] volatile

```
    __asm__ volatile("dsb sy");
    __asm__ volatile("tlbi vmalle1");
    __asm__ volatile("dsb sy");
    __asm__ volatile("isb");
}

void tlb_flush_all_including_global(void)
{
    tlb_flush_all();
}

void tlb_flush_page_addr(uint64_t addr)
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.c (line 19) - [public] volatile

```
    __asm__ volatile("tlbi vmalle1");
    __asm__ volatile("dsb sy");
    __asm__ volatile("isb");
}

void tlb_flush_all_including_global(void)
{
    tlb_flush_all();
}

void tlb_flush_page_addr(uint64_t addr)
{
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.c (line 20) - [public] volatile

```
    __asm__ volatile("dsb sy");
    __asm__ volatile("isb");
}

void tlb_flush_all_including_global(void)
{
    tlb_flush_all();
}

void tlb_flush_page_addr(uint64_t addr)
{
    __asm__ volatile("dsb ishst");
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.c (line 21) - [public] volatile

```
    __asm__ volatile("isb");
}

void tlb_flush_all_including_global(void)
{
    tlb_flush_all();
}

void tlb_flush_page_addr(uint64_t addr)
{
    __asm__ volatile("dsb ishst");
    __asm__ volatile("tlbi vaae1is, %0" :: "r"(addr >> 12));
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.c (line 31) - [public] volatile

```
    __asm__ volatile("dsb ishst");
    __asm__ volatile("tlbi vaae1is, %0" :: "r"(addr >> 12));
    __asm__ volatile("dsb ish");
    __asm__ volatile("isb");
}

void tlb_flush_page(const void *addr)
{
    __tlb_flush_page_addr((uint64_t)(uintptr_t)addr);
}

void tlb_flush_range_addr(uint64_t addr, size_t pages)
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.c (line 32) - [public] volatile

```
    __asm__ volatile("tlbi vaae1is, %0" :: "r"(addr >> 12));
    __asm__ volatile("dsb ish");
    __asm__ volatile("isb");
}

void tlb_flush_page(const void *addr)
{
    __tlb_flush_page_addr((uint64_t)(uintptr_t)addr);
}

void tlb_flush_range_addr(uint64_t addr, size_t pages)
{
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.c (line 33) - [public] volatile

```
    __asm__ volatile("dsb ish");
    __asm__ volatile("isb");
}

void tlb_flush_page(const void *addr)
{
    __tlb_flush_page_addr((uint64_t)(uintptr_t)addr);
}

void tlb_flush_range_addr(uint64_t addr, size_t pages)
{
    if (pages > 32) {
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.c (line 34) - [public] volatile

```
    __asm__ volatile("isb");
}

void tlb_flush_page(const void *addr)
{
    __tlb_flush_page_addr((uint64_t)(uintptr_t)addr);
}

void tlb_flush_range_addr(uint64_t addr, size_t pages)
{
    if (pages > 32) {
        tlb_flush_all();
        // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/aarch64/tlb.c (line 45) - [public] tlb_flush_all

```
■ tlb_flush_all();
■ return;
■}

■for (size_t i = 0; i < pages; i++) {
■ tlb_flush_page_addr(addr + i * 4096);
■}
}

void tlb_flush_range(const void *addr, size_t pages)
{
■tlb_flush_range_addr((uint64_t)(uintptr_t)addr, pages);
// ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/aarch64/tlb.c (line 50) - [public] tlb_flush_page_addr

```
■ tlb_flush_page_addr(addr + i * 4096);
■}
}

void tlb_flush_range(const void *addr, size_t pages)
{
■tlb_flush_range_addr((uint64_t)(uintptr_t)addr, pages);
}
```

kernel_workspace/kernel/src/mm/arch/x86_64/memory.c (line 37) - [public] arch_mem_panic

```
■ arch_mem_panic("bad multiboot2 magic");
■}

■if (mb_info < 8) {
■ arch_mem_panic("bad multiboot2 info address");
■}

■static arch_raw_mem_entry_t raw[ARCH_MAX_MEM_REGIONS];
■size_t n = 0;

■uint32_t total_size = *(const uint32_t *)mb_info;

// ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/memory.c (line 41) - [public] arch_mem_panic

```
■ arch_mem_panic("bad multiboot2 info address");
■}

■static arch_raw_mem_entry_t raw[ARCH_MAX_MEM_REGIONS];
■size_t n = 0;

■uint32_t total_size = *(const uint32_t *)mb_info;

■if (total_size < 8) {
■ arch_mem_panic("multiboot2 info too small");
■}

// ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/memory.c (line 50) - [public] arch_mem_panic

```
■ arch_mem_panic("multiboot2 info too small");
■}

■uintptr_t end_addr = mb_info + total_size;
■uintptr_t tag_addr = (mb_info + 8 + 7) & ~(uintptr_t)7;

■while (tag_addr + 8 <= end_addr) {
■ const struct multiboot_tag *tag =
■ (const struct multiboot_tag *)tag_addr;

■ if (tag->size < 8) {
■ break;
■ // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 60) - [public] if

```
■   if (tag->size < 8) {
■       break;
■   }

■   if (tag->type == MULTIBOOT_TAG_TYPE_END) {
■       break;
■   }

■   if (tag->type == MULTIBOOT_TAG_TYPE_MMAP) {
■       const struct multiboot_tag_mmap *mmap =
■           (const struct multiboot_tag_mmap *)tag;

        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 64) - [public] if

```
■   if (tag->type == MULTIBOOT_TAG_TYPE_END) {
■       break;
■   }

■   if (tag->type == MULTIBOOT_TAG_TYPE_MMAP) {
■       const struct multiboot_tag_mmap *mmap =
■           (const struct multiboot_tag_mmap *)tag;

        uintptr_t entry_addr = (uintptr_t)mmap->entries;
        uintptr_t mmap_end = tag_addr + tag->size;

■       if (mmap->entry_size < sizeof(struct multiboot_mmap_entry)) {
            // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 68) - [public] if

```
■   if (tag->type == MULTIBOOT_TAG_TYPE_MMAP) {
■       const struct multiboot_tag_mmap *mmap =
■           (const struct multiboot_tag_mmap *)tag;

        uintptr_t entry_addr = (uintptr_t)mmap->entries;
        uintptr_t mmap_end = tag_addr + tag->size;

■       if (mmap->entry_size < sizeof(struct multiboot_mmap_entry)) {
■           break;
■       }

■       while (entry_addr + sizeof(struct multiboot_mmap_entry) <= mmap_end) {
            // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 75) - [public] if

```
■       if (mmap->entry_size < sizeof(struct multiboot_mmap_entry)) {
■           break;
■       }

■       while (entry_addr + sizeof(struct multiboot_mmap_entry) <= mmap_end) {
■           const struct multiboot_mmap_entry *e =
■               (const struct multiboot_mmap_entry *)entry_addr;

■           if (n < ARCH_MAX_MEM_REGIONS) {
■               raw[n].base = e->addr;
■               raw[n].len = e->len;
■               raw[n].type = e->type;
            // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 79) - [public] while

```
■ while (entry_addr + sizeof(struct multiboot_mmap_entry) <= mmap_end) {
■     const struct multiboot_mmap_entry *e =
■         (const struct multiboot_mmap_entry *)entry_addr;

■     if (n < ARCH_MAX_MEM_REGIONS) {
■         raw[n].base = e->addr;
■         raw[n].len = e->len;
■         raw[n].type = e->type;
■         n++;
■     }

■     entry_addr += mmap->entry_size;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 83) - [public] if

```
■     if (n < ARCH_MAX_MEM_REGIONS) {
■         raw[n].base = e->addr;
■         raw[n].len = e->len;
■         raw[n].type = e->type;
■         n++;
■     }

■     entry_addr += mmap->entry_size;
■ }
■ }

■ tag_addr = (tag_addr + tag->size + 7) & ~(uintptr_t)7;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 106) - [public] kprintf

```
extern void kprintf(const char *fmt, ...);

static void arch_mem_panic(const char *msg) __attribute__((noreturn));
static void arch_mem_panic(const char *msg)
{
■ kprintf("arch/x86_64/memory.c panic: %s\n", msg);

■ for (;;) {
■     __asm__ volatile("cli; hlt");
■ }
}
#define ARCH_U64_MAX        UINT64_MAX
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 108) - [public] arch_mem_panic

```
static void arch_mem_panic(const char *msg) __attribute__((noreturn));
static void arch_mem_panic(const char *msg)
{
■ kprintf("arch/x86_64/memory.c panic: %s\n", msg);

■ for (;;) {
■     __asm__ volatile("cli; hlt");
■ }
}
#define ARCH_U64_MAX        UINT64_MAX
#define ARCH_TMP_REGIONS    (ARCH_MAX_MEM_REGIONS * 3)

    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/memory.c (line 221) - [public] while

```
■ while (j > 0 && entries[j - 1].base > key.base) {
■     entries[j] = entries[j - 1];
■     j--;
■ }

■ entries[j] = key;
■ }
■ }
```

```
static void map_add(arch_mem_region_t *list,
■     size_t *count,
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/memory.c (line 244) - [public] if

```
■ if (r.base < last_end) {
■     uint64_t delta = last_end - r.base;

■     if (r.len <= delta) {
■         return;
■     }

■     r.base = last_end;
■     r.len -= delta;
■ }

■
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/memory.c (line 247) - [public] if

```
■ if (r.len <= delta) {
■     return;
■ }

■ r.base = last_end;
■ r.len -= delta;
■ }

■
■ if (safe_end(last->base, last->len) == r.base &&
■     last->type == r.type) {
■     last->len += r.len;
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/memory.c (line 264) - [public] arch_mem_panic

```
■ arch_mem_panic("too many memory regions");
■ }

■ list[(*count)++] = r;
■ }

static void map_reserve_range(uint64_t base, uint64_t len)
{
■ if (len == 0) {
■     return;
■ }

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 293) - [public] if

```
■   if (r_end <= base || r.base >= end) {
■       map_add(tmp_regions, &tcnt, ARCH_TMP_REGIONS, r);
■       continue;
■   }

■
■   if (r.base < base) {
■       arch_mem_region_t before;
■       before.base = r.base;
■       before.len = base - r.base;
■       before.type = r.type;

        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 294) - [public] map_add

```
■       map_add(tmp_regions, &tcnt, ARCH_TMP_REGIONS, r);
■       continue;
■   }

■
■   if (r.base < base) {
■       arch_mem_region_t before;
■       before.base = r.base;
■       before.len = base - r.base;
■       before.type = r.type;

■       map_add(tmp_regions, &tcnt, ARCH_TMP_REGIONS, before);
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 299) - [public] if

```
■   if (r.base < base) {
■       arch_mem_region_t before;
■       before.base = r.base;
■       before.len = base - r.base;
■       before.type = r.type;

■       map_add(tmp_regions, &tcnt, ARCH_TMP_REGIONS, before);
■   }

■
■   uint64_t ov_base = u64_max(r.base, base);
■   uint64_t ov_end = u64_min(r_end, end);
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 305) - [public] map_add

```
■       map_add(tmp_regions, &tcnt, ARCH_TMP_REGIONS, before);
■   }

■
■   uint64_t ov_base = u64_max(r.base, base);
■   uint64_t ov_end = u64_min(r_end, end);

■   if (ov_end > ov_base) {
■       arch_mem_region_t reserved;
■       reserved.base = ov_base;
■       reserved.len = ov_end - ov_base;
■       reserved.type = ARCH_MEM_TYPE_RESERVED;
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 312) - [public] if

```
■ if (ov_end > ov_base) {
■     arch_mem_region_t reserved;
■     reserved.base = ov_base;
■     reserved.len = ov_end - ov_base;
■     reserved.type = ARCH_MEM_TYPE_RESERVED;

■     map_add(tmp_regions, &tcount, ARCH_TMP_REGIONS, reserved);
■ }

■
■ if (r_end > end) {
■     arch_mem_region_t after;
■     // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 318) - [public] map_add

```
■     map_add(tmp_regions, &tcount, ARCH_TMP_REGIONS, reserved);
■ }

■
■ if (r_end > end) {
■     arch_mem_region_t after;
■     after.base = end;
■     after.len = r_end - end;
■     after.type = r.type;

■     map_add(tmp_regions, &tcount, ARCH_TMP_REGIONS, after);
■ }
■ // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 322) - [public] if

```
■ if (r_end > end) {
■     arch_mem_region_t after;
■     after.base = end;
■     after.len = r_end - end;
■     after.type = r.type;

■     map_add(tmp_regions, &tcount, ARCH_TMP_REGIONS, after);
■ }
■ }

■ if (tcount > ARCH_MAX_MEM_REGIONS) {
■     arch_mem_panic("memory reserve produced too many regions");
■     // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 328) - [public] map_add

```
■     map_add(tmp_regions, &tcount, ARCH_TMP_REGIONS, after);
■ }
■ }

■ if (tcount > ARCH_MAX_MEM_REGIONS) {
■     arch_mem_panic("memory reserve produced too many regions");
■ }

■ mem.count = tcount;

■ for (size_t i = 0; i < tcount; i++) {
■     mem.regions[i] = tmp_regions[i];
■     // ... (truncated, see source file for more)
```


kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 333) - [public] arch_mem_panic

```
■ arch_mem_panic("memory reserve produced too many regions");
■ }

■ mem.count = tcount;

■ for (size_t i = 0; i < tcount; i++) {
■     mem.regions[i] = tmp_regions[i];
■ }
}

static void map_align_usable_regions(void)
{
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 351) - [public] if

```
■ if (r.type != ARCH_MEM_TYPE_USABLE) {
■     map_add(tmp_regions, &tcount, ARCH_TMP_REGIONS, r);
■     continue;
■ }

■ uint64_t aligned_base = align_up_page_safe(r.base);
■ uint64_t aligned_end = align_down_page(r_end);

■ if (aligned_end <= aligned_base) {
■     arch_mem_region_t tiny;
■     tiny.base = r.base;
■     tiny.len = r.len;
■     // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 352) - [public] map_add

```
■     map_add(tmp_regions, &tcount, ARCH_TMP_REGIONS, r);
■     continue;
■ }

■ uint64_t aligned_base = align_up_page_safe(r.base);
■ uint64_t aligned_end = align_down_page(r_end);

■ if (aligned_end <= aligned_base) {
■     arch_mem_region_t tiny;
■     tiny.base = r.base;
■     tiny.len = r.len;
■     tiny.type = ARCH_MEM_TYPE_RESERVED;
■     // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 359) - [public] if

```
■ if (aligned_end <= aligned_base) {
■     arch_mem_region_t tiny;
■     tiny.base = r.base;
■     tiny.len = r.len;
■     tiny.type = ARCH_MEM_TYPE_RESERVED;

■     map_add(tmp_regions, &tcount, ARCH_TMP_REGIONS, tiny);
■     continue;
■ }

■ if (r.base < aligned_base) {
■     // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 365) - [public] map_add

```
■    map_add(tmp_regions, &tcnt, ARCH_TMP_REGIONS, tiny);
■    continue;
■ }

■
■    if (r.base < aligned_base) {
■        arch_mem_region_t before;
■        before.base = r.base;
■        before.len = aligned_base - r.base;
■        before.type = ARCH_MEM_TYPE_RESERVED;

■        map_add(tmp_regions, &tcnt, ARCH_TMP_REGIONS, before);
■        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 370) - [public] if

```
■    if (r.base < aligned_base) {
■        arch_mem_region_t before;
■        before.base = r.base;
■        before.len = aligned_base - r.base;
■        before.type = ARCH_MEM_TYPE_RESERVED;

■        map_add(tmp_regions, &tcnt, ARCH_TMP_REGIONS, before);
■    }

■
■    {
■        arch_mem_region_t usable;
■        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 376) - [public] map_add

```
■    map_add(tmp_regions, &tcnt, ARCH_TMP_REGIONS, before);
■ }

■
■ {
■     arch_mem_region_t usable;
■     usable.base = aligned_base;
■     usable.len = aligned_end - aligned_base;
■     usable.type = ARCH_MEM_TYPE_USABLE;

■     map_add(tmp_regions, &tcnt, ARCH_TMP_REGIONS, usable);
■ }
■ // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 386) - [public] map_add

```
■    map_add(tmp_regions, &tcnt, ARCH_TMP_REGIONS, usable);
■ }

■
■    if (aligned_end < r_end) {
■        arch_mem_region_t after;
■        after.base = aligned_end;
■        after.len = r_end - aligned_end;
■        after.type = ARCH_MEM_TYPE_RESERVED;

■        map_add(tmp_regions, &tcnt, ARCH_TMP_REGIONS, after);
■    }
■    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/memory.c (line 390) - [public] if

```
■    if (aligned_end < r_end) {
■        arch_mem_region_t after;
■        after.base = aligned_end;
■        after.len = r_end - aligned_end;
■        after.type = ARCH_MEM_TYPE_RESERVED;

■        map_add(tmp_regions, &tcount, ARCH_TMP_REGIONS, after);
■    }
■}

■if (tcount > ARCH_MAX_MEM_REGIONS) {
■    arch_mem_panic("memory alignment produced too many regions");
■    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/memory.c (line 396) - [public] map_add

```
■        map_add(tmp_regions, &tcount, ARCH_TMP_REGIONS, after);
■    }
■}

■if (tcount > ARCH_MAX_MEM_REGIONS) {
■    arch_mem_panic("memory alignment produced too many regions");
■}

■mem.count = tcount;

■for (size_t i = 0; i < tcount; i++) {
■    mem.regions[i] = tmp_regions[i];
■    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/memory.c (line 401) - [public] arch_mem_panic

```
■    arch_mem_panic("memory alignment produced too many regions");
■}

■mem.count = tcount;

■for (size_t i = 0; i < tcount; i++) {
■    mem.regions[i] = tmp_regions[i];
■}
}

static void recalc_stats(void)
{
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/memory.c (line 421) - [public] if

```
■    if (end > max_address) {
■        max_address = end;
■    }

■    if (r->type == ARCH_MEM_TYPE_USABLE) {
■        if (total_usable > ARCH_U64_MAX - r->len) {
■            total_usable = ARCH_U64_MAX;
■        } else {
■            total_usable += r->len;
■        }
■    }

■    if (end > max_usable_address) {
■        // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/memory.c (line 425) - [public] if

```
■ if (r->type == ARCH_MEM_TYPE_USABLE) {
■     if (total_usable > ARCH_U64_MAX - r->len) {
■         total_usable = ARCH_U64_MAX;
■     } else {
■         total_usable += r->len;
■     }

■     if (end > max_usable_address) {
■         max_usable_address = end;
■     }
■ }
■ // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/memory.c (line 426) - [public] if

```
■ if (total_usable > ARCH_U64_MAX - r->len) {
■     total_usable = ARCH_U64_MAX;
■ } else {
■     total_usable += r->len;
■ }

■ if (end > max_usable_address) {
■     max_usable_address = end;
■ }
■ }
■ // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/memory.c (line 432) - [public] if

```
■ if (end > max_usable_address) {
■     max_usable_address = end;
■ }
■ }
■ mem.total_usable = total_usable;
■ mem.max_address = max_address;
■ mem.max_usable_address = max_usable_address;
■ mem.direct_map_base = ARCH_DIRECT_MAP_BASE;
■ }

■ // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/memory.c (line 452) - [public] arch_mem_panic

```
■ arch_mem_panic("arch_memory_init() called twice");
■ }

■ if (entries == NULL || count == 0) {
■     arch_mem_panic("empty memory map");
■ }

■ if (count > ARCH_MAX_MEM_REGIONS) {
■     arch_mem_panic("memory map too large");
■ }

■ mem.count = 0;
■ // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/memory.c (line 456) - [public] arch_mem_panic

```
■ arch_mem_panic("empty memory map");
■}

■if (count > ARCH_MAX_MEM_REGIONS) {
■ arch_mem_panic("memory map too large");
■}

■mem.count = 0;

■for (size_t i = 0; i < count; i++) {
■ raw_sorted[i] = entries[i];
■}

// ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/memory.c (line 460) - [public] arch_mem_panic

```
■ arch_mem_panic("memory map too large");
■}

■mem.count = 0;

■for (size_t i = 0; i < count; i++) {
■ raw_sorted[i] = entries[i];
■}

■sort_raw_entries(raw_sorted, count);

■for (size_t i = 0; i < count; i++) {
// ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/memory.c (line 475) - [public] if

```
■ if (base >= end) {
■ continue;
■ }

■ if (mem.count > 0) {
■ const arch_mem_region_t *last = &mem.regions[mem.count - 1];
■ uint64_t last_end = safe_end(last->base, last->len);

■ if (base < last_end) {
■ base = last_end;
■ }

// ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/memory.c (line 479) - [public] if

```
■ if (mem.count > 0) {
■ const arch_mem_region_t *last = &mem.regions[mem.count - 1];
■ uint64_t last_end = safe_end(last->base, last->len);

■ if (base < last_end) {
■ base = last_end;
■ }

■ if (base >= end) {
■ continue;
■ }

■ }

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 483) - [public] if

```
■      if (base < last_end) {
■          base = last_end;
■      }

■      if (base >= end) {
■          continue;
■      }
■  }

■      arch_mem_region_t r;
■      r.base = base;
■      r.len = end - base;
■      // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 487) - [public] if

```
■      if (base >= end) {
■          continue;
■      }
■  }

■      arch_mem_region_t r;
■      r.base = base;
■      r.len = end - base;
■      r.type = raw_type_to_arch(raw_sorted[i].type);

■      map_add(mem.regions, &mem.count, ARCH_MAX_MEM_REGIONS, r);
■  }
■  // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 497) - [public] map_add

```
■      map_add(mem.regions, &mem.count, ARCH_MAX_MEM_REGIONS, r);
■  }

■  if (mem.count == 0) {
■      arch_mem_panic("no valid memory regions after parsing");
■  }

■  map_reserve_range(0, 0x1000000);

■  if (kernel_phys_end > kernel_phys_start) {
■      map_reserve_range(kernel_phys_start,
■          // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 501) - [public] arch_mem_panic

```
■      arch_mem_panic("no valid memory regions after parsing");
■  }

■  map_reserve_range(0, 0x1000000);

■  if (kernel_phys_end > kernel_phys_start) {
■      map_reserve_range(kernel_phys_start,
■          kernel_phys_end - kernel_phys_start);
■  }

■  if (initrd_phys_end > initrd_phys_start) {
■      // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 571) - [public] if

```
■ if (r->type != ARCH_MEM_TYPE_USABLE) {
■     continue;
■ }

■ uint64_t r_end = safe_end(r->base, r->len);

■ if (base >= r->base && end <= r_end) {
■     return true;
■ }
■}

■return false;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 577) - [public] if

```
■ if (base >= r->base && end <= r_end) {
■     return true;
■ }
■}

■return false;
}

void arch_memory_reserve_range(uint64_t base, uint64_t len)
{
■map_reserve_range(base, len);
■recalc_stats();
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 619) - [public] if

```
■ if (r->type != ARCH_MEM_TYPE_USABLE) {
■     continue;
■ }

■ uint64_t r_end = safe_end(r->base, r->len);
■ uint64_t candidate = align_up_generic(r->base, align);

■ if (candidate < r->base) {
■     continue;
■ }

■ if (candidate <= r_end && (r_end - candidate) >= len) {
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 626) - [public] if

```
■ if (candidate < r->base) {
■     continue;
■ }

■ if (candidate <= r_end && (r_end - candidate) >= len) {
■     *out_base = candidate;
■     return true;
■ }
■}

■return false;
}

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 630) - [public] if

```
    if (candidate <= r_end && (r_end - candidate) >= len) {
        *out_base = candidate;
        return true;
    }

return false;
}
```

```
static const char *region_type_name(arch_mem_type_t type)
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 6) - [public] kprintf

```
extern void kprintf(const char *fmt, ...);

static void paging_panic(const char *msg) __attribute__((noreturn));
static void paging_panic(const char *msg)
{
    kprintf("arch/x86_64/paging.c panic: %s\n", msg);

    for (;;) {
        __asm__ volatile("cli; hlt");
    }

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 8) - [public] paging_panic

```
static void paging_panic(const char *msg) __attribute__((noreturn));
static void paging_panic(const char *msg)
{
    kprintf("arch/x86_64/paging.c panic: %s\n", msg);

    for (;;) {
        __asm__ volatile("cli; hlt");
    }

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 11) - [public] kprintf

```
kprintf("arch/x86_64/paging.c panic: %s\n", msg);

for (;;) {
    __asm__ volatile("cli; hlt");
}

}
```

```
static uint64_t boot_phys_offset = 0;
static bool boot_phys_offset_valid = false;

// ... (truncated, see source file for more)
```


kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 25) - [public] if

```
    if (!boot_phys_offset_valid) {
        paging_panic("boot physical offset not set");
    }

    return (void *)(uintptr_t)(phys + boot_phys_offset);
}

static void zero_page_phys(uint64_t phys)
{
    uint64_t *p = (uint64_t *)phys_to_ptr(phys);

    for (size_t i = 0; i < 512; i++) {
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 26) - [public] paging_panic

```
    paging_panic("boot physical offset not set");
}

    return (void *)(uintptr_t)(phys + boot_phys_offset);
}

static void zero_page_phys(uint64_t phys)
{
    uint64_t *p = (uint64_t *)phys_to_ptr(phys);

    for (size_t i = 0; i < 512; i++) {
        p[i] = 0;
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 29) - [public] return

```
    return (void *)(uintptr_t)(phys + boot_phys_offset);
}

static void zero_page_phys(uint64_t phys)
{
    uint64_t *p = (uint64_t *)phys_to_ptr(phys);

    for (size_t i = 0; i < 512; i++) {
        p[i] = 0;
    }

    __asm__ volatile("" ::: "memory");
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 40) - [public] volatile

```
    __asm__ volatile("" ::: "memory");
}

static inline size_t pml4_index(uint64_t virt)
{
    return (size_t)((virt >> 39) & 0x1FF);
}

static inline size_t pdpt_index(uint64_t virt)
{
    return (size_t)((virt >> 30) & 0x1FF);
}

    // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 45) - [public] return
    return (size_t)((virt >> 39) & 0x1FF);
}

static inline size_t pdpt_index(uint64_t virt)
{
    return (size_t)((virt >> 30) & 0x1FF);
}

static inline size_t pd_index(uint64_t virt)
{
    return (size_t)((virt >> 21) & 0x1FF);
}
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 50) - [public] return
    return (size_t)((virt >> 30) & 0x1FF);
}

static inline size_t pd_index(uint64_t virt)
{
    return (size_t)((virt >> 21) & 0x1FF);
}

static inline size_t pt_index(uint64_t virt)
{
    return (size_t)((virt >> 12) & 0x1FF);
}
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 55) - [public] return
    return (size_t)((virt >> 21) & 0x1FF);
}

static inline size_t pt_index(uint64_t virt)
{
    return (size_t)((virt >> 12) & 0x1FF);
}

uint64_t paging_read_cr3(void)
{
    uint64_t cr3;

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 60) - [public] return
    return (size_t)((virt >> 12) & 0x1FF);
}

uint64_t paging_read_cr3(void)
{
    uint64_t cr3;

    __asm__ volatile("mov %%cr3, %0" : "=r"(cr3));

    return cr3 & PAGING_ADDR_MASK;
}

// ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 67) - [public] volatile

```
__asm__ volatile("mov %%cr3, %0" : "=r"(cr3));
```

```
return cr3 & PAGING_ADDR_MASK;
```

```
}
```

```
void paging_flush_tlb_all(void)
```

```
{
```

```
    tlb_flush_all();
```

```
}
```

```
void paging_flush_page(uint64_t addr)
```

```
{
```

```
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 74) - [public] tlb_flush_all

```
    tlb_flush_all();
```

```
}
```

```
void paging_flush_page(uint64_t addr)
```

```
{
```

```
    tlb_flush_page_addr(addr);
```

```
}
```

```
static uint64_t alloc_zeroed_table_page(void)
```

```
{
```

```
    uint64_t phys = 0;
```

```
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 79) - [public] tlb_flush_page_addr

```
    tlb_flush_page_addr(addr);
```

```
}
```

```
static uint64_t alloc_zeroed_table_page(void)
```

```
{
```

```
    uint64_t phys = 0;
```

```
/*
```

```
 * Once the physical frame allocator is up, page-table pages must come
```

```
 * from the PMM so that `free_tables`/`paging_aspace_destroy` can hand
```

```
 * them back with `pmm_free_frame` without corrupting the PMM's bookkeeping
```

```
 * (the boot allocator has no "free", so freeing a boot-allocated page
```

```
 * through the PMM double-frees it and later produces a MALFORMED_TABLE).
```

```
 * ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 93) - [public] if

```
/*
```

```
 * Once the physical frame allocator is up, page-table pages must come
```

```
 * from the PMM so that `free_tables`/`paging_aspace_destroy` can hand
```

```
 * them back with `pmm_free_frame` without corrupting the PMM's bookkeeping
```

```
 * (the boot allocator has no "free", so freeing a boot-allocated page
```

```
 * through the PMM double-frees it and later produces a MALFORMED_TABLE).
```

```
*/
```

```
if (pmm_ready()) {
```

```
    if (!pmm_alloc_zero_frame(&phys)) {
```

```
        paging_panic("cannot allocate page table page (pmm)");
```

```
    }
```

```
    return phys;
```

```
}
```

```
if (!arch_memory_boot_alloc(PAGING_PAGE_SIZE,
```

```
    PAGING_PAGE_SIZE,
```

```
    &phys)) {
```

```
    paging_panic("cannot allocate page table page");
```

```
}
```

```
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 94) - [public] if

```
    if (!pmm_alloc_zero_frame(&phys)) {
        paging_panic("cannot allocate page table page (pmm)");
    }
    return phys;
}

if (!arch_memory_boot_alloc(PAGING_PAGE_SIZE,
                            PAGING_PAGE_SIZE,
                            &phys)) {
    paging_panic("cannot allocate page table page");
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 95) - [public] paging_panic

```
    paging_panic("cannot allocate page table page (pmm)");
}
return phys;
}

if (!arch_memory_boot_alloc(PAGING_PAGE_SIZE,
                            PAGING_PAGE_SIZE,
                            &phys)) {
    paging_panic("cannot allocate page table page");
}

zero_page_phys(phys);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 103) - [public] paging_panic

```
    paging_panic("cannot allocate page table page");
}

zero_page_phys(phys);

return phys;
}

static uint64_t ensure_table_entry(uint64_t *table,
                                   size_t index,
                                   uint64_t flags)
{
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 106) - [public] zero_page_phys

```
    zero_page_phys(phys);

    return phys;
}

static uint64_t ensure_table_entry(uint64_t *table,
                                   size_t index,
                                   uint64_t flags)
{
    uint64_t entry = table[index];

    if (entry & PTE_PRESENT) {
        // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 117) - [public] if
    if (entry & PTE_PRESENT) {
        if (entry & PTE_PAGE_SIZE) {
            /*
             * Large page (2MB or 1GB) found where we need a page table.
             * Split it into smaller 4KB page entries.
             */
            uint64_t large_phys = entry & PAGING_ADDR_MASK;
            /* Only keep the real flag bits (low 12 bits + high NX bit), not
             * the physical-address bits that live in the middle of the entry. */
            uint64_t large_flags = entry & ~PAGING_ADDR_MASK;

            /* Allocate a new page table */
            // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 118) - [public] if
        if (entry & PTE_PAGE_SIZE) {
            /*
             * Large page (2MB or 1GB) found where we need a page table.
             * Split it into smaller 4KB page entries.
             */
            uint64_t large_phys = entry & PAGING_ADDR_MASK;
            /* Only keep the real flag bits (low 12 bits + high NX bit), not
             * the physical-address bits that live in the middle of the entry. */
            uint64_t large_flags = entry & ~PAGING_ADDR_MASK;

            /* Allocate a new page table */
            uint64_t table_phys = alloc_zeroed_table_page();
            // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 138) - [public] if
            if (i < pages) {
                uint64_t page_phys = large_phys + (uint64_t)i * PAGING_PAGE_SIZE;
                new_table[i] = (page_phys & PAGING_ADDR_MASK) |
                    (large_flags & ~(PTE_PAGE_SIZE));
            } else {
                new_table[i] = 0;
            }
        }

        /*
         * Replace the large page entry with the new page table entry.
         * Preserve the access flags from the large page.
         // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 152) - [public] if
            if (large_flags & PTE_USER) {
                table_flags |= PTE_USER;
            }
            table[index] = (table_phys & PAGING_ADDR_MASK) | table_flags;

            return table_phys;
        }

        return entry & PAGING_ADDR_MASK;
    }

    uint64_t table_phys = alloc_zeroed_table_page();
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 171) - [public] if
    if (flags & PTE_USER) {
        table_flags |= PTE_USER;
    }

    table[index] = (table_phys & PAGING_ADDR_MASK) | table_flags;

    return table_phys;
}

bool paging_map_page(uint64_t virt, uint64_t phys, uint64_t flags)
{
    if (!boot_phys_offset_valid) {
        // ... (truncated, see source file for more)
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 182) - [public] if
        if (!boot_phys_offset_valid) {
            return false;
        }

        if ((virt & PAGING_PAGE_MASK) != 0 ||
            (phys & PAGING_PAGE_MASK) != 0) {
            return false;
        }

        /*
         * Tymczasowo nie u■ywamy NX, dopóki EFER.NXE nie jest w■■czony.
         */
        // ... (truncated, see source file for more)
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 214) - [public] paging_flush_page
        paging_flush_page(virt);

        return true;
    }

    static bool map_page_2m(uint64_t virt, uint64_t phys, uint64_t flags)
    {
        if (!boot_phys_offset_valid) {
            return false;
        }

        // ... (truncated, see source file for more)
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 222) - [public] if
        if (!boot_phys_offset_valid) {
            return false;
        }

        if ((virt & PAGING_2M_PAGE_MASK) != 0 ||
            (phys & PAGING_2M_PAGE_MASK) != 0) {
            return false;
        }

        flags &= ~PTE_NX;
        flags &= ~PTE_PAGE_SIZE;

        // ... (truncated, see source file for more)

```

kernel_workspace/kernel/src/mm/arch/x86_64/paging.c (line 250) - [public] if

```
/*
 * Je■li istnieje ju■ tablica 4 KiB w tym miejscu,
 * nie nadpisujemy jej 2 MiB stron■.
 */

if ((old & PTE_PRESENT) && !(old & PTE_PAGE_SIZE)) {
    paging_panic("cannot map 2M page over existing 4K table");
}

pd[idx] =
    (phys & PAGING_ADDR_MASK) |
    flags |
    PTE_PRESENT |
    PTE_PAGE_SIZE;

return true;
}

// ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/paging.c (line 251) - [public] paging_panic

```
    paging_panic("cannot map 2M page over existing 4K table");
}

pd[idx] =
    (phys & PAGING_ADDR_MASK) |
    flags |
    PTE_PRESENT |
    PTE_PAGE_SIZE;

return true;
}

// ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/paging.c (line 265) - [public] if

```
if (!boot_phys_offset_valid) {
    return false;
}

if (len == 0) {
    return true;
}

if ((virt & PAGING_PAGE_MASK) != 0 ||
    (phys & PAGING_PAGE_MASK) != 0 ||
    (len & PAGING_PAGE_MASK) != 0) {
    return false;
}

// ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/paging.c (line 269) - [public] if

```
if (len == 0) {
    return true;
}

if ((virt & PAGING_PAGE_MASK) != 0 ||
    (phys & PAGING_PAGE_MASK) != 0 ||
    (len & PAGING_PAGE_MASK) != 0) {
    return false;
}

if (len > UINT64_MAX - virt || len > UINT64_MAX - phys) {
    return false;
}

// ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 279) - [public] if
    if (len > UINT64_MAX - virt || len > UINT64_MAX - phys) {
        return false;
    }

    uint64_t virt_end = virt + len;
    uint64_t phys_end = phys + len;

    while (virt < virt_end) {
        uint64_t remaining = virt_end - virt;

        bool can_use_2m =
            (remaining >= PAGING_2M_PAGE_SIZE) &&
            // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 286) - [public] while
    while (virt < virt_end) {
        uint64_t remaining = virt_end - virt;

        bool can_use_2m =
            (remaining >= PAGING_2M_PAGE_SIZE) &&
            ((virt & PAGING_2M_PAGE_MASK) == 0) &&
            ((phys & PAGING_2M_PAGE_MASK) == 0);

        if (can_use_2m) {
            if (!map_page_2m(virt, phys, flags)) {
                return false;
            }
        }
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 294) - [public] if
    if (can_use_2m) {
        if (!map_page_2m(virt, phys, flags)) {
            return false;
        }

        virt += PAGING_2M_PAGE_SIZE;
        phys += PAGING_2M_PAGE_SIZE;
    } else {
        if (!paging_map_page(virt, phys, flags)) {
            return false;
        }
    }

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 295) - [public] if
    if (!map_page_2m(virt, phys, flags)) {
        return false;
    }

    virt += PAGING_2M_PAGE_SIZE;
    phys += PAGING_2M_PAGE_SIZE;
} else {
    if (!paging_map_page(virt, phys, flags)) {
        return false;
    }

    virt += PAGING_PAGE_SIZE;
    // ... (truncated, see source file for more)

```


kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 302) - [public] if

```
        if (!paging_map_page(virt, phys, flags)) {
            return false;
        }

        virt += PAGING_PAGE_SIZE;
        phys += PAGING_PAGE_SIZE;
    }

    paging_flush_tlb_all();

    return true;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 311) - [public] paging_flush_tlb_all

```
    paging_flush_tlb_all();

    return true;
}
```

```
void paging_set_boot_phys_offset(uint64_t phys_offset)
{
    if ((phys_offset & PAGING_PAGE_MASK) != 0) {
        paging_panic("boot physical offset must be page aligned");
    }

    boot_phys_offset = phys_offset;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 318) - [public] if

```
    if ((phys_offset & PAGING_PAGE_MASK) != 0) {
        paging_panic("boot physical offset must be page aligned");
    }

    boot_phys_offset = phys_offset;
    boot_phys_offset_valid = true;
}

bool paging_boot_phys_offset_valid(void)
{
    return boot_phys_offset_valid;
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 319) - [public] paging_panic

```
    paging_panic("boot physical offset must be page aligned");
}

    boot_phys_offset = phys_offset;
    boot_phys_offset_valid = true;
}

bool paging_boot_phys_offset_valid(void)
{
    return boot_phys_offset_valid;
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 333) - [public] if

```
if (!arch_memory_ready()) {
    paging_panic("arch memory not initialized");
}

if (!boot_phys_offset_valid) {
    paging_panic("boot physical offset not set");
}

/*
 * Je■li bootloader ju■ mapuje fizyczn■ pami■ dok■adnie pod
 * ARCH_DIRECT_MAP_BASE, nie musimy nic robi■.
 */
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 334) - [public] paging_panic

```
    paging_panic("arch memory not initialized");
}

if (!boot_phys_offset_valid) {
    paging_panic("boot physical offset not set");
}

/*
 * Je■li bootloader ju■ mapuje fizyczn■ pami■ dok■adnie pod
 * ARCH_DIRECT_MAP_BASE, nie musimy nic robi■.
 */
if (boot_phys_offset == ARCH_DIRECT_MAP_BASE) {
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 337) - [public] if

```
if (!boot_phys_offset_valid) {
    paging_panic("boot physical offset not set");
}

/*
 * Je■li bootloader ju■ mapuje fizyczn■ pami■ dok■adnie pod
 * ARCH_DIRECT_MAP_BASE, nie musimy nic robi■.
 */
if (boot_phys_offset == ARCH_DIRECT_MAP_BASE) {
    return;
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 338) - [public] paging_panic

```
    paging_panic("boot physical offset not set");
}

/*
 * Je■li bootloader ju■ mapuje fizyczn■ pami■ dok■adnie pod
 * ARCH_DIRECT_MAP_BASE, nie musimy nic robi■.
 */
if (boot_phys_offset == ARCH_DIRECT_MAP_BASE) {
    return;
}

const arch_mem_info_t *info = arch_memory_get();
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 345) - [public] if

```
/*  
 * Jeśli bootloader już mapuje fizyczny pamięć, dokonanie pod  
 * ARCH_DIRECT_MAP_BASE, nie musimy nic robić.  
 */  
  
    if (boot_phys_offset == ARCH_DIRECT_MAP_BASE) {  
        return;  
    }  
  
    const arch_mem_info_t *info = arch_memory_get();  
  
    if (info == NULL) {  
        paging_panic("no arch memory info");  
    }  
  
    uint64_t phys_limit = arch_page_align_up(info->max_address);  
  
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 351) - [public] if

```
    if (info == NULL) {  
        paging_panic("no arch memory info");  
    }  
  
    uint64_t phys_limit = arch_page_align_up(info->max_address);  
  
    if (phys_limit == 0 || phys_limit == UINT64_MAX) {  
        paging_panic("invalid physical memory limit");  
    }  
  
    if (!paging_map_range(ARCH_DIRECT_MAP_BASE,  
                          0,  
                          // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 352) - [public] paging_panic

```
        paging_panic("no arch memory info");  
    }  
  
    uint64_t phys_limit = arch_page_align_up(info->max_address);  
  
    if (phys_limit == 0 || phys_limit == UINT64_MAX) {  
        paging_panic("invalid physical memory limit");  
    }  
  
    if (!paging_map_range(ARCH_DIRECT_MAP_BASE,  
                          0,  
                          phys_limit,  
                          // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 357) - [public] if

```
    if (phys_limit == 0 || phys_limit == UINT64_MAX) {  
        paging_panic("invalid physical memory limit");  
    }  
  
    if (!paging_map_range(ARCH_DIRECT_MAP_BASE,  
                          0,  
                          phys_limit,  
                          PAGING_KERNEL_RW)) {  
        paging_panic("failed to create direct map");  
    }  
  
    paging_flush_tlb_all();  
    // ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 358) - [public] paging_panic
    paging_panic("invalid physical memory limit");
}

if (!paging_map_range(ARCH_DIRECT_MAP_BASE,
                      0,
                      phys_limit,
                      PAGING_KERNEL_RW)) {
    paging_panic("failed to create direct map");
}

paging_flush_tlb_all();
}

// ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 365) - [public] paging_panic
    paging_panic("failed to create direct map");
}

paging_flush_tlb_all();
}

void paging_init(uint64_t phys_offset)
{
    paging_set_boot_phys_offset(phys_offset);
    paging_init_direct_map();
}

// ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 368) - [public] paging_flush_tlb_all
    paging_flush_tlb_all();
}

void paging_init(uint64_t phys_offset)
{
    paging_set_boot_phys_offset(phys_offset);
    paging_init_direct_map();
}

/* ----- */
/* Page-table walk helper */
/* ----- */
// ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 373) - [public]
paging_set_boot_phys_offset
    paging_set_boot_phys_offset(phys_offset);
    paging_init_direct_map();
}

/* ----- */
/* Page-table walk helper */
/* ----- */

/*
 * Walk the 4-level page table rooted at `pml4_phys` and return a pointer
 * to the PTE for `virt`. When `create` is true, missing intermediate
 * tables are allocated (with kernel RW flags); otherwise the walk stops
 * // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 374) - [public]

paging_init_direct_map

```
    paging_init_direct_map();
}

/* ----- */
/* Page-table walk helper */
/* ----- */

/*
 * Walk the 4-level page table rooted at `pml4_phys` and return a pointer
 * to the PTE for `virt`. When `create` is true, missing intermediate
 * tables are allocated (with kernel RW flags); otherwise the walk stops
 * and returns NULL on the first absent entry.
 * // ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 384) - [public] allocated

```
/*
 * Walk the 4-level page table rooted at `pml4_phys` and return a pointer
 * to the PTE for `virt`. When `create` is true, missing intermediate
 * tables are allocated (with kernel RW flags); otherwise the walk stops
 * and returns NULL on the first absent entry.
 *
 * Returns NULL on absent entries (create == false), allocation failure,
 * or unexpected large pages.
 */
static uint64_t *resolve_pte(uint64_t pml4_phys, uint64_t virt, bool create)
{
    uint64_t *pml4 = phys_to_ptr(pml4_phys);
    uint64_t *pdpt, *pd, *pt;
    uint64_t entry;
    size_t idx;
    // ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 399) - [public] if

```
    if (!(entry & PTE_PRESENT)) {
        if (!create) {
            return NULL;
        }
        ensure_table_entry(pml4, idx, PTE_PRESENT | PTE_WRITABLE);
        entry = pml4[idx];
    }
    pdpt = phys_to_ptr(entry & PAGING_ADDR_MASK);

    idx = pdpt_index(virt);
    entry = pdpt[idx];
    if (!(entry & PTE_PRESENT)) {
        // ... (truncated, see source file for more)
    }

```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 400) - [public] if

```
        if (!create) {
            return NULL;
        }
        ensure_table_entry(pml4, idx, PTE_PRESENT | PTE_WRITABLE);
        entry = pml4[idx];
    }
    pdpt = phys_to_ptr(entry & PAGING_ADDR_MASK);

    idx = pdpt_index(virt);
    entry = pdpt[idx];
    if (!(entry & PTE_PRESENT)) {
        if (!create) {
            // ... (truncated, see source file for more)
        }
    }

```

```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 403) - [public] ensure_table_entry
    ensure_table_entry(pml4, idx, PTE_PRESENT | PTE_WRITABLE);
    entry = pml4[idx];
}
pdpt = phys_to_ptr(entry & PAGING_ADDR_MASK);

idx = pdpt_index(virt);
entry = pdpt[idx];
if (!(entry & PTE_PRESENT)) {
    if (!create) {
        return NULL;
    }
    ensure_table_entry(pdpt, idx, PTE_PRESENT | PTE_WRITABLE);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 410) - [public] if
    if (!(entry & PTE_PRESENT)) {
        if (!create) {
            return NULL;
        }
        ensure_table_entry(pdpt, idx, PTE_PRESENT | PTE_WRITABLE);
        entry = pdpt[idx];
    }
    if (entry & PTE_PAGE_SIZE) {
        return NULL; /* 1G large page - not handled here */
    }
    pd = phys_to_ptr(entry & PAGING_ADDR_MASK);

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 411) - [public] if
    if (!create) {
        return NULL;
    }
    ensure_table_entry(pdpt, idx, PTE_PRESENT | PTE_WRITABLE);
    entry = pdpt[idx];
}
if (entry & PTE_PAGE_SIZE) {
    return NULL; /* 1G large page - not handled here */
}
pd = phys_to_ptr(entry & PAGING_ADDR_MASK);

idx = pd_index(virt);
// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 414) - [public] ensure_table_entry
    ensure_table_entry(pdpt, idx, PTE_PRESENT | PTE_WRITABLE);
    entry = pdpt[idx];
}
if (entry & PTE_PAGE_SIZE) {
    return NULL; /* 1G large page - not handled here */
}
pd = phys_to_ptr(entry & PAGING_ADDR_MASK);

idx = pd_index(virt);
entry = pd[idx];
if (!(entry & PTE_PRESENT)) {
    if (!create) {
        // ... (truncated, see source file for more)

```

```
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 417) - [public] if
    if (entry & PTE_PAGE_SIZE) {
        return NULL; /* 1G large page - not handled here */
    }
    pd = phys_to_ptr(entry & PAGING_ADDR_MASK);

    idx = pd_index(virt);
    entry = pd[idx];
    if (!(entry & PTE_PRESENT)) {
        if (!create) {
            return NULL;
        }
        ensure_table_entry(pd, idx, PTE_PRESENT | PTE_WRITABLE);
    }
    // ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 424) - [public] if
    if (!(entry & PTE_PRESENT)) {
        if (!create) {
            return NULL;
        }
        ensure_table_entry(pd, idx, PTE_PRESENT | PTE_WRITABLE);
        entry = pd[idx];
    }
    if (entry & PTE_PAGE_SIZE) {
        return NULL; /* 2M large page - not handled here */
    }
    pt = phys_to_ptr(entry & PAGING_ADDR_MASK);

    // ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 425) - [public] if
    if (!create) {
        return NULL;
    }
    ensure_table_entry(pd, idx, PTE_PRESENT | PTE_WRITABLE);
    entry = pd[idx];
}
if (entry & PTE_PAGE_SIZE) {
    return NULL; /* 2M large page - not handled here */
}
pt = phys_to_ptr(entry & PAGING_ADDR_MASK);

return &pt[pt_index(virt)];
// ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 428) - [public] ensure_table_entry
    ensure_table_entry(pd, idx, PTE_PRESENT | PTE_WRITABLE);
    entry = pd[idx];
}
if (entry & PTE_PAGE_SIZE) {
    return NULL; /* 2M large page - not handled here */
}
pt = phys_to_ptr(entry & PAGING_ADDR_MASK);

return &pt[pt_index(virt)];
}

/* ----- */
// ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 431) - [public] if
    if (entry & PTE_PAGE_SIZE) {
        return NULL; /* 2M large page - not handled here */
    }
    pt = phys_to_ptr(entry & PAGING_ADDR_MASK);

    return &pt[pt_index(virt)];
}

/* ----- */
/* Arch-specific operations used by the generic paging layer */
/* ----- */

// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 447) - [public] if
    if (!boot_phys_offset_valid) {
        return false;
    }
    if ((virt & PAGING_PAGE_MASK) != 0 || (phys & PAGING_PAGE_MASK) != 0) {
        return false;
    }
    flags &= ~PTE_PAGE_SIZE;

    pte = resolve_pte(pml4, virt, true);
    if (pte == NULL) {
        return false;
    }
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 450) - [public] if
    if ((virt & PAGING_PAGE_MASK) != 0 || (phys & PAGING_PAGE_MASK) != 0) {
        return false;
    }
    flags &= ~PTE_PAGE_SIZE;

    pte = resolve_pte(pml4, virt, true);
    if (pte == NULL) {
        return false;
    }
    *pte = (phys & PAGING_ADDR_MASK) | flags | PTE_PRESENT;

    paging_flush_page(virt);
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 456) - [public] if
    if (pte == NULL) {
        return false;
    }
    *pte = (phys & PAGING_ADDR_MASK) | flags | PTE_PRESENT;

    paging_flush_page(virt);
    return true;
}

bool paging_unmap_page_in(uint64_t pml4, uint64_t virt)
{
    uint64_t *pte = resolve_pte(pml4, virt, false);
    // ... (truncated, see source file for more)

```


kernel_workspace/kernel/src/mm/arch/x86_64/paging.c (line 461) - [public] paging_flush_page

```
*pte = (phys & PAGING_ADDR_MASK) | flags | PTE_PRESENT;
```

```
    paging_flush_page(virt);  
    return true;  
}
```

```
bool paging_unmap_page_in(uint64_t pml4, uint64_t virt)
```

```
{  
    uint64_t *pte = resolve_pte(pml4, virt, false);  
  
    if (pte == NULL || !(*pte & PTE_PRESENT)) {  
        return false;  
    }  
    *pte = 0;  
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/paging.c (line 469) - [public] if

```
    if (pte == NULL || !(*pte & PTE_PRESENT)) {  
        return false;  
    }
```

```
    *pte = 0;  
    paging_flush_page(virt);  
    return true;  
}
```

```
uint64_t paging_translate_in(uint64_t pml4, uint64_t virt)
```

```
{  
    uint64_t *pte = resolve_pte(pml4, virt, false);  
  
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/paging.c (line 473) - [public] paging_flush_page

```
*pte = 0;
```

```
    paging_flush_page(virt);  
    return true;  
}
```

```
uint64_t paging_translate_in(uint64_t pml4, uint64_t virt)
```

```
{  
    uint64_t *pte = resolve_pte(pml4, virt, false);  
  
    if (pte == NULL || !(*pte & PTE_PRESENT)) {  
        return 0;  
    }  
    return (*pte & PAGING_ADDR_MASK) | (virt & PAGING_PAGE_MASK);  
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/paging.c (line 481) - [public] if

```
    if (pte == NULL || !(*pte & PTE_PRESENT)) {  
        return 0;  
    }  
    return (*pte & PAGING_ADDR_MASK) | (virt & PAGING_PAGE_MASK);  
}
```

```
bool paging_is_mapped_in(uint64_t pml4, uint64_t virt)
```

```
{  
    const uint64_t *pte = resolve_pte(pml4, virt, false);  
  
    return pte != NULL && (*pte & PTE_PRESENT) != 0;  
}  
// ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 484) - [public] return
    return (*pte & PAGING_ADDR_MASK) | (virt & PAGING_PAGE_MASK);
}

bool paging_is_mapped_in(uint64_t pml4, uint64_t virt)
{
    const uint64_t *pte = resolve_pte(pml4, virt, false);

    return pte != NULL && (*pte & PTE_PRESENT) != 0;
}

bool paging_set_flags_in(uint64_t pml4, uint64_t virt, uint64_t flags)
{
    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 498) - [public] if
    if (pte == NULL || !(*pte & PTE_PRESENT)) {
        return false;
    }
    flags &= ~PTE_PAGE_SIZE;
    *pte = (*pte & PAGING_ADDR_MASK) | flags | PTE_PRESENT;
    paging_flush_page(virt);
    return true;
}

bool paging_get_flags_in(uint64_t pml4, uint64_t virt, uint64_t *out_flags)
{
    const uint64_t *pte = resolve_pte(pml4, virt, false);
    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 503) - [public] paging_flush_page
    *pte = (*pte & PAGING_ADDR_MASK) | flags | PTE_PRESENT;

    paging_flush_page(virt);
    return true;
}

bool paging_get_flags_in(uint64_t pml4, uint64_t virt, uint64_t *out_flags)
{
    const uint64_t *pte = resolve_pte(pml4, virt, false);

    if (pte == NULL || !(*pte & PTE_PRESENT)) {
        return false;
    }
    if (out_flags != NULL) {
        // ... (truncated, see source file for more)
    }
}

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 511) - [public] if
    if (pte == NULL || !(*pte & PTE_PRESENT)) {
        return false;
    }
    if (out_flags != NULL) {
        *out_flags = *pte & ~PAGING_ADDR_MASK;
    }
    return true;
}

bool paging_is_mapped(uint64_t virt)
{
    return paging_is_mapped_in(paging_read_cr3(), virt);
    // ... (truncated, see source file for more)
}

```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 514) - [public] if

```
    if (out_flags != NULL) {
        *out_flags = *pte & ~PAGING_ADDR_MASK;
    }
    return true;
}

bool paging_is_mapped(uint64_t virt)
{
    return paging_is_mapped_in(paging_read_cr3(), virt);
}

uint64_t paging_translate(uint64_t virt)
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 522) - [public] paging_is_mapped_in

```
    return paging_is_mapped_in(paging_read_cr3(), virt);
}

uint64_t paging_translate(uint64_t virt)
{
    return paging_translate_in(paging_read_cr3(), virt);
}

bool paging_unmap_page(uint64_t virt)
{
    return paging_unmap_page_in(paging_read_cr3(), virt);
}
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 527) - [public] paging_translate_in

```
    return paging_translate_in(paging_read_cr3(), virt);
}

bool paging_unmap_page(uint64_t virt)
{
    return paging_unmap_page_in(paging_read_cr3(), virt);
}

bool paging_set_flags(uint64_t virt, uint64_t flags)
{
    return paging_set_flags_in(paging_read_cr3(), virt, flags);
}
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 532) - [public] paging_unmap_page_in

```
    return paging_unmap_page_in(paging_read_cr3(), virt);
}

bool paging_set_flags(uint64_t virt, uint64_t flags)
{
    return paging_set_flags_in(paging_read_cr3(), virt, flags);
}

bool paging_get_flags(uint64_t virt, uint64_t *out_flags)
{
    return paging_get_flags_in(paging_read_cr3(), virt, out_flags);
}
    // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 537) - [public] paging_set_flags_in
    return paging_set_flags_in(paging_read_cr3(), virt, flags);
}

bool paging_get_flags(uint64_t virt, uint64_t *out_flags)
{
    return paging_get_flags_in(paging_read_cr3(), virt, out_flags);
}

/* NX is enabled once via the EFER MSR; this is idempotent. */
static bool nx_enabled = false;

void paging_enable_nx(void)
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 542) - [public] paging_get_flags_in
    return paging_get_flags_in(paging_read_cr3(), virt, out_flags);
}

/* NX is enabled once via the EFER MSR; this is idempotent. */
static bool nx_enabled = false;

void paging_enable_nx(void)
{
    uint64_t efer;

    if (nx_enabled) {
        return;
        // ... (truncated, see source file for more)
    }

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 552) - [public] if
    if (nx_enabled) {
        return;
    }
    __asm__ volatile("rdmsr" : "=A"(efer) : "c"(0xC0000080));
    efer |= (1ULL << 11); /* NXE bit */
    __asm__ volatile("wrmsr" : : "A"(efer), "c"(0xC0000080));
    nx_enabled = true;
}

bool paging_nx_enabled(void)
{
    return nx_enabled;
    // ... (truncated, see source file for more)
}

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 555) - [public] volatile
    __asm__ volatile("rdmsr" : "=A"(efer) : "c"(0xC0000080));
    efer |= (1ULL << 11); /* NXE bit */
    __asm__ volatile("wrmsr" : : "A"(efer), "c"(0xC0000080));
    nx_enabled = true;
}

bool paging_nx_enabled(void)
{
    return nx_enabled;
}

void paging_write_cr3(uint64_t pml4_phys)
    // ... (truncated, see source file for more)

```

kernel_workspace/kernel/src/mm/arch/x86_64/paging.c (line 557) - [public] volatile

```
if (nx_enabled) {
    return;
}
__asm__ volatile("rdmsr" : "=A"(efer) : "c"(0xC0000080));
efer |= (1ULL << 11); /* NXE bit */

__asm__ volatile("wrmsr" : : "A"(efer), "c"(0xC0000080));
nx_enabled = true;
}

bool paging_nx_enabled(void)
{
    return nx_enabled;
}

void paging_write_cr3(uint64_t pml4_phys)
{
    __asm__ volatile("mov %0, %%cr3" : : "r"(pml4_phys) : "memory");
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/paging.c (line 568) - [public] volatile

```
__asm__ volatile("mov %0, %%cr3" : : "r"(pml4_phys) : "memory");
}
```

kernel_workspace/kernel/src/mm/arch/x86_64/tlb.c (line 77) - [public] tlb_asm_invpcid

```
■ tlb_asm_invpcid(INVPCID_ALL, &desc);
■ return;
■}

■tlb_asm_write_cr3(tlb_asm_read_cr3());
}

void tlb_flush_all_including_global(void)
{
    ■if (has_invpcid) {
    ■    tlb_flush_all();
    ■    return;
    ■    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/tlb.c (line 87) - [public] tlb_flush_all

```
■ tlb_flush_all();
■ return;
■}

■uint64_t cr4 = tlb_asm_read_cr4();

■if (cr4 & CR4_PGE) {
    ■ tlb_asm_write_cr4(cr4 & ~CR4_PGE);
    ■ tlb_asm_write_cr4(cr4);
    ■} else {
    ■ tlb_flush_all();
    ■}
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/arch/x86_64/tlb.c (line 94) - [public] tlb_asm_write_cr4

```
■ tlb_asm_write_cr4(cr4 & ~CR4_PGE);
■ tlb_asm_write_cr4(cr4);
    ■} else {
    ■ tlb_flush_all();
    ■}
}

void tlb_flush_page_addr(uint64_t addr)
{
    ■tlb_asm_invlpg(addr & ~TLB_PAGE_MASK);
}

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.c (line 95) - [public] tlb_asm_write_cr4

```
    tlb_asm_write_cr4(cr4);
} else {
    tlb_flush_all();
}

void tlb_flush_page_addr(uint64_t addr)
{
    tlb_asm_invlpg(addr & ~TLB_PAGE_MASK);
}

void tlb_flush_page(const void *addr)
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.c (line 97) - [public] tlb_flush_all

```
    tlb_flush_all();
}

void tlb_flush_page_addr(uint64_t addr)
{
    tlb_asm_invlpg(addr & ~TLB_PAGE_MASK);
}

void tlb_flush_page(const void *addr)
{
    if (addr == NULL) {
        // ... (truncated, see source file for more)
    }
}
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.c (line 109) - [public] tlb_flush_all

```
    tlb_flush_all();
    return;
}

tlb_flush_page_addr((uint64_t)(uintptr_t)addr);
}

void tlb_flush_range_addr(uint64_t addr, size_t pages)
{
    if (pages == 0) {
        return;
    }
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.c (line 123) - [public] tlb_flush_all

```
    tlb_flush_all();
    return;
}

uint64_t a = addr & ~TLB_PAGE_MASK;

if (pages > (UINT64_MAX - a) / TLB_PAGE_SIZE) {
    tlb_flush_all();
    return;
}

for (size_t i = 0; i < pages; i++) {
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.c (line 130) - [public] tlb_flush_all

```
■ tlb_flush_all();
■ return;
■}

■for (size_t i = 0; i < pages; i++) {
■ tlb_asm_invlpg(a);
■ a += TLB_PAGE_SIZE;
■}
}

void tlb_flush_range(const void *addr, size_t pages)
{
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.c (line 135) - [public] tlb_asm_invlpg

```
■ tlb_asm_invlpg(a);
■ a += TLB_PAGE_SIZE;
■}
}

void tlb_flush_range(const void *addr, size_t pages)
{
■if (addr == NULL) {
■ tlb_flush_all();
■ return;
■}

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.c (line 143) - [public] tlb_flush_all

```
■ tlb_flush_all();
■ return;
■}

■tlb_flush_range_addr((uint64_t)(uintptr_t)addr, pages);
}

void tlb_flush_pcid(uint16_t pcid)
{
■if (has_invpcid) {
■ invpcid_desc_t desc = { pcid, 0 };
■ tlb_asm_invpcid(INVPCID_SINGLE_PCID, &desc);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.c (line 154) - [public] tlb_asm_invpcid

```
■ tlb_asm_invpcid(INVPCID_SINGLE_PCID, &desc);
■ return;
■}

■if (pcid == TLB_PCID_KERNEL) {
■ tlb_asm_write_cr3(tlb_asm_read_cr3());
■}
}

void tlb_flush_pcid_addr(uint16_t pcid, uint64_t addr)
{
■if (has_invpcid) {
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.c (line 159) - [public] tlb_asm_write_cr3

```
    tlb_asm_write_cr3(tlb_asm_read_cr3());
}

void tlb_flush_pcid_addr(uint16_t pcid, uint64_t addr)
{
    if (has_invpcid) {
        invpcid_desc_t desc = { pcid, addr & ~TLB_PAGE_MASK };
        tlb_asm_invpcid(INVPCID_ADDR_IN_PCID, &desc);
        return;
    }

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.c (line 167) - [public] tlb_asm_invpcid

```
    tlb_asm_invpcid(INVPCID_ADDR_IN_PCID, &desc);
    return;
}

tlb_asm_invlpg(addr & ~TLB_PAGE_MASK);
}

void tlb_wbinvd(void)
{
    tlb_asm_wbinvd();
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/cache/cache.c (line 3) - [public] kprintf

```
extern void kprintf(const char *fmt, ...);

static kcache_t *cache_registry = NULL;
static bool cache_initialized = false;

bool cache_init(void)
{
    if (cache_initialized) {
        return true;
    }

    per_cpu_init();
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/cache/cache.c (line 10) - [public] if

```
    if (cache_initialized) {
        return true;
    }

    per_cpu_init();

    cache_registry = NULL;
    cache_initialized = true;

    return true;
}

// ... (truncated, see source file for more)
```



```

kernel_Workspace/kernel/src/mm/cache/cache.c (line 14) - [public] per_cpu_init
    per_cpu_init();

    cache_registry = NULL;
    cache_initialized = true;

    return true;
}

bool cache_ready(void)
{
    return cache_initialized;
}

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/cache/cache.c (line 29) - [public] if
    if (c == NULL) {
        return;
    }

    c->next = cache_registry;
    cache_registry = c;
}

kcache_t *cache_find(const char *name)
{
    kcache_t *c = cache_registry;

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/cache/cache.c (line 41) - [public] while
    while (c != NULL) {
        const char *a = c->name;
        const char *b = name;

        while (*a != '\0' && *a == *b) {
            a++;
            b++;
        }

        if (*a == '\0' && *b == '\0') {
            return c;
        }
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/cache/cache.c (line 45) - [public] while
    while (*a != '\0' && *a == *b) {
        a++;
        b++;
    }

    if (*a == '\0' && *b == '\0') {
        return c;
    }

    c = c->next;
}

// ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/mm/cache/cache.c (line 50) - [public] if
    if (*a == '\0' && *b == '\0') {
        return c;
    }

    c = c->next;
}

return NULL;
}

void cache_dump(void)
{
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/cache/cache.c (line 62) - [public] if
    if (!cache_initialized) {
        kprintf("CACHE: not initialized\n");
        return;
    }

    kprintf("CACHE:\n");

    kcache_t *c = cache_registry;

    while (c != NULL) {
        kprintf(" %-16s obj=%u free=%u total=%u\n",
            c->name,
            // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/cache/cache.c (line 63) - [public] kprintf
        kprintf("CACHE: not initialized\n");
        return;
    }

    kprintf("CACHE:\n");

    kcache_t *c = cache_registry;

    while (c != NULL) {
        kprintf(" %-16s obj=%u free=%u total=%u\n",
            c->name,
            (unsigned int)c->object_size,
            // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/cache/cache.c (line 67) - [public] kprintf
    kprintf("CACHE:\n");

    kcache_t *c = cache_registry;

    while (c != NULL) {
        kprintf(" %-16s obj=%u free=%u total=%u\n",
            c->name,
            (unsigned int)c->object_size,
            (unsigned int)c->free_count,
            (unsigned int)c->total_objects);

        c = c->next;
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/cache/cache.c (line 71) - [public] while
    while (c != NULL) {
        kprintf(" %-16s obj=%u free=%u total=%u\n",
            c->name,
            (unsigned int)c->object_size,
            (unsigned int)c->free_count,
            (unsigned int)c->total_objects);

        c = c->next;
    }
}

```

kernel_workspace/kernel/src/mm/cache/object_cache.c (line 82) - [public] buddy_free

```
■ buddy_free(va);
■ return false;
■}

■ch->va = (uint64_t)(uintptr_t)va;
■ch->next = c->chunks;
■c->chunks = ch;

■size_t n = c->chunk_size / c->object_size;

■for (size_t i = 0; i < n; i++) {
■    uint8_t *obj =
        // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/cache/object_cache.c (line 96) - [public] if

```
■    if (c->ctor != NULL) {
■        c->ctor(obj);
■    }

■    *(void **)obj = c->free_head;
■    c->free_head = obj;
■}

■c->free_count += n;
■c->total_objects += n;

■return true;
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/cache/object_cache.c (line 180) - [public] buddy_free

```
■    buddy_free((void *) (uintptr_t)ch->va);
■    heap_free(ch);

■    ch = next;
■}

■c->chunks = NULL;
■c->free_head = NULL;
■c->free_count = 0;
■c->total_objects = 0;

■cache_unlock();
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/cache/object_cache.c (line 181) - [public] heap_free

```
■    heap_free(ch);

■    ch = next;
■}

■c->chunks = NULL;
■c->free_head = NULL;
■c->free_count = 0;
■c->total_objects = 0;

■cache_unlock();

    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/cache/object_cache.c (line 208) - [public] cache_unlock

```
■ cache_unlock();
■ return obj;
■}

■if (c->free_head == NULL) {
■    if (!kcache_grow(c)) {
■        cache_unlock();
■        return NULL;
■    }
■}

■void *obj = c->free_head;
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/cache/object_cache.c (line 213) - [public] if

```
■    if (!kcache_grow(c)) {
■        cache_unlock();
■        return NULL;
■    }
■}

■void *obj = c->free_head;
■c->free_head = *(void **)obj;
■c->free_count--;

■cache_unlock();

    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/cache/object_cache.c (line 214) - [public] cache_unlock

```
■        cache_unlock();
■        return NULL;
■    }
■}

■void *obj = c->free_head;
■c->free_head = *(void **)obj;
■c->free_count--;

■cache_unlock();

■return obj;
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/cache/object_cache.c (line 253) - [public] cache_unlock

```
■    cache_unlock();
■    return;
■}

■*(void **)ptr = c->free_head;
■c->free_head = ptr;
■c->free_count++;

■cache_unlock();
}

size_t kcache_object_size(kcache_t *c)
    // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/mm/core/mm.c (line 15) - [public] kprintf
extern void kprintf(const char *fmt, ...);

static bool mm_initialized = false;

bool mm_init(const mm_boot_params_t *params)
{
    if (mm_initialized) {
        return true;
    }

    if (params == NULL ||
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/core/mm.c (line 29) - [public] kprintf
    kprintf("mm_init: bad params\n");
    return false;
}

    kprintf("mm_init: arch_memory_init...\n");
    arch_memory_init(params->memmap,
        params->memmap_count,
        params->kernel_phys_start,
        params->kernel_phys_end,
        params->initrd_phys_start,
        params->initrd_phys_end);

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/core/mm.c (line 46) - [public] kprintf
    kprintf("mm_init: pmm_init FAILED\n");
    return false;
}

    kprintf("mm_init: vmm_init...\n");
    if (!vmm_init()) {
        kprintf("mm_init: vmm_init FAILED\n");
        return false;
    }

    kprintf("mm_init: page_init...\n");
    if (!page_init()) {
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/core/mm.c (line 52) - [public] kprintf
    kprintf("mm_init: vmm_init FAILED\n");
    return false;
}

    kprintf("mm_init: page_init...\n");
    if (!page_init()) {
        kprintf("mm_init: page_init FAILED\n");
        return false;
    }

    mapping_init();

    // ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/mm/core/mm.c (line 58) - [public] kprintf

```
■ kprintf("mm_init: page_init FAILED\n");
■ return false;
■ }
```

```
■ mapping_init();
```

```
■ kprintf("mm_init: heap_init...\n");
■ if (!heap_init()) {
■     kprintf("mm_init: heap_init FAILED\n");
■     return false;
■ }
```

```
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/core/mm.c (line 66) - [public] kprintf

```
■ kprintf("mm_init: heap_init FAILED\n");
■ return false;
■ }
```

```
■ kprintf("mm_init: cache_init...\n");
■ if (!cache_init()) {
■     kprintf("mm_init: cache_init FAILED\n");
■     return false;
■ }
```

```
■ kprintf("mm_init: paging_subsystem_init...\n");
■ if (!paging_subsystem_init()) {
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/core/mm.c (line 72) - [public] kprintf

```
■ kprintf("mm_init: cache_init FAILED\n");
■ return false;
■ }

■ kprintf("mm_init: paging_subsystem_init...\n");
■ if (!paging_subsystem_init()) {
■     kprintf("mm_init: paging_subsystem_init FAILED\n");
■     return false;
■ }
```

```
■ isolation_init();
```

```
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/core/mm.c (line 78) - [public] kprintf

```
■ kprintf("mm_init: paging_subsystem_init FAILED\n");
■ return false;
■ }
```

```
■ isolation_init();
```

```
■ kprintf("mm_init: aspace_subsystem_init...\n");
■ if (!aspace_subsystem_init()) {
■     kprintf("mm_init: aspace_subsystem_init FAILED\n");
■     return false;
■ }
```

```
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/core/mm.c (line 86) - [public] kprintf

```
■ kprintf("mm_init: aspace_subsystem_init FAILED\n");
■ return false;
■}

■mm_initialized = true;

■kprintf("mm_init: OK\n");

■return true;
}

bool mm_ready(void)
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/core/mm.c (line 104) - [public] pmm_stat_total_bytes

```
■return pmm_stat_total_bytes();
}

uint64_t mm_free_ram(void)
{
■return pmm_stat_free_bytes();
}

void mm_dump(void)
{
■kprintf("MM: ready=%d total=%llu MiB free=%llu MiB\n",
■      (int)mm_initialized,
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/core/mm.c (line 109) - [public] pmm_stat_free_bytes

```
■return pmm_stat_free_bytes();
}

void mm_dump(void)
{
■kprintf("MM: ready=%d total=%llu MiB free=%llu MiB\n",
■      (int)mm_initialized,
■      (unsigned long long)(mm_total_ram() >> 20),
■      (unsigned long long)(mm_free_ram() >> 20));

■kalloc_dump();
}
```

kernel_Workspace/kernel/src/mm/core/range.c (line 53) - [public] if

```
■ if (raw_end > UINT64_MAX - pad) {
■     return false;
■ }

■ end = raw_end + pad;
■}

■if (end <= start) {
■     return false;
■}

■if (require_canonical) {
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/core/range.c (line 65) - [public] if

```
■ if (!range_is_canonical(start) || !range_is_canonical(end - 1)) {
■     return false;
■ }
■ }

■ if (start < limit_min) {
■     return false;
■ }

■ if (end > limit_max) {
■     return false;
■ }

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/core/region.c (line 64) - [public] if

```
■ if (region_valid(before)) {
■     out[n++] = before;
■ }
■ }

■ if (cut.end < r.end && n < max_out) {
■     region_t after;

■     after.start = cut.end > r.start ? cut.end : r.start;
■     after.end = r.end;

■     if (region_valid(after)) {
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/core/region.c (line 75) - [public] if

```
■ if (region_valid(after)) {
■     out[n++] = after;
■ }
■ }

■ return n;
}
```

kernel_Workspace/kernel/src/mm/core/smp_lock.c (line 5) - [public] volatile

```
■ __asm__ volatile("pause" ::: "memory");
}

static inline uint64_t irq_save_disable(void)
{
■ uint64_t flags;

■ __asm__ volatile(
■     "pushfq\n"
■     "popq %0\n"
■     "cli"
■     : "=r"(flags)
        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/core/smp_lock.c (line 64) - [public] __atomic_load_n

```
■ __atomic_load_n(&lock->owner_cpu, __ATOMIC_ACQUIRE) == me) {
■     lock->depth++;
■     return;
■ }

■ uint32_t ticket = __atomic_fetch_add(&lock->next_ticket, 1,
■                                     __ATOMIC_RELAXED);

■ while (__atomic_load_n(&lock->now_serving, __ATOMIC_ACQUIRE) != ticket) {
■     cpu_pause();
■ }
■ lock->owner_cpu = me;
    // ... (truncated, see source file for more)
```


kernel_Workspace/kernel/src/mm/core/smp_lock.c (line 73) - [public] cpu_pause

```
■    cpu_pause();
■}
■lock->owner_cpu = me;
■lock->depth = 1;
■lock->saved_flags = flags;
■}

bool smp_lock_release(smp_ticket_lock_t *lock)
{
■if (lock->depth == 0) {
■    return false;
■}
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/kprintf.c (line 18) - [public] volatile

```
■__asm__ volatile("inb %%dx, %0" : "=a"(v) : "d"(port));

■return v;
■}

static inline void outb(uint16_t port, uint8_t val)
{
■__asm__ volatile("outb %0, %%dx" : : "a"(val), "d"(port));
■}

static bool uart_ready = false;

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/kprintf.c (line 25) - [public] volatile

```
■__asm__ volatile("outb %0, %%dx" : : "a"(val), "d"(port));
■}

static bool uart_ready = false;

static void serial_init(void)
{
■if (uart_ready) {
■    return;
■}

■outb(COM1 + 1, 0x00);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/kprintf.c (line 58) - [public] serial_putc

```
■    serial_putc(*s++);
■}
■}

static void serial_put_uint(uint64_t v, unsigned base, bool upper)
{
■char buf[32];
■int i = 32;

■if (v == 0) {
■    serial_putc('0');
■    return;
■}
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/kprintf.c (line 68) - [public] serial_putc

```
■    serial_putc('0');
■    return;
■}

■while (v > 0) {
■    unsigned d = (unsigned)(v % base);

■    buf[--i] = (char)(d < 10 ? '0' + d : (upper ? 'A' : 'a') + d - 10);
■    v /= base;
■}

■while (i < 32) {
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/kprintf.c (line 80) - [public] serial_putc

```
■    serial_putc(buf[i++]);
■}
}

void kprintf(const char *fmt, ...)
{
■serial_init();

■va_list ap;

■va_start(ap, fmt);

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/kprintf.c (line 93) - [public] if

```
■    if (*p != '%') {
■        serial_putc(*p);
■        continue;
■    }

■    p++;
■    if (*p == '\\0') {
■        break;
■    }

■    bool ll = false;
■    bool l = false;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/kprintf.c (line 94) - [public] serial_putc

```
■        serial_putc(*p);
■        continue;
■    }

■    p++;
■    if (*p == '\\0') {
■        break;
■    }

■    bool ll = false;
■    bool l = false;
■    bool z = false;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/kprintf.c (line 99) - [public] if

```
■   if (*p == '\0') {
■       break;
■   }

■   bool ll = false;
■   bool l = false;
■   bool z = false;

■   while (*p == 'l' || *p == 'z' || *p == 'h') {
■       if (*p == 'l') {
■           if (p[1] == 'l') {
■               ll = true;
■           }
■           // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/kprintf.c (line 107) - [public] while

```
■   while (*p == 'l' || *p == 'z' || *p == 'h') {
■       if (*p == 'l') {
■           if (p[1] == 'l') {
■               ll = true;
■               p++;
■           } else {
■               l = true;
■           }
■       } else if (*p == 'z') {
■           z = true;
■       }
■       p++;
■       // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/kprintf.c (line 108) - [public] if

```
■       if (*p == 'l') {
■           if (p[1] == 'l') {
■               ll = true;
■               p++;
■           } else {
■               l = true;
■           }
■       } else if (*p == 'z') {
■           z = true;
■       }
■       p++;
■       // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/kprintf.c (line 109) - [public] if

```
■           if (p[1] == 'l') {
■               ll = true;
■               p++;
■           } else {
■               l = true;
■           }
■       } else if (*p == 'z') {
■           z = true;
■       }
■       p++;
■       // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/kprintf.c (line 121) - [public] switch

```
■ switch (*p) {
■ case 's': {
■     const char *s = va_arg(ap, const char *);

■     serial_puts(s ? s : "(null)");
■     break;
■ }
■ case 'c':
■     serial_putc((char)va_arg(ap, int));
■     break;
■ case 'd':
■ case 'i': {
■     // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/kprintf.c (line 125) - [public] serial_puts

```
■     serial_puts(s ? s : "(null)");
■     break;
■ }
■ case 'c':
■     serial_putc((char)va_arg(ap, int));
■     break;
■ case 'd':
■ case 'i': {
■     int64_t v = ll ? (int64_t)va_arg(ap, long long)
■                  : (l ? (int64_t)va_arg(ap, long)
■                     : (int64_t)va_arg(ap, int));

■     // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/kprintf.c (line 129) - [public] serial_putc

```
■     serial_putc((char)va_arg(ap, int));
■     break;
■ case 'd':
■ case 'i': {
■     int64_t v = ll ? (int64_t)va_arg(ap, long long)
■                  : (l ? (int64_t)va_arg(ap, long)
■                     : (int64_t)va_arg(ap, int));

■     if (v < 0) {
■         serial_putc('-');
■         serial_put_uint((uint64_t)(-v), 10, false);
■     } else {
■         // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/kprintf.c (line 137) - [public] if

```
■     if (v < 0) {
■         serial_putc('-');
■         serial_put_uint((uint64_t)(-v), 10, false);
■     } else {
■         serial_put_uint((uint64_t)v, 10, false);
■     }
■     break;
■ }
■ case 'u': {
■     uint64_t v = ll ? (uint64_t)va_arg(ap, unsigned long long)
■                     : (z ? (uint64_t)va_arg(ap, size_t)
■                        : (l ? (uint64_t)va_arg(ap, unsigned long)
■                          ));
■     // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/kprintf.c (line 138) - [public] serial_putc

```
■      serial_putc('-');
■      serial_put_uint((uint64_t)(-v), 10, false);
■  } else {
■      serial_put_uint((uint64_t)v, 10, false);
■  }
■  break;
■ }
■ case 'u': {
■     uint64_t v = ll ? (uint64_t)va_arg(ap, unsigned long long)
■                   : (z ? (uint64_t)va_arg(ap, size_t)
■                   : (l ? (uint64_t)va_arg(ap, unsigned long)
■                   : (uint64_t)va_arg(ap, unsigned int)));
■
■ // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/kprintf.c (line 139) - [public] serial_put_uint

```
■      serial_put_uint((uint64_t)(-v), 10, false);
■  } else {
■      serial_put_uint((uint64_t)v, 10, false);
■  }
■  break;
■ }
■ case 'u': {
■     uint64_t v = ll ? (uint64_t)va_arg(ap, unsigned long long)
■                   : (z ? (uint64_t)va_arg(ap, size_t)
■                   : (l ? (uint64_t)va_arg(ap, unsigned long)
■                   : (uint64_t)va_arg(ap, unsigned int)));
■
■ // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/kprintf.c (line 141) - [public] serial_put_uint

```
■      serial_put_uint((uint64_t)v, 10, false);
■  }
■  break;
■ }
■ case 'u': {
■     uint64_t v = ll ? (uint64_t)va_arg(ap, unsigned long long)
■                   : (z ? (uint64_t)va_arg(ap, size_t)
■                   : (l ? (uint64_t)va_arg(ap, unsigned long)
■                   : (uint64_t)va_arg(ap, unsigned int)));
■
■     serial_put_uint(v, 10, false);
■     break;
■ // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/kprintf.c (line 151) - [public] serial_put_uint

```
■      serial_put_uint(v, 10, false);
■     break;
■ }
■ case 'x':
■ case 'X': {
■     uint64_t v = ll ? (uint64_t)va_arg(ap, unsigned long long)
■                   : (z ? (uint64_t)va_arg(ap, size_t)
■                   : (l ? (uint64_t)va_arg(ap, unsigned long)
■                   : (uint64_t)va_arg(ap, unsigned int)));
■
■     serial_put_uint(v, 16, *p == 'X');
■     break;
■ // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/kprintf.c (line 161) - [public] serial_put_uint

```
■    serial_put_uint(v, 16, *p == 'X');
■    break;
■    }
■    case 'p': {
■        void *ptr = va_arg(ap, void *);

■        serial_puts("0x");
■        serial_put_uint((uint64_t)(uintptr_t)ptr, 16, false);
■        break;
■    }
■    case '%':
■        serial_putc('%');
■    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/kprintf.c (line 167) - [public] serial_puts

```
■    serial_puts("0x");
■    serial_put_uint((uint64_t)(uintptr_t)ptr, 16, false);
■    break;
■    }
■    case '%':
■        serial_putc('%');
■        break;
■    default:
■        serial_putc('%');
■        serial_putc(*p);
■        break;
■    }
■    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/kprintf.c (line 168) - [public] serial_put_uint

```
■    serial_put_uint((uint64_t)(uintptr_t)ptr, 16, false);
■    break;
■    }
■    case '%':
■        serial_putc('%');
■        break;
■    default:
■        serial_putc('%');
■        serial_putc(*p);
■        break;
■    }
■}
■    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/kprintf.c (line 172) - [public] serial_putc

```
■    serial_putc('%');
■    break;
■    default:
■        serial_putc('%');
■        serial_putc(*p);
■        break;
■    }
■}

■va_end(ap);
}
```

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/kprintf.c (line 175) - [public] serial_putc

```
■    serial_putc('%');
■    serial_putc(*p);
■    break;
■    }
■}

■va_end(ap);
}

void serial_write_str(const char *s)
{
■serial_init();
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/kprintf.c (line 176) - [public] serial_putc

```
■    serial_putc(*p);
■    break;
■    }
■}

■va_end(ap);
}

void serial_write_str(const char *s)
{
■serial_init();
■serial_puts(s);
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/paging/paging.c (line 68) - [public] if

```
■    if (pml_entry_present(e) && !pml_entry_large(e)) {
■        free_tables(pml_entry_addr(e), level - 1);
■    }
■}

■pmm_free_frame(table_phys);
}

address_space_t *paging_aspace_create(void)
{
■if (!paging_subsystem_ready) {
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/paging/paging.c (line 69) - [public] free_tables

```
■        free_tables(pml_entry_addr(e), level - 1);
■    }
■}

■pmm_free_frame(table_phys);
}

address_space_t *paging_aspace_create(void)
{
■if (!paging_subsystem_ready) {
■    return NULL;
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/paging/paging.c (line 99) - [public] pmm_free_frame

```
■    pmm_free_frame(pml4_phys);  
■    return NULL;  
■}
```

```
■as->pml4_phys = pml4_phys;  
■as->kernel = false;
```

```
■return as;  
}
```

```
void paging_aspace_destroy(address_space_t *as)  
{  
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/paging/paging.c (line 120) - [public] if

```
■    if (pml_entry_present(e) && !pml_entry_large(e)) {  
■        free_tables(pml_entry_addr(e), PML_LEVEL_PDPT);  
■    }  
■}
```

```
■pmm_free_frame(as->pml4_phys);
```

```
■heap_free(as);  
}
```

```
void paging_aspace_switch(address_space_t *as)  
{  
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/paging/paging.c (line 121) - [public] free_tables

```
■        free_tables(pml_entry_addr(e), PML_LEVEL_PDPT);  
■    }  
■}
```

```
■pmm_free_frame(as->pml4_phys);
```

```
■heap_free(as);  
}
```

```
void paging_aspace_switch(address_space_t *as)  
{  
■if (as == NULL) {  
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/paging/paging.c (line 173) - [public] if

```
■    if (!paging_map_page_in(pml4, v, p, pte)) {  
■        for (size_t j = 0; j < mapped; j++) {  
■            paging_unmap_page_in(pml4, virt + (uint64_t)j * ARCH_PAGE_SIZE);  
■        }  
■    }  
■    return false;  
■    }
```

```
■    mapped++;  
■}
```

```
■return true;
```

```
    // ... (truncated, see source file for more)
```



```

kernel_Workspace/kernel/src/mm/paging/paging.c (line 175) - [public] paging_unmap_page_in
    ■         paging_unmap_page_in(pml4, virt + (uint64_t)j * ARCH_PAGE_SIZE);
    ■     }

    ■     return false;
    ■ }

    ■ mapped++;
    ■}

    ■return true;
    ■}

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/paging/paging.c (line 207) - [public] if
    ■     if (paging_unmap_page_in(pml4, v)) {
    ■         tlb_batch_add(&batch, v);
    ■     }
    ■}

    ■tlb_batch_commit(&batch);

    ■return true;
    ■}

    bool paging_aspace_protect(address_space_t *as,
    ■         uint64_t virt,
    ■         // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/paging/paging.c (line 208) - [public] tlb_batch_add
    ■         tlb_batch_add(&batch, v);
    ■     }
    ■}

    ■tlb_batch_commit(&batch);

    ■return true;
    ■}

    bool paging_aspace_protect(address_space_t *as,
    ■         uint64_t virt,
    ■         size_t len,
    ■         // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/paging/paging.c (line 238) - [public] if
    ■     if (!paging_is_mapped_in(pml4, v)) {
    ■         return false;
    ■     }
    ■}

    ■tlb_batch_t batch;
    ■tlb_batch_begin(&batch);

    ■for (size_t i = 0; i < pages; i++) {
    ■     uint64_t v = virt + (uint64_t)i * ARCH_PAGE_SIZE;

    ■     if (!paging_set_flags_in(pml4, v, pte)) {
    ■         // ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/mm/paging/paging.c (line 249) - [public] if

```
■    if (!paging_set_flags_in(pml4, v, pte)) {
■        tlb_batch_commit(&batch);
■        return false;
■    }

■    tlb_batch_add(&batch, v);
■}

■tlb_batch_commit(&batch);

■return true;
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/paging/paging.c (line 250) - [public] tlb_batch_commit

```
■        tlb_batch_commit(&batch);
■        return false;
■    }

■    tlb_batch_add(&batch, v);
■}

■tlb_batch_commit(&batch);

■return true;
}
```

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/paging/paging.c (line 254) - [public] tlb_batch_add

```
■    tlb_batch_add(&batch, v);
■}

■tlb_batch_commit(&batch);

■return true;
}

uint64_t paging_aspace_translate(address_space_t *as, uint64_t virt)
{
■if (as == NULL) {
■    return paging_translate_in(paging_kernel_pml4, virt);
■    // ... (truncated, see source file for more)
}
```

kernel_Workspace/kernel/src/mm/paging/paging.c (line 265) - [public] paging_translate_in

```
■    return paging_translate_in(paging_kernel_pml4, virt);
■}

■return paging_translate_in(as->pml4_phys, virt);
}

bool paging_kernel_map(uint64_t virt, uint64_t phys, size_t len, uint32_t prot)
{
■if (!paging_subsystem_ready) {
■    return false;
■}

// ... (truncated, see source file for more)
```

```

kernel_workspace/kernel/src/mm/paging/paging.c (line 268) - [public] paging_translate_in
    ■return paging_translate_in(as->pml4_phys, virt);
    ■}

    bool paging_kernel_map(uint64_t virt, uint64_t phys, size_t len, uint32_t prot)
    {
    ■if (!paging_subsystem_ready) {
    ■    return false;
    ■}

    ■return paging_aspace_map((address_space_t *)&(address_space_t){
    ■    .pml4_phys = paging_kernel_pml4,
    ■    .kernel = true,
    ■    // ... (truncated, see source file for more)
    ■});

kernel_workspace/kernel/src/mm/paging/paging.c (line 277) - [public] paging_aspace_map
    ■return paging_aspace_map((address_space_t *)&(address_space_t){
    ■    .pml4_phys = paging_kernel_pml4,
    ■    .kernel = true,
    ■    },
    ■    virt, phys, len, prot);
    ■}

    bool paging_kernel_unmap(uint64_t virt, size_t len)
    {
    ■if (!paging_subsystem_ready) {
    ■    return false;
    ■}
    ■// ... (truncated, see source file for more)

kernel_workspace/kernel/src/mm/paging/paging.c (line 290) - [public] paging_aspace_unmap
    ■return paging_aspace_unmap((address_space_t *)&(address_space_t){
    ■    .pml4_phys = paging_kernel_pml4,
    ■    .kernel = true,
    ■    },
    ■    virt, len);
    ■}

kernel_workspace/kernel/src/mm/paging/tlb.c (line 33) - [public] tlb_flush_all
    ■    tlb_flush_all();
    ■    batch->count = 0;
    ■    batch->full_flush = false;
    ■    return;
    ■}

    ■for (size_t i = 0; i < batch->count; i++) {
    ■    tlb_flush_page_addr(batch->pages[i]);
    ■}

    ■batch->count = 0;
    ■}

kernel_workspace/kernel/src/mm/paging/tlb.c (line 40) - [public] tlb_flush_page_addr
    ■    tlb_flush_page_addr(batch->pages[i]);
    ■}

    ■batch->count = 0;
    ■}

```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 63) - [public] if

```
■ if (!pmm_alloc_zero_frame(&phys)) {
■     goto rollback;
■ }

■ if (!paging_aspace_map(pa->as, v, phys, ARCH_PAGE_SIZE,
■                         prot | PROT_USER)) {
■     pmm_free_frame(phys);
■     goto rollback;
■ }

■ mapped_end = v + ARCH_PAGE_SIZE;
■ }

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 69) - [public] pmm_free_frame

```
■     pmm_free_frame(phys);
■     goto rollback;
■ }

■ mapped_end = v + ARCH_PAGE_SIZE;
■ }

■ return true;

rollback:
■ for (uint64_t v = start; v < mapped_end; v += ARCH_PAGE_SIZE) {
■     uint64_t phys = paging_aspace_translate(pa->as, v);
■     // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 82) - [public]
paging_aspace_unmap

```
■     paging_aspace_unmap(pa->as, v, ARCH_PAGE_SIZE);

■     if (phys != UINT64_MAX) {
■         pmm_free_frame(phys);
■     }
■ }

■ return false;
}

static void unmap_pages_free(proc_aspace_t *pa, uint64_t start, uint64_t end)
{
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 84) - [public] if

```
■     if (phys != UINT64_MAX) {
■         pmm_free_frame(phys);
■     }
■ }

■ return false;
}

static void unmap_pages_free(proc_aspace_t *pa, uint64_t start, uint64_t end)
{
■ for (uint64_t v = start; v < end; v += ARCH_PAGE_SIZE) {
■     uint64_t phys = paging_aspace_translate(pa->as, v);
■     // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 85) - [public] pmm_free_frame

```
■    pmm_free_frame(phys);
■    }
■}

■return false;
}

static void unmap_pages_free(proc_aspace_t *pa, uint64_t start, uint64_t end)
{
■for (uint64_t v = start; v < end; v += ARCH_PAGE_SIZE) {
■    uint64_t phys = paging_aspace_translate(pa->as, v);

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 97) - [public]
paging_aspace_unmap

```
■    paging_aspace_unmap(pa->as, v, ARCH_PAGE_SIZE);

■    if (phys != UINT64_MAX) {
■        pmm_free_frame(phys);
■    }
■}

static void vma_insert(proc_aspace_t *pa, vma_t *v)
{
■vma_t **p = &pa->vmass;

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 99) - [public] if

```
■    if (phys != UINT64_MAX) {
■        pmm_free_frame(phys);
■    }
■}

static void vma_insert(proc_aspace_t *pa, vma_t *v)
{
■vma_t **p = &pa->vmass;

■while (*p != NULL && (*p)->start < v->start) {
■    p = &(*p)->next;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 100) - [public] pmm_free_frame

```
■    pmm_free_frame(phys);
■    }
■}

static void vma_insert(proc_aspace_t *pa, vma_t *v)
{
■vma_t **p = &pa->vmass;

■while (*p != NULL && (*p)->start < v->start) {
■    p = &(*p)->next;
■}

    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/process/address_space.c (line 122) - [public] if

```
■   if (*p == v) {
■       *p = v->next;
■       return;
■   }

■   p = &(*p)->next;
■}
}

vma_t *aspace_vma_find(proc_aspace_t *pa, uint64_t addr)
{
■vma_t *v = pa->vmass;
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/process/address_space.c (line 136) - [public] if

```
■   if (addr >= v->start && addr < v->end) {
■       return v;
■   }

■   v = v->next;
■}

■return NULL;
}

static uint64_t find_gap(proc_aspace_t *pa, uint64_t len, uint64_t hint)
{
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/process/address_space.c (line 163) - [public] if

```
■   if (v->end <= cand) {
■       v = v->next;
■       continue;
■   }

■   if (cand <= USER_MMAP_MAX - len && cand + len <= v->start) {
■       return cand;
■   }

■   cand = arch_page_align_up(v->end);

■   if (cand > USER_MMAP_MAX - len) {
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/process/address_space.c (line 168) - [public] if

```
■   if (cand <= USER_MMAP_MAX - len && cand + len <= v->start) {
■       return cand;
■   }

■   cand = arch_page_align_up(v->end);

■   if (cand > USER_MMAP_MAX - len) {
■       return 0;
■   }

■   v = v->next;
■}

    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/process/address_space.c (line 174) - [public] if

```
■    if (cand > USER_MMAP_MAX - len) {  
■        return 0;  
■    }  
  
■    v = v->next;  
■}  
  
■return cand;  
}
```

```
proc_aspace_t *aspace_create(void)  
{  
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/process/address_space.c (line 199) - [public] heap_free

```
■    heap_free(pa);  
■    return NULL;  
■}  
  
■pa->brk_base = USER_BRK_BASE;  
■pa->brk = USER_BRK_BASE;  
■pa->brk_max = USER_BRK_MAX;  
  
■uint64_t stack_start = USER_STACK_TOP - USER_STACK_SIZE;  
  
■vma_t *sv = (vma_t *)heap_alloc(sizeof(vma_t));  
  
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/process/address_space.c (line 212) - [public]
paging_aspace_destroy

```
■    paging_aspace_destroy(pa->as);  
■    heap_free(pa);  
■    return NULL;  
■}  
  
■sv->start = stack_start;  
■sv->end = USER_STACK_TOP;  
■sv->prot = PROT_READ | PROT_WRITE;  
■sv->flags = VMA_FLAG_ANON | VMA_FLAG_PRIVATE;  
■sv->next = NULL;  
  
■pa->vmass = sv;  
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/process/address_space.c (line 213) - [public] heap_free

```
■    heap_free(pa);  
■    return NULL;  
■}  
  
■sv->start = stack_start;  
■sv->end = USER_STACK_TOP;  
■sv->prot = PROT_READ | PROT_WRITE;  
■sv->flags = VMA_FLAG_ANON | VMA_FLAG_PRIVATE;  
■sv->next = NULL;  
  
■pa->vmass = sv;  
  
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 226) - [public] heap_free

```
■ heap_free(sv);
■ paging_aspace_destroy(pa->as);
■ heap_free(pa);
■ return NULL;
■}

■return pa;
}

void aspace_destroy(proc_aspace_t *pa)
{
■if (pa == NULL) {
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 227) - [public]
paging_aspace_destroy

```
■ paging_aspace_destroy(pa->as);
■ heap_free(pa);
■ return NULL;
■}

■return pa;
}

void aspace_destroy(proc_aspace_t *pa)
{
■if (pa == NULL) {
■    return;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 228) - [public] heap_free

```
■ heap_free(pa);
■ return NULL;
■}

■return pa;
}

void aspace_destroy(proc_aspace_t *pa)
{
■if (pa == NULL) {
■    return;
■}
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 246) - [public] unmap_pages_free

```
■ unmap_pages_free(pa, v->start, v->end);

■ pa->vmass = v->next;
■ heap_free(v);
■}

■paging_aspace_destroy(pa->as);

■heap_free(pa);

■as_unlock();
}
    // ... (truncated, see source file for more)
```


kernel_workspace/kernel/src/mm/process/address_space.c (line 249) - [public] heap_free

```
■ heap_free(v);
■}

■paging_aspace_destroy(pa->as);

■heap_free(pa);

■as_unlock();
}

address_space_t *aspace_paging_handle(proc_aspace_t *pa)
{
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/process/address_space.c (line 281) - [public] as_unlock

```
■ as_unlock();
■ return 0;
■}

■vma_t *v = (vma_t *)heap_alloc(sizeof(vma_t));

■if (v == NULL) {
■    as_unlock();
■    return 0;
■}

■v->start = at;
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/process/address_space.c (line 288) - [public] as_unlock

```
■ as_unlock();
■ return 0;
■}

■v->start = at;
■v->end = at + bytes;
■v->prot = prot;
■v->flags = VMA_FLAG_ANON | VMA_FLAG_PRIVATE;

■vma_insert(pa, v);

■if (!map_anon_pages(pa, at, at + bytes, prot)) {
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/process/address_space.c (line 300) - [public] vma_remove

```
■ vma_remove(pa, v);
■ heap_free(v);
■ as_unlock();
■ return 0;
■}

■as_unlock();

■return at;
}

uint64_t aspace_map_at(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t prot)
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/process/address_space.c (line 301) - [public] heap_free

```
■ heap_free(v);
■ as_unlock();
■ return 0;
■}

■as_unlock();

■return at;
}

uint64_t aspace_map_at(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t prot)
{
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/process/address_space.c (line 302) - [public] as_unlock

```
■ as_unlock();
■ return 0;
■}

■as_unlock();

■return at;
}

uint64_t aspace_map_at(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t prot)
{
    ■if (pa == NULL || len == 0) {
        // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/process/address_space.c (line 330) - [public] as_unlock

```
■ as_unlock();
■ return 0;
■}

■v->start = a;
■v->end = b;
■v->prot = prot;
■v->flags = VMA_FLAG_ANON | VMA_FLAG_PRIVATE;

■vma_insert(pa, v);

■if (!map_anon_pages(pa, a, b, prot)) {
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/process/address_space.c (line 342) - [public] vma_remove

```
■ vma_remove(pa, v);
■ heap_free(v);
■ as_unlock();
■ return 0;
■}

■as_unlock();

■return a;
}

bool aspace_unmap(proc_aspace_t *pa, uint64_t addr, size_t len)
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/process/address_space.c (line 343) - [public] heap_free

```
■ heap_free(v);
■ as_unlock();
■ return 0;
■ }
```

```
■ as_unlock();
```

```
■ return a;
}
```

```
bool aspace_unmap(proc_aspace_t *pa, uint64_t addr, size_t len)
{
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/process/address_space.c (line 344) - [public] as_unlock

```
■ as_unlock();
■ return 0;
■ }
```

```
■ as_unlock();
```

```
■ return a;
}
```

```
bool aspace_unmap(proc_aspace_t *pa, uint64_t addr, size_t len)
{
    if (pa == NULL || len == 0) {
        // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/process/address_space.c (line 371) - [public] while

```
■ while (v != NULL) {
■     if (v->start < b && v->end > a) {
■         hit = v;
■         break;
■     }

■     v = v->next;
■ }

■ if (hit == NULL) {
■     break;
■ }
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/process/address_space.c (line 372) - [public] if

```
■ if (v->start < b && v->end > a) {
■     hit = v;
■     break;
■ }

■ v = v->next;
■ }

■ if (hit == NULL) {
■     break;
■ }

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 380) - [public] if

```
■   if (hit == NULL) {
■       break;
■   }

■   uint64_t s = hit->start > a ? hit->start : a;
■   uint64_t e = hit->end < b ? hit->end : b;

■   bool needs_split = (hit->start < s && hit->end > e);
■   vma_t *tail = NULL;

■   if (needs_split) {
■       tail = (vma_t *)heap_alloc(sizeof(vma_t));
■       // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 390) - [public] if

```
■   if (needs_split) {
■       tail = (vma_t *)heap_alloc(sizeof(vma_t));

■       if (tail == NULL) {
■           as_unlock();
■           return false;
■       }
■   }

■   unmap_pages_free(pa, s, e);

■   if (needs_split) {
■       // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 393) - [public] if

```
■       if (tail == NULL) {
■           as_unlock();
■           return false;
■       }
■   }

■   unmap_pages_free(pa, s, e);

■   if (needs_split) {
■       tail->start = e;
■       tail->end = hit->end;
■       tail->prot = hit->prot;
■       // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 394) - [public] as_unlock

```
■       as_unlock();
■       return false;
■   }
■   }

■   unmap_pages_free(pa, s, e);

■   if (needs_split) {
■       tail->start = e;
■       tail->end = hit->end;
■       tail->prot = hit->prot;
■       tail->flags = hit->flags;
■       // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 399) - [public] unmap_pages_free

```
■ unmap_pages_free(pa, s, e);

■ if (needs_split) {
■     tail->start = e;
■     tail->end = hit->end;
■     tail->prot = hit->prot;
■     tail->flags = hit->flags;
■     tail->next = hit->next;

■     hit->next = tail;
■     hit->end = s;
■ } else if (hit->start < s) {
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 401) - [public] if

```
■ if (needs_split) {
■     tail->start = e;
■     tail->end = hit->end;
■     tail->prot = hit->prot;
■     tail->flags = hit->flags;
■     tail->next = hit->next;

■     hit->next = tail;
■     hit->end = s;
■ } else if (hit->start < s) {
■     hit->end = s;
■ } else if (hit->end > e) {
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 415) - [public] vma_remove

```
■ vma_remove(pa, hit);
■ heap_free(hit);
■ }
■}

■as_unlock();

■return true;
}

bool aspace_protect_checked(proc_aspace_t *pa, uint64_t addr, size_t len,
■     uint32_t checked_prot,
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 416) - [public] heap_free

```
■ heap_free(hit);
■ }
■}

■as_unlock();

■return true;
}

bool aspace_protect_checked(proc_aspace_t *pa, uint64_t addr, size_t len,
■     uint32_t checked_prot,
■     uint32_t apply_prot,
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 445) - [public] as_unlock

```
■ as_unlock();
■ return false;
■}

■if (allowed != NULL && !allowed(v->prot, checked_prot)) {
■ as_unlock();
■ return false;
■}

■if (!paging_aspace_protect(pa->as, a, b - a, apply_prot)) {
■ as_unlock();
■ return false;
■ // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 450) - [public] as_unlock

```
■ as_unlock();
■ return false;
■}

■if (!paging_aspace_protect(pa->as, a, b - a, apply_prot)) {
■ as_unlock();
■ return false;
■}

■v->prot = apply_prot;

■as_unlock();
■ // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 455) - [public] as_unlock

```
■ as_unlock();
■ return false;
■}

■v->prot = apply_prot;

■as_unlock();

■return true;
}

bool aspace_protect(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t prot)
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 468) - [public]

aspace_protect_checked

```
■return aspace_protect_checked(pa, addr, len, prot, prot, NULL);
}

uint64_t aspace_stack_base(void)
{
■return USER_STACK_TOP - USER_STACK_SIZE;
}

uint64_t aspace_reserve_at(proc_aspace_t *pa,
■ uint64_t addr,
■ size_t len,
■ uint32_t flags)
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 496) - [public] if

```
■    if (v->start < b && v->end > a) {  
■        as_unlock();  
■        return 0;  
■    }  
  
■    v = v->next;  
■}  
  
■vma_t *nv = (vma_t *)heap_alloc(sizeof(vma_t));  
  
■if (nv == NULL) {  
■    as_unlock();  
■    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 497) - [public] as_unlock

```
■        as_unlock();  
■        return 0;  
■    }  
  
■    v = v->next;  
■}  
  
■vma_t *nv = (vma_t *)heap_alloc(sizeof(vma_t));  
  
■if (nv == NULL) {  
■    as_unlock();  
■    return 0;  
■    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 507) - [public] as_unlock

```
■    as_unlock();  
■    return 0;  
■}  
  
■nv->start = a;  
■nv->end = b;  
■nv->prot = 0;  
■nv->flags = flags;  
  
■vma_insert(pa, nv);  
  
■as_unlock();  
■    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 533) - [public] as_unlock

```
■    as_unlock();  
■    return cur;  
■}  
  
■if (new_brk < pa->brk_base || new_brk > pa->brk_max) {  
■    as_unlock();  
■    return pa->brk;  
■}  
  
■uint64_t nb = arch_page_align_up(new_brk);  
  
■if (nb > pa->brk) {  
■    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/process/address_space.c (line 538) - [public] as_unlock

```
■ as_unlock();
■ return pa->brk;
■ }

■ uint64_t nb = arch_page_align_up(new_brk);

■ if (nb > pa->brk) {
■     if (!map_anon_pages(pa, pa->brk, nb, PROT_READ | PROT_WRITE)) {
■         as_unlock();
■         return pa->brk;
■     }

    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/process/address_space.c (line 545) - [public] if

```
■ if (!map_anon_pages(pa, pa->brk, nb, PROT_READ | PROT_WRITE)) {
■     as_unlock();
■     return pa->brk;
■ }

■ vma_t *v = aspace_vma_find(pa, pa->brk_base);

■ if (v == NULL) {
■     v = (vma_t *)heap_alloc(sizeof(vma_t));

    if (v == NULL) {
        unmap_pages_free(pa, pa->brk, nb);
        // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/process/address_space.c (line 546) - [public] as_unlock

```
■ as_unlock();
■ return pa->brk;
■ }

■ vma_t *v = aspace_vma_find(pa, pa->brk_base);

■ if (v == NULL) {
■     v = (vma_t *)heap_alloc(sizeof(vma_t));

    if (v == NULL) {
        unmap_pages_free(pa, pa->brk, nb);
        as_unlock();
        // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/process/address_space.c (line 552) - [public] if

```
■ if (v == NULL) {
■     v = (vma_t *)heap_alloc(sizeof(vma_t));

    if (v == NULL) {
        unmap_pages_free(pa, pa->brk, nb);
        as_unlock();
        return pa->brk;
    }

    v->start = pa->brk_base;
    v->end = nb;
    v->prot = PROT_READ | PROT_WRITE;
    // ... (truncated, see source file for more)
```


kernel_Workspace/kernel/src/mm/process/address_space.c (line 555) - [public] if

```
■      if (v == NULL) {
■          unmap_pages_free(pa, pa->brk, nb);
■          as_unlock();
■          return pa->brk;
■      }

■      v->start = pa->brk_base;
■      v->end = nb;
■      v->prot = PROT_READ | PROT_WRITE;
■      v->flags = VMA_FLAG_ANON | VMA_FLAG_PRIVATE;

■      vma_insert(pa, v);
      // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 556) - [public] unmap_pages_free

```
■      unmap_pages_free(pa, pa->brk, nb);
■      as_unlock();
■      return pa->brk;
■  }

■      v->start = pa->brk_base;
■      v->end = nb;
■      v->prot = PROT_READ | PROT_WRITE;
■      v->flags = VMA_FLAG_ANON | VMA_FLAG_PRIVATE;

■      vma_insert(pa, v);
■  } else {
      // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 557) - [public] as_unlock

```
■      as_unlock();
■      return pa->brk;
■  }

■      v->start = pa->brk_base;
■      v->end = nb;
■      v->prot = PROT_READ | PROT_WRITE;
■      v->flags = VMA_FLAG_ANON | VMA_FLAG_PRIVATE;

■      vma_insert(pa, v);
■  } else {
■      v->end = nb;
      // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 566) - [public] vma_insert

```
■      vma_insert(pa, v);
■  } else {
■      v->end = nb;
■  }

■      pa->brk = nb;
■  } else if (nb < pa->brk) {
■      aspace_unmap(pa, nb, pa->brk - nb);
■      pa->brk = nb;
■  }

■  uint64_t cur = pa->brk;
      // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 573) - [public] aspace_unmap

```
■      aspace_unmap(pa, nb, pa->brk - nb);
■      pa->brk = nb;
■  }

■  uint64_t cur = pa->brk;

■  as_unlock();

■  return cur;
}
```

kernel_Workspace/kernel/src/mm/process/mmap.c (line 17) - [public] aspace_map_at

```
    return aspace_map_at(pa, addr, len, prot);
}
```

```
return aspace_map_anon(pa, addr, len, prot);
}
```

```
bool munmap(proc_aspace_t *pa, uint64_t addr, size_t len)
{
    return aspace_unmap(pa, addr, len);
}
```

```
bool mprotect(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t prot)
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/mmap.c (line 20) - [public] aspace_map_anon

```
return aspace_map_anon(pa, addr, len, prot);
}
```

```
bool munmap(proc_aspace_t *pa, uint64_t addr, size_t len)
{
    return aspace_unmap(pa, addr, len);
}
```

```
bool mprotect(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t prot)
{
    return aspace_protect_checked(pa, addr, len,
    prot, perm_sanitiz(e prot),
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/process/mmap.c (line 25) - [public] aspace_unmap

```
return aspace_unmap(pa, addr, len);
}
```

```
bool mprotect(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t prot)
{
    return aspace_protect_checked(pa, addr, len,
    prot, perm_sanitiz(e prot),
    perm_mprotect_allowed);
}
```

kernel_Workspace/kernel/src/mm/protection/guard.c (line 6) - [public] aspace_reserve_at

```
return aspace_reserve_at(pa, addr, len, VMA_FLAG_GUARD);
}
```

```
bool guard_user_stack(proc_aspace_t *pa)
{
    uint64_t stack_base = aspace_stack_base();

```

```
    if (stack_base < ARCH_PAGE_SIZE) {
        return false;
    }

```

```
return aspace_reserve_at(pa,
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/protection/isolation.c (line 5) - [public] kprintf

```
extern void kprintf(const char *fmt, ...);
```

```
static void cpuid(uint32_t leaf,
    uint32_t subleaf,
    uint32_t *a,
    uint32_t *b,
    uint32_t *c,
    uint32_t *d)
{
    __asm__ volatile(
    "cpuid"
    : "=a"(*a), "=b"(*b), "=c"(*c), "=d"(*d)
    // ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/mm/protection/isolation.c (line 24) - [public] volatile
    ■__asm__ volatile("mov %%cr4, %0" : "=r"(v));

    ■return v;
    }

    static void write_cr4(uint64_t v)
    {
    ■__asm__ volatile("mov %0, %%cr4" :: "r"(v) : "memory");
    }

    bool isolation_has_smep(void)
    {
        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/protection/isolation.c (line 31) - [public] volatile
    ■__asm__ volatile("mov %0, %%cr4" :: "r"(v) : "memory");
    }

    bool isolation_has_smep(void)
    {
    ■uint32_t a, b, c, d;

    ■cpuid(7, 0, &a, &b, &c, &d);

    ■return (b >> 7) & 1;
    }

        // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/protection/isolation.c (line 81) - [public] if
    ■    if (!pml_entry_present(e)) {
    ■        continue;
    ■    }

    ■    if (e & PTE_USER) {
    ■        violations++;
    ■    }

    ■    if (!pml_entry_large(e) && level > PML_LEVEL_PT) {
    ■        violations =
    ■            audit_tables(pml_entry_addr(e), level - 1, violations);
    ■    }
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/protection/isolation.c (line 85) - [public] if
    ■    if (e & PTE_USER) {
    ■        violations++;
    ■    }

    ■    if (!pml_entry_large(e) && level > PML_LEVEL_PT) {
    ■        violations =
    ■            audit_tables(pml_entry_addr(e), level - 1, violations);
    ■    }
    ■}

    ■return violations;
    }
    // ... (truncated, see source file for more)

```

kernel_workspace/kernel/src/mm/protection/isolation.c (line 89) - [public] if

```
■ if (!pml_entry_large(e) && level > PML_LEVEL_PT) {  
■     violations =  
■         audit_tables(pml_entry_addr(e), level - 1, violations);  
■ }  
■ }
```

```
■ return violations;  
}
```

```
size_t isolation_audit_kernel(void)
```

```
{  
■ uint64_t pml4 = paging_read_cr3();  
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/protection/isolation.c (line 91) - [public] audit_tables

```
■         audit_tables(pml_entry_addr(e), level - 1, violations);  
■     }  
■ }
```

```
■ return violations;  
}
```

```
size_t isolation_audit_kernel(void)
```

```
{  
■ uint64_t pml4 = paging_read_cr3();  
■ uint64_t *t = pml_table_ptr(pml4);  
  
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/protection/isolation.c (line 108) - [public] if

```
■ if (!pml_entry_present(e)) {  
■     continue;  
■ }  
  
■ if (e & PTE_USER) {  
■     violations++;  
■ }  
  
■ if (!pml_entry_large(e)) {  
■     violations =  
■         audit_tables(pml_entry_addr(e), PML_LEVEL_PDPT, violations);  
■ }  
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/protection/isolation.c (line 112) - [public] if

```
■ if (e & PTE_USER) {  
■     violations++;  
■ }  
  
■ if (!pml_entry_large(e)) {  
■     violations =  
■         audit_tables(pml_entry_addr(e), PML_LEVEL_PDPT, violations);  
■ }  
■ }
```

```
■ return violations;  
}  
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/protection/isolation.c (line 116) - [public] if

```
■    if (!pml_entry_large(e)) {
■        violations =
■            audit_tables(pml_entry_addr(e), PML_LEVEL_PDPT, violations);
■    }
■}

■return violations;
}

bool isolation_init(void)
{
■paging_enable_nx();
    // ... (truncated, see source file for more)
```

kernel_workspace/kernel/src/mm/protection/isolation.c (line 118) - [public] audit_tables

```
■        audit_tables(pml_entry_addr(e), PML_LEVEL_PDPT, violations);
■    }
■}

■return violations;
}

bool isolation_init(void)
{
■paging_enable_nx();
■isolation_enable_smep();
■isolation_enable_smap();
    // ... (truncated, see source file for more)
```

kstd/src/kernel/audio.c (line 3) - [public] k_audio_play

```
extern int32_t k_audio_play(uint64_t phys, uint32_t len);
extern int32_t k_audio_stop(void);
extern int32_t k_audio_jack(void);
extern int32_t k_audio_amp(int32_t on);
extern uint64_t kvirt_to_phys(void *ptr);

tr_status_t tr_audio_play(const void *data, uint32_t len)
{
    uint64_t phys = kvirt_to_phys((void *)data);

    if (phys == (uint64_t)-1) {
        return TR_ERR_INVALID;
        // ... (truncated, see source file for more)
```

kstd/src/kernel/audio.c (line 4) - [public] k_audio_stop

```
extern int32_t k_audio_stop(void);
extern int32_t k_audio_jack(void);
extern int32_t k_audio_amp(int32_t on);
extern uint64_t kvirt_to_phys(void *ptr);

tr_status_t tr_audio_play(const void *data, uint32_t len)
{
    uint64_t phys = kvirt_to_phys((void *)data);

    if (phys == (uint64_t)-1) {
        return TR_ERR_INVALID;
    }
    // ... (truncated, see source file for more)
```

```

kstd/src/kernel/audio.c (line 5) - [public] k_audio_jack
extern int32_t k_audio_jack(void);
extern int32_t k_audio_amp(int32_t on);
extern uint64_t kvirt_to_phys(void *ptr);

tr_status_t tr_audio_play(const void *data, uint32_t len)
{
    uint64_t phys = kvirt_to_phys((void *)data);

    if (phys == (uint64_t)-1) {
        return TR_ERR_INVALID;
    }

    // ... (truncated, see source file for more)
}

kstd/src/kernel/audio.c (line 6) - [public] k_audio_amp
extern int32_t k_audio_amp(int32_t on);
extern uint64_t kvirt_to_phys(void *ptr);

tr_status_t tr_audio_play(const void *data, uint32_t len)
{
    uint64_t phys = kvirt_to_phys((void *)data);

    if (phys == (uint64_t)-1) {
        return TR_ERR_INVALID;
    }

    return k_audio_play(phys, len) == 0 ? TR_OK : TR_ERR_IO;
    // ... (truncated, see source file for more)
}

kstd/src/kernel/audio.c (line 7) - [public] kvirt_to_phys
extern uint64_t kvirt_to_phys(void *ptr);

tr_status_t tr_audio_play(const void *data, uint32_t len)
{
    uint64_t phys = kvirt_to_phys((void *)data);

    if (phys == (uint64_t)-1) {
        return TR_ERR_INVALID;
    }

    return k_audio_play(phys, len) == 0 ? TR_OK : TR_ERR_IO;
}
// ... (truncated, see source file for more)

kstd/src/kernel/audio.c (line 13) - [public] if
    if (phys == (uint64_t)-1) {
        return TR_ERR_INVALID;
    }

    return k_audio_play(phys, len) == 0 ? TR_OK : TR_ERR_IO;
}

tr_status_t tr_audio_stop(void)
{
    k_audio_stop();
    return TR_OK;
}
// ... (truncated, see source file for more)

```

kstd/src/kernel/audio.c (line 22) - [public] k_audio_stop

```
    k_audio_stop();
    return TR_OK;
}

tr_status_t tr_audio_jack(bool *present)
{
    if (!present) {
        return TR_ERR_INVALID;
    }

    *present = k_audio_jack() == 1;
    return TR_OK;
    // ... (truncated, see source file for more)
```

kstd/src/kernel/audio.c (line 28) - [public] if

```
    if (!present) {
        return TR_ERR_INVALID;
    }

    *present = k_audio_jack() == 1;
    return TR_OK;
}
```

```
tr_status_t tr_audio_amp(bool on)
{
    k_audio_amp(on ? 1 : 0);
    return TR_OK;
    // ... (truncated, see source file for more)
```

kstd/src/kernel/audio.c (line 38) - [public] k_audio_amp

```
    k_audio_amp(on ? 1 : 0);
    return TR_OK;
}
```

kstd/src/kernel/bt.c (line 3) - [public] bt_ready

```
extern bool bt_ready(void);
extern void bt_info(uint8_t *ver, uint8_t *addr);
extern bool bt_hci_cmd(uint16_t op, const uint8_t *p, uint8_t len);
extern bool bt_event_poll(uint8_t *buf, uint8_t *len);
extern bool bt_acl_send(const uint8_t *d, uint16_t len);
extern bool bt_acl_recv(uint8_t *d, uint16_t *len);

tr_status_t tr_bt_info(uint8_t *hci_ver, uint8_t *addr6)
{
    if (!bt_ready()) {
        return TR_ERR_IO;
    }
    // ... (truncated, see source file for more)
```

kstd/src/kernel/bt.c (line 4) - [public] bt_info

```
extern void bt_info(uint8_t *ver, uint8_t *addr);
extern bool bt_hci_cmd(uint16_t op, const uint8_t *p, uint8_t len);
extern bool bt_event_poll(uint8_t *buf, uint8_t *len);
extern bool bt_acl_send(const uint8_t *d, uint16_t len);
extern bool bt_acl_recv(uint8_t *d, uint16_t *len);

tr_status_t tr_bt_info(uint8_t *hci_ver, uint8_t *addr6)
{
    if (!bt_ready()) {
        return TR_ERR_IO;
    }

    // ... (truncated, see source file for more)
```

```
kstd/src/kernel/bt.c (line 5) - [public] bt_hci_cmd
extern bool bt_hci_cmd(uint16_t op, const uint8_t *p, uint8_t len);
extern bool bt_event_poll(uint8_t *buf, uint8_t *len);
extern bool bt_acl_send(const uint8_t *d, uint16_t len);
extern bool bt_acl_recv(uint8_t *d, uint16_t *len);

tr_status_t tr_bt_info(uint8_t *hci_ver, uint8_t *addr6)
{
    if (!bt_ready()) {
        return TR_ERR_IO;
    }

    bt_info(hci_ver, addr6);
    // ... (truncated, see source file for more)
}
```

```
kstd/src/kernel/bt.c (line 6) - [public] bt_event_poll
extern bool bt_event_poll(uint8_t *buf, uint8_t *len);
extern bool bt_acl_send(const uint8_t *d, uint16_t len);
extern bool bt_acl_recv(uint8_t *d, uint16_t *len);

tr_status_t tr_bt_info(uint8_t *hci_ver, uint8_t *addr6)
{
    if (!bt_ready()) {
        return TR_ERR_IO;
    }

    bt_info(hci_ver, addr6);
    return TR_OK;
    // ... (truncated, see source file for more)
}
```

```
kstd/src/kernel/bt.c (line 7) - [public] bt_acl_send
extern bool bt_acl_send(const uint8_t *d, uint16_t len);
extern bool bt_acl_recv(uint8_t *d, uint16_t *len);

tr_status_t tr_bt_info(uint8_t *hci_ver, uint8_t *addr6)
{
    if (!bt_ready()) {
        return TR_ERR_IO;
    }

    bt_info(hci_ver, addr6);
    return TR_OK;
}
// ... (truncated, see source file for more)
```

```
kstd/src/kernel/bt.c (line 8) - [public] bt_acl_recv
extern bool bt_acl_recv(uint8_t *d, uint16_t *len);

tr_status_t tr_bt_info(uint8_t *hci_ver, uint8_t *addr6)
{
    if (!bt_ready()) {
        return TR_ERR_IO;
    }

    bt_info(hci_ver, addr6);
    return TR_OK;
}
// ... (truncated, see source file for more)
```


kstd/src/kernel/bt.c (line 12) - [public] if

```
    if (!bt_ready()) {
        return TR_ERR_IO;
    }

    bt_info(hci_ver, addr6);
    return TR_OK;
}

tr_status_t tr_bt_cmd(uint16_t opcode, const void *params, uint8_t len)
{
    return bt_hci_cmd(opcode, params, len) ? TR_OK : TR_ERR_IO;
}

// ... (truncated, see source file for more)
```

kstd/src/kernel/bt.c (line 16) - [public] bt_info

```
    bt_info(hci_ver, addr6);
    return TR_OK;
}

tr_status_t tr_bt_cmd(uint16_t opcode, const void *params, uint8_t len)
{
    return bt_hci_cmd(opcode, params, len) ? TR_OK : TR_ERR_IO;
}

tr_status_t tr_bt_evt(void *buf, uint8_t *len)
{
    return bt_event_poll(buf, len) ? TR_OK : TR_ERR_TIMEOUT;
}

// ... (truncated, see source file for more)
```

kstd/src/kernel/fs.c (line 3) - [public] k_fs_read

```
extern int32_t k_fs_read(const char *path, void *buf, uint32_t cap);
extern int32_t k_fs_exists(const char *path);

tr_status_t tr_fs_read(const char *path, void *buf, uint32_t cap, uint32_t *out)
{
    int32_t n = k_fs_read(path, buf, cap);

    if (n < 0) {
        return TR_ERR_NOTFOUND;
    }

    if (out) {
        // ... (truncated, see source file for more)
    }
}
```

kstd/src/kernel/fs.c (line 4) - [public] k_fs_exists

```
extern int32_t k_fs_exists(const char *path);

tr_status_t tr_fs_read(const char *path, void *buf, uint32_t cap, uint32_t *out)
{
    int32_t n = k_fs_read(path, buf, cap);

    if (n < 0) {
        return TR_ERR_NOTFOUND;
    }

    if (out) {
        *out = (uint32_t)n;
        // ... (truncated, see source file for more)
    }
}
```

```

kstd/src/kernel/fs.c (line 10) - [public] if
    if (n < 0) {
        return TR_ERR_NOTFOUND;
    }

    if (out) {
        *out = (uint32_t)n;
    }

    return TR_OK;
}

tr_status_t tr_fs_exists(const char *path)
    // ... (truncated, see source file for more)

kstd/src/kernel/fs.c (line 14) - [public] if
    if (out) {
        *out = (uint32_t)n;
    }

    return TR_OK;
}

tr_status_t tr_fs_exists(const char *path)
{
    return k_fs_exists(path) == 1 ? TR_OK : TR_ERR_NOTFOUND;
}

kstd/src/kernel/input.c (line 3) - [public] k_input_key
extern int32_t k_input_key(void);

int32_t tr_input_key(void)
{
    return k_input_key();
}

kstd/src/kernel/input.c (line 7) - [public] k_input_key
    return k_input_key();
}

kstd/src/kernel/log.c (line 3) - [public] kprintf
extern void kprintf(const char *fmt, ...);

void tr_log(const char *s)
{
    kprintf("%s\n", s);
}

kstd/src/kernel/log.c (line 7) - [public] kprintf
    kprintf("%s\n", s);
}

kstd/src/kernel/mem.c (line 4) - [public] kfree
extern void kfree(void *ptr);
extern bool vmm_map_device(uint64_t phys, unsigned long len, uint64_t *virt);
extern uint64_t kvirt_to_phys(void *ptr);

void *tr_alloc(uint32_t bytes)
{
    return kmalloc(bytes);
}

void tr_free(void *ptr, uint32_t bytes)
{
    (void)bytes;
    // ... (truncated, see source file for more)
}

```

```

kstd/src/kernel/mem.c (line 5) - [public] vmm_map_device
extern bool vmm_map_device(uint64_t phys, unsigned long len, uint64_t *virt);
extern uint64_t kvirt_to_phys(void *ptr);

void *tr_alloc(uint32_t bytes)
{
    return kmalloc(bytes);
}

void tr_free(void *ptr, uint32_t bytes)
{
    (void)bytes;
    kfree(ptr);
    // ... (truncated, see source file for more)
}

kstd/src/kernel/mem.c (line 6) - [public] kvirt_to_phys
extern uint64_t kvirt_to_phys(void *ptr);

void *tr_alloc(uint32_t bytes)
{
    return kmalloc(bytes);
}

void tr_free(void *ptr, uint32_t bytes)
{
    (void)bytes;
    kfree(ptr);
}
// ... (truncated, see source file for more)

kstd/src/kernel/mem.c (line 10) - [public] kmalloc
return kmalloc(bytes);
}

void tr_free(void *ptr, uint32_t bytes)
{
    (void)bytes;
    kfree(ptr);
}

tr_status_t tr_map_mmio(uint64_t phys, uint32_t len, void **out_va)
{
    uint64_t v = 0;
    // ... (truncated, see source file for more)
}

kstd/src/kernel/mem.c (line 16) - [public] kfree
kfree(ptr);
}

tr_status_t tr_map_mmio(uint64_t phys, uint32_t len, void **out_va)
{
    uint64_t v = 0;

    if (vmm_map_device(phys, len, &v)) {
        *out_va = (void *)v;
        return TR_OK;
    }

    // ... (truncated, see source file for more)
}

kstd/src/kernel/mem.c (line 23) - [public] if
if (vmm_map_device(phys, len, &v)) {
    *out_va = (void *)v;
    return TR_OK;
}

return TR_ERR_IO;
}

```

kstd/src/kernel/video.c (line 8) - [public] tr_video_init

```
tr_status_t tr_video_init(void) {
    if (dp_ready()) {
        use_dp = true;
        return TR_OK;
    }
    if (hdmi_ready()) {
        use_dp = false;
        return TR_OK;
    }
    return TR_ERR_IO;
}
```

// ... (truncated, see source file for more)

kstd/src/kernel/video.c (line 9) - [public] if

```
if (dp_ready()) {
    use_dp = true;
    return TR_OK;
}
if (hdmi_ready()) {
    use_dp = false;
    return TR_OK;
}
return TR_ERR_IO;
}
```

```
bool tr_video_is_ready(void) {
    // ... (truncated, see source file for more)
```

kstd/src/kernel/video.c (line 13) - [public] if

```
if (hdmi_ready()) {
    use_dp = false;
    return TR_OK;
}
return TR_ERR_IO;
}
```

```
bool tr_video_is_ready(void) {
    return use_dp ? dp_ready() : hdmi_ready();
}
```

```
tr_status_t tr_video_get_caps(tr_video_caps_t *caps) {
    // ... (truncated, see source file for more)
```

kstd/src/kernel/video.c (line 20) - [public] tr_video_is_ready

```
bool tr_video_is_ready(void) {
    return use_dp ? dp_ready() : hdmi_ready();
}
```

```
tr_status_t tr_video_get_caps(tr_video_caps_t *caps) {
    if (!caps) return TR_ERR_INVALID;

    if (use_dp) {
        dp_caps(&caps->fb_phys, &caps->width, &caps->height, &caps->stride);
        caps->ready = dp_ready();
    } else {
        hdmi_caps_raw(&caps->width, &caps->height, &caps->stride, &caps->fb_phys);
        // ... (truncated, see source file for more)
```

kstd/src/kernel/video.c (line 24) - [public] tr_video_get_caps

```
tr_status_t tr_video_get_caps(tr_video_caps_t *caps) {
    if (!caps) return TR_ERR_INVALID;

    if (use_dp) {
        dp_caps(&caps->fb_phys, &caps->width, &caps->height, &caps->stride);
        caps->ready = dp_ready();
    } else {
        hdmi_caps_raw(&caps->width, &caps->height, &caps->stride, &caps->fb_phys);
        caps->ready = hdmi_ready();
    }
    return TR_OK;
}

// ... (truncated, see source file for more)
```

kstd/src/kernel/video.c (line 27) - [public] if

```
if (use_dp) {
    dp_caps(&caps->fb_phys, &caps->width, &caps->height, &caps->stride);
    caps->ready = dp_ready();
} else {
    hdmi_caps_raw(&caps->width, &caps->height, &caps->stride, &caps->fb_phys);
    caps->ready = hdmi_ready();
}
return TR_OK;
}

tr_status_t tr_video_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h) {
    if (!tr_video_is_ready()) return TR_ERR_IO;
    // ... (truncated, see source file for more)
```

kstd/src/kernel/video.c (line 28) - [public] dp_caps

```
dp_caps(&caps->fb_phys, &caps->width, &caps->height, &caps->stride);
caps->ready = dp_ready();
} else {
    hdmi_caps_raw(&caps->width, &caps->height, &caps->stride, &caps->fb_phys);
    caps->ready = hdmi_ready();
}
return TR_OK;
}

tr_status_t tr_video_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h) {
    if (!tr_video_is_ready()) return TR_ERR_IO;

    // ... (truncated, see source file for more)
```

kstd/src/kernel/video.c (line 31) - [public] hdmi_caps_raw

```
hdmi_caps_raw(&caps->width, &caps->height, &caps->stride, &caps->fb_phys);
caps->ready = hdmi_ready();
}
return TR_OK;
}

tr_status_t tr_video_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h) {
    if (!tr_video_is_ready()) return TR_ERR_IO;

    uint64_t seq = 0;
    if (use_dp) {
        seq = dp_submit_fill(color, x, y, w, h);
    }
    // ... (truncated, see source file for more)
```

```

kstd/src/kernel/video.c (line 37) - [public] tr_video_fill
    tr_status_t tr_video_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h) {
        if (!tr_video_is_ready()) return TR_ERR_IO;

        uint64_t seq = 0;
        if (use_dp) {
            seq = dp_submit_fill(color, x, y, w, h);
        } else {
            seq = hdmi_submit_fill(color, x, y, w, h);
        }

        return (seq > 0) ? TR_OK : TR_ERR_BUSY;
    }
    // ... (truncated, see source file for more)

kstd/src/kernel/video.c (line 41) - [public] if
    if (use_dp) {
        seq = dp_submit_fill(color, x, y, w, h);
    } else {
        seq = hdmi_submit_fill(color, x, y, w, h);
    }

    return (seq > 0) ? TR_OK : TR_ERR_BUSY;
}

/* ... reszta funkcji mapuje 1:1 na hdmi/dp ... */

kstd/src/user/audio.c (line 7) - [public] ds_poll
extern void ds_poll(void);
extern int ds_take(uint64_t id, ds_msg_t *out);

static void *bounce = TR_NULL;
static uint64_t bounce_phys = 0;

tr_status_t tr_audio_play(const void *data, uint32_t len)
{
    if (bounce == TR_NULL) {
        bounce = tr_alloc(4096 * 4);

        if (bounce == TR_NULL) {
            // ... (truncated, see source file for more)
        }
    }
}

kstd/src/user/audio.c (line 8) - [public] ds_take
extern int ds_take(uint64_t id, ds_msg_t *out);

static void *bounce = TR_NULL;
static uint64_t bounce_phys = 0;

tr_status_t tr_audio_play(const void *data, uint32_t len)
{
    if (bounce == TR_NULL) {
        bounce = tr_alloc(4096 * 4);

        if (bounce == TR_NULL) {
            return TR_ERR_NOMEM;
        }
        // ... (truncated, see source file for more)
    }
}

kstd/src/user/audio.c (line 15) - [public] if
    if (bounce == TR_NULL) {
        bounce = tr_alloc(4096 * 4);

        if (bounce == TR_NULL) {
            return TR_ERR_NOMEM;
        }
    }

    uint64_t id = ds_call(SVC_SYS, OP_PAGEPHYS, (uint64_t)bounce, 0, 0);
    ds_poll();

    ds_msg_t r;

    // ... (truncated, see source file for more)

```

```

kstd/src/user/audio.c (line 18) - [public] if
    if (bounce == TR_NULL) {
        return TR_ERR_NOMEM;
    }

    uint64_t id = ds_call(SVC_SYS, OP_PAGEPHYS, (uint64_t)bounce, 0, 0);
    ds_poll();

    ds_msg_t r;

    if (!ds_take(id, &r) || r.status != 0) {
        return TR_ERR_IO;
    }
    // ... (truncated, see source file for more)

kstd/src/user/audio.c (line 23) - [public] ds_poll
    ds_poll();

    ds_msg_t r;

    if (!ds_take(id, &r) || r.status != 0) {
        return TR_ERR_IO;
    }

    bounce_phys = r.arg0;
}

if (len > 4096 * 4) {
    // ... (truncated, see source file for more)

kstd/src/user/audio.c (line 27) - [public] if
    if (!ds_take(id, &r) || r.status != 0) {
        return TR_ERR_IO;
    }

    bounce_phys = r.arg0;
}

if (len > 4096 * 4) {
    len = 4096 * 4;
}

uint8_t *dst = (uint8_t *)bounce;
// ... (truncated, see source file for more)

kstd/src/user/audio.c (line 34) - [public] if
    if (len > 4096 * 4) {
        len = 4096 * 4;
    }

    uint8_t *dst = (uint8_t *)bounce;
    const uint8_t *src = (const uint8_t *)data;

    for (uint32_t i = 0; i < len; i++) {
        dst[i] = src[i];
    }

    uint64_t id = ds_call(SVC_AUDIO, AUD_PLAY, bounce_phys, len, 0);
    // ... (truncated, see source file for more)

```

```

kstd/src/user/audio.c (line 46) - [public] ds_poll
    ds_poll();

    ds_msg_t r;

    if (ds_take(id, &r) && r.status == 0) {
        return TR_OK;
    }

    return TR_ERR_DENIED;
}

tr_status_t tr_audio_stop(void)
    // ... (truncated, see source file for more)

kstd/src/user/audio.c (line 50) - [public] if
    if (ds_take(id, &r) && r.status == 0) {
        return TR_OK;
    }

    return TR_ERR_DENIED;
}

tr_status_t tr_audio_stop(void)
{
    ds_call(SVC_AUDIO, AUD_STOP, 0, 0, 0);
    return TR_OK;
}
    // ... (truncated, see source file for more)

kstd/src/user/audio.c (line 59) - [public] ds_call
    ds_call(SVC_AUDIO, AUD_STOP, 0, 0, 0);
    return TR_OK;
}

tr_status_t tr_audio_jack(bool *present)
{
    if (!present) {
        return TR_ERR_INVALID;
    }

    uint64_t id = ds_call(SVC_AUDIO, AUD_JACK, 0, 0, 0);
    ds_poll();
    // ... (truncated, see source file for more)

kstd/src/user/audio.c (line 65) - [public] if
    if (!present) {
        return TR_ERR_INVALID;
    }

    uint64_t id = ds_call(SVC_AUDIO, AUD_JACK, 0, 0, 0);
    ds_poll();

    ds_msg_t r;

    if (ds_take(id, &r) && r.status == 0) {
        *present = (r.arg0 & 1) != 0;
        return TR_OK;
    }
    // ... (truncated, see source file for more)

```



```

kstd/src/user/audio.c (line 70) - [public] ds_poll
    ds_poll();

    ds_msg_t r;

    if (ds_take(id, &r) && r.status == 0) {
        *present = (r.arg0 & 1) != 0;
        return TR_OK;
    }

    return TR_ERR_IO;
}

// ... (truncated, see source file for more)

kstd/src/user/audio.c (line 74) - [public] if
    if (ds_take(id, &r) && r.status == 0) {
        *present = (r.arg0 & 1) != 0;
        return TR_OK;
    }

    return TR_ERR_IO;
}

tr_status_t tr_audio_amp(bool on)
{
    ds_call(SVC_AUDIO, AUD_AMP, on ? 1 : 0, 0, 0);
    return TR_OK;
    // ... (truncated, see source file for more)

kstd/src/user/audio.c (line 84) - [public] ds_call
    ds_call(SVC_AUDIO, AUD_AMP, on ? 1 : 0, 0, 0);
    return TR_OK;
}

kstd/src/user/bt.c (line 6) - [public] ds_poll
extern void ds_poll(void);
extern int ds_take(uint64_t id, ds_msg_t *out);

static volatile uint8_t *scratch = (volatile uint8_t *)DS_SCRATCH_VA;

tr_status_t tr_bt_info(uint8_t *hci_ver, uint8_t *addr6)
{
    uint64_t id = ds_call(SVC_BT, BT_INFO, 0, 0, 0);
    ds_poll();

    ds_msg_t r;

    // ... (truncated, see source file for more)

kstd/src/user/bt.c (line 7) - [public] ds_take
extern int ds_take(uint64_t id, ds_msg_t *out);

static volatile uint8_t *scratch = (volatile uint8_t *)DS_SCRATCH_VA;

tr_status_t tr_bt_info(uint8_t *hci_ver, uint8_t *addr6)
{
    uint64_t id = ds_call(SVC_BT, BT_INFO, 0, 0, 0);
    ds_poll();

    ds_msg_t r;

    if (!ds_take(id, &r) || r.status != 0) {
        // ... (truncated, see source file for more)
    }
}

```

```

kstd/src/user/bt.c (line 14) - [public] ds_poll
    ds_poll();

    ds_msg_t r;

    if (!ds_take(id, &r) || r.status != 0) {
        return TR_ERR_IO;
    }

    *hci_ver = (uint8_t)r.arg0;

    for (int i = 0; i < 6; i++) {
        addr6[i] = (uint8_t)(r.arg1 >> (i * 8));
    }
    // ... (truncated, see source file for more)

```

```

kstd/src/user/bt.c (line 18) - [public] if
    if (!ds_take(id, &r) || r.status != 0) {
        return TR_ERR_IO;
    }

    *hci_ver = (uint8_t)r.arg0;

    for (int i = 0; i < 6; i++) {
        addr6[i] = (uint8_t)(r.arg1 >> (i * 8));
    }

    return TR_OK;
}
// ... (truncated, see source file for more)

```

```

kstd/src/user/bt.c (line 40) - [public] ds_poll
    ds_poll();

    ds_msg_t r;

    if (ds_take(id, &r) && r.status == 0) {
        return TR_OK;
    }

    return TR_ERR_DENIED;
}

tr_status_t tr_bt_evt(void *buf, uint8_t *len)
    // ... (truncated, see source file for more)

```

```

kstd/src/user/bt.c (line 44) - [public] if
    if (ds_take(id, &r) && r.status == 0) {
        return TR_OK;
    }

    return TR_ERR_DENIED;
}

tr_status_t tr_bt_evt(void *buf, uint8_t *len)
{
    uint64_t id = ds_call(SVC_BT, BT_EVT, 0, 0, 0);
    ds_poll();

    // ... (truncated, see source file for more)

```

```

kstd/src/user/bt.c (line 54) - [public] ds_poll
    ds_poll();

    ds_msg_t r;

    if (!ds_take(id, &r) || r.status != 0) {
        return TR_ERR_TIMEOUT;
    }

    uint8_t n = (uint8_t)r.arg0;

    uint8_t *dst = (uint8_t *)buf;

    // ... (truncated, see source file for more)

kstd/src/user/bt.c (line 58) - [public] if
    if (!ds_take(id, &r) || r.status != 0) {
        return TR_ERR_TIMEOUT;
    }

    uint8_t n = (uint8_t)r.arg0;

    uint8_t *dst = (uint8_t *)buf;

    for (uint8_t i = 0; i < n; i++) {
        dst[i] = scratch[i];
    }

    // ... (truncated, see source file for more)

kstd/src/user/bt.c (line 83) - [public] ds_poll
    ds_poll();

    ds_msg_t r;

    if (ds_take(id, &r) && r.status == 0) {
        return TR_OK;
    }

    return TR_ERR_BUSY;
}

tr_status_t tr_bt_acl_recv(void *buf, uint16_t *len)
    // ... (truncated, see source file for more)

kstd/src/user/bt.c (line 87) - [public] if
    if (ds_take(id, &r) && r.status == 0) {
        return TR_OK;
    }

    return TR_ERR_BUSY;
}

tr_status_t tr_bt_acl_recv(void *buf, uint16_t *len)
{
    uint64_t id = ds_call(SVC_BT, BT_ACL_IN, 0, 0, 0);
    ds_poll();

    // ... (truncated, see source file for more)

```

```

kstd/src/user/bt.c (line 97) - [public] ds_poll
    ds_poll();

    ds_msg_t r;

    if (!ds_take(id, &r) || r.status != 0) {
        return TR_ERR_TIMEOUT;
    }

    uint16_t n = (uint16_t)r.arg0;

    uint8_t *dst = (uint8_t *)buf;

    // ... (truncated, see source file for more)

```

```

kstd/src/user/bt.c (line 101) - [public] if
    if (!ds_take(id, &r) || r.status != 0) {
        return TR_ERR_TIMEOUT;
    }

    uint16_t n = (uint16_t)r.arg0;

    uint8_t *dst = (uint8_t *)buf;

    for (uint16_t i = 0; i < n; i++) {
        dst[i] = scratch[i];
    }

    // ... (truncated, see source file for more)

```

```

kstd/src/user/fs.c (line 6) - [public] ds_poll
extern void ds_poll(void);
extern int ds_take(uint64_t id, ds_msg_t *out);

static volatile uint8_t *scratch = (volatile uint8_t *)DS_SCRATCH_VA;

tr_status_t tr_fs_read(const char *path, void *buf, uint32_t cap, uint32_t *out)
{
    uint32_t n = 0;

    while (path[n] && n < 255) {
        scratch[n] = (uint8_t)path[n];
        n++;
    }
    // ... (truncated, see source file for more)

```

```

kstd/src/user/fs.c (line 7) - [public] ds_take
extern int ds_take(uint64_t id, ds_msg_t *out);

static volatile uint8_t *scratch = (volatile uint8_t *)DS_SCRATCH_VA;

tr_status_t tr_fs_read(const char *path, void *buf, uint32_t cap, uint32_t *out)
{
    uint32_t n = 0;

    while (path[n] && n < 255) {
        scratch[n] = (uint8_t)path[n];
        n++;
    }
    // ... (truncated, see source file for more)

```

```

kstd/src/user/fs.c (line 15) - [public] while
    while (path[n] && n < 255) {
        scratch[n] = (uint8_t)path[n];
        n++;
    }

    scratch[n] = 0;

    uint64_t id = ds_call(SVC_FS, FS_READ, n, cap, 0);
    ds_poll();

    ds_msg_t r;

    // ... (truncated, see source file for more)

```

```

kstd/src/user/fs.c (line 23) - [public] ds_poll
    ds_poll();

    ds_msg_t r;

    if (!ds_take(id, &r) || r.status != 0) {
        return TR_ERR_NOTFOUND;
    }

    uint32_t got = (uint32_t)r.arg0;

    if (got > cap) {
        got = cap;
    }
    // ... (truncated, see source file for more)

```

```

kstd/src/user/fs.c (line 27) - [public] if
    if (!ds_take(id, &r) || r.status != 0) {
        return TR_ERR_NOTFOUND;
    }

    uint32_t got = (uint32_t)r.arg0;

    if (got > cap) {
        got = cap;
    }

    uint8_t *dst = (uint8_t *)buf;

    // ... (truncated, see source file for more)

```

```

kstd/src/user/fs.c (line 33) - [public] if
    if (got > cap) {
        got = cap;
    }

    uint8_t *dst = (uint8_t *)buf;

    for (uint32_t i = 0; i < got; i++) {
        dst[i] = scratch[256 + i];
    }

    if (out) {
        *out = got;
    }
    // ... (truncated, see source file for more)

```

```

kstd/src/user/fs.c (line 43) - [public] if
    if (out) {
        *out = got;
    }

    return TR_OK;
}

tr_status_t tr_fs_exists(const char *path)
{
    uint32_t n = 0;

    while (path[n] && n < 255) {
        // ... (truncated, see source file for more)

kstd/src/user/fs.c (line 54) - [public] while
    while (path[n] && n < 255) {
        scratch[n] = (uint8_t)path[n];
        n++;
    }

    scratch[n] = 0;

    uint64_t id = ds_call(SVC_FS, FS_EXISTS, n, 0, 0);
    ds_poll();

    ds_msg_t r;

    // ... (truncated, see source file for more)

kstd/src/user/fs.c (line 62) - [public] ds_poll
    ds_poll();

    ds_msg_t r;

    if (ds_take(id, &r) && r.status == 0 && r.arg0 == 1) {
        return TR_OK;
    }

    return TR_ERR_NOTFOUND;
}

kstd/src/user/fs.c (line 66) - [public] if
    if (ds_take(id, &r) && r.status == 0 && r.arg0 == 1) {
        return TR_OK;
    }

    return TR_ERR_NOTFOUND;
}

kstd/src/user/input.c (line 6) - [public] ds_poll
extern void ds_poll(void);
extern int ds_take(uint64_t id, ds_msg_t *out);

int32_t tr_input_key(void)
{
    uint64_t id = ds_call(SVC_INPUT, IN_KEY_POLL, 0, 0, 0);
    ds_poll();

    ds_msg_t r;

    if (ds_take(id, &r) && r.status == 0 && r.arg0 != 0) {
        return (int32_t)r.arg0;
        // ... (truncated, see source file for more)
    }
}

```

```

kstd/src/user/input.c (line 7) - [public] ds_take
extern int ds_take(uint64_t id, ds_msg_t *out);

int32_t tr_input_key(void)
{
    uint64_t id = ds_call(SVC_INPUT, IN_KEY_POLL, 0, 0, 0);
    ds_poll();

    ds_msg_t r;

    if (ds_take(id, &r) && r.status == 0 && r.arg0 != 0) {
        return (int32_t)r.arg0;
    }
    // ... (truncated, see source file for more)

kstd/src/user/input.c (line 12) - [public] ds_poll
    ds_poll();

    ds_msg_t r;

    if (ds_take(id, &r) && r.status == 0 && r.arg0 != 0) {
        return (int32_t)r.arg0;
    }

    return -1;
}

kstd/src/user/input.c (line 16) - [public] if
    if (ds_take(id, &r) && r.status == 0 && r.arg0 != 0) {
        return (int32_t)r.arg0;
    }

    return -1;
}

kstd/src/user/log.c (line 10) - [public] while
    while (s[n] && n < DS_SCRATCH_SIZE - 1) {
        scratch[n] = s[n];
        n++;
    }

    ds_call(SVC_SYS, OP_LOG, n, 0, 0);
}

kstd/src/user/log.c (line 15) - [public] ds_call
    ds_call(SVC_SYS, OP_LOG, n, 0, 0);
}

kstd/src/user/mem.c (line 6) - [public] ds_poll
extern void ds_poll(void);
extern int ds_take(uint64_t id, ds_msg_t *out);

static uint64_t next_va = 0x44000000ULL;

void *tr_alloc(uint32_t bytes)
{
    uint32_t pages = (bytes + 4095) / 4096;

    uint64_t id = ds_call(SVC_SYS, OP_ALLOC, pages, 0, 0);
    ds_poll();

    // ... (truncated, see source file for more)

```

```

kstd/src/user/mem.c (line 7) - [public] ds_take
extern int ds_take(uint64_t id, ds_msg_t *out);

static uint64_t next_va = 0x44000000ULL;

void *tr_alloc(uint32_t bytes)
{
    uint32_t pages = (bytes + 4095) / 4096;

    uint64_t id = ds_call(SVC_SYS, OP_ALLOC, pages, 0, 0);
    ds_poll();

    ds_msg_t r;
    // ... (truncated, see source file for more)

kstd/src/user/mem.c (line 16) - [public] ds_poll
    ds_poll();

    ds_msg_t r;

    if (ds_take(id, &r) && r.status == 0) {
        return (void *)r.arg0;
    }

    return TR_NULL;
}

void tr_free(void *ptr, uint32_t bytes)
    // ... (truncated, see source file for more)

kstd/src/user/mem.c (line 20) - [public] if
    if (ds_take(id, &r) && r.status == 0) {
        return (void *)r.arg0;
    }

    return TR_NULL;
}

void tr_free(void *ptr, uint32_t bytes)
{
    (void)bytes;
    ds_call(SVC_SYS, OP_FREE, (uint64_t)ptr, 0, 0);
}
    // ... (truncated, see source file for more)

kstd/src/user/mem.c (line 30) - [public] ds_call
    ds_call(SVC_SYS, OP_FREE, (uint64_t)ptr, 0, 0);
}

tr_status_t tr_map_mmio(uint64_t phys, uint32_t len, void **out_va)
{
    uint64_t va = next_va;
    next_va += (len + 4095) & ~4095ULL;

    uint64_t id = ds_call(SVC_SYS, OP_MAPMMIO, phys, len, va);
    ds_poll();

    ds_msg_t r;
    // ... (truncated, see source file for more)

```



```

kstd/src/user/mem.c (line 39) - [public] ds_poll
    ds_poll();

    ds_msg_t r;

    if (ds_take(id, &r) && r.status == 0) {
        *out_va = (void *)va;
        return TR_OK;
    }

    return TR_ERR_DENIED;
}

```

```

kstd/src/user/mem.c (line 43) - [public] if
    if (ds_take(id, &r) && r.status == 0) {
        *out_va = (void *)va;
        return TR_OK;
    }

    return TR_ERR_DENIED;
}

```

```

kstd/src/user/video.c (line 6) - [public] tr_video_init
/* Zakładamy, że ds_call i ds_take są dostępne z dsclient.c */

tr_status_t tr_video_init(void) {
    /* W user-space init robi kernel, tu tylko sprawdzamy stan */
    return tr_video_is_ready() ? TR_OK : TR_ERR_IO;
}

bool tr_video_is_ready(void) {
    uint64_t id = ds_call(SVC_VIDEO, VID_HDMI_CAPS, 0, 0, 0);
    ds_poll();
    ds_msg_t r;
    if (ds_take(id, &r) && r.status == 0) {
        return true;
    }
    // ... (truncated, see source file for more)
}

```

```

kstd/src/user/video.c (line 11) - [public] tr_video_is_ready

bool tr_video_is_ready(void) {
    uint64_t id = ds_call(SVC_VIDEO, VID_HDMI_CAPS, 0, 0, 0);
    ds_poll();
    ds_msg_t r;
    if (ds_take(id, &r) && r.status == 0) {
        return true;
    }
    return false;
}

tr_status_t tr_video_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h) {
    uint64_t arg1 = ((uint64_t)x) | ((uint64_t)y << 16);
    // ... (truncated, see source file for more)
}

```

```

kstd/src/user/video.c (line 13) - [public] ds_poll

    ds_poll();
    ds_msg_t r;
    if (ds_take(id, &r) && r.status == 0) {
        return true;
    }
    return false;
}

tr_status_t tr_video_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h) {
    uint64_t arg1 = ((uint64_t)x) | ((uint64_t)y << 16);
    uint64_t arg2 = ((uint64_t)w) | ((uint64_t)h << 16);

    // ... (truncated, see source file for more)
}

```

```

kstd/src/user/video.c (line 15) - [public] if
    if (ds_take(id, &r) && r.status == 0) {
        return true;
    }
    return false;
}

tr_status_t tr_video_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h) {
    uint64_t arg1 = ((uint64_t)x) | ((uint64_t)y << 16);
    uint64_t arg2 = ((uint64_t)w) | ((uint64_t)h << 16);

    uint64_t id = ds_call(SVC_VIDEO, VID_HDMI_FILL, color, arg1, arg2);
    ds_poll();
    // ... (truncated, see source file for more)

```

```

kstd/src/user/video.c (line 21) - [public] tr_video_fill
tr_status_t tr_video_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h) {
    uint64_t arg1 = ((uint64_t)x) | ((uint64_t)y << 16);
    uint64_t arg2 = ((uint64_t)w) | ((uint64_t)h << 16);

    uint64_t id = ds_call(SVC_VIDEO, VID_HDMI_FILL, color, arg1, arg2);
    ds_poll();

    ds_msg_t r;
    if (ds_take(id, &r)) {
        if (r.status == 0) return TR_OK;
        if (r.status == -1) return TR_ERR_DENIED; // Autoryzacja/Budzet
        return TR_ERR_IO;
    }
    // ... (truncated, see source file for more)

```

```

kstd/src/user/video.c (line 26) - [public] ds_poll
ds_poll();

ds_msg_t r;
if (ds_take(id, &r)) {
    if (r.status == 0) return TR_OK;
    if (r.status == -1) return TR_ERR_DENIED; // Autoryzacja/Budzet
    return TR_ERR_IO;
}
return TR_ERR_TIMEOUT;
}

/* ... reszta funkcji serializuje argumenty do DsMsg ... */

```

```

kstd/src/user/video.c (line 29) - [public] if
    if (ds_take(id, &r)) {
        if (r.status == 0) return TR_OK;
        if (r.status == -1) return TR_ERR_DENIED; // Autoryzacja/Budzet
        return TR_ERR_IO;
    }
    return TR_ERR_TIMEOUT;
}

/* ... reszta funkcji serializuje argumenty do DsMsg ... */

```

```

libs/dsclient.c (line 30) - [public] if
    if (next == d2k->tail) {
        return 0;
    }

    volatile ds_msg_t *m = &d2k_msgs[head];

    m->id = id;
    m->cmd = svc_cmd(class, op);
    m->flags = 1;
    m->arg0 = a0;
    m->arg1 = a1;
    m->arg2 = a2;
    // ... (truncated, see source file for more)

```

```

libs/dsclient.c (line 51) - [public] while
    while (k2d->tail != k2d->head) {
        volatile ds_msg_t *m = &k2d_msgs[k2d->tail];
        ds_msg_t copy = *m;

        k2d->tail = (k2d->tail + 1) % k2d->cap;

        for (int i = 0; i < 16; i++) {
            if (!cache_used[i]) {
                cache[i] = copy;
                cache_used[i] = 1;
                break;
            }
        }
        // ... (truncated, see source file for more)

libs/dsclient.c (line 58) - [public] if
        if (!cache_used[i]) {
            cache[i] = copy;
            cache_used[i] = 1;
            break;
        }
    }
}

int ds_take(uint64_t id, ds_msg_t *out)
{
    for (int i = 0; i < 16; i++) {
        // ... (truncated, see source file for more)

libs/dsclient.c (line 70) - [public] if
        if (cache_used[i] && cache[i].id == id) {
            *out = cache[i];
            cache_used[i] = 0;
            return 1;
        }
    }

    return 0;
}

// ds_init(params_va);
// uint64_t id = ds_call(SVC_VIDEO, VID_FB_INFO, 0, 0, 0);
// ... (truncated, see source file for more)

triang-lang/out/x86_64/prog.c (line 15) - [public] memset
    memset(buffer, (int)0ull, 16);
    buffer[0] = (uint8_t)65ull;
    buffer[1] = (uint8_t)66ull;
    buffer[2] = (uint8_t)67ull;
    r0 = buffer[0];
    r1 = buffer[1];
    r2 = r0 + r1;
    if (r2 != 131ull) goto else_1;
    r0 = 1ull;
    goto end_2;
else_1:
    r0 = 0ull;
    // ... (truncated, see source file for more)

```

C (header)

```

drivers/vgpu/vgpu.h (line 22) - [public] vgpu_init
    bool vgpu_init(uint32_t w, uint32_t h);
    vgpu_info_t vgpu_get_info(void);

    void vgpu_clear(uint32_t color);
    void vgpu_pixel(uint32_t x, uint32_t y, uint32_t color);

    int32_t vgpu_surface_create(uint32_t w, uint32_t h, uint64_t *phys_out);
    bool vgpu_surface_present(int32_t id, uint32_t x, uint32_t y);

    void vgpu_process_ring(volatile void *ring);

#endif

drivers/vgpu/vgpu.h (line 23) - [public] vgpu_get_info
    vgpu_info_t vgpu_get_info(void);

    void vgpu_clear(uint32_t color);
    void vgpu_pixel(uint32_t x, uint32_t y, uint32_t color);

    int32_t vgpu_surface_create(uint32_t w, uint32_t h, uint64_t *phys_out);
    bool vgpu_surface_present(int32_t id, uint32_t x, uint32_t y);

    void vgpu_process_ring(volatile void *ring);

#endif

drivers/vgpu/vgpu.h (line 25) - [public] vgpu_clear
    void vgpu_clear(uint32_t color);
    void vgpu_pixel(uint32_t x, uint32_t y, uint32_t color);

    int32_t vgpu_surface_create(uint32_t w, uint32_t h, uint64_t *phys_out);
    bool vgpu_surface_present(int32_t id, uint32_t x, uint32_t y);

    void vgpu_process_ring(volatile void *ring);

#endif

drivers/vgpu/vgpu.h (line 26) - [public] vgpu_pixel
    void vgpu_pixel(uint32_t x, uint32_t y, uint32_t color);

    int32_t vgpu_surface_create(uint32_t w, uint32_t h, uint64_t *phys_out);
    bool vgpu_surface_present(int32_t id, uint32_t x, uint32_t y);

    void vgpu_process_ring(volatile void *ring);

#endif

drivers/vgpu/vgpu.h (line 28) - [public] vgpu_surface_create
    int32_t vgpu_surface_create(uint32_t w, uint32_t h, uint64_t *phys_out);
    bool vgpu_surface_present(int32_t id, uint32_t x, uint32_t y);

    void vgpu_process_ring(volatile void *ring);

#endif

drivers/vgpu/vgpu.h (line 29) - [public] vgpu_surface_present
    bool vgpu_surface_present(int32_t id, uint32_t x, uint32_t y);

    void vgpu_process_ring(volatile void *ring);

#endif

drivers/vgpu/vgpu.h (line 31) - [public] vgpu_process_ring
    void vgpu_process_ring(volatile void *ring);

#endif

```

kernel_Workspace/kernel/src/battery/battery.h (line 21) - [public] battery_init

```
bool battery_init(void);
bool battery_present(void);
bool battery_status(battery_status_t *out);
bool battery_status_packed(uint64_t *a0, uint64_t *a1, uint64_t *a2);
bool battery_set_threshold(uint32_t low_pct);
uint32_t battery_threshold(void);
bool battery_low_pending(void);
void battery_sim_tick(void);

#endif
```

kernel_Workspace/kernel/src/battery/battery.h (line 22) - [public] battery_present

```
bool battery_present(void);
bool battery_status(battery_status_t *out);
bool battery_status_packed(uint64_t *a0, uint64_t *a1, uint64_t *a2);
bool battery_set_threshold(uint32_t low_pct);
uint32_t battery_threshold(void);
bool battery_low_pending(void);
void battery_sim_tick(void);

#endif
```

kernel_Workspace/kernel/src/battery/battery.h (line 23) - [public] battery_status

```
bool battery_status(battery_status_t *out);
bool battery_status_packed(uint64_t *a0, uint64_t *a1, uint64_t *a2);
bool battery_set_threshold(uint32_t low_pct);
uint32_t battery_threshold(void);
bool battery_low_pending(void);
void battery_sim_tick(void);

#endif
```

kernel_Workspace/kernel/src/battery/battery.h (line 24) - [public] battery_status_packed

```
bool battery_status_packed(uint64_t *a0, uint64_t *a1, uint64_t *a2);
bool battery_set_threshold(uint32_t low_pct);
uint32_t battery_threshold(void);
bool battery_low_pending(void);
void battery_sim_tick(void);

#endif
```

kernel_Workspace/kernel/src/battery/battery.h (line 25) - [public] battery_set_threshold

```
bool battery_set_threshold(uint32_t low_pct);
uint32_t battery_threshold(void);
bool battery_low_pending(void);
void battery_sim_tick(void);

#endif
```

kernel_Workspace/kernel/src/battery/battery.h (line 26) - [public] battery_threshold

```
uint32_t battery_threshold(void);
bool battery_low_pending(void);
void battery_sim_tick(void);

#endif
```

kernel_Workspace/kernel/src/battery/battery.h (line 27) - [public] battery_low_pending

```
bool battery_low_pending(void);
void battery_sim_tick(void);

#endif
```

kernel_Workspace/kernel/src/battery/battery.h (line 28) - [public] battery_sim_tick

```
void battery_sim_tick(void);

#endif
```

kernel_Workspace/kernel/src/battery/operation.h (line 7) - [public] bool

```
    bool (*present)(void);
    bool (*status)(battery_status_t *out);
} battery_backend_t;

void battery_register_backend(const battery_backend_t *b);

#endif
```

kernel_Workspace/kernel/src/battery/operation.h (line 8) - [public] bool

```
    bool (*status)(battery_status_t *out);
} battery_backend_t;

void battery_register_backend(const battery_backend_t *b);

#endif
```

kernel_Workspace/kernel/src/battery/operation.h (line 11) - [public] battery_register_backend

```
void battery_register_backend(const battery_backend_t *b);

#endif
```

kernel_Workspace/kernel/src/bluetooth/bt.h (line 8) - [public] bt_init

```
bool bt_init(void);
bool bt_ready(void);
void bt_info(uint8_t *hci_ver, uint8_t *bdaddr);
bool bt_hci_cmd(uint16_t opcode, const uint8_t *params, uint8_t len);
bool bt_event_poll(uint8_t *buf, uint8_t cap, uint8_t *len);
bool bt_acl_send(const uint8_t *data, uint16_t len);
bool bt_acl_recv(uint8_t *data, uint16_t cap, uint16_t *len);

#endif
```

kernel_Workspace/kernel/src/bluetooth/bt.h (line 9) - [public] bt_ready

```
bool bt_ready(void);
void bt_info(uint8_t *hci_ver, uint8_t *bdaddr);
bool bt_hci_cmd(uint16_t opcode, const uint8_t *params, uint8_t len);
bool bt_event_poll(uint8_t *buf, uint8_t cap, uint8_t *len);
bool bt_acl_send(const uint8_t *data, uint16_t len);
bool bt_acl_recv(uint8_t *data, uint16_t cap, uint16_t *len);

#endif
```

kernel_Workspace/kernel/src/bluetooth/bt.h (line 10) - [public] bt_info

```
void bt_info(uint8_t *hci_ver, uint8_t *bdaddr);
bool bt_hci_cmd(uint16_t opcode, const uint8_t *params, uint8_t len);
bool bt_event_poll(uint8_t *buf, uint8_t cap, uint8_t *len);
bool bt_acl_send(const uint8_t *data, uint16_t len);
bool bt_acl_recv(uint8_t *data, uint16_t cap, uint16_t *len);

#endif
```

kernel_Workspace/kernel/src/bluetooth/bt.h (line 11) - [public] bt_hci_cmd

```
bool bt_hci_cmd(uint16_t opcode, const uint8_t *params, uint8_t len);
bool bt_event_poll(uint8_t *buf, uint8_t cap, uint8_t *len);
bool bt_acl_send(const uint8_t *data, uint16_t len);
bool bt_acl_recv(uint8_t *data, uint16_t cap, uint16_t *len);

#endif
```

kernel_Workspace/kernel/src/bluetooth/bt.h (line 12) - [public] bt_event_poll

```
bool bt_event_poll(uint8_t *buf, uint8_t cap, uint8_t *len);
bool bt_acl_send(const uint8_t *data, uint16_t len);
bool bt_acl_recv(uint8_t *data, uint16_t cap, uint16_t *len);

#endif
```

```

kernel_Workspace/kernel/src/bluetooth/bt.h (line 13) - [public] bt_acl_send
    bool bt_acl_send(const uint8_t *data, uint16_t len);
    bool bt_acl_rcv(uint8_t *data, uint16_t cap, uint16_t *len);

    #endif

kernel_Workspace/kernel/src/bluetooth/bt.h (line 14) - [public] bt_acl_rcv
    bool bt_acl_rcv(uint8_t *data, uint16_t cap, uint16_t *len);

    #endif

kernel_Workspace/kernel/src/bluetooth/operation.h (line 11) - [public] bt_op_model_init
    void bt_op_model_init(void);
    bool bt_op_model_run_cmd(uint16_t opcode, const uint8_t *params, uint8_t len);

    #endif

kernel_Workspace/kernel/src/bluetooth/operation.h (line 12) - [public] bt_op_model_run_cmd
    bool bt_op_model_run_cmd(uint16_t opcode, const uint8_t *params, uint8_t len);

    #endif

kernel_Workspace/kernel/src/camera/camera.h (line 19) - [public] camera_init
    bool camera_init(void);
    bool camera_present(void);
    bool camera_caps_get(camera_caps_t *out);
    bool camera_start(void);
    bool camera_stop(void);
    bool camera_frame(void *buf, uint32_t cap, uint64_t *frame_id);
    bool camera_frame_to_phys(uint64_t phys, uint32_t cap, uint64_t *frame_id);

    #endif

kernel_Workspace/kernel/src/camera/camera.h (line 20) - [public] camera_present
    bool camera_present(void);
    bool camera_caps_get(camera_caps_t *out);
    bool camera_start(void);
    bool camera_stop(void);
    bool camera_frame(void *buf, uint32_t cap, uint64_t *frame_id);
    bool camera_frame_to_phys(uint64_t phys, uint32_t cap, uint64_t *frame_id);

    #endif

kernel_Workspace/kernel/src/camera/camera.h (line 21) - [public] camera_caps_get
    bool camera_caps_get(camera_caps_t *out);
    bool camera_start(void);
    bool camera_stop(void);
    bool camera_frame(void *buf, uint32_t cap, uint64_t *frame_id);
    bool camera_frame_to_phys(uint64_t phys, uint32_t cap, uint64_t *frame_id);

    #endif

kernel_Workspace/kernel/src/camera/camera.h (line 22) - [public] camera_start
    bool camera_start(void);
    bool camera_stop(void);
    bool camera_frame(void *buf, uint32_t cap, uint64_t *frame_id);
    bool camera_frame_to_phys(uint64_t phys, uint32_t cap, uint64_t *frame_id);

    #endif

kernel_Workspace/kernel/src/camera/camera.h (line 23) - [public] camera_stop
    bool camera_stop(void);
    bool camera_frame(void *buf, uint32_t cap, uint64_t *frame_id);
    bool camera_frame_to_phys(uint64_t phys, uint32_t cap, uint64_t *frame_id);

    #endif

```

```

kernel_Workspace/kernel/src/camera/camera.h (line 24) - [public] camera_frame
    bool camera_frame(void *buf, uint32_t cap, uint64_t *frame_id);
    bool camera_frame_to_phys(uint64_t phys, uint32_t cap, uint64_t *frame_id);

    #endif

kernel_Workspace/kernel/src/camera/camera.h (line 25) - [public] camera_frame_to_phys
    bool camera_frame_to_phys(uint64_t phys, uint32_t cap, uint64_t *frame_id);

    #endif

kernel_Workspace/kernel/src/camera/operation.h (line 7) - [public] bool
    bool (*present)(void);
    bool (*start)(void);
    bool (*stop)(void);
    bool (*frame)(void *buf, uint32_t cap, uint64_t *fid);
} camera_backend_t;

void camera_register_backend(const camera_backend_t *b);

#endif

kernel_Workspace/kernel/src/camera/operation.h (line 8) - [public] bool
    bool (*start)(void);
    bool (*stop)(void);
    bool (*frame)(void *buf, uint32_t cap, uint64_t *fid);
} camera_backend_t;

void camera_register_backend(const camera_backend_t *b);

#endif

kernel_Workspace/kernel/src/camera/operation.h (line 9) - [public] bool
    bool (*stop)(void);
    bool (*frame)(void *buf, uint32_t cap, uint64_t *fid);
} camera_backend_t;

void camera_register_backend(const camera_backend_t *b);

#endif

kernel_Workspace/kernel/src/camera/operation.h (line 10) - [public] bool
    bool (*frame)(void *buf, uint32_t cap, uint64_t *fid);
} camera_backend_t;

void camera_register_backend(const camera_backend_t *b);

#endif

kernel_Workspace/kernel/src/camera/operation.h (line 13) - [public] camera_register_backend
void camera_register_backend(const camera_backend_t *b);

#endif

kernel_Workspace/kernel/src/caps/caps.h (line 10) - [public] caps_self_bits
unsigned int caps_self_bits(void);

unsigned int caps_world_bits(unsigned int world_id);

int caps_has(unsigned int world_id, unsigned char cap_id);

int caps_name(unsigned char cap_id, unsigned char *buf, unsigned int len);

int caps_request(unsigned int target, unsigned char cap_id);

int caps_release(unsigned int world_id, unsigned char cap_id);

    // ... (truncated, see source file for more)

```



```

kernel_Workspace/kernel/src/caps/caps.h (line 12) - [public] caps_world_bits
    unsigned int caps_world_bits(unsigned int world_id);

    int caps_has(unsigned int world_id, unsigned char cap_id);

    int caps_name(unsigned char cap_id, unsigned char *buf, unsigned int len);

    int caps_request(unsigned int target, unsigned char cap_id);

    int caps_release(unsigned int world_id, unsigned char cap_id);

    unsigned int caps_world_count(void);

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/caps/caps.h (line 14) - [public] caps_has
    int caps_has(unsigned int world_id, unsigned char cap_id);

    int caps_name(unsigned char cap_id, unsigned char *buf, unsigned int len);

    int caps_request(unsigned int target, unsigned char cap_id);

    int caps_release(unsigned int world_id, unsigned char cap_id);

    unsigned int caps_world_count(void);

    unsigned long long caps_audit_count(void);

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/caps/caps.h (line 16) - [public] caps_name
    int caps_name(unsigned char cap_id, unsigned char *buf, unsigned int len);

    int caps_request(unsigned int target, unsigned char cap_id);

    int caps_release(unsigned int world_id, unsigned char cap_id);

    unsigned int caps_world_count(void);

    unsigned long long caps_audit_count(void);

#endif

kernel_Workspace/kernel/src/caps/caps.h (line 18) - [public] caps_request
    int caps_request(unsigned int target, unsigned char cap_id);

    int caps_release(unsigned int world_id, unsigned char cap_id);

    unsigned int caps_world_count(void);

    unsigned long long caps_audit_count(void);

#endif

kernel_Workspace/kernel/src/caps/caps.h (line 20) - [public] caps_release
    int caps_release(unsigned int world_id, unsigned char cap_id);

    unsigned int caps_world_count(void);

    unsigned long long caps_audit_count(void);

#endif

kernel_Workspace/kernel/src/caps/caps.h (line 22) - [public] caps_world_count
    unsigned int caps_world_count(void);

    unsigned long long caps_audit_count(void);

#endif

```

```

kernel_Workspace/kernel/src/caps/caps.h (line 24) - [public] caps_audit_count
    unsigned long long caps_audit_count(void);

    #endif

kernel_Workspace/kernel/src/core-lang/ast.h (line 50) - [public] cl_arena_init
    void cl_arena_init(arena_t *a, uint8_t *buf, size_t cap);
    void *cl_arena_alloc(arena_t *a, size_t n);
    ast_node_t *cl_node(arena_t *ar, uint8_t kind, uint32_t line);

    #endif

kernel_Workspace/kernel/src/core-lang/bridge.h (line 6) - [public] cl_bridge_init
    int cl_bridge_init(cl_vm_t *vm, uint8_t ring);

    int cl_bridge_add(cl_vm_t *vm, const char *name, cl_ext_fn fn);

    #endif

kernel_Workspace/kernel/src/core-lang/bridge.h (line 8) - [public] cl_bridge_add
    int cl_bridge_add(cl_vm_t *vm, const char *name, cl_ext_fn fn);

    #endif

kernel_Workspace/kernel/src/core-lang/lexer.h (line 16) - [public] cl_lexer_init
    void cl_lexer_init(lexer_t *l, const char *src, size_t len);

    bool cl_lex_next(lexer_t *l, token_t *out);

    int cl_lex_all(lexer_t *l, token_t *buf, size_t cap, size_t *count);

    #endif

kernel_Workspace/kernel/src/core-lang/lexer.h (line 18) - [public] cl_lex_next
    bool cl_lex_next(lexer_t *l, token_t *out);

    int cl_lex_all(lexer_t *l, token_t *buf, size_t cap, size_t *count);

    #endif

kernel_Workspace/kernel/src/core-lang/lexer.h (line 20) - [public] cl_lex_all
    int cl_lex_all(lexer_t *l, token_t *buf, size_t cap, size_t *count);

    #endif

kernel_Workspace/kernel/src/core-lang/loader.h (line 15) - [public] cl_save_bc
    size_t cl_save_bc(cl_prog_t *P, uint8_t *buf, size_t cap);
    cl_prog_t *cl_load_bc(const uint8_t *buf, size_t len, arena_t *ar);

    cl_module_t *cl_load_path(const char *path, arena_t *ar, bool try_native);

    #endif

kernel_Workspace/kernel/src/core-lang/native.h (line 9) - [public] uint64_t
    /* signature wygenerowanej funkcji:
    uint64_t fn(uint64_t *slots, uint64_t *globals, cl_ext_fn *externs) */

    typedef uint64_t (*cl_native_fn)(uint64_t *, uint64_t *, cl_ext_fn *);

    /* Zwraca bufor z kodem (RW) albo NULL gdy arch nie wspiera subsetu.
    Wywołujący robi go executable (mprotect / mapowanie WX w kernelu). */
    void *cl_native_compile(cl_prog_t *P);
    size_t cl_native_size(void);

    #endif

kernel_Workspace/kernel/src/core-lang/native.h (line 14) - [public] cl_native_size
    size_t cl_native_size(void);

    #endif

```

kernel_Workspace/kernel/src/core-lang/sema.h (line 16) - [public] cl_sema_run

```
int cl_sema_run(sema_ctx_t *s, ast_node_t *prog);

size_t cl_sema_diag_count(const sema_ctx_t *s);
const cl_diag_t *cl_sema_diag(const sema_ctx_t *s, size_t i);

#endif
```

kernel_Workspace/kernel/src/core-lang/sema.h (line 18) - [public] cl_sema_diag_count

```
size_t cl_sema_diag_count(const sema_ctx_t *s);
const cl_diag_t *cl_sema_diag(const sema_ctx_t *s, size_t i);

#endif
```

kernel_Workspace/kernel/src/core-lang/vm.h (line 44) - [public] cl_vm_init

```
void cl_vm_init(cl_vm_t *vm, cl_prog_t *prog);
int cl_vm_register_extern(cl_vm_t *vm, const char *name, cl_ext_fn fn);
cl_vm_err_t cl_vm_run(cl_vm_t *vm);

int cl_wbytes(int bits);
void cl_mask(uint8_t *r, int w, int bits);
void cl_add(uint8_t *r, const uint8_t *a, const uint8_t *b, int w);
void cl_sub(uint8_t *r, const uint8_t *a, const uint8_t *b, int w);
void cl_mul(uint8_t *r, const uint8_t *a, const uint8_t *b, int w);
int cl_cmp(const uint8_t *a, const uint8_t *b, int w, bool sign);
void cl_shl(uint8_t *r, const uint8_t *a, uint32_t n, int w);
void cl_shr(uint8_t *r, const uint8_t *a, uint32_t n, int w);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/vm.h (line 45) - [public] cl_vm_register_extern

```
int cl_vm_register_extern(cl_vm_t *vm, const char *name, cl_ext_fn fn);
cl_vm_err_t cl_vm_run(cl_vm_t *vm);

int cl_wbytes(int bits);
void cl_mask(uint8_t *r, int w, int bits);
void cl_add(uint8_t *r, const uint8_t *a, const uint8_t *b, int w);
void cl_sub(uint8_t *r, const uint8_t *a, const uint8_t *b, int w);
void cl_mul(uint8_t *r, const uint8_t *a, const uint8_t *b, int w);
int cl_cmp(const uint8_t *a, const uint8_t *b, int w, bool sign);
void cl_shl(uint8_t *r, const uint8_t *a, uint32_t n, int w);
void cl_shr(uint8_t *r, const uint8_t *a, uint32_t n, int w);
void cl_divmod(const uint8_t *n, const uint8_t *d, int w,
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/vm.h (line 46) - [public] cl_vm_run

```
cl_vm_err_t cl_vm_run(cl_vm_t *vm);

int cl_wbytes(int bits);
void cl_mask(uint8_t *r, int w, int bits);
void cl_add(uint8_t *r, const uint8_t *a, const uint8_t *b, int w);
void cl_sub(uint8_t *r, const uint8_t *a, const uint8_t *b, int w);
void cl_mul(uint8_t *r, const uint8_t *a, const uint8_t *b, int w);
int cl_cmp(const uint8_t *a, const uint8_t *b, int w, bool sign);
void cl_shl(uint8_t *r, const uint8_t *a, uint32_t n, int w);
void cl_shr(uint8_t *r, const uint8_t *a, uint32_t n, int w);
void cl_divmod(const uint8_t *n, const uint8_t *d, int w,
uint8_t *q, uint8_t *m);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/core-lang/vm.h (line 48) - [public] cl_wbytes

```
int cl_wbytes(int bits);
void cl_mask(uint8_t *r, int w, int bits);
void cl_add(uint8_t *r, const uint8_t *a, const uint8_t *b, int w);
void cl_sub(uint8_t *r, const uint8_t *a, const uint8_t *b, int w);
void cl_mul(uint8_t *r, const uint8_t *a, const uint8_t *b, int w);
int cl_cmp(const uint8_t *a, const uint8_t *b, int w, bool sign);
void cl_shl(uint8_t *r, const uint8_t *a, uint32_t n, int w);
void cl_shr(uint8_t *r, const uint8_t *a, uint32_t n, int w);
void cl_divmod(const uint8_t *n, const uint8_t *d, int w,
               uint8_t *q, uint8_t *m);

#endif
```

kernel_Workspace/kernel/src/core-lang/vm.h (line 49) - [public] cl_mask

```
void cl_mask(uint8_t *r, int w, int bits);
void cl_add(uint8_t *r, const uint8_t *a, const uint8_t *b, int w);
void cl_sub(uint8_t *r, const uint8_t *a, const uint8_t *b, int w);
void cl_mul(uint8_t *r, const uint8_t *a, const uint8_t *b, int w);
int cl_cmp(const uint8_t *a, const uint8_t *b, int w, bool sign);
void cl_shl(uint8_t *r, const uint8_t *a, uint32_t n, int w);
void cl_shr(uint8_t *r, const uint8_t *a, uint32_t n, int w);
void cl_divmod(const uint8_t *n, const uint8_t *d, int w,
               uint8_t *q, uint8_t *m);

#endif
```

kernel_Workspace/kernel/src/core-lang/vm.h (line 50) - [public] cl_add

```
void cl_add(uint8_t *r, const uint8_t *a, const uint8_t *b, int w);
void cl_sub(uint8_t *r, const uint8_t *a, const uint8_t *b, int w);
void cl_mul(uint8_t *r, const uint8_t *a, const uint8_t *b, int w);
int cl_cmp(const uint8_t *a, const uint8_t *b, int w, bool sign);
void cl_shl(uint8_t *r, const uint8_t *a, uint32_t n, int w);
void cl_shr(uint8_t *r, const uint8_t *a, uint32_t n, int w);
void cl_divmod(const uint8_t *n, const uint8_t *d, int w,
               uint8_t *q, uint8_t *m);

#endif
```

kernel_Workspace/kernel/src/core-lang/vm.h (line 51) - [public] cl_sub

```
void cl_sub(uint8_t *r, const uint8_t *a, const uint8_t *b, int w);
void cl_mul(uint8_t *r, const uint8_t *a, const uint8_t *b, int w);
int cl_cmp(const uint8_t *a, const uint8_t *b, int w, bool sign);
void cl_shl(uint8_t *r, const uint8_t *a, uint32_t n, int w);
void cl_shr(uint8_t *r, const uint8_t *a, uint32_t n, int w);
void cl_divmod(const uint8_t *n, const uint8_t *d, int w,
               uint8_t *q, uint8_t *m);

#endif
```

kernel_Workspace/kernel/src/core-lang/vm.h (line 52) - [public] cl_mul

```
void cl_mul(uint8_t *r, const uint8_t *a, const uint8_t *b, int w);
int cl_cmp(const uint8_t *a, const uint8_t *b, int w, bool sign);
void cl_shl(uint8_t *r, const uint8_t *a, uint32_t n, int w);
void cl_shr(uint8_t *r, const uint8_t *a, uint32_t n, int w);
void cl_divmod(const uint8_t *n, const uint8_t *d, int w,
               uint8_t *q, uint8_t *m);

#endif
```

kernel_Workspace/kernel/src/core-lang/vm.h (line 53) - [public] cl_cmp

```
int cl_cmp(const uint8_t *a, const uint8_t *b, int w, bool sign);
void cl_shl(uint8_t *r, const uint8_t *a, uint32_t n, int w);
void cl_shr(uint8_t *r, const uint8_t *a, uint32_t n, int w);
void cl_divmod(const uint8_t *n, const uint8_t *d, int w,
               uint8_t *q, uint8_t *m);

#endif
```

```

kernel_Workspace/kernel/src/core-lang/vm.h (line 54) - [public] cl_shl
    void cl_shl(uint8_t *r, const uint8_t *a, uint32_t n, int w);
    void cl_shr(uint8_t *r, const uint8_t *a, uint32_t n, int w);
    void cl_divmod(const uint8_t *n, const uint8_t *d, int w,
                  uint8_t *q, uint8_t *m);

    #endif

kernel_Workspace/kernel/src/core-lang/vm.h (line 55) - [public] cl_shr
    void cl_shr(uint8_t *r, const uint8_t *a, uint32_t n, int w);
    void cl_divmod(const uint8_t *n, const uint8_t *d, int w,
                  uint8_t *q, uint8_t *m);

    #endif

kernel_Workspace/kernel/src/displayport/aux.h (line 17) - [public] dp_aux_sink_init
    void dp_aux_sink_init(void);
    bool dp_aux_read(uint32_t addr, void *buf, uint32_t len);
    bool dp_aux_write(uint32_t addr, const void *buf, uint32_t len);
    bool dp_aux_read_edid(void *buf, uint32_t len);

    #endif

kernel_Workspace/kernel/src/displayport/aux.h (line 18) - [public] dp_aux_read
    bool dp_aux_read(uint32_t addr, void *buf, uint32_t len);
    bool dp_aux_write(uint32_t addr, const void *buf, uint32_t len);
    bool dp_aux_read_edid(void *buf, uint32_t len);

    #endif

kernel_Workspace/kernel/src/displayport/aux.h (line 19) - [public] dp_aux_write
    bool dp_aux_write(uint32_t addr, const void *buf, uint32_t len);
    bool dp_aux_read_edid(void *buf, uint32_t len);

    #endif

kernel_Workspace/kernel/src/displayport/aux.h (line 20) - [public] dp_aux_read_edid
    bool dp_aux_read_edid(void *buf, uint32_t len);

    #endif

kernel_Workspace/kernel/src/displayport/dp.h (line 10) - [public] dp_init
    bool dp_init(void);
    bool dp_ready(void);

    void dp_link_info(uint32_t *rate_mbps, uint32_t *lanes);

    uint32_t dp_mode_count(void);
    bool dp_mode_at(uint32_t i, uint32_t *id, uint32_t *w,
                   uint32_t *h, uint32_t *r);
    bool dp_mode_set_by_id(uint32_t id);

    void dp_caps(uint64_t *fb_phys, uint32_t *w, uint32_t *h, uint32_t *stride);
    bool dp_fb_set(uint64_t phys, uint32_t w, uint32_t h, uint32_t stride);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/displayport/dp.h (line 11) - [public] dp_ready
    bool dp_ready(void);

    void dp_link_info(uint32_t *rate_mbps, uint32_t *lanes);

    uint32_t dp_mode_count(void);
    bool dp_mode_at(uint32_t i, uint32_t *id, uint32_t *w,
                   uint32_t *h, uint32_t *r);
    bool dp_mode_set_by_id(uint32_t id);

    void dp_caps(uint64_t *fb_phys, uint32_t *w, uint32_t *h, uint32_t *stride);
    bool dp_fb_set(uint64_t phys, uint32_t w, uint32_t h, uint32_t stride);

    // ... (truncated, see source file for more)

```

```
kernel_Workspace/kernel/src/displayport/dp.h (line 13) - [public] dp_link_info
void dp_link_info(uint32_t *rate_mbps, uint32_t *lanes);

uint32_t dp_mode_count(void);
bool dp_mode_at(uint32_t i, uint32_t *id, uint32_t *w,
                uint32_t *h, uint32_t *r);
bool dp_mode_set_by_id(uint32_t id);

void dp_caps(uint64_t *fb_phys, uint32_t *w, uint32_t *h, uint32_t *stride);
bool dp_fb_set(uint64_t phys, uint32_t w, uint32_t h, uint32_t stride);

uint64_t dp_submit_fill(uint32_t color, uint32_t x, uint32_t y,
                        uint32_t w, uint32_t h);
// ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/displayport/dp.h (line 15) - [public] dp_mode_count
uint32_t dp_mode_count(void);
bool dp_mode_at(uint32_t i, uint32_t *id, uint32_t *w,
                uint32_t *h, uint32_t *r);
bool dp_mode_set_by_id(uint32_t id);

void dp_caps(uint64_t *fb_phys, uint32_t *w, uint32_t *h, uint32_t *stride);
bool dp_fb_set(uint64_t phys, uint32_t w, uint32_t h, uint32_t stride);

uint64_t dp_submit_fill(uint32_t color, uint32_t x, uint32_t y,
                        uint32_t w, uint32_t h);

#endif
```

```
kernel_Workspace/kernel/src/displayport/dp.h (line 18) - [public] dp_mode_set_by_id
bool dp_mode_set_by_id(uint32_t id);

void dp_caps(uint64_t *fb_phys, uint32_t *w, uint32_t *h, uint32_t *stride);
bool dp_fb_set(uint64_t phys, uint32_t w, uint32_t h, uint32_t stride);

uint64_t dp_submit_fill(uint32_t color, uint32_t x, uint32_t y,
                        uint32_t w, uint32_t h);

#endif
```

```
kernel_Workspace/kernel/src/displayport/dp.h (line 20) - [public] dp_caps
void dp_caps(uint64_t *fb_phys, uint32_t *w, uint32_t *h, uint32_t *stride);
bool dp_fb_set(uint64_t phys, uint32_t w, uint32_t h, uint32_t stride);

uint64_t dp_submit_fill(uint32_t color, uint32_t x, uint32_t y,
                        uint32_t w, uint32_t h);

#endif
```

```
kernel_Workspace/kernel/src/displayport/dp.h (line 21) - [public] dp_fb_set
bool dp_fb_set(uint64_t phys, uint32_t w, uint32_t h, uint32_t stride);

uint64_t dp_submit_fill(uint32_t color, uint32_t x, uint32_t y,
                        uint32_t w, uint32_t h);

#endif
```

```
kernel_Workspace/kernel/src/displayport/link.h (line 7) - [public] dp_link_train
bool dp_link_train(void);
void dp_link_info(uint32_t *rate_mbps, uint32_t *lanes);

#endif
```

```
kernel_Workspace/kernel/src/displayport/link.h (line 8) - [public] dp_link_info
void dp_link_info(uint32_t *rate_mbps, uint32_t *lanes);

#endif
```

```

kernel_Workspace/kernel/src/displayport/operation.h (line 6) - [public] dp_op_set_fb
    void dp_op_set_fb(uint64_t phys, uint32_t w, uint32_t h, uint32_t stride);
    void dp_op_state(uint64_t *phys, uint32_t *w, uint32_t *h, uint32_t *stride);
    void dp_op_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h);

    #endif

kernel_Workspace/kernel/src/displayport/operation.h (line 7) - [public] dp_op_state
    void dp_op_state(uint64_t *phys, uint32_t *w, uint32_t *h, uint32_t *stride);
    void dp_op_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h);

    #endif

kernel_Workspace/kernel/src/displayport/operation.h (line 8) - [public] dp_op_fill
    void dp_op_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h);

    #endif

kernel_Workspace/kernel/src/editor/editor.h (line 21) - [public] editor_run
    int editor_run(const char *path);

    #endif

kernel_Workspace/kernel/src/editor/mouse.h (line 6) - [public] mouse_init
    void mouse_init(void);

    int mouse_poll(int *dx, int *dy, int *dz, int *buttons);

    #endif

kernel_Workspace/kernel/src/editor/mouse.h (line 8) - [public] mouse_poll
    int mouse_poll(int *dx, int *dy, int *dz, int *buttons);

    #endif

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 41) - [public] hdmi_init_with
    bool hdmi_init_with(uint64_t fb_phys, uint32_t w, uint32_t h, uint32_t stride);
    bool hdmi_ready(void);
    void hdmi_caps(hdmi_caps_t *out);
    bool hdmi_fb_set(uint64_t phys, uint32_t w, uint32_t h, uint32_t stride);
    bool hdmi_mode_set_by_id(uint32_t id);
    bool hdmi_mode_current_raw(uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r);
    uint64_t hdmi_submit(const hdmi_transfer_t *t);
    bool hdmi_poll(uint64_t *out_seq);
    uint32_t hdmi_pending(void);
    uint64_t hdmi_submit_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h);
    void hdmi_caps_raw(uint32_t *w, uint32_t *h, uint32_t *s, uint64_t *phys);

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 42) - [public] hdmi_ready
    bool hdmi_ready(void);
    void hdmi_caps(hdmi_caps_t *out);
    bool hdmi_fb_set(uint64_t phys, uint32_t w, uint32_t h, uint32_t stride);
    bool hdmi_mode_set_by_id(uint32_t id);
    bool hdmi_mode_current_raw(uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r);
    uint64_t hdmi_submit(const hdmi_transfer_t *t);
    bool hdmi_poll(uint64_t *out_seq);
    uint32_t hdmi_pending(void);
    uint64_t hdmi_submit_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h);
    void hdmi_caps_raw(uint32_t *w, uint32_t *h, uint32_t *s, uint64_t *phys);

    bool hdmi_fb_grant(void);
    // ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 43) - [public] hdmi_caps

```
void hdmi_caps(hdmi_caps_t *out);
bool hdmi_fb_set(uint64_t phys, uint32_t w, uint32_t h, uint32_t stride);
bool hdmi_mode_set_by_id(uint32_t id);
bool hdmi_mode_current_raw(uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r);
uint64_t hdmi_submit(const hdmi_transfer_t *t);
bool hdmi_poll(uint64_t *out_seq);
uint32_t hdmi_pending(void);
uint64_t hdmi_submit_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h);
void hdmi_caps_raw(uint32_t *w, uint32_t *h, uint32_t *s, uint64_t *phys);

bool hdmi_fb_grant(void);
void hdmi_fb_revoke(void);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 44) - [public] hdmi_fb_set

```
bool hdmi_fb_set(uint64_t phys, uint32_t w, uint32_t h, uint32_t stride);
bool hdmi_mode_set_by_id(uint32_t id);
bool hdmi_mode_current_raw(uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r);
uint64_t hdmi_submit(const hdmi_transfer_t *t);
bool hdmi_poll(uint64_t *out_seq);
uint32_t hdmi_pending(void);
uint64_t hdmi_submit_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h);
void hdmi_caps_raw(uint32_t *w, uint32_t *h, uint32_t *s, uint64_t *phys);

bool hdmi_fb_grant(void);
void hdmi_fb_revoke(void);
bool hdmi_fb_granted(void);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 45) - [public] hdmi_mode_set_by_id

```
bool hdmi_mode_set_by_id(uint32_t id);
bool hdmi_mode_current_raw(uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r);
uint64_t hdmi_submit(const hdmi_transfer_t *t);
bool hdmi_poll(uint64_t *out_seq);
uint32_t hdmi_pending(void);
uint64_t hdmi_submit_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h);
void hdmi_caps_raw(uint32_t *w, uint32_t *h, uint32_t *s, uint64_t *phys);

bool hdmi_fb_grant(void);
void hdmi_fb_revoke(void);
bool hdmi_fb_granted(void);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 46) - [public] hdmi_mode_current_raw

```
bool hdmi_mode_current_raw(uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r);
uint64_t hdmi_submit(const hdmi_transfer_t *t);
bool hdmi_poll(uint64_t *out_seq);
uint32_t hdmi_pending(void);
uint64_t hdmi_submit_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h);
void hdmi_caps_raw(uint32_t *w, uint32_t *h, uint32_t *s, uint64_t *phys);

bool hdmi_fb_grant(void);
void hdmi_fb_revoke(void);
bool hdmi_fb_granted(void);

#endif
```

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 47) - [public] hdmi_submit

```
uint64_t hdmi_submit(const hdmi_transfer_t *t);
bool hdmi_poll(uint64_t *out_seq);
uint32_t hdmi_pending(void);
uint64_t hdmi_submit_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h);
void hdmi_caps_raw(uint32_t *w, uint32_t *h, uint32_t *s, uint64_t *phys);

bool hdmi_fb_grant(void);
void hdmi_fb_revoke(void);
bool hdmi_fb_granted(void);

#endif
```


kernel_Workspace/kernel/src/hdmi/hdmi.h (line 48) - [public] hdmi_poll

```
bool hdmi_poll(uint64_t *out_seq);
uint32_t hdmi_pending(void);
uint64_t hdmi_submit_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h);
void hdmi_caps_raw(uint32_t *w, uint32_t *h, uint32_t *s, uint64_t *phys);

bool hdmi_fb_grant(void);
void hdmi_fb_revoke(void);
bool hdmi_fb_granted(void);

#endif
```

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 49) - [public] hdmi_pending

```
uint32_t hdmi_pending(void);
uint64_t hdmi_submit_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h);
void hdmi_caps_raw(uint32_t *w, uint32_t *h, uint32_t *s, uint64_t *phys);

bool hdmi_fb_grant(void);
void hdmi_fb_revoke(void);
bool hdmi_fb_granted(void);

#endif
```

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 50) - [public] hdmi_submit_fill

```
uint64_t hdmi_submit_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h);
void hdmi_caps_raw(uint32_t *w, uint32_t *h, uint32_t *s, uint64_t *phys);

bool hdmi_fb_grant(void);
void hdmi_fb_revoke(void);
bool hdmi_fb_granted(void);

#endif
```

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 51) - [public] hdmi_caps_raw

```
void hdmi_caps_raw(uint32_t *w, uint32_t *h, uint32_t *s, uint64_t *phys);

bool hdmi_fb_grant(void);
void hdmi_fb_revoke(void);
bool hdmi_fb_granted(void);

#endif
```

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 53) - [public] hdmi_fb_grant

```
bool hdmi_fb_grant(void);
void hdmi_fb_revoke(void);
bool hdmi_fb_granted(void);

#endif
```

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 54) - [public] hdmi_fb_revoke

```
void hdmi_fb_revoke(void);
bool hdmi_fb_granted(void);

#endif
```

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 55) - [public] hdmi_fb_granted

```
bool hdmi_fb_granted(void);

#endif
```

kernel_Workspace/kernel/src/hdmi/interface.h (line 9) - [public] hdmi_iface_acquire

```
bool hdmi_iface_acquire(uint32_t owner);
bool hdmi_iface_release(uint32_t owner);
uint32_t hdmi_iface_owner(void);
bool hdmi_iface_check(uint32_t owner);

bool hdmi_iface_init(uint32_t owner, uint64_t fb_phys, uint32_t w, uint32_t h, uint32_t stride);
bool hdmi_iface_ready(void);

bool hdmi_iface_mode_set(uint32_t owner, uint32_t id);
bool hdmi_iface_mode_current(uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r);
bool hdmi_iface_mode_at(uint32_t i, uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/interface.h (line 10) - [public] hdmi_iface_release

```
bool hdmi_iface_release(uint32_t owner);
uint32_t hdmi_iface_owner(void);
bool hdmi_iface_check(uint32_t owner);

bool hdmi_iface_init(uint32_t owner, uint64_t fb_phys, uint32_t w, uint32_t h, uint32_t stride);
bool hdmi_iface_ready(void);

bool hdmi_iface_mode_set(uint32_t owner, uint32_t id);
bool hdmi_iface_mode_current(uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r);
bool hdmi_iface_mode_at(uint32_t i, uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r);

uint64_t hdmi_iface_submit_fill(uint32_t owner, uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/interface.h (line 11) - [public] hdmi_iface_owner

```
uint32_t hdmi_iface_owner(void);
bool hdmi_iface_check(uint32_t owner);

bool hdmi_iface_init(uint32_t owner, uint64_t fb_phys, uint32_t w, uint32_t h, uint32_t stride);
bool hdmi_iface_ready(void);

bool hdmi_iface_mode_set(uint32_t owner, uint32_t id);
bool hdmi_iface_mode_current(uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r);
bool hdmi_iface_mode_at(uint32_t i, uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r);

uint64_t hdmi_iface_submit_fill(uint32_t owner, uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h);
bool hdmi_iface_poll(uint32_t owner, uint64_t *out_seq);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/interface.h (line 12) - [public] hdmi_iface_check

```
bool hdmi_iface_check(uint32_t owner);

bool hdmi_iface_init(uint32_t owner, uint64_t fb_phys, uint32_t w, uint32_t h, uint32_t stride);
bool hdmi_iface_ready(void);

bool hdmi_iface_mode_set(uint32_t owner, uint32_t id);
bool hdmi_iface_mode_current(uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r);
bool hdmi_iface_mode_at(uint32_t i, uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r);

uint64_t hdmi_iface_submit_fill(uint32_t owner, uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h);
bool hdmi_iface_poll(uint32_t owner, uint64_t *out_seq);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/interface.h (line 14) - [public] hdmi_iface_init

```
bool hdmi_iface_init(uint32_t owner, uint64_t fb_phys, uint32_t w, uint32_t h, uint32_t stride);
bool hdmi_iface_ready(void);

bool hdmi_iface_mode_set(uint32_t owner, uint32_t id);
bool hdmi_iface_mode_current(uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r);
bool hdmi_iface_mode_at(uint32_t i, uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r);

uint64_t hdmi_iface_submit_fill(uint32_t owner, uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h, uint32_t *r);
bool hdmi_iface_poll(uint32_t owner, uint64_t *out_seq);

void hdmi_iface_caps(uint32_t *w, uint32_t *h, uint32_t *s, uint64_t *phys);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/interface.h (line 15) - [public] hdmi_iface_ready

```
bool hdmi_iface_ready(void);

bool hdmi_iface_mode_set(uint32_t owner, uint32_t id);
bool hdmi_iface_mode_current(uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r);
bool hdmi_iface_mode_at(uint32_t i, uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r);

uint64_t hdmi_iface_submit_fill(uint32_t owner, uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h, uint32_t *r);
bool hdmi_iface_poll(uint32_t owner, uint64_t *out_seq);

void hdmi_iface_caps(uint32_t *w, uint32_t *h, uint32_t *s, uint64_t *phys);

bool hdmi_iface_fb_grant(uint32_t owner, uint64_t *phys, uint32_t *w, uint32_t *h, uint32_t *s);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/interface.h (line 17) - [public] hdmi_iface_mode_set

```
bool hdmi_iface_mode_set(uint32_t owner, uint32_t id);
bool hdmi_iface_mode_current(uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r);
bool hdmi_iface_mode_at(uint32_t i, uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r);

uint64_t hdmi_iface_submit_fill(uint32_t owner, uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h, uint32_t *r);
bool hdmi_iface_poll(uint32_t owner, uint64_t *out_seq);

void hdmi_iface_caps(uint32_t *w, uint32_t *h, uint32_t *s, uint64_t *phys);

bool hdmi_iface_fb_grant(uint32_t owner, uint64_t *phys, uint32_t *w, uint32_t *h, uint32_t *s);
bool hdmi_iface_fb_revoke(uint32_t owner);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/hdmi/interface.h (line 18) - [public] hdmi_iface_mode_current

```
bool hdmi_iface_mode_current(uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r);
bool hdmi_iface_mode_at(uint32_t i, uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r);

uint64_t hdmi_iface_submit_fill(uint32_t owner, uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h, uint32_t *r);
bool hdmi_iface_poll(uint32_t owner, uint64_t *out_seq);

void hdmi_iface_caps(uint32_t *w, uint32_t *h, uint32_t *s, uint64_t *phys);

bool hdmi_iface_fb_grant(uint32_t owner, uint64_t *phys, uint32_t *w, uint32_t *h, uint32_t *s);
bool hdmi_iface_fb_revoke(uint32_t owner);

#endif
```

kernel_Workspace/kernel/src/hdmi/interface.h (line 19) - [public] hdmi_iface_mode_at

```
bool hdmi_iface_mode_at(uint32_t i, uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r);

uint64_t hdmi_iface_submit_fill(uint32_t owner, uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h, uint32_t *r);
bool hdmi_iface_poll(uint32_t owner, uint64_t *out_seq);

void hdmi_iface_caps(uint32_t *w, uint32_t *h, uint32_t *s, uint64_t *phys);

bool hdmi_iface_fb_grant(uint32_t owner, uint64_t *phys, uint32_t *w, uint32_t *h, uint32_t *s);
bool hdmi_iface_fb_revoke(uint32_t owner);

#endif
```

```

kernel_Workspace/kernel/src/hdmi/interface.h (line 21) - [public] hdmi_iface_submit_fill
    uint64_t hdmi_iface_submit_fill(uint32_t owner, uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h,
    bool hdmi_iface_poll(uint32_t owner, uint64_t *out_seq);

    void hdmi_iface_caps(uint32_t *w, uint32_t *h, uint32_t *s, uint64_t *phys);

    bool hdmi_iface_fb_grant(uint32_t owner, uint64_t *phys, uint32_t *w, uint32_t *h, uint32_t *s);
    bool hdmi_iface_fb_revoke(uint32_t owner);

#endif

kernel_Workspace/kernel/src/hdmi/interface.h (line 22) - [public] hdmi_iface_poll
    bool hdmi_iface_poll(uint32_t owner, uint64_t *out_seq);

    void hdmi_iface_caps(uint32_t *w, uint32_t *h, uint32_t *s, uint64_t *phys);

    bool hdmi_iface_fb_grant(uint32_t owner, uint64_t *phys, uint32_t *w, uint32_t *h, uint32_t *s);
    bool hdmi_iface_fb_revoke(uint32_t owner);

#endif

kernel_Workspace/kernel/src/hdmi/interface.h (line 24) - [public] hdmi_iface_caps
    void hdmi_iface_caps(uint32_t *w, uint32_t *h, uint32_t *s, uint64_t *phys);

    bool hdmi_iface_fb_grant(uint32_t owner, uint64_t *phys, uint32_t *w, uint32_t *h, uint32_t *s);
    bool hdmi_iface_fb_revoke(uint32_t owner);

#endif

kernel_Workspace/kernel/src/hdmi/interface.h (line 26) - [public] hdmi_iface_fb_grant
    bool hdmi_iface_fb_grant(uint32_t owner, uint64_t *phys, uint32_t *w, uint32_t *h, uint32_t *s);
    bool hdmi_iface_fb_revoke(uint32_t owner);

#endif

kernel_Workspace/kernel/src/hdmi/interface.h (line 27) - [public] hdmi_iface_fb_revoke
    bool hdmi_iface_fb_revoke(uint32_t owner);

#endif

kernel_Workspace/kernel/src/hdmi/mode.h (line 23) - [public] hdmi_mode_count
    uint32_t hdmi_mode_count(void);
    const hdmi_mode_t *hdmi_mode_at(uint32_t i);
    const hdmi_mode_t *hdmi_mode_find(uint32_t w, uint32_t h, uint32_t refresh);
    const hdmi_mode_t *hdmi_mode_by_id(uint32_t id);
    bool hdmi_mode_valid(const hdmi_mode_t *m);
    bool hdmi_mode_apply(const hdmi_mode_t *m);
    const hdmi_mode_t *hdmi_mode_current(void);
    bool hdmi_mode_at_raw(uint32_t i, uint32_t *id, uint32_t *w,
        uint32_t *h, uint32_t *r);

#endif

kernel_Workspace/kernel/src/hdmi/mode.h (line 27) - [public] hdmi_mode_valid
    bool hdmi_mode_valid(const hdmi_mode_t *m);
    bool hdmi_mode_apply(const hdmi_mode_t *m);
    const hdmi_mode_t *hdmi_mode_current(void);
    bool hdmi_mode_at_raw(uint32_t i, uint32_t *id, uint32_t *w,
        uint32_t *h, uint32_t *r);

#endif

kernel_Workspace/kernel/src/hdmi/mode.h (line 28) - [public] hdmi_mode_apply
    bool hdmi_mode_apply(const hdmi_mode_t *m);
    const hdmi_mode_t *hdmi_mode_current(void);
    bool hdmi_mode_at_raw(uint32_t i, uint32_t *id, uint32_t *w,
        uint32_t *h, uint32_t *r);

#endif

```

kernel_Workspace/kernel/src/hdmi/operation.h (line 7) - [public] hdmi_op_set_mode

```
void hdmi_op_set_mode(const hdmi_mode_t *m);
void hdmi_op_exec(hdmi_transfer_t *t);
void hdmi_op_state(hdmi_caps_t *out);
void hdmi_op_set_fb(uint64_t phys, uint32_t w, uint32_t h, uint32_t stride);
bool hdmi_op_ready(void);
```

#endif

kernel_Workspace/kernel/src/hdmi/operation.h (line 8) - [public] hdmi_op_exec

```
void hdmi_op_exec(hdmi_transfer_t *t);
void hdmi_op_state(hdmi_caps_t *out);
void hdmi_op_set_fb(uint64_t phys, uint32_t w, uint32_t h, uint32_t stride);
bool hdmi_op_ready(void);
```

#endif

kernel_Workspace/kernel/src/hdmi/operation.h (line 9) - [public] hdmi_op_state

```
void hdmi_op_state(hdmi_caps_t *out);
void hdmi_op_set_fb(uint64_t phys, uint32_t w, uint32_t h, uint32_t stride);
bool hdmi_op_ready(void);
```

#endif

kernel_Workspace/kernel/src/hdmi/operation.h (line 10) - [public] hdmi_op_set_fb

```
void hdmi_op_set_fb(uint64_t phys, uint32_t w, uint32_t h, uint32_t stride);
bool hdmi_op_ready(void);
```

#endif

kernel_Workspace/kernel/src/hdmi/operation.h (line 11) - [public] hdmi_op_ready

```
bool hdmi_op_ready(void);
```

#endif

kernel_Workspace/kernel/src/mm/alloc/api/alloc.h (line 13) - [public] kfree

```
void kfree(void *ptr);

size_t kmalloc_usable_size(void *ptr);

void *kalloc_pages(size_t pages);
void kfree_pages(void *ptr, size_t pages);

char *kstrdup(const char *s);

uint64_t kvirt_to_phys(void *ptr);

void kalloc_dump(void);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/api/alloc.h (line 15) - [public] kmalloc_usable_size

```
size_t kmalloc_usable_size(void *ptr);

void *kalloc_pages(size_t pages);
void kfree_pages(void *ptr, size_t pages);

char *kstrdup(const char *s);

uint64_t kvirt_to_phys(void *ptr);

void kalloc_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/alloc/api/alloc.h (line 18) - [public] kfree_pages
void kfree_pages(void *ptr, size_t pages);

char *kstrdup(const char *s);

uint64_t kvirt_to_phys(void *ptr);

void kalloc_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/alloc/api/alloc.h (line 22) - [public] kvirt_to_phys
uint64_t kvirt_to_phys(void *ptr);

void kalloc_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/alloc/api/alloc.h (line 24) - [public] kalloc_dump
void kalloc_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/alloc/debug/alloc_debug.h (line 9) - [public] dbg_free
void dbg_free(void *ptr);

bool dbg_verify(void *ptr);
size_t dbg_usable_size(void *ptr);
void mm_debug_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/alloc/debug/alloc_debug.h (line 11) - [public] dbg_verify
bool dbg_verify(void *ptr);
size_t dbg_usable_size(void *ptr);
void mm_debug_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/alloc/debug/alloc_debug.h (line 12) - [public] dbg_usable_size
size_t dbg_usable_size(void *ptr);
void mm_debug_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/alloc/debug/alloc_debug.h (line 13) - [public] mm_debug_dump
void mm_debug_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/alloc/debug/leak.h (line 8) - [public] leak_track
bool leak_track(void *ptr, size_t size, uint64_t caller);

bool leak_track_id(void *ptr, size_t size, uint64_t caller, uint64_t *out_id);

bool leak_untrack(void *ptr, size_t *out_size, uint64_t *out_caller);
bool leak_contains(void *ptr);
size_t leak_count(void);

void leak_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/alloc/debug/leak.h (line 10) - [public] leak_track_id
    bool leak_track_id(void *ptr, size_t size, uint64_t caller, uint64_t *out_id);

    bool leak_untrack(void *ptr, size_t *out_size, uint64_t *out_caller);
    bool leak_contains(void *ptr);
    size_t leak_count(void);

    void leak_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/alloc/debug/leak.h (line 12) - [public] leak_untrack
    bool leak_untrack(void *ptr, size_t *out_size, uint64_t *out_caller);
    bool leak_contains(void *ptr);
    size_t leak_count(void);

    void leak_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/alloc/debug/leak.h (line 13) - [public] leak_contains
    bool leak_contains(void *ptr);
    size_t leak_count(void);

    void leak_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/alloc/debug/leak.h (line 14) - [public] leak_count
    size_t leak_count(void);

    void leak_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/alloc/debug/leak.h (line 16) - [public] leak_dump
    void leak_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/alloc/debug/stats.h (line 8) - [public] alloc_stats_note_alloc
    void alloc_stats_note_alloc(size_t size);
    void alloc_stats_note_free(size_t size);

    size_t alloc_stats_live_count(void);
    size_t alloc_stats_live_bytes(void);
    size_t alloc_stats_peak_bytes(void);
    size_t alloc_stats_total_allocs(void);
    size_t alloc_stats_total_frees(void);

    void alloc_stats_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/alloc/debug/stats.h (line 9) - [public] alloc_stats_note_free
    void alloc_stats_note_free(size_t size);

    size_t alloc_stats_live_count(void);
    size_t alloc_stats_live_bytes(void);
    size_t alloc_stats_peak_bytes(void);
    size_t alloc_stats_total_allocs(void);
    size_t alloc_stats_total_frees(void);

    void alloc_stats_dump(void);

#endif
```

```

kernel_Workspace/kernel/src/mm/alloc/debug/stats.h (line 11) - [public] alloc_stats_live_count
    size_t alloc_stats_live_count(void);
    size_t alloc_stats_live_bytes(void);
    size_t alloc_stats_peak_bytes(void);
    size_t alloc_stats_total_allocs(void);
    size_t alloc_stats_total_frees(void);

    void alloc_stats_dump(void);

#endif

kernel_Workspace/kernel/src/mm/alloc/debug/stats.h (line 12) - [public] alloc_stats_live_bytes
    size_t alloc_stats_live_bytes(void);
    size_t alloc_stats_peak_bytes(void);
    size_t alloc_stats_total_allocs(void);
    size_t alloc_stats_total_frees(void);

    void alloc_stats_dump(void);

#endif

kernel_Workspace/kernel/src/mm/alloc/debug/stats.h (line 13) - [public] alloc_stats_peak_bytes
    size_t alloc_stats_peak_bytes(void);
    size_t alloc_stats_total_allocs(void);
    size_t alloc_stats_total_frees(void);

    void alloc_stats_dump(void);

#endif

kernel_Workspace/kernel/src/mm/alloc/debug/stats.h (line 14) - [public]
alloc_stats_total_allocs
    size_t alloc_stats_total_allocs(void);
    size_t alloc_stats_total_frees(void);

    void alloc_stats_dump(void);

#endif

kernel_Workspace/kernel/src/mm/alloc/debug/stats.h (line 15) - [public]
alloc_stats_total_frees
    size_t alloc_stats_total_frees(void);

    void alloc_stats_dump(void);

#endif

kernel_Workspace/kernel/src/mm/alloc/debug/stats.h (line 17) - [public] alloc_stats_dump
    void alloc_stats_dump(void);

#endif

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.h (line 8) - [public] bool
typedef bool (*buddy_map_cb)(uint64_t virt, size_t size);
typedef void (*buddy_unmap_cb)(uint64_t virt, size_t size);

bool buddy_init(uint64_t base,
                size_t size,
                buddy_map_cb map_cb,
                buddy_unmap_cb unmap_cb);

bool buddy_ready(void);

void *buddy_alloc(size_t size);
void *buddy_alloc_aligned(size_t size, size_t align);
// ... (truncated, see source file for more)

```


kernel_Workspace/kernel/src/mm/alloc/heap/buddy.h (line 9) - [public] void

```
typedef void (*buddy_unmap_cb)(uint64_t virt, size_t size);
```

```
bool buddy_init(uint64_t base,  
                size_t size,  
                buddy_map_cb map_cb,  
                buddy_unmap_cb unmap_cb);
```

```
bool buddy_ready(void);
```

```
void *buddy_alloc(size_t size);  
void *buddy_alloc_aligned(size_t size, size_t align);  
void buddy_free(void *ptr);  
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.h (line 16) - [public] buddy_ready

```
bool buddy_ready(void);
```

```
void *buddy_alloc(size_t size);  
void *buddy_alloc_aligned(size_t size, size_t align);  
void buddy_free(void *ptr);
```

```
size_t buddy_block_size(void *ptr);
```

```
size_t buddy_stat_used_bytes(void);  
size_t buddy_stat_free_bytes(void);
```

```
void buddy_dump(void);  
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.h (line 20) - [public] buddy_free

```
void buddy_free(void *ptr);
```

```
size_t buddy_block_size(void *ptr);
```

```
size_t buddy_stat_used_bytes(void);  
size_t buddy_stat_free_bytes(void);
```

```
void buddy_dump(void);
```

```
#endif
```

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.h (line 22) - [public] buddy_block_size

```
size_t buddy_block_size(void *ptr);
```

```
size_t buddy_stat_used_bytes(void);  
size_t buddy_stat_free_bytes(void);
```

```
void buddy_dump(void);
```

```
#endif
```

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.h (line 24) - [public] buddy_stat_used_bytes

```
size_t buddy_stat_used_bytes(void);  
size_t buddy_stat_free_bytes(void);
```

```
void buddy_dump(void);
```

```
#endif
```

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.h (line 25) - [public] buddy_stat_free_bytes

```
size_t buddy_stat_free_bytes(void);
```

```
void buddy_dump(void);
```

```
#endif
```

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.h (line 27) - [public] buddy_dump

```
void buddy_dump(void);
```

```
#endif
```

kernel_Workspace/kernel/src/mm/alloc/heap/heap.h (line 8) - [public] heap_init

```
bool heap_init(void);
bool heap_ready(void);

void *heap_alloc(size_t size);
void *heap_alloc_aligned(size_t size, size_t align);
void *heap_zalloc(size_t size);

void heap_free(void *ptr);

size_t heap_usable_size(void *ptr);

void heap_dump(void);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/heap.h (line 9) - [public] heap_ready

```
bool heap_ready(void);

void *heap_alloc(size_t size);
void *heap_alloc_aligned(size_t size, size_t align);
void *heap_zalloc(size_t size);

void heap_free(void *ptr);

size_t heap_usable_size(void *ptr);

void heap_dump(void);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/heap.h (line 15) - [public] heap_free

```
void heap_free(void *ptr);

size_t heap_usable_size(void *ptr);

void heap_dump(void);

#endif
```

kernel_Workspace/kernel/src/mm/alloc/heap/heap.h (line 17) - [public] heap_usable_size

```
size_t heap_usable_size(void *ptr);

void heap_dump(void);

#endif
```

kernel_Workspace/kernel/src/mm/alloc/heap/heap.h (line 19) - [public] heap_dump

```
void heap_dump(void);

#endif
```

kernel_Workspace/kernel/src/mm/alloc/heap/slab.h (line 10) - [public] slab_init

```
bool slab_init(uint64_t base, size_t size);
bool slab_ready(void);

void *slab_alloc(size_t size);
void slab_free(void *ptr);

size_t slab_usable_size(void *ptr);

size_t slab_stat_used_bytes(void);
size_t slab_stat_free_bytes(void);

void slab_dump(void);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/slab.h (line 11) - [public] slab_ready

```
bool slab_ready(void);

void *slab_alloc(size_t size);
void slab_free(void *ptr);

size_t slab_usable_size(void *ptr);

size_t slab_stat_used_bytes(void);
size_t slab_stat_free_bytes(void);

void slab_dump(void);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/heap/slab.h (line 14) - [public] slab_free

```
void slab_free(void *ptr);

size_t slab_usable_size(void *ptr);

size_t slab_stat_used_bytes(void);
size_t slab_stat_free_bytes(void);

void slab_dump(void);

#endif
```

kernel_Workspace/kernel/src/mm/alloc/heap/slab.h (line 16) - [public] slab_usable_size

```
size_t slab_usable_size(void *ptr);

size_t slab_stat_used_bytes(void);
size_t slab_stat_free_bytes(void);

void slab_dump(void);

#endif
```

kernel_Workspace/kernel/src/mm/alloc/heap/slab.h (line 18) - [public] slab_stat_used_bytes

```
size_t slab_stat_used_bytes(void);
size_t slab_stat_free_bytes(void);

void slab_dump(void);

#endif
```

kernel_Workspace/kernel/src/mm/alloc/heap/slab.h (line 19) - [public] slab_stat_free_bytes

```
size_t slab_stat_free_bytes(void);

void slab_dump(void);

#endif
```

kernel_Workspace/kernel/src/mm/alloc/heap/slab.h (line 21) - [public] slab_dump

```
void slab_dump(void);

#endif
```

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 30) - [public]

bitmap_words_for_bits

```
size_t bitmap_words_for_bits(size_t bit_count);
size_t bitmap_bytes_for_bits(size_t bit_count);

void bitmap_init_virt(bitmap_t *bm, void *storage, size_t bit_count);

bool bitmap_init_phys(bitmap_t *bm, uint64_t storage_phys, size_t bit_count);

void bitmap_fill(bitmap_t *bm, bool value);

void bitmap_set(bitmap_t *bm, size_t bit);
void bitmap_clear(bitmap_t *bm, size_t bit);
bool bitmap_test(const bitmap_t *bm, size_t bit);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 31) - [public]

bitmap_bytes_for_bits

```
size_t bitmap_bytes_for_bits(size_t bit_count);

void bitmap_init_virt(bitmap_t *bm, void *storage, size_t bit_count);

bool bitmap_init_phys(bitmap_t *bm, uint64_t storage_phys, size_t bit_count);

void bitmap_fill(bitmap_t *bm, bool value);

void bitmap_set(bitmap_t *bm, size_t bit);
void bitmap_clear(bitmap_t *bm, size_t bit);
bool bitmap_test(const bitmap_t *bm, size_t bit);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 33) - [public] bitmap_init_virt

```
void bitmap_init_virt(bitmap_t *bm, void *storage, size_t bit_count);

bool bitmap_init_phys(bitmap_t *bm, uint64_t storage_phys, size_t bit_count);

void bitmap_fill(bitmap_t *bm, bool value);

void bitmap_set(bitmap_t *bm, size_t bit);
void bitmap_clear(bitmap_t *bm, size_t bit);
bool bitmap_test(const bitmap_t *bm, size_t bit);

void bitmap_fill(bitmap_t *bm, bool value);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 35) - [public] bitmap_init_phys

```
bool bitmap_init_phys(bitmap_t *bm, uint64_t storage_phys, size_t bit_count);

void bitmap_fill(bitmap_t *bm, bool value);

void bitmap_set(bitmap_t *bm, size_t bit);
void bitmap_clear(bitmap_t *bm, size_t bit);
bool bitmap_test(const bitmap_t *bm, size_t bit);

void bitmap_fill(bitmap_t *bm, bool value);

void bitmap_set(bitmap_t *bm, size_t bit);
void bitmap_clear(bitmap_t *bm, size_t bit);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 37) - [public] bitmap_fill

```
void bitmap_fill(bitmap_t *bm, bool value);

void bitmap_set(bitmap_t *bm, size_t bit);
void bitmap_clear(bitmap_t *bm, size_t bit);
bool bitmap_test(const bitmap_t *bm, size_t bit);

void bitmap_fill(bitmap_t *bm, bool value);

void bitmap_set(bitmap_t *bm, size_t bit);
void bitmap_clear(bitmap_t *bm, size_t bit);
bool bitmap_test(const bitmap_t *bm, size_t bit);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 39) - [public] bitmap_set

```
void bitmap_set(bitmap_t *bm, size_t bit);
void bitmap_clear(bitmap_t *bm, size_t bit);
bool bitmap_test(const bitmap_t *bm, size_t bit);

void bitmap_fill(bitmap_t *bm, bool value);

void bitmap_set(bitmap_t *bm, size_t bit);
void bitmap_clear(bitmap_t *bm, size_t bit);
bool bitmap_test(const bitmap_t *bm, size_t bit);

void bitmap_set_range(bitmap_t *bm, size_t start, size_t count);
void bitmap_clear_range(bitmap_t *bm, size_t start, size_t count);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 40) - [public] bitmap_clear

```
void bitmap_clear(bitmap_t *bm, size_t bit);
bool bitmap_test(const bitmap_t *bm, size_t bit);

void bitmap_fill(bitmap_t *bm, bool value);

void bitmap_set(bitmap_t *bm, size_t bit);
void bitmap_clear(bitmap_t *bm, size_t bit);
bool bitmap_test(const bitmap_t *bm, size_t bit);

void bitmap_set_range(bitmap_t *bm, size_t start, size_t count);
void bitmap_clear_range(bitmap_t *bm, size_t start, size_t count);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 41) - [public] bitmap_test

```
bool bitmap_test(const bitmap_t *bm, size_t bit);

void bitmap_fill(bitmap_t *bm, bool value);

void bitmap_set(bitmap_t *bm, size_t bit);
void bitmap_clear(bitmap_t *bm, size_t bit);
bool bitmap_test(const bitmap_t *bm, size_t bit);

void bitmap_set_range(bitmap_t *bm, size_t start, size_t count);
void bitmap_clear_range(bitmap_t *bm, size_t start, size_t count);

bool bitmap_test_range_free(const bitmap_t *bm, size_t start, size_t count);
// ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 43) - [public] bitmap_fill
    void bitmap_fill(bitmap_t *bm, bool value);

    void bitmap_set(bitmap_t *bm, size_t bit);
    void bitmap_clear(bitmap_t *bm, size_t bit);
    bool bitmap_test(const bitmap_t *bm, size_t bit);

    void bitmap_set_range(bitmap_t *bm, size_t start, size_t count);
    void bitmap_clear_range(bitmap_t *bm, size_t start, size_t count);

    bool bitmap_test_range_free(const bitmap_t *bm, size_t start, size_t count);

    size_t bitmap_alloc(bitmap_t *bm);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 45) - [public] bitmap_set
    void bitmap_set(bitmap_t *bm, size_t bit);
    void bitmap_clear(bitmap_t *bm, size_t bit);
    bool bitmap_test(const bitmap_t *bm, size_t bit);

    void bitmap_set_range(bitmap_t *bm, size_t start, size_t count);
    void bitmap_clear_range(bitmap_t *bm, size_t start, size_t count);

    bool bitmap_test_range_free(const bitmap_t *bm, size_t start, size_t count);

    size_t bitmap_alloc(bitmap_t *bm);

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 46) - [public] bitmap_clear
    void bitmap_clear(bitmap_t *bm, size_t bit);
    bool bitmap_test(const bitmap_t *bm, size_t bit);

    void bitmap_set_range(bitmap_t *bm, size_t start, size_t count);
    void bitmap_clear_range(bitmap_t *bm, size_t start, size_t count);

    bool bitmap_test_range_free(const bitmap_t *bm, size_t start, size_t count);

    size_t bitmap_alloc(bitmap_t *bm);

    size_t bitmap_alloc_from(bitmap_t *bm, size_t min_bit);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 47) - [public] bitmap_test
    bool bitmap_test(const bitmap_t *bm, size_t bit);

    void bitmap_set_range(bitmap_t *bm, size_t start, size_t count);
    void bitmap_clear_range(bitmap_t *bm, size_t start, size_t count);

    bool bitmap_test_range_free(const bitmap_t *bm, size_t start, size_t count);

    size_t bitmap_alloc(bitmap_t *bm);

    size_t bitmap_alloc_from(bitmap_t *bm, size_t min_bit);

    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 49) - [public] bitmap_set_range
    void bitmap_set_range(bitmap_t *bm, size_t start, size_t count);
    void bitmap_clear_range(bitmap_t *bm, size_t start, size_t count);

    bool bitmap_test_range_free(const bitmap_t *bm, size_t start, size_t count);

    size_t bitmap_alloc(bitmap_t *bm);

    size_t bitmap_alloc_from(bitmap_t *bm, size_t min_bit);

    size_t bitmap_alloc_range(bitmap_t *bm, size_t count, size_t align);

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 50) - [public] bitmap_clear_range
    void bitmap_clear_range(bitmap_t *bm, size_t start, size_t count);

    bool bitmap_test_range_free(const bitmap_t *bm, size_t start, size_t count);

    size_t bitmap_alloc(bitmap_t *bm);

    size_t bitmap_alloc_from(bitmap_t *bm, size_t min_bit);

    size_t bitmap_alloc_range(bitmap_t *bm, size_t count, size_t align);

    void bitmap_free(bitmap_t *bm, size_t bit);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 52) - [public]
bitmap_test_range_free
    bool bitmap_test_range_free(const bitmap_t *bm, size_t start, size_t count);

    size_t bitmap_alloc(bitmap_t *bm);

    size_t bitmap_alloc_from(bitmap_t *bm, size_t min_bit);

    size_t bitmap_alloc_range(bitmap_t *bm, size_t count, size_t align);

    void bitmap_free(bitmap_t *bm, size_t bit);
    void bitmap_free_range(bitmap_t *bm, size_t start, size_t count);

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 54) - [public] bitmap_alloc
    size_t bitmap_alloc(bitmap_t *bm);

    size_t bitmap_alloc_from(bitmap_t *bm, size_t min_bit);

    size_t bitmap_alloc_range(bitmap_t *bm, size_t count, size_t align);

    void bitmap_free(bitmap_t *bm, size_t bit);
    void bitmap_free_range(bitmap_t *bm, size_t start, size_t count);

    size_t bitmap_count_free(const bitmap_t *bm);

    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 57) - [public] bitmap_alloc_from
    size_t bitmap_alloc_from(bitmap_t *bm, size_t min_bit);

    size_t bitmap_alloc_range(bitmap_t *bm, size_t count, size_t align);

    void bitmap_free(bitmap_t *bm, size_t bit);
    void bitmap_free_range(bitmap_t *bm, size_t start, size_t count);

    size_t bitmap_count_free(const bitmap_t *bm);

#endif

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 59) - [public] bitmap_alloc_range
    size_t bitmap_alloc_range(bitmap_t *bm, size_t count, size_t align);

    void bitmap_free(bitmap_t *bm, size_t bit);
    void bitmap_free_range(bitmap_t *bm, size_t start, size_t count);

    size_t bitmap_count_free(const bitmap_t *bm);

#endif

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 61) - [public] bitmap_free
    void bitmap_free(bitmap_t *bm, size_t bit);
    void bitmap_free_range(bitmap_t *bm, size_t start, size_t count);

    size_t bitmap_count_free(const bitmap_t *bm);

#endif

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 62) - [public] bitmap_free_range
    void bitmap_free_range(bitmap_t *bm, size_t start, size_t count);

    size_t bitmap_count_free(const bitmap_t *bm);

#endif

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 64) - [public] bitmap_count_free
    size_t bitmap_count_free(const bitmap_t *bm);

#endif

kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 12) - [public] frame_init
    bool frame_init(uint64_t bitmap_phys, size_t bit_count);

    bool frame_init_from_memory(void);

    bool frame_ready(void);

    bool frame_alloc(frame_t *out);
    bool frame_alloc_zero(frame_t *out);

    bool frame_alloc_contiguous(size_t count,
                                size_t align_frames,
                                frame_t *out);
    // ... (truncated, see source file for more)

```


kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 14) - [public]

frame_init_from_memory

```
bool frame_init_from_memory(void);

bool frame_ready(void);

bool frame_alloc(frame_t *out);
bool frame_alloc_zero(frame_t *out);

bool frame_alloc_contiguous(size_t count,
                             size_t align_frames,
                             frame_t *out);
bool frame_alloc_below(size_t count,
                        size_t align_frames,
                        // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 16) - [public] frame_ready

```
bool frame_ready(void);

bool frame_alloc(frame_t *out);
bool frame_alloc_zero(frame_t *out);

bool frame_alloc_contiguous(size_t count,
                             size_t align_frames,
                             frame_t *out);
bool frame_alloc_below(size_t count,
                        size_t align_frames,
                        uint64_t max_phys,
                        frame_t *out);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 18) - [public] frame_alloc

```
bool frame_alloc(frame_t *out);
bool frame_alloc_zero(frame_t *out);

bool frame_alloc_contiguous(size_t count,
                             size_t align_frames,
                             frame_t *out);
bool frame_alloc_below(size_t count,
                        size_t align_frames,
                        uint64_t max_phys,
                        frame_t *out);

bool frame_free(frame_t frame);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 19) - [public] frame_alloc_zero

```
bool frame_alloc_zero(frame_t *out);

bool frame_alloc_contiguous(size_t count,
                             size_t align_frames,
                             frame_t *out);
bool frame_alloc_below(size_t count,
                        size_t align_frames,
                        uint64_t max_phys,
                        frame_t *out);

bool frame_free(frame_t frame);
bool frame_free_contiguous(frame_t start, size_t count);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 29) - [public] frame_free

```
bool frame_free(frame_t frame);
bool frame_free_contiguous(frame_t start, size_t count);

bool frame_is_valid(frame_t frame);
bool frame_zero(frame_t frame);

void *frame_virt(frame_t frame);
uint64_t frame_phys(frame_t frame);

size_t frame_to_pfn(frame_t frame);
frame_t frame_from_pfn(size_t pfn);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 30) - [public]
frame_free_contiguous

```
bool frame_free_contiguous(frame_t start, size_t count);

bool frame_is_valid(frame_t frame);
bool frame_zero(frame_t frame);

void *frame_virt(frame_t frame);
uint64_t frame_phys(frame_t frame);

size_t frame_to_pfn(frame_t frame);
frame_t frame_from_pfn(size_t pfn);

size_t frame_total(void);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 32) - [public] frame_is_valid

```
bool frame_is_valid(frame_t frame);
bool frame_zero(frame_t frame);

void *frame_virt(frame_t frame);
uint64_t frame_phys(frame_t frame);

size_t frame_to_pfn(frame_t frame);
frame_t frame_from_pfn(size_t pfn);

size_t frame_total(void);
size_t frame_allocated(void);
size_t frame_free_count(void);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 33) - [public] frame_zero

```
bool frame_zero(frame_t frame);

void *frame_virt(frame_t frame);
uint64_t frame_phys(frame_t frame);

size_t frame_to_pfn(frame_t frame);
frame_t frame_from_pfn(size_t pfn);

size_t frame_total(void);
size_t frame_allocated(void);
size_t frame_free_count(void);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 36) - [public] frame_phys

```
uint64_t frame_phys(frame_t frame);

size_t frame_to_pfn(frame_t frame);
frame_t frame_from_pfn(size_t pfn);

size_t frame_total(void);
size_t frame_allocated(void);
size_t frame_free_count(void);

size_t frame_zone_dma32_total(void);
size_t frame_zone_dma32_free(void);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 38) - [public] frame_to_pfn

```
size_t frame_to_pfn(frame_t frame);
frame_t frame_from_pfn(size_t pfn);

size_t frame_total(void);
size_t frame_allocated(void);
size_t frame_free_count(void);

size_t frame_zone_dma32_total(void);
size_t frame_zone_dma32_free(void);
size_t frame_zone_normal_total(void);
size_t frame_zone_normal_free(void);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 39) - [public] frame_from_pfn

```
frame_t frame_from_pfn(size_t pfn);

size_t frame_total(void);
size_t frame_allocated(void);
size_t frame_free_count(void);

size_t frame_zone_dma32_total(void);
size_t frame_zone_dma32_free(void);
size_t frame_zone_normal_total(void);
size_t frame_zone_normal_free(void);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 41) - [public] frame_total

```
size_t frame_total(void);
size_t frame_allocated(void);
size_t frame_free_count(void);

size_t frame_zone_dma32_total(void);
size_t frame_zone_dma32_free(void);
size_t frame_zone_normal_total(void);
size_t frame_zone_normal_free(void);

#endif
```

kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 42) - [public] frame_allocated

```
size_t frame_allocated(void);
size_t frame_free_count(void);

size_t frame_zone_dma32_total(void);
size_t frame_zone_dma32_free(void);
size_t frame_zone_normal_total(void);
size_t frame_zone_normal_free(void);

#endif
```

```

kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 43) - [public] frame_free_count
    size_t frame_free_count(void);

    size_t frame_zone_dma32_total(void);
    size_t frame_zone_dma32_free(void);
    size_t frame_zone_normal_total(void);
    size_t frame_zone_normal_free(void);

    #endif

kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 46) - [public]
frame_zone_dma32_total
    size_t frame_zone_dma32_total(void);
    size_t frame_zone_dma32_free(void);
    size_t frame_zone_normal_total(void);
    size_t frame_zone_normal_free(void);

    #endif

kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 47) - [public]
frame_zone_dma32_free
    size_t frame_zone_dma32_free(void);
    size_t frame_zone_normal_total(void);
    size_t frame_zone_normal_free(void);

    #endif

kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 48) - [public]
frame_zone_normal_total
    size_t frame_zone_normal_total(void);
    size_t frame_zone_normal_free(void);

    #endif

kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 49) - [public]
frame_zone_normal_free
    size_t frame_zone_normal_free(void);

    #endif

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 10) - [public] pmm_init
    bool pmm_init(void);

    bool pmm_ready(void);

    bool pmm_alloc_frame(uint64_t *out_phys);

    bool pmm_alloc_zero_frame(uint64_t *out_phys);

    bool pmm_alloc_frames(size_t count, uint64_t *out_phys);

    bool pmm_alloc_frames_aligned(size_t count,
                                   size_t align_frames,
                                   // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 12) - [public] pmm_ready
    bool pmm_ready(void);

    bool pmm_alloc_frame(uint64_t *out_phys);

    bool pmm_alloc_zero_frame(uint64_t *out_phys);

    bool pmm_alloc_frames(size_t count, uint64_t *out_phys);

    bool pmm_alloc_frames_aligned(size_t count,
                                   size_t align_frames,
                                   uint64_t *out_phys);
    bool pmm_alloc_bytes(size_t bytes, uint64_t *out_phys);
    // ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 14) - [public] pmm_alloc_frame

```
bool pmm_alloc_frame(uint64_t *out_phys);

bool pmm_alloc_zero_frame(uint64_t *out_phys);

bool pmm_alloc_frames(size_t count, uint64_t *out_phys);

bool pmm_alloc_frames_aligned(size_t count,
                              size_t align_frames,
                              uint64_t *out_phys);
bool pmm_alloc_bytes(size_t bytes, uint64_t *out_phys);
bool pmm_alloc_zero_bytes(size_t bytes, uint64_t *out_phys);
bool pmm_alloc_contiguous_bytes(size_t bytes,
                                // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 16) - [public] pmm_alloc_zero_frame

```
bool pmm_alloc_zero_frame(uint64_t *out_phys);

bool pmm_alloc_frames(size_t count, uint64_t *out_phys);

bool pmm_alloc_frames_aligned(size_t count,
                              size_t align_frames,
                              uint64_t *out_phys);
bool pmm_alloc_bytes(size_t bytes, uint64_t *out_phys);
bool pmm_alloc_zero_bytes(size_t bytes, uint64_t *out_phys);
bool pmm_alloc_contiguous_bytes(size_t bytes,
                                size_t align_bytes,
                                uint64_t *out_phys);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 18) - [public] pmm_alloc_frames

```
bool pmm_alloc_frames(size_t count, uint64_t *out_phys);

bool pmm_alloc_frames_aligned(size_t count,
                              size_t align_frames,
                              uint64_t *out_phys);
bool pmm_alloc_bytes(size_t bytes, uint64_t *out_phys);
bool pmm_alloc_zero_bytes(size_t bytes, uint64_t *out_phys);
bool pmm_alloc_contiguous_bytes(size_t bytes,
                                size_t align_bytes,
                                uint64_t *out_phys);

bool pmm_free_frame(uint64_t phys);
bool pmm_alloc_frames_below(size_t count,
                            // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 23) - [public] pmm_alloc_bytes

```
bool pmm_alloc_bytes(size_t bytes, uint64_t *out_phys);
bool pmm_alloc_zero_bytes(size_t bytes, uint64_t *out_phys);
bool pmm_alloc_contiguous_bytes(size_t bytes,
                                size_t align_bytes,
                                uint64_t *out_phys);

bool pmm_free_frame(uint64_t phys);
bool pmm_alloc_frames_below(size_t count,
                            size_t align_frames,
                            uint64_t max_phys,
                            uint64_t *out_phys);
bool pmm_free_frames(uint64_t phys, size_t count);
bool pmm_free_bytes(uint64_t phys, size_t bytes);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 24) - [public] pmm_alloc_zero_bytes

```
bool pmm_alloc_zero_bytes(size_t bytes, uint64_t *out_phys);
bool pmm_alloc_contiguous_bytes(size_t bytes,
                                size_t align_bytes,
                                uint64_t *out_phys);
bool pmm_free_frame(uint64_t phys);
bool pmm_alloc_frames_below(size_t count,
                            size_t align_frames,
                            uint64_t max_phys,
                            uint64_t *out_phys);
bool pmm_free_frames(uint64_t phys, size_t count);
bool pmm_free_bytes(uint64_t phys, size_t bytes);
size_t pmm_stat_total_frames(void);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 28) - [public] pmm_free_frame

```
bool pmm_free_frame(uint64_t phys);
bool pmm_alloc_frames_below(size_t count,
                            size_t align_frames,
                            uint64_t max_phys,
                            uint64_t *out_phys);
bool pmm_free_frames(uint64_t phys, size_t count);
bool pmm_free_bytes(uint64_t phys, size_t bytes);
size_t pmm_stat_total_frames(void);
size_t pmm_stat_free_frames(void);
size_t pmm_stat_allocated_frames(void);

uint64_t pmm_stat_total_bytes(void);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 33) - [public] pmm_free_frames

```
bool pmm_free_frames(uint64_t phys, size_t count);
bool pmm_free_bytes(uint64_t phys, size_t bytes);
size_t pmm_stat_total_frames(void);
size_t pmm_stat_free_frames(void);
size_t pmm_stat_allocated_frames(void);

uint64_t pmm_stat_total_bytes(void);
uint64_t pmm_stat_free_bytes(void);

size_t pmm_stat_dma32_total_frames(void);
size_t pmm_stat_dma32_free_frames(void);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 34) - [public] pmm_free_bytes

```
bool pmm_free_bytes(uint64_t phys, size_t bytes);
size_t pmm_stat_total_frames(void);
size_t pmm_stat_free_frames(void);
size_t pmm_stat_allocated_frames(void);

uint64_t pmm_stat_total_bytes(void);
uint64_t pmm_stat_free_bytes(void);

size_t pmm_stat_dma32_total_frames(void);
size_t pmm_stat_dma32_free_frames(void);
size_t pmm_stat_normal_total_frames(void);
// ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 35) - [public] pmm_stat_total_frames
    size_t pmm_stat_total_frames(void);
    size_t pmm_stat_free_frames(void);
    size_t pmm_stat_allocated_frames(void);

    uint64_t pmm_stat_total_bytes(void);
    uint64_t pmm_stat_free_bytes(void);

    size_t pmm_stat_dma32_total_frames(void);
    size_t pmm_stat_dma32_free_frames(void);
    size_t pmm_stat_normal_total_frames(void);
    size_t pmm_stat_normal_free_frames(void);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 36) - [public] pmm_stat_free_frames
    size_t pmm_stat_free_frames(void);
    size_t pmm_stat_allocated_frames(void);

    uint64_t pmm_stat_total_bytes(void);
    uint64_t pmm_stat_free_bytes(void);

    size_t pmm_stat_dma32_total_frames(void);
    size_t pmm_stat_dma32_free_frames(void);
    size_t pmm_stat_normal_total_frames(void);
    size_t pmm_stat_normal_free_frames(void);

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 37) - [public]
pmm_stat_allocated_frames
    size_t pmm_stat_allocated_frames(void);

    uint64_t pmm_stat_total_bytes(void);
    uint64_t pmm_stat_free_bytes(void);

    size_t pmm_stat_dma32_total_frames(void);
    size_t pmm_stat_dma32_free_frames(void);
    size_t pmm_stat_normal_total_frames(void);
    size_t pmm_stat_normal_free_frames(void);

    void pmm_dump(void);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 39) - [public] pmm_stat_total_bytes
    uint64_t pmm_stat_total_bytes(void);
    uint64_t pmm_stat_free_bytes(void);

    size_t pmm_stat_dma32_total_frames(void);
    size_t pmm_stat_dma32_free_frames(void);
    size_t pmm_stat_normal_total_frames(void);
    size_t pmm_stat_normal_free_frames(void);

    void pmm_dump(void);

#ifdef PMM_DEBUG
    // ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 40) - [public] pmm_stat_free_bytes

```
uint64_t pmm_stat_free_bytes(void);
```

```
size_t pmm_stat_dma32_total_frames(void);  
size_t pmm_stat_dma32_free_frames(void);  
size_t pmm_stat_normal_total_frames(void);  
size_t pmm_stat_normal_free_frames(void);
```

```
void pmm_dump(void);
```

```
#ifdef PMM_DEBUG  
bool pmm_self_test(void);  
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 43) - [public]

pmm_stat_dma32_total_frames

```
size_t pmm_stat_dma32_total_frames(void);  
size_t pmm_stat_dma32_free_frames(void);  
size_t pmm_stat_normal_total_frames(void);  
size_t pmm_stat_normal_free_frames(void);
```

```
void pmm_dump(void);
```

```
#ifdef PMM_DEBUG  
bool pmm_self_test(void);  
#endif
```

```
#endif
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 44) - [public]

pmm_stat_dma32_free_frames

```
size_t pmm_stat_dma32_free_frames(void);  
size_t pmm_stat_normal_total_frames(void);  
size_t pmm_stat_normal_free_frames(void);
```

```
void pmm_dump(void);
```

```
#ifdef PMM_DEBUG  
bool pmm_self_test(void);  
#endif
```

```
#endif
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 45) - [public]

pmm_stat_normal_total_frames

```
size_t pmm_stat_normal_total_frames(void);  
size_t pmm_stat_normal_free_frames(void);
```

```
void pmm_dump(void);
```

```
#ifdef PMM_DEBUG  
bool pmm_self_test(void);  
#endif
```

```
#endif
```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 46) - [public]

pmm_stat_normal_free_frames

```
size_t pmm_stat_normal_free_frames(void);
```

```
void pmm_dump(void);
```

```
#ifdef PMM_DEBUG  
bool pmm_self_test(void);  
#endif
```

```
#endif
```



```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 48) - [public] pmm_dump
    void pmm_dump(void);

    #ifdef PMM_DEBUG
    bool pmm_self_test(void);
    #endif

    #endif

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 51) - [public] pmm_self_test
    bool pmm_self_test(void);
    #endif

    #endif

kernel_Workspace/kernel/src/mm/alloc/special/contiguous.h (line 13) - [public] contig_free
    void contig_free(uint64_t phys, size_t bytes);

    #endif

kernel_Workspace/kernel/src/mm/alloc/special/dma.h (line 16) - [public] dma_free_coherent
    void dma_free_coherent(uint64_t phys, void *virt, size_t bytes);

    void dma_sync_for_device(void *virt, size_t len);
    void dma_sync_for_cpu(void *virt, size_t len);

    #endif

kernel_Workspace/kernel/src/mm/alloc/special/dma.h (line 18) - [public] dma_sync_for_device
    void dma_sync_for_device(void *virt, size_t len);
    void dma_sync_for_cpu(void *virt, size_t len);

    #endif

kernel_Workspace/kernel/src/mm/alloc/special/dma.h (line 19) - [public] dma_sync_for_cpu
    void dma_sync_for_cpu(void *virt, size_t len);

    #endif

kernel_Workspace/kernel/src/mm/alloc/virtual/mapping.h (line 12) - [public] mapping_init
    void mapping_init(void);
    mapping_space_t mapping_kernel_space(void);

    bool mapping_map_range(mapping_space_t space,
                           uint64_t virt,
                           uint64_t phys,
                           size_t len,
                           uint64_t pte_flags);

    bool mapping_unmap_range(mapping_space_t space,
                             uint64_t virt,
                             size_t len);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/alloc/virtual/mapping.h (line 13) - [public]
mapping_kernel_space
    mapping_space_t mapping_kernel_space(void);

    bool mapping_map_range(mapping_space_t space,
                           uint64_t virt,
                           uint64_t phys,
                           size_t len,
                           uint64_t pte_flags);

    bool mapping_unmap_range(mapping_space_t space,
                             uint64_t virt,
                             size_t len);

    // ... (truncated, see source file for more)

```

```
kernel_Workspace/kernel/src/mm/alloc/virtual/mapping.h (line 30) - [public] mapping_translate
uint64_t mapping_translate(mapping_space_t space, uint64_t virt);
bool mapping_is_mapped(mapping_space_t space, uint64_t virt);

bool mapping_copy_range(mapping_space_t dst,
                        uint64_t dst_virt,
                        mapping_space_t src,
                        uint64_t src_virt,
                        size_t len);

#endif
```

```
kernel_Workspace/kernel/src/mm/alloc/virtual/mapping.h (line 31) - [public] mapping_is_mapped
bool mapping_is_mapped(mapping_space_t space, uint64_t virt);

bool mapping_copy_range(mapping_space_t dst,
                        uint64_t dst_virt,
                        mapping_space_t src,
                        uint64_t src_virt,
                        size_t len);

#endif
```

```
kernel_Workspace/kernel/src/mm/alloc/virtual/page.h (line 22) - [public] page_init
bool page_init(void);
bool page_ready(void);

page_t *page_alloc(uint32_t type);
page_t *page_alloc_zero(uint32_t type);

bool page_attach(uint64_t phys, uint32_t type);

bool page_get(page_t *page);
bool page_put(page_t *page);

uint64_t page_phys(const page_t *page);
// ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/mm/alloc/virtual/page.h (line 23) - [public] page_ready
bool page_ready(void);

page_t *page_alloc(uint32_t type);
page_t *page_alloc_zero(uint32_t type);

bool page_attach(uint64_t phys, uint32_t type);

bool page_get(page_t *page);
bool page_put(page_t *page);

uint64_t page_phys(const page_t *page);
void *page_kva(const page_t *page);
// ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/mm/alloc/virtual/page.h (line 28) - [public] page_attach
bool page_attach(uint64_t phys, uint32_t type);

bool page_get(page_t *page);
bool page_put(page_t *page);

uint64_t page_phys(const page_t *page);
void *page_kva(const page_t *page);
size_t page_pfn(const page_t *page);

page_t *page_from_phys(uint64_t phys);
page_t *page_from_pfn(size_t pfn);
page_t *page_from_kva(const void *va);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/page.h (line 30) - [public] page_get

```
bool page_get(page_t *page);
bool page_put(page_t *page);

uint64_t page_phys(const page_t *page);
void *page_kva(const page_t *page);
size_t page_pfn(const page_t *page);

page_t *page_from_phys(uint64_t phys);
page_t *page_from_pfn(size_t pfn);
page_t *page_from_kva(const void *va);

uint32_t page_refcount(const page_t *page);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/page.h (line 31) - [public] page_put

```
bool page_put(page_t *page);

uint64_t page_phys(const page_t *page);
void *page_kva(const page_t *page);
size_t page_pfn(const page_t *page);

page_t *page_from_phys(uint64_t phys);
page_t *page_from_pfn(size_t pfn);
page_t *page_from_kva(const void *va);

uint32_t page_refcount(const page_t *page);
uint32_t page_type(const page_t *page);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/page.h (line 33) - [public] page_phys

```
uint64_t page_phys(const page_t *page);
void *page_kva(const page_t *page);
size_t page_pfn(const page_t *page);

page_t *page_from_phys(uint64_t phys);
page_t *page_from_pfn(size_t pfn);
page_t *page_from_kva(const void *va);

uint32_t page_refcount(const page_t *page);
uint32_t page_type(const page_t *page);
void page_set_type(page_t *page, uint32_t type);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/page.h (line 35) - [public] page_pfn

```
size_t page_pfn(const page_t *page);

page_t *page_from_phys(uint64_t phys);
page_t *page_from_pfn(size_t pfn);
page_t *page_from_kva(const void *va);

uint32_t page_refcount(const page_t *page);
uint32_t page_type(const page_t *page);
void page_set_type(page_t *page, uint32_t type);

size_t page_stat_total(void);
size_t page_stat_referenced(void);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/page.h (line 41) - [public] page_refcount

```
uint32_t page_refcount(const page_t *page);
uint32_t page_type(const page_t *page);
void page_set_type(page_t *page, uint32_t type);

size_t page_stat_total(void);
size_t page_stat_referenced(void);

void page_dump(void);

#endif
```

kernel_Workspace/kernel/src/mm/alloc/virtual/page.h (line 42) - [public] page_type

```
uint32_t page_type(const page_t *page);
void page_set_type(page_t *page, uint32_t type);

size_t page_stat_total(void);
size_t page_stat_referenced(void);

void page_dump(void);

#endif
```

kernel_Workspace/kernel/src/mm/alloc/virtual/page.h (line 43) - [public] page_set_type

```
void page_set_type(page_t *page, uint32_t type);

size_t page_stat_total(void);
size_t page_stat_referenced(void);

void page_dump(void);

#endif
```

kernel_Workspace/kernel/src/mm/alloc/virtual/page.h (line 45) - [public] page_stat_total

```
size_t page_stat_total(void);
size_t page_stat_referenced(void);

void page_dump(void);

#endif
```

kernel_Workspace/kernel/src/mm/alloc/virtual/page.h (line 46) - [public] page_stat_referenced

```
size_t page_stat_referenced(void);

void page_dump(void);

#endif
```

kernel_Workspace/kernel/src/mm/alloc/virtual/page.h (line 48) - [public] page_dump

```
void page_dump(void);

#endif
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 18) - [public] vmm_init

```
bool vmm_init(void);
bool vmm_ready(void);

bool vmm_alloc(size_t bytes, uint32_t flags, uint64_t *out_virt);
bool vmm_alloc_aligned(size_t bytes, size_t align, uint32_t flags, uint64_t *out_virt);
bool vmm_map_device(uint64_t phys, size_t len, uint64_t *out_virt);

bool vmm_free(uint64_t virt, size_t bytes);
bool vmm_unmap_device(uint64_t virt, size_t len);

uint64_t vmm_translate(uint64_t virt);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 19) - [public] vmm_ready

```
bool vmm_ready(void);

bool vmm_alloc(size_t bytes, uint32_t flags, uint64_t *out_virt);
bool vmm_alloc_aligned(size_t bytes, size_t align, uint32_t flags, uint64_t *out_virt);
bool vmm_map_device(uint64_t phys, size_t len, uint64_t *out_virt);

bool vmm_free(uint64_t virt, size_t bytes);
bool vmm_unmap_device(uint64_t virt, size_t len);

uint64_t vmm_translate(uint64_t virt);

size_t vmm_stat_total_pages(void);
// ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 21) - [public] vmm_alloc
    bool vmm_alloc(size_t bytes, uint32_t flags, uint64_t *out_virt);
    bool vmm_alloc_aligned(size_t bytes, size_t align, uint32_t flags, uint64_t *out_virt);
    bool vmm_map_device(uint64_t phys, size_t len, uint64_t *out_virt);

    bool vmm_free(uint64_t virt, size_t bytes);
    bool vmm_unmap_device(uint64_t virt, size_t len);

    uint64_t vmm_translate(uint64_t virt);

    size_t vmm_stat_total_pages(void);
    size_t vmm_stat_free_pages(void);
    size_t vmm_stat_allocated_pages(void);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 22) - [public] vmm_alloc_aligned
    bool vmm_alloc_aligned(size_t bytes, size_t align, uint32_t flags, uint64_t *out_virt);
    bool vmm_map_device(uint64_t phys, size_t len, uint64_t *out_virt);

    bool vmm_free(uint64_t virt, size_t bytes);
    bool vmm_unmap_device(uint64_t virt, size_t len);

    uint64_t vmm_translate(uint64_t virt);

    size_t vmm_stat_total_pages(void);
    size_t vmm_stat_free_pages(void);
    size_t vmm_stat_allocated_pages(void);
    uint64_t vmm_stat_total_bytes(void);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 23) - [public] vmm_map_device
    bool vmm_map_device(uint64_t phys, size_t len, uint64_t *out_virt);

    bool vmm_free(uint64_t virt, size_t bytes);
    bool vmm_unmap_device(uint64_t virt, size_t len);

    uint64_t vmm_translate(uint64_t virt);

    size_t vmm_stat_total_pages(void);
    size_t vmm_stat_free_pages(void);
    size_t vmm_stat_allocated_pages(void);
    uint64_t vmm_stat_total_bytes(void);
    uint64_t vmm_stat_free_bytes(void);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 25) - [public] vmm_free
    bool vmm_free(uint64_t virt, size_t bytes);
    bool vmm_unmap_device(uint64_t virt, size_t len);

    uint64_t vmm_translate(uint64_t virt);

    size_t vmm_stat_total_pages(void);
    size_t vmm_stat_free_pages(void);
    size_t vmm_stat_allocated_pages(void);
    uint64_t vmm_stat_total_bytes(void);
    uint64_t vmm_stat_free_bytes(void);

    void vmm_dump(void);
    // ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 26) - [public] vmm_unmap_device

```
bool vmm_unmap_device(uint64_t virt, size_t len);

uint64_t vmm_translate(uint64_t virt);

size_t vmm_stat_total_pages(void);
size_t vmm_stat_free_pages(void);
size_t vmm_stat_allocated_pages(void);
uint64_t vmm_stat_total_bytes(void);
uint64_t vmm_stat_free_bytes(void);

void vmm_dump(void);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 28) - [public] vmm_translate

```
uint64_t vmm_translate(uint64_t virt);

size_t vmm_stat_total_pages(void);
size_t vmm_stat_free_pages(void);
size_t vmm_stat_allocated_pages(void);
uint64_t vmm_stat_total_bytes(void);
uint64_t vmm_stat_free_bytes(void);

void vmm_dump(void);

#endif
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 30) - [public] vmm_stat_total_pages

```
size_t vmm_stat_total_pages(void);
size_t vmm_stat_free_pages(void);
size_t vmm_stat_allocated_pages(void);
uint64_t vmm_stat_total_bytes(void);
uint64_t vmm_stat_free_bytes(void);

void vmm_dump(void);

#endif
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 31) - [public] vmm_stat_free_pages

```
size_t vmm_stat_free_pages(void);
size_t vmm_stat_allocated_pages(void);
uint64_t vmm_stat_total_bytes(void);
uint64_t vmm_stat_free_bytes(void);

void vmm_dump(void);

#endif
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 32) - [public]

vmm_stat_allocated_pages

```
size_t vmm_stat_allocated_pages(void);
uint64_t vmm_stat_total_bytes(void);
uint64_t vmm_stat_free_bytes(void);

void vmm_dump(void);

#endif
```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 33) - [public] vmm_stat_total_bytes

```
uint64_t vmm_stat_total_bytes(void);
uint64_t vmm_stat_free_bytes(void);

void vmm_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 34) - [public] vmm_stat_free_bytes
    uint64_t vmm_stat_free_bytes(void);

    void vmm_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 36) - [public] vmm_dump
    void vmm_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/arch/aarch64/memory.h (line 25) - [public] arch_memory_ready
    bool arch_memory_ready(void);

    bool arch_memory_boot_alloc(uint64_t len, uint64_t align, uint64_t *out);
    void arch_memory_reserve_range(uint64_t base, uint64_t len);

    bool arch_is_page_aligned(uint64_t a);
    uint64_t arch_page_align_up(uint64_t a);
    uint64_t arch_page_align_down(uint64_t a);

    void *arch_phys_to_virt(uint64_t phys);
    uint64_t arch_virt_to_phys(void *virt);

    // ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/mm/arch/aarch64/memory.h (line 27) - [public]
arch_memory_boot_alloc
    bool arch_memory_boot_alloc(uint64_t len, uint64_t align, uint64_t *out);
    void arch_memory_reserve_range(uint64_t base, uint64_t len);

    bool arch_is_page_aligned(uint64_t a);
    uint64_t arch_page_align_up(uint64_t a);
    uint64_t arch_page_align_down(uint64_t a);

    void *arch_phys_to_virt(uint64_t phys);
    uint64_t arch_virt_to_phys(void *virt);

    uint64_t arch_memory_max_phys(void);
    void arch_memory_dump(void);
    // ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/mm/arch/aarch64/memory.h (line 28) - [public]
arch_memory_reserve_range
    void arch_memory_reserve_range(uint64_t base, uint64_t len);

    bool arch_is_page_aligned(uint64_t a);
    uint64_t arch_page_align_up(uint64_t a);
    uint64_t arch_page_align_down(uint64_t a);

    void *arch_phys_to_virt(uint64_t phys);
    uint64_t arch_virt_to_phys(void *virt);

    uint64_t arch_memory_max_phys(void);
    void arch_memory_dump(void);

    // ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/mm/arch/aarch64/memory.h (line 30) - [public] arch_is_page_aligned
    bool arch_is_page_aligned(uint64_t a);
    uint64_t arch_page_align_up(uint64_t a);
    uint64_t arch_page_align_down(uint64_t a);

    void *arch_phys_to_virt(uint64_t phys);
    uint64_t arch_virt_to_phys(void *virt);

    uint64_t arch_memory_max_phys(void);
    void arch_memory_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/arch/aarch64/memory.h (line 31) - [public] arch_page_align_up
uint64_t arch_page_align_up(uint64_t a);
uint64_t arch_page_align_down(uint64_t a);

void *arch_phys_to_virt(uint64_t phys);
uint64_t arch_virt_to_phys(void *virt);

uint64_t arch_memory_max_phys(void);
void arch_memory_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/arch/aarch64/memory.h (line 32) - [public] arch_page_align_down
uint64_t arch_page_align_down(uint64_t a);

void *arch_phys_to_virt(uint64_t phys);
uint64_t arch_virt_to_phys(void *virt);

uint64_t arch_memory_max_phys(void);
void arch_memory_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/arch/aarch64/memory.h (line 35) - [public] arch_virt_to_phys
uint64_t arch_virt_to_phys(void *virt);

uint64_t arch_memory_max_phys(void);
void arch_memory_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/arch/aarch64/memory.h (line 37) - [public] arch_memory_max_phys
uint64_t arch_memory_max_phys(void);
void arch_memory_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/arch/aarch64/memory.h (line 38) - [public] arch_memory_dump
void arch_memory_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.h (line 17) - [public] paging_init
void paging_init(uint64_t boot_phys_offset);

bool paging_enable_nx(void);

uint64_t paging_read_cr3(void);
void paging_write_cr3(uint64_t pml4_phys);

bool paging_map_page_in(uint64_t pml4, uint64_t virt, uint64_t phys, uint64_t flags);
bool paging_unmap_page_in(uint64_t pml4, uint64_t virt);
bool paging_set_flags_in(uint64_t pml4, uint64_t virt, uint64_t flags);
bool paging_is_mapped_in(uint64_t pml4, uint64_t virt);
uint64_t paging_translate_in(uint64_t pml4, uint64_t virt);
// ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.h (line 19) - [public] paging_enable_nx
bool paging_enable_nx(void);

uint64_t paging_read_cr3(void);
void paging_write_cr3(uint64_t pml4_phys);

bool paging_map_page_in(uint64_t pml4, uint64_t virt, uint64_t phys, uint64_t flags);
bool paging_unmap_page_in(uint64_t pml4, uint64_t virt);
bool paging_set_flags_in(uint64_t pml4, uint64_t virt, uint64_t flags);
bool paging_is_mapped_in(uint64_t pml4, uint64_t virt);
uint64_t paging_translate_in(uint64_t pml4, uint64_t virt);

#endif
```


kernel_Workspace/kernel/src/mm/arch/aarch64/paging.h (line 21) - [public] paging_read_cr3

```
uint64_t paging_read_cr3(void);
void paging_write_cr3(uint64_t pml4_phys);

bool paging_map_page_in(uint64_t pml4, uint64_t virt, uint64_t phys, uint64_t flags);
bool paging_unmap_page_in(uint64_t pml4, uint64_t virt);
bool paging_set_flags_in(uint64_t pml4, uint64_t virt, uint64_t flags);
bool paging_is_mapped_in(uint64_t pml4, uint64_t virt);
uint64_t paging_translate_in(uint64_t pml4, uint64_t virt);

#endif
```

kernel_Workspace/kernel/src/mm/arch/aarch64/paging.h (line 22) - [public] paging_write_cr3

```
void paging_write_cr3(uint64_t pml4_phys);

bool paging_map_page_in(uint64_t pml4, uint64_t virt, uint64_t phys, uint64_t flags);
bool paging_unmap_page_in(uint64_t pml4, uint64_t virt);
bool paging_set_flags_in(uint64_t pml4, uint64_t virt, uint64_t flags);
bool paging_is_mapped_in(uint64_t pml4, uint64_t virt);
uint64_t paging_translate_in(uint64_t pml4, uint64_t virt);

#endif
```

kernel_Workspace/kernel/src/mm/arch/aarch64/paging.h (line 24) - [public] paging_map_page_in

```
bool paging_map_page_in(uint64_t pml4, uint64_t virt, uint64_t phys, uint64_t flags);
bool paging_unmap_page_in(uint64_t pml4, uint64_t virt);
bool paging_set_flags_in(uint64_t pml4, uint64_t virt, uint64_t flags);
bool paging_is_mapped_in(uint64_t pml4, uint64_t virt);
uint64_t paging_translate_in(uint64_t pml4, uint64_t virt);

#endif
```

kernel_Workspace/kernel/src/mm/arch/aarch64/paging.h (line 25) - [public] paging_unmap_page_in

```
bool paging_unmap_page_in(uint64_t pml4, uint64_t virt);
bool paging_set_flags_in(uint64_t pml4, uint64_t virt, uint64_t flags);
bool paging_is_mapped_in(uint64_t pml4, uint64_t virt);
uint64_t paging_translate_in(uint64_t pml4, uint64_t virt);

#endif
```

kernel_Workspace/kernel/src/mm/arch/aarch64/paging.h (line 26) - [public] paging_set_flags_in

```
bool paging_set_flags_in(uint64_t pml4, uint64_t virt, uint64_t flags);
bool paging_is_mapped_in(uint64_t pml4, uint64_t virt);
uint64_t paging_translate_in(uint64_t pml4, uint64_t virt);

#endif
```

kernel_Workspace/kernel/src/mm/arch/aarch64/paging.h (line 27) - [public] paging_is_mapped_in

```
bool paging_is_mapped_in(uint64_t pml4, uint64_t virt);
uint64_t paging_translate_in(uint64_t pml4, uint64_t virt);

#endif
```

kernel_Workspace/kernel/src/mm/arch/aarch64/paging.h (line 28) - [public] paging_translate_in

```
uint64_t paging_translate_in(uint64_t pml4, uint64_t virt);

#endif
```

kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.h (line 8) - [public] tlb_init

```
bool tlb_init(void);
bool tlb_ready(void);

void tlb_flush_all(void);
void tlb_flush_all_including_global(void);

void tlb_flush_page(const void *addr);
void tlb_flush_page_addr(uint64_t addr);

void tlb_flush_range(const void *addr, size_t pages);
void tlb_flush_range_addr(uint64_t addr, size_t pages);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.h (line 9) - [public] tlb_ready

```
bool tlb_ready(void);

void tlb_flush_all(void);
void tlb_flush_all_including_global(void);

void tlb_flush_page(const void *addr);
void tlb_flush_page_addr(uint64_t addr);

void tlb_flush_range(const void *addr, size_t pages);
void tlb_flush_range_addr(uint64_t addr, size_t pages);

#endif
```

kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.h (line 11) - [public] tlb_flush_all

```
void tlb_flush_all(void);
void tlb_flush_all_including_global(void);

void tlb_flush_page(const void *addr);
void tlb_flush_page_addr(uint64_t addr);

void tlb_flush_range(const void *addr, size_t pages);
void tlb_flush_range_addr(uint64_t addr, size_t pages);

#endif
```

kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.h (line 12) - [public]

tlb_flush_all_including_global

```
void tlb_flush_all_including_global(void);

void tlb_flush_page(const void *addr);
void tlb_flush_page_addr(uint64_t addr);

void tlb_flush_range(const void *addr, size_t pages);
void tlb_flush_range_addr(uint64_t addr, size_t pages);

#endif
```

kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.h (line 14) - [public] tlb_flush_page

```
void tlb_flush_page(const void *addr);
void tlb_flush_page_addr(uint64_t addr);

void tlb_flush_range(const void *addr, size_t pages);
void tlb_flush_range_addr(uint64_t addr, size_t pages);

#endif
```

kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.h (line 15) - [public] tlb_flush_page_addr

```
void tlb_flush_page_addr(uint64_t addr);

void tlb_flush_range(const void *addr, size_t pages);
void tlb_flush_range_addr(uint64_t addr, size_t pages);

#endif
```

kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.h (line 17) - [public] tlb_flush_range

```
void tlb_flush_range(const void *addr, size_t pages);
void tlb_flush_range_addr(uint64_t addr, size_t pages);

#endif
```

kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.h (line 18) - [public] tlb_flush_range_addr

```
void tlb_flush_range_addr(uint64_t addr, size_t pages);

#endif
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.h (line 71) - [public] arch_memory_ready

```
bool arch_memory_ready(void);

bool arch_memory_boot_alloc(uint64_t len, uint64_t align, uint64_t *out_base);

const arch_mem_info_t *arch_memory_get(void);
size_t arch_memory_regions(const arch_mem_region_t **out);
uint64_t arch_memory_total_usable(void);
bool arch_memory_range_is_usable(uint64_t base, uint64_t len);
void arch_memory_reserve_range(uint64_t base, uint64_t len);
bool arch_memory_find_usable(uint64_t len,
                             uint64_t align,
                             uint64_t *out_base);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.h (line 73) - [public]

arch_memory_boot_alloc

```
bool arch_memory_boot_alloc(uint64_t len, uint64_t align, uint64_t *out_base);

const arch_mem_info_t *arch_memory_get(void);
size_t arch_memory_regions(const arch_mem_region_t **out);
uint64_t arch_memory_total_usable(void);
bool arch_memory_range_is_usable(uint64_t base, uint64_t len);
void arch_memory_reserve_range(uint64_t base, uint64_t len);
bool arch_memory_find_usable(uint64_t len,
                             uint64_t align,
                             uint64_t *out_base);

void arch_memory_dump(void);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.h (line 76) - [public] arch_memory_regions

```
size_t arch_memory_regions(const arch_mem_region_t **out);
uint64_t arch_memory_total_usable(void);
bool arch_memory_range_is_usable(uint64_t base, uint64_t len);
void arch_memory_reserve_range(uint64_t base, uint64_t len);
bool arch_memory_find_usable(uint64_t len,
                             uint64_t align,
                             uint64_t *out_base);

void arch_memory_dump(void);

static inline uint64_t arch_page_align_down(uint64_t v)
{
    return v & ~ARCH_PAGE_MASK;
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.h (line 77) - [public]

arch_memory_total_usable

```
uint64_t arch_memory_total_usable(void);
bool arch_memory_range_is_usable(uint64_t base, uint64_t len);
void arch_memory_reserve_range(uint64_t base, uint64_t len);
bool arch_memory_find_usable(uint64_t len,
                             uint64_t align,
                             uint64_t *out_base);

void arch_memory_dump(void);

static inline uint64_t arch_page_align_down(uint64_t v)
{
    return v & ~ARCH_PAGE_MASK;
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.h (line 78) - [public]

arch_memory_range_is_usable

```
bool arch_memory_range_is_usable(uint64_t base, uint64_t len);
void arch_memory_reserve_range(uint64_t base, uint64_t len);
bool arch_memory_find_usable(uint64_t len,
                             uint64_t align,
                             uint64_t *out_base);

void arch_memory_dump(void);

static inline uint64_t arch_page_align_down(uint64_t v)
{
    return v & ~ARCH_PAGE_MASK;
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.h (line 79) - [public]

arch_memory_reserve_range

```
void arch_memory_reserve_range(uint64_t base, uint64_t len);
bool arch_memory_find_usable(uint64_t len,
                             uint64_t align,
                             uint64_t *out_base);

void arch_memory_dump(void);

static inline uint64_t arch_page_align_down(uint64_t v)
{
    return v & ~ARCH_PAGE_MASK;
}

static inline uint64_t arch_page_align_up(uint64_t v)
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.h (line 83) - [public] arch_memory_dump

```
void arch_memory_dump(void);

static inline uint64_t arch_page_align_down(uint64_t v)
{
    return v & ~ARCH_PAGE_MASK;
}

static inline uint64_t arch_page_align_up(uint64_t v)
{
    if (v > UINT64_MAX - ARCH_PAGE_MASK) {
        return UINT64_MAX;
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.h (line 92) - [public] if

```
    if (v > UINT64_MAX - ARCH_PAGE_MASK) {
        return UINT64_MAX;
    }

    return (v + ARCH_PAGE_MASK) & ~ARCH_PAGE_MASK;
}

static inline bool arch_is_page_aligned(uint64_t v)
{
    return (v & ARCH_PAGE_MASK) == 0;
}

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.h (line 107) - [public] return

```
    return (void *)((uintptr_t)(phys + ARCH_DIRECT_MAP_BASE);
}

static inline uint64_t arch_virt_to_phys(const void *virt)
{
    return (uint64_t)((uintptr_t)virt - ARCH_DIRECT_MAP_BASE);
}

#endif
```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.h (line 112) - [public] return

```
    return (uint64_t)((uintptr_t)virt - ARCH_DIRECT_MAP_BASE);
}

#endif
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 35) - [public]

paging_set_boot_phys_offset

```
void paging_set_boot_phys_offset(uint64_t phys_offset);

bool paging_boot_phys_offset_valid(void);

void paging_init_direct_map(void);

void paging_init(uint64_t boot_phys_offset);

uint64_t paging_read_cr3(void);
void paging_flush_tlb_all(void);
void paging_flush_page(uint64_t addr);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 37) - [public]

paging_boot_phys_offset_valid

```
bool paging_boot_phys_offset_valid(void);

void paging_init_direct_map(void);

void paging_init(uint64_t boot_phys_offset);

uint64_t paging_read_cr3(void);
void paging_flush_tlb_all(void);
void paging_flush_page(uint64_t addr);

bool paging_map_page(uint64_t virt, uint64_t phys, uint64_t flags);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 39) - [public]

paging_init_direct_map

```
void paging_init_direct_map(void);

void paging_init(uint64_t boot_phys_offset);

uint64_t paging_read_cr3(void);
void paging_flush_tlb_all(void);
void paging_flush_page(uint64_t addr);

bool paging_map_page(uint64_t virt, uint64_t phys, uint64_t flags);

bool paging_map_range(uint64_t virt, uint64_t phys, uint64_t len, uint64_t flags);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 41) - [public] paging_init

```
void paging_init(uint64_t boot_phys_offset);

uint64_t paging_read_cr3(void);
void paging_flush_tlb_all(void);
void paging_flush_page(uint64_t addr);

bool paging_map_page(uint64_t virt, uint64_t phys, uint64_t flags);

bool paging_map_range(uint64_t virt, uint64_t phys, uint64_t len, uint64_t flags);

uint64_t paging_translate(uint64_t virt);
bool paging_is_mapped(uint64_t virt);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 43) - [public] paging_read_cr3

```
uint64_t paging_read_cr3(void);
void paging_flush_tlb_all(void);
void paging_flush_page(uint64_t addr);

bool paging_map_page(uint64_t virt, uint64_t phys, uint64_t flags);

bool paging_map_range(uint64_t virt, uint64_t phys, uint64_t len, uint64_t flags);

uint64_t paging_translate(uint64_t virt);
bool paging_is_mapped(uint64_t virt);
bool paging_unmap_page(uint64_t virt);
bool paging_set_flags(uint64_t virt, uint64_t flags);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 44) - [public] paging_flush_tlb_all

```
void paging_flush_tlb_all(void);
void paging_flush_page(uint64_t addr);

bool paging_map_page(uint64_t virt, uint64_t phys, uint64_t flags);

bool paging_map_range(uint64_t virt, uint64_t phys, uint64_t len, uint64_t flags);

uint64_t paging_translate(uint64_t virt);
bool paging_is_mapped(uint64_t virt);
bool paging_unmap_page(uint64_t virt);
bool paging_set_flags(uint64_t virt, uint64_t flags);
bool paging_get_flags(uint64_t virt, uint64_t *out_flags);
// ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 45) - [public] paging_flush_page
    void paging_flush_page(uint64_t addr);

    bool paging_map_page(uint64_t virt, uint64_t phys, uint64_t flags);

    bool paging_map_range(uint64_t virt, uint64_t phys, uint64_t len, uint64_t flags);

    uint64_t paging_translate(uint64_t virt);
    bool paging_is_mapped(uint64_t virt);
    bool paging_unmap_page(uint64_t virt);
    bool paging_set_flags(uint64_t virt, uint64_t flags);
    bool paging_get_flags(uint64_t virt, uint64_t *out_flags);
    void paging_enable_nx(void);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 47) - [public] paging_map_page
    bool paging_map_page(uint64_t virt, uint64_t phys, uint64_t flags);

    bool paging_map_range(uint64_t virt, uint64_t phys, uint64_t len, uint64_t flags);

    uint64_t paging_translate(uint64_t virt);
    bool paging_is_mapped(uint64_t virt);
    bool paging_unmap_page(uint64_t virt);
    bool paging_set_flags(uint64_t virt, uint64_t flags);
    bool paging_get_flags(uint64_t virt, uint64_t *out_flags);
    void paging_enable_nx(void);
    bool paging_nx_enabled(void);

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 49) - [public] paging_map_range
    bool paging_map_range(uint64_t virt, uint64_t phys, uint64_t len, uint64_t flags);

    uint64_t paging_translate(uint64_t virt);
    bool paging_is_mapped(uint64_t virt);
    bool paging_unmap_page(uint64_t virt);
    bool paging_set_flags(uint64_t virt, uint64_t flags);
    bool paging_get_flags(uint64_t virt, uint64_t *out_flags);
    void paging_enable_nx(void);
    bool paging_nx_enabled(void);

    bool paging_map_page_in(uint64_t pml4,
                            uint64_t virt,
                            // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 51) - [public] paging_translate
    uint64_t paging_translate(uint64_t virt);
    bool paging_is_mapped(uint64_t virt);
    bool paging_unmap_page(uint64_t virt);
    bool paging_set_flags(uint64_t virt, uint64_t flags);
    bool paging_get_flags(uint64_t virt, uint64_t *out_flags);
    void paging_enable_nx(void);
    bool paging_nx_enabled(void);

    bool paging_map_page_in(uint64_t pml4,
                            uint64_t virt,
                            uint64_t phys,
                            uint64_t flags);
    // ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 52) - [public] paging_is_mapped

```
bool paging_is_mapped(uint64_t virt);
bool paging_unmap_page(uint64_t virt);
bool paging_set_flags(uint64_t virt, uint64_t flags);
bool paging_get_flags(uint64_t virt, uint64_t *out_flags);
void paging_enable_nx(void);
bool paging_nx_enabled(void);

bool paging_map_page_in(uint64_t pml4,
                        uint64_t virt,
                        uint64_t phys,
                        uint64_t flags);
bool paging_unmap_page_in(uint64_t pml4, uint64_t virt);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 53) - [public] paging_unmap_page

```
bool paging_unmap_page(uint64_t virt);
bool paging_set_flags(uint64_t virt, uint64_t flags);
bool paging_get_flags(uint64_t virt, uint64_t *out_flags);
void paging_enable_nx(void);
bool paging_nx_enabled(void);

bool paging_map_page_in(uint64_t pml4,
                        uint64_t virt,
                        uint64_t phys,
                        uint64_t flags);
bool paging_unmap_page_in(uint64_t pml4, uint64_t virt);
uint64_t paging_translate_in(uint64_t pml4, uint64_t virt);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 54) - [public] paging_set_flags

```
bool paging_set_flags(uint64_t virt, uint64_t flags);
bool paging_get_flags(uint64_t virt, uint64_t *out_flags);
void paging_enable_nx(void);
bool paging_nx_enabled(void);

bool paging_map_page_in(uint64_t pml4,
                        uint64_t virt,
                        uint64_t phys,
                        uint64_t flags);
bool paging_unmap_page_in(uint64_t pml4, uint64_t virt);
uint64_t paging_translate_in(uint64_t pml4, uint64_t virt);
bool paging_is_mapped_in(uint64_t pml4, uint64_t virt);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 55) - [public] paging_get_flags

```
bool paging_get_flags(uint64_t virt, uint64_t *out_flags);
void paging_enable_nx(void);
bool paging_nx_enabled(void);

bool paging_map_page_in(uint64_t pml4,
                        uint64_t virt,
                        uint64_t phys,
                        uint64_t flags);
bool paging_unmap_page_in(uint64_t pml4, uint64_t virt);
uint64_t paging_translate_in(uint64_t pml4, uint64_t virt);
bool paging_is_mapped_in(uint64_t pml4, uint64_t virt);
bool paging_set_flags_in(uint64_t pml4, uint64_t virt, uint64_t flags);
// ... (truncated, see source file for more)
```


kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 56) - [public] paging_enable_nx

```
void paging_enable_nx(void);
bool paging_nx_enabled(void);

bool paging_map_page_in(uint64_t pml4,
                        uint64_t virt,
                        uint64_t phys,
                        uint64_t flags);
bool paging_unmap_page_in(uint64_t pml4, uint64_t virt);
uint64_t paging_translate_in(uint64_t pml4, uint64_t virt);
bool paging_is_mapped_in(uint64_t pml4, uint64_t virt);
bool paging_set_flags_in(uint64_t pml4, uint64_t virt, uint64_t flags);
bool paging_get_flags_in(uint64_t pml4, uint64_t virt, uint64_t *out_flags);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 57) - [public] paging_nx_enabled

```
bool paging_nx_enabled(void);

bool paging_map_page_in(uint64_t pml4,
                        uint64_t virt,
                        uint64_t phys,
                        uint64_t flags);
bool paging_unmap_page_in(uint64_t pml4, uint64_t virt);
uint64_t paging_translate_in(uint64_t pml4, uint64_t virt);
bool paging_is_mapped_in(uint64_t pml4, uint64_t virt);
bool paging_set_flags_in(uint64_t pml4, uint64_t virt, uint64_t flags);
bool paging_get_flags_in(uint64_t pml4, uint64_t virt, uint64_t *out_flags);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 63) - [public] paging_unmap_page_in

```
bool paging_unmap_page_in(uint64_t pml4, uint64_t virt);
uint64_t paging_translate_in(uint64_t pml4, uint64_t virt);
bool paging_is_mapped_in(uint64_t pml4, uint64_t virt);
bool paging_set_flags_in(uint64_t pml4, uint64_t virt, uint64_t flags);
bool paging_get_flags_in(uint64_t pml4, uint64_t virt, uint64_t *out_flags);

uint64_t paging_create_pml4(void);
void paging_destroy_pml4(uint64_t pml4_phys);
void paging_switch_pml4(uint64_t pml4_phys);

bool paging_map_mmio(uint64_t virt, uint64_t phys, uint64_t len);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 64) - [public] paging_translate_in

```
uint64_t paging_translate_in(uint64_t pml4, uint64_t virt);
bool paging_is_mapped_in(uint64_t pml4, uint64_t virt);
bool paging_set_flags_in(uint64_t pml4, uint64_t virt, uint64_t flags);
bool paging_get_flags_in(uint64_t pml4, uint64_t virt, uint64_t *out_flags);

uint64_t paging_create_pml4(void);
void paging_destroy_pml4(uint64_t pml4_phys);
void paging_switch_pml4(uint64_t pml4_phys);

bool paging_map_mmio(uint64_t virt, uint64_t phys, uint64_t len);

void paging_enable_write_protect(void);
// ... (truncated, see source file for more)
```

```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 65) - [public] paging_is_mapped_in
    bool paging_is_mapped_in(uint64_t pml4, uint64_t virt);
    bool paging_set_flags_in(uint64_t pml4, uint64_t virt, uint64_t flags);
    bool paging_get_flags_in(uint64_t pml4, uint64_t virt, uint64_t *out_flags);

    uint64_t paging_create_pml4(void);
    void paging_destroy_pml4(uint64_t pml4_phys);
    void paging_switch_pml4(uint64_t pml4_phys);

    bool paging_map_mmio(uint64_t virt, uint64_t phys, uint64_t len);

    void paging_enable_write_protect(void);
    void paging_disable_write_protect(void);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 66) - [public] paging_set_flags_in
    bool paging_set_flags_in(uint64_t pml4, uint64_t virt, uint64_t flags);
    bool paging_get_flags_in(uint64_t pml4, uint64_t virt, uint64_t *out_flags);

    uint64_t paging_create_pml4(void);
    void paging_destroy_pml4(uint64_t pml4_phys);
    void paging_switch_pml4(uint64_t pml4_phys);

    bool paging_map_mmio(uint64_t virt, uint64_t phys, uint64_t len);

    void paging_enable_write_protect(void);
    void paging_disable_write_protect(void);
    bool paging_write_protect_enabled(void);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 67) - [public] paging_get_flags_in
    bool paging_get_flags_in(uint64_t pml4, uint64_t virt, uint64_t *out_flags);

    uint64_t paging_create_pml4(void);
    void paging_destroy_pml4(uint64_t pml4_phys);
    void paging_switch_pml4(uint64_t pml4_phys);

    bool paging_map_mmio(uint64_t virt, uint64_t phys, uint64_t len);

    void paging_enable_write_protect(void);
    void paging_disable_write_protect(void);
    bool paging_write_protect_enabled(void);

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 69) - [public] paging_create_pml4
    uint64_t paging_create_pml4(void);
    void paging_destroy_pml4(uint64_t pml4_phys);
    void paging_switch_pml4(uint64_t pml4_phys);

    bool paging_map_mmio(uint64_t virt, uint64_t phys, uint64_t len);

    void paging_enable_write_protect(void);
    void paging_disable_write_protect(void);
    bool paging_write_protect_enabled(void);

    void paging_enable_smep(void);
    void paging_enable_smap(void);
    // ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 70) - [public] paging_destroy_pml4

```
void paging_destroy_pml4(uint64_t pml4_phys);
void paging_switch_pml4(uint64_t pml4_phys);

bool paging_map_mmio(uint64_t virt, uint64_t phys, uint64_t len);

void paging_enable_write_protect(void);
void paging_disable_write_protect(void);
bool paging_write_protect_enabled(void);

void paging_enable_smep(void);
void paging_enable_smap(void);
bool paging_smep_enabled(void);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 71) - [public] paging_switch_pml4

```
void paging_switch_pml4(uint64_t pml4_phys);

bool paging_map_mmio(uint64_t virt, uint64_t phys, uint64_t len);

void paging_enable_write_protect(void);
void paging_disable_write_protect(void);
bool paging_write_protect_enabled(void);

void paging_enable_smep(void);
void paging_enable_smap(void);
bool paging_smep_enabled(void);
bool paging_smap_enabled(void);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 73) - [public] paging_map_mmio

```
bool paging_map_mmio(uint64_t virt, uint64_t phys, uint64_t len);

void paging_enable_write_protect(void);
void paging_disable_write_protect(void);
bool paging_write_protect_enabled(void);

void paging_enable_smep(void);
void paging_enable_smap(void);
bool paging_smep_enabled(void);
bool paging_smap_enabled(void);

bool paging_pcid_supported(void);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 75) - [public]

paging_enable_write_protect

```
void paging_enable_write_protect(void);
void paging_disable_write_protect(void);
bool paging_write_protect_enabled(void);

void paging_enable_smep(void);
void paging_enable_smap(void);
bool paging_smep_enabled(void);
bool paging_smap_enabled(void);

bool paging_pcid_supported(void);
void paging_enable_pcid(void);
bool paging_pcid_enabled(void);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 76) - [public]

paging_disable_write_protect

```
void paging_disable_write_protect(void);
bool paging_write_protect_enabled(void);

void paging_enable_smep(void);
void paging_enable_smmap(void);
bool paging_smep_enabled(void);
bool paging_smmap_enabled(void);

bool paging_pcid_supported(void);
void paging_enable_pcid(void);
bool paging_pcid_enabled(void);
void paging_invpaid(uint64_t type, uint64_t pcid, uint64_t addr);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 77) - [public]

paging_write_protect_enabled

```
bool paging_write_protect_enabled(void);

void paging_enable_smep(void);
void paging_enable_smmap(void);
bool paging_smep_enabled(void);
bool paging_smmap_enabled(void);

bool paging_pcid_supported(void);
void paging_enable_pcid(void);
bool paging_pcid_enabled(void);
void paging_invpaid(uint64_t type, uint64_t pcid, uint64_t addr);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 79) - [public] paging_enable_smep

```
void paging_enable_smep(void);
void paging_enable_smmap(void);
bool paging_smep_enabled(void);
bool paging_smmap_enabled(void);

bool paging_pcid_supported(void);
void paging_enable_pcid(void);
bool paging_pcid_enabled(void);
void paging_invpaid(uint64_t type, uint64_t pcid, uint64_t addr);

bool paging_la57_supported(void);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 80) - [public] paging_enable_smmap

```
void paging_enable_smmap(void);
bool paging_smep_enabled(void);
bool paging_smmap_enabled(void);

bool paging_pcid_supported(void);
void paging_enable_pcid(void);
bool paging_pcid_enabled(void);
void paging_invpaid(uint64_t type, uint64_t pcid, uint64_t addr);

bool paging_la57_supported(void);
void paging_assert_4level_paging(void);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 81) - [public] paging_smep_enabled

```
bool paging_smep_enabled(void);
bool paging_smap_enabled(void);

bool paging_pcid_supported(void);
void paging_enable_pcid(void);
bool paging_pcid_enabled(void);
void paging_invpcid(uint64_t type, uint64_t pcid, uint64_t addr);

bool paging_la57_supported(void);
void paging_assert_4level_paging(void);
void paging_write_cr3(uint64_t pml4_phys);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 82) - [public] paging_smap_enabled

```
bool paging_smap_enabled(void);

bool paging_pcid_supported(void);
void paging_enable_pcid(void);
bool paging_pcid_enabled(void);
void paging_invpcid(uint64_t type, uint64_t pcid, uint64_t addr);

bool paging_la57_supported(void);
void paging_assert_4level_paging(void);
void paging_write_cr3(uint64_t pml4_phys);
#endif
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 84) - [public] paging_pcid_supported

```
bool paging_pcid_supported(void);
void paging_enable_pcid(void);
bool paging_pcid_enabled(void);
void paging_invpcid(uint64_t type, uint64_t pcid, uint64_t addr);

bool paging_la57_supported(void);
void paging_assert_4level_paging(void);
void paging_write_cr3(uint64_t pml4_phys);
#endif
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 85) - [public] paging_enable_pcid

```
void paging_enable_pcid(void);
bool paging_pcid_enabled(void);
void paging_invpcid(uint64_t type, uint64_t pcid, uint64_t addr);

bool paging_la57_supported(void);
void paging_assert_4level_paging(void);
void paging_write_cr3(uint64_t pml4_phys);
#endif
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 86) - [public] paging_pcid_enabled

```
bool paging_pcid_enabled(void);
void paging_invpcid(uint64_t type, uint64_t pcid, uint64_t addr);

bool paging_la57_supported(void);
void paging_assert_4level_paging(void);
void paging_write_cr3(uint64_t pml4_phys);
#endif
```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 87) - [public] paging_invpcid

```
void paging_invpcid(uint64_t type, uint64_t pcid, uint64_t addr);

bool paging_la57_supported(void);
void paging_assert_4level_paging(void);
void paging_write_cr3(uint64_t pml4_phys);
#endif
```

```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 90) - [public] paging_la57_supported
    bool paging_la57_supported(void);
    void paging_assert_4level_paging(void);
    void paging_write_cr3(uint64_t pml4_phys);
    #endif

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 91) - [public]
paging_assert_4level_paging
    void paging_assert_4level_paging(void);
    void paging_write_cr3(uint64_t pml4_phys);
    #endif

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 92) - [public] paging_write_cr3
    void paging_write_cr3(uint64_t pml4_phys);
    #endif

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 14) - [public] tlb_init
    bool tlb_init(void);
    bool tlb_ready(void);
    bool tlb_has_pcid(void);
    bool tlb_has_invpcid(void);

    void tlb_flush_all(void);
    void tlb_flush_all_including_global(void);

    void tlb_flush_page(const void *addr);
    void tlb_flush_page_addr(uint64_t addr);

    void tlb_flush_range(const void *addr, size_t pages);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 15) - [public] tlb_ready
    bool tlb_ready(void);
    bool tlb_has_pcid(void);
    bool tlb_has_invpcid(void);

    void tlb_flush_all(void);
    void tlb_flush_all_including_global(void);

    void tlb_flush_page(const void *addr);
    void tlb_flush_page_addr(uint64_t addr);

    void tlb_flush_range(const void *addr, size_t pages);
    void tlb_flush_range_addr(uint64_t addr, size_t pages);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 16) - [public] tlb_has_pcid
    bool tlb_has_pcid(void);
    bool tlb_has_invpcid(void);

    void tlb_flush_all(void);
    void tlb_flush_all_including_global(void);

    void tlb_flush_page(const void *addr);
    void tlb_flush_page_addr(uint64_t addr);

    void tlb_flush_range(const void *addr, size_t pages);
    void tlb_flush_range_addr(uint64_t addr, size_t pages);

    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 17) - [public] tlb_has_invpcid
    bool tlb_has_invpcid(void);

    void tlb_flush_all(void);
    void tlb_flush_all_including_global(void);

    void tlb_flush_page(const void *addr);
    void tlb_flush_page_addr(uint64_t addr);

    void tlb_flush_range(const void *addr, size_t pages);
    void tlb_flush_range_addr(uint64_t addr, size_t pages);

    void tlb_flush_pcid(uint16_t pcid);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 19) - [public] tlb_flush_all
    void tlb_flush_all(void);
    void tlb_flush_all_including_global(void);

    void tlb_flush_page(const void *addr);
    void tlb_flush_page_addr(uint64_t addr);

    void tlb_flush_range(const void *addr, size_t pages);
    void tlb_flush_range_addr(uint64_t addr, size_t pages);

    void tlb_flush_pcid(uint16_t pcid);
    void tlb_flush_pcid_addr(uint16_t pcid, uint64_t addr);

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 20) - [public]
tlb_flush_all_including_global
    void tlb_flush_all_including_global(void);

    void tlb_flush_page(const void *addr);
    void tlb_flush_page_addr(uint64_t addr);

    void tlb_flush_range(const void *addr, size_t pages);
    void tlb_flush_range_addr(uint64_t addr, size_t pages);

    void tlb_flush_pcid(uint16_t pcid);
    void tlb_flush_pcid_addr(uint16_t pcid, uint64_t addr);

    void tlb_wbinvd(void);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 22) - [public] tlb_flush_page
    void tlb_flush_page(const void *addr);
    void tlb_flush_page_addr(uint64_t addr);

    void tlb_flush_range(const void *addr, size_t pages);
    void tlb_flush_range_addr(uint64_t addr, size_t pages);

    void tlb_flush_pcid(uint16_t pcid);
    void tlb_flush_pcid_addr(uint16_t pcid, uint64_t addr);

    void tlb_wbinvd(void);
    void tlb_clflush(uint64_t addr);

    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 23) - [public] tlb_flush_page_addr
    void tlb_flush_page_addr(uint64_t addr);

    void tlb_flush_range(const void *addr, size_t pages);
    void tlb_flush_range_addr(uint64_t addr, size_t pages);

    void tlb_flush_pcid(uint16_t pcid);
    void tlb_flush_pcid_addr(uint16_t pcid, uint64_t addr);

    void tlb_wbinvd(void);
    void tlb_clflush(uint64_t addr);

#ifdef

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 25) - [public] tlb_flush_range
    void tlb_flush_range(const void *addr, size_t pages);
    void tlb_flush_range_addr(uint64_t addr, size_t pages);

    void tlb_flush_pcid(uint16_t pcid);
    void tlb_flush_pcid_addr(uint16_t pcid, uint64_t addr);

    void tlb_wbinvd(void);
    void tlb_clflush(uint64_t addr);

#ifdef

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 26) - [public] tlb_flush_range_addr
    void tlb_flush_range_addr(uint64_t addr, size_t pages);

    void tlb_flush_pcid(uint16_t pcid);
    void tlb_flush_pcid_addr(uint16_t pcid, uint64_t addr);

    void tlb_wbinvd(void);
    void tlb_clflush(uint64_t addr);

#ifdef

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 28) - [public] tlb_flush_pcid
    void tlb_flush_pcid(uint16_t pcid);
    void tlb_flush_pcid_addr(uint16_t pcid, uint64_t addr);

    void tlb_wbinvd(void);
    void tlb_clflush(uint64_t addr);

#ifdef

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 29) - [public] tlb_flush_pcid_addr
    void tlb_flush_pcid_addr(uint16_t pcid, uint64_t addr);

    void tlb_wbinvd(void);
    void tlb_clflush(uint64_t addr);

#ifdef

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 31) - [public] tlb_wbinvd
    void tlb_wbinvd(void);
    void tlb_clflush(uint64_t addr);

#ifdef

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 32) - [public] tlb_clflush
    void tlb_clflush(uint64_t addr);

#ifdef

```


kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.h (line 6) - [public] tlb_asm_invlpg

```
void tlb_asm_invlpg(uint64_t addr);

uint64_t tlb_asm_read_cr3(void);
void tlb_asm_write_cr3(uint64_t value);

uint64_t tlb_asm_read_cr4(void);
void tlb_asm_write_cr4(uint64_t value);

void tlb_asm_invpaid(uint64_t type, const void *desc);

void tlb_asm_wbinvd(void);
void tlb_asm_clflush(uint64_t addr);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.h (line 8) - [public] tlb_asm_read_cr3

```
uint64_t tlb_asm_read_cr3(void);
void tlb_asm_write_cr3(uint64_t value);

uint64_t tlb_asm_read_cr4(void);
void tlb_asm_write_cr4(uint64_t value);

void tlb_asm_invpaid(uint64_t type, const void *desc);

void tlb_asm_wbinvd(void);
void tlb_asm_clflush(uint64_t addr);

#endif
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.h (line 9) - [public] tlb_asm_write_cr3

```
void tlb_asm_write_cr3(uint64_t value);

uint64_t tlb_asm_read_cr4(void);
void tlb_asm_write_cr4(uint64_t value);

void tlb_asm_invpaid(uint64_t type, const void *desc);

void tlb_asm_wbinvd(void);
void tlb_asm_clflush(uint64_t addr);

#endif
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.h (line 11) - [public] tlb_asm_read_cr4

```
uint64_t tlb_asm_read_cr4(void);
void tlb_asm_write_cr4(uint64_t value);

void tlb_asm_invpaid(uint64_t type, const void *desc);

void tlb_asm_wbinvd(void);
void tlb_asm_clflush(uint64_t addr);

#endif
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.h (line 12) - [public] tlb_asm_write_cr4

```
void tlb_asm_write_cr4(uint64_t value);

void tlb_asm_invpaid(uint64_t type, const void *desc);

void tlb_asm_wbinvd(void);
void tlb_asm_clflush(uint64_t addr);

#endif
```

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.h (line 14) - [public] tlb_asm_invpaid

```
void tlb_asm_invpaid(uint64_t type, const void *desc);

void tlb_asm_wbinvd(void);
void tlb_asm_clflush(uint64_t addr);

#endif
```

```
kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.h (line 16) - [public] tlb_asm_wbinvd
    void tlb_asm_wbinvd(void);
    void tlb_asm_clflush(uint64_t addr);

#endif
```

```
kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.h (line 17) - [public] tlb_asm_clflush
    void tlb_asm_clflush(uint64_t addr);

#endif
```

```
kernel_Workspace/kernel/src/mm/cache/cache.h (line 7) - [public] cache_init
    bool cache_init(void);
    bool cache_ready(void);

    void cache_register(kcache_t *c);
    kcache_t *cache_find(const char *name);

    void cache_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/cache/cache.h (line 8) - [public] cache_ready
    bool cache_ready(void);

    void cache_register(kcache_t *c);
    kcache_t *cache_find(const char *name);

    void cache_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/cache/cache.h (line 10) - [public] cache_register
    void cache_register(kcache_t *c);
    kcache_t *cache_find(const char *name);

    void cache_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/cache/cache.h (line 13) - [public] cache_dump
    void cache_dump(void);

#endif
```

```
kernel_Workspace/kernel/src/mm/cache/object_cashe.h (line 12) - [public] void
typedef void (*kcache_ctor_t)(void *obj);

typedef struct kcache_chunk {
    struct kcache_chunk *next;
    uint64_t va;
} kcache_chunk_t;

typedef struct kcache_mag {
    void *objs[KCACHE_MAG_SIZE];
    size_t count;
} kcache_mag_t;

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/cache/object_cashe.h (line 47) - [public] kcache_destroy

```
void kcache_destroy(kcache_t *c);

void *kcache_alloc(kcache_t *c);
void *kcache_zalloc(kcache_t *c);
void kcache_free(kcache_t *c, void *ptr);

size_t kcache_object_size(kcache_t *c);
size_t kcache_free_count(kcache_t *c);
size_t kcache_total_count(kcache_t *c);

#endif
```

kernel_Workspace/kernel/src/mm/cache/object_cashe.h (line 51) - [public] kcache_free

```
void kcache_free(kcache_t *c, void *ptr);

size_t kcache_object_size(kcache_t *c);
size_t kcache_free_count(kcache_t *c);
size_t kcache_total_count(kcache_t *c);

#endif
```

kernel_Workspace/kernel/src/mm/cache/object_cashe.h (line 53) - [public] kcache_object_size

```
size_t kcache_object_size(kcache_t *c);
size_t kcache_free_count(kcache_t *c);
size_t kcache_total_count(kcache_t *c);

#endif
```

kernel_Workspace/kernel/src/mm/cache/object_cashe.h (line 54) - [public] kcache_free_count

```
size_t kcache_free_count(kcache_t *c);
size_t kcache_total_count(kcache_t *c);

#endif
```

kernel_Workspace/kernel/src/mm/cache/object_cashe.h (line 55) - [public] kcache_total_count

```
size_t kcache_total_count(kcache_t *c);

#endif
```

kernel_Workspace/kernel/src/mm/cache/per_cpu.h (line 12) - [public] per_cpu_init

```
bool per_cpu_init(void);
size_t per_cpu_count(void);
size_t per_cpu_id(void);

#endif
```

kernel_Workspace/kernel/src/mm/cache/per_cpu.h (line 13) - [public] per_cpu_count

```
size_t per_cpu_count(void);
size_t per_cpu_id(void);

#endif
```

kernel_Workspace/kernel/src/mm/cache/per_cpu.h (line 14) - [public] per_cpu_id

```
size_t per_cpu_id(void);

#endif
```

kernel_Workspace/kernel/src/mm/core/address.h (line 10) - [public] addr_is_canonical

```
bool addr_is_canonical(uint64_t a);
bool addr_is_user(uint64_t a);
bool addr_is_kernel(uint64_t a);
bool addr_is_direct_map(uint64_t a);

uint64_t addr_align_up(uint64_t a, uint64_t align);
uint64_t addr_align_down(uint64_t a, uint64_t align);

uint64_t addr_phys_to_direct(uint64_t phys);
uint64_t addr_direct_to_phys(uint64_t va);

const char *addr_describe(uint64_t a);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/core/address.h (line 11) - [public] addr_is_user

```
bool addr_is_user(uint64_t a);
bool addr_is_kernel(uint64_t a);
bool addr_is_direct_map(uint64_t a);

uint64_t addr_align_up(uint64_t a, uint64_t align);
uint64_t addr_align_down(uint64_t a, uint64_t align);

uint64_t addr_phys_to_direct(uint64_t phys);
uint64_t addr_direct_to_phys(uint64_t va);

const char *addr_describe(uint64_t a);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/core/address.h (line 12) - [public] addr_is_kernel

```
bool addr_is_kernel(uint64_t a);
bool addr_is_direct_map(uint64_t a);

uint64_t addr_align_up(uint64_t a, uint64_t align);
uint64_t addr_align_down(uint64_t a, uint64_t align);

uint64_t addr_phys_to_direct(uint64_t phys);
uint64_t addr_direct_to_phys(uint64_t va);

const char *addr_describe(uint64_t a);

#endif
```

kernel_Workspace/kernel/src/mm/core/address.h (line 13) - [public] addr_is_direct_map

```
bool addr_is_direct_map(uint64_t a);

uint64_t addr_align_up(uint64_t a, uint64_t align);
uint64_t addr_align_down(uint64_t a, uint64_t align);

uint64_t addr_phys_to_direct(uint64_t phys);
uint64_t addr_direct_to_phys(uint64_t va);

const char *addr_describe(uint64_t a);

#endif
```

kernel_Workspace/kernel/src/mm/core/address.h (line 15) - [public] addr_align_up

```
uint64_t addr_align_up(uint64_t a, uint64_t align);
uint64_t addr_align_down(uint64_t a, uint64_t align);

uint64_t addr_phys_to_direct(uint64_t phys);
uint64_t addr_direct_to_phys(uint64_t va);

const char *addr_describe(uint64_t a);

#endif
```

```

kernel_Workspace/kernel/src/mm/core/address.h (line 16) - [public] addr_align_down
    uint64_t addr_align_down(uint64_t a, uint64_t align);

    uint64_t addr_phys_to_direct(uint64_t phys);
    uint64_t addr_direct_to_phys(uint64_t va);

    const char *addr_describe(uint64_t a);

#endif

kernel_Workspace/kernel/src/mm/core/address.h (line 18) - [public] addr_phys_to_direct
    uint64_t addr_phys_to_direct(uint64_t phys);
    uint64_t addr_direct_to_phys(uint64_t va);

    const char *addr_describe(uint64_t a);

#endif

kernel_Workspace/kernel/src/mm/core/address.h (line 19) - [public] addr_direct_to_phys
    uint64_t addr_direct_to_phys(uint64_t va);

    const char *addr_describe(uint64_t a);

#endif

kernel_Workspace/kernel/src/mm/core/mm.h (line 26) - [public] mm_init
    bool mm_init(const mm_boot_params_t *params);
    bool mm_ready(void);

    uint64_t mm_total_ram(void);
    uint64_t mm_free_ram(void);

    void mm_dump(void);

#endif

kernel_Workspace/kernel/src/mm/core/mm.h (line 27) - [public] mm_ready
    bool mm_ready(void);

    uint64_t mm_total_ram(void);
    uint64_t mm_free_ram(void);

    void mm_dump(void);

#endif

kernel_Workspace/kernel/src/mm/core/mm.h (line 29) - [public] mm_total_ram
    uint64_t mm_total_ram(void);
    uint64_t mm_free_ram(void);

    void mm_dump(void);

#endif

kernel_Workspace/kernel/src/mm/core/mm.h (line 30) - [public] mm_free_ram
    uint64_t mm_free_ram(void);

    void mm_dump(void);

#endif

kernel_Workspace/kernel/src/mm/core/mm.h (line 32) - [public] mm_dump
    void mm_dump(void);

#endif

kernel_Workspace/kernel/src/mm/core/range.h (line 18) - [public] range_is_canonical
    bool range_is_canonical(uint64_t addr);

#endif

```

kernel_Workspace/kernel/src/mm/core/region.h (line 13) - [public] region_make

```
region_t region_make(uint64_t start, uint64_t len);
bool region_valid(region_t r);
uint64_t region_len(region_t r);

bool region_contains(region_t r, uint64_t addr);
bool region_overlaps(region_t a, region_t b);
region_t region_intersect(region_t a, region_t b);

size_t region_subtract(region_t r,
                       region_t cut,
                       region_t *out,
                       size_t max_out);
// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/core/region.h (line 14) - [public] region_valid

```
bool region_valid(region_t r);
uint64_t region_len(region_t r);

bool region_contains(region_t r, uint64_t addr);
bool region_overlaps(region_t a, region_t b);
region_t region_intersect(region_t a, region_t b);

size_t region_subtract(region_t r,
                       region_t cut,
                       region_t *out,
                       size_t max_out);

// ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/mm/core/region.h (line 15) - [public] region_len

```
uint64_t region_len(region_t r);

bool region_contains(region_t r, uint64_t addr);
bool region_overlaps(region_t a, region_t b);
region_t region_intersect(region_t a, region_t b);

size_t region_subtract(region_t r,
                       region_t cut,
                       region_t *out,
                       size_t max_out);

#endif
```

kernel_Workspace/kernel/src/mm/core/region.h (line 17) - [public] region_contains

```
bool region_contains(region_t r, uint64_t addr);
bool region_overlaps(region_t a, region_t b);
region_t region_intersect(region_t a, region_t b);

size_t region_subtract(region_t r,
                       region_t cut,
                       region_t *out,
                       size_t max_out);

#endif
```

kernel_Workspace/kernel/src/mm/core/region.h (line 18) - [public] region_overlaps

```
bool region_overlaps(region_t a, region_t b);
region_t region_intersect(region_t a, region_t b);

size_t region_subtract(region_t r,
                       region_t cut,
                       region_t *out,
                       size_t max_out);

#endif
```

```

kernel_Workspace/kernel/src/mm/core/region.h (line 19) - [public] region_intersect
    region_t region_intersect(region_t a, region_t b);

    size_t region_subtract(region_t r,
                           region_t cut,
                           region_t *out,
                           size_t max_out);

#endif

kernel_Workspace/kernel/src/mm/core/sizeutil.h (line 8) - [public] size_is_pow2
    bool size_is_pow2(size_t v);

    bool size_round_up_pow2(size_t v, size_t *out);

    bool kfree_pages_validate(size_t pages,
                              size_t page_size,
                              size_t usable_bytes,
                              size_t *out_requested_bytes);

    bool size_bytes_to_pages_checked(size_t bytes,
                                     // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/core/sizeutil.h (line 11) - [public] size_round_up_pow2
    bool size_round_up_pow2(size_t v, size_t *out);

    bool kfree_pages_validate(size_t pages,
                              size_t page_size,
                              size_t usable_bytes,
                              size_t *out_requested_bytes);

    bool size_bytes_to_pages_checked(size_t bytes,
                                     size_t page_size,
                                     size_t *out_pages);

    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/core/smp_lock.h (line 21) - [public] smp_lock_init
    void smp_lock_init(smp_ticket_lock_t *lock);

    void smp_lock_acquire(smp_ticket_lock_t *lock);

    bool smp_lock_release(smp_ticket_lock_t *lock);

    int64_t smp_current_cpu_id(void);

#endif

kernel_Workspace/kernel/src/mm/core/smp_lock.h (line 24) - [public] smp_lock_acquire
    void smp_lock_acquire(smp_ticket_lock_t *lock);

    bool smp_lock_release(smp_ticket_lock_t *lock);

    int64_t smp_current_cpu_id(void);

#endif

```

```

kernel_Workspace/kernel/src/mm/core/smp_lock.h (line 27) - [public] smp_lock_release
    bool smp_lock_release(smp_ticket_lock_t *lock);

    int64_t smp_current_cpu_id(void);

#endif

kernel_Workspace/kernel/src/mm/core/smp_lock.h (line 30) - [public] smp_current_cpu_id
    int64_t smp_current_cpu_id(void);

#endif

kernel_Workspace/kernel/src/mm/paging/paging.h (line 19) - [public] paging_subsystem_init
    bool paging_subsystem_init(void);

    address_space_t *paging_aspace_create(void);
    void paging_aspace_destroy(address_space_t *as);
    void paging_aspace_switch(address_space_t *as);
    uint64_t paging_aspace_cr3(const address_space_t *as);

    bool paging_aspace_map(address_space_t *as,
                           uint64_t virt,
                           uint64_t phys,
                           size_t len,
                           uint32_t prot);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/paging/paging.h (line 22) - [public] paging_aspace_destroy
    void paging_aspace_destroy(address_space_t *as);
    void paging_aspace_switch(address_space_t *as);
    uint64_t paging_aspace_cr3(const address_space_t *as);

    bool paging_aspace_map(address_space_t *as,
                           uint64_t virt,
                           uint64_t phys,
                           size_t len,
                           uint32_t prot);

    bool paging_aspace_unmap(address_space_t *as, uint64_t virt, size_t len);
    // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/paging/paging.h (line 23) - [public] paging_aspace_switch
    void paging_aspace_switch(address_space_t *as);
    uint64_t paging_aspace_cr3(const address_space_t *as);

    bool paging_aspace_map(address_space_t *as,
                           uint64_t virt,
                           uint64_t phys,
                           size_t len,
                           uint32_t prot);

    bool paging_aspace_unmap(address_space_t *as, uint64_t virt, size_t len);

    bool paging_aspace_protect(address_space_t *as,
    // ... (truncated, see source file for more)

```



```

kernel_Workspace/kernel/src/mm/paging/paging.h (line 24) - [public] paging_aspace_cr3
uint64_t paging_aspace_cr3(const address_space_t *as);

bool paging_aspace_map(address_space_t *as,
                        uint64_t virt,
                        uint64_t phys,
                        size_t len,
                        uint32_t prot);

bool paging_aspace_unmap(address_space_t *as, uint64_t virt, size_t len);

bool paging_aspace_protect(address_space_t *as,
                            uint64_t virt,
                            // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/paging/paging.h (line 32) - [public] paging_aspace_unmap
bool paging_aspace_unmap(address_space_t *as, uint64_t virt, size_t len);

bool paging_aspace_protect(address_space_t *as,
                            uint64_t virt,
                            size_t len,
                            uint32_t prot);

uint64_t paging_aspace_translate(address_space_t *as, uint64_t virt);

bool paging_kernel_map(uint64_t virt, uint64_t phys, size_t len, uint32_t prot);
bool paging_kernel_unmap(uint64_t virt, size_t len);

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/paging/paging.h (line 39) - [public] paging_aspace_translate
uint64_t paging_aspace_translate(address_space_t *as, uint64_t virt);

bool paging_kernel_map(uint64_t virt, uint64_t phys, size_t len, uint32_t prot);
bool paging_kernel_unmap(uint64_t virt, size_t len);

#endif

kernel_Workspace/kernel/src/mm/paging/paging.h (line 41) - [public] paging_kernel_map
bool paging_kernel_map(uint64_t virt, uint64_t phys, size_t len, uint32_t prot);
bool paging_kernel_unmap(uint64_t virt, size_t len);

#endif

kernel_Workspace/kernel/src/mm/paging/paging.h (line 42) - [public] paging_kernel_unmap
bool paging_kernel_unmap(uint64_t virt, size_t len);

#endif

kernel_Workspace/kernel/src/mm/paging/pml.h (line 14) - [public] pml_level_shift
uint64_t pml_level_shift(int level);
uint64_t pml_index(uint64_t virt, int level);

bool pml_entry_present(uint64_t entry);
bool pml_entry_large(uint64_t entry);
uint64_t pml_entry_addr(uint64_t entry);
uint64_t *pml_table_ptr(uint64_t table_phys);

#endif

kernel_Workspace/kernel/src/mm/paging/pml.h (line 15) - [public] pml_index
uint64_t pml_index(uint64_t virt, int level);

bool pml_entry_present(uint64_t entry);
bool pml_entry_large(uint64_t entry);
uint64_t pml_entry_addr(uint64_t entry);
uint64_t *pml_table_ptr(uint64_t table_phys);

#endif

```

```

kernel_Workspace/kernel/src/mm/paging/pml.h (line 17) - [public] pml_entry_present
    bool pml_entry_present(uint64_t entry);
    bool pml_entry_large(uint64_t entry);
    uint64_t pml_entry_addr(uint64_t entry);
    uint64_t *pml_table_ptr(uint64_t table_phys);

    #endif

kernel_Workspace/kernel/src/mm/paging/pml.h (line 18) - [public] pml_entry_large
    bool pml_entry_large(uint64_t entry);
    uint64_t pml_entry_addr(uint64_t entry);
    uint64_t *pml_table_ptr(uint64_t table_phys);

    #endif

kernel_Workspace/kernel/src/mm/paging/pml.h (line 19) - [public] pml_entry_addr
    uint64_t pml_entry_addr(uint64_t entry);
    uint64_t *pml_table_ptr(uint64_t table_phys);

    #endif

kernel_Workspace/kernel/src/mm/paging/tlb.h (line 16) - [public] tlb_batch_begin
    void tlb_batch_begin(tlb_batch_t *batch);
    void tlb_batch_add(tlb_batch_t *batch, uint64_t virt);
    void tlb_batch_full(tlb_batch_t *batch);
    void tlb_batch_commit(tlb_batch_t *batch);

    #endif

kernel_Workspace/kernel/src/mm/paging/tlb.h (line 17) - [public] tlb_batch_add
    void tlb_batch_add(tlb_batch_t *batch, uint64_t virt);
    void tlb_batch_full(tlb_batch_t *batch);
    void tlb_batch_commit(tlb_batch_t *batch);

    #endif

kernel_Workspace/kernel/src/mm/paging/tlb.h (line 18) - [public] tlb_batch_full
    void tlb_batch_full(tlb_batch_t *batch);
    void tlb_batch_commit(tlb_batch_t *batch);

    #endif

kernel_Workspace/kernel/src/mm/paging/tlb.h (line 19) - [public] tlb_batch_commit
    void tlb_batch_commit(tlb_batch_t *batch);

    #endif

kernel_Workspace/kernel/src/mm/process/address_space.h (line 30) - [public]
aspace_subsystem_init
    bool aspace_subsystem_init(void);

    proc_aspace_t *aspace_create(void);
    void aspace_destroy(proc_aspace_t *pa);

    address_space_t *aspace_paging_handle(proc_aspace_t *pa);

    uint64_t aspace_map_anon(proc_aspace_t *pa, uint64_t hint, size_t len, uint32_t prot);
    uint64_t aspace_map_at(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t prot);
    uint64_t aspace_reserve_at(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t flags);
    bool aspace_unmap(proc_aspace_t *pa, uint64_t addr, size_t len);
    bool aspace_protect(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t prot);
    // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/mm/process/address_space.h (line 33) - [public] aspace_destroy
void aspace_destroy(proc_aspace_t *pa);

address_space_t *aspace_paging_handle(proc_aspace_t *pa);

uint64_t aspace_map_anon(proc_aspace_t *pa, uint64_t hint, size_t len, uint32_t prot);
uint64_t aspace_map_at(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t prot);
uint64_t aspace_reserve_at(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t flags);
bool aspace_unmap(proc_aspace_t *pa, uint64_t addr, size_t len);
bool aspace_protect(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t prot);

bool aspace_protect_checked(proc_aspace_t *pa, uint64_t addr, size_t len,
                           uint32_t checked_prot,
                           // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/process/address_space.h (line 37) - [public] aspace_map_anon
uint64_t aspace_map_anon(proc_aspace_t *pa, uint64_t hint, size_t len, uint32_t prot);
uint64_t aspace_map_at(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t prot);
uint64_t aspace_reserve_at(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t flags);
bool aspace_unmap(proc_aspace_t *pa, uint64_t addr, size_t len);
bool aspace_protect(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t prot);

bool aspace_protect_checked(proc_aspace_t *pa, uint64_t addr, size_t len,
                           uint32_t checked_prot,
                           uint32_t apply_prot,
                           bool (*allowed)(uint32_t old_prot,
                                             uint32_t checked_prot));

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/process/address_space.h (line 38) - [public] aspace_map_at
uint64_t aspace_map_at(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t prot);
uint64_t aspace_reserve_at(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t flags);
bool aspace_unmap(proc_aspace_t *pa, uint64_t addr, size_t len);
bool aspace_protect(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t prot);

bool aspace_protect_checked(proc_aspace_t *pa, uint64_t addr, size_t len,
                           uint32_t checked_prot,
                           uint32_t apply_prot,
                           bool (*allowed)(uint32_t old_prot,
                                             uint32_t checked_prot));

// ... (truncated, see source file for more)

kernel_Workspace/kernel/src/mm/process/address_space.h (line 39) - [public] aspace_reserve_at
uint64_t aspace_reserve_at(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t flags);
bool aspace_unmap(proc_aspace_t *pa, uint64_t addr, size_t len);
bool aspace_protect(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t prot);

bool aspace_protect_checked(proc_aspace_t *pa, uint64_t addr, size_t len,
                           uint32_t checked_prot,
                           uint32_t apply_prot,
                           bool (*allowed)(uint32_t old_prot,
                                             uint32_t checked_prot));

vma_t *aspace_vma_find(proc_aspace_t *pa, uint64_t addr);
// ... (truncated, see source file for more)

```

```
kernel_Workspace/kernel/src/mm/process/address_space.h (line 40) - [public] aspace_unmap
    bool aspace_unmap(proc_aspace_t *pa, uint64_t addr, size_t len);
    bool aspace_protect(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t prot);

    bool aspace_protect_checked(proc_aspace_t *pa, uint64_t addr, size_t len,
                                uint32_t checked_prot,
                                uint32_t apply_prot,
                                bool (*allowed)(uint32_t old_prot,
                                                  uint32_t checked_prot));

    vma_t *aspace_vma_find(proc_aspace_t *pa, uint64_t addr);
    uint64_t aspace_stack_base(void);
    // ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/mm/process/address_space.h (line 41) - [public] aspace_protect
    bool aspace_protect(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t prot);

    bool aspace_protect_checked(proc_aspace_t *pa, uint64_t addr, size_t len,
                                uint32_t checked_prot,
                                uint32_t apply_prot,
                                bool (*allowed)(uint32_t old_prot,
                                                  uint32_t checked_prot));

    vma_t *aspace_vma_find(proc_aspace_t *pa, uint64_t addr);
    uint64_t aspace_stack_base(void);

    // ... (truncated, see source file for more)
```

```
kernel_Workspace/kernel/src/mm/process/address_space.h (line 51) - [public] aspace_stack_base
    uint64_t aspace_stack_base(void);

    uint64_t aspace_brk(proc_aspace_t *pa, uint64_t new_brk);

#endif
```

```
kernel_Workspace/kernel/src/mm/process/address_space.h (line 53) - [public] aspace_brk
    uint64_t aspace_brk(proc_aspace_t *pa, uint64_t new_brk);

#endif
```

```
kernel_Workspace/kernel/src/mm/process/mmap.h (line 20) - [public] munmap
    bool munmap(proc_aspace_t *pa, uint64_t addr, size_t len);
    bool mprotect(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t prot);

#endif
```

```
kernel_Workspace/kernel/src/mm/process/mmap.h (line 21) - [public] mprotect
    bool mprotect(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t prot);

#endif
```

```
kernel_Workspace/kernel/src/mm/protection/guard.h (line 9) - [public] guard_install
    uint64_t guard_install(proc_aspace_t *pa, uint64_t addr, size_t len);
    bool guard_user_stack(proc_aspace_t *pa);

#endif
```

```
kernel_Workspace/kernel/src/mm/protection/guard.h (line 10) - [public] guard_user_stack
    bool guard_user_stack(proc_aspace_t *pa);

#endif
```

kernel_Workspace/kernel/src/mm/protection/isolation.h (line 8) - [public] isolation_init

```
bool isolation_init(void);

bool isolation_has_smep(void);
bool isolation_has_smap(void);
bool isolation_enable_smep(void);
bool isolation_enable_smap(void);

size_t isolation_audit_kernel(void);

#endif
```

kernel_Workspace/kernel/src/mm/protection/isolation.h (line 10) - [public] isolation_has_smep

```
bool isolation_has_smep(void);
bool isolation_has_smap(void);
bool isolation_enable_smep(void);
bool isolation_enable_smap(void);

size_t isolation_audit_kernel(void);

#endif
```

kernel_Workspace/kernel/src/mm/protection/isolation.h (line 11) - [public] isolation_has_smap

```
bool isolation_has_smap(void);
bool isolation_enable_smep(void);
bool isolation_enable_smap(void);

size_t isolation_audit_kernel(void);

#endif
```

kernel_Workspace/kernel/src/mm/protection/isolation.h (line 12) - [public]

isolation_enable_smep

```
bool isolation_enable_smep(void);
bool isolation_enable_smap(void);

size_t isolation_audit_kernel(void);

#endif
```

kernel_Workspace/kernel/src/mm/protection/isolation.h (line 13) - [public]

isolation_enable_smap

```
bool isolation_enable_smap(void);

size_t isolation_audit_kernel(void);

#endif
```

kernel_Workspace/kernel/src/mm/protection/isolation.h (line 15) - [public]

isolation_audit_kernel

```
size_t isolation_audit_kernel(void);

#endif
```

kernel_Workspace/kernel/src/mm/protection/permissions.h (line 8) - [public] perm_sanitize

```
uint32_t perm_sanitize(uint32_t prot);
bool perm_is_wx(uint32_t prot);
bool perm_mprotect_allowed(uint32_t old_prot, uint32_t new_prot);
void perm_set_strict_wx(bool on);

uint32_t perm_kernel_default(void);
uint32_t perm_user_default(void);

#endif
```

kernel_Workspace/kernel/src/mm/protection/permissions.h (line 9) - [public] perm_is_wx

```
bool perm_is_wx(uint32_t prot);
bool perm_mprotect_allowed(uint32_t old_prot, uint32_t new_prot);
void perm_set_strict_wx(bool on);

uint32_t perm_kernel_default(void);
uint32_t perm_user_default(void);

#endif
```

kernel_Workspace/kernel/src/mm/protection/permissions.h (line 10) - [public] perm_mprotect_allowed

```
bool perm_mprotect_allowed(uint32_t old_prot, uint32_t new_prot);
void perm_set_strict_wx(bool on);

uint32_t perm_kernel_default(void);
uint32_t perm_user_default(void);

#endif
```

kernel_Workspace/kernel/src/mm/protection/permissions.h (line 11) - [public] perm_set_strict_wx

```
void perm_set_strict_wx(bool on);

uint32_t perm_kernel_default(void);
uint32_t perm_user_default(void);

#endif
```

kernel_Workspace/kernel/src/mm/protection/permissions.h (line 13) - [public] perm_kernel_default

```
uint32_t perm_kernel_default(void);
uint32_t perm_user_default(void);

#endif
```

kernel_Workspace/kernel/src/mm/protection/permissions.h (line 14) - [public] perm_user_default

```
uint32_t perm_user_default(void);

#endif
```

kstd/include/kstd.h (line 12) - [public] tr_log

```
void tr_log(const char *s);

#endif
```

kstd/include/kstd_audio.h (line 7) - [public] tr_free

```
void tr_free(void *ptr, uint32_t bytes);
tr_status_t tr_map_mmio(uint64_t phys, uint32_t len, void **out_va);

#endif
```

kstd/include/kstd_audio.h (line 8) - [public] tr_map_mmio

```
tr_status_t tr_map_mmio(uint64_t phys, uint32_t len, void **out_va);

#endif
```

kstd/include/kstd_bt.h (line 6) - [public] tr_bt_info

```
tr_status_t tr_bt_info(uint8_t *hci_ver, uint8_t *addr6);
tr_status_t tr_bt_cmd(uint16_t opcode, const void *params, uint8_t len);
tr_status_t tr_bt_evt(void *buf, uint8_t *len);
tr_status_t tr_bt_acl_send(const void *data, uint16_t len);
tr_status_t tr_bt_acl_recv(void *buf, uint16_t *len);

#endif
```

```

kstd/include/kstd_bt.h (line 7) - [public] tr_bt_cmd
    tr_status_t tr_bt_cmd(uint16_t opcode, const void *params, uint8_t len);
    tr_status_t tr_bt_evt(void *buf, uint8_t *len);
    tr_status_t tr_bt_acl_send(const void *data, uint16_t len);
    tr_status_t tr_bt_acl_recv(void *buf, uint16_t *len);

    #endif

kstd/include/kstd_bt.h (line 8) - [public] tr_bt_evt
    tr_status_t tr_bt_evt(void *buf, uint8_t *len);
    tr_status_t tr_bt_acl_send(const void *data, uint16_t len);
    tr_status_t tr_bt_acl_recv(void *buf, uint16_t *len);

    #endif

kstd/include/kstd_bt.h (line 9) - [public] tr_bt_acl_send
    tr_status_t tr_bt_acl_send(const void *data, uint16_t len);
    tr_status_t tr_bt_acl_recv(void *buf, uint16_t *len);

    #endif

kstd/include/kstd_bt.h (line 10) - [public] tr_bt_acl_recv
    tr_status_t tr_bt_acl_recv(void *buf, uint16_t *len);

    #endif

kstd/include/kstd_fs.h (line 6) - [public] tr_fs_read
    tr_status_t tr_fs_read(const char *path, void *buf, uint32_t cap, uint32_t *out);
    tr_status_t tr_fs_exists(const char *path);

    #endif

kstd/include/kstd_fs.h (line 7) - [public] tr_fs_exists
    tr_status_t tr_fs_exists(const char *path);

    #endif

kstd/include/kstd_input.h (line 6) - [public] tr_input_key
    int32_t tr_input_key(void);

    #endif

kstd/include/kstd_mem.h (line 7) - [public] tr_free
    void tr_free(void *ptr, uint32_t bytes);
    tr_status_t tr_map_mmio(uint64_t phys, uint32_t len, void **out_va);

    #endif

kstd/include/kstd_mem.h (line 8) - [public] tr_map_mmio
    tr_status_t tr_map_mmio(uint64_t phys, uint32_t len, void **out_va);

    #endif

kstd/include/kstd_video.h (line 22) - [public] tr_video_init
    /* Inicjalizacja i stan */
    tr_status_t tr_video_init(void);
    bool tr_video_is_ready(void);
    tr_status_t tr_video_get_caps(tr_video_caps_t *caps);

    /* Zarz■dzenie trybami */
    uint32_t tr_video_mode_count(void);
    tr_status_t tr_video_mode_get(uint32_t index, tr_video_mode_t *mode);
    tr_status_t tr_video_mode_set(uint32_t mode_id);

    /* Operacje na framebufferze */
    tr_status_t tr_video_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h);
    tr_status_t tr_video_blit(const void *src, uint32_t src_stride,
        // ... (truncated, see source file for more)

```

kstd/include/kstd_video.h (line 23) - [public] tr_video_is_ready

```
bool tr_video_is_ready(void);
tr_status_t tr_video_get_caps(tr_video_caps_t *caps);

/* Zarządzenie trybami */
uint32_t tr_video_mode_count(void);
tr_status_t tr_video_mode_get(uint32_t index, tr_video_mode_t *mode);
tr_status_t tr_video_mode_set(uint32_t mode_id);

/* Operacje na framebufferze */
tr_status_t tr_video_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h);
tr_status_t tr_video_blit(const void *src, uint32_t src_stride,
                        uint32_t dst_x, uint32_t dst_y, uint32_t w, uint32_t h);
// ... (truncated, see source file for more)
```

kstd/include/kstd_video.h (line 24) - [public] tr_video_get_caps

```
tr_status_t tr_video_get_caps(tr_video_caps_t *caps);

/* Zarządzenie trybami */
uint32_t tr_video_mode_count(void);
tr_status_t tr_video_mode_get(uint32_t index, tr_video_mode_t *mode);
tr_status_t tr_video_mode_set(uint32_t mode_id);

/* Operacje na framebufferze */
tr_status_t tr_video_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h);
tr_status_t tr_video_blit(const void *src, uint32_t src_stride,
                        uint32_t dst_x, uint32_t dst_y, uint32_t w, uint32_t h);

// ... (truncated, see source file for more)
```

kstd/include/kstd_video.h (line 27) - [public] tr_video_mode_count

```
/* Zarządzenie trybami */

uint32_t tr_video_mode_count(void);
tr_status_t tr_video_mode_get(uint32_t index, tr_video_mode_t *mode);
tr_status_t tr_video_mode_set(uint32_t mode_id);

/* Operacje na framebufferze */
tr_status_t tr_video_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h);
tr_status_t tr_video_blit(const void *src, uint32_t src_stride,
                        uint32_t dst_x, uint32_t dst_y, uint32_t w, uint32_t h);

#endif
```

kstd/include/kstd_video.h (line 28) - [public] tr_video_mode_get

```
tr_status_t tr_video_mode_get(uint32_t index, tr_video_mode_t *mode);
tr_status_t tr_video_mode_set(uint32_t mode_id);

/* Operacje na framebufferze */
tr_status_t tr_video_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h);
tr_status_t tr_video_blit(const void *src, uint32_t src_stride,
                        uint32_t dst_x, uint32_t dst_y, uint32_t w, uint32_t h);

#endif
```

kstd/include/kstd_video.h (line 29) - [public] tr_video_mode_set

```
tr_status_t tr_video_mode_set(uint32_t mode_id);

/* Operacje na framebufferze */
tr_status_t tr_video_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h);
tr_status_t tr_video_blit(const void *src, uint32_t src_stride,
                        uint32_t dst_x, uint32_t dst_y, uint32_t w, uint32_t h);

#endif
```


kstd/include/kstd_video.h (line 32) - [public] tr_video_fill

/ Operacje na framebufferze */*

```
tr_status_t tr_video_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h);
tr_status_t tr_video_blit(const void *src, uint32_t src_stride,
                        uint32_t dst_x, uint32_t dst_y, uint32_t w, uint32_t h);
```

#endif

libs/dsabi.h (line 93) - [public] volatile

```
__asm__ volatile("int $0x80" :: "a"(1));
}
```

```
typedef struct {
    uint64_t id;
    uint32_t cmd;
    uint32_t flags;
    uint64_t arg0;
    uint64_t arg1;
    uint64_t arg2;
    int32_t status;
    uint32_t pad;
    // ... (truncated, see source file for more)
```

libs/dsabi.h (line 124) - [public] ds_init

```
void ds_init(uint64_t params_va);
uint64_t ds_call(uint32_t cls, uint32_t op,
                uint64_t a0, uint64_t a1, uint64_t a2);
int ds_take(uint64_t id, ds_msg_t *out);
void ds_poll(void);
```

#endif

libs/dsabi.h (line 127) - [public] ds_take

```
int ds_take(uint64_t id, ds_msg_t *out);
void ds_poll(void);
```

#endif

libs/dsabi.h (line 128) - [public] ds_poll

```
void ds_poll(void);
```

#endif

libs/dsasync.h (line 21) - [public] if

```
if (r->done) {
    return 1;
}

ds_poll();

if (r->id != 0 && ds_take(r->id, &r->msg)) {
    r->done = 1;
}

return r->done;
}

// ... (truncated, see source file for more)
```

```

libs/dsasync.h (line 25) - [public] ds_poll
    ds_poll();

    if (r->id != 0 && ds_take(r->id, &r->msg)) {
        r->done = 1;
    }

    return r->done;
}

static inline int ds_req_ok(ds_req_t *r)
{
    return r->done && r->msg.status == 0;
    // ... (truncated, see source file for more)

libs/dsasync.h (line 27) - [public] if
    if (r->id != 0 && ds_take(r->id, &r->msg)) {
        r->done = 1;
    }

    return r->done;
}

static inline int ds_req_ok(ds_req_t *r)
{
    return r->done && r->msg.status == 0;
}

    // ... (truncated, see source file for more)

libs/dsaudio.h (line 7) - [public] ds_call
    return ds_call(SVC_AUDIO, AUD_PLAY, phys, len, 0);
}

static inline void ds_audio_stop(void)
{
    ds_call(SVC_AUDIO, AUD_STOP, 0, 0, 0);
}

static inline uint64_t ds_audio_jack_req(void)
{
    return ds_call(SVC_AUDIO, AUD_JACK, 0, 0, 0);
}

    // ... (truncated, see source file for more)

libs/dsaudio.h (line 12) - [public] ds_call
    ds_call(SVC_AUDIO, AUD_STOP, 0, 0, 0);
}

static inline uint64_t ds_audio_jack_req(void)
{
    return ds_call(SVC_AUDIO, AUD_JACK, 0, 0, 0);
}

static inline void ds_audio_amp(int on)
{
    ds_call(SVC_AUDIO, AUD_AMP, on ? 1 : 0, 0, 0);
}

    // ... (truncated, see source file for more)

libs/dsaudio.h (line 17) - [public] ds_call
    return ds_call(SVC_AUDIO, AUD_JACK, 0, 0, 0);
}

static inline void ds_audio_amp(int on)
{
    ds_call(SVC_AUDIO, AUD_AMP, on ? 1 : 0, 0, 0);
}

#endif

```

```

libs/dsaudio.h (line 22) - [public] ds_call
    ds_call(SVC_AUDIO, AUD_AMP, on ? 1 : 0, 0, 0);
}

#endif

libs/dsinput.h (line 7) - [public] ds_call
    return ds_call(SVC_INPUT, IN_KEY_POLL, 0, 0, 0);
}

static inline int ds_key_take(uint64_t id, uint8_t *out)
{
    ds_msg_t m;

    if (!ds_take(id, &m) || m.arg0 == 0) {
        return 0;
    }

    *out = (uint8_t)m.arg0;
    // ... (truncated, see source file for more)

libs/dsinput.h (line 14) - [public] if
    if (!ds_take(id, &m) || m.arg0 == 0) {
        return 0;
    }

    *out = (uint8_t)m.arg0;
    return 1;
}

#endif

libs/dslock.h (line 7) - [public] ds_call
    return ds_call(SVC_BLOCK, BLK_COUNT, disk, 0, 0);
}

static inline uint64_t ds_blk_read(uint64_t disk, uint64_t block)
{
    return ds_call(SVC_BLOCK, BLK_READ, disk, block, 0);
}

static inline uint64_t ds_blk_write(uint64_t disk, uint64_t block)
{
    return ds_call(SVC_BLOCK, BLK_WRITE, disk, block, 0);
}

// ... (truncated, see source file for more)

libs/dslock.h (line 12) - [public] ds_call
    return ds_call(SVC_BLOCK, BLK_READ, disk, block, 0);
}

static inline uint64_t ds_blk_write(uint64_t disk, uint64_t block)
{
    return ds_call(SVC_BLOCK, BLK_WRITE, disk, block, 0);
}

static inline volatile uint8_t *ds_blk_scratch(void)
{
    return (volatile uint8_t *)DS_SCRATCH_VA;
}

// ... (truncated, see source file for more)

```

```

libs/dslock.h (line 17) - [public] ds_call
    return ds_call(SVC_BLOCK, BLK_WRITE, disk, block, 0);
}

static inline volatile uint8_t *ds_blk_scratch(void)
{
    return (volatile uint8_t *)DS_SCRATCH_VA;
}

#endif

libs/dslog.h (line 10) - [public] while
    while (s[n] != '\0' && n < DS_SCRATCH_SIZE - 1) {
        scratch[n] = s[n];
        n++;
    }

    ds_call(SVC_SYS, OP_LOG, n, 0, 0);
}

#endif

libs/dslog.h (line 15) - [public] ds_call
    ds_call(SVC_SYS, OP_LOG, n, 0, 0);
}

#endif

libs/dsmem.h (line 7) - [public] ds_call
    return ds_call(SVC_SYS, OP_ALLOC, pages, 0, 0);
}

static inline uint64_t ds_free_pages(uint64_t va)
{
    return ds_call(SVC_SYS, OP_FREE, va, 0, 0);
}

static inline uint64_t ds_map_mmio(uint64_t phys, uint64_t len, uint64_t va)
{
    return ds_call(SVC_SYS, OP_MAPMMIO, phys, len, va);
}

// ... (truncated, see source file for more)

libs/dsmem.h (line 12) - [public] ds_call
    return ds_call(SVC_SYS, OP_FREE, va, 0, 0);
}

static inline uint64_t ds_map_mmio(uint64_t phys, uint64_t len, uint64_t va)
{
    return ds_call(SVC_SYS, OP_MAPMMIO, phys, len, va);
}

static inline uint64_t ds_page_phys(uint64_t va)
{
    return ds_call(SVC_SYS, OP_PAGEPHYS, va, 0, 0);
}

// ... (truncated, see source file for more)

libs/dsmem.h (line 17) - [public] ds_call
    return ds_call(SVC_SYS, OP_MAPMMIO, phys, len, va);
}

static inline uint64_t ds_page_phys(uint64_t va)
{
    return ds_call(SVC_SYS, OP_PAGEPHYS, va, 0, 0);
}

#endif

```

```

libs/dsmem.h (line 22) - [public] ds_call
    return ds_call(SVC_SYS, OP_PAGEPHYS, va, 0, 0);
}

#endif

libs/dsvgpu.h (line 6) - [public] ds_call
extern uint64_t ds_call(uint32_t, uint32_t, uint64_t, uint64_t, uint64_t);
extern void ds_poll(void);
extern int ds_take(uint64_t, ds_msg_t *);

static inline int dsvgpu_surface_create(uint32_t w, uint32_t h,
                                         uint64_t *phys, int32_t *sid)
{
    uint64_t id = ds_call(SVC_VGPU, VGPU_SURF_CREATE, w, h, 0);
    ds_poll();

    ds_msg_t r;

    // ... (truncated, see source file for more)
}

libs/dsvgpu.h (line 7) - [public] ds_poll
extern void ds_poll(void);
extern int ds_take(uint64_t, ds_msg_t *);

static inline int dsvgpu_surface_create(uint32_t w, uint32_t h,
                                         uint64_t *phys, int32_t *sid)
{
    uint64_t id = ds_call(SVC_VGPU, VGPU_SURF_CREATE, w, h, 0);
    ds_poll();

    ds_msg_t r;

    if (!ds_take(id, &r) || r.status != 0) {
        // ... (truncated, see source file for more)
    }
}

libs/dsvgpu.h (line 8) - [public] ds_take
extern int ds_take(uint64_t, ds_msg_t *);

static inline int dsvgpu_surface_create(uint32_t w, uint32_t h,
                                         uint64_t *phys, int32_t *sid)
{
    uint64_t id = ds_call(SVC_VGPU, VGPU_SURF_CREATE, w, h, 0);
    ds_poll();

    ds_msg_t r;

    if (!ds_take(id, &r) || r.status != 0) {
        return 0;
        // ... (truncated, see source file for more)
    }
}

libs/dsvgpu.h (line 14) - [public] ds_poll
    ds_poll();

    ds_msg_t r;

    if (!ds_take(id, &r) || r.status != 0) {
        return 0;
    }

    *sid = (int32_t)r.arg0;
    *phys = r.arg1;
    return 1;
}

// ... (truncated, see source file for more)

```

```

libs/dsvgpu.h (line 18) - [public] if
    if (!ds_take(id, &r) || r.status != 0) {
        return 0;
    }

    *sid = (int32_t)r.arg0;
    *phys = r.arg1;
    return 1;
}

static inline int dsvgpu_present(int32_t sid, uint32_t x, uint32_t y)
{
    uint64_t id = ds_call(SVC_VGPU, VGPU_PRESENT,
        // ... (truncated, see source file for more)

libs/dsvgpu.h (line 31) - [public] ds_poll
    ds_poll();

    ds_msg_t r;
    return ds_take(id, &r) && r.status == 0;
}

#endif

libs/dsvideo.h (line 14) - [public] ds_call
    return ds_call(SVC_VIDEO, VID_FB_INFO, 0, 0, 0);
}

static inline int ds_fb_info_take(uint64_t id, ds_fb_t *fb)
{
    ds_msg_t m;

    if (!ds_take(id, &m) || m.status != 0) {
        return 0;
    }

    fb->w = (uint32_t)(m.arg0 >> 16);
    // ... (truncated, see source file for more)

libs/dsvideo.h (line 21) - [public] if
    if (!ds_take(id, &m) || m.status != 0) {
        return 0;
    }

    fb->w = (uint32_t)(m.arg0 >> 16);
    fb->h = (uint32_t)(m.arg0 & 0xFFFF);
    fb->stride = (uint32_t)m.arg1;
    fb->phys = m.arg2;

    return 1;
}

    // ... (truncated, see source file for more)

libs/dsvideo.h (line 35) - [public] ds_call
    ds_call(SVC_VIDEO, VID_FB_TAKEOVER, 0, 0, 0);
}

static inline void ds_fb_release(void)
{
    ds_call(SVC_VIDEO, VID_FB_RELEASE, 0, 0, 0);
}

static inline void ds_px(volatile uint32_t *fb, uint32_t stride,
    uint32_t x, uint32_t y, uint32_t c)
{
    fb[(uint64_t)y * stride + x] = c;
    // ... (truncated, see source file for more)

```

```

libs/dsvideo.h (line 40) - [public] ds_call
    ds_call(SVC_VIDEO, VID_FB_RELEASE, 0, 0, 0);
}

static inline void ds_px(volatile uint32_t *fb, uint32_t stride,
                        uint32_t x, uint32_t y, uint32_t c)
{
    fb[(uint64_t)y * stride + x] = c;
}

static inline uint32_t ds_rgb(uint8_t r, uint8_t g, uint8_t b)
{
    return 0xFF000000u | ((uint32_t)r << 16) | ((uint32_t)g << 8) | b;
    // ... (truncated, see source file for more)

```

Nim

kernel_Workspace/kernel/src/nimcore/banner.nim (line 13) - [internal] nim_banner

```

proc nim_banner(buf: ptr uint8, cap: uint32): uint32 {.exportc, cdecl.} =
    var o = 0u32

    for ch in ART:
        if o >= cap:
            break

        buf[o] = uint8(ch)
        o += 1

    return o

```

kernel_Workspace/kernel/src/nimcore/format.nim (line 1) - [internal] nim_u64_to_str

```

proc nim_u64_to_str(v: uint64, base: uint8, buf: ptr uint8, cap: uint32): uint32
    {.exportc, cdecl.} =
        var tmp: array[64, uint8]
        var n = 0u32
        var x = v

        let b = if base < 2: 10u64 else: uint64(base)

        if x == 0:
            tmp[0] = uint8('0')
            n = 1
        else:
            // ... (truncated, see source file for more)

```

kernel_Workspace/kernel/src/nimcore/format.nim (line 29) - [internal] nim_hex_dump

```

proc nim_hex_dump(src: ptr uint8, len: uint32, buf: ptr uint8, cap: uint32): uint32
    {.exportc, cdecl.} =
        const hexd = "0123456789abcdef"
        var o = 0u32
        var i = 0u32

        while i < len and o + 2 < cap:
            let b = src[i]
            buf[o] = uint8(hexd[int(b shr 4)])
            buf[o + 1] = uint8(hexd[int(b and 15)])
            o += 2
            i += 1
            // ... (truncated, see source file for more)

```

```

kernel_Workspace/kernel/src/nimcore/parse.nim (line 1) - [internal] nim_parse_u64
proc nim_parse_u64(s: ptr uint8, len: uint32, base: uint8, out_v: ptr uint64): uint8
  {.exportc, cdecl.} =
    var v = 0u64
    var i = 0u32
    let b = uint64(base)

    if len == 0: return 0

    if base == 16 and len > 2 and s[0] == uint8('0') and (s[1] == uint8('x') or s[1] == uint8('X')):
      i = 2

    if i >= len: return 0
      // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/nimcore/ringbuf.nim (line 1) - [internal] Rb
type Rb = object
  data: array[1024, uint8]
  head: uint32
  tail: uint32

var key_rb: Rb

proc nim_rb_push(b: uint8): uint8 {.exportc, cdecl.} =
  let next = (key_rb.head + 1) mod 1024
  if next == key_rb.tail: return 0
  key_rb.data[key_rb.head] = b
  key_rb.head = next
  // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/nimcore/ringbuf.nim (line 8) - [internal] nim_rb_push
proc nim_rb_push(b: uint8): uint8 {.exportc, cdecl.} =
  let next = (key_rb.head + 1) mod 1024
  if next == key_rb.tail: return 0
  key_rb.data[key_rb.head] = b
  key_rb.head = next
  return 1

proc nim_rb_pop(): int32 {.exportc, cdecl.} =
  if key_rb.head == key_rb.tail: return -1
  let b = key_rb.data[key_rb.tail]
  key_rb.tail = (key_rb.tail + 1) mod 1024
  return int32(b)
  // ... (truncated, see source file for more)

kernel_Workspace/kernel/src/nimcore/ringbuf.nim (line 15) - [internal] nim_rb_pop
proc nim_rb_pop(): int32 {.exportc, cdecl.} =
  if key_rb.head == key_rb.tail: return -1
  let b = key_rb.data[key_rb.tail]
  key_rb.tail = (key_rb.tail + 1) mod 1024
  return int32(b)

proc nim_rb_len(): uint32 {.exportc, cdecl.} =
  return (key_rb.head + 1024 - key_rb.tail) mod 1024

kernel_Workspace/kernel/src/nimcore/ringbuf.nim (line 21) - [internal] nim_rb_len
proc nim_rb_len(): uint32 {.exportc, cdecl.} =
  return (key_rb.head + 1024 - key_rb.tail) mod 1024

```


kernel_Workspace/kernel/src/nimcore/shell.nim (line 1) - [internal] Handler

```
type Handler = proc(arg: ptr uint8, len: uint32): int32 {.cdecl.}

type Cmd = object
  name: array[24, uint8]
  nlen: uint32
  used: bool
  h: Handler

var cmds: array[32, Cmd]

proc nim_shell_register(name: ptr uint8, nlen: uint32, h: Handler): uint8
  {.exportc, cdecl.} =
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/nimcore/shell.nim (line 3) - [internal] Cmd

```
type Cmd = object
  name: array[24, uint8]
  nlen: uint32
  used: bool
  h: Handler

var cmds: array[32, Cmd]

proc nim_shell_register(name: ptr uint8, nlen: uint32, h: Handler): uint8
  {.exportc, cdecl.} =
    if nlen == 0 or nlen >= 24: return 0

    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/nimcore/shell.nim (line 11) - [internal] nim_shell_register

```
proc nim_shell_register(name: ptr uint8, nlen: uint32, h: Handler): uint8
  {.exportc, cdecl.} =
    if nlen == 0 or nlen >= 24: return 0

    for i in 0..<32:
      if not cmds[i].used:
        var k = 0u32
        while k < nlen:
          cmds[i].name[k] = name[k]
          k += 1
        cmds[i].nlen = nlen
        cmds[i].used = true
    // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/nimcore/shell.nim (line 28) - [internal] eq_name

```
proc eq_name(c: var Cmd, line: ptr uint8, nlen: uint32): bool =
  if c.nlen != nlen: return false
  var k = 0u32
  while k < nlen:
    if c.name[k] != line[k]: return false
    k += 1
  return true

proc nim_shell_run(line: ptr uint8, len: uint32): int32
  {.exportc, cdecl.} =
    var sp = 0u32
    while sp < len and line[sp] != uint8(' '):
      // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/nimcore/shell.nim (line 36) - [internal] nim_shell_run

```
proc nim_shell_run(line: ptr uint8, len: uint32): int32
  {.exportc, cdecl.} =
    var sp = 0u32
    while sp < len and line[sp] != uint8(' '):
      sp += 1

    if sp == 0: return -1

    var arg: ptr uint8 = nil
    var alen = 0u32

    if sp < len:
      // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/policy/nim/policynim.nim (line 10) - [internal] nim_policy_log

```
proc nim_policy_log(ring, cls, op, dec: uint8) {.exportc, cdecl.} =
  logbuf[head] = Entry(ring: ring, cls: cls, op: op, dec: dec)
  head = (head + 1) mod 256

  if dec == 1:
    denies += 1

proc nim_policy_denies(): uint64 {.exportc, cdecl.} =
  denies

proc nim_policy_get(idx: uint32,
  out_ring, out_cls, out_op, out_dec: ptr uint8): uint8
  // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/policy/nim/policynim.nim (line 17) - [internal] nim_policy_denies

```
proc nim_policy_denies(): uint64 {.exportc, cdecl.} =
  denies

proc nim_policy_get(idx: uint32,
  out_ring, out_cls, out_op, out_dec: ptr uint8): uint8
  {.exportc, cdecl.} =
    if idx >= 256:
      return 0

    let e = logbuf[idx]

    if out_ring != nil: out_ring[] = e.ring
      // ... (truncated, see source file for more)
```

kernel_Workspace/kernel/src/policy/nim/policynim.nim (line 20) - [internal] nim_policy_get

```
proc nim_policy_get(idx: uint32,
  out_ring, out_cls, out_op, out_dec: ptr uint8): uint8
  {.exportc, cdecl.} =
    if idx >= 256:
      return 0

    let e = logbuf[idx]

    if out_ring != nil: out_ring[] = e.ring
    if out_cls != nil: out_cls[] = e.cls
    if out_op != nil: out_op[] = e.op
    if out_dec != nil: out_dec[] = e.dec
      // ... (truncated, see source file for more)
```

Odin

drivers/audiodriver/src/odin/driver.odin (line 8) - [public] Bdl_Entry

```
Bdl_Entry :: struct {
  ■addr: u32,
  ■ctrl: u32,
}

BDL_N :: 8

// mixer (NAM) — 16-bit
nam16 :: proc(off: u32) -> ^u16 {
  ■return transmute(^u16)(nam_base + u64(off))
}

// ... (truncated, see source file for more)
```

drivers/audiodriver/src/odin/driver.odin (line 16) - [public] nam16

```
// mixer (NAM) — 16-bit

nam16 :: proc(off: u32) -> ^u16 {
  ■return transmute(^u16)(nam_base + u64(off))
}

// bus master — 8/16/32
bm8 :: proc(off: u32) -> ^u8 {
  ■return transmute(^u8)(bm_base + u64(off))
}

bm16 :: proc(off: u32) -> ^u16 {
  ■return transmute(^u16)(bm_base + u64(off))
}

// ... (truncated, see source file for more)
```

drivers/audiodriver/src/odin/driver.odin (line 21) - [public] bm8

```
// bus master — 8/16/32

bm8 :: proc(off: u32) -> ^u8 {
  ■return transmute(^u8)(bm_base + u64(off))
}

bm16 :: proc(off: u32) -> ^u16 {
  ■return transmute(^u16)(bm_base + u64(off))
}

bm32 :: proc(off: u32) -> ^u32 {
  ■return transmute(^u32)(bm_base + u64(off))
}

// ... (truncated, see source file for more)
```

drivers/audiodriver/src/odin/driver.odin (line 25) - [public] bm16

```
bm16 :: proc(off: u32) -> ^u16 {
  ■return transmute(^u16)(bm_base + u64(off))
}

bm32 :: proc(off: u32) -> ^u32 {
  ■return transmute(^u32)(bm_base + u64(off))
}

GLOB_CNT  :: 0x2C
GLOB_STA  :: 0x30

PI_BDBAR  :: 0x00
// ... (truncated, see source file for more)
```

drivers/audiodriver/src/odin/driver.odin (line 29) - [public] bm32

```
bm32 :: proc(off: u32) -> ^u32 {  
  ■return transmute(^u32)(bm_base + u64(off))  
}
```

```
GLOB_CNT  :: 0x2C  
GLOB_STA  :: 0x30
```

```
PI_BDBAR  :: 0x00  
PI_LVI    :: 0x05  
PI_SR     :: 0x06  
PI_CR     :: 0x0B
```

// ... (truncated, see source file for more)

drivers/audiodriver/src/odin/driver.odin (line 50) - [public] ad_init

```
ad_init :: proc "C" (nam_va: u64, bm_va: u64) -> i32 {  
  ■nam_base = nam_va  
  ■bm_base = bm_va  
  
  ■// cold reset  
  ■mem.volatile_store(bm32(GLOB_CNT), 0x04000000)  
  
  ■for _ in 0..<100000 {  
    ■■if mem.volatile_load(bm32(GLOB_STA)) & 0x04 != 0 {  
      ■■■break  
    }  
  }  
}
```

// ... (truncated, see source file for more)

drivers/audiodriver/src/odin/driver.odin (line 73) - [public] ad_stop

```
ad_stop :: proc "C" () -> i32 {  
  ■mem.volatile_store(bm8(P0_CR), 0)  
  ■mem.volatile_store(bm8(PI_CR), 0)  
  ■return 0  
}  
  
@(export)  
ad_position :: proc "C" () -> u32 {  
  ■return u32(mem.volatile_load(bm16(P0_PICB)))  
}
```

drivers/audiodriver/src/odin/driver.odin (line 80) - [public] ad_position

```
ad_position :: proc "C" () -> u32 {  
  ■return u32(mem.volatile_load(bm16(P0_PICB)))  
}
```

drivers/audiodriver/src/odin/jack.odin (line 8) - [public] ad_jack_present

```
ad_jack_present :: proc "C" () -> i32 {  
  ■pwr := mem.volatile_load(nam16(PWR_CTRL_STA))  
  
  ■// bity ready w powerdown status = tor aktywny  
  ■if pwr & 0x000F != 0 {  
    ■■return 1  
  }  
  
  ■// QEMU i tak nie zdejmuj jacka – fallback  
  ■return 1  
}
```

// ... (truncated, see source file for more)

drivers/audiodriver/src/odin/jack.odin (line 21) - [public] ad_set_amp

```
ad_set_amp :: proc "C" (on: i32) -> i32 {  
  ■if on == 0 {  
    ■■mem.volatile_store(nam16(0x02), 0x8000) // mute  
  } else {  
    ■■mem.volatile_store(nam16(0x02), 0x0000)  
  }  
  ■return 0  
}
```

drivers/audiodriver/src/odin/microphone.odin (line 6) - [public] ad_capture

```
ad_capture :: proc "C" (data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) -> i32 {
    ■ bdl := transmute(^Bdl_Entry) bdl_va

    ■ n := 0
    ■ off: u64 = 0

    ■ for off < u64(len) && n < BDL_N {
        ■ ■ chunk: u32 = 4096
        ■ ■ rem := u32(u64(len) - off)

        ■ ■ if rem < chunk {
            ■ ■ ■ chunk = rem
            // ... (truncated, see source file for more)
        }
    }
}
```

drivers/audiodriver/src/odin/speaker.odin (line 6) - [public] ad_play

```
ad_play :: proc "C" (data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) -> i32 {
    ■ bdl := transmute(^Bdl_Entry) bdl_va

    ■ n := 0
    ■ off: u64 = 0

    ■ for off < u64(len) && n < BDL_N {
        ■ ■ chunk: u32 = 4096
        ■ ■ rem := u32(u64(len) - off)

        ■ ■ if rem < chunk {
            ■ ■ ■ chunk = rem
            // ... (truncated, see source file for more)
        }
    }
}
```

userspace-ctrl/src/abi/message.odin (line 15) - [public] MessageHeader

```
MessageHeader :: struct #packed {
    magic:      u32,
    sender_pid: util.PID,
    target_pid: util.PID,
    opcode:     Opcode,
    flags:      MessageFlag,
    payload_sz: u32,
    seq_num:    u64,
}

MessageEnvelope :: struct {
    header: MessageHeader,
    // ... (truncated, see source file for more)
}
```

userspace-ctrl/src/abi/message.odin (line 25) - [public] MessageEnvelope

```
MessageEnvelope :: struct {
    header: MessageHeader,
    payload: [dynamic]u8,
}

make_request_header :: proc(sender: util.PID, target: util.PID, op: Opcode, payload_size: u32, seq: u64)
    return MessageHeader{
        magic      = MESSAGE_MAGIC,
        sender_pid = sender,
        target_pid = target,
        opcode      = op,
        flags       = {MessageFlag.None},
        // ... (truncated, see source file for more)
    }
```

userspace-ctrl/src/abi/message.odin (line 30) - [public] make_request_header

```
make_request_header :: proc(sender: util.PID, target: util.PID, op: Opcode, payload_size: u32, seq: u64)
  return MessageHeader{
    magic      = MESSAGE_MAGIC,
    sender_pid = sender,
    target_pid = target,
    opcode     = op,
    flags      = {MessageFlag.None},
    payload_sz = payload_size,
    seq_num    = seq,
  }
}
```

userspace-ctrl/src/abi/opcodes.odin (line 5) - [public] Opcode

```
Opcode :: enum u32 {
  Sys_Ping      = 0x00000001,
  Sys_GetInfo   = 0x00000002,
  Sys_Shutdown  = 0x000000FF,
  Proc_Create   = 0x01000001,
  Proc_Exit     = 0x01000002,
  Proc_Kill     = 0x01000003,
  Proc_Wait     = 0x01000004,
  Mem_Alloc     = 0x02000001,
  Mem_Free      = 0x02000002,
  Mem_Protect   = 0x02000003,
  Ipc_PortCreate = 0x03000001,
  // ... (truncated, see source file for more)
```

userspace-ctrl/src/abi/opcodes.odin (line 27) - [public] is_proc_opcode

```
is_proc_opcode :: proc(op: Opcode) -> bool {
  return (cast(u32, op) & 0xFF000000) == 0x01000000
}

is_mem_opcode :: proc(op: Opcode) -> bool {
  return (cast(u32, op) & 0xFF000000) == 0x02000000
}

is_ipc_opcode :: proc(op: Opcode) -> bool {
  return (cast(u32, op) & 0xFF000000) == 0x03000000
}

// ... (truncated, see source file for more)
```

userspace-ctrl/src/abi/opcodes.odin (line 31) - [public] is_mem_opcode

```
is_mem_opcode :: proc(op: Opcode) -> bool {
  return (cast(u32, op) & 0xFF000000) == 0x02000000
}

is_ipc_opcode :: proc(op: Opcode) -> bool {
  return (cast(u32, op) & 0xFF000000) == 0x03000000
}

is_fs_opcode :: proc(op: Opcode) -> bool {
  return (cast(u32, op) & 0xFF000000) == 0x04000000
}
```

userspace-ctrl/src/abi/opcodes.odin (line 35) - [public] is_ipc_opcode

```
is_ipc_opcode :: proc(op: Opcode) -> bool {
  return (cast(u32, op) & 0xFF000000) == 0x03000000
}

is_fs_opcode :: proc(op: Opcode) -> bool {
  return (cast(u32, op) & 0xFF000000) == 0x04000000
}
```

userspace-ctrl/src/abi/opcodes.odin (line 39) - [public] is_fs_opcode

```
is_fs_opcode :: proc(op: Opcode) -> bool {
  return (cast(u32, op) & 0xFF000000) == 0x04000000
}
```

```

userspace-ctrl/src/log/logger.odin (line 6) - [public] Level
    Level :: enum {
        TRACE,
        DEBUG,
        INFO,
        WARN,
        ERROR,
        FATAL,
    }

    pub init :: proc() {}

    format_msg :: proc(level: Level, module: string, msg: string, args: ..any) -> string {
        // ... (truncated, see source file for more)
    }

userspace-ctrl/src/log/logger.odin (line 17) - [public] format_msg
    format_msg :: proc(level: Level, module: string, msg: string, args: ..any) -> string {
        lvl_str := ""
        switch level {
        case .TRACE: lvl_str = "TRACE"
        case .DEBUG: lvl_str = "DEBUG"
        case .INFO:  lvl_str = "INFO"
        case .WARN:  lvl_str = "WARN"
        case .ERROR: lvl_str = "ERROR"
        case .FATAL: lvl_str = "FATAL"
        }
        return fmt.Sprintf("[%s] %s: %s\n", lvl_str, module, msg, args)
    }
    // ... (truncated, see source file for more)

userspace-ctrl/src/main.odin (line 7) - [public] main
    main :: proc() -> int {
        status := core.early_boot()
        if status != util.ERR_OK {
            return cast(int, status)
        }

        log.info("main", "Zarządca wystartowała pomyślnie. Wersja: 0.1.0-alpha")
        core.run_main_loop()
        core.shutdown()

        return 0
    }

userspace-ctrl/src/util/error.odin (line 3) - [public] Error
    Error :: struct {
        code:   Errno,
        message: string,
    }

    ERR_OK          :: Errno(0)
    ERR_NO_MEMORY    :: Errno(1)
    ERR_INVALID_ARG  :: Errno(2)
    ERR_NOT_FOUND    :: Errno(3)
    ERR_PERMISSION   :: Errno(4)
    ERR_BUSY         :: Errno(5)
    ERR_TIMEOUT      :: Errno(6)
    // ... (truncated, see source file for more)

userspace-ctrl/src/util/error.odin (line 20) - [public] make_error
    make_error :: proc(code: Errno, msg: string) -> Error {
        return Error{code = code, message = msg}
    }

```

Python

doc.py (line 131) - [public] Definition

```
class Definition:
    path: str          # path relative to the scanned root
    lineno: int
    kind: str          # "function" / "class" / "method" / "type" / "const" / "export"
    name: str
    signature: str      # signature line (short, used in section 1/2)
    is_public: bool
    comment: str = ""   # comment/docstring directly above the definition
    body_snippet: str = "" # first few lines of code (section 3)
    language: str = ""
```

doc.py (line 144) - [public] MarkdownFile

```
class MarkdownFile:
    path: str
    content: str
```

doc.py (line 153) - [public] iter_source_files

```
# ----- #
# Collecting files
# ----- #

def iter_source_files(root: str, exts: set[str] | None, exclude_dirs: set[str]):
    for dirpath, dirnames, filenames in os.walk(root):
        dirnames[:] = [d for d in dirnames if d not in exclude_dirs and not d.startswith(".")
                        or d in (".",)]
        for fn in filenames:
            ext = os.path.splitext(fn)[1].lower()
            full = os.path.join(dirpath, fn)
            rel = os.path.relpath(full, root)
            if ext in MD_EXTS:
                yield rel, full, ext
            elif exts is not None:
                if ext in exts:
                    # ... (truncated, see source file for more)
```

doc.py (line 170) - [public] read_text

```
def read_text(path: str) -> str:
    for enc in ("utf-8", "utf-8-sig", "latin-1"):
        try:
            with open(path, "r", encoding=enc) as f:
                return f.read()
        except (UnicodeDecodeError, LookupError):
            continue
    with open(path, "r", encoding="utf-8", errors="replace") as f:
        return f.read()
```

doc.py (line 185) - [public] extract_preceding_comment

```
# ----- #
# Extracting the comment directly above line `lineno` (1-indexed)
# ----- #

def extract_preceding_comment(lines: list[str], lineno_0based: int, ext: str) -> str:
    prefixes = COMMENT_PREFIXES.get(ext, ("#", "//"))
    collected = []
    i = lineno_0based - 1
    in_block_close = False
    while i >= 0:
        line = lines[i].rstrip("\n")
        stripped = line.strip()
        if stripped == "":
            i -= 1
            if collected:
                # a blank line ends the comment block (unless we haven't
                # ... (truncated, see source file for more)
```


doc.py (line 222) - [public] parse_python

```
# ----- #
# PYTHON - parsed via ast (no code execution)
# ----- #

def parse_python(rel_path: str, full_path: str, source: str) -> list[Definition]:
    defs: list[Definition] = []
    try:
        tree = ast.parse(source, filename=rel_path)
    except SyntaxError:
        return defs

    lines = source.splitlines()

    all_list = None
    for node in ast.walk(tree):
        if isinstance(node, ast.Assign):
            # ... (truncated, see source file for more)
```

doc.py (line 298) - [internal] _rules_js_ts

```
# Each rule: (kind, regex, group_index, decide_public)
# decide_public(match) -> bool, or None when visibility depends on context
# (tracked separately in parse_generic, e.g. Ruby's private/public keyword,
# Ada's "private" section, Odin's @(private) attribute)
```

```
def _rules_js_ts():
    return [
        ("function", re.compile(r"^\s*export\s+default\s+(async\s+)?function\s*\*\s*(\w+)"), 2, lambda m: True),
        ("function", re.compile(r"^\s*export\s+(async\s+)?function\s*\*\s*(\w+)"), 2, lambda m: True),
        ("class", re.compile(r"^\s*export\s+(default\s+)?class\s*(\w+)"), 2, lambda m: True),
        ("const", re.compile(r"^\s*export\s+(const|let|var)\s*(\w+)"), 2, lambda m: True),
        ("interface", re.compile(r"^\s*export\s+interface\s*(\w+)"), 1, lambda m: True),
        ("type", re.compile(r"^\s*export\s+type\s*(\w+)"), 1, lambda m: True),
        ("export", re.compile(r"^\s*module\.exports\.(.\w+)?\s*="), 2, lambda m: True),
        ("function", re.compile(r"^\s*(async\s+)?function\s*\*\s*(\w+)"), 2, lambda m: False),
        ("const", re.compile(r"^\s*(const|let|var)\s*(\w+)\s*=\s*(\(|async\s*(\|function)"), 2, lambda m: True),
        ("method", re.compile(r"^\s*(private|#)\s*(\w+)\s*\("), 2, lambda m: False),
        # ... (truncated, see source file for more)
```

doc.py (line 313) - [internal] _rules_java_cs

```
def _rules_java_cs():
    return [
        ("class", re.compile(r"^\s*public\s+(static\s+)?(final\s+)?(abstract\s+)?(class|interface|enum|record)"), 1, lambda m: True),
        ("method", re.compile(r"^\s*public\s+(static\s+)?(final\s+)?[\w<>\[\],\s]+?\s(\w+)\s*\("), 2, lambda m: True),
        ("class", re.compile(r"^\s*(private|protected|internal)\s+(static\s+)?(class|interface|enum|record)"), 1, lambda m: True),
        ("method", re.compile(r"^\s*(private|protected|internal)\s+(static\s+)?[\w<>\[\],\s]+?\s(\w+)\s*\("), 2, lambda m: True),
    ]
```

doc.py (line 322) - [internal] _rules_go

```
def _rules_go():
    return [
        ("type", re.compile(r"^\s*type\s+([A-Z]\w*)\s+(struct|interface)", 1, lambda m: True),
        ("type", re.compile(r"^\s*type\s+([a-z]\w*)\s+(struct|interface)", 1, lambda m: False),
        ("function", re.compile(r"^\s*func\s+(\([^\)]*\))\s*?([A-Z]\w*)\s*\(", 2, lambda m: True),
        ("function", re.compile(r"^\s*func\s+(\([^\)]*\))\s*?([a-z]\w*)\s*\(", 2, lambda m: False),
    ]
```

doc.py (line 331) - [internal] _rules_rust

```
def _rules_rust():
    return [
        ("function", re.compile(r"^\s*pub\(\([^\)]*\)\)\s+fn\s*(\w+)"), 2, lambda m: True),
        ("struct", re.compile(r"^\s*pub\(\([^\)]*\)\)\s+struct\s*(\w+)"), 2, lambda m: True),
        ("enum", re.compile(r"^\s*pub\(\([^\)]*\)\)\s+enum\s*(\w+)"), 2, lambda m: True),
        ("trait", re.compile(r"^\s*pub\(\([^\)]*\)\)\s+trait\s*(\w+)"), 2, lambda m: True),
        ("function", re.compile(r"^\s*fn\s*(\w+)"), 1, lambda m: False),
        ("struct", re.compile(r"^\s*struct\s*(\w+)"), 1, lambda m: False),
    ]
```

```

doc.py (line 342) - [internal] _rules_c_cpp
def _rules_c_cpp():
    return [
        ("function", re.compile(r"^\\s*static\\s+[\\w:\\*\\<\\>\\s]+?\\s(\\w+)\\s*\\([\\^;{}]*\\)\\s*\\{\\}", 1, lambda m: F
        ("function", re.compile(r"^\\s*[\\w:\\*\\<\\>\\s]+?\\s(\\w+)\\s*\\([\\^;{}]*\\)\\s*\\{\\}", 1, lambda m: True),
        ("declaration", re.compile(r"^\\s*[\\w:\\*\\<\\>\\s]+?\\s(\\w+)\\s*\\([\\^;{}]*\\)\\s*;", 1, lambda m: True),
    ]

doc.py (line 350) - [internal] _rules_php
def _rules_php():
    return [
        ("method", re.compile(r"^\\s*public\\s+(static\\s+)?function\\s+(\\w+)\"", 2, lambda m: True),
        ("method", re.compile(r"^\\s*(private|protected)\\s+(static\\s+)?function\\s+(\\w+)\"", 3, lambda m: Fa
        ("function", re.compile(r"^\\s*function\\s+(\\w+)\"", 1, lambda m: True),
        ("class", re.compile(r"^\\s*class\\s+(\\w+)\"", 1, lambda m: True),
    ]

doc.py (line 359) - [internal] _rules_ruby
def _rules_ruby():
    return [
        ("method", re.compile(r"^\\s*def\\s+(self\\.)?(\\w+[\\?!]?\"", 2, None), # visibility resolved contex
        ("class", re.compile(r"^\\s*class\\s+(\\w+)\"", 1, lambda m: True),
        ("module", re.compile(r"^\\s*module\\s+(\\w+)\"", 1, lambda m: True),
    ]

doc.py (line 367) - [internal] _rules_kotlin_swift
def _rules_kotlin_swift():
    return [
        ("function", re.compile(r"^\\s*(public|open)\\s+fun\\s+(\\w+)\"", 2, lambda m: True),
        ("function", re.compile(r"^\\s*(private|internal|fileprivate)\\s+fun\\s+(\\w+)\"", 2, lambda m: False),
        ("function", re.compile(r"^\\s*fun\\s+(\\w+)\"", 1, lambda m: True),
        ("class", re.compile(r"^\\s*(public|open)\\s+class\\s+(\\w+)\"", 2, lambda m: True),
        ("class", re.compile(r"^\\s*(private|internal|fileprivate)\\s+class\\s+(\\w+)\"", 2, lambda m: False),
        ("class", re.compile(r"^\\s*class\\s+(\\w+)\"", 1, lambda m: True),
    ]

doc.py (line 378) - [internal] _rules_d
def _rules_d():
    return [
        ("class", re.compile(r"^\\s*private\\s+(class|struct|interface|enum|union)\\s+(\\w+)\"", 2, lambda m:
        ("class", re.compile(r"^\\s*(class|struct|interface|enum|union)\\s+(\\w+)\"", 2, lambda m: True),
        ("function", re.compile(r"^\\s*private\\s+[\\w!\\[\\]\\*\\<\\>\\.\\s]+?\\s(\\w+)\\s*\\([\\^;{}]*\\)\\s*\\{\\}", 1, lambd
        ("function", re.compile(r"^\\s*[\\w!\\[\\]\\*\\<\\>\\.\\s]+?\\s(\\w+)\\s*\\([\\^;{}]*\\)\\s*\\{\\}", 1, lambda m: True),
    ]

doc.py (line 387) - [internal] _rules_nim
def _rules_nim():
    return [
        ("proc", re.compile(r"^\\s*(proc|func|template|macro|method|iterator)\\s+(\\w+)\\*\"", 2, lambda m: Tr
        ("proc", re.compile(r"^\\s*(proc|func|template|macro|method|iterator)\\s+(\\w+)\\b\"", 2, lambda m: Fa
        ("type", re.compile(r"^\\s*type\\s+(\\w+)\\*\"", 1, lambda m: True),
        ("type", re.compile(r"^\\s*type\\s+(\\w+)\\b\"", 1, lambda m: False),
    ]

doc.py (line 399) - [internal] _rules_odin
def _rules_odin():
    # visibility resolved contextually (an @(private) attribute on the line
    # above) - see parse_generic
    return [
        ("proc", re.compile(r"^\\s*(\\w+)\\s*:\\s*proc\\b\"", 1, None),
        ("struct", re.compile(r"^\\s*(\\w+)\\s*:\\s*struct\\b\"", 1, None),
        ("enum", re.compile(r"^\\s*(\\w+)\\s*:\\s*enum\\b\"", 1, None),
        ("union", re.compile(r"^\\s*(\\w+)\\s*:\\s*union\\b\"", 1, None),
    ]

```

doc.py (line 410) - [internal] _rules_ada_spec

```
def _rules_ada_spec():
    # in a .ads file (package specification): visibility depends on whether
    # the declaration comes before or after the "private" section
    return [
        ("package", re.compile(r"^s*package\s+(\w+)"), 1, lambda m: True),
        ("procedure", re.compile(r"^s*procedure\s+(\w+)"), 1, None),
        ("function", re.compile(r"^s*function\s+(\w+)"), 1, None),
    ]
```

doc.py (line 420) - [internal] _rules_ada_body

```
def _rules_ada_body():
    # in a .adb file (implementation) - treated as internal detail
    return [
        ("procedure", re.compile(r"^s*procedure\s+(\w+)"), 1, lambda m: False),
        ("function", re.compile(r"^s*function\s+(\w+)"), 1, lambda m: False),
    ]
```

doc.py (line 428) - [internal] _rules_asm

```
def _rules_asm():
    return [
        ("label", re.compile(r"^s*\.(global|globl)\s+(\w+)"), 2, lambda m: True),
        ("label", re.compile(r"^[A-Za-z_]\w*\s*:"), 1, lambda m: False),
    ]
```

doc.py (line 457) - [public] parse_generic

```
def parse_generic(rel_path: str, ext: str, source: str, max_body_lines: int) -> list[Definition]:
    rules = RULES_BY_EXT.get(ext)
    if not rules:
        return []
    lines = source.splitlines()
    lang = LANG_BY_EXT.get(ext, ext)
    defs: list[Definition] = []

    ruby_private_mode = False # .rb only
    ada_private_mode = False # .ads only

    for idx, raw_line in enumerate(lines):
        # ... (truncated, see source file for more)
```

doc.py (line 526) - [public] collect

```
# ----- #
# Collecting everything
# ----- #

def collect(root: str, exts: set[str] | None, exclude_dirs: set[str], max_body_lines: int):
    all_defs: list[Definition] = []
    md_files: list[MarkdownFile] = []

    for rel_path, full_path, ext in iter_source_files(root, exts, exclude_dirs):
        source = read_text(full_path)
        if ext in MD_EXTS:
            md_files.append(MarkdownFile(path=rel_path.replace(os.sep, "/"), content=source))
            continue
        rel_norm = rel_path.replace(os.sep, "/")
        if ext == ".py":
            all_defs.extend(parse_python(rel_norm, full_path, source))
        # ... (truncated, see source file for more)
```

doc.py (line 550) - [public] build_styles

```
# ----- #
# Building the PDF
# ----- #

def build_styles():
    styles = getSampleStyleSheet()
    styles.add(ParagraphStyle(
        name="SectionTitle", parent=styles["Heading1"],
        fontSize=18, spaceAfter=14, spaceBefore=6, textColor=colors.HexColor("#1a1a2e"),
    ))
    styles.add(ParagraphStyle(
        name="LangHeading", parent=styles["Heading2"],
        fontSize=13, spaceBefore=14, spaceAfter=6, textColor=colors.HexColor("#16213e"),
    ))
    styles.add(ParagraphStyle(
        name="ItemPath", parent=styles["Normal"],
        # ... (truncated, see source file for more)
```

doc.py (line 587) - [public] group_by_language

```
def group_by_language(defs: list[Definition]):
    groups: dict[str, list[Definition]] = {}
    for d in defs:
        groups.setdefault(d.language, []).append(d)
    return dict(sorted(groups.items()))
```

doc.py (line 594) - [public] path_line_label

```
def path_line_label(d: Definition) -> str:
    return f"{d.path} (line {d.lineno})"
```

doc.py (line 598) - [public] render_signature_section

```
def render_signature_section(story, styles, title: str, defs: list[Definition], max_items: int):
    story.append(Paragraph(escape(title), styles["SectionTitle"]))
    if not defs:
        story.append(Paragraph("No items in this section.", styles["Normal"]))
        return
    truncated = len(defs) > max_items
    defs = defs[:max_items]
    for lang, items in group_by_language(defs).items():
        story.append(Paragraph(escape(lang), styles["LangHeading"]))
        for d in items:
            block = [
                Paragraph(escape(path_line_label(d)), styles["ItemPath"]),
            ]
        # ... (truncated, see source file for more)
```

doc.py (line 622) - [public] render_fragments_section

```
def render_fragments_section(story, styles, defs: list[Definition], max_items: int):
    story.append(Paragraph("3. Code fragments with comments", styles["SectionTitle"]))
    if not defs:
        story.append(Paragraph("No items.", styles["Normal"]))
        return
    truncated = len(defs) > max_items
    defs = defs[:max_items]
    for lang, items in group_by_language(defs).items():
        story.append(Paragraph(escape(lang), styles["LangHeading"]))
        for d in items:
            block = [Paragraph(escape(path_line_label(d)) + f" - [{ 'public' if d.is_public else 'internal' }]", styles["ItemPath"]),
                Paragraph(escape(d.comment), styles["Text"]),
            ]
        # ... (truncated, see source file for more)
```

doc.py (line 647) - [public] render_markdown_section

```
def render_markdown_section(story, styles, md_files: list[MarkdownFile]):
    story.append(Paragraph("4. .md files (in order)", styles["SectionTitle"]))
    if not md_files:
        story.append(Paragraph("No .md files in the project.", styles["Normal"]))
        return
    for mf in md_files:
        story.append(Paragraph(escape(mf.path), styles["MdHeading"]))
        content = mf.content
        if len(content) > 20000:
            content = content[:20000] + "\n\n... (truncated, file is very long)"
        story.append(Preformatted(content, styles["CodeBlock"]))
```

doc.py (line 660) - [internal] _new_doc

```
def _new_doc(path: str):
    return SimpleDocTemplate(
        path, pagesize=A4,
        leftMargin=18 * mm, rightMargin=18 * mm, topMargin=16 * mm, bottomMargin=16 * mm,
    )
```

doc.py (line 667) - [internal] _summary_paragraph

```
def _summary_paragraph(styles, root, public_defs, internal_defs, md_files):
    block = [
        Paragraph("Project technical documentation", styles["Title"]),
        Paragraph(escape(f"Source directory: {os.path.abspath(root)}"), styles["Normal"]),
        Spacer(1, 4),
        Paragraph(
            f"Public API/ABI: {len(public_defs)} | Internal: {len(internal_defs)} | "
            f".md files: {len(md_files)}", styles["Normal"],
        ),
    ]
    return block
```

doc.py (line 680) - [public] build_pdf

```
def build_pdf(output_path: str, root: str, public_defs, internal_defs, all_defs_for_fragments,
              md_files, max_items: int, split: bool):
    styles = build_styles()

    if not split:
        doc = _new_doc(output_path)
        story = []
        story += _summary_paragraph(styles, root, public_defs, internal_defs, md_files)
        story.append(PageBreak())
        render_signature_section(story, styles, "1. Public API / ABI", public_defs, max_items)
        story.append(PageBreak())
        render_signature_section(story, styles, "2. Internal functions", internal_defs, max_items)
        # ... (truncated, see source file for more)
```

doc.py (line 734) - [public] main

```
# -----#
# main
# -----#
```

```
def main():
    parser = argparse.ArgumentParser(description="PDF documentation generator for a whole repo (no compilation)")
    parser.add_argument("root", help="Root directory of the project to scan")
    parser.add_argument("-o", "--output", default="documentation.pdf", help="Output PDF file")
    parser.add_argument("--max-body-lines", type=int, default=12,
                        help="How many lines of code to show in the fragments section (default: 12)")
    parser.add_argument("--max-items-per-section", type=int, default=4000,
                        help="Cap on entries per section (default: 4000)")
    parser.add_argument("--exclude", default="", help="Extra directories to skip, comma-separated")
    parser.add_argument("--ext", default="", help="Restrict to these extensions only, e.g.: .py,.js,.go")
    parser.add_argument("--split", action="store_true",
                        help="Instead of one PDF, generate a separate file per section ")
    # ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/tools/pci_ids.py (line 63) - [internal]
_pascal_tokens

PascalCase a list of tokens. Runs of digits stay as-is; letter groups get their leading character uppercased.

```
def _pascal_tokens(tokens: list[str], max_tokens: int | None = None) -> str:
    """PascalCase a list of tokens. Runs of digits stay as-is; letter
    groups get their leading character uppercased."""
    if max_tokens is not None:
        tokens = tokens[:max_tokens]
    parts: list[str] = []
    for t in tokens:
        if not t:
            continue
        if t[0].isdigit():
            parts.append(t.lower())
        else:
            # ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/tools/pci_ids.py (line 80) - [public]
sanitize_vendor_name

Turn a pci.ids vendor description into a compact PascalCase identifier.

Heuristics (in priority order):

- 1. If a trailing `[ALIAS]` bracketed tag exists, use its contents — vendors commonly use that for short forms (e.g. AMD, AMI).*
- 2. Strip corporate suffixes ("Corporation", " Inc.", etc.).*
- 3. Pascal-case the remaining tokens.*

Falls back to "Unknown" for empty strings (shouldn't happen on real pci.ids entries but keeps the pipeline total).

```
def sanitize_vendor_name(raw: str) -> str:
    """Turn a pci.ids vendor description into a compact PascalCase
    identifier.

    Heuristics (in priority order):
    1. If a trailing `[ALIAS]` bracketed tag exists, use its contents
       – vendors commonly use that for short forms (e.g. AMD, AMI).
    2. Strip corporate suffixes ("Corporation", " Inc.", etc.).
    3. Pascal-case the remaining tokens.

    Falls back to "Unknown" for empty strings (shouldn't happen on
    real pci.ids entries but keeps the pipeline total).
    # ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-Trangorge0S/tools/pci_ids.py (line 111) - [public]

Turn a pci.ids device description into a compact PascalCase identifier.

Heuristics:

1. If the description contains a `[CODE]` bracketed tag, use just the tag (model codes like "AD102" or "GeForce RTX 4090").
2. Otherwise keep the first `max\_tokens` tokens.
3. Pascal-case; prefix a digit-leading result with `D`.

```
def sanitize_device_name(raw: str, max_tokens: int = 4) -> str:
    """Turn a pci.ids device description into a compact PascalCase
    identifier.
```

Heuristics:

1. If the description contains a `[CODE]` bracketed tag, use just the tag (model codes like "AD102" or "GeForce RTX 4090").
2. Otherwise keep the first `max\_tokens` tokens.
3. Pascal-case; prefix a digit-leading result with `D`.

|| || ||

```
s = raw.strip()
```

```
# ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/tools/pci_ids.py (line 140) - [public]

Class/subclass names are typically short English phrases — pascal the whole thing with no token cap.

```
def sanitize_class_name(raw: str) -> str:
    """Class/subclass names are typically short English phrases –
    pascal the whole thing with no token cap."""
    s = raw.strip()
    tokens = re.split(r"^[^A-Za-z0-9]+", s)
    tokens = [t for t in tokens if t]
    pascal = _pascal_tokens(tokens)
    if not pascal:
        return "Unknown"
    if pascal[0].isdigit():
        pascal = "C" + pascal
    return pascal
```

driverspace_workspace/drivers/iommu-TrangorgeOS/tools/pci_ids.py (line 157) - [public]

```
# Fetch + parse pci.ids
```

```
def fetch_pci_ids(cache: Path, refresh: bool) -> str:
    if cache.exists() and not refresh:
        return cache.read_text(encoding="utf-8", errors="replace")
    print(f"[pci_ids] fetching {UPSTREAM_URL}", file=sys.stderr)
    req = urllib.request.Request(UPSTREAM_URL, headers={"User-Agent": "kraph-pci-ids-generator"})
    with urllib.request.urlopen(req, timeout=30) as r:
        data = r.read().decode("utf-8", errors="replace")
    cache.write_text(data, encoding="utf-8")
    return data
```

driverspace_workspace/drivers/iommu-TrangorgeOS/tools/pci_ids.py (line 168) - [public] PciIds
Parsed pci.ids tree.

Attributes:

vendors: {vid (u16): pretty name}
devices: {vid: {pid (u16): pretty name}}
classes: {cid (u8): pretty name}
subclasses: {cid: {scid (u8): pretty name}}
progifs: {(cid, scid): {piid (u8): pretty name}}

```
class PciIds:
    """Parsed pci.ids tree.

    Attributes:
        vendors:    {vid (u16): pretty name}
        devices:    {vid: {pid (u16): pretty name}}
        classes:    {cid (u8): pretty name}
        subclasses: {cid: {scid (u8): pretty name}}
        progifs:    {(cid, scid): {piid (u8): pretty name}}
    """

    def __init__(self) -> None:
        # ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/tools/pci_ids.py (line 179) - [internal]
PciIds.__init__

```
def __init__(self) -> None:
    self.vendors: dict[int, str] = {}
    self.devices: dict[int, dict[int, str]] = defaultdict(dict)
    self.classes: dict[int, str] = {}
    self.subclasses: dict[int, dict[int, str]] = defaultdict(dict)
    self.progifs: dict[tuple[int, int], dict[int, str]] = defaultdict(dict)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/tools/pci_ids.py (line 186) - [public]
PciIds.merge

*Merge `other` into `self` — later entries win, so the
overlay file's values override upstream where both define the
same ID.*

```
def merge(self, other: "PciIds") -> None:
    """Merge `other` into `self` — later entries win, so the  
overlay file's values override upstream where both define the  
same ID."""
    self.vendors.update(other.vendors)
    for vid, devs in other.devices.items():
        self.devices[vid].update(devs)
    self.classes.update(other.classes)
    for cid, subs in other.subclasses.items():
        self.subclasses[cid].update(subs)
    for key, pis in other.progifs.items():
        self.progifs[key].update(pis)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/tools/pci_ids.py (line 200) - [public]
parse_pci_ids

```
def parse_pci_ids(text: str) -> PciIds:
    db = PciIds()

    current_vendor: int | None = None
    current_device: int | None = None
    current_class: int | None = None
    current_subclass: int | None = None
    # Tracks whether the current indentation context is under a vendor
    # (so tab-2 means subsystem, ignored) or under a class (so tab-2
    # means progif, stored).
    mode: str | None = None # "vendor" | "class" | None

    # ... (truncated, see source file for more)
```


driverspace_workspace/drivers/iommu-TrangorgeOS/tools/pci_ids.py (line 284) - [public]
dedup_by_value

*Split `items` into canonical (first-seen) and aliases for
duplicate IDs. Matches the old generator's dedup behaviour.*

```
def dedup_by_value(
    items: list[tuple[str, int]],
) -> tuple[list[tuple[str, int]], dict[int, list[str]]]:
    """Split `items` into canonical (first-seen) and aliases for
    duplicate IDs. Matches the old generator's dedup behaviour."""
    seen: dict[int, str] = {}
    canon: list[tuple[str, int]] = []
    aliases: dict[int, list[str]] = defaultdict(list)
    for name, val in items:
        if val not in seen:
            seen[val] = name
            canon.append((name, val))
    # ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/tools/pci_ids.py (line 301) - [public]
resolve_name_collisions

```
def resolve_name_collisions(
    variants: list[tuple[str, int]],
    repr_ty: str,
) -> list[tuple[str, int]]:
    digits = 2 if repr_ty == "u8" else 4
    name_counts: dict[str, int] = defaultdict(int)
    for pname, _ in variants:
        name_counts[pname] += 1
    colliding = {n for n, c in name_counts.items() if c > 1}
    if not colliding:
        return variants
    return [
    # ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/tools/pci_ids.py (line 318) - [public]
dedup_name_collisions_across_enums

Ensure none of `variants` collide with any name in `reserved`.

```
def dedup_name_collisions_across_enums(
    variants: list[tuple[str, int]],
    reserved: set[str],
    repr_ty: str,
) -> list[tuple[str, int]]:
    """Ensure none of `variants` collide with any name in `reserved`."""
    digits = 2 if repr_ty == "u8" else 4
    out: list[tuple[str, int]] = []
    for pname, val in variants:
        safe_name = pname
        if pname in reserved:
            safe_name = f"{pname}X{val:0{digits}x}"
    # ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/tools/pci_ids.py (line 351) - [public]
emit_open_enum

```
def emit_open_enum(
    name: str,
    repr_ty: str,
    variants: list[tuple[str, int]],
    const_aliases: list[tuple[str, str]],
    *,
    doc: str = "",
) -> str:
    digits = 2 if repr_ty == "u8" else 4

    def h(v: int) -> str:
        return f"0x{v:0{digits}x}"
    # ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-Trangorge0S/tools/pci_ids.py (line 435) - [public]

```
emit_pci_device_enum
```

```
def emit_pci_device_enum(
    vendor_variants: list[tuple[str, int]],
    device_enum_names: list[str],
) -> str:
    lines: list[str] = []

    vendor_to_device_enum: dict[str, str] = {}
    for ename in device_enum_names:
        vendor_to_device_enum[ename.removesuffix("Device")] = ename

    lines.append("/// Unified PCI device identity, resolved from `(Vendor, device_id)`.")
    lines.append("#[derive(Debug, Clone, Copy, PartialEq, Eq)]")
    # ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-Trangorge0S/tools/pci_ids.py (line 503) - [public]

emit_subclass_enum

```
def emit_subclass_enum(
    class_variants: list[tuple[str, int]], sub_enum_names: list[str]
) -> str:
    lines: list[str] = []

    class_to_sub: dict[str, str] = {}
    for ename in sub_enum_names:
        class_to_sub[ename.removesuffix("Subclass")] = ename

    lines.append("/// Unified PCI subclass, resolved from `(Class, subclass_id)`.")
    lines.append("#[derive(Debug, Clone, Copy, PartialEq, Eq)]")
    lines.append("pub enum Subclass {")
    # ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-Trangorge0S/tools/pci_ids.py (line 569) - [public]

emit_progif_enum

```
def emit_progif_enum(progif_info: list[tuple[str, str, str]]) -> str:
    lines: list[str] = []

    lines.append("/// Unified PCI programming interface, resolved from `(Subclass, prog_if_id)`.")
    lines.append("#[derive(Debug, Clone, Copy, PartialEq, Eq)]")
    lines.append("pub enum ProgIf {")
    for class_p, sub_p, ename in progif_info:
        variant = f"{class_p}{sub_p}"
        lines.append(f"    {variant}({ename}),")
    lines.append("    /// Unrecognized or no programming interface defined.")
    lines.append("    Unknown(u8),")
    lines.append("}")
    # ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/tools/pci_ids.py (line 628) - [public]

```
build variants from db
```

Run the full emit pipeline against a parsed database.

Returns the full `bdf.rs` source text.

[illegible]

driverspace_workspace/drivers/iommu-TrangorgeOS/tools/pci_ids.py (line 801) - [public] main

```
def main() -> None:
    ap = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.RawDescriptionHelpFormatter)
    ap.add_argument(
        "--source",
        type=Path,
        help="Local pci.ids file to use instead of fetching upstream.",
    )
    ap.add_argument(
        "--overlay",
        type=Path,
        help="Extra pci.ids-format file whose entries overlay upstream (later-wins).",
    )
    # ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/tools/ra-target.py (line 30) - [public]
check_command

Build the cargo check command RA should use for a kernel target.

When targeting a freestanding triple, only the root `iommu` package can compile — workspace members like kspec/kmain depend on std. Limiting the check to `-p iommu` prevents RA from failing on those crates.

```
def check_command(target: str) -> list[str]:
    """Build the cargo check command RA should use for a kernel target.

    When targeting a freestanding triple, only the root `iommu` package can
    compile — workspace members like kspec/kmain depend on std. Limiting
    the check to `-p iommu` prevents RA from failing on those crates.
    """
    return [
        "cargo",
        "check",
        "-p",
        "iommu",
    ]
    # ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/tools/ra-target.py (line 48) - [public] main

```
def main():
    project_root = Path(__file__).resolve().parent.parent
    settings_path = project_root / ".vscode" / "settings.json"

    if not settings_path.exists():
        settings_path.parent.mkdir(parents=True, exist_ok=True)
        settings = {}
    else:
        settings = json.loads(settings_path.read_text())

    if len(sys.argv) < 2:
        current = settings.get(RA_TARGET_KEY)
    # ... (truncated, see source file for more)
```

Rust

c.rs (line 57) - [internal] BlockHeader

```
/* Struktura nagłówka bloku pamięci */
typedef struct BlockHeader {
    size_t size; /* Rozmiar danych użytkownika */
    int is_free; /* Czy blok jest wolny */

    struct BlockHeader* next; /* Wskaźnik do następnego bloku */
    struct BlockHeader* prev; /* Wskaźnik do poprzedniego bloku */
} BlockHeader;

/* Globalna pula pamięci i lista bloków */
static uint8_t memory_pool[MEMORY_POOL_SIZE];
static BlockHeader* free_list = NULL;
static int allocator_initialized = 0;

/* Makro do wyrównywania rozmiaru */
#define ALIGN(size) (((size) + (ALIGNMENT - 1)) & ~(ALIGNMENT - 1))

// ... (truncated, see source file for more)
```

c.rs (line 58) - [internal] BlockHeader

```
/* Struktura nagłówka bloku pamięci */
typedef struct BlockHeader {
    size_t size; /* Rozmiar danych użytkownika */
    int is_free; /* Czy blok jest wolny */
    struct BlockHeader* next; /* Wskaźnik do następnego bloku */

    struct BlockHeader* prev; /* Wskaźnik do poprzedniego bloku */
} BlockHeader;

/* Globalna pula pamięci i lista bloków */
static uint8_t memory_pool[MEMORY_POOL_SIZE];
static BlockHeader* free_list = NULL;
static int allocator_initialized = 0;

/* Makro do wyrównywania rozmiaru */
#define ALIGN(size) (((size) + (ALIGNMENT - 1)) & ~(ALIGNMENT - 1))

/* Inicjalizacja alokatora */
// ... (truncated, see source file for more)
```

comgrub/build.rs (line 1) - [internal] main

```
fn main() {
    nasm_rs::compile_library("boot.asm", &["src/boot.asm"]).unwrap();

    // Ścieżka bezwzględna: `cargo:rustc-link-arg` przekazuje argument do linkera
    // w katalogu docelowym, więc relatywna ścieżka by się nie rozwinęła.
    let manifest = std::env::var("CARGO_MANIFEST_DIR").unwrap();

    println!("cargo:rustc-link-arg=-T{/linker.ld", manifest);
    println!("cargo:rustc-link-lib=static=boot");

    println!("cargo:rerun-if-changed=src/boot.asm");
    println!("cargo:rerun-if-changed=linker.ld");
    // ... (truncated, see source file for more)
```

comgrub/src/main.rs (line 11) - [internal] kernel_main

```
fn kernel_main(magic: u32, info: *const u8) -> !;
}

#[panic_handler]
fn panic(_info: &PanicInfo) -> ! {
    loop {
        unsafe { core::arch::asm!("hlt", options(nomem, nostack, preserves_flags)) }
    }
}

/// Punkt wejścia Rust wywołany z `boot.asm` (`call _rust_entry`).
#[unsafe(no_mangle)]
// ... (truncated, see source file for more)
```

```
comgrub/src/main.rs (line 15) - [internal] panic
    fn panic(_info: &PanicInfo) -> ! {
        loop {
            unsafe { core::arch::asm!("hlt", options(nomem, nostack, preserves_flags)) }
        }
    }

    /// Punkt wejścia Rust wywołany z `boot.asm` (`call _rust_entry`).
    #[unsafe(no_mangle)]
    pub extern "C" fn _rust_entry(magic: u32, info: *const u8) -> ! {
        unsafe { kernel_main(magic, info) }
    }
}
```

```
comlimine/build.rs (line 1) - [internal] main
fn main() {
    cc::Build::new()
        .file("src/main.c")
        .include("src")
        .no_default_flags(true)
        .flag("-ffreestanding")
        .flag("-nostdlib")
        .flag("-mno-red-zone")
        .flag("-mno-mmx")
        .flag("-mno-sse")
        .flag("-mno-sse2")
        .compile("limine_entry");
    // ... (truncated, see source file for more)
}
```

```
comlimine/src/main.rs (line 1) - [internal] main
fn main() {}
```

```
drivers/amdgpu_driver/src/main.rs (line 1) - [internal] main
fn main() {
    println!("Hello, world!");
}
```

```
drivers/audiodriver/build.rs (line 12) - [internal] main
fn main() {
    let manifest = PathBuf::from(env::var("CARGO_MANIFEST_DIR").unwrap());
    let out = PathBuf::from(env::var("OUT_DIR").unwrap());

    let odin_dir = manifest.join("src/odin");
    let lib_name = "audiodriver_odin";
    let lib_path = out.join(format!("lib{}.a", lib_name));

    println!("cargo:rerun-if-changed=build.rs");
    println!("cargo:rerun-if-changed={}", odin_dir.display());

    let odin = env::var("ODIN").unwrap_or_else(|_| "odin".to_string());
    // ... (truncated, see source file for more)
}
```

```
drivers/audiodriver/src/jacklib.rs (line 4) - [public] JackMgr
pub struct JackMgr {
    present: bool,
    amp_on: bool,
}

impl JackMgr {
    pub const fn new() -> Self {
        Self { present: false, amp_on: false }
    }

    pub fn tick(&mut self) {
        let now = ad::jack_present();
        // ... (truncated, see source file for more)
    }
}
```

drivers/audiodriver/src/jacklib.rs (line 14) - [public] tick

```
pub fn tick(&mut self) {
    let now = ad::jack_present();

    if now != self.present {
        self.present = now;

        if now {
            ds::log::ds_log("jack: present");
            self.set_amp(true);
        } else {
            ds::log::ds_log("jack: gone");
            self.set_amp(false);
        }
    }
    // ... (truncated, see source file for more)
```

drivers/audiodriver/src/jacklib.rs (line 30) - [public] set_amp

```
pub fn set_amp(&mut self, on: bool) {
    self.amp_on = on;
    ad::set_amp(on);
}

pub fn present(&self) -> bool {
    self.present
}

/// Czy wzmacniacz (amp) jest aktualnie wlaczony.
pub fn amp_enabled(&self) -> bool {
    self.amp_on
}
// ... (truncated, see source file for more)
```

drivers/audiodriver/src/jacklib.rs (line 35) - [public] present

```
pub fn present(&self) -> bool {
    self.present
}

/// Czy wzmacniacz (amp) jest aktualnie wlaczony.
pub fn amp_enabled(&self) -> bool {
    self.amp_on
}
}
```

drivers/audiodriver/src/jacklib.rs (line 40) - [public] amp_enabled

```
/// Czy wzmacniacz (amp) jest aktualnie wlaczony.
pub fn amp_enabled(&self) -> bool {
    self.amp_on
}
}
```

drivers/audiodriver/src/lib.rs (line 7) - [internal] ad_init

```
fn ad_init(nam_va: u64, bm_va: u64) -> i32;
fn ad_play(data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) -> i32;
fn ad_capture(data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) -> i32;
fn ad_stop() -> i32;
fn ad_jack_present() -> i32;
fn ad_set_amp(on: i32) -> i32;
fn ad_position() -> u32;
}

pub fn init(nam_va: u64, bm_va: u64) -> bool {
    unsafe { ad_init(nam_va, bm_va) == 0 }
}

// ... (truncated, see source file for more)
```

drivers/audiodriver/src/lib.rs (line 8) - [internal] ad_play

```
fn ad_play(data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) -> i32;
fn ad_capture(data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) -> i32;
fn ad_stop() -> i32;
fn ad_jack_present() -> i32;
fn ad_set_amp(on: i32) -> i32;
fn ad_position() -> u32;
}

pub fn init(nam_va: u64, bm_va: u64) -> bool {
    unsafe { ad_init(nam_va, bm_va) == 0 }
}

// ... (truncated, see source file for more)
```

drivers/audiodriver/src/lib.rs (line 9) - [internal] ad_capture

```
fn ad_capture(data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) -> i32;
fn ad_stop() -> i32;
fn ad_jack_present() -> i32;
fn ad_set_amp(on: i32) -> i32;
fn ad_position() -> u32;
}

pub fn init(nam_va: u64, bm_va: u64) -> bool {
    unsafe { ad_init(nam_va, bm_va) == 0 }
}

pub fn play(data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) -> bool {
    // ... (truncated, see source file for more)
```

drivers/audiodriver/src/lib.rs (line 10) - [internal] ad_stop

```
fn ad_stop() -> i32;
fn ad_jack_present() -> i32;
fn ad_set_amp(on: i32) -> i32;
fn ad_position() -> u32;
}

pub fn init(nam_va: u64, bm_va: u64) -> bool {
    unsafe { ad_init(nam_va, bm_va) == 0 }
}

pub fn play(data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) -> bool {
    unsafe { ad_play(data_phys, len, bdl_va, bdl_phys) == 0 }
    // ... (truncated, see source file for more)
```

drivers/audiodriver/src/lib.rs (line 11) - [internal] ad_jack_present

```
fn ad_jack_present() -> i32;
fn ad_set_amp(on: i32) -> i32;
fn ad_position() -> u32;
}

pub fn init(nam_va: u64, bm_va: u64) -> bool {
    unsafe { ad_init(nam_va, bm_va) == 0 }
}

pub fn play(data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) -> bool {
    unsafe { ad_play(data_phys, len, bdl_va, bdl_phys) == 0 }
}

// ... (truncated, see source file for more)
```

drivers/audiodriver/src/lib.rs (line 12) - [internal] ad_set_amp

```
fn ad_set_amp(on: i32) -> i32;
fn ad_position() -> u32;
}

pub fn init(nam_va: u64, bm_va: u64) -> bool {
  unsafe { ad_init(nam_va, bm_va) == 0 }
}

pub fn play(data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) -> bool {
  unsafe { ad_play(data_phys, len, bdl_va, bdl_phys) == 0 }
}

// ... (truncated, see source file for more)
```

drivers/audiodriver/src/lib.rs (line 13) - [internal] ad_position

```
fn ad_position() -> u32;
}

pub fn init(nam_va: u64, bm_va: u64) -> bool {
  unsafe { ad_init(nam_va, bm_va) == 0 }
}

pub fn play(data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) -> bool {
  unsafe { ad_play(data_phys, len, bdl_va, bdl_phys) == 0 }
}

pub fn capture(data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) -> bool {
  // ... (truncated, see source file for more)
```

drivers/audiodriver/src/lib.rs (line 16) - [public] init

```
pub fn init(nam_va: u64, bm_va: u64) -> bool {
  unsafe { ad_init(nam_va, bm_va) == 0 }
}

pub fn play(data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) -> bool {
  unsafe { ad_play(data_phys, len, bdl_va, bdl_phys) == 0 }
}

pub fn capture(data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) -> bool {
  unsafe { ad_capture(data_phys, len, bdl_va, bdl_phys) == 0 }
}

// ... (truncated, see source file for more)
```

drivers/audiodriver/src/lib.rs (line 20) - [public] play

```
pub fn play(data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) -> bool {
  unsafe { ad_play(data_phys, len, bdl_va, bdl_phys) == 0 }
}

pub fn capture(data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) -> bool {
  unsafe { ad_capture(data_phys, len, bdl_va, bdl_phys) == 0 }
}

pub fn stop() {
  unsafe { let _ = ad_stop(); }
}

// ... (truncated, see source file for more)
```



```
drivers/audiodriver/src/lib.rs (line 24) - [public] capture
pub fn capture(data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) -> bool {
    unsafe { ad_capture(data_phys, len, bdl_va, bdl_phys) == 0 }
}

pub fn stop() {
    unsafe { let _ = ad_stop(); }
}

pub fn jack_present() -> bool {
    unsafe { ad_jack_present() != 0 }
}

// ... (truncated, see source file for more)
```

```
drivers/audiodriver/src/lib.rs (line 28) - [public] stop
pub fn stop() {
    unsafe { let _ = ad_stop(); }
}

pub fn jack_present() -> bool {
    unsafe { ad_jack_present() != 0 }
}

pub fn set_amp(on: bool) {
    unsafe { let _ = ad_set_amp(on as i32); }
}

// ... (truncated, see source file for more)
```

```
drivers/audiodriver/src/lib.rs (line 32) - [public] jack_present
pub fn jack_present() -> bool {
    unsafe { ad_jack_present() != 0 }
}

pub fn set_amp(on: bool) {
    unsafe { let _ = ad_set_amp(on as i32); }
}

pub fn position() -> u32 {
    unsafe { ad_position() }
}
```

```
drivers/audiodriver/src/lib.rs (line 36) - [public] set_amp
pub fn set_amp(on: bool) {
    unsafe { let _ = ad_set_amp(on as i32); }
}

pub fn position() -> u32 {
    unsafe { ad_position() }
}
```

```
drivers/audiodriver/src/lib.rs (line 40) - [public] position
pub fn position() -> u32 {
    unsafe { ad_position() }
}
```

drivers/audiodriver/src/main.rs (line 8) - [internal] main

```
/// Punkt wejścia binarki pomocniczej.
///
/// Właściwy sterownik działa w driver space przez `ds_entry` (patrz
/// `driverspace/src/main.rs`); ta binarka jest ciekim wrapperem CLI, który
/// pozwala sprawdzić stan gniazda jack z poziomu hosta.
```

```
fn main() {
    let mut mgr = ad::jacklib::JackMgr::new();
    mgr.tick();

    println!(
        "audiodriver: jack present = {}, amp on = {}",
        mgr.present(),
        mgr.amp_enabled()
    );
}
```

drivers/audiodriver/src/tone.rs (line 1) - [public] fill_square

```
pub fn fill_square(buf: &mut [u8], periods: u32) {
    let half = buf.len() as u32 / (periods * 2);

    for i in 0..buf.len() as u32 {
        let v = if (i / half.max(1)) % 2 == 0 { 0xC0 } else { 0x40 };
        buf[i as usize] = v;
    }
}
```

drivers/intelgpu_driver/src/main.rs (line 1) - [internal] main

```
fn main() {
    println!("Hello, world!");
}
```

drivers/netcam_driver/src/main.rs (line 1) - [internal] main

```
fn main() {
    println!("Hello, world!");
}
```

drivers/wacomgraphic_driver/src/main.rs (line 1) - [internal] main

```
fn main() {
    println!("Hello, world!");
}
```

driverspace/src/drivers/storage.rs (line 9) - [public] StorageDrv

```
/// Storage driver stub. It only satisfies the driver registry so that
/// driver space can boot; a real disk driver can slot in later.
```

```
pub struct StorageDrv {
    registered: bool,
}

impl StorageDrv {
    pub const fn new() -> Self {
        Self { registered: false }
    }
}

impl Driver for StorageDrv {
    fn init(&mut self, _info: &DeviceInfo) -> Result<(), DsError> {
        // ... (truncated, see source file for more)
    }
}
```

driverspace/src/drivers/storage.rs (line 20) - [internal] init

```
fn init(&mut self, _info: &DeviceInfo) -> Result<(), DsError> {
    self.registered = true;
    Ok(())
}
```

driverspace/src/main.rs (line 32) - [internal] panic

```
fn panic(_info: &core::panic::PanicInfo) -> ! {  
    loop {}  
}
```

driverspace_workspace/crates/ds-manager/src/main.rs (line 1) - [internal] main

```
fn main() {  
    println!("Hello, world!");  
}
```

driverspace_workspace/crates/ds-registry/src/database.rs (line 4) - [public] RegistryEntry

```
pub struct RegistryEntry {  
    pub vendor_id: u16,  
    pub device_id: u16,  
    pub driver_id: DriverId,  
    pub driver_name: &'static str,  
}  
  
pub struct Registry {  
    entries: &'static [RegistryEntry],  
}  
  
impl Registry {  
    // ... (truncated, see source file for more)
```

driverspace_workspace/crates/ds-registry/src/database.rs (line 11) - [public] Registry

```
pub struct Registry {  
    entries: &'static [RegistryEntry],  
}  
  
impl Registry {  
    pub const fn new(entries: &'static [RegistryEntry]) -> Self {  
        Self { entries }  
    }  
  
    pub fn entries(&self) -> &[RegistryEntry] {  
        self.entries  
    }  
    // ... (truncated, see source file for more)
```

driverspace_workspace/crates/ds-registry/src/database.rs (line 20) - [public] entries

```
pub fn entries(&self) -> &[RegistryEntry] {  
    self.entries  
}  
  
pub fn find(&self, vendor_id: u16, device_id: u16) -> Option<&RegistryEntry> {  
    self.entries.iter().find(|e| e.vendor_id == vendor_id && e.device_id == device_id)  
}  
  
pub static DEFAULT_REGISTRY: Registry = Registry::new(&[  
    RegistryEntry {  
        vendor_id: 0x1234,  
        // ... (truncated, see source file for more)
```

driverspace_workspace/crates/ds-registry/src/database.rs (line 24) - [public] find

```
pub fn find(&self, vendor_id: u16, device_id: u16) -> Option<&RegistryEntry> {  
    self.entries.iter().find(|e| e.vendor_id == vendor_id && e.device_id == device_id)  
}  
  
pub static DEFAULT_REGISTRY: Registry = Registry::new(&[  
    RegistryEntry {  
        vendor_id: 0x1234,  
        device_id: 0x1111,  
        driver_id: 1,  
        driver_name: "qemu-vga",  
    },  
    // ... (truncated, see source file for more)
```

driverspace_workspace/crates/ds-registry/src/matching.rs (line 5) - [public]

find_driver_for_device

```
pub fn find_driver_for_device(vendor_id: u16, device_id: u16) -> Option<DriverId> {
    DEFAULT_REGISTRY.find(vendor_id, device_id).map(|entry| entry.driver_id)
}

/// Zwraca nazw[?] sterownika dla danego ID.
pub fn get_driver_name(driver_id: DriverId) -> Option<&'static str> {
    DEFAULT_REGISTRY.entries().iter()
        .find(|e| e.driver_id == driver_id)
        .map(|e| e.driver_name)
}
```

driverspace_workspace/crates/ds-registry/src/matching.rs (line 10) - [public] get_driver_name

/// Zwraca nazw[?] sterownika dla danego ID.

```
pub fn get_driver_name(driver_id: DriverId) -> Option<&'static str> {
    DEFAULT_REGISTRY.entries().iter()
        .find(|e| e.driver_id == driver_id)
        .map(|e| e.driver_name)
}
```

driverspace_workspace/crates/ds-registry/src/state.rs (line 5) - [public] DeviceStatus

```
pub enum DeviceStatus {
    /// Urz[?]dzenie wykryte, ale sterownik jeszcze nie za[?]adowany.
    Discovered,
    /// Sterownik za[?]adowany i zainicjalizowany.
    Attached,
    /// Urz[?]dzenie od[?]czone (hotunplug lub b[?]ad).
    Detached,
    /// Wyst[?]pi[?] b[?]ad krytyczny (sterownik si[?] wysypa[?], sprz[?]t nie odpowiada).
    Error,
}

/// Struktura przechowuj[?]ca stan konkretnego urz[?]dzenia.
// ... (truncated, see source file for more)
```

driverspace_workspace/crates/ds-registry/src/state.rs (line 17) - [public] DeviceState

/// Struktura przechowuj[?]ca stan konkretnego urz[?]dzenia.

```
pub struct DeviceState {
    pub vendor_id: u16,
    pub device_id: u16,
    pub status: DeviceStatus,
    pub driver_id: Option<u32>, // ID sterownika, który obs[?]uguje to urz[?]dzenie
}

impl DeviceState {
    pub const fn new(vendor_id: u16, device_id: u16) -> Self {
        Self {
            vendor_id,
            device_id,
        }
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/crates/ds-registry/src/state.rs (line 34) - [public] set_attached

```
pub fn set_attached(&mut self, driver_id: u32) {
    self.status = DeviceStatus::Attached;
    self.driver_id = Some(driver_id);
}

pub fn set_detached(&mut self) {
    self.status = DeviceStatus::Detached;
    self.driver_id = None;
}

pub fn set_error(&mut self) {
    self.status = DeviceStatus::Error;
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/crates/ds-registry/src/state.rs (line 39) - [public] set_detached

```
pub fn set_detached(&mut self) {
    self.status = DeviceStatus::Detached;
    self.driver_id = None;
}

pub fn set_error(&mut self) {
    self.status = DeviceStatus::Error;
}
}
```

driverspace_workspace/crates/ds-registry/src/state.rs (line 44) - [public] set_error

```
pub fn set_error(&mut self) {
    self.status = DeviceStatus::Error;
}
}
```

driverspace_workspace/drivers/amdgpu-driver/src/main.rs (line 1) - [internal] main

```
fn main() {
    println!("Hello, world!");
}
```

driverspace_workspace/drivers/audiodriver/src/main.rs (line 1) - [internal] main

```
fn main() {
    println!("Hello, world!");
}
```

driverspace_workspace/drivers/intelgpu-driver/src/main.rs (line 1) - [internal] main

```
fn main() {
    println!("Hello, world!");
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 9) - [public] Unsigned

/// Unsigned integer types that can be used as the underlying representation
/// for MMIO and IOVA addresses.

```
pub trait Unsigned: Copy + Ord + Sized {
    const MAX: usize;

    fn checked_add(self, offset: usize) -> Option<Self>;
}
```

```
impl Unsigned for u32 {
    const MAX: usize = u32::MAX as usize;

    #[inline]
    fn checked_add(self, offset: usize) -> Option<Self> {
        self.checked_add(offset as u32)
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 12) - [internal] checked_add

```
fn checked_add(self, offset: usize) -> Option<Self>;
}

impl Unsigned for u32 {
    const MAX: usize = u32::MAX as usize;

    #[inline]
    fn checked_add(self, offset: usize) -> Option<Self> {
        self.checked_add(offset as u32)
    }
}

// ... (truncated, see source file for more)
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 19) - [internal] checked_add
    fn checked_add(self, offset: usize) -> Option<Self> {
        self.checked_add(offset as u32)
    }
}

impl Unsigned for u64 {
    const MAX: usize = u64::MAX as usize;

    #[inline]
    fn checked_add(self, offset: usize) -> Option<Self> {
        self.checked_add(offset as u64)
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 28) - [internal] checked_add
    fn checked_add(self, offset: usize) -> Option<Self> {
        self.checked_add(offset as u64)
    }
}

impl Unsigned for usize {
    const MAX: usize = usize::MAX;

    #[inline]
    fn checked_add(self, offset: usize) -> Option<Self> {
        self.checked_add(offset)
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 37) - [internal] checked_add
    fn checked_add(self, offset: usize) -> Option<Self> {
        self.checked_add(offset)
    }
}

#[repr(C)]
union Cast<T: Copy, U: Copy> {
    from: T,
    to: U,
}

/// Convert from an unsigned integer type to usize for address manipulation.
// ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 79) - [public] MmioSpace
pub enum MmioSpace {}

/// I/O virtual address-space marker.
#[derive(Copy, Clone, Ord, PartialOrd, Eq, PartialEq, Hash)]
pub enum IoviSpace {}

#[repr(transparent)]
#[derive(Copy, Clone, Ord, PartialOrd, Eq, PartialEq, Hash)]
pub struct IoAddr<T: Unsigned, Space>(T, PhantomData<fn() -> Space>);

impl<T: Unsigned, Space> Into<usize> for IoAddr<T, Space> {
    #[inline]
    // ... (truncated, see source file for more)

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 83) - [public] IoviSpace
    pub enum IoviSpace {}

    #[repr(transparent)]
    #[derive(Copy, Clone, Ord, PartialOrd, Eq, PartialEq, Hash)]
    pub struct IoAddr<T: Unsigned, Space>(T, PhantomData<fn() -> Space>);

    impl<T: Unsigned, Space> Into<usize> for IoAddr<T, Space> {
        #[inline]
        fn into(self) -> usize {
            into_usize(self.0)
        }
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 87) - [public] IoAddr
    pub struct IoAddr<T: Unsigned, Space>(T, PhantomData<fn() -> Space>);

    impl<T: Unsigned, Space> Into<usize> for IoAddr<T, Space> {
        #[inline]
        fn into(self) -> usize {
            into_usize(self.0)
        }
    }

    impl<T: Unsigned, Space> From<usize> for IoAddr<T, Space> {
        #[inline]
        fn from(value: usize) -> Self {
            // ... (truncated, see source file for more)
        }
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 91) - [internal] into
    fn into(self) -> usize {
        into_usize(self.0)
    }
}

impl<T: Unsigned, Space> From<usize> for IoAddr<T, Space> {
    #[inline]
    fn from(value: usize) -> Self {
        Self(from_usize(value), PhantomData)
    }
}

// ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 98) - [internal] from
    fn from(value: usize) -> Self {
        Self(from_usize(value), PhantomData)
    }
}

impl<T: Unsigned, Space> IoAddr<T, Space> {
    #[inline]
    pub fn as_usize(&self) -> usize {
        into_usize(self.0)
    }
}

#[inline]
// ... (truncated, see source file for more)

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 105) - [public] as_usize

```
pub fn as_usize(&self) -> usize {
    into_usize(self.0)
}

#[inline]
pub fn from_usize(addr: usize) -> Self {
    Self(from_usize(addr), PhantomData)
}

#[inline]
pub fn checked_mul(&self, factor: usize) -> Option<Self> {
    let current: usize = self.as_usize();
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 110) - [public] from_usize

```
pub fn from_usize(addr: usize) -> Self {
    Self(from_usize(addr), PhantomData)
}

#[inline]
pub fn checked_mul(&self, factor: usize) -> Option<Self> {
    let current: usize = self.as_usize();
    current.checked_mul(factor).map(Self::from_usize)
}

impl<T: Unsigned> IoAddr<T, MmioSpace> {
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 115) - [public] checked_mul

```
pub fn checked_mul(&self, factor: usize) -> Option<Self> {
    let current: usize = self.as_usize();
    current.checked_mul(factor).map(Self::from_usize)
}

impl<T: Unsigned> IoAddr<T, MmioSpace> {
    #[inline]
    pub const fn from_phys(addr: PhysAddr) -> Self {
        Self(from_usize(addr.as_usize()), PhantomData)
    }

    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 134) - [internal] fmt

```
fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
    write!(f, "IoAddr({:#x})", self.as_usize())
}

impl<T: Unsigned, Space> fmt::Display for IoAddr<T, Space> {
    fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
        write!(f, "{:#x}", self.as_usize())
    }
}

impl<T: Unsigned, Space> fmt::LowerHex for IoAddr<T, Space> {
    // ... (truncated, see source file for more)
```


driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 140) - [internal] fmt

```
    fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
        write!(f, "{:#x}", self.as_usize())
    }
}

impl<T: Unsigned, Space> fmt::LowerHex for IoAddr<T, Space> {
    fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
        fmt::LowerHex::fmt(&self.as_usize(), f)
    }
}

impl<T: Unsigned, Space> fmt::UpperHex for IoAddr<T, Space> {
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 146) - [internal] fmt

```
    fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
        fmt::LowerHex::fmt(&self.as_usize(), f)
    }
}

impl<T: Unsigned, Space> fmt::UpperHex for IoAddr<T, Space> {
    fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
        fmt::UpperHex::fmt(&self.as_usize(), f)
    }
}

/// Memory-mapped I/O address.
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 152) - [internal] fmt

```
    fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
        fmt::UpperHex::fmt(&self.as_usize(), f)
    }
}

/// Memory-mapped I/O address.
pub type MmioAddr<T = usize> = IoAddr<T, MmioSpace>;

/// I/O virtual address.
pub type IoviAddr<T = usize> = IoAddr<T, IoviSpace>;

/// Range of MMIO addresses. `end` is exclusive, matching the convention used
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 180) - [public] MmioRange

*/// Extension methods for MMIO address ranges, providing convenient methods for
/// constructing and accessing registers at fixed offsets from a base address.
/// These methods return `Option` to reflect the possibility of out-of-bounds
/// accesses when the offset is too large for the range.*

```
pub trait MmioRange<T: Unsigned>: Sized {
    fn from_start_size(start: MmioAddr<T>, size: usize) -> Option<Self>;
    fn from_phys_range(range: PhysAddrRange) -> Self;
    fn from_phys_start_size(start: PhysAddr, size: usize) -> Option<Self>;
    fn to_phys_range(self) -> PhysAddrRange;
    fn as_phys_range(self) -> PhysAddrRange {
        self.to_phys_range()
    }
    fn reg<const W: usize>(self, offset: usize) -> Option<MmioAddr<T>>;
    fn reg8(self, offset: usize) -> Option<MmioAddr<T>>;
    fn reg16(self, offset: usize) -> Option<MmioAddr<T>>;
    fn reg32(self, offset: usize) -> Option<MmioAddr<T>>;
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 181) - [internal]
from_start_size

```
fn from_start_size(start: MmioAddr<T>, size: usize) -> Option<Self>;
fn from_phys_range(range: PhysAddrRange) -> Self;
fn from_phys_start_size(start: PhysAddr, size: usize) -> Option<Self>;
fn to_phys_range(self) -> PhysAddrRange;
fn as_phys_range(self) -> PhysAddrRange {
    self.to_phys_range()
}
fn reg<const W: usize>(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg8(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg16(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg32(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg64(self, offset: usize) -> Option<MmioAddr<T>>;
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 182) - [internal]
from_phys_range

```
fn from_phys_range(range: PhysAddrRange) -> Self;
fn from_phys_start_size(start: PhysAddr, size: usize) -> Option<Self>;
fn to_phys_range(self) -> PhysAddrRange;
fn as_phys_range(self) -> PhysAddrRange {
    self.to_phys_range()
}
fn reg<const W: usize>(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg8(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg16(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg32(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg64(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg16_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 183) - [internal]
from_phys_start_size

```
fn from_phys_start_size(start: PhysAddr, size: usize) -> Option<Self>;
fn to_phys_range(self) -> PhysAddrRange;
fn as_phys_range(self) -> PhysAddrRange {
    self.to_phys_range()
}
fn reg<const W: usize>(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg8(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg16(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg32(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg64(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg16_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg32_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 184) - [internal]
to_phys_range

```
fn to_phys_range(self) -> PhysAddrRange;
fn as_phys_range(self) -> PhysAddrRange {
    self.to_phys_range()
}
fn reg<const W: usize>(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg8(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg16(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg32(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg64(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg16_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg32_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg64_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 185) - [internal]
as_phys_range

```
fn as_phys_range(self) -> PhysAddrRange {
    self.to_phys_range()
}
fn reg<const W: usize>(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg8(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg16(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg32(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg64(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg16_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg32_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg64_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
}
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 188) - [internal] reg

```
fn reg<const W: usize>(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg8(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg16(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg32(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg64(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg16_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg32_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg64_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
}
```

```
impl<T: Unsigned> MmioRange<T> for MmioAddrRange<T> {
    #[inline]
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 189) - [internal] reg8

```
fn reg8(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg16(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg32(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg64(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg16_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg32_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg64_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
}

impl<T: Unsigned> MmioRange<T> for MmioAddrRange<T> {
    #[inline]
    fn from_start_size(start: MmioAddr<T>, size: usize) -> Option<Self> {
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 190) - [internal] reg16

```
fn reg16(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg32(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg64(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg16_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg32_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg64_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
}

impl<T: Unsigned> MmioRange<T> for MmioAddrRange<T> {
    #[inline]
    fn from_start_size(start: MmioAddr<T>, size: usize) -> Option<Self> {
        Self::try_from_start_size(start, size)
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 191) - [internal] reg32

```
fn reg32(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg64(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg16_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg32_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg64_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
}

impl<T: Unsigned> MmioRange<T> for MmioAddrRange<T> {
    #[inline]
    fn from_start_size(start: MmioAddr<T>, size: usize) -> Option<Self> {
        Self::try_from_start_size(start, size)
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 192) - [internal] reg64

```
fn reg64(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg16_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg32_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg64_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
}

impl<T: Unsigned> MmioRange<T> for MmioAddrRange<T> {
    #[inline]
    fn from_start_size(start: MmioAddr<T>, size: usize) -> Option<Self> {
        Self::try_from_start_size(start, size)
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 193) - [internal]
reg16_aligned

```
fn reg16_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg32_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg64_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
}

impl<T: Unsigned> MmioRange<T> for MmioAddrRange<T> {
    #[inline]
    fn from_start_size(start: MmioAddr<T>, size: usize) -> Option<Self> {
        Self::try_from_start_size(start, size)
    }

    #[inline]
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 194) - [internal]
reg32_aligned

```
fn reg32_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
fn reg64_aligned(self, offset: usize) -> Option<MmioAddr<T>>;
}

impl<T: Unsigned> MmioRange<T> for MmioAddrRange<T> {
    #[inline]
    fn from_start_size(start: MmioAddr<T>, size: usize) -> Option<Self> {
        Self::try_from_start_size(start, size)
    }

    #[inline]
    fn from_phys_range(range: PhysAddrRange) -> Self {
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 195) - [internal]
reg64_aligned

```
    fn reg64_aligned(self, offset: usize) -> Option<MmioAddr<T>>;  
}  
  
impl<T: Unsigned> MmioRange<T> for MmioAddrRange<T> {  
    #[inline]  
    fn from_start_size(start: MmioAddr<T>, size: usize) -> Option<Self> {  
        Self::try_from_start_size(start, size)  
    }  
  
    #[inline]  
    fn from_phys_range(range: PhysAddrRange) -> Self {  
        Self {  
            // ... (truncated, see source file for more)  
        }  
    }  
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 200) - [internal]
from_start_size

```
    fn from_start_size(start: MmioAddr<T>, size: usize) -> Option<Self> {  
        Self::try_from_start_size(start, size)  
    }  
  
    #[inline]  
    fn from_phys_range(range: PhysAddrRange) -> Self {  
        Self {  
            start: MmioAddr::from_phys(range.start),  
            end: MmioAddr::from_phys(range.end),  
        }  
    }  
  
    // ... (truncated, see source file for more)  
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 205) - [internal]
from_phys_range

```
    fn from_phys_range(range: PhysAddrRange) -> Self {  
        Self {  
            start: MmioAddr::from_phys(range.start),  
            end: MmioAddr::from_phys(range.end),  
        }  
    }  
  
    #[inline]  
    fn from_phys_start_size(start: PhysAddr, size: usize) -> Option<Self> {  
        <Self as MmioRange<T>>::from_start_size(MmioAddr::from_phys(start), size)  
    }  
  
    // ... (truncated, see source file for more)  
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 213) - [internal]
from_phys_start_size

```
    fn from_phys_start_size(start: PhysAddr, size: usize) -> Option<Self> {  
        <Self as MmioRange<T>>::from_start_size(MmioAddr::from_phys(start), size)  
    }  
  
    #[inline]  
    fn to_phys_range(self) -> PhysAddrRange {  
        PhysAddrRange {  
            start: self.start.as_phys(),  
            end: self.end.as_phys(),  
        }  
    }  
  
    // ... (truncated, see source file for more)  
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 218) - [internal] to_phys_range

```
fn to_phys_range(self) -> PhysAddrRange {
    PhysAddrRange {
        start: self.start.as_phys(),
        end: self.end.as_phys(),
    }
}

#[inline]
fn reg<const W: usize>(self, offset: usize) -> Option<MmioAddr<T>> {
    let addr = self.start.checked_add(offset)?;
    let end = addr.checked_add(W)?;
    if end > self.end { None } else { Some(addr) }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 226) - [internal] reg

```
fn reg<const W: usize>(self, offset: usize) -> Option<MmioAddr<T>> {
    let addr = self.start.checked_add(offset)?;
    let end = addr.checked_add(W)?;
    if end > self.end { None } else { Some(addr) }
}

#[inline]
fn reg8(self, offset: usize) -> Option<MmioAddr<T>> {
    self.reg::<1>(offset)
}

#[inline]
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 233) - [internal] reg8

```
fn reg8(self, offset: usize) -> Option<MmioAddr<T>> {
    self.reg::<1>(offset)
}

#[inline]
fn reg16(self, offset: usize) -> Option<MmioAddr<T>> {
    self.reg::<2>(offset)
}

#[inline]
fn reg32(self, offset: usize) -> Option<MmioAddr<T>> {
    self.reg::<4>(offset)
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 238) - [internal] reg16

```
fn reg16(self, offset: usize) -> Option<MmioAddr<T>> {
    self.reg::<2>(offset)
}

#[inline]
fn reg32(self, offset: usize) -> Option<MmioAddr<T>> {
    self.reg::<4>(offset)
}

#[inline]
fn reg64(self, offset: usize) -> Option<MmioAddr<T>> {
    self.reg::<8>(offset)
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 243) - [internal] reg32

```
fn reg32(self, offset: usize) -> Option<MmioAddr<T>> {
    self.reg::<4>(offset)
}

#[inline]
fn reg64(self, offset: usize) -> Option<MmioAddr<T>> {
    self.reg::<8>(offset)
}

#[inline]
fn reg16_aligned(self, offset: usize) -> Option<MmioAddr<T>> {
    let addr = self.reg16(offset)?;
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 248) - [internal] reg64

```
fn reg64(self, offset: usize) -> Option<MmioAddr<T>> {
    self.reg::<8>(offset)
}

#[inline]
fn reg16_aligned(self, offset: usize) -> Option<MmioAddr<T>> {
    let addr = self.reg16(offset)?;
    addr.is_aligned(2usize).then_some(addr)
}

#[inline]
fn reg32_aligned(self, offset: usize) -> Option<MmioAddr<T>> {
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 253) - [internal]
reg16_aligned

```
fn reg16_aligned(self, offset: usize) -> Option<MmioAddr<T>> {
    let addr = self.reg16(offset)?;
    addr.is_aligned(2usize).then_some(addr)
}

#[inline]
fn reg32_aligned(self, offset: usize) -> Option<MmioAddr<T>> {
    let addr = self.reg32(offset)?;
    addr.is_aligned(4usize).then_some(addr)
}

#[inline]
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 259) - [internal]
reg32_aligned

```
fn reg32_aligned(self, offset: usize) -> Option<MmioAddr<T>> {
    let addr = self.reg32(offset)?;
    addr.is_aligned(4usize).then_some(addr)
}

#[inline]
fn reg64_aligned(self, offset: usize) -> Option<MmioAddr<T>> {
    let addr = self.reg64(offset)?;
    addr.is_aligned(8usize).then_some(addr)
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 265) - [internal]

reg64_aligned

```
fn reg64_aligned(self, offset: usize) -> Option<MmioAddr<T>> {
    let addr = self.reg64(offset)?;
    addr.is_aligned(8usize).then_some(addr)
}

/// Port I/O address. x86 legacy, occasionally still in use for
/// firmware-era devices (COM serial, CF8/CFC configuration access,
/// PIT/PIC registers) on platforms that haven't migrated to MMIO.
///
/// Deliberately **not** built on [def_usize_addr]: the port I/O
/// address space is genuinely 16-bit (the `in`/`out` instructions
/// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 287) - [public] IoPort

```
pub struct IoPort(u16);

impl From<u16> for IoPort {
    #[inline]
    fn from(port: u16) -> Self {
        Self(port)
    }
}

impl From<IoPort> for u16 {
    #[inline]
    fn from(port: IoPort) -> Self {
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 291) - [internal] from

```
fn from(port: u16) -> Self {
    Self(port)
}

impl From<IoPort> for u16 {
    #[inline]
    fn from(port: IoPort) -> Self {
        port.0
    }
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 298) - [internal] from

```
fn from(port: IoPort) -> Self {
    port.0
}

impl IoPort {
    #[inline]
    pub const fn new(port: u16) -> Self {
        Self(port)
    }

    #[inline]
    // ... (truncated, see source file for more)
}
```



```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 315) - [public] checked_add
    pub fn checked_add(self, offset: u16) -> Option<Self> {
        self.0.checked_add(offset).map(Self)
    }

    #[inline]
    pub fn checked_sub(self, offset: u16) -> Option<Self> {
        self.0.checked_sub(offset).map(Self)
    }

    #[inline]
    pub fn checked_mul(self, factor: u16) -> Option<Self> {
        self.0.checked_mul(factor).map(Self)
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 320) - [public] checked_sub
    pub fn checked_sub(self, offset: u16) -> Option<Self> {
        self.0.checked_sub(offset).map(Self)
    }

    #[inline]
    pub fn checked_mul(self, factor: u16) -> Option<Self> {
        self.0.checked_mul(factor).map(Self)
    }

    #[inline]
    pub fn checked_div(self, divisor: u16) -> Option<Self> {
        self.0.checked_div(divisor).map(Self)
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 325) - [public] checked_mul
    pub fn checked_mul(self, factor: u16) -> Option<Self> {
        self.0.checked_mul(factor).map(Self)
    }

    #[inline]
    pub fn checked_div(self, divisor: u16) -> Option<Self> {
        self.0.checked_div(divisor).map(Self)
    }
}

impl fmt::Debug for IoPort {
    fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
        // ... (truncated, see source file for more)
    }
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 330) - [public] checked_div
    pub fn checked_div(self, divisor: u16) -> Option<Self> {
        self.0.checked_div(divisor).map(Self)
    }
}

impl fmt::Debug for IoPort {
    fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
        write!(f, "PORT:{:#06x}", self.0)
    }
}

impl fmt::Display for IoPort {
    // ... (truncated, see source file for more)
}

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 336) - [internal] fmt

```
fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
    write!(f, "PORT: {:#06x}", self.0)
}

impl fmt::Display for IoPort {
    fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
        write!(f, "{: #06x}", self.0)
    }
}

impl fmt::LowerHex for IoPort {
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 342) - [internal] fmt

```
fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
    write!(f, "{: #06x}", self.0)
}

impl fmt::LowerHex for IoPort {
    fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
        fmt::LowerHex::fmt(&self.0, f)
    }
}

impl fmt::UpperHex for IoPort {
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 348) - [internal] fmt

```
fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
    fmt::LowerHex::fmt(&self.0, f)
}

impl fmt::UpperHex for IoPort {
    fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
        fmt::UpperHex::fmt(&self.0, f)
    }
}

/// Range of port I/O addresses. `end` is exclusive, matching the
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 354) - [internal] fmt

```
fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
    fmt::UpperHex::fmt(&self.0, f)
}

/// Range of port I/O addresses. `end` is exclusive, matching the
/// convention used by [AddrRange] for MMIO / IOVA.
#[derive(Copy, Clone, Eq, PartialEq, Hash)]
#[cfg_attr(
    not(target_arch = "x86_64"),
    deprecated(
        note = "Io ports are only meaningful on x86_64 targets, but they're being used on a different arch",
        // ... (truncated, see source file for more)
    )
)]
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 368) - [public] IoPortRange

```
pub struct IoPortRange {
    start: IoPort,
    end: IoPort,
}

impl IoPortRange {
    #[inline]
    pub const fn new(start: IoPort, end: IoPort) -> Self {
        Self { start, end }
    }

    /// Build from a base and byte-count. Returns `None` when the range
    /// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 382) - [public]
from_start_size

```
pub fn from_start_size(start: IoPort, size: u16) -> Option<Self> {
    let end = start.checked_add(size)?;
    Some(Self { start, end })
}

#[inline]
pub const fn start(self) -> IoPort {
    self.start
}

#[inline]
pub const fn end(self) -> IoPort {
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 417) - [internal] fmt

```
fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
    write!(f, "PORT[{:#06x}..{:#06x}]", self.start.0, self.end.0)
}

#[cfg(test)]
mod tests {
    use super::*;

    fn assert_mmio_round_trips<T: Unsigned>(samples: &[usize]) {
        for &value in samples {
            let addr = MmioAddr::<T>::from(value);
            // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 426) - [internal]
assert_mmio_round_trips

```
fn assert_mmio_round_trips<T: Unsigned>(samples: &[usize]) {
    for &value in samples {
        let addr = MmioAddr::<T>::from(value);
        let round_trip: usize = addr.into();
        assert_eq!(round_trip, value);

        let phys = PhysAddr::from_usize(value);
        let addr = MmioAddr::<T>::from_phys(phys);
        assert_eq!(addr.as_phys().as_usize(), value);
    }
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 438) - [internal]
assert_iovi_round_trips

```
fn assert_iovi_round_trips<T: Unsigned>(samples: &[usize]) {  
    for &value in samples {  
        let addr = IoviAddr::::from(value);  
        let round_trip: usize = addr.into();  
        assert_eq!(round_trip, value);  
    }  
}  
  
fn mmio_value<T: Unsigned>(addr: MmioAddr<T>) -> usize {  
    addr.into()  
}  
  
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 446) - [internal] mmio_value

```
fn mmio_value<T: Unsigned>(addr: MmioAddr<T>) -> usize {  
    addr.into()  
}  
  
#[test]  
fn mmio_addr_u32_round_trips_representable_values() {  
    assert_mmio_round_trips::(&[0, 1, 0x1000, 0xffff, u32::MAX as usize]);  
}  
  
#[test]  
fn mmio_addr_u64_round_trips_representable_values() {  
    assert_mmio_round_trips::(&[0, 1, 0x1000, 0xffff_ffff, usize::MAX]);  
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 451) - [internal]
mmio_addr_u32_round_trips_representable_values

```
fn mmio_addr_u32_round_trips_representable_values() {  
    assert_mmio_round_trips::(&[0, 1, 0x1000, 0xffff, u32::MAX as usize]);  
}  
  
#[test]  
fn mmio_addr_u64_round_trips_representable_values() {  
    assert_mmio_round_trips::(&[0, 1, 0x1000, 0xffff_ffff, usize::MAX]);  
}  
  
#[test]  
fn iovi_addr_u32_round_trips_representable_values() {  
    assert_iovi_round_trips::(&[0, 1, 0x1000, 0xffff, u32::MAX as usize]);  
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 456) - [internal]
mmio_addr_u64_round_trips_representable_values

```
fn mmio_addr_u64_round_trips_representable_values() {  
    assert_mmio_round_trips::(&[0, 1, 0x1000, 0xffff_ffff, usize::MAX]);  
}  
  
#[test]  
fn iovi_addr_u32_round_trips_representable_values() {  
    assert_iovi_round_trips::(&[0, 1, 0x1000, 0xffff, u32::MAX as usize]);  
}  
  
#[test]  
fn iovi_addr_u64_round_trips_representable_values() {  
    assert_iovi_round_trips::(&[0, 1, 0x1000, 0xffff_ffff, usize::MAX]);  
// ... (truncated, see source file for more)
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 461) - [internal]
iovi_addr_u32_round_trips_representable_values
    fn ioivi_addr_u32_round_trips_representable_values() {
        assert_ioivi_round_trips::<u32>(&[0, 1, 0x1000, 0xffff, u32::MAX as usize]);
    }

    #[test]
    fn ioivi_addr_u64_round_trips_representable_values() {
        assert_ioivi_round_trips::<u64>(&[0, 1, 0x1000, 0xffff_ffff, usize::MAX]);
    }

    #[cfg(target_pointer_width = "64")]
    #[test]
    #[should_panic]
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 466) - [internal]
ioivi_addr_u64_round_trips_representable_values
    fn ioivi_addr_u64_round_trips_representable_values() {
        assert_ioivi_round_trips::<u64>(&[0, 1, 0x1000, 0xffff_ffff, usize::MAX]);
    }

    #[cfg(target_pointer_width = "64")]
    #[test]
    #[should_panic]
    fn mmio_addr_u32_rejects_out_of_range_values() {
        let _ = MmioAddr::<u32>::from(u32::MAX as usize + 1);
    }

    #[cfg(target_pointer_width = "64")]
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 473) - [internal]
mmio_addr_u32_rejects_out_of_range_values
    fn mmio_addr_u32_rejects_out_of_range_values() {
        let _ = MmioAddr::<u32>::from(u32::MAX as usize + 1);
    }

    #[cfg(target_pointer_width = "64")]
    #[test]
    #[should_panic]
    fn mmio_addr_u32_checked_add_and_mul_reject_overflows() {
        let addr = MmioAddr::<u32>::from(0x1000);
        assert_eq!(addr.checked_add(0x1000).unwrap().as_usize(), 0x2000usize);
        assert!(addr.checked_add(u32::MAX as usize).is_none());
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 480) - [internal]
mmio_addr_u32_checked_add_and_mul_reject_overflows
    fn mmio_addr_u32_checked_add_and_mul_reject_overflows() {
        let addr = MmioAddr::<u32>::from(0x1000);
        assert_eq!(addr.checked_add(0x1000).unwrap().as_usize(), 0x2000usize);
        assert!(addr.checked_add(u32::MAX as usize).is_none());
    }

    #[cfg(target_pointer_width = "64")]
    #[test]
    #[should_panic]
    fn ioivi_addr_u32_rejects_out_of_range_values() {
        let _ = IoviAddr::<u32>::from(u32::MAX as usize + 1);
    }
    // ... (truncated, see source file for more)

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 489) - [internal]

iovi_addr_u32_rejects_out_of_range_values

```
fn iovi_addr_u32_rejects_out_of_range_values() {
    let _ = IoviAddr::<u32>::from(u32::MAX as usize + 1);
}

#[test]
fn mmio_range_register_helpers_respect_bounds_and_alignment() {
    let range = MmioAddrRange::<u32>::new(MmioAddr::from(0x1000), MmioAddr::from(0x1040));

    assert_eq!(mmio_value(range.reg8(0).unwrap()), 0x1000);
    assert_eq!(mmio_value(range.reg16(0x10).unwrap()), 0x1010);
    assert_eq!(mmio_value(range.reg32(0x20).unwrap()), 0x1020);
    assert_eq!(mmio_value(range.reg64(0x38).unwrap()), 0x1038);
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/addr.rs (line 494) - [internal]

mmio_range_register_helpers_respect_bounds_and_alignment

```
fn mmio_range_register_helpers_respect_bounds_and_alignment() {
    let range = MmioAddrRange::<u32>::new(MmioAddr::from(0x1000), MmioAddr::from(0x1040));

    assert_eq!(mmio_value(range.reg8(0).unwrap()), 0x1000);
    assert_eq!(mmio_value(range.reg16(0x10).unwrap()), 0x1010);
    assert_eq!(mmio_value(range.reg32(0x20).unwrap()), 0x1020);
    assert_eq!(mmio_value(range.reg64(0x38).unwrap()), 0x1038);
    assert!(range.reg64(0x39).is_none());
    assert_eq!(
        range.to_phys_range(),
        PhysAddrRange::from_start_size(PhysAddr::from_usize(0x1000), 0x40)
    );
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 46) - [public]

IortError

```
pub enum IortError {
    Acpi(AcpiError),
    BdfRanges(BdfRangeSetError),
    Mapping(PagingError),
    Malformed(&'static str),
}

impl From<AcpiError> for IortError {
    #[inline]
    fn from(value: AcpiError) -> Self {
        Self::Acpi(value)
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 55) -

[internal] from

```
fn from(value: AcpiError) -> Self {
    Self::Acpi(value)
}

impl From<BdfRangeSetError> for IortError {
    #[inline]
    fn from(value: BdfRangeSetError) -> Self {
        Self::BdfRanges(value)
    }
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 62) -

```
[internal] from
    fn from(value: BdfRangeSetError) -> Self {
        Self::BdfRanges(value)
    }
}

impl From<PagingError> for IortError {
    #[inline]
    fn from(value: PagingError) -> Self {
        Self::Mapping(value)
    }
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 69) -

```
[internal] from
    fn from(value: PagingError) -> Self {
        Self::Mapping(value)
    }
}

impl fmt::Display for IortError {
    fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
        match self {
            Self::Acpi(error) => write!(f, "{error:?}"),
            Self::BdfRanges(error) => write!(f, "{error}"),
            Self::Mapping(error) => write!(f, "{error:?}"),
            Self::Malformed(message) => f.write_str(message),
        }
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 75) -

```
[internal] fmt
    fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
        match self {
            Self::Acpi(error) => write!(f, "{error:?}"),
            Self::BdfRanges(error) => write!(f, "{error}"),
            Self::Mapping(error) => write!(f, "{error:?}"),
            Self::Malformed(message) => f.write_str(message),
        }
    }
}

fn checked_offset(base: usize, offset: usize) -> Result<usize, IortError> {
    base.checked_add(offset)
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 85) -

```
[internal] checked_offset
    fn checked_offset(base: usize, offset: usize) -> Result<usize, IortError> {
        base.checked_add(offset)
        .ok_or(IortError::Malformed("IORT offset overflow"))
    }

    fn mmio_address(base: u64) -> Result<MmioAddr, IortError> {
        let base = usize::try_from(base)
        .map_err(|_| IortError::Malformed("IORT MMIO address cannot fit in usize"))?;
        Ok(MmioAddr::from(base))
    }

    fn mmio_range_from_start_size(start: u64, size: u64) -> Result<MmioAddrRange, IortError> {
        // ... (truncated, see source file for more)
    }
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 90) -

```
[internal] mmio_address

fn mmio_address(base: u64) -> Result<MmioAddr, IortError> {
    let base = usize::try_from(base)
        .map_err(|_| IortError::Malformed("IORT MMIO address cannot fit in usize"))?;
    Ok(MmioAddr::from(base))
}

fn mmio_range_from_start_size(start: u64, size: u64) -> Result<MmioAddrRange, IortError> {
    let start = mmio_address(start)?;
    let size = usize::try_from(size)
        .map_err(|_| IortError::Malformed("IORT MMIO range size cannot fit in usize"))?;
    if size == 0 {
        return Err(IortError::Malformed("IORT MMIO range is empty"));
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 96) -

```
[internal] mmio_range_from_start_size

fn mmio_range_from_start_size(start: u64, size: u64) -> Result<MmioAddrRange, IortError> {
    let start = mmio_address(start)?;
    let size = usize::try_from(size)
        .map_err(|_| IortError::Malformed("IORT MMIO range size cannot fit in usize"))?;
    if size == 0 {
        return Err(IortError::Malformed("IORT MMIO range is empty"));
    }
    <MmioAddrRange as MmioRange<usize>>::from_start_size(start, size)
        .ok_or(IortError::Malformed("IORT MMIO range overflows"))
}

fn phys_range_from_start_size(start: u64, size: u64) -> Result<PhysAddrRange, IortError> {
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 107) -

```
[internal] phys_range_from_start_size

fn phys_range_from_start_size(start: u64, size: u64) -> Result<PhysAddrRange, IortError> {
    let start = usize::try_from(start)
        .map_err(|_| IortError::Malformed("IORT physical address cannot fit in usize"))?;
    let size = usize::try_from(size)
        .map_err(|_| IortError::Malformed("IORT physical range size cannot fit in usize"))?;
    if size == 0 {
        return Err(IortError::Malformed("IORT physical range is empty"));
    }
    PhysAddrRange::try_from_start_size(PhysAddr::from_usize(start), size)
        .ok_or(IortError::Malformed("IORT physical range overflows"))
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 121) - [public]

```
IortAcpiTable

pub struct IortAcpiTable {
    pub header: SdtHeader,
}

unsafe impl AcpiTable for IortAcpiTable {
    const SIGNATURE: Signature = Signature::IORT;

    #[inline]
    fn header(&self) -> &SdtHeader {
        &self.header
    }
}

// ... (truncated, see source file for more)
```



```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 129) -
[internal] header
    fn header(&self) -> &SdtHeader {
        &self.header
    }
}

fn table_bytes<'a, T: AcpiTable>(
    bytes: &'a [u8],
    short_header: &'static str,
    short_table: &'static str,
) -> Result<&'a [u8], IortError> {
    if bytes.len() < size_of::<SdtHeader>() {
        return Err(IortError::Malformed(short_header));
    }
    // ... (truncated, see source file for more)
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 134) -
[internal] table_bytes
    fn table_bytes<'a, T: AcpiTable>(
        bytes: &'a [u8],
        short_header: &'static str,
        short_table: &'static str,
    ) -> Result<&'a [u8], IortError> {
        if bytes.len() < size_of::<SdtHeader>() {
            return Err(IortError::Malformed(short_header));
        }

        let header = unsafe { &*bytes.as_ptr().cast::<SdtHeader>() };
        if header.signature != T::SIGNATURE {
            return Err(IortError::Acpi(AcpiError::SdtInvalidSignature(
                // ... (truncated, see source file for more)
            )));
        }
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 169) -
[internal] read_array
    fn read_array<const N: usize>(
        bytes: &[u8],
        offset: usize,
        message: &'static str,
    ) -> Result<[u8; N], IortError> {
        let end = offset
            .checked_add(N)
            .ok_or(IortError::Malformed("IORT read offset overflow"))?;
        let bytes = bytes
            .get(offset..end)
            .ok_or(IortError::Malformed(message))?;
        let mut out = [0; N];
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 186) -
[internal] read_u8
    fn read_u8(bytes: &[u8], offset: usize) -> Result<u8, IortError> {
        bytes
            .get(offset)
            .copied()
            .ok_or(IortError::Malformed("IORT table read is out of bounds"))
    }

#[inline]
fn read_u16(bytes: &[u8], offset: usize) -> Result<u16, IortError> {
    Ok(u16::from_le_bytes(read_array(
        bytes,
        offset,
        // ... (truncated, see source file for more)
    )))
}

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 194) -
[internal] read_u16

```
fn read_u16(bytes: &[u8], offset: usize) -> Result<u16, IortError> {
    Ok(u16::from_le_bytes(read_array(
        bytes,
        offset,
        "IORT table read is out of bounds",
    )))
}

#[inline]
fn read_u32(bytes: &[u8], offset: usize) -> Result<u32, IortError> {
    Ok(u32::from_le_bytes(read_array(
        bytes,
        // ... (truncated, see source file for more)
    )))
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 203) -
[internal] read_u32

```
fn read_u32(bytes: &[u8], offset: usize) -> Result<u32, IortError> {
    Ok(u32::from_le_bytes(read_array(
        bytes,
        offset,
        "IORT table read is out of bounds",
    )))
}

#[inline]
fn read_u64(bytes: &[u8], offset: usize) -> Result<u64, IortError> {
    Ok(u64::from_le_bytes(read_array(
        bytes,
        // ... (truncated, see source file for more)
    )))
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 212) -
[internal] read_u64

```
fn read_u64(bytes: &[u8], offset: usize) -> Result<u64, IortError> {
    Ok(u64::from_le_bytes(read_array(
        bytes,
        offset,
        "IORT table read is out of bounds",
    )))
}

#[derive(Clone, Copy, Debug)]
pub struct IortTable<'a> {
    sdt: &'a [u8],
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 221) - [public]
IortTable

```
pub struct IortTable<'a> {
    sdt: &'a [u8],
}

impl<'a> IortTable<'a> {
    fn parse(bytes: &'a [u8]) -> Result<Self, IortError> {
        let sdt = table_bytes::<IortAcpiTable>(
            bytes,
            "IORT table shorter than SDT header",
            "IORT table length is invalid",
        )?;
        if sdt.len() < IORT_TABLE_MIN_LENGTH {
            // ... (truncated, see source file for more)
        }
    }
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 226) -
[internal] parse

```
fn parse(bytes: &'a [u8]) -> Result<Self, IortError> {
    let sdt = table_bytes::<IortAcpiTable>(
        bytes,
        "IORT table shorter than SDT header",
        "IORT table length is invalid",
    )?;
    if sdt.len() < IORT_TABLE_MIN_LENGTH {
        return Err(IortError::Malformed("IORT table shorter than fixed header"));
    }
    Ok(Self { sdt })
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 261) - [public]
node_count

```
pub fn node_count(self) -> Result<u32, IortError> {
    read_u32(self.sdt, IORT_NODE_COUNT_OFFSET)
}

#[inline]
pub fn node_offset(self) -> Result<u32, IortError> {
    read_u32(self.sdt, IORT_NODE_OFFSET_OFFSET)
}

pub fn nodes(self) -> Result<IortNodes<'a>, IortError> {
    let offset = usize::try_from(self.node_offset())?
        .map_err(|_| IortError::Malformed("IORT node offset cannot fit in usize"))?;
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 266) - [public]
node_offset

```
pub fn node_offset(self) -> Result<u32, IortError> {
    read_u32(self.sdt, IORT_NODE_OFFSET_OFFSET)
}

pub fn nodes(self) -> Result<IortNodes<'a>, IortError> {
    let offset = usize::try_from(self.node_offset())?
        .map_err(|_| IortError::Malformed("IORT node offset cannot fit in usize"))?;
    if offset < IORT_TABLE_MIN_LENGTH || offset > self.sdt.len() {
        return Err(IortError::Malformed("IORT node array offset is invalid"));
    }
    Ok(IortNodes {
        table: self,
    })
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 270) - [public]
nodes

```
pub fn nodes(self) -> Result<IortNodes<'a>, IortError> {
    let offset = usize::try_from(self.node_offset())?
        .map_err(|_| IortError::Malformed("IORT node offset cannot fit in usize"))?;
    if offset < IORT_TABLE_MIN_LENGTH || offset > self.sdt.len() {
        return Err(IortError::Malformed("IORT node array offset is invalid"));
    }
    Ok(IortNodes {
        table: self,
        offset,
        remaining: self.node_count()?,
    })
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 283) - [public]
node_by_reference

```
pub fn node_by_reference(self, reference: u32) -> Result<Option<IortNode<'a>>, IortError> {  
    let offset = usize::try_from(reference)  
        .map_err(|_| IortError::Malformed("IORT node reference cannot fit in usize"))?;  
    if offset < self.node_offset()? as usize || offset >= self.sdt.len() {  
        return Ok(None);  
    }  
  
    for node in self.nodes()? {  
        let node = node?;  
        if node.header().offset == offset {  
            return Ok(Some(node));  
        }  
    }  
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 299) - [public]
id_mappings

```
pub fn id_mappings(self, node: IortNode<'a>) -> Result<IortIdMappings<'a>, IortError> {  
    self.id_mappings_for_header(node.header())  
}  
  
pub fn id_mappings_for(self, header: IortNodeHeader) -> Result<IortIdMappings<'a>, IortError> {  
    self.id_mappings_for_header(header)  
}  
  
pub fn its_identifiers(self, group: IortItsGroup) -> Result<IortItsIdentifiers<'a>, IortError> {  
    let start = group.header.offset + ITS_GROUP_MIN_LENGTH;  
    let bytes = usize::try_from(group.its_count)  
        .ok()  
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 303) - [public]
id_mappings_for

```
pub fn id_mappings_for(self, header: IortNodeHeader) -> Result<IortIdMappings<'a>, IortError> {  
    self.id_mappings_for_header(header)  
}  
  
pub fn its_identifiers(self, group: IortItsGroup) -> Result<IortItsIdentifiers<'a>, IortError> {  
    let start = group.header.offset + ITS_GROUP_MIN_LENGTH;  
    let bytes = usize::try_from(group.its_count)  
        .ok()  
        .and_then(|count| count.checked_mul(4))  
        .ok_or(IortError::Malformed(  
            "IORT ITS identifier array length overflows",  
        ))?;  
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 307) - [public]
its_identifiers

```
pub fn its_identifiers(self, group: IortItsGroup) -> Result<IortItsIdentifiers<'a>, IortError> {  
    let start = group.header.offset + ITS_GROUP_MIN_LENGTH;  
    let bytes = usize::try_from(group.its_count)  
        .ok()  
        .and_then(|count| count.checked_mul(4))  
        .ok_or(IortError::Malformed(  
            "IORT ITS identifier array length overflows",  
        ))?;  
    let end = checked_offset(start, bytes)?;  
    if end > group.header.end_offset() {  
        return Err(IortError::Malformed(  
            "IORT ITS identifier array extends past node",  
        ));  
    }  
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 328) - [public]
rmr_descriptors

```
pub fn rmr_descriptors(
    self,
    memory: IortReservedMemory,
) -> Result<IortRmrDescriptors<'a>, IortError> {
    let start = checked_offset(memory.header.offset, memory.descriptor_offset as usize)?;
    let bytes = usize::try_from(memory.descriptor_count)
        .ok()
        .and_then(|count| count.checked_mul(RMR_DESCRIPTOR_LENGTH))
        .ok_or(IortError::Malformed(
            "IORT RMR descriptor array length overflows",
        ))?;
    let end = checked_offset(start, bytes)?;
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 353) - [public]
smmu_context_interrupts

```
pub fn smmu_context_interrupts(
    self,
    smmu: IortSmmu,
) -> Result<IortSmmuInterrupts<'a>, IortError> {
    self.smmu_interrupts(
        smmu.header,
        smmu.context_interrupt_offset,
        smmu.context_interrupt_count,
    )
}

pub fn smmu_pmu_interrupts(self, smmu: IortSmmu) -> Result<IortSmmuInterrupts<'a>, IortError> {
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 364) - [public]
smmu_pmu_interrupts

```
pub fn smmu_pmu_interrupts(self, smmu: IortSmmu) -> Result<IortSmmuInterrupts<'a>, IortError> {
    self.smmu_interrupts(
        smmu.header,
        smmu.pmu_interrupt_offset,
        smmu.pmu_interrupt_count,
    )
}

pub fn smmu_global_interrupt(
    self,
    smmu: IortSmmu,
) -> Result<Option<IortSmmuGlobalInterrupt>, IortError> {
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 372) - [public]
smmu_global_interrupt

```
pub fn smmu_global_interrupt(
    self,
    smmu: IortSmmu,
) -> Result<Option<IortSmmuGlobalInterrupt>, IortError> {
    if smmu.global_interrupt_offset == 0 {
        return Ok(None);
    }
    let offset = checked_offset(smmu.header.offset, smmu.global_interrupt_offset as usize)?;
    if checked_offset(offset, 16)? > smmu.header.end_offset() {
        return Err(IortError::Malformed(
            "IORT SMMU global interrupt extends past node",
        ));
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 393) - [public]

root_complex_for_segment

```
pub fn root_complex_for_segment(
    self,
    segment: u16,
) -> Result<Option<IortRootComplex>, IortError> {
    let mut found = None;
    self.for_each_root_complex(|root| {
        if root.pci_segment_number == u32::from(segment) {
            found = Some(root);
        }
    })?;
    Ok(found)
}
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 407) - [public]

root_complex_bdf_ranges

```
pub fn root_complex_bdf_ranges(
    self,
    segment: u16,
) -> Result<Option<IortBdfRangeSet>, IortError> {
    let Some(root) = self.root_complex_for_segment(segment)? else {
        return Ok(None);
    };

    let mut ranges = IortBdfRangeSet::empty();
    for mapping in self.id_mappings_for(root.header)? {
        if let Some(range) = mapping?.input_bdf_range()? {
            ranges.insert::<IortError>(range)?;
        }
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 427) - [public]

root_complex_requester_match

```
pub fn root_complex_requester_match(
    self,
    requester: PciDevice,
) -> Result<Option<IortRequesterMatch>, IortError> {
    let Some(root) = self.root_complex_for_segment(requester.segment())? else {
        return Ok(None);
    };

    let mut covered = false;
    let mut unresolved = false;
    for mapping in self.id_mappings_for(root.header)? {
        let mapping = mapping?;
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 455) - [public]

map_id_to_kind

```
pub fn map_id_to_kind(
    self,
    node: IortNode<'a>,
    input_id: u32,
    target: IortNodeKind,
) -> Result<Option<IortIdTranslation<'a>>, IortError> {
    let mut current = node;
    let mut id = input_id;
    let mut remaining = self.node_count()?;

    while remaining != 0 {
        if current.header().kind == target {
            // ... (truncated, see source file for more)
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 486) - [public]
map_pci_requester_to_kind
    pub fn map_pci_requester_to_kind(
        self,
        requester: PciDevice,
        target: IortNodeKind,
    ) -> Result<Option<IortIdTranslation<'a>>, IortError> {
        let Some(root) = self.root_complex_for_segment(requester.segment())? else {
            return Ok(None);
        };
        self.map_id_to_kind(
            IortNode::RootComplex(root),
            requester.bdf().as_u32(),
            target,
        )
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 501) - [public]
iommu_for_pci_requester
    pub fn iommu_for_pci_requester(
        self,
        requester: PciDevice,
    ) -> Result<Option<IortIdTranslation<'a>>, IortError> {
        if let Some(translation) =
            self.map_pci_requester_to_kind(requester, IortNodeKind::SmmuV3)?
        {
            return Ok(Some(translation));
        }
        self.map_pci_requester_to_kind(requester, IortNodeKind::Smmu)
    }

    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 513) - [public]
rmr_bdf_ranges
    pub fn rmr_bdf_ranges(
        self,
        memory: PhysAddrRange,
    ) -> Result<Option<IortBdfRangeSet>, IortError> {
        let mut ranges = IortBdfRangeSet::empty();
        let mut found_memory = false;
        let mut unresolved = false;

        self.for_each_reserved_memory(|rmr| {
            let mut rmr_matches = false;
            for descriptor in self.rmr_descriptors(rmr)? {
                if descriptor.memory == memory {
                    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 548) - [public]
for_each_smmu_v3
    pub fn for_each_smmu_v3<F>(self, mut f: F) -> Result<(), IortError>
    where
        F: FnMut(IortSmmuV3) -> Result<(), IortError>,
    {
        self.for_each_node(|node| {
            if let IortNode::SmmuV3(smmu) = node {
                f(smmu)?;
            }
            Ok(())
        })
    }

    // ... (truncated, see source file for more)

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 560) - [public]
for_each_root_complex

```
pub fn for_each_root_complex<F>(self, mut f: F) -> Result<(), IortError>
where
    F: FnMut(IortRootComplex) -> Result<(), IortError>,
{
    self.for_each_node(|node| {
        if let IortNode::RootComplex(root) = node {
            f(root)?;
        }
        Ok(())
    })
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 572) - [public]
for_each_reserved_memory

```
pub fn for_each_reserved_memory<F>(self, mut f: F) -> Result<(), IortError>
where
    F: FnMut(IortReservedMemory) -> Result<(), IortError>,
{
    self.for_each_node(|node| {
        if let IortNode::ReservedMemory(memory) = node {
            f(memory)?;
        }
        Ok(())
    })
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 584) -
[internal] for_each_node

```
fn for_each_node<F>(self, mut f: F) -> Result<(), IortError>
where
    F: FnMut(IortNode<'a>) -> Result<(), IortError>,
{
    for node in self.nodes()? {
        f(node)?;
    }
    Ok(())
}

fn translate_once(
    self,
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 594) -
[internal] translate_once

```
fn translate_once(
    self,
    node: IortNode<'a>,
    input_id: u32,
) -> Result<Option<IortSingleStepTranslation>, IortError> {
    let mut fallback = None;
    for mapping in self.id_mappings(node)? {
        let mapping = mapping?;
        let Some(output_id) = mapping.translate_for_node_kind(node.header().kind, input_id)
        else {
            continue;
        };
    };
    // ... (truncated, see source file for more)
```


driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 624) -

[internal] parse_node

```
fn parse_node(self, offset: usize) -> Result<IortNode<'a>, IortError> {
    let header = self.parse_node_header(offset)?;
    match header.kind {
        IortNodeKind::ItsGroup => Ok(IortNode::ItsGroup(self.parse_its_group(header)?)),
        IortNodeKind::NamedComponent => Ok(IortNode::NamedComponent(
            self.parse_named_component(header)?,
        )),
        IortNodeKind::RootComplex => {
            Ok(IortNode::RootComplex(self.parse_root_complex(header)?))
        }
        IortNodeKind::Smmu => Ok(IortNode::Smmu(self.parse_smmu(header)?)),
        IortNodeKind::SmmuV3 => Ok(IortNode::SmmuV3(self.parse_smmu_v3(header)?)),
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 644) -

[internal] parse_node_header

```
fn parse_node_header(self, offset: usize) -> Result<IortNodeHeader, IortError> {
    if checked_offset(offset, IORT_NODE_HEADER_LENGTH)? > self.sdt.len() {
        return Err(IortError::Malformed("IORT node truncated before header"));
    }
    let length = usize::from(read_u16(self.sdt, offset + 1)?);
    if length < IORT_NODE_HEADER_LENGTH {
        return Err(IortError::Malformed("IORT node length smaller than header"));
    }
    if checked_offset(offset, length)? > self.sdt.len() {
        return Err(IortError::Malformed("IORT node extends past table length"));
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 667) -

[internal] parse_its_group

```
fn parse_its_group(self, header: IortNodeHeader) -> Result<IortItsGroup, IortError> {
    if header.length < ITS_GROUP_MIN_LENGTH {
        return Err(IortError::Malformed(
            "IORT ITS group node shorter than minimum",
        ));
    }
    Ok(IortItsGroup {
        its_count: read_u32(self.sdt, header.offset + 16)?,
        header,
    })
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 679) -

[internal] parse_named_component

```
fn parse_named_component(
    self,
    header: IortNodeHeader,
) -> Result<IortNamedComponent<'a>, IortError> {
    if header.length < NAMED_COMPONENT_MIN_LENGTH {
        return Err(IortError::Malformed(
            "IORT named component node shorter than minimum",
        ));
    }
    let name_start = header.offset + NAMED_COMPONENT_MIN_LENGTH;
    let name_end = if header.mapping_count == 0 {
        header.end_offset()
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 709) -

[internal] parse_root_complex

```
fn parse_root_complex(self, header: IortNodeHeader) -> Result<IortRootComplex, IortError> {
    if header.length < ROOT_COMPLEX_MIN_LENGTH {
        return Err(IortError::Malformed(
            "IORT root complex node shorter than minimum",
        ));
    }
    Ok(IortRootComplex {
        memory_properties: self.read_memory_access(header.offset + 16)?,
        ats_attribute: read_u32(self.sdt, header.offset + 24)?,
        pci_segment_number: read_u32(self.sdt, header.offset + 28)?,
        memory_address_limit: read_u8(self.sdt, header.offset + 32)?,
        pasid_capabilities: read_u16(self.sdt, header.offset + 33)?,
    })
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 725) -

[internal] parse_smmu

```
fn parse_smmu(self, header: IortNodeHeader) -> Result<IortSmmu, IortError> {
    if header.length < SMMU_MIN_LENGTH {
        return Err(IortError::Malformed("IORT SMMU node shorter than minimum"));
    }
    Ok(IortSmmu {
        registers: mmio_range_from_start_size(
            read_u64(self.sdt, header.offset + 16)?,
            read_u64(self.sdt, header.offset + 24)?,
        )?,
        model: read_u32(self.sdt, header.offset + 32)?,
        flags: read_u32(self.sdt, header.offset + 36)?,
        global_interrupt_offset: read_u32(self.sdt, header.offset + 40)?,
    })
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 745) -

[internal] parse_smmu_v3

```
fn parse_smmu_v3(self, header: IortNodeHeader) -> Result<IortSmmuV3, IortError> {
    if header.length < SMMU_V3_MIN_LENGTH {
        return Err(IortError::Malformed(
            "IORT SMMUv3 node shorter than minimum",
        ));
    }
    let vatos_address = read_u64(self.sdt, header.offset + 32)?;
    Ok(IortSmmuV3 {
        base_address: mmio_address(read_u64(self.sdt, header.offset + 16)?),
        flags: read_u32(self.sdt, header.offset + 24)?,
        vatos_address: if vatos_address == 0 {
            None
        } else {
            Some(vatos_address)
        },
    })
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 771) -

[internal] parse_pmcg

```
fn parse_pmcg(self, header: IortNodeHeader) -> Result<IortPmcg, IortError> {
    if header.length < PMCG_MIN_LENGTH {
        return Err(IortError::Malformed("IORT PMCG node shorter than minimum"));
    }
    let page1 = read_u64(self.sdt, header.offset + 32)?;
    Ok(IortPmcg {
        page0_base_address: mmio_address(read_u64(self.sdt, header.offset + 16)?),
        overflow_gsv: read_u32(self.sdt, header.offset + 24)?,
        node_reference: read_u32(self.sdt, header.offset + 28)?,
        page1_base_address: if page1 == 0 {
            None
        } else {
            Some(page1)
        },
    })
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 789) -

```
[internal] parse_reserved_memory
    fn parse_reserved_memory(
        self,
        header: IortNodeHeader,
    ) -> Result<IortReservedMemory, IortError> {
        if header.length < RMR_MIN_LENGTH {
            return Err(IortError::Malformed("IORT RMR node shorter than minimum"));
        }
        Ok(IortReservedMemory {
            flags: read_u32(self.sdt, header.offset + 16)?,
            descriptor_count: read_u32(self.sdt, header.offset + 20)?,
            descriptor_offset: read_u32(self.sdt, header.offset + 24)?,
            header,
        })
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 804) -

```
[internal] id_mappings_for_header
    fn id_mappings_for_header(
        self,
        header: IortNodeHeader,
    ) -> Result<IortIdMappings<'a>, IortError> {
        if header.mapping_count == 0 {
            return Ok(IortIdMappings {
                table: self,
                offset: header.end_offset(),
                remaining: 0,
            });
        }
        let mapping_offset = header.mapping_offset as usize;
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 836) -

```
[internal] smmu_interrupts
    fn smmu_interrupts(
        self,
        header: IortNodeHeader,
        offset: u32,
        count: u32,
    ) -> Result<IortSmmuInterrupts<'a>, IortError> {
        if count == 0 {
            return Ok(IortSmmuInterrupts {
                table: self,
                offset: header.end_offset(),
                remaining: 0,
            });
        }
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 870) -

```
[internal] read_memory_access
    fn read_memory_access(self, offset: usize) -> Result<IortMemoryAccess, IortError> {
        Ok(IortMemoryAccess {
            cache_coherency: read_u32(self.sdt, offset)?,
            hints: read_u8(self.sdt, offset + 4)?,
            memory_flags: read_u8(self.sdt, offset + 7)?,
        })
    }
}

pub struct IortNodes<'a> {
    table: IortTable<'a>,
    offset: usize,
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 879) - [public]

IortNodes

```
pub struct IortNodes<'a> {
    table: IortTable<'a>,
    offset: usize,
    remaining: u32,
}

impl<'a> Iterator for IortNodes<'a> {
    type Item = Result<IortNode<'a>, IortError>;

    fn next(&mut self) -> Option<Self::Item> {
        if self.remaining == 0 {
            return None;
        }
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 888) -

[internal] next

```
fn next(&mut self) -> Option<Self::Item> {
    if self.remaining == 0 {
        return None;
    }
    if self.offset >= self.table.sdt.len() {
        self.remaining = 0;
        return Some(Err(IortError::Malformed(
            "IORT node count exceeds table contents",
        )));
    }

    let offset = self.offset;
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 931) - [public]

IortNode

```
pub enum IortNode<'a> {
    ItsGroup(IortItsGroup),
    NamedComponent(IortNamedComponent<'a>),
    RootComplex(IortRootComplex),
    Smmu(IortSmmu),
    SmmuV3(IortSmmuV3),
    Pmcg(IortPmcg),
    ReservedMemory(IortReservedMemory),
    Unknown(IortUnknownNode),
}

impl IortNode<'_> {
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 959) - [public]

IortNodeHeader

```
pub struct IortNodeHeader {
    pub kind: IortNodeKind,
    pub revision: u8,
    pub identifier: u32,
    pub mapping_count: u32,
    pub mapping_offset: u32,
    pub offset: usize,
    pub length: usize,
}

impl IortNodeHeader {
    #[inline]
    // ... (truncated, see source file for more)
}
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 977) - [public]
IortNodeKind

pub enum IortNodeKind {
    ItsGroup,
    NamedComponent,
    RootComplex,
    Smmu,
    SmmuV3,
    Pmcg,
    ReservedMemory,
    Unknown(u8),
}

impl IortNodeKind {
    // ... (truncated, see source file for more)
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1005) -
[public] IortRequesterMatch

pub enum IortRequesterMatch {
    Covered,
    NotCovered,
    Unresolved,
}

#[derive(Clone, Copy, Debug, PartialEq, Eq)]
pub struct IortIdTranslation<'a> {
    pub node: IortNode<'a>,
    pub id: u32,
}

// ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1012) -
[public] IortIdTranslation

pub struct IortIdTranslation<'a> {
    pub node: IortNode<'a>,
    pub id: u32,
}

#[derive(Clone, Copy, Debug, PartialEq, Eq)]
struct IortSingleStepTranslation {
    id: u32,
    output_reference: u32,
}

#[derive(Clone, Copy, Debug, PartialEq, Eq)]
// ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1018) -
[internal] IortSingleStepTranslation

struct IortSingleStepTranslation {
    id: u32,
    output_reference: u32,
}

#[derive(Clone, Copy, Debug, PartialEq, Eq)]
pub struct IortUnknownNode {
    pub kind: u8,
    pub header: IortNodeHeader,
}

#[derive(Clone, Copy, Debug, PartialEq, Eq)]
// ... (truncated, see source file for more)

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1024) -

```
[public] IortUnknownNode
    pub struct IortUnknownNode {
        pub kind: u8,
        pub header: IortNodeHeader,
    }

    #[derive(Clone, Copy, Debug, PartialEq, Eq)]
    pub struct IortMemoryAccess {
        pub cache_coherency: u32,
        pub hints: u8,
        pub memory_flags: u8,
    }

    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1030) -

```
[public] IortMemoryAccess
    pub struct IortMemoryAccess {
        pub cache_coherency: u32,
        pub hints: u8,
        pub memory_flags: u8,
    }

    impl IortMemoryAccess {
        #[inline]
        pub const fn is_coherent(self) -> bool {
            (self.cache_coherency & 1) != 0
        }
    }

    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1044) -

```
[public] IortIdMapping
    pub struct IortIdMapping {
        pub input_base: u32,
        pub id_count: u32,
        pub output_base: u32,
        pub output_reference: u32,
        pub flags: u32,
    }

    impl IortIdMapping {
        #[inline]
        pub const fn is_single_mapping(self) -> bool {
            (self.flags & 1) != 0
        }
        // ... (truncated, see source file for more)
    }
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1058) -

```
[public] input_bdf_range
    pub fn input_bdf_range(self) -> Result<Option<BdfRange>, IortError> {
        if self.is_single_mapping() {
            return Ok(BdfRange::inclusive(Bdf::from_u16(0), Bdf::from_u16(u16::MAX)).ok());
        }

        let Some(end) = self.input_base.checked_add(self.id_count) else {
            return Ok(None);
        };
        if self.input_base > u32::from(u16::MAX) || end > u32::from(u16::MAX) {
            return Ok(None);
        }
        BdfRange::inclusive(
        // ... (truncated, see source file for more)
    }
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1077) -
[internal] translate_for_node_kind
    fn translate_for_node_kind(self, kind: IortNodeKind, input_id: u32) -> Option<u32> {
        if self.is_single_mapping() {
            return kind.single_mapping_allowed().then_some(self.output_base);
        }

        let end = self.input_base.checked_add(self.id_count)?;
        if input_id < self.input_base || input_id > end {
            return None;
        }
        self.output_base
            .checked_add(input_id.checked_sub(self.input_base)?)
    }
    // ... (truncated, see source file for more)

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1101) -
[public] IortIdMappings
    pub struct IortIdMappings<'a> {
        table: IortTable<'a>,
        offset: usize,
        remaining: u32,
    }

    impl Iterator for IortIdMappings<'_> {
        type Item = Result<IortIdMapping, IortError>;

        fn next(&mut self) -> Option<Self::Item> {
            if self.remaining == 0 {
                return None;
            }
            // ... (truncated, see source file for more)
        }
    }

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1110) -
[internal] next
    fn next(&mut self) -> Option<Self::Item> {
        if self.remaining == 0 {
            return None;
        }
        let offset = self.offset;
        self.offset += IORT_ID_MAPPING_LENGTH;
        self.remaining -= 1;
        Some(
            (|| -> Result<IortIdMapping, IortError> {
                Ok(IortIdMapping {
                    input_base: read_u32(self.table.sdt, offset)?,
                    id_count: read_u32(self.table.sdt, offset + 4)?,
                })
            })
        )
        // ... (truncated, see source file for more)
    }

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1133) -
[public] IortItsGroup
    pub struct IortItsGroup {
        pub its_count: u32,
        pub header: IortNodeHeader,
    }

    #[derive(Clone, Copy, Debug, PartialEq, Eq)]
    pub struct IortItsIdentifier {
        pub identifier: u32,
    }

    pub struct IortItsIdentifiers<'a> {
        table: IortTable<'a>,
        // ... (truncated, see source file for more)
    }

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1139) -

```
[public] IortItsIdentifier
    pub struct IortItsIdentifier {
        pub identifier: u32,
    }

    pub struct IortItsIdentifiers<'a> {
        table: IortTable<'a>,
        offset: usize,
        remaining: u32,
    }

    impl Iterator for IortItsIdentifiers<'_> {
        type Item = Result<IortItsIdentifier, IortError>;
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1143) -

```
[public] IortItsIdentifiers
    pub struct IortItsIdentifiers<'a> {
        table: IortTable<'a>,
        offset: usize,
        remaining: u32,
    }

    impl Iterator for IortItsIdentifiers<'_> {
        type Item = Result<IortItsIdentifier, IortError>;

        fn next(&mut self) -> Option<Self::Item> {
            if self.remaining == 0 {
                return None;
            }
            // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1152) -

```
[internal] next
    fn next(&mut self) -> Option<Self::Item> {
        if self.remaining == 0 {
            return None;
        }
        let offset = self.offset;
        self.offset += 4;
        self.remaining -= 1;
        Some(
            (|| -> Result<IortItsIdentifier, IortError> {
                Ok(IortItsIdentifier {
                    identifier: read_u32(self.table.sdt, offset)?,
                })
            })
        )
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1171) -

```
[public] IortNamedComponent
    pub struct IortNamedComponent<'a> {
        pub node_flags: u32,
        pub memory_properties: IortMemoryAccess,
        pub memory_address_limit: u8,
        pub device_name: &'a [u8],
        pub header: IortNodeHeader,
    }

    #[derive(Clone, Copy, Debug, PartialEq, Eq)]
    pub struct IortRootComplex {
        pub memory_properties: IortMemoryAccess,
        pub ats_attribute: u32,
        // ... (truncated, see source file for more)
```


driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1180) -

```
[public] IortRootComplex
    pub struct IortRootComplex {
        pub memory_properties: IortMemoryAccess,
        pub ats_attribute: u32,
        pub pci_segment_number: u32,
        pub memory_address_limit: u8,
        pub pasid_capabilities: u16,
        pub header: IortNodeHeader,
    }

    impl IortRootComplex {
        #[inline]
        pub const fn ats_supported(self) -> bool {
            // ... (truncated, see source file for more)
        }
    }
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1212) -

```
[public] IortSmmu
    pub struct IortSmmu {
        pub registers: MmioAddrRange,
        pub model: u32,
        pub flags: u32,
        pub global_interrupt_offset: u32,
        pub context_interrupt_count: u32,
        pub context_interrupt_offset: u32,
        pub pmu_interrupt_count: u32,
        pub pmu_interrupt_offset: u32,
        pub header: IortNodeHeader,
    }

    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1225) -

```
[public] IortSmmuGlobalInterrupt
    pub struct IortSmmuGlobalInterrupt {
        pub nsg_irpt: u32,
        pub nsg_irpt_flags: u32,
        pub nsg_cfg_irpt: u32,
        pub nsg_cfg_irpt_flags: u32,
    }

    #[derive(Clone, Copy, Debug, PartialEq, Eq)]
    pub struct IortSmmuInterrupt {
        pub gsiv: u32,
        pub flags: u32,
    }

    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1233) -

```
[public] IortSmmuInterrupt
    pub struct IortSmmuInterrupt {
        pub gsiv: u32,
        pub flags: u32,
    }

    pub struct IortSmmuInterrupts<'a> {
        table: IortTable<'a>,
        offset: usize,
        remaining: u32,
    }

    impl Iterator for IortSmmuInterrupts<'_> {
        // ... (truncated, see source file for more)
    }
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1238) -

```
[public] IortSmmuInterrupts
    pub struct IortSmmuInterrupts<'a> {
        table: IortTable<'a>,
        offset: usize,
        remaining: u32,
    }

    impl Iterator for IortSmmuInterrupts<'_> {
        type Item = Result<IortSmmuInterrupt, IortError>;

        fn next(&mut self) -> Option<Self::Item> {
            if self.remaining == 0 {
                return None;
            }
            // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1247) -

```
[internal] next
    fn next(&mut self) -> Option<Self::Item> {
        if self.remaining == 0 {
            return None;
        }
        let offset = self.offset;
        self.offset += 8;
        self.remaining -= 1;
        Some(
            || -> Result<IortSmmuInterrupt, IortError> {
                Ok(IortSmmuInterrupt {
                    gsiv: read_u32(self.table.sdt, offset)?,
                    flags: read_u32(self.table.sdt, offset + 4)?,
                })
            }
        )
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1267) -

```
[public] IortSmmuV3
    pub struct IortSmmuV3 {
        pub base_address: MmioAddr,
        pub flags: u32,
        pub vatos_address: Option<MmioAddr>,
        pub model: u32,
        pub event_gsiv: u32,
        pub pri_gsiv: u32,
        pub gerr_gsiv: u32,
        pub sync_gsiv: u32,
        pub pxm: u32,
        pub id_mapping_index: u32,
        pub header: IortNodeHeader,
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1294) -

```
[public] IortPmcg
    pub struct IortPmcg {
        pub page0_base_address: MmioAddr,
        pub overflow_gsiv: u32,
        pub node_reference: u32,
        pub page1_base_address: Option<MmioAddr>,
        pub header: IortNodeHeader,
    }

    #[derive(Clone, Copy, Debug, PartialEq, Eq)]
    pub struct IortReservedMemory {
        pub flags: u32,
        pub descriptor_count: u32,
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1303) -

```
[public] IortReservedMemory
    pub struct IortReservedMemory {
        pub flags: u32,
        pub descriptor_count: u32,
        pub descriptor_offset: u32,
        pub header: IortNodeHeader,
    }

    impl IortReservedMemory {
        #[inline]
        pub const fn remap_permitted(self) -> bool {
            (self.flags & 1) != 0
        }
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1328) -

```
[public] IortRmrDescriptor
    pub struct IortRmrDescriptor {
        pub memory: PhysAddrRange,
        pub reserved: u32,
    }

    pub struct IortRmrDescriptors<'a> {
        table: IortTable<'a>,
        offset: usize,
        remaining: u32,
    }

    impl Iterator for IortRmrDescriptors<'_> {
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1333) -

```
[public] IortRmrDescriptors
    pub struct IortRmrDescriptors<'a> {
        table: IortTable<'a>,
        offset: usize,
        remaining: u32,
    }

    impl Iterator for IortRmrDescriptors<'_> {
        type Item = Result<IortRmrDescriptor, IortError>;

        fn next(&mut self) -> Option<Self::Item> {
            if self.remaining == 0 {
                return None;
            }
            // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1342) -

```
[internal] next
    fn next(&mut self) -> Option<Self::Item> {
        if self.remaining == 0 {
            return None;
        }
        let offset = self.offset;
        self.offset += RMR_DESCRIPTOR_LENGTH;
        self.remaining -= 1;
        Some(|| {
            let base = read_u64(self.table.sdt, offset)?;
            let length = read_u64(self.table.sdt, offset + 8)?;
            Ok(IortRmrDescriptor {
                memory: phys_range_from_start_size(base, length)?,
            })
        })
        // ... (truncated, see source file for more)
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1371) -
[internal] TestEntry
    struct TestEntry;

    impl PageTableEntry for TestEntry {
        type Flags = ();

        fn new_leaf(_paddr: PhysAddr, _flags: Self::Flags, _size: PageSize) -> Self {
            Self
        }

        fn new_table(_paddr: PhysAddr, _level: u8) -> Self {
            Self
        }
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1376) -
[internal] new_leaf
    fn new_leaf(_paddr: PhysAddr, _flags: Self::Flags, _size: PageSize) -> Self {
        Self
    }

    fn new_table(_paddr: PhysAddr, _level: u8) -> Self {
        Self
    }

    fn paddr(&self) -> PhysAddr {
        PhysAddr::from_usize(0)
    }

    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1380) -
[internal] new_table
    fn new_table(_paddr: PhysAddr, _level: u8) -> Self {
        Self
    }

    fn paddr(&self) -> PhysAddr {
        PhysAddr::from_usize(0)
    }

    fn flags(&self) -> Self::Flags {}

    fn is_present(&self) -> bool {
        true
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1384) -
[internal] paddr
    fn paddr(&self) -> PhysAddr {
        PhysAddr::from_usize(0)
    }

    fn flags(&self) -> Self::Flags {}

    fn is_present(&self) -> bool {
        true
    }

    fn entry_kind(&self, _level: u8) -> PageTableEntryKind {
        PageTableEntryKind::Leaf
    }
    // ... (truncated, see source file for more)

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1388) -
[internal] flags
    fn flags(&self) -> Self::Flags {}

    fn is_present(&self) -> bool {
        true
    }

    fn entry_kind(&self, _level: u8) -> PageTableEntryKind {
        PageTableEntryKind::Leaf
    }

    fn clear(&mut self) {}

    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1390) -
[internal] is_present
    fn is_present(&self) -> bool {
        true
    }

    fn entry_kind(&self, _level: u8) -> PageTableEntryKind {
        PageTableEntryKind::Leaf
    }

    fn clear(&mut self) {}

    fn bits(&self) -> u64 {
        0
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1394) -
[internal] entry_kind
    fn entry_kind(&self, _level: u8) -> PageTableEntryKind {
        PageTableEntryKind::Leaf
    }

    fn clear(&mut self) {}

    fn bits(&self) -> u64 {
        0
    }

    fn from_bits(_bits: u64) -> Self {
        Self
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1398) -
[internal] clear
    fn clear(&mut self) {}

    fn bits(&self) -> u64 {
        0
    }

    fn from_bits(_bits: u64) -> Self {
        Self
    }
}

fn write_sdt_header(bytes: &mut [u8], signature: Signature, length: usize) {
    // ... (truncated, see source file for more)

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1400) -
[internal] bits
    fn bits(&self) -> u64 {
        0
    }

    fn from_bits(_bits: u64) -> Self {
        Self
    }
}

fn write_sdt_header(bytes: &mut [u8], signature: Signature, length: usize) {
    bytes[0..4].copy_from_slice(signature.as_str().as_bytes());
    bytes[4..8].copy_from_slice(&(length as u32).to_le_bytes());
    // ... (truncated, see source file for more)
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1404) -
[internal] from_bits
    fn from_bits(_bits: u64) -> Self {
        Self
    }
}

fn write_sdt_header(bytes: &mut [u8], signature: Signature, length: usize) {
    bytes[0..4].copy_from_slice(signature.as_str().as_bytes());
    bytes[4..8].copy_from_slice(&(length as u32).to_le_bytes());
    bytes[8] = 1;
}

fn finish_sdt_checksum(bytes: &mut [u8]) {
    // ... (truncated, see source file for more)
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1409) -
[internal] write_sdt_header
    fn write_sdt_header(bytes: &mut [u8], signature: Signature, length: usize) {
        bytes[0..4].copy_from_slice(signature.as_str().as_bytes());
        bytes[4..8].copy_from_slice(&(length as u32).to_le_bytes());
        bytes[8] = 1;
    }

    fn finish_sdt_checksum(bytes: &mut [u8]) {
        bytes[9] = 0;
        bytes[9] = 0u8.wrapping_sub(
            bytes
                .iter()
                .fold(0, |sum: u8, byte| sum.wrapping_add(*byte)),
        );
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1415) -
[internal] finish_sdt_checksum
    fn finish_sdt_checksum(bytes: &mut [u8]) {
        bytes[9] = 0;
        bytes[9] = 0u8.wrapping_sub(
            bytes
                .iter()
                .fold(0, |sum: u8, byte| sum.wrapping_add(*byte)),
        );
    }

    fn table_mapping(bytes: &[u8]) -> Mapping<TestEntry, VirtAddr> {
        let start = VirtAddr::from_usize(bytes.as_ptr() as usize);
        Mapping::new(
            // ... (truncated, see source file for more)
        )
    }
}

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1424) -
[internal] table_mapping

```
fn table_mapping(bytes: &[u8]) -> Mapping<TestEntry, VirtAddr> {  
    let start = VirtAddr::from_usize(bytes.as_ptr() as usize);  
    Mapping::new(  
        VirtAddrRange::from_start_size(start, bytes.len()),  
        PhysAddr::from_usize(0),  
        (),  
    )  
}  
  
fn write_node_header(  
    bytes: &mut [u8],  
    offset: usize,  
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1433) -
[internal] write_node_header

```
fn write_node_header(  
    bytes: &mut [u8],  
    offset: usize,  
    kind: u8,  
    length: usize,  
    identifier: u32,  
    mapping_count: u32,  
    mapping_offset: u32,  
) {  
    bytes[offset] = kind;  
    bytes[offset + 1..offset + 3].copy_from_slice(&(length as u16).to_le_bytes());  
    bytes[offset + 4..offset + 8].copy_from_slice(&identifier.to_le_bytes());  
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1449) -
[internal] write_id_mapping

```
fn write_id_mapping(  
    bytes: &mut [u8],  
    offset: usize,  
    input_base: u32,  
    id_count: u32,  
    output_base: u32,  
    output_reference: u32,  
    flags: u32,  
) {  
    bytes[offset..offset + 4].copy_from_slice(&input_base.to_le_bytes());  
    bytes[offset + 4..offset + 8].copy_from_slice(&id_count.to_le_bytes());  
    bytes[offset + 8..offset + 12].copy_from_slice(&output_base.to_le_bytes());  
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1466) -
[internal] parses_root_complex_smmuv3_and_id_mapping

```
fn parses_root_complex_smmuv3_and_id_mapping() {  
    const ROOT: usize = 48;  
    const ROOT_LEN: usize = 56;  
    const SMMU: usize = ROOT + ROOT_LEN;  
    const SMMU_LEN: usize = 68;  
    const LEN: usize = SMMU + SMMU_LEN;  
  
    let mut iort = [0u8; LEN];  
    write_sdt_header(&mut iort, Signature::IORT, LEN);  
    iort[36..40].copy_from_slice(&2u32.to_le_bytes());  
    iort[40..44].copy_from_slice(&(ROOT as u32).to_le_bytes());  
  
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/iort.rs (line 1586) -

```
[internal] parses_its_group_and_reserved_memory_descriptors
    fn parses_its_group_and_reserved_memory_descriptors() {
        const ITS: usize = 48;
        const ITS_LEN: usize = 28;
        const RMR: usize = ITS + ITS_LEN;
        const RMR_LEN: usize = 68;
        const LEN: usize = RMR + RMR_LEN;

        let mut iort = [0u8; LEN];
        write_sdt_header(&mut iort, Signature::IORT, LEN);
        iort[36..40].copy_from_slice(&2u32.to_le_bytes());
        iort[40..44].copy_from_slice(&(ITS as u32).to_le_bytes());

        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/smmu3/mod.rs (line 26) -

```
[internal] level_shift

    fn level_shift(level: u8) -> u32 {
        12 + ((level as u32) - 1) * 9
    }

    #[inline]
    fn level_index_bits(level: u8) -> u32 {
        if Self::LEVELS == 5 && level == 5 {
            4
        } else {
            9
        }
    }

    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/smmu3/mod.rs (line 31) -

```
[internal] level_index_bits

    fn level_index_bits(level: u8) -> u32 {
        if Self::LEVELS == 5 && level == 5 {
            4
        } else {
            9
        }
    }

    #[inline]
    fn level_supports_leaf(level: u8, size: PageSize) -> bool {
        matches!(
            (level, size),
            // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/smmu3/mod.rs (line 40) -

```
[internal] level_supports_leaf

    fn level_supports_leaf(level: u8, size: PageSize) -> bool {
        matches!(
            (level, size),
            (1, PageSize::Size4K) | (2, PageSize::Size2M) | (3, PageSize::Size1G)
        )
    }
}

};

}

macro_rules! define_smmu_vmsa_meta_16k {
    ($name:ident, $va_bits:expr, $pa_bits:expr, $root_bits:expr) => {
        // ... (truncated, see source file for more)
```


driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/smmu3/mod.rs (line 66) -
[internal] level_shift

```
fn level_shift(level: u8) -> u32 {
    14 + ((level as u32) - 1) * 11
}

#[inline]
fn level_index_bits(level: u8) -> u32 {
    if level == 4 { $root_bits } else { 11 }
}

#[inline]
fn level_supports_leaf(level: u8, size: PageSize) -> bool {
    matches!(
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/smmu3/mod.rs (line 71) -
[internal] level_index_bits

```
fn level_index_bits(level: u8) -> u32 {
    if level == 4 { $root_bits } else { 11 }
}

#[inline]
fn level_supports_leaf(level: u8, size: PageSize) -> bool {
    matches!(
        (level, size),
        (1, PageSize::Size16K) | (2, PageSize::Size32M) | (3, PageSize::Size64G)
    )
}
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/smmu3/mod.rs (line 76) -
[internal] level_supports_leaf

```
fn level_supports_leaf(level: u8, size: PageSize) -> bool {
    matches!(
        (level, size),
        (1, PageSize::Size16K) | (2, PageSize::Size32M) | (3, PageSize::Size64G)
    )
}
};
```

```
macro_rules! define_smmu_vmsa_meta_64k {
    ($name:ident, $va_bits:expr, $pa_bits:expr, $root_bits:expr) => {
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/smmu3/mod.rs (line 102) -
[internal] level_shift

```
fn level_shift(level: u8) -> u32 {
    16 + ((level as u32) - 1) * 13
}

#[inline]
fn level_index_bits(level: u8) -> u32 {
    if level == 3 { $root_bits } else { 13 }
}

#[inline]
fn level_supports_leaf(level: u8, size: PageSize) -> bool {
    matches!(
// ... (truncated, see source file for more)
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/smmu3/mod.rs (line 107) -
[internal] level_index_bits
    fn level_index_bits(level: u8) -> u32 {
        if level == 3 { $root_bits } else { 13 }
    }

    #[inline]
    fn level_supports_leaf(level: u8, size: PageSize) -> bool {
        matches!(
            (level, size),
            (1, PageSize::Size64K) | (2, PageSize::Size512M) | (3, PageSize::Size4T)
        )
    }
}
// ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/smmu3/mod.rs (line 112) -
[internal] level_supports_leaf
    fn level_supports_leaf(level: u8, size: PageSize) -> bool {
        matches!(
            (level, size),
            (1, PageSize::Size64K) | (2, PageSize::Size512M) | (3, PageSize::Size4T)
        )
    }
}
};
}

define_smmu_vmsa_meta_4k!(SmmuVmsaMeta4K48, 4, 48, 48);
define_smmu_vmsa_meta_4k!(SmmuVmsaMeta4K52, 5, 52, 52);
// ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/smmu3/mod.rs (line 145) -
[internal] smmu_vmsa_uses_iova_addresses
    fn smmu_vmsa_uses_iova_addresses() {
        let iova = IoviAddr::<u64>::from(0x1000);
        assert!(SmmuVmsaMeta4K48::vaddr_is_valid(iova));
    }

    #[test]
    fn root_index_widths_match_granule_and_va_width() {
        assert_eq!(SmmuVmsaMeta16K48::level_index_bits(4), 1);
        assert_eq!(SmmuVmsaMeta16K52::level_index_bits(4), 5);
        assert_eq!(SmmuVmsaMeta64K48::level_index_bits(3), 6);
        assert_eq!(SmmuVmsaMeta64K52::level_index_bits(3), 10);
    }
// ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/aarch64/smmu3/mod.rs (line 151) -
[internal] root_index_widths_match_granule_and_va_width
    fn root_index_widths_match_granule_and_va_width() {
        assert_eq!(SmmuVmsaMeta16K48::level_index_bits(4), 1);
        assert_eq!(SmmuVmsaMeta16K52::level_index_bits(4), 5);
        assert_eq!(SmmuVmsaMeta64K48::level_index_bits(3), 6);
        assert_eq!(SmmuVmsaMeta64K52::level_index_bits(3), 10);
    }
}

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/riscv64/riscv/paging.rs (line 24) -
[internal] level_shift
    fn level_shift(level: u8) -> u32 {
        12 + ((level as u32) - 1) * 9
    }

    #[inline]
    fn level_supports_leaf(level: u8, size: PageSize) -> bool {
        matches!(
            (level, size),
            (1, PageSize::Size4K)
            | (2, PageSize::Size2M)
            | (3, PageSize::Size1G)
            | (4, PageSize::Size512G)
        )
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/riscv64/riscv/paging.rs (line 29) -
[internal] level_supports_leaf
    fn level_supports_leaf(level: u8, size: PageSize) -> bool {
        matches!(
            (level, size),
            (1, PageSize::Size4K)
            | (2, PageSize::Size2M)
            | (3, PageSize::Size1G)
            | (4, PageSize::Size512G)
        ) && (level as usize) <= Self::LEVELS
    }
}
// ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/riscv64/riscv/paging.rs (line 59) -
[internal] riscv_iommu_uses_iova_addresses
    fn riscv_iommu_uses_iova_addresses() {
        let iova = IoviAddr::<u64>::from(0x1000);
        assert!(RvIommuMeta39::vaddr_is_valid(iova));
    }

    #[test]
    fn sv_widths_match_expected_levels() {
        assert_eq!(RvIommuMeta39::LEVELS, 3);
        assert_eq!(RvIommuMeta48::LEVELS, 4);
        assert_eq!(RvIommuMeta57::LEVELS, 5);
    }
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/riscv64/riscv/paging.rs (line 65) -
[internal] sv_widths_match_expected_levels
    fn sv_widths_match_expected_levels() {
        assert_eq!(RvIommuMeta39::LEVELS, 3);
        assert_eq!(RvIommuMeta48::LEVELS, 4);
        assert_eq!(RvIommuMeta57::LEVELS, 5);
    }
}

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 71) -
[public] IvrsError
    pub enum IvrsError {
        Acpi(AcpiError),
        BdfRanges(BdfRangeSetError),
        Mapping(PagingError),
        Malformed(&'static str),
    }

    impl From<AcpiError> for IvrsError {
        #[inline]
        fn from(value: AcpiError) -> Self {
            Self::Acpi(value)
        }
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 80) -
[internal] from
    fn from(value: AcpiError) -> Self {
        Self::Acpi(value)
    }
}

    impl From<BdfRangeSetError> for IvrsError {
        #[inline]
        fn from(value: BdfRangeSetError) -> Self {
            Self::BdfRanges(value)
        }
    }

    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 87) -
[internal] from
    fn from(value: BdfRangeSetError) -> Self {
        Self::BdfRanges(value)
    }
}

    impl From<PagingError> for IvrsError {
        #[inline]
        fn from(value: PagingError) -> Self {
            Self::Mapping(value)
        }
    }

    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 94) -
[internal] from
    fn from(value: PagingError) -> Self {
        Self::Mapping(value)
    }
}

    impl fmt::Display for IvrsError {
        fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
            match self {
                Self::Acpi(error) => write!(f, "{error:?}"),
                Self::BdfRanges(error) => write!(f, "{error}"),
                Self::Mapping(error) => write!(f, "{error:?}"),
                Self::Malformed(message) => f.write_str(message),
            }
            // ... (truncated, see source file for more)

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 100) -
[internal] fmt
    fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
        match self {
            Self::Acpi(error) => write!(f, "{error:?}"),
            Self::BdfRanges(error) => write!(f, "{error}"),
            Self::Mapping(error) => write!(f, "{error:?}"),
            Self::Malformed(message) => f.write_str(message),
        }
    }
}

fn mmio_register_window(
    base: u64,
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 110) -
[internal] mmio_register_window
    fn mmio_register_window(
        base: u64,
        size: usize,
        addr_error: &'static str,
        range_error: &'static str,
    ) -> Result<MmioAddrRange, IvrsError> {
        let base = usize::try_from(base).map_err(|_| IvrsError::Malformed(addr_error))?;
        <MmioAddrRange as MmioRange<usize>>::from_start_size(MmioAddr::from(base), size)
            .ok_or(IvrsError::Malformed(range_error))
    }

    fn phys_range_from_start_size(
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 121) -
[internal] phys_range_from_start_size
    fn phys_range_from_start_size(
        start: u64,
        size: u64,
        addr_error: &'static str,
        range_error: &'static str,
    ) -> Result<PhysAddrRange, IvrsError> {
        let start = usize::try_from(start).map_err(|_| IvrsError::Malformed(addr_error))?;
        let size = usize::try_from(size).map_err(|_| IvrsError::Malformed(addr_error))?;
        if size == 0 {
            return Err(IvrsError::Malformed(range_error));
        }
        PhysAddrRange::try_from_start_size(PhysAddr::from_usize(start), size)
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 138) -
[public] IvrsAcpiTable
    pub struct IvrsAcpiTable {
        pub header: SdtHeader,
    }

    unsafe impl AcpiTable for IvrsAcpiTable {
        const SIGNATURE: Signature = Signature::IVRS;

        #[inline]
        fn header(&self) -> &SdtHeader {
            &self.header
        }
    }

    // ... (truncated, see source file for more)

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 146) -
[internal] header
    fn header(&self) -> &SdtHeader {
        &self.header
    }
}

fn table_bytes<'a, T: AcpiTable>(
    bytes: &'a [u8],
    short_header: &'static str,
    short_table: &'static str,
) -> Result<&'a [u8], IvrsError> {
    if bytes.len() < size_of::<SdtHeader>() {
        return Err(IvrsError::Malformed(short_header));
    }
    // ... (truncated, see source file for more)
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 151) -
[internal] table_bytes
    fn table_bytes<'a, T: AcpiTable>(
        bytes: &'a [u8],
        short_header: &'static str,
        short_table: &'static str,
    ) -> Result<&'a [u8], IvrsError> {
        if bytes.len() < size_of::<SdtHeader>() {
            return Err(IvrsError::Malformed(short_header));
        }

        let header = unsafe { &*bytes.as_ptr().cast::<SdtHeader>() };
        if header.signature != T::SIGNATURE {
            return Err(IvrsError::Acpi(AcpiError::SdtInvalidSignature(
                // ... (truncated, see source file for more)
            )));
        }
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 186) -
[internal] read_array
    fn read_array<const N: usize>(
        bytes: &[u8],
        offset: usize,
        message: &'static str,
    ) -> Result<[u8; N], IvrsError> {
        let end = offset
            .checked_add(N)
            .ok_or(IvrsError::Malformed("IVRS read offset overflow"))?;
        let bytes = bytes
            .get(offset..end)
            .ok_or(IvrsError::Malformed(message))?;
        let mut out = [0; N];
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 203) -
[internal] read_u8
    fn read_u8(bytes: &[u8], offset: usize) -> Result<u8, IvrsError> {
        bytes
            .get(offset)
            .copied()
            .ok_or(IvrsError::Malformed("IVRS table read is out of bounds"))
    }

#[inline]
fn read_u16(bytes: &[u8], offset: usize) -> Result<u16, IvrsError> {
    Ok(u16::from_le_bytes(read_array(
        bytes,
        offset,
        // ... (truncated, see source file for more)
    )))
}

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 211) -
[internal] read_u16
    fn read_u16(bytes: &[u8], offset: usize) -> Result<u16, IvrsError> {
        Ok(u16::from_le_bytes(read_array(
            bytes,
            offset,
            "IVRS table read is out of bounds",
        )))
    }

#[inline]
fn read_u32(bytes: &[u8], offset: usize) -> Result<u32, IvrsError> {
    Ok(u32::from_le_bytes(read_array(
        bytes,
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 220) -
[internal] read_u32
    fn read_u32(bytes: &[u8], offset: usize) -> Result<u32, IvrsError> {
        Ok(u32::from_le_bytes(read_array(
            bytes,
            offset,
            "IVRS table read is out of bounds",
        )))
    }

#[inline]
fn read_u64(bytes: &[u8], offset: usize) -> Result<u64, IvrsError> {
    Ok(u64::from_le_bytes(read_array(
        bytes,
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 229) -
[internal] read_u64
    fn read_u64(bytes: &[u8], offset: usize) -> Result<u64, IvrsError> {
        Ok(u64::from_le_bytes(read_array(
            bytes,
            offset,
            "IVRS table read is out of bounds",
        )))
    }

#[derive(Clone, Copy, Debug)]
pub struct IvrsTable<'a> {
    sdt: &'a [u8],
}
// ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 238) -
[public] IvrsTable
pub struct IvrsTable<'a> {
    sdt: &'a [u8],
}

impl<'a> IvrsTable<'a> {
    fn parse(bytes: &'a [u8]) -> Result<Self, IvrsError> {
        let sdt = table_bytes::<IvrsAcpiTable>(
            bytes,
            "IVRS table shorter than SDT header",
            "IVRS table length is invalid",
        );
        if sdt.len() < IVRS_BLOCKS_OFFSET {
            // ... (truncated, see source file for more)

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 243) -
[internal] parse

```
fn parse(bytes: &'a [u8]) -> Result<Self, IvrsError> {
    let sdt = table_bytes::<IvrsAcpiTable>(
        bytes,
        "IVRS table shorter than SDT header",
        "IVRS table length is invalid",
    )?;
    if sdt.len() < IVRS_BLOCKS_OFFSET {
        return Err(IvrsError::Malformed("IVRS table shorter than fixed header"));
    }
    Ok(Self { sdt })
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 278) -
[public] ivinfo

```
pub fn ivinfo(self) -> Result<u32, IvrsError> {
    read_u32(self.sdt, IVRS_IVINFO_OFFSET)
}

#[inline]
pub fn blocks(self) -> IvrsBlocks<'a> {
    IvrsBlocks {
        table: self,
        offset: IVRS_BLOCKS_OFFSET,
    }
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 283) -
[public] blocks

```
pub fn blocks(self) -> IvrsBlocks<'a> {
    IvrsBlocks {
        table: self,
        offset: IVRS_BLOCKS_OFFSET,
    }
}

pub fn for_each_ivhd<F>(self, mut f: F) -> Result<(), IvrsError>
where
    F: FnMut(u32, Ivhd) -> Result<(), IvrsError>,
{
    let ivinfo = self.ivinfo()?;
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 290) -
[public] for_each_ivhd

```
pub fn for_each_ivhd<F>(self, mut f: F) -> Result<(), IvrsError>
where
    F: FnMut(u32, Ivhd) -> Result<(), IvrsError>,
{
    let ivinfo = self.ivinfo()?;
    self.for_each_block(|block| {
        if let IvrsBlock::Ivhd(unit) = block {
            f(ivinfo, unit)?;
        }
    })
    Ok(())
}

// ... (truncated, see source file for more)
```


driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 303) -
[public] for_each_ivmd

```
pub fn for_each_ivmd<F>(self, mut f: F) -> Result<(), IvrsError>
where
    F: FnMut(Ivmd) -> Result<(), IvrsError>,
{
    self.for_each_block(|block| {
        if let IvrsBlock::Ivmd(memory) = block {
            f(memory)?;
        }
        Ok(())
    })
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 315) -
[public] ivhd_bdf_ranges

```
pub fn ivhd_bdf_ranges(
    self,
    registers: MmioAddrRange,
) -> Result<Option<IvrsBdfRangeSet>, IvrsError> {
    let mut found_unit = false;
    let mut ranges = IvrsBdfRangeSet::empty();
    let mut saw_all = false;

    self.for_each_block(|block| {
        let IvrsBlock::Ivhd(unit) = block else {
            return Ok(());
        };
    });
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 370) -
[internal] for_each_block

```
fn for_each_block<F>(self, mut f: F) -> Result<(), IvrsError>
where
    F: FnMut(IvrsBlock) -> Result<(), IvrsError>,
{
    for block in self.blocks() {
        f(block)?;
    }
    Ok(())
}

fn parse_block(self, offset: usize, length: usize, kind: u8) -> Result<IvrsBlock, IvrsError> {
    match kind {
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 380) -
[internal] parse_block

```
fn parse_block(self, offset: usize, length: usize, kind: u8) -> Result<IvrsBlock, IvrsError> {
    match kind {
        IVHD_TYPE_10H | IVHD_TYPE_11H | IVHD_TYPE_40H => {
            Ok(IvrsBlock::Ivhd(self.parse_ivhd(offset, length, kind)))
        }
        IVMD_TYPE_ALL | IVMD_TYPE_SPECIFIED | IVMD_TYPE_RANGE => {
            Ok(IvrsBlock::Ivmd(self.parse_ivmd(offset, length, kind)))
        }
        other => Ok(IvrsBlock::Unknown(IvrsUnknown {
            kind: other,
            offset,
            length,
        }
    ));
    // ... (truncated, see source file for more)
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 396) -
[internal] parse_ivhd
    fn parse_ivhd(self, offset: usize, length: usize, block_type: u8) -> Result<Ivhd, IvrsError> {
        let min_length = match block_type {
            IVHD_TYPE_10H => IVHD_10H_MIN_LENGTH,
            IVHD_TYPE_11H | IVHD_TYPE_40H => IVHD_11H_MIN_LENGTH,
            _ => return Err(IvrsError::Malformed("unsupported IVHD block type")),
        };
        if length < min_length {
            return Err(IvrsError::Malformed(
                "IVHD block shorter than minimum for its type",
            ));
        }

        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 461) -
[internal] parse_ivmd
    fn parse_ivmd(self, offset: usize, length: usize, block_type: u8) -> Result<Ivmd, IvrsError> {
        if length < IVMD_MIN_LENGTH {
            return Err(IvrsError::Malformed("IVMD block too short"));
        }
        Ok(Ivmd {
            ivmd_type: block_type,
            flags: read_u8(self.sdt, offset + IVMD_FLAGS_OFFSET)?,
            device_id: Bdf::from_u16(read_u16(self.sdt, offset + IVMD_DEVICE_ID_OFFSET)?),
            aux_device_id: Bdf::from_u16(read_u16(self.sdt, offset + IVMD_AUX_OFFSET)?),
            memory: phys_range_from_start_size(
                read_u64(self.sdt, offset + IVMD_START_ADDR_OFFSET)?,
                read_u64(self.sdt, offset + IVMD_MEM_LENGTH_OFFSET)?,
            ),
        })
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 479) -
[internal] walk_device_entries
    fn walk_device_entries<F>(
        self,
        block_offset: usize,
        entries_offset: usize,
        block_length: usize,
        mut f: F,
    ) -> Result<(), IvrsError>
    where
        F: FnMut(IvhdDeviceEntry<'a>) -> Result<(), IvrsError>,
    {
        let mut offset = entries_offset;
        let mut pending_range_start = None;
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 621) -
[public] IvrsBlocks
    pub struct IvrsBlocks<'a> {
        table: IvrsTable<'a>,
        offset: usize,
    }

    impl Iterator for IvrsBlocks<'_> {
        type Item = Result<IvrsBlock, IvrsError>;

        fn next(&mut self) -> Option<Self::Item> {
            if self.offset >= self.table.sdt.len() {
                return None;
            }
            // ... (truncated, see source file for more)

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 629) -
[internal] next

```
fn next(&mut self) -> Option<Self::Item> {
    if self.offset >= self.table.sdt.len() {
        return None;
    }
    if self.offset + 4 > self.table.sdt.len() {
        self.offset = self.table.sdt.len();
        return Some(Err(IvrsError::Malformed(
            "IVRS block truncated before header",
        )));
    }

    let offset = self.offset;
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 668) -
[public] IvrsBlock

```
pub enum IvrsBlock {
    Ivhd(Ivhd),
    Ivmd(Ivmd),
    Unknown(IvrsUnknown),
}

#[derive(Clone, Copy, Debug, PartialEq, Eq)]
pub struct IvrsUnknown {
    pub kind: u8,
    pub offset: usize,
    pub length: usize,
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 675) -
[public] IvrsUnknown

```
pub struct IvrsUnknown {
    pub kind: u8,
    pub offset: usize,
    pub length: usize,
}

#[derive(Clone, Copy, Debug, PartialEq, Eq)]
pub struct Ivhd {
    pub block_type: u8,
    pub flags: u8,
    pub registers: MmioAddrRange,
    pub device: PciDevice,
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 682) -
[public] Ivhd

```
pub struct Ivhd {
    pub block_type: u8,
    pub flags: u8,
    pub registers: MmioAddrRange,
    pub device: PciDevice,
    pub capability_offset: u16,
    pub iommu_info: u16,
    pub feature_info: Option<u32>,
    pub efr: Option<u64>,
    pub include_all: bool,
    pub has_device_scopes: bool,
    pub block_offset: usize,
    // ... (truncated, see source file for more)
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 730) -
[public] Ivmd
    pub struct Ivmd {
        pub ivmd_type: u8,
        pub flags: u8,
        pub device_id: Bdf,
        pub aux_device_id: Bdf,
        pub memory: PhysAddrRange,
    }

    impl Ivmd {
        #[inline]
        pub const fn is_all_devices(self) -> bool {
            self.ivmd_type == IVMD_TYPE_ALL
        }
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 760) -
[public] limit_address
    pub fn limit_address(self) -> PhysAddr {
        PhysAddr::from_usize(self.memory.end.as_usize() - 1)
    }
}

#[derive(Clone, Copy, Debug, PartialEq, Eq)]
struct PendingIvhdRange {
    start: Bdf,
    kind: PendingIvhdRangeKind,
    data_setting: u8,
}

// ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 766) -
[internal] PendingIvhdRange
    struct PendingIvhdRange {
        start: Bdf,
        kind: PendingIvhdRangeKind,
        data_setting: u8,
    }

    #[derive(Clone, Copy, Debug, PartialEq, Eq)]
    enum PendingIvhdRangeKind {
        Plain,
        Alias { source: Bdf },
        Extended { extended_data_setting: u32 },
    }

    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 780) -
[public] IvhdDeviceEntry
    pub enum IvhdDeviceEntry<'a> {
        Pad,
        All {
            data_setting: u8,
        },
        Select {
            device_id: Bdf,
            data_setting: u8,
        },
        Range {
            start: Bdf,
            end: Bdf,
        }
        // ... (truncated, see source file for more)
    }

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 850) -
[internal] TestEntry
    struct TestEntry;

    impl PageTableEntry for TestEntry {
        type Flags = ();

        fn new_leaf(_paddr: PhysAddr, _flags: Self::Flags, _size: PageSize) -> Self {
            Self
        }

        fn new_table(_paddr: PhysAddr, _level: u8) -> Self {
            Self
        }
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 855) -
[internal] new_leaf
    fn new_leaf(_paddr: PhysAddr, _flags: Self::Flags, _size: PageSize) -> Self {
        Self
    }

    fn new_table(_paddr: PhysAddr, _level: u8) -> Self {
        Self
    }

    fn paddr(&self) -> PhysAddr {
        PhysAddr::from_usize(0)
    }

    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 859) -
[internal] new_table
    fn new_table(_paddr: PhysAddr, _level: u8) -> Self {
        Self
    }

    fn paddr(&self) -> PhysAddr {
        PhysAddr::from_usize(0)
    }

    fn flags(&self) -> Self::Flags {}

    fn is_present(&self) -> bool {
        true
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 863) -
[internal] paddr
    fn paddr(&self) -> PhysAddr {
        PhysAddr::from_usize(0)
    }

    fn flags(&self) -> Self::Flags {}

    fn is_present(&self) -> bool {
        true
    }

    fn entry_kind(&self, _level: u8) -> PageTableEntryKind {
        PageTableEntryKind::Leaf
    }
    // ... (truncated, see source file for more)

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 867) -
[internal] flags
    fn flags(&self) -> Self::Flags {}

    fn is_present(&self) -> bool {
        true
    }

    fn entry_kind(&self, _level: u8) -> PageTableEntryKind {
        PageTableEntryKind::Leaf
    }

    fn clear(&mut self) {}

    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 869) -
[internal] is_present
    fn is_present(&self) -> bool {
        true
    }

    fn entry_kind(&self, _level: u8) -> PageTableEntryKind {
        PageTableEntryKind::Leaf
    }

    fn clear(&mut self) {}

    fn bits(&self) -> u64 {
        0
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 873) -
[internal] entry_kind
    fn entry_kind(&self, _level: u8) -> PageTableEntryKind {
        PageTableEntryKind::Leaf
    }

    fn clear(&mut self) {}

    fn bits(&self) -> u64 {
        0
    }

    fn from_bits(_bits: u64) -> Self {
        Self
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 877) -
[internal] clear
    fn clear(&mut self) {}

    fn bits(&self) -> u64 {
        0
    }

    fn from_bits(_bits: u64) -> Self {
        Self
    }
}

fn registers(base: usize) -> MmioAddrRange {
    // ... (truncated, see source file for more)

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 879) -
[internal] bits
    fn bits(&self) -> u64 {
        0
    }

    fn from_bits(_bits: u64) -> Self {
        Self
    }
}

fn registers(base: usize) -> MmioAddrRange {
    <MmioAddrRange as MmioRange<usize>>::from_start_size(
        MmioAddr::from(base),
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 883) -
[internal] from_bits
    fn from_bits(_bits: u64) -> Self {
        Self
    }
}

fn registers(base: usize) -> MmioAddrRange {
    <MmioAddrRange as MmioRange<usize>>::from_start_size(
        MmioAddr::from(base),
        super::AMD_VI_REGISTER_WINDOW_SIZE,
    )
    .unwrap()
}
// ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 888) -
[internal] registers
    fn registers(base: usize) -> MmioAddrRange {
        <MmioAddrRange as MmioRange<usize>>::from_start_size(
            MmioAddr::from(base),
            super::AMD_VI_REGISTER_WINDOW_SIZE,
        )
        .unwrap()
    }

    fn write_sdt_header(bytes: &mut [u8], signature: Signature, length: usize) {
        bytes[0..4].copy_from_slice(signature.as_str().as_bytes());
        bytes[4..8].copy_from_slice(&(length as u32).to_le_bytes());
        bytes[8] = 1;
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 896) -
[internal] write_sdt_header
    fn write_sdt_header(bytes: &mut [u8], signature: Signature, length: usize) {
        bytes[0..4].copy_from_slice(signature.as_str().as_bytes());
        bytes[4..8].copy_from_slice(&(length as u32).to_le_bytes());
        bytes[8] = 1;
    }

    fn finish_sdt_checksum(bytes: &mut [u8]) {
        bytes[9] = 0;
        bytes[9] = 0u8.wrapping_sub(
            bytes
                .iter()
                .fold(0, |sum: u8, byte| sum.wrapping_add(*byte)),
    // ... (truncated, see source file for more)

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 902) -
[internal] finish_sdt_checksum
    fn finish_sdt_checksum(bytes: &mut [u8]) {
        bytes[9] = 0;
        bytes[9] = 0u8.wrapping_sub(
            bytes
                .iter()
                .fold(0, |sum: u8, byte| sum.wrapping_add(*byte)),
        );
    }

    fn table_mapping(bytes: &[u8]) -> Mapping<TestEntry, VirtAddr> {
        let start = VirtAddr::from_usize(bytes.as_ptr() as usize);
        Mapping::new(
            // ... (truncated, see source file for more)
        )
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 911) -
[internal] table_mapping
    fn table_mapping(bytes: &[u8]) -> Mapping<TestEntry, VirtAddr> {
        let start = VirtAddr::from_usize(bytes.as_ptr() as usize);
        Mapping::new(
            VirtAddrRange::from_start_size(start, bytes.len()),
            PhysAddr::from_usize(0),
            (),
        )
    }

    #[test]
    fn parses_ivhd_and_bdf_ranges() {
        let mut ivrs = [0u8; 80];
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 921) -
[internal] parses_ivhd_and_bdf_ranges
    fn parses_ivhd_and_bdf_ranges() {
        let mut ivrs = [0u8; 80];
        write_sdt_header(&mut ivrs, Signature::IVRS, 80);
        ivrs[36..40].copy_from_slice(&0x1234u32.to_le_bytes());

        ivrs[48] = 0x10;
        ivrs[49] = 1 << 5;
        ivrs[50..52].copy_from_slice(&32u16.to_le_bytes());
        ivrs[52..54].copy_from_slice(&0x0100u16.to_le_bytes());
        ivrs[54..56].copy_from_slice(&0x40u16.to_le_bytes());
        ivrs[56..64].copy_from_slice(&(0xfed8_0000u64).to_le_bytes());
        ivrs[64..66].copy_from_slice(&2u16.to_le_bytes());
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 963) -
[internal] parses_ivmd_reserved_memory
    fn parses_ivmd_reserved_memory() {
        let mut ivrs = [0u8; 72];
        write_sdt_header(&mut ivrs, Signature::IVRS, 72);
        ivrs[48] = 0x21;
        ivrs[50..52].copy_from_slice(&24u16.to_le_bytes());
        ivrs[52..54].copy_from_slice(&0x320u16.to_le_bytes());
        ivrs[56..64].copy_from_slice(&(0x2000_0000u64).to_le_bytes());
        ivrs[64..72].copy_from_slice(&(0x2000u64).to_le_bytes());

        finish_sdt_checksum(&mut ivrs);

        let mapping = table_mapping(&ivrs);
        // ... (truncated, see source file for more)
    }

```



```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/ivrs.rs (line 991) -
[internal] parses_extended_and_acpi_named_ivhd_entries
    fn parses_extended_and_acpi_named_ivhd_entries() {
        let mut ivrs = [0u8; 122];
        write_sdt_header(&mut ivrs, Signature::IVRS, 122);

        ivrs[48] = 0x11;
        ivrs[49] = 1 << 5;
        ivrs[50..52].copy_from_slice(&74u16.to_le_bytes());
        ivrs[52..54].copy_from_slice(&0x0100u16.to_le_bytes());
        ivrs[54..56].copy_from_slice(&0x40u16.to_le_bytes());
        ivrs[56..64].copy_from_slice(&(0xfed8_0000u64).to_le_bytes());
        ivrs[64..66].copy_from_slice(&2u16.to_le_bytes());
        ivrs[66..68].copy_from_slice(&0x55u16.to_le_bytes());
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/paging.rs (line 36) -
[public] AmdViFlags
    pub struct AmdViFlags {
        pub access: AccessFlags,
        pub attrs: MemoryAttributes,
    }

    impl AmdViFlags {
        #[inline]
        pub const fn new(access: AccessFlags) -> Self {
            Self {
                access,
                attrs: MemoryAttributes::writeback(),
            }
        }
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/paging.rs (line 63) -
[internal] to_leaf_bits
    fn to_leaf_bits(self, size: PageSize) -> u64 {
        let mut bits = PRESENT;
        if self.access.contains(AccessFlags::READ) {
            bits |= IO_READ;
        }
        if self.access.contains(AccessFlags::WRITE) {
            bits |= IO_WRITE;
        }
        if !matches!(size, PageSize::Size4K) {
            bits |= level_enc(LARGE_LEAF_LEVEL);
        }
        bits
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/paging.rs (line 78) -
[internal] from_bits
    fn from_bits(bits: u64) -> Self {
        let mut access = AccessFlags::empty();
        if (bits & IO_READ) != 0 {
            access |= AccessFlags::READ;
        }
        if (bits & IO_WRITE) != 0 {
            access |= AccessFlags::WRITE;
        }
        Self {
            access,
            attrs: MemoryAttributes::writeback(),
        }
        // ... (truncated, see source file for more)
    }

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/paging.rs (line 96) -
[public] AmdViPte
    pub struct AmdViPte(u64);

    impl fmt::Debug for AmdViPte {
        fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
            f.debug_tuple("AmdViPte")
                .field(&format_args!("{:#018x}", self.0))
                .finish()
            }
        }

    impl PageTableEntry for AmdViPte {
        type Flags = AmdViFlags;
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/paging.rs (line 99) -
[internal] fmt
    fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
        f.debug_tuple("AmdViPte")
            .field(&format_args!("{:#018x}", self.0))
            .finish()
        }
    }

    impl PageTableEntry for AmdViPte {
        type Flags = AmdViFlags;

        #[inline]
        fn new_leaf(paddr: PhysAddr, flags: Self::Flags, size: PageSize) -> Self {
            // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/paging.rs (line 110) -
[internal] new_leaf
    fn new_leaf(paddr: PhysAddr, flags: Self::Flags, size: PageSize) -> Self {
        Self(((paddr.as_usize() as u64) & ADDR_MASK) | flags.to_leaf_bits(size))
    }

    #[inline]
    fn new_table(paddr: PhysAddr, level: u8) -> Self {
        let child_level = level.saturating_sub(1);
        Self(
            ((paddr.as_usize() as u64) & ADDR_MASK)
                | PRESENT
                | IO_READ
                | IO_WRITE
            // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/paging.rs (line 115) -
[internal] new_table
    fn new_table(paddr: PhysAddr, level: u8) -> Self {
        let child_level = level.saturating_sub(1);
        Self(
            ((paddr.as_usize() as u64) & ADDR_MASK)
                | PRESENT
                | IO_READ
                | IO_WRITE
                | level_enc(child_level),
            )
        }
    }

    #[inline]
    // ... (truncated, see source file for more)

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/paging.rs (line 127) -
[internal] paddr
    fn paddr(&self) -> PhysAddr {
        PhysAddr::from_usize((self.0 & ADDR_MASK) as usize)
    }

    #[inline]
    fn flags(&self) -> Self::Flags {
        AmdViFlags::from_bits(self.0)
    }

    #[inline]
    fn is_present(&self) -> bool {
        (self.0 & PRESENT) != 0
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/paging.rs (line 132) -
[internal] flags
    fn flags(&self) -> Self::Flags {
        AmdViFlags::from_bits(self.0)
    }

    #[inline]
    fn is_present(&self) -> bool {
        (self.0 & PRESENT) != 0
    }

    #[inline]
    fn entry_kind(&self, level: u8) -> PageTableEntryKind {
        if !self.is_present() {
            // ... (truncated, see source file for more)
        }

        let next = next_level(self.0);
        if level == 1 {
            // ... (truncated, see source file for more)
        }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/paging.rs (line 137) -
[internal] is_present
    fn is_present(&self) -> bool {
        (self.0 & PRESENT) != 0
    }

    #[inline]
    fn entry_kind(&self, level: u8) -> PageTableEntryKind {
        if !self.is_present() {
            return PageTableEntryKind::Invalid;
        }

        let next = next_level(self.0);
        if level == 1 {
            PageTableEntryKind::Leaf
        } else if next == level.saturating_sub(1) {
            PageTableEntryKind::Table
        } else if next == 0 || next == LARGE_LEAF_LEVEL {
            PageTableEntryKind::Leaf
        }
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/paging.rs (line 142) -
[internal] entry_kind
    fn entry_kind(&self, level: u8) -> PageTableEntryKind {
        if !self.is_present() {
            return PageTableEntryKind::Invalid;
        }

        let next = next_level(self.0);
        if level == 1 {
            PageTableEntryKind::Leaf
        } else if next == level.saturating_sub(1) {
            PageTableEntryKind::Table
        } else if next == 0 || next == LARGE_LEAF_LEVEL {
            PageTableEntryKind::Leaf
        }
        // ... (truncated, see source file for more)

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/paging.rs (line 160) -

```
[internal] clear
    fn clear(&mut self) {
        self.0 = 0;
    }

    #[inline]
    fn bits(&self) -> u64 {
        self.0
    }

    #[inline]
    fn from_bits(bits: u64) -> Self {
        Self(bits)
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/paging.rs (line 165) -

```
[internal] bits
    fn bits(&self) -> u64 {
        self.0
    }

    #[inline]
    fn from_bits(bits: u64) -> Self {
        Self(bits)
    }
}

macro_rules! define_aml_vi_meta {
    ($name:ident, $levels:expr, $va_bits:expr) => {
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/paging.rs (line 170) -

```
[internal] from_bits
    fn from_bits(bits: u64) -> Self {
        Self(bits)
    }
}

macro_rules! define_aml_vi_meta {
    ($name:ident, $levels:expr, $va_bits:expr) => {
        #[derive(Clone, Copy, Debug)]
        pub struct $name;

        impl PagingMetaData for $name {
            const LEVELS: usize = $levels;
        }
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/paging.rs (line 188) -

```
[internal] level_shift
    fn level_shift(level: u8) -> u32 {
        12 + ((level as u32) - 1) * 9
    }

    #[inline]
    fn level_supports_leaf(level: u8, size: PageSize) -> bool {
        matches!(
            (level, size),
            (1, PageSize::Size4K) | (2, PageSize::Size2M) | (3, PageSize::Size1G)
        )
    }
}
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/paging.rs (line 193) -

```
[internal] level_supports_leaf
    fn level_supports_leaf(level: u8, size: PageSize) -> bool {
        matches!(
            (level, size),
            (1, PageSize::Size4K) | (2, PageSize::Size2M) | (3, PageSize::Size1G)
        )
    }
};

define_aml_vi_meta!(AmlViMeta39, 3, 39);
define_aml_vi_meta!(AmlViMeta48, 4, 48);
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/paging.rs (line 216) -

```
[internal] table_entry_encodes_child_level
    fn table_entry_encodes_child_level() {
        let pte = AmlViPte::new_table(PhysAddr::from_usize(0x8000), 4);
        assert_eq!(next_level(pte.bits()), 3);
        assert_eq!(pte.entry_kind(4), PageTableEntryKind::Table);
        assert_eq!(pte.paddr(), PhysAddr::from_usize(0x8000));
    }

#[test]
fn large_leaf_uses_large_level_marker() {
    let flags = AmlViFlags::new(AccessFlags::READ | AccessFlags::WRITE);
    let pte = AmlViPte::new_leaf(PhysAddr::from_usize(0x4000_0000), flags, PageSize::Size1G);

    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/amd/paging.rs (line 224) -

```
[internal] large_leaf_uses_large_level_marker
    fn large_leaf_uses_large_level_marker() {
        let flags = AmlViFlags::new(AccessFlags::READ | AccessFlags::WRITE);
        let pte = AmlViPte::new_leaf(PhysAddr::from_usize(0x4000_0000), flags, PageSize::Size1G);

        assert_eq!(next_level(pte.bits()), LARGE_LEAF_LEVEL);
        assert_eq!(pte.entry_kind(3), PageTableEntryKind::Leaf);
        assert!(pte.flags().access.contains(AccessFlags::READ));
        assert!(pte.flags().access.contains(AccessFlags::WRITE));
    }
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/caps.rs (line 63) -

```
[public] VtdCapability
    pub struct VtdCapability(u64);

impl VtdCapability {
    #[inline]
    pub const fn from_bits(bits: u64) -> Self {
        Self(bits)
    }

    #[inline]
    pub const fn bits(self) -> u64 {
        self.0
    }
    // ... (truncated, see source file for more)
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/caps.rs (line 188) -
[internal] from
    fn from(value: u64) -> Self {
        Self::from_bits(value)
    }
}

impl From<VtdCapability> for u64 {
    #[inline]
    fn from(value: VtdCapability) -> Self {
        value.bits()
    }
}

// ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/caps.rs (line 195) -
[internal] from
    fn from(value: VtdCapability) -> Self {
        value.bits()
    }
}

/// Raw VT-d Extended Capability register.
#[derive(Clone, Copy, Debug, Default, PartialEq, Eq, Hash)]
#[repr(transparent)]
pub struct VtdExtendedCapability(u64);

impl VtdExtendedCapability {
    #[inline]
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/caps.rs (line 203) -
[public] VtdExtendedCapability
    pub struct VtdExtendedCapability(u64);

    impl VtdExtendedCapability {
        #[inline]
        pub const fn from_bits(bits: u64) -> Self {
            Self(bits)
        }

        #[inline]
        pub const fn bits(self) -> u64 {
            self.0
        }
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/caps.rs (line 337) -
[public] capability_flags
    pub fn capability_flags(self) -> CapabilityFlags {
        let mut flags = CapabilityFlags::TRANSLATION;

        if self.pasid() {
            flags |= CapabilityFlags::PASID;
        }
        if self.device_tlb() {
            flags |= CapabilityFlags::DEVICE_TLB;
        }
        if self.interrupt_remapping() {
            flags |= CapabilityFlags::INTERRUPT_REMAPPING;
        }
        // ... (truncated, see source file for more)

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/caps.rs (line 365) -
[internal] from
    fn from(value: u64) -> Self {
        Self::from_bits(value)
    }
}

impl From<VtdExtendedCapability> for u64 {
    #[inline]
    fn from(value: VtdExtendedCapability) -> Self {
        value.bits()
    }
}

// ... (truncated, see source file for more)

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/caps.rs (line 372) -
[internal] from
    fn from(value: VtdExtendedCapability) -> Self {
        value.bits()
    }
}

#[cfg(test)]
mod tests {
    use super::*;

    #[test]
    fn cap_decodes_address_widths_and_superpages() {
        let cap = VtdCapability::from_bits(
            // ... (truncated, see source file for more)

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/caps.rs (line 382) -
[internal] cap_decodes_address_widths_and_superpages
    fn cap_decodes_address_widths_and_superpages() {
        let cap = VtdCapability::from_bits(
            (0b111_u64 << CAP_SAGAW_SHIFT)
            | (47_u64 << CAP_MGAW_SHIFT)
            | (0b0011_u64 << CAP_SLLPS_SHIFT),
        );

        assert_eq!(cap.max_guest_address_width(), 48);
        assert!(cap.supports_address_width(VtdSecondLevelAddressWidth::Bits39));
        assert!(cap.supports_address_width(VtdSecondLevelAddressWidth::Bits48));
        assert!(cap.supports_address_width(VtdSecondLevelAddressWidth::Bits57));
        assert_eq!(
            // ... (truncated, see source file for more)

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/caps.rs (line 402) -
[internal] ecap_lifts_common_flags
    fn ecap_lifts_common_flags() {
        let ecap = VtdExtendedCapability::from_bits(ECAP_QI | ECAP_DT | ECAP_IR | ECAP_PASID);
        let flags = ecap.capability_flags();

        assert!(ecap.queued_invalidation());
        assert!(flags.contains(CapabilityFlags::TRANSLATION));
        assert!(flags.contains(CapabilityFlags::PASID));
        assert!(flags.contains(CapabilityFlags::DEVICE_TLB));
        assert!(flags.contains(CapabilityFlags::INTERRUPT_REMAPPING));
    }
}

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 161) -
[public] VtdRootTableBacking
    pub struct VtdRootTableBacking {
        backing: DescriptorTableBacking<ROOT_ENTRY_BYTES>,
    }

    impl VtdRootTableBacking {
        /// # Safety
        ///
        /// `virt` must be a live writable mapping of `phys` for one root-entry
        /// table page, and the memory must remain owned by the VT-d unit while
        /// requester lookup can consult it.
        #[inline]
        pub unsafe fn new(phys: PhysAddrRange, virt: VirtAddrRange) -> Result<Self> {
            // ... (truncated, see source file for more)
        }
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 215) -
[internal] entry_vaddr
    fn entry_vaddr(self, bus: u8) -> Result<VirtAddr> {
        self.backing.entry_vaddr(Self::bus_entry_index(bus))
    }
}

/// Caller-provided backing for one VT-d context-entry table.
///
/// A context table is one 4 KiB page containing 256 128-bit entries, one per
/// device/function slot under a single PCI bus. Context entries select the
/// translation target for a PCI requester; they are not IOVA page-table leaves.
#[derive(Clone, Copy, Debug, PartialEq, Eq)]
pub struct VtdContextTableBacking {
    // ... (truncated, see source file for more)
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 226) -
[public] VtdContextTableBacking
    pub struct VtdContextTableBacking {
        backing: DescriptorTableBacking<CONTEXT_ENTRY_BYTES>,
    }

    impl VtdContextTableBacking {
        /// # Safety
        ///
        /// `virt` must be a live writable mapping of `phys` for one context-entry
        /// table page, and the memory must remain owned by the VT-d unit while
        /// requester lookup can consult it.
        #[inline]
        pub unsafe fn new(phys: PhysAddrRange, virt: VirtAddrRange) -> Result<Self> {
            // ... (truncated, see source file for more)
        }
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 280) -
[internal] entry_vaddr
    fn entry_vaddr(self, index: u8) -> Result<VirtAddr> {
        self.backing.entry_vaddr(index as usize)
    }
}

#[inline]
fn client_entry_vaddr(self, client: PciDevice) -> Result<VirtAddr> {
    self.entry_vaddr(vtd_context_entry_index(client))
}
}

/// VT-d root-entry descriptor for routing a PCI bus to a context table.
#[derive(Clone, Copy, Debug, Default, PartialEq, Eq, Hash)]
// ... (truncated, see source file for more)

```



```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 285) -
[internal] client_entry_vaddr
    fn client_entry_vaddr(self, client: PciDevice) -> Result<VirtAddr> {
        self.entry_vaddr(vtd_context_entry_index(client))
    }
}

/// VT-d root-entry descriptor for routing a PCI bus to a context table.
#[derive(Clone, Copy, Debug, Default, PartialEq, Eq, Hash)]
#[repr(C)]
pub struct VtdRootEntry {
    low: u64,
    high: u64,
}
// ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 293) -
[public] VtdRootEntry
    pub struct VtdRootEntry {
        low: u64,
        high: u64,
    }

    impl VtdRootEntry {
        #[inline]
        pub const fn from_words(low: u64, high: u64) -> Self {
            Self { low, high }
        }

        #[inline]
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 310) -
[public] from_context_table
    pub fn from_context_table(table: VtdContextTableBacking) -> Result<Self> {
        let root = table.phys().start;
        if !root.is_aligned(PageSize::Size4K.bytes()) {
            return Err(Error::InvalidAddress);
        }
        Ok(Self::from_words(
            ((root.as_usize() as u64) & TABLE_ADDR_MASK) | ROOT_ENTRY_PRESENT,
            0,
        ))
    }

    #[inline]
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 337) -
[public] context_table_root
    pub fn context_table_root(self) -> Option<PhysAddr> {
        self.present()
            .then(|| PhysAddr::from_usize((self.low & TABLE_ADDR_MASK) as usize))
    }
}

/// VT-d legacy context-entry descriptor for routing one requester.
///
/// A context entry either aborts the requester, lets it pass through when
/// hardware supports that mode, or points at a second-level DMA translation
/// root represented by [VtdDomain].
#[derive(Clone, Copy, Debug, Default, PartialEq, Eq, Hash)]
// ... (truncated, see source file for more)

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 350) -

```
[public] VtdContextEntry
    pub struct VtdContextEntry {
        low: u64,
        high: u64,
    }

    impl VtdContextEntry {
        #[inline]
        pub const fn from_words(low: u64, high: u64) -> Self {
            Self { low, high }
        }

        #[inline]
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 367) -

```
[public] from_domain
    pub fn from_domain(domain: VtdDomain) -> Result<Self> {
        let root = domain.root();
        if !root.is_aligned(PageSize::Size4K.bytes()) {
            return Err(Error::InvalidAddress);
        }

        Ok(Self::from_words(
            ((root.as_usize() as u64) & TABLE_ADDR_MASK)
            | CONTEXT_PRESENT
            | (CONTEXT_TRANSLATION_TYPE_MULTI_LEVEL << CONTEXT_TRANSLATION_TYPE_SHIFT),
            vtd_agaw_bits(domain.width()) | ((domain.id().id() as u64) << CONTEXT_DOMAIN_ID_SHIFT),
        ))
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 391) -

```
[public] from_binding
    pub fn from_binding(
        binding: Binding<PciDevice, VtdDomain>,
        cap: VtdExtendedCapability,
    ) -> Result<Self> {
        if binding.selector() != BindingSelector::Default {
            return Err(Error::FeatureUnavailable);
        }

        match binding.target() {
            BindingTarget::Abort => Ok(Self::disabled()),
            BindingTarget::PassThrough => {
                if cap.pass_through() {
                    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 448) -

```
[public] page_table_root
    pub fn page_table_root(self) -> Option<PhysAddr> {
        (self.present() && self.translation_type() == CONTEXT_TRANSLATION_TYPE_MULTI_LEVEL as u8)
        .then(|| PhysAddr::from_usize((self.low & TABLE_ADDR_MASK) as usize))
    }
}

/// Caller-provided backing for a VT-d interrupt-remapping table.
///
/// The backing must contain `entry_count` contiguous 128-bit IRTE slots and
/// must remain writable while interrupt remapping can consult it.
#[derive(Clone, Copy, Debug, PartialEq, Eq)]
pub struct VtdInterruptRemapTableBacking {
    // ... (truncated, see source file for more)
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 459) -
[public] VtdInterruptRemapTableBacking
    pub struct VtdInterruptRemapTableBacking {
        backing: DescriptorTableBacking<INTERRUPT_REMAP_ENTRY_BYTES>,
    }

    impl VtdInterruptRemapTableBacking {
        /// # Safety
        ///
        /// `virt` must be a live writable mapping of `phys` for at least
        /// `entry_count * 16` bytes, and the table memory must stay owned by the
        /// VT-d unit while interrupt remapping is enabled.
        #[inline]
        pub unsafe fn new(
            // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 513) -
[public] byte_len
    pub fn byte_len(self) -> usize {
        self.backing.byte_len()
    }

    #[inline]
    fn irta_size(self) -> u64 {
        u64::from(self.entry_count().trailing_zeros() - 1) & IRTA_SIZE_MASK
    }

    #[inline]
    fn entry_vaddr(self, entry: VtdInterruptEntry) -> Result<VirtAddr> {
        let index = usize::from(entry.as_u16());
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 518) -
[internal] irta_size
    fn irta_size(self) -> u64 {
        u64::from(self.entry_count().trailing_zeros() - 1) & IRTA_SIZE_MASK
    }

    #[inline]
    fn entry_vaddr(self, entry: VtdInterruptEntry) -> Result<VirtAddr> {
        let index = usize::from(entry.as_u16());
        self.backing.entry_vaddr(index)
    }
}

/// VT-d remapped MSI/MSI-X message produced for a device-side MSI source.
// ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 523) -
[internal] entry_vaddr
    fn entry_vaddr(self, entry: VtdInterruptEntry) -> Result<VirtAddr> {
        let index = usize::from(entry.as_u16());
        self.backing.entry_vaddr(index)
    }
}

/// VT-d remapped MSI/MSI-X message produced for a device-side MSI source.
#[derive(Clone, Copy, Debug, PartialEq, Eq, Hash)]
pub struct VtdRemappedMsi {
    entry: VtdInterruptEntry,
    subhandle: u16,
    message: MsiMessage,
    // ... (truncated, see source file for more)

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 531) -

```
[public] VtdRemappedMsi
    pub struct VtdRemappedMsi {
        entry: VtdInterruptEntry,
        subhandle: u16,
        message: MsiMessage,
    }

    impl VtdRemappedMsi {
        #[inline]
        pub const fn new(entry: VtdInterruptEntry, subhandle: u16) -> Self {
            Self {
                entry,
                subhandle,
                // ... (truncated, see source file for more)
            }
        }
    }
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 565) -

```
[internal] from
    fn from(value: VtdRemappedMsi) -> Self {
        value.message()
    }
}

/// VT-d interrupt-remapping delivery target.
#[derive(Clone, Copy, Debug, PartialEq, Eq, Hash)]
pub struct VtdInterruptRemapTarget {
    destination: u32,
    vector: X86InterruptVector,
    delivery_mode: X86MsiDeliveryMode,
    trigger_mode: X86MsiTriggerMode,
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 572) -

```
[public] VtdInterruptRemapTarget
    pub struct VtdInterruptRemapTarget {
        destination: u32,
        vector: X86InterruptVector,
        delivery_mode: X86MsiDeliveryMode,
        trigger_mode: X86MsiTriggerMode,
        destination_mode_logical: bool,
        redirection_hint: bool,
    }

    impl VtdInterruptRemapTarget {
        #[inline]
        pub const fn new(
            // ... (truncated, see source file for more)
        )
    }
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 662) -

```
[public] VtdInterruptRemapEntry
    pub struct VtdInterruptRemapEntry {
        low: u64,
        high: u64,
    }

    impl VtdInterruptRemapEntry {
        #[inline]
        pub const fn from_words(low: u64, high: u64) -> Self {
            Self { low, high }
        }

        #[inline]
        // ... (truncated, see source file for more)
    }
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 743) -

```
[public] VtdRegisterWindow
    pub struct VtdRegisterWindow<Entry>
    where
        Entry: PageTableEntry,
    {
        mapping: Mapping<Entry, VirtAddr>,
    }

    impl<Entry> VtdRegisterWindow<Entry>
    where
        Entry: PageTableEntry,
    {
        #[inline]
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 770) -

```
[public] mmio_range
    pub fn mmio_range(&self) -> MmioAddrRange {
        <MmioAddrRange as MmioRange<usize>>::from_phys_range(PhysAddrRange::from_start_size(
            self.mapping.paddr,
            self.mapping.range.size(),
        ))
    }

    #[inline]
    pub fn read32(&self, offset: usize) -> Result<u32> {
        self.mapping.read_vo32(offset).map_err(Into::into)
    }

    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 778) -

```
[public] read32
    pub fn read32(&self, offset: usize) -> Result<u32> {
        self.mapping.read_vo32(offset).map_err(Into::into)
    }

    #[inline]
    pub fn write32(&mut self, offset: usize, value: u32) -> Result {
        self.mapping.write_vo32(offset, value).map_err(Into::into)
    }

    #[inline]
    pub fn modify32(&mut self, offset: usize, f: impl FnOnce(u32) -> u32) -> Result<u32> {
        self.mapping.modify_vo32(offset, f).map_err(Into::into)
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 783) -

```
[public] write32
    pub fn write32(&mut self, offset: usize, value: u32) -> Result {
        self.mapping.write_vo32(offset, value).map_err(Into::into)
    }

    #[inline]
    pub fn modify32(&mut self, offset: usize, f: impl FnOnce(u32) -> u32) -> Result<u32> {
        self.mapping.modify_vo32(offset, f).map_err(Into::into)
    }

    #[inline]
    pub fn read64(&self, offset: usize) -> Result<u64> {
        self.mapping.read_vo64(offset).map_err(Into::into)
    }
    // ... (truncated, see source file for more)
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 788) -
[public] modify32
    pub fn modify32(&mut self, offset: usize, f: impl FnOnce(u32) -> u32) -> Result<u32> {
        self.mapping.modify_vo32(offset, f).map_err(Into::into)
    }

    #[inline]
    pub fn read64(&self, offset: usize) -> Result<u64> {
        self.mapping.read_vo64(offset).map_err(Into::into)
    }

    #[inline]
    pub fn write64(&mut self, offset: usize, value: u64) -> Result {
        self.mapping.write_vo64(offset, value).map_err(Into::into)
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 793) -
[public] read64
    pub fn read64(&self, offset: usize) -> Result<u64> {
        self.mapping.read_vo64(offset).map_err(Into::into)
    }

    #[inline]
    pub fn write64(&mut self, offset: usize, value: u64) -> Result {
        self.mapping.write_vo64(offset, value).map_err(Into::into)
    }

    #[inline]
    fn write64_shared(&self, offset: usize, value: u64) -> Result {
        let end = offset
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 798) -
[public] write64
    pub fn write64(&mut self, offset: usize, value: u64) -> Result {
        self.mapping.write_vo64(offset, value).map_err(Into::into)
    }

    #[inline]
    fn write64_shared(&self, offset: usize, value: u64) -> Result {
        let end = offset
            .checked_add(size_of::<u64>())
            .ok_or(Error::AddressOverflow)?;
        if end > self.mapping.range.size() {
            return Err(Error::InvalidRange);
        }
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 803) -
[internal] write64_shared
    fn write64_shared(&self, offset: usize, value: u64) -> Result {
        let end = offset
            .checked_add(size_of::<u64>())
            .ok_or(Error::AddressOverflow)?;
        if end > self.mapping.range.size() {
            return Err(Error::InvalidRange);
        }
        let addr = self
            .mapping
            .range
            .start
            .checked_add(offset)
    }
    // ... (truncated, see source file for more)

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 821) -
[public] modify64
    pub fn modify64(&mut self, offset: usize, f: impl FnOnce(u64) -> u64) -> Result<u64> {
        self.mapping.modify_vo64(offset, f).map_err(Into::into)
    }

    #[inline]
    pub fn version(&self) -> Result<VtdVersion> {
        self.read32(REG_VERSION).map(VtdVersion::from_bits)
    }

    #[inline]
    pub fn capability(&self) -> Result<VtdCapability> {
        self.read64(REG_CAP).map(VtdCapability::from_bits)
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 826) -
[public] version
    pub fn version(&self) -> Result<VtdVersion> {
        self.read32(REG_VERSION).map(VtdVersion::from_bits)
    }

    #[inline]
    pub fn capability(&self) -> Result<VtdCapability> {
        self.read64(REG_CAP).map(VtdCapability::from_bits)
    }

    #[inline]
    pub fn extended_capability(&self) -> Result<VtdExtendedCapability> {
        self.read64(REG_ECAP).map(VtdExtendedCapability::from_bits)
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 831) -
[public] capability
    pub fn capability(&self) -> Result<VtdCapability> {
        self.read64(REG_CAP).map(VtdCapability::from_bits)
    }

    #[inline]
    pub fn extended_capability(&self) -> Result<VtdExtendedCapability> {
        self.read64(REG_ECAP).map(VtdExtendedCapability::from_bits)
    }

    #[inline]
    pub fn global_status(&self) -> Result<u32> {
        self.read32(REG_GSTS)
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 836) -
[public] extended_capability
    pub fn extended_capability(&self) -> Result<VtdExtendedCapability> {
        self.read64(REG_ECAP).map(VtdExtendedCapability::from_bits)
    }

    #[inline]
    pub fn global_status(&self) -> Result<u32> {
        self.read32(REG_GSTS)
    }

    #[inline]
    pub fn global_command(&mut self, command: u32) -> Result {
        self.write32(REG_GCMD, command)
    }
    // ... (truncated, see source file for more)

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 841) -

```
[public] global_status
    pub fn global_status(&self) -> Result<u32> {
        self.read32(REG_GSTS)
    }

    #[inline]
    pub fn global_command(&mut self, command: u32) -> Result {
        self.write32(REG_GCMD, command)
    }

    #[inline]
    pub fn fault_status(&self) -> Result<u32> {
        self.read32(REG_FSTS)
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 846) -

```
[public] global_command
    pub fn global_command(&mut self, command: u32) -> Result {
        self.write32(REG_GCMD, command)
    }

    #[inline]
    pub fn fault_status(&self) -> Result<u32> {
        self.read32(REG_FSTS)
    }

    #[inline]
    pub fn invalidate_queue_head(&self) -> Result<usize> {
        self.read64(REG_IQH).map(|head| head as usize)
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 851) -

```
[public] fault_status
    pub fn fault_status(&self) -> Result<u32> {
        self.read32(REG_FSTS)
    }

    #[inline]
    pub fn invalidate_queue_head(&self) -> Result<usize> {
        self.read64(REG_IQH).map(|head| head as usize)
    }

    #[inline]
    pub fn invalidate_queue_tail(&self) -> Result<usize> {
        self.read64(REG_IQT).map(|tail| tail as usize)
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 856) -

```
[public] invalidate_queue_head
    pub fn invalidate_queue_head(&self) -> Result<usize> {
        self.read64(REG_IQH).map(|head| head as usize)
    }

    #[inline]
    pub fn invalidate_queue_tail(&self) -> Result<usize> {
        self.read64(REG_IQT).map(|tail| tail as usize)
    }

    #[inline]
    pub fn write_invalidate_queue_tail(&mut self, tail: usize) -> Result {
        self.write64(REG_IQT, tail as u64)
    }
    // ... (truncated, see source file for more)
```



```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 861) -
[public] invalidate_queue_tail
    pub fn invalidate_queue_tail(&self) -> Result<usize> {
        self.read64(REG_IQT).map(|tail| tail as usize)
    }

    #[inline]
    pub fn write_invalidate_queue_tail(&mut self, tail: usize) -> Result {
        self.write64(REG_IQT, tail as u64)
    }

    #[inline]
    pub fn write_invalidate_queue_tail_shared(&self, tail: usize) -> Result {
        self.write64_shared(REG_IQT, tail as u64)
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 866) -
[public] write_invalidate_queue_tail
    pub fn write_invalidate_queue_tail(&mut self, tail: usize) -> Result {
        self.write64(REG_IQT, tail as u64)
    }

    #[inline]
    pub fn write_invalidate_queue_tail_shared(&self, tail: usize) -> Result {
        self.write64_shared(REG_IQT, tail as u64)
    }

    #[inline]
    fn poll_transition() {
        spin_loop();
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 871) -
[public] write_invalidate_queue_tail_shared
    pub fn write_invalidate_queue_tail_shared(&self, tail: usize) -> Result {
        self.write64_shared(REG_IQT, tail as u64)
    }

    #[inline]
    fn poll_transition() {
        spin_loop();
    }

    pub fn wait_global_status(&self, mask: u32, expected: u32) -> Result {
        for _ in 0..REGISTER_TRANSITION_POLL_LIMIT {
            if (self.global_status()? & mask) == (expected & mask) {
                // ... (truncated, see source file for more)
            }
        }
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 876) -
[internal] poll_transition
    fn poll_transition() {
        spin_loop();
    }

    pub fn wait_global_status(&self, mask: u32, expected: u32) -> Result {
        for _ in 0..REGISTER_TRANSITION_POLL_LIMIT {
            if (self.global_status()? & mask) == (expected & mask) {
                return Ok(());
            }
            Self::poll_transition();
        }
    }

    // ... (truncated, see source file for more)

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 880) -
[public] wait_global_status

```
pub fn wait_global_status(&self, mask: u32, expected: u32) -> Result {
    for _ in 0..REGISTER_TRANSITION_POLL_LIMIT {
        if (self.global_status()? & mask) == (expected & mask) {
            return Ok(());
        }
        Self::poll_transition();
    }

    Err(VtdError::RegisterTransitionTimeout.into())
}

pub fn set_root_entry_table(&mut self, root: PhysAddr) -> Result {
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 891) -
[public] set_root_entry_table

```
pub fn set_root_entry_table(&mut self, root: PhysAddr) -> Result {
    if !root.is_aligned(PageSize::Size4K.bytes()) {
        return Err(VtdError::UnsupportedGranule.into());
    }

    without_interrupts(|| {
        self.write64(REG_RTADDR, root.as_usize() as u64)?;
        let status = self.global_status()?;
        self.global_command(status | GCMD_SRTP);
    });
    self.wait_global_status(GSTS_RTPS, GSTS_RTPS)
}
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 904) -
[public] set_queued_invalidation_table

```
pub fn set_queued_invalidation_table(&mut self, backing: CommandQueueBacking) -> Result {
    let iqa = queued_invalidation_table_address(backing)?;
    without_interrupts(|| {
        self.write64(REG_IQA, iqa)?;
        self.write64(REG_IQT, 0)
    })
}

#[inline]
pub fn set_root_table(&mut self, table: VtdRootTableBacking) -> Result {
    self.set_root_entry_table(table.phys().start)
}
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 913) -
[public] set_root_table

```
pub fn set_root_table(&mut self, table: VtdRootTableBacking) -> Result {
    self.set_root_entry_table(table.phys().start)
}

#[inline]
pub fn write_root_entry(
    &mut self,
    table: VtdRootTableBacking,
    bus: u8,
    entry: VtdRootEntry,
) -> Result {
    write_root_entry(table, bus, entry)
}
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 918) -

```
[public] write_root_entry
    pub fn write_root_entry(
        &mut self,
        table: VtdRootTableBacking,
        bus: u8,
        entry: VtdRootEntry,
    ) -> Result {
        write_root_entry(table, bus, entry)
    }

#[inline]
pub fn read_root_entry(&self, table: VtdRootTableBacking, bus: u8) -> Result<VtdRootEntry> {
    read_root_entry(table, bus)
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 928) -

```
[public] read_root_entry
    pub fn read_root_entry(&self, table: VtdRootTableBacking, bus: u8) -> Result<VtdRootEntry> {
        read_root_entry(table, bus)
    }

#[inline]
pub fn clear_root_entry(&mut self, table: VtdRootTableBacking, bus: u8) -> Result {
    self.write_root_entry(table, bus, VtdRootEntry::disabled())
}

#[inline]
pub fn write_context_entry(
    &mut self,
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 933) -

```
[public] clear_root_entry
    pub fn clear_root_entry(&mut self, table: VtdRootTableBacking, bus: u8) -> Result {
        self.write_root_entry(table, bus, VtdRootEntry::disabled())
    }

#[inline]
pub fn write_context_entry(
    &mut self,
    table: VtdContextTableBacking,
    client: PciDevice,
    entry: VtdContextEntry,
) -> Result {
    write_context_entry(table, client, entry)
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 938) -

```
[public] write_context_entry
    pub fn write_context_entry(
        &mut self,
        table: VtdContextTableBacking,
        client: PciDevice,
        entry: VtdContextEntry,
    ) -> Result {
        write_context_entry(table, client, entry)
    }

#[inline]
pub fn read_context_entry(
    &self,
    // ... (truncated, see source file for more)
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 948) -
[public] read_context_entry
    pub fn read_context_entry(
        &self,
        table: VtdContextTableBacking,
        client: PciDevice,
    ) -> Result<VtdContextEntry> {
        read_context_entry(table, client)
    }

    #[inline]
    pub fn clear_context_entry(
        &mut self,
        table: VtdContextTableBacking,
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 957) -
[public] clear_context_entry
    pub fn clear_context_entry(
        &mut self,
        table: VtdContextTableBacking,
        client: PciDevice,
    ) -> Result {
        self.write_context_entry(table, client, VtdContextEntry::disabled())
    }

    #[inline]
    pub fn write_binding_context(
        &mut self,
        table: VtdContextTableBacking,
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 966) -
[public] write_binding_context
    pub fn write_binding_context(
        &mut self,
        table: VtdContextTableBacking,
        binding: Binding<PciDevice, VtdDomain>,
    ) -> Result {
        self.write_context_entry(
            table,
            binding.client(),
            VtdContextEntry::from_binding(binding, self.extended_capability())?,
        )
    }

    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 978) -
[public] set_translation_enabled
    pub fn set_translation_enabled(&mut self, enabled: bool) -> Result {
        let status_bit = if enabled { GSTS_TES } else { 0 };

        without_interrupts(|| {
            let status = self.global_status()?;
            let command = if enabled {
                status | GCMD_TE
            } else {
                status & !GCMD_TE
            };
            self.global_command(command)
        })?;
        // ... (truncated, see source file for more)

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 994) -

[public] enable_translation

```
pub fn enable_translation(&mut self) -> Result {
    self.set_translation_enabled(true)
}

#[inline]
pub fn disable_translation(&mut self) -> Result {
    self.set_translation_enabled(false)
}

pub fn set_queued_invalidation_enabled(&mut self, enabled: bool) -> Result {
    let status_bit = if enabled { GSTS_QIES } else { 0 };

    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 999) -

[public] disable_translation

```
pub fn disable_translation(&mut self) -> Result {
    self.set_translation_enabled(false)
}

pub fn set_queued_invalidation_enabled(&mut self, enabled: bool) -> Result {
    let status_bit = if enabled { GSTS_QIES } else { 0 };

    without_interrupts(|| {
        let status = self.global_status()?;
        let command = if enabled {
            status | GCMD_QIE
        } else {
            status & !GCMD_QIE
        };
        self.global_command(command)
    });
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1003) -

[public] set_queued_invalidation_enabled

```
pub fn set_queued_invalidation_enabled(&mut self, enabled: bool) -> Result {
    let status_bit = if enabled { GSTS_QIES } else { 0 };

    without_interrupts(|| {
        let status = self.global_status()?;
        let command = if enabled {
            status | GCMD_QIE
        } else {
            status & !GCMD_QIE
        };
        self.global_command(command)
    });
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1019) -

[public] enable_queued_invalidation

```
pub fn enable_queued_invalidation(&mut self) -> Result {
    self.set_queued_invalidation_enabled(true)
}

#[inline]
pub fn disable_queued_invalidation(&mut self) -> Result {
    self.set_queued_invalidation_enabled(false)
}

pub fn set_interrupt_remapping_enabled(&mut self, enabled: bool) -> Result {
    let status_bit = if enabled { GSTS_IRES } else { 0 };

    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1024) -

[public] disable_queued_invalidation

```
pub fn disable_queued_invalidation(&mut self) -> Result {
    self.set_queued_invalidation_enabled(false)
}

pub fn set_interrupt_remapping_enabled(&mut self, enabled: bool) -> Result {
    let status_bit = if enabled { GSTS_IRES } else { 0 };

    without_interrupts(|| {
        let status = self.global_status()?;
        let command = if enabled {
            status | GCMD_IRES
        } else {
            status & !GCMD_IRES
        };
        self.global_command(command)
    });
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1028) -

[public] set_interrupt_remapping_enabled

```
pub fn set_interrupt_remapping_enabled(&mut self, enabled: bool) -> Result {
    let status_bit = if enabled { GSTS_IRES } else { 0 };

    without_interrupts(|| {
        let status = self.global_status()?;
        let command = if enabled {
            status | GCMD_IRES
        } else {
            status & !GCMD_IRES
        };
        self.global_command(command)
    });
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1044) -

[public] enable_interrupt_remapping

```
pub fn enable_interrupt_remapping(&mut self) -> Result {
    self.set_interrupt_remapping_enabled(true)
}

#[inline]
pub fn disable_interrupt_remapping(&mut self) -> Result {
    self.set_interrupt_remapping_enabled(false)
}

pub fn set_interrupt_remap_table(
    &mut self,
    table: VtdInterruptRemapTableBacking,
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1049) -

[public] disable_interrupt_remapping

```
pub fn disable_interrupt_remapping(&mut self) -> Result {
    self.set_interrupt_remapping_enabled(false)
}

pub fn set_interrupt_remap_table(
    &mut self,
    table: VtdInterruptRemapTableBacking,
    extended_interrupt_mode: bool,
) -> Result {
    let mut value = (table.phys().start.as_usize() as u64 & PAGE_ADDR_MASK) | table.irta_size();
    if extended_interrupt_mode {
        value |= IRTA_EIME;
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1053) -

```
[public] set_interrupt_remap_table
    pub fn set_interrupt_remap_table(
        &mut self,
        table: VtdInterruptRemapTableBacking,
        extended_interrupt_mode: bool,
    ) -> Result {
        let mut value = (table.phys().start.as_usize() as u64 & PAGE_ADDR_MASK) | table.irta_size();
        if extended_interrupt_mode {
            value |= IRTA_EIME;
        }

        without_interrupts(|| {
            self.write64(REG_IRTA, value)?;
        })
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1072) -

```
[public] write_interrupt_remap_entry
    pub fn write_interrupt_remap_entry(
        &mut self,
        table: VtdInterruptRemapTableBacking,
        entry: VtdInterruptEntry,
        remap: VtdInterruptRemapEntry,
    ) -> Result {
        write_interrupt_remap_entry(table, entry, remap)
    }

    #[inline]
    pub fn clear_interrupt_remap_entry(
        &mut self,
    )
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1082) -

```
[public] clear_interrupt_remap_entry
    pub fn clear_interrupt_remap_entry(
        &mut self,
        table: VtdInterruptRemapTableBacking,
        entry: VtdInterruptEntry,
    ) -> Result {
        self.write_interrupt_remap_entry(table, entry, VtdInterruptRemapEntry::disabled())
    }

    pub fn compose_interrupt_remap<const N: usize>(
        &mut self,
        table: VtdInterruptRemapTableBacking,
        queue: &VtdQueuedInvalidationQueue<N>,
    )
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1090) -

```
[public] compose_interrupt_remap
    pub fn compose_interrupt_remap<const N: usize>(
        &mut self,
        table: VtdInterruptRemapTableBacking,
        queue: &VtdQueuedInvalidationQueue<N>,
        entry: VtdInterruptEntry,
        subhandle: u16,
        remap: VtdInterruptRemapEntry,
    ) -> Result<VtdRemappedMsi> {
        self.write_interrupt_remap_entry(table, entry, remap)?;
        self.submit_queued_invalidation(
            queue,
            VtdQueuedInvalidationDescriptor::interrupt_entry_cache(entry),
        )
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1106) -
[public] configure_fault_event

```
pub fn configure_fault_event(&mut self, route: InterruptRoute) -> Result {
    without_interrupts(|| match route {
        InterruptRoute::Disabled => self.write32(REG_FECTL, FECTL_INTERRUPT_MASK),
        InterruptRoute::Msi(message) => {
            self.write32(REG_FEDATA, message.data());
            self.write32(REG_FEADDR, message.address() as u32);
            self.write32(REG_FEUADDR, (message.address() >> 32) as u32);
            self.write32(REG_FECTL, 0)
        }
    })
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1119) -
[public] fault_event_message

```
pub fn fault_event_message(&self) -> Result<MsiMessage> {
    let data = self.read32(REG_FEDATA)?;
    let low = self.read32(REG_FEADDR)? as u64;
    let high = self.read32(REG_FEUADDR)? as u64;
    Ok(MsiMessage::new(low | (high << 32), data))
}

#[inline]
pub fn fault_event_route(&self) -> Result<InterruptRoute> {
    if (self.read32(REG_FECTL)? & FECTL_INTERRUPT_MASK) != 0 {
        Ok(InterruptRoute::Disabled)
    } else {
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1127) -
[public] fault_event_route

```
pub fn fault_event_route(&self) -> Result<InterruptRoute> {
    if (self.read32(REG_FECTL)? & FECTL_INTERRUPT_MASK) != 0 {
        Ok(InterruptRoute::Disabled)
    } else {
        self.fault_event_message().map(InterruptRoute::Msi)
    }
}

pub fn submit_queued_invalidation<const N: usize>(
    &mut self,
    queue: &VtdQueuedInvalidationQueue<N>,
    descriptor: VtdQueuedInvalidationDescriptor,
    // ... (truncated, see source file for more)
)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1135) -
[public] submit_queued_invalidation

```
pub fn submit_queued_invalidation<const N: usize>(
    &mut self,
    queue: &VtdQueuedInvalidationQueue<N>,
    descriptor: VtdQueuedInvalidationDescriptor,
) -> Result {
    let mut tail_result = Ok(());
    let this = self as *mut Self;
    descriptor.submit_to(
        queue,
        || unsafe { (*this).invalidate_queue_head() },
        |tail| unsafe { tail_result = (&mut *this).write_invalidate_queue_tail(tail) },
        || unsafe { (*this).fault_status().map(|_| ()) },
    // ... (truncated, see source file for more)
)
```


driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1152) -

```
[public] invalidate_context_global
    pub fn invalidate_context_global<const N: usize>(
        &mut self,
        queue: &VtdQueuedInvalidationQueue<N>,
    ) -> Result {
        self.submit_queued_invalidation(queue, VtdQueuedInvalidationDescriptor::context_global())
    }

#[inline]
pub fn invalidate_context_device<const N: usize>(
    &mut self,
    queue: &VtdQueuedInvalidationQueue<N>,
    client: PciDevice,
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1160) -

```
[public] invalidate_context_device
    pub fn invalidate_context_device<const N: usize>(
        &mut self,
        queue: &VtdQueuedInvalidationQueue<N>,
        client: PciDevice,
    ) -> Result {
        self.submit_queued_invalidation(
            queue,
            VtdQueuedInvalidationDescriptor::context_device(client),
        )
    }

#[inline]
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1172) -

```
[public] invalidate_iotlb_global
    pub fn invalidate_iotlb_global<const N: usize>(
        &mut self,
        queue: &VtdQueuedInvalidationQueue<N>,
        cap: VtdCapability,
    ) -> Result {
        self.submit_queued_invalidation(queue, VtdQueuedInvalidationDescriptor::iotlb_global(cap))
    }

#[inline]
pub fn invalidate_iotlb_domain<const N: usize>(
    &mut self,
    queue: &VtdQueuedInvalidationQueue<N>,
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1181) -

```
[public] invalidate_iotlb_domain
    pub fn invalidate_iotlb_domain<const N: usize>(
        &mut self,
        queue: &VtdQueuedInvalidationQueue<N>,
        domain: VtdIoDomain,
        cap: VtdCapability,
    ) -> Result {
        self.submit_queued_invalidation(
            queue,
            VtdQueuedInvalidationDescriptor::iotlb_domain(domain, cap),
        )
    }

    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1193) -

```
[public] invalidate_iotlb_page
    pub fn invalidate_iotlb_page<const N: usize>(
        &mut self,
        queue: &VtdQueuedInvalidationQueue<N>,
        domain: VtdIoDomain,
        iova: IoviAddr<u64>,
        granule: PageSize,
        cap: VtdCapability,
    ) -> Result {
        let descriptor = VtdQueuedInvalidationDescriptor::iotlb_page(domain, iova, granule, cap)?;
        self.submit_queued_invalidation(queue, descriptor)
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1208) -

```
[public] VtdFaultReason
    pub enum VtdFaultReason {
        Primary { raw: u8 },
        PrimaryFaultOverflow,
        AdvancedFaultOverflow,
        AdvancedPendingFault,
        InvalidationQueueError,
        InvalidationCompletionError,
        InvalidationTimeout,
        PageRequestOverflow,
    }

    /// One decoded VT-d fault observation.
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1221) -

```
[public] VtdFault
    pub struct VtdFault {
        reason: VtdFaultReason,
        source_id: Option<u16>,
        client: Option<PciDevice>,
        iova: Option<IoviAddr<u64>>,
        access: DmaAccess,
        record_index: Option<u16>,
        overflow: bool,
        pending: bool,
    }

    impl VtdFault {
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1298) -

```
[public] VtdUnit
    /// Runtime state for one Intel VT-d remapping unit.
    pub struct VtdUnit<Entry, const QN: usize>
    where
        Entry: PageTableEntry,
    {
        info: VtdInfo,
        registers: VtdRegisterWindow<Entry>,
        root_table: VtdRootTableBacking,
        context_tables: [Option<VtdContextTableBacking>; ROOT_ENTRY_COUNT],
        queue: VtdQueuedInvalidationQueue<QN>,
        interrupt_remap_table: Option<VtdInterruptRemapTableBacking>,
    }

    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1351) -

```
[public] queue
    pub fn queue(&self) -> &VtdQueuedInvalidationQueue<QN> {
        &self.queue
    }

    #[inline]
    pub fn interrupt_remap_table(&self) -> Option<VtdInterruptRemapTableBacking> {
        self.interrupt_remap_table
    }

    #[inline]
    pub fn install_interrupt_remap_table(&mut self, table: VtdInterruptRemapTableBacking) {
        self.interrupt_remap_table = Some(table);
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1356) -

```
[public] interrupt_remap_table
    pub fn interrupt_remap_table(&self) -> Option<VtdInterruptRemapTableBacking> {
        self.interrupt_remap_table
    }

    #[inline]
    pub fn install_interrupt_remap_table(&mut self, table: VtdInterruptRemapTableBacking) {
        self.interrupt_remap_table = Some(table);
    }

    pub fn install_context_table(&mut self, bus: u8, table: VtdContextTableBacking) -> Result {
        self.registers.write_root_entry(
            self.root_table,
            // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1361) -

```
[public] install_interrupt_remap_table
    pub fn install_interrupt_remap_table(&mut self, table: VtdInterruptRemapTableBacking) {
        self.interrupt_remap_table = Some(table);
    }

    pub fn install_context_table(&mut self, bus: u8, table: VtdContextTableBacking) -> Result {
        self.registers.write_root_entry(
            self.root_table,
            bus,
            VtdRootEntry::from_context_table(table)?,
        );
        self.context_tables[VtdRootTableBacking::bus_entry_index(bus)] = Some(table);
        Ok(())
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1365) -

```
[public] install_context_table
    pub fn install_context_table(&mut self, bus: u8, table: VtdContextTableBacking) -> Result {
        self.registers.write_root_entry(
            self.root_table,
            bus,
            VtdRootEntry::from_context_table(table)?,
        );
        self.context_tables[VtdRootTableBacking::bus_entry_index(bus)] = Some(table);
        Ok(())
    }

    #[inline]
    pub fn context_table(&self, bus: u8) -> Option<VtdContextTableBacking> {
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1376) -
[public] context_table

```
pub fn context_table(&self, bus: u8) -> Option<VtdContextTableBacking> {
    self.context_tables[VtdRootTableBacking::bus_entry_index(bus)]
}

pub fn install_queued_invalidation(&mut self, backing: CommandQueueBacking) -> Result {
    queued_invalidation_table_address(backing)?;
    self.queue.init(backing)?;
    self.registers.set_queued_invalidation_table(backing)?;
    self.registers.enable_queued_invalidation()
}

pub fn enable_translation(&mut self) -> Result {
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1380) -
[public] install_queued_invalidation

```
pub fn install_queued_invalidation(&mut self, backing: CommandQueueBacking) -> Result {
    queued_invalidation_table_address(backing)?;
    self.queue.init(backing)?;
    self.registers.set_queued_invalidation_table(backing)?;
    self.registers.enable_queued_invalidation()
}

pub fn enable_translation(&mut self) -> Result {
    self.registers.set_root_table(self.root_table)?;
    if !self.queue.is_active() {
        return Err(Error::ControllerUnavailable);
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1387) -
[public] enable_translation

```
pub fn enable_translation(&mut self) -> Result {
    self.registers.set_root_table(self.root_table)?;
    if !self.queue.is_active() {
        return Err(Error::ControllerUnavailable);
    }
    if (self.registers.global_status()? & GSTS_QIES) == 0 {
        self.registers.enable_queued_invalidation()?;
    }
    self.registers.enable_translation()
}

#[inline]
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1399) -
[internal] validate_client

```
fn validate_client(&self, client: PciDevice) -> Result {
    if self
        .info
        .base()
        .segment()
        .is_some_and(|segment| segment != client.segment())
    {
        return Err(Error::InvalidClient);
    }
    Ok(())
}

// ... (truncated, see source file for more)
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1412) -
[internal] context_table_for_client
    fn context_table_for_client(&self, client: PciDevice) -> Result<VtdContextTableBacking> {
        self.context_table(client.bus())
            .ok_or(Error::ControllerUnavailable)
    }

    fn submit_queued_invalidation(&self, descriptor: VtdQueuedInvalidationDescriptor) -> Result {
        let mut tail_result = Ok(());
        descriptor.submit_to(
            &self.queue,
            || self.registers.invalidate_queue_head(),
            |tail| tail_result = self.registers.write_invalidate_queue_tail_shared(tail),
            || self.registers.fault_status().map(|_| ()),
        );
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1417) -
[internal] submit_queued_invalidation
    fn submit_queued_invalidation(&self, descriptor: VtdQueuedInvalidationDescriptor) -> Result {
        let mut tail_result = Ok(());
        descriptor.submit_to(
            &self.queue,
            || self.registers.invalidate_queue_head(),
            |tail| tail_result = self.registers.write_invalidate_queue_tail_shared(tail),
            || self.registers.fault_status().map(|_| ()),
        );
        tail_result
    }

    #[inline]
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1429) -
[internal] invalidate_context_device
    fn invalidate_context_device(&self, client: PciDevice) -> Result {
        self.submit_queued_invalidation(VtdQueuedInvalidationDescriptor::context_device(client))
    }

    #[inline]
    fn invalidate_iotlb_domain(&self, domain: VtdIoDomain) -> Result {
        self.submit_queued_invalidation(VtdQueuedInvalidationDescriptor::iotlb_domain(
            domain,
            self.info.cap(),
        ))
    }

    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1434) -
[internal] invalidate_iotlb_domain
    fn invalidate_iotlb_domain(&self, domain: VtdIoDomain) -> Result {
        self.submit_queued_invalidation(VtdQueuedInvalidationDescriptor::iotlb_domain(
            domain,
            self.info.cap(),
        ))
    }

    #[inline]
    fn invalidate_iotlb_global(&self) -> Result {
        self.submit_queued_invalidation(VtdQueuedInvalidationDescriptor::iotlb_global(
            self.info.cap(),
        ))
    }

    // ... (truncated, see source file for more)

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1442) -

```
[internal] invalidate_iotlb_global

fn invalidate_iotlb_global(&self) -> Result {
    self.submit_queued_invalidation(VtdQueuedInvalidationDescriptor::iotlb_global(
        self.info.cap(),
    ))
}

fn invalidate_iotlb_page(
    &self,
    domain: VtdIoDomain,
    iova: IoviAddr<u64>,
    granule: PageSize,
) -> Result<InvalidateScope> {
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1448) -

```
[internal] invalidate_iotlb_page

fn invalidate_iotlb_page(
    &self,
    domain: VtdIoDomain,
    iova: IoviAddr<u64>,
    granule: PageSize,
) -> Result<InvalidateScope> {
    match VtdQueuedInvalidationDescriptor::iotlb_page(domain, iova, granule, self.info.cap()) {
        Ok(descriptor) => {
            self.submit_queued_invalidation(descriptor)?;
            Ok(InvalidateScope::Leaf)
        }
        Err(Error::FeatureUnavailable) => {
            // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1467) -

```
[internal] invalidate_iotlb_range

fn invalidate_iotlb_range(
    &self,
    domain: VtdIoDomain,
    start: IoviAddr<u64>,
    page_size: PageSize,
    count_pages: usize,
) -> Result<InvalidateScope> {
    if count_pages == 0 {
        return Ok(InvalidateScope::Leaf);
    }

    let cap = self.info.cap();
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1525) -

```
[internal] bind_context

fn bind_context(
    &mut self,
    active_domain: VtdDomain,
    binding: Binding<PciDevice, VtdDomain>,
) -> Result {
    if binding.selector() != BindingSelector::Default {
        return Err(Error::FeatureUnavailable);
    }
    self.validate_client(binding.client())?;

    if let BindingTarget::Domain(domain) = binding.target() {
        if domain != active_domain {
            // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1552) -
[internal] unbind_context

```
fn unbind_context(
    &mut self,
    domain: VtdDomain,
    client: PciDevice,
    selector: BindingSelector,
) -> Result {
    if selector != BindingSelector::Default {
        return Err(Error::FeatureUnavailable);
    }
    self.validate_client(client)?;
    let table = self.context_table_for_client(client)?;
    self.registers.clear_context_entry(table, client)?;
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1569) -
[public] poll_fault

```
pub fn poll_fault(&mut self) -> Result<Option<VtdFault>> {
    let status = self.registers.fault_status()?;
    if (status & FSTS_PPF) != 0 {
        return self.poll_primary_fault(status);
    }
    if let Some(fault) = decode_fault_status(status) {
        self.clear_fault_status(status);
        return Ok(Some(fault));
    }
    Ok(None)
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1581) -
[internal] clear_fault_status

```
fn clear_fault_status(&mut self, status: u32) {
    let bits =
        status & (FSTS_PFO | FSTS_AFO | FSTS_APF | FSTS_IQE | FSTS_ICE | FSTS_ITE | FSTS_PRO);
    if bits != 0 {
        let _ = self.registers.write32(REG_FSTS, bits);
    }
}

fn poll_primary_fault(&mut self, status: u32) -> Result<Option<VtdFault>> {
    let index = ((status >> FSTS_FRI_SHIFT) & FSTS_FRI_MASK) as u16;
    if index >= self.info.cap().fault_record_count() {
        return Err(Error::ControllerUnavailable);
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1589) -
[internal] poll_primary_fault

```
fn poll_primary_fault(&mut self, status: u32) -> Result<Option<VtdFault>> {
    let index = ((status >> FSTS_FRI_SHIFT) & FSTS_FRI_MASK) as u16;
    if index >= self.info.cap().fault_record_count() {
        return Err(Error::ControllerUnavailable);
    }

    let fro = self.info.cap().fault_record_register_offset() as usize;
    let record = fro
        .checked_add(usize::from(index) * FRCD_ENTRY_STRIDE)
        .ok_or(Error::AddressOverflow)?;
    let low = self.registers.read64(
        record
    )
    // ... (truncated, see source file for more)
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1622) -
[public] VtdQueuedInvalidator
    pub struct VtdQueuedInvalidator<Entry, const QN: usize>
    where
        Entry: PageTableEntry,
    {
        unit: *const VtdUnit<Entry, QN>,
        domain: VtdIoDomain,
        _unit: PhantomData<fn() -> VtdUnit<Entry, QN>>,
    }

    unsafe impl<Entry, const QN: usize> Send for VtdQueuedInvalidator<Entry, QN> where
        Entry: PageTableEntry
    {
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1660) -
[internal] unit
    fn unit(&self) -> &VtdUnit<Entry, QN> {
        unsafe { &*self.unit }
    }

    #[inline]
    pub fn invalidate_domain(&self) -> Result {
        self.unit().invalidate_iotlb_domain(self.domain)
    }

    #[inline]
    pub fn invalidate_page(
        &self,
        // ... (truncated, see source file for more)
    ) -> Result<InvalidateScope> {
        self.unit()
            .invalidate_iotlb_page(self.domain, iova, granule)
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1665) -
[public] invalidate_domain
    pub fn invalidate_domain(&self) -> Result {
        self.unit().invalidate_iotlb_domain(self.domain)
    }

    #[inline]
    pub fn invalidate_page(
        &self,
        iova: IoviAddr<u64>,
        granule: PageSize,
    ) -> Result<InvalidateScope> {
        self.unit()
            .invalidate_iotlb_page(self.domain, iova, granule)
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1670) -
[public] invalidate_page
    pub fn invalidate_page(
        &self,
        iova: IoviAddr<u64>,
        granule: PageSize,
    ) -> Result<InvalidateScope> {
        self.unit()
            .invalidate_iotlb_page(self.domain, iova, granule)
    }

    #[inline]
    pub fn invalidate_range(
        &self,
        // ... (truncated, see source file for more)
    ) -> Result<InvalidateScope> {
        self.unit()
            .invalidate_range(self.domain, start, end, granule)
        // ... (truncated, see source file for more)
    }

```


driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1680) -

```
[public] invalidate_range
    pub fn invalidate_range(
        &self,
        start: IoviAddr<u64>,
        page_size: PageSize,
        count_pages: usize,
    ) -> Result<InvalidateScope> {
        self.unit()
            .invalidate_iotlb_range(self.domain, start, page_size, count_pages)
    }
}

impl<Entry, const QN: usize> TlbInvalidation<IoviAddr<u64>> for VtdQueuedInvalidator<Entry, QN>
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1696) -

```
[internal] flush_tlb_local
    fn flush_tlb_local(&self, vaddr: IoviAddr<u64>) {
        let _ = self.invalidate_page(vaddr, PageSize::Size4K);
    }

    #[inline]
    fn flush_tlb_all_local(&self) {
        let _ = self.invalidate_domain();
    }

    #[inline]
    fn flush_tlb_range_local(&self, start: IoviAddr<u64>, page_size: PageSize, count_pages: usize) {
        let _ = self.invalidate_range(start, page_size, count_pages);
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1701) -

```
[internal] flush_tlb_all_local
    fn flush_tlb_all_local(&self) {
        let _ = self.invalidate_domain();
    }

    #[inline]
    fn flush_tlb_range_local(&self, start: IoviAddr<u64>, page_size: PageSize, count_pages: usize) {
        let _ = self.invalidate_range(start, page_size, count_pages);
    }

    #[inline]
    fn prefer_full_flush(&self, pending_count: usize) -> bool {
        pending_count > 32
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1706) -

```
[internal] flush_tlb_range_local
    fn flush_tlb_range_local(&self, start: IoviAddr<u64>, page_size: PageSize, count_pages: usize) {
        let _ = self.invalidate_range(start, page_size, count_pages);
    }

    #[inline]
    fn prefer_full_flush(&self, pending_count: usize) -> bool {
        pending_count > 32
    }
}

impl<Entry, const QN: usize> IoTlbInvalidation<IoviAddr<u64>> for VtdQueuedInvalidator<Entry, QN>
    where
        // ... (truncated, see source file for more)
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1711) -
[internal] prefer_full_flush
    fn prefer_full_flush(&self, pending_count: usize) -> bool {
        pending_count > 32
    }
}

impl<Entry, const QN: usize> IoTlbInvalidation<IoviAddr<u64>> for VtdQueuedInvalidator<Entry, QN>
where
    Entry: PageTableEntry,
{
    type Client = PciDevice;

    #[inline]
    // ... (truncated, see source file for more)
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1723) -
[internal] flush_iotlb
    fn flush_iotlb(&self, iova: IoviAddr<u64>) {
        let _ = self.invalidate_page(iova, PageSize::Size4K);
    }

    #[inline]
    fn flush_iotlb_all(&self) {
        let _ = self.invalidate_domain();
    }

    #[inline]
    fn flush_iotlb_range(&self, start: IoviAddr<u64>, page_size: PageSize, count_pages: usize) {
        let _ = self.invalidate_range(start, page_size, count_pages);
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1728) -
[internal] flush_iotlb_all
    fn flush_iotlb_all(&self) {
        let _ = self.invalidate_domain();
    }

    #[inline]
    fn flush_iotlb_range(&self, start: IoviAddr<u64>, page_size: PageSize, count_pages: usize) {
        let _ = self.invalidate_range(start, page_size, count_pages);
    }

    #[inline]
    fn flush_device_tlb(&self, _client: PciDevice, _iova: IoviAddr<u64>) {}

    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1733) -
[internal] flush_iotlb_range
    fn flush_iotlb_range(&self, start: IoviAddr<u64>, page_size: PageSize, count_pages: usize) {
        let _ = self.invalidate_range(start, page_size, count_pages);
    }

    #[inline]
    fn flush_device_tlb(&self, _client: PciDevice, _iova: IoviAddr<u64>) {}

    #[inline]
    fn flush_device_tlb_all(&self, _client: PciDevice) {}

    #[inline]
    fn prefer_full_iotlb_flush(&self, pending_count: usize) -> bool {
        // ... (truncated, see source file for more)
    }
}

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1738) -
[internal] flush_device_tlb
    fn flush_device_tlb(&self, _client: PciDevice, _iova: IoviAddr<u64>) {}

    #[inline]
    fn flush_device_tlb_all(&self, _client: PciDevice) {}

    #[inline]
    fn prefer_full_iotlb_flush(&self, pending_count: usize) -> bool {
        pending_count > 32
    }
}

/// Per-domain VT-d controller facade.
// ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1741) -
[internal] flush_device_tlb_all
    fn flush_device_tlb_all(&self, _client: PciDevice) {}

    #[inline]
    fn prefer_full_iotlb_flush(&self, pending_count: usize) -> bool {
        pending_count > 32
    }
}

/// Per-domain VT-d controller facade.
pub struct VtdDomainController<'unit, Pt, Entry, const QN: usize>
where
    Entry: PageTableEntry,
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1744) -
[internal] prefer_full_iotlb_flush
    fn prefer_full_iotlb_flush(&self, pending_count: usize) -> bool {
        pending_count > 32
    }
}

/// Per-domain VT-d controller facade.
pub struct VtdDomainController<'unit, Pt, Entry, const QN: usize>
where
    Entry: PageTableEntry,
    {
        domain: VtdDomain,
        page_table: Pt,
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1750) -
[public] VtdDomainController
    /// Per-domain VT-d controller facade.

    pub struct VtdDomainController<'unit, Pt, Entry, const QN: usize>
    where
        Entry: PageTableEntry,
        {
            domain: VtdDomain,
            page_table: Pt,
            unit: &'unit mut VtdUnit<Entry, QN>,
            invalidator: VtdQueuedInvalidator<Entry, QN>,
        }

    impl<'unit, Pt, Entry, const QN: usize> VtdDomainController<'unit, Pt, Entry, QN>
    where
        // ... (truncated, see source file for more)

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1765) -

```
[public] new
    pub fn new(domain: VtdDomain, page_table: Pt, unit: &'unit mut VtdUnit<Entry, QN>) -> Self {
        let unit_ptr = unit as *const VtdUnit<Entry, QN>;
        Self {
            domain,
            page_table,
            unit,
            invalidator: VtdQueuedInvalidator::new(unit_ptr, domain.id()),
        }
    }

    #[inline]
    pub const fn page_table(&self) -> &Pt {
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1808) -

```
[internal] root
    fn root(&self) -> PhysAddr {
        self.page_table.root()
    }

    #[inline]
    fn query(&self, vaddr: IoviAddr<u64>) -> PagingResult<Mapping<Self::Entry, IoviAddr<u64>>> {
        self.page_table.query(vaddr)
    }

    #[inline]
    fn map<'a, B, F, Tlb>(
        &mut self,
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1813) -

```
[internal] query
    fn query(&self, vaddr: IoviAddr<u64>) -> PagingResult<Mapping<Self::Entry, IoviAddr<u64>>> {
        self.page_table.query(vaddr)
    }

    #[inline]
    fn map<'a, B, F, Tlb>(
        &mut self,
        range: AddrRange<IoviAddr<u64>>,
        backing: B,
        flags: F,
        tlb: &Tlb,
    ) -> PagingResult
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1818) -

```
[internal] map
    fn map<'a, B, F, Tlb>(
        &mut self,
        range: AddrRange<IoviAddr<u64>>,
        backing: B,
        flags: F,
        tlb: &Tlb,
    ) -> PagingResult
    where
        B: IntoMapBacking<'a>,
        F: Into<MappingFlags<<Self::Entry as PageTableEntry>::Flags>>,
        Tlb: TlbInvalidation<IoviAddr<u64>>,
    {
        // ... (truncated, see source file for more)
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1834) -
[internal] remap
    fn remap<Tlb>(
        &mut self,
        range: AddrRange<IoviAddr<u64>>,
        paddr: PhysAddr,
        flags: <Self::Entry as PageTableEntry>::Flags,
        tlb: &Tlb,
    ) -> PagingResult<Mapping<Self::Entry, IoviAddr<u64>>>
    where
        Tlb: TlbInvalidation<IoviAddr<u64>>,
    {
        self.page_table.remap(range, paddr, flags, tlb)
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1848) -
[internal] protect
    fn protect<Tlb>(
        &mut self,
        range: AddrRange<IoviAddr<u64>>,
        flags: <Self::Entry as PageTableEntry>::Flags,
        tlb: &Tlb,
    ) -> PagingResult<Mapping<Self::Entry, IoviAddr<u64>>>
    where
        Tlb: TlbInvalidation<IoviAddr<u64>>,
    {
        self.page_table.protect(range, flags, tlb)
    }

    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1861) -
[internal] unmap
    fn unmap<Tlb>(
        &mut self,
        range: AddrRange<IoviAddr<u64>>,
        tlb: &Tlb,
    ) -> PagingResult<Mapping<Self::Entry, IoviAddr<u64>>>
    where
        Tlb: TlbInvalidation<IoviAddr<u64>>,
    {
        self.page_table.unmap(range, tlb)
    }

    #[inline]
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1873) -
[internal] split_at
    fn split_at<Tlb>(
        &mut self,
        range: AddrRange<IoviAddr<u64>>,
        tlb: &Tlb,
    ) -> PagingResult<PageSize>
    where
        Tlb: TlbInvalidation<IoviAddr<u64>>,
    {
        self.page_table.split_at(range, tlb)
    }

    #[inline]
    // ... (truncated, see source file for more)

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1885) -
[internal] merge_at
    fn merge_at<Tlb>(<
        &mut self,
        range: AddrRange<IoviAddr<u64>>,
        tlb: &Tlb,
    ) -> PagingResult<PageSize>
    where
        Tlb: TlbInvalidation<IoviAddr<u64>>,
    {
        self.page_table.merge_at(range, tlb)
    }
}

// ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1910) -
[internal] info
    fn info(&self) -> &Self::Info {
        self.unit.info()
    }

    #[inline]
    fn domain(&self) -> Self::Domain {
        self.domain
    }

    #[inline]
    fn stage(&self) -> crate::TranslationStage {
        self.domain.stage()
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1915) -
[internal] domain
    fn domain(&self) -> Self::Domain {
        self.domain
    }

    #[inline]
    fn stage(&self) -> crate::TranslationStage {
        self.domain.stage()
    }

    #[inline]
    fn enable(&mut self) -> Result {
        self.unit.enable_translation()
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1920) -
[internal] stage
    fn stage(&self) -> crate::TranslationStage {
        self.domain.stage()
    }

    #[inline]
    fn enable(&mut self) -> Result {
        self.unit.enable_translation()
    }

    #[inline]
    fn bind(&mut self, binding: Binding<Self::Client, Self::Domain>) -> Result {
        self.unit.bind_context(self.domain, binding)
    }
    // ... (truncated, see source file for more)

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1925) -
[internal] enable
    fn enable(&mut self) -> Result {
        self.unit.enable_translation()
    }

    #[inline]
    fn bind(&mut self, binding: Binding<Self::Client, Self::Domain>) -> Result {
        self.unit.bind_context(self.domain, binding)
    }

    #[inline]
    fn unbind(&mut self, client: Self::Client, selector: BindingSelector) -> Result {
        self.unit.unbind_context(self.domain, client, selector)
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1930) -
[internal] bind
    fn bind(&mut self, binding: Binding<Self::Client, Self::Domain>) -> Result {
        self.unit.bind_context(self.domain, binding)
    }

    #[inline]
    fn unbind(&mut self, client: Self::Client, selector: BindingSelector) -> Result {
        self.unit.unbind_context(self.domain, client, selector)
    }

    #[inline]
    fn invalidator(&self) -> &Self::Invalidator {
        &self.invalidator
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1935) -
[internal] unbind
    fn unbind(&mut self, client: Self::Client, selector: BindingSelector) -> Result {
        self.unit.unbind_context(self.domain, client, selector)
    }

    #[inline]
    fn invalidator(&self) -> &Self::Invalidator {
        &self.invalidator
    }

    fn invalidate(
        &mut self,
        request: Invalidate<Self::Client, IoviAddr<u64>>,
    ) -> Result<InvalidateOutcome> {
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1940) -
[internal] invalidator
    fn invalidator(&self) -> &Self::Invalidator {
        &self.invalidator
    }

    fn invalidate(
        &mut self,
        request: Invalidate<Self::Client, IoviAddr<u64>>,
    ) -> Result<InvalidateOutcome> {
        let scope = match request {
            Invalidate::Global => {
                self.unit.invalidate_iotlb_global()?;
                InvalidateScope::Global
            }
            // ... (truncated, see source file for more)

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1944) -
[internal] invalidate
    fn invalidate(
        &mut self,
        request: Invalidate<Self::Client, IoviAddr<u64>>,
    ) -> Result<InvalidateOutcome> {
        let scope = match request {
            Invalidate::Global => {
                self.unit.invalidate_iotlb_global()?;
                InvalidateScope::Global
            }
            Invalidate::AddressSpace => {
                self.unit.invalidate_iotlb_domain(self.domain.id());
                InvalidateScope::Domain
            }
        }
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1969) -
[internal] configure_fault_event
    fn configure_fault_event(&mut self, route: InterruptRoute) -> Result {
        self.unit.registers.configure_fault_event(route)
    }

    #[inline]
    fn poll_fault(&mut self) -> Result<Option<Self::Fault>> {
        self.unit.poll_fault()
    }
}

/// Raw queued-invalidation descriptor, ready to be written to a VT-d QI ring.
#[derive(Clone, Copy, Debug, Default, PartialEq, Eq)]
// ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1974) -
[internal] poll_fault
    fn poll_fault(&mut self) -> Result<Option<Self::Fault>> {
        self.unit.poll_fault()
    }
}

/// Raw queued-invalidation descriptor, ready to be written to a VT-d QI ring.
#[derive(Clone, Copy, Debug, Default, PartialEq, Eq)]
#[repr(C)]
pub struct VtdQueuedInvalidationDescriptor {
    low: u64,
    high: u64,
}
// ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 1982) -
[public] VtdQueuedInvalidationDescriptor
pub struct VtdQueuedInvalidationDescriptor {
    low: u64,
    high: u64,
}

impl VtdQueuedInvalidationDescriptor {
    #[inline]
    pub const fn from_words(low: u64, high: u64) -> Self {
        Self { low, high }
    }
}

#[inline]
// ... (truncated, see source file for more)

```



```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2004) -
[public] submit_to
    pub fn submit_to<const N: usize, RH, WT, CE>(
        self,
        queue: &VtdQueuedInvalidationQueue<N>,
        read_head: RH,
        write_tail: WT,
        check_error: CE,
    ) -> Result
    where
        RH: FnMut() -> Result<usize>,
        WT: FnMut(usize),
        CE: FnMut() -> Result,
    {
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2052) -
[public] iotlb_page
    pub fn iotlb_page(
        domain: VtdIoDomain,
        iova: IoviAddr<u64>,
        granule: PageSize,
        cap: VtdCapability,
    ) -> Result<Self> {
        let address_mask = cap
            .page_selective_address_mask(granule)
            .ok_or(VtdError::PageSelectiveInvalidationUnavailable)?;

        Self::iotlb_page_with_address_mask(domain, iova, address_mask, cap)
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2066) -
[public] iotlb_page_with_address_mask
    pub fn iotlb_page_with_address_mask(
        domain: VtdIoDomain,
        iova: IoviAddr<u64>,
        address_mask: u8,
        cap: VtdCapability,
    ) -> Result<Self> {
        if !cap.page_selective_invalidation()
            || address_mask > cap.max_address_mask_value()
            || address_mask > IVA_AM_MASK as u8
        {
            return Err(VtdError::PageSelectiveInvalidationUnavailable.into());
        }
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2101) -
[public] VtdInterruptEntry
    pub struct VtdInterruptEntry(u16);

impl VtdInterruptEntry {
    #[inline]
    pub const fn new(index: u16) -> Self {
        Self(index)
    }

    #[inline]
    pub const fn as_u16(self) -> u16 {
        self.0
    }
    // ... (truncated, see source file for more)

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2139) -
[internal] queued_invalidation_table_address

fn queued_invalidation_table_address(backing: CommandQueueBacking) -> Result<u64> {
    if backing.entry_bytes() != QI_ENTRY_BYTES {
        return Err(Error::InvalidGranule);
    }
    if backing.entry_count() < QI_MIN_ENTRIES || !backing.entry_count().is_power_of_two() {
        return Err(Error::InvalidRange);
    }
    if !backing.phys().start.is_aligned(PageSize::Size4K.bytes()) {
        return Err(Error::InvalidAddress);
    }

    let size = backing.entry_count() / QI_MIN_ENTRIES;
    // ... (truncated, see source file for more)
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2174) -
[internal] decode_fault_status

fn decode_fault_status(status: u32) -> Option<VtdFault> {
    let reason = if (status & FSTS_PFO) != 0 {
        VtdFaultReason::PrimaryFaultOverflow
    } else if (status & FSTS_AFO) != 0 {
        VtdFaultReason::AdvancedFaultOverflow
    } else if (status & FSTS_APF) != 0 {
        VtdFaultReason::AdvancedPendingFault
    } else if (status & FSTS_IQE) != 0 {
        VtdFaultReason::InvalidationQueueError
    } else if (status & FSTS_ICE) != 0 {
        VtdFaultReason::InvalidationCompletionError
    } else if (status & FSTS_ITE) != 0 {
        // ... (truncated, see source file for more)
    }

    // ... (truncated, see source file for more)
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2209) -
[internal] decode_primary_fault

fn decode_primary_fault(segment: u16, index: u16, low: u64, high: u64) -> VtdFault {
    let raw_reason = ((high >> FRCDH_REASON_SHIFT) & FRCDH_REASON_MASK) as u8;
    let source_id = (high & FRCDH_SOURCE_ID_MASK) as u16;
    let client = PciDevice::from_segment_bdf(segment, Bdf::from_u16(source_id));
    let iova = IoviAddr::<u64>::from((low & FRCDL_FAULT_INFO_MASK) as usize);
    let mut access = DmaAccess::empty();

    if (high & (FRCDH_TYPE_1 | FRCDH_TYPE_2)) != 0 {
        access |= DmaAccess::READ;
    }
    if (high & FRCDH_EXECUTE) != 0 {
        access |= DmaAccess::READ | DmaAccess::EXECUTE;
    }
    // ... (truncated, see source file for more)
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2242) -
[internal] VtdIotlbRangeBlock

struct VtdIotlbRangeBlock {
    page: usize,
    address_mask: u8,
}

#[derive(Clone, Copy, Debug, PartialEq, Eq)]
struct VtdIotlbRangePlan {
    next_page: usize,
    remaining_pages: usize,
    min_mask: u8,
    max_mask: u8,
}

// ... (truncated, see source file for more)

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2248) -

[internal] VtdIotlbRangePlan

```
struct VtdIotlbRangePlan {
    next_page: usize,
    remaining_pages: usize,
    min_mask: u8,
    max_mask: u8,
}
```

```
impl VtdIotlbRangePlan {
```

```
    #[inline]
```

```
    fn new(start_page: usize, page_count: usize, min_mask: u8, max_mask: u8) -> Result<Self> {
        if page_count == 0 {
            return Ok(Self {
                // ... (truncated, see source file for more)
            })
        }
    }
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2257) -

[internal] new

```
fn new(start_page: usize, page_count: usize, min_mask: u8, max_mask: u8) -> Result<Self> {
    if page_count == 0 {
        return Ok(Self {
            next_page: start_page,
            remaining_pages: 0,
            min_mask,
            max_mask,
        });
    }
    if max_mask < min_mask {
        return Err(Error::FeatureUnavailable);
    }
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2285) -

[internal] descriptor_count

```
fn descriptor_count(self) -> usize {
    let mut cursor = self;
    let mut count = 0usize;
    while cursor.next_block().is_some() {
        count = count.saturating_add(1);
    }
    count
}
```

```
#[inline]
```

```
fn next_block(&mut self) -> Option<VtdIotlbRangeBlock> {
    if self.remaining_pages == 0 {
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2295) -

[internal] next_block

```
fn next_block(&mut self) -> Option<VtdIotlbRangeBlock> {
    if self.remaining_pages == 0 {
        return None;
    }

    let mask = self.next_address_mask();
    let pages = 1usize << mask;
    let block = VtdIotlbRangeBlock {
        page: self.next_page,
        address_mask: mask,
    };
    self.next_page = self.next_page.saturating_add(pages);
    // ... (truncated, see source file for more)
}
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2312) -
[internal] next_address_mask

    fn next_address_mask(self) -> u8 {
        let remaining_mask = floor_log2_usize(self.remaining_pages);
        let alignment_mask = if self.next_page == 0 {
            (usize::BITS - 1) as u8
        } else {
            self.next_page.trailing_zeros().min(usize::BITS - 1) as u8
        };

        remaining_mask
            .min(alignment_mask)
            .min(self.max_mask)
            .max(self.min_mask)
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2343) -
[internal] write_128

    fn write_128(vaddr: VirtAddr, low: u64, high: u64) {
        unsafe {
            let ptr = vaddr.as_mut_ptr_of::<u64>();
            ptr.write_volatile(low.to_le());
            ptr.add(1).write_volatile(high.to_le());
        }
    }

    #[inline]
    fn read_128(vaddr: VirtAddr) -> (u64, u64) {
        unsafe {
            let ptr = vaddr.as_ptr_of::<u64>();
            // ... (truncated, see source file for more)
        }
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2352) -
[internal] read_128

    fn read_128(vaddr: VirtAddr) -> (u64, u64) {
        unsafe {
            let ptr = vaddr.as_ptr_of::<u64>();
            (
                u64::from_le(ptr.read_volatile()),
                u64::from_le(ptr.add(1).read_volatile()),
            )
        }
    }

    #[inline]
    fn write_root_entry(table: VtdRootTableBacking, bus: u8, entry: VtdRootEntry) -> Result {
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2363) -
[internal] write_root_entry

    fn write_root_entry(table: VtdRootTableBacking, bus: u8, entry: VtdRootEntry) -> Result {
        let vaddr = table.entry_vaddr(bus)?;
        write_128(vaddr, entry.low(), entry.high());
        Ok(())
    }

    #[inline]
    fn read_root_entry(table: VtdRootTableBacking, bus: u8) -> Result<VtdRootEntry> {
        let vaddr = table.entry_vaddr(bus)?;
        let (low, high) = read_128(vaddr);
        Ok(VtdRootEntry::from_words(low, high))
    }

    // ... (truncated, see source file for more)

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2370) -

[internal] read_root_entry

```
fn read_root_entry(table: VtdRootTableBacking, bus: u8) -> Result<VtdRootEntry> {
    let vaddr = table.entry_vaddr(bus)?;
    let (low, high) = read_128(vaddr);
    Ok(VtdRootEntry::from_words(low, high))
}
```

#[inline]

```
fn write_context_entry(
    table: VtdContextTableBacking,
    client: PciDevice,
    entry: VtdContextEntry,
) -> Result {
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2377) -

[internal] write_context_entry

```
fn write_context_entry(
    table: VtdContextTableBacking,
    client: PciDevice,
    entry: VtdContextEntry,
) -> Result {
    let vaddr = table.client_entry_vaddr(client)?;
    write_128(vaddr, entry.low(), entry.high());
    Ok(())
}
```

#[inline]

```
fn read_context_entry(table: VtdContextTableBacking, client: PciDevice) -> Result<VtdContextEntry> {
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2388) -

[internal] read_context_entry

```
fn read_context_entry(table: VtdContextTableBacking, client: PciDevice) -> Result<VtdContextEntry> {
    let vaddr = table.client_entry_vaddr(client)?;
    let (low, high) = read_128(vaddr);
    Ok(VtdContextEntry::from_words(low, high))
}
```

#[inline]

```
fn write_interrupt_remap_entry(
    table: VtdInterruptRemapTableBacking,
    entry: VtdInterruptEntry,
    remap: VtdInterruptRemapEntry,
) -> Result {
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2395) -

[internal] write_interrupt_remap_entry

```
fn write_interrupt_remap_entry(
    table: VtdInterruptRemapTableBacking,
    entry: VtdInterruptEntry,
    remap: VtdInterruptRemapEntry,
) -> Result {
    let vaddr = table.entry_vaddr(entry)?;
    write_128(vaddr, remap.low(), remap.high());
    Ok(())
}
```

#[inline]

```
const fn iotlb_drain_bits(cap: VtdCapability) -> u64 {
    // ... (truncated, see source file for more)
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2428) -
[internal] mmio_range
    fn mmio_range(base: usize, size: usize) -> MmioAddrRange {
        <MmioAddrRange as MmioRange<usize>>::from_start_size(crate::MmioAddr::from(base), size)
            .unwrap()
    }

    fn table_range(buffer: &mut [u8], phys_base: usize) -> (PhysAddrRange, VirtAddrRange) {
        (
            PhysAddrRange::from_start_size(PhysAddr::from_usize(phys_base), buffer.len()),
            VirtAddrRange::from_start_size(
                VirtAddr::from_usize(buffer.as_mut_ptr() as usize),
                buffer.len(),
            ),
        ),
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2433) -
[internal] table_range
    fn table_range(buffer: &mut [u8], phys_base: usize) -> (PhysAddrRange, VirtAddrRange) {
        (
            PhysAddrRange::from_start_size(PhysAddr::from_usize(phys_base), buffer.len()),
            VirtAddrRange::from_start_size(
                VirtAddr::from_usize(buffer.as_mut_ptr() as usize),
                buffer.len(),
            ),
        ),
    }

    fn queue_backing(buffer: &mut [u8], phys_base: usize, entries: usize) -> CommandQueueBacking {
        let (phys, virt) = table_range(buffer, phys_base);
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2443) -
[internal] queue_backing
    fn queue_backing(buffer: &mut [u8], phys_base: usize, entries: usize) -> CommandQueueBacking {
        let (phys, virt) = table_range(buffer, phys_base);
        unsafe { CommandQueueBacking::new(phys, virt, entries, QI_ENTRY_BYTES) }.unwrap()
    }

    fn register_window(buffer: &mut [u8]) -> VtdRegisterWindow<VtdSecondLevelPte> {
        let range = VirtAddrRange::from_start_size(
            VirtAddr::from_usize(buffer.as_mut_ptr() as usize),
            buffer.len(),
        );
        VtdRegisterWindow::new(Mapping::new(
            range,
        ),
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2448) -
[internal] register_window
    fn register_window(buffer: &mut [u8]) -> VtdRegisterWindow<VtdSecondLevelPte> {
        let range = VirtAddrRange::from_start_size(
            VirtAddr::from_usize(buffer.as_mut_ptr() as usize),
            buffer.len(),
        );
        VtdRegisterWindow::new(Mapping::new(
            range,
            PhysAddr::from_usize(0xf000_0000),
            super::super::paging::VtdSecondLevelFlags::default(),
        ),
    }

    // ... (truncated, see source file for more)

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2460) -
[internal] unit_info

```
fn unit_info(segment: Option<u16>, cap: u64, ecap: u64) -> VtdInfo {  
    VtdInfo::from_registers(  
        segment,  
        mmio_range(0xf000_0000, PageSize::Size4K.bytes()),  
        0x10,  
        Some(48),  
        cap,  
        ecap,  
        true,  
    )  
}  
  
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2472) -
[internal] root_table

```
fn root_table(buffer: &mut [u8]) -> VtdRootTableBacking {  
    let (phys, virt) = table_range(buffer, 0x4000);  
    unsafe { VtdRootTableBacking::new(phys, virt) }.unwrap()  
}  
  
fn context_table(buffer: &mut [u8]) -> VtdContextTableBacking {  
    let (phys, virt) = table_range(buffer, 0x8000);  
    unsafe { VtdContextTableBacking::new(phys, virt) }.unwrap()  
}  
  
#[test]  
fn root_entry_points_at_context_table() {  
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2477) -
[internal] context_table

```
fn context_table(buffer: &mut [u8]) -> VtdContextTableBacking {  
    let (phys, virt) = table_range(buffer, 0x8000);  
    unsafe { VtdContextTableBacking::new(phys, virt) }.unwrap()  
}  
  
#[test]  
fn root_entry_points_at_context_table() {  
    let mut root_page = vec![0u8; PageSize::Size4K.bytes()];  
    let mut context_page = vec![0u8; PageSize::Size4K.bytes()];  
    let (root_phys, root_virt) = table_range(&mut root_page, 0x4000);  
    let (context_phys, context_virt) = table_range(&mut context_page, 0x8000);  
    let root_table = unsafe { VtdRootTableBacking::new(root_phys, root_virt) }.unwrap();  
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2483) -
[internal] root_entry_points_at_context_table

```
fn root_entry_points_at_context_table() {  
    let mut root_page = vec![0u8; PageSize::Size4K.bytes()];  
    let mut context_page = vec![0u8; PageSize::Size4K.bytes()];  
    let (root_phys, root_virt) = table_range(&mut root_page, 0x4000);  
    let (context_phys, context_virt) = table_range(&mut context_page, 0x8000);  
    let root_table = unsafe { VtdRootTableBacking::new(root_phys, root_virt) }.unwrap();  
    let context_table =  
        unsafe { VtdContextTableBacking::new(context_phys, context_virt) }.unwrap();  
    let entry = VtdRootEntry::from_context_table(context_table).unwrap();  
  
    write_root_entry(root_table, 0x2a, entry).unwrap();  
  
    // ... (truncated, see source file for more)
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2501) -
[internal] context_entry_encodes_domain_root_width_and_id
    fn context_entry_encodes_domain_root_width_and_id() {
        let domain = VtdDomain::new(
            VtdIoDomain::from_asid(0x1234).unwrap(),
            PhysAddr::from_usize(0x20_0000),
            super::super::paging::VtdSecondLevelAddressWidth::Bits48,
            crate::TranslationStage::Stage2,
        );
        let entry = VtdContextEntry::from_domain(domain).unwrap();

        assert!(entry.present());
        assert_eq!(
            entry.translation_type(),
            // ... (truncated, see source file for more)
        );
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2521) -
[internal] context_table_writes_device_function_slot
    fn context_table_writes_device_function_slot() {
        let mut context_page = vec![0u8; PageSize::Size4K.bytes()];
        let (context_phys, context_virt) = table_range(&mut context_page, 0x8000);
        let context_table =
            unsafe { VtdContextTableBacking::new(context_phys, context_virt) }.unwrap();
        let client = PciDevice::new(0, 4, 5, 3).unwrap();
        let entry = VtdContextEntry::pass_through();

        write_context_entry(context_table, client, entry).unwrap();

        assert_eq!(VtdContextTableBacking::client_entry_index(client), 0x2b);
        assert_eq!(read_context_entry(context_table, client).unwrap(), entry);
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2536) -
[internal] queued_invalidation_table_address_encodes_queue_size
    fn queued_invalidation_table_address_encodes_queue_size() {
        for (entries, encoded) in [(256, 0), (512, 1), (1024, 2)] {
            let mut buffer = vec![0u8; entries * QI_ENTRY_BYTES];
            let backing = queue_backing(&mut buffer, 0x20_0000, entries);

            assert_eq!(
                queued_invalidation_table_address(backing).unwrap(),
                0x20_0000 | encoded
            );
        }
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2549) -
[internal] queued_invalidation_table_address_rejects_invalid_backing
    fn queued_invalidation_table_address_rejects_invalid_backing() {
        let mut small = vec![0u8; 128 * QI_ENTRY_BYTES];
        let small = unsafe {
            CommandQueueBacking::new(
                table_range(&mut small, 0x20_0000).0,
                table_range(&mut small, 0x20_0000).1,
                128,
                QI_ENTRY_BYTES,
            )
        }
        .unwrap();
        assert_eq!(
            // ... (truncated, see source file for more)
        );
    }

```



```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2582) -
[internal] unit_installs_context_table_for_bus
    fn unit_installs_context_table_for_bus() {
        let mut regs = vec![0u8; PageSize::Size4K.bytes()];
        let mut root_page = vec![0u8; PageSize::Size4K.bytes()];
        let mut context_page = vec![0u8; PageSize::Size4K.bytes()];
        let root = root_table(&mut root_page);
        let context = context_table(&mut context_page);
        let mut unit: VtdUnit<VtdSecondLevelPte, 256> =
            VtdUnit::new(unit_info(Some(0), 0, 0), register_window(&mut regs), root);

        unit.install_context_table(7, context).unwrap();

        assert_eq!(unit.context_table(7), Some(context));
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2601) -
[internal] bind_validation_rejects_unsupported_or_wrong_contexts
    fn bind_validation_rejects_unsupported_or_wrong_contexts() {
        let mut regs = vec![0u8; PageSize::Size4K.bytes()];
        let mut root_page = vec![0u8; PageSize::Size4K.bytes()];
        let root = root_table(&mut root_page);
        let mut unit: VtdUnit<VtdSecondLevelPte, 256> =
            VtdUnit::new(unit_info(Some(3), 0, 0), register_window(&mut regs), root);
        let active = VtdDomain::new(
            VtdIoDomain::from_asid(1).unwrap(),
            PhysAddr::from_usize(0x20_0000),
            super::super::paging::VtdSecondLevelAddressWidth::Bits48,
            crate::TranslationStage::Stage2,
        );
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2668) -
[internal] page_invalidation_falls_back_to_domain_without_psi
    fn page_invalidation_falls_back_to_domain_without_psi() {
        let mut regs = vec![0u8; PageSize::Size4K.bytes()];
        let mut root_page = vec![0u8; PageSize::Size4K.bytes()];
        let mut queue = vec![0u8; 256 * QI_ENTRY_BYTES];
        let root = root_table(&mut root_page);
        let mut registers = register_window(&mut regs);
        registers.write64(REG_IQH, QI_ENTRY_BYTES as u64).unwrap();
        let mut unit: VtdUnit<VtdSecondLevelPte, 256> =
            VtdUnit::new(unit_info(Some(0), 0, 0), registers, root);
        unit.queue
            .init(queue_backing(&mut queue, 0x20_0000, 256))
            .unwrap();
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2697) -
[internal] range_invalidation_coalesces_to_largest_supported_address_mask
    fn range_invalidation_coalesces_to_largest_supported_address_mask() {
        let mut regs = vec![0u8; PageSize::Size4K.bytes()];
        let mut root_page = vec![0u8; PageSize::Size4K.bytes()];
        let mut queue = vec![0u8; 256 * QI_ENTRY_BYTES];
        let root = root_table(&mut root_page);
        let mut registers = register_window(&mut regs);
        registers.write64(REG_IQH, QI_ENTRY_BYTES as u64).unwrap();
        let cap = CAP_PSI | (18_u64 << CAP_MAMV_SHIFT);
        let mut unit: VtdUnit<VtdSecondLevelPte, 256> =
            VtdUnit::new(unit_info(Some(0), cap, 0), registers, root);
        unit.queue
            .init(queue_backing(&mut queue, 0x20_0000, 256))
            .unwrap();
        // ... (truncated, see source file for more)
    }

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2734) -
[internal] range_invalidation_uses_domain_when_descriptor_count_is_too_high

```
fn range_invalidation_uses_domain_when_descriptor_count_is_too_high() {  
    let mut regs = vec![0u8; PageSize::Size4K.bytes()];  
    let mut root_page = vec![0u8; PageSize::Size4K.bytes()];  
    let mut queue = vec![0u8; 256 * QI_ENTRY_BYTES];  
    let root = root_table(&mut root_page);  
    let mut registers = register_window(&mut regs);  
    registers.write64(REG_IQH, QI_ENTRY_BYTES as u64).unwrap();  
    let cap = CAP_PSI;  
    let mut unit: VtdUnit<VtdSecondLevelPte, 256> =  
        VtdUnit::new(unit_info(Some(0), cap, 0), registers, root);  
    unit.queue  
        .init(queue_backing(&mut queue, 0x20_0000, 256))  
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2765) -
[internal] range_plan_splits_by_alignment_and_capability

```
fn range_plan_splits_by_alignment_and_capability() {  
    let mut plan = VtdIotlbRangePlan::new(512, 1024, 0, 18).unwrap();  
  
    assert_eq!(  
        plan.next_block(),  
        Some(VtdIotlbRangeBlock {  
            page: 512,  
            address_mask: 9,  
        })  
    );  
    assert_eq!(  
        plan.next_block(),  
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2786) -
[internal] fault_decoders_preserve_primary_record_details

```
fn fault_decoders_preserve_primary_record_details() {  
    let fault = decode_primary_fault(  
        2,  
        3,  
        0x1234_5678,  
        FRCDH_FAULT | FRCDH_EXECUTE | (0x2a << FRCDH_REASON_SHIFT) | 0x0408,  
    );  
  
    assert_eq!(fault.reason(), VtdFaultReason::Primary { raw: 0x2a });  
    assert_eq!(fault.record_index(), Some(3));  
    assert_eq!(fault.source_id(), Some(0x0408));  
    assert_eq!(  
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2812) -
[internal] binding_context_rejects_non_default_selector_for_legacy_contexts

```
fn binding_context_rejects_non_default_selector_for_legacy_contexts() {  
    let client = PciDevice::new(0, 0, 1, 0).unwrap();  
    let binding = Binding::new(  
        client,  
        BindingSelector::from_substream(7),  
        BindingTarget::<VtdDomain>::Abort,  
    );  
  
    assert_eq!(  
        VtdContextEntry::from_binding(binding, VtdExtendedCapability::from_bits(ECAP_PT)),  
        Err(Error::FeatureUnavailable)  
    );  
    // ... (truncated, see source file for more)
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2827) -
[internal] binding_context_checks_pass_through_capability
    fn binding_context_checks_pass_through_capability() {
        let client = PciDevice::new(0, 0, 1, 0).unwrap();
        let binding = Binding::new(client, BindingSelector::Default, BindingTarget::PassThrough);

        assert_eq!(
            VtdContextEntry::from_binding(binding, VtdExtendedCapability::from_bits(0)),
            Err(Error::FeatureUnavailable)
        );

        let entry =
            VtdContextEntry::from_binding(binding, VtdExtendedCapability::from_bits(ECAP_PT))
                .unwrap();
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2847) -
[internal] context_device_uses_pci_source_id
    fn context_device_uses_pci_source_id() {
        let client = PciDevice::new(0, 0x2a, 5, 3).unwrap();
        let desc = VtdQueuedInvalidationDescriptor::context_device(client);

        assert_eq!(desc.low() >> QI_CC_SID_SHIFT, 0x2a2b);
    }

#[test]
fn page_iotlb_descriptor_encodes_address_mask() {
    let cap = VtdCapability::from_bits(CAP_PSI | (18_u64 << CAP_MAMV_SHIFT));
    let desc = VtdQueuedInvalidationDescriptor::iotlb_page(
        VtdIoDomain::from_asid(7).unwrap(),
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2855) -
[internal] page_iotlb_descriptor_encodes_address_mask
    fn page_iotlb_descriptor_encodes_address_mask() {
        let cap = VtdCapability::from_bits(CAP_PSI | (18_u64 << CAP_MAMV_SHIFT));
        let desc = VtdQueuedInvalidationDescriptor::iotlb_page(
            VtdIoDomain::from_asid(7).unwrap(),
            IoviAddr::<u64>::from(0x4000_1234),
            PageSize::Size1G,
            cap,
        )
        .unwrap();

        assert_eq!((desc.low() >> QI_IOTLB_DID_SHIFT) & 0xffff, 7);
        assert_eq!(desc.high() & PAGE_ADDR_MASK, 0x4000_1000);
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2871) -
[internal] interrupt_remap_entry_encodes_x86_delivery_and_source
    fn interrupt_remap_entry_encodes_x86_delivery_and_source() {
        let source = PciDevice::new(0, 0x2a, 5, 3).unwrap();
        let delivery = X86MsiDelivery::fixed(
            X86MsiDestination::Physical(0x42),
            crate::arch::x86_64::X86InterruptVector::new(0x45).unwrap(),
        );
        let entry = VtdInterruptRemapEntry::from_x86_delivery(source, delivery);

        assert!(entry.present());
        assert_eq!((entry.low() >> IRTE_VECTOR_SHIFT) & 0xff, 0x45);
        assert_eq!((entry.low() >> IRTE_DESTINATION_SHIFT) & 0xffff_ffff, 0x42);
        assert_eq!(entry.low() & IRTE_DESTINATION_MODE_LOGICAL, 0);
        // ... (truncated, see source file for more)

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2892) -
[internal] interrupt_remap_entry_keeps_delivery_mode_bits

```
fn interrupt_remap_entry_keeps_delivery_mode_bits() {  
    let source = PciDevice::new(0, 0, 1, 0).unwrap();  
    let delivery = X86MsiDelivery::new(  
        X86MsiDestination::Logical(0x03),  
        crate::arch::x86_64::X86InterruptVector::new(0x80).unwrap(),  
        X86MsiDeliveryMode::LowestPriority,  
    )  
    .with_trigger_mode(X86MsiTriggerMode::Level)  
    .with_redirection_hint(true);  
    let entry = VtdInterruptRemapEntry::from_x86_delivery(source, delivery);  
  
    assert_ne!(entry.low() & IRTE_DESTINATION_MODE_LOGICAL, 0);  
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2910) -
[internal] interrupt_remap_target_preserves_extended_destination_id

```
fn interrupt_remap_target_preserves_extended_destination_id() {  
    let source = PciDevice::new(0, 0, 2, 0).unwrap();  
    let target = VtdInterruptRemapTarget::new(  
        0x1234_5678,  
        crate::arch::x86_64::X86InterruptVector::new(0x90).unwrap(),  
        X86MsiDeliveryMode::Fixed,  
    );  
    let entry = VtdInterruptRemapEntry::from_target(source, target);  
  
    assert_eq!(  
        (entry.low() >> IRTE_DESTINATION_SHIFT) & 0xffff_ffff,  
        0x1234_5678  
    );  
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2926) -
[internal] remapped_msi_message_encodes_index_and_subhandle

```
fn remapped_msi_message_encodes_index_and_subhandle() {  
    let message = VtdInterruptEntry::new(0x1234).remapped_msi(7);  
  
    assert_eq!(message.message().address(), 0xfe2_4698);  
    assert_eq!(message.message().data(), 7);  
}  
  
#[test]  
fn interrupt_remap_table_writes_128_bit_entry() {  
    let mut buffer = vec![0u128; 2];  
    let phys = PhysAddrRange::from_start_size(PhysAddr::from_usize(0x4000), 32);  
    let virt =  
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2934) -
[internal] interrupt_remap_table_writes_128_bit_entry

```
fn interrupt_remap_table_writes_128_bit_entry() {  
    let mut buffer = vec![0u128; 2];  
    let phys = PhysAddrRange::from_start_size(PhysAddr::from_usize(0x4000), 32);  
    let virt =  
        VirtAddrRange::from_start_size(VirtAddr::from_usize(buffer.as_mut_ptr() as usize), 32);  
    let table = unsafe { VtdInterruptRemapTableBacking::new(phys, virt, 2) }.unwrap();  
    let remap =  
        VtdInterruptRemapEntry::from_words(0x1122_3344_5566_7788, 0x99aa_bbcc_ddee_ff00);  
  
    write_interrupt_remap_entry(table, VtdInterruptEntry::new(1), remap).unwrap();  
  
    let words = unsafe { core::slice::from_raw_parts(buffer.as_ptr() as *const u64, 4) };  
    // ... (truncated, see source file for more)
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/ctrl.rs (line 2951) -
[internal] queued_invalidation_descriptor_submits_to_command_queue
    fn queued_invalidation_descriptor_submits_to_command_queue() {
        const ENTRIES: usize = 4;
        const ENTRY_BYTES: usize = 16;

        let buffer = vec![0u8; ENTRIES * ENTRY_BYTES];
        let phys = PhysAddrRange::from_start_size(
            PhysAddr::from_usize(buffer.as_ptr() as usize),
            buffer.len(),
        );
        let virt = VirtAddrRange::from_start_size(
            VirtAddr::from_usize(buffer.as_ptr() as usize),
            buffer.len(),
        );
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 50) -
[public] DmarError
    pub enum DmarError {
        Acpi(AcpiError),
        BdfRanges(BdfRangeSetError),
        Mapping(PagingError),
        Malformed(&'static str),
    }

    impl From<AcpiError> for DmarError {
        #[inline]
        fn from(value: AcpiError) -> Self {
            Self::Acpi(value)
        }
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 59) -
[internal] from
    fn from(value: AcpiError) -> Self {
        Self::Acpi(value)
    }
}

impl From<BdfRangeSetError> for DmarError {
    #[inline]
    fn from(value: BdfRangeSetError) -> Self {
        Self::BdfRanges(value)
    }
}

// ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 66) -
[internal] from
    fn from(value: BdfRangeSetError) -> Self {
        Self::BdfRanges(value)
    }
}

impl From<PagingError> for DmarError {
    #[inline]
    fn from(value: PagingError) -> Self {
        Self::Mapping(value)
    }
}

// ... (truncated, see source file for more)

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 73) -
[internal] from
    fn from(value: PagingError) -> Self {
        Self::Mapping(value)
    }
}

impl fmt::Display for DmarError {
    fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
        match self {
            Self::Acpi(error) => write!(f, "{error:?}"),
            Self::BdfRanges(error) => write!(f, "{error}"),
            Self::Mapping(error) => write!(f, "{error:?}"),
            Self::Malformed(message) => f.write_str(message),
            // ... (truncated, see source file for more)
        }
    }
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 79) -
[internal] fmt
    fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
        match self {
            Self::Acpi(error) => write!(f, "{error:?}"),
            Self::BdfRanges(error) => write!(f, "{error}"),
            Self::Mapping(error) => write!(f, "{error:?}"),
            Self::Malformed(message) => f.write_str(message),
        }
    }
}

fn mmio_register_window(
    base: u64,
    // ... (truncated, see source file for more)
)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 89) -
[internal] mmio_register_window
    fn mmio_register_window(
        base: u64,
        size: usize,
        addr_error: &'static str,
        range_error: &'static str,
    ) -> Result<MmioAddrRange, DmarError> {
        let base = usize::try_from(base).map_err(|_| DmarError::Malformed(addr_error))?;
        <MmioAddrRange as MmioRange<usize>>::from_start_size(MmioAddr::from(base), size)
            .ok_or(DmarError::Malformed(range_error))
    }

    fn inclusive_phys_range(
        // ... (truncated, see source file for more)
    )

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 100) -
[internal] inclusive_phys_range
    fn inclusive_phys_range(
        base: u64,
        limit: u64,
        addr_error: &'static str,
        limit_error: &'static str,
    ) -> Result<PhysAddrRange, DmarError> {
        let base = usize::try_from(base).map_err(|_| DmarError::Malformed(addr_error))?;
        let end = limit
            .checked_add(1)
            .ok_or(DmarError::Malformed(limit_error))?;
        let end = usize::try_from(end).map_err(|_| DmarError::Malformed(addr_error))?;
        Ok(PhysAddrRange::new(
            // ... (truncated, see source file for more)
        ))
    }
}

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 119) -
[public] DmarAcpiTable
    pub struct DmarAcpiTable {
        pub header: SdtHeader,
    }

    unsafe impl AcpiTable for DmarAcpiTable {
        const SIGNATURE: Signature = Signature::DMAR;

        #[inline]
        fn header(&self) -> &SdtHeader {
            &self.header
        }
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 127) -
[internal] header
    fn header(&self) -> &SdtHeader {
        &self.header
    }
}

fn table_bytes<'a, T: AcpiTable>(
    bytes: &'a [u8],
    short_header: &'static str,
    short_table: &'static str,
) -> Result<&'a [u8], DmarError> {
    if bytes.len() < size_of::<SdtHeader>() {
        return Err(DmarError::Malformed(short_header));
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 132) -
[internal] table_bytes
    fn table_bytes<'a, T: AcpiTable>(
        bytes: &'a [u8],
        short_header: &'static str,
        short_table: &'static str,
    ) -> Result<&'a [u8], DmarError> {
        if bytes.len() < size_of::<SdtHeader>() {
            return Err(DmarError::Malformed(short_header));
        }

        let header = unsafe { &*bytes.as_ptr().cast::<SdtHeader>() };
        if header.signature != T::SIGNATURE {
            return Err(DmarError::Acpi(AcpiError::SdtInvalidSignature(
                // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 167) -
[internal] read_array
    fn read_array<const N: usize>(
        bytes: &[u8],
        offset: usize,
        message: &'static str,
    ) -> Result<[u8; N], DmarError> {
        let end = offset
            .checked_add(N)
            .ok_or(DmarError::Malformed("DMAR read offset overflow"))?;
        let bytes = bytes
            .get(offset..end)
            .ok_or(DmarError::Malformed(message))?;
        let mut out = [0; N];
        // ... (truncated, see source file for more)

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 184) -
[internal] read_u8
    fn read_u8(bytes: &[u8], offset: usize) -> Result<u8, DmarError> {
        bytes
            .get(offset)
            .copied()
            .ok_or(DmarError::Malformed("DMAR table read is out of bounds"))
    }

#[inline]
fn read_u16(bytes: &[u8], offset: usize) -> Result<u16, DmarError> {
    Ok(u16::from_le_bytes(read_array(
        bytes,
        offset,
        // ... (truncated, see source file for more)
    )))
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 192) -
[internal] read_u16
    fn read_u16(bytes: &[u8], offset: usize) -> Result<u16, DmarError> {
        Ok(u16::from_le_bytes(read_array(
            bytes,
            offset,
            "DMAR table read is out of bounds",
        )))
    }

#[inline]
fn read_u32(bytes: &[u8], offset: usize) -> Result<u32, DmarError> {
    Ok(u32::from_le_bytes(read_array(
        bytes,
        // ... (truncated, see source file for more)
    )))
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 201) -
[internal] read_u32
    fn read_u32(bytes: &[u8], offset: usize) -> Result<u32, DmarError> {
        Ok(u32::from_le_bytes(read_array(
            bytes,
            offset,
            "DMAR table read is out of bounds",
        )))
    }

#[inline]
fn read_u64(bytes: &[u8], offset: usize) -> Result<u64, DmarError> {
    Ok(u64::from_le_bytes(read_array(
        bytes,
        // ... (truncated, see source file for more)
    )))
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 210) -
[internal] read_u64
    fn read_u64(bytes: &[u8], offset: usize) -> Result<u64, DmarError> {
        Ok(u64::from_le_bytes(read_array(
            bytes,
            offset,
            "DMAR table read is out of bounds",
        )))
    }

#[derive(Clone, Copy, Debug)]
pub struct DmarTable<'a> {
    sdt: &'a [u8],
}

// ... (truncated, see source file for more)

```



```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 219) -
[public] DmarTable
    pub struct DmarTable<'a> {
        sdt: &'a [u8],
    }

    impl<'a> DmarTable<'a> {
        fn parse(bytes: &'a [u8]) -> Result<Self, DmarError> {
            let sdt = table_bytes::<DmarAcpiTable>(
                bytes,
                "DMAR table shorter than SDT header",
                "DMAR table length is invalid",
            )?;
            if sdt.len() < DMAR_STRUCTURES_OFFSET {
                // ... (truncated, see source file for more)
            }
        }
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 224) -
[internal] parse
    fn parse(bytes: &'a [u8]) -> Result<Self, DmarError> {
        let sdt = table_bytes::<DmarAcpiTable>(
            bytes,
            "DMAR table shorter than SDT header",
            "DMAR table length is invalid",
        )?;
        if sdt.len() < DMAR_STRUCTURES_OFFSET {
            return Err(DmarError::Malformed("DMAR table shorter than fixed header"));
        }
        Ok(Self { sdt })
    }

    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 259) -
[public] host_address_width
    pub fn host_address_width(self) -> Result<u8, DmarError> {
        read_u8(self.sdt, DMAR_HOST_ADDRESS_WIDTH_OFFSET)
    }

    #[inline]
    pub fn flags(self) -> Result<u8, DmarError> {
        read_u8(self.sdt, DMAR_FLAGS_OFFSET)
    }

    #[inline]
    pub fn structures(self) -> DmarStructures<'a> {
        DmarStructures {
            // ... (truncated, see source file for more)
        }
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 264) -
[public] flags
    pub fn flags(self) -> Result<u8, DmarError> {
        read_u8(self.sdt, DMAR_FLAGS_OFFSET)
    }

    #[inline]
    pub fn structures(self) -> DmarStructures<'a> {
        DmarStructures {
            table: self,
            offset: DMAR_STRUCTURES_OFFSET,
        }
    }

    // ... (truncated, see source file for more)

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 269) -
[public] structures

```
pub fn structures(self) -> DmarStructures<'a> {
    DmarStructures {
        table: self,
        offset: DMAR_STRUCTURES_OFFSET,
    }
}

pub fn for_each_drhd<F>(self, mut f: F) -> Result<(), DmarError>
where
    F: FnMut(u8, DmarDrhd) -> Result<(), DmarError>,
{
    let host_width = self.host_address_width()?;
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 276) -
[public] for_each_drhd

```
pub fn for_each_drhd<F>(self, mut f: F) -> Result<(), DmarError>
where
    F: FnMut(u8, DmarDrhd) -> Result<(), DmarError>,
{
    let host_width = self.host_address_width()?;
    self.for_each_structure(|structure| {
        if let DmarStructure::Drhd(unit) = structure {
            f(host_width, unit)?;
        }
        Ok(())
    })
}
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 289) -
[public] for_each_rmrr

```
pub fn for_each_rmrr<F>(self, mut f: F) -> Result<(), DmarError>
where
    F: FnMut(DmarRmrr) -> Result<(), DmarError>,
{
    self.for_each_structure(|structure| {
        if let DmarStructure::Rmrr(region) = structure {
            f(region)?;
        }
        Ok(())
    })
}
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 301) -
[public] for_each_atrs

```
pub fn for_each_atrs<F>(self, mut f: F) -> Result<(), DmarError>
where
    F: FnMut(DmarAtrs) -> Result<(), DmarError>,
{
    self.for_each_structure(|structure| {
        if let DmarStructure::Atrs(unit) = structure {
            f(unit)?;
        }
        Ok(())
    })
}
// ... (truncated, see source file for more)
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 313) -
[public] for_each_rhsa
    pub fn for_each_rhsa<F>(self, mut f: F) -> Result<(), DmarError>
    where
        F: FnMut(DmarRhsa) -> Result<(), DmarError>,
    {
        self.for_each_structure(|structure| {
            if let DmarStructure::Rhsa(affinity) = structure {
                f(affinity)?;
            }
            Ok(())
        })
    }

    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 325) -
[public] for_each_andd
    pub fn for_each_andd<F>(self, mut f: F) -> Result<(), DmarError>
    where
        F: FnMut(DmarAndd<'a>) -> Result<(), DmarError>,
    {
        self.for_each_structure(|structure| {
            if let DmarStructure::Andd(device) = structure {
                f(device)?;
            }
            Ok(())
        })
    }

    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 337) -
[public] for_each_satc
    pub fn for_each_satc<F>(self, mut f: F) -> Result<(), DmarError>
    where
        F: FnMut(DmarSatc) -> Result<(), DmarError>,
    {
        self.for_each_structure(|structure| {
            if let DmarStructure::Satc(unit) = structure {
                f(unit)?;
            }
            Ok(())
        })
    }

    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 349) -
[public] for_each_sidp
    pub fn for_each_sidp<F>(self, mut f: F) -> Result<(), DmarError>
    where
        F: FnMut(DmarSidp) -> Result<(), DmarError>,
    {
        self.for_each_structure(|structure| {
            if let DmarStructure::Sidp(unit) = structure {
                f(unit)?;
            }
            Ok(())
        })
    }

    // ... (truncated, see source file for more)

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 361) -

[public] has_sidp

```
pub fn has_sidp(self) -> Result<bool, DmarError> {
    let mut found = false;
    self.for_each_sidp(|_| {
        found = true;
        Ok(())
    })?;
    Ok(found)
}

pub fn for_each_drhd_device_scope<F>(self, unit: DmarDrhd, f: F) -> Result<(), DmarError>
where
    F: FnMut(DmarDeviceScope<'a>) -> Result<(), DmarError>,
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 370) -

[public] for_each_drhd_device_scope

```
pub fn for_each_drhd_device_scope<F>(self, unit: DmarDrhd, f: F) -> Result<(), DmarError>
where
    F: FnMut(DmarDeviceScope<'a>) -> Result<(), DmarError>,
{
    self.walk_device_scopes(unit.structure_offset, DRHD_MIN_LENGTH, unit.length, f)
}

pub fn for_each_rmrr_device_scope<F>(self, region: DmarRmrr, f: F) -> Result<(), DmarError>
where
    F: FnMut(DmarDeviceScope<'a>) -> Result<(), DmarError>,
{
    self.walk_device_scopes(region.structure_offset, RMRR_MIN_LENGTH, region.length, f)
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 377) -

[public] for_each_rmrr_device_scope

```
pub fn for_each_rmrr_device_scope<F>(self, region: DmarRmrr, f: F) -> Result<(), DmarError>
where
    F: FnMut(DmarDeviceScope<'a>) -> Result<(), DmarError>,
{
    self.walk_device_scopes(region.structure_offset, RMRR_MIN_LENGTH, region.length, f)
}

pub fn for_each_atrs_device_scope<F>(self, unit: DmarAtsr, f: F) -> Result<(), DmarError>
where
    F: FnMut(DmarDeviceScope<'a>) -> Result<(), DmarError>,
{
    self.walk_device_scopes(unit.structure_offset, ATSR_MIN_LENGTH, unit.length, f)
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 384) -

[public] for_each_atrs_device_scope

```
pub fn for_each_atrs_device_scope<F>(self, unit: DmarAtsr, f: F) -> Result<(), DmarError>
where
    F: FnMut(DmarDeviceScope<'a>) -> Result<(), DmarError>,
{
    self.walk_device_scopes(unit.structure_offset, ATSR_MIN_LENGTH, unit.length, f)
}

pub fn for_each_satc_device_scope<F>(self, unit: DmarSatc, f: F) -> Result<(), DmarError>
where
    F: FnMut(DmarDeviceScope<'a>) -> Result<(), DmarError>,
{
    self.walk_device_scopes(unit.structure_offset, SATC_MIN_LENGTH, unit.length, f)
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 391) -

```
[public] for_each_satc_device_scope
    pub fn for_each_satc_device_scope<F>(self, unit: DmarSatc, f: F) -> Result<(), DmarError>
    where
        F: FnMut(DmarDeviceScope<'a>) -> Result<(), DmarError>,
    {
        self.walk_device_scopes(unit.structure_offset, SATC_MIN_LENGTH, unit.length, f)
    }

    pub fn for_each_sidp_device_scope<F>(self, unit: DmarSidp, f: F) -> Result<(), DmarError>
    where
        F: FnMut(DmarDeviceScope<'a>) -> Result<(), DmarError>,
    {
        self.walk_device_scopes(unit.structure_offset, SIDP_MIN_LENGTH, unit.length, f)
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 398) -

```
[public] for_each_sidp_device_scope
    pub fn for_each_sidp_device_scope<F>(self, unit: DmarSidp, f: F) -> Result<(), DmarError>
    where
        F: FnMut(DmarDeviceScope<'a>) -> Result<(), DmarError>,
    {
        self.walk_device_scopes(unit.structure_offset, SIDP_MIN_LENGTH, unit.length, f)
    }

    pub fn drhd_bdf_ranges(
        self,
        registers: MmioAddrRange,
    ) -> Result<Option<DmarBdfRangeSet>, DmarError> {
        let mut found_unit = false;
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 405) -

```
[public] drhd_bdf_ranges
    pub fn drhd_bdf_ranges(
        self,
        registers: MmioAddrRange,
    ) -> Result<Option<DmarBdfRangeSet>, DmarError> {
        let mut found_unit = false;
        let mut ranges = DmarBdfRangeSet::empty();
        let mut unresolved = false;

        self.for_each_structure(|structure| {
            let DmarStructure::Drhd(unit) = structure else {
                return Ok(());
            };
        });
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 447) -

```
[public] rmrr_bdf_ranges
    pub fn rmrr_bdf_ranges(
        self,
        segment: u16,
        memory: PhysAddrRange,
    ) -> Result<Option<DmarBdfRangeSet>, DmarError> {
        let mut found_region = false;
        let mut ranges = DmarBdfRangeSet::empty();
        let mut unresolved = false;

        self.for_each_structure(|structure| {
            let DmarStructure::Rmrr(region) = structure else {
                return Ok(());
            };
        });
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 486) -
[public] drhd_requester_match

```
pub fn drhd_requester_match(
    self,
    registers: MmioAddrRange,
    requester: PciDevice,
) -> Result<Option<DmarRequesterMatch>, DmarError> {
    let mut match_result = None;

    self.for_each_structure(|structure| {
        let DmarStructure::Drhd(unit) = structure else {
            return Ok(());
        };
        if unit.registers != registers {
            // ... (truncated, see source file for more)
        }
    });
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 543) -
[public] satc_device_ats_required

```
pub fn satc_device_ats_required(self, requester: PciDevice) -> Result<bool, DmarError> {
    if self.has_sidp()? {
        return Ok(false);
    }

    let mut required = false;
    self.for_each_structure(|structure| {
        let DmarStructure::Satc(unit) = structure else {
            return Ok(());
        };
        if unit.segment != requester.segment() || !unit.atc_required || !unit.has_device_scopes {
            // ... (truncated, see source file for more)
        }
    });
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 576) -
[public] rhsa_proximity_domain_for_registers

```
pub fn rhsa_proximity_domain_for_registers(
    self,
    registers: MmioAddrRange,
) -> Result<Option<u32>, DmarError> {
    let mut proximity_domain = None;
    self.for_each_rhsa(|rhsa| {
        if rhsa.registers == registers {
            proximity_domain = Some(rhsa.proximity_domain);
        }
    });
    Ok(proximity_domain)
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 590) -
[public] andd_name_for_device_number

```
pub fn andd_name_for_device_number(
    self,
    acpi_device_number: u8,
) -> Result<Option<&'a [u8]>, DmarError> {
    let mut object_name = None;
    self.for_each_andd(|andd| {
        if andd.acpi_device_number == acpi_device_number {
            object_name = Some(andd.object_name);
        }
    });
    Ok(object_name)
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 604) -
[internal] for_each_structure

```
fn for_each_structure<F>(self, mut f: F) -> Result<(), DmarError>
where
    F: FnMut(DmarStructure<'a>) -> Result<(), DmarError>,
{
    for structure in self.structures() {
        f(structure)?;
    }
    Ok(())
}

fn parse_structure(
    self,
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 614) -
[internal] parse_structure

```
fn parse_structure(
    self,
    offset: usize,
    length: usize,
    kind: u16,
) -> Result<DmarStructure<'a>, DmarError> {
    match kind {
        DMAR_DRHD => Ok(DmarStructure::Drhd(self.parse_drhd(offset, length)?)),
        DMAR_RMRR => Ok(DmarStructure::Rmrr(self.parse_rmrr(offset, length)?)),
        DMAR_ATSR => Ok(DmarStructure::Atsr(self.parse_atsr(offset, length)?)),
        DMAR_RHSA => Ok(DmarStructure::Rhsa(self.parse_rhsa(offset, length)?)),
        DMAR_ANDD => Ok(DmarStructure::Andd(self.parse_andd(offset, length)?)),
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 636) -
[internal] parse_drhd

```
fn parse_drhd(self, offset: usize, length: usize) -> Result<DmarDrhd, DmarError> {
    if length < DRHD_MIN_LENGTH {
        return Err(DmarError::Malformed(
            "DMAR DRHD structure shorter than minimum",
        ));
    }
    let flags = read_u8(self.sdt, offset + 4)?;
    Ok(DmarDrhd {
        flags,
        segment: read_u16(self.sdt, offset + 6)?,
        registers: mmio_register_window(
            read_u64(self.sdt, offset + 8)?,
        ),
    })
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 659) -
[internal] parse_andd

```
fn parse_andd(self, offset: usize, length: usize) -> Result<DmarAndd<'a>, DmarError> {
    if length < ANDD_MIN_LENGTH {
        return Err(DmarError::Malformed(
            "DMAR ANDD structure shorter than minimum",
        ));
    }
    Ok(DmarAndd {
        acpi_device_number: read_u8(self.sdt, offset + 7)?,
        object_name: &self.sdt[offset + ANDD_MIN_LENGTH..offset + length],
    })
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 671) -
[internal] parse_rmrr

```
fn parse_rmrr(self, offset: usize, length: usize) -> Result<DmarRmrr, DmarError> {  
    if length < RMRR_MIN_LENGTH {  
        return Err(DmarError::Malformed(  
            "DMAR RMRR structure shorter than minimum",  
        ));  
    }  
    let base = read_u64(self.sdt, offset + 8)?;  
    let limit = read_u64(self.sdt, offset + 16)?;  
    if limit < base {  
        return Err(DmarError::Malformed("DMAR RMRR limit smaller than base"));  
    }  
    Ok(DmarRmrr {  
        // ... (truncated, see source file for more)  
    })  
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 696) -
[internal] parse_atrsr

```
fn parse_atrsr(self, offset: usize, length: usize) -> Result<DmarAtrs, DmarError> {  
    if length < ATSR_MIN_LENGTH {  
        return Err(DmarError::Malformed(  
            "DMAR ATSR structure shorter than minimum",  
        ));  
    }  
    let flags = read_u8(self.sdt, offset + 4)?;  
    Ok(DmarAtrs {  
        flags,  
        segment: read_u16(self.sdt, offset + 6)?,  
        include_all: (flags & 0x01) != 0,  
        has_device_scopes: length > ATSR_MIN_LENGTH,  
        // ... (truncated, see source file for more)  
    })  
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 713) -
[internal] parse_rhsa

```
fn parse_rhsa(self, offset: usize, length: usize) -> Result<DmarRhsa, DmarError> {  
    if length < RHSA_MIN_LENGTH {  
        return Err(DmarError::Malformed(  
            "DMAR RHSA structure shorter than minimum",  
        ));  
    }  
    Ok(DmarRhsa {  
        registers: mmio_register_window(  
            read_u64(self.sdt, offset + 8)?,  
            VTD_REGISTER_WINDOW_SIZE,  
            "DMAR RHSA register base cannot fit in usize",  
            "DMAR RHSA register window overflows",  
        ),  
        // ... (truncated, see source file for more)  
    })  
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 730) -
[internal] parse_satc

```
fn parse_satc(self, offset: usize, length: usize) -> Result<DmarSatc, DmarError> {  
    if length < SATC_MIN_LENGTH {  
        return Err(DmarError::Malformed(  
            "DMAR SATC structure shorter than minimum",  
        ));  
    }  
    let flags = read_u8(self.sdt, offset + 4)?;  
    Ok(DmarSatc {  
        flags,  
        segment: read_u16(self.sdt, offset + 6)?,  
        atc_required: (flags & 0x01) != 0,  
        has_device_scopes: length > SATC_MIN_LENGTH,  
        // ... (truncated, see source file for more)  
    })  
}
```



```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 747) -
[internal] parse_sidp
    fn parse_sidp(self, offset: usize, length: usize) -> Result<DmarSidp, DmarError> {
        if length < SIDP_MIN_LENGTH {
            return Err(DmarError::Malformed(
                "DMAR SIDP structure shorter than minimum",
            ));
        }
        Ok(DmarSidp {
            segment: read_u16(self.sdt, offset + 6)?,
            has_device_scopes: length > SIDP_MIN_LENGTH,
            structure_offset: offset,
            length,
        })
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 761) -
[internal] walk_device_scopes
    fn walk_device_scopes<F>(
        self,
        structure_offset: usize,
        scopes_offset: usize,
        structure_length: usize,
        mut f: F,
    ) -> Result<(), DmarError>
    where
        F: FnMut(DmarDeviceScope<'a>) -> Result<(), DmarError>,
    {
        let mut offset = scopes_offset;
        while offset < structure_length {
            // ... (truncated, see source file for more)
        }
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 822) -
[public] DmarStructures
    pub struct DmarStructures<'a> {
        table: DmarTable<'a>,
        offset: usize,
    }

    impl<'a> Iterator for DmarStructures<'a> {
        type Item = Result<DmarStructure<'a>, DmarError>;

        fn next(&mut self) -> Option<Self::Item> {
            if self.offset >= self.table.sdt.len() {
                return None;
            }
            // ... (truncated, see source file for more)
        }
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 830) -
[internal] next
    fn next(&mut self) -> Option<Self::Item> {
        if self.offset >= self.table.sdt.len() {
            return None;
        }
        if self.offset + 4 > self.table.sdt.len() {
            self.offset = self.table.sdt.len();
            return Some(Err(DmarError::Malformed(
                "DMAR structure truncated before header",
            )));
        }
        let offset = self.offset;
        // ... (truncated, see source file for more)
    }

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 869) -
[public] DmarStructure
    pub enum DmarStructure<'a> {
        Drhd(DmarDrhd),
        Rmrr(DmarRmrr),
        Atsr(DmarAtsr),
        Rhsa(DmarRhsa),
        Andd(DmarAndd<'a>),
        Satc(DmarSatc),
        Sidp(DmarSidp),
        Unknown(DmarUnknown),
    }

    #[derive(Clone, Copy, Debug, PartialEq, Eq)]
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 881) -
[public] DmarUnknown
    pub struct DmarUnknown {
        pub kind: u16,
        pub offset: usize,
        pub length: usize,
    }

    #[derive(Clone, Copy, Debug, PartialEq, Eq)]
    pub struct DmarDrhd {
        pub flags: u8,
        pub segment: u16,
        pub registers: MmioAddrRange,
        pub include_all: bool,
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 888) -
[public] DmarDrhd
    pub struct DmarDrhd {
        pub flags: u8,
        pub segment: u16,
        pub registers: MmioAddrRange,
        pub include_all: bool,
        pub has_device_scopes: bool,
        structure_offset: usize,
        length: usize,
    }

    #[derive(Clone, Copy, Debug, PartialEq, Eq)]
    pub struct DmarRmrr {
        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 899) -
[public] DmarRmrr
    pub struct DmarRmrr {
        pub segment: u16,
        pub memory: PhysAddrRange,
        pub has_device_scopes: bool,
        structure_offset: usize,
        length: usize,
    }

    #[derive(Clone, Copy, Debug, PartialEq, Eq)]
    pub struct DmarAtsr {
        pub flags: u8,
        pub segment: u16,
        // ... (truncated, see source file for more)

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 908) -
[public] DmarAtsr
    pub struct DmarAtsr {
        pub flags: u8,
        pub segment: u16,
        pub include_all: bool,
        pub has_device_scopes: bool,
        structure_offset: usize,
        length: usize,
    }

#[derive(Clone, Copy, Debug, PartialEq, Eq)]
pub struct DmarRhsa {
    pub registers: MmioAddrRange,
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 918) -
[public] DmarRhsa
    pub struct DmarRhsa {
        pub registers: MmioAddrRange,
        pub proximity_domain: u32,
    }

#[derive(Clone, Copy, Debug, PartialEq, Eq)]
pub struct DmarAndd<'a> {
    pub acpi_device_number: u8,
    pub object_name: &'a [u8],
}

#[derive(Clone, Copy, Debug, PartialEq, Eq)]
// ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 924) -
[public] DmarAndd
    pub struct DmarAndd<'a> {
        pub acpi_device_number: u8,
        pub object_name: &'a [u8],
    }

#[derive(Clone, Copy, Debug, PartialEq, Eq)]
pub struct DmarSatc {
    pub flags: u8,
    pub segment: u16,
    pub atc_required: bool,
    pub has_device_scopes: bool,
    structure_offset: usize,
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 930) -
[public] DmarSatc
    pub struct DmarSatc {
        pub flags: u8,
        pub segment: u16,
        pub atc_required: bool,
        pub has_device_scopes: bool,
        structure_offset: usize,
        length: usize,
    }

#[derive(Clone, Copy, Debug, PartialEq, Eq)]
pub struct DmarSidp {
    pub segment: u16,
    // ... (truncated, see source file for more)

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 940) -

```
[public] DmarSidp
    pub struct DmarSidp {
        pub segment: u16,
        pub has_device_scopes: bool,
        pub structure_offset: usize,
        pub length: usize,
    }

    #[derive(Clone, Copy, Debug, PartialEq, Eq)]
    pub struct DmarDeviceScope<'a> {
        pub kind: DmarDeviceScopeKind,
        pub flags: u8,
        pub enumeration_id: u8,
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 948) -

```
[public] DmarDeviceScope
    pub struct DmarDeviceScope<'a> {
        pub kind: DmarDeviceScopeKind,
        pub flags: u8,
        pub enumeration_id: u8,
        pub start_bus: u8,
        pub path: DmarDeviceScopePath<'a>,
        pub requester: Option<BdfRange>,
    }

    #[derive(Clone, Copy, Debug, PartialEq, Eq)]
    pub struct DmarDeviceScopePath<'a> {
        bytes: &'a [u8],
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 958) -

```
[public] DmarDeviceScopePath
    pub struct DmarDeviceScopePath<'a> {
        bytes: &'a [u8],
    }

    impl<'a> DmarDeviceScopePath<'a> {
        #[inline]
        pub const fn as_bytes(self) -> &'a [u8] {
            self.bytes
        }

        #[inline]
        pub const fn depth(self) -> usize {
            // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 974) -

```
[public] entry
    pub fn entry(self, index: usize) -> Option<DmarDeviceScopePathEntry> {
        let offset = index.checked_mul(2)?;
        let device = *self.bytes.get(offset)?;
        let function = *self.bytes.get(offset + 1)?;
        Some(DmarDeviceScopePathEntry { device, function })
    }

    #[inline]
    pub fn last_entry(self) -> Option<DmarDeviceScopePathEntry> {
        self.depth()
            .checked_sub(1)
            .and_then(|index| self.entry(index))
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 982) -

```
[public] last_entry
    pub fn last_entry(self) -> Option<DmarDeviceScopePathEntry> {
        self.depth()
            .checked_sub(1)
            .and_then(|index| self.entry(index))
    }

    #[inline]
    pub fn single_requester(self, start_bus: u8) -> Result<Bdf, DmarError> {
        if self.depth() != 1 {
            return Err(DmarError::Malformed(
                "DMAR scope path is not a single requester",
            ));
        }
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 989) -

```
[public] single_requester
    pub fn single_requester(self, start_bus: u8) -> Result<Bdf, DmarError> {
        if self.depth() != 1 {
            return Err(DmarError::Malformed(
                "DMAR scope path is not a single requester",
            ));
        }
        let entry = self
            .entry(0)
            .ok_or(DmarError::Malformed("DMAR scope path missing requester"))?;
        Bdf::new(start_bus, entry.device, entry.function)
            .map_err(|_| DmarError::Malformed("DMAR scope requester BDF is invalid"))
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 1004) -

```
[public] DmarDeviceScopePathEntry
    pub struct DmarDeviceScopePathEntry {
        pub device: u8,
        pub function: u8,
    }

    #[derive(Clone, Copy, Debug, PartialEq, Eq)]
    pub enum DmarDeviceScopeKind {
        Endpoint,
        Bridge,
        IoApic,
        Hpet,
        NamespaceDevice,
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 1010) -

```
[public] DmarDeviceScopeKind
    pub enum DmarDeviceScopeKind {
        Endpoint,
        Bridge,
        IoApic,
        Hpet,
        NamespaceDevice,
        Unknown(u8),
    }

    impl DmarDeviceScopeKind {
        #[inline]
        const fn from_raw(raw: u8) -> Self {
            // ... (truncated, see source file for more)
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 1034) -
[public] DmarRequesterMatch
    pub enum DmarRequesterMatch {
        Covered,
        NotCovered,
        Unresolved,
    }

    pub use crate::firm::pcie::BdfRange as DmarBdfRange;

    #[cfg(test)]
    mod tests {
        use super::{DmarBdfRange, DmarRequesterMatch, DmarStructure, DmarTable};
        use crate::firm::pcie::{Bdf, PciDevice};
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 1052) -
[internal] TestEntry
    struct TestEntry;

    impl PageTableEntry for TestEntry {
        type Flags = ();

        fn new_leaf(_paddr: PhysAddr, _flags: Self::Flags, _size: PageSize) -> Self {
            Self
        }

        fn new_table(_paddr: PhysAddr, _level: u8) -> Self {
            Self
        }
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 1057) -
[internal] new_leaf
    fn new_leaf(_paddr: PhysAddr, _flags: Self::Flags, _size: PageSize) -> Self {
        Self
    }

    fn new_table(_paddr: PhysAddr, _level: u8) -> Self {
        Self
    }

    fn paddr(&self) -> PhysAddr {
        PhysAddr::from_usize(0)
    }

    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 1061) -
[internal] new_table
    fn new_table(_paddr: PhysAddr, _level: u8) -> Self {
        Self
    }

    fn paddr(&self) -> PhysAddr {
        PhysAddr::from_usize(0)
    }

    fn flags(&self) -> Self::Flags {}

    fn is_present(&self) -> bool {
        true
    }
    // ... (truncated, see source file for more)

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 1065) -
[internal] paddr
    fn paddr(&self) -> PhysAddr {
        PhysAddr::from_usize(0)
    }

    fn flags(&self) -> Self::Flags {}

    fn is_present(&self) -> bool {
        true
    }

    fn entry_kind(&self, _level: u8) -> PageTableEntryKind {
        PageTableEntryKind::Leaf
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 1069) -
[internal] flags
    fn flags(&self) -> Self::Flags {}

    fn is_present(&self) -> bool {
        true
    }

    fn entry_kind(&self, _level: u8) -> PageTableEntryKind {
        PageTableEntryKind::Leaf
    }

    fn clear(&mut self) {}

    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 1071) -
[internal] is_present
    fn is_present(&self) -> bool {
        true
    }

    fn entry_kind(&self, _level: u8) -> PageTableEntryKind {
        PageTableEntryKind::Leaf
    }

    fn clear(&mut self) {}

    fn bits(&self) -> u64 {
        0
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 1075) -
[internal] entry_kind
    fn entry_kind(&self, _level: u8) -> PageTableEntryKind {
        PageTableEntryKind::Leaf
    }

    fn clear(&mut self) {}

    fn bits(&self) -> u64 {
        0
    }

    fn from_bits(_bits: u64) -> Self {
        Self
    }
    // ... (truncated, see source file for more)

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 1079) -
[internal] clear
    fn clear(&mut self) {}

    fn bits(&self) -> u64 {
        0
    }

    fn from_bits(_bits: u64) -> Self {
        Self
    }
}

fn registers(base: usize) -> MmioAddrRange {
    // ... (truncated, see source file for more)
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 1081) -
[internal] bits
    fn bits(&self) -> u64 {
        0
    }

    fn from_bits(_bits: u64) -> Self {
        Self
    }
}

fn registers(base: usize) -> MmioAddrRange {
    <MmioAddrRange as MmioRange<usize>>::from_start_size(
        MmioAddr::from(base),
    // ... (truncated, see source file for more)
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 1085) -
[internal] from_bits
    fn from_bits(_bits: u64) -> Self {
        Self
    }
}

fn registers(base: usize) -> MmioAddrRange {
    <MmioAddrRange as MmioRange<usize>>::from_start_size(
        MmioAddr::from(base),
        super::VTD_REGISTER_WINDOW_SIZE,
    )
    .unwrap()
}
// ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 1090) -
[internal] registers
    fn registers(base: usize) -> MmioAddrRange {
        <MmioAddrRange as MmioRange<usize>>::from_start_size(
            MmioAddr::from(base),
            super::VTD_REGISTER_WINDOW_SIZE,
        )
        .unwrap()
    }

    fn write_sdt_header(bytes: &mut [u8], signature: Signature, length: usize) {
        bytes[0..4].copy_from_slice(signature.as_str().as_bytes());
        bytes[4..8].copy_from_slice(&(length as u32).to_le_bytes());
        bytes[8] = 1;
    // ... (truncated, see source file for more)
}

```



```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 1098) -
[internal] write_sdt_header

fn write_sdt_header(bytes: &mut [u8], signature: Signature, length: usize) {
    bytes[0..4].copy_from_slice(signature.as_str().as_bytes());
    bytes[4..8].copy_from_slice(&(length as u32).to_le_bytes());
    bytes[8] = 1;
}

fn finish_sdt_checksum(bytes: &mut [u8]) {
    bytes[9] = 0;
    bytes[9] = 0u8.wrapping_sub(
        bytes
            .iter()
            .fold(0, |sum: u8, byte| sum.wrapping_add(*byte)),
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 1104) -
[internal] finish_sdt_checksum

fn finish_sdt_checksum(bytes: &mut [u8]) {
    bytes[9] = 0;
    bytes[9] = 0u8.wrapping_sub(
        bytes
            .iter()
            .fold(0, |sum: u8, byte| sum.wrapping_add(*byte)),
    );
}

fn table_mapping(bytes: &[u8]) -> Mapping<TestEntry, VirtAddr> {
    let start = VirtAddr::from_usize(bytes.as_ptr() as usize);
    Mapping::new(
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 1113) -
[internal] table_mapping

fn table_mapping(bytes: &[u8]) -> Mapping<TestEntry, VirtAddr> {
    let start = VirtAddr::from_usize(bytes.as_ptr() as usize);
    Mapping::new(
        VirtAddrRange::from_start_size(start, bytes.len()),
        PhysAddr::from_usize(0),
        (),
    )
}

#[test]
fn parses_drhd_and_exact_bdf_ranges() {
    let mut dmar = [0u8; 88];
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 1123) -
[internal] parses_drhd_and_exact_bdf_ranges

fn parses_drhd_and_exact_bdf_ranges() {
    let mut dmar = [0u8; 88];
    write_sdt_header(&mut dmar, Signature::DMAR, 88);
    dmar[36] = 47;

    dmar[48..50].copy_from_slice(&0u16.to_le_bytes());
    dmar[50..52].copy_from_slice(&40u16.to_le_bytes());
    dmar[54..56].copy_from_slice(&3u16.to_le_bytes());
    dmar[56..64].copy_from_slice(&(0xfed9_0000u64).to_le_bytes());

    for (offset, function) in [(64, 0u8), (72, 1u8), (80, 2u8)] {
        dmar[offset] = 1;
    // ... (truncated, see source file for more)

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 1174) -
[internal] parses_reserved_memory_affinity_and_satc
    fn parses_reserved_memory_affinity_and_satc() {
        let mut dmar = [0u8; 100];
        write_sdt_header(&mut dmar, Signature::DMAR, 100);
        dmar[36] = 47;

        dmar[48..50].copy_from_slice(&1u16.to_le_bytes());
        dmar[50..52].copy_from_slice(&24u16.to_le_bytes());
        dmar[54..56].copy_from_slice(&2u16.to_le_bytes());
        dmar[56..64].copy_from_slice(&(0x1000_0000u64).to_le_bytes());
        dmar[64..72].copy_from_slice(&(0x1000_ffffu64).to_le_bytes());

        dmar[72..74].copy_from_slice(&3u16.to_le_bytes());
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 1225) -
[internal] parses_andd_and_preserves_multi_hop_scope_path
    fn parses_andd_and_preserves_multi_hop_scope_path() {
        let mut dmar = [0u8; 86];
        write_sdt_header(&mut dmar, Signature::DMAR, 86);
        dmar[36] = 47;

        dmar[48..50].copy_from_slice(&0u16.to_le_bytes());
        dmar[50..52].copy_from_slice(&26u16.to_le_bytes());
        dmar[56..64].copy_from_slice(&(0xfed9_0000u64).to_le_bytes());

        dmar[64] = 5;
        dmar[65] = 10;
        dmar[68] = 7;
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/dmar.rs (line 1284) -
[internal] parses_sidp_and_ignores_satc_policy_when_present
    fn parses_sidp_and_ignores_satc_policy_when_present() {
        let mut dmar = [0u8; 80];
        write_sdt_header(&mut dmar, Signature::DMAR, 80);
        dmar[36] = 47;

        dmar[48..50].copy_from_slice(&5u16.to_le_bytes());
        dmar[50..52].copy_from_slice(&16u16.to_le_bytes());
        dmar[52] = 1;
        dmar[54..56].copy_from_slice(&2u16.to_le_bytes());
        dmar[56] = 1;
        dmar[57] = 8;
        dmar[61] = 0x2a;
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/error.rs (line 8) -
[public] VtdError
    pub enum VtdError {
        UnsupportedAddressWidth,
        UnsupportedGranule,
        PageSelectiveInvalidationUnavailable,
        InvalidDomainId,
        InvalidSourceId,
        RegisterTransitionTimeout,
    }

    impl From<VtdError> for Error {
        #[inline]
        fn from(value: VtdError) -> Self {
            // ... (truncated, see source file for more)
        }
    }

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/error.rs (line 19) -
[internal] from

```
fn from(value: VtdError) -> Self {
    match value {
        VtdError::UnsupportedAddressWidth => Self::InvalidWidth,
        VtdError::UnsupportedGranule => Self::InvalidGranule,
        VtdError::PageSelectiveInvalidationUnavailable => Self::FeatureUnavailable,
        VtdError::InvalidDomainId => Self::InvalidAddressSpace,
        VtdError::InvalidSourceId => Self::InvalidClient,
        VtdError::RegisterTransitionTimeout => Self::ControllerUnavailable,
    }
}
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/info.rs (line 21) -
[public] VtdVersion

```
pub struct VtdVersion(u32);

impl VtdVersion {
    #[inline]
    pub const fn from_bits(bits: u32) -> Self {
        Self(bits)
    }

    #[inline]
    pub const fn bits(self) -> u32 {
        self.0
    }
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/info.rs (line 47) -
[public] VtdDomain

```
pub struct VtdDomain {
    id: VtdIoDomain,
    root: PhysAddr,
    width: VtdSecondLevelAddressWidth,
    stage: TranslationStage,
}

impl VtdDomain {
    #[inline]
    pub const fn new(
        id: VtdIoDomain,
        root: PhysAddr,
    ) // ... (truncated, see source file for more)
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/info.rs (line 98) -
[internal] from

```
fn from(value: VtdDomainToken) -> Self {
    value.domain()
}

/// VT-d second-level address-space installation controls.
#[derive(Clone, Copy, Debug, PartialEq, Eq, Hash)]
pub struct VtdDomainControls {
    id: VtdIoDomain,
    width: VtdSecondLevelAddressWidth,
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/info.rs (line 105) -

```
[public] VtdDomainControls
    pub struct VtdDomainControls {
        id: VtdIoDomain,
        width: VtdSecondLevelAddressWidth,
    }

    impl VtdDomainControls {
        #[inline]
        pub const fn new(id: VtdIoDomain, width: VtdSecondLevelAddressWidth) -> Self {
            Self { id, width }
        }

        #[inline]
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/info.rs (line 134) -

```
[public] VtdDomainToken
    pub struct VtdDomainToken {
        root: PhysAddr,
        controls: VtdDomainControls,
    }

    impl VtdDomainToken {
        #[inline]
        pub const fn new(root: PhysAddr, controls: VtdDomainControls) -> Self {
            Self { root, controls }
        }

        #[inline]
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/info.rs (line 173) -

```
[internal] root
    fn root(self) -> PhysAddr {
        self.root
    }
}

/// Stateless VT-d domain installer for `PageTable::install_with`.
///
/// This encodes the second-level page-table root into a domain token. Hardware
/// activation is controller-specific because VT-d still has to publish that
/// token through root/context tables for concrete requesters.
#[derive(Clone, Copy, Debug, Default)]
pub struct VtdDomainActivation;
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/info.rs (line 184) -

```
[public] VtdDomainActivation
    pub struct VtdDomainActivation;

    impl AddrSpaceActivation for VtdDomainActivation {
        type Token = VtdDomainToken;
        type Controls = VtdDomainControls;

        #[inline]
        fn install(&self, root: PhysAddr, controls: Self::Controls) -> PagingResult<Self::Token> {
            if !root.is_aligned(PageSize::Size4K.bytes()) {
                return Err(PagingError::NotAligned);
            }
            Ok(VtdDomainToken::new(root, controls))
            // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/info.rs (line 191) -
[internal] install

```
fn install(&self, root: PhysAddr, controls: Self::Controls) -> PagingResult<Self::Token> {
    if !root.is_aligned(PageSize::Size4K.bytes()) {
        return Err(PagingError::NotAligned);
    }
    Ok(VtdDomainToken::new(root, controls))
}

#[inline]
unsafe fn activate(&self, _token: Self::Token) -> PagingResult {
    Err(PagingError::InvalidMappingShape)
}
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/info.rs (line 206) -
[public] VtdInfo

```
pub struct VtdInfo {
    base: IommuInfo,
    version: VtdVersion,
    host_address_width: Option<u8>,
    cap: VtdCapability,
    ecap: VtdExtendedCapability,
    explicit_device_scopes: bool,
}

impl VtdInfo {
    #[inline]
    pub const fn new(
        // ... (truncated, see source file for more)
    )
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/info.rs (line 236) -
[public] from_registers

```
pub fn from_registers(
    segment: Option<u16>,
    mmio: MmioAddrRange,
    version: u32,
    host_address_width: Option<u8>,
    cap: u64,
    ecap: u64,
    explicit_device_scopes: bool,
) -> Self {
    let cap = VtdCapability::from_bits(cap);
    let ecap = VtdExtendedCapability::from_bits(ecap);
    let base = IommuInfo::new(
        // ... (truncated, see source file for more)
    )
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/info.rs (line 311) -
[internal] version_splits_major_and_minor

```
fn version_splits_major_and_minor() {
    let version = VtdVersion::from_bits(0x23);
    assert_eq!(version.major(), 2);
    assert_eq!(version.minor(), 3);
}

#[test]
fn vtd_domain_is_the_common_domain_value() {
    let domain = VtdIoDomain::new(0x1234, 7).unwrap();

    assert_eq!(domain.id(), 0x1234);
    assert_eq!(domain.generation(), 7);
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/info.rs (line 318) -

```
[internal] vtd_domain_is_the_common_domain_value
fn vtd_domain_is_the_common_domain_value() {
    let domain = VtdIoDomain::new(0x1234, 7).unwrap();

    assert_eq!(domain.id(), 0x1234);
    assert_eq!(domain.generation(), 7);
    assert_eq!(VTD_DEFAULT_DOMAIN.id(), 1);
}

#[test]
fn domain_activation_installs_second_level_root_token() {
    let activation = VtdDomainActivation;
    let controls =
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/info.rs (line 327) -

```
[internal] domain_activation_installs_second_level_root_token
```

```
fn domain_activation_installs_second_level_root_token() {
    let activation = VtdDomainActivation;
    let controls =
        VtdDomainControls::new(VTD_DEFAULT_DOMAIN, VtdSecondLevelAddressWidth::Bits48);
    let token = activation
        .install(PhysAddr::from(0x4000usize), controls)
        .unwrap();
    let domain = token.domain();

    assert_eq!(token.root(), PhysAddr::from(0x4000usize));
    assert_eq!(token.controls(), controls);
    assert_eq!(domain.id(), VTD_DEFAULT_DOMAIN);
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/info.rs (line 345) -

```
[internal] domain_activation_rejects_unaligned_roots
```

```
fn domain_activation_rejects_unaligned_roots() {
    let activation = VtdDomainActivation;
    let controls =
        VtdDomainControls::new(VTD_DEFAULT_DOMAIN, VtdSecondLevelAddressWidth::Bits39);

    assert_eq!(
        activation.install(PhysAddr::from(0x1234usize), controls),
        Err(PagingError::NotAligned)
    );
}

#[test]
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/info.rs (line 357) -

```
[internal] stateless_domain_activation_does_not_publish_to_hardware
```

```
fn stateless_domain_activation_does_not_publish_to_hardware() {
    let activation = VtdDomainActivation;
    let controls =
        VtdDomainControls::new(VTD_DEFAULT_DOMAIN, VtdSecondLevelAddressWidth::Bits39);
    let token = activation
        .install(PhysAddr::from(0x4000usize), controls)
        .unwrap();

    assert_eq!(
        unsafe { activation.activate(token) },
        Err(PagingError::InvalidMappingShape)
    );
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/paging.rs (line 26) -

```
[public] VtdSecondLevelAddressWidth

pub enum VtdSecondLevelAddressWidth {
    Bits39 = 39,
    Bits48 = 48,
    Bits57 = 57,
}

impl VtdSecondLevelAddressWidth {
    #[inline]
    pub const fn bits(self) -> u8 {
        self as u8
    }

    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/paging.rs (line 73) -

```
[public] VtdSecondLevelFlags

pub struct VtdSecondLevelFlags {
    access: AccessFlags,
    attrs: MemoryAttributes,
}

impl VtdSecondLevelFlags {
    #[inline]
    pub const fn new(access: AccessFlags) -> Self {
        Self {
            access,
            attrs: MemoryAttributes::writeback(),
        }
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/paging.rs (line 110) -

```
[internal] to_leaf_bits

fn to_leaf_bits(self, size: PageSize) -> u64 {
    let mut bits = 0;
    if self.access.contains(AccessFlags::READ) {
        bits |= READ;
    }
    if self.access.contains(AccessFlags::WRITE) {
        bits |= WRITE;
    }
    if !matches!(size, PageSize::Size4K) {
        bits |= SUPER_PAGE;
    }
    bits
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/paging.rs (line 125) -

```
[internal] from_bits

fn from_bits(bits: u64) -> Self {
    let mut access = AccessFlags::empty();
    if (bits & READ) != 0 {
        access |= AccessFlags::READ;
    }
    if (bits & WRITE) != 0 {
        access |= AccessFlags::WRITE;
    }
    Self {
        access,
        attrs: MemoryAttributes::writeback(),
    }
    // ... (truncated, see source file for more)
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/paging.rs (line 142) -
[internal] from
    fn from(value: DmaAttrs) -> Self {
        Self::new(value.access()).with_attrs(value.memory())
    }
}

impl From<VtdSecondLevelFlags> for DmaAttrs {
    #[inline]
    fn from(value: VtdSecondLevelFlags) -> Self {
        Self::new(value.access, value.attrs)
    }
}

// ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/paging.rs (line 149) -
[internal] from
    fn from(value: VtdSecondLevelFlags) -> Self {
        Self::new(value.access, value.attrs)
    }
}

/// Intel VT-d second-level page-table entry.
#[derive(Clone, Copy, Default, PartialEq, Eq)]
#[repr(transparent)]
pub struct VtdSecondLevelPte(u64);

impl fmt::Debug for VtdSecondLevelPte {
    fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
        // ... (truncated, see source file for more)
    }
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/paging.rs (line 157) -
[public] VtdSecondLevelPte
pub struct VtdSecondLevelPte(u64);

impl fmt::Debug for VtdSecondLevelPte {
    fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
        f.debug_tuple("VtdSecondLevelPte")
            .field(&format_args!("{:#018x}", self.0))
            .finish()
    }
}

impl PageTableEntry for VtdSecondLevelPte {
    type Flags = VtdSecondLevelFlags;
    // ... (truncated, see source file for more)
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/paging.rs (line 160) -
[internal] fmt
    fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
        f.debug_tuple("VtdSecondLevelPte")
            .field(&format_args!("{:#018x}", self.0))
            .finish()
    }
}

impl PageTableEntry for VtdSecondLevelPte {
    type Flags = VtdSecondLevelFlags;

    #[inline]
    fn new_leaf(paddr: PhysAddr, flags: Self::Flags, size: PageSize) -> Self {
        // ... (truncated, see source file for more)
    }
}

```



```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/paging.rs (line 171) -
[internal] new_leaf
    fn new_leaf(paddr: PhysAddr, flags: Self::Flags, size: PageSize) -> Self {
        Self(((paddr.as_usize() as u64) & ADDR_MASK) | flags.to_leaf_bits(size))
    }

    #[inline]
    fn new_table(paddr: PhysAddr, _level: u8) -> Self {
        Self(((paddr.as_usize() as u64) & ADDR_MASK) | PRESENT)
    }

    #[inline]
    fn paddr(&self) -> PhysAddr {
        PhysAddr::from_usize((self.0 & ADDR_MASK) as usize)
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/paging.rs (line 176) -
[internal] new_table
    fn new_table(paddr: PhysAddr, _level: u8) -> Self {
        Self(((paddr.as_usize() as u64) & ADDR_MASK) | PRESENT)
    }

    #[inline]
    fn paddr(&self) -> PhysAddr {
        PhysAddr::from_usize((self.0 & ADDR_MASK) as usize)
    }

    #[inline]
    fn flags(&self) -> Self::Flags {
        VtdSecondLevelFlags::from_bits(self.0)
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/paging.rs (line 181) -
[internal] paddr
    fn paddr(&self) -> PhysAddr {
        PhysAddr::from_usize((self.0 & ADDR_MASK) as usize)
    }

    #[inline]
    fn flags(&self) -> Self::Flags {
        VtdSecondLevelFlags::from_bits(self.0)
    }

    #[inline]
    fn is_present(&self) -> bool {
        (self.0 & PRESENT) != 0
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/paging.rs (line 186) -
[internal] flags
    fn flags(&self) -> Self::Flags {
        VtdSecondLevelFlags::from_bits(self.0)
    }

    #[inline]
    fn is_present(&self) -> bool {
        (self.0 & PRESENT) != 0
    }

    #[inline]
    fn entry_kind(&self, level: u8) -> PageTableEntryKind {
        if !self.is_present() {
            // ... (truncated, see source file for more)
        }
    }

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/paging.rs (line 191) -
[internal] is_present
    fn is_present(&self) -> bool {
        (self.0 & PRESENT) != 0
    }

    #[inline]
    fn entry_kind(&self, level: u8) -> PageTableEntryKind {
        if !self.is_present() {
            PageTableEntryKind::Invalid
        } else if level == 1 || (self.0 & SUPER_PAGE) != 0 {
            PageTableEntryKind::Leaf
        } else {
            PageTableEntryKind::Table
        }
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/paging.rs (line 196) -
[internal] entry_kind
    fn entry_kind(&self, level: u8) -> PageTableEntryKind {
        if !self.is_present() {
            PageTableEntryKind::Invalid
        } else if level == 1 || (self.0 & SUPER_PAGE) != 0 {
            PageTableEntryKind::Leaf
        } else {
            PageTableEntryKind::Table
        }
    }

    #[inline]
    fn clear(&mut self) {
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/paging.rs (line 207) -
[internal] clear
    fn clear(&mut self) {
        self.0 = 0;
    }

    #[inline]
    fn bits(&self) -> u64 {
        self.0
    }

    #[inline]
    fn from_bits(bits: u64) -> Self {
        Self(bits)
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/paging.rs (line 212) -
[internal] bits
    fn bits(&self) -> u64 {
        self.0
    }

    #[inline]
    fn from_bits(bits: u64) -> Self {
        Self(bits)
    }
}

macro_rules! define_vtd_meta {
    ($name:ident, $levels:expr, $va_bits:expr) => {
        // ... (truncated, see source file for more)
    }
}

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/paging.rs (line 217) -
[internal] from_bits
    fn from_bits(bits: u64) -> Self {
        Self(bits)
    }
}

macro_rules! define_vtd_meta {
    ($name:ident, $levels:expr, $va_bits:expr) => {
        #[derive(Clone, Copy, Debug)]
        pub struct $name;

        impl PagingMetaData for $name {
            const LEVELS: usize = $levels;
            // ... (truncated, see source file for more)
        }
    }
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/paging.rs (line 235) -
[internal] level_shift
    fn level_shift(level: u8) -> u32 {
        12 + ((level as u32) - 1) * 9
    }

    #[inline]
    fn level_supports_leaf(level: u8, size: PageSize) -> bool {
        matches!(
            (level, size),
            (1, PageSize::Size4K) | (2, PageSize::Size2M) | (3, PageSize::Size1G)
        )
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/paging.rs (line 240) -
[internal] level_supports_leaf
    fn level_supports_leaf(level: u8, size: PageSize) -> bool {
        matches!(
            (level, size),
            (1, PageSize::Size4K) | (2, PageSize::Size2M) | (3, PageSize::Size1G)
        )
    }
}

define_vtd_meta!(VtdSecondLevelMeta39, 3, 39);
define_vtd_meta!(VtdSecondLevelMeta48, 4, 48);
// ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/paging.rs (line 266) -
[internal] leaf_2m_sets_super_bit
    fn leaf_2m_sets_super_bit() {
        let flags = VtdSecondLevelFlags::new(AccessFlags::READ | AccessFlags::WRITE);
        let pte =
            VtdSecondLevelPte::new_leaf(PhysAddr::from_usize(0x20_0000), flags, PageSize::Size2M);

        assert_eq!(pte.entry_kind(2), PageTableEntryKind::Leaf);
        assert_eq!(pte.bits() & SUPER_PAGE, SUPER_PAGE);
        assert_eq!(pte.paddr(), PhysAddr::from_usize(0x20_0000));
        assert!(pte.flags().access().contains(AccessFlags::READ));
        assert!(pte.flags().access().contains(AccessFlags::WRITE));
    }
    // ... (truncated, see source file for more)

```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/paging.rs (line 279) -
[internal] table_entry_is_intermediate
    fn table_entry_is_intermediate() {
        let pte = VtdSecondLevelPte::new_table(PhysAddr::from_usize(0x4000), 4);
        assert_eq!(pte.entry_kind(4), PageTableEntryKind::Table);
        assert_eq!(pte.paddr(), PhysAddr::from_usize(0x4000));
    }

    #[test]
    fn address_widths_match_vtd_agaw_shapes() {
        assert_eq!(VtdSecondLevelAddressWidth::Bits39.levels(), 3);
        assert_eq!(VtdSecondLevelAddressWidth::Bits48.levels(), 4);
        assert_eq!(VtdSecondLevelAddressWidth::Bits57.levels(), 5);
        assert_eq!(VtdSecondLevelAddressWidth::Bits39.sagaw_bit(), 0b0010);
        // ... (truncated, see source file for more)
    }
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/intel/paging.rs (line 286) -
[internal] address_widths_match_vtd_agaw_shapes
    fn address_widths_match_vtd_agaw_shapes() {
        assert_eq!(VtdSecondLevelAddressWidth::Bits39.levels(), 3);
        assert_eq!(VtdSecondLevelAddressWidth::Bits48.levels(), 4);
        assert_eq!(VtdSecondLevelAddressWidth::Bits57.levels(), 5);
        assert_eq!(VtdSecondLevelAddressWidth::Bits39.sagaw_bit(), 0b0010);
        assert_eq!(VtdSecondLevelAddressWidth::Bits48.sagaw_bit(), 0b0100);
        assert_eq!(VtdSecondLevelAddressWidth::Bits57.sagaw_bit(), 0b1000);
    }
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/interrupt.rs (line 17) -
[public] X86InterruptVector
    pub struct X86InterruptVector(u8);

    impl X86InterruptVector {
        pub const MIN_EXTERNAL: u8 = 32;

        #[inline]
        pub const fn new(vector: u8) -> Result<Self> {
            if vector < Self::MIN_EXTERNAL {
                return Err(Error::InvalidRange);
            }
            Ok(Self(vector))
        }
        // ... (truncated, see source file for more)
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/interrupt.rs (line 38) -
[public] X86MsiDestination
    pub enum X86MsiDestination {
        Physical(u8),
        Logical(u8),
    }

    impl X86MsiDestination {
        #[inline]
        pub const fn id(self) -> u8 {
            match self {
                Self::Physical(id) | Self::Logical(id) => id,
            }
        }
        // ... (truncated, see source file for more)
    }

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/interrupt.rs (line 60) -

```
[public] X86MsiDeliveryMode
    pub enum X86MsiDeliveryMode {
        Fixed = 0b000,
        LowestPriority = 0b001,
        Smi = 0b010,
        Nmi = 0b100,
        Init = 0b101,
        ExtInt = 0b111,
    }

    /// x86 MSI trigger mode.
    #[derive(Clone, Copy, Debug, Default, PartialEq, Eq, Hash)]
    #[repr(u8)]
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/interrupt.rs (line 72) -

```
[public] X86MsiTriggerMode
    pub enum X86MsiTriggerMode {
        #[default]
        Edge,
        Level,
    }

    /// x86 MSI level bit.
    #[derive(Clone, Copy, Debug, PartialEq, Eq, Hash)]
    #[repr(u8)]
    pub enum X86MsiLevel {
        Deassert,
        Assert,
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/interrupt.rs (line 81) -

```
[public] X86MsiLevel
    pub enum X86MsiLevel {
        Deassert,
        Assert,
    }

    /// Builder for an x86 APIC MSI/MSI-X delivery message.
    #[derive(Clone, Copy, Debug, PartialEq, Eq, Hash)]
    pub struct X86MsiDelivery {
        destination: X86MsiDestination,
        vector: X86InterruptVector,
        delivery_mode: X86MsiDeliveryMode,
        trigger_mode: X86MsiTriggerMode,
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/interrupt.rs (line 88) -

```
[public] X86MsiDelivery
    pub struct X86MsiDelivery {
        destination: X86MsiDestination,
        vector: X86InterruptVector,
        delivery_mode: X86MsiDeliveryMode,
        trigger_mode: X86MsiTriggerMode,
        level: X86MsiLevel,
        redirection_hint: bool,
    }

    impl X86MsiDelivery {
        #[inline]
        pub const fn fixed(destination: X86MsiDestination, vector: X86InterruptVector) -> Self {
            // ... (truncated, see source file for more)
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/interrupt.rs (line 200) -
[internal] from
    fn from(value: X86MsiDelivery) -> Self {
        value.message()
    }
}

#[cfg(test)]
mod tests {
    use super::*;

    #[test]
    fn rejects_exception_vectors_for_msi_delivery() {
        assert_eq!(X86InterruptVector::new(31), Err(Error::InvalidRange));
        // ... (truncated, see source file for more)
    }
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/interrupt.rs (line 210) -
[internal] rejects_exception_vectors_for_msi_delivery
    fn rejects_exception_vectors_for_msi_delivery() {
        assert_eq!(X86InterruptVector::new(31), Err(Error::InvalidRange));
        assert_eq!(X86InterruptVector::new(32).unwrap().get(), 32);
    }

    #[test]
    fn fixed_physical_delivery_encodes_apic_msi_message() {
        let message = X86MsiDelivery::fixed(
            X86MsiDestination::Physical(0x2a),
            X86InterruptVector::new(0x45).unwrap(),
        )
        .message();
        // ... (truncated, see source file for more)
    }
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/interrupt.rs (line 216) -
[internal] fixed_physical_delivery_encodes_apic_msi_message
    fn fixed_physical_delivery_encodes_apic_msi_message() {
        let message = X86MsiDelivery::fixed(
            X86MsiDestination::Physical(0x2a),
            X86InterruptVector::new(0x45).unwrap(),
        )
        .message();

        assert_eq!(message.address(), 0xfe2_a000);
        assert_eq!(message.data(), 0x4045);
    }

    #[test]
    // ... (truncated, see source file for more)
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/arch/x86_64/interrupt.rs (line 228) -
[internal] logical_lowest_priority_delivery_sets_route_bits
    fn logical_lowest_priority_delivery_sets_route_bits() {
        let message = X86MsiDelivery::new(
            X86MsiDestination::Logical(0x03),
            X86InterruptVector::new(0x80).unwrap(),
            X86MsiDeliveryMode::LowestPriority,
        )
        .with_trigger_mode(X86MsiTriggerMode::Level)
        .with_redirection_hint(true)
        .message();

        assert_eq!(message.address(), 0xfe0_300c);
        assert_eq!(message.data(), 0xc180);
        // ... (truncated, see source file for more)
    }
}

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/caps.rs (line 14) - [public] DmaAttrs

```
pub struct DmaAttrs {
    access: DmaAccess,
    memory: MemoryAttributes,
}

impl DmaAttrs {
    #[inline]
    pub const fn new(access: DmaAccess, memory: MemoryAttributes) -> Self {
        Self { access, memory }
    }

    #[inline]
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/caps.rs (line 39) - [public]
TranslationStage

```
pub enum TranslationStage {
    Stage1,
    Stage2,
    Nested,
}

/// Client-local binding slot.
#[derive(Clone, Copy, Debug, PartialEq, Eq, Hash, Default)]
pub enum BindingSelector {
    #[default]
    Default,
    AddrSpace(u32),
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/caps.rs (line 47) - [public]
BindingSelector

```
pub enum BindingSelector {
    #[default]
    Default,
    AddrSpace(u32),
    Substream(u32),
}

impl BindingSelector {
    #[inline]
    pub const fn from_addr_space(id: u32) -> Self {
        Self::AddrSpace(id)
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/caps.rs (line 68) - [public] BindingTarget

```
pub enum BindingTarget<Domain> {
    Abort,
    PassThrough,
    Domain(Domain),
}

/// One client binding request.
#[derive(Clone, Copy, Debug, PartialEq, Eq)]
pub struct Binding<Client, Domain> {
    client: Client,
    selector: BindingSelector,
    target: BindingTarget<Domain>,
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/caps.rs (line 76) - [public] Binding

```
pub struct Binding<Client, Domain> {
    client: Client,
    selector: BindingSelector,
    target: BindingTarget<Domain>,
}

impl<Client, Domain> Binding<Client, Domain> {
    #[inline]
    pub const fn new(
        client: Client,
        selector: BindingSelector,
        target: BindingTarget<Domain>,
        // ... (truncated, see source file for more)
    )
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/caps.rs (line 97) - [public] client

```
pub fn client(self) -> Client
where
    Client: Copy,
{
    self.client
}

#[inline]
pub fn selector(self) -> BindingSelector {
    self.selector
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/caps.rs (line 105) - [public] selector

```
pub fn selector(self) -> BindingSelector {
    self.selector
}

#[inline]
pub fn target(self) -> BindingTarget<Domain>
where
    Domain: Copy,
{
    self.target
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/caps.rs (line 110) - [public] target

```
pub fn target(self) -> BindingTarget<Domain>
where
    Domain: Copy,
{
    self.target
}

/// Coarse feature flags common to modern DMA-remapping units.
#[derive(Clone, Copy, Debug, Default, PartialEq, Eq, Hash)]
pub struct CapabilityFlags(u64);

// ... (truncated, see source file for more)
```


driverspace_workspace/drivers/iommu-TrangorgeOS/src/caps.rs (line 120) - [public]

CapabilityFlags

```
pub struct CapabilityFlags(u64);

impl CapabilityFlags {
    pub const TRANSLATION: Self = Self(1 << 0);
    pub const PASID: Self = Self(1 << 1);
    pub const ATS: Self = Self(1 << 2);
    pub const DEVICE_TLB: Self = Self(1 << 3);
    pub const INTERRUPT_REMAPPING: Self = Self(1 << 4);
    pub const NESTED_TRANSLATION: Self = Self(1 << 5);
    pub const PAGE_REQUESTS: Self = Self(1 << 6);
    pub const DIRTY_TRACKING: Self = Self(1 << 7);

    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/caps.rs (line 152) - [internal] bitor

```
fn bitor(self, rhs: Self) -> Self::Output {
    Self(self.0 | rhs.0)
}

impl BitOrAssign for CapabilityFlags {
    #[inline]
    fn bitor_assign(&mut self, rhs: Self) {
        self.0 |= rhs.0;
    }

    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/caps.rs (line 159) - [internal]

bitor_assign

```
fn bitor_assign(&mut self, rhs: Self) {
    self.0 |= rhs.0;
}

impl BitAnd for CapabilityFlags {
    type Output = Self;

    #[inline]
    fn bitand(self, rhs: Self) -> Self::Output {
        Self(self.0 & rhs.0)
    }

    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/caps.rs (line 168) - [internal] bitand

```
fn bitand(self, rhs: Self) -> Self::Output {
    Self(self.0 & rhs.0)
}

impl BitAndAssign for CapabilityFlags {
    #[inline]
    fn bitand_assign(&mut self, rhs: Self) {
        self.0 &= rhs.0;
    }
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/caps.rs (line 175) - [internal]

bitand_assign

```
fn bitand_assign(&mut self, rhs: Self) {
    self.0 &= rhs.0;
}
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 24) - [public] MsiMessage

```
pub struct MsiMessage {
    address: u64,
    data: u32,
}

impl MsiMessage {
    #[inline]
    pub const fn new(address: u64, data: u32) -> Self {
        Self { address, data }
    }

    #[inline]
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 48) - [public]

InterruptRoute

```
pub enum InterruptRoute {
    Disabled,
    Msi(MsiMessage),
}

impl InterruptRoute {
    #[inline]
    pub const fn msi(message: MsiMessage) -> Self {
        Self::Msi(message)
    }

    #[inline]
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 67) - [internal] from

```
fn from(value: MsiMessage) -> Self {
    Self::Msi(value)
}

/// Caller-provided backing for one hardware command queue.
///
/// The backing must be physically contiguous and mapped writable at `virt`.
/// Construction is unsafe because the crate cannot prove that the virtual
/// range is live, writable, and aliases the supplied physical range.
#[derive(Clone, Copy, Debug, PartialEq, Eq)]
pub struct CommandQueueBacking {
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 78) - [public]

CommandQueueBacking

```
pub struct CommandQueueBacking {
    phys: PhysAddrRange,
    virt: VirtAddrRange,
    entry_count: usize,
    entry_bytes: usize,
}

impl CommandQueueBacking {
    /// # Safety
    ///
    /// `virt` must be a live writable mapping of `phys` for at least
    /// `entry_count * entry_bytes` bytes, and the memory must remain owned by
    /// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 143) - [public] byte_len

```
pub fn byte_len(self) -> usize {
    self.entry_count * self.entry_bytes
}

#[inline]
fn entry_vaddr(self, slot: usize) -> Result<VirtAddr> {
    let offset = slot
        .checked_mul(self.entry_bytes)
        .ok_or(Error::AddressOverflow)?;
    self.virt
        .start
        .checked_add(offset)
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 148) - [internal] entry_vaddr

```
fn entry_vaddr(self, slot: usize) -> Result<VirtAddr> {
    let offset = slot
        .checked_mul(self.entry_bytes)
        .ok_or(Error::AddressOverflow)?;
    self.virt
        .start
        .checked_add(offset)
        .ok_or(Error::AddressOverflow)
}

/// Caller-provided backing for a fixed-width hardware descriptor table.
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 171) - [public] DescriptorTableBacking

```
pub struct DescriptorTableBacking<const ENTRY_BYTES: usize> {
    phys: PhysAddrRange,
    virt: VirtAddrRange,
    entry_count: usize,
}

impl<const ENTRY_BYTES: usize> DescriptorTableBacking<ENTRY_BYTES> {
    /// # Safety
    ///
    /// `virt` must be a live writable mapping of `phys` for at least
    /// `entry_count * ENTRY_BYTES` bytes, and the memory must remain owned by
    /// the hardware table while the controller can read it.
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 251) - [public] byte_len

```
pub fn byte_len(self) -> usize {
    self.entry_count * ENTRY_BYTES
}

#[inline]
pub fn entry_vaddr(self, index: usize) -> Result<VirtAddr> {
    if index >= self.entry_count {
        return Err(Error::InvalidRange);
    }
    let offset = index
        .checked_mul(ENTRY_BYTES)
        .ok_or(Error::AddressOverflow)?;
    // ... (truncated, see source file for more)
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 256) - [public] entry_vaddr
    pub fn entry_vaddr(self, index: usize) -> Result<VirtAddr> {
        if index >= self.entry_count {
            return Err(Error::InvalidRange);
        }
        let offset = index
            .checked_mul(ENTRY_BYTES)
            .ok_or(Error::AddressOverflow)?;
        self.virt
            .start
            .checked_add(offset)
            .ok_or(Error::AddressOverflow)
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 278) - [public] CommandQueue
    pub struct CommandQueue<const N: usize> {
        backing: Option<CommandQueueBacking>,
        claim: AtomicUsize,
        committed: AtomicUsize,
        retired: AtomicUsize,
        slot_state: [AtomicU8; N],
    }

    impl<const N: usize> CommandQueue<N> {
        #[inline]
        pub const fn new() -> Self {
            Self {
                // ... (truncated, see source file for more)
            }
        }
    }

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 299) - [public] init
    pub fn init(&mut self, backing: CommandQueueBacking) -> Result {
        if N == 0 || backing.entry_count() > N {
            return Err(Error::InvalidRange);
        }
        self.backing = Some(backing);
        self.claim.store(0, Ordering::Release);
        self.committed.store(0, Ordering::Release);
        self.retired.store(0, Ordering::Release);
        for slot in &self.slot_state[..backing.entry_count()] {
            slot.store(QueueSlotEmpty, Ordering::Release);
        }
        Ok(())
    }
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 314) - [public] backing
    pub fn backing(&self) -> Option<CommandQueueBacking> {
        self.backing
    }

    #[inline]
    pub fn is_active(&self) -> bool {
        self.backing.is_some()
    }

    #[inline]
    pub fn submit<const W: usize, RH, WT, CE>(
        &self,
        // ... (truncated, see source file for more)
    ) {

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 319) - [public] is_active

```
pub fn is_active(&self) -> bool {
    self.backing.is_some()
}

#[inline]
pub fn submit<const W: usize, RH, WT, CE>(
    &self,
    bytes: [u8; W],
    read_head: RH,
    mut write_tail: WT,
    mut check_error: CE,
) -> Result
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 324) - [public] submit

```
pub fn submit<const W: usize, RH, WT, CE>(
    &self,
    bytes: [u8; W],
    read_head: RH,
    mut write_tail: WT,
    mut check_error: CE,
) -> Result
where
    RH: FnMut() -> Result<usize>,
    WT: FnMut(usize),
    CE: FnMut() -> Result,
{
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 363) - [internal]

wait_for_capacity

```
fn wait_for_capacity(&self, ticket: usize, entry_count: usize) -> Result {
    for _ in 0..COMMAND_QUEUE_POLL_LIMIT {
        let retired = self.retired.load(Ordering::Acquire);
        if ticket.wrapping_sub(retired) < entry_count {
            return Ok(());
        }
        spin_loop();
    }
    Err(Error::ControllerUnavailable)
}

#[inline]
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 375) - [internal]

wait_for_empty

```
fn wait_for_empty(&self, slot: usize) -> Result {
    for _ in 0..COMMAND_QUEUE_POLL_LIMIT {
        if self.slot_state[slot].load(Ordering::Acquire) == QUEUE_SLOT_EMPTY {
            return Ok(());
        }
        spin_loop();
    }
    Err(Error::ControllerUnavailable)
}

#[inline]
fn wait_to_publish(&self, ticket: usize) -> Result {
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 386) - [internal]

wait_to_publish

```
fn wait_to_publish(&self, ticket: usize) -> Result {
    for _ in 0..COMMAND_QUEUE_POLL_LIMIT {
        if self.committed.load(Ordering::Acquire) == ticket {
            return Ok(());
        }
        spin_loop();
    }
    Err(Error::ControllerUnavailable)
}

#[inline]
fn wait_for_consumed<RH>(&self, slot: usize, entry_bytes: usize, mut read_head: RH) -> Result
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 397) - [internal]

wait_for_consumed

```
fn wait_for_consumed<RH>(&self, slot: usize, entry_bytes: usize, mut read_head: RH) -> Result
where
    RH: FnMut() -> Result<usize>,
{
    for _ in 0..COMMAND_QUEUE_POLL_LIMIT {
        let head_slot = read_head()? / entry_bytes;
        if head_slot != slot {
            return Ok(());
        }
        spin_loop();
    }
    Err(Error::ControllerUnavailable)
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 412) - [internal]

advance_retired

```
fn advance_retired(&self, next: usize) {
    let mut current = self.retired.load(Ordering::Acquire);
    while current < next {
        match self.retired.compare_exchange_weak(
            current,
            next,
            Ordering::AcqRel,
            Ordering::Acquire,
        ) {
            Ok(_) => break,
            Err(observed) => current = observed,
        }
    }
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 428) - [internal] write_slot

```
fn write_slot<const W: usize>(
    &self,
    backing: CommandQueueBacking,
    slot: usize,
    bytes: &[u8; W],
) -> Result {
    let vaddr = backing.entry_vaddr(slot)?;
    unsafe {
        let base = vaddr.as_mut_ptr_of::<u8>();
        for offset in 0..backing.entry_bytes() {
            let value = bytes.get(offset).copied().unwrap_or(0);
            base.add(offset).write_volatile(value);
        }
    }
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 448) - [internal] default

```
fn default() -> Self {
    Self::new()
}

/// IOTLB invalidation contract.
///
/// The supertrait is the CPU-side `kore_memory` invalidation hook. Device-facing
/// implementations can therefore be passed directly into a `kore_memory` map/unmap
/// call while also carrying IOTLB and device-TLB invalidation primitives for
/// the controller's post-write hardware synchronization.
pub trait IoTlbInvalidation<V: MemoryAddr = IoviAddr>: TlbInvalidation<V> {
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 459) - [public]

IoTlbInvalidation

```
/// IOTLB invalidation contract.
///
/// The supertrait is the CPU-side `kore_memory` invalidation hook. Device-facing
/// implementations can therefore be passed directly into a `kore_memory` map/unmap
/// call while also carrying IOTLB and device-TLB invalidation primitives for
/// the controller's post-write hardware synchronization.

pub trait IoTlbInvalidation<V: MemoryAddr = IoviAddr>: TlbInvalidation<V> {
    type Client: Copy + Debug + Eq;

    fn flush_iotlb(&self, iova: V);
    fn flush_iotlb_all(&self);

    fn flush_iotlb_range(&self, start: V, page_size: PageSize, count_pages: usize) {
        let stride = page_size.bytes();
        let mut base: usize = start.into();
        for _ in 0..count_pages {
            self.flush_iotlb(<V as From<usize>>::from(base));
            base = base.saturating_add(stride);
        }
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 462) - [internal]

flush_iotlb

```
fn flush_iotlb(&self, iova: V);
fn flush_iotlb_all(&self);

fn flush_iotlb_range(&self, start: V, page_size: PageSize, count_pages: usize) {
    let stride = page_size.bytes();
    let mut base: usize = start.into();
    for _ in 0..count_pages {
        self.flush_iotlb(<V as From<usize>>::from(base));
        base = base.saturating_add(stride);
    }
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 463) - [internal]

flush_iotlb_all

```
fn flush_iotlb_all(&self);

fn flush_iotlb_range(&self, start: V, page_size: PageSize, count_pages: usize) {
    let stride = page_size.bytes();
    let mut base: usize = start.into();
    for _ in 0..count_pages {
        self.flush_iotlb(<V as From<usize>>::from(base));
        base = base.saturating_add(stride);
    }
}

fn flush_device_tlb(&self, client: Self::Client, iova: V);
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 465) - [internal]

flush_iotlb_range

```
fn flush_iotlb_range(&self, start: V, page_size: PageSize, count_pages: usize) {
    let stride = page_size.bytes();
    let mut base: usize = start.into();
    for _ in 0..count_pages {
        self.flush_iotlb(<V as From<usize>>::from(base));
        base = base.saturating_add(stride);
    }
}

fn flush_device_tlb(&self, client: Self::Client, iova: V);
fn flush_device_tlb_all(&self, client: Self::Client);

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 474) - [internal]

flush_device_tlb

```
fn flush_device_tlb(&self, client: Self::Client, iova: V);
fn flush_device_tlb_all(&self, client: Self::Client);

fn flush_device_tlb_range(
    &self,
    client: Self::Client,
    start: V,
    page_size: PageSize,
    count_pages: usize,
) {
    let stride = page_size.bytes();
    let mut base: usize = start.into();
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 475) - [internal]

flush_device_tlb_all

```
fn flush_device_tlb_all(&self, client: Self::Client);

fn flush_device_tlb_range(
    &self,
    client: Self::Client,
    start: V,
    page_size: PageSize,
    count_pages: usize,
) {
    let stride = page_size.bytes();
    let mut base: usize = start.into();
    for _ in 0..count_pages {
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 477) - [internal]

flush_device_tlb_range

```
fn flush_device_tlb_range(
    &self,
    client: Self::Client,
    start: V,
    page_size: PageSize,
    count_pages: usize,
) {
    let stride = page_size.bytes();
    let mut base: usize = start.into();
    for _ in 0..count_pages {
        self.flush_device_tlb(client, <V as From<usize>>::from(base));
        base = base.saturating_add(stride);
    }
    // ... (truncated, see source file for more)
```


driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 492) - [internal]

prefer_full_iotlb_flush

```
    fn prefer_full_iotlb_flush(&self, pending_count: usize) -> bool {
        self.prefer_full_flush(pending_count)
    }
}

/// Unit requester identity for APIs that need a client type but ignore it.
#[derive(Clone, Copy, Debug, Default, PartialEq, Eq, Hash)]
pub struct NoClient;

/// No-op IOTLB invalidator for host tests and pre-hardware table construction.
#[derive(Clone, Copy, Debug, Default)]
pub struct NoIoTlbFlush<Client = NoClient>(core::marker::PhantomData<fn() -> Client>);
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 499) - [public] NoClient

```
pub struct NoClient;

/// No-op IOTLB invalidator for host tests and pre-hardware table construction.
#[derive(Clone, Copy, Debug, Default)]
pub struct NoIoTlbFlush<Client = NoClient>(core::marker::PhantomData<fn() -> Client>);

impl<Client> NoIoTlbFlush<Client> {
    #[inline]
    pub const fn new() -> Self {
        Self(core::marker::PhantomData)
    }
}
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 503) - [public] NoIoTlbFlush

```
pub struct NoIoTlbFlush<Client = NoClient>(core::marker::PhantomData<fn() -> Client>);

impl<Client> NoIoTlbFlush<Client> {
    #[inline]
    pub const fn new() -> Self {
        Self(core::marker::PhantomData)
    }
}

impl<V, Client> TlbInvalidation<V> for NoIoTlbFlush<Client>
where
    V: MemoryAddr,
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 518) - [internal]

flush_tlb_local

```
    fn flush_tlb_local(&self, _vaddr: V) {}

    #[inline(always)]
    fn flush_tlb_all_local(&self) {}

    #[inline(always)]
    fn flush_tlb_range_local(&self, _start: V, _page_size: PageSize, _count_pages: usize) {}

    #[inline(always)]
    fn prefer_full_flush(&self, _pending_count: usize) -> bool {
        false
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 521) - [internal]

```
flush_tlb_all_local
    fn flush_tlb_all_local(&self) {}

    #[inline(always)]
    fn flush_tlb_range_local(&self, _start: V, _page_size: PageSize, _count_pages: usize) {}

    #[inline(always)]
    fn prefer_full_flush(&self, _pending_count: usize) -> bool {
        false
    }
}

impl<V, Client> IoTlbInvalidation<V> for NoIoTlbFlush<Client>
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 524) - [internal]

```
flush_tlb_range_local
    fn flush_tlb_range_local(&self, _start: V, _page_size: PageSize, _count_pages: usize) {}

    #[inline(always)]
    fn prefer_full_flush(&self, _pending_count: usize) -> bool {
        false
    }
}

impl<V, Client> IoTlbInvalidation<V> for NoIoTlbFlush<Client>
where
    V: MemoryAddr,
    Client: Copy + Debug + Eq + 'static,
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 527) - [internal]

```
prefer_full_flush
    fn prefer_full_flush(&self, _pending_count: usize) -> bool {
        false
    }
}

impl<V, Client> IoTlbInvalidation<V> for NoIoTlbFlush<Client>
where
    V: MemoryAddr,
    Client: Copy + Debug + Eq + 'static,
{
    type Client = Client;

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 540) - [internal]

```
flush_iotlb
    fn flush_iotlb(&self, _iova: V) {}

    #[inline(always)]
    fn flush_iotlb_all(&self) {}

    #[inline(always)]
    fn flush_iotlb_range(&self, _start: V, _page_size: PageSize, _count_pages: usize) {}

    #[inline(always)]
    fn flush_device_tlb(&self, _client: Client, _iova: V) {}

    #[inline(always)]
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 543) - [internal]

flush_iotlb_all

```
fn flush_iotlb_all(&self) {}

#[inline(always)]
fn flush_iotlb_range(&self, _start: V, _page_size: PageSize, _count_pages: usize) {}

#[inline(always)]
fn flush_device_tlb(&self, _client: Client, _iova: V) {}

#[inline(always)]
fn flush_device_tlb_all(&self, _client: Client) {}

#[inline(always)]
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 546) - [internal]

flush_iotlb_range

```
fn flush_iotlb_range(&self, _start: V, _page_size: PageSize, _count_pages: usize) {}

#[inline(always)]
fn flush_device_tlb(&self, _client: Client, _iova: V) {}

#[inline(always)]
fn flush_device_tlb_all(&self, _client: Client) {}

#[inline(always)]
fn flush_device_tlb_range(
    &self,
    _client: Client,
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 549) - [internal]

flush_device_tlb

```
fn flush_device_tlb(&self, _client: Client, _iova: V) {}

#[inline(always)]
fn flush_device_tlb_all(&self, _client: Client) {}

#[inline(always)]
fn flush_device_tlb_range(
    &self,
    _client: Client,
    _start: V,
    _page_size: PageSize,
    _count_pages: usize,
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 552) - [internal]

flush_device_tlb_all

```
fn flush_device_tlb_all(&self, _client: Client) {}

#[inline(always)]
fn flush_device_tlb_range(
    &self,
    _client: Client,
    _start: V,
    _page_size: PageSize,
    _count_pages: usize,
) {
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 555) - [internal]

flush_device_tlb_range

```
fn flush_device_tlb_range(
    &self,
    _client: Client,
    _start: V,
    _page_size: PageSize,
    _count_pages: usize,
) {
}

#[inline(always)]
fn prefer_full_iotlb_flush(&self, _pending_count: usize) -> bool {
    false
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 565) - [internal]

prefer_full_iotlb_flush

```
fn prefer_full_iotlb_flush(&self, _pending_count: usize) -> bool {
    false
}
}
```

/// Explicit invalidation request against one controller.

#[derive(Clone, Copy, Debug, PartialEq, Eq)]

```
pub enum Invalidate<Client, V = IoviAddr> {
    Global,
    AddressSpace,
    Leaf {
        iova: V,
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 572) - [public] Invalidate

```
pub enum Invalidate<Client, V = IoviAddr> {
    Global,
    AddressSpace,
    Leaf {
        iova: V,
        granule: PageSize,
    },
    Device {
        client: Client,
    },
    DeviceLeaf {
        client: Client,
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 591) - [public]

InvalidateScope

```
pub enum InvalidateScope {
    Global,
    Domain,
    Leaf,
    Device,
    DeviceLeaf,
}
```

/// Result summary for one invalidate request.

#[derive(Clone, Copy, Debug, PartialEq, Eq)]

```
pub struct InvalidateOutcome {
    scope: InvalidateScope,
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 601) - [public]

InvalidateOutcome

```
pub struct InvalidateOutcome {
    scope: InvalidateScope,
    queued: bool,
}

impl InvalidateOutcome {
    #[inline]
    pub const fn new(scope: InvalidateScope, queued: bool) -> Self {
        Self { scope, queued }
    }

    #[inline]
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 629) - [public] Controller

/// Live IOMMU controller contract.

///

/// A controller is also a typed IOVA page table. Implementations can expose

/// a concrete inner `kore\_memory::PageTableWalker` and get most of the mapping

/// surface mechanically through the [PageTable] trait, while this trait

/// adds the controller-specific attachment, invalidation, and fault paths.

```
pub trait Controller<V: MemoryAddr = IoviAddr>: PageTable<V> {
    type Info;
    type Client: Copy + Debug + Eq;
    type Domain: Copy + Debug + Eq;
    type Invalidator: IoTlbInvalidation<V, Client = Self::Client>;
    type Fault;

    fn info(&self) -> &Self::Info;
    fn domain(&self) -> Self::Domain;
    fn stage(&self) -> TranslationStage;
    fn enable(&mut self) -> Result;
    fn bind(&mut self, binding: Binding<Self::Client, Self::Domain>) -> Result;
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 636) - [internal] info

```
fn info(&self) -> &Self::Info;
fn domain(&self) -> Self::Domain;
fn stage(&self) -> TranslationStage;
fn enable(&mut self) -> Result;
fn bind(&mut self, binding: Binding<Self::Client, Self::Domain>) -> Result;
fn unbind(&mut self, client: Self::Client, selector: BindingSelector) -> Result;
fn invalidator(&self) -> &Self::Invalidator;
fn invalidate(&mut self, request: Invalidate<Self::Client, V>) -> Result<InvalidateOutcome>;
fn configure_fault_event(&mut self, route: InterruptRoute) -> Result;
fn poll_fault(&mut self) -> Result<Option<Self::Fault>>;
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 637) - [internal] domain

```
fn domain(&self) -> Self::Domain;
fn stage(&self) -> TranslationStage;
fn enable(&mut self) -> Result;
fn bind(&mut self, binding: Binding<Self::Client, Self::Domain>) -> Result;
fn unbind(&mut self, client: Self::Client, selector: BindingSelector) -> Result;
fn invalidator(&self) -> &Self::Invalidator;
fn invalidate(&mut self, request: Invalidate<Self::Client, V>) -> Result<InvalidateOutcome>;
fn configure_fault_event(&mut self, route: InterruptRoute) -> Result;
fn poll_fault(&mut self) -> Result<Option<Self::Fault>>;
}

#[cfg(test)]
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 638) - [internal] stage

```
fn stage(&self) -> TranslationStage;
fn enable(&mut self) -> Result;
fn bind(&mut self, binding: Binding<Self::Client, Self::Domain>) -> Result;
fn unbind(&mut self, client: Self::Client, selector: BindingSelector) -> Result;
fn invalidator(&self) -> &Self::Invalidator;
fn invalidate(&mut self, request: Invalidate<Self::Client, V>) -> Result<InvalidateOutcome>;
fn configure_fault_event(&mut self, route: InterruptRoute) -> Result;
fn poll_fault(&mut self) -> Result<Option<Self::Fault>>;
}

#[cfg(test)]
mod tests {
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 639) - [internal] enable

```
fn enable(&mut self) -> Result;
fn bind(&mut self, binding: Binding<Self::Client, Self::Domain>) -> Result;
fn unbind(&mut self, client: Self::Client, selector: BindingSelector) -> Result;
fn invalidator(&self) -> &Self::Invalidator;
fn invalidate(&mut self, request: Invalidate<Self::Client, V>) -> Result<InvalidateOutcome>;
fn configure_fault_event(&mut self, route: InterruptRoute) -> Result;
fn poll_fault(&mut self) -> Result<Option<Self::Fault>>;
}

#[cfg(test)]
mod tests {
    use super::*;
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 640) - [internal] bind

```
fn bind(&mut self, binding: Binding<Self::Client, Self::Domain>) -> Result;
fn unbind(&mut self, client: Self::Client, selector: BindingSelector) -> Result;
fn invalidator(&self) -> &Self::Invalidator;
fn invalidate(&mut self, request: Invalidate<Self::Client, V>) -> Result<InvalidateOutcome>;
fn configure_fault_event(&mut self, route: InterruptRoute) -> Result;
fn poll_fault(&mut self) -> Result<Option<Self::Fault>>;
}

#[cfg(test)]
mod tests {
    use super::*;
    use memory_addr::{PhysAddr, VirtAddr};
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 641) - [internal] unbind

```
fn unbind(&mut self, client: Self::Client, selector: BindingSelector) -> Result;
fn invalidator(&self) -> &Self::Invalidator;
fn invalidate(&mut self, request: Invalidate<Self::Client, V>) -> Result<InvalidateOutcome>;
fn configure_fault_event(&mut self, route: InterruptRoute) -> Result;
fn poll_fault(&mut self) -> Result<Option<Self::Fault>>;
}

#[cfg(test)]
mod tests {
    use super::*;
    use memory_addr::{PhysAddr, VirtAddr};
    use std::{
        // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 642) - [internal]

invalidator

```
fn invalidator(&self) -> &Self::Invalidator;
fn invalidate(&mut self, request: Invalidate<Self::Client, V>) -> Result<InvalidateOutcome>;
fn configure_fault_event(&mut self, route: InterruptRoute) -> Result;
fn poll_fault(&mut self) -> Result<Option<Self::Fault>>;
}

#[cfg(test)]
mod tests {
    use super::*;
    use memory_addr::{PhysAddr, VirtAddr};
    use std::{
        sync::atomic::{AtomicUsize, Ordering},
        // ... (truncated, see source file for more)
    }
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 643) - [internal] invalidate

```
fn invalidate(&mut self, request: Invalidate<Self::Client, V>) -> Result<InvalidateOutcome>;
fn configure_fault_event(&mut self, route: InterruptRoute) -> Result;
fn poll_fault(&mut self) -> Result<Option<Self::Fault>>;
}

#[cfg(test)]
mod tests {
    use super::*;
    use memory_addr::{PhysAddr, VirtAddr};
    use std::{
        sync::atomic::{AtomicUsize, Ordering},
        vec,
        // ... (truncated, see source file for more)
    }
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 644) - [internal]

configure_fault_event

```
fn configure_fault_event(&mut self, route: InterruptRoute) -> Result;
fn poll_fault(&mut self) -> Result<Option<Self::Fault>>;
}

#[cfg(test)]
mod tests {
    use super::*;
    use memory_addr::{PhysAddr, VirtAddr};
    use std::{
        sync::atomic::{AtomicUsize, Ordering},
        vec,
    };
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 645) - [internal] poll_fault

```
fn poll_fault(&mut self) -> Result<Option<Self::Fault>>;
}

#[cfg(test)]
mod tests {
    use super::*;
    use memory_addr::{PhysAddr, VirtAddr};
    use std::{
        sync::atomic::{AtomicUsize, Ordering},
        vec,
    };
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 660) - [internal] test_queue

```
fn test_queue() -> (CommandQueue<ENTRIES>, std::vec::Vec<u8>, AtomicUsize) {
    let buffer = vec![0u8; ENTRIES * ENTRY_BYTES];
    let phys = PhysAddrRange::from_start_size(
        PhysAddr::from_usize(buffer.as_ptr() as usize),
        buffer.len(),
    );
    let virt = VirtAddrRange::from_start_size(
        VirtAddr::from_usize(buffer.as_ptr() as usize),
        buffer.len(),
    );
    let backing = unsafe { CommandQueueBacking::new(phys, virt, ENTRIES, ENTRY_BYTES) }
        .expect("valid queue backing");
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 677) - [internal] read_slot

```
fn read_slot(buffer: &[u8], slot: usize) -> (u64, u64) {
    let offset = slot * ENTRY_BYTES;
    let low = u64::from_ne_bytes(buffer[offset..offset + 8].try_into().unwrap());
    let high = u64::from_ne_bytes(buffer[offset + 8..offset + 16].try_into().unwrap());
    (low, high)
}
```

```
#[test]
fn interrupt_route_tracks_msi_delivery_state() {
    let message = MsiMessage::new(0xfe0_0000, 0x45);
    let route = InterruptRoute::from(message);

    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 685) - [internal]

interrupt_route_tracks_msi_delivery_state

```
fn interrupt_route_tracks_msi_delivery_state() {
    let message = MsiMessage::new(0xfe0_0000, 0x45);
    let route = InterruptRoute::from(message);

    assert!(route.is_enabled());
    assert_eq!(route, InterruptRoute::Msi(message));
    assert!(!InterruptRoute::Disabled.is_enabled());
}

#[test]
fn command_queue_backing_rejects_non_power_of_two_entries() {
    let buffer = vec![0u8; ENTRIES * ENTRY_BYTES];
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 695) - [internal]

command_queue_backing_rejects_non_power_of_two_entries

```
fn command_queue_backing_rejects_non_power_of_two_entries() {
    let buffer = vec![0u8; ENTRIES * ENTRY_BYTES];
    let phys = PhysAddrRange::from_start_size(
        PhysAddr::from_usize(buffer.as_ptr() as usize),
        buffer.len(),
    );
    let virt = VirtAddrRange::from_start_size(
        VirtAddr::from_usize(buffer.as_ptr() as usize),
        buffer.len(),
    );

    let result = unsafe { CommandQueueBacking::new(phys, virt, 7, ENTRY_BYTES) };
    // ... (truncated, see source file for more)
```


driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 711) - [internal]
command_queue_submit_writes_entry_and_advances_tail

```
fn command_queue_submit_writes_entry_and_advances_tail() {
    let (queue, buffer, head) = test_queue();
    let tail = AtomicUsize::new(0);
    let low = 0xaaaa_bbbb_cccc_ddddu64;
    let high = 0x1111_2222_3333_4444u64;
    let mut bytes = [0u8; 16];
    bytes[..8].copy_from_slice(&low.to_ne_bytes());
    bytes[8..].copy_from_slice(&high.to_ne_bytes());

    queue
        .submit(
            bytes,
            // ... (truncated, see source file for more)
        )
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/ctrl.rs (line 737) - [internal]
command_queue_sequential_submits_wrap_slots

```
fn command_queue_sequential_submits_wrap_slots() {
    let (queue, buffer, head) = test_queue();

    for i in 0..(ENTRIES * 2) {
        let marker = i as u64;
        let mut bytes = [0u8; 16];
        bytes[..8].copy_from_slice(&marker.to_ne_bytes());
        bytes[8..].copy_from_slice(&(!marker).to_ne_bytes());
        queue
            .submit(
                bytes,
                || Ok(head.load(Ordering::Acquire)),
                // ... (truncated, see source file for more)
            )
    }
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/error.rs (line 8) - [public] Error

```
pub enum Error {
    Unsupported,
    ControllerUnavailable,
    InvalidController,
    InvalidAddressSpace,
    InvalidClient,
    InvalidBinding,
    InvalidAddress,
    InvalidRange,
    InvalidGranule,
    InvalidWidth,
    AlreadyMapped,
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/error.rs (line 28) - [internal] from

```
fn from(value: PagingError) -> Self {
    Self::PageTable(value)
}

#[cfg(target_arch = "x86_64")]
impl From<crate::arch::x86_64::intel::dmar::DmarError> for Error {
    #[inline]
    fn from(value: crate::arch::x86_64::intel::dmar::DmarError) -> Self {
        match value {
            crate::arch::x86_64::intel::dmar::DmarError::Acpi(_) => Self::InvalidController,
            crate::arch::x86_64::intel::dmar::DmarError::Mapping(error) => Self::PageTable(error),
            // ... (truncated, see source file for more)
        }
    }
}
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/error.rs (line 36) - [internal] from
    fn from(value: crate::arch::x86_64::intel::dmar::DmarError) -> Self {
        match value {
            crate::arch::x86_64::intel::dmar::DmarError::Acpi(_) => Self::InvalidController,
            crate::arch::x86_64::intel::dmar::DmarError::Mapping(error) => Self::PageTable(error),
            crate::arch::x86_64::intel::dmar::DmarError::BdfRanges(_)
            | crate::arch::x86_64::intel::dmar::DmarError::Malformed(_) => Self::InvalidRange,
        }
    }
}

#[cfg(target_arch = "x86_64")]
impl From<crate::arch::x86_64::amd::ivrs::IvrsError> for Error {
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/error.rs (line 49) - [internal] from
    fn from(value: crate::arch::x86_64::amd::ivrs::IvrsError) -> Self {
        match value {
            crate::arch::x86_64::amd::ivrs::IvrsError::Acpi(_) => Self::InvalidController,
            crate::arch::x86_64::amd::ivrs::IvrsError::Mapping(error) => Self::PageTable(error),
            crate::arch::x86_64::amd::ivrs::IvrsError::BdfRanges(_)
            | crate::arch::x86_64::amd::ivrs::IvrsError::Malformed(_) => Self::InvalidRange,
        }
    }
}

#[cfg(target_arch = "aarch64")]
impl From<crate::arch::aarch64::iort::IortError> for Error {
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/error.rs (line 62) - [internal] from
    fn from(value: crate::arch::aarch64::iort::IortError) -> Self {
        match value {
            crate::arch::aarch64::iort::IortError::Acpi(_) => Self::InvalidController,
            crate::arch::aarch64::iort::IortError::Mapping(error) => Self::PageTable(error),
            crate::arch::aarch64::iort::IortError::BdfRanges(_)
            | crate::arch::aarch64::iort::IortError::Malformed(_) => Self::InvalidRange,
        }
    }
}

driverspace_workspace/drivers/iommu-TrangorgeOS/src/firm/pcie.rs (line 7) - [public] Bdf
pub struct Bdf(u16);

impl Bdf {
    #[inline]
    pub const fn new(bus: u8, device: u8, function: u8) -> Result<Self, PciDeviceError> {
        if device >= 32 {
            return Err(PciDeviceError::DeviceOutOfRange);
        }
        if function >= 8 {
            return Err(PciDeviceError::FunctionOutOfRange);
        }

        // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/firm/pcie.rs (line 79) - [internal] from
    fn from(value: u16) -> Self {
        Self::from_u16(value)
    }
}

impl From<Bdf> for u16 {
    #[inline]
    fn from(value: Bdf) -> Self {
        value.as_u16()
    }
}

// ... (truncated, see source file for more)

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/firm/pcie.rs (line 86) - [internal] from

```
    fn from(value: Bdf) -> Self {
        value.as_u16()
    }
}

#[derive(Clone, Copy, Debug, PartialEq, Eq)]
pub enum PciDeviceError {
    DeviceOutOfRange,
    FunctionOutOfRange,
}

#[derive(Clone, Copy, Debug, PartialEq, Eq, PartialOrd, Ord, Hash)]
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/firm/pcie.rs (line 92) - [public]

PciDeviceError

```
pub enum PciDeviceError {
    DeviceOutOfRange,
    FunctionOutOfRange,
}

#[derive(Clone, Copy, Debug, PartialEq, Eq, PartialOrd, Ord, Hash)]
#[repr(transparent)]
pub struct PciDevice(u32);

impl PciDevice {
    #[inline]
    pub const fn new(
        // ... (truncated, see source file for more)
    )
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/firm/pcie.rs (line 99) - [public]

PciDevice

```
pub struct PciDevice(u32);

impl PciDevice {
    #[inline]
    pub const fn new(
        segment: u16,
        bus: u8,
        device: u8,
        function: u8,
    ) -> Result<Self, PciDeviceError> {
        Ok(Self::from_segment_bdf(
            segment,
            // ... (truncated, see source file for more)
        ))
    }
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/firm/pcie.rs (line 124) - [public]

from_addr

```
pub fn from_addr(address: PciAddress) -> Result<Self, PciDeviceError> {
    Self::new(
        address.segment(),
        address.bus(),
        address.device(),
        address.function(),
    )
}

#[inline]
pub fn as_addr(self) -> PciAddress {
    PciAddress::new(self.segment(), self.bus(), self.device(), self.function())
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/firm/pcie.rs (line 134) - [public] as_addr

```
pub fn as_addr(self) -> PciAddress {
    PciAddress::new(self.segment(), self.bus(), self.device(), self.function())
}

#[inline]
pub const fn bdf(self) -> Bdf {
    Bdf::from_u16((self.0 & 0xffff) as u16)
}

#[inline]
pub const fn segment(self) -> u16 {
    (self.0 >> 16) as u16
}
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/firm/pcie.rs (line 165) - [public]
BdfRange

```
pub struct BdfRange {
    start: Bdf,
    end_exclusive: u32,
}

impl BdfRange {
    pub const EMPTY: Self = Self {
        start: Bdf::from_u16(0),
        end_exclusive: 0,
    };

    #[inline]
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/firm/pcie.rs (line 243) - [public]
BdfRangeSet

```
pub struct BdfRangeSet<const N: usize> {
    windows: [BdfRange; N],
    count: usize,
}

impl<const N: usize> BdfRangeSet<N> {
    #[inline]
    pub const fn empty() -> Self {
        Self {
            windows: [BdfRange::EMPTY; N],
            count: 0,
        }
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/firm/pcie.rs (line 268) - [public]
as_slice

```
pub fn as_slice(&self) -> &[BdfRange] {
    &self.windows[..self.count]
}

#[inline]
pub fn insert_single<E>(&mut self, bdf: Bdf) -> Result<(), E>
where
    E: From<BdfRangeSetError>,
{
    self.insert(BdfRange::single(bdf))
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/firm/pcie.rs (line 273) - [public]

insert_single

```
pub fn insert_single<E>(&mut self, bdf: Bdf) -> Result<(), E>
where
    E: From<BdfRangeSetError>,
{
    self.insert(BdfRange::single(bdf))
}

pub fn insert<E>(&mut self, range: BdfRange) -> Result<(), E>
where
    E: From<BdfRangeSetError>,
{
    if range.is_empty() {
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/firm/pcie.rs (line 280) - [public] insert

```
pub fn insert<E>(&mut self, range: BdfRange) -> Result<(), E>
where
    E: From<BdfRangeSetError>,
{
    if range.is_empty() {
        return Ok(());
    }

    let mut index = 0;
    while index < self.count {
        let current = self.windows[index];
        let current_end = current.end_exclusive;
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/firm/pcie.rs (line 317) - [internal]

insert_at

```
fn insert_at<E>(&mut self, index: usize, range: BdfRange) -> Result<(), E>
where
    E: From<BdfRangeSetError>,
{
    if self.count == N {
        return Err(E::from(BdfRangeSetError::Full));
    }

    let mut cursor = self.count;
    while cursor > index {
        self.windows[cursor] = self.windows[cursor - 1];
        cursor -= 1;
    }
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/firm/pcie.rs (line 335) - [internal]

merge_around

```
fn merge_around<E>(&mut self, mut index: usize) -> Result<(), E>
where
    E: From<BdfRangeSetError>,
{
    while index > 0 {
        if !self.merge_pair::<E>(index - 1, index)? {
            break;
        }
        index -= 1;
    }

    while index + 1 < self.count {
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/firm/pcie.rs (line 355) - [internal]
merge_pair

```
fn merge_pair<E>(&mut self, left: usize, right: usize) -> Result<bool, E>
where
    E: From<BdfRangeSetError>,
{
    let left_range = self.windows[left];
    let right_range = self.windows[right];
    let left_end = left_range.end_exclusive;
    if right_range.start.as_u32() > left_end {
        return Ok(false);
    }

    let right_end = right_range.end_exclusive;
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/firm/pcie.rs (line 375) - [internal]
remove

```
fn remove(&mut self, index: usize) {
    let mut cursor = index;
    while cursor + 1 < self.count {
        self.windows[cursor] = self.windows[cursor + 1];
        cursor += 1;
    }
    self.count -= 1;
    self.windows[self.count] = BdfRange::EMPTY;
}

impl<const N: usize> Default for BdfRangeSet<N> {
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/firm/pcie.rs (line 388) - [internal]
default

```
fn default() -> Self {
    Self::empty()
}

#[derive(Clone, Copy, Debug, PartialEq, Eq)]
pub enum BdfRangeSetError {
    Full,
    Overflow,
    InvalidRange(BdfRangeError),
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/firm/pcie.rs (line 394) - [public]
BdfRangeSetError

```
pub enum BdfRangeSetError {
    Full,
    Overflow,
    InvalidRange(BdfRangeError),
}

impl From<BdfRangeError> for BdfRangeSetError {
    #[inline]
    fn from(value: BdfRangeError) -> Self {
        Self::InvalidRange(value)
    }
}

// ... (truncated, see source file for more)
```

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/firm/pcie.rs (line 402) - [internal] from
    fn from(value: BdfRangeError) -> Self {
        Self::InvalidRange(value)
    }
}

#[derive(Clone, Copy, Debug, PartialEq, Eq)]
pub enum BdfRangeError {
    Reversed,
    OutOfRange,
}

impl fmt::Display for BdfRangeSetError {
    // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/firm/pcie.rs (line 408) - [public]
BdfRangeError

    pub enum BdfRangeError {
        Reversed,
        OutOfRange,
    }

    impl fmt::Display for BdfRangeSetError {
        fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
            match self {
                Self::Full => f.write_str("too many BDF ranges"),
                Self::Overflow => f.write_str("BDF range overflow"),
                Self::InvalidRange(error) => write!(f, "{error}"),
            }
            // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/firm/pcie.rs (line 414) - [internal] fmt
    fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
        match self {
            Self::Full => f.write_str("too many BDF ranges"),
            Self::Overflow => f.write_str("BDF range overflow"),
            Self::InvalidRange(error) => write!(f, "{error}"),
        }
    }
}

    impl fmt::Display for BdfRangeError {
        fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
            match self {
                // ... (truncated, see source file for more)

driverspace_workspace/drivers/iommu-TrangorgeOS/src/firm/pcie.rs (line 424) - [internal] fmt
    fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
        match self {
            Self::Reversed => f.write_str("BDF range end is before start"),
            Self::OutOfRange => f.write_str("BDF range bound is outside the PCI requester space"),
        }
    }
}

#[cfg(test)]
mod tests {
    use super::{Bdf, BdfRange, BdfRangeSet};

    // ... (truncated, see source file for more)

```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/firm/pcie.rs (line 437) - [internal]
bdf_encodes_requester_id

```
fn bdf_encodes_requester_id() {
    let requester = Bdf::from_bus_device_function(0x2a, 5, 3).unwrap();
    assert_eq!(requester.as_u32(), (0x2a_u32 << 8) | (5_u32 << 3) | 3);
    assert_eq!(requester.bus(), 0x2a);
    assert_eq!(requester.device(), 5);
    assert_eq!(requester.function(), 3);
}

#[test]
fn bdf_range_set_sorts_and_merges_adjacent_ranges() {
    let mut ranges = BdfRangeSet::<4>::empty();
    ranges
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/firm/pcie.rs (line 446) - [internal]
bdf_range_set_sorts_and_merges_adjacent_ranges

```
fn bdf_range_set_sorts_and_merges_adjacent_ranges() {
    let mut ranges = BdfRangeSet::<4>::empty();
    ranges
        .insert::<super::BdfRangeSetError>(
            BdfRange::inclusive(Bdf::from_u16(12), Bdf::from_u16(13)).unwrap(),
        )
        .unwrap();
    ranges
        .insert::<super::BdfRangeSetError>(
            BdfRange::inclusive(Bdf::from_u16(10), Bdf::from_u16(11)).unwrap(),
        )
        .unwrap();
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/info.rs (line 7) - [public] ControllerKind

```
pub enum ControllerKind {
    IntelVtd,
    AmdVi,
    ArmSmmuV2,
    ArmSmmuV3,
    RiscvIommu,
    Unknown(u16),
}
```

```
/// Immutable descriptor for one discovered IOMMU unit.
#[derive(Clone, Copy, Debug, PartialEq, Eq)]
pub struct IommuInfo {
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/info.rs (line 18) - [public] IommuInfo

```
pub struct IommuInfo {
    kind: ControllerKind,
    segment: Option<u16>,
    mmio: MmioAddrRange,
    caps: CapabilityFlags,
    stage: TranslationStage,
}
```

```
impl IommuInfo {
    #[inline]
    pub const fn new(
        kind: ControllerKind,
        // ... (truncated, see source file for more)
```


driverspace_workspace/drivers/iommu-TrangorgeOS/src/info.rs (line 79) - [public] IoDomain

```
pub struct IoDomain<const MASK: u32 = 0x0000_ffff>(u32);

impl<const MASK: u32> IoDomain<MASK> {
    const ID_BITS: u32 = MASK.count_ones();
    const GENERATION_MASK: u32 = if Self::ID_BITS >= u32::BITS {
        0
    } else {
        u32::MAX >> Self::ID_BITS
    };

    #[inline]
    pub const fn new(id: u32, generation: u32) -> Option<Self> {
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/info.rs (line 144) - [public] ReservedRegion

```
pub struct ReservedRegion<Client> {
    range: PhysAddrRange,
    client: Option<Client>,
}

impl<Client> ReservedRegion<Client> {
    #[inline]
    pub const fn new(range: PhysAddrRange, client: Option<Client>) -> Self {
        Self { range, client }
    }

    #[inline]
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/info.rs (line 156) - [public] range

```
pub fn range(self) -> PhysAddrRange {
    self.range
}

#[inline]
pub fn client(self) -> Option<Client>
where
    Client: Copy,
{
    self.client
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/info.rs (line 161) - [public] client

```
pub fn client(self) -> Option<Client>
where
    Client: Copy,
{
    self.client
}

#[cfg(test)]
mod tests {
    use super::*;

    // ... (truncated, see source file for more)
}
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/info.rs (line 174) - [internal]
io_domain_keeps_asid_and_generation_separate

```
fn io_domain_keeps_asid_and_generation_separate() {
    let domain = IoDomain:::<0x0000_ffff>::new(0x1234, 0xabcd).unwrap();

    assert_eq!(domain.id(), 0x1234);
    assert_eq!(domain.generation(), 0xabcd);
    assert_eq!(domain.bits(), 0xabcd_1234);
    assert_eq!(IoDomain:::<0x0000_ffff>::capacity(), 0x1_0000);
}

#[test]
fn io_domain_rejects_ids_outside_mask() {
    assert!(IoDomain:::<0x0000_ffff>::from_asid(0xffff).is_some());
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/iommu-TrangorgeOS/src/info.rs (line 184) - [internal]
io_domain_rejects_ids_outside_mask

```
fn io_domain_rejects_ids_outside_mask() {
    assert!(IoDomain:::<0x0000_ffff>::from_asid(0xffff).is_some());
    assert!(IoDomain:::<0x0000_ffff>::from_asid(0x1_0000).is_none());
}
}
```

driverspace_workspace/drivers/netcam_driver/src/main.rs (line 1) - [internal] main

```
fn main() {
    println!("Hello, world!");
}
```

driverspace_workspace/drivers/vgpu/src/main.rs (line 42) - [internal] VgpuInfo

```
struct VgpuInfo {
    fb: *mut u32,
    width: u32,
    height: u32,
    bpp: u32,
    stride: u32,
}

#[no_mangle]
pub extern "C" fn driver_main() {
    ds_info!("VGPU", "Virtual GPU driver starting...");

    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/vgpu/src/main.rs (line 67) - [internal] vgpu_init

```
fn vgpu_init(w: u32, h: u32) -> bool {
    unsafe {
        if G_READY {
            return true;
        }
    }

    ds_info!("VGPU", "Searching for VGA device...");

    let bdf = match find_vga_device() {
        Some(bdf) => bdf,
        None => {
            // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/vgpu/src/main.rs (line 138) - [internal] find_vga_device

```
fn find_vga_device() -> Option<u64> {
    let mut msg = kapi_abi::DsMsg::new(DsCmd::PciFind as u32);
    msg.arg0 = PCI_CLASS_VGA as u64;

    let reply = kapi_syscall::sys_ipc_call(DsCmd::PciFind, PCI_CLASS_VGA as u64, 0, 0);

    if reply.is_ok() && reply.arg0 != 0 {
        Some(reply.arg0)
    } else {
        None
    }
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/vgpu/src/main.rs (line 151) - [internal] read_pci_config

```
fn read_pci_config(bdf: u64, offset: u32) -> u32 {
    let reply = kapi_syscall::sys_ipc_call(DsCmd::PciRead, bdf, offset as u64, 0);

    if reply.is_ok() {
        reply.arg0 as u32
    } else {
        0xFFFFFFFF
    }
}

fn read_pci_bar(bdf: u64, bar: u32) -> u64 {
    let offset = 0x10 + (bar * 4);
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/vgpu/src/main.rs (line 161) - [internal] read_pci_bar

```
fn read_pci_bar(bdf: u64, bar: u32) -> u64 {
    let offset = 0x10 + (bar * 4);
    let val = read_pci_config(bdf, offset);

    if val == 0 || val == 0xFFFFFFFF {
        return 0;
    }

    (val & 0xFFFFFFFF0) as u64
}

fn enable_pci_device(bdf: u64) {
    // ... (truncated, see source file for more)
```

driverspace_workspace/drivers/vgpu/src/main.rs (line 172) - [internal] enable_pci_device

```
fn enable_pci_device(bdf: u64) {
    let cmd_offset = 0x04;
    let mut cmd = read_pci_config(bdf, cmd_offset);
    cmd |= 0x07;

    let mut msg = kapi_abi::DsMsg::new(DsCmd::PciWrite as u32);
    msg.arg0 = bdf;
    msg.arg1 = cmd_offset as u64;
    msg.arg2 = cmd as u64;

    kapi_syscall::sys_ipc_call(DsCmd::PciWrite, bdf, cmd_offset as u64, cmd as u64);
}

// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/vgpu/src/main.rs (line 185) - [internal] vbe_write

```
fn vbe_write(idx: u16, val: u16) {
    unsafe {
        core::arch::asm!("out dx, ax", in("dx") VBE_INDEX_PORT, in("ax") idx, options(nomem, nostack, preserve_flags),
        core::arch::asm!("out dx, ax", in("dx") VBE_DATA_PORT, in("ax") val, options(nomem, nostack, preserve_flags));
    }
}

fn vgpu_clear(color: u32) {
    unsafe {
        if !G_READY {
            return;
        }
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/drivers/vgpu/src/main.rs (line 192) - [internal] vgpu_clear

```
fn vgpu_clear(color: u32) {
    unsafe {
        if !G_READY {
            return;
        }

        let fb = G_INFO.fb;
        let n = (G_INFO.width * G_INFO.height) as usize;

        for i in 0..n {
            core::ptr::write_volatile(fb.add(i), color);
        }
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/drivers/vgpu/src/main.rs (line 207) - [internal] vgpu_pixel

```
fn vgpu_pixel(x: u32, y: u32, color: u32) {
    unsafe {
        if !G_READY || x >= G_INFO.width || y >= G_INFO.height {
            return;
        }

        let offset = (y * G_INFO.stride + x) as usize;
        core::ptr::write_volatile(G_INFO.fb.add(offset), color);
    }
}

#[panic_handler]
// ... (truncated, see source file for more)
```

driverspace_workspace/drivers/vgpu/src/main.rs (line 219) - [internal] panic

```
fn panic(_info: &core::panic::PanicInfo) -> ! {
    ds_error!("VGPU", "PANIC!");
    loop {
        kapi_syscall::sys_yield();
    }
}
```

driverspace_workspace/drivers/wacomgraphic_driver/src/main.rs (line 1) - [internal] main

```
fn main() {
    println!("Hello, world!");
}
```

driverspace_workspace/lib/ds-formal-ffi/build.rs (line 10) - [internal] main

```
fn main() {  
    // ■cie■ka do projektu Ada SPARK  
    let spark_project_dir = PathBuf::from("../..formal/ds-spark-core");  
  
    println!("cargo:rerun-if-changed=../..formal/ds-spark-core/src");  
  
    let gprbuild_available = Command::new("gprbuild")  
        .arg("--version")  
        .status()  
        .map(|s| s.success())  
        .unwrap_or(false);  
  
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/ds-formal-ffi/src/lib.rs (line 7) - [public] ds_spark_safe_add_32

// Odpowiednik: function Safe_Add_32 (A, B : UInt32) return UInt32

```
pub fn ds_spark_safe_add_32(a: u32, b: u32) -> u32;  
  
// Odpowiednik: function Safe_Array_Index ...  
pub fn ds_spark_safe_array_index(index: u32, max_size: u32) -> u32;  
}  
  
// Bezpieczne wrappery dla reszty kodu Rust w workspace  
pub mod math {  
    use super::*;  
  
    /// Bezpieczne dodawanie. W trybie debug Rust i tak to sprawdzi,  
    /// ale w release mode (gdzie overflow to undefined behavior w C/Ada),  
    /// ... (truncated, see source file for more)
```

driverspace_workspace/lib/ds-formal-ffi/src/lib.rs (line 10) - [public]

ds_spark_safe_array_index

// Odpowiednik: function Safe_Array_Index ...

```
pub fn ds_spark_safe_array_index(index: u32, max_size: u32) -> u32;  
}  
  
// Bezpieczne wrappery dla reszty kodu Rust w workspace  
pub mod math {  
    use super::*;  
  
    /// Bezpieczne dodawanie. W trybie debug Rust i tak to sprawdzi,  
    /// ale w release mode (gdzie overflow to undefined behavior w C/Ada),  
    /// ta funkcja ma matematyczny dowód z SPARK, ■e nigdy nie overflownie.  
    #[inline(always)]  
    pub fn safe_add(a: u32, b: u32) -> u32 {  
        // ... (truncated, see source file for more)
```

driverspace_workspace/lib/ds-formal-ffi/src/lib.rs (line 21) - [public] safe_add

```
pub fn safe_add(a: u32, b: u32) -> u32 {  
    // W produkcji wo■amy udowodniony kod z Ady  
    unsafe { ds_spark_safe_add_32(a, b) }  
}  
  
unsafe extern "C" {  
    pub fn ds_spark_buddy_get_buddy(addr: u64, size_exp: u64) -> u64;  
    pub fn ds_spark_safe_dma_end(base: u64, len: u64) -> u64;  
}  
  
pub mod pmm {  
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/ds-formal-ffi/src/lib.rs (line 28) - [public]

ds_spark_buddy_get_buddy

```
pub fn ds_spark_buddy_get_buddy(addr: u64, size_exp: u64) -> u64;
pub fn ds_spark_safe_dma_end(base: u64, len: u64) -> u64;

}

pub mod pmm {
    use super::*;
    #[inline(always)]
    pub fn get_buddy(addr: u64, exp: u64) -> u64 {
        // Wywołanie kodu, który ma MATEMATYCZNY DOWÓD poprawności
        unsafe { ds_spark_buddy_get_buddy(addr, exp) }
    }
}
```

driverspace_workspace/lib/ds-formal-ffi/src/lib.rs (line 29) - [public] ds_spark_safe_dma_end

```
pub fn ds_spark_safe_dma_end(base: u64, len: u64) -> u64;

}

pub mod pmm {
    use super::*;
    #[inline(always)]
    pub fn get_buddy(addr: u64, exp: u64) -> u64 {
        // Wywołanie kodu, który ma MATEMATYCZNY DOWÓD poprawności
        unsafe { ds_spark_buddy_get_buddy(addr, exp) }
    }
}
```

driverspace_workspace/lib/ds-formal-ffi/src/lib.rs (line 35) - [public] get_buddy

```
pub fn get_buddy(addr: u64, exp: u64) -> u64 {
    // Wywołanie kodu, który ma MATEMATYCZNY DOWÓD poprawności
    unsafe { ds_spark_buddy_get_buddy(addr, exp) }
}

}
```

driverspace_workspace/lib/ds-fw-audio/src/lib.rs (line 1) - [public] add

```
pub fn add(left: u64, right: u64) -> u64 {
    left + right
}

#[cfg(test)]
mod tests {
    use super::*;

    #[test]
    fn it_works() {
        let result = add(2, 2);
        assert_eq!(result, 4);
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/lib/ds-fw-audio/src/lib.rs (line 10) - [internal] it_works

```
fn it_works() {
    let result = add(2, 2);
    assert_eq!(result, 4);
}

}
```

driverspace_workspace/lib/ds-fw-block/src/request.rs (line 7) - [public] RequestState

```
pub enum RequestState {
    Pending = 0,
    InProgress = 1,
    Completed = 2,
    Failed = 3,
}

/// High-level I/O request with metadata.
pub struct IoRequest {
    pub id: u64,
    pub lba: u64,
    pub block_count: u32,
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/lib/ds-fw-block/src/request.rs (line 15) - [public] IoRequest

/// High-level I/O request with metadata.

```
pub struct IoRequest {
    pub id: u64,
    pub lba: u64,
    pub block_count: u32,
    pub buffer_ptr: *mut u8,
    pub buffer_len: usize,
    pub is_write: bool,
    pub priority: u8,
    state: AtomicU32,
    pub callback: Option<fn(&IoRequest)>,
}
```

// ... (truncated, see source file for more)

driverspace_workspace/lib/ds-fw-block/src/request.rs (line 28) - [public] new

```
pub fn new(id: u64, lba: u64, block_count: u32, buffer: *mut u8, len: usize, is_write: bool) -> Self {
    Self {
        id,
        lba,
        block_count,
        buffer_ptr: buffer,
        buffer_len: len,
        is_write,
        priority: 0,
        state: AtomicU32::new(RequestState::Pending as u32),
        callback: None,
    }
}
```

// ... (truncated, see source file for more)

driverspace_workspace/lib/ds-fw-block/src/request.rs (line 42) - [public] state

```
pub fn state(&self) -> RequestState {
    match self.state.load(Ordering::Acquire) {
        0 => RequestState::Pending,
        1 => RequestState::InProgress,
        2 => RequestState::Completed,
        3 => RequestState::Failed,
        _ => RequestState::Failed,
    }
}
```

```
pub fn mark_in_progress(&self) {
    self.state.store(RequestState::InProgress as u32, Ordering::Release);
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/lib/ds-fw-block/src/request.rs (line 52) - [public] mark_in_progress

```
pub fn mark_in_progress(&self) {
    self.state.store(RequestState::InProgress as u32, Ordering::Release);
}
```

```
pub fn mark_completed(&self) {
    self.state.store(RequestState::Completed as u32, Ordering::Release);
    if let Some(cb) = self.callback {
        cb(self);
    }
}
```

```
pub fn mark_failed(&self) {
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/lib/ds-fw-block/src/request.rs (line 56) - [public] mark_completed

```
pub fn mark_completed(&self) {
    self.state.store(RequestState::Completed as u32, Ordering::Release);
    if let Some(cb) = self.callback {
        cb(self);
    }
}

pub fn mark_failed(&self) {
    self.state.store(RequestState::Failed as u32, Ordering::Release);
}
}
```

driverspace_workspace/lib/ds-fw-block/src/request.rs (line 63) - [public] mark_failed

```
pub fn mark_failed(&self) {
    self.state.store(RequestState::Failed as u32, Ordering::Release);
}
}
```

driverspace_workspace/lib/ds-fw-block/src/traits.rs (line 6) - [public] BlockRequest

/// Represents a single I/O request to a block device.

```
pub struct BlockRequest {
    pub lba: u64,
    pub block_count: u32,
    pub buffer_ptr: *mut u8,
    pub buffer_len: usize,
    pub is_write: bool,
}
```

/// Core trait for all block devices.

/// Drivers must implement this to integrate with the block subsystem.

```
pub trait BlockDevice {
    /// Read blocks from the device starting at LBA.
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/lib/ds-fw-block/src/traits.rs (line 16) - [public] BlockDevice

/// Core trait for all block devices.

/// Drivers must implement this to integrate with the block subsystem.

```
pub trait BlockDevice {
    /// Read blocks from the device starting at LBA.
    fn read_blocks(&mut self, lba: u64, block_count: u32, buffer: &mut [u8]) -> Result<(), DsError>;

    /// Write blocks to the device starting at LBA.
    fn write_blocks(&mut self, lba: u64, block_count: u32, buffer: &[u8]) -> Result<(), DsError>;

    /// Flush internal caches to physical media.
    fn flush(&mut self) -> Result<(), DsError>;

    /// Get device geometry (block size, total blocks).
    fn geometry(&self) -> BlockGeometry;
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/lib/ds-fw-block/src/traits.rs (line 18) - [internal] read_blocks

/// Read blocks from the device starting at LBA.

```
fn read_blocks(&mut self, lba: u64, block_count: u32, buffer: &mut [u8]) -> Result<(), DsError>;

/// Write blocks to the device starting at LBA.
fn write_blocks(&mut self, lba: u64, block_count: u32, buffer: &[u8]) -> Result<(), DsError>;

/// Flush internal caches to physical media.
fn flush(&mut self) -> Result<(), DsError>;

/// Get device geometry (block size, total blocks).
fn geometry(&self) -> BlockGeometry;

/// Check if device supports TRIM/discard.
// ... (truncated, see source file for more)
```


driverspace_workspace/lib/ds-fw-block/src/traits.rs (line 21) - [internal] write_blocks

/// Write blocks to the device starting at LBA.

```
fn write_blocks(&mut self, lba: u64, block_count: u32, buffer: &[u8]) -> Result<(), DsError>;

/// Flush internal caches to physical media.
fn flush(&mut self) -> Result<(), DsError>;

/// Get device geometry (block size, total blocks).
fn geometry(&self) -> BlockGeometry;

/// Check if device supports TRIM/discard.
fn supports_trim(&self) -> bool { false }

/// Optional: Submit async request (for advanced schedulers).
// ... (truncated, see source file for more)
```

driverspace_workspace/lib/ds-fw-block/src/traits.rs (line 24) - [internal] flush

/// Flush internal caches to physical media.

```
fn flush(&mut self) -> Result<(), DsError>;

/// Get device geometry (block size, total blocks).
fn geometry(&self) -> BlockGeometry;

/// Check if device supports TRIM/discard.
fn supports_trim(&self) -> bool { false }

/// Optional: Submit async request (for advanced schedulers).
fn submit_request(&mut self, _req: BlockRequest) -> Result<(), DsError> {
    Err(DsError::Unknown)
}
// ... (truncated, see source file for more)
```

driverspace_workspace/lib/ds-fw-block/src/traits.rs (line 27) - [internal] geometry

/// Get device geometry (block size, total blocks).

```
fn geometry(&self) -> BlockGeometry;

/// Check if device supports TRIM/discard.
fn supports_trim(&self) -> bool { false }

/// Optional: Submit async request (for advanced schedulers).
fn submit_request(&mut self, _req: BlockRequest) -> Result<(), DsError> {
    Err(DsError::Unknown)
}

}

/// Physical geometry of a block device.
// ... (truncated, see source file for more)
```

driverspace_workspace/lib/ds-fw-block/src/traits.rs (line 30) - [internal] supports_trim

/// Check if device supports TRIM/discard.

```
fn supports_trim(&self) -> bool { false }

/// Optional: Submit async request (for advanced schedulers).
fn submit_request(&mut self, _req: BlockRequest) -> Result<(), DsError> {
    Err(DsError::Unknown)
}

}

/// Physical geometry of a block device.
#[derive(Debug, Clone, Copy)]
pub struct BlockGeometry {
    pub block_size: u32,
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/ds-fw-block/src/traits.rs (line 33) - [internal] submit_request

/// Optional: Submit async request (for advanced schedulers).

```
    fn submit_request(&mut self, _req: BlockRequest) -> Result<(), DsError> {
        Err(DsError::Unknown)
    }
}

/// Physical geometry of a block device.
#[derive(Debug, Clone, Copy)]
pub struct BlockGeometry {
    pub block_size: u32,
    pub total_blocks: u64,
    pub max_transfer_blocks: u32,
}

// ... (truncated, see source file for more)
```

driverspace_workspace/lib/ds-fw-block/src/traits.rs (line 40) - [public] BlockGeometry

```
pub struct BlockGeometry {
    pub block_size: u32,
    pub total_blocks: u64,
    pub max_transfer_blocks: u32,
}

impl BlockGeometry {
    pub fn total_bytes(&self) -> u64 {
        self.total_blocks * self.block_size as u64
    }
}
```

driverspace_workspace/lib/ds-fw-block/src/traits.rs (line 47) - [public] total_bytes

```
    pub fn total_bytes(&self) -> u64 {
        self.total_blocks * self.block_size as u64
    }
}
```

driverspace_workspace/lib/ds-fw-gpu/src/lib.rs (line 1) - [public] add

```
pub fn add(left: u64, right: u64) -> u64 {
    left + right
}

#[cfg(test)]
mod tests {
    use super::*;

    #[test]
    fn it_works() {
        let result = add(2, 2);
        assert_eq!(result, 4);
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/lib/ds-fw-gpu/src/lib.rs (line 10) - [internal] it_works

```
    fn it_works() {
        let result = add(2, 2);
        assert_eq!(result, 4);
    }
}
```

driverspace_workspace/lib/ds-fw-input/src/lib.rs (line 1) - [public] add

```
pub fn add(left: u64, right: u64) -> u64 {
    left + right
}

#[cfg(test)]
mod tests {
    use super::*;

    #[test]
    fn it_works() {
        let result = add(2, 2);
        assert_eq!(result, 4);
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/lib/ds-fw-input/src/lib.rs (line 10) - [internal] it_works

```
fn it_works() {
    let result = add(2, 2);
    assert_eq!(result, 4);
}
}
```

driverspace_workspace/lib/ds-ipc/src/channel.rs (line 6) - [public] Channel

```
pub struct Channel {
    k2d_ring: *mut DsRing,
    d2k_ring: *mut DsRing,
    k2d_msgs: *mut DsMsg,
    d2k_msgs: *mut DsMsg,
    next_id: AtomicU64,
}

unsafe impl Send for Channel {}
unsafe impl Sync for Channel {}

impl Channel {
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/lib/ds-ipc/src/channel.rs (line 33) - [internal] next_id

```
fn next_id(&self) -> u64 {
    self.next_id.fetch_add(1, Ordering::Relaxed)
}

pub unsafe fn send(&self, cmd: u16, arg0: u64, arg1: u64, arg2: u64) -> u64 {
    let ring = &*self.d2k_ring;
    if ring.is_full() {
        return 0;
    }

    let id = self.next_id();
    let slot = (ring.head % ring.cap) as usize;
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/lib/ds-ipc/src/client.rs (line 6) - [public] ManagerClient

```
pub struct ManagerClient {
    manager_ep: Handle,
}

impl ManagerClient {
    pub const fn new(ep: Handle) -> Self {
        Self { manager_ep: ep }
    }

    pub fn request_mmio(&self, phys_base: u64, size: u64) -> Result<u64, DsError> {
        let reply = sys_ipc_call(DsCmd::MemMapMmio, phys_base, size, 0);
        if reply.is_ok() { Ok(reply.arg0) } else { Err(reply.error_status()) }
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/lib/ds-ipc/src/client.rs (line 15) - [public] request_mmio

```
pub fn request_mmio(&self, phys_base: u64, size: u64) -> Result<u64, DsError> {
    let reply = sys_ipc_call(DsCmd::MemMapMmio, phys_base, size, 0);
    if reply.is_ok() { Ok(reply.arg0) } else { Err(reply.error_status()) }
}

pub fn bind_irq(&self, irq_num: u32) -> Result<(), DsError> {
    let reply = sys_ipc_call(DsCmd::BindIrq, irq_num as u64, 0, 0);
    if reply.is_ok() { Ok(()) } else { Err(reply.error_status()) }
}

pub fn alloc_dma(&self, size: u64, flags: u32) -> Result<(u64, u64), DsError> {
    let reply = sys_ipc_call(DsCmd::MemAllocDma, size, flags as u64, 0);
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/ds-ipc/src/client.rs (line 20) - [public] bind_irq

```
pub fn bind_irq(&self, irq_num: u32) -> Result<(), DsError> {
    let reply = sys_ipc_call(DsCmd::BindIrq, irq_num as u64, 0, 0);
    if reply.is_ok() { Ok(()) } else { Err(reply.error_status()) }
}

pub fn alloc_dma(&self, size: u64, flags: u32) -> Result<(u64, u64), DsError> {
    let reply = sys_ipc_call(DsCmd::MemAllocDma, size, flags as u64, 0);
    if reply.is_ok() { Ok((reply.arg0, reply.arg1)) } else { Err(reply.error_status()) }
}

pub fn pci_find(&self, class_code: u32) -> Result<u64, DsError> {
    let reply = sys_ipc_call(DsCmd::PciFind, class_code as u64, 0, 0);
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/ds-ipc/src/client.rs (line 25) - [public] alloc_dma

```
pub fn alloc_dma(&self, size: u64, flags: u32) -> Result<(u64, u64), DsError> {
    let reply = sys_ipc_call(DsCmd::MemAllocDma, size, flags as u64, 0);
    if reply.is_ok() { Ok((reply.arg0, reply.arg1)) } else { Err(reply.error_status()) }
}

pub fn pci_find(&self, class_code: u32) -> Result<u64, DsError> {
    let reply = sys_ipc_call(DsCmd::PciFind, class_code as u64, 0, 0);
    if reply.is_ok() { Ok(reply.arg0) } else { Err(reply.error_status()) }
}

pub fn get_page_phys(&self, virt_addr: u64) -> Result<u64, DsError> {
    let reply = sys_ipc_call(DsCmd::PagePhys, virt_addr, 0, 0);
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/ds-ipc/src/client.rs (line 30) - [public] pci_find

```
pub fn pci_find(&self, class_code: u32) -> Result<u64, DsError> {
    let reply = sys_ipc_call(DsCmd::PciFind, class_code as u64, 0, 0);
    if reply.is_ok() { Ok(reply.arg0) } else { Err(reply.error_status()) }
}

pub fn get_page_phys(&self, virt_addr: u64) -> Result<u64, DsError> {
    let reply = sys_ipc_call(DsCmd::PagePhys, virt_addr, 0, 0);
    if reply.is_ok() { Ok(reply.arg0) } else { Err(reply.error_status()) }
}
}
```

driverspace_workspace/lib/ds-ipc/src/client.rs (line 35) - [public] get_page_phys

```
pub fn get_page_phys(&self, virt_addr: u64) -> Result<u64, DsError> {
    let reply = sys_ipc_call(DsCmd::PagePhys, virt_addr, 0, 0);
    if reply.is_ok() { Ok(reply.arg0) } else { Err(reply.error_status()) }
}
}
```

driverspace_workspace/lib/ds-ipc/src/dispatcher.rs (line 10) - [public] MessageDispatcher

```
pub struct MessageDispatcher {
    handlers: [Option<HandlerFn>; 64],
}

impl MessageDispatcher {
    pub const fn new() -> Self {
        Self {
            handlers: [None; 64],
        }
    }

    pub fn register(&mut self, opcode: Opcode, handler: HandlerFn) -> Result<(), DsError> {
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/lib/ds-ipc/src/dispatcher.rs (line 21) - [public] register

```
pub fn register(&mut self, opcode: Opcode, handler: HandlerFn) -> Result<(), DsError> {
    let idx = (opcode as u32 & 0x3F) as usize;
    if self.handlers[idx].is_some() {
        return Err(DsError::DeviceBusy);
    }
    self.handlers[idx] = Some(handler);
    Ok(())
}

pub fn dispatch(&self, sender: Handle, opcode_raw: u32, payload_ptr: *const u8, payload_len: u32) ->
    let opcode = Opcode::from_u32(opcode_raw).ok_or(DsError::InvalidMessage)?;
    let idx = (opcode_raw & 0x3F) as usize;
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/lib/ds-ipc/src/dispatcher.rs (line 30) - [public] dispatch

```
pub fn dispatch(&self, sender: Handle, opcode_raw: u32, payload_ptr: *const u8, payload_len: u32) ->
    let opcode = Opcode::from_u32(opcode_raw).ok_or(DsError::InvalidMessage)?;
    let idx = (opcode_raw & 0x3F) as usize;

    if let Some(handler) = self.handlers[idx] {
        handler(sender, payload_ptr, payload_len)
    } else {
        ds_error!("Unhandled opcode: {:?}", opcode);
        Err(DsError::InvalidMessage)
    }
}

}
```

driverspace_workspace/lib/ds-ipc/src/port.rs (line 11) - [public] PortState

```
pub enum PortState {
    Closed = 0,
    Listening = 1,
    Connected = 2,
    Error = 3,
}

#[repr(C)]
pub struct Port {
    pub handle: Handle,
    pub state: AtomicU32,
    pub name: [u8; PORT_NAME_LEN],
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/lib/ds-ipc/src/port.rs (line 19) - [public] Port

```
pub struct Port {
    pub handle: Handle,
    pub state: AtomicU32,
    pub name: [u8; PORT_NAME_LEN],
    pub owner: Handle,
    pub max_queue: u32,
    pub pending: AtomicU32,
    pub peer: Handle,
    pub flags: u32,
}
```

```
impl Port {
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/ds-ipc/src/port.rs (line 44) - [public] set_name

```
pub fn set_name(&mut self, name: &str) {
    self.name = [0; PORT_NAME_LEN];
    let bytes = name.as_bytes();
    let len = if bytes.len() < PORT_NAME_LEN { bytes.len() } else { PORT_NAME_LEN - 1 };
    self.name[..len].copy_from_slice(&bytes[..len]);
}
```

```
pub fn state(&self) -> PortState {
    match self.state.load(Ordering::Acquire) {
        0 => PortState::Closed,
        1 => PortState::Listening,
        2 => PortState::Connected,
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/ds-ipc/src/port.rs (line 51) - [public] state

```
pub fn state(&self) -> PortState {
    match self.state.load(Ordering::Acquire) {
        0 => PortState::Closed,
        1 => PortState::Listening,
        2 => PortState::Connected,
        3 => PortState::Error,
        _ => PortState::Error,
    }
}
```

```
pub fn set_state(&self, state: PortState) {
    self.state.store(state as u32, Ordering::Release);
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/ds-ipc/src/port.rs (line 61) - [public] set_state

```
pub fn set_state(&self, state: PortState) {
    self.state.store(state as u32, Ordering::Release);
}

pub fn is_open(&self) -> bool {
    matches!(self.state(), PortState::Listening | PortState::Connected)
}

pub fn try_acquire(&self) -> bool {
    self.state
        .compare_exchange(
            PortState::Listening as u32,
            // ... (truncated, see source file for more)
```

driverspace_workspace/lib/ds-ipc/src/port.rs (line 65) - [public] is_open

```
pub fn is_open(&self) -> bool {
    matches!(self.state(), PortState::Listening | PortState::Connected)
}

pub fn try_acquire(&self) -> bool {
    self.state
        .compare_exchange(
            PortState::Listening as u32,
            PortState::Connected as u32,
            Ordering::AcqRel,
            Ordering::Acquire,
        )
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/ds-ipc/src/port.rs (line 69) - [public] try_acquire

```
pub fn try_acquire(&self) -> bool {
    self.state
        .compare_exchange(
            PortState::Listening as u32,
            PortState::Connected as u32,
            Ordering::AcqRel,
            Ordering::Acquire,
        )
        .is_ok()
}

// ... (truncated, see source file for more)
```

driverspace_workspace/lib/ds-ipc/src/port.rs (line 81) - [public] PortRegistry

```
pub struct PortRegistry {
    ports: [Port; MAX_PORTS],
    next_handle: AtomicU32,
}

impl PortRegistry {
    pub const fn new() -> Self {
        const EMPTY_PORT: Port = Port::new(Handle::INVALID, Handle::INVALID);
        Self {
            ports: [EMPTY_PORT; MAX_PORTS],
            next_handle: AtomicU32::new(2),
        }
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/ds-ipc/src/port.rs (line 95) - [public] alloc_port

```
pub fn alloc_port(&self, owner: Handle) -> Option<Handle> {
    let handle_val = self.next_handle.fetch_add(1, Ordering::Relaxed);
    if handle_val as usize >= MAX_PORTS {
        return None;
    }
    let handle = Handle(handle_val);
    self.ports[handle_val as usize] = Port::new(handle, owner);
    Some(handle)
}

pub fn get(&self, handle: Handle) -> Option<&Port> {
    let idx = handle.raw() as usize;
    // ... (truncated, see source file for more)
```

```

driverspace_workspace/lib/ds-ipc/src/port.rs (line 105) - [public] get
    pub fn get(&self, handle: Handle) -> Option<&Port> {
        let idx = handle.raw() as usize;
        if idx >= MAX_PORTS { return None; }
        let port = &self.ports[idx];
        if port.handle != handle { return None; }
        Some(port)
    }

    pub fn get_mut(&mut self, handle: Handle) -> Option<&mut Port> {
        let idx = handle.raw() as usize;
        if idx >= MAX_PORTS { return None; }
        let port = &mut self.ports[idx];
        // ... (truncated, see source file for more)

driverspace_workspace/lib/ds-ipc/src/port.rs (line 113) - [public] get_mut
    pub fn get_mut(&mut self, handle: Handle) -> Option<&mut Port> {
        let idx = handle.raw() as usize;
        if idx >= MAX_PORTS { return None; }
        let port = &mut self.ports[idx];
        if port.handle != handle { return None; }
        Some(port)
    }

    pub fn close(&mut self, handle: Handle) {
        if let Some(port) = self.get_mut(handle) {
            port.set_state(PortState::Closed);
            port.peer = Handle::INVALID;
            // ... (truncated, see source file for more)

driverspace_workspace/lib/ds-ipc/src/port.rs (line 121) - [public] close
    pub fn close(&mut self, handle: Handle) {
        if let Some(port) = self.get_mut(handle) {
            port.set_state(PortState::Closed);
            port.peer = Handle::INVALID;
        }
    }
}

driverspace_workspace/lib/ds-log/src/formatter.rs (line 7) - [public] LogBuffer
    pub struct LogBuffer {
        buf: [u8; MAX_LOG_LINE_LEN],
        pos: usize,
    }

    impl LogBuffer {
        pub const fn new() -> Self {
            Self {
                buf: [0; MAX_LOG_LINE_LEN],
                pos: 0,
            }
        }
        // ... (truncated, see source file for more)

driverspace_workspace/lib/ds-log/src/formatter.rs (line 20) - [public] as_bytes
    pub fn as_bytes(&self) -> &[u8] {
        &self.buf[..self.pos]
    }

    pub fn reset(&mut self) {
        self.pos = 0;
    }
}

impl Write for LogBuffer {
    fn write_str(&mut self, s: &str) -> fmt::Result {
        let bytes = s.as_bytes();
        // ... (truncated, see source file for more)

```


driverspace_workspace/lib/ds-log/src/formatter.rs (line 24) - [public] reset

```
pub fn reset(&mut self) {
    self.pos = 0;
}

impl Write for LogBuffer {
    fn write_str(&mut self, s: &str) -> fmt::Result {
        let bytes = s.as_bytes();
        let remaining = self.buf.len() - self.pos;
        let copy_len = if bytes.len() > remaining { remaining } else { bytes.len() };

        self.buf[self.pos..self.pos + copy_len].copy_from_slice(&bytes[..copy_len]);
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/lib/ds-log/src/formatter.rs (line 30) - [internal] write_str

```
fn write_str(&mut self, s: &str) -> fmt::Result {
    let bytes = s.as_bytes();
    let remaining = self.buf.len() - self.pos;
    let copy_len = if bytes.len() > remaining { remaining } else { bytes.len() };

    self.buf[self.pos..self.pos + copy_len].copy_from_slice(&bytes[..copy_len]);
    self.pos += copy_len;
    Ok(())
}

}
```

driverspace_workspace/lib/ds-log/src/levels.rs (line 6) - [public] LogLevel

```
pub enum LogLevel {
    Trace = 0,
    Info = 1,
    Warn = 2,
    Error = 3,
    Fatal = 4,
}

impl LogLevel {
    pub const fn as_str(&self) -> &'static str {
        match self {
            Self::Trace => "TRACE",
            // ... (truncated, see source file for more)
        }
    }
}
```

driverspace_workspace/lib/ds-log/src/lib.rs (line 15) - [public] EndpointRef

```
pub struct EndpointRef;

impl EndpointRef {
    pub const fn new() -> Self { Self }

    pub fn is_valid(&self) -> bool {
        MANAGER_ENDPOINT_RAW.load(Ordering::Acquire) != 0
    }

    /// Handle endpointu Managera (0 = niezarejestrowany).
    pub fn handle(&self) -> u32 {
        MANAGER_ENDPOINT_RAW.load(Ordering::Acquire)
    }
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/lib/ds-log/src/lib.rs (line 20) - [public] is_valid

```
pub fn is_valid(&self) -> bool {
    MANAGER_ENDPOINT_RAW.load(Ordering::Acquire) != 0
}

/// Handle endpointu Managera (0 = niezarejestrowany).
pub fn handle(&self) -> u32 {
    MANAGER_ENDPOINT_RAW.load(Ordering::Acquire)
}

pub fn set(&self, handle: Handle) {
    MANAGER_ENDPOINT_RAW.store(handle.0, Ordering::Release);
}
// ... (truncated, see source file for more)
```

driverspace_workspace/lib/ds-log/src/lib.rs (line 25) - [public] handle

/// Handle endpointu Managera (0 = niezarejestrowany).

```
pub fn handle(&self) -> u32 {
    MANAGER_ENDPOINT_RAW.load(Ordering::Acquire)
}

pub fn set(&self, handle: Handle) {
    MANAGER_ENDPOINT_RAW.store(handle.0, Ordering::Release);
}

pub static MANAGER_ENDPOINT: EndpointRef = EndpointRef::new();
```

driverspace_workspace/lib/ds-log/src/lib.rs (line 29) - [public] set

```
pub fn set(&self, handle: Handle) {
    MANAGER_ENDPOINT_RAW.store(handle.0, Ordering::Release);
}

pub static MANAGER_ENDPOINT: EndpointRef = EndpointRef::new();
```

driverspace_workspace/lib/ds-log/src/transport.rs (line 8) - [internal] uart_putchar

```
fn uart_putchar(c: u8) {
    #[cfg(target_arch = "x86_64")]
    unsafe {
        core::arch::asm!("out dx, al", in("dx") 0x3F8u16, in("al") c, options(nomem, nostack, preserves_flags));
    }
}

pub fn emit_log(level: LogLevel, module: &str, buf: &LogBuffer) {
    let bytes = buf.as_bytes();

    for &b in bytes {
        uart_putchar(b);
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/ds-log/src/transport.rs (line 15) - [public] emit_log

```
pub fn emit_log(level: LogLevel, module: &str, buf: &LogBuffer) {
    let bytes = buf.as_bytes();

    for &b in bytes {
        uart_putchar(b);
    }
    uart_putchar(b'\n');

    let _ = sys_ipc_call(
        DsCmd::Log,
        bytes.as_ptr() as u64,
        bytes.len() as u64,
    );
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/ds-mem/src/alloc/buddy.rs (line 5) - [public] BuddyAllocator

```
pub struct BuddyAllocator {
    base_addr: usize,
    total_pages: usize,
    free_lists: [usize; MAX_ORDER + 1],
}

impl BuddyAllocator {
    pub const fn new(base_addr: usize, total_pages: usize) -> Self {
        Self {
            base_addr,
            total_pages,
            free_lists: [0; MAX_ORDER + 1],
        }
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/ds-mem/src/alloc/buddy.rs (line 20) - [public] alloc_pages

```
pub fn alloc_pages(&mut self, order: usize) -> Option<usize> {
    if order > MAX_ORDER {
        return None;
    }
    // TODO: Implement actual buddy allocation (find free block, split if necessary)
    Some(self.base_addr)
}

pub fn free_pages(&mut self, addr: usize, order: usize) {
    // TODO: Implement buddy merge logic
}
}
```

driverspace_workspace/lib/ds-mem/src/alloc/buddy.rs (line 28) - [public] free_pages

```
pub fn free_pages(&mut self, addr: usize, order: usize) {
    // TODO: Implement buddy merge logic
}
}
```

driverspace_workspace/lib/ds-mem/src/alloc/heap.rs (line 6) - [public] DriverHeap

```
pub struct DriverHeap {
    buddy: BuddyAllocator,
    // TODO: Add array of SlabAllocator for sizes: 16, 32, 64, 128, 256, 512, 1024
}

impl DriverHeap {
    pub const fn new() -> Self {
        Self {
            buddy: BuddyAllocator::new(0, 0),
        }
    }

    // ... (truncated, see source file for more)
}
```

driverspace_workspace/lib/ds-mem/src/alloc/heap.rs (line 18) - [public] init

```
pub fn init(&mut self, base_addr: usize, total_pages: usize) {
    self.buddy = BuddyAllocator::new(base_addr, total_pages);
}

unsafe impl GlobalAlloc for DriverHeap {
    unsafe fn alloc(&self, layout: Layout) -> *mut u8 {
        ds_error!("DriverHeap::alloc called before full initialization or unimplemented");
        null_mut()
    }

    unsafe fn dealloc(&self, ptr: *mut u8, layout: Layout) {
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/lib/ds-mem/src/alloc/slab.rs (line 1) - [public] SlabAllocator

```
pub struct SlabAllocator {
    object_size: usize,
    free_head: usize,
}

impl SlabAllocator {
    pub const fn new(object_size: usize) -> Self {
        Self {
            object_size,
            free_head: 0,
        }
    }

    // ... (truncated, see source file for more)
}
```

driverspace_workspace/lib/ds-mem/src/alloc/slab.rs (line 14) - [public] alloc

```
pub fn alloc(&mut self) -> Option<usize> {
    // TODO: Pop address from free_head linked list
    Some(0)
}

pub fn free(&mut self, ptr: usize) {
    // TODO: Push address to free_head linked list
}
}
```

driverspace_workspace/lib/ds-mem/src/alloc/slab.rs (line 19) - [public] free

```
pub fn free(&mut self, ptr: usize) {
    // TODO: Push address to free_head linked list
}
}
```

driverspace_workspace/lib/ds-mem/src/dma/buffer.rs (line 7) - [public] DmaFlags

```
pub struct DmaFlags(pub u32);

impl DmaFlags {
    pub const COHERENT: Self = Self(1 << 0);
    pub const HIGH_MEM: Self = Self(1 << 1);
    pub const CONTIGUOUS: Self = Self(1 << 2);
}
```

```
pub struct DmaBuffer {
    pub phys_addr: u64,
    pub virt_addr: u64,
    pub size: u64,
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/lib/ds-mem/src/dma/buffer.rs (line 15) - [public] DmaBuffer

```
pub struct DmaBuffer {
    pub phys_addr: u64,
    pub virt_addr: u64,
    pub size: u64,
    pub handle: Handle,
}

impl DmaBuffer {
    pub fn allocate(manager_ep: Handle, size: u64, flags: DmaFlags) -> Result<Self, DsError> {
        let reply = sys_ipc_call(DsCmd::MemAllocDma, size, flags.0 as u64, 0);
        if reply.is_ok() {
            Ok(Self { phys_addr: reply.arg0, virt_addr: reply.arg1, size, handle: manager_ep })
        } else {
            Err(reply.error_status())
        }
    }
}
```

driverspace_workspace/lib/ds-mem/src/dma/buffer.rs (line 23) - [public] allocate

```
pub fn allocate(manager_ep: Handle, size: u64, flags: DmaFlags) -> Result<Self, DsError> {
    let reply = sys_ipc_call(DsCmd::MemAllocDma, size, flags.0 as u64, 0);
    if reply.is_ok() {
        Ok(Self { phys_addr: reply.arg0, virt_addr: reply.arg1, size, handle: manager_ep })
    } else {
        Err(reply.error_status())
    }
}

impl Drop for DmaBuffer {
    fn drop(&mut self) {
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/lib/ds-mem/src/dma/buffer.rs (line 34) - [internal] drop

```
fn drop(&mut self) {
    let _ = sys_ipc_call(DsCmd::MemFreeDma, self.phys_addr, self.size, 0);
}
}
```

driverspace_workspace/lib/ds-mem/src/dma/mapping.rs (line 6) - [public] MmioRegion

```
pub struct MmioRegion {
    pub phys_base: u64,
    pub virt_base: u64,
    pub size: u64,
    pub handle: Handle,
}

impl MmioRegion {
    pub fn map(manager_ep: Handle, phys_base: u64, size: u64) -> Result<Self, DsError> {
        let reply = sys_ipc_call(DsCmd::MemMapMmio, phys_base, size, 0);
        if reply.is_ok() {
            Ok(Self { phys_base, virt_base: reply.arg0, size, handle: manager_ep })
        }
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/lib/ds-mem/src/dma/mapping.rs (line 14) - [public] map

```
pub fn map(manager_ep: Handle, phys_base: u64, size: u64) -> Result<Self, DsError> {
    let reply = sys_ipc_call(DsCmd::MemMapMmio, phys_base, size, 0);
    if reply.is_ok() {
        Ok(Self { phys_base, virt_base: reply.arg0, size, handle: manager_ep })
    } else {
        Err(reply.error_status())
    }
}

impl Drop for MmioRegion {
    fn drop(&mut self) {
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/lib/ds-mem/src/dma/mapping.rs (line 25) - [internal] drop

```
fn drop(&mut self) {
    let _ = sys_ipc_call(DsCmd::MemUnmapMmio, self.virt_base, self.size, 0);
}

pub struct Volatile<T>(pub T);
```

```
impl<T: Copy> Volatile<T> {
    #[inline(always)]
    pub fn read(&self) -> T {
        unsafe { core::ptr::read_volatile(&self.0) }
    }
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/lib/ds-mem/src/dma/mapping.rs (line 30) - [public] Volatile

```
pub struct Volatile<T>(pub T);

impl<T: Copy> Volatile<T> {
    #[inline(always)]
    pub fn read(&self) -> T {
        unsafe { core::ptr::read_volatile(&self.0) }
    }

    #[inline(always)]
    pub fn write(&mut self, value: T) {
        unsafe { core::ptr::write_volatile(&mut self.0, value); }
    }
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/lib/ds-mem/src/dma/mapping.rs (line 34) - [public] read

```
pub fn read(&self) -> T {
    unsafe { core::ptr::read_volatile(&self.0) }
}

#[inline(always)]
pub fn write(&mut self, value: T) {
    unsafe { core::ptr::write_volatile(&mut self.0, value); }
}

}
```

driverspace_workspace/lib/ds-mem/src/dma/mapping.rs (line 39) - [public] write

```
pub fn write(&mut self, value: T) {
    unsafe { core::ptr::write_volatile(&mut self.0, value); }
}
```

driverspace_workspace/lib/ds-mem/src/vmm.rs (line 6) - [public] map_memory

```
pub fn map_memory(
    _manager_ep: Handle,
    phys_addr: u64,
    _virt_addr: u64,
    size: u64,
    flags: u64,
) -> Result<(), DsError> {
    let reply = sys_ipc_call(DsCmd::MemMapMmio, phys_addr, size, flags);
    if reply.is_ok() { Ok(()) } else { Err(reply.error_status()) }
}

pub fn unmap_memory(_manager_ep: Handle, virt_addr: u64, size: u64) -> Result<(), DsError> {
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/lib/ds-mem/src/vmm.rs (line 17) - [public] unmap_memory

```
pub fn unmap_memory(_manager_ep: Handle, virt_addr: u64, size: u64) -> Result<(), DsError> {
    let reply = sys_ipc_call(DsCmd::MemUnmapMmio, virt_addr, size, 0);
    if reply.is_ok() { Ok(()) } else { Err(reply.error_status()) }
}
```

driverspace_workspace/lib/kapi-abi/build.rs (line 9) - [internal] main

```
/// Build script dla `kapi-abi`.
///
/// Crate jest w 100% `no_std` i nie generuje kodu — skrypt istnieje tylko po to,
/// aby Cargo miało poprawny entry point (pusty `build.rs` powoduje błąd
/// "main` function not found in crate `build_script_build`").
///
/// Gdy pojawi się generowanie nagłówek C/Odin z typów ABI, kod trafi tutaj.

fn main() {
    println!("cargo:rerun-if-changed=src");
}
```

driverspace_workspace/lib/kapi-abi/src/errors.rs (line 7) - [public] DsError

```
pub enum DsError {
    /// Success
    Success = 0,

    /// Unknown
    Unknown = 1,

    /// InvalidHandle
    InvalidHandle = 2,

    /// OutOfMemory
    OutOfMemory = 3,
    // ... (truncated, see source file for more)
}
```

driverspace_workspace/lib/kapi-abi/src/errors.rs (line 44) - [public] from_u32

```
/// u32 to DsError Unknown

pub fn from_u32(val: u32) -> Self {
    match val {
        0 => Self::Success,
        1 => Self::Unknown,
        2 => Self::InvalidHandle,
        3 => Self::OutOfMemory,
        4 => Self::PermissionDenied,
        5 => Self::DeviceNotFound,
        6 => Self::DeviceBusy,
        7 => Self::InvalidMessage,
        8 => Self::BufferTooSmall,
        9 => Self::DeviceFault,
        // ... (truncated, see source file for more)
    }
}
```


driverspace_workspace/lib/kapi-abi/src/payloads/mem.rs (line 5) - [public] DmaAllocPayload

```
pub struct DmaAllocPayload {
    pub size: u64,
    pub flags: u32,
    pub _pad: u32,
}

#[repr(C)]
#[derive(Debug, Clone, Copy)]
pub struct MmioMapPayload {
    pub phys_base: u64,
    pub size: u64,
}
```

driverspace_workspace/lib/kapi-abi/src/payloads/mem.rs (line 13) - [public] MmioMapPayload

```
pub struct MmioMapPayload {
    pub phys_base: u64,
    pub size: u64,
}
```

driverspace_workspace/lib/kapi-abi/src/payloads/sys.rs (line 9) - [public] LogPayload

```
pub struct LogPayload {
    /// ■■■■■■■■■■ `ds-log` ■ `LogLevel` ■
    pub level: u8,

    /// ■■■■■■■■■■
    pub module_len: u16,

    /// ■■■■■■■■■■
    pub msg_len: u16,

    /// ■■■■■■■■■■
    pub _pad: u32,
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/primitives.rs (line 5) - [public] Handle

```
pub struct Handle(pub u32);

impl Handle {
    pub const INVALID: Self = Self(u32::MAX);
    pub const KERNEL: Self = Self(0);
    pub const MANAGER: Self = Self(1);

    #[inline]
    pub const fn is_valid(self) -> bool {
        self.0 != u32::MAX
    }

    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/primitives.rs (line 25) - [public] CapId

```
pub struct CapId(pub u64);

impl CapId {
    pub const NONE: Self = Self(0);
    pub const KERNEL_ALL: Self = Self(u64::MAX);

    pub const MEM_ALLOC: Self = Self(1 << 0);
    pub const MEM_MAP: Self = Self(1 << 1);
    pub const MEM_DMA: Self = Self(1 << 2);
    pub const IPC_CREATE: Self = Self(1 << 8);
    pub const IPC_SEND: Self = Self(1 << 9);
    pub const DEV_REGISTER: Self = Self(1 << 16);
    // ... (truncated, see source file for more)
```


driverspace_workspace/lib/kapi-abi/src/primitives.rs (line 62) - [public] Status

```
pub enum Status {
    Ok = 0,
    Unknown = -1,
    InvalidArg = -2,
    NoMemory = -3,
    PermissionDenied = -4,
    NotFound = -5,
    Busy = -6,
    Timeout = -7,
    InvalidHandle = -8,
    BufferTooSmall = -9,
    DeviceFault = -10,
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/primitives.rs (line 113) - [public] PhysAddr

```
pub struct PhysAddr(pub u64);

#[derive(Debug, Clone, Copy, PartialEq, Eq)]
#[repr(transparent)]
pub struct VirtAddr(pub u64);

#[derive(Debug, Clone, Copy, PartialEq, Eq)]
#[repr(transparent)]
pub struct DeviceId(pub u64);

impl DeviceId {
    #[inline]
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/primitives.rs (line 117) - [public] VirtAddr

```
pub struct VirtAddr(pub u64);

#[derive(Debug, Clone, Copy, PartialEq, Eq)]
#[repr(transparent)]
pub struct DeviceId(pub u64);

impl DeviceId {
    #[inline]
    pub const fn pci(vendor: u16, device: u16) -> Self {
        Self(((vendor as u64) << 16) | (device as u64))
    }

    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/primitives.rs (line 121) - [public] DeviceId

```
pub struct DeviceId(pub u64);

impl DeviceId {
    #[inline]
    pub const fn pci(vendor: u16, device: u16) -> Self {
        Self(((vendor as u64) << 16) | (device as u64))
    }

    #[inline]
    pub const fn usb(vendor: u16, product: u16) -> Self {
        Self(0x0100_0000_0000_0000 | ((vendor as u64) << 16) | (product as u64))
    }

    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/primitives.rs (line 152) - [public] BusType

```
pub enum BusType {
    Pci = 0,
    Usb = 1,
    Acpi = 2,
    Platform = 3,
    Virtio = 4,
}

#[derive(Debug, Clone, Copy, PartialEq, Eq)]
#[repr(u8)]
pub enum DriverState {
    Unloaded = 0,
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/primitives.rs (line 162) - [public] DriverState

```
pub enum DriverState {
    Unloaded = 0,
    Loading = 1,
    Running = 2,
    Suspended = 3,
    Error = 4,
    Unloading = 5,
}

#[derive(Debug, Clone, Copy, PartialEq, Eq)]
#[repr(u8)]
pub enum DeviceState {
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/primitives.rs (line 173) - [public] DeviceState

```
pub enum DeviceState {
    Discovered = 0,
    Attached = 1,
    Detached = 2,
    Error = 3,
}
```

driverspace_workspace/lib/kapi-abi/src/wire/decode.rs (line 5) - [public] Decoder

```
pub struct Decoder<'a> {
    buf: &'a [u8],
    pos: usize,
}

impl<'a> Decoder<'a> {
    pub fn new(buf: &'a [u8]) -> Self {
        Self { buf, pos: 0 }
    }

    #[inline]
    pub fn remaining(&self) -> usize {
        // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/wire/decode.rs (line 11) - [public] new

```
pub fn new(buf: &'a [u8]) -> Self {
    Self { buf, pos: 0 }
}

#[inline]
pub fn remaining(&self) -> usize {
    self.buf.len() - self.pos
}

#[inline]
pub fn read_u8(&mut self) -> Option<u8> {
    if self.pos >= self.buf.len() { return None; }
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/wire/decode.rs (line 16) - [public] remaining

```
pub fn remaining(&self) -> usize {
    self.buf.len() - self.pos
}

#[inline]
pub fn read_u8(&mut self) -> Option<u8> {
    if self.pos >= self.buf.len() { return None; }
    let v = self.buf[self.pos];
    self.pos += 1;
    Some(v)
}

// ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/wire/decode.rs (line 21) - [public] read_u8

```
pub fn read_u8(&mut self) -> Option<u8> {
    if self.pos >= self.buf.len() { return None; }
    let v = self.buf[self.pos];
    self.pos += 1;
    Some(v)
}

#[inline]
pub fn read_u16(&mut self) -> Option<u16> {
    if self.pos + 2 > self.buf.len() { return None; }
    let v = u16::from_le_bytes([self.buf[self.pos], self.buf[self.pos + 1]]);
    self.pos += 2;
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/wire/decode.rs (line 29) - [public] read_u16

```
pub fn read_u16(&mut self) -> Option<u16> {
    if self.pos + 2 > self.buf.len() { return None; }
    let v = u16::from_le_bytes([self.buf[self.pos], self.buf[self.pos + 1]]);
    self.pos += 2;
    Some(v)
}

#[inline]
pub fn read_u32(&mut self) -> Option<u32> {
    if self.pos + 4 > self.buf.len() { return None; }
    let mut bytes = [0u8; 4];
    bytes.copy_from_slice(&self.buf[self.pos..self.pos + 4]);
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/wire/decode.rs (line 37) - [public] read_u32

```
pub fn read_u32(&mut self) -> Option<u32> {
    if self.pos + 4 > self.buf.len() { return None; }
    let mut bytes = [0u8; 4];
    bytes.copy_from_slice(&self.buf[self.pos..self.pos + 4]);
    self.pos += 4;
    Some(u32::from_le_bytes(bytes))
}

#[inline]
pub fn read_u64(&mut self) -> Option<u64> {
    if self.pos + 8 > self.buf.len() { return None; }
    let mut bytes = [0u8; 8];
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/wire/decode.rs (line 46) - [public] read_u64

```
pub fn read_u64(&mut self) -> Option<u64> {
    if self.pos + 8 > self.buf.len() { return None; }
    let mut bytes = [0u8; 8];
    bytes.copy_from_slice(&self.buf[self.pos..self.pos + 8]);
    self.pos += 8;
    Some(u64::from_le_bytes(bytes))
}

pub fn read_msg(&mut self) -> Option<DsMsg> {
    if self.remaining() < super::MSG_SIZE { return None; }
    let magic = self.read_u32()?;
    if magic != DS_MAGIC { return None; }
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/wire/decode.rs (line 54) - [public] read_msg

```
pub fn read_msg(&mut self) -> Option<DsMsg> {
    if self.remaining() < super::MSG_SIZE { return None; }
    let magic = self.read_u32()?;
    if magic != DS_MAGIC { return None; }
    let version = self.read_u16()?;
    if version != DS_VERSION as u16 { return None; }
    let cmd = self.read_u16()?;
    let id = self.read_u64()?;
    let flags = self.read_u32()?;
    let status = self.read_u32()? as i32;
    let arg0 = self.read_u64()?;
    let arg1 = self.read_u64()?;
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/wire/decode.rs (line 80) - [public] read_bytes

```
pub fn read_bytes(&mut self, len: usize) -> Option<&'a [u8]> {
    if self.pos + len > self.buf.len() { return None; }
    let slice = &self.buf[self.pos..self.pos + len];
    self.pos += len;
    Some(slice)
}
}
```

driverspace_workspace/lib/kapi-abi/src/wire/encode.rs (line 5) - [public] Encoder

```
pub struct Encoder<'a> {
    buf: &'a mut [u8],
    pos: usize,
}

impl<'a> Encoder<'a> {
    pub fn new(buf: &'a mut [u8]) -> Self {
        Self { buf, pos: 0 }
    }

    #[inline]
    pub fn remaining(&self) -> usize {
        // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/wire/encode.rs (line 11) - [public] new

```
pub fn new(buf: &'a mut [u8]) -> Self {
    Self { buf, pos: 0 }
}

#[inline]
pub fn remaining(&self) -> usize {
    self.buf.len() - self.pos
}

#[inline]
pub fn write_u8(&mut self, v: u8) -> bool {
    if self.pos >= self.buf.len() { return false; }
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/wire/encode.rs (line 16) - [public] remaining

```
pub fn remaining(&self) -> usize {
    self.buf.len() - self.pos
}

#[inline]
pub fn write_u8(&mut self, v: u8) -> bool {
    if self.pos >= self.buf.len() { return false; }
    self.buf[self.pos] = v;
    self.pos += 1;
    true
}

// ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/wire/encode.rs (line 21) - [public] write_u8

```
pub fn write_u8(&mut self, v: u8) -> bool {
    if self.pos >= self.buf.len() { return false; }
    self.buf[self.pos] = v;
    self.pos += 1;
    true
}

#[inline]
pub fn write_u16(&mut self, v: u16) -> bool {
    if self.pos + 2 > self.buf.len() { return false; }
    let bytes = v.to_le_bytes();
    self.buf[self.pos] = bytes[0];
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/wire/encode.rs (line 29) - [public] write_u16

```
pub fn write_u16(&mut self, v: u16) -> bool {
    if self.pos + 2 > self.buf.len() { return false; }
    let bytes = v.to_le_bytes();
    self.buf[self.pos] = bytes[0];
    self.buf[self.pos + 1] = bytes[1];
    self.pos += 2;
    true
}

#[inline]
pub fn write_u32(&mut self, v: u32) -> bool {
    if self.pos + 4 > self.buf.len() { return false; }
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/wire/encode.rs (line 39) - [public] write_u32

```
pub fn write_u32(&mut self, v: u32) -> bool {
    if self.pos + 4 > self.buf.len() { return false; }
    let bytes = v.to_le_bytes();
    self.buf[self.pos..self.pos + 4].copy_from_slice(&bytes);
    self.pos += 4;
    true
}

#[inline]
pub fn write_u64(&mut self, v: u64) -> bool {
    if self.pos + 8 > self.buf.len() { return false; }
    let bytes = v.to_le_bytes();
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/wire/encode.rs (line 48) - [public] write_u64

```
pub fn write_u64(&mut self, v: u64) -> bool {
    if self.pos + 8 > self.buf.len() { return false; }
    let bytes = v.to_le_bytes();
    self.buf[self.pos..self.pos + 8].copy_from_slice(&bytes);
    self.pos += 8;
    true
}

pub fn write_msg(&mut self, msg: &DsMsg) -> bool {
    if self.remaining() < super::MSG_SIZE { return false; }
    self.write_u32(msg.magic);
    self.write_u16(msg.version);
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/wire/encode.rs (line 56) - [public] write_msg

```
pub fn write_msg(&mut self, msg: &DsMsg) -> bool {
    if self.remaining() < super::MSG_SIZE { return false; }
    self.write_u32(msg.magic);
    self.write_u16(msg.version);
    self.write_u16(msg.cmd);
    self.write_u64(msg.id);
    self.write_u32(msg.flags);
    self.write_u32(msg.status as u32);
    self.write_u64(msg.arg0);
    self.write_u64(msg.arg1);
    self.write_u64(msg.arg2);
    true
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/wire/encode.rs (line 70) - [public] write_bytes

```
pub fn write_bytes(&mut self, data: &[u8]) -> bool {
    if self.pos + data.len() > self.buf.len() { return false; }
    self.buf[self.pos..self.pos + data.len()].copy_from_slice(data);
    self.pos += data.len();
    true
}

pub fn pos(&self) -> usize {
    self.pos
}
}
```

driverspace_workspace/lib/kapi-abi/src/wire/encode.rs (line 77) - [public] pos

```
pub fn pos(&self) -> usize {
    self.pos
}
}
```

driverspace_workspace/lib/kapi-abi/src/wire/mod.rs (line 15) - [public] DsMsg

```
pub struct DsMsg {
    pub magic: u32,
    pub version: u16,
    pub cmd: u16,
    pub id: u64,
    pub flags: u32,
    pub status: i32,
    pub arg0: u64,
    pub arg1: u64,
    pub arg2: u64,
}

// ... (truncated, see source file for more)
```

```

driverspace_workspace/lib/kapi-abi/src/wire/mod.rs (line 28) - [internal] default
    fn default() -> Self {
        Self::new(0)
    }
}

impl DsMsg {
    pub const fn new(cmd: u32) -> Self {
        Self {
            magic: DS_MAGIC,
            version: DS_VERSION as u16,
            cmd: cmd as u16,
            id: 0,
            // ... (truncated, see source file for more)
        }
    }
}

driverspace_workspace/lib/kapi-abi/src/wire/mod.rs (line 49) - [public] is_valid
    pub fn is_valid(&self) -> bool {
        self.magic == DS_MAGIC && self.version == DS_VERSION as u16
    }

    #[inline]
    pub fn is_ok(&self) -> bool {
        self.status == 0
    }

    #[inline]
    pub fn error(&self) -> super::Status {
        super::Status::from_i32(self.status)
        // ... (truncated, see source file for more)
    }

driverspace_workspace/lib/kapi-abi/src/wire/mod.rs (line 54) - [public] is_ok
    pub fn is_ok(&self) -> bool {
        self.status == 0
    }

    #[inline]
    pub fn error(&self) -> super::Status {
        super::Status::from_i32(self.status)
    }

    #[inline]
    pub fn error_status(&self) -> crate::DsError {
        // status is i32-based; map negative/unknown codes to DsError::Unknown
        // ... (truncated, see source file for more)
    }

driverspace_workspace/lib/kapi-abi/src/wire/mod.rs (line 59) - [public] error
    pub fn error(&self) -> super::Status {
        super::Status::from_i32(self.status)
    }

    #[inline]
    pub fn error_status(&self) -> crate::DsError {
        // status is i32-based; map negative/unknown codes to DsError::Unknown
        // via truncation. Success (0) maps to DsError::Success.
        crate::DsError::from_u32(self.status as u32)
    }
}

// ... (truncated, see source file for more)

```

driverspace_workspace/lib/kapi-abi/src/wire/mod.rs (line 64) - [public] error_status

```
pub fn error_status(&self) -> crate::DsError {
    // status is i32-based; map negative/unknown codes to DsError::Unknown
    // via truncation. Success (0) maps to DsError::Success.
    crate::DsError::from_u32(self.status as u32)
}

#[repr(C)]
pub struct DsRing {
    pub head: u64,
    pub tail: u64,
    pub cap: u64,
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/wire/mod.rs (line 72) - [public] DsRing

```
pub struct DsRing {
    pub head: u64,
    pub tail: u64,
    pub cap: u64,
    pub _pad: u64,
}

impl DsRing {
    pub const fn new(cap: u64) -> Self {
        Self { head: 0, tail: 0, cap, _pad: 0 }
    }

    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/wire/mod.rs (line 85) - [public] is_full

```
pub fn is_full(&self) -> bool {
    self.head.wrapping_sub(self.tail) >= self.cap
}

#[inline]
pub fn is_empty(&self) -> bool {
    self.head == self.tail
}

#[inline]
pub fn available(&self) -> u64 {
    self.head.wrapping_sub(self.tail)
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/wire/mod.rs (line 90) - [public] is_empty

```
pub fn is_empty(&self) -> bool {
    self.head == self.tail
}

#[inline]
pub fn available(&self) -> u64 {
    self.head.wrapping_sub(self.tail)
}
}
```

driverspace_workspace/lib/kapi-abi/src/wire/mod.rs (line 95) - [public] available

```
pub fn available(&self) -> u64 {
    self.head.wrapping_sub(self.tail)
}
}
```


driverspace_workspace/lib/kapi-abi/src/wire/wire.rs (line 6) - [public] DsMsg

```
pub struct DsMsg {
    pub id: u64,
    pub cmd: u32,
    pub flags: u32,
    pub arg0: u64,
    pub arg1: u64,
    pub arg2: u64,
    pub status: i32,
    pub pad: u32,
}

impl DsMsg {
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-abi/src/wire/wire.rs (line 31) - [public] is_ok

```
pub fn is_ok(&self) -> bool {
    self.status == 0
}

pub fn error_status(&self) -> crate::DsError {
    crate::DsError::from_u32(self.status as u32)
}
}
```

driverspace_workspace/lib/kapi-abi/src/wire/wire.rs (line 35) - [public] error_status

```
pub fn error_status(&self) -> crate::DsError {
    crate::DsError::from_u32(self.status as u32)
}
}
```

driverspace_workspace/lib/kapi-syscall/src/channel.rs (line 5) - [public] RingBuffer

```
pub struct RingBuffer {
    head: AtomicU64,
    tail: AtomicU64,
    capacity: u32,
    _pad: u32,
}

impl RingBuffer {
    pub unsafe fn from_ptr(ptr: *mut u8) -> &'static mut Self {
        unsafe { &mut *(ptr as *mut Self) }
    }

    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-syscall/src/channel.rs (line 17) - [internal] slot_ptr

```
fn slot_ptr(&self, index: u64) -> *mut DsMsg {
    let base = self as *const _ as *const u8;
    let offset = core::mem::size_of::<Self>()
        + (index as usize % self.capacity as usize) * core::mem::size_of::<DsMsg>();
    unsafe { base.add(offset) as *mut DsMsg }
}

pub fn push(&self, msg: &DsMsg) -> bool {
    let head = self.head.load(Ordering::Acquire);
    let tail = self.tail.load(Ordering::Acquire);

    if head - tail >= self.capacity as u64 {
        // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-syscall/src/channel.rs (line 24) - [public] push

```
pub fn push(&self, msg: &DsMsg) -> bool {
    let head = self.head.load(Ordering::Acquire);
    let tail = self.tail.load(Ordering::Acquire);

    if head - tail >= self.capacity as u64 {
        return false;
    }

    unsafe {
        self.slot_ptr(head).write(*msg);
    }

    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-syscall/src/channel.rs (line 40) - [public] pop

```
pub fn pop(&self) -> Option<DsMsg> {
    let head = self.head.load(Ordering::Acquire);
    let tail = self.tail.load(Ordering::Acquire);

    if tail >= head {
        return None;
    }

    let msg = unsafe { self.slot_ptr(tail).read() };
    self.tail.store(tail + 1, Ordering::Release);
    Some(msg)
}

// ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-syscall/src/channel.rs (line 53) - [public] is_empty

```
pub fn is_empty(&self) -> bool {
    let head = self.head.load(Ordering::Acquire);
    let tail = self.tail.load(Ordering::Acquire);
    tail >= head
}

}
```

driverspace_workspace/lib/kapi-syscall/src/lib.rs (line 21) - [public] sys_ipc_call

```
pub fn sys_ipc_call(cmd: DsCmd, arg0: u64, arg1: u64, arg2: u64) -> DsMsg {
    let mut msg = DsMsg::new(cmd as u32);
    msg.arg0 = arg0;
    msg.arg1 = arg1;
    msg.arg2 = arg2;

    unsafe {
        while !(*K2D_RING).push(&msg) {
            sys_yield();
        }
    }

    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-syscall/src/lib.rs (line 40) - [public] sys_poll

```
pub fn sys_poll() {
    unsafe {
        while (*D2K_RING).is_empty() {
            sys_yield();
        }
    }
}

pub fn sys_yield() {
    #[cfg(target_arch = "x86_64")]
    unsafe {
        core::arch::asm!("pause", options(nomem, nostack, preserves_flags));
    }
    // ... (truncated, see source file for more)
```

driverspace_workspace/lib/kapi-syscall/src/lib.rs (line 48) - [public] sys_yield

```
pub fn sys_yield() {
    #[cfg(target_arch = "x86_64")]
    unsafe {
        core::arch::asm!("pause", options(nomem, nostack, preserves_flags));
    }
}
```

driverspace_workspace/src/drivers/vgpu.rs (line 8) - [public] init

```
pub fn init() -> Result<(), Status> {
    let pci_msg = crate::ipc::Message {
        cmd: 0x20,
        arg0: VENDOR_ID as u64,
        arg1: DEVICE_ID as u64,
    };

    crate::ipc::send(pci_msg)?;

    match crate::ipc::receive() {
        Some(reply) if reply.cmd == 0 => {
            let bar0_phys = reply.arg0;
            // ... (truncated, see source file for more)
        }
    }
}
```

driverspace_workspace/src/ipc.rs (line 7) - [public] Message

```
pub struct Message {
    pub cmd: u32,
    pub arg0: u64,
    pub arg1: u64,
}

static RING_BUFFER: [AtomicU64; 1024] = {
    const INIT: AtomicU64 = AtomicU64::new(0);
    [INIT; 1024]
};

static HEAD: AtomicU64 = AtomicU64::new(0);
// ... (truncated, see source file for more)
```

driverspace_workspace/src/ipc.rs (line 21) - [public] send

```
pub fn send(msg: Message) -> Result<(), Status> {
    let head = HEAD.load(Ordering::Acquire);
    let tail = TAIL.load(Ordering::Acquire);

    if (head + 1) % 1024 == tail {
        return Err(Status::Busy);
    }

    let idx = (head * 2) as usize;
    RING_BUFFER[idx].store(msg.cmd as u64, Ordering::Release);
    RING_BUFFER[idx + 1].store(msg.arg0, Ordering::Release);

    // ... (truncated, see source file for more)
}
```

driverspace_workspace/src/ipc.rs (line 38) - [public] receive

```
pub fn receive() -> Option<Message> {
    let head = HEAD.load(Ordering::Acquire);
    let tail = TAIL.load(Ordering::Acquire);

    if head == tail {
        return None;
    }

    let idx = (tail * 2) as usize;
    let cmd = RING_BUFFER[idx].load(Ordering::Acquire) as u32;
    let arg0 = RING_BUFFER[idx + 1].load(Ordering::Acquire);

    // ... (truncated, see source file for more)
}
```

driverspace_workspace/src/ipc.rs (line 55) - [public] reply

```
pub fn reply(status: Status, data: u64) {
    let msg = Message {
        cmd: status as u32,
        arg0: data,
        arg1: 0,
    };
    send(msg).ok();
}
```

driverspace_workspace/src/memory.rs (line 5) - [public] alloc

```
pub fn alloc(size: usize) -> Result<*mut u8, Status> {
    let msg = crate::ipc::Message {
        cmd: 0x10,
        arg0: size as u64,
        arg1: 0,
    };

    crate::ipc::send(msg)?;

    match crate::ipc::receive() {
        Some(reply) if reply.cmd == 0 => {
            Ok(reply.arg0 as *mut u8)
        }
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/src/memory.rs (line 23) - [public] free

```
pub fn free(ptr: *mut u8, size: usize) {
    let msg = crate::ipc::Message {
        cmd: 0x11,
        arg0: ptr as u64,
        arg1: size as u64,
    };

    crate::ipc::send(msg).ok();
}
```

driverspace_workspace/src/self_test.rs (line 6) - [public] run_all

```
pub fn run_all() -> Result<usize, Status> {
    let mut passed = 0;

    test_ipc_ring()?;
    passed += 1;

    test_memory_alloc()?;
    passed += 1;

    test_capability_check()?;
    passed += 1;

    // ... (truncated, see source file for more)
}
```

driverspace_workspace/src/self_test.rs (line 21) - [internal] test_ipc_ring

```
fn test_ipc_ring() -> Result<(), Status> {
    let msg = ipc::Message {
        cmd: 0xFF,
        arg0: 0x12345678,
        arg1: 0xABCDEF00,
    };

    ipc::send(msg)?;

    let received = ipc::receive().ok_or(Status::IoError)?;

    if received.cmd != msg.cmd || received.arg0 != msg.arg0 {
        // ... (truncated, see source file for more)
    }
}
```

driverspace_workspace/src/self_test.rs (line 39) - [internal] test_memory_alloc

```
fn test_memory_alloc() -> Result<(), Status> {
    let ptr = memory::alloc(4096)?;

    if ptr.is_null() {
        return Err(Status::OutOfMemory);
    }

    unsafe {
        core::ptr::write_bytes(ptr, 0xAA, 4096);

        for i in 0..4096 {
            if core::ptr::read(ptr.add(i)) != 0xAA {
                // ... (truncated, see source file for more)
            }
        }
    }
}
```

driverspace_workspace/src/self_test.rs (line 62) - [internal] test_capability_check

```
fn test_capability_check() -> Result<(), Status> {
    let caps = crate::capabilities::current();

    if !caps.has(crate::capabilities::Capability::MMIO) {
        return Err(Status::PermissionDenied);
    }

    Ok(())
}
```

driverspacelib/src/abi.rs (line 53) - [public] DsCmd

```
pub enum DsCmd {
    None = 0,
    Init = 1,
    Caps = 2,
    Ready = 3,
    RegisterDriver = 4,
    AttachDevice = 5,
    DetachDevice = 6,
    MapDeviceMemory = 7,
    BindIrq = 8,
    Shutdown = 9,
    Log = 10,
    // ... (truncated, see source file for more)
}
```

driverspacelib/src/abi.rs (line 83) - [public] DsRing

```
pub struct DsRing {
    pub head: u64,
    pub tail: u64,
    pub cap: u64,
}

pub const DS_RING_HDR_SIZE: usize = core::mem::size_of::<DsRing>();

/// A single message exchanged between kernel and driver space.
#[repr(C)]
#[derive(Clone, Copy, Default)]
pub struct DsMsg {
    // ... (truncated, see source file for more)
}
```

driverspacelib/src/abi.rs (line 94) - [public] DsMsg

```
pub struct DsMsg {
    pub id: u64,
    pub cmd: u32,
    pub flags: u32,
    pub arg0: u64,
    pub arg1: u64,
    pub arg2: u64,
    pub status: i32,
    pub pad: u32,
}

pub const DS_MSG_SIZE: usize = core::mem::size_of::<DsMsg>();
// ... (truncated, see source file for more)
```

driverspacelib/src/abi.rs (line 110) - [public] DsInitParams

```
pub struct DsInitParams {
    pub magic: u64,
    pub version: u32,
    pub pad: u32,
    pub k2d_va: u64,
    pub d2k_va: u64,
    pub ring_cap: u64,
    pub ds_va_base: u64,
    pub ds_va_size: u64,
}

/// Library-side error type for failed driver-space calls.
// ... (truncated, see source file for more)
```

driverspacelib/src/abi.rs (line 123) - [public] DsError

```
pub enum DsError {
    Timeout,
    QueueFull,
    BadStatus(i32),
    NoAspace,
    NoMemory,
    NotPrepared,
}

impl From<i32> for DsError {
    fn from(v: i32) -> Self {
        DsError::BadStatus(v)
        // ... (truncated, see source file for more)
    }
}
```

driverspacelib/src/abi.rs (line 133) - [internal] from

```
fn from(v: i32) -> Self {
    DsError::BadStatus(v)
}

}
```

driverspacelib/src/audio.rs (line 4) - [public] AudioBars

```
pub struct AudioBars {
    pub nam_phys: u64,
    pub bm_phys: u64,
}

pub fn info_req() -> u64 {
    request(DsCmd::AudioInfo, 0, 0, 0)
}

pub fn info_take(id: u64) -> Option<AudioBars> {
    let r = take_resp(id)?;

    // ... (truncated, see source file for more)
}
```

driverspacelib/src/audio.rs (line 9) - [public] info_req

```
pub fn info_req() -> u64 {
    request(DsCmd::AudioInfo, 0, 0, 0)
}

pub fn info_take(id: u64) -> Option<AudioBars> {
    let r = take_resp(id)?;

    if r.status != 0 {
        return None;
    }

    Some(AudioBars { nam_phys: r.arg0, bm_phys: r.arg1 })
    // ... (truncated, see source file for more)
}
```

driverspacelib/src/audio.rs (line 13) - [public] info_take

```
pub fn info_take(id: u64) -> Option<AudioBars> {
    let r = take_resp(id)?;

    if r.status != 0 {
        return None;
    }

    Some(AudioBars { nam_phys: r.arg0, bm_phys: r.arg1 })
}

pub fn page_phys_req(va: u64) -> u64 {
    request(DsCmd::PagePhys, va, 0, 0)
    // ... (truncated, see source file for more)
```

driverspacelib/src/audio.rs (line 23) - [public] page_phys_req

```
pub fn page_phys_req(va: u64) -> u64 {
    request(DsCmd::PagePhys, va, 0, 0)
}

pub fn page_phys_take(id: u64) -> Option<u64> {
    let r = take_resp(id)?;

    if r.status != 0 {
        return None;
    }

    Some(r.arg0)
    // ... (truncated, see source file for more)
```

driverspacelib/src/audio.rs (line 27) - [public] page_phys_take

```
pub fn page_phys_take(id: u64) -> Option<u64> {
    let r = take_resp(id)?;

    if r.status != 0 {
        return None;
    }

    Some(r.arg0)
}
```

driverspacelib/src/driver.rs (line 6) - [public] DeviceInfo

```
pub struct DeviceInfo {
    pub device_id: u64,
    pub vendor_id: u64,
    pub class_code: u64,
}

/// Any driver that lives in driver space and is registered with the kernel
/// driver-space manager.
pub trait Driver {
    fn init(&mut self, info: &DeviceInfo) -> Result<(), DsError>;
}
```

driverspacelib/src/driver.rs (line 14) - [public] Driver

```
/// Any driver that lives in driver space and is registered with the kernel
/// driver-space manager.

pub trait Driver {
    fn init(&mut self, info: &DeviceInfo) -> Result<(), DsError>;
}
```

driverspacelib/src/driver.rs (line 15) - [internal] init

```
fn init(&mut self, info: &DeviceInfo) -> Result<(), DsError>;
}
```

driverspacelib/src/input.rs (line 4) - [public] key_req

```
pub fn key_req() -> u64 {
    svc::call(SVC_INPUT, IN_KEY_POLL, 0, 0, 0)
}

pub fn key_take(id: u64) -> Option<u8> {
    let r = svc::take(id)?;

    if r.arg0 == 0 { None } else { Some(r.arg0 as u8) }
}
```

driverspacelib/src/input.rs (line 8) - [public] key_take

```
pub fn key_take(id: u64) -> Option<u8> {
    let r = svc::take(id)?;

    if r.arg0 == 0 { None } else { Some(r.arg0 as u8) }
}
```

driverspacelib/src/jack.rs (line 4) - [public] JackInfo

```
pub struct JackInfo {
    pub present: bool,
    pub amp_on: bool,
}

pub fn query_req() -> u64 {
    request(DsCmd::JackQuery, 0, 0, 0)
}

pub fn query_take(id: u64) -> Option<JackInfo> {
    let r = take_resp(id)?;

    // ... (truncated, see source file for more)
}
```

driverspacelib/src/jack.rs (line 9) - [public] query_req

```
pub fn query_req() -> u64 {
    request(DsCmd::JackQuery, 0, 0, 0)
}

pub fn query_take(id: u64) -> Option<JackInfo> {
    let r = take_resp(id)?;

    Some(JackInfo {
        present: r.arg0 & 1 != 0,
        amp_on: r.arg0 & 2 != 0,
    })
}

// ... (truncated, see source file for more)
```

driverspacelib/src/jack.rs (line 13) - [public] query_take

```
pub fn query_take(id: u64) -> Option<JackInfo> {
    let r = take_resp(id)?;

    Some(JackInfo {
        present: r.arg0 & 1 != 0,
        amp_on: r.arg0 & 2 != 0,
    })
}

pub fn set_amp(on: bool) {
    request(DsCmd::JackSetAmp, on as u64, 0, 0);
}

// ... (truncated, see source file for more)
```


driverspacelib/src/jack.rs (line 22) - [public] set_amp

```
pub fn set_amp(on: bool) {
    request(DsCmd::JackSetAmp, on as u64, 0, 0);
}

pub fn play(va: u64, len: u32) -> u64 {
    request(DsCmd::AudioPlay, va, len as u64, 0)
}

pub fn stop() {
    request(DsCmd::AudioStop, 0, 0, 0);
}
```

driverspacelib/src/jack.rs (line 26) - [public] play

```
pub fn play(va: u64, len: u32) -> u64 {
    request(DsCmd::AudioPlay, va, len as u64, 0)
}

pub fn stop() {
    request(DsCmd::AudioStop, 0, 0, 0);
}
```

driverspacelib/src/jack.rs (line 30) - [public] stop

```
pub fn stop() {
    request(DsCmd::AudioStop, 0, 0, 0);
}
```

driverspacelib/src/log.rs (line 10) - [public] ds_log

/// Wysyła tekst do logu i dra (komenda `DsCmd::Log`).

```
pub fn ds_log(msg: &str) {
    let _ = request(DsCmd::Log, msg.as_ptr() as u64, msg.len() as u64, 0);
}

/// To samo, ale dla surowego wskaźnika i długości (np. bufora bajtów).
pub fn ds_log_raw(ptr: *const u8, len: usize) {
    let _ = request(DsCmd::Log, ptr as u64, len as u64, 0);
}
```

driverspacelib/src/log.rs (line 15) - [public] ds_log_raw

/// To samo, ale dla surowego wskaźnika i długości (np. bufora bajtów).

```
pub fn ds_log_raw(ptr: *const u8, len: usize) {
    let _ = request(DsCmd::Log, ptr as u64, len as u64, 0);
}
```

driverspacelib/src/runtime.rs (line 38) - [public] init_once

/// Reads the boot parameters block and points our ring views at the

/// kernel-provided buffers.

```
pub fn init_once(params_va: u64) {
    unsafe {
        let p = &*(params_va as *const DsInitParams);

        K2D = p.k2d_va as *mut DsRing;
        D2K = p.d2k_va as *mut DsRing;
        K2D_MSGS = (p.k2d_va + DS_RING_HDR_SIZE as u64) as *mut DsMsg;
        D2K_MSGS = (p.d2k_va + DS_RING_HDR_SIZE as u64) as *mut DsMsg;
    }
}

/// Registers a driver (by calling its `Driver::init` hook) and notifies
// ... (truncated, see source file for more)
```

driverspacelib/src/runtime.rs (line 51) - [public] register

*/// Registers a driver (by calling its `Driver::init` hook) and notifies
/// the kernel that this driver space is now managed.*

```
pub fn register<D: Driver>(drv: &mut D) {  
    let _ = drv.init(&DeviceInfo::default());  
    let _ = request(DsCmd::RegisterDriver, 0, 0, 0);  
}  
  
unsafe fn push_raw(cmd: u32, a0: u64, a1: u64, a2: u64) -> u64 {  
    let ring = D2K;  
    if ring.is_null() {  
        return 0;  
    }  
  
    let id = NEXT_ID;  
    // ... (truncated, see source file for more)
```

driverspacelib/src/runtime.rs (line 94) - [public] request

*/// Sends a driver-space command to the kernel. Returns the id used later
/// with [take_resp].*

```
pub fn request(cmd: DsCmd, a0: u64, a1: u64, a2: u64) -> u64 {  
    unsafe { push_raw(cmd as u32, a0, a1, a2) }  
}  
  
/// Sends a raw (service-class packed) command to the kernel.  
pub fn request_raw(cmd: u32, a0: u64, a1: u64, a2: u64) -> u64 {  
    unsafe { push_raw(cmd, a0, a1, a2) }  
}  
  
/// Drains pending kernel->driver responses into the local cache.  
pub fn tick() {  
    unsafe {  
        // ... (truncated, see source file for more)
```

driverspacelib/src/runtime.rs (line 99) - [public] request_raw

/// Sends a raw (service-class packed) command to the kernel.

```
pub fn request_raw(cmd: u32, a0: u64, a1: u64, a2: u64) -> u64 {  
    unsafe { push_raw(cmd, a0, a1, a2) }  
}  
  
/// Drains pending kernel->driver responses into the local cache.  
pub fn tick() {  
    unsafe {  
        let ring = K2D;  
        if ring.is_null() {  
            return;  
        }  
  
        // ... (truncated, see source file for more)
```

driverspacelib/src/runtime.rs (line 104) - [public] tick

/// Drains pending kernel->driver responses into the local cache.

```
pub fn tick() {  
    unsafe {  
        let ring = K2D;  
        if ring.is_null() {  
            return;  
        }  
  
        while (*ring).tail != (*ring).head {  
            let slot = K2D_MSGS.add((*ring).tail as usize);  
            let copy = *slot;  
            (*ring).tail = ((*ring).tail + 1) % (*ring).cap;  
        }  
  
        // ... (truncated, see source file for more)
```

driverspacelib/src/runtime.rs (line 128) - [public] take_resp

/// Returns (and consumes) the response with the given request id.

```
pub fn take_resp(id: u64) -> Option<DsMsg> {
    unsafe {
        for i in 0..RESP_CACHE {
            if CACHE_USED[i] && CACHE_MSG[i].id == id {
                let m = CACHE_MSG[i];
                CACHE_USED[i] = false;
                return Some(m);
            }
        }
    }
    None
}
```

// ... (truncated, see source file for more)

driverspacelib/src/svc.rs (line 4) - [public] call

```
pub fn call(class: u32, op: u32, a0: u64, a1: u64, a2: u64) -> u64 {
    request_raw(svc_cmd(class, op), a0, a1, a2)
}

pub fn take(id: u64) -> Option<DsMsg> {
    take_resp(id)
}
```

driverspacelib/src/svc.rs (line 8) - [public] take

```
pub fn take(id: u64) -> Option<DsMsg> {
    take_resp(id)
}
```

driverspacelib/src/video.rs (line 4) - [public] FbInfo

```
pub struct FbInfo {
    pub width: u32,
    pub height: u32,
    pub stride: u32,
    pub phys: u64,
}

pub fn fb_info_req() -> u64 {
    svc::call(SVC_VIDEO, VID_FB_INFO, 0, 0, 0)
}

pub fn fb_info_take(id: u64) -> Option<FbInfo> {
    // ... (truncated, see source file for more)
}
```

driverspacelib/src/video.rs (line 11) - [public] fb_info_req

```
pub fn fb_info_req() -> u64 {
    svc::call(SVC_VIDEO, VID_FB_INFO, 0, 0, 0)
}

pub fn fb_info_take(id: u64) -> Option<FbInfo> {
    let r = svc::take(id)?;

    Some(FbInfo {
        width: (r.arg0 >> 16) as u32,
        height: (r.arg0 & 0xFFFF) as u32,
        stride: r.arg1 as u32,
        phys: r.arg2,
    })
    // ... (truncated, see source file for more)
}
```

driverspacelib/src/video.rs (line 15) - [public] fb_info_take

```
pub fn fb_info_take(id: u64) -> Option<FbInfo> {
    let r = svc::take(id)?;

    Some(FbInfo {
        width: (r.arg0 >> 16) as u32,
        height: (r.arg0 & 0xFFFF) as u32,
        stride: r.arg1 as u32,
        phys: r.arg2,
    })
}

pub fn takeover() -> u64 {
    // ... (truncated, see source file for more)
```

driverspacelib/src/video.rs (line 26) - [public] takeover

```
pub fn takeover() -> u64 {
    svc::call(SVC_VIDEO, VID_FB_TAKEOVER, 0, 0, 0)
}

pub fn release() -> u64 {
    svc::call(SVC_VIDEO, VID_FB_RELEASE, 0, 0, 0)
}
```

driverspacelib/src/video.rs (line 30) - [public] release

```
pub fn release() -> u64 {
    svc::call(SVC_VIDEO, VID_FB_RELEASE, 0, 0, 0)
}
```

gfx_protocol_workspace/crates/gfx-api/src/lib.rs (line 6) - [public] GfxContext

```
pub struct GfxContext {
    server_endpoint: u64,
}

impl GfxContext {
    pub fn init() -> Self {
        Self { server_endpoint: 0x8000 }
    }

    pub fn create_surface(&self, width: u32, height: u32) -> Option<SurfaceId> {
        let config = SurfaceConfig {
            width,
            height,
            format: PixelFormat::XRGB8888,
            flags: gfx_protocol::surfaces::SURFACE_FLAG_VISIBLE,
        };
        // ... (truncated, see source file for more)
```

gfx_protocol_workspace/crates/gfx-api/src/lib.rs (line 11) - [public] init

```
pub fn init() -> Self {
    Self { server_endpoint: 0x8000 }
}

pub fn create_surface(&self, width: u32, height: u32) -> Option<SurfaceId> {
    let config = SurfaceConfig {
        width,
        height,
        format: PixelFormat::XRGB8888,
        flags: gfx_protocol::surfaces::SURFACE_FLAG_VISIBLE,
    };
    // ... (truncated, see source file for more)
```

gfx_protocol_workspace/crates/gfx-api/src/lib.rs (line 15) - [public] create_surface

```
pub fn create_surface(&self, width: u32, height: u32) -> Option<SurfaceId> {
    let config = SurfaceConfig {
        width,
        height,
        format: PixelFormat::XRGB8888,
        flags: gfx_protocol::surfaces::SURFACE_FLAG_VISIBLE,
    };

    let mut msg = GfxIpcMsg::default();
    msg.cmd = GfxCmd::CreateSurface as u32;
    msg.arg0 = &config as *const _ as u64;

    // ... (truncated, see source file for more)
```

gfx_protocol_workspace/crates/gfx-api/src/lib.rs (line 35) - [public] commit

```
pub fn commit(&self, surface: SurfaceId) {
    let mut msg = GfxIpcMsg::default();
    msg.cmd = GfxCmd::Commit as u32;
    msg.arg0 = surface.0 as u64;
    self.send_ipc(msg);
}

fn send_ipc(&self, msg: GfxIpcMsg) -> GfxIpcMsg {
    // Stub: Wywołanie syscalla jdra (np. int 0x80 lub SVC)
    // przekazujące wiadomość do gfx-server
    GfxIpcMsg::default()
}

// ... (truncated, see source file for more)
```

gfx_protocol_workspace/crates/gfx-api/src/lib.rs (line 42) - [internal] send_ipc

```
fn send_ipc(&self, msg: GfxIpcMsg) -> GfxIpcMsg {
    // Stub: Wywołanie syscalla jdra (np. int 0x80 lub SVC)
    // przekazujące wiadomość do gfx-server
    GfxIpcMsg::default()
}

}
```

gfx_protocol_workspace/crates/gfx-server/src/compositor.rs (line 4) - [internal] Surface

```
struct Surface {
    id: u32,
    x: i32,
    y: i32,
    width: u32,
    height: u32,
    z_order: i32,
    buffer: *mut u32,
}

pub struct DisplayServer {
    pub width: u32,
    // ... (truncated, see source file for more)
```

gfx_protocol_workspace/crates/gfx-server/src/compositor.rs (line 14) - [public] DisplayServer

```
pub struct DisplayServer {
    pub width: u32,
    pub height: u32,
    fb: *mut u32,
    surfaces: [Option<Surface>; 64],
    next_id: u32,
}

impl DisplayServer {
    pub fn new(width: u32, height: u32) -> Self {
        let fb_size = (width * height * 4) as usize;
        let layout = core::alloc::Layout::from_size_align(fb_size, 4096).unwrap();
        // ... (truncated, see source file for more)
```

gfx_protocol_workspace/crates/gfx-server/src/compositor.rs (line 23) - [public] new

```
pub fn new(width: u32, height: u32) -> Self {
    let fb_size = (width * height * 4) as usize;
    let layout = core::alloc::Layout::from_size_align(fb_size, 4096).unwrap();
    let fb = unsafe { kstd_alloc::KernelHeap.alloc(layout) } as *mut u32;

    Self { width, height, fb, surfaces: core::array::from_fn(|_| None), next_id: 1 }
}

pub fn create_surface(&mut self, width: u32, height: u32) -> u32 {
    let id = self.next_id;
    self.next_id += 1;

    // ... (truncated, see source file for more)
```

gfx_protocol_workspace/crates/gfx-server/src/compositor.rs (line 31) - [public] create_surface

```
pub fn create_surface(&mut self, width: u32, height: u32) -> u32 {
    let id = self.next_id;
    self.next_id += 1;

    let buf_size = (width * height * 4) as usize;
    let layout = core::alloc::Layout::from_size_align(buf_size, 4096).unwrap();
    let buffer = unsafe { kstd_alloc::KernelHeap.alloc(layout) } as *mut u32;

    for slot in self.surfaces.iter_mut() {
        if slot.is_none() {
            *slot = Some(Surface { id, x: 0, y: 0, width, height, z_order: 0, buffer });
            break;
        }
    }
    // ... (truncated, see source file for more)
```

gfx_protocol_workspace/crates/gfx-server/src/compositor.rs (line 48) - [public] set_position

```
pub fn set_position(&mut self, id: u32, x: i32, y: i32) {
    for slot in self.surfaces.iter_mut() {
        if let Some(s) = slot {
            if s.id == id { s.x = x; s.y = y; break; }
        }
    }
}

pub fn commit(&mut self, _id: u32) {}

pub fn composite(&mut self) {
    unsafe {
        // ... (truncated, see source file for more)
```

gfx_protocol_workspace/crates/gfx-server/src/compositor.rs (line 56) - [public] commit

```
pub fn commit(&mut self, _id: u32) {}

pub fn composite(&mut self) {
    unsafe {
        for i in 0..(self.width * self.height) {
            self.fb.add(i as usize).write_volatile(0xFF101020);
        }
    }

    for slot in self.surfaces.iter() {
        if let Some(s) = slot {
            if s.buffer.is_null() { continue; }
        }
    }
    // ... (truncated, see source file for more)
```

gfx_protocol_workspace/crates/gfx-server/src/compositor.rs (line 58) - [public] composite

```
pub fn composite(&mut self) {
    unsafe {
        for i in 0..(self.width * self.height) {
            self.fb.add(i as usize).write_volatile(0xFF101020);
        }

        for slot in self-surfaces.iter() {
            if let Some(s) = slot {
                if s.buffer.is_null() { continue; }
                unsafe {
                    for row in 0..s.height {
                        // ... (truncated, see source file for more)
                    }
                }
            }
        }
    }
}
```

gfx_protocol_workspace/crates/gfx-server/src/compositor.rs (line 84) - [public] framebuffer_ptr

```
pub fn framebuffer_ptr(&self) -> *const u32 { self.fb }
pub fn fb_size(&self) -> usize { (self.width * self.height * 4) as usize }
}
```

gfx_protocol_workspace/crates/gfx-server/src/compositor.rs (line 85) - [public] fb_size

```
pub fn fb_size(&self) -> usize { (self.width * self.height * 4) as usize }
}
```

gfx_protocol_workspace/crates/gfx-server/src/gpu_bridge.rs (line 5) - [public] GpuBridge

```
pub struct GpuBridge {
    fb_phys: u64,
    fb_virt: u64,
    fb_size: usize,
    width: u32,
    height: u32,
}

impl GpuBridge {
    pub fn new() -> Self {
        Self { fb_phys: 0, fb_virt: 0, fb_size: 0, width: 0, height: 0 }
    }
    // ... (truncated, see source file for more)
}
```

gfx_protocol_workspace/crates/gfx-server/src/gpu_bridge.rs (line 14) - [public] new

```
pub fn new() -> Self {
    Self { fb_phys: 0, fb_virt: 0, fb_size: 0, width: 0, height: 0 }
}

pub fn init(&mut self) -> bool {
    ds_info!("GPU-Bridge", "Requesting framebuffer info from vgpu via DS IPC");

    let reply = kapi_syscall::sys_ipc_call(DsCmd::VideoFbInfo, 0, 0, 0);
    if !reply.is_ok() {
        ds_error!("GPU-Bridge", "Failed to get FB info from vgpu");
        return false;
    }
    // ... (truncated, see source file for more)
}
```

gfx_protocol_workspace/crates/gfx-server/src/gpu_bridge.rs (line 18) - [public] init

```
pub fn init(&mut self) -> bool {
    ds_info!("GPU-Bridge", "Requesting framebuffer info from vgpu via DS IPC");

    let reply = kapi_syscall::sys_ipc_call(DsCmd::VideoFbInfo, 0, 0, 0);
    if !reply.is_ok() {
        ds_error!("GPU-Bridge", "Failed to get FB info from vgpu");
        return false;
    }

    self.fb_phys = reply.arg0;
    self.width = reply.arg1 as u32;
    self.height = reply.arg2 as u32;
    // ... (truncated, see source file for more)
}
```

gfx_protocol_workspace/crates/gfx-server/src/gpu_bridge.rs (line 47) - [public] present

```
pub fn present(&self, src_buffer: *const u32, size: usize) {
    if self.fb_virt == 0 { return; }
    unsafe {
        core::ptr::copy_nonoverlapping(
            src_buffer,
            self.fb_virt as *mut u32,
            size / 4,
        );
    }
}
```

gfx_protocol_workspace/crates/gfx-server/src/main.rs (line 49) - [internal] wait_for_ipc

```
fn wait_for_ipc() -> GfxIpcMsg {
    // Block on kernel IPC ring (k2d) for messages from user-space apps
    GfxIpcMsg::default()
}

fn reply(cmd: u32, status: i32, data: u64) {
    // Push reply to kernel IPC ring (d2k)
}
```

gfx_protocol_workspace/crates/gfx-server/src/main.rs (line 54) - [internal] reply

```
fn reply(cmd: u32, status: i32, data: u64) {
    // Push reply to kernel IPC ring (d2k)
}
```

gfx_protocol_workspace/lib/gfx-protocol/src/commands.rs (line 7) - [public] GfxCmd

```
pub enum GfxCmd {
    CreateSurface = 0x01,
    DestroySurface = 0x02,
    ResizeSurface = 0x03,

    AllocBuffer = 0x10,
    AttachBuffer = 0x11,
    Commit = 0x12,

    SetPosition = 0x20,
    SetZOrder = 0x21,

    // ... (truncated, see source file for more)
```

gfx_protocol_workspace/lib/gfx-protocol/src/commands.rs (line 24) - [public] GfxIpcMsg

```
pub struct GfxIpcMsg {
    pub cmd: u32,
    pub status: i32,
    pub arg0: u64,
    pub arg1: u64,
    pub arg2: u64,
}
```

gfx_protocol_workspace/lib/gfx-protocol/src/surfaces.rs (line 5) - [public] SurfaceId

```
pub struct SurfaceId(pub u32);

#[derive(Debug, Clone, Copy, PartialEq, Eq)]
#[repr(transparent)]
pub struct BufferId(pub u32);

#[derive(Debug, Clone, Copy, PartialEq, Eq)]
#[repr(u32)]
pub enum PixelFormat {
    XRGB8888 = 0,
    ARGB8888 = 1,
}

// ... (truncated, see source file for more)
```


gfx_protocol_workspace/lib/gfx-protocol/src/surfaces.rs (line 9) - [public] BufferId

```
pub struct BufferId(pub u32);

#[derive(Debug, Clone, Copy, PartialEq, Eq)]
#[repr(u32)]
pub enum PixelFormat {
    XRGB8888 = 0,
    ARGB8888 = 1,
}

#[derive(Debug, Clone, Copy)]
#[repr(C)]
pub struct SurfaceConfig {
    // ... (truncated, see source file for more)
```

gfx_protocol_workspace/lib/gfx-protocol/src/surfaces.rs (line 13) - [public] PixelFormat

```
pub enum PixelFormat {
    XRGB8888 = 0,
    ARGB8888 = 1,
}

#[derive(Debug, Clone, Copy)]
#[repr(C)]
pub struct SurfaceConfig {
    pub width: u32,
    pub height: u32,
    pub format: PixelFormat,
    pub flags: u32,
    // ... (truncated, see source file for more)
```

gfx_protocol_workspace/lib/gfx-protocol/src/surfaces.rs (line 20) - [public] SurfaceConfig

```
pub struct SurfaceConfig {
    pub width: u32,
    pub height: u32,
    pub format: PixelFormat,
    pub flags: u32,
}

pub const SURFACE_FLAG_VISIBLE: u32 = 1 << 0;
pub const SURFACE_FLAG_FULLSCREEN: u32 = 1 << 1;
```

kernel-drivers/audio/src/lib.rs (line 9) - [internal] ad_init

```
fn ad_init(base: u32) -> bool;
fn ad_jack_present() -> bool;
fn ad_set_amp(on: bool);
fn ad_play_phys(phys: u64, len: u32) -> bool;
fn ad_stop();
}

static mut AMP_ON: bool = false;
static mut PRESENT: bool = false;

pub fn init(base: u32) -> bool {
    unsafe { ad_init(base) }
    // ... (truncated, see source file for more)
```

kernel-drivers/audio/src/lib.rs (line 10) - [internal] ad_jack_present

```
fn ad_jack_present() -> bool;
fn ad_set_amp(on: bool);
fn ad_play_phys(phys: u64, len: u32) -> bool;
fn ad_stop();
}

static mut AMP_ON: bool = false;
static mut PRESENT: bool = false;

pub fn init(base: u32) -> bool {
    unsafe { ad_init(base) }
}

// ... (truncated, see source file for more)
```

kernel-drivers/audio/src/lib.rs (line 11) - [internal] ad_set_amp

```
fn ad_set_amp(on: bool);
fn ad_play_phys(phys: u64, len: u32) -> bool;
fn ad_stop();
}

static mut AMP_ON: bool = false;
static mut PRESENT: bool = false;

pub fn init(base: u32) -> bool {
    unsafe { ad_init(base) }
}

// ... (truncated, see source file for more)
```

kernel-drivers/audio/src/lib.rs (line 12) - [internal] ad_play_phys

```
fn ad_play_phys(phys: u64, len: u32) -> bool;
fn ad_stop();
}

static mut AMP_ON: bool = false;
static mut PRESENT: bool = false;

pub fn init(base: u32) -> bool {
    unsafe { ad_init(base) }
}

pub fn poll_jack() -> bool {
    // ... (truncated, see source file for more)
```

kernel-drivers/audio/src/lib.rs (line 13) - [internal] ad_stop

```
fn ad_stop();
}

static mut AMP_ON: bool = false;
static mut PRESENT: bool = false;

pub fn init(base: u32) -> bool {
    unsafe { ad_init(base) }
}

pub fn poll_jack() -> bool {
    let now = unsafe { ad_jack_present() };
    // ... (truncated, see source file for more)
```

kernel-drivers/audio/src/lib.rs (line 19) - [public] init

```
pub fn init(base: u32) -> bool {
    unsafe { ad_init(base) }
}

pub fn poll_jack() -> bool {
    let now = unsafe { ad_jack_present() };

    unsafe {
        if now != PRESENT {
            PRESENT = now;
            return true;
        }
    }
    // ... (truncated, see source file for more)
```

kernel-drivers/audio/src/lib.rs (line 23) - [public] poll_jack

```
pub fn poll_jack() -> bool {
    let now = unsafe { ad_jack_present() };

    unsafe {
        if now != PRESENT {
            PRESENT = now;
            return true;
        }
    }

    false
}

// ... (truncated, see source file for more)
```

kernel-drivers/audio/src/lib.rs (line 36) - [public] query

```
pub fn query() -> u32 {
    unsafe { (PRESENT as u32) | ((AMP_ON as u32) << 1) }
}

pub fn set_amp(on: bool) {
    unsafe { AMP_ON = on; }
    unsafe { ad_set_amp(on); }
}

pub fn play_phys(phys: u64, len: u32) -> bool {
    unsafe { ad_play_phys(phys, len) }
}

// ... (truncated, see source file for more)
```

kernel-drivers/audio/src/lib.rs (line 40) - [public] set_amp

```
pub fn set_amp(on: bool) {
    unsafe { AMP_ON = on; }
    unsafe { ad_set_amp(on); }
}

pub fn play_phys(phys: u64, len: u32) -> bool {
    unsafe { ad_play_phys(phys, len) }
}

pub fn stop() {
    unsafe { ad_stop(); }
}
```

kernel-drivers/audio/src/lib.rs (line 45) - [public] play_phys

```
pub fn play_phys(phys: u64, len: u32) -> bool {
    unsafe { ad_play_phys(phys, len) }
}

pub fn stop() {
    unsafe { ad_stop(); }
}
```

kernel-drivers/audio/src/lib.rs (line 49) - [public] stop

```
pub fn stop() {
    unsafe { ad_stop(); }
}
```

kernel-drivers/battery/src/lib.rs (line 6) - [internal] battery_init

```
fn battery_init() -> bool;
}
```

```
pub fn init() -> bool {
    unsafe { battery_init() }
}
```

kernel-drivers/battery/src/lib.rs (line 9) - [public] init

```
pub fn init() -> bool {
    unsafe { battery_init() }
}
```

kernel-drivers/bluetooth/src/lib.rs (line 7) - [internal] bt_init

```
    fn bt_init() -> bool;
    fn bt_ready() -> bool;
}

pub fn init() -> bool {
    unsafe { bt_init() }
}

pub fn ready() -> bool {
    unsafe { bt_ready() }
}
```

kernel-drivers/bluetooth/src/lib.rs (line 8) - [internal] bt_ready

```
    fn bt_ready() -> bool;
}

pub fn init() -> bool {
    unsafe { bt_init() }
}

pub fn ready() -> bool {
    unsafe { bt_ready() }
}
```

kernel-drivers/bluetooth/src/lib.rs (line 11) - [public] init

```
pub fn init() -> bool {
    unsafe { bt_init() }
}

pub fn ready() -> bool {
    unsafe { bt_ready() }
}
```

kernel-drivers/bluetooth/src/lib.rs (line 15) - [public] ready

```
pub fn ready() -> bool {
    unsafe { bt_ready() }
}
```

kernel-drivers/camera/src/lib.rs (line 6) - [internal] camera_init

```
    fn camera_init() -> bool;
}

pub fn init() -> bool {
    unsafe { camera_init() }
}
```

kernel-drivers/camera/src/lib.rs (line 9) - [public] init

```
pub fn init() -> bool {
    unsafe { camera_init() }
}
```

kernel-drivers/displayport/src/lib.rs (line 6) - [internal] dp_init

```
    fn dp_init() -> bool;
    fn dp_ready() -> bool;
}

pub fn init() -> bool {
    unsafe { dp_init() }
}

pub fn ready() -> bool {
    unsafe { dp_ready() }
}
```

kernel-drivers/displayport/src/lib.rs (line 7) - [internal] dp_ready

```
    fn dp_ready() -> bool;
}

pub fn init() -> bool {
    unsafe { dp_init() }
}

pub fn ready() -> bool {
    unsafe { dp_ready() }
}
```

kernel-drivers/displayport/src/lib.rs (line 10) - [public] init

```
pub fn init() -> bool {
    unsafe { dp_init() }
}

pub fn ready() -> bool {
    unsafe { dp_ready() }
}
```

kernel-drivers/displayport/src/lib.rs (line 14) - [public] ready

```
pub fn ready() -> bool {
    unsafe { dp_ready() }
}
```

kernel-drivers/gfx/src/lib.rs (line 9) - [public] FramebufferInfo

```
pub struct FramebufferInfo {
    pub width: u32,
    pub height: u32,
    pub stride: u32,
    pub phys: u64,
}

pub fn fb_info() -> Option<FramebufferInfo> {
    let (mut w, mut h, mut s, mut p) = (0u32, 0u32, 0u32, 0u64);

    unsafe {
        if hw_sys::fb_info(&mut w, &mut h, &mut s, &mut p) {
            // ... (truncated, see source file for more)
        }
    }
}
```

kernel-drivers/gfx/src/lib.rs (line 16) - [public] fb_info

```
pub fn fb_info() -> Option<FramebufferInfo> {
    let (mut w, mut h, mut s, mut p) = (0u32, 0u32, 0u32, 0u64);

    unsafe {
        if hw_sys::fb_info(&mut w, &mut h, &mut s, &mut p) {
            Some(FramebufferInfo { width: w, height: h, stride: s, phys: p })
        } else {
            None
        }
    }
}
```

kernel-drivers/hdmi/src/lib.rs (line 6) - [internal] hdmi_init_with

```
fn hdmi_init_with(fb_phys: u64, w: u32, h: u32, stride: u32) -> bool;
fn hdmi_ready() -> bool;
}

pub fn init() -> bool {
    let (mut w, mut h, mut stride, mut phys) = (0u32, 0u32, 0u32, 0u64);

    unsafe {
        if !hw_sys::fb_info(&mut w, &mut h, &mut stride, &mut phys) {
            return false;
        }
    }

    // ... (truncated, see source file for more)
}
```

```

kernel-drivers/hdmi/src/lib.rs (line 7) - [internal] hdmi_ready
    fn hdmi_ready() -> bool;
}

pub fn init() -> bool {
    let (mut w, mut h, mut stride, mut phys) = (0u32, 0u32, 0u32, 0u64);

    unsafe {
        if !hw_sys::fb_info(&mut w, &mut h, &mut stride, &mut phys) {
            return false;
        }

        hdmi_init_with(phys, w, h, stride)
        // ... (truncated, see source file for more)
    }
}

kernel-drivers/hdmi/src/lib.rs (line 10) - [public] init
pub fn init() -> bool {
    let (mut w, mut h, mut stride, mut phys) = (0u32, 0u32, 0u32, 0u64);

    unsafe {
        if !hw_sys::fb_info(&mut w, &mut h, &mut stride, &mut phys) {
            return false;
        }

        hdmi_init_with(phys, w, h, stride)
    }
}

// ... (truncated, see source file for more)

kernel-drivers/hdmi/src/lib.rs (line 22) - [public] ready
pub fn ready() -> bool {
    unsafe { hdmi_ready() }
}

kernel-drivers/hw-sys/src/lib.rs (line 13) - [public] kprintf
/// printf-style logging (mirrors the kernel `kprintf`).
pub fn kprintf(fmt: *const u8, ...);

/// Allocate a DMA-coherent buffer; returns phys + virt.
pub fn dma_alloc_coherent(
    bytes: usize,
    zone_max: u64,
    out_phys: *mut u64,
    out_virt: *mut *mut u8,
) -> bool;

pub fn dma_free_coherent(phys: u64, virt: *mut u8, bytes: usize);

// ... (truncated, see source file for more)

kernel-drivers/hw-sys/src/lib.rs (line 16) - [public] dma_alloc_coherent
/// Allocate a DMA-coherent buffer; returns phys + virt.
pub fn dma_alloc_coherent(
    bytes: usize,
    zone_max: u64,
    out_phys: *mut u64,
    out_virt: *mut *mut u8,
) -> bool;

pub fn dma_free_coherent(phys: u64, virt: *mut u8, bytes: usize);

/// Map device MMIO into the higher half; returns a virtual address.
pub fn mmio_map(phys: u64, len: usize, out_virt: *mut u64) -> bool;

// ... (truncated, see source file for more)

```

```

kernel-drivers/hw-sys/src/lib.rs (line 23) - [public] dma_free_coherent
    pub fn dma_free_coherent(phys: u64, virt: *mut u8, bytes: usize);

    /// Map device MMIO into the higher half; returns a virtual address.
    pub fn mmio_map(phys: u64, len: usize, out_virt: *mut u64) -> bool;

    pub fn mmio_unmap(virt: u64, len: usize);

    /// Is `virt` actually present in the current page table?
    pub fn paging_is_mapped(virt: u64) -> bool;

    /// Query the primary framebuffer (used by `hdmi`/`gfx`).
    pub fn fb_info(out_w: *mut u32, out_h: *mut u32, out_stride: *mut u32, out_phys: *mut u64) -> bool;
    // ... (truncated, see source file for more)

kernel-drivers/hw-sys/src/lib.rs (line 26) - [public] mmio_map
    /// Map device MMIO into the higher half; returns a virtual address.

    pub fn mmio_map(phys: u64, len: usize, out_virt: *mut u64) -> bool;

    pub fn mmio_unmap(virt: u64, len: usize);

    /// Is `virt` actually present in the current page table?
    pub fn paging_is_mapped(virt: u64) -> bool;

    /// Query the primary framebuffer (used by `hdmi`/`gfx`).
    pub fn fb_info(out_w: *mut u32, out_h: *mut u32, out_stride: *mut u32, out_phys: *mut u64) -> bool;
}

pub mod log {
    // ... (truncated, see source file for more)

kernel-drivers/hw-sys/src/lib.rs (line 28) - [public] mmio_unmap
    pub fn mmio_unmap(virt: u64, len: usize);

    /// Is `virt` actually present in the current page table?
    pub fn paging_is_mapped(virt: u64) -> bool;

    /// Query the primary framebuffer (used by `hdmi`/`gfx`).
    pub fn fb_info(out_w: *mut u32, out_h: *mut u32, out_stride: *mut u32, out_phys: *mut u64) -> bool;
}

pub mod log {
    /// Print a plain (already formatted) message, `%s`-style.
    pub fn info(msg: &str) {
        // ... (truncated, see source file for more)

kernel-drivers/hw-sys/src/lib.rs (line 31) - [public] paging_is_mapped
    /// Is `virt` actually present in the current page table?

    pub fn paging_is_mapped(virt: u64) -> bool;

    /// Query the primary framebuffer (used by `hdmi`/`gfx`).
    pub fn fb_info(out_w: *mut u32, out_h: *mut u32, out_stride: *mut u32, out_phys: *mut u64) -> bool;
}

pub mod log {
    /// Print a plain (already formatted) message, `%s`-style.
    pub fn info(msg: &str) {
        let mut buf = [0u8; 256];
        let len = msg.len().min(255);
        buf[..len].copy_from_slice(&msg.as_bytes()[..len]);
        // ... (truncated, see source file for more)

```

kernel-drivers/hw-sys/src/lib.rs (line 34) - [public] fb_info

/// Query the primary framebuffer (used by `hdmî/gfx`).

```
pub fn fb_info(out_w: *mut u32, out_h: *mut u32, out_stride: *mut u32, out_phys: *mut u64) -> bool;
}

pub mod log {
    /// Print a plain (already formatted) message, `%s`-style.
    pub fn info(msg: &str) {
        let mut buf = [0u8; 256];
        let len = msg.len().min(255);
        buf[..len].copy_from_slice(&msg.as_bytes()[..len]);
        buf[len] = 0;
        unsafe { super::kprintf(b"%s\n\0".as_ptr(), buf.as_ptr()); }
    }
    // ... (truncated, see source file for more)
}
```

kernel-drivers/hw-sys/src/lib.rs (line 39) - [public] info

/// Print a plain (already formatted) message, `%s`-style.

```
pub fn info(msg: &str) {
    let mut buf = [0u8; 256];
    let len = msg.len().min(255);
    buf[..len].copy_from_slice(&msg.as_bytes()[..len]);
    buf[len] = 0;
    unsafe { super::kprintf(b"%s\n\0".as_ptr(), buf.as_ptr()); }
}
}
```

kernel-drivers/pcie/src/lib.rs (line 20) - [public] PciDev

```
pub struct PciDev {
    pub bus: u8,
    pub dev: u8,
    pub func: u8,
}

/// A fully-probed device (as returned by [`enumerate`]).
#[derive(Debug, Clone, Copy)]
pub struct PciDevice {
    pub address: PciDev,
    pub vendor_id: u16,
    pub device_id: u16,
    // ... (truncated, see source file for more)
}
```

kernel-drivers/pcie/src/lib.rs (line 28) - [public] PciDevice

```
pub struct PciDevice {
    pub address: PciDev,
    pub vendor_id: u16,
    pub device_id: u16,
    pub class_code: u8,
    pub subclass: u8,
    pub prog_if: u8,
    pub header_type: u8,
}

// -----
// Port I/O (config mechanism #1)
// ... (truncated, see source file for more)
```



```

kernel-drivers/pcie/src/lib.rs (line 54) - [internal] cfg_addr
    fn cfg_addr(bus: u32, dev: u32, func: u32, off: u32) -> u32 {
        0x8000_0000 | (bus << 16) | (dev << 11) | (func << 8) | (off & 0xFC)
    }

    // -----
    // Config-space access
    // -----

    pub fn read32(d: PciDev, off: u32) -> u32 {
        unsafe {
            outl(0xCF8, cfg_addr(d.bus as u32, d.dev as u32, d.func as u32, off));
            inl(0xCFC)
        }
        // ... (truncated, see source file for more)
    }

kernel-drivers/pcie/src/lib.rs (line 62) - [public] read32
    // -----
    // Config-space access
    // -----

    pub fn read32(d: PciDev, off: u32) -> u32 {
        unsafe {
            outl(0xCF8, cfg_addr(d.bus as u32, d.dev as u32, d.func as u32, off));
            inl(0xCFC)
        }
    }

    pub fn write32(d: PciDev, off: u32, v: u32) {
        unsafe {
            outl(0xCF8, cfg_addr(d.bus as u32, d.dev as u32, d.func as u32, off));
            outl(0xCFC, v);
        }
        // ... (truncated, see source file for more)
    }

kernel-drivers/pcie/src/lib.rs (line 69) - [public] write32
    pub fn write32(d: PciDev, off: u32, v: u32) {
        unsafe {
            outl(0xCF8, cfg_addr(d.bus as u32, d.dev as u32, d.func as u32, off));
            outl(0xCFC, v);
        }
    }

    pub fn read16(d: PciDev, off: u32) -> u16 {
        (read32(d, off) >> ((off & 2) * 8)) as u16
    }

    pub fn read8(d: PciDev, off: u32) -> u8 {
        // ... (truncated, see source file for more)
    }

kernel-drivers/pcie/src/lib.rs (line 76) - [public] read16
    pub fn read16(d: PciDev, off: u32) -> u16 {
        (read32(d, off) >> ((off & 2) * 8)) as u16
    }

    pub fn read8(d: PciDev, off: u32) -> u8 {
        (read32(d, off) >> ((off & 3) * 8)) as u8
    }

    impl PciDev {
        pub fn vendor(self) -> u16 {
            read16(self, 0x00)
        }
        // ... (truncated, see source file for more)
    }

```

kernel-drivers/pcie/src/lib.rs (line 80) - [public] read8

```
pub fn read8(d: PciDev, off: u32) -> u8 {
    (read32(d, off) >> ((off & 3) * 8)) as u8
}
```

```
impl PciDev {
    pub fn vendor(self) -> u16 {
        read16(self, 0x00)
    }

    pub fn device_id(self) -> u16 {
        read16(self, 0x02)
    }
    // ... (truncated, see source file for more)
```

kernel-drivers/pcie/src/lib.rs (line 85) - [public] vendor

```
pub fn vendor(self) -> u16 {
    read16(self, 0x00)
}

pub fn device_id(self) -> u16 {
    read16(self, 0x02)
}

pub fn class(self) -> u8 {
    (read32(self, 0x08) >> 24) as u8
}

// ... (truncated, see source file for more)
```

kernel-drivers/pcie/src/lib.rs (line 89) - [public] device_id

```
pub fn device_id(self) -> u16 {
    read16(self, 0x02)
}

pub fn class(self) -> u8 {
    (read32(self, 0x08) >> 24) as u8
}

pub fn subclass(self) -> u8 {
    (read32(self, 0x08) >> 16) as u8
}

// ... (truncated, see source file for more)
```

kernel-drivers/pcie/src/lib.rs (line 93) - [public] class

```
pub fn class(self) -> u8 {
    (read32(self, 0x08) >> 24) as u8
}

pub fn subclass(self) -> u8 {
    (read32(self, 0x08) >> 16) as u8
}

pub fn prog_if(self) -> u8 {
    (read32(self, 0x08) >> 8) as u8
}

// ... (truncated, see source file for more)
```

kernel-drivers/pcie/src/lib.rs (line 97) - [public] subclass

```
pub fn subclass(self) -> u8 {
    (read32(self, 0x08) >> 16) as u8
}

pub fn prog_if(self) -> u8 {
    (read32(self, 0x08) >> 8) as u8
}

/// Read BAR `idx`, handling both 32-bit and 64-bit BARs.
pub fn bar(self, idx: u32) -> u64 {
    let lo = read32(self, 0x10 + idx * 4) as u64;
    if lo & 0x04 != 0 {
        // ... (truncated, see source file for more)
    }
}
```

kernel-drivers/pcie/src/lib.rs (line 101) - [public] prog_if

```
pub fn prog_if(self) -> u8 {
    (read32(self, 0x08) >> 8) as u8
}

/// Read BAR `idx`, handling both 32-bit and 64-bit BARs.
pub fn bar(self, idx: u32) -> u64 {
    let lo = read32(self, 0x10 + idx * 4) as u64;
    if lo & 0x04 != 0 {
        let hi = read32(self, 0x10 + (idx + 1) * 4) as u64;
        (hi << 32) | (lo & 0xFFFF_FFF0)
    } else {
        lo & 0xFFFF_FFF0
    }
    // ... (truncated, see source file for more)
}
```

kernel-drivers/pcie/src/lib.rs (line 106) - [public] bar

/// Read BAR `idx`, handling both 32-bit and 64-bit BARs.

```
pub fn bar(self, idx: u32) -> u64 {
    let lo = read32(self, 0x10 + idx * 4) as u64;
    if lo & 0x04 != 0 {
        let hi = read32(self, 0x10 + (idx + 1) * 4) as u64;
        (hi << 32) | (lo & 0xFFFF_FFF0)
    } else {
        lo & 0xFFFF_FFF0
    }
}

/// Enable memory-space access and bus mastering.
pub fn enable_mmio(self) {
    // ... (truncated, see source file for more)
}
```

kernel-drivers/pcie/src/lib.rs (line 117) - [public] enable_mmio

/// Enable memory-space access and bus mastering.

```
pub fn enable_mmio(self) {
    let cmd = read16(self, 0x04) as u32;
    write32(self, 0x04, cmd | 0x0006);
}

/// Find the first device matching class/subclass/prog-if.
pub fn find_class(class: u8, subclass: u8, prog_if: u8) -> Option<PciDev> {
    for bus in 0u32..256 {
        for dev in 0u32..32 {
            let mut func = 0u32;
            let d = PciDev { bus: bus as u8, dev: dev as u8, func: 0 };
            // ... (truncated, see source file for more)
        }
    }
}
```

kernel-drivers/pcie/src/lib.rs (line 124) - [public] find_class

/// Find the first device matching class/subclass/prog-if.

```
pub fn find_class(class: u8, subclass: u8, prog_if: u8) -> Option<PciDev> {
    for bus in 0u32..256 {
        for dev in 0u32..32 {
            let mut func = 0u32;
            let d = PciDev { bus: bus as u8, dev: dev as u8, func: 0 };
            if d.vendor() == 0xFFFF {
                continue;
            }
            let multifunction = (read8(d, 0x0E) & 0x80) != 0;
            while func < 8 {
                let f = PciDev { bus: bus as u8, dev: dev as u8, func: func as u8 };
                if f.vendor() != 0xFFFF
                // ... (truncated, see source file for more)
```

kernel-drivers/pcie/src/lib.rs (line 152) - [internal] probe

```
fn probe(bus: u8, dev: u8, func: u8) -> Option<PciDevice> {
    let address = PciDev { bus, dev, func };
    let vendor_id = read16(address, 0x00);
    if vendor_id == 0xFFFF {
        return None;
    }
    Some(PciDevice {
        address,
        vendor_id,
        device_id: read16(address, 0x02),
        class_code: read8(address, 0x0B),
        subclass: read8(address, 0x0A),
        // ... (truncated, see source file for more)
```

kernel-drivers/pcie/src/lib.rs (line 170) - [public] enumerate

/// Enumerate the whole bus into a `Vec` of probed devices.

```
pub fn enumerate() -> Vec<PciDevice> {
    let mut devices = Vec::new();
    for bus in 0u8..=255 {
        for dev in 0u8..32 {
            if let Some(d0) = probe(bus, dev, 0) {
                let multifunction = d0.header_type & 0x80 != 0;
                devices.push(d0);
                if multifunction {
                    for func in 1u8..8 {
                        if let Some(dn) = probe(bus, dev, func) {
                            devices.push(dn);
                        }
                    }
                }
            }
        }
        // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/class/hid/keyboard.rs (line 18) - [public] key_to_ascii

```
pub fn key_to_ascii(key: u8, shift: bool) -> Option<u8> {
    match key {
        0x04..0x1D => {
            let t = if shift { KEY_SHIFT } else { KEY_NORMAL };
            Some(t[key as usize])
        }
        0x1E..0x26 => {
            let t = if shift { NUM_SHIFT } else { NUM_NORMAL };
            Some(t[(key - 0x1E) as usize])
        }
        0x27 => {
            let t = if shift { NUM_SHIFT } else { NUM_NORMAL };
            // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/class/hid/keyboard.rs (line 44) - [public] push_char

```
pub fn push_char(c: u8) {
    unsafe {
        let next = (IN_HEAD + 1) % 64;

        if next != IN_TAIL {
            INBUF[IN_HEAD] = c;
            IN_HEAD = next;
        }
    }
}

pub fn take_char() -> Option<u8> {
    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/class/hid/keyboard.rs (line 55) - [public] take_char

```
pub fn take_char() -> Option<u8> {
    unsafe {
        if IN_HEAD == IN_TAIL {
            return None;
        }

        let c = INBUF[IN_TAIL];
        IN_TAIL = (IN_TAIL + 1) % 64;
        Some(c)
    }
}
```

kernel-drivers/usb/src/class/hid/mod.rs (line 16) - [public] HidKeyboard

```
pub struct HidKeyboard {
    pub slot: u8,
    pub ep_idx: u32,
    ring: TransferRing,
    data: DmaBuf,
    prev: [u8; 6],
}

static mut KEYS: [Option<HidKeyboard>; 2] = [None, None];

fn submit(x: &mut Xhci, kb: &mut HidKeyboard) {
    kb.ring.enqueue(Trb::normal(kb.data.phys, 8));
    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/class/hid/mod.rs (line 26) - [internal] submit

```
fn submit(x: &mut Xhci, kb: &mut HidKeyboard) {
    kb.ring.enqueue(Trb::normal(kb.data.phys, 8));
    x.regs.doorbell(kb.slot as u32, kb.ep_idx, 0);
}

pub fn attach(x: &mut Xhci, dev: &mut UsbDevice) -> Result<bool, UsbError> {
    let mut iface = 0u8;
    let mut found = false;

    for i in 0..dev.iface_count {
        let f = &dev.ifaces[i];

        // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/class/hid/mod.rs (line 31) - [public] attach

```
pub fn attach(x: &mut Xhci, dev: &mut UsbDevice) -> Result<bool, UsbError> {
    let mut iface = 0u8;
    let mut found = false;

    for i in 0..dev.iface_count {
        let f = &dev.ifaces[i];

        if f.class == 3 && f.subclass == 1 && f.protocol == 1 {
            iface = f.number;
            found = true;
            break;
        }
    }
    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/class/hid/mod.rs (line 105) - [public] poll

```
pub fn poll(x: &mut Xhci) {
    while let Some(t) = x.ev.pending() {
        let t = t;
        x.ev.pop();
        x.regs.rt_write(super::super::host::xhci::init::RT_ERDP, x.ev.erdp() as u32);

        if t.typ() != TRB_TRANSFER_EVENT {
            continue;
        }

        let slot = t.slot_id();
        let ep = t.ep_id();
        // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/class/hid/report.rs (line 1) - [public] BootReport

```
pub struct BootReport {
    pub shift: bool,
    pub keys: [u8; 6],
}

pub fn parse_boot_keyboard(report: &[u8]) -> Option<BootReport> {
    if report.len() < 8 {
        return None;
    }

    let mut keys = [0u8; 6];
    keys.copy_from_slice(&report[2..8]);
    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/class/hid/report.rs (line 6) - [public] parse_boot_keyboard

```
pub fn parse_boot_keyboard(report: &[u8]) -> Option<BootReport> {
    if report.len() < 8 {
        return None;
    }

    let mut keys = [0u8; 6];
    keys.copy_from_slice(&report[2..8]);

    Some(BootReport {
        shift: report[0] & 0x22 != 0,
        keys,
    })
    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/class/mass/blockdev.rs (line 6) - [public] DriverError

```
pub enum DriverError {
    Io,
}

pub trait BlockDevice {
    fn name(&self) -> &'static str;
    fn block_size(&self) -> usize;
    fn block_count(&self) -> u64;
    fn read_block(&self, block: u64, buf: &mut [u8]) -> Result<(), DriverError>;
    fn write_block(&self, block: u64, buf: &[u8]) -> Result<(), DriverError>;
}

// ... (truncated, see source file for more)
```

kernel-drivers/usb/src/class/mass/blockdev.rs (line 10) - [public] BlockDevice

```
pub trait BlockDevice {
    fn name(&self) -> &'static str;
    fn block_size(&self) -> usize;
    fn block_count(&self) -> u64;
    fn read_block(&self, block: u64, buf: &mut [u8]) -> Result<(), DriverError>;
    fn write_block(&self, block: u64, buf: &[u8]) -> Result<(), DriverError>;
}

impl BlockDevice for UsbMass {
    fn name(&self) -> &'static str {
        "usb0"
    }
    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/class/mass/blockdev.rs (line 11) - [internal] name

```
fn name(&self) -> &'static str;
fn block_size(&self) -> usize;
fn block_count(&self) -> u64;
fn read_block(&self, block: u64, buf: &mut [u8]) -> Result<(), DriverError>;
fn write_block(&self, block: u64, buf: &[u8]) -> Result<(), DriverError>;
}

impl BlockDevice for UsbMass {
    fn name(&self) -> &'static str {
        "usb0"
    }
    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/class/mass/blockdev.rs (line 12) - [internal] block_size

```
fn block_size(&self) -> usize;
fn block_count(&self) -> u64;
fn read_block(&self, block: u64, buf: &mut [u8]) -> Result<(), DriverError>;
fn write_block(&self, block: u64, buf: &[u8]) -> Result<(), DriverError>;
}

impl BlockDevice for UsbMass {
    fn name(&self) -> &'static str {
        "usb0"
    }
    fn block_size(&self) -> usize {
        // ... (truncated, see source file for more)
```

```

kernel-drivers/usb/src/class/mass/blockdev.rs (line 13) - [internal] block_count
    fn block_count(&self) -> u64;
    fn read_block(&self, block: u64, buf: &mut [u8]) -> Result<(), DriverError>;
    fn write_block(&self, block: u64, buf: &[u8]) -> Result<(), DriverError>;
}

impl BlockDevice for UsbMass {
    fn name(&self) -> &'static str {
        "usb0"
    }

    fn block_size(&self) -> usize {
        self.block_len as usize
        // ... (truncated, see source file for more)
    }
}

kernel-drivers/usb/src/class/mass/blockdev.rs (line 14) - [internal] read_block
    fn read_block(&self, block: u64, buf: &mut [u8]) -> Result<(), DriverError>;
    fn write_block(&self, block: u64, buf: &[u8]) -> Result<(), DriverError>;
}

impl BlockDevice for UsbMass {
    fn name(&self) -> &'static str {
        "usb0"
    }

    fn block_size(&self) -> usize {
        self.block_len as usize
    }
    // ... (truncated, see source file for more)
}

kernel-drivers/usb/src/class/mass/blockdev.rs (line 15) - [internal] write_block
    fn write_block(&self, block: u64, buf: &[u8]) -> Result<(), DriverError>;
}

impl BlockDevice for UsbMass {
    fn name(&self) -> &'static str {
        "usb0"
    }

    fn block_size(&self) -> usize {
        self.block_len as usize
    }
    // ... (truncated, see source file for more)
}

kernel-drivers/usb/src/class/mass/blockdev.rs (line 19) - [internal] name
    fn name(&self) -> &'static str {
        "usb0"
    }

    fn block_size(&self) -> usize {
        self.block_len as usize
    }

    fn block_count(&self) -> u64 {
        self.sectors as u64
    }
    // ... (truncated, see source file for more)
}

```



```

kernel-drivers/usb/src/class/mass/blockdev.rs (line 23) - [internal] block_size
    fn block_size(&self) -> usize {
        self.block_len as usize
    }

    fn block_count(&self) -> u64 {
        self.sectors as u64
    }

    fn read_block(&self, block: u64, buf: &mut [u8]) -> Result<(), DriverError> {
        let m = unsafe { super::MASS0.as_mut().ok_or(DriverError::Io)? };

        super::with_controller(|x| {
            // ... (truncated, see source file for more)
        })

kernel-drivers/usb/src/class/mass/blockdev.rs (line 27) - [internal] block_count
    fn block_count(&self) -> u64 {
        self.sectors as u64
    }

    fn read_block(&self, block: u64, buf: &mut [u8]) -> Result<(), DriverError> {
        let m = unsafe { super::MASS0.as_mut().ok_or(DriverError::Io)? };

        super::with_controller(|x| {
            scsi::read10(x, m, block as u32, 1, m.data.phys)
        }).map_err(|_| DriverError::Io)?;

        unsafe {
            // ... (truncated, see source file for more)
        }

kernel-drivers/usb/src/class/mass/blockdev.rs (line 31) - [internal] read_block
    fn read_block(&self, block: u64, buf: &mut [u8]) -> Result<(), DriverError> {
        let m = unsafe { super::MASS0.as_mut().ok_or(DriverError::Io)? };

        super::with_controller(|x| {
            scsi::read10(x, m, block as u32, 1, m.data.phys)
        }).map_err(|_| DriverError::Io)?;

        unsafe {
            core::ptr::copy_nonoverlapping(m.data.virt, buf.as_mut_ptr(),
                                           m.block_len as usize);
        }

        // ... (truncated, see source file for more)

kernel-drivers/usb/src/class/mass/blockdev.rs (line 46) - [internal] write_block
    fn write_block(&self, block: u64, buf: &[u8]) -> Result<(), DriverError> {
        let m = unsafe { super::MASS0.as_mut().ok_or(DriverError::Io)? };

        unsafe {
            core::ptr::copy_nonoverlapping(buf.as_ptr(), m.data.virt,
                                           m.block_len as usize);
        }

        super::with_controller(|x| {
            scsi::write10(x, m, block as u32, 1, m.data.phys)
        }).map_err(|_| DriverError::Io)?;

        // ... (truncated, see source file for more)

```

kernel-drivers/usb/src/class/mass/mod.rs (line 18) - [public] UsbMass

```
pub struct UsbMass {
    pub slot: u8,
    pub out_idx: u32,
    pub in_idx: u32,
    pub out_ring: TransferRing,
    pub in_ring: TransferRing,
    pub data: DmaBuf,
    pub cmd: DmaBuf,
    pub tag: u32,
    pub sectors: u32,
    pub block_len: u32,
}

// ... (truncated, see source file for more)
```

kernel-drivers/usb/src/class/mass/mod.rs (line 32) - [public] bulk_out

```
pub fn bulk_out(&mut self, x: &mut Xhci, phys: u64, len: u32) -> Result<(), UsbError> {
    self.out_ring.enqueue(Trb::normal(phys, len));
    x.regs.doorbell(self.slot as u32, self.out_idx, 0);

    let cc = control::wait_transfer_ep(x, self.slot, self.out_idx as u8)?;

    if cc != CC_SUCCESS {
        return Err(UsbError::Transfer(cc));
    }

    Ok(())
}

// ... (truncated, see source file for more)
```

kernel-drivers/usb/src/class/mass/mod.rs (line 45) - [public] bulk_in

```
pub fn bulk_in(&mut self, x: &mut Xhci, phys: u64, len: u32) -> Result<(), UsbError> {
    self.in_ring.enqueue(Trb::normal(phys, len));
    x.regs.doorbell(self.slot as u32, self.in_idx, 0);

    let cc = control::wait_transfer_ep(x, self.slot, self.in_idx as u8)?;

    if cc != CC_SUCCESS {
        return Err(UsbError::Transfer(cc));
    }

    Ok(())
}

// ... (truncated, see source file for more)
```

kernel-drivers/usb/src/class/mass/mod.rs (line 58) - [public] scsi_cmd

```
pub fn scsi_cmd(&mut self, x: &mut Xhci, cdb: &[u8],
    data_phys: u64, data_len: u32, dir_in: bool) -> Result<(), UsbError> {
    self.tag += 1;

    let mut cbw = [0u8; 31];

    cbw[0..4].copy_from_slice(&CBW_SIG.to_le_bytes());
    cbw[4..8].copy_from_slice(&self.tag.to_le_bytes());
    cbw[8..12].copy_from_slice(&data_len.to_le_bytes());
    cbw[12] = if dir_in { 0x80 } else { 0x00 };
    cbw[13] = 0;
    cbw[14] = cdb.len() as u8;
}

// ... (truncated, see source file for more)
```

kernel-drivers/usb/src/class/mass/mod.rs (line 102) - [public] attach

```
pub fn attach(x: &mut Xhci, dev: &mut UsbDevice) -> Result<bool, UsbError> {
    let mut found = false;

    for i in 0..dev.iface_count {
        let f = &dev.ifaces[i];

        if f.class == 8 && f.subclass == 6 && f.protocol == 0x50 {
            found = true;
            break;
        }
    }

    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/class/mass/mod.rs (line 182) - [public] mass0

/// Borrow the attached mass-storage device (implements [blockdev::BlockDevice]).

```
pub fn mass0() -> Option<&'static mut UsbMass> {
    unsafe { MASS0.as_mut() }
}

pub fn with_controller<F: FnOnce(&mut Xhci) -> R, R>(f: F) -> R {
    let x = unsafe { &mut *XHCI_PTR };
    f(x)
}
```

kernel-drivers/usb/src/class/mass/mod.rs (line 186) - [public] with_controller

```
pub fn with_controller<F: FnOnce(&mut Xhci) -> R, R>(f: F) -> R {
    let x = unsafe { &mut *XHCI_PTR };
    f(x)
}
```

kernel-drivers/usb/src/class/mass/scsi.rs (line 5) - [public] read_capacity

```
pub fn read_capacity(x: &mut Xhci, m: &mut UsbMass) -> Result<(u32, u32), UsbError> {
    let cdb = [0x25u8, 0, 0, 0, 0, 0, 0, 0, 0, 0];

    m.scsi_cmd(x, &cdb, m.data.phys, 8, true)?;

    unsafe {
        let p = m.data.virt;

        let last = u32::from_be_bytes([*p.add(0), *p.add(1), *p.add(2), *p.add(3)]);
        let blen = u32::from_be_bytes([*p.add(4), *p.add(5), *p.add(6), *p.add(7)]);

        Ok((last + 1, blen))
    }
    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/class/mass/scsi.rs (line 20) - [public] read10

```
pub fn read10(x: &mut Xhci, m: &mut UsbMass,
              lba: u32, count: u16, phys: u64) -> Result<(), UsbError> {
    let cdb = [
        0x28, 0,
        (lba >> 24) as u8, (lba >> 16) as u8, (lba >> 8) as u8, lba as u8,
        0,
        (count >> 8) as u8, count as u8,
        0,
    ];

    m.scsi_cmd(x, &cdb, phys, count as u32 * m.block_len, true)

}
// ... (truncated, see source file for more)
```

```

kernel-drivers/usb/src/class/mass/scsi.rs (line 33) - [public] write10
    pub fn write10(x: &mut Xhci, m: &mut UsbMass,
                   lba: u32, count: u16, phys: u64) -> Result<(), UsbError> {
        let cdb = [
            0x2A, 0,
            (lba >> 24) as u8, (lba >> 16) as u8, (lba >> 8) as u8, lba as u8,
            0,
            (count >> 8) as u8, count as u8,
            0,
        ];

        m.scsi_cmd(x, &cdb, phys, count as u32 * m.block_len, false)
    }

kernel-drivers/usb/src/class/mod.rs (line 6) - [public] ClassDriver
    pub trait ClassDriver {
        fn probe(&self, dev: &UsbDevice) -> bool;
    }

kernel-drivers/usb/src/class/mod.rs (line 7) - [internal] probe
    fn probe(&self, dev: &UsbDevice) -> bool;
}

kernel-drivers/usb/src/core/descriptor.rs (line 2) - [public] DeviceDesc
    pub struct DeviceDesc {
        pub max_packet0: u8,
        pub vendor: u16,
        pub product: u16,
        pub class: u8,
        pub subclass: u8,
        pub protocol: u8,
        pub num_configs: u8,
    }

    #[derive(Clone, Copy, Default)]
    pub struct InterfaceDesc {
        // ... (truncated, see source file for more)
    }

kernel-drivers/usb/src/core/descriptor.rs (line 13) - [public] InterfaceDesc
    pub struct InterfaceDesc {
        pub number: u8,
        pub class: u8,
        pub subclass: u8,
        pub protocol: u8,
        pub num_eps: u8,
    }

    #[derive(Clone, Copy, Default)]
    pub struct EndpointDesc {
        pub address: u8,
        pub attributes: u8,
        // ... (truncated, see source file for more)
    }

kernel-drivers/usb/src/core/descriptor.rs (line 22) - [public] EndpointDesc
    pub struct EndpointDesc {
        pub address: u8,
        pub attributes: u8,
        pub max_packet: u16,
        pub interval: u8,
    }

    pub fn parse_device(buf: &[u8]) -> Option<DeviceDesc> {
        if buf.len() < 18 || buf[1] != 1 {
            return None;
        }

        // ... (truncated, see source file for more)
    }

```

kernel-drivers/usb/src/core/descriptor.rs (line 29) - [public] parse_device

```
pub fn parse_device(buf: &[u8]) -> Option<DeviceDesc> {
    if buf.len() < 18 || buf[1] != 1 {
        return None;
    }

    Some(DeviceDesc {
        max_packet0: buf[7],
        vendor: u16::from_le_bytes([buf[8], buf[9]]),
        product: u16::from_le_bytes([buf[10], buf[11]]),
        class: buf[4],
        subclass: buf[5],
        protocol: buf[6],
    })
    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/core/descriptor.rs (line 45) - [public] parse_config

```
pub fn parse_config(buf: &[u8],
                    ifaces: &mut [InterfaceDesc],
                    eps: &mut [EndpointDesc])
    -> Option<(u8, u16, usize, usize)> {
    if buf.len() < 9 || buf[1] != 2 {
        return None;
    }

    let total = u16::from_le_bytes([buf[2], buf[3]]);
    let config_value = buf[5];

    let mut off = 9usize;
    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/core/device.rs (line 7) - [public] UsbDevice

```
pub struct UsbDevice {
    pub slot: u8,
    pub port: u32,
    pub speed: u32,
    pub ep0_mps: u16,
    pub desc: DeviceDesc,
    pub config_value: u8,
    pub ifaces: [InterfaceDesc; 4],
    pub iface_count: usize,
    pub eps: [EndpointDesc; 8],
    pub ep_count: usize,
    pub ctx: Contexts,
    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/core/device.rs (line 24) - [public] new

```
pub fn new(slot: u8, speed: u32, ctx_size: usize) -> Result<Self, UsbError> {
    Ok(Self {
        slot,
        port: 0,
        speed,
        ep0_mps: 8,
        desc: DeviceDesc::default(),
        config_value: 0,
        ifaces: [InterfaceDesc::default(); 4],
        iface_count: 0,
        eps: [EndpointDesc::default(); 8],
        ep_count: 0,
    })
    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/core/enumerate.rs (line 11) - [public] enumerate

```
pub fn enumerate(x: &mut Xhci, port: u32) -> Result<UsbDevice, UsbError> {
    let slot = x.command(Trb::enable_slot())?.slot_id();
    let speed = x.regs.port_speed(port);

    let mut dev = UsbDevice::new(slot, speed, x.ctx_size?);

    dev.port = port;

    let mps = default_ep0_mps(speed);
    dev.ep0_mps = mps;

    dev.ctx.setup_address_device(speed, port, mps, dev.ep0.phys());
    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/core/speed.rs (line 6) - [public] default_ep0_mps

```
pub fn default_ep0_mps(speed: u32) -> u16 {
    match speed {
        SPEED_HIGH => 64,
        SPEED_SUPER => 512,
        _ => 8,
    }
}

pub const EP_CONTROL: u8 = 0;
pub const EP_ISO: u8 = 1;
pub const EP_BULK: u8 = 2;
pub const EP_INTERRUPT: u8 = 3;
```

kernel-drivers/usb/src/dma.rs (line 4) - [public] DmaBuf

/// A DMA-coherent buffer (physical + virtual address).

```
pub struct DmaBuf {
    pub phys: u64,
    pub virt: *mut u8,
    pub len: usize,
}

impl DmaBuf {
    pub fn new(len: usize) -> Result<Self, UsbError> {
        let mut phys = 0u64;
        let mut virt: *mut u8 = core::ptr::null_mut();

        let ok = unsafe { hw_sys::dma_alloc_coherent(len, 0xFFFF_FFFF, &mut phys, &mut virt) };
        // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/dma.rs (line 11) - [public] new

```
pub fn new(len: usize) -> Result<Self, UsbError> {
    let mut phys = 0u64;
    let mut virt: *mut u8 = core::ptr::null_mut();

    let ok = unsafe { hw_sys::dma_alloc_coherent(len, 0xFFFF_FFFF, &mut phys, &mut virt) };

    if !ok || virt.is_null() {
        return Err(UsbError::Invalid);
    }

    Ok(Self { phys, virt, len })
}
// ... (truncated, see source file for more)
```

kernel-drivers/usb/src/dma.rs (line 24) - [public] zero

```
pub fn zero(&mut self) {
    unsafe { core::ptr::write_bytes(self.virt, 0, self.len) }
}

impl Drop for DmaBuf {
    fn drop(&mut self) {
        unsafe { hw_sys::dma_free_coherent(self.phys, self.virt, self.len) };
    }
}
```

kernel-drivers/usb/src/dma.rs (line 30) - [internal] drop

```
fn drop(&mut self) {
    unsafe { hw_sys::dma_free_coherent(self.phys, self.virt, self.len) };
}
```

kernel-drivers/usb/src/host/xhci/context.rs (line 4) - [public] Contexts

```
pub struct Contexts {
    pub input: DmaBuf,
    pub output: DmaBuf,
    pub ctx_size: usize,
}
```

```
impl Contexts {
    pub fn setup_configure_ep(&mut self, speed: u32, port: u32,
                             epidx: u32, ep_type: u32, mps: u16,
                             interval: u32, ring_phys: u64) {
        self.input.zero();
        self.in_add_drop(1 | (1 << epidx), 0);
        // ... (truncated, see source file for more)
    }
}
```

kernel-drivers/usb/src/host/xhci/context.rs (line 11) - [public] setup_configure_ep

```
pub fn setup_configure_ep(&mut self, speed: u32, port: u32,
                          epidx: u32, ep_type: u32, mps: u16,
                          interval: u32, ring_phys: u64) {
    self.input.zero();
    self.in_add_drop(1 | (1 << epidx), 0);

    let s = self.slot_ptr();

    unsafe {
        s.add(0).write_volatile((speed << 20) | (2 << 27));
        s.add(1).write_volatile(port << 16);
    }
    // ... (truncated, see source file for more)
}
```

kernel-drivers/usb/src/host/xhci/context.rs (line 37) - [public] setup_configure_bulk_pair

```
pub fn setup_configure_bulk_pair(&mut self, speed: u32, port: u32,
                                 out_idx: u32, in_idx: u32,
                                 mps_out: u16, mps_in: u16,
                                 ring_out: u64, ring_in: u64) {
    self.input.zero();
    self.in_add_drop(1 | (1 << out_idx) | (1 << in_idx), 0);

    let s = self.slot_ptr();

    unsafe {
        s.add(0).write_volatile((speed << 20) | (3 << 27));
        s.add(1).write_volatile(port << 16);
    }
    // ... (truncated, see source file for more)
}
```

kernel-drivers/usb/src/host/xhci/context.rs (line 71) - [public] new

```
pub fn new(ctx_size: usize) -> Result<Self, UsbError> {
    let mut input = DmaBuf::new(ctx_size * 3)?;
    let output = DmaBuf::new(ctx_size * 2)?;

    input.zero();

    Ok(Self { input, output, ctx_size })
}

fn in_add_drop(&mut self, add: u32, drop: u32) {
    let p = self.input.virt as *mut u32;

    // ... (truncated, see source file for more)
}
```

kernel-drivers/usb/src/host/xhci/context.rs (line 80) - [internal] in_add_drop

```
fn in_add_drop(&mut self, add: u32, drop: u32) {
    let p = self.input.virt as *mut u32;

    unsafe {
        p.add(0).write_volatile(drop);
        p.add(1).write_volatile(add);
    }
}

fn slot_ptr(&mut self) -> *mut u32 {
    unsafe { self.input.virt.add(self.ctx_size) as *mut u32 }
}
// ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/context.rs (line 89) - [internal] slot_ptr

```
fn slot_ptr(&mut self) -> *mut u32 {
    unsafe { self.input.virt.add(self.ctx_size) as *mut u32 }
}

fn ep0_ptr(&mut self) -> *mut u32 {
    unsafe { self.input.virt.add(self.ctx_size * 2) as *mut u32 }
}

pub fn setup_address_device(&mut self, speed: u32, port: u32,
                           mps: u16, ring_phys: u64) {
    self.input.zero();
    self.in_add_drop(3, 0);
    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/context.rs (line 93) - [internal] ep0_ptr

```
fn ep0_ptr(&mut self) -> *mut u32 {
    unsafe { self.input.virt.add(self.ctx_size * 2) as *mut u32 }
}

pub fn setup_address_device(&mut self, speed: u32, port: u32,
                           mps: u16, ring_phys: u64) {
    self.input.zero();
    self.in_add_drop(3, 0);

    let s = self.slot_ptr();

    unsafe {
        // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/context.rs (line 97) - [public] setup_address_device

```
pub fn setup_address_device(&mut self, speed: u32, port: u32,
                           mps: u16, ring_phys: u64) {
    self.input.zero();
    self.in_add_drop(3, 0);

    let s = self.slot_ptr();

    unsafe {
        s.add(0).write_volatile((speed << 20) | (1 << 27));
        s.add(1).write_volatile(port << 16);
    }

    // ... (truncated, see source file for more)
```


kernel-drivers/usb/src/host/xhci/context.rs (line 120) - [public] setup_evaluate_mps

```
pub fn setup_evaluate_mps(&mut self, mps: u16) {
    self.input.zero();
    self.in_add_drop(2, 0);

    let e = self.ep0_ptr();

    unsafe {
        e.add(1).write_volatile((mps as u32) << 16);
    }
}
```

kernel-drivers/usb/src/host/xhci/control.rs (line 6) - [public] wait_transfer

```
pub fn wait_transfer(x: &mut Xhci, slot: u8) -> Result<u8, UsbError> {
    for _ in 0..2_000_000 {
        if let Some(t) = x.ev.pending() {
            let t = t;
            x.ev.pop();
            super::init::rt_write64(&x.regs, super::init::RT_ERDP, x.ev.erdp());

            if t.typ() == TRB_TRANSFER_EVENT && t.slot_id() == slot {
                return Ok(t.completion_code());
            }

            continue;
        }
        // ... (truncated, see source file for more)
    }
}
```

kernel-drivers/usb/src/host/xhci/control.rs (line 26) - [public] wait_transfer_ep

```
pub fn wait_transfer_ep(x: &mut Xhci, slot: u8, ep: u8) -> Result<u8, UsbError> {
    for _ in 0..2_000_000 {
        if let Some(t) = x.ev.pending() {
            let t = t;
            x.ev.pop();
            super::init::rt_write64(&x.regs, super::init::RT_ERDP, x.ev.erdp());

            if t.typ() == TRB_TRANSFER_EVENT && t.slot_id() == slot && t.ep_id() == ep {
                return Ok(t.completion_code());
            }

            continue;
        }
        // ... (truncated, see source file for more)
    }
}
```

kernel-drivers/usb/src/host/xhci/control.rs (line 46) - [public] control

```
pub fn control(x: &mut Xhci,
    dev: &mut UsbDevice,
    setup: u64,
    trt: u32,
    data: Option<(&mut [u8], usize)>,
    out_len: usize) -> Result<usize, UsbError> {
    dev.ep0.enqueue(Trb::setup_stage(setup, trt));

    let (dir_in, len) = match &data {
        Some( (_, l)) => (trt == 3, *l),
        None => (false, 0),
    };
    // ... (truncated, see source file for more)
}
```

kernel-drivers/usb/src/host/xhci/control.rs (line 94) - [public] control_in

```
pub fn control_in(x: &mut Xhci, dev: &mut UsbDevice,
    req: u8, value: u16, index: u16,
    buf: &mut [u8]) -> Result<usize, UsbError> {
    let len = buf.len();
    let setup = pack_setup(0x80, req, value, index, len as u16);
    control(x, dev, setup, 3, Some((buf, len)), 0)
}

pub fn control_out(x: &mut Xhci, dev: &mut UsbDevice,
    req: u8, value: u16, index: u16,
    data: &[u8]) -> Result<usize, UsbError> {
    let setup = pack_setup(0x00, req, value, index, data.len() as u16);
    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/control.rs (line 102) - [public] control_out

```
pub fn control_out(x: &mut Xhci, dev: &mut UsbDevice,
    req: u8, value: u16, index: u16,
    data: &[u8]) -> Result<usize, UsbError> {
    let setup = pack_setup(0x00, req, value, index, data.len() as u16);

    if data.is_empty() {
        control(x, dev, setup, 0, None, 0)
    } else {
        let mut tmp = [0u8; 512];
        tmp[..data.len()].copy_from_slice(data);
        control(x, dev, setup, 2, Some((&mut tmp, data.len()))), 0)
    }
    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/event.rs (line 6) - [public] drain_events

```
pub fn drain_events(&mut self) {
    while let Some(t) = self.ev.pending() {
        let t = t;
        self.ev.pop();
        super::init::rt_write64(&self.regs, super::init::RT_ERDP, self.ev.erdp());

        match t.typ() {
            TRB_PORT_STATUS => unsafe {
                kprintf(b"usb: port status change, port=%d\n\0".as_ptr(),
                    (t.param >> 24) as u32);
            },
            TRB_HC_EVENT => unsafe {
                // ... (truncated, see source file for more)
            }
        }
    }
}
```

kernel-drivers/usb/src/host/xhci/init.rs (line 14) - [public] Xhci

```
pub struct Xhci {
    pub regs: XhciRegs,
    pub cmd: CmdRing,
    pub ev: EventRing,
    pub dcbaa: DmaBuf,
    pub ctx_size: usize,
    pub slots: u32,
    pub ports: u32,
}

fn spin_wait<F: Fn() -> bool>(f: F) -> Result<(), UsbError> {
    for _ in 0..2_000_000 {
        // ... (truncated, see source file for more)
    }
}
```

kernel-drivers/usb/src/host/xhci/init.rs (line 24) - [internal] spin_wait

```
fn spin_wait<F: Fn() -> bool>(f: F) -> Result<(), UsbError> {
    for _ in 0..2_000_000 {
        if f() {
            return Ok(());
        }

        core::hint::spin_loop();
    }

    Err(UsbError::Timeout)
}

// ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/init.rs (line 36) - [internal] op_write64

```
fn op_write64(regs: &XhciRegs, off: usize, v: u64) {
    regs.op_write(off, (v & 0xFFFF_FFFF) as u32);
    regs.op_write(off + 4, (v >> 32) as u32);
}

pub(super) fn rt_write64(regs: &XhciRegs, off: usize, v: u64) {
    regs.rt_write(off, (v & 0xFFFF_FFFF) as u32);
    regs.rt_write(off + 4, (v >> 32) as u32);
}

pub fn init(regs: XhciRegs) -> Result<Xhci, UsbError> {
    let cmd0 = regs.op_read(OP_USBCMD);
    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/init.rs (line 41) - [public] rt_write64

```
pub(super) fn rt_write64(regs: &XhciRegs, off: usize, v: u64) {
    regs.rt_write(off, (v & 0xFFFF_FFFF) as u32);
    regs.rt_write(off + 4, (v >> 32) as u32);
}

pub fn init(regs: XhciRegs) -> Result<Xhci, UsbError> {
    let cmd0 = regs.op_read(OP_USBCMD);

    if cmd0 & CMD_RS != 0 {
        regs.op_write(OP_USBCMD, cmd0 & !CMD_RS);
        spin_wait(|| regs.op_read(OP_USBSTS) & STS_HCH != 0)?;
    }

    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/init.rs (line 46) - [public] init

```
pub fn init(regs: XhciRegs) -> Result<Xhci, UsbError> {
    let cmd0 = regs.op_read(OP_USBCMD);

    if cmd0 & CMD_RS != 0 {
        regs.op_write(OP_USBCMD, cmd0 & !CMD_RS);
        spin_wait(|| regs.op_read(OP_USBSTS) & STS_HCH != 0)?;
    }

    regs.op_write(OP_USBCMD, CMD_HCRST);

    spin_wait(|| regs.op_read(OP_USBCMD) & CMD_HCRST == 0)?;
    spin_wait(|| regs.op_read(OP_USBSTS) & STS_CNR == 0)?;
    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/init.rs (line 100) - [public] command

```
pub fn command(&mut self, trb: Trb) -> Result<Trb, UsbError> {
    self.cmd.enqueue(trb);
    self.regs.doorbell(0, 0, 0);

    for _ in 0..2_000_000 {
        if let Some(t) = self.ev.pending() {
            let t = t;
            self.ev.pop();
            rt_write64(&self.regs, RT_ERDP, self.ev.erdp());

            if t.typ() == TRB_CMD_COMPLETION {
                if t.completion_code() == CC_SUCCESS {
                    // ... (truncated, see source file for more)
                }
            }
        }
    }
}
```

kernel-drivers/usb/src/host/xhci/init.rs (line 127) - [public] scan_ports

```
pub fn scan_ports(&mut self) {
    for p in 1..=self.ports {
        let sc = self.regs.port_sc(p);

        if sc & PORTSC_CCS == 0 {
            continue;
        }

        let speed = (sc & PORTSC_SPEED) >> 10;

        unsafe {
            kprintf(b"usb: port %d connected, speed=%d\n\0".as_ptr(), p, speed);
        }
        // ... (truncated, see source file for more)
    }
}
```

kernel-drivers/usb/src/host/xhci/init.rs (line 170) - [internal] attach_port

```
fn attach_port(&mut self, p: u32) {
    let mut dev = match crate::core::enumerate::enumerate(self, p) {
        Ok(d) => d,
        Err(_) => {
            unsafe {
                kprintf(b"usb: port %d enumeration failed\n\0".as_ptr(), p);
            }
            return;
        }
    };
    use crate::class::{hid, mass};
    // ... (truncated, see source file for more)
}
```

kernel-drivers/usb/src/host/xhci/regs.rs (line 35) - [public] XhciRegs

```
pub struct XhciRegs {
    base: *mut u8,
    pub cap_len: usize,
    pub db_off: usize,
    pub rt_off: usize,
    pub max_slots: u32,
    pub max_intrs: u32,
    pub max_ports: u32,
    pub addr64: bool,
    pub csz64: bool,
}

// ... (truncated, see source file for more)
```

```

kernel-drivers/usb/src/host/xhci/regs.rs (line 48) - [public] new
pub fn new(phys: u64) -> Result<Self, UsbError> {
    let ptr = if phys >= 0xFFFF800000000000 {
        phys
    } else {
        let mut virt = 0u64;

        if !unsafe { hw_sys::mmio_map(phys, MAP_SIZE, &mut virt) } {
            return Err(UsbError::MapFailed);
        }

        // The runtime can report success yet leave the MMIO unmapped;
        // reading it would page-fault, so verify.
        // ... (truncated, see source file for more)
}

kernel-drivers/usb/src/host/xhci/regs.rs (line 90) - [internal] op
fn op(&self) -> *mut u8 {
    unsafe { self.base.add(self.cap_len) }
}

fn rt(&self) -> *mut u8 {
    unsafe { self.base.add(self.rt_off) }
}

fn db(&self) -> *mut u8 {
    unsafe { self.base.add(self.db_off) }
}

// ... (truncated, see source file for more)

kernel-drivers/usb/src/host/xhci/regs.rs (line 94) - [internal] rt
fn rt(&self) -> *mut u8 {
    unsafe { self.base.add(self.rt_off) }
}

fn db(&self) -> *mut u8 {
    unsafe { self.base.add(self.db_off) }
}

pub fn op_read(&self, off: usize) -> u32 {
    unsafe { r32(self.op().add(off)) }
}

// ... (truncated, see source file for more)

kernel-drivers/usb/src/host/xhci/regs.rs (line 98) - [internal] db
fn db(&self) -> *mut u8 {
    unsafe { self.base.add(self.db_off) }
}

pub fn op_read(&self, off: usize) -> u32 {
    unsafe { r32(self.op().add(off)) }
}

pub fn op_write(&self, off: usize, v: u32) {
    unsafe { w32(self.op().add(off), v) }
}

// ... (truncated, see source file for more)

```

```

kernel-drivers/usb/src/host/xhci/regs.rs (line 102) - [public] op_read
    pub fn op_read(&self, off: usize) -> u32 {
        unsafe { r32(self.op().add(off)) }
    }

    pub fn op_write(&self, off: usize, v: u32) {
        unsafe { w32(self.op().add(off), v) }
    }

    pub fn rt_read(&self, off: usize) -> u32 {
        unsafe { r32(self.rt().add(off)) }
    }

    // ... (truncated, see source file for more)

kernel-drivers/usb/src/host/xhci/regs.rs (line 106) - [public] op_write
    pub fn op_write(&self, off: usize, v: u32) {
        unsafe { w32(self.op().add(off), v) }
    }

    pub fn rt_read(&self, off: usize) -> u32 {
        unsafe { r32(self.rt().add(off)) }
    }

    pub fn rt_write(&self, off: usize, v: u32) {
        unsafe { w32(self.rt().add(off), v) }
    }

    // ... (truncated, see source file for more)

kernel-drivers/usb/src/host/xhci/regs.rs (line 110) - [public] rt_read
    pub fn rt_read(&self, off: usize) -> u32 {
        unsafe { r32(self.rt().add(off)) }
    }

    pub fn rt_write(&self, off: usize, v: u32) {
        unsafe { w32(self.rt().add(off), v) }
    }

    pub fn doorbell(&self, slot: u32, target: u32, task: u32) {
        unsafe {
            w32(self.db().add(slot as usize * 4),
                (target & 0xFF) | ((task & 0xFFFF) << 16));
        }
        // ... (truncated, see source file for more)
    }

kernel-drivers/usb/src/host/xhci/regs.rs (line 114) - [public] rt_write
    pub fn rt_write(&self, off: usize, v: u32) {
        unsafe { w32(self.rt().add(off), v) }
    }

    pub fn doorbell(&self, slot: u32, target: u32, task: u32) {
        unsafe {
            w32(self.db().add(slot as usize * 4),
                (target & 0xFF) | ((task & 0xFFFF) << 16));
        }
    }

    pub fn port_sc(&self, port: u32) -> u32 {
        // ... (truncated, see source file for more)
    }

```

kernel-drivers/usb/src/host/xhci/regs.rs (line 118) - [public] doorbell

```
pub fn doorbell(&self, slot: u32, target: u32, task: u32) {
    unsafe {
        w32(self.db().add(slot as usize * 4),
            (target & 0xFF) | ((task & 0xFFFF) << 16));
    }
}

pub fn port_sc(&self, port: u32) -> u32 {
    self.op_read(OP_PORTSC + (port as usize - 1) * 0x10)
}

pub fn port_sc_write(&self, port: u32, v: u32) {
    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/regs.rs (line 125) - [public] port_sc

```
pub fn port_sc(&self, port: u32) -> u32 {
    self.op_read(OP_PORTSC + (port as usize - 1) * 0x10)
}

pub fn port_sc_write(&self, port: u32, v: u32) {
    self.op_write(OP_PORTSC + (port as usize - 1) * 0x10, v);
}

pub fn port_speed(&self, port: u32) -> u32 {
    (self.port_sc(port) & PORTSC_SPEED) >> 10
}
}
```

kernel-drivers/usb/src/host/xhci/regs.rs (line 129) - [public] port_sc_write

```
pub fn port_sc_write(&self, port: u32, v: u32) {
    self.op_write(OP_PORTSC + (port as usize - 1) * 0x10, v);
}

pub fn port_speed(&self, port: u32) -> u32 {
    (self.port_sc(port) & PORTSC_SPEED) >> 10
}
}
```

kernel-drivers/usb/src/host/xhci/regs.rs (line 133) - [public] port_speed

```
pub fn port_speed(&self, port: u32) -> u32 {
    (self.port_sc(port) & PORTSC_SPEED) >> 10
}
}
```

kernel-drivers/usb/src/host/xhci/ring.rs (line 5) - [public] CmdRing

```
pub struct CmdRing {
    buf: DmaBuf,
    len: usize,
    enqueue: usize,
    cycle: bool,
}

impl CmdRing {
    pub fn new(count: usize) -> Result<Self, UsbError> {
        let len = count + 1;
        let buf = DmaBuf::new(len * 16)?;

        // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/ring.rs (line 13) - [public] new

```
pub fn new(count: usize) -> Result<Self, UsbError> {
    let len = count + 1;
    let buf = DmaBuf::new(len * 16)?;

    unsafe {
        (buf.virt as *mut Trb).add(count).write_volatile(Trb::link(buf.phys));
    }

    Ok(Self { buf, len, enqueue: 0, cycle: true })
}

pub fn phys(&self) -> u64 {
    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/ring.rs (line 24) - [public] phys

```
pub fn phys(&self) -> u64 {
    self.buf.phys
}

pub fn enqueue(&mut self, mut trb: Trb) {
    if self.enqueue == self.len - 1 {
        self.enqueue = 0;
        self.cycle = !self.cycle;
    }

    if self.cycle {
        trb.control |= 1;
    }
    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/ring.rs (line 28) - [public] enqueue

```
pub fn enqueue(&mut self, mut trb: Trb) {
    if self.enqueue == self.len - 1 {
        self.enqueue = 0;
        self.cycle = !self.cycle;
    }

    if self.cycle {
        trb.control |= 1;
    } else {
        trb.control &= !1;
    }

    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/ring.rs (line 48) - [public] EventRing

```
pub struct EventRing {
    buf: DmaBuf,
    erst: DmaBuf,
    len: usize,
    dequeue: usize,
    cycle: bool,
}

pub struct TransferRing {
    buf: DmaBuf,
    len: usize,
    enqueue: usize,
    // ... (truncated, see source file for more)
```


kernel-drivers/usb/src/host/xhci/ring.rs (line 56) - [public] TransferRing

```
pub struct TransferRing {
    buf: DmaBuf,
    len: usize,
    enqueue: usize,
    cycle: bool,
}

impl TransferRing {
    pub fn new(count: usize) -> Result<Self, UsbError> {
        let len = count + 1;
        let buf = DmaBuf::new(len * 16)?;

        // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/ring.rs (line 64) - [public] new

```
pub fn new(count: usize) -> Result<Self, UsbError> {
    let len = count + 1;
    let buf = DmaBuf::new(len * 16)?;

    unsafe {
        (buf.virt as *mut Trb).add(count).write_volatile(Trb::link(buf.phys));
    }

    Ok(Self { buf, len, enqueue: 0, cycle: true })
}

pub fn phys(&self) -> u64 {
    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/ring.rs (line 75) - [public] phys

```
pub fn phys(&self) -> u64 {
    self.buf.phys
}

pub fn enqueue(&mut self, mut trb: Trb) {
    if self.enqueue == self.len - 1 {
        self.enqueue = 0;
        self.cycle = !self.cycle;
    }

    if self.cycle {
        trb.control |= 1;
    }
    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/ring.rs (line 79) - [public] enqueue

```
pub fn enqueue(&mut self, mut trb: Trb) {
    if self.enqueue == self.len - 1 {
        self.enqueue = 0;
        self.cycle = !self.cycle;
    }

    if self.cycle {
        trb.control |= 1;
    } else {
        trb.control &= !1;
    }

    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/ring.rs (line 100) - [public] new

```
pub fn new(count: usize) -> Result<Self, UsbError> {
    let buf = DmaBuf::new(count * 16)?;
    let erst = DmaBuf::new(16)?;

    unsafe {
        let e = erst.virt as *mut u64;
        e.add(0).write_volatile(buf.phys);
        e.add(1).write_volatile(count as u64);
    }

    Ok(Self { buf, erst, len: count, dequeue: 0, cycle: true })
}
// ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/ring.rs (line 113) - [public] erst_phys

```
pub fn erst_phys(&self) -> u64 {
    self.erst.phys
}

pub fn pending(&self) -> Option<Trb> {
    let t = unsafe {
        (self.buf.virt as *const Trb).add(self.dequeue).read_volatile()
    };

    if t.cycle() == self.cycle {
        Some(t)
    } else {
        None
    }
}
// ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/ring.rs (line 117) - [public] pending

```
pub fn pending(&self) -> Option<Trb> {
    let t = unsafe {
        (self.buf.virt as *const Trb).add(self.dequeue).read_volatile()
    };

    if t.cycle() == self.cycle {
        Some(t)
    } else {
        None
    }
}

// ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/ring.rs (line 129) - [public] pop

```
pub fn pop(&mut self) {
    self.dequeue += 1;

    if self.dequeue == self.len {
        self.dequeue = 0;
        self.cycle = !self.cycle;
    }
}

pub fn erdp(&self) -> u64 {
    (self.buf.phys + (self.dequeue as u64) * 16) & 0xFFFF_FFFF_FFFF_FF00
}
// ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/ring.rs (line 138) - [public] erdp

```
pub fn erdp(&self) -> u64 {
    (self.buf.phys + (self.dequeue as u64) * 16) & 0xFFFF_FFFF_FFFF_FF00
}
}
```

kernel-drivers/usb/src/host/xhci/trb.rs (line 29) - [public] Trb

```
pub struct Trb {
    pub param: u64,
    pub status: u32,
    pub control: u32,
}

impl Trb {
    pub fn typ(&self) -> u32 {
        (self.control >> 10) & 0x3F
    }

    pub fn cycle(&self) -> bool {
        // ... (truncated, see source file for more)
    }
}
```

kernel-drivers/usb/src/host/xhci/trb.rs (line 36) - [public] typ

```
pub fn typ(&self) -> u32 {
    (self.control >> 10) & 0x3F
}

pub fn cycle(&self) -> bool {
    self.control & 1 != 0
}

pub fn completion_code(&self) -> u8 {
    (self.status >> 24) as u8
}

// ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/trb.rs (line 40) - [public] cycle

```
pub fn cycle(&self) -> bool {
    self.control & 1 != 0
}

pub fn completion_code(&self) -> u8 {
    (self.status >> 24) as u8
}

pub fn slot_id(&self) -> u8 {
    (self.control >> 24) as u8
}

// ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/trb.rs (line 44) - [public] completion_code

```
pub fn completion_code(&self) -> u8 {
    (self.status >> 24) as u8
}

pub fn slot_id(&self) -> u8 {
    (self.control >> 24) as u8
}

pub fn ep_id(&self) -> u8 {
    ((self.control >> 16) & 0x1F) as u8
}

// ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/trb.rs (line 48) - [public] slot_id

```
pub fn slot_id(&self) -> u8 {
    (self.control >> 24) as u8
}

pub fn ep_id(&self) -> u8 {
    ((self.control >> 16) & 0x1F) as u8
}

pub fn transfer_len(&self) -> u32 {
    self.status & 0x00FF_FFFF
}

// ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/trb.rs (line 52) - [public] ep_id

```
pub fn ep_id(&self) -> u8 {
    ((self.control >> 16) & 0x1F) as u8
}

pub fn transfer_len(&self) -> u32 {
    self.status & 0x00FF_FFFF
}

pub fn link(addr: u64) -> Self {
    Trb {
        param: addr & ADDR_MASK,
        status: 0,
    }
    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/trb.rs (line 56) - [public] transfer_len

```
pub fn transfer_len(&self) -> u32 {
    self.status & 0x00FF_FFFF
}

pub fn link(addr: u64) -> Self {
    Trb {
        param: addr & ADDR_MASK,
        status: 0,
        control: (TRB_LINK << 10) | (1 << 1),
    }
}

// ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/trb.rs (line 60) - [public] link

```
pub fn link(addr: u64) -> Self {
    Trb {
        param: addr & ADDR_MASK,
        status: 0,
        control: (TRB_LINK << 10) | (1 << 1),
    }
}

pub fn enable_slot() -> Self {
    Trb { control: TRB_ENABLE_SLOT << 10, ..Default::default() }
}

// ... (truncated, see source file for more)
```

```

kernel-drivers/usb/src/host/xhci/trb.rs (line 68) - [public] enable_slot
    pub fn enable_slot() -> Self {
        Trb { control: TRB_ENABLE_SLOT << 10, ..Default::default() }
    }

    pub fn address_device(slot: u8, ctx_phys: u64) -> Self {
        Trb {
            param: ctx_phys & ADDR_MASK,
            status: 0,
            control: (TRB_ADDRESS_DEVICE << 10) | ((slot as u32) << 24),
        }
    }

    // ... (truncated, see source file for more)

kernel-drivers/usb/src/host/xhci/trb.rs (line 72) - [public] address_device
    pub fn address_device(slot: u8, ctx_phys: u64) -> Self {
        Trb {
            param: ctx_phys & ADDR_MASK,
            status: 0,
            control: (TRB_ADDRESS_DEVICE << 10) | ((slot as u32) << 24),
        }
    }

    pub fn configure_ep(slot: u8, ctx_phys: u64) -> Self {
        Trb {
            param: ctx_phys & ADDR_MASK,
            status: 0,
        }
        // ... (truncated, see source file for more)

kernel-drivers/usb/src/host/xhci/trb.rs (line 80) - [public] configure_ep
    pub fn configure_ep(slot: u8, ctx_phys: u64) -> Self {
        Trb {
            param: ctx_phys & ADDR_MASK,
            status: 0,
            control: (TRB_CONFIGURE_EP << 10) | ((slot as u32) << 24),
        }
    }

    pub fn noop_cmd() -> Self {
        Trb { control: TRB_NOOP_CMD << 10, ..Default::default() }
    }

    // ... (truncated, see source file for more)

kernel-drivers/usb/src/host/xhci/trb.rs (line 88) - [public] noop_cmd
    pub fn noop_cmd() -> Self {
        Trb { control: TRB_NOOP_CMD << 10, ..Default::default() }
    }

    pub fn setup_stage(raw_setup: u64, trt: u32) -> Self {
        Trb {
            param: raw_setup,
            status: 0,
            control: (TRB_SETUP << 10) | (1 << 6) | ((trt & 3) << 16),
        }
    }

    // ... (truncated, see source file for more)

```

kernel-drivers/usb/src/host/xhci/trb.rs (line 92) - [public] setup_stage

```
pub fn setup_stage(raw_setup: u64, trt: u32) -> Self {
    Trb {
        param: raw_setup,
        status: 0,
        control: (TRB_SETUP << 10) | (1 << 6) | ((trt & 3) << 16),
    }
}

pub fn data_stage(addr: u64, len: u32, dir_in: bool) -> Self {
    Trb {
        param: addr & ADDR_MASK,
        status: len & 0xFFFF,
        // ... (truncated, see source file for more)
    }
}
```

kernel-drivers/usb/src/host/xhci/trb.rs (line 100) - [public] data_stage

```
pub fn data_stage(addr: u64, len: u32, dir_in: bool) -> Self {
    Trb {
        param: addr & ADDR_MASK,
        status: len & 0xFFFF,
        control: (TRB_DATA << 10) | ((dir_in as u32) << 16),
    }
}

pub fn status_stage(dir_in: bool) -> Self {
    Trb {
        param: 0,
        status: 0,
        // ... (truncated, see source file for more)
    }
}
```

kernel-drivers/usb/src/host/xhci/trb.rs (line 108) - [public] status_stage

```
pub fn status_stage(dir_in: bool) -> Self {
    Trb {
        param: 0,
        status: 0,
        control: (TRB_STATUS << 10) | ((dir_in as u32) << 16) | (1 << 5),
    }
}

pub fn normal(addr: u64, len: u32) -> Self {
    Trb {
        param: addr & ADDR_MASK,
        status: len & 0xFFFF,
        // ... (truncated, see source file for more)
    }
}
```

kernel-drivers/usb/src/host/xhci/trb.rs (line 116) - [public] normal

```
pub fn normal(addr: u64, len: u32) -> Self {
    Trb {
        param: addr & ADDR_MASK,
        status: len & 0xFFFF,
        control: (TRB_NORMAL << 10) | (1 << 5),
    }
}

pub fn evaluate_ctx(slot: u8, ctx_phys: u64) -> Self {
    Trb {
        param: ctx_phys & ADDR_MASK,
        status: 0,
        // ... (truncated, see source file for more)
    }
}
```

kernel-drivers/usb/src/host/xhci/trb.rs (line 124) - [public] evaluate_ctx

```
pub fn evaluate_ctx(slot: u8, ctx_phys: u64) -> Self {
    Trb {
        param: ctx_phys & ADDR_MASK,
        status: 0,
        control: (TRB_EVALUATE_CTX << 10) | ((slot as u32) << 24),
    }
}

pub fn pack_setup(bm_request: u8, b_request: u8, value: u16,
    index: u16, length: u16) -> u64 {
    (bm_request as u64)
    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/host/xhci/trb.rs (line 133) - [public] pack_setup

```
pub fn pack_setup(bm_request: u8, b_request: u8, value: u16,
    index: u16, length: u16) -> u64 {
    (bm_request as u64)
    | ((b_request as u64) << 8)
    | ((value as u64) << 16)
    | ((index as u64) << 32)
    | ((length as u64) << 48)
}
```

kernel-drivers/usb/src/lib.rs (line 23) - [public] UsbError

```
pub enum UsbError {
    NoController,
    MapFailed,
    Timeout,
    NotReady,
    BadDescriptor,
    Invalid,
    Transfer(u8),
}

static mut CONTROLLER: Option<Xhci> = None;

// ... (truncated, see source file for more)
```

kernel-drivers/usb/src/lib.rs (line 36) - [public] init

/// Probe the PCI bus for an xHCI controller and bring it online.

```
pub fn init() -> Result<(), UsbError> {
    if unsafe { CONTROLLER.is_some() } {
        return Ok(());
    }

    let xhci_pci = pci_glue::find_xhci()?;
    hw_sys::log::info("usb: xHCI controller found");

    let regs = host::xhci::regs::XhciRegs::new(xhci_pci.bar0_phys?);

    let mut xhci = host::xhci::init::init(regs)?;
    xhci.scan_ports();
    // ... (truncated, see source file for more)
```

kernel-drivers/usb/src/lib.rs (line 57) - [public] poll

/// Poll attached HID devices and drain controller events.

```
pub fn poll() {
    if let Some(x) = unsafe { CONTROLLER.as_mut() } {
        class::hid::poll(x);
    }
}

/// Borrow the active controller (when it is online).
pub fn with_controller<F: FnOnce(&mut Xhci) -> R, R>(f: F) -> Option<R> {
    unsafe { CONTROLLER.as_mut().map(f) }
}
```

kernel-drivers/usb/src/lib.rs (line 64) - [public] with_controller

/// Borrow the active controller (when it is online).

```
pub fn with_controller<F: FnOnce(&mut Xhci) -> R, R>(f: F) -> Option<R> {
    unsafe { CONTROLLER.as_mut().map(f) }
}
```

kernel-drivers/usb/src/pci_glue.rs (line 8) - [public] XhciPci

```
pub struct XhciPci {
    pub dev: PciDev,
    pub bar0_phys: u64,
}

pub fn find_xhci() -> Result<XhciPci, UsbError> {
    let dev = pcie::find_class(XHCI_CLASS, XHCI_SUBCLASS, XHCI_PROGIF)
        .ok_or(UsbError::NoController)?;

    let bar0 = dev.bar(0);

    if bar0 == 0 {
        // ... (truncated, see source file for more)
    }
}
```

kernel-drivers/usb/src/pci_glue.rs (line 13) - [public] find_xhci

```
pub fn find_xhci() -> Result<XhciPci, UsbError> {
    let dev = pcie::find_class(XHCI_CLASS, XHCI_SUBCLASS, XHCI_PROGIF)
        .ok_or(UsbError::NoController)?;

    let bar0 = dev.bar(0);

    if bar0 == 0 {
        return Err(UsbError::NoController);
    }

    dev.enable_mmio();

    // ... (truncated, see source file for more)
}
```

kernel_Workspace/C_base/kc-abi-menager/src/main.rs (line 2) - [internal] main

```
fn main() {
    println!("Hello, world!");
}
```

kernel_Workspace/C_base/kc-abi/src/lib.rs (line 2) - [public] add

```
pub fn add(left: u64, right: u64) -> u64 {
    left + right
}

#[cfg(test)]
mod tests {
    use super::*;

    #[test]
    fn it_works() {
        let result = add(2, 2);
        assert_eq!(result, 4);
        // ... (truncated, see source file for more)
    }
}
```

kernel_Workspace/C_base/kc-abi/src/lib.rs (line 11) - [internal] it_works

```
fn it_works() {
    let result = add(2, 2);
    assert_eq!(result, 4);
}

}
```


kernel_Workspace/Odin_base/odin-abi-bridge/src/lib.rs (line 2) - [public] add

```
pub fn add(left: u64, right: u64) -> u64 {
    left + right
}

#[cfg(test)]
mod tests {
    use super::*;

    #[test]
    fn it_works() {
        let result = add(2, 2);
        assert_eq!(result, 4);
        // ... (truncated, see source file for more)
    }
}
```

kernel_Workspace/Odin_base/odin-abi-bridge/src/lib.rs (line 11) - [internal] it_works

```
fn it_works() {
    let result = add(2, 2);
    assert_eq!(result, 4);
}
```

kernel_Workspace/Odin_base/odin-abi/src/lib.rs (line 2) - [public] add

```
pub fn add(left: u64, right: u64) -> u64 {
    left + right
}

#[cfg(test)]
mod tests {
    use super::*;

    #[test]
    fn it_works() {
        let result = add(2, 2);
        assert_eq!(result, 4);
        // ... (truncated, see source file for more)
    }
}
```

kernel_Workspace/Odin_base/odin-abi/src/lib.rs (line 11) - [internal] it_works

```
fn it_works() {
    let result = add(2, 2);
    assert_eq!(result, 4);
}
```

kernel_Workspace/Odin_base/odin-bin-loader/src/main.rs (line 2) - [internal] main

```
fn main() {
    println!("Hello, world!");
}
```

kernel_Workspace/base/kw-C-abi/src/lib.rs (line 2) - [public] add

```
pub fn add(left: u64, right: u64) -> u64 {
    left + right
}

#[cfg(test)]
mod tests {
    use super::*;

    #[test]
    fn it_works() {
        let result = add(2, 2);
        assert_eq!(result, 4);
        // ... (truncated, see source file for more)
    }
}
```

kernel_Workspace/base/kw-C-abi/src/lib.rs (line 11) - [internal] it_works

```
fn it_works() {
    let result = add(2, 2);
    assert_eq!(result, 4);
}
```

kernel_workspace/base/kw-base/src/api/dispatch.rs (line 21) - [public] dispatch

// Main syscall dispatcher. Called from the trampoline/assembly entry point.

// Returns i64: positive for success/data, negative for KernelError codes.

```
pub fn dispatch(num: u64, a0: u64, a1: u64, a2: u64, a3: u64) -> i64 {
    let res: Result<i64, KernelError> = match num {
        SYS_READ => fs_ops::read(a0 as u32, a1 as *mut u8, a2 as usize).map(|v| v as i64),
        SYS_WRITE => fs_ops::write(a0 as u32, a1 as *const u8, a2 as usize).map(|v| v as i64),
        SYS_OPEN => fs_ops::open(a0 as *const u8, a1 as u32).map(|v| v as i64),
        SYS_CLOSE => fs_ops::close(a0 as u32).map(|_| 0),

        SYS_MMAP => memory::mmap(a0, a1 as usize, a2 as u32, a3 as u32).map(|v| v as i64),
        SYS_MUNMAP => memory::munmap(a0, a1 as usize).map(|_| 0),

        SYS_SPAWN => process::spawn(a0 as u32, a1, a2).map(|v| v as i64),
        SYS_EXIT => { process::exit(a0 as i32); },
        // ... (truncated, see source file for more)
    }
}
```

kernel_workspace/base/kw-base/src/api/fs_ops.rs (line 4) - [public] open

```
pub fn open(path_ptr: *const u8, flags: u32) -> KResult<u32> {
    if path_ptr.is_null() { return Err(KernelError::BadAddress); }
    let path = unsafe { core::ffi::CStr::from_ptr(path_ptr as *const i8) }
        .to_str().map_err(|_| KernelError::InvalidArg)?;
    let handle = Vfs::open(path, flags)?;
    Ok(handle.0)
}

pub fn read(fd: u32, buf: *mut u8, len: usize) -> KResult<usize> {
    if buf.is_null() { return Err(KernelError::BadAddress); }
    let slice = unsafe { core::slice::from_raw_parts_mut(buf, len) };
    Vfs::read(Handle(fd as i32), slice)
    // ... (truncated, see source file for more)
}
```

kernel_workspace/base/kw-base/src/api/fs_ops.rs (line 12) - [public] read

```
pub fn read(fd: u32, buf: *mut u8, len: usize) -> KResult<usize> {
    if buf.is_null() { return Err(KernelError::BadAddress); }
    let slice = unsafe { core::slice::from_raw_parts_mut(buf, len) };
    Vfs::read(Handle(fd as i32), slice)
}

pub fn write(fd: u32, buf: *const u8, len: usize) -> KResult<usize> {
    if buf.is_null() { return Err(KernelError::BadAddress); }
    let slice = unsafe { core::slice::from_raw_parts(buf, len) };
    Vfs::write(Handle(fd as i32), slice)
}

// ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/api/fs_ops.rs (line 18) - [public] write

```
pub fn write(fd: u32, buf: *const u8, len: usize) -> KResult<usize> {
    if buf.is_null() { return Err(KernelError::BadAddress); }
    let slice = unsafe { core::slice::from_raw_parts(buf, len) };
    Vfs::write(Handle(fd as i32), slice)
}

pub fn close(fd: u32) -> KResult<()> {
    extern "C" { fn k_fs_close(fd: i32) -> i32; }
    let rc = unsafe { k_fs_close(fd as i32) };
    if rc < 0 { Err(KernelError::from_code(rc)) } else { Ok(()) }
}
```

kernel_workspace/base/kw-base/src/api/fs_ops.rs (line 24) - [public] close

```
pub fn close(fd: u32) -> KResult<()> {
    extern "C" { fn k_fs_close(fd: i32) -> i32; }
    let rc = unsafe { k_fs_close(fd as i32) };
    if rc < 0 { Err(KernelError::from_code(rc)) } else { Ok(()) }
}
```

kernel_workspace/base/kw-base/src/api/memory.rs (line 8) - [public] mmap

```
pub fn mmap(addr: u64, len: usize, prot: u32, _flags: u32) -> KResult<u64> {
    if len == 0 { return Err(KernelError::InvalidArg); }

    let mut vmm_flags = 0;
    if prot & PROT_READ != 0 { vmm_flags |= vmm::VMM_FLAG_READ; }
    if prot & PROT_WRITE != 0 { vmm_flags |= vmm::VMM_FLAG_WRITE; }
    if prot & PROT_EXEC != 0 { vmm_flags |= vmm::VMM_FLAG_EXEC; }

    let virt = vmm::alloc(len, vmm_flags).map_err(KernelError::from)?;
    Ok(virt.0)
}

// ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/api/memory.rs (line 20) - [public] munmap

```
pub fn munmap(addr: u64, len: usize) -> KResult<()> {
    if len == 0 { return Err(KernelError::InvalidArg); }
    vmm::free(kstd_base::VirtAddr(addr), len).map_err(KernelError::from)
}
```

kernel_workspace/base/kw-base/src/api/net_ops.rs (line 4) - [internal] k_net_socket

```
fn k_net_socket(domain: u32, sock_type: u32, proto: u32) -> i32;
fn k_net_bind(fd: i32, addr: *const u8, len: u32) -> i32;
fn k_net_send(fd: i32, buf: *const u8, len: u32, flags: u32) -> i32;
fn k_net_recv(fd: i32, buf: *mut u8, len: u32, flags: u32) -> i32;

pub fn socket(domain: u32, sock_type: u32, proto: u32) -> KResult<u32> {
    let rc = unsafe { k_net_socket(domain, sock_type, proto) };
    if rc < 0 { Err(KernelError::from_code(rc)) } else { Ok(rc as u32) }
}

pub fn send(fd: u32, buf: *const u8, len: u32, flags: u32) -> KResult<usize> {
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/api/net_ops.rs (line 5) - [internal] k_net_bind

```
fn k_net_bind(fd: i32, addr: *const u8, len: u32) -> i32;
fn k_net_send(fd: i32, buf: *const u8, len: u32, flags: u32) -> i32;
fn k_net_recv(fd: i32, buf: *mut u8, len: u32, flags: u32) -> i32;

pub fn socket(domain: u32, sock_type: u32, proto: u32) -> KResult<u32> {
    let rc = unsafe { k_net_socket(domain, sock_type, proto) };
    if rc < 0 { Err(KernelError::from_code(rc)) } else { Ok(rc as u32) }
}

pub fn send(fd: u32, buf: *const u8, len: u32, flags: u32) -> KResult<usize> {
    if buf.is_null() { return Err(KernelError::BadAddress); }
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/api/net_ops.rs (line 6) - [internal] k_net_send

```
fn k_net_send(fd: i32, buf: *const u8, len: u32, flags: u32) -> i32;
fn k_net_recv(fd: i32, buf: *mut u8, len: u32, flags: u32) -> i32;

pub fn socket(domain: u32, sock_type: u32, proto: u32) -> KResult<u32> {
    let rc = unsafe { k_net_socket(domain, sock_type, proto) };
    if rc < 0 { Err(KernelError::from_code(rc)) } else { Ok(rc as u32) }
}

pub fn send(fd: u32, buf: *const u8, len: u32, flags: u32) -> KResult<usize> {
    if buf.is_null() { return Err(KernelError::BadAddress); }
    let rc = unsafe { k_net_send(fd as i32, buf, len, flags) };
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/api/net_ops.rs (line 7) - [internal] k_net_recv

```
fn k_net_recv(fd: i32, buf: *mut u8, len: u32, flags: u32) -> i32;
}

pub fn socket(domain: u32, sock_type: u32, proto: u32) -> KResult<u32> {
    let rc = unsafe { k_net_socket(domain, sock_type, proto) };
    if rc < 0 { Err(KernelError::from_code(rc)) } else { Ok(rc as u32) }
}

pub fn send(fd: u32, buf: *const u8, len: u32, flags: u32) -> KResult<usize> {
    if buf.is_null() { return Err(KernelError::BadAddress); }
    let rc = unsafe { k_net_send(fd as i32, buf, len, flags) };
    if rc < 0 { Err(KernelError::from_code(rc)) } else { Ok(rc as usize) }
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/base/kw-base/src/api/net_ops.rs (line 10) - [public] socket

```
pub fn socket(domain: u32, sock_type: u32, proto: u32) -> KResult<u32> {
    let rc = unsafe { k_net_socket(domain, sock_type, proto) };
    if rc < 0 { Err(KernelError::from_code(rc)) } else { Ok(rc as u32) }
}

pub fn send(fd: u32, buf: *const u8, len: u32, flags: u32) -> KResult<usize> {
    if buf.is_null() { return Err(KernelError::BadAddress); }
    let rc = unsafe { k_net_send(fd as i32, buf, len, flags) };
    if rc < 0 { Err(KernelError::from_code(rc)) } else { Ok(rc as usize) }
}

pub fn recv(fd: u32, buf: *mut u8, len: u32, flags: u32) -> KResult<usize> {
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/base/kw-base/src/api/net_ops.rs (line 15) - [public] send

```
pub fn send(fd: u32, buf: *const u8, len: u32, flags: u32) -> KResult<usize> {
    if buf.is_null() { return Err(KernelError::BadAddress); }
    let rc = unsafe { k_net_send(fd as i32, buf, len, flags) };
    if rc < 0 { Err(KernelError::from_code(rc)) } else { Ok(rc as usize) }
}

pub fn recv(fd: u32, buf: *mut u8, len: u32, flags: u32) -> KResult<usize> {
    if buf.is_null() { return Err(KernelError::BadAddress); }
    let rc = unsafe { k_net_recv(fd as i32, buf, len, flags) };
    if rc < 0 { Err(KernelError::from_code(rc)) } else { Ok(rc as usize) }
}
```

kernel_Workspace/base/kw-base/src/api/net_ops.rs (line 21) - [public] recv

```
pub fn recv(fd: u32, buf: *mut u8, len: u32, flags: u32) -> KResult<usize> {
    if buf.is_null() { return Err(KernelError::BadAddress); }
    let rc = unsafe { k_net_recv(fd as i32, buf, len, flags) };
    if rc < 0 { Err(KernelError::from_code(rc)) } else { Ok(rc as usize) }
}
```

kernel_Workspace/base/kw-base/src/api/process.rs (line 4) - [internal] k_proc_spawn

```
fn k_proc_spawn(world_id: u32, entry: u64, stack: u64) -> i64;
fn k_proc_exit(code: i32);
fn k_proc_yield();
fn k_proc_get_pid() -> u32;
}

pub fn spawn(world_id: u32, entry: u64, stack: u64) -> KResult<u64> {
    let rc = unsafe { k_proc_spawn(world_id, entry, stack) };
    if rc < 0 { Err(KernelError::from_code(rc as i32)) } else { Ok(rc as u64) }
}

pub fn exit(code: i32) -> ! {
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/base/kw-base/src/api/process.rs (line 5) - [internal] k_proc_exit

```
fn k_proc_exit(code: i32);
fn k_proc_yield();
fn k_proc_get_pid() -> u32;
}

pub fn spawn(world_id: u32, entry: u64, stack: u64) -> KResult<u64> {
    let rc = unsafe { k_proc_spawn(world_id, entry, stack) };
    if rc < 0 { Err(KernelError::from_code(rc as i32)) } else { Ok(rc as u64) }
}

pub fn exit(code: i32) -> ! {
    unsafe { k_proc_exit(code); }
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/api/process.rs (line 6) - [internal] k_proc_yield

```
fn k_proc_yield();
fn k_proc_get_pid() -> u32;
}

pub fn spawn(world_id: u32, entry: u64, stack: u64) -> KResult<u64> {
    let rc = unsafe { k_proc_spawn(world_id, entry, stack) };
    if rc < 0 { Err(KernelError::from_code(rc as i32)) } else { Ok(rc as u64) }
}

pub fn exit(code: i32) -> ! {
    unsafe { k_proc_exit(code); }
    loop { core::hint::spin_loop(); }
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/api/process.rs (line 7) - [internal] k_proc_get_pid

```
fn k_proc_get_pid() -> u32;
}

pub fn spawn(world_id: u32, entry: u64, stack: u64) -> KResult<u64> {
    let rc = unsafe { k_proc_spawn(world_id, entry, stack) };
    if rc < 0 { Err(KernelError::from_code(rc as i32)) } else { Ok(rc as u64) }
}

pub fn exit(code: i32) -> ! {
    unsafe { k_proc_exit(code); }
    loop { core::hint::spin_loop(); }
}

// ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/api/process.rs (line 10) - [public] spawn

```
pub fn spawn(world_id: u32, entry: u64, stack: u64) -> KResult<u64> {
    let rc = unsafe { k_proc_spawn(world_id, entry, stack) };
    if rc < 0 { Err(KernelError::from_code(rc as i32)) } else { Ok(rc as u64) }
}

pub fn exit(code: i32) -> ! {
    unsafe { k_proc_exit(code); }
    loop { core::hint::spin_loop(); }
}

pub fn yield_now() {
    unsafe { k_proc_yield(); }
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/api/process.rs (line 15) - [public] exit

```
pub fn exit(code: i32) -> ! {
    unsafe { k_proc_exit(code); }
    loop { core::hint::spin_loop(); }
}

pub fn yield_now() {
    unsafe { k_proc_yield(); }
}

pub fn get_pid() -> u32 {
    unsafe { k_proc_get_pid() }
}
```

kernel_workspace/base/kw-base/src/api/process.rs (line 20) - [public] yield_now

```
pub fn yield_now() {
    unsafe { k_proc_yield(); }
}

pub fn get_pid() -> u32 {
    unsafe { k_proc_get_pid() }
}
```

kernel_workspace/base/kw-base/src/api/process.rs (line 24) - [public] get_pid

```
pub fn get_pid() -> u32 {
    unsafe { k_proc_get_pid() }
}
```

kernel_workspace/base/kw-base/src/arch/interrupts.rs (line 4) - [public] disable

```
pub fn disable() {
    #[cfg(target_arch = "x86_64")]
    unsafe { core::arch::asm!("cli", options(nomem, nostack)); }
}

#[inline(always)]
pub fn enable() {
    #[cfg(target_arch = "x86_64")]
    unsafe { core::arch::asm!("sti", options(nomem, nostack)); }
}

#[inline(always)]
// ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/arch/interrupts.rs (line 10) - [public] enable

```
pub fn enable() {
    #[cfg(target_arch = "x86_64")]
    unsafe { core::arch::asm!("sti", options(nomem, nostack)); }
}

#[inline(always)]
pub fn halt() {
    unsafe { core::arch::asm!("hlt", options(nomem, nostack)); }
}

#[inline(always)]
pub fn are_enabled() -> bool {
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/arch/interrupts.rs (line 16) - [public] halt

```
pub fn halt() {
    unsafe { core::arch::asm!("hlt", options(nomem, nostack)); }
}

#[inline(always)]
pub fn are_enabled() -> bool {
    #[cfg(target_arch = "x86_64")]
    {
        let flags: u64;
        unsafe { core::arch::asm!("pushfq; pop {}", out(reg) flags, options(nomem)); }
        (flags & (1 << 9)) != 0
    }
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/base/kw-base/src/arch/interrupts.rs (line 21) - [public] are_enabled

```
pub fn are_enabled() -> bool {
    #[cfg(target_arch = "x86_64")]
    {
        let flags: u64;
        unsafe { core::arch::asm!("pushfq; pop {}", out(reg) flags, options(nomem)); }
        (flags & (1 << 9)) != 0
    }
    #[cfg(not(target_arch = "x86_64"))]
    { false }
}

pub struct InterruptGuard {
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/base/kw-base/src/arch/interrupts.rs (line 32) - [public] InterruptGuard

```
pub struct InterruptGuard {
    was_enabled: bool,
}

impl InterruptGuard {
    pub fn disable() -> Self {
        let was_enabled = are_enabled();
        disable();
        Self { was_enabled }
    }
}

// ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/arch/interrupts.rs (line 37) - [public] disable

```
pub fn disable() -> Self {
    let was_enabled = are_enabled();
    disable();
    Self { was_enabled }
}

impl Drop for InterruptGuard {
    fn drop(&mut self) {
        if self.was_enabled { enable(); }
    }
}
```

kernel_Workspace/base/kw-base/src/arch/interrupts.rs (line 45) - [internal] drop

```
fn drop(&mut self) {
    if self.was_enabled { enable(); }
}

}
```

kernel_Workspace/base/kw-base/src/arch/registers.rs (line 3) - [public] CpuContext

```
pub struct CpuContext {
    pub r15: u64, pub r14: u64, pub r13: u64, pub r12: u64,
    pub rbp: u64, pub rbx: u64,
    pub r11: u64, pub r10: u64, pub r9: u64, pub r8: u64,
    pub rax: u64, pub rcx: u64, pub rdx: u64, pub rsi: u64,
    pub rdi: u64,
    pub rip: u64,
    pub cs: u64,
    pub rflags: u64,
    pub rsp: u64,
    pub ss: u64,
}
```

kernel_Workspace/base/kw-base/src/arch/x86_64.rs (line 30) - [public] read_cr3

```
pub fn read_cr3() -> u64 {
    let val: u64;
    unsafe { asm!("mov {}, cr3", out(reg) val, options(nomem, nostack, preserves_flags)); }
    val
}

#[inline(always)]
pub fn write_cr3(val: u64) {
    unsafe { asm!("mov cr3, {}", in(reg) val, options(nomem, nostack, preserves_flags)); }
}
```

kernel_Workspace/base/kw-base/src/arch/x86_64.rs (line 37) - [public] write_cr3

```
pub fn write_cr3(val: u64) {
    unsafe { asm!("mov cr3, {}", in(reg) val, options(nomem, nostack, preserves_flags)); }
}
```

kernel_Workspace/base/kw-base/src/boot/early_print.rs (line 5) - [public] init_serial

```
pub fn init_serial() {
    unsafe {
        outb(COM1 + 1, 0x00); // Disable interrupts
        outb(COM1 + 3, 0x80); // Enable DLAB
        outb(COM1 + 0, 0x03); // 38400 baud
        outb(COM1 + 1, 0x00);
        outb(COM1 + 3, 0x03); // 8 bits, no parity, one stop bit
        outb(COM1 + 2, 0xC7); // Enable FIFO
        outb(COM1 + 4, 0x0B); // IRQs enabled, RTS/DSR set
    }

    // ... (truncated, see source file for more)
}
```

kernel_Workspace/base/kw-base/src/boot/early_print.rs (line 17) - [internal]

is_transmitter_empty

```
fn is_transmitter_empty() -> bool {
    unsafe { (inb(COM1 + 5) & 0x20) != 0 }
}

pub fn print_char(c: u8) {
    while !is_transmitter_empty() { core::hint::spin_loop(); }
    unsafe { outb(COM1, c); }
}

pub fn print_str(s: &str) {
    for b in s.bytes() {
        if b == b'\n' { print_char(b'\r'); }
    }
    // ... (truncated, see source file for more)
}
```


kernel_workspace/base/kw-base/src/boot/early_print.rs (line 21) - [public] print_char

```
pub fn print_char(c: u8) {
    while !is_transmitter_empty() { core::hint::spin_loop(); }
    unsafe { outb(COM1, c); }
}

pub fn print_str(s: &str) {
    for b in s.bytes() {
        if b == b'\n' { print_char(b'\r'); }
        print_char(b);
    }
}
```

kernel_workspace/base/kw-base/src/boot/early_print.rs (line 26) - [public] print_str

```
pub fn print_str(s: &str) {
    for b in s.bytes() {
        if b == b'\n' { print_char(b'\r'); }
        print_char(b);
    }
}
```

kernel_workspace/base/kw-base/src/boot/memory_detect.rs (line 5) - [public] MemoryRegion

```
pub struct MemoryRegion {
    pub base: PhysAddr,
    pub size: Size,
    pub is_usable: bool,
}

pub fn extract_memory_map(info: &BootInfo) -> (usize, [MemoryRegion; 32]) {
    let mut regions = [MemoryRegion { base: PhysAddr(0), size: Size(0), is_usable: false }; 32];
    let mut count = 0;

    let tag_ptr = info.find_tag(TAG_TYPE_MMAP);
    if tag_ptr.is_null() { return (0, regions); }
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/boot/memory_detect.rs (line 11) - [public]

extract_memory_map

```
pub fn extract_memory_map(info: &BootInfo) -> (usize, [MemoryRegion; 32]) {
    let mut regions = [MemoryRegion { base: PhysAddr(0), size: Size(0), is_usable: false }; 32];
    let mut count = 0;

    let tag_ptr = info.find_tag(TAG_TYPE_MMAP);
    if tag_ptr.is_null() { return (0, regions); }

    let mmap = unsafe { &*(tag_ptr as *const MmapTag) };
    let mut entry_ptr = unsafe { (tag_ptr as *const u8).add(core::mem::size_of::<MmapTag>()) };
    let end_ptr = unsafe { (tag_ptr as *const u8).add(mmap.tag.size as usize) };

    while entry_ptr < end_ptr && count < 32 {
        // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/boot/multiboot.rs (line 6) - [public] Multiboot2Info

```
pub struct Multiboot2Info {
    pub total_size: u32,
    pub reserved: u32,
}

#[repr(C)]
pub struct Multiboot2Tag {
    pub tag_type: u32,
    pub size: u32,
}

pub const TAG_TYPE_END: u32 = 0;
// ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/boot/multiboot.rs (line 12) - [public] Multiboot2Tag

```
pub struct Multiboot2Tag {
    pub tag_type: u32,
    pub size: u32,
}

pub const TAG_TYPE_END: u32 = 0;
pub const TAG_TYPE_CMDLINE: u32 = 1;
pub const TAG_TYPE_BOOTLOADER: u32 = 2;
pub const TAG_TYPE_MODULE: u32 = 3;
pub const TAG_TYPE_BASIC_MEMINFO: u32 = 4;
pub const TAG_TYPE_BOOTDEV: u32 = 5;
pub const TAG_TYPE_MMAP: u32 = 6;
// ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/boot/multiboot.rs (line 27) - [public] MmapTag

```
pub struct MmapTag {
    pub tag: Multiboot2Tag,
    pub entry_size: u32,
    pub entry_version: u32,
}

#[repr(C)]
pub struct MmapEntry {
    pub base_addr: u64,
    pub length: u64,
    pub entry_type: u32,
    pub reserved: u32,
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/boot/multiboot.rs (line 34) - [public] MmapEntry

```
pub struct MmapEntry {
    pub base_addr: u64,
    pub length: u64,
    pub entry_type: u32,
    pub reserved: u32,
}

pub struct BootInfo {
    pub ptr: *const Multiboot2Info,
}

unsafe impl Send for BootInfo {}
// ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/boot/multiboot.rs (line 41) - [public] BootInfo

```
pub struct BootInfo {
    pub ptr: *const Multiboot2Info,
}

unsafe impl Send for BootInfo {}
unsafe impl Sync for BootInfo {}

impl BootInfo {
    pub unsafe fn from_ptr(ptr: *const u8) -> Self {
        Self { ptr: ptr as *const Multiboot2Info }
    }

    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/boot/multiboot.rs (line 53) - [public] first_tag

```
pub fn first_tag(&self) -> *const Multiboot2Tag {
    unsafe { (self.ptr as *const u8).add(8) as *const Multiboot2Tag }
}

pub fn next_tag(tag: *const Multiboot2Tag) -> *const Multiboot2Tag {
    unsafe {
        let addr = tag as usize;
        let next = (addr + (*tag).size as usize + 7) & !7;
        next as *const Multiboot2Tag
    }
}

// ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/boot/multiboot.rs (line 57) - [public] next_tag

```
pub fn next_tag(tag: *const Multiboot2Tag) -> *const Multiboot2Tag {
    unsafe {
        let addr = tag as usize;
        let next = (addr + (*tag).size as usize + 7) & !7;
        next as *const Multiboot2Tag
    }
}

pub fn find_tag(&self, tag_type: u32) -> *const Multiboot2Tag {
    let mut tag = self.first_tag();
    unsafe {
        while (*tag).tag_type != TAG_TYPE_END {
            // ... (truncated, see source file for more)
        }
    }
}
```

kernel_Workspace/base/kw-base/src/boot/multiboot.rs (line 65) - [public] find_tag

```
pub fn find_tag(&self, tag_type: u32) -> *const Multiboot2Tag {
    let mut tag = self.first_tag();
    unsafe {
        while (*tag).tag_type != TAG_TYPE_END {
            if (*tag).tag_type == tag_type { return tag; }
            tag = Self::next_tag(tag);
        }
    }
    core::ptr::null()
}
```

kernel_Workspace/base/kw-base/src/boot/stack.rs (line 2) - [public] BootStack

```
pub struct BootStack {
    pub data: [u8; 65536],
}

impl BootStack {
    pub const fn new() -> Self {
        Self { data: [0; 65536] }
    }

    pub fn top(&self) -> u64 {
        (self.data.as_ptr() as u64) + (self.data.len() as u64)
    }
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/base/kw-base/src/boot/stack.rs (line 11) - [public] top

```
pub fn top(&self) -> u64 {
    (self.data.as_ptr() as u64) + (self.data.len() as u64)
}

}
```

kernel_Workspace/base/kw-base/src/core/error.rs (line 5) - [public] KernelError

```
pub enum KernelError {
    Ok = 0,
    NoMemory = -1,
    InvalidArg = -2,
    NotFound = -3,
    AlreadyExists = -4,
    PermissionDenied = -5,
    Busy = -6,
    Timeout = -7,
    NotSupported = -8,
    IOError = -9,
    BadAddress = -10,
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/core/error.rs (line 65) - [internal] from

```
fn from(s: kstd_base::Status) -> Self {
    match s {
        kstd_base::Status::Success => Self::Ok,
        kstd_base::Status::OutOfMemory => Self::NoMemory,
        kstd_base::Status::InvalidArgument => Self::InvalidArg,
        kstd_base::Status::NotFound => Self::NotFound,
        kstd_base::Status::AlreadyExists => Self::AlreadyExists,
        kstd_base::Status::PermissionDenied => Self::PermissionDenied,
        kstd_base::Status::Busy => Self::Busy,
        kstd_base::Status::TimedOut => Self::Timeout,
        kstd_base::Status::NotSupported => Self::NotSupported,
        kstd_base::Status::IOError => Self::IOError,
        // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/core/error.rs (line 84) - [internal] fmt

```
fn fmt(&self, f: &mut fmt::Formatter<'_>) -> fmt::Result {
    write!(f, "{:?}", self)
}
}
```

kernel_Workspace/base/kw-base/src/core/result.rs (line 5) - [public] IntoKResult

```
pub trait IntoKResult<T> {
    fn into_kresult(self) -> KResult<T>;
}

impl<T> IntoKResult<T> for Option<T> {
    fn into_kresult(self) -> KResult<T> {
        self.ok_or(KernelError::NotFound)
    }
}

impl<T> IntoKResult<T> for Result<T, kstd_base::Status> {
    fn into_kresult(self) -> KResult<T> {
        // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/core/result.rs (line 6) - [internal] into_kresult

```
fn into_kresult(self) -> KResult<T>;
}

impl<T> IntoKResult<T> for Option<T> {
    fn into_kresult(self) -> KResult<T> {
        self.ok_or(KernelError::NotFound)
    }
}

impl<T> IntoKResult<T> for Result<T, kstd_base::Status> {
    fn into_kresult(self) -> KResult<T> {
        self.map_err(KernelError::from)
        // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/core/result.rs (line 10) - [internal] into_kresult

```
    fn into_kresult(self) -> KResult<T> {
        self.ok_or(KernelError::NotFound)
    }
}

impl<T> IntoKResult<T> for Result<T, kstd_base::Status> {
    fn into_kresult(self) -> KResult<T> {
        self.map_err(KernelError::from)
    }
}
```

kernel_workspace/base/kw-base/src/core/result.rs (line 16) - [internal] into_kresult

```
    fn into_kresult(self) -> KResult<T> {
        self.map_err(KernelError::from)
    }
}
```

kernel_workspace/base/kw-base/src/core/traits.rs (line 4) - [public] BlockDevice

```
pub trait BlockDevice {
    fn read_block(&self, block: u64, buf: &mut [u8]) -> Result<(), KernelError>;
    fn write_block(&self, block: u64, buf: &[u8]) -> Result<(), KernelError>;
    fn block_size(&self) -> usize;
    fn block_count(&self) -> u64;
}

pub trait CharDevice {
    fn read(&self, buf: &mut [u8]) -> Result<usize, KernelError>;
    fn write(&self, buf: &[u8]) -> Result<usize, KernelError>;
}

// ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/core/traits.rs (line 5) - [internal] read_block

```
    fn read_block(&self, block: u64, buf: &mut [u8]) -> Result<(), KernelError>;
    fn write_block(&self, block: u64, buf: &[u8]) -> Result<(), KernelError>;
    fn block_size(&self) -> usize;
    fn block_count(&self) -> u64;
}

pub trait CharDevice {
    fn read(&self, buf: &mut [u8]) -> Result<usize, KernelError>;
    fn write(&self, buf: &[u8]) -> Result<usize, KernelError>;
}

pub trait NetworkDevice {
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/core/traits.rs (line 6) - [internal] write_block

```
    fn write_block(&self, block: u64, buf: &[u8]) -> Result<(), KernelError>;
    fn block_size(&self) -> usize;
    fn block_count(&self) -> u64;
}

pub trait CharDevice {
    fn read(&self, buf: &mut [u8]) -> Result<usize, KernelError>;
    fn write(&self, buf: &[u8]) -> Result<usize, KernelError>;
}

pub trait NetworkDevice {
    fn send(&self, frame: &[u8]) -> Result<(), KernelError>;
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/core/traits.rs (line 7) - [internal] block_size

```
    fn block_size(&self) -> usize;
    fn block_count(&self) -> u64;
}

pub trait CharDevice {
    fn read(&self, buf: &mut [u8]) -> Result<usize, KernelError>;
    fn write(&self, buf: &[u8]) -> Result<usize, KernelError>;
}

pub trait NetworkDevice {
    fn send(&self, frame: &[u8]) -> Result<(), KernelError>;
    fn recv(&self, buf: &mut [u8]) -> Result<usize, KernelError>;
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/core/traits.rs (line 8) - [internal] block_count

```
    fn block_count(&self) -> u64;
}

pub trait CharDevice {
    fn read(&self, buf: &mut [u8]) -> Result<usize, KernelError>;
    fn write(&self, buf: &[u8]) -> Result<usize, KernelError>;
}

pub trait NetworkDevice {
    fn send(&self, frame: &[u8]) -> Result<(), KernelError>;
    fn recv(&self, buf: &mut [u8]) -> Result<usize, KernelError>;
    fn mac_address(&self) -> [u8; 6];
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/core/traits.rs (line 11) - [public] CharDevice

```
pub trait CharDevice {
    fn read(&self, buf: &mut [u8]) -> Result<usize, KernelError>;
    fn write(&self, buf: &[u8]) -> Result<usize, KernelError>;
}

pub trait NetworkDevice {
    fn send(&self, frame: &[u8]) -> Result<(), KernelError>;
    fn recv(&self, buf: &mut [u8]) -> Result<usize, KernelError>;
    fn mac_address(&self) -> [u8; 6];
}

pub trait FileSystem {
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/core/traits.rs (line 12) - [internal] read

```
    fn read(&self, buf: &mut [u8]) -> Result<usize, KernelError>;
    fn write(&self, buf: &[u8]) -> Result<usize, KernelError>;
}

pub trait NetworkDevice {
    fn send(&self, frame: &[u8]) -> Result<(), KernelError>;
    fn recv(&self, buf: &mut [u8]) -> Result<usize, KernelError>;
    fn mac_address(&self) -> [u8; 6];
}

pub trait FileSystem {
    fn open(&self, path: &str, flags: u32) -> Result<Handle, KernelError>;
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/core/traits.rs (line 13) - [internal] write

```
    fn write(&self, buf: &[u8]) -> Result<usize, KernelError>;
}

pub trait NetworkDevice {
    fn send(&self, frame: &[u8]) -> Result<(), KernelError>;
    fn recv(&self, buf: &mut [u8]) -> Result<usize, KernelError>;
    fn mac_address(&self) -> [u8; 6];
}

pub trait FileSystem {
    fn open(&self, path: &str, flags: u32) -> Result<Handle, KernelError>;
    fn close(&self, handle: Handle) -> Result<(), KernelError>;
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/core/traits.rs (line 16) - [public] NetworkDevice

```
pub trait NetworkDevice {
    fn send(&self, frame: &[u8]) -> Result<(), KernelError>;
    fn recv(&self, buf: &mut [u8]) -> Result<usize, KernelError>;
    fn mac_address(&self) -> [u8; 6];
}

pub trait FileSystem {
    fn open(&self, path: &str, flags: u32) -> Result<Handle, KernelError>;
    fn close(&self, handle: Handle) -> Result<(), KernelError>;
    fn read(&self, handle: Handle, buf: &mut [u8], offset: u64) -> Result<usize, KernelError>;
    fn write(&self, handle: Handle, buf: &[u8], offset: u64) -> Result<usize, KernelError>;
    fn stat(&self, path: &str) -> Result<FileStat, KernelError>;
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/core/traits.rs (line 17) - [internal] send

```
    fn send(&self, frame: &[u8]) -> Result<(), KernelError>;
    fn recv(&self, buf: &mut [u8]) -> Result<usize, KernelError>;
    fn mac_address(&self) -> [u8; 6];
}

pub trait FileSystem {
    fn open(&self, path: &str, flags: u32) -> Result<Handle, KernelError>;
    fn close(&self, handle: Handle) -> Result<(), KernelError>;
    fn read(&self, handle: Handle, buf: &mut [u8], offset: u64) -> Result<usize, KernelError>;
    fn write(&self, handle: Handle, buf: &[u8], offset: u64) -> Result<usize, KernelError>;
    fn stat(&self, path: &str) -> Result<FileStat, KernelError>;
}

// ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/core/traits.rs (line 18) - [internal] recv

```
    fn recv(&self, buf: &mut [u8]) -> Result<usize, KernelError>;
    fn mac_address(&self) -> [u8; 6];
}

pub trait FileSystem {
    fn open(&self, path: &str, flags: u32) -> Result<Handle, KernelError>;
    fn close(&self, handle: Handle) -> Result<(), KernelError>;
    fn read(&self, handle: Handle, buf: &mut [u8], offset: u64) -> Result<usize, KernelError>;
    fn write(&self, handle: Handle, buf: &[u8], offset: u64) -> Result<usize, KernelError>;
    fn stat(&self, path: &str) -> Result<FileStat, KernelError>;
}

// ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/core/traits.rs (line 19) - [internal] mac_address

```
fn mac_address(&self) -> [u8; 6];
}

pub trait FileSystem {
    fn open(&self, path: &str, flags: u32) -> Result<Handle, KernelError>;
    fn close(&self, handle: Handle) -> Result<(), KernelError>;
    fn read(&self, handle: Handle, buf: &mut [u8], offset: u64) -> Result<usize, KernelError>;
    fn write(&self, handle: Handle, buf: &[u8], offset: u64) -> Result<usize, KernelError>;
    fn stat(&self, path: &str) -> Result<FileStat, KernelError>;
}

#[derive(Debug, Clone, Copy)]
// ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/core/traits.rs (line 22) - [public] FileSystem

```
pub trait FileSystem {
    fn open(&self, path: &str, flags: u32) -> Result<Handle, KernelError>;
    fn close(&self, handle: Handle) -> Result<(), KernelError>;
    fn read(&self, handle: Handle, buf: &mut [u8], offset: u64) -> Result<usize, KernelError>;
    fn write(&self, handle: Handle, buf: &[u8], offset: u64) -> Result<usize, KernelError>;
    fn stat(&self, path: &str) -> Result<FileStat, KernelError>;
}

#[derive(Debug, Clone, Copy)]
pub struct FileStat {
    pub size: u64,
    pub mode: u32,
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/core/traits.rs (line 23) - [internal] open

```
fn open(&self, path: &str, flags: u32) -> Result<Handle, KernelError>;
fn close(&self, handle: Handle) -> Result<(), KernelError>;
fn read(&self, handle: Handle, buf: &mut [u8], offset: u64) -> Result<usize, KernelError>;
fn write(&self, handle: Handle, buf: &[u8], offset: u64) -> Result<usize, KernelError>;
fn stat(&self, path: &str) -> Result<FileStat, KernelError>;
}

#[derive(Debug, Clone, Copy)]
pub struct FileStat {
    pub size: u64,
    pub mode: u32,
    pub uid: u32,
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/core/traits.rs (line 24) - [internal] close

```
fn close(&self, handle: Handle) -> Result<(), KernelError>;
fn read(&self, handle: Handle, buf: &mut [u8], offset: u64) -> Result<usize, KernelError>;
fn write(&self, handle: Handle, buf: &[u8], offset: u64) -> Result<usize, KernelError>;
fn stat(&self, path: &str) -> Result<FileStat, KernelError>;
}

#[derive(Debug, Clone, Copy)]
pub struct FileStat {
    pub size: u64,
    pub mode: u32,
    pub uid: u32,
    pub gid: u32,
    // ... (truncated, see source file for more)
```


kernel_Workspace/base/kw-base/src/core/traits.rs (line 25) - [internal] read

```
fn read(&self, handle: Handle, buf: &mut [u8], offset: u64) -> Result<usize, KernelError>;
fn write(&self, handle: Handle, buf: &[u8], offset: u64) -> Result<usize, KernelError>;
fn stat(&self, path: &str) -> Result<FileStat, KernelError>;
}

#[derive(Debug, Clone, Copy)]
pub struct FileStat {
    pub size: u64,
    pub mode: u32,
    pub uid: u32,
    pub gid: u32,
    pub atime: u64,
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/core/traits.rs (line 26) - [internal] write

```
fn write(&self, handle: Handle, buf: &[u8], offset: u64) -> Result<usize, KernelError>;
fn stat(&self, path: &str) -> Result<FileStat, KernelError>;
}

#[derive(Debug, Clone, Copy)]
pub struct FileStat {
    pub size: u64,
    pub mode: u32,
    pub uid: u32,
    pub gid: u32,
    pub atime: u64,
    pub mtime: u64,
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/core/traits.rs (line 27) - [internal] stat

```
fn stat(&self, path: &str) -> Result<FileStat, KernelError>;
}

#[derive(Debug, Clone, Copy)]
pub struct FileStat {
    pub size: u64,
    pub mode: u32,
    pub uid: u32,
    pub gid: u32,
    pub atime: u64,
    pub mtime: u64,
    pub ctime: u64,
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/core/traits.rs (line 31) - [public] FileStat

```
pub struct FileStat {
    pub size: u64,
    pub mode: u32,
    pub uid: u32,
    pub gid: u32,
    pub atime: u64,
    pub mtime: u64,
    pub ctime: u64,
    pub is_dir: bool,
}

pub trait Scheduler {
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/core/traits.rs (line 42) - [public] Scheduler

```
pub trait Scheduler {
    fn schedule(&mut self) -> Option<TaskId>;
    fn enqueue(&mut self, task: TaskId);
    fn dequeue(&mut self, task: TaskId);
    fn yield_current(&mut self);
}

pub trait InterruptController {
    fn enable_irq(&self, irq: u32);
    fn disable_irq(&self, irq: u32);
    fn ack_irq(&self, irq: u32);
    fn end_of_interrupt(&self);
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/core/traits.rs (line 43) - [internal] schedule

```
    fn schedule(&mut self) -> Option<TaskId>;
    fn enqueue(&mut self, task: TaskId);
    fn dequeue(&mut self, task: TaskId);
    fn yield_current(&mut self);
}

pub trait InterruptController {
    fn enable_irq(&self, irq: u32);
    fn disable_irq(&self, irq: u32);
    fn ack_irq(&self, irq: u32);
    fn end_of_interrupt(&self);
}
```

kernel_workspace/base/kw-base/src/core/traits.rs (line 44) - [internal] enqueue

```
    fn enqueue(&mut self, task: TaskId);
    fn dequeue(&mut self, task: TaskId);
    fn yield_current(&mut self);
}

pub trait InterruptController {
    fn enable_irq(&self, irq: u32);
    fn disable_irq(&self, irq: u32);
    fn ack_irq(&self, irq: u32);
    fn end_of_interrupt(&self);
}
```

kernel_workspace/base/kw-base/src/core/traits.rs (line 45) - [internal] dequeue

```
    fn dequeue(&mut self, task: TaskId);
    fn yield_current(&mut self);
}

pub trait InterruptController {
    fn enable_irq(&self, irq: u32);
    fn disable_irq(&self, irq: u32);
    fn ack_irq(&self, irq: u32);
    fn end_of_interrupt(&self);
}
```

kernel_workspace/base/kw-base/src/core/traits.rs (line 46) - [internal] yield_current

```
    fn yield_current(&mut self);
}

pub trait InterruptController {
    fn enable_irq(&self, irq: u32);
    fn disable_irq(&self, irq: u32);
    fn ack_irq(&self, irq: u32);
    fn end_of_interrupt(&self);
}
```

kernel_workspace/base/kw-base/src/core/traits.rs (line 49) - [public] InterruptController

```
pub trait InterruptController {
    fn enable_irq(&self, irq: u32);
    fn disable_irq(&self, irq: u32);
    fn ack_irq(&self, irq: u32);
    fn end_of_interrupt(&self);
}
```

kernel_workspace/base/kw-base/src/core/traits.rs (line 50) - [internal] enable_irq

```
fn enable_irq(&self, irq: u32);
fn disable_irq(&self, irq: u32);
fn ack_irq(&self, irq: u32);
fn end_of_interrupt(&self);
}
```

kernel_workspace/base/kw-base/src/core/traits.rs (line 51) - [internal] disable_irq

```
fn disable_irq(&self, irq: u32);
fn ack_irq(&self, irq: u32);
fn end_of_interrupt(&self);
}
```

kernel_workspace/base/kw-base/src/core/traits.rs (line 52) - [internal] ack_irq

```
fn ack_irq(&self, irq: u32);
fn end_of_interrupt(&self);
}
```

kernel_workspace/base/kw-base/src/core/traits.rs (line 53) - [internal] end_of_interrupt

```
fn end_of_interrupt(&self);
}
```

kernel_workspace/base/kw-base/src/core/types.rs (line 5) - [public] Pid

```
pub struct Pid(pub u32);

#[derive(Debug, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Default)]
#[repr(transparent)]
pub struct Tid(pub u32);

#[derive(Debug, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Default)]
#[repr(transparent)]
pub struct Fd(pub i32);

impl Fd {
    pub const INVALID: Self = Self(-1);
    // ... (truncated, see source file for more)
}
```

kernel_workspace/base/kw-base/src/core/types.rs (line 9) - [public] Tid

```
pub struct Tid(pub u32);

#[derive(Debug, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Default)]
#[repr(transparent)]
pub struct Fd(pub i32);

impl Fd {
    pub const INVALID: Self = Self(-1);
    pub const STDIN: Self = Self(0);
    pub const STDOUT: Self = Self(1);
    pub const STDERR: Self = Self(2);

    // ... (truncated, see source file for more)
}
```

kernel_workspace/base/kw-base/src/core/types.rs (line 13) - [public] Fd

```
pub struct Fd(pub i32);

impl Fd {
    pub const INVALID: Self = Self(-1);
    pub const STDIN: Self = Self(0);
    pub const STDOUT: Self = Self(1);
    pub const STDERR: Self = Self(2);

    #[inline]
    pub const fn is_valid(self) -> bool {
        self.0 >= 0
    }
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/core/types.rs (line 29) - [public] PrivilegeLevel

```
pub enum PrivilegeLevel {
    Kernel = 0,
    Driver = 1,
    System = 2,
    User = 3,
}

#[derive(Debug, Clone, Copy, PartialEq, Eq)]
#[repr(u8)]
pub enum Arch {
    X86_64 = 0,
    AArch64 = 1,
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/core/types.rs (line 38) - [public] Arch

```
pub enum Arch {
    X86_64 = 0,
    AArch64 = 1,
    Riscv64 = 2,
}
```

kernel_workspace/base/kw-base/src/crypto/ciphers.rs (line 4) - [internal] k_aes256_enc

```
fn k_aes256_enc(key: *const u8, iv: *const u8, data: *mut u8, len: usize) -> bool;
fn k_aes256_dec(key: *const u8, iv: *const u8, data: *mut u8, len: usize) -> bool;
}

pub trait BlockCipher {
    const KEY_LEN: usize;
    const IV_LEN: usize;
    const BLOCK_LEN: usize;

    fn encrypt(key: &[u8], iv: &[u8], data: &mut [u8]) -> KResult<>;
    fn decrypt(key: &[u8], iv: &[u8], data: &mut [u8]) -> KResult<>;
}
// ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/crypto/ciphers.rs (line 5) - [internal] k_aes256_dec

```
fn k_aes256_dec(key: *const u8, iv: *const u8, data: *mut u8, len: usize) -> bool;
}

pub trait BlockCipher {
    const KEY_LEN: usize;
    const IV_LEN: usize;
    const BLOCK_LEN: usize;

    fn encrypt(key: &[u8], iv: &[u8], data: &mut [u8]) -> KResult<>;
    fn decrypt(key: &[u8], iv: &[u8], data: &mut [u8]) -> KResult<>;
}

// ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/crypto/ciphers.rs (line 8) - [public] BlockCipher

```
pub trait BlockCipher {
    const KEY_LEN: usize;
    const IV_LEN: usize;
    const BLOCK_LEN: usize;

    fn encrypt(key: &[u8], iv: &[u8], data: &mut [u8]) -> KResult<>;
    fn decrypt(key: &[u8], iv: &[u8], data: &mut [u8]) -> KResult<>;
}

pub struct Aes256Cbc;
impl BlockCipher for Aes256Cbc {
    const KEY_LEN: usize = 32;
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/crypto/ciphers.rs (line 13) - [internal] encrypt

```
    fn encrypt(key: &[u8], iv: &[u8], data: &mut [u8]) -> KResult<>;
    fn decrypt(key: &[u8], iv: &[u8], data: &mut [u8]) -> KResult<>;
}

pub struct Aes256Cbc;
impl BlockCipher for Aes256Cbc {
    const KEY_LEN: usize = 32;
    const IV_LEN: usize = 16;
    const BLOCK_LEN: usize = 16;

    fn encrypt(key: &[u8], iv: &[u8], data: &mut [u8]) -> KResult<> {
        if key.len() != Self::KEY_LEN || iv.len() != Self::IV_LEN {
            // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/crypto/ciphers.rs (line 14) - [internal] decrypt

```
    fn decrypt(key: &[u8], iv: &[u8], data: &mut [u8]) -> KResult<>;
}

pub struct Aes256Cbc;
impl BlockCipher for Aes256Cbc {
    const KEY_LEN: usize = 32;
    const IV_LEN: usize = 16;
    const BLOCK_LEN: usize = 16;

    fn encrypt(key: &[u8], iv: &[u8], data: &mut [u8]) -> KResult<> {
        if key.len() != Self::KEY_LEN || iv.len() != Self::IV_LEN {
            return Err(KernelError::InvalidArg);
        }
        // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/crypto/ciphers.rs (line 17) - [public] Aes256Cbc

```
pub struct Aes256Cbc;
impl BlockCipher for Aes256Cbc {
    const KEY_LEN: usize = 32;
    const IV_LEN: usize = 16;
    const BLOCK_LEN: usize = 16;

    fn encrypt(key: &[u8], iv: &[u8], data: &mut [u8]) -> KResult<> {
        if key.len() != Self::KEY_LEN || iv.len() != Self::IV_LEN {
            return Err(KernelError::InvalidArg);
        }
        if data.len() % Self::BLOCK_LEN != 0 {
            return Err(KernelError::InvalidArg);
        }
        // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/crypto/ciphers.rs (line 23) - [internal] encrypt

```
fn encrypt(key: &[u8], iv: &[u8], data: &mut [u8]) -> KResult<()> {
    if key.len() != Self::KEY_LEN || iv.len() != Self::IV_LEN {
        return Err(KernelError::InvalidArg);
    }
    if data.len() % Self::BLOCK_LEN != 0 {
        return Err(KernelError::InvalidArg);
    }
    if unsafe { k_aes256_enc(key.as_ptr(), iv.as_ptr(), data.as_mut_ptr(), data.len()) } {
        Ok(())
    } else {
        Err(KernelError::IoError)
    }
}
// ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/crypto/ciphers.rs (line 37) - [internal] decrypt

```
fn decrypt(key: &[u8], iv: &[u8], data: &mut [u8]) -> KResult<()> {
    if key.len() != Self::KEY_LEN || iv.len() != Self::IV_LEN {
        return Err(KernelError::InvalidArg);
    }
    if data.len() % Self::BLOCK_LEN != 0 {
        return Err(KernelError::InvalidArg);
    }
    if unsafe { k_aes256_dec(key.as_ptr(), iv.as_ptr(), data.as_mut_ptr(), data.len()) } {
        Ok(())
    } else {
        Err(KernelError::IoError)
    }
}
// ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/crypto/hashes.rs (line 4) - [internal] k_sha256

```
fn k_sha256(data: *const u8, len: usize, out: *mut u8) -> bool;
fn k_sha512(data: *const u8, len: usize, out: *mut u8) -> bool;
}

pub trait Hasher {
    const OUTPUT_LEN: usize;
    fn hash(data: &[u8], out: &mut [u8]) -> KResult<()>;
}

pub struct Sha256;
impl Hasher for Sha256 {
    const OUTPUT_LEN: usize = 32;
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/crypto/hashes.rs (line 5) - [internal] k_sha512

```
fn k_sha512(data: *const u8, len: usize, out: *mut u8) -> bool;
}

pub trait Hasher {
    const OUTPUT_LEN: usize;
    fn hash(data: &[u8], out: &mut [u8]) -> KResult<()>;
}

pub struct Sha256;
impl Hasher for Sha256 {
    const OUTPUT_LEN: usize = 32;
    fn hash(data: &[u8], out: &mut [u8]) -> KResult<()> {
        // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/crypto/hashes.rs (line 8) - [public] Hasher

```
pub trait Hasher {
    const OUTPUT_LEN: usize;
    fn hash(data: &[u8], out: &mut [u8]) -> KResult<>;
}

pub struct Sha256;
impl Hasher for Sha256 {
    const OUTPUT_LEN: usize = 32;
    fn hash(data: &[u8], out: &mut [u8]) -> KResult<> {
        if out.len() < Self::OUTPUT_LEN { return Err(KernelError::InvalidArg); }
        if unsafe { k_sha256(data.as_ptr(), data.len(), out.as_mut_ptr()) } {
            Ok(())
        }
        // ... (truncated, see source file for more)
    }
}
```

kernel_workspace/base/kw-base/src/crypto/hashes.rs (line 10) - [internal] hash

```
fn hash(data: &[u8], out: &mut [u8]) -> KResult<>;
}

pub struct Sha256;
impl Hasher for Sha256 {
    const OUTPUT_LEN: usize = 32;
    fn hash(data: &[u8], out: &mut [u8]) -> KResult<> {
        if out.len() < Self::OUTPUT_LEN { return Err(KernelError::InvalidArg); }
        if unsafe { k_sha256(data.as_ptr(), data.len(), out.as_mut_ptr()) } {
            Ok(())
        } else {
            Err(KernelError::IoError)
        }
        // ... (truncated, see source file for more)
    }
}
```

kernel_workspace/base/kw-base/src/crypto/hashes.rs (line 13) - [public] Sha256

```
pub struct Sha256;
impl Hasher for Sha256 {
    const OUTPUT_LEN: usize = 32;
    fn hash(data: &[u8], out: &mut [u8]) -> KResult<> {
        if out.len() < Self::OUTPUT_LEN { return Err(KernelError::InvalidArg); }
        if unsafe { k_sha256(data.as_ptr(), data.len(), out.as_mut_ptr()) } {
            Ok(())
        } else {
            Err(KernelError::IoError)
        }
    }
}

// ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/crypto/hashes.rs (line 16) - [internal] hash

```
fn hash(data: &[u8], out: &mut [u8]) -> KResult<> {
    if out.len() < Self::OUTPUT_LEN { return Err(KernelError::InvalidArg); }
    if unsafe { k_sha256(data.as_ptr(), data.len(), out.as_mut_ptr()) } {
        Ok(())
    } else {
        Err(KernelError::IoError)
    }
}

pub struct Sha512;
impl Hasher for Sha512 {
    // ... (truncated, see source file for more)
}
```

kernel_workspace/base/kw-base/src/crypto/hashes.rs (line 26) - [public] Sha512

```
pub struct Sha512;
impl Hasher for Sha512 {
    const OUTPUT_LEN: usize = 64;
    fn hash(data: &[u8], out: &mut [u8]) -> KResult<> {
        if out.len() < Self::OUTPUT_LEN { return Err(KernelError::InvalidArg); }
        if unsafe { k_sha512(data.as_ptr(), data.len(), out.as_mut_ptr()) } {
            Ok(())
        } else {
            Err(KernelError::IoError)
        }
    }
}
```

kernel_workspace/base/kw-base/src/crypto/hashes.rs (line 29) - [internal] hash

```
fn hash(data: &[u8], out: &mut [u8]) -> KResult<> {
    if out.len() < Self::OUTPUT_LEN { return Err(KernelError::InvalidArg); }
    if unsafe { k_sha512(data.as_ptr(), data.len(), out.as_mut_ptr()) } {
        Ok(())
    } else {
        Err(KernelError::IoError)
    }
}
```

kernel_workspace/base/kw-base/src/crypto/keys.rs (line 4) - [public] SymmetricKey

```
pub struct SymmetricKey {
    inner: SecureBuffer,
}

impl SymmetricKey {
    pub fn generate(len: usize) -> KResult<Self> {
        let mut buf = SecureBuffer::new(len).ok_or(KernelError::NoMemory)?;
        super::random::fill_random(buf.as_mut_slice());
        Ok(Self { inner: buf })
    }

    pub fn from_bytes(bytes: &[u8]) -> KResult<Self> {
        // ... (truncated, see source file for more)
    }
}
```

kernel_workspace/base/kw-base/src/crypto/keys.rs (line 9) - [public] generate

```
pub fn generate(len: usize) -> KResult<Self> {
    let mut buf = SecureBuffer::new(len).ok_or(KernelError::NoMemory)?;
    super::random::fill_random(buf.as_mut_slice());
    Ok(Self { inner: buf })
}

pub fn from_bytes(bytes: &[u8]) -> KResult<Self> {
    let mut buf = SecureBuffer::new(bytes.len()).ok_or(KernelError::NoMemory)?;
    buf.as_mut_slice().copy_from_slice(bytes);
    Ok(Self { inner: buf })
}

// ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/crypto/keys.rs (line 15) - [public] from_bytes

```
pub fn from_bytes(bytes: &[u8]) -> KResult<Self> {
    let mut buf = SecureBuffer::new(bytes.len()).ok_or(KernelError::NoMemory)?;
    buf.as_mut_slice().copy_from_slice(bytes);
    Ok(Self { inner: buf })
}

pub fn as_slice(&self) -> &[u8] {
    self.inner.as_slice()
}

}
```


kernel_Workspace/base/kw-base/src/crypto/keys.rs (line 21) - [public] as_slice

```
pub fn as_slice(&self) -> &[u8] {
    self.inner.as_slice()
}
```

kernel_Workspace/base/kw-base/src/crypto/random.rs (line 4) - [internal] k_get_random

```
fn k_get_random(buf: *mut u8, len: usize) -> bool;
fn k_get_hw_random(buf: *mut u8, len: usize) -> bool;

pub fn fill_random(buf: &mut [u8]) -> KResult<()> {
    if unsafe { k_get_random(buf.as_mut_ptr(), buf.len()) } {
        Ok(())
    } else {
        Err(KernelError::IoError)
    }
}

// ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/crypto/random.rs (line 5) - [internal] k_get_hw_random

```
fn k_get_hw_random(buf: *mut u8, len: usize) -> bool;

pub fn fill_random(buf: &mut [u8]) -> KResult<()> {
    if unsafe { k_get_random(buf.as_mut_ptr(), buf.len()) } {
        Ok(())
    } else {
        Err(KernelError::IoError)
    }
}

pub fn fill_hw_random(buf: &mut [u8]) -> KResult<()> {
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/crypto/random.rs (line 8) - [public] fill_random

```
pub fn fill_random(buf: &mut [u8]) -> KResult<()> {
    if unsafe { k_get_random(buf.as_mut_ptr(), buf.len()) } {
        Ok(())
    } else {
        Err(KernelError::IoError)
    }
}

pub fn fill_hw_random(buf: &mut [u8]) -> KResult<()> {
    if unsafe { k_get_hw_random(buf.as_mut_ptr(), buf.len()) } {
        Ok(())
    } else {
        // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/crypto/random.rs (line 16) - [public] fill_hw_random

```
pub fn fill_hw_random(buf: &mut [u8]) -> KResult<()> {
    if unsafe { k_get_hw_random(buf.as_mut_ptr(), buf.len()) } {
        Ok(())
    } else {
        Err(KernelError::NotSupported)
    }
}

pub fn next_u32() -> KResult<u32> {
    let mut val = 0u32;
    fill_random(unsafe { core::slice::from_raw_parts_mut(&mut val as *mut u32 as *mut u8, 4) })?;
    Ok(val)
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/crypto/random.rs (line 24) - [public] next_u32

```
pub fn next_u32() -> KResult<u32> {
    let mut val = 0u32;
    fill_random(unsafe { core::slice::from_raw_parts_mut(&mut val as *mut u32 as *mut u8, 4) })?;
    Ok(val)
}
```

kernel_workspace/base/kw-base/src/crypto/secure_mem.rs (line 5) - [public] secure_zero

*// Securely zero out memory. Uses volatile writes to prevent the compiler
// from optimizing the memset away (crucial for wiping keys).*

```
pub fn secure_zero(ptr: *mut u8, len: usize) {
    unsafe {
        for i in 0..len {
            ptr::write_volatile(ptr.add(i), 0);
        }
        core::sync::atomic::compiler_fence(core::sync::atomic::Ordering::SeqCst);
    }
}

pub struct SecureBuffer {
    ptr: *mut u8,
    len: usize,
    // ... (truncated, see source file for more)
}
```

kernel_workspace/base/kw-base/src/crypto/secure_mem.rs (line 14) - [public] SecureBuffer

```
pub struct SecureBuffer {
    ptr: *mut u8,
    len: usize,
}

impl SecureBuffer {
    pub fn new(len: usize) -> Option<Self> {
        let layout = core::alloc::Layout::from_size_align(len, 8).ok()?;
        let ptr = unsafe { kstd_alloc::KernelHeap.alloc(layout) }.ok()?;
        secure_zero(ptr, len);
        Some(Self { ptr, len })
    }
    // ... (truncated, see source file for more)
}
```

kernel_workspace/base/kw-base/src/crypto/secure_mem.rs (line 20) - [public] new

```
pub fn new(len: usize) -> Option<Self> {
    let layout = core::alloc::Layout::from_size_align(len, 8).ok()?;
    let ptr = unsafe { kstd_alloc::KernelHeap.alloc(layout) }.ok()?;
    secure_zero(ptr, len);
    Some(Self { ptr, len })
}

pub fn as_slice(&self) -> &[u8] {
    unsafe { core::slice::from_raw_parts(self.ptr, self.len) }
}

pub fn as_mut_slice(&mut self) -> &mut [u8] {
    // ... (truncated, see source file for more)
}
```

kernel_workspace/base/kw-base/src/crypto/secure_mem.rs (line 27) - [public] as_slice

```
pub fn as_slice(&self) -> &[u8] {
    unsafe { core::slice::from_raw_parts(self.ptr, self.len) }
}

pub fn as_mut_slice(&mut self) -> &mut [u8] {
    unsafe { core::slice::from_raw_parts_mut(self.ptr, self.len) }
}

impl Drop for SecureBuffer {
    fn drop(&mut self) {
        secure_zero(self.ptr, self.len);
        // ... (truncated, see source file for more)
    }
}
```

kernel_workspace/base/kw-base/src/crypto/secure_mem.rs (line 31) - [public] as_mut_slice

```
pub fn as_mut_slice(&mut self) -> &mut [u8] {
    unsafe { core::slice::from_raw_parts_mut(self.ptr, self.len) }
}

impl Drop for SecureBuffer {
    fn drop(&mut self) {
        secure_zero(self.ptr, self.len);
        let layout = core::alloc::Layout::from_size_align(self.len, 8).unwrap();
        unsafe { kstd_alloc::KernelHeap.dealloc(self.ptr, layout); }
    }
}
```

kernel_workspace/base/kw-base/src/crypto/secure_mem.rs (line 37) - [internal] drop

```
fn drop(&mut self) {
    secure_zero(self.ptr, self.len);
    let layout = core::alloc::Layout::from_size_align(self.len, 8).unwrap();
    unsafe { kstd_alloc::KernelHeap.dealloc(self.ptr, layout); }
}
```

kernel_workspace/base/kw-base/src/drivers/ps2.rs (line 7) - [public] Ps2Controller

```
pub struct Ps2Controller;

impl Ps2Controller {
    pub fn init() {
        // Disable devices
        unsafe {
            outb(PS2_CMD, 0xAD);
            outb(PS2_CMD, 0xA7);

            // Flush output buffer
            while (inb(PS2_STATUS) & 0x01) != 0 { inb(PS2_DATA); }

            // ... (truncated, see source file for more)
        }
    }
}
```

kernel_workspace/base/kw-base/src/drivers/ps2.rs (line 10) - [public] init

```
pub fn init() {
    // Disable devices
    unsafe {
        outb(PS2_CMD, 0xAD);
        outb(PS2_CMD, 0xA7);

        // Flush output buffer
        while (inb(PS2_STATUS) & 0x01) != 0 { inb(PS2_DATA); }

        // Enable devices
        outb(PS2_CMD, 0xAE);
        outb(PS2_CMD, 0xA8);
    }
    // ... (truncated, see source file for more)
}
```

kernel_workspace/base/kw-base/src/drivers/ps2.rs (line 25) - [public] read_scancode

```
pub fn read_scancode() -> Option<u8> {
    unsafe {
        if (inb(PS2_STATUS) & 0x01) != 0 {
            Some(inb(PS2_DATA))
        } else {
            None
        }
    }
}
```

kernel_Workspace/base/kw-base/src/drivers/uart.rs (line 4) - [public] Uart16550

```
pub struct Uart16550 {
    port: u16,
}

impl Uart16550 {
    pub const COM1: u16 = 0x3F8;

    pub fn new(port: u16) -> Self {
        let mut uart = Self { port };
        uart.init();
        uart
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/drivers/uart.rs (line 11) - [public] new

```
pub fn new(port: u16) -> Self {
    let mut uart = Self { port };
    uart.init();
    uart
}

fn init(&mut self) {
    unsafe {
        outb(self.port + 1, 0x00); // Disable interrupts
        outb(self.port + 3, 0x80); // Enable DLAB
        outb(self.port + 0, 0x03); // 38400 baud
        outb(self.port + 1, 0x00);
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/drivers/uart.rs (line 17) - [internal] init

```
fn init(&mut self) {
    unsafe {
        outb(self.port + 1, 0x00); // Disable interrupts
        outb(self.port + 3, 0x80); // Enable DLAB
        outb(self.port + 0, 0x03); // 38400 baud
        outb(self.port + 1, 0x00);
        outb(self.port + 3, 0x03); // 8 bits, no parity, one stop bit
        outb(self.port + 2, 0xC7); // Enable FIFO
        outb(self.port + 4, 0x0B); // IRQs enabled, RTS/DSR set
    }
}

// ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/drivers/uart.rs (line 29) - [internal] is_transmitter_empty

```
fn is_transmitter_empty(&self) -> bool {
    unsafe { (inb(self.port + 5) & 0x20) != 0 }
}

pub fn write_byte(&mut self, b: u8) {
    while !self.is_transmitter_empty() { core::hint::spin_loop(); }
    unsafe { outb(self.port, b); }
}

impl Write for Uart16550 {
    fn write_str(&mut self, s: &str) -> Result<(), ()> {
        // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/drivers/uart.rs (line 33) - [public] write_byte

```
pub fn write_byte(&mut self, b: u8) {
    while !self.is_transmitter_empty() { core::hint::spin_loop(); }
    unsafe { outb(self.port, b); }
}

impl Write for Uart16550 {
    fn write_str(&mut self, s: &str) -> Result<(), ()> {
        for b in s.bytes() {
            if b == b'\n' { self.write_byte(b'\r'); }
            self.write_byte(b);
        }
        // ... (truncated, see source file for more)
    }
}
```

kernel_Workspace/base/kw-base/src/drivers/uart.rs (line 40) - [internal] write_str

```
fn write_str(&mut self, s: &str) -> Result<(), ()> {
    for b in s.bytes() {
        if b == b'\n' { self.write_byte(b'\r'); }
        self.write_byte(b);
    }
    Ok(())
}
```

kernel_Workspace/base/kw-base/src/fs/vfs.rs (line 5) - [public] FileType

```
pub enum FileType { Regular, Directory, CharDevice, BlockDevice, Pipe, Socket }

pub trait Inode {
    fn stat(&self) -> KResult<FileStat>;
    fn read(&self, offset: u64, buf: &mut [u8]) -> KResult<usize>;
    fn write(&self, offset: u64, buf: &[u8]) -> KResult<usize>;
    fn lookup(&self, name: &str) -> KResult<Handle>;
}

pub trait File {
    fn inode(&self) -> &dyn Inode;
    fn flags(&self) -> u32;
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/base/kw-base/src/fs/vfs.rs (line 7) - [public] Inode

```
pub trait Inode {
    fn stat(&self) -> KResult<FileStat>;
    fn read(&self, offset: u64, buf: &mut [u8]) -> KResult<usize>;
    fn write(&self, offset: u64, buf: &[u8]) -> KResult<usize>;
    fn lookup(&self, name: &str) -> KResult<Handle>;
}

pub trait File {
    fn inode(&self) -> &dyn Inode;
    fn flags(&self) -> u32;
}

// ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/fs/vfs.rs (line 8) - [internal] stat

```
fn stat(&self) -> KResult<FileStat>;
fn read(&self, offset: u64, buf: &mut [u8]) -> KResult<usize>;
fn write(&self, offset: u64, buf: &[u8]) -> KResult<usize>;
fn lookup(&self, name: &str) -> KResult<Handle>;
}

pub trait File {
    fn inode(&self) -> &dyn Inode;
    fn flags(&self) -> u32;
}

pub trait FileSystem {
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/base/kw-base/src/fs/vfs.rs (line 9) - [internal] read

```
    fn read(&self, offset: u64, buf: &mut [u8]) -> KResult<usize>;
    fn write(&self, offset: u64, buf: &[u8]) -> KResult<usize>;
    fn lookup(&self, name: &str) -> KResult<Handle>;
}

pub trait File {
    fn inode(&self) -> &dyn Inode;
    fn flags(&self) -> u32;
}

pub trait FileSystem {
    fn name(&self) -> &str;
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/fs/vfs.rs (line 10) - [internal] write

```
    fn write(&self, offset: u64, buf: &[u8]) -> KResult<usize>;
    fn lookup(&self, name: &str) -> KResult<Handle>;
}

pub trait File {
    fn inode(&self) -> &dyn Inode;
    fn flags(&self) -> u32;
}

pub trait FileSystem {
    fn name(&self) -> &str;
    fn root_inode(&self) -> Handle;
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/fs/vfs.rs (line 11) - [internal] lookup

```
    fn lookup(&self, name: &str) -> KResult<Handle>;
}

pub trait File {
    fn inode(&self) -> &dyn Inode;
    fn flags(&self) -> u32;
}

pub trait FileSystem {
    fn name(&self) -> &str;
    fn root_inode(&self) -> Handle;
    fn sync(&self) -> KResult<()>;
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/fs/vfs.rs (line 14) - [public] File

```
pub trait File {
    fn inode(&self) -> &dyn Inode;
    fn flags(&self) -> u32;
}

pub trait FileSystem {
    fn name(&self) -> &str;
    fn root_inode(&self) -> Handle;
    fn sync(&self) -> KResult<()>;
}

pub struct Vfs;
// ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/fs/vfs.rs (line 15) - [internal] inode

```
    fn inode(&self) -> &dyn Inode;
    fn flags(&self) -> u32;
}

pub trait FileSystem {
    fn name(&self) -> &str;
    fn root_inode(&self) -> Handle;
    fn sync(&self) -> KResult<>;
}

pub struct Vfs;

    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/fs/vfs.rs (line 16) - [internal] flags

```
    fn flags(&self) -> u32;
}

pub trait FileSystem {
    fn name(&self) -> &str;
    fn root_inode(&self) -> Handle;
    fn sync(&self) -> KResult<>;
}

pub struct Vfs;

impl Vfs {
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/fs/vfs.rs (line 19) - [public] FileSystem

```
pub trait FileSystem {
    fn name(&self) -> &str;
    fn root_inode(&self) -> Handle;
    fn sync(&self) -> KResult<>;
}

pub struct Vfs;

impl Vfs {
    pub fn open(path: &str, flags: u32) -> KResult<Handle> {
        // Route to legacy kernel VFS via FFI if needed, or internal registry
        extern "C" { fn k_fs_open(path: *const u8, flags: u32) -> i32; }
        // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/fs/vfs.rs (line 20) - [internal] name

```
    fn name(&self) -> &str;
    fn root_inode(&self) -> Handle;
    fn sync(&self) -> KResult<>;
}

pub struct Vfs;

impl Vfs {
    pub fn open(path: &str, flags: u32) -> KResult<Handle> {
        // Route to legacy kernel VFS via FFI if needed, or internal registry
        extern "C" { fn k_fs_open(path: *const u8, flags: u32) -> i32; }
        let rc = unsafe { k_fs_open(path.as_ptr(), flags) };
        // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/fs/vfs.rs (line 21) - [internal] root_inode

```
fn root_inode(&self) -> Handle;
fn sync(&self) -> KResult<()>;
}

pub struct Vfs;

impl Vfs {
    pub fn open(path: &str, flags: u32) -> KResult<Handle> {
        // Route to legacy kernel VFS via FFI if needed, or internal registry
        extern "C" { fn k_fs_open(path: *const u8, flags: u32) -> i32; }
        let rc = unsafe { k_fs_open(path.as_ptr(), flags) };
        if rc < 0 { Err(KernelError::from_code(rc)) } else { Ok(Handle(rc as u32)) }
        // ... (truncated, see source file for more)
    }
}
```

kernel_Workspace/base/kw-base/src/fs/vfs.rs (line 22) - [internal] sync

```
fn sync(&self) -> KResult<()>;
}

pub struct Vfs;

impl Vfs {
    pub fn open(path: &str, flags: u32) -> KResult<Handle> {
        // Route to legacy kernel VFS via FFI if needed, or internal registry
        extern "C" { fn k_fs_open(path: *const u8, flags: u32) -> i32; }
        let rc = unsafe { k_fs_open(path.as_ptr(), flags) };
        if rc < 0 { Err(KernelError::from_code(rc)) } else { Ok(Handle(rc as u32)) }
    }
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/base/kw-base/src/fs/vfs.rs (line 25) - [public] Vfs

```
pub struct Vfs;

impl Vfs {
    pub fn open(path: &str, flags: u32) -> KResult<Handle> {
        // Route to legacy kernel VFS via FFI if needed, or internal registry
        extern "C" { fn k_fs_open(path: *const u8, flags: u32) -> i32; }
        let rc = unsafe { k_fs_open(path.as_ptr(), flags) };
        if rc < 0 { Err(KernelError::from_code(rc)) } else { Ok(Handle(rc as u32)) }
    }

    pub fn read(fd: Handle, buf: &mut [u8]) -> KResult<usize> {
        extern "C" { fn k_fs_read(fd: i32, buf: *mut u8, cap: u32) -> i32; }
        // ... (truncated, see source file for more)
    }
}
```

kernel_Workspace/base/kw-base/src/fs/vfs.rs (line 28) - [public] open

```
pub fn open(path: &str, flags: u32) -> KResult<Handle> {
    // Route to legacy kernel VFS via FFI if needed, or internal registry
    extern "C" { fn k_fs_open(path: *const u8, flags: u32) -> i32; }
    let rc = unsafe { k_fs_open(path.as_ptr(), flags) };
    if rc < 0 { Err(KernelError::from_code(rc)) } else { Ok(Handle(rc as u32)) }
}

pub fn read(fd: Handle, buf: &mut [u8]) -> KResult<usize> {
    extern "C" { fn k_fs_read(fd: i32, buf: *mut u8, cap: u32) -> i32; }
    let rc = unsafe { k_fs_read(fd.0 as i32, buf.as_mut_ptr(), buf.len() as u32) };
    if rc < 0 { Err(KernelError::from_code(rc)) } else { Ok(rc as usize) }
}
// ... (truncated, see source file for more)
```


kernel_workspace/base/kw-base/src/fs/vfs.rs (line 35) - [public] read

```
pub fn read(fd: Handle, buf: &mut [u8]) -> KResult<usize> {
    extern "C" { fn k_fs_read(fd: i32, buf: *mut u8, cap: u32) -> i32; }
    let rc = unsafe { k_fs_read(fd.0 as i32, buf.as_mut_ptr(), buf.len() as u32) };
    if rc < 0 { Err(KernelError::from_code(rc)) } else { Ok(rc as usize) }
}

pub fn write(fd: Handle, buf: &[u8]) -> KResult<usize> {
    extern "C" { fn k_fs_write(fd: i32, buf: *const u8, len: u32) -> i32; }
    let rc = unsafe { k_fs_write(fd.0 as i32, buf.as_ptr(), buf.len() as u32) };
    if rc < 0 { Err(KernelError::from_code(rc)) } else { Ok(rc as usize) }
}
}
```

kernel_workspace/base/kw-base/src/fs/vfs.rs (line 41) - [public] write

```
pub fn write(fd: Handle, buf: &[u8]) -> KResult<usize> {
    extern "C" { fn k_fs_write(fd: i32, buf: *const u8, len: u32) -> i32; }
    let rc = unsafe { k_fs_write(fd.0 as i32, buf.as_ptr(), buf.len() as u32) };
    if rc < 0 { Err(KernelError::from_code(rc)) } else { Ok(rc as usize) }
}
}
```

kernel_workspace/base/kw-base/src/hal/acpi.rs (line 4) - [internal] k_acpi_get_rsdp

```
fn k_acpi_get_rsdp() -> *const u8;
fn k_acpi_find_table(sig: *const u8, idx: u32) -> *const u8;
}

pub fn get_rsdp() -> Option<*const u8> {
    let ptr = unsafe { k_acpi_get_rsdp() };
    if ptr.is_null() { None } else { Some(ptr) }
}

pub fn find_table(sig: &[u8; 4], index: u32) -> Option<*const u8> {
    let ptr = unsafe { k_acpi_find_table(sig.as_ptr(), index) };
    if ptr.is_null() { None } else { Some(ptr) }
    // ... (truncated, see source file for more)
}
```

kernel_workspace/base/kw-base/src/hal/acpi.rs (line 5) - [internal] k_acpi_find_table

```
fn k_acpi_find_table(sig: *const u8, idx: u32) -> *const u8;
}

pub fn get_rsdp() -> Option<*const u8> {
    let ptr = unsafe { k_acpi_get_rsdp() };
    if ptr.is_null() { None } else { Some(ptr) }
}

pub fn find_table(sig: &[u8; 4], index: u32) -> Option<*const u8> {
    let ptr = unsafe { k_acpi_find_table(sig.as_ptr(), index) };
    if ptr.is_null() { None } else { Some(ptr) }
}
}
```

kernel_workspace/base/kw-base/src/hal/acpi.rs (line 8) - [public] get_rsdp

```
pub fn get_rsdp() -> Option<*const u8> {
    let ptr = unsafe { k_acpi_get_rsdp() };
    if ptr.is_null() { None } else { Some(ptr) }
}

pub fn find_table(sig: &[u8; 4], index: u32) -> Option<*const u8> {
    let ptr = unsafe { k_acpi_find_table(sig.as_ptr(), index) };
    if ptr.is_null() { None } else { Some(ptr) }
}
}
```

kernel_workspace/base/kw-base/src/hal/acpi.rs (line 13) - [public] find_table

```
pub fn find_table(sig: &[u8; 4], index: u32) -> Option<*const u8> {
    let ptr = unsafe { k_acpi_find_table(sig.as_ptr(), index) };
    if ptr.is_null() { None } else { Some(ptr) }
}
}
```

kernel_workspace/base/kw-base/src/hal/dma.rs (line 5) - [public] DmaBuffer

```
pub struct DmaBuffer {
    frames: ContiguousFrames,
    virt: VirtAddr,
}

impl DmaBuffer {
    pub fn allocate(pages: usize) -> KResult<Self> {
        let frames = ContiguousFrames::alloc(pages)?;
        let virt = gluecore::bridge::mm::vmm::alloc(
            frames.byte_size(),
            gluecore::bridge::mm::vmm::VMM_FLAG_READ | gluecore::bridge::mm::vmm::VMM_FLAG_WRITE | gluecore::bridge::mm::vmm::VMM_FLAG_EXEC,
        ).map_err(KernelError::from)?;
        // ... (truncated, see source file for more)
    }
}
```

kernel_workspace/base/kw-base/src/hal/dma.rs (line 11) - [public] allocate

```
pub fn allocate(pages: usize) -> KResult<Self> {
    let frames = ContiguousFrames::alloc(pages)?;
    let virt = gluecore::bridge::mm::vmm::alloc(
        frames.byte_size(),
        gluecore::bridge::mm::vmm::VMM_FLAG_READ | gluecore::bridge::mm::vmm::VMM_FLAG_WRITE | gluecore::bridge::mm::vmm::VMM_FLAG_EXEC,
    ).map_err(KernelError::from)?;

    Ok(Self { frames, virt })
}

pub fn phys_addr(&self) -> PhysAddr { self.frames.phys_addr() }
pub fn virt_addr(&self) -> VirtAddr { self.virt }
// ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/hal/dma.rs (line 21) - [public] phys_addr

```
pub fn phys_addr(&self) -> PhysAddr { self.frames.phys_addr() }
pub fn virt_addr(&self) -> VirtAddr { self.virt }
pub fn size(&self) -> usize { self.frames.byte_size() }

pub fn as_slice(&self) -> &[u8] {
    unsafe { core::slice::from_raw_parts(self.virt.0 as *const u8, self.size()) }
}

pub fn as_mut_slice(&mut self) -> &mut [u8] {
    unsafe { core::slice::from_raw_parts_mut(self.virt.0 as *mut u8, self.size()) }
}
}
```

kernel_workspace/base/kw-base/src/hal/dma.rs (line 22) - [public] virt_addr

```
pub fn virt_addr(&self) -> VirtAddr { self.virt }
pub fn size(&self) -> usize { self.frames.byte_size() }

pub fn as_slice(&self) -> &[u8] {
    unsafe { core::slice::from_raw_parts(self.virt.0 as *const u8, self.size()) }
}

pub fn as_mut_slice(&mut self) -> &mut [u8] {
    unsafe { core::slice::from_raw_parts_mut(self.virt.0 as *mut u8, self.size()) }
}
}
```

kernel_workspace/base/kw-base/src/hal/dma.rs (line 23) - [public] size

```
pub fn size(&self) -> usize { self.frames.byte_size() }

pub fn as_slice(&self) -> &[u8] {
    unsafe { core::slice::from_raw_parts(self.virt.0 as *const u8, self.size()) }
}

pub fn as_mut_slice(&mut self) -> &mut [u8] {
    unsafe { core::slice::from_raw_parts_mut(self.virt.0 as *mut u8, self.size()) }
}
}
```

kernel_workspace/base/kw-base/src/hal/dma.rs (line 25) - [public] as_slice

```
pub fn as_slice(&self) -> &[u8] {
    unsafe { core::slice::from_raw_parts(self.virt.0 as *const u8, self.size()) }
}

pub fn as_mut_slice(&mut self) -> &mut [u8] {
    unsafe { core::slice::from_raw_parts_mut(self.virt.0 as *mut u8, self.size()) }
}
}
```

kernel_workspace/base/kw-base/src/hal/dma.rs (line 29) - [public] as_mut_slice

```
pub fn as_mut_slice(&mut self) -> &mut [u8] {
    unsafe { core::slice::from_raw_parts_mut(self.virt.0 as *mut u8, self.size()) }
}
}
```

kernel_workspace/base/kw-base/src/hal/pci.rs (line 7) - [public] PciAddress

```
pub struct PciAddress {
    pub bus: u8,
    pub device: u8,
    pub function: u8,
}

impl PciAddress {
    pub const fn new(bus: u8, device: u8, function: u8) -> Self {
        Self { bus, device, function }
    }

    fn config_addr(self, offset: u8) -> u32 {
        // ... (truncated, see source file for more)
    }
}
```

kernel_workspace/base/kw-base/src/hal/pci.rs (line 18) - [internal] config_addr

```
fn config_addr(self, offset: u8) -> u32 {
    0x8000_0000
    | ((self.bus as u32) << 16)
    | ((self.device as u32 & 0x1F) << 11)
    | ((self.function as u32 & 0x07) << 8)
    | (offset as u32 & 0xFC)
}
}
```

```
#[derive(Debug, Clone, Copy)]
pub struct PciDevice {
    pub addr: PciAddress,
    // ... (truncated, see source file for more)
}
```

kernel_workspace/base/kw-base/src/hal/pci.rs (line 28) - [public] PciDevice

```
pub struct PciDevice {
    pub addr: PciAddress,
    pub vendor_id: u16,
    pub device_id: u16,
    pub class_code: u8,
    pub subclass: u8,
}

pub fn read_config32(addr: PciAddress, offset: u8) -> u32 {
    unsafe {
        outl(PCI_CONFIG_ADDR, addr.config_addr(offset));
        inl(PCI_CONFIG_DATA)
    }
    // ... (truncated, see source file for more)
}
```

kernel_workspace/base/kw-base/src/hal/pci.rs (line 36) - [public] read_config32

```
pub fn read_config32(addr: PciAddress, offset: u8) -> u32 {
    unsafe {
        outl(PCI_CONFIG_ADDR, addr.config_addr(offset));
        inl(PCI_CONFIG_DATA)
    }
}

pub fn write_config32(addr: PciAddress, offset: u8, value: u32) {
    unsafe {
        outl(PCI_CONFIG_ADDR, addr.config_addr(offset));
        outl(PCI_CONFIG_DATA, value);
    }
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/hal/pci.rs (line 43) - [public] write_config32

```
pub fn write_config32(addr: PciAddress, offset: u8, value: u32) {
    unsafe {
        outl(PCI_CONFIG_ADDR, addr.config_addr(offset));
        outl(PCI_CONFIG_DATA, value);
    }
}

pub fn scan_bus() -> kstd_data::Vec<PciDevice> {
    let mut devices = kstd_data::Vec::new();
    for bus in 0..=255 {
        for dev in 0..32 {
            let addr = PciAddress::new(bus, dev, 0);
            // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/hal/pci.rs (line 50) - [public] scan_bus

```
pub fn scan_bus() -> kstd_data::Vec<PciDevice> {
    let mut devices = kstd_data::Vec::new();
    for bus in 0..=255 {
        for dev in 0..32 {
            let addr = PciAddress::new(bus, dev, 0);
            let id = read_config32(addr, 0);
            let vendor = id as u16;
            if vendor == 0xFFFF { continue; }

            devices.push(PciDevice {
                addr,
                vendor_id: vendor,
            });
            // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/hal/timers.rs (line 4) - [internal] k_timer_get_ticks

```
fn k_timer_get_ticks() -> u64;
fn k_timer_get_frequency() -> u64;
fn k_timer_sleep_ms(ms: u32);
fn k_timer_set_oneshot(ns: u64) -> bool;
}

pub fn get_ticks() -> u64 {
    unsafe { k_timer_get_ticks() }
}

pub fn get_frequency() -> u64 {
    unsafe { k_timer_get_frequency() }
    // ... (truncated, see source file for more)
```

```

kernel_Workspace/base/kw-base/src/hal/timers.rs (line 5) - [internal] k_timer_get_frequency
    fn k_timer_get_frequency() -> u64;
    fn k_timer_sleep_ms(ms: u32);
    fn k_timer_set_oneshot(ns: u64) -> bool;
}

pub fn get_ticks() -> u64 {
    unsafe { k_timer_get_ticks() }
}

pub fn get_frequency() -> u64 {
    unsafe { k_timer_get_frequency() }
}
// ... (truncated, see source file for more)

kernel_Workspace/base/kw-base/src/hal/timers.rs (line 6) - [internal] k_timer_sleep_ms
    fn k_timer_sleep_ms(ms: u32);
    fn k_timer_set_oneshot(ns: u64) -> bool;
}

pub fn get_ticks() -> u64 {
    unsafe { k_timer_get_ticks() }
}

pub fn get_frequency() -> u64 {
    unsafe { k_timer_get_frequency() }
}
// ... (truncated, see source file for more)

kernel_Workspace/base/kw-base/src/hal/timers.rs (line 7) - [internal] k_timer_set_oneshot
    fn k_timer_set_oneshot(ns: u64) -> bool;
}

pub fn get_ticks() -> u64 {
    unsafe { k_timer_get_ticks() }
}

pub fn get_frequency() -> u64 {
    unsafe { k_timer_get_frequency() }
}

pub fn sleep_ms(ms: u32) {
    // ... (truncated, see source file for more)

kernel_Workspace/base/kw-base/src/hal/timers.rs (line 10) - [public] get_ticks
    pub fn get_ticks() -> u64 {
        unsafe { k_timer_get_ticks() }
    }

    pub fn get_frequency() -> u64 {
        unsafe { k_timer_get_frequency() }
    }

    pub fn sleep_ms(ms: u32) {
        unsafe { k_timer_sleep_ms(ms); }
    }

    // ... (truncated, see source file for more)

```

kernel_Workspace/base/kw-base/src/hal/timers.rs (line 14) - [public] get_frequency

```
pub fn get_frequency() -> u64 {
    unsafe { k_timer_get_frequency() }
}

pub fn sleep_ms(ms: u32) {
    unsafe { k_timer_sleep_ms(ms); }
}

pub fn set_oneshot(ns: u64) -> KResult<()> {
    if unsafe { k_timer_set_oneshot(ns) } {
        Ok(())
    } else {
        // ... (truncated, see source file for more)
    }
}
```

kernel_Workspace/base/kw-base/src/hal/timers.rs (line 18) - [public] sleep_ms

```
pub fn sleep_ms(ms: u32) {
    unsafe { k_timer_sleep_ms(ms); }
}

pub fn set_oneshot(ns: u64) -> KResult<()> {
    if unsafe { k_timer_set_oneshot(ns) } {
        Ok(())
    } else {
        Err(KernelError::IoError)
    }
}
```

kernel_Workspace/base/kw-base/src/hal/timers.rs (line 22) - [public] set_oneshot

```
pub fn set_oneshot(ns: u64) -> KResult<()> {
    if unsafe { k_timer_set_oneshot(ns) } {
        Ok(())
    } else {
        Err(KernelError::IoError)
    }
}
```

kernel_Workspace/base/kw-base/src/hal/usb.rs (line 4) - [internal] k_usb_init

```
fn k_usb_init() -> bool;
fn k_usb_enum_devices() -> u32;
fn k_usb_get_device_count() -> u32;

pub fn init() -> KResult<()> {
    if unsafe { k_usb_init() } {
        Ok(())
    } else {
        Err(KernelError::HardwareFault)
    }
}

// ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/hal/usb.rs (line 5) - [internal] k_usb_enum_devices

```
fn k_usb_enum_devices() -> u32;
fn k_usb_get_device_count() -> u32;

pub fn init() -> KResult<()> {
    if unsafe { k_usb_init() } {
        Ok(())
    } else {
        Err(KernelError::HardwareFault)
    }
}

// ... (truncated, see source file for more)
```

```

kernel_Workspace/base/kw-base/src/hal/usb.rs (line 6) - [internal] k_usb_get_device_count
    fn k_usb_get_device_count() -> u32;
}

pub fn init() -> KResult<()> {
    if unsafe { k_usb_init() } {
        Ok(())
    } else {
        Err(KernelError::HardwareFault)
    }
}

pub fn enumerate_devices() -> KResult<u32> {
    // ... (truncated, see source file for more)
}

kernel_Workspace/base/kw-base/src/hal/usb.rs (line 9) - [public] init
pub fn init() -> KResult<()> {
    if unsafe { k_usb_init() } {
        Ok(())
    } else {
        Err(KernelError::HardwareFault)
    }
}

pub fn enumerate_devices() -> KResult<u32> {
    let count = unsafe { k_usb_enum_devices() };
    if count == 0 && unsafe { k_usb_get_device_count() } == 0 {
        Err(KernelError::NotFound)
    }
    // ... (truncated, see source file for more)
}

kernel_Workspace/base/kw-base/src/hal/usb.rs (line 17) - [public] enumerate_devices
pub fn enumerate_devices() -> KResult<u32> {
    let count = unsafe { k_usb_enum_devices() };
    if count == 0 && unsafe { k_usb_get_device_count() } == 0 {
        Err(KernelError::NotFound)
    } else {
        Ok(count)
    }
}

kernel_Workspace/base/kw-base/src/ipc/message_queue.rs (line 4) - [internal] k_ipc_send
    fn k_ipc_send(dst: u32, a0: u64, a1: u64) -> i32;
    fn k_ipc_recv(a0: *mut u64, a1: *mut u64) -> i32;
}

#[derive(Debug, Clone, Copy, Default)]
pub struct IpcMessage {
    pub from: u32,
    pub arg0: u64,
    pub arg1: u64,
}

pub fn send(target_pid: u32, arg0: u64, arg1: u64) -> KResult<()> {
    // ... (truncated, see source file for more)
}

kernel_Workspace/base/kw-base/src/ipc/message_queue.rs (line 5) - [internal] k_ipc_recv
    fn k_ipc_recv(a0: *mut u64, a1: *mut u64) -> i32;
}

#[derive(Debug, Clone, Copy, Default)]
pub struct IpcMessage {
    pub from: u32,
    pub arg0: u64,
    pub arg1: u64,
}

pub fn send(target_pid: u32, arg0: u64, arg1: u64) -> KResult<()> {
    let rc = unsafe { k_ipc_send(target_pid, arg0, arg1) };
    // ... (truncated, see source file for more)
}

```

kernel_workspace/base/kw-base/src/ipc/message_queue.rs (line 9) - [public] IpcMessage

```
pub struct IpcMessage {
    pub from: u32,
    pub arg0: u64,
    pub arg1: u64,
}

pub fn send(target_pid: u32, arg0: u64, arg1: u64) -> KResult<()> {
    let rc = unsafe { k_ipc_send(target_pid, arg0, arg1) };
    if rc == 0 { Ok(()) } else { Err(KernelError::from_code(rc)) }
}

pub fn recv() -> KResult<IpcMessage> {
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/ipc/message_queue.rs (line 15) - [public] send

```
pub fn send(target_pid: u32, arg0: u64, arg1: u64) -> KResult<()> {
    let rc = unsafe { k_ipc_send(target_pid, arg0, arg1) };
    if rc == 0 { Ok(()) } else { Err(KernelError::from_code(rc)) }
}

pub fn recv() -> KResult<IpcMessage> {
    let mut a0: u64 = 0;
    let mut a1: u64 = 0;
    let from = unsafe { k_ipc_recv(&mut a0, &mut a1) };
    if from < 0 {
        Err(KernelError::from_code(from))
    } else {
        // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/ipc/message_queue.rs (line 20) - [public] recv

```
pub fn recv() -> KResult<IpcMessage> {
    let mut a0: u64 = 0;
    let mut a1: u64 = 0;
    let from = unsafe { k_ipc_recv(&mut a0, &mut a1) };
    if from < 0 {
        Err(KernelError::from_code(from))
    } else {
        Ok(IpcMessage { from: from as u32, arg0: a0, arg1: a1 })
    }
}
```

kernel_workspace/base/kw-base/src/ipc/shared_mem.rs (line 5) - [public] SharedMemory

```
pub struct SharedMemory {
    frames: ContiguousFrames,
    virt: VirtAddr,
}

impl SharedMemory {
    pub fn allocate(pages: usize) -> KResult<Self> {
        let frames = ContiguousFrames::alloc(pages)?;
        let virt = gluecore::bridge::mm::vmm::alloc(
            frames.byte_size(),
            gluecore::bridge::mm::vmm::VMM_FLAG_READ | gluecore::bridge::mm::vmm::VMM_FLAG_WRITE
        ).map_err(KernelError::from)?;
        // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/ipc/shared_mem.rs (line 11) - [public] allocate

```
pub fn allocate(pages: usize) -> KResult<Self> {
    let frames = ContiguousFrames::alloc(pages)?;
    let virt = gluecore::bridge::mm::vmm::alloc(
        frames.byte_size(),
        gluecore::bridge::mm::vmm::VMM_FLAG_READ | gluecore::bridge::mm::vmm::VMM_FLAG_WRITE
    ).map_err(KernelError::from)?;

    for i in 0..pages {
        let phys = PhysAddr(frames.phys_addr().0 + (i as u64 * 4096));
        let v = VirtAddr(virt.0 + (i as u64 * 4096));
        paging::map_page(v, phys, paging::PAGE_PRESENT | paging::PAGE_WRITABLE)?;
    }
    // ... (truncated, see source file for more)
```



```

kernel_Workspace/base/kw-base/src/ipc/shared_mem.rs (line 27) - [public] virt_addr
    pub fn virt_addr(&self) -> VirtAddr { self.virt }
    pub fn size(&self) -> usize { self.frames.byte_size() }
}

impl Drop for SharedMemory {
    fn drop(&mut self) {
        let _ = gluecore::bridge::mm::vmm::free(self.virt, self.frames.byte_size());
    }
}

kernel_Workspace/base/kw-base/src/ipc/shared_mem.rs (line 28) - [public] size
    pub fn size(&self) -> usize { self.frames.byte_size() }
}

impl Drop for SharedMemory {
    fn drop(&mut self) {
        let _ = gluecore::bridge::mm::vmm::free(self.virt, self.frames.byte_size());
    }
}

kernel_Workspace/base/kw-base/src/ipc/shared_mem.rs (line 32) - [internal] drop
    fn drop(&mut self) {
        let _ = gluecore::bridge::mm::vmm::free(self.virt, self.frames.byte_size());
    }
}

kernel_Workspace/base/kw-base/src/mem/buddy.rs (line 4) - [public] ContiguousFrames
pub struct ContiguousFrames {
    addr: PhysAddr,
    count: usize,
}

impl ContiguousFrames {
    pub fn alloc(count: usize) -> KResult<Self> {
        let addr = pmm::alloc_frames(count).map_err(KernelError::from)?;
        Ok(Self { addr, count })
    }

    pub fn alloc_aligned(count: usize, align_frames: usize) -> KResult<Self> {
        // ... (truncated, see source file for more)
    }
}

kernel_Workspace/base/kw-base/src/mem/buddy.rs (line 10) - [public] alloc
    pub fn alloc(count: usize) -> KResult<Self> {
        let addr = pmm::alloc_frames(count).map_err(KernelError::from)?;
        Ok(Self { addr, count })
    }

    pub fn alloc_aligned(count: usize, align_frames: usize) -> KResult<Self> {
        let addr = pmm::alloc_frames_aligned(count, align_frames).map_err(KernelError::from)?;
        Ok(Self { addr, count })
    }

    pub fn phys_addr(&self) -> PhysAddr { self.addr }
    pub fn frame_count(&self) -> usize { self.count }
    // ... (truncated, see source file for more)
}

kernel_Workspace/base/kw-base/src/mem/buddy.rs (line 15) - [public] alloc_aligned
    pub fn alloc_aligned(count: usize, align_frames: usize) -> KResult<Self> {
        let addr = pmm::alloc_frames_aligned(count, align_frames).map_err(KernelError::from)?;
        Ok(Self { addr, count })
    }

    pub fn phys_addr(&self) -> PhysAddr { self.addr }
    pub fn frame_count(&self) -> usize { self.count }
    pub fn byte_size(&self) -> usize { self.count * 4096 }

    pub fn into_raw(self) -> (PhysAddr, usize) {
        let addr = self.addr;
        let count = self.count;
        // ... (truncated, see source file for more)
    }
}

```

kernel_Workspace/base/kw-base/src/mem/buddy.rs (line 20) - [public] phys_addr

```
pub fn phys_addr(&self) -> PhysAddr { self.addr }
pub fn frame_count(&self) -> usize { self.count }
pub fn byte_size(&self) -> usize { self.count * 4096 }

pub fn into_raw(self) -> (PhysAddr, usize) {
    let addr = self.addr;
    let count = self.count;
    core::mem::forget(self);
    (addr, count)
}

pub unsafe fn from_raw(addr: PhysAddr, count: usize) -> Self {
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/mem/buddy.rs (line 21) - [public] frame_count

```
pub fn frame_count(&self) -> usize { self.count }
pub fn byte_size(&self) -> usize { self.count * 4096 }

pub fn into_raw(self) -> (PhysAddr, usize) {
    let addr = self.addr;
    let count = self.count;
    core::mem::forget(self);
    (addr, count)
}

pub unsafe fn from_raw(addr: PhysAddr, count: usize) -> Self {
    Self { addr, count }
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/mem/buddy.rs (line 22) - [public] byte_size

```
pub fn byte_size(&self) -> usize { self.count * 4096 }

pub fn into_raw(self) -> (PhysAddr, usize) {
    let addr = self.addr;
    let count = self.count;
    core::mem::forget(self);
    (addr, count)
}

pub unsafe fn from_raw(addr: PhysAddr, count: usize) -> Self {
    Self { addr, count }
}
// ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/mem/buddy.rs (line 24) - [public] into_raw

```
pub fn into_raw(self) -> (PhysAddr, usize) {
    let addr = self.addr;
    let count = self.count;
    core::mem::forget(self);
    (addr, count)
}

pub unsafe fn from_raw(addr: PhysAddr, count: usize) -> Self {
    Self { addr, count }
}
}

// ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/mem/buddy.rs (line 37) - [internal] drop

```
fn drop(&mut self) {
    let _ = pmm::free_frames(self.addr, self.count);
}
}
```

kernel_Workspace/base/kw-base/src/mem/frame.rs (line 4) - [public] Frame

```
pub struct Frame {
    addr: PhysAddr,
}

impl Frame {
    pub fn alloc() -> KResult<Self> {
        let addr = pmm::alloc_frame().map_err(KernelError::from)?;
        Ok(Self { addr })
    }

    pub fn alloc_zeroed() -> KResult<Self> {
        let addr = pmm::alloc_zero_frame().map_err(KernelError::from)?;
        // ... (truncated, see source file for more)
    }
}
```

kernel_Workspace/base/kw-base/src/mem/frame.rs (line 9) - [public] alloc

```
pub fn alloc() -> KResult<Self> {
    let addr = pmm::alloc_frame().map_err(KernelError::from)?;
    Ok(Self { addr })
}

pub fn alloc_zeroed() -> KResult<Self> {
    let addr = pmm::alloc_zero_frame().map_err(KernelError::from)?;
    Ok(Self { addr })
}

pub fn phys_addr(&self) -> PhysAddr {
    self.addr
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/base/kw-base/src/mem/frame.rs (line 14) - [public] alloc_zeroed

```
pub fn alloc_zeroed() -> KResult<Self> {
    let addr = pmm::alloc_zero_frame().map_err(KernelError::from)?;
    Ok(Self { addr })
}

pub fn phys_addr(&self) -> PhysAddr {
    self.addr
}

pub fn into_raw(self) -> PhysAddr {
    let addr = self.addr;
    core::mem::forget(self);
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/base/kw-base/src/mem/frame.rs (line 19) - [public] phys_addr

```
pub fn phys_addr(&self) -> PhysAddr {
    self.addr
}

pub fn into_raw(self) -> PhysAddr {
    let addr = self.addr;
    core::mem::forget(self);
    addr
}

pub unsafe fn from_raw(addr: PhysAddr) -> Self {
    Self { addr }
    // ... (truncated, see source file for more)
}
```

kernel_workspace/base/kw-base/src/mem/frame.rs (line 23) - [public] into_raw

```
pub fn into_raw(self) -> PhysAddr {
    let addr = self.addr;
    core::mem::forget(self);
    addr
}

pub unsafe fn from_raw(addr: PhysAddr) -> Self {
    Self { addr }
}

impl Drop for Frame {
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/mem/frame.rs (line 35) - [internal] drop

```
fn drop(&mut self) {
    let _ = pmm::free_frame(self.addr);
}
```

kernel_workspace/base/kw-base/src/mem/heap.rs (line 4) - [public] alloc

```
pub fn alloc<T>() -> KResult<*mut T> {
    let layout = Layout::new(core::mem::size_of::<T>(), core::mem::align_of::<T>());
    let ptr = KernelHeap.alloc(layout).map_err(KernelError::from)?;
    Ok(ptr as *mut T)
}

pub fn alloc_zeroed<T>() -> KResult<*mut T> {
    let ptr = alloc::<T>()?;
    unsafe { core::ptr::write_bytes(ptr, 0, 1); }
    Ok(ptr)
}

// ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/mem/heap.rs (line 10) - [public] alloc_zeroed

```
pub fn alloc_zeroed<T>() -> KResult<*mut T> {
    let ptr = alloc::<T>()?;
    unsafe { core::ptr::write_bytes(ptr, 0, 1); }
    Ok(ptr)
}

pub unsafe fn free<T>(ptr: *mut T) {
    let layout = Layout::new(core::mem::size_of::<T>(), core::mem::align_of::<T>());
    KernelHeap.dealloc(ptr as *mut u8, layout);
}

pub fn alloc_slice<T>(len: usize) -> KResult<*mut T> {
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/mem/heap.rs (line 21) - [public] alloc_slice

```
pub fn alloc_slice<T>(len: usize) -> KResult<*mut T> {
    let layout = Layout::new(core::mem::size_of::<T>() * len, core::mem::align_of::<T>());
    let ptr = KernelHeap.alloc(layout).map_err(KernelError::from)?;
    Ok(ptr as *mut T)
}

pub unsafe fn free_slice<T>(ptr: *mut T, len: usize) {
    let layout = Layout::new(core::mem::size_of::<T>() * len, core::mem::align_of::<T>());
    KernelHeap.dealloc(ptr as *mut u8, layout);
}
```

kernel_Workspace/base/kw-base/src/mem/paging.rs (line 14) - [internal] paging_map_page

```
fn paging_map_page(virt: u64, phys: u64, flags: u64) -> bool;
fn paging_unmap_page(virt: u64) -> bool;
}

pub fn map_page(virt: VirtAddr, phys: PhysAddr, flags: u64) -> KResult<()> {
    if unsafe { paging_map_page(virt.0, phys.0, flags) } {
        Ok(())
    } else {
        Err(KernelError::NoMemory)
    }
}

// ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/mem/paging.rs (line 15) - [internal] paging_unmap_page

```
fn paging_unmap_page(virt: u64) -> bool;
}

pub fn map_page(virt: VirtAddr, phys: PhysAddr, flags: u64) -> KResult<()> {
    if unsafe { paging_map_page(virt.0, phys.0, flags) } {
        Ok(())
    } else {
        Err(KernelError::NoMemory)
    }
}

pub fn unmap_page(virt: VirtAddr) -> KResult<()> {
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/mem/paging.rs (line 18) - [public] map_page

```
pub fn map_page(virt: VirtAddr, phys: PhysAddr, flags: u64) -> KResult<()> {
    if unsafe { paging_map_page(virt.0, phys.0, flags) } {
        Ok(())
    } else {
        Err(KernelError::NoMemory)
    }
}

pub fn unmap_page(virt: VirtAddr) -> KResult<()> {
    if unsafe { paging_unmap_page(virt.0) } {
        Ok(())
    } else {
        // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/mem/paging.rs (line 26) - [public] unmap_page

```
pub fn unmap_page(virt: VirtAddr) -> KResult<()> {
    if unsafe { paging_unmap_page(virt.0) } {
        Ok(())
    } else {
        Err(KernelError::InvalidArg)
    }
}

pub fn flush_tlb_page(virt: VirtAddr) {
    context::flush_page(virt);
}

// ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/mem/paging.rs (line 34) - [public] flush_tlb_page

```
pub fn flush_tlb_page(virt: VirtAddr) {
    context::flush_page(virt);
}

pub fn flush_tlb_all() {
    context::flush_tlb_all();
}
```

```

kernel_Workspace/base/kw-base/src/mem/paging.rs (line 38) - [public] flush_tlb_all
    pub fn flush_tlb_all() {
        context::flush_tlb_all();
    }

kernel_Workspace/base/kw-base/src/mem/slab.rs (line 4) - [internal] slab_cache_create
    fn slab_cache_create(name: *const u8, obj_size: usize, align: usize) -> *mut u8;
    fn slab_cache_destroy(cache: *mut u8);
    fn slab_alloc(cache: *mut u8) -> *mut u8;
    fn slab_free(cache: *mut u8, ptr: *mut u8);
}

pub struct SlabCache {
    inner: *mut u8,
}

unsafe impl Send for SlabCache {}
unsafe impl Sync for SlabCache {}
// ... (truncated, see source file for more)

kernel_Workspace/base/kw-base/src/mem/slab.rs (line 5) - [internal] slab_cache_destroy
    fn slab_cache_destroy(cache: *mut u8);
    fn slab_alloc(cache: *mut u8) -> *mut u8;
    fn slab_free(cache: *mut u8, ptr: *mut u8);
}

pub struct SlabCache {
    inner: *mut u8,
}

unsafe impl Send for SlabCache {}
unsafe impl Sync for SlabCache {}

// ... (truncated, see source file for more)

kernel_Workspace/base/kw-base/src/mem/slab.rs (line 6) - [internal] slab_alloc
    fn slab_alloc(cache: *mut u8) -> *mut u8;
    fn slab_free(cache: *mut u8, ptr: *mut u8);
}

pub struct SlabCache {
    inner: *mut u8,
}

unsafe impl Send for SlabCache {}
unsafe impl Sync for SlabCache {}

impl SlabCache {
    // ... (truncated, see source file for more)

kernel_Workspace/base/kw-base/src/mem/slab.rs (line 7) - [internal] slab_free
    fn slab_free(cache: *mut u8, ptr: *mut u8);
}

pub struct SlabCache {
    inner: *mut u8,
}

unsafe impl Send for SlabCache {}
unsafe impl Sync for SlabCache {}

impl SlabCache {
    pub fn new(name: &str, obj_size: usize, align: usize) -> KResult<Self> {
        // ... (truncated, see source file for more)
    }
}

```

kernel_workspace/base/kw-base/src/mem/slab.rs (line 10) - [public] SlabCache

```
pub struct SlabCache {
    inner: *mut u8,
}

unsafe impl Send for SlabCache {}
unsafe impl Sync for SlabCache {}

impl SlabCache {
    pub fn new(name: &str, obj_size: usize, align: usize) -> KResult<Self> {
        let c_name = name.as_ptr();
        let cache = unsafe { slab_cache_create(c_name, obj_size, align) };
        if cache.is_null() {
            // ... (truncated, see source file for more)
        }
    }
}
```

kernel_workspace/base/kw-base/src/mem/slab.rs (line 18) - [public] new

```
pub fn new(name: &str, obj_size: usize, align: usize) -> KResult<Self> {
    let c_name = name.as_ptr();
    let cache = unsafe { slab_cache_create(c_name, obj_size, align) };
    if cache.is_null() {
        Err(KernelError::NoMemory)
    } else {
        Ok(Self { inner: cache })
    }
}

pub fn alloc<T>(&self) -> KResult<*mut T> {
    let ptr = unsafe { slab_alloc(self.inner) };
    // ... (truncated, see source file for more)
}
```

kernel_workspace/base/kw-base/src/mem/slab.rs (line 28) - [public] alloc

```
pub fn alloc<T>(&self) -> KResult<*mut T> {
    let ptr = unsafe { slab_alloc(self.inner) };
    if ptr.is_null() {
        Err(KernelError::NoMemory)
    } else {
        Ok(ptr as *mut T)
    }
}

pub fn free<T>(&self, ptr: *mut T) {
    if !ptr.is_null() {
        unsafe { slab_free(self.inner, ptr as *mut u8) };
    }
    // ... (truncated, see source file for more)
}
```

kernel_workspace/base/kw-base/src/mem/slab.rs (line 37) - [public] free

```
pub fn free<T>(&self, ptr: *mut T) {
    if !ptr.is_null() {
        unsafe { slab_free(self.inner, ptr as *mut u8) };
    }
}

impl Drop for SlabCache {
    fn drop(&mut self) {
        if !self.inner.is_null() {
            unsafe { slab_cache_destroy(self.inner) };
        }
        // ... (truncated, see source file for more)
    }
}
```

kernel_workspace/base/kw-base/src/mem/slab.rs (line 45) - [internal] drop

```
fn drop(&mut self) {
    if !self.inner.is_null() {
        unsafe { slab_cache_destroy(self.inner) };
    }
}
```

kernel_workspace/base/kw-base/src/sync/atomic.rs (line 5) - [public] compiler_fence

```
pub fn compiler_fence() {
    core::sync::atomic::compiler_fence(Ordering::SeqCst);
}

#[inline(always)]
pub fn memory_barrier() {
    #[cfg(target_arch = "x86_64")]
    unsafe { core::arch::asm!("mfence", options(nostack, preserves_flags)); }
}
```

kernel_workspace/base/kw-base/src/sync/atomic.rs (line 10) - [public] memory_barrier

```
pub fn memory_barrier() {
    #[cfg(target_arch = "x86_64")]
    unsafe { core::arch::asm!("mfence", options(nostack, preserves_flags)); }
}
```

kernel_workspace/base/kw-base/src/sync/futex.rs (line 4) - [internal] k_futex_wait

```
fn k_futex_wait(uaddr: *const u32, expected_val: u32, timeout_ns: u64) -> i32;
fn k_futex_wake(uaddr: *const u32, wake_count: u32) -> i32;
}

pub const FUTEX_WAIT: u32 = 0;
pub const FUTEX_WAKE: u32 = 1;

pub fn wait(uaddr: *const u32, expected: u32, timeout_ns: u64) -> KResult<()> {
    let rc = unsafe { k_futex_wait(uaddr, expected, timeout_ns) };
    if rc == 0 { Ok(()) } else { Err(KernelError::from_code(rc)) }
}

// ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/sync/futex.rs (line 5) - [internal] k_futex_wake

```
fn k_futex_wake(uaddr: *const u32, wake_count: u32) -> i32;
}

pub const FUTEX_WAIT: u32 = 0;
pub const FUTEX_WAKE: u32 = 1;

pub fn wait(uaddr: *const u32, expected: u32, timeout_ns: u64) -> KResult<()> {
    let rc = unsafe { k_futex_wait(uaddr, expected, timeout_ns) };
    if rc == 0 { Ok(()) } else { Err(KernelError::from_code(rc)) }
}

pub fn wake(uaddr: *const u32, count: u32) -> KResult<u32> {
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/sync/futex.rs (line 11) - [public] wait

```
pub fn wait(uaddr: *const u32, expected: u32, timeout_ns: u64) -> KResult<()> {
    let rc = unsafe { k_futex_wait(uaddr, expected, timeout_ns) };
    if rc == 0 { Ok(()) } else { Err(KernelError::from_code(rc)) }
}

pub fn wake(uaddr: *const u32, count: u32) -> KResult<u32> {
    let rc = unsafe { k_futex_wake(uaddr, count) };
    if rc < 0 { Err(KernelError::from_code(rc)) } else { Ok(rc as u32) }
}
```

kernel_workspace/base/kw-base/src/sync/futex.rs (line 16) - [public] wake

```
pub fn wake(uaddr: *const u32, count: u32) -> KResult<u32> {
    let rc = unsafe { k_futex_wake(uaddr, count) };
    if rc < 0 { Err(KernelError::from_code(rc)) } else { Ok(rc as u32) }
}
```


kernel_Workspace/base/kw-base/src/sync/mutex.rs (line 11) - [public] Mutex

// A sleeping mutex. Uses atomics for the fast path.

// In a full implementation, the slow path would enqueue the thread

// into a WaitQueue and call the scheduler. Here we use yield as a placeholder.

```
pub struct Mutex<T> {
    state: AtomicU8,
    data: UnsafeCell<T>,
}

unsafe impl<T: Send> Sync for Mutex<T> {}
unsafe impl<T: Send> Send for Mutex<T> {}

impl<T> Mutex<T> {
    pub const fn new(data: T) -> Self {
        Self {
            state: AtomicU8::new(UNLOCKED),
            // ... (truncated, see source file for more)
        }
    }
}
```

kernel_Workspace/base/kw-base/src/sync/mutex.rs (line 27) - [public] lock

```
pub fn lock(&self) -> MutexGuard<'_, T> {
    loop {
        if self.state.compare_exchange_weak(
            UNLOCKED, LOCKED, Ordering::Acquire, Ordering::Relaxed
        ).is_ok() {
            return MutexGuard { lock: self };
        }
        // Slow path: yield CPU instead of busy-spinning
        crate::task::scheduler::yield_now();
    }
}

// ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/sync/mutex.rs (line 39) - [public] try_lock

```
pub fn try_lock(&self) -> Option<MutexGuard<'_, T>> {
    if self.state.compare_exchange(
        UNLOCKED, LOCKED, Ordering::Acquire, Ordering::Relaxed
    ).is_ok() {
        Some(MutexGuard { lock: self })
    } else {
        None
    }
}

fn unlock(&self) {
    self.state.store(UNLOCKED, Ordering::Release);
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/base/kw-base/src/sync/mutex.rs (line 49) - [internal] unlock

```
fn unlock(&self) {
    self.state.store(UNLOCKED, Ordering::Release);
    // TODO: Wake up one thread from the waitqueue if any
}

pub struct MutexGuard<'a, T> {
    lock: &'a Mutex<T>,
}

impl<T> Deref for MutexGuard<'_, T> {
    type Target = T;
    // ... (truncated, see source file for more)
}
```

kernel_workspace/base/kw-base/src/sync/mutex.rs (line 55) - [public] MutexGuard

```
pub struct MutexGuard<'a, T> {
    lock: &'a Mutex<T>,
}

impl<T> Deref for MutexGuard<'_, T> {
    type Target = T;
    fn deref(&self) -> &T { unsafe { &*self.lock.data.get() } }
}

impl<T> DerefMut for MutexGuard<'_, T> {
    fn deref_mut(&mut self) -> &mut T { unsafe { &mut *self.lock.data.get() } }
}

// ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/sync/mutex.rs (line 61) - [internal] deref

```
fn deref(&self) -> &T { unsafe { &*self.lock.data.get() } }
}

impl<T> DerefMut for MutexGuard<'_, T> {
    fn deref_mut(&mut self) -> &mut T { unsafe { &mut *self.lock.data.get() } }
}

impl<T> Drop for MutexGuard<'_, T> {
    fn drop(&mut self) { self.lock.unlock(); }
}
```

kernel_workspace/base/kw-base/src/sync/mutex.rs (line 65) - [internal] deref_mut

```
fn deref_mut(&mut self) -> &mut T { unsafe { &mut *self.lock.data.get() } }
}

impl<T> Drop for MutexGuard<'_, T> {
    fn drop(&mut self) { self.lock.unlock(); }
}
```

kernel_workspace/base/kw-base/src/sync/mutex.rs (line 69) - [internal] drop

```
fn drop(&mut self) { self.lock.unlock(); }
}
```

kernel_workspace/base/kw-base/src/sync/rwlock.rs (line 5) - [public] RwLock

```
pub struct RwLock<T> {
    // -1 = write locked, >0 = number of readers, 0 = unlocked
    state: AtomicIsz,
    data: UnsafeCell<T>,
}

unsafe impl<T: Send + Sync> Sync for RwLock<T> {}
unsafe impl<T: Send> Send for RwLock<T> {}

impl<T> RwLock<T> {
    pub const fn new(data: T) -> Self {
        Self { state: AtomicIsz::new(0), data: UnsafeCell::new(data) }
    }
    // ... (truncated, see source file for more)
}
```

kernel_workspace/base/kw-base/src/sync/rwlock.rs (line 19) - [public] read

```
pub fn read(&self) -> RwLockReadGuard<'_, T> {
    loop {
        let val = self.state.load(Ordering::Relaxed);
        if val >= 0 {
            if self.state.compare_exchange_weak(
                val, val + 1, Ordering::Acquire, Ordering::Relaxed
            ).is_ok() {
                return RwLockReadGuard { lock: self };
            }
        }
        crate::task::scheduler::yield_now();
    }
}

// ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/sync/rwlock.rs (line 33) - [public] write

```
pub fn write(&self) -> RwLockWriteGuard<'_, T> {
    loop {
        if self.state.compare_exchange_weak(
            0, -1, Ordering::Acquire, Ordering::Relaxed
        ).is_ok() {
            return RwLockWriteGuard { lock: self };
        }
        crate::task::scheduler::yield_now();
    }
}

// ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/sync/rwlock.rs (line 45) - [public] RwLockReadGuard

```
pub struct RwLockReadGuard<'a, T> { lock: &'a RwLock<T> }
impl<T> Deref for RwLockReadGuard<'_, T> {
    type Target = T;
    fn deref(&self) -> &T { unsafe { &*self.lock.data.get() } }
}
impl<T> Drop for RwLockReadGuard<'_, T> {
    fn drop(&mut self) { self.lock.state.fetch_sub(1, Ordering::Release); }
}

pub struct RwLockWriteGuard<'a, T> { lock: &'a RwLock<T> }
impl<T> Deref for RwLockWriteGuard<'_, T> {
    type Target = T;
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/sync/rwlock.rs (line 48) - [internal] deref

```
fn deref(&self) -> &T { unsafe { &*self.lock.data.get() } }
}
impl<T> Drop for RwLockReadGuard<'_, T> {
    fn drop(&mut self) { self.lock.state.fetch_sub(1, Ordering::Release); }
}

pub struct RwLockWriteGuard<'a, T> { lock: &'a RwLock<T> }
impl<T> Deref for RwLockWriteGuard<'_, T> {
    type Target = T;
    fn deref(&self) -> &T { unsafe { &*self.lock.data.get() } }
}
impl<T> DerefMut for RwLockWriteGuard<'_, T> {
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/sync/rwlock.rs (line 51) - [internal] drop

```
fn drop(&mut self) { self.lock.state.fetch_sub(1, Ordering::Release); }
}

pub struct RwLockWriteGuard<'a, T> { lock: &'a RwLock<T> }
impl<T> Deref for RwLockWriteGuard<'_, T> {
    type Target = T;
    fn deref(&self) -> &T { unsafe { &*self.lock.data.get() } }
}
impl<T> DerefMut for RwLockWriteGuard<'_, T> {
    fn deref_mut(&mut self) -> &mut T { unsafe { &mut *self.lock.data.get() } }
}
impl<T> Drop for RwLockWriteGuard<'_, T> {
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/sync/rwlock.rs (line 54) - [public] RwLockWriteGuard

```
pub struct RwLockWriteGuard<'a, T> { lock: &'a RwLock<T> }
impl<T> Deref for RwLockWriteGuard<'_, T> {
    type Target = T;
    fn deref(&self) -> &T { unsafe { &*self.lock.data.get() } }
}
impl<T> DerefMut for RwLockWriteGuard<'_, T> {
    fn deref_mut(&mut self) -> &mut T { unsafe { &mut *self.lock.data.get() } }
}
impl<T> Drop for RwLockWriteGuard<'_, T> {
    fn drop(&mut self) { self.lock.state.store(0, Ordering::Release); }
}
```

kernel_workspace/base/kw-base/src/sync/rwlock.rs (line 57) - [internal] deref

```
fn deref(&self) -> &T { unsafe { &*self.lock.data.get() } }
}
impl<T> DerefMut for RwLockWriteGuard<'_, T> {
    fn deref_mut(&mut self) -> &mut T { unsafe { &mut *self.lock.data.get() } }
}
impl<T> Drop for RwLockWriteGuard<'_, T> {
    fn drop(&mut self) { self.lock.state.store(0, Ordering::Release); }
}
```

kernel_workspace/base/kw-base/src/sync/rwlock.rs (line 60) - [internal] deref_mut

```
fn deref_mut(&mut self) -> &mut T { unsafe { &mut *self.lock.data.get() } }
}
impl<T> Drop for RwLockWriteGuard<'_, T> {
    fn drop(&mut self) { self.lock.state.store(0, Ordering::Release); }
}
```

kernel_workspace/base/kw-base/src/sync/rwlock.rs (line 63) - [internal] drop

```
fn drop(&mut self) { self.lock.state.store(0, Ordering::Release); }
}
```

kernel_workspace/base/kw-base/src/sync/semaphore.rs (line 3) - [public] Semaphore

```
pub struct Semaphore {
    count: AtomicI32,
}

impl Semaphore {
    pub const fn new(initial: i32) -> Self {
        Self { count: AtomicI32::new(initial) }
    }

    pub fn acquire(&self) {
        loop {
            let val = self.count.load(Ordering::Relaxed);
            // ... (truncated, see source file for more)
        }
    }
}
```

kernel_workspace/base/kw-base/src/sync/semaphore.rs (line 12) - [public] acquire

```
pub fn acquire(&self) {
    loop {
        let val = self.count.load(Ordering::Relaxed);
        if val > 0 {
            if self.count.compare_exchange_weak(
                val, val - 1, Ordering::Acquire, Ordering::Relaxed
            ).is_ok() {
                return;
            }
        }
        crate::task::scheduler::yield_now();
    }
    // ... (truncated, see source file for more)
}
```

kernel_workspace/base/kw-base/src/sync/semaphore.rs (line 26) - [public] try_acquire

```
pub fn try_acquire(&self) -> bool {
    let val = self.count.load(Ordering::Relaxed);
    if val > 0 {
        self.count.compare_exchange(
            val, val - 1, Ordering::Acquire, Ordering::Relaxed
        ).is_ok()
    } else {
        false
    }
}

pub fn release(&self) {
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/sync/semaphore.rs (line 37) - [public] release

```
pub fn release(&self) {
    self.count.fetch_add(1, Ordering::Release);
    // TODO: Wake up one thread from the waitqueue
}
}
```

kernel_workspace/base/kw-base/src/task/context.rs (line 6) - [public] context_switch

```
pub fn context_switch(old_ctx: *mut TaskContext, new_ctx: *mut TaskContext);
}

pub unsafe fn switch(old: &mut TaskContext, new: &TaskContext) {
    context_switch(old, new as *const _ as *mut _);
}
}
```

kernel_workspace/base/kw-base/src/task/process.rs (line 6) - [public] ProcessState

```
pub enum ProcessState {
    Created,
    Running,
    Sleeping,
    Zombie,
    Dead,
}

pub struct Process {
    pub pid: Pid,
    pub world_id: WorldId,
    pub state: ProcessState,
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/task/process.rs (line 14) - [public] Process

```
pub struct Process {
    pub pid: Pid,
    pub world_id: WorldId,
    pub state: ProcessState,
    pub cr3_phys: u64,
}

impl Process {
    pub fn new(pid: Pid, world_id: WorldId) -> KResult<Self> {
        let cr3 = pmm::alloc_zero_frame().map_err(KernelError::from)?;
        Ok(Self {
            pid,
            // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/task/process.rs (line 22) - [public] new

```
pub fn new(pid: Pid, world_id: WorldId) -> KResult<Self> {
    let cr3 = pmm::alloc_zero_frame().map_err(KernelError::from)?;
    Ok(Self {
        pid,
        world_id,
        state: ProcessState::Created,
        cr3_phys: cr3.0,
    })
}

pub fn transition(&mut self, new_state: ProcessState) -> KResult<()> {
    match (self.state, new_state) {
        // ... (truncated, see source file for more)
    }
}
```

kernel_workspace/base/kw-base/src/task/process.rs (line 32) - [public] transition

```
pub fn transition(&mut self, new_state: ProcessState) -> KResult<()> {
    match (self.state, new_state) {
        (ProcessState::Created, ProcessState::Running) |
        (ProcessState::Running, ProcessState::Sleeping) |
        (ProcessState::Sleeping, ProcessState::Running) |
        (ProcessState::Running, ProcessState::Zombie) |
        (ProcessState::Zombie, ProcessState::Dead) => {
            self.state = new_state;
            Ok(())
        }
        _ => Err(KernelError::InvalidArg),
    }
    // ... (truncated, see source file for more)
}
```

kernel_workspace/base/kw-base/src/task/process.rs (line 48) - [internal] drop

```
fn drop(&mut self) {
    if self.cr3_phys != 0 {
        let _ = pmm::free_frame(kstd_base::PhysAddr(self.cr3_phys));
    }
}
}
```

kernel_workspace/base/kw-base/src/task/queue.rs (line 5) - [public] WaitQueue

```
pub struct WaitQueue {
    lock: SpinLock<()>,
    head: ListHead,
}

impl WaitQueue {
    pub const fn new() -> Self {
        Self {
            lock: SpinLock::new(),
            head: ListHead::new(),
        }
    }
    // ... (truncated, see source file for more)
}
```

kernel_workspace/base/kw-base/src/task/queue.rs (line 18) - [public] init

```
pub fn init(&mut self) {
    self.head.init();
}

pub fn sleep(&self, thread: &mut Thread) {
    let _guard = self.lock.lock();
    thread.state = super::thread::ThreadState::Blocked;
    // In a real implementation, we would link thread's list_node into self.head
    // and then call context::switch() to the scheduler.
}

pub fn wake_one(&self) {
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/base/kw-base/src/task/queue.rs (line 22) - [public] sleep

```
pub fn sleep(&self, thread: &mut Thread) {
    let _guard = self.lock.lock();
    thread.state = super::thread::ThreadState::Blocked;
    // In a real implementation, we would link thread's list_node into self.head
    // and then call context::switch() to the scheduler.
}

pub fn wake_one(&self) {
    let _guard = self.lock.lock();
    if self.head.is_empty() { return; }
    // In a real implementation, we would pop the first thread from self.head,
    // set its state to Ready, and add it to the scheduler runqueue.
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/task/queue.rs (line 29) - [public] wake_one

```
pub fn wake_one(&self) {
    let _guard = self.lock.lock();
    if self.head.is_empty() { return; }
    // In a real implementation, we would pop the first thread from self.head,
    // set its state to Ready, and add it to the scheduler runqueue.
}

pub fn wake_all(&self) {
    let _guard = self.lock.lock();
    // Wake all threads in the list
}
}
```

kernel_Workspace/base/kw-base/src/task/queue.rs (line 36) - [public] wake_all

```
pub fn wake_all(&self) {
    let _guard = self.lock.lock();
    // Wake all threads in the list
}
}
```

kernel_Workspace/base/kw-base/src/task/scheduler.rs (line 4) - [public] yield_now

```
pub fn yield_now() {
    unsafe {
        core::arch::asm!("int 0x30", options(nomem, nostack)); // Trigger scheduler IPI/interrupt
    }
}

pub fn halt_cpu() {
    loop {
        crate::arch::interrupts::enable();
        crate::arch::interrupts::halt();
    }
}
// ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/task/scheduler.rs (line 10) - [public] halt_cpu

```
pub fn halt_cpu() {
    loop {
        crate::arch::interrupts::enable();
        crate::arch::interrupts::halt();
    }
}

pub fn set_current_priority(prio: i32) -> KResult<()> {
    if prio < -20 || prio > 19 {
        return Err(KernelError::InvalidArg);
    }
    // Hook into legacy kernel scheduler via gluecore if needed
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/task/scheduler.rs (line 17) - [public] set_current_priority

```
pub fn set_current_priority(prio: i32) -> KResult<()> {  
    if prio < -20 || prio > 19 {  
        return Err(KernelError::InvalidArg);  
    }  
    // Hook into legacy kernel scheduler via gluecore if needed  
    Ok(())  
}
```

kernel_workspace/base/kw-base/src/task/thread.rs (line 5) - [public] ThreadState

```
pub enum ThreadState {  
    Ready,  
    Running,  
    Blocked,  
    Terminated,  
}  
  
pub struct Thread {  
    pub tid: Tid,  
    pub state: ThreadState,  
    pub context: TaskContext,  
    pub kernel_stack_top: u64,  
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/task/thread.rs (line 12) - [public] Thread

```
pub struct Thread {  
    pub tid: Tid,  
    pub state: ThreadState,  
    pub context: TaskContext,  
    pub kernel_stack_top: u64,  
}  
  
impl Thread {  
    pub fn new(tid: Tid, entry_point: u64, stack_top: u64) -> Self {  
        let mut ctx = TaskContext::default();  
        ctx.rip = entry_point;  
        ctx.rsp = stack_top;  
        // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/task/thread.rs (line 20) - [public] new

```
pub fn new(tid: Tid, entry_point: u64, stack_top: u64) -> Self {  
    let mut ctx = TaskContext::default();  
    ctx.rip = entry_point;  
    ctx.rsp = stack_top;  
    ctx.rflags = 0x202; // Interrupts enabled  
  
    Self {  
        tid,  
        state: ThreadState::Ready,  
        context: ctx,  
        kernel_stack_top: stack_top,  
    }  
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/utils/bitmap.rs (line 3) - [public] Bitmap

```
pub struct Bitmap {  
    data: *mut u64,  
    bits: usize,  
    owned: bool,  
}  
  
impl Bitmap {  
    pub fn new_external(ptr: *mut u64, bits: usize) -> Self {  
        Self { data: ptr, bits, owned: false }  
    }  
  
    pub fn words_needed(bits: usize) -> usize {  
        // ... (truncated, see source file for more)
```



```

kernel_Workspace/base/kw-base/src/utils/bitmap.rs (line 10) - [public] new_external
    pub fn new_external(ptr: *mut u64, bits: usize) -> Self {
        Self { data: ptr, bits, owned: false }
    }

    pub fn words_needed(bits: usize) -> usize {
        (bits + 63) / 64
    }

    #[inline]
    fn word_idx(&self, bit: usize) -> usize {
        bit / 64
    }
    // ... (truncated, see source file for more)

kernel_Workspace/base/kw-base/src/utils/bitmap.rs (line 14) - [public] words_needed
    pub fn words_needed(bits: usize) -> usize {
        (bits + 63) / 64
    }

    #[inline]
    fn word_idx(&self, bit: usize) -> usize {
        bit / 64
    }

    #[inline]
    fn bit_mask(&self, bit: usize) -> u64 {
        1u64 << (bit % 64)
    }
    // ... (truncated, see source file for more)

kernel_Workspace/base/kw-base/src/utils/bitmap.rs (line 19) - [internal] word_idx
    fn word_idx(&self, bit: usize) -> usize {
        bit / 64
    }

    #[inline]
    fn bit_mask(&self, bit: usize) -> u64 {
        1u64 << (bit % 64)
    }

    pub fn set(&mut self, bit: usize) {
        if bit < self.bits {
            unsafe { *self.data.add(self.word_idx(bit)) |= self.bit_mask(bit); }
        }
    }
    // ... (truncated, see source file for more)

kernel_Workspace/base/kw-base/src/utils/bitmap.rs (line 24) - [internal] bit_mask
    fn bit_mask(&self, bit: usize) -> u64 {
        1u64 << (bit % 64)
    }

    pub fn set(&mut self, bit: usize) {
        if bit < self.bits {
            unsafe { *self.data.add(self.word_idx(bit)) |= self.bit_mask(bit); }
        }
    }

    pub fn clear(&mut self, bit: usize) {
        if bit < self.bits {
            // ... (truncated, see source file for more)
        }
    }

```

kernel_Workspace/base/kw-base/src/utils/bitmap.rs (line 28) - [public] set

```
pub fn set(&mut self, bit: usize) {
    if bit < self.bits {
        unsafe { *self.data.add(self.word_idx(bit)) |= self.bit_mask(bit); }
    }
}

pub fn clear(&mut self, bit: usize) {
    if bit < self.bits {
        unsafe { *self.data.add(self.word_idx(bit)) &= !self.bit_mask(bit); }
    }
}

// ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/utils/bitmap.rs (line 34) - [public] clear

```
pub fn clear(&mut self, bit: usize) {
    if bit < self.bits {
        unsafe { *self.data.add(self.word_idx(bit)) &= !self.bit_mask(bit); }
    }
}

pub fn test(&self, bit: usize) -> bool {
    if bit >= self.bits { return false; }
    unsafe { (*self.data.add(self.word_idx(bit)) & self.bit_mask(bit)) != 0 }
}

pub fn find_first_zero(&self) -> Option<usize> {
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/utils/bitmap.rs (line 40) - [public] test

```
pub fn test(&self, bit: usize) -> bool {
    if bit >= self.bits { return false; }
    unsafe { (*self.data.add(self.word_idx(bit)) & self.bit_mask(bit)) != 0 }
}

pub fn find_first_zero(&self) -> Option<usize> {
    for i in 0..Self::words_needed(self.bits) {
        let word = unsafe { *self.data.add(i) };
        if word != u64::MAX {
            let bit = (!word).trailing_zeros() as usize;
            let idx = i * 64 + bit;
            if idx < self.bits { return Some(idx); }
        }
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/utils/bitmap.rs (line 45) - [public] find_first_zero

```
pub fn find_first_zero(&self) -> Option<usize> {
    for i in 0..Self::words_needed(self.bits) {
        let word = unsafe { *self.data.add(i) };
        if word != u64::MAX {
            let bit = (!word).trailing_zeros() as usize;
            let idx = i * 64 + bit;
            if idx < self.bits { return Some(idx); }
        }
    }
    None
}

// ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/utils/bitmap.rs (line 57) - [public] find_first_set

```
pub fn find_first_set(&self) -> Option<usize> {
    for i in 0..Self::words_needed(self.bits) {
        let word = unsafe { *self.data.add(i) };
        if word != 0 {
            let bit = word.trailing_zeros() as usize;
            let idx = i * 64 + bit;
            if idx < self.bits { return Some(idx); }
        }
    }
    None
}

// ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/utils/bitmap.rs (line 69) - [public] alloc_first_zero

```
pub fn alloc_first_zero(&mut self) -> Option<usize> {
    let idx = self.find_first_zero()?;
    self.set(idx);
    Some(idx)
}

pub fn count_set(&self) -> usize {
    let mut count = 0;
    for i in 0..Self::words_needed(self.bits) {
        count += unsafe { (*self.data.add(i)).count_ones() as usize };
    }
    count
}

// ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/utils/bitmap.rs (line 75) - [public] count_set

```
pub fn count_set(&self) -> usize {
    let mut count = 0;
    for i in 0..Self::words_needed(self.bits) {
        count += unsafe { (*self.data.add(i)).count_ones() as usize };
    }
    count
}

pub fn count_free(&self) -> usize {
    self.bits - self.count_set()
}

}
```

kernel_Workspace/base/kw-base/src/utils/bitmap.rs (line 83) - [public] count_free

```
pub fn count_free(&self) -> usize {
    self.bits - self.count_set()
}

}
```

kernel_Workspace/base/kw-base/src/utils/btree.rs (line 9) - [internal] Node

```
struct Node<K, V> {
    keys: [mem::MaybeUninit<K>; MAX_KEYS],
    vals: [mem::MaybeUninit<V>; MAX_KEYS],
    children: [*mut Node<K, V>; MAX_KEYS + 1],
    len: usize,
    leaf: bool,
}

impl<K: Ord, V> Node<K, V> {
    fn new_leaf() -> *mut Self {
        let layout = Layout::new(mem::size_of::<Self>(), mem::align_of::<Self>());
        let ptr = KernelHeap.alloc(layout).unwrap() as *mut Self;
        // ... (truncated, see source file for more)
    }
}
```

kernel_workspace/base/kw-base/src/utils/btree.rs (line 18) - [internal] new_leaf

```
fn new_leaf() -> *mut Self {
    let layout = Layout::new(mem::size_of::<Self>(), mem::align_of::<Self>());
    let ptr = KernelHeap.alloc(layout).unwrap() as *mut Self;
    unsafe {
        (*ptr).len = 0;
        (*ptr).leaf = true;
        (*ptr).children = [core::ptr::null_mut(); MAX_KEYS + 1];
    }
    ptr
}

fn new_internal() -> *mut Self {
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/utils/btree.rs (line 29) - [internal] new_internal

```
fn new_internal() -> *mut Self {
    let ptr = Self::new_leaf();
    unsafe { (*ptr).leaf = false; }
    ptr
}

pub struct BTreeMap<K, V> {
    root: *mut Node<K, V>,
    len: usize,
}
```

// ... (truncated, see source file for more)

kernel_workspace/base/kw-base/src/utils/btree.rs (line 36) - [public] BTreeMap

```
pub struct BTreeMap<K, V> {
    root: *mut Node<K, V>,
    len: usize,
}

unsafe impl<K: Send, V: Send> Send for BTreeMap<K, V> {}
unsafe impl<K: Sync, V: Sync> Sync for BTreeMap<K, V> {}

impl<K: Ord, V> BTreeMap<K, V> {
    pub fn new() -> Self {
        Self { root: core::ptr::null_mut(), len: 0 }
    }
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/utils/btree.rs (line 45) - [public] new

```
pub fn new() -> Self {
    Self { root: core::ptr::null_mut(), len: 0 }
}

pub fn insert(&mut self, key: K, val: V) {
    if self.root.is_null() {
        self.root = Node::new_leaf();
    }

    let r = self.root;
    if unsafe { (*r).len == MAX_KEYS } {
        let s = Node::new_internal();
        // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/utils/btree.rs (line 49) - [public] insert

```
pub fn insert(&mut self, key: K, val: V) {
    if self.root.is_null() {
        self.root = Node::new_leaf();
    }

    let r = self.root;
    if unsafe { (*r).len == MAX_KEYS } {
        let s = Node::new_internal();
        unsafe {
            (*s).children[0] = r;
            self.split_child(s, 0);
        }
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/utils/btree.rs (line 69) - [internal] split_child

```
fn split_child(&mut self, parent: *mut Node<K, V>, i: usize) {
    let child = unsafe { (*parent).children[i] };
    let new_node = if unsafe { (*child).leaf } { Node::new_leaf() } else { Node::new_internal() };
    let mid = MIN_KEYS;

    unsafe {
        for j in 0..MIN_KEYS {
            (*new_node).keys[j] = mem::MaybeUninit::new((*child).keys[mid + 1 + j].assume_init_read());
            (*new_node).vals[j] = mem::MaybeUninit::new((*child).vals[mid + 1 + j].assume_init_read());
        }
        if !(*child).leaf {
            for j in 0..B {
                // ... (truncated, see source file for more)
            }
        }
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/utils/btree.rs (line 102) - [internal] insert_non_full

```
fn insert_non_full(&mut self, node: *mut Node<K, V>, key: K, val: V) {
    let mut i = unsafe { (*node).len };
    unsafe {
        if (*node).leaf {
            while i > 0 && key < (*node).keys[i - 1].assume_init_ref() {
                (*node).keys[i] = mem::MaybeUninit::new((*node).keys[i - 1].assume_init_read());
                (*node).vals[i] = mem::MaybeUninit::new((*node).vals[i - 1].assume_init_read());
                i -= 1;
            }
            (*node).keys[i] = mem::MaybeUninit::new(key);
            (*node).vals[i] = mem::MaybeUninit::new(val);
            (*node).len += 1;
        }
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/utils/btree.rs (line 129) - [public] get

```
pub fn get(&self, key: &K) -> Option<V> {
    if self.root.is_null() { return None; }
    let mut node = self.root;
    loop {
        let n = unsafe { &*node };
        let mut i = 0;
        while i < n.len && *key > *unsafe { n.keys[i].assume_init_ref() } {
            i += 1;
        }
        if i < n.len && *key == *unsafe { n.keys[i].assume_init_ref() } {
            return Some(unsafe { n.vals[i].assume_init_ref() });
        }
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/utils/btree.rs (line 146) - [public] len

```
pub fn len(&self) -> usize { self.len }
pub fn is_empty(&self) -> bool { self.len == 0 }
}
```

kernel_Workspace/base/kw-base/src/utils/btree.rs (line 147) - [public] is_empty

```
pub fn is_empty(&self) -> bool { self.len == 0 }
}
```

kernel_Workspace/base/kw-base/src/utils/hashmap.rs (line 6) - [internal] FnvHasher

```
struct FnvHasher(u64);
impl Hasher for FnvHasher {
    fn finish(&self) -> u64 { self.0 }
    fn write(&mut self, bytes: &[u8]) {
        for &b in bytes {
            self.0 ^= b as u64;
            self.0 = self.0.wrapping_mul(0x1000000001b3);
        }
    }
}

fn hash_key<K: Hash>(key: &K) -> usize {
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/utils/hashmap.rs (line 8) - [internal] finish

```
fn finish(&self) -> u64 { self.0 }
fn write(&mut self, bytes: &[u8]) {
    for &b in bytes {
        self.0 ^= b as u64;
        self.0 = self.0.wrapping_mul(0x1000000001b3);
    }
}

fn hash_key<K: Hash>(key: &K) -> usize {
    let mut h = FnvHasher(0xcbf29ce484222325);
    key.hash(&mut h);
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/utils/hashmap.rs (line 9) - [internal] write

```
fn write(&mut self, bytes: &[u8]) {
    for &b in bytes {
        self.0 ^= b as u64;
        self.0 = self.0.wrapping_mul(0x1000000001b3);
    }
}

fn hash_key<K: Hash>(key: &K) -> usize {
    let mut h = FnvHasher(0xcbf29ce484222325);
    key.hash(&mut h);
    h.finish() as usize
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/utils/hashmap.rs (line 17) - [internal] hash_key

```
fn hash_key<K: Hash>(key: &K) -> usize {
    let mut h = FnvHasher(0xcbf29ce484222325);
    key.hash(&mut h);
    h.finish() as usize
}

enum Slot<K, V> { Empty, Occupied(K, V), Tombstone }

pub struct HashMap<K, V> {
    slots: *mut Slot<K, V>,
    cap: usize,
    len: usize,
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/utils/hashmap.rs (line 25) - [public] HashMap

```
pub struct HashMap<K, V> {
    slots: *mut Slot<K, V>,
    cap: usize,
    len: usize,
}

unsafe impl<K: Send, V: Send> Send for HashMap<K, V> {}
unsafe impl<K: Sync, V: Sync> Sync for HashMap<K, V> {}

impl<K: Hash + Eq, V> HashMap<K, V> {
    pub fn new() -> Self {
        Self { slots: core::ptr::null_mut(), cap: 0, len: 0 }
        // ... (truncated, see source file for more)
    }
}
```

kernel_workspace/base/kw-base/src/utils/hashmap.rs (line 35) - [public] new

```
pub fn new() -> Self {
    Self { slots: core::ptr::null_mut(), cap: 0, len: 0 }
}

fn grow(&mut self, new_cap: usize) {
    let layout = Layout::new(new_cap * mem::size_of::<Slot<K, V>>(), mem::align_of::<Slot<K, V>>());
    let new_slots = KernelHeap.alloc(layout).unwrap() as *mut Slot<K, V>;
    for i in 0..new_cap {
        unsafe { new_slots.add(i).write(Slot::Empty); }
    }

    let old_slots = self.slots;
    // ... (truncated, see source file for more)
}
```

kernel_workspace/base/kw-base/src/utils/hashmap.rs (line 39) - [internal] grow

```
fn grow(&mut self, new_cap: usize) {
    let layout = Layout::new(new_cap * mem::size_of::<Slot<K, V>>(), mem::align_of::<Slot<K, V>>());
    let new_slots = KernelHeap.alloc(layout).unwrap() as *mut Slot<K, V>;
    for i in 0..new_cap {
        unsafe { new_slots.add(i).write(Slot::Empty); }
    }

    let old_slots = self.slots;
    let old_cap = self.cap;
    self.slots = new_slots;
    self.cap = new_cap;
    self.len = 0;
    // ... (truncated, see source file for more)
}
```

kernel_workspace/base/kw-base/src/utils/hashmap.rs (line 64) - [public] insert

```
pub fn insert(&mut self, key: K, val: V) {
    if self.cap == 0 || self.len * 4 >= self.cap * 3 {
        self.grow(if self.cap == 0 { 8 } else { self.cap * 2 });
    }
    let h = hash_key(&key);
    let mut idx = h % self.cap;
    loop {
        let slot = unsafe { &mut *self.slots.add(idx) };
        match slot {
            Slot::Empty | Slot::Tombstone => {
                *slot = Slot::Occupied(key, val);
                self.len += 1;
            }
        }
    }
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/base/kw-base/src/utils/hashmap.rs (line 87) - [public] get

```
pub fn get(&self, key: &K) -> Option<&V> {
    if self.cap == 0 { return None; }
    let h = hash_key(key);
    let mut idx = h % self.cap;
    loop {
        let slot = unsafe { &*self.slots.add(idx) };
        match slot {
            Slot::Empty => return None,
            Slot::Occupied(k, v) if *k == *key => return Some(v),
            _ => { idx = (idx + 1) % self.cap; }
        }
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/utils/hashmap.rs (line 101) - [public] get_mut

```
pub fn get_mut(&mut self, key: &K) -> Option<&mut V> {
    if self.cap == 0 { return None; }
    let h = hash_key(key);
    let mut idx = h % self.cap;
    loop {
        let slot = unsafe { &mut *self.slots.add(idx) };
        match slot {
            Slot::Empty => return None,
            Slot::Occupied(k, v) if *k == *key => return Some(v),
            _ => { idx = (idx + 1) % self.cap; }
        }
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/utils/hashmap.rs (line 115) - [public] remove

```
pub fn remove(&mut self, key: &K) -> Option<V> {
    if self.cap == 0 { return None; }
    let h = hash_key(key);
    let mut idx = h % self.cap;
    loop {
        let slot = unsafe { &mut *self.slots.add(idx) };
        match slot {
            Slot::Empty => return None,
            Slot::Occupied(k, _) if *k == *key => {
                let old = mem::replace(slot, Slot::Tombstone);
                self.len -= 1;
                if let Slot::Occupied(_, v) = old { return Some(v); }
            }
        }
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-base/src/utils/hashmap.rs (line 133) - [public] len

```
pub fn len(&self) -> usize { self.len }
pub fn is_empty(&self) -> bool { self.len == 0 }
}

impl<K, V> Drop for HashMap<K, V> {
    fn drop(&mut self) {
        if !self.slots.is_null() {
            for i in 0..self.cap {
                unsafe { self.slots.add(i).drop_in_place(); }
            }
            let layout = Layout::new(self.cap * mem::size_of::<Slot<K, V>>(), mem::align_of::<Slot<K, V>>());
            KernelHeap.dealloc(self.slots as *mut u8, layout);
        }
        // ... (truncated, see source file for more)
```


kernel_workspace/base/kw-base/src/utils/hashmap.rs (line 134) - [public] is_empty

```
pub fn is_empty(&self) -> bool { self.len == 0 }
}

impl<K, V> Drop for HashMap<K, V> {
    fn drop(&mut self) {
        if !self.slots.is_null() {
            for i in 0..self.cap {
                unsafe { self.slots.add(i).drop_in_place(); }
            }
            let layout = Layout::new(self.cap * mem::size_of::<Slot<K, V>>(), mem::align_of::<Slot<K, V>>(),
                KernelHeap.dealloc(self.slots as *mut u8, layout));
        }
        // ... (truncated, see source file for more)
    }
}
```

kernel_workspace/base/kw-base/src/utils/hashmap.rs (line 138) - [internal] drop

```
fn drop(&mut self) {
    if !self.slots.is_null() {
        for i in 0..self.cap {
            unsafe { self.slots.add(i).drop_in_place(); }
        }
        let layout = Layout::new(self.cap * mem::size_of::<Slot<K, V>>(), mem::align_of::<Slot<K, V>>(),
            KernelHeap.dealloc(self.slots as *mut u8, layout));
    }
}
}
```

kernel_workspace/base/kw-base/src/utils/ring_buffer.rs (line 1) - [public] RingBuffer

```
pub struct RingBuffer<T, const N: usize> {
    data: [core::mem::MaybeUninit<T>; N],
    head: usize,
    tail: usize,
    count: usize,
}

impl<T, const N: usize> RingBuffer<T, N> {
    pub const fn new() -> Self {
        Self {
            data: unsafe { core::mem::MaybeUninit::uninit().assume_init() },
            head: 0,
            // ... (truncated, see source file for more)
        }
    }
}
```

kernel_workspace/base/kw-base/src/utils/ring_buffer.rs (line 18) - [public] push

```
pub fn push(&mut self, val: T) -> bool {
    if self.count == N { return false; }
    self.data[self.head] = core::mem::MaybeUninit::new(val);
    self.head = (self.head + 1) % N;
    self.count += 1;
    true
}

pub fn pop(&mut self) -> Option<T> {
    if self.count == 0 { return None; }
    let val = unsafe { self.data[self.tail].as_ptr().read() };
    self.tail = (self.tail + 1) % N;
    // ... (truncated, see source file for more)
}
}
```

kernel_workspace/base/kw-base/src/utils/ring_buffer.rs (line 26) - [public] pop

```
pub fn pop(&mut self) -> Option<T> {
    if self.count == 0 { return None; }
    let val = unsafe { self.data[self.tail].as_ptr().read() };
    self.tail = (self.tail + 1) % N;
    self.count -= 1;
    Some(val)
}

pub fn peek(&self) -> Option<&T> {
    if self.count == 0 { return None; }
    Some(unsafe { self.data[self.tail].assume_init_ref() })
}
// ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/utils/ring_buffer.rs (line 34) - [public] peek

```
pub fn peek(&self) -> Option<T> {
    if self.count == 0 { return None; }
    Some(unsafe { self.data[self.tail].assume_init_ref() })
}

pub fn len(&self) -> usize { self.count }
pub fn is_empty(&self) -> bool { self.count == 0 }
pub fn is_full(&self) -> bool { self.count == N }
pub fn capacity(&self) -> usize { N }
}

impl<T, const N: usize> Drop for RingBuffer<T, N> {
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-base/src/utils/ring_buffer.rs (line 39) - [public] len

```
pub fn len(&self) -> usize { self.count }
pub fn is_empty(&self) -> bool { self.count == 0 }
pub fn is_full(&self) -> bool { self.count == N }
pub fn capacity(&self) -> usize { N }
}

impl<T, const N: usize> Drop for RingBuffer<T, N> {
    fn drop(&mut self) {
        while self.pop().is_some() {}
    }
}
```

kernel_workspace/base/kw-base/src/utils/ring_buffer.rs (line 40) - [public] is_empty

```
pub fn is_empty(&self) -> bool { self.count == 0 }
pub fn is_full(&self) -> bool { self.count == N }
pub fn capacity(&self) -> usize { N }
}

impl<T, const N: usize> Drop for RingBuffer<T, N> {
    fn drop(&mut self) {
        while self.pop().is_some() {}
    }
}
```

kernel_workspace/base/kw-base/src/utils/ring_buffer.rs (line 41) - [public] is_full

```
pub fn is_full(&self) -> bool { self.count == N }
pub fn capacity(&self) -> usize { N }
}

impl<T, const N: usize> Drop for RingBuffer<T, N> {
    fn drop(&mut self) {
        while self.pop().is_some() {}
    }
}
```

kernel_workspace/base/kw-base/src/utils/ring_buffer.rs (line 42) - [public] capacity

```
pub fn capacity(&self) -> usize { N }
}

impl<T, const N: usize> Drop for RingBuffer<T, N> {
    fn drop(&mut self) {
        while self.pop().is_some() {}
    }
}
```

kernel_workspace/base/kw-base/src/utils/ring_buffer.rs (line 46) - [internal] drop

```
fn drop(&mut self) {
    while self.pop().is_some() {}
}
}
```

kernel_workspace/base/kw-lib/src/kernel/syscall.rs (line 7) - [public] syscall0

```
pub fn syscall0(num: u64) -> i64 {
    let ret: i64;
    unsafe {
        core::arch::asm!(
            "int 0x80",
            in("rax") num,
            out("rax") ret,
            options(nostack, preserves_flags)
        );
    }
    ret
}
// ... (truncated, see source file for more)
```

kernel_workspace/base/kw-lib/src/kernel/syscall.rs (line 22) - [public] syscall3

```
pub fn syscall3(num: u64, a1: u64, a2: u64, a3: u64) -> i64 {
    let ret: i64;
    unsafe {
        core::arch::asm!(
            "int 0x80",
            in("rax") num,
            in("rdi") a1,
            in("rsi") a2,
            in("rdx") a3,
            out("rax") ret,
            options(nostack, preserves_flags)
        );
    }
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-lib/src/kernel/time.rs (line 3) - [public] ticks

// High-level time API for libraries and drivers.

```
pub fn ticks() -> u64 {
    kw_base::hal::timers::get_ticks()
}

pub fn frequency() -> u64 {
    kw_base::hal::timers::get_frequency()
}

pub fn sleep_ms(ms: u32) {
    kw_base::hal::timers::sleep_ms(ms);
}

// ... (truncated, see source file for more)
```

kernel_workspace/base/kw-lib/src/kernel/time.rs (line 7) - [public] frequency

```
pub fn frequency() -> u64 {
    kw_base::hal::timers::get_frequency()
}

pub fn sleep_ms(ms: u32) {
    kw_base::hal::timers::sleep_ms(ms);
}

pub fn sleep_ticks(ticks: u64) {
    let start = ticks();
    while ticks() - start < ticks {
        core::hint::spin_loop();
    }
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-lib/src/kernel/time.rs (line 11) - [public] sleep_ms

```
pub fn sleep_ms(ms: u32) {
    kw_base::hal::timers::sleep_ms(ms);
}

pub fn sleep_ticks(ticks: u64) {
    let start = ticks();
    while ticks() - start < ticks {
        core::hint::spin_loop();
    }
}
```

kernel_workspace/base/kw-lib/src/kernel/time.rs (line 15) - [public] sleep_ticks

```
pub fn sleep_ticks(ticks: u64) {
    let start = ticks();
    while ticks() - start < ticks {
        core::hint::spin_loop();
    }
}
```

kernel_workspace/base/kw-lib/src/mem/allocators/slab.rs (line 8) - [internal] SlabPage

```
struct SlabPage {
    next: *mut SlabPage,
    free_head: *mut u8,
    in_use: u32,
}

pub struct SlabCache {
    obj_size: usize,
    align: usize,
    objs_per_page: usize,
    partial_pages: *mut SlabPage,
    full_pages: *mut SlabPage,
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-lib/src/mem/allocators/slab.rs (line 14) - [public] SlabCache

```
pub struct SlabCache {
    obj_size: usize,
    align: usize,
    objs_per_page: usize,
    partial_pages: *mut SlabPage,
    full_pages: *mut SlabPage,
}

unsafe impl Send for SlabCache {}
unsafe impl Sync for SlabCache {}

impl SlabCache {
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-lib/src/mem/allocators/slab.rs (line 26) - [public] new

```
pub fn new(obj_size: usize, align: usize) -> Self {
    let actual_size = if obj_size < core::mem::size_of::<mut u8>() {
        core::mem::size_of::<mut u8>()
    } else {
        obj_size
    };

    let aligned_size = (actual_size + align - 1) & !(align - 1);
    let objs_per_page = (4096 - core::mem::size_of::<SlabPage>()) / aligned_size;

    Self {
        obj_size: aligned_size,
        // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-lib/src/mem/allocators/slab.rs (line 45) - [internal] alloc_page

```
fn alloc_page(&self) -> KResult<*mut SlabPage> {
    let layout = core::alloc::Layout::from_size_align(4096, 4096).unwrap();
    let ptr = unsafe { kstd_alloc::KernelHeap.alloc(layout) }.map_err(|_| KernelError::NoMemory)?;

    let page = ptr as *mut SlabPage;
    unsafe {
        (*page).next = core::ptr::null_mut();
        (*page).in_use = 0;

        // Build free list inside the page
        let data_start = ptr.add(core::mem::size_of::<SlabPage>());
        let mut current = data_start;
    }
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-lib/src/mem/allocators/slab.rs (line 69) - [public] alloc

```
pub fn alloc(&mut self) -> KResult<*mut u8> {
    if self.partial_pages.is_null() {
        let page = self.alloc_page()?;
        unsafe {
            (*page).next = self.partial_pages;
            self.partial_pages = page;
        }
    }

    let page = self.partial_pages;
    let obj = unsafe { (*page).free_head };

    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-lib/src/mem/allocators/slab.rs (line 99) - [public] free

```
pub fn free(&mut self, obj: *mut u8) {
    // Find the page this object belongs to
    let page_addr = (obj as usize) & !(4096 - 1);
    let page = page_addr as *mut SlabPage;

    unsafe {
        *(obj as *mut *mut u8) = (*page).free_head;
        (*page).free_head = obj;
        (*page).in_use -= 1;

        if (*page).in_use == 0 {
            // TODO: Remove from partial/full list and free page
        }
    }
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-lib/src/mem/physical/buddy.rs (line 9) - [internal] FreeList

```
struct FreeList {
    head: u64, // Physical address of the first free block, 0 if empty
}

impl FreeList {
    const fn new() -> Self { Self { head: 0 } }
}

pub struct BuddyAllocator {
    base_phys: PhysAddr,
    total_frames: usize,
    free_lists: [FreeList; MAX_ORDER],
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-lib/src/mem/physical/buddy.rs (line 17) - [public] BuddyAllocator

```
pub struct BuddyAllocator {
    base_phys: PhysAddr,
    total_frames: usize,
    free_lists: [FreeList; MAX_ORDER],
    // Bitmap to track if a block is split or free
    // For simplicity, we use a basic tracking mechanism here.
    // In a full implementation, this would be a per-order bitmap.
    split_bitmap: *mut u64,
}

unsafe impl Send for BuddyAllocator {}
unsafe impl Sync for BuddyAllocator {}
// ... (truncated, see source file for more)
```

kernel_workspace/base/kw-lib/src/mem/physical/buddy.rs (line 40) - [public] init

```
pub fn init(&mut self, base: PhysAddr, frames: usize, bitmap_ptr: *mut u64) {
    self.base_phys = base;
    self.total_frames = frames;
    self.split_bitmap = bitmap_ptr;

    // Clear bitmap
    let words = kw_base::utils::Bitmap::words_needed(frames);
    unsafe {
        core::ptr::write_bytes(bitmap_ptr, 0, words * 8);
    }

    // Add all frames to the highest possible order blocks
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-lib/src/mem/physical/buddy.rs (line 78) - [internal] frame_to_phys

```
fn frame_to_phys(&self, frame: usize) -> PhysAddr {
    PhysAddr(self.base_phys.0 + (frame as u64 * 4096))
}

fn phys_to_frame(&self, phys: PhysAddr) -> usize {
    ((phys.0 - self.base_phys.0) / 4096) as usize
}

fn free_block(&mut self, frame: usize, order: usize) {
    // Mark as not split in bitmap (simplified logic)
    let mut bitmap = Bitmap::new_external(self.split_bitmap, self.total_frames);
    bitmap.clear(frame);
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-lib/src/mem/physical/buddy.rs (line 82) - [internal] phys_to_frame

```
fn phys_to_frame(&self, phys: PhysAddr) -> usize {
    ((phys.0 - self.base_phys.0) / 4096) as usize
}

fn free_block(&mut self, frame: usize, order: usize) {
    // Mark as not split in bitmap (simplified logic)
    let mut bitmap = Bitmap::new_external(self.split_bitmap, self.total_frames);
    bitmap.clear(frame);

    // Push to free list
    let phys = self.frame_to_phys(frame);
    unsafe {
        // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-lib/src/mem/physical/buddy.rs (line 86) - [internal] free_block

```
fn free_block(&mut self, frame: usize, order: usize) {  
    // Mark as not split in bitmap (simplified logic)  
    let mut bitmap = Bitmap::new_external(self.split_bitmap, self.total_frames);  
    bitmap.clear(frame);  
  
    // Push to free list  
    let phys = self.frame_to_phys(frame);  
    unsafe {  
        let ptr = phys.0 as *mut u64;  
        *ptr = self.free_lists[order].head;  
    }  
    self.free_lists[order].head = phys.0;  
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-lib/src/mem/physical/buddy.rs (line 100) - [public] alloc

```
pub fn alloc(&mut self, order: usize) -> KResult<PhysAddr> {  
    if order >= MAX_ORDER { return Err(KernelError::InvalidArg); }  
  
    // Find the smallest available order >= requested  
    let mut current_order = order;  
    while current_order < MAX_ORDER && self.free_lists[current_order].head == 0 {  
        current_order += 1;  
    }  
  
    if current_order == MAX_ORDER {  
        return Err(KernelError::NoMemory);  
    }  
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-lib/src/mem/physical/buddy.rs (line 133) - [public] free

```
pub fn free(&mut self, phys: PhysAddr, order: usize) {  
    let mut frame = self.phys_to_frame(phys);  
    let mut current_order = order;  
  
    let mut bitmap = Bitmap::new_external(self.split_bitmap, self.total_frames);  
  
    // Merge with buddy if possible  
    while current_order < MAX_ORDER - 1 {  
        let buddy_frame = frame ^ (1 << current_order);  
        if buddy_frame >= self.total_frames { break; }  
  
        // Check if buddy is free and of the same order  
    }  
    // ... (truncated, see source file for more)
```

kernel_workspace/base/kw-lib/src/mem/virtual/aspace.rs (line 7) - [public] AddressSpace

```
pub struct AddressSpace {  
    pub cr3_phys: u64,  
    vma_head: *mut Vma,  
    vma_count: usize,  
}  
  
unsafe impl Send for AddressSpace {}  
unsafe impl Sync for AddressSpace {}  
  
impl AddressSpace {  
    pub fn new(cr3: u64) -> Self {  
        Self {  
            // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-lib/src/mem/virtual/aspace.rs (line 17) - [public] new

```
pub fn new(cr3: u64) -> Self {
    Self {
        cr3_phys: cr3,
        vma_head: core::ptr::null_mut(),
        vma_count: 0,
    }
}

pub fn map_region(&mut self, start: VirtAddr, size: usize, flags: u32, vma_type: VmaType) -> KResult<Vma> {
    let end = VirtAddr(start.0 + size as u64);

    // Check for overlaps
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-lib/src/mem/virtual/aspace.rs (line 25) - [public] map_region

```
pub fn map_region(&mut self, start: VirtAddr, size: usize, flags: u32, vma_type: VmaType) -> KResult<Vma> {
    let end = VirtAddr(start.0 + size as u64);

    // Check for overlaps
    let mut current = self.vma_head;
    while !current.is_null() {
        let vma = unsafe { &*current };
        if vma.overlaps(start, end) {
            return Err(KernelError::AlreadyExists);
        }
        current = vma.next;
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-lib/src/mem/virtual/aspace.rs (line 53) - [public] find_vma

```
pub fn find_vma(&self, addr: VirtAddr) -> Option<&Vma> {
    let mut current = self.vma_head;
    while !current.is_null() {
        let vma = unsafe { &*current };
        if vma.contains(addr) {
            return Some(vma);
        }
        current = vma.next;
    }
    None
}

// ... (truncated, see source file for more)
```

kernel_Workspace/base/kw-lib/src/mem/virtual/aspace.rs (line 67) - [internal] drop

```
fn drop(&mut self) {
    let mut current = self.vma_head;
    while !current.is_null() {
        let next = unsafe { (*current).next };
        let layout = core::alloc::Layout::new::(<Vma>());
        unsafe { kstd_alloc::KernelHeap.dealloc(current as *mut u8, layout); }
        current = next;
    }
}

}
```

kernel_Workspace/base/kw-lib/src/mem/virtual/vma.rs (line 6) - [public] VmaType

```
pub enum VmaType {
    Anonymous,
    FileBacked,
    DeviceMapped,
    Stack,
    Heap,
}

pub struct Vma {
    pub start: VirtAddr,
    pub end: VirtAddr,
    pub flags: u32, // PROT_READ | PROT_WRITE | PROT_EXEC
    // ... (truncated, see source file for more)
```


kernel_workspace/base/kw-lib/src/mem/virtual/vma.rs (line 14) - [public] Vma

```
pub struct Vma {
    pub start: VirtAddr,
    pub end: VirtAddr,
    pub flags: u32, // PROT_READ | PROT_WRITE | PROT_EXEC
    pub vma_type: VmaType,
    pub file_offset: u64, // If FileBacked
    pub next: *mut Vma,    // For linked list in AddressSpace
}

impl Vma {
    pub fn new(start: VirtAddr, end: VirtAddr, flags: u32, vma_type: VmaType) -> Self {
        Self {
            // ... (truncated, see source file for more)
        }
    }
}
```

kernel_workspace/base/kw-lib/src/mem/virtual/vma.rs (line 24) - [public] new

```
pub fn new(start: VirtAddr, end: VirtAddr, flags: u32, vma_type: VmaType) -> Self {
    Self {
        start,
        end,
        flags,
        vma_type,
        file_offset: 0,
        next: core::ptr::null_mut(),
    }
}

pub fn size(&self) -> usize {
    // ... (truncated, see source file for more)
}
```

kernel_workspace/base/kw-lib/src/mem/virtual/vma.rs (line 35) - [public] size

```
pub fn size(&self) -> usize {
    (self.end.0 - self.start.0) as usize
}

pub fn contains(&self, addr: VirtAddr) -> bool {
    addr.0 >= self.start.0 && addr.0 < self.end.0
}

pub fn overlaps(&self, start: VirtAddr, end: VirtAddr) -> bool {
    self.start.0 < end.0 && start.0 < self.end.0
}
}
```

kernel_workspace/base/kw-lib/src/mem/virtual/vma.rs (line 39) - [public] contains

```
pub fn contains(&self, addr: VirtAddr) -> bool {
    addr.0 >= self.start.0 && addr.0 < self.end.0
}

pub fn overlaps(&self, start: VirtAddr, end: VirtAddr) -> bool {
    self.start.0 < end.0 && start.0 < self.end.0
}
}
```

kernel_workspace/base/kw-lib/src/mem/virtual/vma.rs (line 43) - [public] overlaps

```
pub fn overlaps(&self, start: VirtAddr, end: VirtAddr) -> bool {
    self.start.0 < end.0 && start.0 < self.end.0
}
}
```

kernel_workspace/base/kw-lib/src/Utils/string.rs (line 20) - [public] StrBuffer

// Simple fixed-capacity string buffer for formatting paths/logs without heap allocation.

```
pub struct StrBuffer<const N: usize> {
    buf: [u8; N],
    len: usize,
}

impl<const N: usize> StrBuffer<N> {
    pub const fn new() -> Self {
        Self { buf: [0; N], len: 0 }
    }

    pub fn push_str(&mut self, s: &str) -> bool {
        let bytes = s.as_bytes();
        // ... (truncated, see source file for more)
    }
}
```

kernel_workspace/base/kw-lib/src/Utils/string.rs (line 30) - [public] push_str

```
pub fn push_str(&mut self, s: &str) -> bool {
    let bytes = s.as_bytes();
    if self.len + bytes.len() > N { return false; }
    self.buf[self.len..self.len + bytes.len()].copy_from_slice(bytes);
    self.len += bytes.len();
    true
}

pub fn as_str(&self) -> &str {
    unsafe { core::str::from_utf8_unchecked(&self.buf[..self.len]) }
}

// ... (truncated, see source file for more)
```

kernel_workspace/base/kw-lib/src/Utils/string.rs (line 38) - [public] as_str

```
pub fn as_str(&self) -> &str {
    unsafe { core::str::from_utf8_unchecked(&self.buf[..self.len]) }
}

pub fn as_cstr_ptr(&mut self) -> *const u8 {
    if self.len < N {
        self.buf[self.len] = 0;
    }
    self.buf.as_ptr()
}

}
```

kernel_workspace/base/kw-lib/src/Utils/string.rs (line 42) - [public] as_cstr_ptr

```
pub fn as_cstr_ptr(&mut self) -> *const u8 {
    if self.len < N {
        self.buf[self.len] = 0;
    }
    self.buf.as_ptr()
}

}
```

kernel_workspace/base/kw-odin-abi/src/lib.rs (line 2) - [public] add

```
pub fn add(left: u64, right: u64) -> u64 {
    left + right
}

#[cfg(test)]
mod tests {
    use super::*;

    #[test]
    fn it_works() {
        let result = add(2, 2);
        assert_eq!(result, 4);
        // ... (truncated, see source file for more)
    }
}
```

kernel_workspace/base/kw-odin-abi/src/lib.rs (line 11) - [internal] it_works

```
fn it_works() {  
    let result = add(2, 2);  
    assert_eq!(result, 4);  
}
```

kernel_workspace/core/gluecore/build.rs (line 1) - [internal] main

```
fn main() {}
```

kernel_workspace/core/gluecore/src/bindings.rs (line 6) - [public] CPhysRange

```
pub struct CPhysRange {  
    pub start: u64,  
    pub end: u64,  
}  
  
impl From<CPhysRange> for (PhysAddr, usize) {  
    fn from(val: CPhysRange) -> Self {  
        (PhysAddr(val.start), (val.end - val.start) as usize)  
    }  
}
```

kernel_workspace/core/gluecore/src/bindings.rs (line 12) - [internal] from

```
fn from(val: CPhysRange) -> Self {  
    (PhysAddr(val.start), (val.end - val.start) as usize)  
}
```

kernel_workspace/core/gluecore/src/boot.rs (line 4) - [internal] boot_get_memory_map

```
fn boot_get_memory_map(out_ptr: *mut u64, out_size: *mut usize) -> bool;  
fn boot_get_cmdline(out_ptr: *mut u64, out_len: *mut usize) -> bool;  
fn boot_get_framebuffer(out_phys: *mut u64, out_w: *mut u32, out_h: *mut u32, out_pitch: *mut u32) -> bool;  
  
pub struct BootMemoryMap {  
    pub ptr: u64,  
    pub size: usize,  
}  
  
pub fn get_memory_map() -> Option<BootMemoryMap> {  
    let mut ptr: u64 = 0;  
    // ... (truncated, see source file for more)
```

kernel_workspace/core/gluecore/src/boot.rs (line 5) - [internal] boot_get_cmdline

```
fn boot_get_cmdline(out_ptr: *mut u64, out_len: *mut usize) -> bool;  
fn boot_get_framebuffer(out_phys: *mut u64, out_w: *mut u32, out_h: *mut u32, out_pitch: *mut u32) -> bool;  
  
pub struct BootMemoryMap {  
    pub ptr: u64,  
    pub size: usize,  
}  
  
pub fn get_memory_map() -> Option<BootMemoryMap> {  
    let mut ptr: u64 = 0;  
    let mut size: usize = 0;  
    // ... (truncated, see source file for more)
```

kernel_Workspace/core/gluecore/src/boot.rs (line 6) - [internal] boot_get_framebuffer

```
fn boot_get_framebuffer(out_phys: *mut u64, out_w: *mut u32, out_h: *mut u32, out_pitch: *mut u32) ->
}

pub struct BootMemoryMap {
    pub ptr: u64,
    pub size: usize,
}

pub fn get_memory_map() -> Option<BootMemoryMap> {
    let mut ptr: u64 = 0;
    let mut size: usize = 0;
    if unsafe { boot_get_memory_map(&mut ptr, &mut size) } {
        // ... (truncated, see source file for more)
    }
```

kernel_Workspace/core/gluecore/src/boot.rs (line 9) - [public] BootMemoryMap

```
pub struct BootMemoryMap {
    pub ptr: u64,
    pub size: usize,
}

pub fn get_memory_map() -> Option<BootMemoryMap> {
    let mut ptr: u64 = 0;
    let mut size: usize = 0;
    if unsafe { boot_get_memory_map(&mut ptr, &mut size) } {
        Some(BootMemoryMap { ptr, size })
    } else {
        None
    }
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/core/gluecore/src/boot.rs (line 14) - [public] get_memory_map

```
pub fn get_memory_map() -> Option<BootMemoryMap> {
    let mut ptr: u64 = 0;
    let mut size: usize = 0;
    if unsafe { boot_get_memory_map(&mut ptr, &mut size) } {
        Some(BootMemoryMap { ptr, size })
    } else {
        None
    }
}

pub struct BootFramebuffer {
    pub phys: PhysAddr,
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/core/gluecore/src/boot.rs (line 24) - [public] BootFramebuffer

```
pub struct BootFramebuffer {
    pub phys: PhysAddr,
    pub width: u32,
    pub height: u32,
    pub pitch: u32,
}

pub fn get_framebuffer() -> Option<BootFramebuffer> {
    let mut phys: u64 = 0;
    let mut w: u32 = 0;
    let mut h: u32 = 0;
    let mut pitch: u32 = 0;
    // ... (truncated, see source file for more)
}
```

kernel_workspace/core/gluecore/src/boot.rs (line 31) - [public] get_framebuffer

```
pub fn get_framebuffer() -> Option<BootFramebuffer> {
    let mut phys: u64 = 0;
    let mut w: u32 = 0;
    let mut h: u32 = 0;
    let mut pitch: u32 = 0;
    if unsafe { boot_get_framebuffer(&mut phys, &mut w, &mut h, &mut pitch) } {
        Some(BootFramebuffer { phys: PhysAddr(phys), width: w, height: h, pitch })
    } else {
        None
    }
}
```

kernel_workspace/core/gluecore/src/bridge/caps/check.rs (line 6) - [public] has_cap

```
pub fn has_cap(world_id: u32, cap: CapId) -> bool {
    unsafe { extern_block::caps_has(world_id, cap.0 as u8) == 1 }
}

pub fn request_cap(target_world: u32, cap: CapId) -> Result<(), Status> {
    let rc = unsafe { extern_block::caps_request(target_world, cap.0 as u8) };
    if rc == 0 {
        Ok(())
    } else {
        Err(Status::PermissionDenied)
    }
}

// ... (truncated, see source file for more)
```

kernel_workspace/core/gluecore/src/bridge/caps/check.rs (line 10) - [public] request_cap

```
pub fn request_cap(target_world: u32, cap: CapId) -> Result<(), Status> {
    let rc = unsafe { extern_block::caps_request(target_world, cap.0 as u8) };
    if rc == 0 {
        Ok(())
    } else {
        Err(Status::PermissionDenied)
    }
}

pub fn release_cap(world_id: u32, cap: CapId) -> Result<(), Status> {
    let rc = unsafe { extern_block::caps_release(world_id, cap.0 as u8) };
    if rc == 0 {
        // ... (truncated, see source file for more)
    }
}
```

kernel_workspace/core/gluecore/src/bridge/caps/check.rs (line 19) - [public] release_cap

```
pub fn release_cap(world_id: u32, cap: CapId) -> Result<(), Status> {
    let rc = unsafe { extern_block::caps_release(world_id, cap.0 as u8) };
    if rc == 0 {
        Ok(())
    } else {
        Err(Status::InvalidArgument)
    }
}
```

kernel_workspace/core/gluecore/src/bridge/mm/pmm.rs (line 7) - [public] alloc_frame

/// Allocates a single physical frame (4KB).

```
pub fn alloc_frame() -> Result<PhysAddr, Status> {
    let mut phys: u64 = 0;
    if unsafe { extern_block::pmm_alloc_frame(&mut phys) } {
        Ok(PhysAddr(phys))
    } else {
        Err(Status::OutOfMemory)
    }
}
```

/// Allocates a single zeroed physical frame.

```
pub fn alloc_zero_frame() -> Result<PhysAddr, Status> {
    let mut phys: u64 = 0;
    // ... (truncated, see source file for more)
}
```

kernel_workspace/core/gluecore/src/bridge/mm/pmm.rs (line 17) - [public] alloc_zero_frame

/// Allocates a single zeroed physical frame.

```
pub fn alloc_zero_frame() -> Result<PhysAddr, Status> {
    let mut phys: u64 = 0;
    if unsafe { extern_block::pmm_alloc_zero_frame(&mut phys) } {
        Ok(PhysAddr(phys))
    } else {
        Err(Status::OutOfMemory)
    }
}

/// Allocates contiguous physical frames.
pub fn alloc_frames(count: usize) -> Result<PhysAddr, Status> {
    let mut phys: u64 = 0;
    // ... (truncated, see source file for more)
```

kernel_workspace/core/gluecore/src/bridge/mm/pmm.rs (line 27) - [public] alloc_frames

/// Allocates contiguous physical frames.

```
pub fn alloc_frames(count: usize) -> Result<PhysAddr, Status> {
    let mut phys: u64 = 0;
    if unsafe { extern_block::pmm_alloc_frames(count, &mut phys) } {
        Ok(PhysAddr(phys))
    } else {
        Err(Status::OutOfMemory)
    }
}

/// Allocates contiguous physical frames with a specific alignment (in frames).
pub fn alloc_frames_aligned(count: usize, align_frames: usize) -> Result<PhysAddr, Status> {
    let mut phys: u64 = 0;
    // ... (truncated, see source file for more)
```

kernel_workspace/core/gluecore/src/bridge/mm/pmm.rs (line 37) - [public] alloc_frames_aligned

/// Allocates contiguous physical frames with a specific alignment (in frames).

```
pub fn alloc_frames_aligned(count: usize, align_frames: usize) -> Result<PhysAddr, Status> {
    let mut phys: u64 = 0;
    if unsafe { extern_block::pmm_alloc_frames_aligned(count, align_frames, &mut phys) } {
        Ok(PhysAddr(phys))
    } else {
        Err(Status::OutOfMemory)
    }
}

/// Frees a single physical frame.
pub fn free_frame(addr: PhysAddr) -> Result<(), Status> {
    if unsafe { extern_block::pmm_free_frame(addr.0) } {
        // ... (truncated, see source file for more)
```

kernel_workspace/core/gluecore/src/bridge/mm/pmm.rs (line 47) - [public] free_frame

/// Frees a single physical frame.

```
pub fn free_frame(addr: PhysAddr) -> Result<(), Status> {
    if unsafe { extern_block::pmm_free_frame(addr.0) } {
        Ok(())
    } else {
        Err(Status::InvalidArgument)
    }
}

/// Frees contiguous physical frames.
pub fn free_frames(addr: PhysAddr, count: usize) -> Result<(), Status> {
    if unsafe { extern_block::pmm_free_frames(addr.0, count) } {
        Ok(())
    }
    // ... (truncated, see source file for more)
```

kernel_workspace/core/gluecore/src/bridge/mm/pmm.rs (line 56) - [public] free_frames

/// Frees contiguous physical frames.

```
pub fn free_frames(addr: PhysAddr, count: usize) -> Result<(), Status> {
    if unsafe { extern_block::pmm_free_frames(addr.0, count) } {
        Ok(())
    } else {
        Err(Status::InvalidArgument)
    }
}
```

kernel_workspace/core/gluecore/src/bridge/mm/vmm.rs (line 14) - [public] alloc

/// Allocates virtual memory and maps it to physical frames.

```
pub fn alloc(bytes: usize, flags: u32) -> Result<VirtAddr, Status> {
    let mut virt: u64 = 0;
    if unsafe { extern_block::vmm_alloc(bytes, flags, &mut virt) } {
        Ok(VirtAddr(virt))
    } else {
        Err(Status::OutOfMemory)
    }
}
```

/// Allocates virtual memory with a specific alignment.

```
pub fn alloc_aligned(bytes: usize, align: usize, flags: u32) -> Result<VirtAddr, Status> {
    let mut virt: u64 = 0;
    // ... (truncated, see source file for more)
```

kernel_workspace/core/gluecore/src/bridge/mm/vmm.rs (line 24) - [public] alloc_aligned

/// Allocates virtual memory with a specific alignment.

```
pub fn alloc_aligned(bytes: usize, align: usize, flags: u32) -> Result<VirtAddr, Status> {
    let mut virt: u64 = 0;
    if unsafe { extern_block::vmm_alloc_aligned(bytes, align, flags, &mut virt) } {
        Ok(VirtAddr(virt))
    } else {
        Err(Status::OutOfMemory)
    }
}
```

/// Frees previously allocated virtual memory.

```
pub fn free(addr: VirtAddr, bytes: usize) -> Result<(), Status> {
    if unsafe { extern_block::vmm_free(addr.0, bytes) } {
        // ... (truncated, see source file for more)
```

kernel_workspace/core/gluecore/src/bridge/mm/vmm.rs (line 34) - [public] free

/// Frees previously allocated virtual memory.

```
pub fn free(addr: VirtAddr, bytes: usize) -> Result<(), Status> {
    if unsafe { extern_block::vmm_free(addr.0, bytes) } {
        Ok(())
    } else {
        Err(Status::InvalidArgument)
    }
}
```

/// Maps a physical device region (MMIO) into the kernel virtual space.

```
pub fn map_device(phys: PhysAddr, len: usize) -> Result<VirtAddr, Status> {
    let mut virt: u64 = 0;
    if unsafe { extern_block::vmm_map_device(phys.0, len, &mut virt) } {
        // ... (truncated, see source file for more)
```

kernel_workspace/core/gluecore/src/bridge/mm/vmm.rs (line 43) - [public] map_device

/// Maps a physical device region (MMIO) into the kernel virtual space.

```
pub fn map_device(phys: PhysAddr, len: usize) -> Result<VirtAddr, Status> {
    let mut virt: u64 = 0;
    if unsafe { extern_block::vmm_map_device(phys.0, len, &mut virt) } {
        Ok(VirtAddr(virt))
    } else {
        Err(Status::BadAddress)
    }
}

/// Unmaps a device region.
pub fn unmap_device(addr: VirtAddr, len: usize) -> Result<(), Status> {
    if unsafe { extern_block::vmm_unmap_device(addr.0, len) } {
        // ... (truncated, see source file for more)
    }
}
```

kernel_workspace/core/gluecore/src/bridge/mm/vmm.rs (line 53) - [public] unmap_device

/// Unmaps a device region.

```
pub fn unmap_device(addr: VirtAddr, len: usize) -> Result<(), Status> {
    if unsafe { extern_block::vmm_unmap_device(addr.0, len) } {
        Ok(())
    } else {
        Err(Status::InvalidArgument)
    }
}

/// Translates a kernel virtual address to a physical address.
pub fn virt_to_phys(vaddr: VirtAddr) -> PhysAddr {
    let ptr = vaddr.0 as *const u8;
    PhysAddr(unsafe { extern_block::kvirt_to_phys(ptr) })
    // ... (truncated, see source file for more)
}
```

kernel_workspace/core/gluecore/src/bridge/mm/vmm.rs (line 62) - [public] virt_to_phys

/// Translates a kernel virtual address to a physical address.

```
pub fn virt_to_phys(vaddr: VirtAddr) -> PhysAddr {
    let ptr = vaddr.0 as *const u8;
    PhysAddr(unsafe { extern_block::kvirt_to_phys(ptr) })
}
```

kernel_workspace/core/gluecore/src/context.rs (line 4) - [internal] paging_read_cr3

```
fn paging_read_cr3() -> u64;
fn paging_write_cr3(pml4_phys: u64);
fn paging_flush_tlb_all();
fn paging_flush_page(addr: u64);
fn context_switch_save_restore(old_ctx: *mut u8, new_ctx: *mut u8);

pub fn read_cr3() -> PhysAddr {
    PhysAddr(unsafe { paging_read_cr3() })
}

pub fn write_cr3(pml4: PhysAddr) {
    // ... (truncated, see source file for more)
}
```

kernel_workspace/core/gluecore/src/context.rs (line 5) - [internal] paging_write_cr3

```
fn paging_write_cr3(pml4_phys: u64);
fn paging_flush_tlb_all();
fn paging_flush_page(addr: u64);
fn context_switch_save_restore(old_ctx: *mut u8, new_ctx: *mut u8);

pub fn read_cr3() -> PhysAddr {
    PhysAddr(unsafe { paging_read_cr3() })
}

pub fn write_cr3(pml4: PhysAddr) {
    unsafe { paging_write_cr3(pml4.0); }
    // ... (truncated, see source file for more)
}
```



```

kernel_Workspace/core/gluecore/src/context.rs (line 6) - [internal] paging_flush_tlb_all
    fn paging_flush_tlb_all();
    fn paging_flush_page(addr: u64);
    fn context_switch_save_restore(old_ctx: *mut u8, new_ctx: *mut u8);
}

pub fn read_cr3() -> PhysAddr {
    PhysAddr(unsafe { paging_read_cr3() })
}

pub fn write_cr3(pml4: PhysAddr) {
    unsafe { paging_write_cr3(pml4.0); }
}

// ... (truncated, see source file for more)

kernel_Workspace/core/gluecore/src/context.rs (line 7) - [internal] paging_flush_page
    fn paging_flush_page(addr: u64);
    fn context_switch_save_restore(old_ctx: *mut u8, new_ctx: *mut u8);
}

pub fn read_cr3() -> PhysAddr {
    PhysAddr(unsafe { paging_read_cr3() })
}

pub fn write_cr3(pml4: PhysAddr) {
    unsafe { paging_write_cr3(pml4.0); }
}

// ... (truncated, see source file for more)

kernel_Workspace/core/gluecore/src/context.rs (line 8) - [internal]
context_switch_save_restore
    fn context_switch_save_restore(old_ctx: *mut u8, new_ctx: *mut u8);
}

pub fn read_cr3() -> PhysAddr {
    PhysAddr(unsafe { paging_read_cr3() })
}

pub fn write_cr3(pml4: PhysAddr) {
    unsafe { paging_write_cr3(pml4.0); }
}

pub fn flush_tlb_all() {
    // ... (truncated, see source file for more)

kernel_Workspace/core/gluecore/src/context.rs (line 11) - [public] read_cr3
pub fn read_cr3() -> PhysAddr {
    PhysAddr(unsafe { paging_read_cr3() })
}

pub fn write_cr3(pml4: PhysAddr) {
    unsafe { paging_write_cr3(pml4.0); }
}

pub fn flush_tlb_all() {
    unsafe { paging_flush_tlb_all(); }
}

// ... (truncated, see source file for more)

```

kernel_workspace/core/gluecore/src/context.rs (line 15) - [public] write_cr3

```
pub fn write_cr3(pml4: PhysAddr) {
    unsafe { paging_write_cr3(pml4.0); }
}

pub fn flush_tlb_all() {
    unsafe { paging_flush_tlb_all(); }
}

pub fn flush_page(addr: VirtAddr) {
    unsafe { paging_flush_page(addr.0); }
}

// ... (truncated, see source file for more)
```

kernel_workspace/core/gluecore/src/context.rs (line 19) - [public] flush_tlb_all

```
pub fn flush_tlb_all() {
    unsafe { paging_flush_tlb_all(); }
}

pub fn flush_page(addr: VirtAddr) {
    unsafe { paging_flush_page(addr.0); }
}

pub unsafe fn switch_context(old_ctx: *mut u8, new_ctx: *mut u8) {
    context_switch_save_restore(old_ctx, new_ctx);
}
```

kernel_workspace/core/gluecore/src/context.rs (line 23) - [public] flush_page

```
pub fn flush_page(addr: VirtAddr) {
    unsafe { paging_flush_page(addr.0); }
}

pub unsafe fn switch_context(old_ctx: *mut u8, new_ctx: *mut u8) {
    context_switch_save_restore(old_ctx, new_ctx);
}
```

kernel_workspace/core/gluecore/src/ffi/c/extern_block.rs (line 8) - [public] pmm_alloc_frame

// --- Physical Memory Manager (PMM) ---

```
pub fn pmm_alloc_frame(out_phys: *mut u64) -> bool;
pub fn pmm_alloc_zero_frame(out_phys: *mut u64) -> bool;
pub fn pmm_alloc_frames(count: usize, out_phys: *mut u64) -> bool;
pub fn pmm_alloc_frames_aligned(count: usize, align_frames: usize, out_phys: *mut u64) -> bool;
pub fn pmm_free_frame(phys: u64) -> bool;
pub fn pmm_free_frames(phys: u64, count: usize) -> bool;

// --- Virtual Memory Manager (VMM) ---
pub fn vmm_alloc(bytes: usize, flags: u32, out_virt: *mut u64) -> bool;
pub fn vmm_alloc_aligned(bytes: usize, align: usize, flags: u32, out_virt: *mut u64) -> bool;
pub fn vmm_free(virt: u64, bytes: usize) -> bool;
pub fn vmm_map_device(phys: u64, len: usize, out_virt: *mut u64) -> bool;
// ... (truncated, see source file for more)
```

kernel_workspace/core/gluecore/src/ffi/c/extern_block.rs (line 9) - [public]

pmm_alloc_zero_frame

```
pub fn pmm_alloc_zero_frame(out_phys: *mut u64) -> bool;
pub fn pmm_alloc_frames(count: usize, out_phys: *mut u64) -> bool;
pub fn pmm_alloc_frames_aligned(count: usize, align_frames: usize, out_phys: *mut u64) -> bool;
pub fn pmm_free_frame(phys: u64) -> bool;
pub fn pmm_free_frames(phys: u64, count: usize) -> bool;

// --- Virtual Memory Manager (VMM) ---
pub fn vmm_alloc(bytes: usize, flags: u32, out_virt: *mut u64) -> bool;
pub fn vmm_alloc_aligned(bytes: usize, align: usize, flags: u32, out_virt: *mut u64) -> bool;
pub fn vmm_free(virt: u64, bytes: usize) -> bool;
pub fn vmm_map_device(phys: u64, len: usize, out_virt: *mut u64) -> bool;
pub fn vmm_unmap_device(virt: u64, len: usize) -> bool;
// ... (truncated, see source file for more)
```

```

kernel_workspace/core/gluecore/src/ffi/c/extern_block.rs (line 10) - [public] pmm_alloc_frames
    pub fn pmm_alloc_frames(count: usize, out_phys: *mut u64) -> bool;
    pub fn pmm_alloc_frames_aligned(count: usize, align_frames: usize, out_phys: *mut u64) -> bool;
    pub fn pmm_free_frame(phys: u64) -> bool;
    pub fn pmm_free_frames(phys: u64, count: usize) -> bool;

    // --- Virtual Memory Manager (VMM) ---
    pub fn vmm_alloc(bytes: usize, flags: u32, out_virt: *mut u64) -> bool;
    pub fn vmm_alloc_aligned(bytes: usize, align: usize, flags: u32, out_virt: *mut u64) -> bool;
    pub fn vmm_free(virt: u64, bytes: usize) -> bool;
    pub fn vmm_map_device(phys: u64, len: usize, out_virt: *mut u64) -> bool;
    pub fn vmm_unmap_device(virt: u64, len: usize) -> bool;

    // ... (truncated, see source file for more)

kernel_workspace/core/gluecore/src/ffi/c/extern_block.rs (line 11) - [public]
pmm_alloc_frames_aligned
    pub fn pmm_alloc_frames_aligned(count: usize, align_frames: usize, out_phys: *mut u64) -> bool;
    pub fn pmm_free_frame(phys: u64) -> bool;
    pub fn pmm_free_frames(phys: u64, count: usize) -> bool;

    // --- Virtual Memory Manager (VMM) ---
    pub fn vmm_alloc(bytes: usize, flags: u32, out_virt: *mut u64) -> bool;
    pub fn vmm_alloc_aligned(bytes: usize, align: usize, flags: u32, out_virt: *mut u64) -> bool;
    pub fn vmm_free(virt: u64, bytes: usize) -> bool;
    pub fn vmm_map_device(phys: u64, len: usize, out_virt: *mut u64) -> bool;
    pub fn vmm_unmap_device(virt: u64, len: usize) -> bool;

    // --- Kernel Heap ---
    // ... (truncated, see source file for more)

kernel_workspace/core/gluecore/src/ffi/c/extern_block.rs (line 12) - [public] pmm_free_frame
    pub fn pmm_free_frame(phys: u64) -> bool;
    pub fn pmm_free_frames(phys: u64, count: usize) -> bool;

    // --- Virtual Memory Manager (VMM) ---
    pub fn vmm_alloc(bytes: usize, flags: u32, out_virt: *mut u64) -> bool;
    pub fn vmm_alloc_aligned(bytes: usize, align: usize, flags: u32, out_virt: *mut u64) -> bool;
    pub fn vmm_free(virt: u64, bytes: usize) -> bool;
    pub fn vmm_map_device(phys: u64, len: usize, out_virt: *mut u64) -> bool;
    pub fn vmm_unmap_device(virt: u64, len: usize) -> bool;

    // --- Kernel Heap ---
    pub fn kmalloc(size: usize) -> *mut u8;
    // ... (truncated, see source file for more)

kernel_workspace/core/gluecore/src/ffi/c/extern_block.rs (line 13) - [public] pmm_free_frames
    pub fn pmm_free_frames(phys: u64, count: usize) -> bool;

    // --- Virtual Memory Manager (VMM) ---
    pub fn vmm_alloc(bytes: usize, flags: u32, out_virt: *mut u64) -> bool;
    pub fn vmm_alloc_aligned(bytes: usize, align: usize, flags: u32, out_virt: *mut u64) -> bool;
    pub fn vmm_free(virt: u64, bytes: usize) -> bool;
    pub fn vmm_map_device(phys: u64, len: usize, out_virt: *mut u64) -> bool;
    pub fn vmm_unmap_device(virt: u64, len: usize) -> bool;

    // --- Kernel Heap ---
    pub fn kmalloc(size: usize) -> *mut u8;
    pub fn kzalloc(size: usize) -> *mut u8;
    // ... (truncated, see source file for more)

```

kernel_Workspace/core/gluecore/src/ffi/c/extern_block.rs (line 16) - [public] vmm_alloc

// --- Virtual Memory Manager (VMM) ---

```
pub fn vmm_alloc(bytes: usize, flags: u32, out_virt: *mut u64) -> bool;
pub fn vmm_alloc_aligned(bytes: usize, align: usize, flags: u32, out_virt: *mut u64) -> bool;
pub fn vmm_free(virt: u64, bytes: usize) -> bool;
pub fn vmm_map_device(phys: u64, len: usize, out_virt: *mut u64) -> bool;
pub fn vmm_unmap_device(virt: u64, len: usize) -> bool;
```

// --- Kernel Heap ---

```
pub fn kmalloc(size: usize) -> *mut u8;
pub fn kzalloc(size: usize) -> *mut u8;
pub fn krealloc(ptr: *mut u8, size: usize) -> *mut u8;
pub fn kfree(ptr: *mut u8);
pub fn kalloc_pages(pages: usize) -> *mut u8;
// ... (truncated, see source file for more)
```

kernel_Workspace/core/gluecore/src/ffi/c/extern_block.rs (line 17) - [public]

vmm_alloc_aligned

```
pub fn vmm_alloc_aligned(bytes: usize, align: usize, flags: u32, out_virt: *mut u64) -> bool;
pub fn vmm_free(virt: u64, bytes: usize) -> bool;
pub fn vmm_map_device(phys: u64, len: usize, out_virt: *mut u64) -> bool;
pub fn vmm_unmap_device(virt: u64, len: usize) -> bool;
```

// --- Kernel Heap ---

```
pub fn kmalloc(size: usize) -> *mut u8;
pub fn kzalloc(size: usize) -> *mut u8;
pub fn krealloc(ptr: *mut u8, size: usize) -> *mut u8;
pub fn kfree(ptr: *mut u8);
pub fn kalloc_pages(pages: usize) -> *mut u8;
pub fn kfree_pages(ptr: *mut u8, pages: usize);
// ... (truncated, see source file for more)
```

kernel_Workspace/core/gluecore/src/ffi/c/extern_block.rs (line 18) - [public] vmm_free

```
pub fn vmm_free(virt: u64, bytes: usize) -> bool;
pub fn vmm_map_device(phys: u64, len: usize, out_virt: *mut u64) -> bool;
pub fn vmm_unmap_device(virt: u64, len: usize) -> bool;
```

// --- Kernel Heap ---

```
pub fn kmalloc(size: usize) -> *mut u8;
pub fn kzalloc(size: usize) -> *mut u8;
pub fn krealloc(ptr: *mut u8, size: usize) -> *mut u8;
pub fn kfree(ptr: *mut u8);
pub fn kalloc_pages(pages: usize) -> *mut u8;
pub fn kfree_pages(ptr: *mut u8, pages: usize);
pub fn kvirt_to_phys(ptr: *const u8) -> u64;
// ... (truncated, see source file for more)
```

kernel_Workspace/core/gluecore/src/ffi/c/extern_block.rs (line 19) - [public] vmm_map_device

```
pub fn vmm_map_device(phys: u64, len: usize, out_virt: *mut u64) -> bool;
pub fn vmm_unmap_device(virt: u64, len: usize) -> bool;
```

// --- Kernel Heap ---

```
pub fn kmalloc(size: usize) -> *mut u8;
pub fn kzalloc(size: usize) -> *mut u8;
pub fn krealloc(ptr: *mut u8, size: usize) -> *mut u8;
pub fn kfree(ptr: *mut u8);
pub fn kalloc_pages(pages: usize) -> *mut u8;
pub fn kfree_pages(ptr: *mut u8, pages: usize);
pub fn kvirt_to_phys(ptr: *const u8) -> u64;
```

// ... (truncated, see source file for more)

kernel_Workspace/core/gluecore/src/ffi/c/extern_block.rs (line 20) - [public] vmm_unmap_device

```
pub fn vmm_unmap_device(virt: u64, len: usize) -> bool;

// --- Kernel Heap ---
pub fn kmalloc(size: usize) -> *mut u8;
pub fn kzalloc(size: usize) -> *mut u8;
pub fn krealloc(ptr: *mut u8, size: usize) -> *mut u8;
pub fn kfree(ptr: *mut u8);
pub fn kalloc_pages(pages: usize) -> *mut u8;
pub fn kfree_pages(ptr: *mut u8, pages: usize);
pub fn kvirt_to_phys(ptr: *const u8) -> u64;

// --- Paging / Address Spaces ---
// ... (truncated, see source file for more)
```

kernel_Workspace/core/gluecore/src/ffi/c/extern_block.rs (line 23) - [public] kmalloc

// --- Kernel Heap ---

```
pub fn kmalloc(size: usize) -> *mut u8;
pub fn kzalloc(size: usize) -> *mut u8;
pub fn krealloc(ptr: *mut u8, size: usize) -> *mut u8;
pub fn kfree(ptr: *mut u8);
pub fn kalloc_pages(pages: usize) -> *mut u8;
pub fn kfree_pages(ptr: *mut u8, pages: usize);
pub fn kvirt_to_phys(ptr: *const u8) -> u64;

// --- Paging / Address Spaces ---
pub fn paging_read_cr3() -> u64;
pub fn paging_write_cr3(pml4_phys: u64);
pub fn paging_map_page(virt: u64, phys: u64, flags: u64) -> bool;
// ... (truncated, see source file for more)
```

kernel_Workspace/core/gluecore/src/ffi/c/extern_block.rs (line 24) - [public] kzalloc

```
pub fn kzalloc(size: usize) -> *mut u8;
pub fn krealloc(ptr: *mut u8, size: usize) -> *mut u8;
pub fn kfree(ptr: *mut u8);
pub fn kalloc_pages(pages: usize) -> *mut u8;
pub fn kfree_pages(ptr: *mut u8, pages: usize);
pub fn kvirt_to_phys(ptr: *const u8) -> u64;

// --- Paging / Address Spaces ---
pub fn paging_read_cr3() -> u64;
pub fn paging_write_cr3(pml4_phys: u64);
pub fn paging_map_page(virt: u64, phys: u64, flags: u64) -> bool;
pub fn paging_unmap_page(virt: u64) -> bool;
// ... (truncated, see source file for more)
```

kernel_Workspace/core/gluecore/src/ffi/c/extern_block.rs (line 25) - [public] krealloc

```
pub fn krealloc(ptr: *mut u8, size: usize) -> *mut u8;
pub fn kfree(ptr: *mut u8);
pub fn kalloc_pages(pages: usize) -> *mut u8;
pub fn kfree_pages(ptr: *mut u8, pages: usize);
pub fn kvirt_to_phys(ptr: *const u8) -> u64;

// --- Paging / Address Spaces ---
pub fn paging_read_cr3() -> u64;
pub fn paging_write_cr3(pml4_phys: u64);
pub fn paging_map_page(virt: u64, phys: u64, flags: u64) -> bool;
pub fn paging_unmap_page(virt: u64) -> bool;
pub fn paging_flush_tlb_all();
// ... (truncated, see source file for more)
```

kernel_Workspace/core/gluecore/src/ffi/c/extern_block.rs (line 26) - [public] kfree

```
pub fn kfree(ptr: *mut u8);
pub fn kalloc_pages(pages: usize) -> *mut u8;
pub fn kfree_pages(ptr: *mut u8, pages: usize);
pub fn kvirt_to_phys(ptr: *const u8) -> u64;

// --- Paging / Address Spaces ---
pub fn paging_read_cr3() -> u64;
pub fn paging_write_cr3(pml4_phys: u64);
pub fn paging_map_page(virt: u64, phys: u64, flags: u64) -> bool;
pub fn paging_unmap_page(virt: u64) -> bool;
pub fn paging_flush_tlb_all();
pub fn paging_flush_page(addr: u64);
// ... (truncated, see source file for more)
```

kernel_Workspace/core/gluecore/src/ffi/c/extern_block.rs (line 27) - [public] kalloc_pages

```
pub fn kalloc_pages(pages: usize) -> *mut u8;
pub fn kfree_pages(ptr: *mut u8, pages: usize);
pub fn kvirt_to_phys(ptr: *const u8) -> u64;

// --- Paging / Address Spaces ---
pub fn paging_read_cr3() -> u64;
pub fn paging_write_cr3(pml4_phys: u64);
pub fn paging_map_page(virt: u64, phys: u64, flags: u64) -> bool;
pub fn paging_unmap_page(virt: u64) -> bool;
pub fn paging_flush_tlb_all();
pub fn paging_flush_page(addr: u64);
}
```

kernel_Workspace/core/gluecore/src/ffi/c/extern_block.rs (line 28) - [public] kfree_pages

```
pub fn kfree_pages(ptr: *mut u8, pages: usize);
pub fn kvirt_to_phys(ptr: *const u8) -> u64;

// --- Paging / Address Spaces ---
pub fn paging_read_cr3() -> u64;
pub fn paging_write_cr3(pml4_phys: u64);
pub fn paging_map_page(virt: u64, phys: u64, flags: u64) -> bool;
pub fn paging_unmap_page(virt: u64) -> bool;
pub fn paging_flush_tlb_all();
pub fn paging_flush_page(addr: u64);
}
```

kernel_Workspace/core/gluecore/src/ffi/c/extern_block.rs (line 29) - [public] kvirt_to_phys

```
pub fn kvirt_to_phys(ptr: *const u8) -> u64;

// --- Paging / Address Spaces ---
pub fn paging_read_cr3() -> u64;
pub fn paging_write_cr3(pml4_phys: u64);
pub fn paging_map_page(virt: u64, phys: u64, flags: u64) -> bool;
pub fn paging_unmap_page(virt: u64) -> bool;
pub fn paging_flush_tlb_all();
pub fn paging_flush_page(addr: u64);
}
```

kernel_Workspace/core/gluecore/src/ffi/c/extern_block.rs (line 32) - [public] paging_read_cr3

```
// --- Paging / Address Spaces ---

pub fn paging_read_cr3() -> u64;
pub fn paging_write_cr3(pml4_phys: u64);
pub fn paging_map_page(virt: u64, phys: u64, flags: u64) -> bool;
pub fn paging_unmap_page(virt: u64) -> bool;
pub fn paging_flush_tlb_all();
pub fn paging_flush_page(addr: u64);
}
```

```

kernel_Workspace/core/gluecore/src/ffi/c/extern_block.rs (line 33) - [public] paging_write_cr3
    pub fn paging_write_cr3(pml4_phys: u64);
    pub fn paging_map_page(virt: u64, phys: u64, flags: u64) -> bool;
    pub fn paging_unmap_page(virt: u64) -> bool;
    pub fn paging_flush_tlb_all();
    pub fn paging_flush_page(addr: u64);
}

kernel_Workspace/core/gluecore/src/ffi/c/extern_block.rs (line 34) - [public] paging_map_page
    pub fn paging_map_page(virt: u64, phys: u64, flags: u64) -> bool;
    pub fn paging_unmap_page(virt: u64) -> bool;
    pub fn paging_flush_tlb_all();
    pub fn paging_flush_page(addr: u64);
}

kernel_Workspace/core/gluecore/src/ffi/c/extern_block.rs (line 35) - [public]
paging_unmap_page
    pub fn paging_unmap_page(virt: u64) -> bool;
    pub fn paging_flush_tlb_all();
    pub fn paging_flush_page(addr: u64);
}

kernel_Workspace/core/gluecore/src/ffi/c/extern_block.rs (line 36) - [public]
paging_flush_tlb_all
    pub fn paging_flush_tlb_all();
    pub fn paging_flush_page(addr: u64);
}

kernel_Workspace/core/gluecore/src/ffi/c/extern_block.rs (line 37) - [public]
paging_flush_page
    pub fn paging_flush_page(addr: u64);
}

kernel_Workspace/core/gluecore/src/interrupts.rs (line 4) - [public] disable
    pub fn disable() {
        unsafe { asm!("cli", options(nomem, nostack)); }
    }

    #[inline(always)]
    pub fn enable() {
        unsafe { asm!("sti", options(nomem, nostack)); }
    }

    #[inline(always)]
    pub fn halt() {
        unsafe { asm!("hlt", options(nomem, nostack)); }
        // ... (truncated, see source file for more)
    }

kernel_Workspace/core/gluecore/src/interrupts.rs (line 9) - [public] enable
    pub fn enable() {
        unsafe { asm!("sti", options(nomem, nostack)); }
    }

    #[inline(always)]
    pub fn halt() {
        unsafe { asm!("hlt", options(nomem, nostack)); }
    }

    #[inline(always)]
    pub fn are_enabled() -> bool {
        let flags: u64;
        // ... (truncated, see source file for more)
    }

```

kernel_Workspace/core/gluecore/src/interrupts.rs (line 14) - [public] halt

```
pub fn halt() {
    unsafe { asm!("hlt", options(nomem, nostack)); }
}

#[inline(always)]
pub fn are_enabled() -> bool {
    let flags: u64;
    unsafe { asm!("pushfq; pop {}", out(reg) flags, options(nomem)); }
    (flags & (1 << 9)) != 0
}

pub struct InterruptGuard {
    // ... (truncated, see source file for more)
```

kernel_Workspace/core/gluecore/src/interrupts.rs (line 19) - [public] are_enabled

```
pub fn are_enabled() -> bool {
    let flags: u64;
    unsafe { asm!("pushfq; pop {}", out(reg) flags, options(nomem)); }
    (flags & (1 << 9)) != 0
}

pub struct InterruptGuard {
    was_enabled: bool,
}

impl InterruptGuard {
    pub fn disable() -> Self {
        // ... (truncated, see source file for more)
```

kernel_Workspace/core/gluecore/src/interrupts.rs (line 25) - [public] InterruptGuard

```
pub struct InterruptGuard {
    was_enabled: bool,
}

impl InterruptGuard {
    pub fn disable() -> Self {
        let was_enabled = are_enabled();
        disable();
        Self { was_enabled }
    }
}

// ... (truncated, see source file for more)
```

kernel_Workspace/core/gluecore/src/interrupts.rs (line 30) - [public] disable

```
pub fn disable() -> Self {
    let was_enabled = are_enabled();
    disable();
    Self { was_enabled }
}

impl Drop for InterruptGuard {
    fn drop(&mut self) {
        if self.was_enabled {
            enable();
        }
        // ... (truncated, see source file for more)
```

kernel_Workspace/core/gluecore/src/interrupts.rs (line 38) - [internal] drop

```
fn drop(&mut self) {
    if self.was_enabled {
        enable();
    }
}
```



```
kernel_Workspace/core/gluecore/src/syscalls.rs (line 4) - [internal] syscall_dispatch
    fn syscall_dispatch(num: u64, arg1: u64, arg2: u64, arg3: u64, arg4: u64) -> i64;
}

pub fn invoke(num: u64, args: [u64; 4]) -> Result<u64, Status> {
    let ret = unsafe {
        syscall_dispatch(num, args[0], args[1], args[2], args[3])
    };

    if ret < 0 {
        Err(Status::from((-ret) as u32))
    } else {
        Ok(ret as u64)
    }
    // ... (truncated, see source file for more)
}
```

```
kernel_Workspace/core/gluecore/src/syscalls.rs (line 7) - [public] invoke
pub fn invoke(num: u64, args: [u64; 4]) -> Result<u64, Status> {
    let ret = unsafe {
        syscall_dispatch(num, args[0], args[1], args[2], args[3])
    };

    if ret < 0 {
        Err(Status::from((-ret) as u32))
    } else {
        Ok(ret as u64)
    }
}

// ... (truncated, see source file for more)
```

```
kernel_Workspace/core/kstd_alloc/src/heap.rs (line 5) - [internal] kmalloc
    fn kmalloc(size: usize) -> *mut u8;
    fn kcalloc(size: usize) -> *mut u8;
    fn kcalloc(n: usize, size: usize) -> *mut u8;
    fn krealloc(ptr: *mut u8, size: usize) -> *mut u8;
    fn kfree(ptr: *mut u8);
    fn kmalloc_aligned(size: usize, align: usize) -> *mut u8;
}

pub struct KernelHeap;

impl Allocator for KernelHeap {
    fn alloc(&self, layout: Layout) -> Result<*mut u8, Status> {
        // ... (truncated, see source file for more)
    }
}
```

```
kernel_Workspace/core/kstd_alloc/src/heap.rs (line 6) - [internal] kcalloc
    fn kcalloc(size: usize) -> *mut u8;
    fn kcalloc(n: usize, size: usize) -> *mut u8;
    fn krealloc(ptr: *mut u8, size: usize) -> *mut u8;
    fn kfree(ptr: *mut u8);
    fn kmalloc_aligned(size: usize, align: usize) -> *mut u8;
}

pub struct KernelHeap;

impl Allocator for KernelHeap {
    fn alloc(&self, layout: Layout) -> Result<*mut u8, Status> {
        let ptr = if layout.align() > 8 {
            // ... (truncated, see source file for more)
        }
    }
}
```

kernel_Workspace/core/kstd_alloc/src/heap.rs (line 7) - [internal] kcalloc

```
fn kcalloc(n: usize, size: usize) -> *mut u8;
fn krealloc(ptr: *mut u8, size: usize) -> *mut u8;
fn kfree(ptr: *mut u8);
fn kmalloc_aligned(size: usize, align: usize) -> *mut u8;
}

pub struct KernelHeap;

impl Allocator for KernelHeap {
    fn alloc(&self, layout: Layout) -> Result<*mut u8, Status> {
        let ptr = if layout.align() > 8 {
            unsafe { kmalloc_aligned(layout.size(), layout.align()) }
        }
        // ... (truncated, see source file for more)
    }
}
```

kernel_Workspace/core/kstd_alloc/src/heap.rs (line 8) - [internal] krealloc

```
fn krealloc(ptr: *mut u8, size: usize) -> *mut u8;
fn kfree(ptr: *mut u8);
fn kmalloc_aligned(size: usize, align: usize) -> *mut u8;
}

pub struct KernelHeap;

impl Allocator for KernelHeap {
    fn alloc(&self, layout: Layout) -> Result<*mut u8, Status> {
        let ptr = if layout.align() > 8 {
            unsafe { kmalloc_aligned(layout.size(), layout.align()) }
        } else {
            unsafe { kmalloc(layout.size()) }
        }
        // ... (truncated, see source file for more)
    }
}
```

kernel_Workspace/core/kstd_alloc/src/heap.rs (line 9) - [internal] kfree

```
fn kfree(ptr: *mut u8);
fn kmalloc_aligned(size: usize, align: usize) -> *mut u8;
}

pub struct KernelHeap;

impl Allocator for KernelHeap {
    fn alloc(&self, layout: Layout) -> Result<*mut u8, Status> {
        let ptr = if layout.align() > 8 {
            unsafe { kmalloc_aligned(layout.size(), layout.align()) }
        } else {
            unsafe { kmalloc(layout.size()) }
        }
        // ... (truncated, see source file for more)
    }
}
```

kernel_Workspace/core/kstd_alloc/src/heap.rs (line 10) - [internal] kmalloc_aligned

```
fn kmalloc_aligned(size: usize, align: usize) -> *mut u8;
}

pub struct KernelHeap;

impl Allocator for KernelHeap {
    fn alloc(&self, layout: Layout) -> Result<*mut u8, Status> {
        let ptr = if layout.align() > 8 {
            unsafe { kmalloc_aligned(layout.size(), layout.align()) }
        } else {
            unsafe { kmalloc(layout.size()) }
        };
        // ... (truncated, see source file for more)
    }
}
```

kernel_Workspace/core/kstd_alloc/src/heap.rs (line 13) - [public] KernelHeap

```
pub struct KernelHeap;

impl Allocator for KernelHeap {
    fn alloc(&self, layout: Layout) -> Result<*mut u8, Status> {
        let ptr = if layout.align() > 8 {
            unsafe { kmalloc_aligned(layout.size(), layout.align()) }
        } else {
            unsafe { kmalloc(layout.size()) }
        };
        if ptr.is_null() {
            Err(Status::OutOfMemory)
        } else {
            // ... (truncated, see source file for more)
        }
    }
}
```

kernel_Workspace/core/kstd_alloc/src/heap.rs (line 16) - [internal] alloc

```
fn alloc(&self, layout: Layout) -> Result<*mut u8, Status> {
    let ptr = if layout.align() > 8 {
        unsafe { kmalloc_aligned(layout.size(), layout.align()) }
    } else {
        unsafe { kmalloc(layout.size()) }
    };
    if ptr.is_null() {
        Err(Status::OutOfMemory)
    } else {
        Ok(ptr)
    }
}
// ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_alloc/src/heap.rs (line 29) - [internal] dealloc

```
fn dealloc(&self, ptr: *mut u8, _layout: Layout) {
    if !ptr.is_null() {
        unsafe { kfree(ptr) };
    }
}

fn realloc(&self, ptr: *mut u8, _old: Layout, new_size: usize) -> Result<*mut u8, Status> {
    let new_ptr = unsafe { krealloc(ptr, new_size) };
    if new_ptr.is_null() {
        Err(Status::OutOfMemory)
    } else {
        Ok(new_ptr)
    }
}
// ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_alloc/src/heap.rs (line 35) - [internal] realloc

```
fn realloc(&self, ptr: *mut u8, _old: Layout, new_size: usize) -> Result<*mut u8, Status> {
    let new_ptr = unsafe { krealloc(ptr, new_size) };
    if new_ptr.is_null() {
        Err(Status::OutOfMemory)
    } else {
        Ok(new_ptr)
    }
}

pub fn alloc_zeroed(size: usize) -> Result<*mut u8, Status> {
    let ptr = unsafe { kzalloc(size) };
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/core/kstd_alloc/src/heap.rs (line 45) - [public] alloc_zeroed

```
pub fn alloc_zeroed(size: usize) -> Result<*mut u8, Status> {
    let ptr = unsafe { kzalloc(size) };
    if ptr.is_null() { Err(Status::OutOfMemory) } else { Ok(ptr) }
}

pub fn alloc_array(n: usize, elem_size: usize) -> Result<*mut u8, Status> {
    let ptr = unsafe { kcalloc(n, elem_size) };
    if ptr.is_null() { Err(Status::OutOfMemory) } else { Ok(ptr) }
}
```

```
kernel_workspace/core/kstd_alloc/src/heap.rs (line 50) - [public] alloc_array
pub fn alloc_array(n: usize, elem_size: usize) -> Result<*mut u8, Status> {
    let ptr = unsafe { kcalloc(n, elem_size) };
    if ptr.is_null() { Err(Status::OutOfMemory) } else { Ok(ptr) }
}
```

```
kernel_workspace/core/kstd_alloc/src/lib.rs (line 15) - [internal] oom
fn oom() -> ! {
    loop {
        core::hint::spin_loop();
    }
}
```

```
pub struct KernelAllocator;

unsafe impl GlobalAlloc for KernelAllocator {
    unsafe fn alloc(&self, layout: CoreLayout) -> *mut u8 {
        let l = Layout::new(layout.size(), layout.align());
        match KernelHeap.alloc(l) {
            // ... (truncated, see source file for more)
        }
    }
}
```

```
kernel_workspace/core/kstd_alloc/src/lib.rs (line 21) - [public] KernelAllocator
pub struct KernelAllocator;

unsafe impl GlobalAlloc for KernelAllocator {
    unsafe fn alloc(&self, layout: CoreLayout) -> *mut u8 {
        let l = Layout::new(layout.size(), layout.align());
        match KernelHeap.alloc(l) {
            Ok(ptr) => ptr,
            Err(_) => oom(),
        }
    }

    unsafe fn dealloc(&self, ptr: *mut u8, layout: CoreLayout) {
        // ... (truncated, see source file for more)
    }
}
```

```
kernel_workspace/core/kstd_alloc/src/omm.rs (line 5) - [public] OomHandler
pub trait OomHandler: Fn() -> ! + Send + Sync {}
impl<T: Fn() -> ! + Send + Sync> OomHandler for T {}

static mut HANDLER: Option<&'static dyn OomHandler> = None;

pub fn set_handler(handler: &'static dyn OomHandler) {
    unsafe { HANDLER = Some(handler) };
}

pub fn invoke() -> ! {
    if OOM_HANDLER_ACTIVE.swap(true, Ordering::SeqCst) {
        loop { core::hint::spin_loop(); }
    }
    // ... (truncated, see source file for more)
}
```

```
kernel_workspace/core/kstd_alloc/src/omm.rs (line 10) - [public] set_handler
pub fn set_handler(handler: &'static dyn OomHandler) {
    unsafe { HANDLER = Some(handler) };
}

pub fn invoke() -> ! {
    if OOM_HANDLER_ACTIVE.swap(true, Ordering::SeqCst) {
        loop { core::hint::spin_loop(); }
    }
    unsafe {
        if let Some(h) = HANDLER {
            h();
        }
    }
    // ... (truncated, see source file for more)
}
```

kernel_workspace/core/kstd_alloc/src/omm.rs (line 14) - [public] invoke

```
pub fn invoke() -> ! {
    if OOM_HANDLER_ACTIVE.swap(true, Ordering::SeqCst) {
        loop { core::hint::spin_loop(); }
    }
    unsafe {
        if let Some(h) = HANDLER {
            h();
        }
    }
    panic!("OOM");
}
```

kernel_workspace/core/kstd_alloc/src/page.rs (line 5) - [internal] kalloc_pages

```
fn kalloc_pages(order: usize) -> *mut u8;
fn kfree_pages(ptr: *mut u8, pages: usize);

pub fn alloc_pages(order: usize) -> Result<VirtAddr, Status> {
    let ptr = unsafe { kalloc_pages(order) };
    if ptr.is_null() {
        Err(Status::OutOfMemory)
    } else {
        Ok(VirtAddr(ptr as u64))
    }
}

// ... (truncated, see source file for more)
```

kernel_workspace/core/kstd_alloc/src/page.rs (line 6) - [internal] kfree_pages

```
fn kfree_pages(ptr: *mut u8, pages: usize);

pub fn alloc_pages(order: usize) -> Result<VirtAddr, Status> {
    let ptr = unsafe { kalloc_pages(order) };
    if ptr.is_null() {
        Err(Status::OutOfMemory)
    } else {
        Ok(VirtAddr(ptr as u64))
    }
}

// ... (truncated, see source file for more)
```

kernel_workspace/core/kstd_alloc/src/page.rs (line 9) - [public] alloc_pages

```
pub fn alloc_pages(order: usize) -> Result<VirtAddr, Status> {
    let ptr = unsafe { kalloc_pages(order) };
    if ptr.is_null() {
        Err(Status::OutOfMemory)
    } else {
        Ok(VirtAddr(ptr as u64))
    }
}

pub fn free_pages(addr: VirtAddr, order: usize) {
    let pages = 1 << order;
    unsafe { kfree_pages(addr.0 as *mut u8, pages) };
    // ... (truncated, see source file for more)
```

kernel_workspace/core/kstd_alloc/src/page.rs (line 18) - [public] free_pages

```
pub fn free_pages(addr: VirtAddr, order: usize) {
    let pages = 1 << order;
    unsafe { kfree_pages(addr.0 as *mut u8, pages) };
}

pub fn alloc_single_page() -> Result<VirtAddr, Status> {
    alloc_pages(0)
}

pub fn free_single_page(addr: VirtAddr) {
    free_pages(addr, 0)
}

// ... (truncated, see source file for more)
```

kernel_workspace/core/kstd_alloc/src/page.rs (line 23) - [public] alloc_single_page

```
pub fn alloc_single_page() -> Result<VirtAddr, Status> {
    alloc_pages(0)
}

pub fn free_single_page(addr: VirtAddr) {
    free_pages(addr, 0)
}

pub fn pages_for_size(size: usize) -> usize {
    (size + PAGE_SIZE - 1) / PAGE_SIZE
}
```

kernel_workspace/core/kstd_alloc/src/page.rs (line 27) - [public] free_single_page

```
pub fn free_single_page(addr: VirtAddr) {
    free_pages(addr, 0)
}

pub fn pages_for_size(size: usize) -> usize {
    (size + PAGE_SIZE - 1) / PAGE_SIZE
}
```

kernel_workspace/core/kstd_alloc/src/page.rs (line 31) - [public] pages_for_size

```
pub fn pages_for_size(size: usize) -> usize {
    (size + PAGE_SIZE - 1) / PAGE_SIZE
}
```

kernel_workspace/core/kstd_alloc/src/slab.rs (line 5) - [internal] slab_alloc

```
fn slab_alloc(cache: *mut u8) -> *mut u8;
fn slab_free(cache: *mut u8, ptr: *mut u8);
fn slab_cache_create(name: *const u8, obj_size: usize, align: usize) -> *mut u8;
fn slab_cache_destroy(cache: *mut u8);
}

pub struct SlabCache {
    inner: *mut u8,
}

unsafe impl Send for SlabCache {}
unsafe impl Sync for SlabCache {}
// ... (truncated, see source file for more)
```

kernel_workspace/core/kstd_alloc/src/slab.rs (line 6) - [internal] slab_free

```
fn slab_free(cache: *mut u8, ptr: *mut u8);
fn slab_cache_create(name: *const u8, obj_size: usize, align: usize) -> *mut u8;
fn slab_cache_destroy(cache: *mut u8);
}

pub struct SlabCache {
    inner: *mut u8,
}

unsafe impl Send for SlabCache {}
unsafe impl Sync for SlabCache {}

// ... (truncated, see source file for more)
```

```

kernel_Workspace/core/kstd_alloc/src/slab.rs (line 7) - [internal] slab_cache_create
    fn slab_cache_create(name: *const u8, obj_size: usize, align: usize) -> *mut u8;
    fn slab_cache_destroy(cache: *mut u8);
}

pub struct SlabCache {
    inner: *mut u8,
}

unsafe impl Send for SlabCache {}
unsafe impl Sync for SlabCache {}

impl SlabCache {
    // ... (truncated, see source file for more)
}

kernel_Workspace/core/kstd_alloc/src/slab.rs (line 8) - [internal] slab_cache_destroy
    fn slab_cache_destroy(cache: *mut u8);
}

pub struct SlabCache {
    inner: *mut u8,
}

unsafe impl Send for SlabCache {}
unsafe impl Sync for SlabCache {}

impl SlabCache {
    pub fn new(name: &str, obj_size: usize, align: usize) -> Result<Self, Status> {
        // ... (truncated, see source file for more)
    }
}

kernel_Workspace/core/kstd_alloc/src/slab.rs (line 11) - [public] SlabCache
pub struct SlabCache {
    inner: *mut u8,
}

unsafe impl Send for SlabCache {}
unsafe impl Sync for SlabCache {}

impl SlabCache {
    pub fn new(name: &str, obj_size: usize, align: usize) -> Result<Self, Status> {
        let c_name = name.as_ptr();
        let cache = unsafe { slab_cache_create(c_name, obj_size, align) };
        if cache.is_null() {
            // ... (truncated, see source file for more)
        }
    }
}

kernel_Workspace/core/kstd_alloc/src/slab.rs (line 19) - [public] new
    pub fn new(name: &str, obj_size: usize, align: usize) -> Result<Self, Status> {
        let c_name = name.as_ptr();
        let cache = unsafe { slab_cache_create(c_name, obj_size, align) };
        if cache.is_null() {
            Err(Status::OutOfMemory)
        } else {
            Ok(Self { inner: cache })
        }
    }
}

impl Allocator for SlabCache {
    // ... (truncated, see source file for more)
}

```

kernel_Workspace/core/kstd_alloc/src/slab.rs (line 31) - [internal] alloc

```
fn alloc(&self, _layout: Layout) -> Result<*mut u8, Status> {
    let ptr = unsafe { slab_alloc(self.inner) };
    if ptr.is_null() { Err(Status::OutOfMemory) } else { Ok(ptr) }
}

fn dealloc(&self, ptr: *mut u8, _layout: Layout) {
    if !ptr.is_null() {
        unsafe { slab_free(self.inner, ptr) };
    }
}

// ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_alloc/src/slab.rs (line 36) - [internal] dealloc

```
fn dealloc(&self, ptr: *mut u8, _layout: Layout) {
    if !ptr.is_null() {
        unsafe { slab_free(self.inner, ptr) };
    }
}

impl Drop for SlabCache {
    fn drop(&mut self) {
        if !self.inner.is_null() {
            unsafe { slab_cache_destroy(self.inner) };
        }
    }
}

// ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_alloc/src/slab.rs (line 44) - [internal] drop

```
fn drop(&mut self) {
    if !self.inner.is_null() {
        unsafe { slab_cache_destroy(self.inner) };
    }
}

}
```

kernel_Workspace/core/kstd_alloc/src/traits.rs (line 3) - [public] Allocator

```
pub trait Allocator {
    fn alloc(&self, layout: Layout) -> Result<*mut u8, Status>;
    fn dealloc(&self, ptr: *mut u8, layout: Layout);
    fn realloc(&self, ptr: *mut u8, old: Layout, new_size: usize) -> Result<*mut u8, Status> {
        let new_ptr = self.alloc(Layout::new(new_size, old.align()))?;
        if !ptr.is_null() {
            unsafe {
                kstd_base::intrinsics::memcpy(new_ptr, ptr, old.size().min(new_size));
            }
            self.dealloc(ptr, old);
        }
        Ok(new_ptr)
    }
}

// ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_alloc/src/traits.rs (line 4) - [internal] alloc

```
fn alloc(&self, layout: Layout) -> Result<*mut u8, Status>;
fn dealloc(&self, ptr: *mut u8, layout: Layout);
fn realloc(&self, ptr: *mut u8, old: Layout, new_size: usize) -> Result<*mut u8, Status> {
    let new_ptr = self.alloc(Layout::new(new_size, old.align()))?;
    if !ptr.is_null() {
        unsafe {
            kstd_base::intrinsics::memcpy(new_ptr, ptr, old.size().min(new_size));
        }
        self.dealloc(ptr, old);
    }
    Ok(new_ptr)
}

// ... (truncated, see source file for more)
```


kernel_Workspace/core/kstd_alloc/src/traits.rs (line 5) - [internal] dealloc

```
fn dealloc(&self, ptr: *mut u8, layout: Layout);
fn realloc(&self, ptr: *mut u8, old: Layout, new_size: usize) -> Result<*mut u8, Status> {
    let new_ptr = self.alloc(Layout::new(new_size, old.align()))?;
    if !ptr.is_null() {
        unsafe {
            kstd_base::intrinsics::memcpy(new_ptr, ptr, old.size().min(new_size));
        }
        self.dealloc(ptr, old);
    }
    Ok(new_ptr)
}
// ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_alloc/src/traits.rs (line 6) - [internal] realloc

```
fn realloc(&self, ptr: *mut u8, old: Layout, new_size: usize) -> Result<*mut u8, Status> {
    let new_ptr = self.alloc(Layout::new(new_size, old.align()))?;
    if !ptr.is_null() {
        unsafe {
            kstd_base::intrinsics::memcpy(new_ptr, ptr, old.size().min(new_size));
        }
        self.dealloc(ptr, old);
    }
    Ok(new_ptr)
}
// ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_alloc/src/traits.rs (line 19) - [public] Layout

```
pub struct Layout {
    size: usize,
    align: usize,
}

impl Layout {
    pub const fn new(size: usize, align: usize) -> Self {
        Self { size, align }
    }

    pub const fn from_size(size: usize) -> Self {
        Self { size, align: 8 }
    }
    // ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_alloc/src/traits.rs (line 41) - [public] padded

```
pub fn padded(&self) -> usize {
    (self.size + self.align - 1) & !(self.align - 1)
}
}
```

kernel_Workspace/core/kstd_base/src/errors.rs (line 3) - [public] Status

```
pub enum Status {
    Success = 0,
    OutOfMemory = 1,
    InvalidArgument = 2,
    PermissionDenied = 3,
    NotFound = 4,
    AlreadyExists = 5,
    Busy = 6,
    TimedOut = 7,
    NotSupported = 8,
    IOError = 9,
    BadAddress = 10,
    // ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_base/src/errors.rs (line 20) - [internal] from

```
fn from(val: u32) -> Self {
    match val {
        0 => Self::Success,
        1 => Self::OutOfMemory,
        2 => Self::InvalidArgument,
        3 => Self::PermissionDenied,
        4 => Self::NotFound,
        5 => Self::AlreadyExists,
        6 => Self::Busy,
        7 => Self::TimedOut,
        8 => Self::NotSupported,
        9 => Self::IOError,
        // ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_base/src/errors.rs (line 40) - [internal] from

```
fn from(val: Status) -> Self {
    val as u32
}

impl Status {
    #[inline]
    pub const fn is_ok(self) -> bool {
        matches!(self, Self::Success)
    }

    #[inline]
    // ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_base/src/intrinsics.rs (line 12) - [public] memory_barrier

```
pub fn memory_barrier() {
    core::sync::atomic::compiler_fence(core::sync::atomic::Ordering::SeqCst);
}

#[inline(always)]
pub unsafe fn memcpy(dst: *mut u8, src: *const u8, n: usize) {
    core::ptr::copy_nonoverlapping(src, dst, n);
}

#[inline(always)]
pub unsafe fn memset(dst: *mut u8, val: u8, n: usize) {
    core::ptr::write_bytes(dst, val, n);
    // ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_base/src/markers.rs (line 3) - [public] SendSync

```
pub struct SendSync<T>(PhantomData<T>);
unsafe impl<T> Send for SendSync<T> {}
unsafe impl<T> Sync for SendSync<T> {}

pub struct Immutable<T>(PhantomData<T>);
impl<T> Immutable<T> {
    pub const fn new() -> Self {
        Self(PhantomData)
    }
}

pub struct Unpin<T>(PhantomData<T>);
// ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_base/src/markers.rs (line 7) - [public] Immutable

```
pub struct Immutable<T>(PhantomData<T>);
impl<T> Immutable<T> {
    pub const fn new() -> Self {
        Self(PhantomData)
    }
}

pub struct Unpin<T>(PhantomData<T>);
impl<T> core::marker::Unpin for Unpin<T> {}
```

kernel_workspace/core/kstd_base/src/markers.rs (line 14) - [public] Unpin

```
pub struct Unpin<T>(PhantomData<T>);
impl<T> core::marker::Unpin for Unpin<T> {}
```

kernel_workspace/core/kstd_base/src/primitives.rs (line 7) - [public] Ring

```
pub enum Ring {
    Kernel = 0,
    Driver = 1,
    User    = 3,
}

#[derive(Debug, Clone, Copy, PartialEq, Eq)]
#[repr(transparent)]
pub struct ProtFlags(pub u32);

impl ProtFlags {
    pub const NONE: Self = Self(0);
    // ... (truncated, see source file for more)
```

kernel_workspace/core/kstd_base/src/primitives.rs (line 15) - [public] ProtFlags

```
pub struct ProtFlags(pub u32);

impl ProtFlags {
    pub const NONE: Self = Self(0);
    pub const READ: Self = Self(1 << 0);
    pub const WRITE: Self = Self(1 << 1);
    pub const EXEC: Self = Self(1 << 2);
    pub const USER: Self = Self(1 << 3);
    pub const DEVICE: Self = Self(1 << 4);

    #[inline]
    pub const fn contains(self, other: Self) -> bool {
        // ... (truncated, see source file for more)
```

kernel_workspace/core/kstd_base/src/primitives.rs (line 39) - [internal] bitor

```
fn bitor(self, rhs: Self) -> Self {
    self.union(rhs)
}

#[derive(Debug, Clone, Copy, PartialEq, Eq)]
#[repr(transparent)]
pub struct MapFlags(pub u32);

impl MapFlags {
    pub const NONE: Self = Self(0);
    pub const FIXED: Self = Self(1 << 0);
    // ... (truncated, see source file for more)
```

kernel_workspace/core/kstd_base/src/primitives.rs (line 46) - [public] MapFlags

```
pub struct MapFlags(pub u32);

impl MapFlags {
    pub const NONE: Self = Self(0);
    pub const FIXED: Self = Self(1 << 0);
    pub const ANONYMOUS: Self = Self(1 << 1);
    pub const SHARED: Self = Self(1 << 2);
    pub const PRIVATE: Self = Self(1 << 3);
}

#[derive(Debug, Clone, Copy, PartialEq, Eq)]
#[repr(transparent)]
// ... (truncated, see source file for more)
```

kernel_workspace/core/kstd_base/src/primitives.rs (line 58) - [public] IrqFlags

```
pub struct IrqFlags(pub u32);

impl IrqFlags {
    pub const NONE: Self = Self(0);
    pub const LEVEL_TRIGGERED: Self = Self(1 << 0);
    pub const ACTIVE_LOW: Self = Self(1 << 1);
    pub const SHARED: Self = Self(1 << 2);
}
```

kernel_workspace/core/kstd_base/src/trails.rs (line 3) - [public] PageAlign

```
pub trait PageAlign {
    fn align_up(self) -> Self;
    fn align_down(self) -> Self;
    fn is_aligned(self) -> bool;
}

impl PageAlign for u64 {
    #[inline]
    fn align_up(self) -> Self {
        (self + PAGE_SIZE as u64 - 1) & !(PAGE_SIZE as u64 - 1)
    }

    // ... (truncated, see source file for more)
}
```

kernel_workspace/core/kstd_base/src/trails.rs (line 4) - [internal] align_up

```
fn align_up(self) -> Self;
fn align_down(self) -> Self;
fn is_aligned(self) -> bool;
}

impl PageAlign for u64 {
    #[inline]
    fn align_up(self) -> Self {
        (self + PAGE_SIZE as u64 - 1) & !(PAGE_SIZE as u64 - 1)
    }

    #[inline]
    // ... (truncated, see source file for more)
}
```

kernel_workspace/core/kstd_base/src/trails.rs (line 5) - [internal] align_down

```
fn align_down(self) -> Self;
fn is_aligned(self) -> bool;
}

impl PageAlign for u64 {
    #[inline]
    fn align_up(self) -> Self {
        (self + PAGE_SIZE as u64 - 1) & !(PAGE_SIZE as u64 - 1)
    }

    #[inline]
    fn align_down(self) -> Self {
        // ... (truncated, see source file for more)
    }
}
```

kernel_workspace/core/kstd_base/src/trails.rs (line 6) - [internal] is_aligned

```
fn is_aligned(self) -> bool;
}

impl PageAlign for u64 {
    #[inline]
    fn align_up(self) -> Self {
        (self + PAGE_SIZE as u64 - 1) & !(PAGE_SIZE as u64 - 1)
    }

    #[inline]
    fn align_down(self) -> Self {
        self & !(PAGE_SIZE as u64 - 1)
    }
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/core/kstd_base/src/trails.rs (line 11) - [internal] align_up

```
fn align_up(self) -> Self {
    (self + PAGE_SIZE as u64 - 1) & !(PAGE_SIZE as u64 - 1)
}

#[inline]
fn align_down(self) -> Self {
    self & !(PAGE_SIZE as u64 - 1)
}

#[inline]
fn is_aligned(self) -> bool {
    self & (PAGE_SIZE as u64 - 1) == 0
}
// ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_base/src/trails.rs (line 16) - [internal] align_down

```
fn align_down(self) -> Self {
    self & !(PAGE_SIZE as u64 - 1)
}

#[inline]
fn is_aligned(self) -> bool {
    self & (PAGE_SIZE as u64 - 1) == 0
}
}

impl PageAlign for usize {
    #[inline]
    // ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_base/src/trails.rs (line 21) - [internal] is_aligned

```
fn is_aligned(self) -> bool {
    self & (PAGE_SIZE as u64 - 1) == 0
}
}

impl PageAlign for usize {
    #[inline]
    fn align_up(self) -> Self {
        (self + PAGE_SIZE - 1) & !(PAGE_SIZE - 1)
    }

    #[inline]
    // ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_base/src/trails.rs (line 28) - [internal] align_up

```
fn align_up(self) -> Self {
    (self + PAGE_SIZE - 1) & !(PAGE_SIZE - 1)
}

#[inline]
fn align_down(self) -> Self {
    self & !(PAGE_SIZE - 1)
}

#[inline]
fn is_aligned(self) -> bool {
    self & (PAGE_SIZE - 1) == 0
}
// ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_base/src/trails.rs (line 33) - [internal] align_down

```
fn align_down(self) -> Self {
    self & !(PAGE_SIZE - 1)
}

#[inline]
fn is_aligned(self) -> bool {
    self & (PAGE_SIZE - 1) == 0
}

pub trait IoPort {
    fn read_u8(&self, port: u16) -> u8;
    // ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_base/src/trails.rs (line 38) - [internal] is_aligned

```
fn is_aligned(self) -> bool {
    self & (PAGE_SIZE - 1) == 0
}

pub trait IoPort {
    fn read_u8(&self, port: u16) -> u8;
    fn write_u8(&self, port: u16, val: u8);
    fn read_u16(&self, port: u16) -> u16;
    fn write_u16(&self, port: u16, val: u16);
    fn read_u32(&self, port: u16) -> u32;
    fn write_u32(&self, port: u16, val: u32);
    // ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_base/src/trails.rs (line 43) - [public] IoPort

```
pub trait IoPort {
    fn read_u8(&self, port: u16) -> u8;
    fn write_u8(&self, port: u16, val: u8);
    fn read_u16(&self, port: u16) -> u16;
    fn write_u16(&self, port: u16, val: u16);
    fn read_u32(&self, port: u16) -> u32;
    fn write_u32(&self, port: u16, val: u32);
}

pub trait MmioRegion {
    fn read_volatile_u32(&self, offset: usize) -> u32;
    fn write_volatile_u32(&self, offset: usize, val: u32);
    // ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_base/src/trails.rs (line 44) - [internal] read_u8

```
fn read_u8(&self, port: u16) -> u8;
fn write_u8(&self, port: u16, val: u8);
fn read_u16(&self, port: u16) -> u16;
fn write_u16(&self, port: u16, val: u16);
fn read_u32(&self, port: u16) -> u32;
fn write_u32(&self, port: u16, val: u32);
}

pub trait MmioRegion {
    fn read_volatile_u32(&self, offset: usize) -> u32;
    fn write_volatile_u32(&self, offset: usize, val: u32);
    fn read_volatile_u64(&self, offset: usize) -> u64;
    // ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_base/src/trails.rs (line 45) - [internal] write_u8

```
fn write_u8(&self, port: u16, val: u8);
fn read_u16(&self, port: u16) -> u16;
fn write_u16(&self, port: u16, val: u16);
fn read_u32(&self, port: u16) -> u32;
fn write_u32(&self, port: u16, val: u32);
}

pub trait MmioRegion {
fn read_volatile_u32(&self, offset: usize) -> u32;
fn write_volatile_u32(&self, offset: usize, val: u32);
fn read_volatile_u64(&self, offset: usize) -> u64;
fn write_volatile_u64(&self, offset: usize, val: u64);
// ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_base/src/trails.rs (line 46) - [internal] read_u16

```
fn read_u16(&self, port: u16) -> u16;
fn write_u16(&self, port: u16, val: u16);
fn read_u32(&self, port: u16) -> u32;
fn write_u32(&self, port: u16, val: u32);
}

pub trait MmioRegion {
fn read_volatile_u32(&self, offset: usize) -> u32;
fn write_volatile_u32(&self, offset: usize, val: u32);
fn read_volatile_u64(&self, offset: usize) -> u64;
fn write_volatile_u64(&self, offset: usize, val: u64);
}
```

kernel_Workspace/core/kstd_base/src/trails.rs (line 47) - [internal] write_u16

```
fn write_u16(&self, port: u16, val: u16);
fn read_u32(&self, port: u16) -> u32;
fn write_u32(&self, port: u16, val: u32);
}

pub trait MmioRegion {
fn read_volatile_u32(&self, offset: usize) -> u32;
fn write_volatile_u32(&self, offset: usize, val: u32);
fn read_volatile_u64(&self, offset: usize) -> u64;
fn write_volatile_u64(&self, offset: usize, val: u64);
}
```

kernel_Workspace/core/kstd_base/src/trails.rs (line 48) - [internal] read_u32

```
fn read_u32(&self, port: u16) -> u32;
fn write_u32(&self, port: u16, val: u32);
}

pub trait MmioRegion {
fn read_volatile_u32(&self, offset: usize) -> u32;
fn write_volatile_u32(&self, offset: usize, val: u32);
fn read_volatile_u64(&self, offset: usize) -> u64;
fn write_volatile_u64(&self, offset: usize, val: u64);
}
```

kernel_Workspace/core/kstd_base/src/trails.rs (line 49) - [internal] write_u32

```
fn write_u32(&self, port: u16, val: u32);
}

pub trait MmioRegion {
fn read_volatile_u32(&self, offset: usize) -> u32;
fn write_volatile_u32(&self, offset: usize, val: u32);
fn read_volatile_u64(&self, offset: usize) -> u64;
fn write_volatile_u64(&self, offset: usize, val: u64);
}
```

kernel_workspace/core/kstd_base/src/trails.rs (line 52) - [public] MmioRegion

```
pub trait MmioRegion {
    fn read_volatile_u32(&self, offset: usize) -> u32;
    fn write_volatile_u32(&self, offset: usize, val: u32);
    fn read_volatile_u64(&self, offset: usize) -> u64;
    fn write_volatile_u64(&self, offset: usize, val: u64);
}
```

kernel_workspace/core/kstd_base/src/trails.rs (line 53) - [internal] read_volatile_u32

```
fn read_volatile_u32(&self, offset: usize) -> u32;
fn write_volatile_u32(&self, offset: usize, val: u32);
fn read_volatile_u64(&self, offset: usize) -> u64;
fn write_volatile_u64(&self, offset: usize, val: u64);
}
```

kernel_workspace/core/kstd_base/src/trails.rs (line 54) - [internal] write_volatile_u32

```
fn write_volatile_u32(&self, offset: usize, val: u32);
fn read_volatile_u64(&self, offset: usize) -> u64;
fn write_volatile_u64(&self, offset: usize, val: u64);
}
```

kernel_workspace/core/kstd_base/src/trails.rs (line 55) - [internal] read_volatile_u64

```
fn read_volatile_u64(&self, offset: usize) -> u64;
fn write_volatile_u64(&self, offset: usize, val: u64);
}
```

kernel_workspace/core/kstd_base/src/trails.rs (line 56) - [internal] write_volatile_u64

```
fn write_volatile_u64(&self, offset: usize, val: u64);
}
```

kernel_workspace/core/kstd_base/src/types.rs (line 3) - [public] PhysAddr

```
pub struct PhysAddr(pub u64);

#[derive(Debug, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Default)]
#[repr(transparent)]
pub struct VirtAddr(pub u64);

#[derive(Debug, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash)]
#[repr(transparent)]
pub struct Handle(pub u32);

impl Handle {
    pub const INVALID: Self = Self(u32::MAX);
    // ... (truncated, see source file for more)
}
```

kernel_workspace/core/kstd_base/src/types.rs (line 7) - [public] VirtAddr

```
pub struct VirtAddr(pub u64);

#[derive(Debug, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash)]
#[repr(transparent)]
pub struct Handle(pub u32);

impl Handle {
    pub const INVALID: Self = Self(u32::MAX);

    #[inline]
    pub const fn is_valid(self) -> bool {
        self.0 != Self::INVALID.0
    }
    // ... (truncated, see source file for more)
}
```


kernel_Workspace/core/kstd_base/src/types.rs (line 11) - [public] Handle

```
pub struct Handle(pub u32);

impl Handle {
    pub const INVALID: Self = Self(u32::MAX);

    #[inline]
    pub const fn is_valid(self) -> bool {
        self.0 != Self::INVALID.0
    }
}

impl Default for Handle {
    // ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_base/src/types.rs (line 23) - [internal] default

```
fn default() -> Self {
    Self::INVALID
}

#[derive(Debug, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Default)]
#[repr(transparent)]
pub struct CapId(pub u64);

#[derive(Debug, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Default)]
#[repr(transparent)]
pub struct WorldId(pub u32);
// ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_base/src/types.rs (line 30) - [public] CapId

```
pub struct CapId(pub u64);

#[derive(Debug, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Default)]
#[repr(transparent)]
pub struct WorldId(pub u32);

#[derive(Debug, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Default)]
#[repr(transparent)]
pub struct TaskId(pub u64);

#[derive(Debug, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Default)]
#[repr(transparent)]
// ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_base/src/types.rs (line 34) - [public] WorldId

```
pub struct WorldId(pub u32);

#[derive(Debug, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Default)]
#[repr(transparent)]
pub struct TaskId(pub u64);

#[derive(Debug, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Default)]
#[repr(transparent)]
pub struct Size(pub usize);

impl PhysAddr {
    #[inline]
    // ... (truncated, see source file for more)
```

kernel_workspace/core/kstd_base/src/types.rs (line 38) - [public] TaskId

```
pub struct TaskId(pub u64);

#[derive(Debug, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Default)]
#[repr(transparent)]
pub struct Size(pub usize);

impl PhysAddr {
    #[inline]
    pub const fn is_aligned_to(self, align: u64) -> bool {
        self.0 & (align - 1) == 0
    }

    // ... (truncated, see source file for more)
```

kernel_workspace/core/kstd_base/src/types.rs (line 42) - [public] Size

```
pub struct Size(pub usize);

impl PhysAddr {
    #[inline]
    pub const fn is_aligned_to(self, align: u64) -> bool {
        self.0 & (align - 1) == 0
    }

    #[inline]
    pub const fn page_offset(self) -> u64 {
        self.0 & 0xFFF
    }

    // ... (truncated, see source file for more)
```

kernel_workspace/core/kstd_core/src/panic.rs (line 5) - [internal] write_all

```
fn write_all(serial: &mut kstd_io::Serial, vga: &mut kstd_io::VgaConsole, s: &str) {
    let _ = serial.write_str(s);
    let _ = vga.write_str(s);
}
```

```
pub fn panic(info: &PanicInfo) -> ! {
    let mut serial = kstd_io::Serial;
    let mut vga = kstd_io::VgaConsole;

    write_all(&mut serial, &mut vga, "\n!!! KERNEL PANIC !!!\n");

    // ... (truncated, see source file for more)
```

kernel_workspace/core/kstd_core/src/panic.rs (line 11) - [public] panic

```
pub fn panic(info: &PanicInfo) -> ! {
    let mut serial = kstd_io::Serial;
    let mut vga = kstd_io::VgaConsole;

    write_all(&mut serial, &mut vga, "\n!!! KERNEL PANIC !!!\n");

    if let Some(location) = info.location() {
        let _ = core::write!(FmtWriter(&mut serial), "At {}:{}:{}\n", location.file(), location.line(), 1);
        let _ = core::write!(FmtWriter(&mut vga), "At {}:{}:{}\n", location.file(), location.line(), loca
    }

    {
        // ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_core/src/panic.rs (line 34) - [internal] FmtWriter

```
struct FmtWriter<'a, W: Write>(&'a mut W);

impl<'a, W: Write> core::fmt::Write for FmtWriter<'a, W> {
    fn write_str(&mut self, s: &str) -> core::fmt::Result {
        let _ = self.0.write_str(s);
        Ok(())
    }
}

// Render the panic message into a fixed buffer without using alloc.
// Note: kstd_io::Serial already truncates to 4096 bytes internally, we pass &str directly.
fn heapless_msg(message: core::panic::PanicMessage) -> kstd_data::string::String {
    // ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_core/src/panic.rs (line 37) - [internal] write_str

```
fn write_str(&mut self, s: &str) -> core::fmt::Result {
    let _ = self.0.write_str(s);
    Ok(())
}

// Render the panic message into a fixed buffer without using alloc.
// Note: kstd_io::Serial already truncates to 4096 bytes internally, we pass &str directly.
fn heapless_msg(message: core::panic::PanicMessage) -> kstd_data::string::String {
    use core::fmt::Write as _;
    let mut s = kstd_data::string::String::new();
    let _ = core::write!(FmtMsg(&mut s), "{}", message);
    // ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_core/src/panic.rs (line 45) - [internal] heapless_msg

// Render the panic message into a fixed buffer without using alloc.

// Note: kstd_io::Serial already truncates to 4096 bytes internally, we pass &str directly.

```
fn heapless_msg(message: core::panic::PanicMessage) -> kstd_data::string::String {
    use core::fmt::Write as _;
    let mut s = kstd_data::string::String::new();
    let _ = core::write!(FmtMsg(&mut s), "{}", message);
    s
}

struct FmtMsg<'a>(&'a mut kstd_data::string::String);

impl<'a> core::fmt::Write for FmtMsg<'a> {
    fn write_str(&mut self, s: &str) -> core::fmt::Result {
        self.0.push_str(s);
        // ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_core/src/panic.rs (line 52) - [internal] FmtMsg

```
struct FmtMsg<'a>(&'a mut kstd_data::string::String);

impl<'a> core::fmt::Write for FmtMsg<'a> {
    fn write_str(&mut self, s: &str) -> core::fmt::Result {
        self.0.push_str(s);
        Ok(())
    }
}
```

kernel_Workspace/core/kstd_core/src/panic.rs (line 55) - [internal] write_str

```
fn write_str(&mut self, s: &str) -> core::fmt::Result {
    self.0.push_str(s);
    Ok(())
}
```

kernel_workspace/core/kstd_core/src/sync.rs (line 5) - [public] SpinLock

```
pub struct SpinLock<T> {
    locked: AtomicBool,
    data: UnsafeCell<T>,
}

unsafe impl<T: Send> Sync for SpinLock<T> {}
unsafe impl<T: Send> Send for SpinLock<T> {}

impl<T> SpinLock<T> {
    pub const fn new(data: T) -> Self {
        Self {
            locked: AtomicBool::new(false),
            // ... (truncated, see source file for more)
        }
    }
}
```

kernel_workspace/core/kstd_core/src/sync.rs (line 21) - [public] lock

```
pub fn lock(&self) -> SpinLockGuard<'_, T> {
    while self.locked.compare_exchange_weak(false, true, Ordering::Acquire, Ordering::Relaxed).is_err() {
        while self.locked.load(Ordering::Relaxed) {
            core::hint::spin_loop();
        }
    }
    SpinLockGuard { lock: self }
}

pub fn try_lock(&self) -> Option<SpinLockGuard<'_, T>> {
    if self.locked.compare_exchange(false, true, Ordering::Acquire, Ordering::Relaxed).is_ok() {
        Some(SpinLockGuard { lock: self })
    }
    // ... (truncated, see source file for more)
}
```

kernel_workspace/core/kstd_core/src/sync.rs (line 30) - [public] try_lock

```
pub fn try_lock(&self) -> Option<SpinLockGuard<'_, T>> {
    if self.locked.compare_exchange(false, true, Ordering::Acquire, Ordering::Relaxed).is_ok() {
        Some(SpinLockGuard { lock: self })
    } else {
        None
    }
}

pub struct SpinLockGuard<'a, T> {
    lock: &'a SpinLock<T>,
}

// ... (truncated, see source file for more)
```

kernel_workspace/core/kstd_core/src/sync.rs (line 39) - [public] SpinLockGuard

```
pub struct SpinLockGuard<'a, T> {
    lock: &'a SpinLock<T>,
}

impl<T> Deref for SpinLockGuard<'_, T> {
    type Target = T;
    fn deref(&self) -> &T {
        unsafe { &*self.lock.data.get() }
    }
}

impl<T> DerefMut for SpinLockGuard<'_, T> {
    // ... (truncated, see source file for more)
}
```

kernel_Workspace/core/kstd_core/src/sync.rs (line 45) - [internal] deref

```
    fn deref(&self) -> &T {
        unsafe { &*self.lock.data.get() }
    }
}

impl<T> DerefMut for SpinLockGuard<'_, T> {
    fn deref_mut(&mut self) -> &mut T {
        unsafe { &mut *self.lock.data.get() }
    }
}

impl<T> Drop for SpinLockGuard<'_, T> {
    // ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_core/src/sync.rs (line 51) - [internal] deref_mut

```
    fn deref_mut(&mut self) -> &mut T {
        unsafe { &mut *self.lock.data.get() }
    }
}

impl<T> Drop for SpinLockGuard<'_, T> {
    fn drop(&mut self) {
        self.lock.locked.store(false, Ordering::Release);
    }
}
```

kernel_Workspace/core/kstd_core/src/sync.rs (line 57) - [internal] drop

```
    fn drop(&mut self) {
        self.lock.locked.store(false, Ordering::Release);
    }
}
```

kernel_Workspace/core/kstd_core/src/task.rs (line 4) - [internal] sched_create_task

```
    fn sched_create_task(world_id: u32, flags: u32, out_task_id: *mut u64) -> bool;
    fn sched_destroy_task(task_id: u64) -> bool;
    fn sched_yield() -> bool;
    fn sched_get_current_task() -> u64;
}

pub struct Task {
    pub id: TaskId,
    pub world: WorldId,
}

impl Task {
    // ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_core/src/task.rs (line 5) - [internal] sched_destroy_task

```
    fn sched_destroy_task(task_id: u64) -> bool;
    fn sched_yield() -> bool;
    fn sched_get_current_task() -> u64;
}

pub struct Task {
    pub id: TaskId,
    pub world: WorldId,
}

impl Task {
    pub fn create(world: WorldId, flags: u32) -> Result<Self, Status> {
        // ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_core/src/task.rs (line 6) - [internal] sched_yield

```
fn sched_yield() -> bool;
fn sched_get_current_task() -> u64;
}

pub struct Task {
    pub id: TaskId,
    pub world: WorldId,
}

impl Task {
    pub fn create(world: WorldId, flags: u32) -> Result<Self, Status> {
        let mut id: u64 = 0;
        // ... (truncated, see source file for more)
    }
}
```

kernel_Workspace/core/kstd_core/src/task.rs (line 7) - [internal] sched_get_current_task

```
fn sched_get_current_task() -> u64;
}

pub struct Task {
    pub id: TaskId,
    pub world: WorldId,
}

impl Task {
    pub fn create(world: WorldId, flags: u32) -> Result<Self, Status> {
        let mut id: u64 = 0;
        if unsafe { sched_create_task(world.0, flags, &mut id) } {
            Ok(Self { id: TaskId(id), world })
        } else {
            Err(Status::OutOfMemory)
        }
        // ... (truncated, see source file for more)
    }
}
```

kernel_Workspace/core/kstd_core/src/task.rs (line 10) - [public] Task

```
pub struct Task {
    pub id: TaskId,
    pub world: WorldId,
}

impl Task {
    pub fn create(world: WorldId, flags: u32) -> Result<Self, Status> {
        let mut id: u64 = 0;
        if unsafe { sched_create_task(world.0, flags, &mut id) } {
            Ok(Self { id: TaskId(id), world })
        } else {
            Err(Status::OutOfMemory)
        }
        // ... (truncated, see source file for more)
    }
}
```

kernel_Workspace/core/kstd_core/src/task.rs (line 16) - [public] create

```
pub fn create(world: WorldId, flags: u32) -> Result<Self, Status> {
    let mut id: u64 = 0;
    if unsafe { sched_create_task(world.0, flags, &mut id) } {
        Ok(Self { id: TaskId(id), world })
    } else {
        Err(Status::OutOfMemory)
    }
}

pub fn destroy(&self) -> Result<(), Status> {
    if unsafe { sched_destroy_task(self.id.0) } {
        Ok(())
    }
    // ... (truncated, see source file for more)
}
```

kernel_workspace/core/kstd_core/src/task.rs (line 25) - [public] destroy

```
pub fn destroy(&self) -> Result<(), Status> {
    if unsafe { sched_destroy_task(self.id.0) } {
        Ok(())
    } else {
        Err(Status::InvalidArgument)
    }
}

pub fn current() -> TaskId {
    TaskId(unsafe { sched_get_current_task() })
}

// ... (truncated, see source file for more)
```

kernel_workspace/core/kstd_core/src/task.rs (line 33) - [public] current

```
pub fn current() -> TaskId {
    TaskId(unsafe { sched_get_current_task() })
}

pub fn yield_now() {
    unsafe { sched_yield(); }
}
}
```

kernel_workspace/core/kstd_core/src/task.rs (line 37) - [public] yield_now

```
pub fn yield_now() {
    unsafe { sched_yield(); }
}
}
```

kernel_workspace/core/kstd_core/src/thread.rs (line 4) - [internal] thread_create

```
fn thread_create(task_id: u64, entry: u64, stack_ptr: u64, arg: u64, out_tid: *mut u64) -> bool;
fn thread_exit() -> !;
fn thread_get_current_id() -> u64;
}

pub struct Thread {
    pub id: u64,
    pub task: TaskId,
}

impl Thread {
    pub fn spawn(task: TaskId, entry: u64, stack_ptr: u64, arg: u64) -> Result<Self, Status> {
        // ... (truncated, see source file for more)
    }
}
```

kernel_workspace/core/kstd_core/src/thread.rs (line 5) - [internal] thread_exit

```
fn thread_exit() -> !;
fn thread_get_current_id() -> u64;
}

pub struct Thread {
    pub id: u64,
    pub task: TaskId,
}

impl Thread {
    pub fn spawn(task: TaskId, entry: u64, stack_ptr: u64, arg: u64) -> Result<Self, Status> {
        let mut tid: u64 = 0;
        // ... (truncated, see source file for more)
    }
}
```

kernel_Workspace/core/kstd_core/src/thread.rs (line 6) - [internal] thread_get_current_id

```
fn thread_get_current_id() -> u64;
}

pub struct Thread {
    pub id: u64,
    pub task: TaskId,
}

impl Thread {
    pub fn spawn(task: TaskId, entry: u64, stack_ptr: u64, arg: u64) -> Result<Self, Status> {
        let mut tid: u64 = 0;
        if unsafe { thread_create(task.0, entry, stack_ptr, arg, &mut tid) } {
            // ... (truncated, see source file for more)
        }
    }
}
```

kernel_Workspace/core/kstd_core/src/thread.rs (line 9) - [public] Thread

```
pub struct Thread {
    pub id: u64,
    pub task: TaskId,
}

impl Thread {
    pub fn spawn(task: TaskId, entry: u64, stack_ptr: u64, arg: u64) -> Result<Self, Status> {
        let mut tid: u64 = 0;
        if unsafe { thread_create(task.0, entry, stack_ptr, arg, &mut tid) } {
            Ok(Self { id: tid, task })
        } else {
            Err(Status::OutOfMemory)
        }
        // ... (truncated, see source file for more)
    }
}
```

kernel_Workspace/core/kstd_core/src/thread.rs (line 15) - [public] spawn

```
pub fn spawn(task: TaskId, entry: u64, stack_ptr: u64, arg: u64) -> Result<Self, Status> {
    let mut tid: u64 = 0;
    if unsafe { thread_create(task.0, entry, stack_ptr, arg, &mut tid) } {
        Ok(Self { id: tid, task })
    } else {
        Err(Status::OutOfMemory)
    }
}

pub fn exit() -> ! {
    unsafe { thread_exit(); }
}
// ... (truncated, see source file for more)
```

kernel_Workspace/core/kstd_core/src/thread.rs (line 24) - [public] exit

```
pub fn exit() -> ! {
    unsafe { thread_exit(); }
}

pub fn current_id() -> u64 {
    unsafe { thread_get_current_id() }
}
}
```

kernel_Workspace/core/kstd_core/src/thread.rs (line 28) - [public] current_id

```
pub fn current_id() -> u64 {
    unsafe { thread_get_current_id() }
}
}
```

(List truncated to 4000 entries to keep the document size manageable.)